

eRules

AIRCREW

(IR + AMC & GM)



Easy Access Rules for Aircrew (Regulation (EU) No 1178/2011)

Published November 2025¹

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This document is issued by the European Union Aviation Safety Agency (referred to as both 'EASA' and 'the Agency') to provide its stakeholders with an updated, consolidated, and easy-to-read publication. It has been prepared by putting together the officially published EU regulations with the related EASA acceptable means of compliance (AMC) and guidance material (GM) (including their amendments) adopted so far. However, this document is not an official publication, and EASA accepts no liability for damage of any kind resulting from the risks inherent in its use.

LIST OF REVISIONS

Published	Reason for revision
May 2019	First Easy Access Rules document powered by eRules.
June 2020	To incorporate: — Commission Implementing Regulation (EU) 2019/1747 as regards requirements for certain flight crew licences and certificates, rules on training organisations and competent authorities; — the <u>applicable content</u> from Commission Implementing Regulation (EU) 2020/359 laying down technical requirements and administrative procedures related to civil aviation aircrew; and — ED Decisions 2019/017/R and 2020/005/R. To incorporate Commission Delegated Regulation (EU) 2020/723 with regard to the acceptance of third-country certification of pilots.
August 2020	To incorporate information on deferred applicability date, omitted table of Airship theoretical knowledge syllabus, and additional corrections in Part-FCL.
December 2021	To incorporate: — Commission Implementing Regulation (EU) 2020/359 as regards basic instrument rating (BIR); — Commission Implementing Regulation (EU) 2020/2193 as regards the requirements for flight crew competence and training methods, and the reporting, analysis and follow-up of occurrences in civil aviation; and — ED Decision 2020/018/R amending the Acceptable Means of Compliance (AMC) and Guidance Material (GM) to Part-FCL and Part-ARA on ‘Basic instrument rating’.
February 2022	To incorporate Commission Implementing Regulation (EU) 2021/2227 amending Regulation (EU) No 1178/2011 as regards the requirements for all-weather operations and for instrument and type rating training in helicopters, as well as ED Decision 2021/002/R amending the AMC & GM to Part-FCL and Part-ARA on evidence-based training (EBT).
August 2023	To incorporate: — ED Decision 2022/014/R updating the AMC & GM to Subpart ORO.FC with regard to a review of crew training provisions; — Commission Implementing Regulation (EU) 2023/203 laying down rules for the application of Regulation (EU) 2018/1139 of the European Parliament and of the Council, as regards requirements for the management of information security risks with a potential impact on aviation safety for organisations covered by Commission Regulation (EU) No 1178/2011, and for competent authorities covered by Commission Regulation (EU) No 1178/2011, and amending Commission Regulation (EU) No 1178/2011; and — ED Decision 2023/010/R ‘Management of information security risks’ that provides for acceptable means of compliance and guidance material to support the Part-IS regulatory package implementation.
December 2024	To incorporate: — Commission Implementing Regulation (EU) 2024/1111 amending Regulation (EU) No 1178/2011, Implementing Regulation (EU) No 923/2012,

	Regulation (EU) No 965/2012 and Implementing Regulation (EU) 2017/373, as regards the establishment of requirements for the operation of manned aircraft with a vertical take-off and landing capability; and — Commission Implementing Regulation (EU) 2024/2076 amending Regulations (EU) No 1178/2011 and (EU) No 965/2012 as regards the clarification of requirements for cruise relief co-pilots, updates of requirements for flight crew licensing and medical certification, and improvements for general aviation.
November 2025	To incorporate: — Commission Implementing Regulation (EU) 2025/134 of 28 January 2025 amending Regulation (EU) No 1178/2011 as regards the introduction of a gyroplane pilot licence; — ED Decision 2025/002/R providing a regular update to the AMC & GM to Regulation (EU) 1178/2011; — ED Decision 2025/011/R introduction a regulatory framework for the operation of drones - enabling innovative air mobility with manned VTOL-capable aircraft; — Commission Implementing Regulation (EU) 2025/2293 of 10 November 2025 amending Implementing Regulation (EU) 2023/203 as regards the requirements applicable to organisations subject to a declaration and correcting Regulations (EU) No 1178/2011, (EU) No 748/2012, (EU) No 965/2012, (EU) No 139/2014, (EU) No 1321/2014, (EU) 2015/340, and Implementing Regulation (EU) 2017/373;¹ and — The corrections to topics AMC1 FCL.720.A(a)(2)(ii)(A) and AMC1 MED.B.045.

¹ Enter into force date - 1 December, 2025.

NOTE FROM THE EDITOR

The content of this document is arranged as follows: the cover regulation (recitals and articles) of the implementing rule (IR) appears first, then the IR annex points, followed by the related acceptable means of compliance (AMC) and guidance material (GM), and, finally, by the cover regulation (recitals and articles) of, and the annex to, the delegated rule (DR).

All elements (i.e. articles, IRs, AMC, GM, and DRs) are colour-coded and can be identified according to the illustration below. The EU regulation or EASA Executive Director (ED) decision through which the article, IR, AMC, GM, or DR was introduced or last amended is indicated below the article, IR, AMC, GM, or DR title in *italics*.

<u>Cover regulation article</u>	<i>EU regulation</i>
Implementing rule annex or delegated rule annex	<i>EU regulation</i>
Acceptable means of compliance	<i>ED decision</i>
Guidance material	<i>ED decision</i>

Note:

Rules that have a future applicability date are marked with purple. The respective applicability date is indicated below the rule text in purple, in square brackets '[]', and in *italics*.

This document will be updated regularly to incorporate further amendments.

The format of this document has been adjusted to make it user-friendly and for reference purposes. Any comments should be sent to erules@easa.europa.eu.

INCORPORATED AMENDMENTS

COMMISSION REGULATIONS

Incorporated Commission Regulation	Affected Part	Applicability date ¹
Regulation (EU) No 1178/2011	Cover regulation	8/4/2012*
	Annex I (Part-FCL)	
	Annex II	
	Annex III	
	Annex IV (Part-MED)	
Regulation (EU) No 290/2012	Cover regulation	8/4/2012*
	Annex V (Part-CC)	
	Annex VI (Part-ARA)	
	Annex VII (Part-ORA)	
Regulation (EU) No 70/2014	Cover regulation	17/2/2014
	Annex VII (Part-ORA)	
Regulation (EU) No 245/2014	Cover regulation	3/4/2014
	Annex I (Part-FCL)	
	Annex II	
	Annex III	
	Annex VI (Part-ARA)	
Regulation (EU) 2015/445	Cover regulation	8/4/2015*
	Annex I (Part-FCL)	
	Annex II	
	Annex III	
	Annex VI (Part-ARA)	
	Annex VII (Part-ORA)	
Regulation (EU) 2016/539	Cover regulation	8/4/2016*
	Annex I (Part-FCL)	
	Annex VII (Part-ORA)	
Regulation (EU) 2018/1065	Cover regulation	19/8/2018
	Annex I (Part-FCL)	
	Annex VI (Part-ARA)	
Regulation (EU) 2018/1119	Cover regulation	2/9/2018
	Annex I (Part-FCL)	
	Annex VI (Part-ARA)	
	Annex VII (Part-ORA)	
	Annex VIII (Part-DTO)	
Regulation (EU) 2018/1974	Cover regulation	20/12/2019*
	Annex I (Part-FCL)	
Regulation (EU) 2019/27	Cover regulation	30/1/2019

¹ This is the earliest date of application (i.e. the date from which an act or a provision in an act produces its full legal effects) as defined in the relevant cover regulation article. Some provisions of the regulations though may be applicable at a later date (deferred applicability). Besides, there may be some opt-outs (derogations from certain provisions) notified by the Member States.

	Annex IV (Part-MED)	
	Annex VI (Part-ARA)	
Regulation (EU) 2019/430	Cover regulation	11/4/2019
Regulation (EU) 2019/1747	Cover regulation	11/11/2019*
	Annex I (Part-FCL)	
	Annex VI (Part-ARA)	
	Annex VIII (Part-DTO)	
Regulation (EU) 2020/359	Cover regulation	8/4/2020*
	Annex I (Part-FCL)	
	Annex IV (Part-MED)	
	Annex VI (Part-ARA)	
	Annex VIII (Part-DTO)	
Regulation (EU) 2020/723	Cover regulation	22/6/2020
	Annex III	
Regulation (EU) 2020/2193	Cover regulation	12/1/2021
	Annex I (Part-FCL)	8/9/2021 — point (1)(r) of Annex I and point (1)(a) of Annex II to Regulation (EU) 2020/2193
	Annex VI (Part-ARA)	31/1/2022 — point (1)(p) of Annex I to Regulation (EU) 2020/2193
	Annex VII (Part-ORA)	
Regulation (EU) 2021/2227	Cover Regulation	4/1/2022 (Article 4(c))
	Annex I (Part-FCL)	30/10/2022
Regulation (EU) 2023/203	Annex VI (Part-ARA)	22/02/2026
	Annex VII (Part-ORA)	
Regulation (EU) 2024/1111	Cover Regulation	1/5/2025
	Annex I (Part-FCL)	1/5/2025
Regulation (EU) 2024/2076	Cover Regulation	14/8/2024
	Annex I (Part-FCL)	14/8/2024 13/2/2025 (as specified by the Article 3 of the Regulation)
	Annex IV (Part-MED)	13/2/2025
	Annex VI (Part-ARA)	14/8/2024 13/2/2025 (as specified by the Article 3 of the Regulation)
	Annex VII (Part-ORA)	14/8/2024 13/2/2025 (as specified by the Article 3 of the Regulation)
	Annex VIII (Part-DTO)	14/8/2024
Regulation (EU) 2025/134	Cover Regulation	

	Annex I (Part-FCL)	18/2/2026
	Annex IV (Part-MED)	
	Annex VI (Part-ARA)	
	Annex VIII (Part-DTO)	
Regulation (EU) 2025/2293	Annex VI (Part-ARA)	22/2/2025

* Refer to Article 12 of the cover regulation.

AMC/GM TO IRs (ED DECISIONS)

Incorporated ED Decision	AMC/GM Issue No, Amendment No	Applicability date ¹
ED Decision 2011/015/R	Initial issue of Annex IV (Part-MED)	22/12/2011
ED Decision 2011/016/R	Initial issue of Annex I (Part-FCL)	22/12/2011
ED Decision 2012/005/R	Initial issue of Annex V (Part-CC)	20/4/2012
ED Decision 2012/006/R	Initial issue of Annex VI (Part-ARA)	20/4/2012
ED Decision 2012/007/R	Initial issue of Annex VII (Part-ORA)	20/4/2012
ED Decision 2013/006/R	Amendment 1 to Annex VI (Part-ARA)	23/4/2013
ED Decision 2013/008/R	Amendment 1 to Annex VII (Part-ORA)	23/4/2013
ED Decision 2013/016/R	Amendment 1 to Annex IV (Part-MED)	10/8/2013
ED Decision 2014/020/R	Amendment 2 to Annex VI (Part-ARA)	3/4/2014
ED Decision 2014/021/R	Amendment 2 to Annex VII (Part-ORA)	3/4/2014
ED Decision 2014/022/R	Amendment 1 to Annex I (Part-FCL)	3/4/2014
ED Decision 2015/011/R	Amendment 3 to Annex VII (Part-ORA)	16/4/2015
ED Decision 2015/023/R	Amendment 1 to Annex V (Part-CC)	1/10/2016
ED Decision 2016/008/R	Amendment 2 to Annex I (Part-FCL)	3/5/2016
	Amendment 3 to Annex VI (Part-ARA)	
ED Decision 2017/022/R	Amendment 3 to Annex I (Part-FCL)	12/12/2017
	Amendment 4 to Annex VI (Part-ARA)	
	Amendment 4 to Annex VII (Part-ORA)	
ED Decision 2018/001/R	Amendment 4 to Annex I (Part-FCL)	31/1/2022
	Amendment 5 to Annex VII (Part-ORA)	
ED Decision 2018/009/R	Amendment 5 to Annex I (Part-FCL)	15/9/2018
	Amendment 5 to Annex VI (Part-ARA)	
	Initial issue of Annex VIII (Part-DTO)	
ED Decision 2018/011/R	Amendment 6 to Annex I (Part-FCL)	7/11/2018 ¹
	Amendment 6 to Annex VI (Part-ARA)	31/1/2022 ¹
ED Decision 2019/002/R	Amendment 7 to Annex VI (Part-ARA)	30/1/2019
	Issue 2 of Annex IV (Part-MED)	
ED Decision 2019/005/R	Amendment 7 to Annex I (Part-FCL)	20/12/2019
	Amendment 6 to Annex VII (Part-ORA)	
ED Decision 2019/017/R	Amendment 8 to Annex I (Part-FCL)	31/1/2022
	Amendment 8 to Annex VI (Part-ARA)	
ED Decision 2020/005/R	Amendment 9 to Annex I (Part-FCL)	19/3/2020
	Amendment 9 to Annex VI (Part-ARA)	
	Amendment 7 to Annex VII (Part-ORA)	
	Amendment 1 to Annex VIII (Part-DTO)	
ED Decision 2020/018/R	Amendment 10 to Annex I (Part-FCL)	8/9/2021
	Amendment 10 to Annex VI (Part-ARA)	
	Amendment 11 to Annex I (Part-FCL)	3/3/2021

¹ Derogation of the applicability date in some amended points.

ED Decision 2021/002/R	Amendment 11 to Annex VI (Part-ARA)	
ED Decision 2022/014/R	Amendment 12 to Issue 1 to Annex I (Part-FCL)	30/10/2022
ED Decision 2023/010/R	Amendment 12 to Issue 1 to Annex VI (Part-ARA)	22/2/2026
ED Decision 2025/002/R	Amendment 1 to Issue 1 to Cover Regulation	13/2/2025
	Amendment 13 to Issue 1 to Annex I (Part-FCL)	
	Amendment 1 to Issue 2 to Annex IV (Part-MED)	
	Amendment 13 to Issue 1 to Annex VI (Part-ARA)	
	Amendment 8 to Issue 1 to Annex VII (Part-ORA)	
	Amendment 2 to Issue 1 to Annex VIII (Part-DTO)	
ED Decision 2025/011/R	Amendment 2 to Issue 1 to Cover Regulation	9/7/2025

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COVER REGULATION

COMMISSION REGULATION (EU) No 1178/2011

of 3 November 2011

laying down technical requirements and administrative procedures related to civil aviation aircrew pursuant to Regulation (EC) No 216/2008 of the European Parliament and of the Council

Regulation (EU) No 1178/2011

THE EUROPEAN COMMISSION,

Having regard to the Treaty on the Functioning of the European Union,

Having regard to Regulation (EC) No 216/2008 of the European Parliament and of the Council of 20 February 2008 on common rules in the field of civil aviation and establishing a European Aviation Safety Agency, and repealing Council Directive 91/670/EEC, Regulation (EC) No 1592/2002 and Directive 2004/36/EC¹, and in particular Articles 7(6), 8(5) and 10(5) thereof,

Whereas:

- (1) Regulation (EC) No 216/2008 aims at establishing and maintaining a high uniform level of civil aviation safety in Europe. That Regulation provides for the means of achieving that objective and other objectives in the field of civil aviation safety.
- (2) Pilots involved in the operation of certain aircraft, as well as flight simulation training devices, persons and organisations involved in training, testing or checking of those pilots, have to comply with the relevant essential requirements set out in Annex III to Regulation (EC) No 216/2008. According to that Regulation pilots as well as persons and organisations involved in their training should be certified once they have been found to comply with essential requirements.
- (3) Similarly, pilots should be issued with a medical certificate and aero-medical examiners, responsible for assessing the medical fitness of pilots, should be certified once they have been found to comply with the relevant essential requirements. However, Regulation (EC) No 216/2008 envisages the possibility of general medical practitioners to act as aero-medical examiners under certain conditions and if permitted under national law.
- (4) Cabin crew involved in the operation of certain aircraft have to comply with the relevant essential requirements set out in Annex IV to Regulation (EC) No 216/2008. According to that Regulation, cabin crew should be periodically assessed for medical fitness to safely exercise their assigned safety duties. Compliance must be shown by an appropriate assessment based on aero-medical best practice.
- (5) Regulation (EC) No 216/2008 requires the Commission to adopt the necessary implementing rules for establishing the conditions for certifying pilots as well as persons involved in their training, testing or checking, for the attestation of cabin crew members and for the assessment of their medical fitness.
- (6) The requirements and procedures for the conversion of national pilot licences and national flight engineer licences into pilot licences should be laid down, to ensure that they are allowed to perform their activities under harmonised conditions; flight test qualifications should also be converted in accordance with this Regulation.

¹ OJ L 79, 19.3.2008, p. 1.

- (7) It should be possible for Member States to accept licences issued by third countries where a level of safety equivalent to that specified by Regulation (EC) No 216/2008 can be guaranteed; Conditions for the acceptance of licences issued by third countries should be laid down.
- (8) In order to ensure that training commenced before the application of this Regulation may be taken into account for the purposes of obtaining pilots' licences, the conditions for recognising training already completed should be laid down; the conditions for recognising military licences should also be laid down.
- (9) It is necessary to provide sufficient time for the aeronautical industry and Member State administrations to adapt to the new regulatory framework, to allow Member States the time to issue specific types of pilot licences and medical certificates not covered by the 'JAR', and to recognise under certain conditions the validity of licences and certificates issued, as well as aero-medical assessment performed, before this Regulation applies.
- (10) Council Directive 91/670/EEC of 16 December 1991 on mutual acceptance of personnel licences for the exercise of functions in civil aviation¹ is repealed in accordance with Article 69(2) of Regulation (EC) No 216/2008. The measures adopted by this Regulation are to be regarded as the corresponding measures.
- (11) In order to ensure a smooth transition and a high uniform level of civil aviation safety in the Union, implementing measures should reflect the state of the art, including best practices, and scientific and technical progress in the field of pilot training and aircrew aero- medical fitness. Accordingly, technical requirements and administrative procedures agreed by the International Civil Aviation Organisation (ICAO) and the Joint Aviation Authorities until 30 June 2009 as well as existing legislation pertaining to a specific national environment, should be considered.
- (12) The Agency prepared draft implementing rules and submitted them as an opinion to the Commission in accordance with Article 19(1) of Regulation (EC) No 216/2008.
- (13) The measures provided for in this Regulation are in accordance with the opinion of the Committee established by Article 65 of Regulation (EC) No 216/2008,

HAS ADOPTED THIS REGULATION:

Article 1 — Subject matter

Regulation (EU) 2020/359

1. This Regulation lays down detailed rules for:
 - (a) different ratings for pilot licences, the conditions for issuing, maintaining, amending, limiting, suspending or revoking pilot licences, the privileges and responsibilities of the holders of pilot licences, as well as the conditions for the conversion of existing national pilot licences and of national flight engineer licences into pilot licences;
 - (b) the certification of persons who are responsible for providing flight training or flight simulation training and for assessing pilots' skills;
 - (c) different medical certificates for pilots, the conditions for issuing, maintaining, amending, limiting, suspending or revoking medical certificates, the privileges and responsibilities of the holders of medical certificates as well as the conditions for the conversion of national medical certificates into mutually recognised medical certificates;

¹ OJ L 373, 31.12.1991, p. 21.

- (d) the certification of aero-medical examiners, as well as the conditions under which general medical practitioners may act as aero-medical examiners;
 - (e) the periodical aero-medical assessment of cabin crew members, as well as the qualification of persons who are responsible for that assessment;
 - (f) the conditions for issuing, maintaining, amending, limiting, suspending or revoking cabin crew attestations, as well as the privileges and responsibilities of the holders of cabin crew attestations;
 - (g) the conditions for issuing, maintaining, amending, limiting, suspending or revoking certificates of pilot training organisations and of aero-medical centres involved in the qualification and aero-medical assessment of civil aviation aircrew;
 - (h) the requirements for the certification of flight simulation training devices and for organisations that operate and use those devices;
 - (i) the requirements for the administration and management system to be fulfilled by the Member States, the European Union Aviation Safety Agency ('EASA') and organisations in relation to the rules referred to in points (a) to (h).
2. Articles 11b and 11c of this Regulation as well as Annex IV (Part-MED), Annex VI (Part-ARA), Annex VII (Part-ORA) and Annex VIII (Part-DTO) to this Regulation shall apply to pilot licences for balloons and sailplanes.

Article 2 — Definitions

Regulation (EU) 2024/2076

For the purposes of this Regulation, the following definitions shall apply:

- (1) 'Part-FCL licence' means a flight crew licence which complies with the requirements of Annex I;
- (2) 'JAR' means joint aviation requirements adopted by the Joint Aviation Authorities as applicable on 30 June 2009;
- (3a) 'Complex aeroplane' means an aeroplane that meets either of the following characteristics:
 - (i) with a maximum certificated take-off mass exceeding 5 700 kg;
 - (ii) certificated for a maximum passenger seating configuration of more than nineteen;
 - (iii) certificated for operation with a minimum crew of two pilots;
 - (iv) equipped with a turbojet engine or turbojet engines or more than one turboprop engine;
- (3b) 'Complex helicopter' means a helicopter that meets either of the following characteristics:
 - (i) with a maximum certificated take-off mass exceeding 3 175 kg; 2/43
 - (ii) certificated for a maximum passenger seating configuration of more than nine;
 - (iii) certificated for operation with a minimum crew of two pilots;
- (5) 'Non-JAR-compliant licence' means the pilot licence issued or recognised by a Member State in accordance with national legislation and not having been recommended for mutual recognition in relation to the relevant JAR;
- (6) 'Credit' means the recognition of prior experience or qualifications;
- (7) 'Credit report' means a report on the basis of which prior experience or qualifications may be recognised;

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- (8) 'Conversion report' means a report on the basis of which a licence may be converted into a Part-FCL licence;
- (8a) "rotorcraft" means a power-driven, heavier-than-air aircraft that depends principally for its support in flight on the lift generated by up to two rotors;
- (8b) "vertical take-off and landing (VTOL)-capable aircraft (VCA)" means a power-driven, heavier-than-air aircraft, other than aeroplane or rotorcraft, capable of performing vertical take-off and landing by means of lift and thrust units used to provide lift during take-off and landing;
- (8c) 'SEP aeroplane' means a single-engine, single-pilot aeroplane for which no type rating is required and whose single centric propulsion unit is operated by a single thrust control and driven by either of the following types of engine:
- (a) a piston engine;
 - (b) an electric engine system which, if so specified following the certification process in accordance with Regulation (EU) No 748/2012, may consist of more than one electric engine;
 - (c) a hybrid engine system that consists of piston and electric engines, if so specified following the certification process in accordance with Regulation (EU) No 748/2012;
- (8d) 'SEP helicopter' means a single-engine, single-pilot helicopter which is powered by a piston engine;'
- (11) 'Cabin crew member' means an appropriately qualified crew member, other than a flight crew or technical crew member, who is assigned by an operator to perform duties related to the safety of passengers and flight during operations;
- (12) 'Aircrew' means flight crew and cabin crew;
- (14) 'acceptable means of compliance (AMC)' means non-binding standards adopted by the Agency to illustrate means to establish compliance with Regulation (EC) No 216/2008 and its implementing rules;
- (15) 'alternative means of compliance (AltMoC)' means those means that propose an alternative to an existing AMC or those that propose new means to establish compliance with Regulation (EC) No 216/2008 and its implementing rules for which no associated AMC have been adopted by the Agency;
- (16) 'approved training organisation (ATO)' means an organisation which is entitled to provide training to pilots on the basis of an approval issued in accordance with the first subparagraph of Article 10a(1);
- (17) 'basic instrument training device (BITD)' means a ground-based training device for the training of pilots representing the student pilot's station of a class of aeroplanes, which may use screen-based instrument panels and spring-loaded flight controls, and providing a training platform for at least the procedural aspects of instrument flight;
- (18) 'certification specifications (CS)' mean technical standards adopted by the Agency indicating means to be used by an organisation for the purpose of certification;
- (19) 'Flight instructor (FI)' means an instructor with the privileges to provide training in an aircraft in accordance with Subpart J of Annex I (Part-FCL) to this Regulation, Subpart FI of Annex III

(Part-BFCL) to [Regulation \(EU\) 2018/395¹](#), or Subpart FI of Annex III (Part-SFCL) to [Implementing Regulation \(EU\) 2018/1976²](#);

- (20) 'flight simulation training device (FSTD)' means a device for the training of pilots which is:
- (a) in the case of aeroplanes, a full flight simulator (FFS), a flight training device (FTD), a flight and navigation procedures trainer (FNPT) or a basic instrument training device (BITD);
 - (b) in the case of helicopters, a full flight simulator (FFS), a flight training device (FTD) or a flight and navigation procedures trainer (FNPT);
- (21) 'FSTD qualification' means the level of technical ability of an FSTD as specified in the certification specifications relating to the FSTD in question;
- (22) 'principal place of business' of an organisation means the head office or registered office of the organisation within which the principal financial functions and operational control of the activities referred to in this Regulation are exercised;
- (22a) 'ARO.RAMP' means the Subpart RAMP of Annex II to the Regulation on Air Operations;
- (22b) 'Automatically validated' means the acceptance, without formalities, by an ICAO contracting State listed in the ICAO attachment of a flight crew licence issued by a State in accordance with Annex 1 to the Chicago Convention;
- (22c) 'ICAO attachment' means an attachment to an automatically validated flight crew licence issued in accordance with Annex 1 to the Chicago Convention, which is mentioned under item XIII of the flight crew licence;
- (23) 'qualification test guide (QTG)' means a document established to demonstrate that the performance and handling qualities of an FSTD represent those of the aircraft, class of aeroplane or type of helicopter, simulated within prescribed limits and that all applicable requirements have been met. The QTG includes both the data of the aircraft, class of aeroplane or type of helicopter and FSTD data used to support the validation;
- (24) 'declared training organisation (DTO)' means an organisation which is entitled to provide training to pilots on the basis of a declaration made in accordance with the second subparagraph of Article 10a(1);
- (25) 'DTO training programme' means a document established by a DTO, describing in detail the training course provided by that DTO.

¹ Commission Regulation (EU) 2018/395 of 13 March 2018 laying down detailed rules for the operation of balloons pursuant to Regulation (EC) No 216/2008 of the European Parliament and of the Council ([OJ L 71, 14.3.2018, p. 10](#)).

² Commission Implementing Regulation (EU) 2018/1976 of 14 December 2018 laying down detailed rules for the operation of sailplanes pursuant to Regulation (EU) 2018/1139 of the European Parliament and of the Council ([OJ L 326, 20.12.2018, p. 64](#)).

GM1 Article 2 Definitions

ED Decision 2018/009/R

Following is a list of acronyms that are used throughout the AMC/GM to Regulation (EU) No 1178/2011:

(A)	aeroplane
(H)	helicopter
A/C	aircraft
ACAS	airborne collision avoidance system
AeMC	aero-medical centre
ALARP	as low as reasonably practicable
AMC	acceptable means of compliance
AME	aero-medical examiner
APU	auxiliary power unit
ARA	authority requirements for aircrew
ATO	approved training organisation
ATPL	airline transport pilot licence
BITD	basic instrument training device
bpm	beats per minute
CAT	category
CC	cabin crew
cm	centimetre
CPL	commercial pilot licence
CS	certification specification
CS-FSTD(A)	Certification Specifications for Aeroplane Flight Simulation Training Devices
CS-FSTD(H)	Certification Specifications for Helicopter Flight Simulation Training Devices
dB	decibel
DH	decision height
DPATO	defined point after take-off
DPBL	decision point before landing
EC	European Community
ECG	electrocardiogram
ENT	ear, nose and throat
EOG	electro-oculography
ETOPS	extended range operations with twin-engined aeroplanes
EU	European Union
FANS	future air navigation system
FD	flight director
FEV1	forced expiratory volume in 1 second
FFS	full flight simulator
FMECA	failure mode, effects and criticality analysis
FMGC	flight management and guidance computer
FMS	flight management system
FNPT	flight navigation and procedures trainer
FSTD	flight simulation training device
FTD	flight training device
FTE	full-time equivalent
ft	foot; feet
FVC	forced vital capacity
GM	guidance material
GPS	global positioning system
HF	human factors
Hg	mercury
HUD/HUGS	head-up display/head-up guidance system
Hz	Herz

IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
IGE	in ground effect
ILS	instrument landing system
IOS	instructor operating station
IR	implementing rule
IR	instrument rating
kg	kilogram
LDP	landing decision point
LVTO	low-visibility take-off
m	metre
mm	millimetre
OGE	out of ground effect
ORA	organisation requirements for aircrew
ORO	organisation requirements for air operations
OSD	operational suitability data
QTG	qualification test guide
POM	proof of match
ROD	rate of descent
RVR	runway visual range
TDP	take-off decision point
VDR	validation data roadmap

GM1 Article 2(8c) Definitions

ED Decision 2025/002/R

MEANING OF THE TERM ‘SINGLE THRUST CONTROL’

The SEP aeroplane class, as defined in [Article 2\(8c\)](#), includes aeroplanes whose single propulsion unit (propeller) is driven by more than one engine (e.g. a hybrid engine system, consisting of a piston and an electric engine). Despite such a ‘multi-engine’ power plant system, such an aircraft can follow the single-engine aeroplane licensing framework since malfunctions of one engine will lead to a (partial) loss of power but never to an asymmetric flight condition. Hence, if during the certification process for a particular aeroplane it is determined that, from a pilot’s perspective, there is no difference between operating a conventional single-engine power plant and the as above described multi-engine single-propeller power plant of the aeroplane under certification, that aeroplane for the purpose of flight crew licensing can be treated as a single-engine aeroplane. In this context, the term ‘single thrust control’ is meant to refer to engine controls of such an innovative power plant which are equivalent to the engine controls of a conventional single-engine power plant.

Article 3 — Pilot licensing and medical certification

Regulation (EU) 2024/2076

1. Without prejudice to Commission Delegated Regulation (EU) 2020/723 ⁽¹⁾, pilots of aircraft referred to in Article 2(1), points (b)(i) and (ii), of Regulation (EU) 2018/1139 shall comply with the technical requirements and administrative procedures laid down in [Annex I](#) and [Annex IV](#) to this Regulation.
2. Notwithstanding the privileges of the holders of licences as defined in Annex I to this Regulation, holders of pilot licences issued in accordance with Subpart B or C of Annex I to this Regulation

¹ Commission Delegated Regulation (EU) 2020/723 of 4 March 2020 laying down detailed rules with regard to the acceptance of third-country certification of pilots and amending Regulation (EU) No 1178/2011 (OJ L 170, 2.6.2020, p. 1, ELI: http://data.europa.eu/eli/reg_del/2020/723/oj).

may carry out flights referred to in Article 6(4a) of Regulation (EU) No 965/2012. This is without prejudice to compliance with any additional requirements for the carriage of passengers or the development of commercial operations defined in Subparts B or C of Annex I to this Regulation.

Article 3a — Transitional measures for medical assessment protocols applied in accordance with point ARA.MED.330 of Annex VI (Part-ARA) and medical certificates issued on the basis thereof

Regulation (EU) 2024/2076

1. Competent authorities involved in medical assessment protocols, or wishing to join an existing medical assessment protocol, in accordance with point [ARA.MED.330](#) of Annex VI (Part-ARA) applicable before 13 February 2025 may continue to apply the, or join the, relevant protocol until the end of its validity, where the validity has been determined before 13 August 2024.
2. Holders of medical certificates issued in accordance with the medical assessment protocols referred to in paragraph 1 may exercise the privileges of their pilot licences on the basis of their medical certificates, as long as the relevant medical assessment protocol continues to apply in accordance with paragraph 1.

Article 4 — Existing national pilots' licences

Regulation (EU) 2025/134

1. [Deleted]
2. Non-JAR-compliant licences including any associated ratings, certificates, authorisations and/or qualifications issued or recognised by a Member State before the applicability of this Regulation shall be converted into Part-FCL licences by the Member State that issued the licence.

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

2. Non-JAR-compliant licences including any associated ratings, certificates, authorisations or qualifications issued or recognised by a Member State before the applicability of this Regulation shall be converted into Part-FCL licences by the Member State that issued the licence. By way of derogation from this paragraph, [Article 4g](#) shall apply to gyroplane pilot licences.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

3. Non-JAR-compliant licences shall be converted into Part-FCL licences and associated ratings or certificates in accordance with:
 - (a) the provisions of Annex II; or
 - (b) the elements laid down in a conversion report.
4. The conversion report shall:
 - (a) be established by the Member State that issued the pilot licence in consultation with the European Aviation Safety Agency (the Agency);
 - (b) describe the national requirements on the basis of which the pilot licences were issued;
 - (c) describe the scope of the privileges that were given to the pilots;
 - (d) indicate for which requirements in Annex I credit is to be given;
 - (e) indicate any limitations that need to be included on the Part-FCL licences and any requirements the pilot has to comply with in order to remove those limitations.

5. The conversion report shall include copies of all documents necessary to demonstrate the elements set out in points (a) to (e) of paragraph 4, including copies of the relevant national requirements and procedures. When developing the conversion report, Member States shall aim at allowing pilots to, as far as possible, maintain their current scope of activities.
6. Notwithstanding paragraph 3, holders of a class rating instructor certificate or an examiner certificate who have privileges for single-pilot high performance complex aircraft shall have those privileges converted into a type rating instructor certificate or an examiner certificate for single-pilot aeroplanes.
7. A Member State may authorise student pilots who follow a LAPL training course to exercise limited privileges without supervision before they meet all the requirements necessary for the issuance of a LAPL, subject to the following conditions:
 - (a) the scope of the privileges shall be based on a safety risk assessment carried out by the Member State, taking into account the extent of training necessary for the intended level of pilot competence to be achieved;
 - (b) the privileges shall be limited to the following:
 - (i) the whole or part of the national territory of the authorising Member State;
 - (ii) aircraft registered in the authorising Member State;
 - (iii) SEP aeroplanes and SEP helicopters, both with a maximum take-off mass not exceeding 2 000 kg;
 - (c) for training conducted under the authorisation, the holder of such an authorisation who applies for the issuance of a LAPL shall receive credits that are determined by the Member State on the basis of a recommendation from an ATO or a DTO;
 - (d) the Member State shall submit periodical reports and safety risk assessments to the Commission and to the Agency every 3 years;
 - (e) the Member States shall monitor the use of authorisations issued under this paragraph to ensure an acceptable level of aviation safety and take appropriate action in case of identifying an increased safety risk or any other safety concerns.
8. Until 8 September 2021, a Member State may issue an authorisation to a pilot to exercise specified limited privileges to fly aeroplanes under instrument flight rules before the pilot complies with all of the requirements necessary for the issue of an instrument rating in accordance with this Regulation, subject to the following conditions:
 - (a) the Member State shall only issue these authorisations when justified by a specific local need which cannot be met by the ratings established under this Regulation;
 - (b) the scope of the privileges granted by the authorisation shall be based on a safety risk assessment carried out by the Member State, taking into account the extent of training necessary for the intended level of pilot competence to be achieved;
 - (c) the privileges of the authorisation shall be limited to the airspace of the Member State's national territory or parts of it;
 - (d) the authorisation shall be issued to applicants having completed appropriate training with qualified instructors and demonstrated the required competencies to a qualified examiner, as determined by the Member State;
 - (e) the Member State shall inform the Commission, EASA and the other Member States of the specificities of this authorisation, including its justification and safety risk assessment.

- (f) the Member State shall monitor the activities associated with the authorisation to ensure an acceptable level of safety and take appropriate action in case of identifying an increased risk or any safety concerns;
 - (g) the Member State shall carry out a review of the safety aspects of the implementation of the authorisation and submit a report to the Commission by 8 April 2017 at the latest.
9. For licences issued before 19 August 2018, Member States shall comply with the requirements laid down in the second paragraph of point (a) of ARA.FCL.200 as amended by Commission Regulation (EU) 2018/1065¹ by 31 December 2022 at the latest.

Article 4a — Performance-based navigation instrument rating privileges

Regulation (EU) 2016/539

1. Pilots may only fly in accordance with performance-based navigation (“PBN”) procedures after they have been granted PBN privileges as an endorsement to their instrument rating (“IR”).
2. A pilot shall be granted PBN privileges where he or she fulfils all of the following requirements:
 - (a) the pilot has successfully completed a course of theoretical knowledge including PBN, in accordance with FCL.615 of Annex I (Part-FCL);
 - (b) the pilot has successfully completed flying training including PBN, in accordance with FCL.615 of Annex I (Part-FCL);
 - (c) the pilot has successfully completed either a skill test in accordance with Appendix 7 to Annex I (Part-FCL) or a skill test or a proficiency check in accordance with Appendix 9 of Annex I (Part-FCL).
3. The requirements of paragraph 2(a) and (b) shall be deemed to have been fulfilled where the competent authority considers that the competence acquired, either through training or from familiarity with PBN operations, is equivalent to the competence acquired through the courses referred to in paragraph 2(a) and (b) and the pilot demonstrates such competence to the satisfaction of the examiner at the proficiency check or skill test referred to in paragraph 2(c).
4. A record of the successful demonstration of competency in PBN shall, upon completion of the skill test or the proficiency check referred to in paragraph 2(c), be entered in the pilot's logbook or equivalent record and signed by the examiner who conducted the test or check.
5. IR pilots without PBN privileges may only fly on routes and approaches that do not require PBN privileges and no PBN items shall be required for the renewal of their IR, until 25 August 2020; after that date, PBN privileges shall be required for every IR.

Article 4b — Upset prevention and recovery training

Regulation (EU) 2018/1974

1. Upset prevention and recovery training shall become a mandatory part of a training course for a multi-crew pilot licence (MPL), an integrated training course for airline transport pilots for aeroplanes (ATP(A)), a training course for a commercial pilot licence for aeroplanes (CPL(A)) and training courses for a class or type rating for:
 - (a) single-pilot aeroplanes operated in multi-pilot operations;

¹ Commission Regulation (EU) 2018/1065 of 27 July 2018 amending Regulation (EU) No 1178/2011 as regards the automatic validation of Union flight crew licences and take-off and landing training (OJ L 192, 30.7.2018, p. 31).

- (b) single-pilot non-high-performance complex aeroplanes;
 - (c) single-pilot high-performance complex aeroplanes; or
 - (d) multi-pilot aeroplanes;
- in accordance with Annex I (Part-FCL).
2. For training courses referred to in paragraph 1 that commence before 20 December 2019 at an approved training organisation (ATO), upset prevention and recovery training shall not be mandatory provided that:
- (a) CPL(A), ATP(A) or MPL training course is otherwise completed in accordance with Annex I (Part-FCL) and the skill test is completed in compliance with points FCL.320 (CPL), FCL.620 (IR) or FCL.415.A (MPL) of Annex I (Part-FCL) by 20 December 2021 at the latest; or
 - (b) class or type rating training course for the aeroplanes is otherwise completed in accordance with Annex I (Part-FCL) and the skill test is completed in compliance with the second subparagraph of paragraph (c) of point FCL.725 of Annex I (Part-FCL) to this Regulation by 20 December 2021 at the latest.

For the purpose of paragraph 1, the competent authority may on its own assessment and pursuant to a recommendation from an ATO give credit for any upset prevention and recovery training completed before 20 December 2019 under national training requirements.

Article 4c — Transitional measures for holders of an en route instrument rating

Regulation (EU) 2021/2227

1. Up to and including 8 September 2022, holders of an en route instrument rating ('EIR') set out in point [FCL.825](#) of Annex I (Part-FCL) shall:
- (a) be entitled to continue to exercise the privileges of their EIR;
 - (b) continue to be entitled to receive revalidation or renewal of their EIR, in accordance with point [FCL.825\(g\)](#) of Annex I (Part-FCL);
 - (c) be entitled to receive full credit towards the training requirements in point [FCL.835\(c\)\(2\)\(i\) and \(iii\)](#) of Annex I (Part-FCL), when applying for the issue of a basic instrument rating (BIR) in accordance with point [FCL.835](#) of Annex I (Part-FCL); and
 - (d) continue to be entitled to receive full credit as established for EIR holders in Annex I (Part-FCL).
2. As from 8 September 2021, training courses for an EIR referred to in paragraph 1, that have commenced prior to that date, can be continued and shall be regarded as training courses for a BIR. Based on an assessment of the applicant, the approved training organisation responsible for the BIR training course shall determine the amount of EIR training to be credited towards the issue of the BIR.
3. Applicants for a BIR who hold an EIR or have passed the theoretical knowledge examination for an EIR in accordance with point [FCL.825\(d\)](#) prior to 8 September 2021 shall receive full credit towards the requirements for the theoretical knowledge instruction and examination for the BIR.

Article 4d — Transitional measures for single-engine instrument rating privileges for helicopters

Regulation (EU) 2021/2227

Without prejudice to point [FCL.630.H](#) of Annex I (Part-FCL) to this Regulation, all of the following shall apply:

1. instrument ratings for helicopters (IR(H)) issued in accordance with [Annex I](#) (Part-FCL) to this Regulation before 30 October 2022 shall be deemed as IR(H) for both single-engine and multi-engine helicopters and shall be reissued as such IR(H), when reissuing a helicopter pilot licence for administrative reasons.
2. Applicants who before 30 October 2022 commenced training for an IR(H) for either single-engine or multi-engine helicopters shall be allowed to complete that training and, in such a case, be issued with an IR(H) for both single-engine and multi-engine helicopters.

Article 4e — Transitional measures for training, testing and checking related to multi-pilot operations in single-pilot helicopters

Regulation (EU) 2021/2227

1. Member States may decide to issue specific privileges for conducting training, skill tests and proficiency checks in multi-pilot operation in single-pilot helicopters to applicants who meet all of the following conditions:
 - (a) hold an instructor or examiner certificate, as applicable, issued in accordance with [Annex I](#) (Part-FCL) to this Regulation, including the privileges to instruct or to examine, as applicable, in the relevant type of helicopter;
 - (b) have completed the training specified in point [FCL.735.H](#) of Part-FCL;
 - (c) have experience in multi-pilot operation in helicopters at a level that is acceptable to the competent authority of that Member State.
2. The privileges issued in accordance with paragraph 1 shall be valid until 30 October 2025. In order to revalidate the privileges, applicants shall comply with the experience requirements for instructor and examiner privileges related to multi-pilot operation in single-pilot helicopters as set out in Part-FCL.

Article 4f — Type ratings for VCA

Regulation (EU) 2024/1111

1. Applicants that hold a commercial pilot licence for aeroplanes (CPL(A)) or helicopters (CPL(H)) in accordance with [Annex I](#) (Part-FCL) shall be entitled to be issued with a type rating for a VCA and shall exercise the privileges of such a type rating, provided they comply with all the following:
 - (a) the prerequisites specified in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012;
 - (b) Section 1 of Subpart H of Annex I (Part-FCL) and the provisions of this Article.
2. The theoretical knowledge examination shall be written, and the number of multiple-choice questions shall depend on the complexity of the aircraft.
3. Type rating training, skill tests and proficiency checks for aircraft specified in paragraph 1 shall:
 - (a) comply with the following requirements of [Appendix 9](#) to Annex I (Part-FCL):

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- (i) Section A;
 - (ii) Sections B, C or D, as determined and unless otherwise specified in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012; and
 - (b) under the conditions and to the extent specified in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012, include additional training and testing to allow applicants to obtain the competence to operate the relevant VCA
4. By way of derogation from the paragraphs above, applicants that hold a CPL(A) or a CPL(H) and that were involved in test flights for a particular type of VCA shall be issued with a type rating for that aircraft, provided they comply with all the following:
- (a) they comply with the flight conditions for acting as test pilot in the relevant VCA type, as established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012;
 - (b) they have completed either 50 hours of total flight time or 10 hours of flight time as pilot-in-command on test flights in the relevant VCA type;
 - (c) they comply with the prerequisites referred to in paragraph 1(a).
5. The validity period of type ratings issued in accordance with this Article shall be 1 year. Holders shall do all the following:
- (a) in order to revalidate the type rating:
 - (i) within the validity period of the rating, complete at least 2 hours of flight time as pilot of the relevant VCA type;
 - (ii) within the 3 months immediately preceding the expiry date of the rating and in the relevant VCA type or an FSTD representing that aircraft, pass a proficiency check in accordance with paragraph 3, the duration of which may be counted towards the flight time specified in paragraph (a)(i). If applicants choose to pass the proficiency check earlier than within these 3 months, the new validity period shall commence from the date of the proficiency check;
 - (b) in order to renew the type rating, comply with point [FCL.740\(b\)](#) of Annex I (Part-FCL).
6. Holders of a licence and a type rating as specified in paragraph 1 shall be entitled to operate the relevant VCA under instrument flight rules, provided they comply with all the following:
- (a) they hold an IR(A) or an IR(H), as applicable;
 - (b) they have, in the relevant VCA type, completed the skill test or the proficiency check, as applicable, in accordance with paragraph 3 including the content relevant for instrument flight.
7. Notwithstanding point [FCL.900\(b\)](#) of [Annex I](#) (Part-FCL), applicants that hold an instructor certificate in accordance with [Annex I](#) (Part-FCL) with privileges to provide training for aeroplane or helicopter type ratings shall be issued with privileges to provide training for type ratings specified in paragraph 1, provided they:
- (a) hold a type rating as per paragraph 1 for the relevant VCA type;
 - (b) unless otherwise specified in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012, have, within the 12 months preceding the application, completed at least 30 route sectors, including take-offs and

landings, as pilot-in-command in the relevant VCA type, of which 15 route sectors may be completed in an FSTD representing that VCA type;

- (c) have completed, at an ATO, theoretical and practical training for extending instructor privileges to that VCA type, including mandatory training elements as specified in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012;
- (d) pass the relevant sections of the assessment of competence in accordance with point [FCL.935](#) of [Annex I](#) (PartFCL).

By way of derogation from paragraphs (b), (c) and (d), applicants that hold a TRI(A) certificate or a TRI(H) certificate and that were issued with a type rating for a VCA in accordance with paragraph 4, shall receive an extension of their TRI privileges to that VCA type.

8. Holders of instructor privileges referred to in paragraph 7 shall receive revalidation or renewal, as applicable, of these privileges when they comply with the relevant revalidation or renewal requirements of Subpart J of [Annex I](#) (Part-FCL), as applicable for the instructor certificate held, and additionally do either of the following:
 - (a) complete, at an ATO, instructor refresher training that focuses on the privileges as per paragraph 7;
 - (b) pass the relevant sections of the assessment of competence in accordance with point [FCL.935](#) of Annex I (PartFCL) in the relevant VCA type specified in paragraph 1 or an FSTD representing that type.
9. Notwithstanding point [FCL.1000](#)(b) of [Annex I](#) (Part-FCL), applicants that hold an examiner certificate in accordance with [Annex I](#) (Part-FCL) with privileges to act as an examiner for aeroplane or helicopter type ratings shall be issued with privileges to conduct skill tests and proficiency checks for an VCA type specified in paragraph 1, provided they hold instructor privileges as per paragraph 7 for the relevant VCA type and comply with all the following in the relevant VCA type or an FSTD representing that type:
 - (a) complete examiner standardisation in accordance with point [FCL.1015](#) of [Annex I](#) (Part-FCL), including the conduct of at least one skill test or proficiency check;
 - (b) pass the relevant sections of the assessment of competence in accordance with point [FCL.1020](#) of [Annex I](#) (PartFCL).
10. Holders of examiner privileges referred to in paragraph 9 shall receive revalidation or renewal, as applicable, of these privileges when they comply with the relevant parts of point [FCL.1025](#) of [Annex I](#) (Part-FCL) and additionally do either of the following:
 - (a) complete an examiner refresher course in accordance with point [FCL.1025](#)(b)(2) of [Annex I](#) (Part-FCL) that focuses on the privileges as per paragraph 9;
 - (b) pass the relevant sections of the assessment of competence in accordance with point [FCL.1020](#) of [Annex I](#) (PartFCL) in the relevant VCA type or an FSTD representing that type.

AMC1 Article 4f(2) and (3)

ED Decision 2025/011/R

TYPE RATING COURSES — VTOL-CAPABLE AIRCRAFT (VCA)

(a) General

The means for compliance demonstration established for the design and the provision of helicopter type rating training courses, as contained in [AMC3 ORA.ATO.125 of](#) ED Decision 2012/007/R¹, should be followed when designing and providing type rating training courses for VTOL-capable aircraft (VCA).

(b) Theoretical knowledge instruction and examination

Unless otherwise specified in the operational suitability data (OSD) established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012, the theoretical knowledge instruction and examination for VCA should follow the syllabi contained in [AMC1 FCL.725\(a\)](#) from which topics should be selected, modified and complemented, as applicable, for the relevant type of VCA. Particularly, all the following should be appropriately addressed:

- (1) engine and propulsion system of the VCA;
- (2) general knowledge about U-space airspace.

GM1 Article 4f(3)(a)(i)

ED Decision 2025/011/R

According to [Article 4f\(3\)\(a\)\(i\)](#) of Regulation (EU) No 1178/2011², the VCA type rating training should comply with the requirements of Section A of [Appendix 9](#) to Annex I (Part-FCL). Inter alia, this means that the general principles for the use of FSTDs for the training, testing and checking for single-pilot aircraft ([Appendix 9](#), Section A, point (1c)) apply.

AMC1 Article 4f(8)(a)

ED Decision 2025/011/R

INSTRUCTOR REFRESHER TRAINING FOR VTOL-CAPABLE AIRCRAFT (VCA) TYPE RATING INSTRUCTIONAL PRIVILEGES

- (a) If applicants for the revalidation of an instructor certificate referred to in [Article 4f\(7\)](#) of Regulation (EU) No 1178/2011 wish to revalidate instructor privileges for VCA as per [Article 4f\(8\)](#), they should complete the instructor refresher training in the form of a seminar.
- (b) If applicants for the renewal of an instructor certificate referred to in [Article 4f\(7\)](#) of Regulation (EU) No 1178/2011 wish to renew instructor privileges for VCA as per [Article 4f\(8\)](#), they should complete the instructor refresher training following an individual training programme that is established by an ATO, after that ATO has determined on a case-by-case basis the amount of refresher training needed, based on an assessment of the applicant. Such individual training programmes should be based on the content of the training referred to in [Article 4f\(7\)\(c\)](#).

¹ [ED Decision 2012/007/R - AMC & GM to Part-ORA | EASA](#)

² Commission Regulation (EU) No 1178/2011 of 3 November 2011 laying down technical requirements and administrative procedures related to civil aviation aircrew pursuant to Regulation (EC) No 216/2008 of the European Parliament and of the Council (OJ L 311, 25.11.2011, p. 1) (<http://data.europa.eu/eli/reg/2011/1178/oj>).

Article 4g — Specific requirements for applicants for a gyroplane pilot licence

Regulation (EU) 2025/134

1. Until 18 February 2028, applicants who hold a gyroplane pilot licence and associated ratings and certificates that were issued in accordance with national gyroplane flight crew licensing requirements, or who commenced training in accordance with such requirements, shall receive credits for the purpose of being issued with a gyroplane pilot licence and associated ratings and certificates in accordance with this Regulation, on the basis of a credit report established by a Member State in consultation with EASA. The credit report shall comply with [Article 9\(3\)](#)
2. When establishing a credit report in accordance with paragraph (1), Member States shall:
 - (a) solely grant credits for flight time that applicants completed in gyroplanes that have a maximum take-off mass of at least 450 kg;
 - (b) for the purpose of issuing a gyroplane pilot licence in accordance with this Regulation, solely grant credits the amount of which does not exceed the credit that may have been granted in accordance with point [FCL.210.G\(c\)](#) of Annex I. In such cases, applicants shall complete the additional flight training at an approved training organisation (ATO) or at a declared training organisation (DTO) in a gyroplane that has been certified in accordance with Regulation (EU) 2018/1139 and the implementing and delegated acts adopted on the basis thereof, before being recommended by the ATO or the DTO for the skill test.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

Article 4h — Transitional measures for holders of a mountain rating

Regulation (EU) 2024/2076

Holders of a mountain rating which was issued before 13 August 2024 with an expiry date endorsed in accordance with point [FCL.815](#) of Annex I shall, in order to continue to exercise their privileges after that date, take the following steps:

- (a) have their mountain rating re-issued by the competent authority without an expiry date;
- (b) comply with point [FCL.815\(d\)](#) of Annex I, unless, within the preceding 2 years, they have revalidated their mountain rating in accordance with point [FCL.815\(e\)](#) of Annex I applicable until 12 August 2024.

Article 5

*Regulation (EU) 2019/1747**[Deleted]*

Article 6 — Conversion of flight test qualifications

Regulation (EU) No 1178/2011

1. Pilots who before this Regulation applies conducted category 1 and 2 flight tests as defined in the Annex to Commission Regulation (EC) No 1702/2003¹, or who provided instruction to flight test pilots, shall have their flight test qualifications converted into flight test ratings in accordance with Annex I to this Regulation and, where applicable, flight test instructor certificates by the Member State that issued the flight test qualifications.

¹ OJ L 243, 27.9.2003, p. 6.

2. This conversion shall be carried out in accordance with the elements established in a conversion report that complies with the requirements set out in [Article 4](#)(4) and (5).

Article 7 — Existing national flight engineers' licences

Regulation (EU) No 1178/2011

1. In order to convert flight engineer licences, issued in accordance with Annex 1 to the Chicago Convention, into Part-FCL licences, holders shall apply to the Member State that issued the licences.
2. Flight engineer licences shall be converted into Part-FCL licences in accordance with a conversion report that complies with the requirements set out in [Article 4](#)(4) and (5).
3. When applying for the airline transport pilot licence (ATPL) for aeroplanes, the provisions on credit in FCL.510.A(c)(2) of Annex I shall be complied with.

Article 8

Regulation (EU) 2020/723

[Deleted by [Commission Delegated Regulation \(EU\) 2020/723](#)]

Article 9 — Credit for training commenced prior to the application of this Regulation

Regulation (EU) 2019/1747

1. In respect of issuing Part-FCL licences in accordance with Annex I, training commenced prior to the application of this Regulation in accordance with the JARs and procedures, under the regulatory oversight of a Member State recommended for mutual recognition within the Joint Aviation Authorities' system in relation to the relevant JARs, shall be given full credit provided that the training and testing were completed by 8 April 2016 at the latest and a Part-FCL licence is issued by 1 April 2020 at the latest.
2. Training commenced prior to the application of this Regulation in accordance with Annex 1 to the Chicago Convention shall be given credit for the purposes of issuing Part-FCL licences on the basis of a credit report established by the Member State in consultation with the Agency.
3. The credit report shall describe the scope of the training, indicate for which requirements of Part-FCL licences credit is given and, if applicable, which requirements applicants need to comply with in order to be issued with Part-FCL licences. It shall include copies of all documents necessary to demonstrate the scope of the training and of the national regulations and procedures in accordance with which the training was commenced.

Article 9a — Type rating training and operational suitability data

Regulation (EU) No 70/2014

1. Where the Annexes to this Regulation make reference to the operational suitability data established in accordance with Regulation (EU) No 748/2012, and that data is not available for the relevant type aircraft, the applicant for a type rating training course shall comply with the provisions of the Annexes of Regulation (EU) No 1178/2011 only.
2. Type rating training courses approved before the approval of the minimum syllabus of pilot type rating training in the operational suitability data for the relevant type of aircraft in accordance with Regulation (EU) No 748/2012 shall include the mandatory training elements not later than 18 December 2017 or within two years after the operational suitability data was approved, whichever is the latest.

Article 10 — Credit for pilot licences obtained during military service

Regulation (EU) No 1178/2011

1. In order for holders of military flight crew licences to obtain Part-FCL licences, they shall apply to the Member State where they served.
2. The knowledge, experience and skill gained in military service shall be given credit for the purposes of the relevant requirements of Annex I in accordance with the elements of a credit report established by the Member State in consultation with the Agency.
3. The credit report shall:
 - (a) describe the national requirements on the basis of which the military licences, ratings, certificates, authorisations and/or qualifications were issued;
 - (b) describe the scope of the privileges that were given to the pilots;
 - (c) indicate for which requirements of Annex I credit is to be given;
 - (d) indicate any limitations that need to be included on the Part-FCL licences and indicate any requirements pilots have to comply with to remove those limitations;
 - (e) include copies of all documents necessary to demonstrate the elements above, accompanied by copies of the relevant national requirements and procedures.

Article 10a — Pilot training organisations

Regulation (EU) 2021/2227

1. Organisations shall, in accordance with Article 24(2) of [Regulation \(EU\) 2018/1139](#), be entitled to provide training to pilots involved in the operation of aircraft referred to in points (b)(i) and (ii) of Article (2)(1) of [Regulation \(EU\) 2018/1139](#) only where those organisations have been issued by the competent authority with an approval confirming that they comply with the essential requirements set out in Annex IV to [Regulation \(EU\) 2018/1139](#) and with the requirements of Annex VII to this Regulation.

However, having regard to Article 24(6) of [Regulation \(EU\) 2018/1139](#), organisations having their principal place of business in a Member State shall be entitled to provide the training referred to in point [DTO.GEN.110](#) of Annex VIII to this Regulation without such approval inside the territory for which Member States are responsible under the Chicago Convention where they have made a declaration to the competent authority in accordance with the requirements laid down in point [DTO.GEN.115](#) of that Annex and, where so required pursuant to point [DTO.GEN.230\(c\)](#) of that Annex, the competent authority has approved the training programme.
2. *[deleted]*
3. *[deleted]*
4. *[deleted]*
5. Pilot training organisations shall ensure that the IR training course they offer include training for PBN privileges compliant with the requirements of [Annex I](#) (Part-FCL) by 25 August 2020 at the latest.
6. Pilot training organisations that provide training for the IR(H) shall adapt their training programme to be compliant with [Annex I](#) by 30 October 2023.

Article 10b — Flight simulation training devices

Regulation (EU) 2019/1747

1. Flight simulation training devices (FSTDs) used for pilot training, testing and checking, with the exception of developmental training devices used for flight test training, shall comply with the technical requirements and administrative procedures laid down in Annexes VI and VII and shall be qualified.

Article 10c — Aero-medical centres

Regulation (EU) 2019/1747

1. Aero-medical centres shall comply with the technical requirements and administrative procedures laid down in Annexes VI and VII and shall be certified.

Article 11 — Cabin crew medical fitness

Regulation (EU) 2019/1747

1. Cabin crew members involved in the operation of aircraft referred to in Article 4(1)(b) and (c) of Regulation (EC) No 216/2008 shall comply with the technical requirements and administrative procedures laid down in Annex IV.

Article 11a — Cabin crew qualifications and related attestations

Regulation (EU) 2019/1747

1. Cabin crew members involved in commercial operation of aircraft referred to in Article 4(1)(b) and (c) of Regulation (EC) No 216/2008 shall be qualified and hold the related attestation in accordance with the technical requirements and administrative procedures laid down in Annexes V and VI.
2. *[deleted]*
3. *[deleted]*
4. Cabin crew members involved in commercial operations of helicopters on the date of application of this Regulation:
 - (a) shall be deemed to be compliant with the initial training requirements of Annex V if they comply with the applicable training, checking and recency provisions of the JARs for commercial air transportation by helicopters; or
 - (b) if they do not comply with the applicable training, checking and recency requirements of the JARs for commercial air transportation by helicopters, they shall complete all relevant training and checking required to operate on helicopter(s), except the initial training, before being deemed to be compliant with this Regulation; or
 - (c) if they have not operated in commercial operations by helicopters for more than 5 years, they shall complete the initial training course and shall pass the related examination as required in Annex V before being deemed to be compliant with this Regulation.
5. Without prejudice to [Article 2](#), cabin crew attestations complying with the format laid down in Annex VI shall be issued to all cabin crew members involved in commercial operations by helicopters by 8 April 2013 at the latest.

Article 11b — Oversight capabilities

Regulation (EU) No 290/2012

1. Member States shall designate one or more entities as the competent authority within that Member State with the necessary powers and allocated responsibilities for the certification and oversight of persons and organisations subject to Regulation (EC) No 216/2008 and its implementing rules.
2. If a Member State designates more than one entity as competent authority:
 - (a) the areas of competence of each competent authority shall be clearly defined in terms of responsibilities and geographic limitation;
 - (b) coordination shall be established between those entities to ensure effective oversight of all organisations and persons subject to Regulation (EC) No 216/2008 and its implementing rules within their respective remits.
3. Member States shall ensure that the competent authority(ies) has/have the necessary capability to ensure the oversight of all persons and organisations covered by their oversight programme, including sufficient resources to fulfil the requirements of this Regulation.
4. Member States shall ensure that competent authority personnel do not perform oversight activities when there is evidence that this could result directly or indirectly in a conflict of interest, in particular when relating to family or financial interest.
5. Personnel authorised by the competent authority to carry out certification and/or oversight tasks shall be empowered to perform at least the following tasks:
 - (a) examine the records, data, procedures and any other material relevant to the execution of the certification and/or oversight task;
 - (b) take copies of or extracts from such records, data, procedures and other material;
 - (c) ask for an oral explanation on site;
 - (d) enter relevant premises, operating sites or means of transport;
 - (e) perform audits, investigations, assessments and inspections, including ramp inspections and unannounced inspections; and
 - (f) take or initiate enforcement measures as appropriate.
6. The tasks under paragraph 5 shall be carried out in compliance with the legal provisions of the relevant Member State.

Article 11c — Transitional measures

Regulation (EU) 2020/359

Member States shall:

- (a) by 8 April 2021 at the latest, transfer to EASA all records related to the oversight of organisations that provide training for pilot licences in accordance with [Regulation \(EU\) 2018/395](#) and Implementing [Regulation \(EU\) 2018/1976](#) and for which EASA is the competent authority in

accordance with Article 78 of [Regulation \(EU\) 2018/1139](#) of the European Parliament and of the Council¹;

- (b) in coordination with EASA, conclude, certification processes initiated before 8 April 2020 and issue the certificate following which EASA assumes all its responsibilities as a competent authority for those certified organisations.

Article 12 — Entry into force and application (of Commission Regulation (EU) No 1178/2011)

Regulation (EU) 2020/2193

1. This Regulation shall enter into force on the 20th day following its publication in the Official Journal of the European Union.
It shall apply from 8 April 2012.
2. [deleted]
3. [deleted]
4. By way of derogation from paragraph 1, Member States may decide not to apply the provisions of this Regulation until 20 June 2022, to pilots holding a licence and associated medical certificate issued by a third country involved in the non-commercial operation of aircraft as specified in Article 2(1)(b), points (i) or (ii), of Regulation (EU) 2018/1139. Member States shall make those decisions publicly available.
5. [deleted]
6. [deleted]
7. When a Member State makes use of the requirements of paragraphs 2a and 4, it shall notify the Commission and the Agency. This notification shall describe the reasons for such derogation as well as the programme for implementation containing actions envisaged and related timing.
8. By way of derogation from paragraph 1, point [FCL.315.A](#), the second sentence of paragraph (a) of point [FCL.410.A](#) and paragraph (c) of point [FCL.725.A](#) of Annex I (Part-FCL) shall apply from 20 December 2019.

Commission Regulation (EU) No 290/2012 of 30 March 2012

Regulation (EU) 2015/445

1. This Regulation shall enter into force on the 20th day following its publication in the *Official Journal of the European Union*.
It shall apply from 8 April 2012.
2. By way of derogation from the second subparagraph of paragraph 1, Member States may decide not to apply the following provisions:
 - (a) Annexes V to VII until 8 April 2013;

¹ Regulation (EU) 2018/1139 of the European Parliament and of the Council of 4 July 2018 on common rules in the field of civil aviation and establishing a European Union Aviation Safety Agency, and amending Regulations (EC) No 2111/2005, (EC) No 1008/2008, (EU) No 996/2010, (EU) No 376/2014 and Directives 2014/30/EU and 2014/53/EU of the European Parliament and of the Council, and repealing Regulations (EC) No 552/2004 and (EC) No 216/2008 of the European Parliament and of the Council and Council Regulation (EEC) No 3922/91 ([OJ L 212, 22.8.2018, p. 1](#)).

- (b) point ORA.GEN.200(a)(3) of Annex VII to FSTD qualification certificate holders not being an approved training organisation and not holding an air operator certificate until 8 April 2014;
 - (c) Annexes VI and VII to non-JAR-compliant approved training organisations and aero-medical centres until 8 April 2014;
 - (d) point CC.GEN.030 of Annex V until 8 April 2015;
 - (e) Annex V to cabin crew members involved in commercial operations by helicopters until 8 April 2015;
 - (f) Annexes VI and VII to training organisations providing training for flight test ratings in accordance with point FCL.820 of Annex I to Regulation (EU) No 1178/2011 until 8 April 2015.
3. When a Member State makes use of the provisions of paragraph 2, it shall notify the Commission and the Agency. This notification shall describe the duration and the reasons for such derogation as well as the programme for implementation containing actions envisaged and related timing.

Commission Regulation (EU) No 70/2014 of 27 January 2014

Regulation (EU) No 70/2014

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

Commission Regulation (EU) No 245/2014 of 13 March 2014

Regulation (EU) No 245/2014

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

Commission Regulation (EU) 2015/445 of 17 March 2015

Regulation (EU) 2015/445

1. This Regulation shall enter into force on 8 April 2015.
2. By way of derogation from paragraph 1, the amendments to the provisions in FCL315.A, FCL.410.A, FCL.725.A of Annex I shall apply from 8 April 2018.
3. By way of derogation from paragraph 1, Member States may decide not to apply the provisions of Annexes VI and VII to a training organisation providing training only for a national licence that is eligible in accordance with [Article 4\(3\)](#) of Regulation (EU) No 1178/2011, for conversion into a Part-FCL light aircraft pilot licence (LAPL), sailplane pilot licence (SPL) or balloon pilot licence (BPL) until 8 April 2018.

Commission Regulation (EU) 2016/539 of 6 April 2016

Regulation (EU) 2016/539

This Regulation shall enter into force on the day of its publication in the *Official Journal of the European Union*.

It shall apply from 8 April 2016.

However, points 1, 2 and 4 of Article 1 shall apply from 25 August 2018, with the exception of point 1(g) of the Annex, which shall apply from 8 April 2016.

Commission Regulation (EU) 2018/1065 of 27 July 2018

Regulation (EU) 2018/1065

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

Commission Regulation (EU) 2018/1119 of 31 July 2018

Regulation (EU) 2018/1119

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

Commission Implementing Regulation (EU) 2018/1974 of 14 December 2018

Regulation (EU) 2018/1974

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

However:

- (a) Article 1(1) shall apply from 20 December 2019.
- (b) Article 1(4) shall apply from 20 December 2019.
- (c) Notwithstanding point (b) above, points (2), (4), (5) and (12) of the Annex to this Regulation shall apply from 31 January 2022.

Commission Implementing Regulation (EU) 2019/27 of 19 December 2018

Regulation (EU) 2019/27

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

Commission Implementing Regulation (EU) 2019/430 of 18 March 2019

Regulation (EU) 2019/430

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

Commission Implementing Regulation (EU) 2019/1747 of 15 October 2019

Regulation (EU) 2019/1747

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

However points (57), (58), (59) and (66) of the Annex to this Regulation shall apply from 21 December 2019.

Commission Implementing Regulation (EU) 2020/359 of 4 March 2020

Regulation (EU) 2020/359

1. This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.
2. This Regulation shall apply from 8 April 2020.
3. By way of derogation from paragraph 2, the following provisions shall apply from 8 September 2021:
 - (a) points (1)(e), (4)(b), (5) to (7), (32), (34), (36)(d), (40)(a), (41); (42), (44), (46) to (48), (52)(f), (53)(a) to (53)(c) (53) (e), (53)(f), (54), (55), (56)(a) to (56)(c) and (57) of Annex I;
 - (b) point (b) of Annex II;
 - (c) point (10)(d)(ii) of Annex III.
4. By way of derogation from paragraph 2, Article 1 point (7) and points (49), (53)(d), (58)(b), (58)(d) and (58)(e) of Annex I shall apply from the day of entry into force of this Regulation.

Commission Implementing Regulation (EU) 2020/2193 of 16 December 2020

Regulation (EU) 2020/2193

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

Point (1)(r) of Annex I and point (1)(a) of Annex II shall apply from 8 September 2021 and point (1)(p) of Annex I shall apply from 31 January 2022.

Commission Implementing Regulation (EU) 2021/2227 of 14 December 2021

Regulation (EU) 2021/2227

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

It shall apply from 30 October 2022. However, point (1) of Article 1 shall apply as of the date of entry into force.

**Commission Implementing Regulation (EU) 2023/203
of 27 October 2022**

Regulation (EU) 2023/203

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

It shall apply from 22 February 2026.

Regulation (EU) No 1178/2011

This Regulation shall be binding in its entirety and directly applicable in all Member States.

Done at Brussels, 3 November 2011.

For the Commission

The President

José Manuel BARROSO

ANNEX I (PART-FCL)

SUBPART A – GENERAL REQUIREMENTS

FCL.001 Competent authority

Regulation (EU) No 1178/2011

For the purpose of this Part, the competent authority shall be an authority designated by the Member State to whom a person applies for the issue of pilot licences or associated ratings or certificates.

FCL.005 Scope

Regulation (EU) No 1178/2011

This Part establishes the requirements for the issue of pilot licences and associated ratings and certificates and the conditions of their validity and use.

GM1 FCL.005 Scope

ED Decision 2020/005/R

INTERPRETATIVE MATERIAL

- (a) Whenever licences, ratings, approvals or certificates are mentioned in Part-FCL, these are meant to be valid licences, ratings, approvals or certificates issued in accordance with Part-FCL. In all other cases, these documents are specified.
- (b) Whenever a reference is made to Member States to mutual recognition of licences, ratings, approvals or certificates, this means a European Union Member State and states associated to the Agency in accordance with Article 55 of the Regulation (EC) No 216/2008 of the European Parliament and of the Council of 20 February 2008.
- (c) Whenever an inclusive or exclusive 'or' is used, it should be understood within the context of the whole meaning of the requirement in which it is used.

FCL.010 Definitions

Regulation (EU) 2025/134

For the purposes of this Annex (Part-FCL), the following definitions shall apply:

- "Accessible" means that a device can be used by:
 - the approved training organisation (ATO) under whose approval a training course for a class or type rating is being conducted; or
 - the examiner conducting the assessment of competence, skill test or proficiency check for the purpose of assessing, testing or checking.
- "Aerobatic flight" means an intentional manoeuvre involving an abrupt change in an aircraft's attitude, an abnormal attitude, or abnormal acceleration, not necessary for normal flight or for instruction for licences, certificates, or ratings other than the aerobatic rating.
- "Aeroplane" means an engine-driven fixed-wing aircraft heavier than air which is supported in flight by the dynamic reaction of the air against its wings.

- "Aeroplane" required to be operated with a co-pilot" means a type of aeroplane which is required to be operated with a co-pilot as specified in the flight manual or by the air operator certificate.
- "Aeroplane upset prevention and recovery training" (UPRT) means training consisting of:
 - aeroplane upset prevention training: a combination of theoretical knowledge and flying training with the aim of providing flight crew with the required competencies to prevent aeroplane upsets; and
 - aeroplane upset recovery training: a combination of theoretical knowledge and flying training with the aim of providing flight crew with the required competencies to recover from aeroplane upsets.
- "Aircraft" means any machine which can derive support in the atmosphere from the reactions of the air other than the reactions of the air against the earth's surface.
- "Airmanship" means the consistent use of good judgement and well-developed knowledge, skills and attitudes to accomplish flight objectives.
- "Airship" means a power-driven lighter-than-air aircraft, with the exception of hot-air airships, which are considered to be balloons in accordance with Article 2(7) of [Commission Regulation \(EU\) 2018/395](#).
- "Available FSTD" means any flight simulation training device (FSTD) that is vacant for use of the FSTD operator or of the customer irrespective of any time considerations.
- "Angular operation" means an instrument approach operation in which the maximum tolerable error/deviation from the planned track is expressed in terms of deflection of the needles on the Course Deviation Indicator (CDI) or equivalent display in the cockpit.
- "Assessment of competence" means the demonstration of skills, knowledge and attitude for the initial issue, revalidation or renewal of an instructor or examiner certificate.
- "Balloon" means a lighter-than-air aircraft which is not engine-driven and sustains flight through the use of either gas or an airborne heater. For the purposes of this Part, a hot-air airship, although engine-driven, is also considered a balloon.
- "Category of aircraft" means a categorisation of aircraft according to specified basic characteristics, for example aeroplane, powered-lift, helicopter, airship, sailplane, free balloon.
- "Class of aeroplane" means a categorisation of single-pilot aeroplanes not requiring a type rating.
- "Commercial air transport" means the transport of passengers, cargo or mail for remuneration or hire.
- "Competency" means a combination of skills, knowledge and attitude required to perform a task to the prescribed standard.
- "Competency element" means an action which constitutes a task that has a triggering event and a terminating event that clearly defines its limits, and an observable outcome.
- "Competency unit" means a discrete function consisting of a number of competency elements.
- "Co-pilot" means a pilot operating other than as pilot-in-command, on an aircraft for which more than one pilot is required, but excluding a pilot who is on board the aircraft for the sole purpose of receiving flight instruction for a licence or rating.

- "Cross-country" means a flight between a point of departure and a point of arrival following a pre-planned route, using standard navigation procedures.
- "Cruise relief co-pilot" means a pilot who relieves the co-pilot of his/her duties at the controls during the cruise phase of a flight in multi-pilot operations above FL 200.
- "Dual instruction time" means flight time or instrument ground time during which a person is receiving flight instruction from a properly authorised instructor.
- "En route IFR flight" means the phase of an IFR flight that commences after the completion of an IFR departure procedure and finishes when commencing an IFR approach procedure.
- "Error" means an action or inaction taken by the flight crew which leads to deviations from organisational or flight intentions or expectations.
- "Error management" means the process of detecting and responding to errors with countermeasures which reduce or eliminate the consequences of errors, and mitigate the probability of errors or undesired aircraft states.
- "Evidence based training (EBT) operator" means an organisation that is holding an air operator certificate (AOC) in accordance with Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#) and that has implemented an EBT programme approved by the competent authority, in accordance with the provisions of that Regulation.
- "EBT practical assessment" means a method for assessing performance that serves to verify the integrated performance of competencies. It takes place in either a simulated or an operational environment.
- "EBT programme" means a pilot assessment and training programme in accordance with point ORO.FC.231 of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#).
- "Gyroplane" means a type of rotorcraft supported in flight by the reactions of the air on up to two rotors, which rotate freely on substantially vertical axes.
[applicable from 18 February 2026 - Regulation (EU) 2025/134]
- "Mixed EBT programme" means an operator's recurrent training and checking programme provided for in point ORO.FC.230 of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#), a portion of which is dedicated to the application of EBT but which does not replace proficiency checks provided for in [Appendix 9](#) to this Annex.
- "Full Flight Simulator" (FFS) means a full size replica of a specific type or make, model and series aircraft flight deck, including the assemblage of all equipment and computer programmes necessary to represent the aircraft in ground and flight operations, a visual system providing an out-of-the-flight deck view, and a force cueing motion system.
- "Flight time":
 - for aeroplanes, touring motor gliders and powered-lift aircraft, it means the total time from the moment an aircraft first moves for the purpose of taking off until the moment it finally comes to rest at the end of the flight;
 - for helicopters, it means the total time from the moment a helicopter's rotor blades start turning until the moment the helicopter finally comes to rest at the end of the flight, and the rotor blades are stopped;
 - for airships, it means the total time from the moment an airship is released from the mast for the purpose of taking off until the moment the airship finally comes to rest at the end of the flight, and is secured on the mast;

- for VTOL-capable aircraft (VCA), it means the total time between the moment the lift and thrust units are powered on for the purpose of taking off until the moment the aircraft finally comes to rest at the end of the flight and the lift and thrust units are powered off.
- "Flight time under Instrument Flight Rules" (IFR) means all flight time during which the aircraft is being operated under the Instrument Flight Rules.
- "Flight Training Device" (FTD) means a full size replica of a specific aircraft type's instruments, equipment, panels and controls in an open flight deck area or an enclosed aircraft flight deck, including the assemblage of equipment and computer software programmes necessary to represent the aircraft in ground and flight conditions to the extent of the systems installed in the device. It does not require a force cueing motion or visual system, except in the case of helicopter FTD levels 2 and 3, where visual systems are required.
- "Flight and Navigation Procedures Trainer" (FNPT) means a training device which represents the flight deck or cockpit environment, including the assemblage of equipment and computer programmes necessary to represent an aircraft type or class in flight operations to the extent that the systems appear to function as in an aircraft.
- "Flown solely by reference to instruments" means that the pilots fly the aircraft without any external visual references, in simulated or actual instrument meteorological conditions (IMC).
- "Helicopter" means a type of rotorcraft supported in flight chiefly by the reactions of the air on up to two power-driven rotors on substantially vertical axes.
- "Instrument flight time" means the time during which a pilot is controlling an aircraft in flight solely by reference to instruments.
- "Instrument ground time" means the time during which a pilot is receiving instruction in simulated instrument flight, in flight simulation training devices (FSTD).
- "Instrument time" means instrument flight time or instrument ground time.
- "Limited panel instrument flight" means attitude interpretation by reference to standby instruments interpretation after the loss of main attitude and heading reference system.
- "Linear operation" means an instrument approach operation in which the maximum tolerable error/deviation from the planned track is expressed in units of length, for instance nautical miles, for cross-track lateral deviation.
- "Line flying under supervision" (LIFUS) means line flying after an approved zero flight time type rating training course or the line flying required by an operational suitability data (OSD) report.
- "LNAV" means Lateral Navigation.
- "LPV" means Localiser Performance with Vertical Guidance.
- "Multi-pilot operation" means an operation requiring at least two pilots using multi-crew cooperation in either a multi-pilot or a single-pilot aircraft.
- "Multi-crew cooperation" (MCC) means the functioning of the flight crew as a team of cooperating members led by the pilot-in-command.
- "Multi-pilot aircraft":
 - for aeroplanes, it means aeroplanes certificated for operation with a minimum crew of at least two pilots,

- or helicopters, airships and powered-lift aircraft, it means an aircraft which is certificated for operation with a minimum crew of at least two pilots or which is required to be operated with at least two pilots in accordance with [Regulation \(EU\) No 965/2012](#).
- "Night" means the period between the end of evening civil twilight and the beginning of morning civil twilight or such other period between sunset and sunrise as may be prescribed by the appropriate authority.
- "OSD" means the operational suitability data established in accordance with Annex I (Part-21) to Regulation (EU) No 748/2012.;
- "Other training devices" (OTD) means training aids other than FSTDs which provide means for training where a complete flight deck environment is not necessary.
- "Performance-Based Navigation (PBN)" means area navigation based on performance requirements for aircraft operating along an ATS route, on an instrument approach procedure or in a designated airspace.
- "Performance criteria" means a simple, evaluative statement on the required outcome of the competency element and a description of the criteria used to judge if the required level of performance has been achieved.
- "Pilot-in-command" (PIC) means the pilot designated as being in command and charged with the safe conduct of the flight.
- "Pilot-in-command under supervision" (PICUS) means a co-pilot performing, under the supervision of the pilot-in-command, the duties and functions of a pilot-in-command.
- "Powered-lift aircraft" means any aircraft deriving vertical lift and in flight propulsion/lift from variable geometry rotors or engines/propulsive devices attached to or contained within the fuselage or wings.
- "Powered sailplane" means a sailplane equipped with one or more engines that has, with engines inoperative, the characteristics of a sailplane.
- "Private pilot" means a pilot who holds a licence which prohibits the piloting of aircraft in operations for which remuneration is given, with the exclusion of instruction or examination activities, as established in this Part.
- "Proficiency check" means the demonstration of skill to revalidate or renew ratings or privileges, and including such oral examination as may be required.
- "Renewal" (of, e.g. a rating or certificate) means the administrative action taken after a rating or certificate has lapsed for the purpose of renewing the privileges of the rating or certificate for a further specified period consequent upon the fulfilment of specified requirements.
- "Revalidation" (of, e.g. a rating or certificate) means the administrative action taken within the period of validity of a rating or certificate which allows the holder to continue to exercise the privileges of a rating or certificate for a further specified period consequent upon the fulfilment of specified requirements.
- "RNP APCH" means a PBN specification used for instrument approach operations.
- "RNP APCH operation down to LNAV minima" means a 2D instrument approach operation for which the lateral guidance is based on GNSS positioning.
- "RNP APCH operation down to LNAV/VNAV minima" means a 3D instrument approach operation for which the lateral guidance is based on GNSS positioning and the vertical guidance is provided either by the Baro VNAV function or by the GNSS positioning including SBAS.

- "RNP APCH operation down to LPV minima" means a 3D instrument approach operation for which both lateral and vertical guidance are based on GNSS positioning including SBAS.
- "RNP AR APCH" means a navigation specification used for instrument approach operations requiring a specific approval.
- "Route sector" means a flight comprising take-off, departure, cruise of not less than 15 minutes, arrival, approach and landing phases.
- "Sailplane" means a heavier-than-air aircraft which is supported in flight by the dynamic reaction of the air against its fixed lifting surfaces, the free flight of which does not depend on an engine.
- "Single-pilot aircraft":
 - for aeroplanes, it means an aircraft certificated for operation by one pilot,
 - for helicopters, airships and powered lift aircraft, it means an aircraft which is certificated for operation by one pilot and which is not required to be operated with at least two pilots by [Regulation \(EU\) No 965/2012](#).
- "Skill test" means the demonstration of skill for a licence or rating issue, including such oral examination as may be required.
- "Solo flight time" means flight time during which a student pilot is the sole occupant of an aircraft.
- "Student pilot-in-command" (SPIC) means a student pilot acting as pilot-in-command on a flight with an instructor where the latter will only observe the student pilot and shall not influence or control the flight of the aircraft.
- "Threat" means events or errors which occur beyond the influence of the flight crew, increase operational complexity and which must be managed to maintain the margin of safety.
- "Threat management" means the process of detecting and responding to the threats with countermeasures which reduce or eliminate the consequences of threats, and mitigate the probability of errors or undesired aircraft states.
- "Three-dimensional (3D) instrument approach operation" means an instrument approach operation using both lateral and vertical navigation guidance.
- "Touring motor glider (TMG)" means, unless otherwise specified following the certification process in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012, a specific class of powered sailplanes that has an integrally mounted, non-retractable engine and a non-retractable propeller. It shall be capable of taking off and climbing under its engine power according to its flight manual.
- "Two-dimensional (2D) instrument approach operation" means an instrument approach operation using lateral navigation guidance only.
- "Type of aircraft" means a categorisation of aircraft requiring a type rating as determined in the operational suitability data established in accordance with Part-21, and which include all aircraft of the same basic design including all modifications thereto except those which result in a change in handling or flight characteristics.
- "Type rating and licence endorsement list" means a list published by the Agency based on the result of the OSD evaluation and containing classes of aeroplanes and types of aircraft for the purpose of flight crew licensing.
- "VNAV" means Vertical Navigation.

GM1 FCL.010 Definitions

ED Decision 2025/002/R

ABBREVIATIONS

The following abbreviations apply to the Acceptable Means of Compliance and Guidance Material to Part-FCL:

A	Aeroplane
AC	Alternating Current
ACAS	Airborne Collision Avoidance System
ADF	Automatic Direction Finding
ADS	Aeronautical Design Standard
AFCS	Automatic Flight Control System
AFM	Aircraft Flight Manual
AGL	Above Ground Level
AIC	Aeronautical Information Circular
AIP	Aeronautical Information Publication
AIRAC	Aeronautical Information Regulation and Control
AIS	Aeronautical Information Services
AMC	Acceptable Means of Compliance
AeMC	Aero-medical Centre
AME	Aero-medical Examiner
AoA	Angle of Attack
AOH	Aircraft Operating Handbook
AOM	Aircraft Operating Manual
APU	Auxiliary Power Unit
As	Airship
ATC	Air Traffic Control
ATIS	Automatic Terminal Information Service
ATO	Approved Training Organisation
ATP	Airline Transport Pilot
ATPL	Airline Transport Pilot Licence
ATS	Air Traffic Service
AUM	All Up Mass
AUPRTA	Airplane Upset Prevention and Recovery Training Aid
B	Balloon
BCAR	British Civil Airworthiness Requirement

BEM	Basic Empty Mass
BIR	Basic instrument rating
BITD	Basic Instrument Training Device
BPL	Balloon Pilot Licence
CAS	Calibrated Airspeed
CAT	Clear Air Turbulence
CB-IR	Competency-based training course for Instrument Rating
CDFA	Constant-Descent Final Approach
CDI	Course Deviation Indicator
CFI	Chief Flight Instructor
CG	Centre of Gravity
CP	Co-pilot
CPL	Commercial Pilot Licence
CRE	Class Rating Examiner
CRI	Class Rating Instructor
CRM	Crew Resource Management
CS	Certification Specification
CTKI	Chief Theoretical Knowledge Instructor
DC	Direct Current
DF	Direction Finding
DME	Distance Measuring Equipment
DPATO	Defined Point After Take-Off
DPBL	Defined Point Before Landing
DR	Dead Reckoning navigation
DTO	Declared Training Organisation
DVE	Degraded Visual Environment
ECQB	European Central Question Bank
EFIS	Electronic Flight Instrument System
EOL	Engine Off Landings
ERPM	Engine Revolutions per Minute
ETA	Estimated Time of Arrival
ETOPS	Extended-range Twin-engine Operation Performance Standard

FAF	Final Approach Fix
FAR	Federal Aviation Regulations
FCL	Flight Crew Licensing
FE	Flight Examiner
F/E	Flight Engineer
FEM	Flight Examiner Manual
FFS	Full-Flight Simulator
FI	Flight Instructor
FIE	Flight Instructor Examiner
FIS	Flight Information Service
FMC	Flight Management Computer
FMS	Flight Management System
FNPT	Flight and Navigation Procedures Trainer
FS	Flight Simulator
FSTD	Flight Simulation Training Device
ft	feet
FTD	Flight Training Device
G	Gravity forces
GLONASS	Global Orbiting Navigation Satellite System
GM	Guidance Material
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
H	Helicopter
HF	High Frequency
HOFCs	High Order Flight Control System
HPA	High-Performance Aeroplane
hrs	Hours
HUMS	Health and Usage Monitoring System
HT	Head of Training
IAS	Indicated Airspeed
ICAO	International Civil Aviation Organization
IGE	In-Ground Effect
IFR	Instrument Flight Rules

ILS	Instrument Landing System
IMC	Instrument Meteorological Conditions
IOS	Instructor Operating Station
IR	Instrument Rating
IRE	Instrument Rating Examiner
IRI	Instrument Rating Instructor
ISA	International Standard Atmosphere
JAR	Joint Aviation Requirements
kg	Kilogram
LAPL	Light Aircraft Pilot Licence
LDP	Landing Decision Point
LMT	Local Mean Time
LO	Learning Objectives
LOC-I	Loss of Control In-flight
LOFT	Line-Orientated Flight Training
m	Meter
MCC	Multi-Crew Cooperation
MCCI	Multi-Crew Cooperation Instructor
ME	Multi-Engine
MEL	Minimum Equipment List
MEP	Multi-Engine Piston
MET	Multi-Engine Turboprop
METAR	Meteorological Aerodrome Report
MI	Mountain Rating Instructor
MP	Multi-Pilot
MPA	Multi-Pilot Aeroplane
MPL	Multi-crew Pilot Licence
MPH	Multi-Pilot Helicopter
MTOM	Maximum Take-Off Mass
NDB	Non-Directional Beacon
NM	Nautical Miles
NOTAM	Notice To Airmen

NOTAR	No Tail Rotor
OAT	Outside Air Temperature
OBS	Omni Bearing Selector
OEI	One Engine Inoperative
OEM	Original Equipment Manufacturer
OGE	Out of Ground Effect
OML	Operational Multi-pilot Limitation
OSL	Operational Safety Pilot Limitation
OTD	Other Training Devices
PAPI	Precision Approach Path Indicator
PBN	Performance-based Navigation
PF	Pilot Flying
PIC	Pilot-In-Command
PICUS	Pilot-In-Command Under Supervision
PL	Powered-lift
PNF	Pilot Not Flying
POM	Pilot Operating Manual
PPL	Private Pilot Licence
QDM	Magnetic Heading (aircraft to station)
QDR	Magnetic Heading (station to aircraft)
QFE	Atmospheric pressure at aerodrome elevation
QNH	Altimeter sub-scale setting to obtain elevation when on the ground
RAIM	Receiver Autonomous Integrity Monitoring
RNAV	Radio Navigation
RPM	Revolutions per Minute
RRPM	Rotor Revolutions per Minute
R/T	Radio-telephony
S	Sailplane
SATCOM	Satellite Communication
SE	Single-Engine
SEP	in the context of aeroplanes: Single-Engine Single-Pilot aeroplane, as defined in Article 2(8c)

SEP	in the context of helicopters: Single-Engine Piston
SET	Single-Engine Turboprop
SFE	Synthetic Flight Examiner
SFI	Synthetic Flight Instructor
SID	Standard Instrument Departure
SIGMET	Significant Meteorological Weather
SLPC	Single Lever Power Control
SOP	Standard Operating Procedure
SP	Single-Pilot
SPA	Single-Pilot Aeroplane
SPH	Single-Pilot Helicopter
SPIC	Student PIC
SPL	Sailplane Pilot Licence
SSR	Secondary Surveillance Radar
STI	Synthetic Training Instructor
TAF	(Terminal Area Forecasts) Aerodrome Forecast
TAS	True Airspeed
TAWS	Terrain Awareness Warning System
TCH	Type Certificate Holder
TDP	Take-off Decision Point
TEM	Threat and Error Management
TK	Theoretical Knowledge
TMG	Touring Motor Glider
TORA	Take-Off Run Available
TODA	Take-Off Distance Available
TR	Type Rating
TRE	Type Rating Examiner
TRI	Type Rating Instructor
UPRT	Upset Prevention and Recovery Training
UTC	Universal Time Coordinated
V	Velocity
VASI	Visual Approach Slope Indicator
VFR	Visual Flight Rules

VHF	Very High Frequency
VMC	Visual Meteorological Conditions
VOLMET	Meteorological Information for Aircraft in-Flight
VOR	VHF Omni-directional Radio Range
ZFTT	Zero Flight Time Training
ZFM	Zero Fuel Mass

GM2 FCL.010 Definitions – lateral and vertical navigation

ED Decision 2016/008/R

Lateral and vertical navigation guidance refers to the guidance provided either by:

- (a) a ground-based radio navigation aid; or
- (b) computer-generated navigation data from ground-based, space-based, self-contained navigation aids or a combination of these.

GM3 FCL.010 Definitions

ED Decision 2019/005/R

UPSET PREVENTION AND RECOVERY TRAINING (UPRT) DEFINITIONS

In the context of UPRT, the following abbreviations apply to the Acceptable Means of Compliance and Guidance Material to Part-FCL:

‘Advanced UPRT’ refers to the advanced UPRT course in accordance with point [FCL.745.A](#).

‘Aeroplane upset’ refers to an undesired aircraft state characterised by unintentional divergences from parameters normally experienced during operations. An aeroplane upset may involve pitch and/or bank angle divergences as well as inappropriate airspeeds for the conditions.

‘Angle of Attack (AoA)’ refers to the angle between the oncoming air, or relative wind, and a defined reference line on the aeroplane or wing.

‘Approach-to-stall’ refers to flight conditions bordered by the stall warning and stall.

‘Basic UPRT’ refers to the UPRT elements and exercises integrated into training courses for the issue of a CPL, MPL or Phases 1 to 3 of the integrated ATP course.

‘Developed upset’ refers to a condition meeting the definition of an aeroplane upset.

‘Developing upset’ refers to any time the aeroplane begins to unintentionally diverge from the intended flight path or airspeed.

‘Energy state’ refers to how much of each kind of energy (kinetic, potential or chemical) the aeroplane has available at any given time.

‘First indication of a stall’ refers to the initial aural, tactile or visual sign of a stall event which can be either naturally or synthetically induced.

‘Flight crew resilience’ refers to the ability of a flight crew member to recognise, absorb and adapt to disruptions.

‘Fidelity level’ refers to the level of realism assigned to each of the defined FSTD features.

‘Flight path’ refers to the trajectory or path of the aeroplane travelling through the air over a given space of time.

‘Flight path management’ refers to active manipulation, using either the aeroplane’s automation or manual handling, to command the aeroplane’s flight controls in order to direct the aeroplane along a desired trajectory.

‘FSTD validation envelope’ refers to the envelope consisting of the following three subdivisions:

(a) Flight test validated region

This is the region of the flight envelope which has been validated with flight test data, typically by comparing the performance of the FSTD against the flight test data through tests incorporated in the qualification test guide (QTG) and other flight test data utilised to further extend the model beyond the minimum requirements. Within this region, there is high confidence that the simulator responds similarly to the aircraft. Note that this region is not strictly limited to what has been tested in the QTG; as long as the aerodynamics mathematical model has been conformed to the flight test results, that portion of the mathematical model can be considered to be within the flight test validated region.

(b) Wind tunnel and/or analytical region

This is the region of the flight envelope for which the FSTD has not been compared to flight test data, but for which there has been wind tunnel testing or the use of other reliable predictive methods (typically by the aircraft manufacturer) to define the aerodynamic model. Any extensions to the aerodynamic model that have been evaluated in accordance with the definition of an exemplar stall model (as described in the stall manoeuvre evaluation section) must be clearly indicated. Within this region, there is moderate confidence that the simulator will respond similarly to the aircraft.

(c) Extrapolated region

This is the region extrapolated beyond the flight test validated and wind tunnel/analytical regions. The extrapolation may be a linear extrapolation, a holding of the last value before the extrapolation began, or some other set of values. Whether this extrapolated data is provided by the aircraft or simulator manufacturer, it is a ‘best guess’ only. Within this region, there is low confidence that the simulator will respond similarly to the aircraft. Brief excursions into this region may still retain a moderate confidence level in FSTD fidelity; however, the instructor should be aware that the FSTD’s response may deviate from that of the actual aircraft.

‘Load factor’ refers to the ratio of a specified load to the weight of the aeroplane, the former being expressed in terms of aerodynamic forces, propulsive forces or ground reactions.

‘Loss of Control In-flight (LOC-I)’ refers to a categorisation of an accident or incident resulting from a deviation from the intended flight path.

‘Manoeuvre-based training’ refers to training that focuses on a single event or manoeuvre in isolation.

‘Negative training’ refers to training which unintentionally introduces incorrect information or invalid concepts, which could actually decrease rather than increase safety.

‘Negative transfer of training’ refers to the application (and ‘transfer’) of what was learned in a training environment (i.e. a classroom, an FSTD) to normal practice, i.e. it describes the degree to which what was learned in training is applied to actual, normal practices. In this context, negative transfer of training refers to the inappropriate generalisation of knowledge and skills to a situation or setting in normal practice that does not equal the training situation or setting.

‘Original Equipment Manufacturer (OEM)’ refers to the original equipment manufacturer of an aircraft or associated parts or equipment or of parts or equipment installed on the basis of a supplemental type certificate (STC).

‘Post-stall regime’ refers to flight conditions at an AoA greater than the critical AoA.

‘Scenario-based training’ refers to training that incorporates manoeuvres into real-world experiences to cultivate practical flying skills in an operational environment.

‘Stall’ refers to loss of lift caused by exceeding the aeroplane’s critical AoA.

Note: A stalled condition can exist at any attitude and airspeed, and may be recognised by continuous stall warning activation accompanied by at least one of the following:

- (a) buffeting, which could be heavy at times;
- (b) lack of pitch authority and/or roll control; and
- (c) inability to arrest the descent rate.

Note: It is possible that in certain conditions the stall warning may not be activated.

‘Stall event’ refers to an occurrence whereby the aeroplane experiences conditions associated with an approach-to-stall or a stall.

‘Stall (event) recovery procedure’ refers to the manufacturer-approved aeroplane-specific stall recovery procedures, such as those contained in the flight crew operations manual (FCOM). If an OEM-approved recovery procedure does not exist, the aeroplane-specific stall recovery procedure developed by the ATO, based on the stall recovery template, may be used.

‘Stall warning’ refers to a natural or synthetic indication provided when approaching a stall that may include one or more of the following indications:

- (a) aerodynamic buffeting (some aeroplanes will buffet more than others);
- (b) reduced roll stability and aileron effectiveness;
- (c) visual or aural cues and warnings;
- (d) reduced elevator (pitch) authority;
- (e) inability to maintain altitude or arrest rate of descent; and
- (f) stick shaker activation (if installed).

Note: A stall warning indicates an immediate need to reduce the AoA.

‘Startle’ refers to the initial, short-term, involuntary physiological and cognitive reactions to an unexpected event that commence the normal human stress response.

‘Stick pusher’ refers to any device that automatically applies a nose-down movement and pitch force to an aeroplane’s control columns to attempt to decrease the aeroplane’s AoA. Device activation may occur before or after aerodynamic stall, depending on the aeroplane type.

Note: A stick pusher is not installed on all aeroplane types.

‘Stick shaker’ refers to a device that automatically vibrates the control column to warn the pilot of an approaching stall.

Note: A stick shaker is not installed on all aeroplane types.

‘Stress (response)’ refers to the response to a threatening event that includes physiological, psychological and cognitive effects. These effects may range from positive to negative and can either enhance or decrease performance.

‘Surprise’ refers to the emotionally based recognition of a difference in what was expected and what is actual.

‘Train-to-proficiency’ refers to approved training designed to achieve end-state performance objectives, providing sufficient assurances that the trained individual is capable of consistently carrying out specific tasks safely and effectively.

Note: In the context of this definition, ‘train-to-proficiency’ can be replaced by ‘training-to-proficiency’.

‘Type-specific UPRT’ refers to UPRT elements and exercises integrated into training courses for the issue of a class or type rating pursuant to Part-FCL or during recurrent or refresher training for a specific aeroplane class or type.

‘Undesired aircraft state’ refers to flight-crew-induced aircraft position or speed deviation, misapplication of controls, or incorrect systems configuration, associated with a reduction in margins of safety.

Note (1): Undesired states can be managed effectively, restoring margins of safety, or flight crew response(s) can induce an additional error, incident or accident.

Note (2): All countermeasures are necessary flight crew actions. However, some countermeasures to threats, errors and undesired aircraft states that flight crew employ are built upon ‘hard’/systemic-based resources provided by the aviation system.

‘Unsafe situation’ refers to a situation which has led to an unacceptable reduction in safety margin.

‘Unusual attitude’ refers to an aircraft in flight intentionally exceeding the parameters normally experienced in line operations or training, as applicable.

‘Incipient spin’ refers to a transient flight condition in the post-stall regime where an initial, uncommanded roll in excess of 45° has resulted from yaw asymmetry during a stall and which, if recovery action is not taken, will lead rapidly to a developing spin. Prompt recovery during this incipient spin stage will normally result in an overall heading change, from pre-stall conditions, of not more than 180°.

‘Developing spin’ refers to a flight condition in the post-stall regime where the aeroplane exhibits abnormal, but varying, rates of yaw and roll, together with changing pitch attitude, following an incipient spin but before the establishment of a developed spin. A developing spin follows an unrecovered incipient spin and will usually persist, in the absence of any recovery action, until a developed spin ensues.

‘Developed spin’ refers to a flight condition in the post-stall regime where the aeroplane has achieved approximately constant pitch attitude, yaw rate and roll rate on a descending flight path. In transition from a stall with significant, persistent yaw, with no recovery action, to attaining a developed spin, the aeroplane is likely to have rolled through at least 540°.

‘FSTD training envelope’ refers to the high and moderate confidence regions of the FSTD validation envelope.

GM4 FCL.010 Definitions

ED Decision 2019/005/R

DEFINITIONS IN GM3 FCL.010 RELATED TO THE POST-STALL REGIME

The definitions for ‘incipient spin’, ‘developing spin’ and ‘developed spin’ in [GM3 FCL.010](#) relate to the post-stall regime in aeroplanes that might typically be used in the context of the advanced UPRT in accordance with point [FCL.745.A](#). The definitions are not intended for application to commercial air transport operations.

GM5 FCL.010 Definitions

ED Decision 2020/005/R

AVAILABLE AND ACCESSIBLE FSTDs

- (a) To determine the availability of an FSTD, the following additional criteria should be taken into account. The FSTD should be:
 - (1) certified by a competent authority within the scope of the Basic Regulation;
 - (2) approved by the competent authority for use within the scope of the Basic Regulation;
 - (3) representative of the operator’s or applicant’s aircraft class or type, and serviceable; and
 - (4) representative of the configuration of the operator’s or applicant’s aircraft.
- (b) To determine the accessibility of an FSTD, the following additional criteria should be taken into account. The FSTD should be:
 - (1) accessible to the instructor or examiner of the applicant;
 - (2) accessible for use within the scope of the candidate’s/operator’s training and checking activities; and
 - (3) accessible to allow normal programming and prevent excessive scheduling disruptions within the operator’s crew roster patterns.
- (c) ‘irrespective of any time considerations’ means that the FSTD may be used at any time during day or night.
- (d) If an FSTD is not available or accessible, mitigating measures to ensure the required level of safety should be agreed with the competent authority before testing or checking the applicant in an aircraft.

GM6 FCL.010 Definitions

ED Decision 2022/014/R

- (a) For the purpose of crediting of a pilot’s prior experience, ‘multi-pilot operation’ can be understood to include State aircraft operations (such as military or search & rescue) where two pilots are required by the applicable national regulations, the relevant operations manual or an equivalent document.
- (b) Operations under Annex VII (Part-NCO) to [Regulation \(EU\) No 965/2012](#) are not under the ‘multi-pilot operation’ category, except for operations at an ATO for the purpose of providing training in multi-pilot operations, in accordance with the training manual of the ATO.

FCL.015 Application and issue, revalidation and renewal of licences, ratings and certificates

Regulation (EU) 2024/2076

- (a) An application for the issue, revalidation or renewal of pilot licences and associated ratings and certificates as well as any amendment thereto shall be submitted to the competent authority in a form and manner established by that authority. The application shall be accompanied by evidence that applicants comply with the requirements for the issue, revalidation or renewal of the licence or certificate as well as associated ratings or endorsements established in this Annex (Part-FCL) and in Annex IV (Part-MED).
- (b) Unless otherwise specified in this Annex, any limitation or extension of the privileges granted by a licence, rating or certificate shall be endorsed in the licence or certificate by the competent authority.
- (c) A person shall not hold at any time more than one licence per category of aircraft issued in accordance with this Part.
- (d) A licence holder shall submit applications in accordance with paragraph (a) to the competent authority designated by the Member State in which his or her licence was issued in accordance with this Annex (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) or with Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), as applicable.
- (e) The holder of a licence that has been issued in accordance with this Annex (Part-FCL) may apply to the competent authority designated by another Member State for a change of competent authority relating to all licences held, as specified in point (d). Holders of a medical certificate issued in accordance with Annex IV (Part-MED) who do not yet hold a pilot licence shall be entitled to make such an application for a change of competent authority relating to all medical records kept by the competent authority.
- (f) For the issue of a licence, rating or certificate the applicant shall apply not later than 6 months after having succeeded at the skill test or assessment of competence.
- (g) Training completed in aircraft or in FSTDs in accordance with Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#) shall be taken into account for the experience and revalidation requirements established in this Annex (Part-FCL).

AMC1 FCL.015 Application for and issue of licences, ratings and certificates

ED Decision 2025/002/R

APPLICATION AND REPORT FORMS

Common application and report forms can be found:

- (a) For skill tests, proficiency checks for the issue, revalidation or renewal of LAPL, PPL, CPL and IR in [AMC1 to Appendix 7](#).
- (b) For training, skill tests or proficiency checks for ATPL, MPL and class and type ratings, in [AMC1 to Appendix 9](#).
- (c) For assessments of competence for instructors, in [AMC5 FCL.935](#).

GM1 FCL.015(a) Application and issue, revalidation and renewal of licences, ratings and certificates

ED Decision 2018/011/R

The application for the addition of the remark on the automatic validation of licences under licence item XIII can be submitted either when the pilot applies for the issue, revalidation or renewal of the licence, or independently at any other time.

FCL.020 Student pilot

Regulation (EU) 2024/2076

- (a) A student pilot shall not fly solo unless he or she meets the following conditions:
 - (1) is authorised to do so and supervised by a flight instructor;
 - (2) prior to receiving the authorisation specified in point (1), has acquired the competence to safely operate the relevant aircraft during the intended solo flight.
- (b) Before his or her first solo flight, a student pilot shall be at least 16 years of age.

GM1 FCL.020(a) Student pilot

ED Decision 2025/002/R

USE OF SEP AEROPLANES WITH DIFFERENT ENGINE TYPES DURING A TRAINING COURSE

A student pilot who undergoes training in an SEP aeroplane with a particular engine type (as specified in [Article 2](#)(8c) of Commission Regulation (EU) No 1178/2011) should be authorised in accordance with point [FCL.020\(a\)](#) to undertake the necessary solo flights in an SEP variant with another engine type only after he or she has acquired the competence to safely operate such SEP variants.

FCL.025 Theoretical knowledge examinations for the issue of licences and ratings

Regulation (EU) 2025/134

- (a) Responsibilities of the applicant
 - (1) Applicants shall take the entire set of theoretical knowledge examinations for a specific licence or rating under the responsibility of the same Member State's competent authority
 - (2) Applicants shall only take the theoretical knowledge examination when recommended by the declared training organisation (DTO) or the approved training organisation (ATO) responsible for their training, once they have completed the appropriate elements of the training course of theoretical knowledge instruction to a satisfactory standard.
 - (3) The recommendation by a DTO or an ATO shall be valid for 12 months. If the applicant has failed to attempt at least one theoretical knowledge examination paper within this period of validity, the need for further training shall be determined by the DTO or the ATO, based on the needs of the applicant.
- (b) Pass standards
 - (1) A pass in a theoretical knowledge examination paper will be awarded to an applicant achieving at least 75 % of the marks allocated to that paper. No penalty marking shall be applied.

- (2) Unless otherwise determined in this Part, an applicant has successfully completed the required theoretical knowledge examination for the appropriate pilot licence or rating if he or she has passed all the required theoretical knowledge examination papers within a period of 18 months counted from the end of the calendar month when the applicant first attempted an examination.
- (3) If an applicant for the ATPL theoretical knowledge examination, or for the issue of a commercial pilot licence (CPL), or an instrument rating (IR) has failed to pass one of the theoretical knowledge examination papers within four attempts, or has failed to pass all papers within either six sittings or within the period mentioned in point (b)(2), he or she shall retake the complete set of theoretical knowledge examination papers.
- (4) If applicants for the issue of a light aircraft pilot licence (LAPL), a private pilot licence (PPL), or a basic instrument rating (BIR) have failed to pass one of the theoretical knowledge examination papers within four attempts or have failed to pass all papers within the period mentioned in point (b)(2), they shall retake the complete set of theoretical knowledge examination papers.

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (4) If applicants for the issue of a light aircraft pilot licence (LAPL), a private pilot licence (PPL), a basic instrument rating (BIR) or a gyroplane pilot licence (GPL) have failed to pass one of the theoretical knowledge examination papers within four attempts or have failed to pass all papers within the period mentioned in point (b)(2), they shall retake the complete set of theoretical knowledge examination papers.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (5) Before retaking the complete set of theoretical knowledge examination papers, applicants shall undertake further training at a DTO or an ATO. The extent and scope of the training needed shall be determined by the DTO or the ATO, based on the needs of the applicants.
- (c) Validity period
- (1) The successful completion of the theoretical knowledge examinations will be valid:
 - (i) for the issue of a light aircraft pilot licence or a private pilot licence, for a period of 24 months;

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (i) for the issue of a light aircraft pilot licence, a private pilot licence or a gyroplane pilot licence, for a period of 24 months;

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (ii) for the issue of a commercial pilot licence, a multi-crew pilot licence or an instrument rating (IR), for a period of 36 months;
- (iii) for the issue of a basic instrument rating (BIR), for an unlimited duration.

The periods in paragraphs (i) and (ii) shall be counted from the day on which the pilots have successfully completed the theoretical knowledge examination, in accordance with paragraph (b)(2).

- (2) The completion of the airline transport pilot licence (ATPL) theoretical knowledge examinations will remain valid for the issue of an ATPL for a period of 7 years from the last validity date of:
- (i) an IR entered in the licence; or
 - (ii) in the case of helicopters, a helicopter's type rating entered in that licence.

GM1 FCL.025 Theoretical knowledge examinations for the issue of licences and ratings

ED Decision 2025/002/R

TERMINOLOGY

The meaning of the following terms used in [FCL.025](#) should be as follows:

- (a) 'Entire set of examinations': an examination in all subjects required by the licence level.
- (b) 'Examination': the demonstration of knowledge in one or more examination papers.
- (c) 'Examination paper': a set of questions, which covers one subject or, in the case of examinations for the BIR, one module, required by the licence level or rating, to be answered by a candidate for examination.
- (d) 'Attempt': a try to pass a specific paper.
- (e) 'Sitting': a period of time established by the competent authority within which a candidate can take an examination. This period should not exceed 10 consecutive days. Only one attempt at each examination paper is allowed in one sitting.

AMC1 FCL.025(a) Theoretical knowledge examinations for the issue of licences and ratings

ED Decision 2025/002/R

ISSUANCE OF THE RECOMMENDATION FOR THEORETICAL KNOWLEDGE EXAMINATION

The ATO or DTO should issue the recommendation as per point [FCL.025\(a\)\(2\)](#) without undue delay, after applicants have finished the relevant part of the theoretical knowledge instruction to a satisfactory standard.

AMC1 FCL.025(a)(2) Theoretical knowledge examinations for the issue of licences and ratings

ED Decision 2018/001/R

COMPLETION OF THE AREA 100 KSA ASSESSMENT BEFORE FINAL EXAMINATION

Before being recommended by an ATO to sit the final examination paper at the first attempt, an applicant for a professional licence should have successfully completed the applicable Area 100 KSA summative assessments and mental maths test at the ATO.

FCL.030 Practical skill test

Regulation (EU) 2020/359

- (a) Before a skill test for the issue of a licence, rating or certificate is taken, the applicant shall have passed the required theoretical knowledge examination, except in the case of applicants undergoing a course of integrated flying training.

In any case, the theoretical knowledge instruction shall always have been completed before the skill tests are taken.

- (b) Except for the issue of an airline transport pilot licence, the applicant for a skill test shall be recommended for the test by the organisation/person responsible for the training, once the training is completed. The training records shall be made available to the examiner.
- (c) For the issue of a BIR, the applicant for a skill test must first complete all training modules and be recommended for the skill test by an ATO. His or her training records shall be made available to the examiner, by the ATO.

FCL.035 Crediting of flight time and theoretical knowledge

Regulation (EU) 2025/134

- (a) Crediting of flight time
- (1) Unless otherwise specified in this Part, flight time to be credited for a licence, rating or certificate shall have been flown in the same category of aircraft for which the licence, rating or certificate is sought.
 - (2) PIC or under instruction.
 - (i) An applicant for a licence, rating or certificate shall be credited in full with all solo, dual instruction or PIC flight time towards the total flight time required for the licence, rating or certificate.
 - (ii) A graduate of an ATP integrated training course is entitled to be credited with up to 50 hours of student pilot-in-command instrument time towards the PIC time required for the issue of the airline transport pilot licence, commercial pilot licence and a multi-engine type or class rating.
 - (iii) A graduate of a CPL/IR integrated training course is entitled to be credited with up to 50 hours of the student pilot-in-command instrument time towards the PIC time required for the issue of the commercial pilot licence and a multi-engine type or class rating.
 - (3) Flight time as co-pilot or PICUS. Unless otherwise determined in this Part, the holder of a pilot licence, when acting as co-pilot or PICUS, is entitled to be credited with all of the co-pilot time towards the total flight time required for a higher grade of pilot licence.
 - (4) All hours flown in aeroplanes or TMGs that are subject to a decision of a Member State taken in accordance with points (a) or (c) of Article 2(8) of [Regulation \(EU\) 2018/1139](#) or that fall within the scope of Annex I to that Regulation shall be credited in full towards fulfilling the flight time requirements of point [FCL.140.A\(a\)\(1\)](#) and point [FCL.740.A\(b\)\(1\)\(ii\)](#) of this Annex, provided that the following conditions are met:
 - (i) the aeroplane or TMG concerned is of the same category and class as the Part-FCL aircraft in respect of which the hours flown are to be credited;

- (ii) in case of training flights with an instructor, the aeroplane or TMG used is subject to an authorisation specified in point [ORA.ATO.135](#) of Annex VII (Part-ORA) or point [DTO.GEN.240](#) of Annex VIII (Part-DTO).
- (5) All hours flown in gyroplanes that fall within the scope of Annex I to Regulation (EU) 2018/1139 and which have a maximum certificated take-off mass of at least 450 kg shall be credited in full towards completing the 12 hours of flight time and 12 take-offs and landings in accordance with point [FCL.240.G\(a\)](#) but shall not be credited towards the requirement in point [FCL.240.G\(a\)\(1\)\(ii\)](#).

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

(b) Crediting of theoretical knowledge

- (1) Applicants that have passed the theoretical knowledge examination for an airline transport pilot licence shall be credited towards the requirements for the theoretical knowledge for the light aircraft pilot licence, the private pilot licence, the commercial pilot licence and, except in the case of helicopters, the IR and the BIR in the same category of aircraft.
- (2) Applicants that have passed the theoretical knowledge examination for a commercial pilot licence shall be credited towards the requirements for the theoretical knowledge for:
 - (i) the light aircraft pilot licence in the same category of aircraft;
 - (ii) the private pilot licence in the same category of aircraft; and
 - (iii) the subject 'communications' for the BIR. This credit shall include the IFR part of the subject 'communications' only if that subject was completed in accordance with point [FCL.310](#), as applicable as of 20 December 2019.
- (3) Holders of an IR or applicants that have passed the IR theoretical knowledge examination for a category of aircraft shall be credited towards the requirements for the theoretical knowledge instruction and examination for:
 - (i) the IR in another category of aircraft; and
 - (ii) the BIR.
- (4) Holders of a pilot licence shall be credited towards the requirements for theoretical knowledge instruction and examination for a licence in another category of aircraft in accordance with Appendix 1 to this Part. This credit also applies to applicants for a pilot licence who have already successfully completed the theoretical knowledge examinations for the issue of that licence in another category of aircraft, as long as the theoretical knowledge examination is within the validity period specified in point [FCL.025\(c\)](#).
- (5) By way of derogation from paragraph (b)(3), holders of an IR(A) who have completed a competency-based modular IR(A) course shall be fully credited towards the requirements for theoretical knowledge instruction and examination for an IR in another category of aircraft only if they have also passed the theoretical knowledge instruction and examination for the IFR part of the course required in accordance with point [FCL.720.A.\(a\)\(2\)\(ii\)\(A\)](#).

- (6) When Appendix 1 includes a credit for the subject Communications, all of the following shall apply:
- (i) such credit shall only be granted for training and examination for that subject if applicants, during previous theoretical knowledge examinations in accordance with point [ARA.FCL.300](#) of Annex VI, have completed either the subject Communications or both subjects VFR communications and IFR communications;
 - (ii) applicants who have completed either only the subject VFR communications or only the subject IFR communications shall complete theoretical knowledge instruction in the subject Communications, the duration of which may be reduced based on an assessment of the applicants by the ATO. For applicants who have completed the subject VFR communications, aspects purely related to VFR communication may be reduced. For applicants who have completed the subject IFR communications, aspects purely related to IFR communication may be reduced. In any case, applicants shall complete the theoretical knowledge examination in the subject Communications. [DxCrossRefBm1631614980](#)

GM1 FCL.035(b)(6)(ii) Crediting of flight time and theoretical knowledge

ED Decision 2025/002/R

CASES OF CREDITS FOR VFR COMMUNICATIONS OR IFR COMMUNICATIONS

Applicants can complete training of reduced duration in the subject Communication in line with point [FCL.035\(b\)\(6\)\(ii\)](#), before completing the theoretical knowledge examinations in subject Communications, in the following cases:

- (a) in the case of applicants who have already passed an ECQB examination for the subject VFR communications:
 - (1) applicants for a CPL who hold a CPL in another category of aircraft;
 - (2) applicants for an ATPL who hold a CPL in the same category of aircraft;
 - (3) applicants for an IR or BIR holding a CPL in the same category of aircraft;
 - (4) applicants for an IR(H) holding an ATPL(H)/VFR;
- (b) in the case of holders of a PPL and an IR who apply for a CPL, an ATPL, or for an IR in another aircraft category, and who have passed the subject IFR communications but who have not passed an ECQB examination for the subject VFR communications.

FCL.040 Exercise of the privileges of licences

Regulation (EU) 2019/1747

The exercise of the privileges granted by a licence shall be dependent upon the validity of the ratings contained therein, if applicable, and of the medical certificate as appropriate to the privileges exercised.

FCL.045 Obligation to carry and present documents

Regulation (EU) 2024/2076

- (a) A valid licence and a valid medical certificate shall always be carried by the pilot when exercising the privileges of the licence.
- (b) The pilot shall also carry a personal identification document containing his/her photo.
- (c) A pilot or a student pilot shall without undue delay present his/her flight time record for inspection upon request by an authorised representative of a competent authority.
- (d) A student pilot shall carry on all solo cross-country flights evidence of the authorisation required by point [FCL.020\(a\)\(1\)](#).
- (e) A pilot intending to fly outside Union territory on an aircraft registered in a Member State other than the one that issued the flight crew licence shall carry, in print or in electronic format, the latest issue of the ICAO attachment, which includes a reference to the ICAO registration number of the agreement that recognises the automatic validation of licences, as well as the list of States which are party to this agreement.

FCL.050 Recording of flight time

Regulation (EU) No 1178/2011

The pilot shall keep a reliable record of the details of all flights flown in a form and manner established by the competent authority.

AMC1 FCL.050 Recording of flight time

ED Decision 2025/002/R

GENERAL

- (a) The record of the flights flown should contain at least the following information:
 - (1) personal details: name(s) and address of the pilot;
 - (2) for each flight:
 - (i) name(s) of PIC;
 - (ii) date of flight;
 - (iii) place and time of departure and arrival;
 - (iv) type, including make, model and variant, and registration of the aircraft;
 - (v) indication if the aircraft is SE or ME, if applicable;
 - (vi) total time of flight;
 - (vii) accumulated total time of flight.
 - (3) for each FSTD session, if applicable:
 - (i) type and qualification number of the training device;
 - (ii) FSTD instruction;
 - (iii) date;
 - (iv) total time of session;
 - (v) accumulated total time.

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- (4) details on pilot function, namely PIC, including solo, SPIC and PICUS time, co-pilot, cruise-relief co-pilot, dual, FI or FE;
 - (5) Operational conditions, namely if the operation takes place at night, or is conducted under instrument flight rules.
- (b) Logging of time:
- (1) PIC flight time:
 - (i) the holder of a licence may log as PIC time all of the flight time during which he or she is the PIC;
 - (ii) the applicant for, or holder of, a pilot licence may log as PIC time all solo flight time, flight time as SPIC, flight time under supervision as well as flight time of successfully completed skill tests, proficiency checks and assessments of competence, provided that such SPIC time, flight time under supervision and tests, checks and assessments are countersigned by the instructor or examiner, as applicable ;
 - (iii) the holder of an instructor certificate may log as PIC all flight time during which he or she acts as an instructor in an aircraft;
 - (iv) the holder of an examiner's certificate may log as PIC all flight time during which he or she occupies a pilot's seat and acts as an examiner in an aircraft;
 - (v) a co-pilot acting as PICUS on an aircraft on which more than one pilot is required under the type certification of the aircraft or as required by operational requirements provided that such PICUS time is countersigned by the PIC;
 - (vi) if the holder of a pilot licence carries out a number of flights upon the same day returning on each occasion to the same place of departure and the interval between successive flights does not exceed 30 minutes, such series of flights may be recorded as a single entry;
 - (vii) where [Regulation \(EU\) No 965/2012](#) requires the pilot to act as PIC under the supervision of another pilot (supervisor), both the pilot and the supervisor may log the flight time as PIC.
 - (2) co-pilot flight time: the holder of a pilot licence occupying a pilot seat as co-pilot may log all flight time as co-pilot flight time on an aircraft on which more than one pilot is required under the type certification of the aircraft, the regulations or the operations manual of the operator under which the flight is conducted;
 - (3) cruise relief co-pilot flight time: a cruise relief co-pilot may log all flight time as co-pilot when occupying a pilot's seat;
 - (4) instruction time: a summary of all time logged by an applicant for a licence or rating as flight instruction, instrument flight instruction, instrument ground time, etc., may be logged if certified by the appropriately rated or authorised instructor from whom it was received;
 - (5) PICUS flight time: provided that the method of supervision is acceptable to the competent authority, a co-pilot may log as PIC flight time flown as PICUS when all the duties and functions of PIC on that flight were carried out in such a way that the intervention of the PIC in the interest of safety was not required.

(c) Format of the record:

- (1) details of flights flown under commercial air transport may be recorded in an electronic format maintained by the operator.

In this case an operator should make the records of all flights operated by the pilot, including differences and familiarisation training, available upon request to the flight crew member concerned;

- (2) for other types of flights in aeroplanes, helicopters and powered-lift aircraft, the pilot should record the details of the flights flown in the following logbook format, which may be kept in electronic format. All data set out in (a) should be included.
- (3) For airships, a suitable format, which may be electronic, should be used. That format should contain the relevant items mentioned in (a) and additional information specific to the type of operation.

PILOT LOGBOOK

Holder's name(s) _____

Holder's licence number _____

HOLDER'S ADDRESS:	
	 <i>[space for address change]</i>
 <i>[space for address change]</i>	 <i>[space for address change]</i>
 <i>[space for address change]</i>	 <i>[space for address change]</i>

1	2		3		4		5			6	7	8	
DATE (dd/mm/yy)	DEPARTURE		ARRIVAL		AIRCRAFT		SINGLE-PILOT TIME		MULTI-PILOT TIME	TOTAL TIME OF FLIGHT	NAME(S) PIC	LANDINGS	
	PLACE	TIME	PLACE	TIME	MAKE, MODEL, VARIANT	REGISTRATION	SE	ME				DAY	NIGHT
						TOTAL THIS PAGE							
						TOTAL FROM PREVIOUS PAGES							
						TOTAL TIME							

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INSTRUCTIONS FOR USE

- (d) [FCL.050](#) requires holders of a pilot licence to record details of all flights flown. This logbook enables pilot licence holders to record flying experience in a manner which will facilitate this process while providing a permanent record of the licence holders flying. Pilots who fly regularly aeroplanes and helicopters or other aircraft categories are recommended to maintain separate logbooks for each aircraft category.
- (e) Flight crew logbook entries should be made as soon as practicable after any flight undertaken. All entries in the flight crew logbook should comply with the following:
 - (1) in case of paper records, they should be made in ink or indelible pencil; or
 - (2) in case of electronic records, they should be made and kept in a way to be readily available at the request of a competent authority, and contain all relevant items that are mentioned in (a), certified by the pilot, and in a format acceptable by the competent authority.
- (f) The particulars of every flight in the course of which the holder of a flight crew licence acts as a member of the operating crew of an aircraft are to be recorded in the appropriate columns using one line for each flight, provided that if an aircraft carries out a number of flights upon the same day returning on each occasion to the same place of departure and the interval between successive flights does not exceed 30 minutes, such series of flights may be recorded as a single entry.
- (g) Flight time is recorded:
 - (1) for aeroplanes, touring motor gliders and powered-lift aircraft, from the moment an aircraft first moves to taking off until the moment it finally comes to rest at the end of the flight;
 - (2) for helicopters, from the moment a helicopter's rotor blades start turning until the moment the helicopter finally comes to rest at the end of the flight, and the rotor blades are stopped;
 - (3) for airships, from the moment an airship is released from the mast to taking off until the moment the airship finally comes to rest at the end of the flight, and is secured on the mast;
- (h) When an aircraft carries two or more pilots as members of the operating crew, one of them shall, before the flight commences, be designated by the operator as the aircraft PIC, according to operational requirements, who may delegate the conduct of the flight to another suitably qualified pilot. All flying carried out as PIC is entered in the logbook as 'PIC'. A pilot flying as 'PICUS' or 'SPIC' enters flying time as 'PIC' but all such entries are to be certified by the PIC or FI in the 'Remarks' column of the logbook.
- (i) Notes on recording of flight time:
 - (1) column 1: enter the date (dd/mm/yy) on which the flight commences;

- (2) column 2 or 3: enter the place of departure and destination either in full or the internationally recognised three or four letter designator. All times should be in UTC;
- (3) column 5: indicate whether the operation was SP or MP, and for SP operation whether SE or ME;

Example:

1	2		3		4		5				6		7	8	
DATE (dd/mm/yy)	DEPARTURE		ARRIVAL		AIRCRAFT		SINGLE PILOT TIME		MULTI- PILOT TIME		TOTAL TIME OF FLIGHT		NAME(S) PIC	LANDINGS	
	PLACE	TIME	PLACE	TIME	MAKE, MODEL, VARIANT	REGISTR ATION	SE	ME						DAY	NIGHT
08/04/12	LFAC	1025	EGBJ	1240	PA34-250	G-SENE		✓			2	15	SELF	1	
09/04/12	EGBJ	1810	EGBJ	1930	C152	G-NONE	✓				1	20	SELF		2
11/04/12	LGW	1645	LAX	0225	B747-400	G-ABCD			9	40	9	40	NAME(S) PIC		1

- (4) column 6: total time of flight may be entered in hours and minutes or decimal notation as desired;
- (5) column 7: enter the name(s) of PIC or SELF as appropriate;
- (6) column 8: indicate the number of landings as pilot flying by day or night;
- (7) column 9: enter flight time undertaken at night or under instrument flight rules if applicable;
- (8) column 10: pilot function time:
 - (i) enter flight time as PIC, SPIC and PICUS as PIC;
 - (ii) all time recorded as SPIC or PICUS is countersigned by the aircraft PIC/PI in the 'remarks' (column 12);
 - (iii) instructor time should be recorded as appropriate and also entered as PIC.

-
- (9) column 11: FSTD:
- (i) for any FSTD enter the type of aircraft and qualification number of the device. For other flight training devices enter either FNPT I or FNPT II as appropriate;
 - (ii) total time of session includes all exercises carried out in the device, including pre- and after-flight checks;
 - (iii) enter the type of exercise performed in the 'remarks' (column 12), for example operator proficiency check, revalidation.
- (10) column 12: the 'remarks' column may be used to record details of the flight at the holder's discretion. The following entries, however, should always be made:
- (i) instrument flight time undertaken as part of the training for a licence or rating;
 - (ii) details of all skill tests and proficiency checks;
 - (iii) name and signature of PIC if the pilot is recording flight time as SPIC or PICUS;
 - (iv) name and signature of the instructor if the flight is part of any of the following:
 - an SEP aeroplane or TMG class rating or single-engine helicopter type rating revalidation;
 - flying activity that is performed by an LAPL holder to comply with recency requirements;
 - (v) in cases where a skill test and proficiency check in single-pilot aircraft is completed in either a combination of single-pilot and multi-pilot operation or solely in multi-pilot operation, information as follows:
 - (A) name and signature of the examiner conducting the skill test or proficiency check or operator proficiency check;
 - (B) the name of the operator in the case of the operator proficiency check; and
 - (C) the form of operation as follows:
 - (1) in the case of a combination of single-pilot operation and multi-pilot operation: 'SP/MP';
 - (2) in the case of solely multi-pilot operation: 'MP only'
 - (vi) 'CRCP' when the pilot acted as cruise-relief co-pilot.

- (j) When each page is completed, accumulated flight time or hours should be entered in the appropriate columns and certified by the pilot in the ‘remarks’ column.

Example:

9				10								11				12
OPERATIONAL CONDITION TIME				PILOT FUNCTION TIME								FSTD SESSION				REMARKS AND ENDORSEMENTS
NIGHT		IFR		PIC		CO-PILOT		DUAL		INSTRUCTOR		DATE (dd/mm/yy)	TYPE	TOTAL TIME OF SESSION		
		2	15	2	15											
1	20			1	20					1	20					Night rating training
												10/04/12	B747-400 (Q1234)	4	10	Revalidation proficiency check
8	10	9	40	9	40											PIC(US): signature of NAME(S) PIC

GM1 FCL.050 Recording of flight time

ED Decision 2022/014/R

EXAMPLES FOR THE USE OF COLUMN 12 'REMARKS' OF THE PILOT LOGBOOK

Pilots may use column 12 'remarks' of the pilot logbook ([AMC1 FCL.050](#)) to record the specific nature of a particular flight in the following cases, since a record of relevant experience might be useful with respect to operational requirements:

- (a) flight time as a pilot in a specialised operation, using the list provided in GM1 NCO.SPEC.100 and GM1 SPO.GEN.100 (see AMC1 ORO.FC.146(e);(f)&(g) of the AMC and GM to Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#) — [ED Decision 2014/017/R](#));
- (b) HEC 1 and 2 cycles, HESLO 1, 2, 3, and 4 cycles, and HHO hoisting cycles by day and night, as pilot flying (see AMC1 SPO.SPEC.HEC.100 and AMC1 SPO.SPEC.HESLO.100 of the AMC and GM to Annex VIII (Part-SPO) to [Regulation \(EU\) No 965/2012](#) — [ED Decision 2014/018/R](#), and point SPA.HHO.130 of Annex V (Part-SPA) to [Regulation \(EU\) No 965/2012](#));
- (c) HHO hours (see point SPA.HHO.130 of Part-SPA);
- (d) offshore landings by day / by night, as pilot flying (see point SPA.HOFO.170 of Part-SPA);
- (e) NVIS flights or hours (see GM1 SPA.NVIS.130 of the AMC and GM to Annex V (Part-SPA) to [Regulation \(EU\) No 965/2012](#) — [ED Decision 2012/019/R](#));
- (f) IFR approaches in the single-pilot role (see point ORO.FC.202 of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#));
- (g) any activity deemed necessary to be recorded for evidence purposes.

Pilots may also use column 12 'remarks' to record IFR approaches exercising PBN privileges and RNP APCH approaches in single-pilot operations (see [Appendix 8](#) to Annex I (Part-FCL)).

FCL.055 Language proficiency

Regulation (EU) 2025/134

- (a) General. Aeroplane, helicopter, powered-lift and airship pilots required to use the radio telephone shall not exercise the privileges of their licences and ratings unless they have a language proficiency endorsement on their licence in either English or the language used for radio communications involved in the flight. The endorsement shall indicate the language, the proficiency level and the validity date, and it shall be obtained in accordance with a procedure established by a competent authority. The minimum acceptable proficiency level is the operational level (Level 4) in accordance with Appendix 2 to this Annex.

[applicable until 17 February 2026 - Regulation (EU) 2020/359]

- (a) General. Aeroplane, helicopter, powered-lift aircraft, airship and gyroplane pilots required to use the radio telephone shall not exercise the privileges of their licences and ratings unless they have a language proficiency endorsement on their licence in either in English or the language used for radio communications involved in the flight. The endorsement shall indicate the language, the proficiency level and the validity date, and it shall be obtained in accordance with a procedure established by a competent authority. The minimum acceptable proficiency level is the operational level (Level 4) in accordance with Appendix 2 to this Annex.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (b) The applicant for a language proficiency endorsement shall demonstrate, in accordance with [Appendix 2](#) to this Annex, at least an operational level of language proficiency both in the use of phraseologies and plain language to an assessor certified by a competent authority or a language-testing body approved by a competent authority as applicable. To do so, the applicant shall demonstrate the ability to:
- (1) communicate effectively in voice-only and in face-to-face situations;
 - (2) communicate on common and work-related topics with accuracy and clarity;
 - (3) use appropriate communicative strategies to exchange messages and to recognise and resolve misunderstandings in a general or work-related context;
 - (4) handle successfully the linguistic challenges presented by a complication or unexpected turn of events which occurs within the context of a routine work situation or communicative task with which they are otherwise familiar; and
 - (5) use a dialect or accent which is intelligible to the aeronautical community.
- (c) Except for pilots who have demonstrated language proficiency at an expert level (level 6) in accordance with Appendix 2 to this Annex, the language proficiency endorsement shall be re-evaluated every:
- (1) 4 years, if the level demonstrated is operational level (level 4); or
 - (2) 6 years, if the level demonstrated is extended level (level 5).
- (d) Specific requirements for holders of an instrument rating (IR). By way of derogation from the paragraphs above, holders of an IR shall have demonstrated the ability to use the English language at the appropriate proficiency level as defined in Appendix 2 to this Annex.
- (e) The demonstration of language proficiency and the use of the English language for IR holders shall be done through a method of assessment established by any competent authority.

AMC1 FCL.055 Language proficiency

ED Decision 2020/005/R

GENERAL

- (a) The method of assessment of the language proficiency level (hereafter: assessment) should be designed to reflect a range of tasks undertaken by pilots but with specific focus on language rather than operational procedures.
- (b) The assessment should determine the applicant's ability to:
- (1) communicate effectively using standard R/T phraseology;
 - (2) deliver and understand messages in plain language in both usual and unusual situations that necessitate departure from standard R/T phraseology.

Note: refer to the 'Manual on the Implementation of ICAO Language Proficiency Requirements' (ICAO Doc 9835), Appendix A Part III and Appendix B for further guidance.

ASSESSMENT

- (c) The assessment may be subdivided into three elements, as follows:
- (1) listening: assessment of comprehension;
 - (2) speaking: assessment of pronunciation, fluency, structure and vocabulary;

- (3) interaction.
- (d) The three elements mentioned above may be combined and they can be covered by using a wide variety of means or technologies.
- (e) Where appropriate, some or all of these elements may be achieved through the use of the R/T testing arrangements.
- (f) When the elements of the testing are assessed separately, the final assessment should be consolidated in the language proficiency endorsement issued by the competent authority.
- (g) The assessment may be conducted during one of the several existing checking or training activities, such as licence issue or rating issue and revalidation, line training, operator line checks or proficiency checks.
- (h) The competent authority may use its own resources in developing or conducting the language proficiency assessment, or may delegate this task to language testing bodies.
- (i) The competent authority should establish an appeal procedure for applicants.
- (j) The holder of a licence should receive a statement containing the level and validity of the language endorsements.
- (k) Where the assessment method for the English language established by the competent authority is equivalent to that established for the assessment of use of the English language in accordance with [AMC2 FCL.055](#), the same assessment may be used for both purposes.

BASIC ASSESSMENT REQUIREMENTS

- (l) The aim of the assessment is to determine the ability of an applicant for a pilot licence or a licence holder to speak and understand the language used for R/T communications.
 - (1) The assessment should determine the ability of the applicant to use both:
 - (i) standard R/T phraseology;
 - (ii) plain language, in situations when standardised phraseology cannot serve an intended transmission.
 - (2) The assessment should include:
 - (i) voice-only and face-to-face situations;
 - (ii) common, concrete and work-related topics for pilots.
 - (3) The applicants should demonstrate their linguistic ability in dealing with an unexpected turn of events, and in solving apparent misunderstandings.
 - (4) The assessment should determine the applicant's speaking and listening abilities. Indirect assessments, of grammatical knowledge, reading and writing, are not appropriate.
 - (5) The assessment should determine the language skills of the applicant in the following areas:
 - (i) pronunciation:
 - (A) the extent to which the pronunciation, stress, rhythm and intonation are influenced by the applicant's first language or national variations;
 - (B) how much they interfere with ease of understanding.
 - (ii) structure:

- (A) the ability of the applicant to use both basic and complex grammatical structures;
- (B) the extent to which the applicant's errors interfere with the meaning.
- (iii) vocabulary:
 - (A) the range and accuracy of the vocabulary used;
 - (B) the ability of the applicant to paraphrase successfully when lacking vocabulary.
- (iv) fluency:
 - (A) tempo;
 - (B) hesitancy;
 - (C) rehearsed versus spontaneous speech;
 - (D) use of discourse markers and connectors.
- (v) comprehension:
 - (A) on common, concrete and work-related topics;
 - (B) when confronted with a linguistic or situational complication or an unexpected turn of events.

Note: the accent or variety of accents used in the test material should be sufficiently intelligible for an international community of users.

- (vi) interactions:
 - (A) quality of response (immediate, appropriate, and informative);
 - (B) the ability to initiate and maintain exchanges:
 - (a) on common, concrete and work-related topics;
 - (b) when dealing with an unexpected turn of events.
 - (C) the ability to deal with apparent misunderstandings by checking, confirming or clarifying.

Note: the assessment of the language skills in the areas mentioned above is conducted using the rating scale in [AMC2 FCL.055](#).

ASSESSORS

- (m) It is essential that the persons responsible for language proficiency assessment ('assessors') are suitably trained and qualified. They should be either aviation specialists (for example current or former flight crew members or air traffic controllers), or language specialists with additional aviation related training. An alternative approach would be to form an assessment team consisting of an operational expert and a language expert.
 - (1) The assessors should be trained on the specific requirements of the assessment.
 - (2) The assessors should not test applicants to whom they have given language training.

CRITERIA FOR THE ACCEPTABILITY OF LANGUAGE-TESTING BODIES

- (n) To ensure an impartial assessment process, the language assessment should be independent of the language training.

- (1) To be accepted, the language-testing bodies should demonstrate:
 - (i) appropriate management and staffing;
 - (ii) quality system established and maintained to ensure compliance with, and adequacy of, assessment requirements, standards and procedures.
- (2) The quality system established by a language-testing body should address the following:
 - (i) management;
 - (ii) policy and strategy;
 - (iii) processes;
 - (iv) the relevant provisions of ICAO or Part-FCL, standards and assessment procedures;
 - (v) organisational structure;
 - (vi) responsibility for the development, establishment and management of the quality system;
 - (vii) documentation;
 - (viii) quality assurance programme;
 - (ix) human resources and training (initial and recurrent);
 - (x) assessment requirements;
 - (xi) customer satisfaction.
- (3) The assessment documentation and records should be kept for a period of time determined by the competent authority and made available to this competent authority, on request.
- (4) The assessment documentation should include at least the following:
 - (i) assessment objectives;
 - (ii) assessment layout, time scale, technologies used, assessment samples, voice samples;
 - (iii) assessment criteria and standards (at least for the levels 4, 5 and 6 of the rating scale mentioned in [AMC2 FCL.055](#));
 - (iv) documentation demonstrating the assessment validity, relevance and reliability;
 - (v) assessment procedures and responsibilities:
 - (A) preparation of individual assessment;
 - (B) administration: location(s), identity check and invigilation, assessment discipline, confidentiality or security;
 - (C) reporting and documentation provided to the competent authority or to the applicant, including sample certificate;
 - (D) retention of documents and records.

Note: refer to the 'Manual on the Implementation of ICAO Language Proficiency Requirements' (ICAO Doc 9835) for further guidance.

AMC2 FCL.055 Language proficiency

ED Decision 2011/016/R

RATING SCALE

The following table describes the different levels of language proficiency:

LEVEL	PRONUNCIATION	STRUCTURE	VOCABULARY	FLUENCY	COMPREHENSION	INTERACTIONS
Expert (Level 6)	Pronunciation, stress, rhythm, and intonation, though possibly influenced by the first language or regional variation, almost never interfere with ease of understanding.	Both basic and complex grammatical structures and sentence patterns are consistently well controlled.	Vocabulary range and accuracy are sufficient to communicate effectively on a wide variety of familiar and unfamiliar topics. Vocabulary is idiomatic, nuanced and sensitive to register.	Able to speak at length with a natural, effortless flow. Varies speech flow for stylistic effect, for example to emphasise a point. Uses appropriate discourse markers and connectors spontaneously.	Comprehension is consistently accurate in nearly all contexts and includes comprehension of linguistic and cultural subtleties.	Interacts with ease in nearly all situations. Is sensitive to verbal and non-verbal cues, and responds to them appropriately.
Extended (Level 5)	Pronunciation, stress, rhythm, and intonation, though influenced by the first language or regional variation, rarely interfere with ease of understanding.	Basic grammatical structures and sentence patterns are consistently well controlled. Complex structures are attempted but with errors which sometimes interfere with meaning.	Vocabulary range and accuracy are sufficient to communicate effectively on common, concrete, and work-related topics. Paraphrases consistently and successfully. Vocabulary is sometimes idiomatic.	Able to speak at length with relative ease on familiar topics, but may not vary speech flow as a stylistic device. Can make use of appropriate discourse markers or connectors.	Comprehension is accurate on common, concrete, and work-related topics and mostly accurate when the speaker is confronted with a linguistic or situational complication or an unexpected turn of events. Is able to comprehend a range of speech varieties (dialect or accent) or registers.	Responses are immediate, appropriate, and informative. Manages the speaker or listener relationship effectively.

LEVEL	PRONUNCIATION Assumes a dialect or accent intelligible to the aeronautical community	STRUCTURE Relevant grammatical structures and sentence patterns are determined by language functions appropriate to the task	VOCABULARY	FLUENCY	COMPREHENSION	INTERACTIONS
Operational (Level 4)	Pronunciation, stress, rhythm, and intonation are influenced by the first language or regional variation but only sometimes interfere with ease of understanding.	Basic grammatical structures and sentence patterns are used creatively and are usually well controlled. Errors may occur, particularly in unusual or unexpected circumstances, but rarely interfere with meaning.	Vocabulary range and accuracy are usually sufficient to communicate effectively on common, concrete, and work-related topics. Can often paraphrase successfully when lacking vocabulary particularly in unusual or unexpected circumstances.	Produces stretches of language at an appropriate tempo. There may be occasional loss of fluency on transition from rehearsed or formulaic speech to spontaneous interaction, but this does not prevent effective communication. Can make limited use of discourse markers and connectors. Fillers are not distracting.	Comprehension is mostly accurate on common, concrete, and work-related topics when the accent or variety used is sufficiently intelligible for an international community of users. When the speaker is confronted with a linguistic or situational complication or an unexpected turn of events, comprehension may be slower or require clarification strategies.	Responses are usually immediate, appropriate, and informative. Initiates and maintains exchanges even when dealing with an unexpected turn of events. Deals adequately with apparent misunderstandings by checking, confirming, or clarifying.
Pre-Operational (Level 3)	Pronunciation, stress, rhythm, and intonation are influenced by the first language or regional variation and frequently interfere with ease of understanding.	Basic grammatical structures and sentence patterns associated with predictable situations are not always well controlled. Errors frequently interfere with meaning.	Vocabulary range and accuracy are often sufficient to communicate effectively on common, concrete, and work-related topics but range is limited and the word choice often inappropriate. Is often unable to paraphrase	Produces stretches of language, but phrasing and pausing are often inappropriate. Hesitations or slowness in language processing may prevent effective communication. Fillers are sometimes distracting.	Comprehension is often accurate on common, concrete, and work-related topics when the accent or variety used is sufficiently intelligible for an international community of users. May fall to understand a linguistic or situational complication or an unexpected turn of events.	Responses are sometimes immediate, appropriate, and informative. Can initiate and maintain exchanges with reasonable ease on familiar topics and in predictable situations. Generally inadequate when

LEVEL	PRONUNCIATION Assumes a dialect or accent intelligible to the aeronautical community	STRUCTURE Relevant grammatical structures and sentence patterns are determined by language functions appropriate to the task	VOCABULARY	FLUENCY	COMPREHENSION	INTERACTIONS
			successfully when lacking vocabulary.			dealing with an unexpected turn of events.
Elementary (Level 2)	Pronunciation, stress, rhythm, and intonation are heavily influenced by the first language or regional variation and usually interfere with ease of understanding.	Shows only limited control of few simple memorised grammatical structures and sentence patterns.	Limited vocabulary range consisting only of isolated words and memorised phrases.	Can produce very short, isolated, memorised utterances with frequent pausing and a distracting use of fillers to search for expressions and articulate less familiar words.	Comprehension is limited to isolated, memorised phrases when they are carefully and slowly articulated.	Response time is slow, and often inappropriate. Interaction is limited to simple routine exchanges.
Pre-Elementary (Level 1)	Performs at a level below the elementary level.	Performs at a level below the elementary level.	Performs at a level below the elementary level.	Performs at a level below the elementary level.	Performs at a level below the elementary level.	Performs at a level below the elementary level.

Note: operational Level (Level 4) is the minimum required proficiency level for R/T communication.

Levels 1 through 3 describe pre-elementary, elementary and pre-operational levels of language proficiency respectively, all of which describe a level below the language proficiency requirement.

Levels 5 and 6 describe extended and expert levels at levels of proficiency more advanced than the minimum required standard.

AMC3 FCL.055 Language proficiency

ED Decision 2011/016/R

SPECIFIC REQUIREMENTS FOR HOLDERS OF AN IR

USE OF ENGLISH LANGUAGE

- (a) The requirement of [FCL.055\(d\)](#) includes the ability to use the English language for the following purposes:
- (1) flight: R/T relevant to all phases of flight, including emergency situations.
 - (2) ground: all information relevant to the accomplishment of a flight:
 - (i) be able to read and demonstrate an understanding of technical manuals written in English, for example an operations manual, a helicopter flight manual, etc.;
 - (ii) pre-flight planning, weather information collection, NOTAMs, ATC flight plan, etc.;
 - (iii) use of all aeronautical en-route, departure and approach charts and associated documents written in English.
 - (3) communication: be able to communicate with other crew members in English during all phases of flight, including flight preparation.
- (b) Alternatively, the items in (a) above may be demonstrated:
- (1) by having passed a specific examination given by the competent authority after having undertaken a course of training enabling the applicant to meet all the objectives listed in (a) above; or
 - (2) the item in (a)(1) above is considered to be fulfilled, if the applicant has passed an IR, MPL or ATPL skill test and proficiency check during which the two-way R/T communication is performed in English;
 - (3) the item in (a)(2) above is considered to be fulfilled if the applicant has graduated from an IR, MPL or ATP course given in English or if he or she has passed the theoretical IR or ATPL examination in English;
 - (4) the item in (a)(3) above is considered to be fulfilled, if the applicant for or the holder of an IR has graduated from an MCC course given in English and is holding a certificate of satisfactory completion of that course or if the applicant has passed a MP skill test and proficiency check for the issue of a class or type rating during which the two-way R/T communication and the communication with other crew members are performed in English.
- (c) Where the examination methods referred to above are equivalent to those established for the language proficiency requirements in accordance with [AMC1 FCL.055](#), the examination may be used to issue a language proficiency endorsement.

FCL.060 Recent experience

Regulation (EU) 2025/134

- (b) Aeroplanes, helicopters, powered-lift aircraft, airships and VTOL-capable aircraft (VCA). A pilot shall not operate an aircraft in commercial air transport or to carry passengers:

[applicable until 17 February 2026 - Regulation (EU) 2024/1111]

- (b) Aeroplanes, helicopters, powered-lift aircraft, airships, VTOL-capable aircraft (VCA) and gyroplanes. A pilot shall not operate an aircraft in commercial air transport or for carrying passengers:

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (1) as PIC or co-pilot unless he or she has carried out, in the preceding 90 days, at least 3 take-offs, approaches and landings as a pilot flying in an aircraft of the same type or class or an FFS representing that type or class. The 3 take-offs and landings shall be performed in either multi-pilot or single-pilot operations, depending on the privileges held by the pilot; and
 - (2) as PIC at night unless he/she:
 - (i) has carried out in the preceding 90 days at least 1 take-off, approach and landing at night as a pilot flying in an aircraft of the same type or class or an FFS representing that type or class; or
 - (ii) holds an IR;
 - (3) as cruise relief co-pilot unless he or she has completed, within the preceding 90 days, either of the following:
 - (i) at least three sectors as cruise relief co-pilot in the relevant type;
 - (ii) training the content of which is determined by the operator and that comprises at least three sectors as cruise relief co-pilot in an FSTD representing the relevant type;
 - (iii) recurrent training in accordance with Subpart FC (ORO.FC) of Annex III (Part-ORO) to Regulation (EU) No 965/2012;
 - (iv) a proficiency check for cruise relief co-pilots in accordance with Appendix 9.
 - (4) When a pilot has the privilege to operate more than one type of aeroplane with similar handling and operation characteristics, the 3 take-offs, approaches and landings required in (1) may be performed as defined in the operational suitability data established in accordance with Part-21.
 - (5) When a pilot has the privilege to operate more than one type of non-complex helicopter with similar handling and operation characteristics, as defined in the operational suitability data established in accordance with Part-21, the 3 take-offs, approaches and landings required in (1) may be performed in just one of the types, provided that the pilot has completed at least 2 hours of flight in each of the types of helicopter, during the preceding 6 months.
- (c) Specific requirements for commercial air transport:
- (1) In the case of commercial air transport, the 90-day period prescribed in subparagraphs (b)(1) and (2) above may be extended up to a maximum of 120 days, as long as the pilot undertakes line flying under the supervision of a type rating instructor or examiner.
 - (2) If the pilot does not comply with the requirement in point (1), he or she shall complete a training flight with an instructor qualified in accordance with Subpart J to instruct for that aircraft type. The training flight shall be performed in the aircraft or an FFS of the aircraft type to be used, and shall include at least the requirements described in points (b)(1) and (2) before he or she can exercise his/her privileges.

AMC1 FCL.060(b)(1) Recent experience

ED Decision 2011/016/R

When a pilot needs to carry out one or more flights with an instructor or an examiner to comply with the requirement of [FCL.060\(b\)\(1\)](#) before the pilot can carry passengers, the instructor or examiner on board those flights will not be considered as a passenger.

GM1 FCL.060(b)(1) Recent experience

ED Decision 2025/002/R

AEROPLANES, HELICOPTERS, POWERED-LIFT AND AIRSHIPS

If a pilot or a PIC operates under the supervision of an instructor to comply with the required three take-offs, approaches and landings, no passengers should be on board.

AMC1 FCL.060(b)(5) Recent experience

ED Decision 2025/002/R

NON-COMPLEX HELICOPTERS

Grouping of non-complex helicopters with similar handling and operational characteristics:

- (a) Group 1: Bell 206/206L, Bell 407;
- (b) Group 2: Hughes 369, MD 500N, MD 520N, MD 600;
- (c) Group 3: SA 341/342, EC 120;
- (d) Group 4: SA 313/318, SA 315/316/319, AS 350, EC 130;
- (e) Group 5: all types listed in [AMC1 FCL.740.H\(b\)](#) and R 22 and R 44.

FCL.065 Curtailment of privileges of licence holders aged 60 years or more in commercial air transport

Regulation (EU) 2024/2076

- (a) Age 60–64. Aeroplanes and helicopters. The holder of a pilot licence who has attained the age of 60 years shall not act as a pilot of an aircraft engaged in commercial air transport except as a member of a multi-pilot crew. By way of derogation, such holders shall be entitled to act as pilots of an aircraft engaged in single-pilot helicopter emergency medical service operations in accordance with Regulation (EU) No 965/2012, provided that they comply with the applicable requirements specified for that purpose in that Regulation and in Annex IV (Part- MED) to this Regulation.
- (b) Age 65. Holders of a pilot licence who has attained the age of 65 years shall not act as a pilot of an aircraft that is engaged in commercial air transport.

FCL.070 Revocation, suspension and limitation of licences, ratings and certificates

Regulation (EU) No 1178/2011

- (a) Licences, ratings and certificates issued in accordance with this Part may be limited, suspended or revoked by the competent authority when the pilot does not comply with the requirements of this Part, Part-Medical or the applicable operational requirements, in accordance with the conditions and procedures laid down in Part-ARA.

- (b) When the pilot has his/her licence suspended or revoked, he/she shall immediately return the licence or certificate to the competent authority.

SUBPART B – LIGHT AIRCRAFT PILOT LICENCE – LAPL

SECTION 1 – COMMON REQUIREMENTS

FCL.100 LAPL – Minimum age

Regulation (EU) 2020/359

Applicants for the LAPL for aeroplanes or helicopters shall be at least 17 years old.

FCL.105 LAPL – Privileges and conditions

Regulation (EU) No 1178/2011

- (a) General. The privileges of the holder of an LAPL are to act without remuneration as PIC in non-commercial operations on the appropriate aircraft category.
- (b) Conditions. Applicants for the LAPL shall have fulfilled the requirements for the relevant aircraft category and, when applicable, for the class or type of aircraft used in the skill test.

FCL.110 LAPL – Crediting for the same aircraft category

Regulation (EU) No 1178/2011

- (a) Applicants for an LAPL who have held another licence in the same category of aircraft shall be fully credited towards the requirements of the LAPL in that category of aircraft.
- (b) Without prejudice to the paragraph above, if the licence has lapsed, the applicant shall have to pass a skill test in accordance with [FCL.125](#) for the issue of an LAPL in the appropriate aircraft category.

FCL.115 LAPL – Training course

Regulation (EU) 2024/2076

- (a) Applicants for an LAPL shall complete a training course at a DTO or an ATO. That training course shall include:
 - (1) theoretical knowledge and flight instruction appropriate to the privileges of the LAPL applied for;
 - (2) additionally, for the LAPL(H), the mandatory training elements for the relevant helicopter type as defined in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012.
- (b) Theoretical knowledge instruction and flight instruction may be completed at a DTO or at an ATO different from the one where applicants have commenced their training.
- (c) For the training for the SEP aeroplanes-sea class privilege, the elements of Section B (Specific requirements for the aeroplane category), point 7 (Class ratings — sea), of Appendix 9 shall be considered.
- (d) Applicants for an LAPL may receive credits for previous PPL training they have undergone in accordance with Subpart C in the same aircraft category, based on an assessment of the applicant by the ATO or the DTO that is responsible for the LAPL training course. In any case,

applicants shall comply with the experience requirements set out in points [FCL.110.A\(a\)](#) or [FCL.110.H\(a\)](#), as applicable.

AMC1 FCL.115 LAPL – Training course

ED Decision 2025/002/R

FLIGHT INSTRUCTION FOR THE LAPL (A)

(a) Entry to training

Before being accepted for training an applicant should be informed that the appropriate medical certificate must be obtained before solo flying is permitted.

(b) Flight instruction

(1) The LAPL (A) flight instruction syllabus should take into account the principles of threat and error management and also cover:

- (i) pre-flight operations, including mass and balance determination, aircraft inspection and servicing;
- (ii) aerodrome and traffic pattern operations, collision avoidance precautions and procedures;
- (iii) control of the aircraft by external visual reference;
- (iv) flight at critically low air speeds, recognition of, and recovery from, incipient and full stalls;
- (v) flight at critically high air speeds, recognition of, and recovery from, spiral dive;
- (vi) normal and crosswind take-offs and landings;
- (vii) maximum performance (short field and obstacle clearance) take-offs, short-field landings;
- (viii) cross-country flying using visual reference, dead reckoning and radio navigation aids;
- (ix) emergency operations, including simulated aeroplane equipment malfunctions;
- (x) operations to, from and transiting controlled aerodromes, compliance with air traffic services procedures and communication procedures.

(2) Before allowing applicants to undertake their first solo flight, the FI should ensure that the applicants can use R/T communication can operate the required systems and equipment.

(c) Syllabus of flight instruction

(1) The numbering of exercises should be used primarily as an exercise reference list and as a broad instructional sequencing guide; therefore the demonstrations and practices need not necessarily be given in the order listed. The actual order and content will depend upon the following interrelated factors:

- (i) the applicant's progress and ability;
- (ii) the weather conditions affecting the flight;
- (iii) the flight time available;
- (iv) instructional technique considerations;

- (v) the local operating environment;
- (vi) applicability of the exercises to the aeroplane or TMG type.
- (2) Each of the exercises involves the need for the applicant to be aware of the needs of good airmanship and look-out, which should be emphasised at all times.
 - (i) Exercise 1a: Familiarisation with the aeroplane or TMG:
 - (A) characteristics of the aeroplane or TMG;
 - (B) cockpit layout;
 - (C) systems;
 - (D) checklists, drills and controls.
 - (ii) Exercise 1b: Emergency drills:
 - (A) action if fire on the ground and in the air;
 - (B) engine cabin and electrical system fire;
 - (C) systems failure;
 - (D) escape drills, location and use of emergency equipment and exits.
 - (iii) Exercise 2: Preparation for and action after flight:
 - (A) flight authorisation and aeroplane or TMG acceptance;
 - (B) serviceability documents;
 - (C) equipment required, maps, etc.;
 - (D) external checks;
 - (E) internal checks;
 - (F) harness, seat or rudder panel adjustments;
 - (G) starting and warm-up checks;
 - (H) power checks;
 - (I) running down system checks and switching off the engine;
 - (J) parking, security and picketing (for example tie down);
 - (K) completion of authorisation sheet and serviceability documents.
 - (iv) Exercise 3: Air experience: flight exercise.
 - (v) Exercise 4: Effects of controls:
 - (A) primary effects when laterally level and when banked;
 - (B) further effects of aileron and rudder;
 - (C) effects of:
 - (a) air speed;
 - (b) slipstream;
 - (c) power;
 - (d) trimming controls;

- (e) flaps;
 - (f) other controls, as applicable.
- (D) operation of:
 - (a) mixture control;
 - (b) carburettor heat;
 - (c) cabin heating or ventilation.
- (vi) Exercise 5a: Taxiing:
 - (A) pre-taxi checks;
 - (B) starting, control of speed and stopping;
 - (C) engine handling;
 - (D) control of direction and turning;
 - (E) turning in confined spaces;
 - (F) parking area procedure and precautions;
 - (G) effects of wind and use of flying controls;
 - (H) effects of ground surface;
 - (I) freedom of rudder movement;
 - (J) marshalling signals;
 - (K) instrument checks;
 - (L) air traffic control procedures.
- (vii) Exercise 5b: Emergencies: brake and steering failure.
- (viii) Exercise 6: Straight and level:
 - (A) at normal cruising power, attaining and maintaining straight and level flight;
 - (B) flight at critically high air speeds;
 - (C) demonstration of inherent stability;
 - (D) control in pitch, including use of trim;
 - (E) lateral level, direction and balance, trim;
 - (F) at selected air speeds (use of power);
 - (G) during speed and configuration changes;
 - (H) use of instruments for precision.
- (ix) Exercise 7: Climbing:
 - (A) entry, maintaining the normal and max rate climb, levelling off;
 - (B) levelling off at selected altitudes;
 - (C) en-route climb (cruise climb);
 - (D) climbing with flap down;
 - (E) recovery to normal climb;

- (F) maximum angle of climb;
- (G) use of instruments for precision.
- (x) Exercise 8: Descending:
 - (A) entry, maintaining and levelling off;
 - (B) levelling off at selected altitudes;
 - (C) glide, powered and cruise descent (including effect of power and air speed);
 - (D) side slipping (on suitable types);
 - (E) use of instruments for precision flight.
- (xi) Exercise 9: Turning:
 - (A) entry and maintaining medium level turns;
 - (B) resuming straight flight;
 - (C) faults in the turn (in correct pitch, bank and balance);
 - (D) climbing turns;
 - (E) descending turns;
 - (F) slipping turns (for suitable types);
 - (G) turns onto selected headings, use of gyro heading indicator and compass;
 - (H) use of instruments for precision.
- (xii) Exercise 10a: Slow flight: Note: the objective is to improve the student's ability to recognise inadvertent flight at critically low speeds and provide practice in maintaining the aeroplane or TMG in balance while returning to normal air speed.
 - (A) safety checks;
 - (B) introduction to slow flight;
 - (C) controlled flight down to critically slow air speed;
 - (D) application of full power with correct attitude and balance to achieve normal climb speed.
- (xiii) Exercise 10b: Stalling:
 - (A) safety checks;
 - (B) symptoms;
 - (C) recognition;
 - (D) clean stall and recovery without power and with power;
 - (E) recovery when a wing drops;
 - (F) approach to stall in the approach and in the landing configurations, with and without power and recovery at the incipient stage.
- (xiv) Exercise 11: Spin avoidance:

Note: The limitations of the training aircraft as set out in the AFM (manoeuvre limitations, mass and balance calculations) need to be carefully considered.

- (A) safety checks;
 - (B) stalling and approaching the incipient spin stage (stall with asymmetric condition, with non-centred slip indicator);
 - (C) instructor-induced distractions during the stall.
- (xv) Exercise 12: Take-off and climb to downwind position:
- (A) pre-take-off checks;
 - (B) into wind take-off;
 - (C) safeguarding the nose wheel (if applicable);
 - (D) crosswind take-off;
 - (E) drills during and after take-off;
 - (F) short take-off and soft field procedure or techniques including performance calculations;
 - (G) noise abatement procedures.
- (xvi) Exercise 13: Circuit, approach and landing:
- (A) circuit procedures, downwind and base leg;
 - (B) powered approach and landing;
 - (C) safeguarding the nose wheel (if applicable);
 - (D) effect of wind on approach and touchdown speeds and use of flaps;
 - (E) crosswind approach and landing;
 - (F) glide approach and landing;
 - (G) short landing and soft field procedures or techniques;
 - (H) flapless approach and landing;
 - (I) wheel landing (tail wheel aeroplanes);
 - (J) missed approach and go-around;
 - (K) noise abatement procedures.
- (xvii) Exercise 12/13: Emergencies:
- (A) abandoned take-off;
 - (B) simulated engine failure after take-off;
 - (C) rejected landing and go-around;
 - (D) missed approach;
 - (E) simulated engine failure and restart procedures in-flight.
- Note: In the interests of safety, it will be necessary for pilots trained on nose wheel aeroplanes or TMGs to undergo dual conversion training before flying tail wheel aeroplanes or TMGs, and vice versa.
- (xviii) Exercise 14: First solo:
- (A) instructor's briefing;

- (B) use of required equipment;
- (C) observation of flight and debriefing by instructor.

Note: During flights immediately following the solo circuit, the consolidation of the following should be revised:

- (1) procedures for leaving and rejoining the circuit;
 - (2) the local area, restrictions, map reading;
 - (3) use of radio aids for homing;
 - (4) turns using magnetic compass, compass errors.
- (xix) Exercise 15: Advanced turning:
- (A) steep turns (45 °), level and descending;
 - (B) stalling in the turn and recovery;
 - (C) recoveries from unusual attitudes, including spiral dives.
- (xx) Exercise 16: Forced landing without power:
- (A) forced landing procedure;
 - (B) choice of landing area, provision for change of plan;
 - (C) gliding distance;
 - (D) descent plan;
 - (E) key positions;
 - (F) engine cooling;
 - (G) engine failure checks;
 - (H) use of radio;
 - (I) base leg;
 - (J) final approach;
 - (K) landing;
 - (L) actions after landing.
- (xxi) Exercise 17: Precautionary landing:
- (A) full procedure away from aerodrome to break-off height;
 - (B) occasions necessitating a precautionary landing;
 - (C) in-flight conditions;
 - (D) landing area selection:
 - (a) normal aerodrome;
 - (b) disused aerodrome;
 - (c) ordinary field.
 - (E) circuit and approach;
 - (F) actions after landing.

(xxii) Exercise 18a: Navigation:

(A) flight planning:

- (a) weather forecast and actuals;
- (b) map selection and preparation:
 - (1) choice of route;
 - (2) airspace structure;
 - (3) safety altitudes.
- (c) calculations:
 - (1) magnetic heading(s) and time(s) en-route;
 - (2) fuel consumption;
 - (3) mass and balance;
 - (4) mass and performance.
- (d) flight information:
 - (1) NOTAMs, etc.;
 - (2) radio frequencies;
 - (3) selection of alternate aerodromes.
- (e) aeroplane or TMG documentation;
- (f) notification of the flight:
 - (1) pre-flight administrative procedures;
 - (2) flight plan form.

(B) departure:

- (a) organisation of cockpit workload;
- (b) departure procedures:
 - (1) altimeter settings;
 - (2) ATC liaison in regulated airspace;
 - (3) setting heading procedure;
 - (4) noting of ETAs.
- (c) maintenance of altitude and heading;
- (d) revisions of ETA and heading;
- (e) log keeping;
- (f) use of radio;
- (g) minimum weather conditions for continuation of flight;
- (h) in-flight decisions;
- (i) transiting controlled or regulated airspace;
- (j) diversion procedures;

- (k) uncertainty of position procedure;
 - (l) lost procedure.
- (C) arrival and aerodrome joining procedure:
 - (a) ATC liaison in regulated airspace;
 - (b) altimeter setting;
 - (c) entering the traffic pattern;
 - (d) circuit procedures;
 - (e) parking;
 - (f) security of aeroplane or TMG;
 - (g) refuelling;
 - (h) closing of flight plan, if appropriate;
 - (i) post-flight administrative procedures.
- (xxiii) Exercise 18b: Navigation problems at lower levels and in reduced visibility:
 - (A) actions before descending;
 - (B) hazards (for example obstacles, and terrain);
 - (C) difficulties of map reading;
 - (D) effects of wind and turbulence;
 - (E) vertical situational awareness (avoidance of controlled flight into terrain);
 - (F) avoidance of noise sensitive areas;
 - (G) joining the circuit;
 - (H) bad weather circuit and landing.
- (xxiv) Exercise 18c: Radio navigation (basics):
 - (A) use of GNSS or VOR/ADF:
 - (a) selection of waypoints or stations;
 - (b) to or from indications and orientation;
 - (c) error messages.
 - (B) use of VHF/DF:
 - (a) availability, AIP and frequencies;
 - (b) R/T procedures and ATC liaison;
 - (c) obtaining a QDM and homing.
 - (C) use of en-route or terminal radar:
 - (a) availability and AIP;
 - (b) procedures and ATC liaison;
 - (c) pilot's responsibilities;
 - (d) secondary surveillance radar:

- (1) transponders;
- (2) code selection;
- (3) interrogation and reply.

(xxv) Exercise 19: Stopping and restarting the engine (in the case of TMGs only):

- (A) engine cooling;
- (B) switching-off procedure;
- (C) restarting of the engine.

AMC2 FCL.115 LAPL — Training course

ED Decision 2025/002/R

FLIGHT INSTRUCTION FOR THE LAPL(H)

(a) Entry to training

Before being accepted for training an applicant should be informed that the appropriate medical certificate must be obtained before solo flying is permitted.

(b) Flight instruction

- (1) The LAPL(H) flight instruction syllabus should take into account the principles of threat and error management and also cover:
 - (i) pre-flight operations, including mass and balance determination, helicopter inspection and servicing;
 - (ii) aerodrome and traffic pattern operations, collision avoidance precautions and procedures;
 - (iii) control of the helicopter by external visual reference;
 - (iv) take-offs, landings, hovering, look-out turns and normal transitions from and to the hover;
 - (v) emergency procedures, basic autorotations, simulated engine failure and ground resonance recovery if relevant to type;
 - (vi) sideways and backwards flight and turns on the spot;
 - (vii) recognition of and recovery from the incipient vortex ring state;
 - (viii) touchdown autorotations, simulated engine-off landings, practice forced landings. Simulated equipment malfunctions and emergency procedures relating to malfunctions of engines, controls, electrical and hydraulic circuits;
 - (ix) steep turns;
 - (x) transitions, quick stops, out of wind manoeuvres, sloping ground landings and take-offs;
 - (xi) limited power and confined area operations including selection of and operations to and from unprepared sites;

- (xii) cross-country flying by using visual reference, dead reckoning and, where available and radio navigation aids;
 - (xiii) operations to and from aerodromes; compliance with air traffic services procedures and communication procedures.
 - (2) Before allowing applicants to undertake their first solo flight, the FI should ensure that the applicants can use R/T communication and can operate the required systems and equipment.
- (c) Syllabus of flight instruction
 - (1) The numbering of exercises should be used primarily as an exercise reference list and as a broad instructional sequencing guide; therefore the demonstrations and practices need not necessarily be given in the order listed. The actual order and content will depend upon the following interrelated factors:
 - (i) the applicant's progress and ability;
 - (ii) the weather conditions affecting the flight;
 - (iii) the flight time available;
 - (iv) instructional technique considerations;
 - (v) the local operating environment;
 - (vi) applicability of the exercises to the helicopter type.
 - (2) Each of the exercises involves the need for the applicant to be aware of the needs of good airmanship and look-out, which should be emphasised at all times.
 - (i) Exercise 1a: Familiarisation with the helicopter:
 - (A) characteristics of the helicopter, external features;
 - (B) cockpit layout;
 - (C) systems;
 - (D) checklists, procedures, controls.
 - (ii) Exercise 1b: Emergency procedures:
 - (A) action if fire on the ground and in the air;
 - (B) engine, cabin and electrical system fire;
 - (C) systems failures;
 - (D) escape drills, location and use of emergency equipment and exits.
 - (iii) Exercise 2: Preparation for and action after flight:
 - (A) flight authorisation and helicopter acceptance;
 - (B) serviceability documents;
 - (C) equipment required, maps, etc.;
 - (D) external checks;
 - (E) internal checks;
 - (F) seat, harness and flight controls adjustments;

- (G) starting and warm-up checks clutch engagement and starting rotors;
- (H) power checks;
- (I) running down system checks and switching off the engine;
- (J) parking, security and picketing;
- (K) completion of authorisation sheet and serviceability documents.
- (iv) Exercise 3: Air experience:
 - (A) to introduce the student to rotary wing flight;
 - (B) flight exercise.
- (v) Exercise 4: Effects of controls:
 - (A) function of flight controls, primary and secondary effect;
 - (B) effect of air speed;
 - (C) effect of power changes (torque);
 - (D) effect of yaw (sideslip);
 - (E) effect of disc loading (bank and flare);
 - (F) effect on controls of selecting hydraulics on/off;
 - (G) effect of control friction;
 - (H) instruments;
 - (I) use of carburettor heat or anti-icing control.
- (vi) Exercise 5: Power and attitude changes:
 - (A) relationship between cyclic control position, disc attitude, fuselage attitude and air speed;
 - (B) flapback;
 - (C) power required diagram in relation to air speed;
 - (D) power and air speed changes in level flight;
 - (E) use of instruments for precision;
 - (F) engine and air speed limitations.
- (vii) Exercise 6a: Straight and level:
 - (A) at normal cruising power, attaining and maintaining straight and level flight;
 - (B) control in pitch, including use of control friction or trim;
 - (C) maintaining direction and balance, (ball or yawstring use);
 - (D) setting power for selected air speeds and speed changes;
 - (E) use of instruments for precision.
- (viii) Exercise 6b: Climbing:
 - (A) optimum climb speed, best angle or rate of climb from power required diagram;

- (B) initiation, maintaining the normal and maximum rate of climb, levelling off;
 - (C) levelling off at selected altitudes or heights;
 - (D) use of instruments for precision.
- (ix) Exercise 6c: Descending:
- (A) optimum descent speed and best angle or rate of descent from power required diagram;
 - (B) initiation, maintaining and levelling off;
 - (C) levelling off at selected altitudes or heights;
 - (D) descent (including effect of power and air speed);
 - (E) use of instruments for precision.
- (x) Exercise 6d: Turning:
- (A) initiation and maintaining medium level turns;
 - (B) resuming straight flight;
 - (C) altitude, bank and coordination;
 - (D) climbing and descending turns and effect on rate of climb or descent;
 - (E) turns onto selected headings, use of gyro heading indicator and compass;
 - (F) use of instruments for precision.
- (xi) Exercise 7: Basic autorotation:
- (A) safety checks, verbal warning and look-out;
 - (B) entry, development and characteristics;
 - (C) control of air speed and RRPM, rotor and engine limitations;
 - (D) effect of AUM, IAS, disc loading, G-forces and density altitude
 - (E) re-engagement and go-around procedures (throttle over-ride or ERPM control);
 - (F) awareness of increased risk of vortex condition during power recovery;
 - (G) gentle and medium turns in autorotation;
 - (H) demonstration of variable flare simulated engine off landing.
- (xii) Exercise 8a: Hovering:
- (A) demonstrate hover IGE, importance of wind effect and attitude, ground cushion, stability in the hover, effects of over controlling;
 - (B) student holding cyclic stick only;
 - (C) student handling collective lever (and throttle) only;
 - (D) student handling collective lever, (throttle) and pedals;
 - (E) student handling all controls;
 - (F) demonstration of ground effect;
 - (G) demonstration of wind effect;

- (H) demonstrate gentle forward running touchdown;
- (I) specific hazards, for example snow, dust and litter.
- (xiii) Exercise 8b: Hover taxiing and spot turns:
 - (A) revise hovering;
 - (B) precise ground speed and height control;
 - (C) effect of wind direction on helicopter attitude and control margin;
 - (D) control and coordination during spot turns;
 - (E) carefully introduce gentle forward running touchdown.
- (xiv) Exercise 8c: Hovering and taxiing emergencies:
 - (A) revise hovering and gentle forward running touchdown, explain (demonstrate where applicable) effect of hydraulics failure in the hover;
 - (B) demonstrate simulated engine failure in the hover and hover taxi.
 - (C) demonstrate dangers of mishandling and over-pitching.
- (xv) Exercise 9: Take-off and landing
 - (A) pre-take-off checks or drills;
 - (B) look-out;
 - (C) lifting to hover;
 - (D) after take-off checks;
 - (E) danger of horizontal movement near ground;
 - (F) danger of mishandling and overpitching;
 - (G) landing (without sideways or backwards movement);
 - (H) after landing checks or drills;
 - (I) take-off and landing crosswind and downwind.
- (xvi) Exercise 10: Transitions from hover to climb and approach to hover:
 - (A) look-out;
 - (B) revise take-off and landing;
 - (C) ground effect, translational lift and its effects;
 - (D) flapback and its effects;
 - (E) effect of wind speed and direction during transitions from or to the hover;
 - (F) the constant angle approach;
 - (G) demonstration of variable flare simulated engine off landing.
- (xvii) Exercise 11a: Circuit, approach and landing:
 - (A) revise transitions from hover to climb and approach to hover;
 - (B) circuit procedures, downwind and base leg;
 - (C) approach and landing with power;

- (D) pre-landing checks;
 - (E) effect of wind on approach and IGE hover
 - (F) crosswind approach and landing;
 - (G) go-around;
 - (H) noise abatement procedures.
- (xviii) Exercise 11b: Steep and limited power approaches and landings:
- (A) revise the constant angle approach;
 - (B) the steep approach (explain danger of high sink rate and low air speed);
 - (C) limited power approach (explain danger of high speed at touch down);
 - (D) use of the ground effect;
 - (E) variable flare simulated engine off landing.
- (xix) Exercise 11c: Emergency procedures:
- (A) abandoned take-off;
 - (B) missed approach and go-around;
 - (C) hydraulic off landing (if applicable);
 - (D) tail rotor control or tail rotor drive failure (briefing only);
 - (E) simulated emergencies in the circuit to include:
 - (F) hydraulics failure;
 - (G) simulated engine failure on take-off, crosswind, downwind and base leg;
 - (H) governor failure.
- (xx) Exercise 12: First solo:
- (A) instructor's briefing, observation of flight and debriefing;
 - (B) warn of change of attitude from reduced and laterally displaced weight;
 - (C) warn of low tail, low skid or wheel during hover and landing;
 - (D) warn of dangers of loss of RRPM and overpitching;
 - (E) pre-take-off checks;
 - (F) into wind take-off;
 - (G) procedures during and after take-off;
 - (H) normal circuit, approaches and landings;
 - (I) action if an emergency.
- (xxi) Exercise 13: Sideways and backwards hover manoeuvring:
- (A) manoeuvring sideways flight heading into wind;
 - (B) manoeuvring backwards flight heading into wind;
 - (C) combination of sideways and backwards manoeuvring;
 - (D) manoeuvring sideways and backwards, heading out of wind;

- (E) stability and weather cocking;
 - (F) recovery from backwards manoeuvring, (pitch nose down);
 - (G) groundspeed limitations for sideways and backwards manoeuvring.
- (xxii) Exercise 14: Spot turns:
- (A) revise hovering into wind and downwind;
 - (B) turn on spot through 360°:
 - (a) around pilots position;
 - (b) around tail rotor;
 - (c) around helicopter geometric centre;
 - (d) square and safe visibility clearing turn.
 - (C) rotor RPM control, torque effect, cyclic limiting stops due to CG position and wind speed and direction.
- (xxiii) Exercise 15: Hover OGE – Vortex ring— unanticipated yaw (LTE):
- (A) establishing hover OGE;
 - (B) drift, height or power control;
 - (C) demonstration of incipient stage of vortex ring, recognition and recovery (from a safe altitude);
 - (D) Demonstration of unanticipated yaw which could lead to a perceived loss of tail rotor effectiveness (LTE).
- (xxiv) Exercise 16: Simulated EOL:
- (A) the effect of weight, disc loading, density attitude and RRPM decay;
 - (B) revise basic autorotation entry;
 - (C) optimum use of cyclic and collective to control speed or RRPM;
 - (D) variable flare simulated EOL;
 - (E) demonstrate constant attitude simulated EOL;
 - (F) demonstrate simulated EOL from hover or hover taxi;
 - (G) demonstrate simulated EOL from transition and low level.
- (xxv) Exercise 17: Advanced autorotation:
- (A) over a selected point at various height and speed;
 - (B) revise basic autorotation: note ground distance covered;
 - (C) range autorotation;
 - (D) low speed autorotation;
 - (E) constant attitude autorotation (terminate at safe altitude);
 - (F) 'S' turns;
 - (G) turns through 180° and 360°;
 - (H) effects on angles of descent, IAS, RRPM and effect of AUM.

(xxvi) Exercise 18: Practice forced landings:

- (A) procedure and choice of the forced landing area;
- (B) forced landing checks and crash action;
- (C) re-engagement and go-around procedures.

(xxvii) Exercise 19: Steep turns:

- (A) steep (level) turns (30° bank);
- (B) maximum rate turns (45° bank if possible);
- (C) steep autorotative turns;
- (D) faults in the turn: balance, attitude, bank and coordination;
- (E) RRPM control and disc loading;
- (F) vibration and control feedback;
- (G) effect of wind at low level.

(xxviii) Exercise 20: Transitions:

- (A) revise ground effect, translational lift and flapback;
- (B) maintaining constant height, (20–30 ft AGL);
- (C) transition from hover to minimum 50 knots IAS and back to hover;
- (D) demonstrate effect of wind.

(xxix) Exercise 21: Quick stops:

- (A) use of power and controls;
- (B) effect of wind;
- (C) quick stops into wind;
- (D) quick stops from crosswind and downwind terminating into wind;
- (E) awareness of the danger of vortex ring;
- (F) danger of high disc loading.

(xxx) Exercise 22a: Navigation:

- (A) Flight planning:
 - (a) weather forecast and actuals;
 - (b) map selection and preparation and use:
 - (1) choice of route;
 - (2) controlled airspace, danger and prohibited areas;
 - (3) safety altitudes and noise abatement considerations.
 - (c) calculations:
 - (1) magnetic heading(s) and time(s) en-route;
 - (2) fuel consumption;
 - (3) mass and balance.

- (d) flight information:
 - (1) NOTAMs, etc.;
 - (2) radio frequencies;
 - (3) selection of alternate landing sites.
- (e) helicopter documentation;
- (f) notification of the flight:
 - (1) pre-flight administrative procedures;
 - (2) flight plan form (where appropriate).
- (B) Departure:
 - (a) organisation of cockpit workload;
 - (b) departure procedures:
 - (1) altimeter settings;
 - (2) ATC liaison in regulated airspace;
 - (3) setting heading procedure;
 - (4) noting of ETAs.
 - (c) maintenance of height or altitude and heading;
 - (d) revisions of ETA and heading:
 - (1) 10° line, double track, track error and closing angle;
 - (2) 1 in 60 rule;
 - (3) amending an ETA.
 - (e) log keeping;
 - (f) use of radio;
 - (g) minimum weather conditions for continuation of flight;
 - (h) in-flight decisions;
 - (i) transiting controlled or regulated airspace;
 - (j) uncertainty of position procedure;
 - (k) lost procedure.
- (C) Arrival and aerodrome joining procedure:
 - (a) ATC liaison in regulated airspace;
 - (b) altimeter setting;
 - (c) entering the traffic pattern;
 - (d) circuit procedures;
 - (e) parking;
 - (f) security of helicopter;
 - (g) refuelling;

- (h) closing of flight plan, (if appropriate);
- (i) post-flight administrative procedures.

(xxxi) Exercise 22b: Navigation problems at low heights and in reduced visibility:

- (A) actions before descending;
- (B) hazards (for example obstacles and other aircraft);
- (C) difficulties in map reading;
- (D) effects of wind and turbulence;
- (E) avoidance of noise-sensitive areas;
- (F) actions in case of DVE;
- (G) decision to divert or make a precautionary landing;
- (H) bad-weather circuit and landing;
- (I) appropriate procedures and choice of landing area;
- (J) precautionary landing.

(xxxii) Exercise 22c: Radio navigation (basics):

- (A) Use of GNNS or VOR/NDB:
 - (a) selection of waypoints;
 - (b) to or from indications or orientation;
 - (c) error messages.
- (B) Use of VHF/DF:
 - (a) availability, AIP and frequencies;
 - (b) R/T procedures and ATC liaison;
 - (c) obtaining a QDM and homing.
- (C) Use of en-route or terminal radar:
 - (a) availability and AIP;
 - (b) procedures and ATC liaison;
 - (c) pilot's responsibilities;
 - (d) secondary surveillance radar:
 - (1) transponders;
 - (2) code selection;
 - (3) interrogation and reply.

(xxxiii) Exercise 23: Advanced take-off, landings and transitions:

- (A) landing and take-off out of wind (performance reduction);
- (B) ground effect, translational lift and directional stability variation when out of wind;
- (C) downwind transitions;

- (D) vertical take-off over obstacles;
- (E) reconnaissance of landing site;
- (F) running landing;
- (G) zero speed landing;
- (H) crosswind and downwind landings;
- (I) steep approach;
- (J) go-around.

(xxxiv) Exercise 24: Sloping ground:

- (A) limitations and assessing slope angle;
- (B) wind and slope relationship: blade and control stops;
- (C) effect of CG when on slope;
- (D) ground effect on slope and power required;
- (E) right skid up slope;
- (F) left skid up slope;
- (G) nose up slope;
- (H) avoidance of dynamic roll over, dangers soft ground and sideways movement on touchdown;
- (I) danger of striking main or tail rotor by harsh control movement near ground.

(xxxv) Exercise 25: Limited power:

- (A) take-off power check;
- (B) vertical take-off over obstacles;
- (C) in-flight power check;
- (D) running landing;
- (E) zero speed landing;
- (F) approach to low hover;
- (G) approach to hover;
- (H) approach to hover OGE;
- (I) steep approach;
- (J) go-around.

(xxxvi) Exercise 26: Confined areas:

- (A) landing capability and performance assessment;
- (B) locating landing site and assessing wind speed and direction;
- (C) reconnaissance of landing site;
- (D) select markers;
- (E) select direction and type of approach;

- (F) circuit;
- (G) approach to committed point and go-around;
- (H) approach;
- (I) clearing turn;
- (J) landing;
- (K) power check and performance assessment in and OGE;
- (L) normal take-off to best angle of climb speed;
- (M) vertical take-off from hover.

AMC1 FCL.115(b) LAPL – Training course

ED Decision 2018/009/R

CHANGE OF TRAINING ORGANISATION

In cases where the applicant completes the training course (theoretical knowledge instruction or flight instruction) at a different DTO or ATO ('completing training organisation') from the one where they have started the training course ('starting training organisation'), the applicant should request from the starting training organisation a copy of the records kept in accordance with point DTO.GEN.220 or point ORA.ATO.120.

FCL.120 LAPL – Theoretical knowledge examination

Regulation (EU) 2020/359

Applicants for an LAPL shall demonstrate a level of theoretical knowledge appropriate to the privileges granted, through examinations on the following:

- (a) common subjects:
 - Air law,
 - Human performance,
 - Meteorology,
 - Communications, and
 - Navigation.
- (b) specific subjects concerning the different aircraft categories:
 - Principles of flight,
 - Operational procedures,
 - Flight performance and planning, and
 - Aircraft general knowledge.

AMC1 FCL.115; FCL.120 LAPL — Training course and theoretical knowledge examination

ED Decision 2025/002/R

SYLLABUS OF THEORETICAL KNOWLEDGE FOR THE LAPL

- (a) The training and examination should cover aspects related to non-technical skills in an integrated manner, taking into account the particular risks associated with the licence and the activity. The DTO or the ATO responsible for the training has to check if all the appropriate elements of the training course of theoretical knowledge instruction have been completed to a satisfactory standard before recommending the applicant for the examination.
- (b) The syllabi for the theoretical knowledge instruction and examination for the PPL(A) and the PPL(H) in [AMC1 FCL.210; FCL.215](#) should be used for the LAPL(A) and the LAPL(H) respectively.

FCL.125 LAPL – Skill test

Regulation (EU) No 1178/2011

- (a) Applicants for an LAPL shall demonstrate through the completion of a skill test the ability to perform, as PIC on the appropriate aircraft category, the relevant procedures and manoeuvres with competency appropriate to the privileges granted.
- (b) Applicants for the skill test shall have received flight instruction on the same class or type of aircraft to be used for the skill test. The privileges will be restricted to the class or type used for the skill test until further extensions are endorsed on the licence, in accordance with this Subpart.
- (c) Pass marks
 - (1) The skill test shall be divided into different sections, representing all the different phases of flight appropriate to the category of aircraft flown.
 - (2) Failure in any item of a section will cause the applicant to fail the entire section. If the applicant fails only 1 section, he/she shall repeat only that section. Failure in more than 1 section will cause the applicant to fail the entire test.
 - (3) When the test needs to be repeated in accordance with (2), failure in any section, including those that have been passed on a previous attempt, will cause the applicant to fail the entire test.
 - (4) Failure to achieve a pass in all sections of the test in 2 attempts will require further practical training.

AMC1 FCL.120; FCL.125

ED Decision 2011/016/R

THEORETICAL KNOWLEDGE EXAMINATION AND SKILL TEST FOR THE LAPL

- (a) Theoretical knowledge examination
 - (1) The examinations should be in written form and should comprise a total of 120 multiple-choice questions covering all the subjects.
 - (2) For the subject 'communication' practical classroom testing may be conducted.
 - (3) The competent authority should inform applicants of the language(s) in which the examinations will be conducted.
- (b) Skill test

Further training may be required following any failed skill test or part thereof. There should be no limit to the number of skill tests that may be attempted.
- (c) Conduct of the test

- (1) If the applicant chooses to terminate a skill test for reasons considered inadequate by the FE, the applicant should retake the entire skill test. If the test is terminated for reasons considered adequate by the FE, only those sections not completed should be tested in a further flight.
- (2) Any manoeuvre or procedure of the test may be repeated once by the applicant. The FE may stop the test at any stage if it is considered that the applicant's demonstration of flying skill requires a complete retest.
- (3) An applicant should be required to fly the aircraft from a position where the PIC functions can be performed and to carry out the test as if there is no other crew member. Responsibility for the flight should be allocated in accordance with national regulations.

AMC1 FCL.125 LAPL – Skill test

ED Decision 2011/016/R

CONTENTS OF THE SKILL TEST FOR THE ISSUE OF A LAPL(A)

- (a) The route to be flown for the skill test should be chosen by the FE. The route should end at the aerodrome of departure or at another aerodrome. The applicant should be responsible for the flight planning and should ensure that all equipment and documentation for the execution of the flight are on board. The navigation section of the test should have a duration of at least 30 minutes which allows the pilot to demonstrate his/her ability to complete a route with at least two identified waypoints and may, as agreed between applicant and FE, be flown as a separate test.
- (b) An applicant should indicate to the FE the checks and duties carried out, including the identification of radio facilities. Checks should be completed in accordance with the flight manual or the authorised checklist for the aeroplane or TMG on which the test is being taken. During pre-flight preparation for the test the applicant should be required to determine power settings and speeds. Performance data for take-off, approach and landing should be calculated by the applicant in compliance with the operations manual or flight manual for the aeroplane or TMG used.

FLIGHT TEST TOLERANCE

- (c) The applicant should demonstrate the ability to:
 - (1) operate the aeroplane or TMG within its limitations;
 - (2) complete all manoeuvres with smoothness and accuracy;
 - (3) exercise good judgment and airmanship;
 - (4) apply aeronautical knowledge;
 - (5) maintain control of the aeroplane or TMG at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.
- (d) The following limits are for general guidance. The FE should make allowance for turbulent conditions and the handling qualities and performance of the aeroplane or TMG used:
 - (1) height: normal flight ± 150 ft
 - (2) speed:
 - (i) take-off and approach $+15/-5$ knots
 - (ii) all other flight regimes ± 15 knots

CONTENT OF THE SKILL TEST

- (e) The skill test contents and sections set out in this AMC should be used for the skill test for the issue of a LAPL(A):

SECTION 1 PRE-FLIGHT OPERATIONS AND DEPARTURE	
Use of checklist, airmanship, control of aeroplane or TMG by external visual reference, anti/de-icing procedures, etc. apply in all sections.	
a	Pre-flight documentation, NOTAM and weather briefing
b	Mass and balance and performance calculation
c	Aeroplane or TMG inspection and servicing
d	Engine starting and after starting procedures
e	Taxiing and aerodrome procedures, pre-take-off procedures
f	Take-off and after take-off checks
g	Aerodrome departure procedures
h	ATC liaison: compliance
SECTION 2 GENERAL AIRWORK	
a	ATC liaison
b	Straight and level flight, with speed changes
c	Climbing: i. best rate of climb; ii. climbing turns iii. levelling off.
d	Medium (30° bank) turns, look-out procedures and collision avoidance
e	Steep (45 ° bank) turns
f	Flight at critically low air speed with and without flaps
g	Stalling: i. clean stall and recover with power; ii. approach to stall descending turn with bank angle 20 °, approach configuration; iii. approach to stall in landing configuration.
h	Descending: i. with and without power; ii. descending turns (steep gliding turns); iii. levelling off.
SECTION 3 EN-ROUTE PROCEDURES	
a	Flight plan, dead reckoning and map reading
b	Maintenance of altitude, heading and speed
c	Orientation, airspace structure, timing and revision of ETAs, log keeping
d	Diversion to alternate aerodrome (planning and implementation)
e	Flight management (checks, fuel systems, carburettor icing, etc.)
f	ATC liaison: compliance
SECTION 4 APPROACH AND LANDING PROCEDURES	
a	Aerodrome arrival procedures
b	Collision avoidance (look-out procedures)
c	Precision landing (short field landing) and crosswind, if suitable conditions available
d	Flapless landing (if applicable)
e	Approach to landing with idle power
f	Touch and go
g	Go-around from low height

h	ATC liaison
i	Actions after flight
SECTION 5 ABNORMAL AND EMERGENCY PROCEDURES	
This section may be combined with Sections 1 through 4	
a	Simulated engine failure after take-off
b	* Simulated forced landing
c	* Simulated precautionary landing
d	Simulated emergencies
e	Oral questions

* These items may be combined, at the discretion of the FE.

AMC2 FCL.125 LAPL – Skill test

ED Decision 2011/016/R

CONTENTS OF THE SKILL TEST FOR THE ISSUE OF A LAPL(H)

- (a) The area and route to be flown for the skill test should be chosen by the FE. The route should end at the aerodrome of departure or at another aerodrome. The applicant should be responsible for the flight planning and should ensure that all equipment and documentation for the execution of the flight are on board. The navigation section of the test should consist of at least two legs, each leg of a minimum duration of 10 minutes. The skill test may be conducted in two flights.
- (b) An applicant should indicate to the FE the checks and duties carried out, including the identification of radio facilities. Checks should be completed in accordance with the flight manual or the authorised checklist or pilot operating handbook for the helicopter on which the test is being taken. During pre-flight preparation for the test the applicant should be required to determine power settings and speeds. Performance data for take-off, approach and landing should be calculated by the applicant in compliance with the operations manual or flight manual for the helicopter used.

FLIGHT TEST TOLERANCE

- (c) The applicant should demonstrate the ability to:
 - (1) operate the helicopter within its limitations;
 - (2) complete all manoeuvres with smoothness and accuracy;
 - (3) exercise good judgment and airmanship;
 - (4) apply aeronautical knowledge;
 - (5) maintain control of the helicopter at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.
- (d) The following limits are for general guidance. The FE should make allowance for turbulent conditions and the handling qualities and performance of the helicopter used:
 - (1) height:
 - (i) normal forward flight ± 150 ft
 - (ii) with simulated major emergency ± 200 ft
 - (iii) hovering IGE flight ± 2 ft
 - (2) speed:

- | | | |
|------------------|--------------------------|----------------------------------|
| (i) | take-off approach | +15 knots /-10 knots |
| (ii) | all other flight regimes | ± 15 knots |
| (3) round drift: | | |
| (i) | take-off hover IGE | ± 3 ft |
| (ii) | landing | no sideways or backward movement |

CONTENT OF THE SKILL TEST

- (e) The skill test contents and sections set out in this AMC should be used for the skill test for the issue of a LAPL(H):

SECTION 1 PRE-FLIGHT OR POST-FLIGHT CHECKS AND PROCEDURES

Use of checklist, airmanship, control of helicopter by external visual reference, anti/de-icing procedures, etc. apply in all sections.

- | | |
|---|---|
| a | Helicopter knowledge (for example technical log, fuel, mass and balance, performance), flight planning, NOTAM, and weather briefing |
| b | Pre-flight inspection or action, location of parts and purpose |
| c | Cockpit inspection, starting procedure |
| d | Communication and navigation equipment checks, selecting and setting frequencies |
| e | Pre-take-off procedure and ATC liaison |
| f | Parking, shutdown and post-flight procedure |

SECTION 2 HOVER MANOEUVRES, ADVANCED HANDLING AND CONFINED AREAS

- | | |
|---|--|
| a | Take-off and landing (lift off and touch down) |
| b | Taxi and hover taxi |
| c | Stationary hover with head, cross and tail wind |
| d | Stationary hover turns, 360° left and right (spot turns) |
| e | Forward, sideways and backwards hover manoeuvring |
| f | Simulated engine failure from the hover |
| g | Quick stops into and downwind |
| h | Sloping ground or unprepared sites landings and take-offs |
| i | Take-offs (various profiles) |
| j | Crosswind and downwind take-off (if practicable) |
| k | Take-off at maximum take-off mass (actual or simulated) |
| l | Approaches (various profiles) |
| m | Limited power take-off and landing |
| n | Autorotations (FE to select two items from the following: basic, range, low speed, and 360° turns) |
| o | Autorotative landing |
| p | Practice forced landing with power recovery |
| q | Power checks, reconnaissance technique, approach and departure technique |

SECTION 3 NAVIGATION AND EN-ROUTE PROCEDURES

- | | |
|---|---|
| a | Navigation and orientation at various altitudes or heights and map reading |
| b | Altitude or height, speed, heading control, observation of airspace and altimeter setting |
| c | Monitoring of flight progress, flight-log, fuel usage, endurance, ETA, assessment of track error, re-establishment of correct track and instrument monitoring |
| d | Observation of weather conditions and diversion planning |
| e | Collision avoidance (look-out procedures) |
| f | ATC liaison with due observance of regulations |

SECTION 4 FLIGHT PROCEDURES AND MANOEUVRES

- | | |
|---|---|
| a | Level flight, control of heading, altitude or height and speed |
| b | Climbing and descending turns to specified headings |
| c | Level turns with up to 30 ° bank, 180 ° to 360 ° left and right |

SECTION 5 ABNORMAL AND EMERGENCY PROCEDURES (SIMULATED WHERE APPROPRIATE)

Note: The FE selects 4 items from the following:

- | | |
|---|---|
| a | Engine malfunctions, including governor failure, carburettor or engine icing and oil system, as appropriate |
| b | Fuel system malfunction |
| c | Electrical system malfunction |
| d | Hydraulic system malfunction, including approach and landing without hydraulics, as applicable |
| e | Main rotor or anti-torque system malfunction (FFS or discussion only) |
| f | Fire drills, including smoke control and removal, as applicable |
| g | Other abnormal and emergency procedures as outlined in appropriate flight manual |

SECTION 2 – SPECIFIC REQUIREMENTS FOR THE LAPL FOR AEROPLANES – LAPL(A)

FCL.105.A LAPL(A) – Privileges and conditions

Regulation (EU) 2024/2076

(a) Privileges

The privileges of the holder of an LAPL for aeroplanes are to act as PIC on SEP aeroplanes-land (SEP(land)), SEP aeroplanes-sea (SEP(sea)) or TMG with a maximum certified take-off mass of 2 000 kg or less, carrying a maximum of three passengers, such that there are never more than four persons on board the aircraft.

(b) Conditions

- (1) Holders of a LAPL(A) shall carry passengers only if they have completed 10 hours of flight time as PIC on aeroplanes or TMG after the issuance of the licence.
- (2) Holders of a LAPL(A) who previously held an ATPL(A), an MPL(A), a CPL(A) or a PPL(A), are exempted from the requirements laid down in point (b)(1).

AMC1 FCL.105.A(b)(2) Privileges and conditions

ED Decision 2020/005/R

In the case of previous MPL(A) holders, only those who extended their MPL(A) to include CPL privileges or PPL privileges in accordance with point [FCL.405.A\(b\)](#) may benefit from the exemption of point [FCL.105.A\(b\)\(2\)](#).

FCL.110.A LAPL(A) – Experience requirements and crediting

Regulation (EU) 2024/2076

(a) Applicants for an LAPL(A) shall have completed at least 30 hours of flight instruction on aeroplanes or TMGs, including at least:

- (1) 15 hours of dual flight instruction in the class in which the skill test will be taken;
- (2) 6 hours of supervised solo flight time, including at least 3 hours of solo cross-country flight time with at least 1 cross-country flight of at least 150 km (80 NM), during which 1 full stop landing at an aerodrome different from the aerodrome of departure shall be made.

(b) Specific requirements for applicants who hold an SPL issued in accordance with Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), including privileges to fly TMGs. Applicants for an LAPL(A) who hold an SPL with the privileges to fly TMGs shall have completed at least 21 hours of flight time on TMGs after the endorsement of the TMG privileges and shall comply with the requirements of point [FCL.135.A\(a\)](#) on aeroplanes.

(c) Crediting. Applicants with prior experience as PIC may be credited towards the requirements of point (a) under the following conditions:

- (1) the amount of credit shall be decided by the DTO or the ATO where the pilot undergoes the training course, on the basis of a pre-entry flight assessment, but shall in any case:
 - (i) not exceed the total flight time as PIC;

- (ii) not exceed 50 % of the hours required in point (a);
- (iii) not include the requirements of point (a)(2);
- (2) previous experience as PIC in aircraft that are subject to a decision of a Member State taken in accordance with Article 2(8), points (a) or (c), of Regulation (EU) 2018/1139 or that fall within the scope of Annex I to that Regulation may be considered, provided that the aircraft match the definition of the respective aircraft category under this Annex (Part-FCL).

AMC1 FCL.110.A(c) LAPL(A) Experience requirements and crediting

ED Decision 2025/002/R

CREDITING: FOR PRIOR EXPERIENCE AS PIC

The pre-entry flight assessment referred to in point [FCL.110.A\(c\)\(1\)](#) should cover the total content of the syllabus of flight instruction for the issuance of the LAPL(A), in accordance with [AMC1 FCL.115](#).

FCL.135.A LAPL(A) – Extension of privileges to another class or variant of aeroplane

Regulation (EU) 2024/2076

- (a) Applicants for an LAPL(A) shall be issued with privileges for the class of aeroplanes or TMGs in which the skill test was taken. In order to extend their privileges to another class, holders of an LAPL(A) shall complete, in that other class, all of the following:
 - (1) 3 hours of flight instruction, including the following:
 - (i) 10 dual take-offs and landings;
 - (ii) 10 supervised solo take-offs and landings;
 - (2) a skill test to demonstrate an adequate level of practical skill in the new class. During that skill test, the applicant shall also demonstrate to the examiner an adequate level of theoretical knowledge for the other class in the following subjects:
 - (i) Operational procedures;
 - (ii) Flight performance and planning;
 - (iii) Aircraft general knowledge.
- (b) In order to extend the privileges to another variant within a class, the pilot shall either undergo differences training or do familiarisation. The differences training shall be entered in the pilot's logbook or in an equivalent record and shall be signed by the instructor. When extending the privileges for an SEP aeroplane class to a variant with another type of engine referred to in Article 2, point (8c), the differences training shall consist of dual flight instruction and theoretical knowledge instruction which shall include, with regard to that other type of engine and related aircraft systems, at least all of the following subjects:
 - (1) Operational procedures;
 - (2) Flight performance and planning;
 - (3) Aircraft general knowledge.
- (c) Applicants for the extension of privileges of the LAPL(A) to TMG who also hold an SPL in accordance with Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#),

including the privileges to fly on TMGs, shall receive full credits towards the requirements in paragraph (a).

AMC1 FCL.135.A(b) LAPL(A) — Extension of privileges to another class or variant of aeroplane

ED Decision 2025/002/R

DIFFERENCES TRAINING FOR VARIANTS WITHIN THE SEP AEROPLANE CLASS WITH AN ELECTRIC ENGINE SYSTEM

As regards variants within the SEP aeroplane class with electric engine, the differences training should follow the content of [AMC1 FCL.710\(a\)](#).

GM1 FCL.135.A; FCL.135.H

ED Decision 2011/016/R

DIFFERENCES AND FAMILIARISATION TRAINING

- (a) Differences training requires the acquisition of additional knowledge and training on an appropriate training device or the aircraft.
- (b) Familiarisation training requires the acquisition of additional knowledge.

FCL.140.A LAPL(A) – Recency requirements

Regulation (EU) 2024/2076

- (a) Holders of a LAPL(A) shall exercise the privileges of their licence only if in the last 2 years they have met any of the following conditions as pilots of aeroplanes or TMGs:
 - (1) they have completed at least 12 hours of flight time as PIC or flying dual or solo under the supervision of an instructor, including:
 - (i) 12 take-offs and landings;
 - (ii) refresher training of at least 1 hour of total flight time with and to the satisfaction of an instructor who shall select those flight exercises that allow the applicant to refresh their competence in safely operating the aircraft and applying normal, abnormal and emergency procedures;
 - (2) they have passed a LAPL(A) proficiency check with an examiner. The proficiency check programme shall be based on the skill test for the LAPL(A);
- (b) If holders of a LAPL(A) hold both a SEP(land) and a SEP(sea) privilege, they may comply with the requirements in point (a)(1) in either class or a combination thereof which shall be valid for both privileges. For this purpose, at least 1 hour of the required flight time and 6 out of the required 12 take-offs and landings shall be completed in each class.
- (c) Holders of an LAPL(A) with privileges for SEP aeroplanes who, in accordance with point [FCL.135.A\(b\)](#), have extended their privileges to a variant with a different type of engine referred to in Article 2, point (8c), if they have not flown that variant within the preceding 2 years, shall complete any of the following in that variant, before exercising their privileges in that variant:
 - (1) further differences training in accordance with point [FCL.135.A\(b\)](#);
 - (2) a proficiency check;
 - (3) refresher training in accordance with point [FCL.140.A\(a\)\(1\)\(ii\)](#).

- (d) The dual flights, flights under supervision referred to in the introductory phrase of point (a)(1), the refresher training referred to in point (a)(1)(ii) and point (c)(3) and the proficiency check referred to in point (a)(2) and point (c)(2) shall be entered in the pilot's logbook or equivalent record and shall be signed by the instructor or examiner, as applicable.

AMC1 FCL.140.A(a)(1)(ii) LAPL(A) — Recency requirements

ED Decision 2025/002/R

CONTENT OF THE REFRESHER TRAINING

- (a) Before the flight training takes place, the instructor should hold a briefing with the candidate. That briefing should include a discussion on all of the following:
- (1) TEM with special emphasis on decision-making when encountering adverse meteorological conditions or unintentional IMC;
 - (2) navigation flight techniques;
 - (3) exercises as specified in point (b), as applicable.
- (b) Flight training items should be based on the exercise items of the proficiency check, as deemed relevant by the instructor, and depending on the experience of the candidate. In any case, the instructor should select scenarios from the following list and include the relevant recognition and recovery exercises in the flight training:
- (1) clean stall;
 - (2) approach to stall in descending turn with bank with approach configuration and power;
 - (3) approach to stall in landing configuration and power;
 - (4) approach to stall, climbing turn with take-off flap and climb power; and
 - (5) simulated loss or partial loss of engine power during different phases of flight.

AMC1 FCL.140.A(a)(2) LAPL(A) — Recency requirements

ED Decision 2025/002/R

The proficiency check should follow the content of the skill test that is set out in [AMC1 FCL.125](#), point (e).

GM1 FCL.140.A LAPL(A) — Recency requirements

ED Decision 2025/002/R

SAFETY AWARENESS BRIEFING BEFORE REFRESHER TRAINING OR PROFICIENCY CHECKS

It is recommended that the pre-flight briefing before refresher training or a proficiency check in accordance with point [FCL.140.A](#) with the pilot, the instructor or examiner, as applicable, includes elements to raise the pilot's safety awareness with regard to safely flying aeroplanes or TMGs (as applicable), as outlined in [GM2 FCL.740.A](#).

SECTION 3 – SPECIFIC REQUIREMENTS FOR THE LAPL FOR HELICOPTERS – LAPL(H)

FCL.105.H LAPL(H) – Privileges

Regulation (EU) No 1178/2011

The privileges of the holder of an LAPL for helicopters are to act as PIC on single-engine helicopters with a maximum certificated take-off mass of 2 000 kg or less, carrying a maximum of 3 passengers, such that there are never more than 4 persons on board.

FCL.110.H LAPL(H) – Experience requirements and crediting

Regulation (EU) 2018/1119

- (a) Applicants for the LAPL(H) shall have completed 40 hours of flight instruction on helicopters. At least 35 hours of which shall be flown on the type of helicopter that is to be used for the skill test. The flight instruction shall include at least:
 - (1) 20 hours of dual flight instruction; and
 - (2) 10 hours of supervised solo flight time, including at least 5 hours of solo cross-country flight time with at least 1 cross-country flight of at least 150 km (80 NM), during which one full stop landing at an aerodrome different from the aerodrome of departure shall be made.
- (b) Crediting. Applicants with prior experience as PIC may be credited towards the requirements of point (a) under the following conditions:
 - (1) the amount of credit shall be decided by the DTO or the ATO where the pilot undergoes the training course, on the basis of a pre-entry flight assessment, but shall in any case:
 - (i) not exceed the total flight time as PIC;
 - (ii) not exceed 50 % of the hours required in point (a);
 - (iii) not include the requirements of point (a)(2);
 - (2) previous experience as PIC in aircraft that are subject to a decision of a Member State taken in accordance with Article 2(8), point (b), of Regulation (EU) 2018/1139 or that fall within the scope of Annex I to that Regulation may be considered, provided that the aircraft match the definition of the helicopter category under this Annex (Part-FCL).

AMC1 FCL.110.H(b) LAPL(H) — Experience requirements and crediting

ED Decision 2025/002/R

CREDITING FOR PRIOR EXPERIENCE AS PIC

The pre-entry flight assessment referred to in point [FCL.110.H\(b\)\(1\)](#) should cover the total content of the syllabus of flight instruction for the issuance of the LAPL(H), in accordance with [AMC2 FCL.115](#).

FCL.135.H LAPL(H) – Extension of privileges to another type or variant of helicopter

Regulation (EU) 2024/2076

- (a) Applicants for an LAPL(H) shall be issued with privileges for the specific type of helicopter in which the skill test was taken. In order to extend their privileges to another type of helicopter, holders of an LAPL(H) shall complete, in that other type, all of the following:
 - (1) unless otherwise specified in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012, at least 5 hours of flight instruction, including:
 - (i) 15 dual take-offs, approaches and landings;
 - (ii) 15 supervised solo take-offs, approaches and landings;
 - (2) a skill test to demonstrate an adequate level of practical skill in the new type. During that skill test, the applicant shall also demonstrate to the examiner an adequate level of theoretical knowledge for the other type in the following subjects:
 - (i) Operational procedures,
 - (ii) Flight performance and planning,
 - (iii) Aircraft general knowledge.
- (b) Before the holder of an LAPL(H) can exercise the privileges of the licence in another variant of helicopter than the one used for the skill test, the pilot shall undergo differences training or do familiarisation, as determined in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012. The differences training shall be entered in the pilot's logbook or equivalent record and shall be signed by the instructor.

GM1 FCL.135.A; FCL.135.H

ED Decision 2011/016/R

DIFFERENCES AND FAMILIARISATION TRAINING

- (a) Differences training requires the acquisition of additional knowledge and training on an appropriate training device or the aircraft.
- (b) Familiarisation training requires the acquisition of additional knowledge.

FCL.140.H LAPL(H) – Recency requirements

Regulation (EU) 2024/2076

- (a) Holders of an LAPL(H) shall exercise the privileges of their licence on a specific type only if in the last 12 months they have, in the relevant type, taken one of the following steps:
 - (1) completed both of the following:
 - (i) at least 6 hours of flight time on helicopters of that type as PIC, or flying dual or solo under the supervision of an instructor, including six take-offs, approaches and landings;
 - (ii) a refresher training of at least 1 hour of total flight time with and to the satisfaction of an instructor who shall select those flight exercises that allow the applicant to

refresh their competence in safely operating the aircraft and applying normal, abnormal and emergency procedures;

- (2) passed a proficiency check with an examiner on the specific type before resuming the exercise of the privileges of their licence. That proficiency check programme shall be based on the skill test for the LAPL(H).
- (b) Applicants may choose to complete the refresher training referred to in point (a)(1)(ii) in the aircraft or an FSTD representing the relevant type, or in a combination of both.
- (c) The dual flights, flights under supervision referred to in point (a)(1)(i), the refresher training referred to in point (a)(1)(ii) and the proficiency check referred to in point (a)(2) shall be entered in the pilot's logbook or equivalent record and shall be signed by the instructor or examiner, as applicable.

GM1 FCL.140.H LAPL(H) — Recency requirements

ED Decision 2025/002/R

SAFETY AWARENESS BRIEFING BEFORE REFRESHER TRAINING OR PROFICIENCY CHECKS

It is recommended that the pre-flight briefing before refresher training or a proficiency check in accordance with point [FCL.140.H](#) with the pilot, the instructor or examiner, as applicable, includes elements to raise the pilot's safety awareness with regard to safely flying helicopters, as outlined in [GM1 FCL.740.H](#).

AMC1 FCL.140.H(a)(1)(ii) LAPL(H) — Recency requirements

ED Decision 2025/002/R

CONTENT OF THE REFRESHER TRAINING

- (a) Before the training takes place, the instructor should hold a briefing with the candidate. That briefing should include a discussion on all of the following:
 - (1) TEM with special emphasis on decision-making when encountering adverse meteorological conditions or unintentional IMC;
 - (2) navigation flight techniques;
 - (3) exercises as specified in point (b), as applicable.
- (b) Training items should be based on the exercise items of the proficiency check, as deemed relevant by the instructor, and depending on the experience of the candidate. In any case, the training flight items should include the following exercises from the LAPL(H) flight training syllabus ([AMC2 FCL.115](#)):
 - (1) Exercise 15: Hover OGE — Vortex ring — unanticipated yaw (LTE);
 - (2) Exercise 18: Practice forced landings;
 - (3) Exercise 26: Confined areas.

AMC1 FCL.140.H(a)(2) LAPL(H) — Recency requirements

ED Decision 2025/002/R

The proficiency check should follow the content of the skill test that is set out in [AMC2 FCL.125](#), point (e).

SUBPART C – PRIVATE PILOT LICENCE (PPL) AND GYROPLANE PILOT LICENCE (GPL)

SECTION 1 – COMMON REQUIREMENTS

FCL.200 Minimum age

Regulation (EU) 2025/134

Applicants for a PPL shall be at least 17 years old.

[applicable until 17 February 2026 - Regulation (EU) 2020/359]

Applicants for a PPL or a GPL shall be at least 17 years old.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

FCL.205 Conditions

Regulation (EU) 2025/134

Applicants for the issue of a PPL shall have fulfilled the requirements for the class or type rating for the aircraft used in the skill test, as established in Subpart H.

[applicable until 17 February 2026 - Regulation (EU) 1178/2011]

- (a) Applicants for the issue of a PPL shall have fulfilled the requirements for the class or type rating for the aircraft used in the skill test, as established in Subpart H.
- (b) Applicants for the issue of a GPL shall have fulfilled the requirements for the class or type of aircraft used in the skill test.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

FCL.210 Training course

Regulation (EU) 2025/134

- (a) Applicants for a PPL shall complete a training course at an ATO or a DTO.
- (b) The course shall include theoretical knowledge and flight instruction appropriate to the privileges of the PPL applied for.
- (c) Theoretical knowledge instruction and flight instruction may be completed at a DTO or at an ATO different from the one where applicants have commenced their training.

[applicable until 17 February 2026 - Regulation (EU) 2020/359]

- (a) Applicants for a PPL or a GPL shall complete a training course at an ATO or a DTO.
- (b) The course shall include theoretical knowledge and flight instruction appropriate to the privileges of the PPL or GPL applied for.
- (c) Theoretical knowledge instruction and flight instruction may be completed at a DTO or at an ATO different from the one where applicants have commenced their training.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

AMC1 FCL.210 PPL(A) Training course

ED Decision 2025/002/R

FLIGHT INSTRUCTION FOR THE PPL(A)**(a) Entry to training**

Before being accepted for training an applicant should be informed that the appropriate medical certificate must be obtained before solo flying is permitted.

(b) Flight instruction

(1) The PPL(A) flight instruction syllabus takes into account the principles of threat and error management and also covers:

- (i) pre-flight operations, including mass and balance determination, aircraft inspection and servicing;
- (ii) aerodrome and traffic pattern operations, collision avoidance precautions and procedures;
- (iii) control of the aircraft by external visual reference;
- (iv) flight at critically low air speeds, recognition of, and recovery from, incipient and full stalls;
- (v) flight at critically high air speeds, recognition of, and recovery from, spiral dive;
- (vi) normal and crosswind take-offs and landings;
- (vii) maximum performance (short field and obstacle clearance) takeoffs, short-field landings;
- (viii) flight by reference solely to instruments, including the completion of a level 180 ° turn;
- (ix) cross-country flying using visual reference, dead reckoning and radio navigation aids;
- (x) emergency operations, including simulated aeroplane equipment malfunctions;
- (xi) operations to, from and transiting controlled aerodromes, compliance with air traffic services procedures, communication procedures and phraseology.

(2) Before allowing applicants for a PPL(A) to undertake their first solo flight, the FI should ensure that the applicants can use R/T communication and can operate the required systems and equipment.

(c) Syllabus of flight instruction

(1) The numbering of exercises should be used primarily as an exercise reference list and as a broad instructional sequencing guide; therefore the demonstrations and practices need not necessarily be given in the order listed. The actual order and content will depend upon the following interrelated factors:

- (i) the applicant's progress and ability;
- (ii) the weather conditions affecting the flight;
- (iii) the flight time available;
- (iv) instructional technique considerations;

- (v) the local operating environment;
- (vi) applicability of the exercises to the aeroplane.
- (2) Each of the exercises involves the need for the applicant to be aware of the needs of good airmanship and look-out, which should be emphasised at all times.
 - (i) Exercise 1a: Familiarisation with the aeroplane:
 - (A) characteristics of the aeroplane;
 - (B) cockpit layout;
 - (C) systems;
 - (D) checklists, drills and controls.
 - (ii) Exercise 1b: Emergency drills:
 - (A) action if fire on the ground and in the air;
 - (B) engine cabin and electrical system fire;
 - (C) systems failure;
 - (D) escape drills, location and use of emergency equipment and exits.
 - (iii) Exercise 2: Preparation for and action after flight:
 - (A) flight authorisation and aeroplane acceptance;
 - (B) serviceability documents;
 - (C) equipment required, maps, etc.;
 - (D) external checks;
 - (E) internal checks;
 - (F) harness, seat or rudder panel adjustments;
 - (G) starting and warm-up checks;
 - (H) power checks;
 - (I) running down system checks and switching off the engine;
 - (J) parking, security and picketing (for example tie down);
 - (K) completion of authorisation sheet and serviceability documents.
 - (iv) Exercise 3: Air experience: flight exercise.
 - (v) Exercise 4: Effects of controls:
 - (A) primary effects when laterally level and when banked;
 - (B) further effects of aileron and rudder;
 - (C) effects of:
 - (a) air speed;
 - (b) slipstream;
 - (c) power;

- (d) trimming controls;
 - (e) flaps;
 - (f) other controls, as applicable.
- (D) operation of:
 - (a) mixture control;
 - (b) carburettor heat;
 - (c) cabin heating or ventilation.
- (vi) Exercise 5a: Taxiing:
 - (A) pre-taxi checks;
 - (B) starting, control of speed and stopping;
 - (C) engine handling;
 - (D) control of direction and turning;
 - (E) turning in confined spaces;
 - (F) parking area procedure and precautions;
 - (G) effects of wind and use of flying controls;
 - (H) effects of ground surface;
 - (I) freedom of rudder movement;
 - (J) marshalling signals;
 - (K) instrument checks;
 - (L) air traffic control procedures.
- (vii) Exercise 5b: Emergencies: brake and steering failure.
- (viii) Exercise 6: Straight and level:
 - (A) at normal cruising power, attaining and maintaining straight and level flight;
 - (B) flight at critically high air speeds;
 - (C) demonstration of inherent stability;
 - (D) control in pitch, including use of trim;
 - (E) lateral level, direction and balance and trim;
 - (F) at selected air speeds (use of power);
 - (G) during speed and configuration changes;
 - (H) use of instruments for precision.
- (ix) Exercise 7: Climbing:
 - (A) entry, maintaining the normal and max rate climb and levelling off;
 - (B) levelling off at selected altitudes;
 - (C) en-route climb (cruise climb);

- (D) climbing with flap down;
 - (E) recovery to normal climb;
 - (F) maximum angle of climb;
 - (G) use of instruments for precision.
- (x) Exercise 8: Descending:
- (A) entry, maintaining and levelling off;
 - (B) levelling off at selected altitudes;
 - (C) glide, powered and cruise descent (including effect of power and air speed);
 - (D) side slipping (on suitable types);
 - (E) use of instruments for precision flight.
- (xi) Exercise 9: Turning:
- (A) entry and maintaining medium level turns;
 - (B) resuming straight flight;
 - (C) faults in the turn (for example in correct pitch, bank and balance);
 - (D) climbing turns;
 - (E) descending turns;
 - (F) faults in the turns (slipping and skidding on suitable types);
 - (G) turns onto selected headings, use of gyro heading indicator and compass;
 - (H) use of instruments for precision.
- (xii) Exercise 10a: Slow flight:
- Note: the objective is to improve the student's ability to recognise inadvertent flight at critically low speeds and provide practice in maintaining the aeroplane in balance while returning to normal air speed.
- (A) safety checks;
 - (B) introduction to slow flight;
 - (C) controlled flight down to critically slow air speed;
 - (D) application of full power with correct attitude and balance to achieve normal climb speed.
- (xiii) Exercise 10b: Stalling:
- (A) safety checks;
 - (B) symptoms;
 - (C) recognition;
 - (D) clean stall and recovery without power and with power;
 - (E) recovery when a wing drops;

- (F) approach to stall in the approach and in the landing configurations, with and without power and recovery at the incipient stage.
- (xiv) Exercise 11: Spin avoidance:

Note: The limitations of the training aircraft as set out in the AFM (manoeuvre limitations, mass and balance calculations) need to be carefully considered.

 - (A) safety checks;
 - (B) stalling and approaching the incipient spin stage (stall with asymmetric condition, with not-centred slip indicator);
 - (C) instructor - induced distractions during the stall.
- (xv) Exercise 12: Take-off and climb to downwind position:
 - (A) pre-take-off checks;
 - (B) into wind take-off;
 - (C) safeguarding the nose wheel;
 - (D) crosswind take-off;
 - (E) drills during and after take-off;
 - (F) short take-off and soft field procedure/techniques including performance calculations;
 - (G) noise abatement procedures.
- (xvi) Exercise 13: Circuit, approach and landing:
 - (A) circuit procedures, downwind and base leg;
 - (B) powered approach and landing;
 - (C) safeguarding the nose wheel;
 - (D) effect of wind on approach and touchdown speeds and use of flaps;
 - (E) crosswind approach and landing;
 - (F) glide approach and landing;
 - (G) short landing and soft field procedures or techniques;
 - (H) flapless approach and landing;
 - (I) wheel landing (tail wheel aeroplanes);
 - (J) missed approach and go-around;
 - (K) noise abatement procedures.
- (xvii) Exercise 12/13: Emergencies:
 - (A) abandoned take-off;
 - (B) simulated engine failure after take-off;
 - (C) rejected and go-around;
 - (D) missed approach;

- (E) simulated engine failure and restart procedures in-flight.

Note: In the interests of safety it will be necessary for pilots that are trained on nose-wheel aeroplanes to undergo dual conversion training before flying tail-wheel aeroplanes, and vice-versa.

(xviii) Exercise 14: First solo:

- (A) instructor's briefing
- (B) use of required equipment;
- (C) observation of flight and debriefing by instructor;

Note: During flights immediately following the solo circuit consolidation, the following should be revised:

- (1) procedures for leaving and rejoining the circuit;
- (2) the local area, restrictions, map reading;
- (3) use of radio aids for homing;
- (4) turns using magnetic compass, compass errors.

(xix) Exercise 15: Advanced turning:

- (A) steep turns (45°), level and descending;
- (B) stalling in the turn and recovery;
- (C) recoveries from unusual attitudes, including spiral dives.

(xx) Exercise 16: Forced landing without power:

- (A) forced landing procedure;
- (B) choice of landing area, provision for change of plan;
- (C) gliding distance;
- (D) descent plan;
- (E) key positions;
- (F) engine cooling;
- (G) engine failure checks;
- (H) use of radio;
- (I) base leg;
- (J) final approach;
- (K) landing;
- (L) actions after landing.

(xxi) Exercise 17: Precautionary landing:

- (A) full procedure away from aerodrome to break-off height;
- (B) occasions necessitating;
- (C) in-flight conditions;

- (D) landing area selection:
 - (a) normal aerodrome;
 - (b) disused aerodrome;
 - (c) ordinary field.
 - (E) circuit and approach;
 - (F) actions after landing.
- (xxii) Exercise 18a: Navigation:
- (A) flight planning:
 - (a) weather forecast and actuals;
 - (b) map selection and preparation:
 - (1) choice of route;
 - (2) controlled airspace;
 - (3) danger, prohibited and restricted areas;
 - (4) safety altitudes.
 - (c) calculations:
 - (1) magnetic heading(s) and time(s) en-route;
 - (2) fuel consumption;
 - (3) mass and balance;
 - (4) mass and performance.
 - (d) flight information:
 - (1) NOTAMs etc.;
 - (2) radio frequencies;
 - (3) selection of alternate aerodromes.
 - (e) aeroplane documentation;
 - (f) notification of the flight:
 - (1) pre-flight administrative procedures;
 - (2) flight plan form.
 - (B) departure:
 - (a) organisation of cockpit workload;
 - (b) departure procedures:
 - (1) altimeter settings;
 - (2) ATC liaison in controlled or regulated airspace;
 - (3) setting heading procedure;
 - (4) noting of ETAs.

- (c) maintenance of altitude and heading;
 - (d) revisions of ETA and heading;
 - (e) log keeping;
 - (f) use of radio;
 - (g) use of navaids;
 - (h) minimum weather conditions for continuation of flight;
 - (i) in-flight decisions;
 - (j) transiting controlled or regulated airspace;
 - (k) diversion procedures;
 - (l) uncertainty of position procedure;
 - (m) lost procedure.
- (C) arrival and aerodrome joining procedure:
 - (a) ATC liaison in controlled or regulated airspace;
 - (b) altimeter setting;
 - (c) entering the traffic pattern;
 - (d) circuit procedures;
 - (e) parking;
 - (f) security of aeroplane;
 - (g) refuelling;
 - (h) closing of flight plan, if appropriate;
 - (i) post-flight administrative procedures.
- (xxiii) Exercise 18b: Navigation problems at lower levels and in reduced visibility:
 - (A) actions before descending;
 - (B) hazards (for example obstacles and terrain);
 - (C) difficulties of map reading;
 - (D) effects of wind and turbulence;
 - (E) vertical situational awareness (avoidance of controlled flight into terrain);
 - (F) avoidance of noise sensitive areas;
 - (G) joining the circuit;
 - (H) bad weather circuit and landing.
- (xxiv) Exercise 18c: Radio navigation:
 - (A) use of GNSS:
 - (a) selection of waypoints;
 - (b) to or from indications and orientation;

- (c) error messages.
 - (B) use of VHF omni range:
 - (a) availability, AIP and frequencies;
 - (b) selection and identification;
 - (c) OBS;
 - (d) to or from indications and orientation;
 - (e) CDI;
 - (f) determination of radial;
 - (g) intercepting and maintaining a radial;
 - (h) VOR passage;
 - (i) obtaining a fix from two VORs.
 - (C) use of ADF equipment: NDBs:
 - (a) availability, AIP and frequencies;
 - (b) selection and identification;
 - (c) orientation relative to the beacon;
 - (d) homing.
 - (D) use of VHF/DF:
 - (a) availability, AIP, frequencies;
 - (b) R/T procedures and ATC liaison;
 - (c) obtaining a QDM and homing.
 - (E) use of en-route or terminal radar:
 - (a) availability and AIP;
 - (b) procedures and ATC liaison;
 - (c) pilot's responsibilities;
 - (d) secondary surveillance radar:
 - (1) transponders;
 - (2) code selection;
 - (3) interrogation and reply.
 - (F) use of DME:
 - (a) station selection and identification;
 - (b) modes of operation: distance, groundspeed and time to run.
- (xxv) Exercise 19: Basic instrument flight:
- (A) physiological sensations;
 - (B) instrument appreciation; attitude instrument flight;

- (C) instrument limitations;
- (D) basic manoeuvres:
 - (a) straight and level at various air speeds and configurations;
 - (b) climbing and descending;
 - (c) standard rate turns, climbing and descending, onto selected headings;
 - (d) recoveries from climbing and descending turns.
- (d) BITD
 - (1) A BITD may be used for flight training for:
 - (i) flight by reference solely to instruments;
 - (ii) navigation using radio navigation aids;
 - (iii) basic instrument flight.
 - (2) The use of the BITD should be subject to the following:
 - (i) the training should be complemented by exercises on an aeroplane;
 - (ii) the record of the parameters of the flight must be available;
 - (iii) A FI(A) or STI(A) should conduct the instruction.

AMC2 FCL.210 PPL(H) – Training course

ED Decision 2025/002/R

FLIGHT INSTRUCTION FOR THE PPL(H)

- (a) Entry to training

Before being accepted for training an applicant should be informed that the appropriate medical certificate must be obtained before solo flying is permitted.
- (b) Ground instruction

Enhanced ground instruction in weather interpretation, planning and route assessment, decision making on encountering DVE including reversing course or conducting a precautionary landing.
- (c) Flight instruction
 - (1) The PPL(H) flight instruction syllabus should take into account the principles of threat and error management and cover:
 - (i) pre-flight operations, including mass and balance determination, helicopter inspection and servicing;
 - (ii) aerodrome and traffic pattern operations, collision avoidance precautions and procedures;
 - (iii) control of the helicopter by external visual reference;
 - (iv) take-offs, landings, hovering, look-out turns and normal transitions from and to the hover;

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- (v) emergency procedures, basic autorotations, simulated engine failure, ground resonance recovery if relevant to type;
 - (vi) sideways and backwards flight, turns on the spot;
 - (vii) recognition of and recovery from the incipient vortex ring state;
 - (viii) touchdown autorotations, simulated engine-off landings, practice forced landings. Simulated equipment malfunctions and emergency procedures relating to malfunctions of engines, controls, electrical and hydraulic circuits;
 - (ix) steep turns;
 - (x) transitions, quick stops, out of wind manoeuvres, sloping ground landings and take-offs;
 - (xi) limited power and confined area operations, including selection of and operations to and from unprepared sites;
 - (xii) flight by sole reference to basic flight instruments, including completion of a level 180° turn and recovery from unusual attitudes to simulate inadvertent entry into cloud (this training may be conducted by an FI(H));
 - (xiii) cross-country flying by using visual reference, DR, GNNS and, where available, radio navigation aids; simulation of deteriorating weather conditions and actions to divert or conduct precautionary landing;
 - (xiv) operations to, from and transiting controlled aerodromes; compliance with air traffic services procedures, communication procedures and phraseology.
- (2) Before allowing applicants for a PPL(H) to undertake their first solo flight, the FI should ensure that the applicants can use R/T communication and can operate the required systems and equipment.
 - (3) Wherever possible, flight simulation should be used to demonstrate to student pilots the effects of flight into DVE and to enhance their understanding and need for avoidance of this potentially fatal flight regime.
- (d) Syllabus of flight instruction
- (1) The numbering of exercises should be used primarily as an exercise reference list and as a broad instructional sequencing guide; therefore the demonstrations and practices need not necessarily be given in the order listed. The actual order and content will depend upon the following interrelated factors:
 - (i) the applicant's progress and ability;
 - (ii) the weather conditions affecting the flight;
 - (iii) the flight time available;
 - (iv) instructional technique considerations;
 - (v) the local operating environment;
 - (vi) applicability of the exercises to the helicopter.
 - (2) Each of the exercises involves the need for the applicant to be aware of the needs of good airmanship and look-out, which should be emphasised at all times.
 - (i) Exercise 1a: Familiarisation with the helicopter:

- (A) characteristics of the helicopter, external features;
 - (B) cockpit layout;
 - (C) systems;
 - (D) checklists, procedures and controls.
- (ii) Exercise 1b: Emergency procedures:
- (A) action if fire on the ground and in the air;
 - (B) engine, cabin and electrical system fire;
 - (C) systems failures;
 - (D) escape drills, location and use of emergency equipment and exits.
- (iii) Exercise 2: Preparation for and action after flight:
- (A) flight authorisation and helicopter acceptance;
 - (B) serviceability documents;
 - (C) equipment required, maps, etc.;
 - (D) external checks;
 - (E) internal checks;
 - (F) seat, harness and flight controls adjustments;
 - (G) starting and warm-up checks clutch engagement and starting rotors;
 - (H) power checks;
 - (I) running down system checks and switching off the engine;
 - (J) parking, security and picketing;
 - (K) completion of authorisation sheet and serviceability documents.
- (iv) Exercise 3: Air experience:
- (A) to introduce the student to rotary wing flight;
 - (B) flight exercise.
- (v) Exercise 4: Effects of controls:
- (A) function of flight controls, primary and secondary effect;
 - (B) effects of:
 - (a) air speed;
 - (b) power changes (torque);
 - (c) yaw (sideslip);
 - (d) disc loading (bank and flare);
 - (e) controls of selecting hydraulics on/off
 - (f) control friction.
 - (C) instruments;

- (D) use of carburettor heat or anti-icing control.
- (vi) Exercise 5: Power and attitude changes:
 - (A) relationship between cyclic control position, disc attitude, fuselage attitude and air speed;
 - (B) flapback;
 - (C) power required diagram in relation to air speed;
 - (D) power and air speed changes in level flight;
 - (E) use of instruments for precision;
 - (F) engine and air speed limitations.
- (vii) Exercise 6: Straight and level:
 - (A) at normal cruising power, attaining and maintaining straight and level flight;
 - (B) control in pitch, including use of control friction or trim;
 - (C) maintaining direction and balance, (ball or yawstring use);
 - (D) setting power for selected air speeds and speed changes;
 - (E) use of instruments for precision.
- (viii) Exercise 7: Climbing:
 - (A) optimum climb speed, best angle or rate of climb from power required diagram;
 - (B) initiation, maintaining the normal and maximum rate of climb, levelling off;
 - (C) levelling off at selected altitudes or heights
 - (D) use of instruments for precision.
- (ix) Exercise 8: Descending:
 - (A) optimum descent speed, best angle or rate of descent from power required diagram;
 - (B) initiation, maintaining and levelling off;
 - (C) levelling off at selected altitudes or heights;
 - (D) descent (including effect of power and air speed);
 - (E) use of instruments for precision.
- (x) Exercise 9: Turning:
 - (A) initiation and maintaining medium level turns;
 - (B) resuming straight flight;
 - (C) altitude, bank and co-ordination;
 - (D) climbing and descending turns and effect on rate of climb or descent;
 - (E) turns onto selected headings, use of gyro heading indicator and compass;
 - (F) use of instruments for precision.

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- (xi) Exercise 10: Basic autorotation:
 - (A) safety checks, verbal warning and look-out;
 - (B) entry, development and characteristics;
 - (C) control of air speed and RRPM, rotor and engine limitations;
 - (D) effect of AUM, IAS, disc loading, G forces and density altitude;
 - (E) re-engagement and go-around procedures (throttle over-ride or ERPM control);
 - (F) awareness of increased risk of vortex condition during power recovery;
 - (G) gentle and medium turns in autorotation;
 - (H) demonstration of variable flare simulated engine off landing.
 - (xii) Exercise 11a: Hovering:
 - (A) demonstrate hover IGE, importance of wind effect and attitude, ground cushion, stability in the hover and effects of over controlling;
 - (B) student holding cyclic stick only;
 - (C) student handling collective lever (and throttle) only;
 - (D) student handling collective lever, (throttle) and pedals;
 - (E) student handling all controls;
 - (F) demonstration of ground effect;
 - (G) demonstration of wind effect;
 - (H) demonstrate gentle forward running touchdown;
 - (I) specific hazards for example snow, dust and litter.
 - (xiii) Exercise 11b: Hover taxiing and spot turns:
 - (A) revise hovering;
 - (B) precise ground speed and height control;
 - (C) effect of wind direction on helicopter attitude and control margin;
 - (D) control and co-ordination during spot turns;
 - (E) carefully introduce gentle forward running touchdown.
 - (xiv) Exercise 11c: Hovering and taxiing emergencies:
 - (A) revise hovering and gentle forward running touchdown, explain (demonstrate where applicable) effect of hydraulics failure in the hover;
 - (B) demonstrate simulated engine failure in the hover and hover taxi;
 - (C) demonstrate dangers of mishandling and over-pitching.
 - (xv) Exercise 12: Take-off and landing:
 - (A) pre-take-off checks or drills;
 - (B) look-out;

- (C) lifting to hover;
 - (D) after take-off checks;
 - (E) danger of horizontal movement near ground;
 - (F) danger of mishandling and overpitching;
 - (G) landing (without sideways or backwards movement);
 - (H) after landing checks or drills;
 - (I) take-off and landing crosswind and downwind.
- (xvi) Exercise 13: Transitions from hover to climb and approach to hover:
- (A) look-out;
 - (B) revise take-off and landing;
 - (C) ground effect, translational lift and its effects;
 - (D) flapback and its effects;
 - (E) effect of wind speed and direction during transitions from or to the hover;
 - (F) the constant angle approach;
 - (G) demonstration of variable flare simulated engine off landing.
- (xvii) Exercise 14a: Circuit, approach and landing:
- (A) revise transitions from hover to climb and approach to hover;
 - (B) circuit procedures, downwind and base leg;
 - (C) approach and landing with power;
 - (D) pre-landing checks;
 - (E) effect of wind on approach and IGE hover;
 - (F) crosswind approach and landing;
 - (G) go-around;
 - (H) noise abatement procedures.
- (xviii) Exercise 14b: Steep and limited power approaches and landings:
- (A) revise the constant angle approach;
 - (B) the steep approach (explain danger of high sink rate and low air speed)
 - (C) limited power approach (explain danger of high speed at touch down);
 - (D) use of the ground effect;
 - (E) variable flare simulated engine off landing.
- (xix) Exercise 14c: Emergency procedures:
- (A) abandoned take-off;
 - (B) missed approach and go-around;
 - (C) hydraulic off landing (if applicable);

- (D) tail rotor control or tail rotor drive failure (briefing only)
- (E) simulated emergencies in the circuit to include:
 - (a) hydraulics failure;
 - (b) simulated engine failure on take-off, crosswind, downwind and base leg;
 - (c) governor failure.
- (xx) Exercise 15: First solo:
 - (A) instructor's briefing, observation of flight and debriefing;
 - (B) warn of change of attitude from reduced and laterally displaced weight;
 - (C) warn of low tail, low skid or wheel during hover, landing;
 - (D) warn of dangers of loss of RRPM and overpitching;
 - (E) pre-take-off checks;
 - (F) into wind take-off;
 - (G) procedures during and after take-off;
 - (H) normal circuit, approaches and landings;
 - (I) action if an emergency.
- (xxi) Exercise 16: Sideways and backwards hover manoeuvring:
 - (A) manoeuvring sideways flight heading into wind;
 - (B) manoeuvring backwards flight heading into wind;
 - (C) combination of sideways and backwards manoeuvring;
 - (D) manoeuvring sideways and backwards and heading out of wind;
 - (E) stability and weather cocking;
 - (F) recovery from backwards manoeuvring (pitch nose down);
 - (G) limitations for sideways and backwards manoeuvring.
- (xxii) Exercise 17: Spot turns:
 - (A) revise hovering into wind and downwind;
 - (B) turn on spot through 360°:
 - (a) around pilots position;
 - (b) around tail rotor;
 - (c) around helicopter geometric centre;
 - (d) square and safe visibility clearing turn.
 - (C) rotor RPM control, torque effect, cyclic limiting stops due to CG position and wind speed and direction.
- (xxiii) Exercise 18: Hover OGE – Vortex ring — unanticipated yaw (LTE):
 - (A) establishing hover OGE;

- (B) drift, height or power control;
- (C) demonstration of incipient stage of vortex ring, recognition and recovery (from a safe altitude);
- (D) Demonstration of unanticipated yaw which could lead to a perceived loss of tail rotor effectiveness (LTE).

(xxiv) Exercise 19: Simulated EOL:

- (A) the effect of weight, disc loading, density altitude and RRPM decay;
- (B) revise basic autorotation entry;
- (C) optimum use of cyclic and collective to control speed or RRPM;
- (D) variable flare simulated EOL;
- (E) demonstrate constant attitude simulated EOL;
- (F) demonstrate simulated EOL from hover or hover taxi;
- (G) demonstrate simulated EOL from transition and low level.

(xxv) Exercise 20: Advanced autorotation:

- (A) over a selected point at various height and speed;
- (B) revise basic autorotation: note ground distance covered;
- (C) range autorotation;
- (D) low speed autorotation;
- (E) constant attitude autorotation (terminate at safe altitude);
- (F) 'S' turns;
- (G) turns through 180° and 360°;
- (H) effects on angles of descent, IAS, RRPM and effect of AUM.

(xxvi) Exercise 21: Practice forced landings:

- (A) procedure and choice of the forced landing area;
- (B) forced landing checks and crash action;
- (C) re-engagement and go-around procedures.

(xxvii) Exercise 22: Steep turns:

- (A) steep (level) turns (30° bank);
- (B) maximum rate turns (45° bank if possible);
- (C) steep autorotative turns;
- (D) faults in the turn: balance, attitude, bank and co-ordination;
- (E) RRPM control and disc loading;
- (F) vibration and control feedback;
- (G) effect of wind at low level.

(xxviii) Exercise 23: Transitions:

- (A) revise ground effect, translational lift and flapback;
 - (B) maintaining constant height, (20-30 ft AGL);
 - (C) transition from hover to minimum 50 knots IAS and back to hover;
 - (D) demonstrate effect of wind.
- (xxix) Exercise 24: Quick stops:
- (A) use of power and controls;
 - (B) effect of wind;
 - (C) quick stops into wind;
 - (D) quick stops from crosswind and downwind terminating into wind;
 - (E) awareness of the danger of vortex ring;
 - (F) danger of high disc loading.
- (xxx) Exercise 25a: Navigation:
- (A) flight planning:
 - (a) weather forecast and actuals;
 - (b) map selection and preparation and use;
 - (1) choice of route;
 - (2) controlled airspace, danger and prohibited areas;
 - (3) safety altitudes and noise abatement considerations.
 - (c) calculations:
 - (1) magnetic heading(s) and time(s) en-route;
 - (2) fuel consumption;
 - (3) mass and balance.
 - (d) flight information:
 - (1) NOTAMs, etc.;
 - (2) radio frequencies;
 - (e) helicopter documentation;
 - (f) notification of the flight:
 - (1) pre-flight administrative procedures;
 - (2) flight plan form (where appropriate).
 - (B) departure:
 - (a) organisation of cockpit workload;
 - (b) departure procedures:
 - (1) altimeter settings;
 - (2) ATC liaison in controlled or regulated airspace;

- (3) setting heading procedure;
 - (4) noting of ETAs.
 - (c) maintenance of height or altitude and heading;
 - (d) revisions of ETA and heading:
 - (1) 10° line, double track and track error and closing angle;
 - (2) 1 in 60 rule;
 - (3) amending an ETA.
 - (e) log keeping;
 - (f) use of radio;
 - (g) use of nav aids (if fitted);
 - (h) minimum weather conditions for continuation of flight;
 - (i) in-flight decisions;
 - (j) transiting controlled or regulated airspace;
 - (k) uncertainty of position procedure;
 - (l) lost procedure.
- (C) arrival and aerodrome joining procedure:
- (a) ATC liaison in controlled or regulated airspace;
 - (b) altimeter setting;
 - (c) entering the traffic pattern;
 - (d) circuit procedures.
 - (e) parking;
 - (f) security of helicopter;
 - (g) refuelling;
 - (h) closing of flight plan (if appropriate);
 - (i) post-flight administrative procedures.
- (xxxi) Exercise 25b: Navigation problems at low heights and in reduced visibility:
- (A) actions before descending;
 - (B) hazards (for example obstacles and other aircraft);
 - (C) difficulties of map reading;
 - (D) effects of wind and turbulence;
 - (E) avoidance of noise sensitive areas;
 - (F) actions in the event of encountering DVE;
 - (G) decision to divert or conduct precautionary landing;
 - (H) bad weather circuit and landing;

- (I) appropriate procedures and choice of landing area;
 - (J) precautionary landing.
- (xxxii) Exercise 25c: Radio navigation:
- (A) use of GNSS:
 - (a) selection of waypoints;
 - (b) to or from indications and orientation;
 - (c) error messages;
 - (d) hazards of over-reliance on the use of GNSS in the continuation of flight in DVE.
 - (B) use of VHF omni range:
 - (a) availability, AIP and frequencies;
 - (b) selection and identification;
 - (c) OBS;
 - (d) to or from indications and orientation;
 - (e) CDI;
 - (f) determination of radial;
 - (g) intercepting and maintaining a radial;
 - (h) VOR passage;
 - (i) obtaining a fix from two VORs.
 - (C) use of ADF equipment: NDBs:
 - (a) availability, AIP and frequencies;
 - (b) selection and identification;
 - (c) orientation relative to the beacon;
 - (d) homing.
 - (D) use of VHF/DF:
 - (a) availability, AIP and frequencies;
 - (b) RTF procedures and ATC liaison;
 - (c) obtaining a QDM and homing.
 - (E) use of en-route or terminal radar:
 - (a) availability and AIP;
 - (b) procedures and ATC liaison;
 - (c) pilots responsibilities;
 - (d) secondary surveillance radar (if transponder fitted):
 - (1) transponders;

- (2) code selection;
 - (3) interrogation and reply.
- (F) use of DME:
 - (a) station selection and identification;
 - (b) modes of operation: distance, groundspeed and time to run.
- (xxxiii) Exercise 26: Advanced take-off, landings and transitions:
 - (A) landing and take-off out of wind (performance reduction);
 - (B) ground effect, translational lift and directional stability variation when out of wind;
 - (C) downwind transitions;
 - (D) vertical take-off over obstacles;
 - (E) running take-off;
 - (F) cushion creep take-off;
 - (G) reconnaissance of landing site;
 - (H) running landing;
 - (I) zero speed landing;
 - (J) crosswind and downwind landings;
 - (K) steep approach;
 - (L) go-around.
- (xxxiv) Exercise 27: Sloping ground:
 - (A) limitations and assessing slope angle;
 - (B) wind and slope relationship: blade and control stops;
 - (C) effect of CG when on slope;
 - (D) ground effect on slope and power required;
 - (E) right skid up slope;
 - (F) left skid up slope;
 - (G) nose up slope;
 - (H) avoidance of dynamic roll over, dangers of soft ground and sideways movement on touchdown;
 - (I) danger of striking main or tail rotor by harsh control movement near ground.
- (xxxv) Exercise 28: Limited power:
 - (A) take-off power check;
 - (B) vertical take-off over obstacles;
 - (C) in-flight power check;
 - (D) running landing;

- (E) zero speed landing;
- (F) approach to low hover;
- (G) approach to hover;
- (H) approach to hover OGE;
- (I) steep approach;
- (J) go-around.

(xxxvi) Exercise 29: Confined areas:

- (A) landing capability and performance assessment;
- (B) locating landing site and assessing wind speed and direction;
- (C) reconnaissance of landing site;
- (D) select markers;
- (E) select direction and type of approach;
- (F) circuit;
- (G) approach to committed point and go-around;
- (H) approach;
- (I) clearing turn;
- (J) landing;
- (K) power check and performance assessment in and out of ground effect;
- (L) normal take-off to best angle of climb speed;
- (M) vertical take-off from hover.

(xxxvii) Exercise 30: Basic instrument flight:

- (A) physiological sensations;
- (B) instrument appreciation:
 - (a) attitude instrument flight;
 - (b) instrument scan.
- (C) instrument limitations;
- (D) basic manoeuvres:
 - (a) straight and level at various air speeds and configurations;
 - (b) climbing and descending;
 - (c) standard rate turns, climbing and descending, onto selected headings.
- (E) recoveries from climbing and descending turns;
- (F) recoveries from unusual attitudes.

AMC1 FCL.210(c) Training course

ED Decision 2018/009/R

CHANGE OF TRAINING ORGANISATION

In cases where the applicant completes the training course (theoretical knowledge instruction or flight instruction) at a different DTO or ATO ('completing training organisation') from the one where they have started the training course ('starting training organisation'), the applicant should request from the starting training organisation a copy of the records kept in accordance with point DTO.GEN.220 or point ORA.ATO.120

FCL.215 Theoretical knowledge examination

Regulation (EU) 2025/134

Applicants for a PPL shall demonstrate a level of theoretical knowledge appropriate to the privileges granted through examinations in the following subjects:

[applicable until 17 February 2026 - Regulation (EU) 2020/359]

Applicants for a PPL or a GPL shall demonstrate a level of theoretical knowledge appropriate to the privileges granted through examinations in the following subjects:

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

(a) common subjects:

- Air law,
- Human performance,
- Meteorology,
- Communications, and
- Navigation.

(b) specific subjects concerning the different aircraft categories:

- Principles of flight,
- Operational procedures,
- Flight performance and planning, and
- Aircraft general knowledge.

AMC1 FCL.210; FCL.215 Training course and theoretical knowledge examination

ED Decision 2020/005/R

SYLLABUS OF THEORETICAL KNOWLEDGE FOR THE PPL(A) AND PPL(H)

The following tables contain the syllabi for the courses of theoretical knowledge, as well as for the theoretical knowledge examinations for the PPL(A) and PPL(H). The training and examination should cover aspects related to non-technical skills in an integrated manner, taking into account the particular risks associated to the licence and the activity.

The DTO or the ATO responsible for the training should check if all the appropriate elements of the training course of theoretical knowledge instruction have been completed to a satisfactory standard before recommending the applicant for the examination.

The applicable items for each licence are marked with 'x'. An 'x' on the main title of a subject means that all the sub-divisions are applicable.

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
1.	AIR LAW AND ATC PROCEDURES				
	International law: conventions, agreements and organisations				
	The Convention on international civil aviation (Chicago) Doc. 7300/6				
	Part I Air Navigation: relevant parts of the following chapters: (a) general principles and application of the convention; (b) flight over territory of Contracting States; (c) nationality of aircraft; (d) measures to facilitate air navigation; (e) conditions to be fulfilled on aircraft; (f) international standards and recommended practices; (g) validity of endorsed certificates and licences; (h) notification of differences.	x		x	
	Part II The International Civil Aviation Organisation (ICAO): objectives and composition	x		x	
	Annex 8: Airworthiness of aircraft				
	Foreword and definitions	x		x	
	Certificate of airworthiness	x		x	
	Annex 7: Aircraft nationality and registration marks				
	Foreword and definitions	x		x	
	Common- and registration marks	x		x	
	Certificate of registration and aircraft nationality	x		x	
	Annex 1: Personnel licensing				
	Definitions	x		x	
	Relevant parts of Annex 1 connected to Part-FCL and Part-Medical	x		x	
	Annex 2: Rules of the air				
	Essential definitions, applicability of the rules of the air, general rules (except water operations), visual flight rules, signals and interception of civil aircraft	x		x	
	Procedures for air navigation: aircraft operations doc. 8168-ops/611, volume 1				
	Altimeter setting procedures (including IACO doc. 7030 – regional supplementary procedures)				
	Basic requirements (except tables), procedures applicable to operators and pilots (except tables)	x		x	
	Secondary surveillance radar transponder operating procedures (including ICAO Doc. 7030 – regional supplementary procedures)				
	Operation of transponders	x		x	
	Phraseology	x		x	
	Annex 11: Doc. 4444 air traffic management				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Definitions	x		x	
	General provisions for air traffic services	x		x	
	Visual separation in the vicinity of aerodromes	x		x	
	Procedures for aerodrome control services	x		x	
	Radar services	x		x	
	Flight information service and alerting service	x		x	
	Phraseologies	x		x	
	Procedures related to emergencies, communication failure and contingencies	x		x	
	Annex 15: Aeronautical information service				
	Introduction, essential definitions	x		x	
	AIP, NOTAM, AIRAC and AIC	x		x	
	Annex 14, volume 1 and 2: Aerodromes				
	Definitions	x		x	
	Aerodrome data: conditions of the movement area and related facilities	x		x	
	Visual aids for navigation: (a) indicators and signalling devices; (b) markings; (c) lights; (d) signs; (e) markers.	x		x	
	Visual aids for denoting obstacles: (a) marking of objects; (b) lighting of objects.	x		x	
	Visual aids for denoting restricted use of areas	x		x	
	Emergency and other services: (a) rescue and fire fighting; (b) apron management service.	x		x	
	Annex 12: Search and rescue				
	Essential definitions	x		x	
	Operating procedures: (a) procedures for PIC at the scene of an accident; (b) procedures for PIC intercepting a distress transmission; (c) search and rescue signals.	x		x	
	Search and rescue signals: (a) signals with surface craft; (b) ground or air visual signal code; (c) air or ground signals.	x		x	
	Annex 17: Security				
	General: aims and objectives	x		x	
	Annex 13: Aircraft accident investigation				
	Essential definitions	x		x	
	Applicability	x		x	
	National law				
	National law and differences to relevant ICAO Annexes and relevant EU regulations.	x		x	
2.	HUMAN PERFORMANCE				
	Human factors: basic concepts				
	Human factors in aviation				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Becoming a competent pilot	x		x	
	Basic aviation physiology and health maintenance				
	The atmosphere: (a) composition; (b) gas laws.	x		x	
	Respiratory and circulatory systems: (a) oxygen requirement of tissues; (b) functional anatomy; (c) main forms of hypoxia (hypoxic and anaemic): (1) sources, effects and countermeasures of carbon monoxide; (2) counter measures and hypoxia; (3) symptoms of hypoxia. (d) hyperventilation; (e) the effects of accelerations on the circulatory system; (f) hypertension and coronary heart disease.	x		x	
	Man and environment				
	Central, peripheral and autonomic nervous systems	x		x	
	Vision: (a) functional anatomy; (b) visual field, foveal and peripheral vision; (c) binocular and monocular vision; (d) monocular vision cues; (e) night vision; (f) visual scanning and detection techniques and importance of 'look-out'; (g) defective vision.	x		x	
	Hearing: (a) descriptive and functional anatomy; (b) flight related hazards to hearing; (c) hearing loss.	x		x	
	Equilibrium: (a) functional anatomy; (b) motion and acceleration; (c) motion sickness.	x		x	
	Integration of sensory inputs: (a) spatial disorientation: forms, recognition and avoidance;	x		x	
	(b) illusions: forms, recognition and avoidance: (1) physical origin; (2) physiological origin; (3) psychological origin. (c) approach and landing problems.				
	Health and hygiene				
	Personal hygiene: personal fitness	x		x	
	Body rhythm and sleep: (a) rhythm disturbances; (b) symptoms, effects and management.	x		x	

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Problem areas for pilots: (a) common minor ailments including cold, influenza and gastro-intestinal upset; (b) entrapped gases and barotrauma, (scuba diving); (c) obesity; (d) food hygiene; (e) infectious diseases; (f) nutrition; (g) various toxic gases and materials.	x		x	
	Intoxication:	x		x	
	(a) prescribed medication; (b) tobacco; (c) alcohol and drugs; (d) caffeine; (e) self-medication.				
	Basic aviation psychology				
	Human information processing				
	Attention and vigilance: (a) selectivity of attention; (b) divided attention.	x		x	
	Perception: (A) perceptual illusions; (B) subjectivity of perception; (C) processes of perception.	x		x	
	Memory: (a) sensory memory; (b) working or short term memory; (c) long term memory to include motor memory (skills).	x		x	
	Human error and reliability				
	Reliability of human behaviour	x		x	
	Error generation: social environment (group, organisation)	x		x	
	Decision making				
	Decision-making concepts:	x		x	
	(a) structure (phases);				
	(b) limits; (c) risk assessment; (d) practical application.				
	Avoiding and managing errors: cockpit management				
	Safety awareness: (a) risk area awareness; (b) situational awareness.	x		x	
	Communication: verbal and non-verbal communication	x		x	
	Human behaviour				
	Personality and attitudes: (a) development; (b) environmental influences.	x		x	
	Identification of hazardous attitudes (error proneness)	x		x	
	Human overload and underload				
	Arousal	x		x	
	Stress:	x		x	

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	(a) definition(s); (b) anxiety and stress; (c) effects of stress.				
	Fatigue and stress management: (a) types, causes and symptoms of fatigue; (b) effects of fatigue; (c) coping strategies; (d) management techniques; (e) health and fitness programmes;	x		x	
3.	METEOROLOGY				
	The atmosphere				
	Composition, extent and vertical division				
	Structure of the atmosphere	x		x	
	Troposphere	x		x	
	Air temperature				
	Definition and units	x		x	
	Vertical distribution of temperature	x		x	
	Transfer of heat	x		x	
	Lapse rates, stability and instability	x		x	
	Development of inversions and types of inversions	x		x	
	Temperature near the earth's surface, surface effects, diurnal and seasonal variation, effect of clouds and effect of wind	x		x	
	Atmospheric pressure				
	Barometric pressure and isobars	x		x	
	Pressure variation with height	x		x	
	Reduction of pressure to mean sea level	x		x	
	Relationship between surface pressure centres and pressure centres aloft	x		x	
	Air density				
	Relationship between pressure, temperature and density	x		x	
	ISA				
	ICAO standard atmosphere	x		x	
	Altimetry				
	Terminology and definitions	x		x	
	Altimeter and altimeter settings	x		x	
	Calculations	x		x	
	Effect of accelerated airflow due to topography	x		x	
	Wind				
	Definition and measurement of wind				
	Definition and measurement	x		x	
	Primary cause of wind				
	Primary cause of wind, pressure gradient, coriolis force and gradient wind	x		x	
	Variation of wind in the friction layer	x		x	
	Effects of convergence and divergence	x		x	
	General global circulation				
	General circulation around the globe	x		x	
	Local winds				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Anabatic and katabatic winds, mountain and valley winds, Venturi effects, land and sea breezes	x		x	
	Mountain waves (standing waves, lee waves)				
	Origin and characteristics	x		x	
	Turbulence				
	Description and types of turbulence	x		x	
	Formation and location of turbulence	x		x	
	THERMODYNAMICS				
	Humidity				
	Water vapour in the atmosphere	x		x	
	Mixing ratio	x		x	
	Temperature/dew point, relative humidity	x		x	
	Change of state of aggregation				
	Condensation, evaporation, sublimation, freezing and melting, latent heat	x		x	
	Adiabatic processes				
	Adiabatic processes, stability of the atmosphere	x		x	
	CLOUDS AND FOG				
	Cloud formation and description				
	Cooling by adiabatic expansion and by advection	x		x	
	Cloud types and cloud classification	x		x	
	Influence of inversions on cloud development	x		x	
	Fog, mist, haze				
	General aspects	x		x	
	Radiation fog	x		x	
	Advection fog	x		x	
	Steaming fog	x		x	
	Frontal fog	x		x	
	Orographic fog (hill fog)	x		x	
	PRECIPITATION				
	Development of precipitation				
	Processes of development of precipitation	x		x	
	Types of precipitation				
	Types of precipitation, relationship with cloud types	x		x	
	AIR MASSES AND FRONTS				
	Air masses				
	Description, classification and source regions of air masses	x		x	
	Modifications of air masses	x		x	
	Fronts				
	General aspects	x		x	
	Warm front, associated clouds, and weather	x		x	
	Cold front, associated clouds, and weather	x		x	
	Warm sector, associated clouds, and weather	x		x	
	Weather behind the cold front	x		x	
	Occlusions, associated clouds, and weather	x		x	
	Stationary front, associated clouds, and weather	x		x	
	Movement of fronts and pressure systems, life cycle	x		x	
	Changes of meteorological elements at a frontal wave	x		x	
	PRESSURE SYSTEMS				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Anticyclone				
	Anticyclones, types, general properties, cold and warm anticyclones, ridges and wedges, subsidence	x		x	
	Non-frontal depressions				
	Thermal, orographic and polar depressions, troughs	x		x	
	CLIMATOLOGY				
	Climatic zones				
	General seasonal circulation in the troposphere	x		x	
	Typical weather situations in the mid-latitudes				
	Westerly situation	x		x	
	High-pressure area	x		x	
	Flat-pressure pattern	x		x	
	Local winds and associated weather				
	<i>e.g. Foehn</i>	x		x	
	FLIGHT HAZARDS				
	Icing				
	Conditions for ice accretion	x		x	
	Types of ice accretion	x		x	
	Hazards of ice accretion, avoidance	x		x	
	Turbulence				
	Effects on flight, avoidance	x		x	
	Wind shear				
	Definition of wind shear	x		x	
	Weather conditions for wind shear	x		x	
	Effects on flight, avoidance	x		x	
	Thunderstorms				
	Conditions for, and process of, development, forecast, location, type specification	x		x	
	Structure of thunderstorms, life cycle, squall lines, electricity in the atmosphere, static charges	x		x	
	Electrical discharges				
	Development and effects of downbursts	x		x	
	Thunderstorm avoidance	x		x	
	Inversions				
	Influence on aircraft performance	x		x	
	Hazards in mountainous areas				
	Influence of terrain on clouds and precipitation, frontal passage	x		x	
	Vertical movements, mountain waves, wind shear, turbulence, ice accretion	x		x	
	Development and effect of valley inversions	x		x	
	Visibility-reducing phenomena				
	Reduction of visibility caused by precipitation and obscuration	x		x	
	Reduction of visibility caused by other phenomena	x		x	
	METEOROLOGICAL INFORMATION				
	Observation				
	Surface observations	x		x	
	Radiosonde observations	x		x	
	Satellite observations	x		x	
	Weather radar observations	x		x	

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Aircraft observations and reporting	x		x	
	Weather charts				
	Significant weather charts	x		x	
	Surface charts	x		x	
	Information for flight planning				
	Aviation weather messages	x		x	
	Meteorological broadcasts for aviation	x		x	
	Use of meteorological documents	x		x	
	Meteorological warnings	x		x	
	Meteorological services				
	World area forecast system (WAFS) and meteorological offices	x		x	
4.	COMMUNICATIONS				
	VFR COMMUNICATIONS				
	Definitions				
	Meanings and significance of associated terms	x		x	
	ATS abbreviations	x		x	
	Q-code groups commonly used in RTF airground communications	x		x	
	Categories of messages	x		x	
	General operating procedures				
	Transmission of letters	x		x	
	Transmission of numbers (including level information)	x		x	
	Transmission of time	x		x	
	Transmission technique	x		x	
	Standard words and phrases (relevant RTF phraseology included)	x		x	
	R/T call signs for aeronautical stations including use of abbreviated call signs	x		x	
	R/T call signs for aircraft including use of abbreviated call signs	x		x	
	Transfer of communication	x		x	
	Test procedures including readability scale	x		x	
	Read back and acknowledgement requirements	x		x	
	Relevant weather information terms (VFR)				
	Aerodrome weather	x		x	
	Weather broadcast	x		x	
	Action required to be taken in case of communication failure	x		x	
	Distress and urgency procedures				
	Distress (definition, frequencies, watch of distress frequencies, distress signal and distress message)	x		x	
	Urgency (definition, frequencies, urgency signal and urgency message)	x		x	
	General principles of VHF propagation and allocation of frequencies	x		x	
5.	PRINCIPLES OF FLIGHT				
5.1.	PRINCIPLES OF FLIGHT: AEROPLANE				
	Subsonic aerodynamics				
	Basics concepts, laws and definitions				
	Laws and definitions:	x	x		

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	(a) conversion of units; (b) Newton's laws; (c) Bernoulli's equation and venture; (d) static pressure, dynamic pressure and total pressure; (e) density; (f) IAS and TAS.				
	Basics about airflow: (a) streamline; (b) two-dimensional airflow; (c) three-dimensional airflow.	x	x		
	Aerodynamic forces on surfaces: (a) resulting airforce; (b) lift; (c) drag; (d) angle of attack.	x	x		
	Shape of an aerofoil section: (a) thickness to chord ratio; (b) chord line; (c) camber line; (d) camber; (e) angle of attack.	x	x		
	The wing shape: (a) aspect ratio; (b) root chord; (c) tip chord; (d) tapered wings; (e) wing planform.	x	x		
	The two-dimensional airflow about an aerofoil				
	Streamline pattern	x	x		
	Stagnation point	x	x		
	Pressure distribution	x	x		
	Centre of pressure	x	x		
	Influence of angle of attack	x	x		
	Flow separation at high angles of attack	x	x		
	The lift – α graph	x	x		
	The coefficients				
	The lift coefficient C_L : the lift formula	x	x		
	The drag coefficient C_D : the drag formula	x	x		
	The three-dimensional airflow round a wing and a fuselage				
	Streamline pattern: (a) span-wise flow and causes; (b) tip vortices and angle of attack; (c) upwash and downwash due to tip vortices; (d) wake turbulence behind an aeroplane (causes, distribution and duration of the phenomenon).	x	x		
	Induced drag: (a) influence of tip vortices on the angle of attack; (b) the induced local α ; (c) influence of induced angle of attack on the direction of the lift vector;	x	x		

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	(d) induced drag and angle of attack.				
	Drag				
	The parasite drag:	x	x		
	(a) pressure drag;				
	(b) interference drag;				
	(c) friction drag.				
	The parasite drag and speed	x	x		
	The induced drag and speed	x	x		
	The total drag	x	x		
	The ground effect				
	Effect on take off and landing characteristics of an aeroplane	x	x		
	The stall				
	Flow separation at increasing angles of attack:	x	x		
	(a) the boundary layer:				
	(1) laminar layer;				
	(2) turbulent layer;				
	(3) transition.				
	(b) separation point;				
	(c) influence of angle of attack;				
	(d) influence on:				
	(1) pressure distribution;				
	(2) location of centre of pressure;				
	(3) C_L ;				
	(4) C_D ;				
	(5) pitch moments.				
	(e) buffet;				
	(f) use of controls.				
	The stall speed:	x	x		
	(a) in the lift formula;				
	(b) 1g stall speed;				
	(c) influence of:				
	(1) the centre of gravity;				
	(2) power setting;				
	(3) altitude (IAS);				
	(4) wing loading;				
	(5) load factor n:				
	(i) definition;				
	(ii) turns;				
	(iii) forces.				
	The initial stall in span-wise direction:	x	x		
	(a) influence of planform;				
	(b) geometric twist (wash out);				
	(c) use of ailerons.				
	Stall warning:	x	x		
	(a) importance of stall warning;				
	(b) speed margin;				
	(c) buffet;				
	(d) stall strip;				
	(e) flapper switch;				
	(f) recovery from stall.				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Special phenomena of stall: (a) the power-on stall; (b) climbing and descending turns; (c) t-tailed aeroplane; (d) avoidance of spins: (1) spin development; (2) spin recognition; (3) spin recovery. (e) ice (in stagnation point and on surface): (1) absence of stall warning; (2) abnormal behaviour of the aircraft during stall.	x	x		
	CL augmentation				
	Trailing edge flaps and the reasons for use in take-off and landing: (a) influence on C_L - α -graph; (b) different types of flaps; (c) flap asymmetry; (d) influence on pitch movement.	x	x		
	Leading edge devices and the reasons for use in take-off and landing	x	x		
	The boundary layer				
	Different types: (a) laminar; (b) turbulent.	x	x		
	Special circumstances				
	Ice and other contamination: (a) ice in stagnation point; (b) ice on the surface (frost, snow and clear ice); (c) rain; (d) contamination of the leading edge; (e) effects on stall; (f) effects on loss of controllability; (g) effects on control surface moment; (h) influence on high lift devices during takeoff, landing and low speeds.	x	x		
	Stability				
	Condition of equilibrium in steady horizontal flight				
	Precondition for static stability	x	x		
	Equilibrium: (a) lift and weight; (b) drag and thrust.	x	x		
	Methods of achieving balance				
	Wing and empennage (tail and canard)	x	x		
	Control surfaces	x	x		
	Ballast or weight trim	x	x		
	Static and dynamic longitudinal stability				
	Basics and definitions: (a) static stability, positive, neutral and negative; (b) precondition for dynamic stability; (c) dynamic stability, positive, neutral and negative.	x	x		

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Location of centre of gravity: (a) aft limit and minimum stability margin; (b) forward position; (c) effects on static and dynamic stability.	x	x		
	Dynamic lateral or directional stability				
	Spiral dive and corrective actions	x	x		
	Control				
	General				
	Basics, the three planes and three axis	x	x		
	Angle of attack change	x	x		
	Pitch control				
	Elevator	x	x		
	Downwash effects	x	x		
	Location of centre of gravity	x	x		
	Yaw control				
	Pedal or rudder	x	x		
	Roll control				
	Ailerons: function in different phases of flight	x	x		
	Adverse yaw	x	x		
	Means to avoid adverse yaw: (a) frise ailerons; (b) differential ailerons deflection.	x	x		
	Means to reduce control forces				
	Aerodynamic balance: (a) balance tab and anti-balance tab; (b) servo tab.	x	x		
	Mass balance				
	Reasons to balance: means	x	x		
	Trimming				
	Reasons to trim	x	x		
	Trim tabs	x	x		
	Limitations				
	Operating limitations				
	Flutter	x	x		
	V _{fe}	x	x		
	V _{no} , V _{ne}	x	x		
	Manoeuvring envelope				
	Manoeuvring load diagram: (a) load factor; (b) accelerated stall speed; (c) V _a ; (d) manoeuvring limit load factor or certification category.	x	x		
	Contribution of mass	x	x		
	Gust envelope				
	Gust load diagram	x	x		
	Factors contributing to gust loads	x	x		
	Propellers				
	Conversion of engine torque to thrust				
	Meaning of pitch	x	x		

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Blade twist	x	x		
	Effects of ice on propeller	x	x		
	Engine failure or engine stop				
	Windmilling drag	x	x		
	Moments due to propeller operation				
	Torque reaction	x	x		
	Asymmetric slipstream effect	x	x		
	Asymmetric blade effect	x	x		
	Flight mechanics				
	Forces acting on an aeroplane				
	Straight horizontal steady flight	x	x		
	Straight steady climb	x	x		
	Straight steady descent	x	x		
	Straight steady glide	x	x		
	Steady coordinated turn: (a) bank angle; (b) load factor; (c) turn radius; (d) rate one turn.	x	x		
5.2.	PRINCIPLES OF FLIGHT: HELICOPTER				
	Subsonic aerodynamics				
	Basic concepts, laws and definitions			x	x
	Conversion of units			x	x
	Definitions and basic concepts about air:			x	x
	(a) the atmosphere and International Standard Atmosphere; (b) density; (c) influence of pressure and temperature on density.				
	Newton's laws: (a) Newton's second law: Momentum equation; (b) Newton's third law: action and reaction.			x	x
	Basic concepts about airflow: (a) steady airflow and unsteady airflow; (b) Bernoulli's equation; (c) static pressure, dynamic pressure, total pressure and stagnation point; (d) TAS and IAS; (e) two-dimensional airflow and three-dimensional airflow; (f) viscosity and boundary layer.			x	x
	Two-dimensional airflow			x	x
	Aerofoil section geometry: (a) aerofoil section; (b) chord line, thickness and thickness to chord ratio of a section; (c) camber line and camber; (d) symmetrical and asymmetrical aerofoils sections.			x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Aerodynamic forces on aerofoil elements: (a) angle of attack; (b) pressure distribution; (c) lift and lift coefficient (d) relation lift coefficient: angle of attack; (e) profile drag and drag coefficient; (f) relation drag coefficient: angle of attack; (g) resulting force, centre of pressure and pitching moment.			x	x
	Stall: (a) boundary layer and reasons for stalling; (b) variation of lift and drag as a function of angle of attack; (c) displacement of the centre of pressure and pitching moment.			x	x
	Disturbances due to profile contamination: (a) ice contamination; (b) ice on the surface (frost, snow and clear ice).			x	x
	The three-dimensional airflow round a wing and a fuselage			x	x
	The wing:			x	x
	(a) planform, rectangular and tapered wings;				
	(b) wing twist.				
	Airflow pattern and influence on lift:			x	x
	(a) span wise flow on upper and lower surface; (b) tip vortices; (c) span-wise lift distribution.				
	Induced drag: causes and vortices			x	x
	The airflow round a fuselage: (a) components of a fuselage; (b) parasite drag; (c) variation with speed.			x	x
	Transonic aerodynamics and compressibility effects				
	Airflow velocities			x	x
	Airflow speeds: (a) speed of sound; (b) subsonic, high subsonic and supersonic flows.			x	x
	Shock waves: (a) compressibility and shock waves; (b) the reasons for their formation at upstream high subsonic airflow; (c) their effect on lift and drag.			x	x
	Influence of wing planform: sweep-angle			x	x
	Rotorcraft types			x	x
	Rotorcraft			x	x
	Rotorcraft types: (a) autogyro; (b) helicopter.			x	x
	Helicopters			x	x
	Helicopters configurations: the single main rotor helicopter			x	x
	The helicopter, characteristics and associated terminology: (a) general lay-out, fuselage, engine and gearbox; (b) tail rotor, fenestron and NOTAR;			x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	(c) engines (reciprocating and turbo shaft engines); (d) power transmission;				
	(e) rotor shaft axis, rotor hub and rotor blades; (f) rotor disc and rotor disc area; (g) teetering rotor (two blades) and rotors with more than two blades;				
	(h) skids and wheels; (i) helicopter axes and fuselage centre line;				
	(j) roll axis, pitch axis and normal or yaw axis; (k) gross mass, gross weight and disc loading.				
	Main rotor aerodynamics			x	x
	Hover flight outside ground effect			x	x
	Airflow through the rotor discs and round the blades: (a) circumferential velocity of the blade sections; (b) induced airflow, through the disc and downstream; (c) downward fuselage drag; (d) equilibrium of rotor thrust, weight and fuselage drag; (e) rotor disc induced power; (f) relative airflow to the blade; (g) pitch angle and angle of attack of a blade section; (h) lift and profile drag on the blade element; (i) resulting lift and thrust on the blade and rotor thrust; (j) collective pitch angle changes and necessity of blade feathering; (k) required total main rotor-torque and rotor-power; (l) influence of the air density.			x	x
	Anti-torque force and tail rotor: (a) force of tail rotor as a function of main rotor-torque; (b) anti-torque rotor power; (c) necessity of blade feathering of tail rotor blades and yaw pedals.			x	x
	Maximum hover altitude OGE: (a) total power required and power available; (b) maximum hover altitude as a function of pressure altitude and OAT.			x	x
	Vertical climb			x	x
	Relative airflow and angles of attack:			x	x
	(a) climb velocity V_C , induced and relative velocity and angle of attack; (b) collective pitch angle and blade feathering.				
	Power and vertical speed: (a) induced power, climb power and profile power; (b) total main rotor power and main rotor torque; (c) tail rotor power; (d) total power requirement in vertical flight.			x	x
	Forward flight			x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Airflow and forces in uniform inflow distribution: (a) assumption of uniform inflow distribution on rotor disc; (b) advancing blade (90°) and retreating blade (270°); (c) airflow velocity relative to the blade sections, area of reverse flow; (d) lift on the advancing and retreating blades at constant pitch angles; (e) necessity of cyclic pitch changes; (f) compressibility effects on the advancing blade tip and speed limitations; (g) high angle of attack on the retreating blade, blade stall and speed limitations; (h) thrust on rotor disc and tilt of thrust vector; (i) vertical component of the thrust vector and gross weight equilibrium; (j) horizontal component of the thrust vector and drag equilibrium.			x	x
	The flare (power flight):			x	x
	(a) thrust reversal and increase in rotor thrust; (b) increase of rotor RPM on non governed rotor.				
	Power and maximum speed: (a) induced power as a function of helicopter speed; (b) rotor profile power as a function of helicopter speed; (c) fuselage drag and parasite power as a function of forward speed; (d) tail rotor power and power ancillary equipment; (e) total power requirement as a function of forward speed; (f) influence of helicopter mass, air density and drag of additional external equipment; (g) translational lift and influence on power required.			x	x
	Hover and forward flight in ground effect			x	x
	Airflow in ground effect and downwash: rotor power decrease as a function of rotor height above the ground at constant helicopter mass			x	x
	Vertical descent			x	x
	Vertical descent, power on:			x	x
	(a) airflow through the rotor, low and moderate descent speeds; (b) vortex ring state, settling with power and consequences.				
	Autorotation: (a) collective lever position after failure; (b) up flow through the rotor, auto-rotation and anti-autorotation rings; (c) tail rotor thrust and yaw control; (d) control of rotor RPM with collective lever; (e) landing after increase of rotor thrust by pulling collective and reduction in vertical speed.			x	x
	Forward flight: Autorotation			x	x
	Airflow through the rotor disc: (a) descent speed and up flow through the disc;			x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	(b) the flare, increase in rotor thrust, reduction of vertical speed and ground speed.				
	Flight and landing:			x	x
	(a) turning; (b) flare; (c) autorotative landing; (d) height or velocity avoidance graph and dead man's curve.				
	Main rotor mechanics			x	x
	Flapping of the blade in hover			x	x
	Forces and stresses on the blade: (a) centrifugal force on the blade and attachments; (b) limits of rotor RPM; (c) lift on the blade and bending stresses on a rigid attachment; (d) the flapping hinge of the articulated rotor and flapping hinge offset; (e) the flapping of the hinge less rotor and flexible element.			x	x
	Coning angle in hover:			x	x
	(a) lift and centrifugal force in hover and blade weight negligible (b) flapping, tip path plane and disc area.				
	Flapping angles of the blade in forward flight			x	x
	Forces on the blade in forward flight without cyclic feathering: (a) aerodynamic forces on the advancing and retreating blades without cyclic feathering; (b) periodic forces and stresses, fatigue and flapping hinge; (c) phase lag between the force and the flapping angle (about 90°); (d) flapping motion of the hinged blades and tilting of the cone and flap back of rotor; (e) rotor disc attitude and thrust vector tilt.			x	x
	Cyclic pitch (feathering) in helicopter mode, forward flight: (a) necessity of forward rotor disc tilt and thrust vector tilt; (b) flapping and tip path plane, virtual rotation axis or no flapping axis and plane of rotation; (c) shaft axis and hub plane; (d) cyclic pitch change (feathering) and rotor thrust vector tilt; (e) collective pitch change, collective lever, swash plate, pitch link and pitch horn; (f) cyclic stick, rotating swash plate and pitch link movement and phase angle.			x	x
	Blade lag motion			x	x
	Forces on the blade in the disc plane (tip path plane) in forward flight: (a) forces due to the Coriolis effect because of the flapping; (b) alternating stresses and the need of the drag or lag hinge.			x	x
	The drag or lag hinge:			x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	(a) the drag hinge in the fully articulated rotor; (b) the lag flexure in the hinge less rotor; (c) drag dampers.				
	Ground resonance: (a) blade lag motion and movement of the centre of gravity of the blades and the rotor; (b) oscillating force on the fuselage; (c) fuselage, undercarriage and resonance.			x	x
	Rotor systems			x	x
	See-saw or teetering rotor			x	x
	Fully articulated rotor: (a) three hinges arrangement; (b) bearings and elastomeric hinges.			x	x
	Hinge less rotor and bearing less rotor			x	x
	Blade sailing: (a) low rotor RPM and effect of adverse wind; (b) minimising the danger; (c) droop stops.			x	x
	Vibrations due to main rotor: (a) origins of the vibrations: in plane and vertical; (b) blade tracking and balancing.			x	x
	Tail rotors			x	x
	Conventional tail rotor			x	x
	Rotor description: (a) two-blades tail rotors with teetering hinge; (b) rotors with more than two blades; (c) feathering bearings and flapping hinges; (d) dangers to people and to the tail rotor, rotor height and safety.			x	x
	Aerodynamics: (a) induced airflow and tail rotor thrust; (b) thrust control by feathering, tail rotor drift and roll; (c) effect of tail rotor failure and vortex ring.			x	x
	The fenestron: technical lay-out			x	x
	The NOTAR: technical lay-out			x	x
	Vibrations: high frequency vibrations due to the tail rotors			x	x
	Equilibrium, stability and control			x	x
	Equilibrium and helicopter attitudes			x	x
	Hover: (a) forces and equilibrium conditions; (b) helicopter pitching moment and pitch angle; (c) helicopter rolling moment and roll angle.			x	x
	Forward flight: (a) forces and equilibrium conditions; (b) helicopter moments and angles; (c) effect of speed on fuselage attitude.			x	x
	Control			x	x
	Control power			x	x
	(a) fully articulated rotor; (b) hinge less rotor;				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	(c) teetering rotor.				
	Static and dynamic roll over			x	x
	Helicopter performances				
	Engine performances			x	x
	Piston engines:			x	x
	(a) power available;				
	(b) effects of density altitude.				
	Turbine engines:			x	x
	(a) power available;				
	(b) effects of ambient pressure and temperature.				
	Helicopter performances			x	x
	Hover and vertical flight:			x	x
	(a) power required and power available;				
	(b) OGE and IGE maximum hover height;				
	(c) influence of AUM, pressure, temperature and density.				
	Forward flight:			x	x
	(a) maximum speed;				
	(b) maximum rate of climb speed;				
	(c) maximum angle of climb speed;				
	(d) range and endurance;				
	(e) influence of AUM, pressure, temperature and density.				
	Manoeuvring:			x	x
	(a) load factor;				
	(b) bank angle and number of g's;				
	(c) manoeuvring limit load factor.				
	Special conditions:			x	x
	(a) operating with limited power;				
	(b) over pitch and over torque.				
6.	OPERATIONAL PROCEDURES				
	General				
	Operation of aircraft: ICAO Annex 6, General requirements				
	Definitions	x	x	x	x
	Applicability	x	x	x	x
	Special operational procedures and hazards (general aspects)	x	x	x	x
	Noise abatement				
	Noise abatement procedures	x	x	x	x
	Influence of the flight procedure (departure, cruise and approach)	x	x	x	x
	Runway incursion awareness (meaning of surface markings and signals)	x	x	x	x
	Fire or smoke				
	Carburettor fire	x	x	x	x
	Engine fire	x	x	x	x
	Fire in the cabin and cockpit, (choice of extinguishing agents according to fire classification and use of the extinguishers)	x	x	x	x
	Smoke in the cockpit and (effects and action to be taken) and smoke in the cockpit and cabin (effects and actions taken)	x	x	x	x
	Windshear and microburst				
	Effects and recognition during departure and approach	x	x	x	x
	Actions to avoid and actions taken during encounter	x	x	x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Wake turbulence				
	Cause	x	x	x	x
	List of relevant parameters	x	x	x	x
	Actions taken when crossing traffic, during take-off and landing	x	x	x	x
	Emergency and precautionary landings				
	Definition	x	x	x	x
	Cause	x	x	x	x
	Passenger information	x	x	x	x
	Evacuation	x	x	x	x
	Action after landing	x	x	x	x
	Contaminated runways				
	Kinds of contamination	x	x		
	Estimated surface friction and friction coefficient	x	x		
	Rotor downwash			x	x
	Operation influence by meteorological conditions (helicopter)				
	White out, sand or dust			x	x
	Strong winds			x	x
	Mountain environment			x	x
	Emergency procedures				
	Influence by technical problems				
	Engine failure			x	x
	Fire in cabin, cockpit or engine			x	x
	Tail, rotor or directional control failure			x	x
	Ground resonance			x	x
	Blade stall			x	x
	Settling with power (vortex ring)			x	x
	Overpitch			x	x
	Overspeed: rotor or engine			x	x
	Dynamic rollover			x	x
	Mast bumping			x	x
7.	FLIGHT PERFORMANCE AND PLANNING				
7.1.	MASS AND BALANCE: AEROPLANES OR HELICOPTERS				
	Purpose of mass and balance considerations				
	Mass limitations				
	Importance in regard to structural limitations	x	x	x	x
	Importance in regard to performance limitations	x	x	x	x
	CG limitations				
	Importance in regard to stability and controllability	x	x	x	x
	Importance in regard to performance	x	x	x	x
	Loading				
	Terminology				
	Mass terms	x	x	x	x
	Load terms (including fuel terms)	x	x	x	x
	Mass limits				
	Structural limitations	x	x	x	x
	Performance limitations	x	x	x	x
	Baggage compartment limitations	x	x	x	x
	Mass calculations				
	Maximum masses for take-off and landing	x	x	x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Use of standard masses for passengers, baggage and crew	x	x	x	x
	Fundamentals of CG calculations				
	Definition of centre of gravity	x	x	x	x
	Conditions of equilibrium (balance of forces and balance of moments)	x	x	x	x
	Basic calculations of CG	x	x	x	x
	Mass and balance details of aircraft				
	Contents of mass and balance documentation				
	Datum and moment arm	x	x	x	x
	CG position as distance from datum	x	x	x	x
	Extraction of basic mass and balance data from aircraft documentation				
	BEM	x	x	x	x
	CG position or moment at BEM	x	x	x	x
	Deviations from standard configuration	x	x	x	x
	Determination of CG position				
	Methods				
	Arithmetic method	x	x	x	x
	Graphic method	x	x	x	x
	Load and trim sheet				
	General considerations	x	x	x	x
	Load sheet and CG envelope for light aeroplanes and for helicopters	x	x	x	x
7.2.	PERFORMANCE: AEROPLANES				
	Introduction				
	Performance classes	x	x		
	Stages of flight	x	x		
	Effect of aeroplane mass, wind, altitude, runway slope and runway conditions	x	x		
	Gradients	x	x		
	SE aeroplanes				
	Definitions of terms and speeds	x	x		
	Take-off and landing performance				
	Use of aeroplane flight manual data	x	x		
	Climb and cruise performance				
	Use of aeroplane flight data	x	x		
	Effect of density altitude and aeroplane mass	x	x		
	Endurance and the effects of the different recommended power or thrust settings	x	x		
	Still air range with various power or thrust settings	x	x		
7.3.	FLIGHT PLANNING AND FLIGHT MONITORING				
	Flight planning for VFR flights				
	VFR navigation plan				
	Routes, airfields, heights and altitudes from VFR charts	x	x	x	x
	Courses and distances from VFR charts	x	x	x	x
	Aerodrome charts and aerodrome directory	x	x	x	x
	Communications and radio navigation planning data	x	x	x	x
	Completion of navigation plan	x	x	x	x
	Fuel planning				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	General knowledge	x	x	x	x
	Pre-flight calculation of fuel required				
	Calculation of extra fuel	x	x	x	x
	Completion of the fuel section of the navigation plan (fuel log) and calculation of total fuel	x	x	x	x
	Pre-flight preparation				
	AIP and NOTAM briefing				
	Ground facilities and services	x	x	x	x
	Departure, destination and alternate aerodromes	x	x	x	x
	Airway routings and airspace structure	x	x	x	x
	Meteorological briefing				
	Extraction and analysis of relevant data from meteorological documents	x	x	x	x
	ICAO flight plan (ATS flight plan)				
	Individual flight plan				
	Format of flight plan	x	x	x	x
	Completion of the flight plan	x	x	x	x
	Submission of the flight plan	x	x	x	x
	Flight monitoring and in-flight replanning				
	Flight monitoring				
	Monitoring of track and time	x	x	x	x
	In-flight fuel management	x	x	x	x
	In-flight re-planning in case of deviation from planned data	x	x	x	x
7.4.	PERFORMANCE: HELICOPTERS				
	General				
	Introduction				
	Stages of flight			x	x
	Effect on performance of atmospheric, airport or heliport and helicopter conditions			x	x
	Applicability of airworthiness requirements			x	x
	Definitions and terminology			x	x
	Performance: SE helicopters				
	Definitions of terms			x	x
	(a) masses;				
	(b) velocities: v_x , v_y ;				
	(c) velocity of best range and of maximum endurance;				
	(d) power limitations;				
	(e) altitudes.				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Take-off, cruise and landing performance Use and interpretation of diagrams and tables: (a) Take-off: (1) take-off run and distance available; (2) take-off and initial climb; (3) effects of mass, wind and density altitude; (4) effects of ground surface and gradient. (b) Landing: (1) effects of mass, wind, density altitude and approach speed; (2) effects of ground surface and gradient. (c) In-flight: (1) relationship between power required and power available; (2) performance diagram; (3) effects of configuration, mass, temperature and altitude; (4) reduction of performance during climbing turns; (5) autorotation; (6) adverse effects (icing, rain and condition of the airframe).			x	x
8.	AIRCRAFT GENERAL KNOWLEDGE				
8.1.	AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT AND EMERGENCY EQUIPMENT				
	System design, loads, stresses, maintenance				
	Loads and combination loadings applied to an aircraft's structure	x	x	x	x
	Airframe				
	Wings, tail surfaces and control surfaces				
	Design and constructions	x	x		
	Structural components and materials	x	x		
	Stresses	x	x		
	Structural limitations	x	x		
	Fuselage, doors, floor, wind-screen and windows				
	Design and constructions	x	x	x	x
	Structural components and materials	x	x	x	x
	Stresses	x	x	x	x
	Structural limitations	x	x	x	x
	Flight and control surfaces				
	Design and constructions			x	x
	Structural components and materials			x	x
	Stresses and aero elastic vibrations			x	x
	Structural limitations			x	x
	Hydraulics				
	Hydromechanics: basic principles	x	x	x	x
	Hydraulic systems	x	x	x	x
	Hydraulic fluids: types and characteristics, limitations	x	x	x	x
	System components: design, operation, degraded modes of operation, indications and warnings	x	x	x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Landing gear, wheels, tyres and brakes				
	Landing gear				
	Types and materials	x	x	x	x
	Nose wheel steering: design and operation	x	x		
	Brakes				
	Types and materials	x	x	x	x
	System components: design, operation, indications and warnings	x	x	x	x
	Wheels and tyres				
	Types and operational limitations	x	x	x	x
	Helicopter equipments			x	x
	Flight controls				
	Mechanical or powered	x	x	x	x
	Control systems and mechanical	x	x	x	x
	System components: design, operation, indications and warnings, degraded modes of operation and jamming	x	x	x	x
	Secondary flight controls				
	System components: design, operation, degraded modes of operation, indications and warnings	x	x		
	Anti-icing systems				
	Types and operation (pitot and windshield)	x	x	x	x
	Fuel system				
	Piston engine				
	System components: design, operation, degraded modes of operation, indications and warnings	x	x	x	x
	Turbine engine				
	System components: design, operation, degraded modes of operation, indications and warnings			x	x
	Electrics				
	Electrics: general and definitions				
	Direct current: voltage, current, resistance, conductivity, Ohm's law, power and work	x	x	x	x
	Alternating current: voltage, current, amplitude, phase, frequency and resistance	x	x	x	x
	Circuits: series and parallel	x	x	x	x
	Magnetic field: effects in an electrical circuit	x	x	x	x
	Batteries				
	Types, characteristics and limitations	x	x	x	x
	Battery chargers, characteristics and limitations	x	x	x	x
	Static electricity: general				
	Basic principles	x	x	x	x
	Static dischargers	x	x	x	x
	Protection against interference	x	x	x	x
	Lightning effects	x	x	x	x
	Generation: production, distribution and use				
	DC generation: types, design, operation, degraded modes of operation, indications and warnings	x	x	x	x
	AC generation: types, design, operation, degraded modes of operation, indications and warnings	x	x	x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Electric components				
	Basic elements: basic principles of switches, circuit-breakers and relays	x	x	x	x
	Distribution				
	General: (a) bus bar, common earth and priority; (b) AC and DC comparison.	x	x	x	x
	Piston engines				
	General				
	Types of internal combustion engine: basic principles and definitions	x	x	x	x
	Engine: design, operation, components and materials	x	x	x	x
	Fuel				
	Types, grades, characteristics and limitations	x	x	x	x
	Alternate fuel: characteristics and limitations	x	x	x	x
	Carburettor or injection system				
	Carburettor: design, operation, degraded modes of operation, indications and warnings	x	x	x	x
	Injection: design, operation, degraded modes of operation, indications and warnings	x	x	x	x
	Icing	x	x	x	x
	Air cooling systems				
	Design, operation, degraded modes of operation, indications and warnings	x	x	x	x
	Lubrication systems				
	Lubricants: types, characteristics and limitations	x	x	x	x
	Design, operation, degraded modes of operation, indications and warnings	x	x	x	x
	Ignition circuits				
	Design, operation, degraded modes of operation	x	x	x	x
	Mixture				
	Definition, characteristic mixtures, control instruments, associated control levers and indications	x	x	x	x
	Propellers				
	Definitions and general: (a) aerodynamic parameters; (b) types; (c) operating modes.	x	x		
	Constant speed propeller: design, operation and system components	x	x		
	Propeller handling: associated control levers, degraded modes of operation, indications and warnings	x	x		
	Performance and engine handling				
	Performance: influence of engine parameters, influence of atmospheric conditions, limitations and power augmentation systems	x	x	x	x
	Engine handling: power and mixture settings during various flight phases and operational limitations	x	x	x	x
	Turbine engines				
	Definitions			x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Coupled turbine engine: design, operation, components and materials			x	x
	Free turbine engine: design, operation, components and materials			x	x
	Fuel				
	Types, characteristics and limitations			x	x
	Main engine components				
	Compressor:			x	x
	(a) types, design, operation, components and materials;				
	(b) stresses and limitations;				
	(c) stall, surge and means of prevention.				
	Combustion chamber:			x	x
	(a) types, design, operation, components and materials;				
	(b) stresses and limitations;				
	(c) emission problems.				
	Turbine:			x	x
	(a) types, design, operation, components and materials;				
	(b) stresses, creep and limitations.				
	Exhaust:			x	x
	(a) design, operation and materials;				
	(b) noise reduction.				
	Fuel control units: types, operation and sensors			x	x
	Helicopter air intake: different types, design, operation, materials and optional equipments			x	x
	Additional components and systems				
	Helicopter additional components and systems: lubrication system, ignition circuit, starter, accessory gearbox, free wheel units: design, operation and components			x	x
	Performance aspects				
	Torque, performance aspects, engine handling and limitations:			x	x
	(a) engine ratings;				
	(b) engine performance and limitations;				
	(c) engine handling.				
	Protection and detection systems				
	Fire detection systems				
	Operation and indications			x	x
	Miscellaneous systems				
	Rotor design			x	x
	Rotor heads				
	Main rotor				
	Types			x	x
	Structural components and materials, stresses and structural limitations			x	x
	Design and construction			x	x
	Adjustment			x	x
	Tail rotor				
	Types			x	x
	Structural components and materials, stresses and structural limitations			x	x
	Design and construction			x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Adjustment			x	x
	Transmission				
	Main gear box				
	Different types, design, operation and limitations			x	x
	Rotor brake				
	Different types, design, operation and limitations			x	x
	Auxiliary systems			x	x
	Drive shaft and associated installation			x	x
	Intermediate and tail gear box				
	Different types, design, operation and limitations			x	x
	Blades				
	Main rotor blade				
	Design and construction			x	x
	Structural components and materials			x	x
	Stresses			x	x
	Structural limitations			x	x
	Adjustment			x	x
	Tip shape			x	x
	Tail rotor blade				
	Design and construction			x	x
	Structural components and materials			x	x
	Stresses			x	x
	Structural limitations			x	x
	Adjustment			x	x
8.2.	INSTRUMENTATION				
	Instrument and indication systems				
	Pressure gauge				
	Different types, design, operation, characteristics and accuracy	x	x	x	x
	Temperature sensing				
	Different types, design, operation, characteristics and accuracy	x	x	x	x
	Fuel gauge				
	Different types, design, operation, characteristics and accuracy	x	x	x	x
	Flow meter				
	Different types, design, operation, characteristics and accuracy	x	x	x	x
	Position transmitter				
	Different types, design, operation, characteristics and accuracy	x	x	x	x
	Torque meter				
	Design, operation, characteristics and accuracy			x	x
	Tachometer				
	Design, operation, characteristics and accuracy	x	x	x	x
	Measurement of aerodynamic parameters				
	Pressure measurement				
	Static pressure, dynamic pressure, density and definitions	x	x	x	x
	Design, operation, errors and accuracy	x	x	x	x
	Temperature measurement: aeroplane				
	Design, operation, errors and accuracy	x	x		
	Displays	x	x		
	Temperature measurement: helicopter				
	Design, operation, errors and accuracy			x	x

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Displays			x	x
	Altimeter				
	Standard atmosphere	x	x	x	x
	The different barometric references (QNH, QFE and 1013.25)	x	x	x	x
	Height, indicated altitude, true altitude, pressure altitude and density altitude	x	x	x	x
	Design, operation, errors and accuracy	x	x	x	x
	Displays	x	x	x	x
	Vertical speed indicator				
	Design, operation, errors and accuracy	x	x	x	x
	Displays	x	x	x	x
	Air speed indicator				
	The different speeds IAS, CAS, TAS: definition, usage and relationships	x	x	x	x
	Design, operation, errors and accuracy	x	x	x	x
	Displays	x	x	x	x
	Magnetism: direct reading compass				
	Earth magnetic field	x	x	x	x
	Direct reading compass				
	Design, operation, data processing, accuracy and deviation	x	x	x	x
	Turning and acceleration errors	x	x	x	x
	Gyroscopic instruments				
	Gyroscope: basic principles				
	Definitions and design	x	x	x	x
	Fundamental properties	x	x	x	x
	Drifts	x	x	x	x
	Turn and bank indicator				
	Design, operation and errors	x	x	x	x
	Attitude indicator				
	Design, operation, errors and accuracy	x	x	x	x
	Directional gyroscope				
	Design, operation, errors and accuracy	x	x	x	x
	Communication systems				
	Transmission modes: VHF, HF and SATCOM				
	Principles, bandwidth, operational limitations and use	x	x	x	x
	Voice communication				
	Definitions, general and applications	x	x	x	x
	Alerting systems and proximity systems				
	Flight warning systems				
	Design, operation, indications and alarms	x	x	x	x
	Stall warning				
	Design, operation, indications and alarms	x	x		
	Radio-altimeter				
	Design, operation, errors, accuracy and indications			x	x
	Rotor or engine over speed alert system				
	Design, operation, displays and alarms			x	x
	Integrated instruments: electronic displays				
	Display units				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Design, different technologies and limitations	x	x	x	x
9.	NAVIGATION				
9.1.	GENERAL NAVIGATION				
	Basics of navigation				
	The solar system				
	Seasonal and apparent movements of the sun	x		x	
	The earth				
	Great circle, small circle and rhumb line	x		x	
	Latitude and difference of latitude	x		x	
	Longitude and difference of longitude	x		x	
	Use of latitude and longitude co-ordinates to locate any specific position	x		x	
	Time and time conversions				
	Apparent time	x		x	
	UTC	x		x	
	LMT	x		x	
	Standard times	x		x	
	Dateline	x		x	
	Definition of sunrise, sunset and civil twilight	x		x	
	Directions				
	True north, magnetic north and compass north	x		x	
	Compass deviation	x		x	
	Magnetic poles, isogonals, relationship between true and magnetic	x		x	
	Distance				
	Units of distance and height used in navigation: nautical miles, statute miles, kilometres, metres and ft	x		x	
	Conversion from one unit to another	x		x	
	Relationship between nautical miles and minutes of latitude and minutes of longitude	x		x	
	Magnetism and compasses				
	General principles				
	Terrestrial magnetism	x		x	
	Resolution of the earth's total magnetic force into vertical and horizontal components	x		x	
	Variation-annual change	x		x	
	Aircraft magnetism				
	The resulting magnetic fields	x		x	
	Keeping magnetic materials clear of the compass	x		x	
	Charts				
	General properties of miscellaneous types of projections				
	Direct Mercator	x		x	
	Lambert conformal conic	x		x	
	The representation of meridians, parallels, great circles and rhumb lines				
	Direct Mercator	x		x	
	Lambert conformal conic	x		x	
	The use of current aeronautical charts				
	Plotting positions	x		x	

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Methods of indicating scale and relief (ICAO topographical chart)	x		x	
	Conventional signs	x		x	
	Measuring tracks and distances	x		x	
	Plotting bearings and distances	x		x	
	DR navigation				
	Basis of DR				
	Track	x		x	
	Heading (compass, magnetic and true)	x		x	
	Wind velocity	x		x	
	Air speed (IAS, CAS and TAS)	x		x	
	Groundspeed	x		x	
	ETA	x		x	
	Drift and wind correction angle	x		x	
	DR position fix	x		x	
	Use of the navigational computer				
	Speed	x		x	
	Time	x		x	
	Distance	x		x	
	Fuel consumption	x		x	
	Conversions	x		x	
	Air speed	x		x	
	Wind velocity	x		x	
	True altitude	x		x	
	The triangle of velocities				
	Heading	x		x	
	Ground speed	x		x	
	Wind velocity	x		x	
	Track and drift angle	x		x	
	Measurement of DR elements				
	Calculation of altitude	x		x	
	Determination of appropriate speed	x		x	
	In-flight navigation				
	Use of visual observations and application to in-flight navigation	x		x	
	Navigation in cruising flight, use of fixes to revise navigation data				
	Ground speed revision	x		x	
	Off-track corrections	x		x	
	Calculation of wind speed and direction	x		x	
	ETA revisions	x		x	
	Flight log	x		x	
9.2.	RADIO NAVIGATION				
	Basic radio propagation theory				
	Antennas				
	Characteristics	x		x	
	Wave propagation				
	Propagation with the frequency bands	x		x	
	Radio aids				

		Aeroplane		Helicopter	
		PPL	Bridge course	PPL	Bridge course
	Ground DF				
	Principles	x		x	
	Presentation and interpretation	x		x	
	Coverage	x		x	
	Range	x		x	
	Errors and accuracy	x		x	
	Factors affecting range and accuracy	x		x	
	NDB/ADF				
	Principles	x		x	
	Presentation and interpretation	x		x	
	Coverage	x		x	
	Range	x		x	
	Errors and accuracy	x		x	
	Factors affecting range and accuracy	x		x	
	VOR				
	Principles	x		x	
	Presentation and interpretation	x		x	
	Coverage	x		x	
	Range	x		x	
	Errors and accuracy	x		x	
	Factors affecting range and accuracy	x		x	
	DME				
	Principles	x		x	
	Presentation and interpretation	x		x	
	Coverage	x		x	
	Range	x		x	
	Errors and accuracy	x		x	
	Factors affecting range and accuracy	x		x	
	Radar				
	Ground radar				
	Principles	x		x	
	Presentation and interpretation	x		x	
	Coverage	x		x	
	Range	x		x	
	Errors and accuracy	x		x	
	Factors affecting range and accuracy	x		x	
	Secondary surveillance radar and transponder				
	Principles	x		x	
	Presentation and interpretation	x		x	
	Modes and codes	x		x	
	GNSS				
	GPS, GLONASS OR GALILEO				
	Principles	x		x	
	Operation	x		x	
	Errors and accuracy	x		x	
	Factors affecting accuracy	x		x	

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SYLLABUS OF THEORETICAL KNOWLEDGE FOR THE PPL(AS)

The following table contains the syllabi for the courses of theoretical knowledge, as well as for the theoretical knowledge examinations for the PPL(As). The training and examination should cover aspects related to non-technical skills in an integrated manner, taking into account the particular risks associated to the licence and the activity.

		PPL
1.	AIR LAW AND ATC PROCEDURES	
	International law: conventions, agreements and organisations	x
	Airworthiness of aircraft	x
	Aircraft nationality and registration marks	x
	Personnel licensing	x
	Rules of the air	x
	Procedures for air navigation services: aircraft operations	x
	Air traffic services and air traffic management	x
	Aeronautical information service	x
	Aerodromes	x
	Search and rescue	x
	Security	x
	Aircraft accident and incident investigation	x
	National law	x

		PPL
2.	HUMAN PERFORMANCE	
	Human factors: basic concepts	x
	Basic aviation physiology and health maintenance	x
	Basic aviation psychology	x

		PPL
3.	METEOROLOGY	
	The atmosphere	x
	Wind	x
	Thermodynamics	x
	Clouds and fog	x
	Precipitation	x
	Air masses and fronts	x
	Pressure systems	x
	Climatology	x
	Flight hazards	x
	Meteorological information	x

		PPL
4.	COMMUNICATIONS	
	VFR COMMUNICATIONS	
	Definitions	x
	General operating procedures	x
	Relevant weather information terms (VFR)	x
	Action required to be taken in case of communication failure	x
	Distress and urgency procedures	x

	General principles of VHF propagation and allocation of frequencies	x
		PPL
5.	PRINCIPLES OF FLIGHT	
	Basics of aerostatics	x
	Basics of subsonic aerodynamics	x
	Aerodynamics of airships	x
	Stability	x
	Controllability	x
	Limitations	x
	Propellers	x
	Basics of airship flight mechanics	x
		PPL
6.	OPERATIONAL PROCEDURES	
	General requirements	x
	Special operational procedures and hazards (general aspects)	x
	Emergency procedures	x
		PPL
7.	FLIGHT PERFORMANCE AND PLANNING	
7.1	MASS AND BALANCE	
	Purpose of mass and balance considerations	x
	Loading	x
	Fundamentals of CG calculations	x
	Mass and balance details of aircraft	x
	Determination of CG position	x
	Passenger, cargo and ballast handling	x
7.2	PERFORMANCE	
	Airworthiness requirements	x
	Basics of airship performance	x
	Definitions and terms	x
	Stages of flight	x
	Use of flight manual	x
7.3	FLIGHT PLANNING AND FLIGHT MONITORING	
	Flight planning for VFR flights	x
	Fuel planning	x
	Pre-flight preparation	x
	ATS flight plan	x
	Flight monitoring and in-flight re-planning	x
		PPL
8.	AIRCRAFT GENERAL KNOWLEDGE	
8.1	ENVELOPE, AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT AND EMERGENCY EQUIPMENT	
	Design, materials, loads and stresses	x
	Envelope and airbags	x
	Framework	x
	Gondola	x
	Flight controls	x
	Landing gear	x
	Hydraulics and pneumatics	x
	Heating and air conditioning	x

	Fuel system	X
	Piston engines (propellers)	X
	Turbine engines (basics)	X
	Electrics	X
	Fire protection and detection systems	X
	Maintenance	X
8.2	INSTRUMENTATION	
	Sensors and instruments	X
	Measurement of air data and gas parameters	X
	Magnetism: direct reading compass and flux valve	X
	Gyroscopic instruments	X
	Communication systems	X
	Alerting systems	X
	Integrated instruments: electronic displays	X
	Flight management system (general basics)	X
	Digital circuits and computers	X

		PPL
9.	NAVIGATION	
9.1.	GENERAL NAVIGATION	
	Basics of navigation	X
	Magnetism and compasses	X
	Charts	X
	DR navigation	X
	In-flight navigation	X
9.2.	RADIO NAVIGATION	
	Basic radio propagation theory	X
	Radio aids	X
	Radar	X
	GNSS	X

FCL.235 Skill test

Regulation (EU) 2025/134

- (a) Through the completion of a skill test, applicants for a PPL shall demonstrate the ability to perform as PIC on the appropriate aircraft category the relevant procedures and manoeuvres with the competency appropriate to the privileges granted.

[applicable until 17 February 2026 - Regulation (EU) 2020/2193]

- (a) Through the completion of a skill test, applicants for a PPL or a GPL shall demonstrate the ability to perform as PIC on the appropriate aircraft category the relevant procedures and manoeuvres with the competency appropriate to the privileges granted.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (b) Applicants for the skill test shall have received flight instruction on the same class or type of aircraft to be used for the skill test.
- (c) Pass marks
- (1) The skill test shall be divided into different sections, representing all the different phases of flight appropriate to the category of aircraft flown.

- (2) Failure in any item of a section will cause the applicant to fail the entire section. If the applicant fails only 1 section, he/she shall repeat only that section. Failure in more than 1 section will cause the applicant to fail the entire test.
- (3) When the test needs to be repeated in accordance with (2), failure in any section, including those that have been passed on a previous attempt, will cause the applicant to fail the entire test.
- (4) Failure to achieve a pass in all sections of the test in 2 attempts will require further training.

AMC1 FCL.125; FCL.235

ED Decision 2011/016/R

CONTENTS OF THE SKILL TEST FOR THE ISSUE OF A LAPL(S) AND OF AN SPL

- (a) An applicant should be responsible for the flight planning and should ensure that all equipment and documentation for the execution of the flight are on board.
- (b) The applicant should indicate to the FE the checks and duties carried out.
Checks should be completed in accordance with the flight manual or the authorised checklist for the sailplane on which the test is being taken.

FLIGHT TEST TOLERANCE

- (c) The applicant should demonstrate the ability to:
 - (1) operate the sailplane within its limitations;
 - (2) complete all manoeuvres with smoothness and accuracy;
 - (3) exercise good judgment and airmanship;
 - (4) apply aeronautical knowledge;
 - (5) maintain control of the sailplane at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.

CONTENT OF THE SKILL TEST

- (d) The skill test contents and sections set out in this AMC should be used for the skill test for the issue of a LAPL(S) and of an SPL:

SECTION 1 PRE-FLIGHT OPERATIONS AND DEPARTURE

Use of checklist, airmanship (control of sailplane by external visual reference), look-out. Apply in all sections.

- | | |
|---|--|
| a | Pre-flight sailplane (daily) inspection, documentation, NOTAM and weather briefing |
| b | Verifying in-limits mass and balance and performance calculation |
| c | Sailplane servicing compliance |
| d | Pre-take-off checks |

SECTION 2 LAUNCH METHOD

Note: at least for one of the three launch methods all the mentioned items are fully exercised during the skill test

SECTION 2 (A) WINCH OR CAR LAUNCH

- | | |
|---|--|
| a | Signals before and during launch, including messages to winch driver |
| b | Adequate profile of winch launch |

c	Simulated launch failure
d	Situational awareness
SECTION 2 (B) AEROTOW LAUNCH	
a	Signals before and during launch, including signals to or communications with tow plane pilot for any problems
b	Initial roll and take-off climb
c	Launch abandonment (simulation only or ‘talk-through’)
d	Correct positioning during straight flight and turns
e	Out of position and recovery
f	Correct release from tow
g	Look-out and airmanship through whole launch phase
SECTION 2 (C) SELF-LAUNCH (powered sailplanes only)	
a	ATC compliance (if applicable)
b	Aerodrome departure procedures
c	Initial roll and take-off climb
d	Look-out and airmanship during the whole take-off
e	Simulated engine failure after take-off
f	Engine shut down and stowage
SECTION 3 GENERAL AIRWORK	
a	Maintain straight flight: attitude and speed control
b	Coordinated medium (30 ° bank) turns, look-out procedures and collision avoidance
c	Turning on to selected headings visually and with use of compass
d	Flight at high angle of attack (critically low air speed)
e	Clean stall and recovery
f	Spin avoidance and recovery
g	Steep (45 ° bank) turns, look-out procedures and collision avoidance
h	Local area navigation and awareness
SECTION 4 CIRCUIT, APPROACH AND LANDING	
a	Aerodrome circuit joining procedure
b	Collision avoidance: look-out procedures
c	Pre-landing checks
d	Circuit, approach control and landing
e	Precision landing (simulation of out-landing and short field)
f	Crosswind landing if suitable conditions available

AMC2 FCL.125; FCL.235

ED Decision 2011/016/R

CONTENTS OF THE SKILL TEST FOR THE ISSUE OF A LAPL(B) AND A BPL

- (a) The take-off site should be chosen by the applicant depending on the actual meteorological conditions, the area which has to be over flown and the possible options for suitable landing sites. The applicant should be responsible for the flight planning and should ensure that all equipment and documentation for the execution of the flight are on board.

- (b) An applicant should indicate to the FE the checks and duties carried out. Checks should be completed in accordance with the flight manual or the authorised checklist for the balloon on which the test is being taken. During pre-flight preparation for the test the applicant should be required to perform crew and passenger briefings and demonstrate crowd control. The load calculation should be performed by the applicant in compliance with the operations manual or flight manual for the balloon used.

FLIGHT TEST TOLERANCE

- (c) The applicant should demonstrate the ability to:
- (1) operate the balloon within its limitations;
 - (2) complete all manoeuvres with smoothness and accuracy
 - (3) exercise good judgment and airmanship;
 - (4) apply aeronautical knowledge;
 - (5) maintain control of the balloon at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.

CONTENT OF THE SKILL TEST

- (d) The skill test contents and sections set out in this paragraph should be used for the skill test for the issue of a LAPL(B) (hot-air balloon) and a BPL (hot-air balloon):

SECTION 1 PRE-FLIGHT OPERATIONS, INFLATION AND TAKE-OFF	
Use of checklist, airmanship, control of balloon by external visual reference, look-out procedures, etc. apply in all sections.	
a	Pre-flight documentation, flight planning, NOTAM and weather briefing
b	Balloon inspection and servicing
c	Load calculation
d	Crowd control, crew and passenger briefings
e	Assembly and layout
f	Inflation and pre-take-off procedures
g	Take-off
h	ATC compliance(if applicable)
SECTION 2 GENERAL AIRWORK	
a	Climb to level flight
b	Level flight
c	Descent to level flight
d	Operating at low level
e	ATC compliance (if applicable)
SECTION 3 EN-ROUTE PROCEDURES	
a	Dead reckoning and map reading
b	Marking positions and time
c	Orientation and airspace structure
d	Maintenance of altitude
e	Fuel management
f	Communication with retrieve crew
g	ATC compliance
SECTION 4 APPROACH AND LANDING PROCEDURES	

a	Approach from low level, missed approach and fly on
b	Approach from high level, missed approach and fly on
c	Pre-landing checks
d	Passenger pre-landing briefing
e	Selection of landing field
f	Landing, dragging and deflation
g	ATC compliance (if applicable)
h	Actions after flight
SECTION 5 ABNORMAL AND EMERGENCY PROCEDURES	
a	Simulated fire on the ground and in the air
b	Simulated pilot light and burner failures
c	Other abnormal and emergency procedures as outlined in the appropriate flight manual.
d	Oral questions

- (e) The skill test contents and sections set out in this paragraph should be used for the skill test for the issue of a LAPL(B) (gas balloon) and a BPL (gas balloon):

SECTION 1 PRE-FLIGHT OPERATIONS, INFLATION AND TAKE-OFF	
Use of checklist, airmanship, control of balloon by external visual reference, look-out procedures, etc. apply in all sections.	
a	Pre-flight documentation, flight planning, NOTAM and weather briefing
b	Balloon inspection and servicing
c	Load calculation
d	Crowd control, crew and passenger briefings
e	Assembly and layout
f	Inflation and pre-take-off procedures
g	Take-off
h	ATC compliance (if applicable)
SECTION 2 GENERAL AIRWORK	
a	Climb to level flight
b	Level flight
c	Descent to level flight
d	Operating at low level
e	ATC compliance (if applicable)
SECTION 3 EN-ROUTE PROCEDURES	
a	Dead reckoning and map reading
b	Marking positions and time
c	Orientation and airspace structure
d	Maintenance of altitude
e	Ballast management
f	Communication with retrieve crew
g	ATC compliance
SECTION 4 APPROACH AND LANDING PROCEDURES	
a	Approach from low level, missed approach and fly on
b	Approach from high level, missed approach and fly on
c	Pre-landing checks

d	Passenger pre-landing briefing
e	Selection of landing field
f	Landing, dragging and deflation
g	ATC compliance (if applicable)
h	Actions after flight
SECTION 5 ABNORMAL AND EMERGENCY PROCEDURES	
a	Simulated closed appendix during take-off and climb
b	Simulated parachute or valve failure
c	Other abnormal and emergency procedures as outlined in the appropriate flight manual
d	Oral questions

AMC1 FCL.215; FCL.235

ED Decision 2011/016/R

THEORETICAL KNOWLEDGE EXAMINATION AND SKILL TEST FOR THE PPL

(a) Theoretical knowledge examination

- (1) The examinations should comprise a total of 120 multiple-choice questions covering all the subjects.
- (2) Communication practical classroom testing may be conducted.
- (3) The period of 18 months mentioned in [FCL.025\(b\)\(2\)](#) should be counted from the end of the calendar month when the applicant first attempted an examination.

(b) Skill test

Further training may be required following any failed skill test or part thereof. There should be no limit to the number of skill tests that may be attempted.

(c) Conduct of the test

- (1) If the applicant chooses to terminate a skill test for reasons considered inadequate by the FE, the applicant should retake the entire skill test. If the test is terminated for reasons considered adequate by the FE, only those sections not completed should be tested in a further flight.
- (2) Any manoeuvre or procedure of the test may be repeated once by the applicant. The FE may stop the test at any stage if it is considered that the applicant's demonstration of flying skill requires a complete retest.
- (3) An applicant should be required to fly the aircraft from a position where the PIC functions can be performed and to carry out the test as if there is no other crew member. Responsibility for the flight should be allocated in accordance with national regulations.

AMC1 FCL.235 Skill test

ED Decision 2011/016/R

CONTENTS OF THE SKILL TEST FOR THE ISSUE OF A PPL(A)

- (a) The route to be flown for the navigation test should be chosen by the FE. The route may end at the aerodrome of departure or at another aerodrome. The applicant should be responsible for the flight planning and should ensure that all equipment and documentation for the execution

of the flight are on board. The navigation section of the test should have a duration that allows the pilot to demonstrate his/her ability to complete a route with at least three identified waypoints and may, as agreed between the applicant and FE, be flown as a separate test.

- (b) An applicant should indicate to the FE the checks and duties carried out, including the identification of radio facilities. Checks should be completed in accordance with the authorised checklist for the aeroplane on which the test is being taken. During pre-flight preparation for the test the applicant should be required to determine power settings and speeds. Performance data for take-off, approach and landing should be calculated by the applicant in compliance with the operations manual or flight manual for the aeroplane used.

FLIGHT TEST TOLERANCE

- (c) The applicant should demonstrate the ability to:
- (1) operate the aeroplane within its limitations;
 - (2) complete all manoeuvres with smoothness and accuracy;
 - (3) exercise good judgment and airmanship;
 - (4) apply aeronautical knowledge;
 - (5) maintain control of the aeroplane at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.
- (d) The following limits are for general guidance. The FE should make allowance for turbulent conditions and the handling qualities and performance of the aeroplane used:
- (1) height:
 - (i) normal flight ± 150 ft
 - (ii) with simulated engine failure ± 200 ft (if ME aeroplane is used)
 - (2) heading or tracking of radio aids:
 - (i) normal flight $\pm 10^\circ$
 - (ii) with simulated engine failure $\pm 15^\circ$ (if ME aeroplane is used)
 - (3) speed:
 - (i) take-off and approach $+15/-5$ knots
 - (ii) all other flight regimes ± 15 knots

CONTENT OF THE SKILL TEST

- (e) The skill test contents and sections set out in this AMC should be used for the skill test for the issue of a PPL(A) on SE and ME aeroplanes or on TMGs.

SECTION 1 PRE-FLIGHT OPERATIONS AND DEPARTURE	
Use of checklist, airmanship, control of aeroplane by external visual reference, anti/de-icing procedures, etc. apply in all sections.	
a	Pre-flight documentation, NOTAM and weather briefing
b	Mass and balance and performance calculation
c	Aeroplane inspection and servicing
d	Engine starting and after starting procedures
e	Taxiing and aerodrome procedures, pre-take-off procedures

f	Take-off and after take-off checks
g	Aerodrome departure procedures
h	ATC compliance and R/T procedures
SECTION 2 GENERAL AIRWORK	
a	ATC compliance and R/T procedures
b	Straight and level flight, with speed changes
c	Climbing: i. best rate of climb; ii. climbing turns; iii. levelling off.
d	Medium (30 ° bank) turns
e	Steep (45 ° bank) turns (including recognition and recovery from a spiral dive)
f	Flight at critically low air speed with and without flaps
g	Stalling: i. clean stall and recover with power; ii. approach to stall descending turn with bank angle 20°, approach configuration; iii. approach to stall in landing configuration.
h	Descending: i. with and without power; ii. descending turns (steep gliding turns); iii. levelling off.
SECTION 3 EN-ROUTE PROCEDURES	
a	Flight plan, dead reckoning and map reading
b	Maintenance of altitude, heading and speed
c	Orientation, timing and revision of ETAs and log keeping
d	Diversion to alternate aerodrome (planning and implementation)
e	Use of radio navigation aids
f	Basic instrument flying check (180° turn in simulated IMC)
g	Flight management (checks, fuel systems and carburettor icing, etc.)
h	ATC compliance and R/T procedures
SECTION 4 APPROACH AND LANDING PROCEDURES	
a	Aerodrome arrival procedures
b	* Precision landing (short field landing), crosswind, if suitable conditions available
c	* Flapless landing
d	* Approach to landing with idle power (SE only)
e	Touch and go
f	Go-around from low height
g	ATC compliance and R/T procedures
h	Actions after flight
SECTION 5 ABNORMAL AND EMERGENCY PROCEDURES	
This section may be combined with sections 1 through 4	
a	Simulated engine failure after take-off (SE only)
b	* Simulated forced landing (SE only)
c	Simulated precautionary landing (SE only)
d	Simulated emergencies
e	Oral questions
SECTION 6 SIMULATED ASYMMETRIC FLIGHT AND RELEVANT CLASS OR TYPE ITEMS	

This section may be combined with sections 1 through 5	
a	Simulated engine failure during take-off (at a safe altitude unless carried out in an FFS)
b	Asymmetric approach and go-around
c	Asymmetric approach and full stop landing
d	Engine shutdown and restart
e	ATC compliance, R/T procedures or airmanship
f	As determined by the FE: any relevant items of the class or type rating skill test to include, if applicable: i. aeroplane systems including handling of auto pilot; ii. operation of pressurisation system; iii. use of de-icing and anti-icing system.
g	Oral questions

* These items may be combined, at the discretion of the FE.

AMC2 FCL.235 Skill test

ED Decision 2011/016/R

CONTENTS OF THE SKILL TEST FOR THE ISSUE OF A PPL(H)

- (a) The area and route to be flown should be chosen by the FE and all low level and hover work should be at an adequate aerodrome or site. Routes used for section 3 may end at the aerodrome of departure or at another aerodrome. The applicant should be responsible for the flight planning and should ensure that all equipment and documentation for the execution of the flight are on board. The navigation section of the test, as set out in this AMC should consist of at least three legs, each leg of a minimum duration of 10 minutes. The skill test may be conducted in two flights.
- (b) An applicant should indicate to the FE the checks and duties carried out, including the identification of radio facilities. Checks should be completed in accordance with the authorised checklist or pilot operating handbook for the helicopter on which the test is being taken. During pre-flight preparation for the test the applicant is required to determine power settings and speeds. Performance data for take-off, approach and landing should be calculated by the applicant in compliance with the operations manual or flight manual for the helicopter used.

FLIGHT TEST TOLERANCE

- (c) The applicant should demonstrate the ability to:
 - (1) operate the helicopter within its limitations;
 - (2) complete all manoeuvres with smoothness and accuracy;
 - (3) exercise good judgement and airmanship;
 - (4) apply aeronautical knowledge;
 - (5) maintain control of the helicopter at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.
- (d) The following limits are for general guidance. The FE should make allowance for turbulent conditions and the handling qualities and performance of the helicopter used.
 - (1) height:
 - (i) normal forward flight ± 150 ft

- (ii) with simulated major emergency ± 200 ft
 - (iii) hovering IGE flight ± 2 ft
- (2) heading or tracking of radio aids:
 - (i) normal flight $\pm 10^\circ$
 - (ii) with simulated major emergency $\pm 15^\circ$
- (3) speed:
 - (i) take-off approach $- 10$ knots/+15 knots
 - (ii) all other flight regimes ± 15 knots
- (4) ground drift:
 - (i) take-off hover IGE ± 3 ft
 - (ii) landing no sideways or backwards movement

CONTENT OF THE SKILL TEST

- (e) The skill test contents and sections set out in this AMC should be used for the skill test for the issue of a PPL(H) on SE or ME helicopters.

SECTION 1 PRE-FLIGHT OR POST-FLIGHT CHECKS AND PROCEDURES	
Use of checklist, airmanship, control of helicopter by external visual reference, anti-icing procedures, etc. apply in all sections	
a	Helicopter knowledge, (for example technical log, fuel, mass and balance, performance), flight planning, NOTAM and weather briefing
b	Pre-flight inspection or action, location of parts and purpose
c	Cockpit inspection and starting procedure
d	Communication and navigation equipment checks, selecting and setting frequencies
e	Pre-take-off procedure, R/T procedure and ATC compliance
f	Parking, shutdown and post-flight procedure
SECTION 2 HOVER MANOEUVRES, ADVANCED HANDLING AND CONFINED AREAS	
a	Take-off and landing (lift-off and touch down)
b	Taxi and hover taxi
c	Stationary hover with head, cross or tail wind
d	Stationary hover turns, 360° left and right (spot turns)
e	Forward, sideways and backwards hover manoeuvring
f	Simulated engine failure from the hover
g	Quick stops into and downwind
h	Sloping ground or unprepared sites landings and take-offs
i	Take-offs (various profiles)
j	Crosswind and downwind take-off (if practicable)
k	Take-off at maximum take-off mass (actual or simulated)
l	Approaches (various profiles)
m	Limited power take-off and landing
n	Autorotations, (FE to select two items from: basic, range, low speed and 360° turns)
o	Autorotative landing
p	Practice forced landing with power recovery

q	Power checks, reconnaissance technique, approach and departure technique
SECTION 3 NAVIGATION - EN ROUTE PROCEDURES	
a	Navigation and orientation at various altitudes or heights and map reading
b	Altitude or height, speed, heading control, observation of airspace and altimeter setting
c	Monitoring of flight progress, flight log, fuel usage, endurance, ETA, assessment of track error and re-establishment of correct track and instrument monitoring
d	Observation of weather conditions and diversion planning
e	Use of navigation aids (where available)
f	ATC liaison with due observance of regulations, etc.
SECTION 4 FLIGHT PROCEDURES AND MANOEUVRES	
a	Level flight, control of heading, altitude or height and speed
b	Climbing and descending turns to specified headings
c	Level turns with up to 30° bank, 180° to 360° left and right
d	Level turns 180° left and right by sole reference to instruments
SECTION 5 ABNORMAL AND EMERGENCY PROCEDURES (SIMULATED WHERE APPROPRIATE)	
Note (1) Where the test is conducted on an ME helicopter, a simulated engine failure drill, including an SE approach and landing should be included in the test.	
Note (2) The FE should select four items from the following:	
a	Engine malfunctions, including governor failure, carburettor or engine icing and oil system, as appropriate
b	Fuel system malfunction
c	Electrical system malfunction
d	Hydraulic system malfunction, including approach and landing without hydraulics, as applicable
e	Main rotor or anti-torque system malfunction (FFS or discussion only)
f	Fire drills, including smoke control and removal, as applicable
g	Other abnormal and emergency procedures as outlined in an appropriate flight manual and with reference to Appendix 9 C to Part-FCL, sections 3 and 4, including for ME helicopters: (a) Simulated engine failure at take-off: (1) rejected take-off at or before TDP or safe forced landing at or before DPATO; (2) shortly after TDP or DPATO. (b) Landing with simulated engine failure: (1) landing or go-around following engine failure before LDP or DPBL; (2) following engine failure after LDP or safe forced landing after DPBL.

AMC3 FCL.235 Skill test

ED Decision 2011/016/R

CONTENT OF THE SKILL TEST FOR THE ISSUE OF THE PPL(AS)

- (a) The area and route to be flown is chosen by the FE. Routes used for section 3 may end at the aerodrome of departure or at another aerodrome and one destination should be a controlled aerodrome. The skill test may be conducted in two flights. The total duration of the flight(s) should be at least 60 minutes.
- (b) The applicant should demonstrate the ability to:
 - (1) operate the airship within its limitations;
 - (2) complete all manoeuvres with smoothness and accuracy;

- (3) exercise good judgement and airmanship;
- (4) apply aeronautical knowledge;
- (5) maintain control of the airship at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.

FLIGHT TEST TOLERANCES

- (c) The following limits should apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the airship used.
- (1) height:
 - (i) normal flight ± 200 ft
 - (ii) simulated major emergency ± 300 ft
 - (2) tracking on radio aids: $\pm 15^\circ$
 - (3) heading:
 - (i) normal flight $\pm 15^\circ$
 - (ii) simulated major emergency $\pm 20^\circ$

CONTENT OF THE TEST

- (d) The skill test contents and sections set out in this AMC should be used for the skill test for the issue of a PPL(As).
- (e) Items in sections 5 and 6 may be performed in an FNPT (As) or a FS (As).

SECTION 1 PRE-FLIGHT OPERATIONS AND DEPARTURE

Use of airship checklists, airmanship, control of airship by external visual reference, anti-icing procedures, and principles of threat and error management, etc. apply in all sections

- | | |
|---|--|
| a | Pre-flight, including:
flight planning, documentation, mass and balance, NOTAM and weather briefing |
| b | Airship inspection and servicing |
| c | Off-mast procedure, ground manoeuvring and take-off |
| d | Performance considerations and trim |
| e | Aerodrome and traffic pattern operations |
| f | Departure procedure, altimeter setting, collision avoidance (look-out) |
| g | ATC compliance and R/T procedures |

SECTION 2 GENERAL AIRWORK

- | | |
|---|---|
| a | Control of the airship by external visual reference, including straight and level, climb, descent and look-out |
| b | Flight close to pressure height |
| c | Turns |
| d | Steep descents and climbs |
| e | Flight by reference solely to instruments, including: <ol style="list-style-type: none"> i. Level flight, control of heading, altitude and air speed; ii. Climbing and descending turns; iii. Recoveries from unusual attitudes. |
| f | ATC compliance and R/T procedures |

SECTION 3 EN-ROUTE PROCEDURES

- | | |
|---|---|
| a | Flight plan, dead reckoning and map reading |
|---|---|

b	Maintenance of altitude, heading and speed and collision avoidance (look-out procedures)
c	Orientation, timing and revision of ETAs and log keeping
d	Observation of weather conditions and diversion to alternate aerodrome (planning and implementation)
e	Use of radio navigation aids
f	Flight management (checks, fuel systems, etc.)
g	ATC compliance and R/T procedures
SECTION 4 APPROACH AND LANDING PROCEDURES	
a	Aerodrome arrival procedures, altimeter setting, checks and look-out
b	ATC compliance and R/T procedures
c	Go-around action
d	Normal landing
e	Short field landing
f	Post-flight actions
SECTION 5 ABNORMAL AND EMERGENCY PROCEDURES	
This section may be combined with sections 1 through 4	
a	Simulated engine failure after take-off (at a safe altitude) and fire drill
b	Equipment malfunctions
c	Forced landing (simulated)
d	ATC compliance and R/T procedures
e	Oral questions
SECTION 6 RELEVANT TYPE ITEMS	
This section may be combined with sections 1 through 5	
a	Simulated engine failure during take-off (at a safe altitude unless carried out in a FFS)
b	Approach and go-around with failed engine(s)
c	Approach and full stop landing with failed engine(s)
d	Malfunctions in the envelope pressure system
e	ATC compliance, R/T procedures and airmanship
f	As determined by the FE: any relevant items of the type rating skill test to include, if applicable: i. Airship systems; ii. Operation of envelope pressure system.
g	Oral questions

SECTION 2 – SPECIFIC REQUIREMENTS FOR THE PPL AEROPLANES – PPL(A)

FCL.205.A PPL(A) – Privileges

Regulation (EU) 2024/2076

- (a) The privileges of the holders of a PPL(A) are to act without remuneration as PIC or co-pilots of aeroplanes or TMGs engaged in non-commercial operations and to exercise all privileges of holders of an LAPL(A).
- (b) Notwithstanding the paragraph above, the holder of a PPL(A) with instructor or examiner privileges may receive remuneration for:
 - (1) the provision of flight instruction for the LAPL(A) or PPL(A);
 - (2) the conduct of skill tests and proficiency checks for the licences specified in point (1);
 - (3) the training, testing and checking for the ratings or certificates associated with the instructor's or examiner's licence, as applicable.

GM1 FCL.205.A(a); FCL.205.H(a); FCL.305(a)(1); FCL.505(a)(1) Privileges and conditions

ED Decision 2025/002/R

EXERCISING LAPL PRIVILEGES WHEN HOLDING A PPL, A CPL OR AN ATPL

- (a) Part-FCL requirements stipulate that the privileges of a PPL, CPL or ATPL include LAPL privileges. LAPL privileges in this context mean the privileges to act as pilot in aircraft specified in point [FCL.105.A\(a\)](#) or point [FCL.105.H\(a\)](#), as applicable, and under the conditions specified in these points. It does however not mean that the holder of a PPL, CPL or ATPL can keep the relevant class or type ratings valid through complying with the LAPL recency requirements. Class or type ratings need to be kept valid in accordance with Subpart H of Part-FCL.
- (b) As an example, when the holder of a PPL and a SEP aeroplane class rating temporarily loses his or her class 2 medical certificate but keeps an LAPL medical certificate, he or she is still holder of a PPL and of that SEP aeroplane class rating in terms of Part-FCL Subpart H. This pilot could still fly SEP aeroplanes under the conditions specified in point [FCL.105.A\(a\)](#), as long as the class rating is still valid. If the class rating is about to expire, it needs to be revalidated in accordance with point [FCL.740.A](#). Only after the PPL is exchanged for an LAPL, that pilot will fall under Subpart B of Part-FCL and can maintain SEP aeroplane class privileges through compliance with LAPL(A) recency requirements.
- (c) In line with the above explanations, a PPL(A) holder who temporarily only holds an LAPL medical certificate can revalidate class ratings that fall within the scope of the LAPL(A) (SEP aeroplane, TMG). However, it would not be possible for such a pilot to revalidate class ratings outside the scope of the LAPL. For example, if the pilot, on his or her PPL, also holds an MEP or SET class rating, he or she can revalidate (or renew) such a rating only after being issued with a class 2 medical certificate again.

AMC1 FCL.210.A(b) PPL(A) — Experience requirements and crediting

ED Decision 2025/002/R

PPL(A) TRAINING FOR APPLICANTS WHO HOLD AN LAPL(A) OR WHO HAVE UNDERGONE LAPL(A) TRAINING

Applicants for a PPL(A) who already hold an LAPL(A) and applicants who change their ongoing LAPL(A) training into PPL(A) training should receive training the content of which should be determined by the head of training of the DTO or the ATO where the applicant undergoes PPL(A) training, after assessing the individual applicant's training needs. In any case, flight training should include the following exercises from the PPL(A) flight instruction syllabus ([AMC1 FCL.210](#)):

- (a) Exercise 11 — Spin avoidance;
- (b) Exercise 18a — Navigation;
- (c) Exercise 18c — Radio navigation;
- (d) Exercise 19 — Basic instrument flight.

FCL.210.A PPL(A) – Experience requirements and crediting

Regulation (EU) 2024/2076

- (a) Applicants for a PPL(A) shall have completed at least 45 hours of flight instruction in aeroplanes or TMGs, of which 5 hours may have been completed in an FSTD, including at least the following:

- (1) 25 hours of dual flight instruction;
- (2) 10 hours of supervised solo flight time, including at least 5 hours of solo cross-country flight time with at least one cross-country flight of at least 270 km (150 NM), during which full-stop landings at two aerodromes different from the aerodrome of departure shall be made.

Applicants may complete the flight time of a night rating training course in accordance with point [FCL.810\(a\)\(1\)\(ii\)](#) as part of the 45 hours of flight instruction, provided that, before commencing night rating training, they have completed basic instrument flight training.

- (b) Specific requirements for applicants that have undergone LAPL(A) training. Applicants for a PPL(A) shall receive credits for previous LAPL(A) training they have undergone and shall be issued with a PPL(A), provided that the LAPL(A) training and the PPL(A) training that they completed in total complies with the experience requirements in point (a). However, in such a case all of the following shall apply:

- (1) if applicants already hold an LAPL(A), all of the following shall apply, provided that applicants have completed at least 45 hours of total flight time in aeroplanes or TMGs:
 - (i) the requirement in the introductory sentence of point (a) is reduced to 40 hours of flight instruction;
 - (ii) the requirement in point (a)(1) is reduced to 21 hours of dual flight instruction;
- (2) applicants shall have completed at least all of the following with an instructor that is qualified to instruct for a PPL(A):
 - (i) 5 hours of dual flight instruction;
 - (ii) solo flight time as specified in point (a)(2).

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- (c) Specific requirements for applicants who hold an SPL issued in accordance with Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), including privileges to fly TMGs. Applicants for a PPL(A) who hold an SPL with the privileges to fly TMGs shall have completed:
- (1) at least 24 hours of flight time on TMGs after endorsement of the TMG privileges; and
 - (2) at least 15 hours of flight instruction in aeroplanes in a training course at a DTO or at an ATO, including at least the requirements of point (a)(2).
- (d) Crediting. Applicants holding a pilot licence for another category of aircraft, with the exception of balloons, shall be credited with 10 % of their total flight time as PIC on such aircraft up to a maximum of 10 hours. The amount of credit given shall in any case not include the requirements in (a)(2).

SECTION 3 – SPECIFIC REQUIREMENTS FOR THE PPL HELICOPTERS – PPL(H)

FCL.205.H PPL(H) – Privileges

Regulation (EU) 2024/2076

- (a) The privileges of the holder of a PPL(H) are to act without remuneration as PIC or co-pilot of helicopters engaged in non-commercial operations and to exercise all privileges of holders of an LAPL(H).
- (b) Notwithstanding the paragraph above, the holder of a PPL(H) with instructor or examiner privileges may receive remuneration for:
 - (1) the provision of flight instruction for the LAPL(H) or the PPL(H);
 - (2) the conduct of skill tests and proficiency checks for the licences specified in point (1);
 - (3) the training, testing and checking for the ratings or certificates associated with the instructor's or examiner's licence, as applicable.

FCL.210.H PPL(H) – Experience requirements and crediting

Regulation (EU) 2024/2076

- (a) Applicants for a PPL(H) shall have completed at least 45 hours of flight instruction on helicopters, of which 5 hours may have been completed in an FSTD, including at least:
 - (1) 25 hours of dual flight instruction; and
 - (2) 10 hours of supervised solo flight time, including at least 5 hours of solo cross-country flight time with at least 1 cross-country flight of at least 185 km (100 NM), with full stop landings at 2 aerodromes different from the aerodrome of departure.
 - (3) 35 of the 45 hours of flight instruction have to be completed on the same type of helicopter as the one used for the skill test.
- (b) Specific requirements for applicants that have undergone LAPL(H) training. Applicants for a PPL(H) shall receive credits for previous LAPL(H) training they have undergone and shall be issued with a PPL(H), provided that the LAPL(H) training and the PPL(H) training that they completed in total complies with the experience requirements in point (a).
- (c) Applicants holding a pilot licence for another category of aircraft, with the exception of balloons, shall be credited with 10 % of their total flight time as PIC on such aircraft up to a maximum of 6 hours. The amount of credit given shall in any case not include the requirements in (a)(2).

AMC1 FCL.210.H(b) PPL(H) — Experience requirements and crediting

ED Decision 2025/002/R

PPL(H) TRAINING FOR APPLICANTS WHO HOLD AN LAPL(H) OR WHO HAVE UNDERGONE LAPL(H) TRAINING

Applicants for a PPL(H) who already hold an LAPL(H) and applicants who change their ongoing LAPL(H) training into PPL(H) training should receive training the content of which should be determined by the head of training of the DTO or the ATO where the applicant undergoes PPL(H) training, after assessing

the individual applicant's training needs. In any case, flight training should include the following exercises from the PPL(H) flight instruction syllabus ([AMC2 FCL.210](#)):

- (a) Exercise 25(c) — Radio navigation;
- (b) Exercise 26 — Advanced take-off, landings and transitions;
- (c) Exercise 30 — Basic instrument flight.

SECTION 4 – SPECIFIC REQUIREMENTS FOR THE PPL AIRSHIPS – PPL(As)

FCL.205.As PPL(As) – Privileges

Regulation (EU) 2024/2076

- (a) The privileges of the holder of a PPL(As) are to act without remuneration as PIC or co-pilot on airships engaged in non-commercial operations.
- (b) Notwithstanding the paragraph above, the holder of a PPL(As) with instructor or examiner privileges may receive remuneration for:
 - (1) the provision of flight instruction for the PPL(As);
 - (2) the conduct of skill tests and proficiency checks for the licence specified in point (1);
 - (3) the training, testing and checking for the ratings or certificates associated with the instructor's or examiner's licence, as applicable.

FCL.210.As PPL(As) – Experience requirements and crediting

Regulation (EU) 2020/359

- (a) Applicants for a PPL(As) shall have completed at least 35 hours of flight instruction in airships, 5 of which may have been completed in an FSTD, including at least:
 - (1) 25 hours of dual flight instruction, including:
 - (i) 3 hours of cross-country flight training, including 1 cross-country flight of at least 65 km (35 NM);
 - (ii) 3 hours of instrument instruction;
 - (2) 8 take-offs and landings at an aerodrome, including masting and unmasting procedures;
 - (3) 8 hours of supervised solo flight time.
- (b) Applicants who hold a BPL issued in accordance with Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and are qualified to fly hot-air airships shall be credited with 10 % of their total flight time as PIC on such airships and up to a maximum of 5 hours.

AMC1 FCL.210.As PPL(As) – Experience requirements and crediting

ED Decision 2011/016/R

FLIGHT INSTRUCTION FOR THE PPL(AS)

- (a) Entry to training

Before being accepted for training an applicant should be informed that the appropriate medical certificate must be obtained before solo flying is permitted.
- (b) Flight instruction
 - (1) The PPL(As) flight instruction syllabus should take into account the principles of threat and error management and cover:
 - (i) pre-flight operations, including mass and balance determination, airship inspection and servicing;

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- (ii) ground manoeuvring, masting and unmasting procedures;
 - (iii) aerodrome and traffic pattern operations, collision avoidance precautions and procedures;
 - (iv) control of the airship by external visual reference;
 - (v) take-offs and landings;
 - (vi) flight by reference solely to instruments, including the completion of a level 180 ° turn;
 - (vii) cross-country flying using visual reference, dead reckoning and radio navigation aids;
 - (viii) emergency operations, including simulated airship equipment malfunctions;
 - (ix) operations to, from and transiting controlled aerodromes, compliance with air traffic services procedures, communication procedures and phraseology.
 - (2) Before allowing the applicant for a PPL(As) to undertake his/her first solo flight, the FI should ensure that the applicant can use R/T communication.
- (c) Syllabus of flight instruction
- (1) The numbering of exercises should be used primarily as an exercise reference list and as a broad instructional sequencing guide; therefore the demonstrations and practices need not necessarily be given in the order listed. The actual order and content will depend upon the following interrelated factors:
 - (i) the applicant's progress and ability;
 - (ii) the weather conditions affecting the flight;
 - (iii) the flight time available;
 - (iv) instructional technique considerations;
 - (v) the local operating environment;
 - (vi) applicability of the exercises to the airship.
 - (2) Each of the exercises involves the need for the applicant to be aware of the needs of good airmanship and look-out, which should be emphasised at all times.
 - (i) Exercise 1a: Familiarisation with the airship:
 - (A) characteristics of the airship;
 - (B) cockpit layout;
 - (C) systems;
 - (D) checklists, drills and controls.
 - (ii) Exercise 1b: Emergency drills:
 - (A) action if fire on the ground and in the air;
 - (B) engine cabin and electrical system fire;
 - (C) systems failure;
 - (D) escape drills, location and use of emergency equipment and exits.

- (iii) Exercise 2: Preparation for and action after flight:
 - (A) flight authorisation and airship acceptance;
 - (B) serviceability documents;
 - (C) equipment required, maps, etc.;
 - (D) mass and balance;
 - (E) external checks;
 - (F) ground crew briefing;
 - (G) internal checks;
 - (H) harness, seat or rudder panel adjustments;
 - (I) starting and warm-up checks;
 - (J) power checks;
 - (K) running down system checks and switching off the engine;
 - (L) parking, security and masting;
 - (M) completion of authorisation sheet and serviceability documents.
- (iv) Exercise 3: Air experience: flight exercise.
- (v) Exercise 4: Effects of controls:
 - (A) primary effects;
 - (B) further effects;
 - (C) effects of:
 - (a) air speed;
 - (b) power;
 - (c) trimming controls;
 - (d) other controls, as applicable.
 - (D) operation of:
 - (a) mixture control;
 - (b) carburettor heat;
 - (c) cabin heating or ventilation.
- (vi) Exercise 5: Ground manoeuvring:
 - (A) pre-taxi checks;
 - (B) starting, control of speed and stopping;
 - (C) engine handling;
 - (D) masting procedures;
 - (E) control of direction and turning;
 - (F) effects of wind;

- (G) effects of ground surface;
- (H) marshalling signals;
- (I) instrument checks;
- (J) air traffic control procedures;
- (K) emergencies.
- (vii) Exercise 6a: Take-off procedures:
 - (A) pre-take-off checks;
 - (B) take-off with different static heaviness;
 - (C) drills during and after take-off;
 - (D) noise abatement procedures.
- (viii) Exercise 6b: Emergencies:
 - (A) abandoned take-off;
 - (B) engine failure after take-off;
 - (C) malfunctions of thrust vector control;
 - (D) aerodynamic control failures;
 - (E) electrical and system failures.
- (ix) Exercise 7: Climbing:
 - (A) entry, maintaining the normal and max rate climb and levelling off;
 - (B) levelling off at selected altitudes;
 - (C) maximum angle of climb;
 - (D) maximum rate of climb.
- (x) Exercise 8: Straight and level:
 - (A) attaining and maintaining straight and level flight;
 - (B) flight at or close to pressure height;
 - (C) control in pitch, including use of trim;
 - (D) at selected air speeds (use of power);
 - (E) during speed changes;
 - (F) use of instruments for precision.
- (xi) Exercise 9: Descending:
 - (A) entry, maintaining and levelling off;
 - (B) levelling off at selected altitudes;
 - (C) maximum rate of descent;
 - (D) maximum angle of descent;
 - (E) use of instruments for precision flight.

- (xii) Exercise 10: Turning:
 - (A) entry and maintaining level turns;
 - (B) resuming straight flight;
 - (C) faults in the turn;
 - (D) climbing turns;
 - (E) descending turns;
 - (F) turns onto selected headings, use of gyro heading indicator and compass;
 - (G) use of instruments for precision.
- (xiii) Exercise 11: Hovering: hovering manoeuvres (as applicable);
- (xiv) Exercise 12a: Approach and landing:
 - (A) effect of wind on approach and touchdown speeds;
 - (B) landing with different static heaviness;
 - (C) missed approach and go-around procedures;
 - (D) noise abatement procedures.
- (xv) Exercise 12b: Emergencies:
 - (A) aborted approach or go-around;
 - (B) malfunction of thrust vector control;
 - (C) envelope emergencies;
 - (D) fire emergencies;
 - (E) aerodynamic control failures;
 - (F) electrical and system failures.
- (xvi) Exercise 13: Precautionary landing:
 - (A) occasions necessitating;
 - (B) in-flight conditions;
 - (C) landing area selection;
 - (D) circuit and approach;
 - (E) actions after landing;
- (xvii) Exercise 14a: Navigation:
 - (A) flight planning:
 - (a) weather forecast and actuals;
 - (b) map selection and preparation:
 - (1) choice of route;
 - (2) airspace structure;
 - (3) sensitive areas;

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- (4) safety altitudes.
 - (c) calculations:
 - (1) magnetic heading(s) and time(s) en-route;
 - (2) fuel consumption;
 - (3) mass and balance;
 - (4) performance.
 - (d) flight information:
 - (1) NOTAMs etc.;
 - (2) radio frequencies;
 - (3) selection of alternate aerodromes.
 - (e) airship documentation;
 - (f) notification of the flight:
 - (1) pre-flight administrative procedures;
 - (2) flight plan form.
 - (B) departure:
 - (a) organisation of cockpit workload;
 - (b) departure procedures:
 - (1) altimeter settings;
 - (2) ATC liaison in controlled or regulated airspace;
 - (3) setting heading procedure;
 - (4) noting of ETAs.
 - (c) maintenance of altitude and heading;
 - (d) revisions of ETA and heading;
 - (e) log keeping;
 - (f) use of radio;
 - (g) use of nav aids;
 - (h) minimum weather conditions for continuation of flight;
 - (i) in-flight decisions;
 - (j) transiting controlled or regulated airspace;
 - (k) diversion procedures;
 - (l) uncertainty of position procedure;
 - (m) lost procedure.
 - (C) arrival, aerodrome joining procedure:
 - (a) ATC liaison in controlled or regulated airspace;

- (b) altimeter setting;
 - (c) entering the traffic pattern;
 - (d) circuit procedures;
 - (e) parking or on masting;
 - (f) security of airship;
 - (g) refuelling;
 - (h) closing of flight plan, if appropriate;
 - (i) post-flight administrative procedures.
- (xviii) Exercise 14b: Navigation problems at lower levels and in reduced visibility:
- (A) actions before descending;
 - (B) hazards (for example obstacles, and terrain);
 - (C) difficulties of map reading;
 - (D) effects of winds, turbulence and precipitation;
 - (E) vertical situational awareness;
 - (F) avoidance of noise sensitive areas;
 - (G) joining the circuit;
 - (H) bad weather circuit and landing.
- (xix) Exercise 14c: Radio navigation:
- (A) use of GNSS
 - (a) selection of waypoints;
 - (b) to or from indications and orientation;
 - (c) error messages.
 - (B) use of VHF omni range (if applicable):
 - (a) availability, AIP and frequencies;
 - (b) selection and identification;
 - (c) OBS;
 - (d) to or from indications and orientation;
 - (e) CDI;
 - (f) determination of radial;
 - (g) intercepting and maintaining a radial;
 - (h) VOR passage;
 - (i) obtaining a fix from two VORs.
 - (C) use of ADF equipment: NDBs (if applicable):
 - (a) availability, AIP and frequencies;

- (b) selection and identification;
 - (c) orientation relative to the beacon;
 - (d) homing.
- (D) use of VHF/DF:
 - (a) availability, AIP and frequencies;
 - (b) R/T procedures and ATC liaison;
 - (c) obtaining a QDM and homing.
- (E) use of en-route or terminal radar:
 - (a) availability and AIP;
 - (b) procedures and ATC liaison;
 - (c) pilot's responsibilities;
 - (d) secondary surveillance radar:
 - (1) transponders;
 - (2) code selection;
 - (3) interrogation and reply.
- (F) use of DME (if applicable);
 - (a) station selection and identification;
 - (b) modes of operation: distance, groundspeed and time to run.
- (xx) Exercise 15: Basic instrument flight:
 - (A) physiological sensations;
 - (B) instrument appreciation: attitude instrument flight;
 - (C) instrument limitations;
 - (D) basic manoeuvres:
 - (a) straight and level;
 - (b) climbing and descending;
 - (c) turns, climbing and descending, onto selected headings;
 - (d) recoveries from climbing and descending turns.

(d) BITD

- (1) A BITD may be used for flight training for:
 - (i) flight by reference solely to instruments;
 - (ii) navigation using radio navigation aids;
 - (iii) basic instrument flight.
- (2) The use of the BITD should be subject to the following:
 - (i) the training should be complemented by exercises on an airship;
 - (ii) the record of the parameters of the flight must be available; and an FI(As) should conduct the instruction.

SECTION 5 - SPECIFIC REQUIREMENTS FOR THE GPL

FCL.205.G GPL – Privileges

Regulation (EU) 2025/134

- (a) The privileges of the holder of a GPL are to act as PIC in gyroplanes:
 - (1) without remuneration in non-commercial operations;
 - (2) performing the carriage of passengers only when they have completed 10 hours of flight time as PIC on gyroplanes after the issuance of the licence.
- (b) Notwithstanding point (a)(1) above, the holder of a GPL with instructor or examiner privileges may receive remuneration for:
 - (1) the provision of flight instruction for the GPL;
 - (2) the conduct of skill tests and proficiency checks for the GPL;
 - (3) the training, testing and checking for the ratings and privileges associated with the instructor's or examiner's licence, as applicable.
- (c) GPL holders shall exercise GPL privileges only if they comply with the applicable recency requirements and only if their medical certificate, appropriate to the privileges exercised, is valid.

[applicable from 18 February 2026 – Regulation (EU) 2025/134]

FCL.210.G GPL – Experience requirements and crediting

Regulation (EU) 2025/134

- (a) An applicant for a GPL shall have completed at least 35 hours of flight instruction in gyroplanes, 5 of which may have been completed in an FSTD, including at least:
 - (1) 20 hours of dual flight instruction; and
 - (2) 8 hours of supervised solo flight time, including at least 4 hours of solo cross-country flight time with at least one cross-country flight of at least 150 km (80 NM), during which full-stop landings at two aerodromes different from the aerodrome of departure shall be made.
- (b) Crediting. An applicant that holds a pilot licence for another category of aircraft, with the exception of balloons, shall be credited with 10 % of their total flight time as PIC on such aircraft up to a maximum of 10 hours. In any case, credits shall not be given for the requirements of point (a)(2).
- (c) Applicants with prior experience as PIC on gyroplanes, that have a maximum take-off mass of at least 450 kg, may receive credits towards the requirements of point (a). The amount of credit shall be decided by the DTO or the ATO where the applicant undergoes the training course, on the basis of a pre-entry flight assessment, but shall in any case not exceed any of the following:
 - (1) the total flight time as PIC;
 - (2) 50 % of the hours required in point (a).

In any case, credits shall not be given for the requirements of point (a)(2).

[applicable from 18 February 2026 – Regulation (EU) 2025/134]

FCL.235.G GPL – Privileges for classes, types and variants of gyroplanes

Regulation (EU) 2025/134

- (a) For the purpose of gyroplane pilot licensing in accordance with this Annex, gyroplanes shall be categorised into the following class and types:
 - (1) single-propeller gyroplane (SPG) class: Single-pilot gyroplanes whose single-centric propulsion unit is operated by a single thrust control and driven by either of the following types of engines:
 - (a) a piston engine system which, if so specified following the certification process in accordance with Regulation (EU) No 748/2012, may consist of more than one piston engine;
 - (b) an electric engine system which, if so specified following the certification process in accordance with Regulation (EU) No 748/2012, may consist of more than one electric engine;
 - (c) if so specified following the certification process in accordance with Regulation (EU) No 748/2012, a hybrid engine system consisting of piston and electric engines;
 - (2) gyroplane types that require privileges for that individual type, as determined following the certification process in accordance with Regulation (EU) No 748/2012.
- (b) Applicants for a GPL shall be issued with privileges for the class or type of gyroplanes in which the skill test was taken. In order to extend their privileges to another class or type, holders of a GPL shall complete, in that other class or type, all of the following:
 - (1) unless specified otherwise in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012, where applicable, 3 hours of flight instruction, including:
 - (i) 10 dual take-offs and landings; and
 - (ii) 10 supervised solo take-offs and landings;
 - (2) a skill test to demonstrate an adequate level of practical skill in the new class or type. During this skill test, the applicant shall also demonstrate to the examiner an adequate level of theoretical knowledge for the other class or type in the following subjects:
 - (i) operational procedures;
 - (ii) flight performance and planning;
 - (iii) aircraft general knowledge.
- (c) In order to extend the privileges to another variant within a class or type, the pilot shall complete differences training or familiarisation. Where applicable, that differences training or familiarisation shall comply with the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012. The differences training shall be entered in the pilot's logbook or into an equivalent record and be signed by the instructor. When extending the privileges for an SPG class to a variant with another type of engine, as specified in point (a)(1), the differences training shall consist of dual flight instruction and theoretical knowledge

instruction which shall include, with regard to that other type of engine and related aircraft systems, at least all of the following:

- (1) operational procedures;
- (2) flight performance and planning;
- (3) aircraft general knowledge.

[applicable from 18 February 2026 – Regulation (EU) 2025/134]

FCL.240.G GPL – Recency requirements

Regulation (EU) 2025/134

- (a) Holders of a GPL shall exercise the privileges of their licence in a specific class or type of gyroplane only if in the last 2 years they have met any of the following conditions as pilots of gyroplanes in the relevant class or type:
 - (1) they have completed at least 12 hours of flight time as PIC or flying dual or solo under the supervision of an instructor, including:
 - (i) 12 take-offs and landings;
 - (ii) refresher training of at least 1 hour of total flight time with and to the satisfaction of an instructor who shall select those flight exercises that allow the applicant to refresh their competence in safely operating the aircraft and applying normal, abnormal and emergency procedures;
 - (2) they have passed a GPL proficiency check with an examiner. The proficiency check programme shall be based on the skill test for the GPL.
- (b) Holders of a GPL with privileges for the SPG class who, in accordance with point [FCL.235.G\(c\)](#), have extended their privileges to a variant with a different type of engine as specified in point [FCL.235.G\(a\)](#) shall, if they have not flown that variant within the preceding 2 years, complete any of the following in that variant, before exercising their privileges in that variant:
 - (1) further differences training in accordance with point [FCL.235.G\(c\)](#);
 - (2) a proficiency check;
 - (3) refresher training in accordance with point (a)(1)(ii).
- (c) The dual flights and the solo flights under supervision as per the introductory phrase of point (a)(1), the refresher training as per points (a)(1)(ii) and (b)(3), and the proficiency check as per point (a)(2) and (b)(2) shall be entered in the pilot's logbook or equivalent record and shall be signed by the instructor or examiner, as applicable.

[applicable from 18 February 2026 – Regulation (EC) 2025/134]

SUBPART D – COMMERCIAL PILOT LICENCE – CPL

SECTION 1 – COMMON REQUIREMENTS

FCL.300 CPL – Minimum age

Regulation (EU) No 1178/2011

An applicant for a CPL shall be at least 18 years of age.

FCL.305 CPL – Privileges and conditions

Regulation (EU) No 1178/2011

- (a) Privileges. The privileges of the holder of a CPL are, within the appropriate aircraft category, to:
- (1) exercise all the privileges of the holder of an LAPL and a PPL;
 - (2) act as PIC or co-pilot of any aircraft engaged in operations other than commercial air transport;
 - (3) act as PIC in commercial air transport of any single-pilot aircraft subject to the restrictions specified in [FCL.060](#) and in this Subpart;
 - (4) act as co-pilot in commercial air transport subject to the restrictions specified in [FCL.060](#).
- (b) Conditions. An applicant for the issue of a CPL shall have fulfilled the requirements for the class or type rating of the aircraft used in the skill test.

FCL.310 CPL – Theoretical knowledge examinations

Regulation (EU) 2018/1974

Applicants for the issue of a CPL shall demonstrate a level of knowledge appropriate to the privileges granted in the following subjects:

- (a) air law;
- (b) aircraft general knowledge — airframe/systems/power plant;
- (c) aircraft general knowledge — instrumentation;
- (d) mass and balance;
- (e) performance;
- (f) flight planning and monitoring;
- (g) human performance;
- (h) meteorology;
- (i) general navigation;
- (j) radio navigation;
- (k) operational procedures;
- (l) principles of flight; and
- (m) communications.

AMC1 FCL.310; FCL.515(b); FCL.615(b); FCL.835(d) Theoretical knowledge examinations

ED Decision 2020/018/R

LEARNING OBJECTIVES FOR ATPL, CPL, IR, CB-IR(A) and BIR

(a) Aeroplanes and helicopters

GENERAL

In the tables of this AMC, the applicable LOs for each ATPL, CPL, IR, CB-IR(A) are marked with an 'X', and for the BIR exam and BIR BK with the number 1, 2 or 3 (corresponding to the modules as mentioned in [FCL.835](#) 'Basic instrument rating (BIR)').

The LOs define the subject knowledge and applied knowledge, skills and attitudes that a student pilot should have assimilated during the theoretical knowledge course.

The LOs are intended to be used by an approved training organisation (ATO) when developing the Part-FCL theoretical knowledge elements of the appropriate course. It should be noted, however, that the LOs do not provide a ready-made ground training syllabus for individual ATOs, and should not be seen by organisations as a substitute for a thorough course design. Adherence to the LOs should become part of the ATO's compliance monitoring scheme as required by [ORA.GEN.200\(a\)\(6\)](#).

ATOs are required to produce a training plan for each of their courses based on the instructional systems design (ISD) methodology as specified in [AMC2 ORA.ATO.230](#).

Additional guidance on the meaning and taxonomy of the verbs used in the LOs can be found in [GM1 FCL.310; FCL.515\(b\); FCL.615\(b\); FCL.835\(d\)](#).

TRAINING AIMS

After completion of the training, a student pilot should:

- be able to understand and apply the subject knowledge in order to be able to identify and manage threats and errors effectively;
- meet at least the Area 100 KSA minimum standard.

INTERPRETATION

The abbreviations used are ICAO abbreviations listed in ICAO Doc 8400 'ICAO Abbreviations and Codes', or those listed in [GM1 FCL.010](#).

Where a LO refers to a definition, e.g. 'Define the following terms' or 'Define and understand' or 'Explain the definitions in ...', candidates are also expected to be able to recognise a given definition.

Below is a table showing the short references to applicable legislation and standards:

Reference	Legislation/Standard
The Basic Regulation	Regulation (EU) 2018/1139 of the European Parliament and of the Council of 4 July 2018
The Aircrew Regulation	Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)
Part-FCL	Annex I to Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)
Part-MED	Annex IV to Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)

Reference	Legislation/Standard
CS-23, AMC & GM to CS-23, CS-25, CS-27, CS-29, CS-E and CS-Definitions	Refer to the respective EASA Certification Specifications / AMC & GM
Single European Sky Regulations	Regulation (EC) No 549/2004 of the European Parliament and of the Council of 10 March 2004 laying down the framework for the creation of the single European sky (the framework Regulation) Regulation (EC) No 550/2004 of the European Parliament and of the Council of 10 March 2004 on the provision of air navigation services in the single European sky (the service provision Regulation) Regulation (EC) No 551/2004 of the European Parliament and of the Council of 10 March 2004 on the organisation and use of the airspace in the single European sky (the airspace Regulation) Regulation (EC) No 552/2004 of the European Parliament and of the Council of 10 March 2004 on the interoperability of the European Air Traffic Management network (the interoperability Regulation)
Passenger Rights Regulation	Regulation (EC) No 261/2004 of the European Parliament and of the Council of 11 February 2004 establishing common rules on compensation and assistance to passengers in the event of denied boarding and of cancellation or long delay of flights, and repealing Regulation (EEC) No 295/91
RTCA/EUROCAE	Refers to correspondingly numbered documents issued by the Radio Technical Commission for Aeronautics/ European Organisation for Civil Aviation Equipment
ITU Radio Regulation	International Telecommunication Union Radio Regulation
NASA TM-85652	National Aeronautics and Space Administration — Technical Memorandum 85652

‘Applicable operational requirements’ refers to, for the ATPL(A), CPL(A), ATPL(H)/IR, ATPL(H)/VFR, CPL(H), IR and CBIR, Annexes I, II, III, IV, V and VIII to Commission Regulation (EU) No 965/2012 of 5 October 2012 (as amended). For the BIR, it refers to Annexes I, II, V and VII to that Regulation.

The General Student Pilot Route Manual (GSPRM) contains planning data plus aerodrome and approach charts that may be used in theoretical knowledge training courses. The guidelines on its content can be found in this AMC, before the LO table for Subject 033 ‘Flight planning and monitoring’.

Excerpts from any aircraft manuals including but not limited to CAP 696, 697 and 698 for aeroplanes, and CAP 758 for helicopters may be used in training. Where questions refer to excerpts from aircraft manuals, the associated aircraft data will be provided in the examinations.

Some numerical data (e.g. speeds, altitudes/levels and masses) used in questions for theoretical knowledge examinations may not be representative for helicopter operations, but the data is satisfactory for the calculations required.

Note: In all subject areas, the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

DETAILED THEORETICAL KNOWLEDGE SYLLABUS AND LOs FOR ATPL, CPL, IR, CB-IR(A) and BIR

GENERAL

The detailed theoretical knowledge syllabus outlines the topics that should be taught and examined in order to meet the theoretical knowledge requirements appropriate to ATPL, MPL, CPL, IR, CB-IR(A) and BIR.

For each topic in the detailed theoretical knowledge syllabus, one or more LOs are set out in the appendices as shown below:

- [Appendix 010 AIR LAW](#)
- [Appendix 021 AIRCRAFT GENERAL KNOWLEDGE – AIRFRAME, SYSTEMS AND POWER PLANT](#)
- [Appendix 022 AIRCRAFT GENERAL KNOWLEDGE – INSTRUMENTATION](#)
- [Appendix 031 FLIGHT PERFORMANCE AND PLANNING – MASS AND BALANCE](#)
- [Appendix 032 FLIGHT PERFORMANCE AND PLANNING – PERFORMANCE – AEROPLANES](#)
- [Appendix 033 FLIGHT PERFORMANCE AND PLANNING – FLIGHT PLANNING AND MONITORING](#)
- [Appendix 034 FLIGHT PERFORMANCE AND PLANNING – PERFORMANCE – HELICOPTERS](#)
- [Appendix 040 HUMAN PERFORMANCE AND LIMITATIONS](#)
- [Appendix 050 METEOROLOGY](#)
- [Appendix 061 NAVIGATION – GENERAL NAVIGATION](#)
- [Appendix 062 NAVIGATION – RADIO NAVIGATION](#)
- [Appendix 070 OPERATIONAL PROCEDURES](#)
- [Appendix 081 PRINCIPLES OF FLIGHT – AEROPLANES](#)
- [Appendix 082 PRINCIPLES OF FLIGHT – HELICOPTERS](#)
- [Appendix 090 RADIO COMMUNICATIONS](#)
- [Appendix AREA 100 KNOWLEDGE, SKILLS AND ATTITUDES \(KSA\)](#)

(b) Airships

SYLLABUS OF THEORETICAL KNOWLEDGE FOR CPL AND IR

The applicable items for each licence or rating are marked with ‘x’. An ‘x’ on the main title of a subject means that all the subdivisions are applicable.’

		CPL	IR
1.	AIR LAW AND ATC PROCEDURES	x	
	INTERNATIONAL LAW: CONVENTIONS, AGREEMENTS AND ORGANISATIONS		

		CPL	IR
	AIRWORTHINESS OF AIRCRAFT		
	AIRCRAFT NATIONALITY AND REGISTRATION MARKS		
	PERSONNEL LICENSING		x
	RULES OF THE AIR		x
	PROCEDURES FOR AIR NAVIGATION SERVICES: AIRCRAFT OPERATIONS		x
	AIR TRAFFIC SERVICES AND AIR TRAFFIC MANAGEMENT		x
	AERONAUTICAL INFORMATION SERVICE		x
	AERODROMES		x
	FACILITATION		
	SEARCH AND RESCUE		
	SECURITY		
	AIRCRAFT ACCIDENT AND INCIDENT INVESTIGATION		
2.	AIRSHIP GENERAL KNOWLEDGE: ENVELOPE, AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT AND EMERGENCY EQUIPMENT	x	
	DESIGN, MATERIALS, LOADS AND STRESSES		
	ENVELOPE AND AIRBAGS		
	FRAMEWORK		
	GONDOLA		
	FLIGHT CONTROLS		
	LANDING GEAR		
	HYDRAULICS AND PNEUMATICS		
	HEATING AND AIR CONDITIONING		
	FUEL SYSTEM		
	PISTON ENGINES		
	TURBINE ENGINES (BASICS)		
	ELECTRICS		
	FIRE PROTECTION AND DETECTION SYSTEMS		
	MAINTENANCE		
3.	AIRSHIP GENERAL KNOWLEDGE: INSTRUMENTATION	x	
	SENSORS AND INSTRUMENTS		
	MEASUREMENT OF AIR DATA AND GAS PARAMETERS		

		CPL	IR
	MAGNETISM: DIRECT READING COMPASS AND FLUX VALVE		
	GYROSCOPIC INSTRUMENTS		
	COMMUNICATION SYSTEMS		
	ALERTING SYSTEMS		
	INTEGRATED INSTRUMENTS: ELECTRONIC DISPLAYS		
	FLIGHT MANAGEMENT SYSTEM (GENERAL BASICS)		
	DIGITAL CIRCUITS AND COMPUTERS		
4.	FLIGHT PERFORMANCE AND PLANNING	x	
4.1.	MASS AND BALANCE: AIRSHIPS	x	
	PURPOSE OF MASS AND BALANCE CONSIDERATIONS		
	LOADING		
	FUNDAMENTALS OF CG CALCULATIONS		
	MASS AND BALANCE DETAILS OF AIRCRAFT		
	DETERMINATION OF CG POSITION		
	PASSENGER, CARGO AND BALLAST HANDLING		
4.2.	FLIGHT PLANNING AND FLIGHT MONITORING		
	FLIGHT PLANNING FOR VFR FLIGHTS	x	
	FLIGHT PLANNING FOR IFR FLIGHTS		x
	FUEL PLANNING	x	x
	PRE-FLIGHT PREPARATION	x	x
	ATS FLIGHT PLAN	x	x
	FLIGHT MONITORING AND IN-FLIGHT RE-PLANNING	x	x
4.3.	PERFORMANCE: AIRSHIPS	x	
	AIRWORTHINESS REQUIREMENTS		
	BASICS OF AIRSHIP PERFORMANCE		
	DEFINITIONS AND TERMS		
	STAGES OF FLIGHT		
	USE OF FLIGHT MANUAL		
5.	HUMAN PERFORMANCE	x	
	HUMAN FACTORS: BASIC CONCEPTS		

		CPL	IR
	BASIC AVIATION PHYSIOLOGY AND HEALTH MAINTENANCE		
	BASIC AVIATION PSYCHOLOGY		
6.	METEOROLOGY	x	
	THE ATMOSPHERE		
	WIND		
	THERMODYNAMICS		
	CLOUDS AND FOG		
	PRECIPITATION		
	AIR MASSES AND FRONTS		
	PRESSURE SYSTEMS		
	CLIMATOLOGY		
	FLIGHT HAZARDS		
	METEOROLOGICAL INFORMATION		
7.	NAVIGATION		
7.1.	GENERAL NAVIGATION	x	
	BASICS OF NAVIGATION		
	MAGNETISM AND COMPASSES		
	CHARTS		
	DR NAVIGATION		
	IN-FLIGHT NAVIGATION		
7.2.	RADIO NAVIGATION		
	BASIC RADIO PROPAGATION THEORY	x	x
	RADIO AIDS	x	x
	RADAR	x	x
	INTENTIONALLY LEFT BLANK		
	AREA NAVIGATION SYSTEMS AND RNAV/FMS		x
	GNSS	x	x
8.	OPERATIONAL PROCEDURES AIRSHIP	x	
	GENERAL REQUIREMENTS		
	SPECIAL OPERATIONAL PROCEDURES AND HAZARDS (GENERAL ASPECTS)		
	EMERGENCY PROCEDURES		
9.	PRINCIPLES OF FLIGHT	x	
9.1.	PRINCIPLES OF FLIGHT: AIRSHIPS	x	
	BASICS OF AEROSTATICS		
	BASICS OF SUBSONIC AERODYNAMICS		
	AERODYNAMICS OF AIRSHIPS		
	STABILITY		

		CPL	IR
	CONTROLLABILITY		
	LIMITATIONS		
	PROPELLERS		
	BASICS OF AIRSHIP FLIGHT MECHANICS		
10.	COMMUNICATIONS		
10.1.	VFR COMMUNICATIONS	x	
	DEFINITIONS	x	
	GENERAL OPERATING PROCEDURES	x	
	RELEVANT WEATHER INFORMATION TERMS (VFR)	x	
	ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE	x	
	DISTRESS AND URGENCY PROCEDURES	x	
	GENERAL PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES	x	
10.2.	IFR COMMUNICATIONS		
	DEFINITIONS		x
	GENERAL OPERATING PROCEDURES		x
	ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE		x
	DISTRESS AND URGENCY PROCEDURES		x
	RELEVANT WEATHER INFORMATION TERMS (IFR)		x
	GENERAL PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES		x
	MORSE CODE		x

Appendix to AMC1 FCL.310; FCL.515(b); FCL.615(b) Theoretical knowledge examinations

SUBJECT 010 – AIR LAW

ED Decision 2020/018/R

Note that the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The professional pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

- (1) The subjects ‘Air law’ and ‘ATC procedures’ are primarily based on ICAO documentation and European Union regulations.
- (2) National law should not be taken into account for theoretical-examination purposes; it should remain relevant though during practical training and operational flying.

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 00 00 00		AIR LAW										
010 01 00 00		INTERNATIONAL LAW: CONVENTIONS, AGREEMENTS AND ORGANISATIONS										
010 01 01 00		The Convention on International Civil Aviation (Chicago) — ICAO Doc 7300/9 Convention on the High Seas (Geneva, 29 April 1958)										
010 01 01 01		<i>The establishment of the Convention on International Civil Aviation, Chicago, 7 December 1944</i>										
(01)	X	Explain the circumstances that led to the establishment of the Convention on International Civil Aviation, Chicago, 7 December 1944.	X	X	X	X	X					
010 01 01 02		<i>Part I — Air navigation</i>										
(01)	X	Recall the general contents of relevant parts of the following chapters:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— general principles and application of the Convention; — flight over territory of Contracting States; — nationality of aircraft; — international standards and recommended practices (SARPs), especially notification of differences and validity of endorsed certificates and licences.										
(02)	X	General principles Describe the application of the following terms in civil aviation: — sovereignty; — territory and high seas according to the UN Convention on the High Seas.	X	X	X	X	X					
(03)		Explain the following terms and how they apply to international air traffic: — right of non-scheduled flight (including the two technical freedoms of the air); — scheduled air services; — cabotage; — landing at customs airports; — Rules of the Air; — search of aircraft.	X	X	X	X	X					
(04)	X	Explain the duties of Contracting States in relation to: — documents carried on board the aircraft:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— certificate of registration; — certificates of airworthiness; — licences of personnel; — recognition of certificates and licences; — cargo restrictions; — photographic apparatus.										
010 01 01 03		Part II — The International Civil Aviation Organization (ICAO)										
(01)	X	Describe the objectives of ICAO.	X	X	X	X	X					
(02)	X	Recognise the organisation and duties of the ICAO Assembly, Council and Air Navigation Commission (ANC).	X	X	X	X	X					
(03)	X	Describe the annexes to the Convention.	X	X	X	X	X					
010 01 02 00		Other conventions and agreements										
010 01 02 01		The International Air Services Transit Agreement (ICAO Doc 7500)										
(01)		Explain the two technical freedoms of the air.	X	X	X	X	X					
010 01 02 02		The International Air Transport Agreement (ICAO Doc 9626)										
(01)		Explain the three commercial freedoms of the air.	X	X	X	X	X					
010 01 02 03		Suppression of Unlawful Acts Against the Safety of Civil Aviation — The Tokyo Convention of 1963										
(01)		Describe the measures and actions to be taken by the pilot-in-command (PIC) of an aircraft in order to	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		suppress unlawful acts against the safety of the aircraft.										
010 01 02 04		<i>Intentionally left blank</i>										
010 01 02 05		<i>Private international law</i>										
(01)		Explain the legal significance of the issue of a passenger ticket or of baggage/cargo documents (that the issue is a form of contract).	X	X	X	X	X					
(02)		Describe the consequences for an airline or the PIC when a document of carriage is not issued (that the contract is unaffected).	X	X	X	X	X					
(03)	X	Explain the consequences for an airline operator of Regulation (EC) No 261/2004 on passenger rights in the event of delay, cancellation or denial of boarding.	X	X	X	X	X					
(04)		Explain the liability limit in relation to destruction, loss, damage or delay of baggage.	X	X	X	X	X					
010 01 03 00		World organisations										
010 01 03 01		<i>The International Air Transport Association (IATA)</i>										
(01)		Describe the objectives of IATA.	X		X	X						
010 01 04 00		European organisations										
010 01 04 01		<i>European Union Aviation Safety Agency (EASA) Regulation (EU) 2018/1139</i>										
(01)	X	Describe the objectives of EASA.	X	X	X	X	X					
(02)		Describe the role of EASA in European civil aviation.	X	X	X	X	X					
(03)		State that the structure of the regulatory material related to EASA involves: — hard law (regulations, delegated acts, implementing acts, and implementing rules);	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— soft law (certification specifications, acceptable means of compliance, and guidance material).										
(04)		State the meaning of the terminology associated with the structure of the regulatory material related to EASA, specifically: regulations, delegated acts, implementing acts, and implementing rules, as applicable until 11 September 2023; and certification specifications, acceptable means of compliance, and guidance material.	X	X	X	X	X					
010 01 04 02		EUROCONTROL										
(01)	X	Describe the Single European Sky (SES) regulations.	X	X	X	X	X					
010 02 00 00		AIRWORTHINESS OF AIRCRAFT, AIRCRAFT NATIONALITY AND REGISTRATION MARKS										
010 02 01 00		<i>Intentionally left blank</i>										
010 02 02 00		Certificate of Airworthiness (CofA)										
010 02 02 01		<i>Certificate of Airworthiness (CofA) — Details</i>										
(01)		State the issuing authority of a CofA.	X	X	X	X	X					
(02)		State the necessity to hold a CofA.	X	X	X	X	X					
(03)	X	Explain the prerequisites for the issue of a CofA according to Commission Regulation (EU) No 748/2012.	X	X	X	X	X					
(04)		State who shall determine an aircraft's continuing airworthiness.	X	X	X	X	X					
(05)		Describe how a CofA can be renewed or may remain valid.	X	X	X	X	X					
010 02 03 00		ICAO Annex 7 — Aircraft Nationality and Registration Marks										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 02 03 01		ICAO Annex 7 — Definitions										
(01)	X	Recall the definition of the following terms: — aircraft; — heavier-than-air aircraft; — State of Registry.	X	X	X	X	X					
010 02 04 00		Nationality marks, common marks and registration marks										
010 02 04 01		Nationality marks, common marks and registration marks — assignment and location										
(01)		State the location of nationality marks, common marks and registration marks.	X		X							
(02)		Explain who is responsible for assigning nationality marks, common marks and registration marks.	X	X	X	X	X					
010 03 00 00		Intentionally left blank										
010 04 00 00		PERSONNEL LICENSING										
010 04 01 00		ICAO Annex 1										
010 04 01 01		Differences between ICAO Annex 1 and Regulation (EU) No 1178/2011 (hereinafter: Aircrew Regulation)										
(01)	X	Describe the relationship and differences between ICAO Annex 1 and the Aircrew Regulation.	X	X	X	X	X	X				
010 04 02 00		Aircrew Regulation — Annex I (Part-FCL) Source: Aircrew Regulation										
010 04 02 01		Definitions										
(01)		Define the following: — Category, class and type of aircraft, cross-country, dual instruction time, flight time, student pilot-in-command (SPIC), instrument time, instrument flight time, instrument ground time, night, private pilot,	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		proficiency check, renewal, revalidation, skill test, solo flight time.										
(02)		Define the following: — multi-crew cooperation (MCC), multi-pilot aircraft, rating.	X	X	X	X	X					
010 04 02 02		Content and structure										
(01)	X	Explain the structure of Part-FCL.	X	X	X	X	X	X	X		1	
(02)		Explain the requirements to act as a flight crew member of a civil aircraft registered in a Member State, and know the general principles of the licensing system (light aircraft pilot licence (LAPL), private pilot licence (PPL), commercial pilot licence (CPL), multi-crew pilot licence (MPL), airline transport pilot licence (ATPL)).	X	X	X	X	X	X				
(03)	X	List the two factors that are relevant to the exercise of the privileges of a licence.	X	X	X	X	X	X				
(04)	X	State the circumstances in which a language proficiency endorsement is required.	X	X	X	X	X	X				
(05)	X	List the restrictions for licence holders with an age of 60 years or more.	X	X	X	X	X					
(06)	X	Explain the term ‘competent authority’.	X	X	X	X	X	X				
(07)		Describe the obligation to carry and present documents (e.g. a flight crew licence) under Part-FCL.	X	X	X	X	X	X				
010 04 02 03		Commercial pilot licence (CPL)										
(01)	X	State the requirements for the issue of a CPL.	X	X	X	X	X					
(02)		State the privileges of a CPL.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 04 02 04		Airline transport pilot licence (ATPL) and multi-crew pilot licence (MPL)										
(01)	X	State the requirements for the issue of an ATPL.	X		X	X						
(02)		State the privileges of an ATPL.	X		X	X						
(03)	X	State the requirements for the issue of an MPL.	X									
(04)		State the privileges of an MPL.	X									
010 04 02 05		Ratings										
(01)		State the requirements for class ratings, their validity and privileges.	X	X								
(02)		State the requirements for type ratings, their validity and privileges.	X	X	X	X	X					
(03)		State the requirements for instrument ratings, their validity and privileges (instrument rating (IR), competency-based instrument rating (CB-IR) and basic instrument rating (BIR)).	X		X			X	X		1	
(04)		State the requirements for other ratings, their validity and privileges according to Part-FCL.	X	X	X	X	X					
010 04 03 00		Aircrew Regulation — Annex IV (Part-MED)										
010 04 03 01		Aircrew Regulation — Annex IV (Part-MED) — Details										
(01)	X	Describe the relevant content of Part-MED — Medical requirements (administrative parts and requirements related to licensing only).	X	X	X	X	X	X				
(02)		State the requirements for the issue of a medical certificate.	X	X	X	X	X	X				
(03)		Name the class of medical certificate required when exercising the privileges of a CPL, MPL or ATPL.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		State the actions to be taken in case of a decrease in medical fitness.	X	X	X	X	X	X				
010 05 00 00		RULES OF THE AIR ACCORDING TO ICAO ANNEX 2 AND SERA										
010 05 01 00		Overview of ICAO Annex 2 and SERA (Commission Implementing Regulation (EU) No 923/2012 and its references and subsequent amendments)										
010 05 01 01		ICAO Annex 2 and SERA — Relationship and content										
(01)		Explain the scope and purpose of ICAO Annex 2.	X	X	X	X	X	X				
(02)		Explain the scope and main content of SERA.	X	X	X	X	X	X				
010 05 02 00		Rules of the Air										
010 05 02 01		Applicability of the Rules of the Air										
(01)		Explain the principle of territorial application of the various Rules of the Air, e.g. ICAO, SERA, national rules.	X	X	X	X	X					
(02)		Explain the necessity to comply with the Rules of the Air.	X	X	X	X	X					
(03)		State the responsibilities of the PIC.	X	X	X	X	X					
(04)		Identify under what circumstances departure from the Rules of the Air may be allowed.	X	X	X	X	X				1, 2, 3	
(05)		Explain the duties of the PIC concerning pre-flight actions in case of an instrument flight rule (IFR) flight.	X		X			X	X	3	1	
(06)		State that the PIC of an aircraft has final authority as to the disposition of the aircraft while in command.	X	X	X	X	X					
(07)		Explain when the use of psychoactive substances, taking into consideration their effects, by flight crew members is prohibited.	X	X	X	X	X	X				
010 05 03 00		General rules										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 05 03 01		General rules — Collision avoidance — SERA										
(01)		Describe the rules for the avoidance of collisions.	X	X	X	X	X					
(02)		Describe the lights, including their angles, to be displayed by aircraft.	X	X	X	X	X					
(03)		Interpret marshalling signals.	X	X	X	X	X					
(04)		State the basic requirements for minimum height (HGT) for the flight over congested areas of cities, towns or settlements, or over an open-air assembly of persons.	X	X	X	X	X					
(05)		Define when the cruising levels shall be expressed in terms of flight levels (FLs).	X	X	X	X	X					
(06)		Define under what circumstances cruising levels shall be expressed in terms of altitude (ALT).	X	X	X	X	X					
(07)		Explain the limitation for proximity to other aircraft and the right-of-way rules, including holding at runway (RWY) holding positions and lighted stop bars.	X	X	X	X	X					
(08)		Describe the meaning of light signals displayed to aircraft and by aircraft.	X	X	X	X	X					
(09)		Describe the requirements when carrying out simulated instrument flights.	X		X			X	X	3	1	
(10)		Explain the basic rules for an aircraft operating on and in the vicinity of an aerodrome (AD).	X	X	X	X	X					
(11)		Explain the requirements for the submission of an air traffic service (ATS) flight plan.	X	X	X	X	X			3	1	
(12)		Explain the actions to be taken in case of flight plan change or delay.	X	X	X	X	X	X		3	1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(13)		State the actions to be taken in case of inadvertent changes to track, true airspeed (TAS) and time estimate affecting the current flight plan.	X	X	X	X	X	X				
(14)		Explain the procedures for closing a flight plan.	X	X	X	X	X			3	1, 2	
(15)		State for which flights an air traffic control (ATC) clearance shall be obtained.	X	X	X	X	X			3	1, 2	
(16)		State how a pilot may request ATC clearance.	X	X	X	X	X			3	1, 2	
(17)		State the action to be taken if an ATC clearance is not satisfactory to a PIC.	X	X	X	X	X					
(18)		Describe the required actions to be carried out if the continuation of a controlled visual flight rule (VFR) flight in visual meteorological conditions (VMC) is not practicable any more.	X		X	X		X	X	3	2	
(19)		Describe the provisions for transmitting a position report to the appropriate ATS unit including time of transmission and normal content of the message.	X	X	X	X	X	X	X	3		
(20)		Describe the necessary action when an aircraft experiences a communication (COM) failure.	X	X	X	X	X	X	X	3		
(21)		State what information an aircraft being subjected to unlawful interference shall give to the appropriate ATS unit.	X	X	X	X	X	X				
010 05 04 00		Visual flight rules (VFR)										
010 05 04 01		Visual flight rules (VFR) — SERA										
(01)		Describe the VFR as contained in Commission Implementing Regulation (EU) No 923/2012.	X	X	X	X	X					
010 05 05 00		Instrument flight rules (IFR)										
010 05 05 01		Instrument flight rules (IFR) — SERA										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the IFR as contained in Commission Implementing Regulation (EU) No 923/2012.	X		X			X	X	3		
010 05 06 00		Interception of civil aircraft										
010 05 06 01		Interception of civil aircraft — SERA										
(01)		List the circumstances in which interception of a civil aircraft may occur.	X	X	X	X	X					
(02)		State what primary action should be carried out by an intercepted aircraft.	X	X	X	X	X					
(03)		State which frequency should primarily be tried in order to contact an intercepting aircraft.	X	X	X	X	X					
(04)		State on which mode and code a transponder on board the intercepted aircraft should be operated.	X	X	X	X	X					
(05)		Recall the interception signals and phrases.	X	X	X	X	X					
010 06 00 00		AIRCRAFT OPERATIONS										
010 06 01 00		Intentionally left blank										
010 06 02 00		Definitions and abbreviations (PANS-OPS Flight Procedures, ICAO Doc 8168, Volume I)										
010 06 02 01		Definitions and abbreviations — ICAO Doc 8168, Volume I										
(01)	X	Recall all definitions included in ICAO Doc 8168, Volume I, Part I, Section 1, Chapter 1.	X		X			X				
(02)	X	Interpret all abbreviations and acronyms as shown in ICAO Doc 8168, Volume I, Part I, Section 1, Chapter 2.	X		X			X				
010 06 03 00		Departure procedures — (ICAO Doc 8168, Volume I)										
010 06 03 01		General criteria (assuming all engines operating)										
(01)	X	State the factors dictating the design of instrument departure procedures.	X		X			X	X		2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain in which situations the criteria for omnidirectional departures are applied.	X		X			X	X	2		
010 06 03 02		Standard instrument departures (SIDs)										
(01)		Explain the terms ‘straight departure’ and ‘turning departure’.	X		X			X	X	2		
010 06 03 03		Omnidirectional departures										
(01)		Explain what is the meaning of an ‘omnidirectional departure’.	X		X			X	X	2		
010 06 03 04		Intentionally left blank										
010 06 03 05		Intentionally left blank										
010 06 04 00		Approach procedures — ICAO Doc 8168, Volume I										
010 06 04 01		General criteria										
(01)		State the general criteria (except ‘Speeds for procedure calculations’) of the approach procedure design: — instrument approach areas; — accuracy of fixes; — fixes formed by intersections; — intersection fix-tolerance factors; — other fix-tolerance factors; — descent gradient.	X		X			X			2	
(02)		Name the five possible segments of an instrument approach procedure.	X		X			X	X	2		
(03)		State the reasons for establishing aircraft categories for the approach.	X		X			X	X	2		
(04)		State the maximum angle between the final approach track and the extended RWY centre line to still	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		consider a non-precision approach as being a 'straight-in approach'.										
(05)		State the minimum obstacle clearance (MOC) provided by the minimum sector altitudes (MSAs) established for an aerodrome.	X		X			X	X	2		
(06)	X	State that a pilot shall apply wind corrections when carrying out an instrument approach procedure.	X		X			X	X		2	
(07)		State the most significant factor influencing the conduct of instrument approach procedures.	X		X			X	X	2		
(08)		Explain why a pilot should not descend below obstacle clearance altitude/height (OCA/H), which are established for: — precision approach procedures; — non-precision approach procedures; — visual (circling) procedures; — APV approach procedures.	X		X			X	X	2		
(09)		Describe in general terms the relevant factors for the calculation of operational minima.	X		X			X	X	2		
(10)		State the following acronyms in plain language: DA, DH, OCA, OCH, MDA, MDH, MOC, DA/H, OCA/H, MDA/H.	X		X			X	X	2		
(11)		Explain the relationship between the terms: DA, DH, OCA, OCH, MDA, MDH, MOC, DA/H, OCA/H, and MDA/H.	X		X			X	X	2		
010 06 04 02		Approach procedure design										
(01)		Describe how the vertical cross section for each of the five approach segments is broken down into the various areas.	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		State within which area of the cross section the minimum obstacle clearance (MOC) is provided for the whole width of the area.	X		X			X	X	2		
(03)		Define the terms 'IAF', 'IF', 'FAF', 'FAP', 'MAPt' and 'TP'.	X		X			X	X	2		
(04)	X	State the accuracy of facilities providing track (VHF omnidirectional radio range (VOR), instrument landing system (ILS), non-directional beacon (NDB)).	X		X			X	X		2	
(05)		State the optimum descent gradient (preferred for a precision approach) in degrees and per cent.	X		X			X	X	2		
010 06 04 03		Arrival and approach segments										
(01)		Name the five standard segments of an instrument approach procedure, and state the beginning and end for each of them.	X		X			X	X	2		
(02)		Describe where an arrival route normally ends.	X		X			X	X	2		
(03)		State the main task of the initial approach segment.	X		X			X	X	2		
(04)		Describe the maximum angle of interception between the initial approach segment and the intermediate approach segment (provided at the intermediate fix) for a precision approach and a non-precision approach.	X		X			X	X	2		
(05)		Describe the main task of the intermediate approach segment.	X		X			X	X	2		
(06)		State the main task of the final approach segment.	X		X			X	X	2		
(07)		Name the two possible aims of a final approach.	X		X			X	X	2		
(08)		Explain the term 'final approach point' in case of an ILS approach.	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(09)		State what happens if an ILS glide path (GP) becomes inoperative during the approach.	X		X			X	X	2		
010 06 04 04		Missed approach										
(01)		Name the three phases of a missed approach procedure and describe their geometric limits.	X		X			X	X	2		
(02)		State the main task of a missed approach procedure.	X		X			X	X	2		
(03)		Define the term ‘missed approach point (MAPt)’.	X		X			X	X	2		
(04)		Describe how an MAPt may be established in an approach procedure.	X		X			X	X	2		
(05)		State the pilot’s action if, upon reaching the MAPt, the required visual reference is not established.	X		X			X	X	2		
(06)		Describe what a pilot is expected to do in the event a missed approach is initiated prior to arriving at the MAPt.	X		X			X	X	2		
(07)		State whether the pilot is obliged to cross the MAPt at the height (HGT)/altitude (ALT) required by the procedure or whether they are allowed to cross the MAPt at a HGT/ALT greater than that required by the procedure.	X		X			X	X	2		
010 06 04 05		Visual manoeuvring (circling) in the vicinity of the aerodrome (AD)										
(01)		Describe what is meant by ‘visual manoeuvring (circling)’.	X		X			X	X	2		
(02)		Describe how a prominent obstacle in the visual manoeuvring (circling) area outside the final approach and missed approach area has to be considered for the visual circling.	X		X			X	X	2		
(03)		State for which category of aircraft the obstacle clearance altitude/height (OCA/H) within an	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		established visual manoeuvring (circling) area is determined.										
(04)		Describe how the minimum descent altitude/height (MDA/H) is specified for visual manoeuvring (circling) if the OCA/H is known.	X		X			X	X	2		
(05)		State the conditions to be fulfilled before descending below MDA/H in a visual manoeuvring (circling) approach.	X		X			X	X	2		
(06)		Explain why there can be no single procedure designed that will cater for conducting a circling approach in every situation.	X		X			X	X	2		
(07)		State how the pilot is expected to act after initial visual contact during a visual manoeuvring (circling).	X		X			X	X	2		
(08)		Describe what the pilot is expected to do if visual reference is lost while circling to land from an instrument approach.	X		X			X	X	2		
010 06 04 06		<i>Intentionally left blank</i> <i>Note: VOR and VOR/DME are covered under 062 02 03 00 and 062 02 04 00.</i>										
010 06 05 00		Holding procedures — ICAO Doc 8168, Volume I										
010 06 05 01		<i>Entry and holding</i>										
(01)		Explain why deviations from the in-flight procedures of a holding established in accordance with ICAO Doc 8168 are dangerous.	X		X			X	X	2		
(02)		State that if for any reason a pilot is unable to conform to the procedures for normal conditions laid down for any particular holding pattern, this pilot should advise ATC as early as possible.	X		X			X	X	2		
(03)		Describe the shape and terminology associated with the holding pattern.	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		State the bank angle and rate of turn to be used whilst flying in a holding pattern.	X		X			X	X	2		
(05)		Explain why a pilot in a holding pattern should attempt to maintain tracks and how this can be achieved.	X		X			X	X	2		
(06)		Describe where outbound timing begins in a holding pattern.	X		X			X	X	2		
(07)		State where the outbound leg in a holding terminates if the outbound leg is based on DME.	X		X			X	X	2		
(08)		Describe the three heading entry sectors for entries into a holding pattern.	X		X			X	X	2		
(09)		Describe the terms ‘parallel entry’, ‘offset entry’ and ‘direct entry’.	X		X			X	X	2		
(10)		Determine the correct entry procedure for a given holding pattern.	X		X			X	X	2		
(11)		State the still-air time for flying the outbound entry heading with or without DME.	X		X			X	X	2		
(12)		Describe what the pilot is expected to do when clearance is received specifying the time of departure from the holding point.	X		X			X	X	2		
010 06 05 02		Obstacle clearance										
(01)	X	Describe the layout of the basic holding area, entry area and buffer area of a holding pattern.	X		X			X	X		2	
(02)	X	State which obstacle clearance is provided by a minimum permissible holding level referring to the holding area, the buffer area (general only) and over high terrain or in mountainous areas.	X		X			X	X		2	
010 06 06 00		Altimeter-setting procedures — ICAO Doc 8168										
010 06 06 01		Basic requirements and procedures										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the two main objectives of altimeter settings.	X	X	X	X	X	X	X	2	3	
(02)		Define the terms 'QNH' and 'QFE'.	X	X	X	X	X	X	X		3	
(03)		Describe the different terms for ALT or flight levels (FLs) respectively, which are the references during climb or descent to change the altimeter settings from QNH to 1013.2 hPa and vice versa.	X	X	X	X	X	X	X	2	3	
(04)		Define the term 'flight level (FL)'.	X	X	X	X	X	X	X	2	3	
(05)		State where FL zero shall be located.	X	X	X	X	X	X	X		3	
(06)		State the interval by which consecutive FLs shall be separated.	X	X	X	X	X	X	X	2	3	
(07)		Describe how FLs are defined.	X	X	X	X	X	X	X		3	
(08)		Define the term 'transition altitude (TA)'.	X	X	X	X	X	X	X	2	3	
(09)		State how TAs shall normally be specified.	X	X	X	X	X	X	X	2	3	
(10)		Explain how the HGT of the TA is calculated and expressed in practice.	X	X	X	X	X	X	X	2	3	
(11)		State where TAs shall be published.	X	X	X	X	X	X	X	2	3	
(12)		Define the term 'transition level (TRL)'.	X	X	X	X	X	X	X	2	3	
(13)		State when the TRL is normally passed on to the aircraft.	X	X	X	X	X	X	X	2	3	
(14)		State how the vertical position of the aircraft shall be expressed at or below the TA and TRL.	X	X	X	X	X	X	X	2	3	
(15)		Define the term 'transition layer'.	X	X	X	X	X	X	X	2	3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(16)		Describe when the vertical position of an aircraft passing through the transition layer shall be expressed in terms of FLs and when in terms of ALT.	X	X	X	X	X	X	X	2	3	
(17)		State when the QNH altimeter setting shall be made available to departing aircraft.	X	X	X	X	X	X	X	2	3	
(18)		Explain when the vertical separation of an aircraft during en-route flight shall be assessed in terms of ALT and when in terms of FLs.	X	X	X	X	X	X	X	2	3	
(19)		Explain when, in air-ground communications during an en-route flight, the vertical position of an aircraft shall be expressed in terms of ALT and when in terms of FLs.	X	X	X	X	X	X	X	2	3	
(20)		Describe why QNH altimeter-setting reports should be provided from sufficient locations.	X	X	X	X	X	X	X	2	3	
(21)		State how a QNH altimeter setting shall be made available to aircraft approaching a controlled aerodrome (AD) for landing.	X	X	X	X	X	X	X	2	3	
(22)		State under which circumstances the vertical position of an aircraft above the TRL may be referenced in ALT.	X	X	X	X	X	X	X	2	3	
010 06 06 02		Procedures for operators and pilots										
(01)		State on which setting at least one altimeter shall be set prior to take-off.	X	X	X	X	X	X	X	2	3	
(02)		State where during the climb the altimeter setting shall be changed from QNH to 1013.2 hPa.	X	X	X	X	X	X	X	2	3	
(03)		Describe when a pilot of an aircraft intending to land at an AD shall obtain the TRL.	X	X	X	X	X	X	X			
(04)		Describe when a pilot of an aircraft intending to land at an AD shall obtain the actual QNH altimeter setting.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		State where the altimeter settings shall be changed from 1013.2 hPa to QNH during descent for landing.	X	X	X	X	X	X	X			
010 06 07 00		Parallel or near-parallel instrument RWYs — ICAO Doc 8168, Volume III										
010 06 07 01		Simultaneous operation on parallel or near-parallel instrument RWYs										
(01)	X	Describe the difference between independent and dependent parallel approaches.	X	X	X	X	X	X	X		2	
(02)		Describe the following different operations: — simultaneous instrument departures; — segregated parallel approaches/departures; — semi-mixed and mixed operations.	X	X	X	X	X	X	X	2		
(03)		Describe the terms ‘normal operating zone (NOZ)’ and ‘no transgression zone (NTZ)’.	X	X	X	X	X	X				
(04)		State the aircraft avionics requirements for conducting parallel instrument approaches.	X	X	X	X	X	X				
(05)		State where guidance material may be located for simultaneous operations on parallel or near-parallel instrument runways.	X	X	X	X	X	X				
(06)		State the radar requirements for simultaneous, independent, and parallel instrument approaches, and how weather conditions effect these.	X	X	X	X	X	X				
(07)		State the maximum angle of interception for an ILS localiser course (CRS) or microwave landing system (MLS) final approach track in case of simultaneous, independent, and parallel instrument approaches.	X	X	X	X	X	X				
(08)		Describe the special conditions for tracks on missed approach procedures and departures in case of simultaneous or parallel operations.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 06 08 00		Secondary surveillance radar (transponder) operating procedures — ICAO Doc 8168										
010 06 08 01		Operation of transponders										
(01)		State when and where the pilot shall operate the transponder.	X	X	X	X	X	X	X		3	
(02)		State the modes and codes that the pilot shall operate in the absence of any ATC directions or regional air navigation agreements.	X	X	X	X	X	X	X		3	
(03)		State when the pilot shall operate Mode C.	X	X	X	X	X	X				
(04)		State when the pilot shall 'SQUAWK IDENT'.	X	X	X	X	X	X	X		3	
(05)		State the transponder code to indicate: — a state of emergency; — a COM failure; — unlawful interference.	X	X	X	X	X	X	X		3	
(06)		Describe the consequences of a transponder failure in flight.	X	X	X	X	X	X	X		3	
(07)		State the primary action of the pilot in the case of an unserviceable transponder before departure when no repair or replacement at the given AD is possible.	X	X	X	X	X	X	X		3	
(08)		State when the pilot shall operate Mode S.							X		3	
010 06 08 02		Operation of airborne collision avoidance system (ACAS) equipment										
(01)		Describe the main reason for using ACAS.	X	X	X	X	X	X	X	2	3	
(02)		State whether the 'use of ACAS indications' described in ICAO Doc 8168 is absolutely mandatory.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the pilots' reaction required to allow ACAS to fulfil its role of assisting pilots in the avoidance of potential collisions.	X	X	X	X	X	X				
(04)		Explain why pilots shall not manoeuvre their aircraft in response to traffic advisories (TAs) only.	X	X	X	X	X	X				
(05)		Explain the significance of TAs in view of possible resolution advisories (RAs).	X	X	X	X	X	X				
(06)		State why a pilot should follow RAs immediately.	X	X	X	X	X	X				
(07)		List the reasons which may force a pilot to disregard an RA.	X	X	X	X	X	X				
(08)		Explain the importance of instructing ATC immediately that an RA has been followed.	X	X	X	X	X	X				
(09)		Explain the duties of a pilot with regard to ATC when an RA situation is resolved.	X	X	X	X	X	X				
010 06 09 00		REGULATION (EU) No 965/2012 ON AIR OPERATIONS										
010 06 09 01		Regulation structure										
(01)		Describe the subject matter and scope of that Regulation.	X	X	X	X	X	X	X			
(02)	X	State that Regulation (EU) No 965/2012 covers all types of commercial and non-commercial operations.	X	X	X	X	X	X	X			
010 06 09 02		Definitions (Annex I)										
(01)		Recall the definitions in the Regulation not already given in ICAO PAN-OPS.	X	X	X	X	X	X	X			
010 06 09 03		Part-SPA (Annex V), Part-NCC (Annex VI) and Part-NCO (Annex VII)										
(01)		Describe the scope of these Parts.	X	X	X	X	X					
(02)	X	Explain the main content of these Parts, except the operational procedures.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 07 00 00		AIR TRAFFIC SERVICES (ATS) AND AIR TRAFFIC MANAGEMENT (ATM)										
010 07 01 00		ICAO Annex 11 — Air Traffic Services										
010 07 01 01		Definitions										
(01)	X	Recall the definitions given in ICAO Annex 11.	X	X	X	X	X	X				
010 07 01 02		General										
(01)	X	State the objectives of ATS.	X	X	X	X	X	X				
(02)	X	Describe the three basic types of ATS.	X	X	X	X	X	X				
(03)	X	Describe the three basic types of ATC services.	X	X	X	X	X	X				
(04)		State on which frequencies a pilot can expect ATC to contact them in case of an emergency.	X	X	X	X	X	X				
(05)		Describe the procedure for the transfer of an aircraft from one ATC unit to another.	X	X	X	X	X					
010 07 01 03		Airspace										
(01)		Describe the purpose for establishing flight information regions (FIRs) including upper flight information regions (UIRs).	X	X	X	X	X	X				
(02)		Describe the various rules and services that apply to the various classes of airspace.	X	X	X	X	X	X	X	3		
(03)		Explain which airspace shall be included in an FIR or UIR.	X	X	X	X	X	X				
(04)		State the designation for those portions of the airspace where flight information service (FIS) and alerting service shall be provided.	X	X	X	X	X	X				
(05)		State the designations for those portions of the airspace where ATC services shall be provided.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Identify whether or not control areas (CTAs) and control zones (CTRs) designated within an FIR shall form part of that FIR.	X	X	X	X	X	X				
(07)		State the lower limit of a CTA as far as ICAO Standards are concerned.	X	X	X	X	X	X				
(08)		State whether or not the lower limit of a CTA has to be established uniformly.	X	X	X	X	X	X				
(09)		Explain why a UIR or upper CTA should be delineated to include the upper airspace within the lateral limits of a number of lower FIRs or CTAs.	X	X	X	X	X	X				
(10)		Describe in general the lateral limits of CTRs.	X	X	X	X	X	X				
(11)		State the minimum extension (in NM) of the lateral limits of a CTR.	X	X	X	X	X	X				
(12)		State the upper limits of a CTR located within the lateral limits of a CTA.	X	X	X	X	X	X				
010 07 01 04		Air traffic control (ATC) services										
(01)		Name all classes of airspace in which ATC services shall be provided.	X	X	X	X	X	X			2, 3	
(02)		Name the ATS units providing ATC services (area control service, approach control service, aerodrome control service).	X	X	X	X	X	X	X		3	
(03)		Describe which unit(s) may be assigned with the task to provide specified services on the apron.	X	X	X	X	X	X	X		3	
(04)		State the purpose of clearances issued by an ATC unit.	X	X	X	X	X	X	X		3	
(05)		List the various (five possible) parts of an ATC clearance.	X	X	X	X	X	X	X	3		
(06)		Explain why the movement of persons, vehicles and towed aircraft on the manoeuvring area of an AD	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		shall be controlled by the aerodrome control tower (TWR) (as necessary).										
010 07 01 05		Flight information service (FIS)										
(01)	X	State for which aircraft FIS shall be provided.	X	X	X	X	X	X				
(02)	X	State whether or not FIS shall include the provision of pertinent significant meteorological information (SIGMET) and air meteorological information report (AIRMET) information.	X	X	X	X	X	X				
(03)	X	State which information FIS shall include in addition to SIGMET and AIRMET information.	X	X	X	X	X	X				
(04)	X	Indicate which other information the FIS shall include in addition to the special information given in Annex 11.	X	X	X	X	X	X				
(05)	X	State the meaning of the acronym 'ATIS' in plain language.	X	X	X	X	X	X				
(06)		List the basic information concerning automatic terminal information service (ATIS) broadcasts (e.g. frequencies used, number of ADs included, updating, identification, acknowledgment of receipt, language and channels, ALT- setting).	X	X	X	X	X	X			2, 3	
(07)		State the content of an ATIS message.	X	X	X	X	X					
(08)		State the reasons and circumstances when an ATIS message shall be updated.	X	X	X	X	X	X				
010 07 01 06		Alerting service										
(01)		State who provides the alerting service.	X	X	X	X	X					
(02)		State who is responsible for initiating the appropriate emergency phase.	X	X	X	X	X					
(03)		State the aircraft to which alerting service shall be provided.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		State which unit shall be notified by the responsible ATS unit immediately when an aircraft is considered to be in a state of emergency.	X	X	X	X	X					
(05)		Name the three stages of emergency and describe the basic conditions for each kind of emergency.	X	X	X	X	X				3	
(06)	X	State the meaning of the expressions 'INCERFA', 'ALERFA' and 'DETRESFA'.	X	X	X	X	X					
(07)	X	State the information to be provided to those aircraft that operate in the vicinity of an aircraft that is either in a state of emergency or unlawful interference.	X	X	X	X	X					
010 07 01 07		Principles governing required navigation performance (RNP) and air traffic service (ATS) route designators										
(01)		State the meaning of the acronym 'RNP'.	X	X	X	X	X					
(02)		State the factors that RNP is based on.	X	X	X	X	X					
(03)	X	Describe the reason for establishing a system of route designators and navigation specifications.	X	X	X	X	X					
(04)		State whether or not a prescribed RNP type is considered an integral part of the ATS route designator.	X	X	X	X	X					
(05)		Explain the composition of an ATS route designator.	X	X	X	X	X					
010 07 02 00		ICAO Doc 4444 — Air Traffic Management										
010 07 02 01		Foreword (Scope and purpose)										
(01)		State which ATS units provide clearances that do, and do not, include the prevention of collision with terrain.	X	X	X	X	X	X	X	3		
010 07 02 02		Definitions										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Recall all definitions given in ICAO Doc 4444 except the following: — accepting unit/controller, AD taxi circuit, aeronautical fixed service (AFS), aeronautical fixed station, air-taxiing, allocation, approach funnel, assignment, data convention, data processing, discrete code, D-value, flight status, ground effect, receiving unit/controller, sending unit/controller, transfer of control point, transferring unit/controller, unmanned free balloon.	X	X	X	X	X	X				
010 07 02 03		ATS system capacity and air traffic flow management (ATFM)										
(01)	X	Explain when and where ATFM services shall be implemented.	X	X	X	X	X	X	X		3	
010 07 02 04		General provisions for air traffic services (ATS)										
(01)	X	Describe who is responsible for the provision of flight information and alerting services within an FIR, within controlled airspace and at controlled ADs.	X	X	X	X	X	X				
010 07 02 05		ATC clearances										
(01)		State which information the issue of an ATC clearance is based on.	X	X	X	X	X	X	X		3	
(02)		Describe what a PIC should do if an ATC clearance is not suitable.	X	X	X	X	X	X	X	3		
(03)		State who bears the responsibility for adhering to the applicable rules and regulations whilst flying under the control of an ATC unit.	X	X	X	X	X	X	X		3	
(04)	X	State the two primary purposes of clearances issued by ATC units.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		State why clearances must be issued ‘early enough’ to aircraft.	X	X	X	X	X	X				
(06)		Explain what is meant by the expression ‘clearance limit’.	X	X	X	X	X	X	X	3		
(07)		Explain the meaning of the phrases ‘cleared via flight planned route’, ‘cleared via (designation) departure’ and ‘cleared via (designation) arrival’ in an ATC clearance.	X	X	X	X	X	X	X	3		
(08)		List which items of an ATC clearance shall always be read back by the flight crew.	X	X	X	X	X	X	X	3		
010 07 02 06		Horizontal speed control instructions										
(01)		Explain the reason for speed control by ATC	X	X	X	X	X	X	X	3		
(02)	X	Define the maximum speed changes that ATC may impose.	X	X	X	X	X	X	X		3	
(03)		State within what distance from the THR the PIC should not expect any kind of speed control.	X	X	X	X	X	X	X	3		
010 07 02 07		Change from IFR to VFR flight										
(01)		Explain how the change from IFR to VFR can be initiated by the PIC.	X		X			X	X	3		
(02)		Describe the expected reaction of the appropriate ATC unit upon a request to change from IFR to VFR.	X		X			X	X	3		
010 07 02 08		Wake turbulence										
(01)		State the wake-turbulence categories of aircraft.	X	X	X	X	X	X		3	2	
(02)		State the wake-turbulence separation minima.	X	X	X	X	X	X		3	2	
(03)		Describe how a ‘heavy’ aircraft shall indicate this in the initial radiotelephony contact with ATS.	X	X	X	X	X	X				
010 07 02 09		Altimeter-setting procedures										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define the following terms: — TRL; — transition layer; and — TA.	X	X	X	X	X	X	X		3	
(02)		Describe how the vertical position of an aircraft in the vicinity of an AD shall be expressed at or below the TA, at or above the TRL, and while climbing or descending through the transition layer.	X	X	X	X	X	X	X		3	
(03)		Describe when the HGT of an aircraft using QFE during an NDB approach is referred to the landing THR instead of the AD elevation.	X	X	X	X	X	X	X	3		
(04)		State in which margin altimeter settings provided to aircraft shall be rounded up or down.	X	X	X	X	X	X	X	3		
(05)		Describe the expression 'lowest usable FL'.	X	X	X	X	X	X	X	3		
(06)		Determine how the vertical position of an aircraft on an en-route flight is expressed at or above the lowest usable FL and below the lowest usable FL.	X	X	X	X	X	X	X	3		
(07)		State who establishes the TRL to be used in the vicinity of an AD.	X	X	X	X	X	X	X	3		
(08)		Decide how and when a flight crew member shall be informed about the TRL.	X	X	X	X	X	X	X	3		
(09)		State whether or not the pilot can request TRL to be included in the approach clearance.	X	X	X	X	X	X	X	3		
010 07 02 10		Position reporting										
(01)		Describe when position reports shall be made by an aircraft flying on routes defined by designated significant points.	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		List the six items that are normally included in a voice position report.	X	X	X	X	X	X	X	3		
(03)	X	State the requirements for using a simplified position report with FL, next position (and time-over) and ensuing significant points omitted.	X	X	X	X	X	X	X		3	
(04)		State the item of a position report which must be forwarded on to ATC with the initial call after changing to a new frequency.	X	X	X	X	X	X	X	3		
(05)		Indicate the item of a position report which may be omitted if secondary surveillance radar (SSR) Mode C is used.	X	X	X	X	X	X	X		3	
(06)		Explain in which circumstances the airspeed should be included in a position report.	X	X	X	X	X	X				
(07)		Explain the meaning of the acronym 'ADS'.	X	X	X	X	X	X				
(08)		Describe which expression shall precede the level figures in a position report if the level is reported in relation to 1013.2 hPa (standard pressure).	X	X	X	X	X	X				
010 07 02 11		Reporting of operational and meteorological information										
(01)		List the occasions when special air-reports shall be made.	X	X	X	X	X	X				
010 07 02 12		Separation methods and minima										
(01)		Explain the general provisions for the separation of controlled air traffic.	X		X			X	X			
(02)	X	Name the different kinds of separation used in aviation.	X		X			X	X		3	
(03)		State the difference between the type of separation provided within the various classes of airspace and the various types of flight.	X		X			X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		State who is responsible for the avoidance of collision with other aircraft when operating in VMC.	X		X			X	X	3		
(05)		Describe how vertical separation is obtained.	X		X			X	X	3		
(06)		State the required vertical separation minimum.	X		X			X	X	3		
(07)		Describe how the cruising levels of aircraft flying to the same destination and in the expected approach sequence are correlated with each other.	X		X			X	X	3		
(08)		Name the conditions that must be adhered to when two aircraft are cleared to maintain a specified vertical separation between them during climb or descent.	X		X			X	X	3		
(09)		State the two main methods for horizontal separation.	X		X			X	X	3		
(10)		Describe how lateral separation of aircraft at the same level may be obtained.	X		X			X	X	3		
(11)		Explain the term 'geographical separation'.	X		X			X	X	3		
(12)		Describe track separation between aircraft using the same navigation aid or method.	X		X			X	X	3		
(13)		Describe the three basic means for the establishment of longitudinal separation.	X		X			X	X	3		
(14)		State the minimum standard horizontal radar separation in NM.	X		X			X	X	3		
(15)		Describe the method of the Mach number technique.	X	X								
010 07 02 13		Separation in the vicinity of aerodromes (ADs)										
(01)		Describe the expression 'essential local traffic'.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		State which possible decision the PIC may choose to take if they are asked to accept take-off in a direction which is not 'into the wind'.	X	X	X	X	X	X				
(03)		State the condition to enable ATC to initiate a visual approach for an IFR flight.	X	X	X	X	X	X	X	3	2	
(04)		State whether or not separation shall be provided by ATC between an aircraft executing a visual approach and other arriving or departing aircraft.	X	X	X	X	X	X	X	3	2	
(05)		State in which case, when the flight crew are not familiar with the instrument approach procedure being carried out, only the final approach track has to be given to them by ATC.	X	X	X	X	X	X	X	3	2	
(06)		Describe which FL should be assigned to an aircraft first arriving over a holding fix for landing.	X	X	X	X	X	X	X	3	2	
(07)		State which kinds of priority can be applied to aircraft for a landing.	X	X	X	X	X	X	X	3	2	
(08)		Describe the situation when a pilot of an aircraft in an approach sequence indicates their intention to hold for weather improvements.	X	X	X	X	X	X	X	3	2	
(09)		Explain the term 'expected approach time' and the procedures for its use.	X	X	X	X	X	X	X	3	2	
(10)		State the reasons which could probably lead to the decision to use another take-off or landing direction than the one into the wind.	X	X	X	X	X	X	X	3	2	
(11)		State the possible consequences for a PIC if the 'RWY-in-use' is not considered suitable for the operation involved.	X	X	X	X	X	X	X			
010 07 02 14		Miscellaneous separation procedures										
(01)		State the minimum separation between departing and arriving aircraft.	X	X	X	X	X	X	X	3	2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		State the non-radar wake-turbulence longitudinal separation minima.	X	X	X	X	X	X	X	3	2	
(03)		Describe the consequences of a clearance to 'maintain own separation' while in VMC.	X	X	X	X	X	X	X	3	2	
(04)		Give a brief description of 'essential traffic' and 'essential traffic information'.	X	X	X	X	X	X	X	3	2	
(05)		Describe the circumstances under which a reduction in separation minima may be allowed.	X	X	X	X	X	X	X	3	2	
010 07 02 15		Arriving and departing aircraft										
(01)		List the elements of information which shall be transmitted to an aircraft as early as practicable if an approach for landing is intended.	X	X	X	X	X	X	X	3	2	
(02)		List the elements of information to be transmitted to an aircraft at the commencement of final approach.	X	X	X	X	X	X	X	3	2	
(03)		List the elements of information to be transmitted to an aircraft during final approach.	X	X	X	X	X	X	X	3	2	
(04)		State the prerequisites for operating on parallel or near-parallel RWYs including the different combinations of parallel arrivals or departures.	X	X	X	X	X	X				
(05)		State the sequence of priority between aircraft landing (or in the final stage of an approach to land) and aircraft intending to depart.	X	X	X	X	X	X	X	3	2	
(06)		State the significant changes in the meteorological conditions in the take-off or climb-out area that shall be transmitted without delay to a departing aircraft.	X	X	X	X	X	X	X	3	2	
(07)		State the significant changes that shall be transmitted as early as practicably possible to an arriving aircraft, particularly changes in the meteorological conditions.	X	X	X	X	X	X	X	3	2	
010 07 02 16		Procedures for aerodrome (AD) control service										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Name the operational failure or irregularity of AD equipment which shall be reported by the TWR immediately.	X	X	X	X	X	X	X	3	2	
(02)		Explain that, after a given period of time, the TWR shall report to the area control centre (ACC) or flight information centre (FIC) if an aircraft does not land as expected.	X	X	X	X	X	X	X	3	2	
(03)		Describe the procedures to be observed by the TWR whenever VFR operations are suspended.	X	X	X	X	X	X	X	3	2	
(04)		Explain the term 'RWY-in-use' and its selection.	X	X	X	X	X	X				
(05)		List the information the TWR should give to an aircraft prior to: — taxiing for take-off; — take-off; — entering the traffic circuit.	X	X	X	X	X	X				
(06)		Explain that a report of surface wind direction given to a pilot by the TWR is magnetic.	X	X	X	X	X	X				
(07)		Explain the exact meaning of the expression 'RWY vacated'.	X	X	X	X	X	X				
010 07 02 17		Radar services										
(01)		State the basic identification procedures used with radar.	X	X	X	X	X	X	X	3	2	
(02)		Define the term 'PSR'.	X	X	X	X	X	X	X	3	2	
(03)		Describe the circumstances under which an aircraft provided with radar service should be informed of its position.	X	X	X	X	X	X	X	3	2	
(04)		List the possible forms of position information passed on to the aircraft by radar services.	X	X	X	X	X	X	X	3	2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Describe the term ‘radar vectoring’.	X	X	X	X	X	X	X	3	2	
(06)		State the aims of radar vectoring as shown in ICAO Doc 4444.	X	X	X	X	X	X	X	3	2	
(07)		Describe how radar vectoring shall be achieved.	X	X	X	X	X	X	X	3	2	
(08)		Describe the information which shall be given to an aircraft when radar vectoring is terminated and the pilot is instructed to resume own navigation.	X	X	X	X	X	X	X	3	2	
(09)		Explain the procedures for the conduct of surveillance radar approaches (SRAs).	X	X	X	X	X	X	X	3	2	
(10)		Describe what kind of action (concerning the transponder) the pilot is expected to perform in case of emergency if they have previously been directed by ATC to operate the transponder on a specific code.	X	X	X	X	X	X	X	3	2	
010 07 02 18		Air traffic advisory service										
(01)		Describe the objective and basic principles of the air traffic advisory service.	X	X	X	X	X	X				
(02)		State to which aircraft air traffic advisory service may be provided.	X	X	X	X	X	X				
(03)		Explain the difference between advisory information and clearances, stating which ATS units are responsible for their issue.	X	X	X	X	X	X				
010 07 02 19		Procedures related to emergencies, communication (COM) failure and contingencies										
(01)		State the mode and code of SSR equipment a pilot might operate in a (general) state of emergency or (specifically) in case the aircraft is subject to unlawful interference.	X	X	X	X	X	X	X		3	
(02)		State the special rights an aircraft in a state of emergency can expect from ATC.	X	X	X	X	X	X	X		3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Describe the expected action of aircraft after receiving a broadcast from ATS concerning the emergency descent of an aircraft.	X	X	X	X	X	X	X	3		
(04)		State how it can be ascertained, in case of a failure of two-way COM, whether the aircraft is able to receive transmissions from the ATS unit.	X	X	X	X	X	X	X	3		
(05)		State on which frequencies appropriate information, for an aircraft encountering two-way COM failure, shall be sent by ATS.	X	X	X	X	X	X	X	3		
(06)		State what is meant by the expressions ‘strayed aircraft’ and ‘unidentified aircraft’.	X	X	X	X	X	X	X	3		
(07)		Explain the reasons for fuel-dumping and state the minimum level.	X	X	X	X	X	X				
(08)		Explain the possible request of ATC to an aircraft to change its radio-telephone (RTF) call sign.	X	X	X	X	X	X				
010 07 02 20		Miscellaneous procedures										
(01)		Explain the meaning of ‘AIRPROX’.	X	X	X	X	X	X				
(02)		Describe the task of an air traffic incident report.	X	X	X	X	X	X				
010 08 00 00		AERONAUTICAL INFORMATION SERVICE (AIS)										
010 08 01 00		Introduction										
010 08 01 01		Introduction to ICAO Annex 15 — Aeronautical Information Service (AIS)										
(01)		State, in general terms, the objective of an AIS.	X	X	X	X	X	X				
010 08 02 00		Definitions of ICAO Annex 15										
010 08 02 01		Definitions of ICAO Annex 15										
(01)		Recall the following definitions: — aeronautical information circular (AIC), aeronautical information publication (AIP), AIP	X	X	X	X	X	X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		amendment, AIP supplement, aeronautical information regulation and control (AIRAC), danger area, aeronautical information management, international airport, international NOTAM office (NOF), manoeuvring area, movement area, NOTAM, pre-flight information bulletin (PIB), prohibited area, restricted area, SNOWTAM, ASHTAM.										
010 08 03 00		General										
010 08 03 01		General — AIS responsibilities and functions										
(01)		State during which period of time an AIS shall be available with reference to an aircraft flying in the area of responsibility of an AIS, provided a 24-hour service is not available.	X	X	X	X	X	X				
(02)		List, in general, the kind of aeronautical information/data which an AIS service shall make available in a suitable form to flight crew.	X	X	X	X	X	X				
(03)		Summarise the duties of an AIS concerning aeronautical information data for the territory of a particular State.	X	X	X	X	X	X				
010 08 04 00		Aeronautical information products and services										
010 08 04 01		Aeronautical information publication (AIP)										
(01)		State the primary purpose of the AIP.	X	X	X	X	X	X				
(02)		Name the different parts of the AIP.	X	X	X	X	X	X				
(03)		State the main parts of the AIP where the following information can be found: — differences from the ICAO Standards, Recommended Practices and Procedures;	X	X	X	X	X	X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- location indicators, AIS, minimum flight ALT, meteorological information for aircraft in flight (VOLMET) service, SIGMET service; - general rules and procedures (especially general rules, VFR, IFR, ALT-setting procedure, interception of civil aircraft, unlawful interference, air traffic incidents); - ATS airspace (especially FIR, UIR, TMA); - ATS routes (especially lower ATS routes, upper ATS routes, area navigation routes); - AD data including aprons, taxiways (TWYs) and check locations/positions data; - navigation warnings (especially prohibited, restricted and danger areas); - aircraft instruments, equipment and flight documents; - AD surface movement guidance and control system and markings; - RWY physical characteristics, declared distances, approach (APP) and RWY lighting; - AD radio navigation and landing aids; - charts related to an AD; - entry, transit and departure of aircraft, passengers, crew and cargo, and the significance of this information to flight crew.										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		State how permanent changes to the AIP shall be published.	X	X	X	X	X	X				
(05)		Explain what kind of information shall be published in the form of AIP Supplements.	X	X	X	X	X	X				
010 08 04 02		Notices to airmen (NOTAMs)										
(01)		Describe how information shall be published which in principle would belong to NOTAMs but includes extensive text or graphics.	X	X	X	X	X	X	X		2	
(02)		Summarise the essential information which leads to the issue of a NOTAM.	X	X	X	X	X	X	X	2		
(03)		State how NOTAMs shall be distributed.	X	X	X	X	X	X				
(04)		Explain how information regarding snow, ice and standing water on AD pavements shall be reported.	X	X	X	X	X	X	X			
(05)		Describe the means by which NOTAMs shall be distributed.	X	X	X	X	X	X				
(06)		Define and state which information an ASHTAM may contain.	X	X	X	X	X	X				
010 08 04 03		Aeronautical information regulation and control (AIRAC)										
(01)	X	List the circumstances under which the information concerned shall or should be distributed as an AIRAC.	X	X	X	X	X	X	X		2	
010 08 04 04		Aeronautical information circulars (AICs)										
(01)	X	Describe the type of information that may be published in AICs.	X	X	X	X	X	X				
(02)		Explain the organisation of AICs.	X	X	X	X	X	X				
010 08 04 05		Pre-flight and post-flight information/data										
(01)		Summarise, in addition to the elements of the integrated AIP and maps/charts, the additional	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		current information relating to the AD of departure that shall be provided as pre-flight information.										
(02)		Describe how a recapitulation of current NOTAM and other information of urgent character shall be made available to flight crew.	X	X	X	X	X	X	X	2		
(03)		State which post-flight information from flight crew shall be submitted to AIS for distribution as required by the circumstances.	X	X	X	X	X	X				
010 08 05 00		ATM service providers										
010 08 05 01		ATM										
(01)		State that Commission Implementing Regulation (EU) 2017/373 provides: — general requirements for the provision of air navigation services; — specific requirements for the provision of air traffic services; — specific requirements for the provision of meteorological services; — specific requirements for the provision of aeronautical information services; — specific requirements for the provision of communication, navigation or surveillance services.	X		X	X						
010 09 00 00		AERODROMES (ICAO Annex 14 and Regulation (EU) No 139/2014)										
010 09 01 00		General										
010 09 01 01		General — AD reference code										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the intent of the AD reference code and state the functions of the two code elements.	X	X								
010 09 02 00		Aerodrome (AD) data										
010 09 02 01		Aerodrome (AD) reference point										
(01)		Describe where the AD reference point shall be located and where it shall normally remain.	X	X	X	X	X	X	X		2	
010 09 02 02		Pavement strengths										
(01)		Explain the terms: ‘pavement classification number (PCN)’ and ‘aircraft classification number (ACN)’, and describe their mutual dependence.	X	X	X	X	X	X				
(02)		Describe how the bearing strength for an aircraft with an apron mass equal to or less than 5 700 kg shall be reported.	X	X	X	X	X	X				
010 09 02 03		Declared distances										
(01)		State that ICAO Annex 14 provides guidance on the calculation of declared distances (TORA, TODA, ASDA, LDA).	X	X	X	X	X	X				
(02)		Recall the definitions for the four main declared distances.	X	X	X	X	X	X				
010 09 02 04		Condition of the movement area and related facilities										
(01)		State the purpose of informing AIS and ATS units about the condition of the movement area and related facilities.	X	X	X	X	X	X				
(02)		List the matters of operational significance or affecting aircraft performance which should be reported to AIS and ATS units to be transmitted to aircraft involved.	X	X	X	X	X	X				
(03)		Describe the different types of contaminant on RWYs.	X	X	X	X	X	X		2	1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the different types of frozen water on the RWY and their impact on aircraft braking performance.	X	X	X	X	X	X				
(05)		Describe the runway condition codes and the associated runway braking action.	X	X	X	X	X			2	1	
010 09 03 00		Physical characteristics										
010 09 03 01		Runways (RWYs)										
(01)		Describe where a THR should normally be located.	X	X	X	X	X	X				
(02)		Describe the general considerations concerning RWYs associated with a stopway (SWY) or clearway (CWY).	X	X	X	X	X	X	X		2	
010 09 03 02		Runway (RWY) strips										
(01)		Explain the term 'runway strip'.	X	X	X	X	X	X	X		2	
010 09 03 03		Runway-end safety area										
(01)		Explain the term 'runway-end safety area'.	X	X	X	X	X	X	X		2	
010 09 03 04		Clearway (CWY)										
(01)		Explain the term 'clearway'.	X	X	X	X	X	X	X		2	
010 09 03 05		Stopway (SWY)										
(01)		Explain the term 'stopway'.	X	X	X	X	X	X	X		2	
010 09 03 06		Intentionally left blank										
010 09 03 07		Taxiways (TWYs)										
(01)		Describe the reasons and the requirements for rapid-exit TWYs.	X	X	X	X	X	X				
(02)		Explain TWY widening in curves.	X	X	X	X	X	X				
(03)		Explain when and where holding bays should be provided.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Describe where RWY holding positions shall be established.	X	X	X	X	X	X	X		2	
(05)		Describe the term ‘road holding position’.	X	X	X	X	X	X				
(06)		Describe where intermediate TWY holding positions should be established.	X	X	X	X	X	X				
010 09 04 00		Visual aids for navigation										
010 09 04 01		Indicators and signalling devices										
(01)		Describe the wind-direction indicators with which ADs shall be equipped.	X	X	X	X	X	X	X			
(02)		Describe a landing-direction indicator.	X	X	X	X	X	X				
(03)		Explain the capabilities of a signalling lamp.	X	X	X	X	X	X	X			
(04)	X	State which characteristics a signal area should have.	X	X	X	X	X	X	X			
(05)	X	Interpret all indications and signals that may be used in a signal area.	X	X	X	X	X	X	X			
010 09 04 02		Markings										
(01)		Name the colours used for the various markings (RWY, TWY, aircraft stands, apron safety lines).	X	X	X	X	X	X	X	2		
(02)		State where a RWY designation marking shall be provided and describe the different layouts (excluding dimensions).	X	X	X	X	X	X	X			
(03)		Describe the application and general characteristics (excluding dimensions) of: — RWY-centre-line markings; — THR markings; — touchdown-zone (TDZ) markings; — RWY-side-stripe markings;	X	X	X	X	X	X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— TWY-centre-line markings; — RWY holding position markings; — intermediate holding position markings; — aircraft-stand markings; — apron safety lines; — road holding position markings; — mandatory instruction markings; — information markings.										
010 09 04 03		Lights										
(01)		Describe the mechanical safety considerations regarding elevated approach lights and elevated RWY, SWY and TWY lights.	X	X	X	X	X	X	X	2		
(02)		List the conditions for the installation of an aerodrome beacon (ABN) and describe its general characteristics.	X	X	X	X	X	X	X		2	
(03)		Describe the different kinds of operations for which a simple approach lighting system shall be used.	X	X	X	X	X	X	X	2		
(04)		Describe the basic installations of a simple approach lighting system including the dimensions and distances normally used.	X	X	X	X	X	X	X	2		
(05)		Describe the principle of a precision approach category I lighting system including information such as location and characteristics.	X	X	X	X	X	X	X	2		
(06)		Describe the principle of a precision approach category II and III lighting system including information such as location and characteristics, especially the inner 300 m of the system.	X					X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Describe the wing bars of the precision approach path indicator (PAPI) and the abbreviated precision approach path indicator (APAPI). Interpret what the pilot will see during the approach using PAPI.	X	X	X	X	X	X	X		2	
(08)		Interpret what the pilot will see during an approach using a helicopter approach path indicator (HAPI).			X	X	X					
(09)		Explain the application and characteristics (as applicable, but limited to colour, intensity, direction and whether fixed or flashing) of: — RWY-edge lights; — RWY-THR and wing-bar lights; — RWY-end lights; — RWY-centre-line lights; — RWY-lead-in lights; — RWY-TDZ lights; — SWY lights; — TWY-centre-line lights; — TWY-edge lights; — stop bars; — intermediate holding position lights; — RWY guard lights; — road holding position lights.	X	X	X	X	X	X	X	2		
(10)		State the timescale within which aeronautical ground lights shall be made available to arriving aircraft.	X	X	X	X	X	X	X			
010 09 04 04		Signs										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain which signs are the only ones on the movement area utilising red.	X	X	X	X	X	X	X		2	
(02)		List the provisions for illuminating signs.	X	X	X	X	X	X	X		2	
(03)		Name the kinds of signs which shall be included in mandatory instruction signs.	X	X	X	X	X	X	X		2	
(04)		Name the colours used for mandatory instruction signs.	X	X	X	X	X	X	X		2	
(05)		Describe by which sign a pattern 'A' RWY holding position (i.e. at an intersection of a TWY and a non-instrument, non-precision approach or take-off RWY) marking shall be supplemented.	X	X	X	X	X	X	X			
(06)		Describe by which sign a pattern 'B' RWY holding position (i.e. at an intersection of a TWY and a precision approach RWY) marking shall be supplemented.	X	X	X	X	X	X	X			
(07)		Describe the location of: — a RWY designation sign at a TWY/RWY intersection; — a 'NO ENTRY' sign; — a RWY holding position sign.	X	X	X	X	X	X	X		2	
(08)		State which sign indicates that a taxiing aircraft is about to infringe an obstacle limitation surface or interfere with the operation of radio navigation aids (e.g. ILS/MLS critical/sensitive area).	X	X	X	X	X	X	X		2	
(09)		Describe the various possible inscriptions on RWY designation signs and on holding position signs.	X	X	X	X	X	X	X		2	
(10)		Describe the colours used in connection with information signs.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(11)		Describe the possible inscriptions on information signs.	X	X	X	X	X	X	X			
(12)		Explain the application, location and characteristics of aircraft stand identification signs.	X	X	X	X	X	X	X			
(13)		Explain the application, location and characteristics of road holding position signs.	X	X	X	X	X	X	X			
010 09 04 05		Markers										
(01)		Explain why markers located near a RWY or TWY shall be HGT limited.	X	X	X	X	X	X	X			
(02)		Explain the application and characteristics (excluding dimensions) of: — unpaved RWY-edge markers; — TWY-edge markers; — TWY-centre-line markers; — unpaved TWY-edge markers; — boundary markers; — SWY-edge markers.	X	X	X	X	X	X	X			
010 09 05 00		Visual aids for denoting obstacles										
010 09 05 01		Marking of objects										
(01)		State how fixed or mobile objects shall be marked if colouring is not practicable.	X	X	X	X	X	X				
(02)		Describe marking by colours (fixed or mobile objects).	X	X	X	X	X	X				
(03)		Explain the use of markers for the marking of objects, overhead wires, cables, etc.	X	X	X	X	X	X				
(04)		Explain the use of flags for the marking of objects.	X	X	X	X	X	X				
010 09 05 02		Lighting of objects										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Name the different types of lights to indicate the presence of objects which must be lighted.	X	X	X	X	X	X				
(02)		Describe (in general terms) the location of obstacle lights.	X	X	X	X	X	X				
(03)		Describe (in general and for normal circumstances) the colour and sequence of low-intensity obstacle lights, medium-intensity obstacle lights and high-intensity obstacle lights.	X	X	X	X	X	X				
(04)		State that information about lights to be displayed by aircraft is provided in both ICAO Annex 2 (Rules of the Air) and SERA.	X	X	X	X	X	X				
010 09 06 00		Visual aids for denoting restricted use of areas										
010 09 06 01		Visual aids for denoting restricted use of areas on RWYs and TWYs										
(01)		Describe the colours and meaning of ‘closed markings’ on RWYs and TWYs.	X	X	X	X	X	X				
(02)		State how the pilot of an aircraft moving on the surface of a TWY, holding bay or apron shall be warned that the shoulders of these surfaces are ‘non-load-bearing’.	X	X	X	X	X	X				
(03)		Describe the pre-THR marking (including colours) when the surface before the THR is not suitable for normal use by aircraft.	X	X	X	X	X	X				
010 09 07 00		Aerodrome (AD) operational services, equipment and installations										
010 09 07 01		Rescue and firefighting (RFF)										
(01)		State the principal objective of RFF services.	X	X	X	X	X	X				
(02)		Explain the basic information the AD category (for RFF) depends upon.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Describe what is meant by the term ‘response time’, and state its normal and maximum limits.	X	X	X	X	X	X				
010 09 07 02		Apron management service										
(01)		State who has a right-of-way against vehicles operating on an apron.	X	X	X	X	X	X				
010 09 07 03		Ground-servicing of aircraft										
(01)		Describe the necessary actions during the ground-servicing of an aircraft with regard to the possible event of a fuel fire.	X	X	X	X	X	X				
010 09 08 00		Supplementary Guidance Material										
010 09 08 01		Declared distances										
(01)		List the four types of ‘declared distances’ on a RWY and also the appropriate abbreviations.	X	X	X	X	X	X			1	
(02)		Explain the circumstances which lead to the situation that the four declared distances on a RWY are equal to the length of the RWY.	X	X	X	X	X	X				
(03)		Describe the influence of a CWY, SWY or displaced THR upon the four ‘declared distances’.	X	X	X	X	X	X				
010 09 08 02		Intentionally left blank										
010 09 08 03		Approach lighting systems										
(01)		Name the two main groups of approach lighting systems.	X	X	X	X	X	X	X	2		
(02)		Describe the two different versions of a simple approach lighting system.	X	X	X	X	X	X	X	2		
(03)		Describe the two different basic versions of precision approach lighting systems for CAT I.	X	X	X	X	X	X	X	2		
(04)		Describe the diagram of the inner 300 m of the precision approach lighting system in the case of CAT II and III.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Describe how the arrangement of an approach lighting system and the location of the appropriate THR are interrelated.	X	X	X	X	X	X	X	2		
010 10 00 00		FACILITATION (ICAO Annex 9)										
010 10 01 00		<i>Intentionally left blank</i>										
010 10 02 00		Entry and departure of aircraft										
010 10 02 01		<i>General declaration</i>										
(01)		Describe the purpose and use of aircraft documents as regards a ‘general declaration’.	X	X	X	X	X					
010 10 02 02		<i>Entry and departure of crew</i>										
(01)		Explain entry requirements for crew.	X	X	X	X	X					
(02)		Explain the reasons for the use of crew member certificates (CMC) for crew members engaged in international air transport.	X	X	X	X	X					
(03)		Explain in which cases Contracting States should accept the CMC as an identity document instead of a passport or visa.	X	X	X	X	X					
010 10 02 03		<i>Entry and departure of passengers and baggage</i>										
(01)		Explain the entry requirements for passengers and their baggage.	X	X	X	X	X					
(02)		Explain the requirements and documentation for unaccompanied baggage.	X	X	X	X	X					
(03)		Identify the documentation required for the departure and entry of passengers and their baggage.	X	X	X	X	X					
(04)		Explain the arrangements in the event of a passenger being declared an inadmissible person.	X	X	X	X	X					
(05)		Describe the pilot’s authority towards unruly passengers.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 10 02 04		Entry and departure of cargo										
(01)		Explain the entry requirements for cargo.	X	X	X	X	X					
010 11 00 00		SEARCH AND RESCUE (SAR)										
010 11 01 00		Essential SAR definitions										
010 11 01 01		Essential SAR definitions — ICAO Annex 12										
(01)		Recall the definitions of the following terms: alert phase, distress phase, emergency phase, operator, PIC, rescue coordination centre, State of Registry, uncertainty phase.	X	X	X	X	X					
010 11 02 00		SAR — Organisation										
010 11 02 01		SAR — Organisation — Establishment and provision										
(01)		Describe how ICAO Contracting States shall arrange for the establishment and prompt provision of SAR services.	X	X	X	X	X					
(02)		Explain the establishment of SAR by Contracting States.	X	X	X	X	X					
(03)		Describe the areas within which SAR services shall be established by Contracting States.	X	X	X	X	X					
(04)		State the period of time per day within which SAR services shall be available.	X	X	X	X	X					
(05)		Describe for which areas rescue coordination centres shall be established.	X	X	X	X	X					
010 11 03 00		Operating procedures for non-SAR crews										
010 11 03 01		Operating procedures for non-SAR crews — PIC										
(01)		Explain the SAR operating procedures for the PIC who arrives first at the scene of an accident.	X	X	X	X	X					
(02)		Explain the SAR operating procedures for the PIC intercepting a distress transmission.	X	X	X	X	X					
010 11 04 00		Search and rescue signals										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 11 04 01		Search and rescue signals — Survivors										
(01)		Explain the ‘ground–air visual signal code’ for use by survivors.	X	X	X	X	X					
(02)		Recognise the SAR ‘air-to-ground signals’ for use by survivors.	X	X	X	X	X					
010 12 00 00		SECURITY — Safeguarding International Civil Aviation against Acts of Unlawful Interference (ICAO Annex 17)										
010 12 01 00		Definitions of ICAO Annex 17										
010 12 01 01		Essential definitions of ICAO Annex 17										
(01)		Recall the definitions of the following terms: — airside, aircraft security check, screening, security, security control, security-restricted area, unidentified baggage.	X	X	X	X	X					
010 12 02 00		General principles										
010 12 02 01		General principles — Objectives of security										
(01)		State the objectives of security.	X	X	X	X	X					
010 12 03 00		Intentionally left blank										
010 12 04 00		Preventive security measures										
010 12 04 01		Preventive security measures										
(01)		Describe the objects not allowed (for reasons of aviation security) on board an aircraft that is engaged in international civil aviation.	X	X	X	X	X					
(02)		State what each Contracting State is supposed to do if passengers subjected to security control have mixed after a security screening point.	X	X	X	X	X					
(03)		Explain what has to be done when passengers who are obliged to travel because of judicial or	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		administrative proceedings are supposed to board an aircraft.										
(04)		Explain what has to be considered if law enforcement officers carry weapons on board.	X	X	X	X	X					
010 12 05 00		Management of response to acts of unlawful interference										
010 12 05 01		Management of response to acts of unlawful interference										
(01)		Describe the assistance each Contracting State shall provide to an aircraft subjected to an act of unlawful seizure.	X	X	X	X	X					
(02)		State the circumstances which could prevent a Contracting State from detaining an aircraft on the ground after being subjected to an act of unlawful seizure.	X	X	X	X	X					
010 12 06 00		Operators' security programme										
010 12 06 01		Operators' security programme — Principles										
(01)		Describe the principles of the written operator's security programme each Contracting State requires from operators.	X	X	X	X	X					
010 12 07 00		Security procedures in other documents, i.e. ICAO Annexes 2, 6 and 14, ICAO Doc 4444, Regulation (EU) No 965/2012 and CS-ADR-DSN										
010 12 07 01		ICAO Annex 2 — Rules of the Air, including Attachment B — Unlawful interference										
(01)		Describe what the PIC should do, in a situation of unlawful interference, unless considerations aboard the aircraft dictate otherwise.	X	X	X	X	X					
(02)		Describe what the PIC, of an aircraft subjected to unlawful interference, should do if:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- the aircraft must depart from its assigned track; - the aircraft must depart from its assigned cruising level; - the aircraft is unable to notify an ATS unit of the unlawful interference.										
(03)		Describe what the PIC should attempt to do with regard to broadcast warnings and the level at which to proceed, in a situation of unlawful interference, if no applicable regional procedures for in-flight contingencies have been established.	X	X	X	X	X					
010 12 07 02		ICAO Annex 6 — Operation of Aircraft — Security										
(01)		Describe the special considerations referring to flight crew compartment doors with regard to aviation security.	X	X	X	X	X					
010 12 07 03		ICAO Annex 14 Volume I — Aerodromes — Physical characteristics										
(01)		Describe what minimum distance an isolated aircraft parking position (after the aircraft has been subjected to unlawful interference) should have from other parking positions, buildings or public areas.	X	X	X	X	X					
010 12 07 04		ICAO Doc 4444 — Air Traffic Management										
(01)		Describe the considerations that must take place with regard to a taxi clearance in case an aircraft is known or believed to have been subjected to unlawful interference.	X	X	X	X	X					
010 13 00 00		AIRCRAFT ACCIDENT AND INCIDENT INVESTIGATION										
010 13 01 00		Essential definitions of ICAO Annex 13										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
010 13 01 01		Definitions and descriptions										
(01)		Recall the definitions of the following terms: — accident, aircraft, flight recorder, incident, investigation, maximum mass, operator, serious incident, serious injury, State of Design, State of Manufacture, State of Occurrence, State of the Operator, State of Registry.	X	X	X	X	X					
(02)		Explain the difference between ‘serious incident’ and ‘accident’.	X	X	X	X	X					
(03)		Determine whether a certain occurrence has to be defined as a serious incident or as an accident.	X	X	X	X	X					
(04)		Recognise the description of an accident or incident.	X	X	X	X	X					
010 13 02 00		Accident and incident investigation in ICAO Annex 13										
010 13 02 01		Objectives and procedures										
(01)		State the objective(s) of the investigation of an accident or incident according to ICAO Annex 13.	X	X	X	X	X					
(02)		Describe the general procedures for the investigation of an accident or incident according to ICAO Annex 13.	X	X	X	X	X					
010 13 03 00		Accident and incident investigation in EU regulations										
010 13 03 01		Occurrences										
(01)		Identify an occurrence as being either an accident, incident or serious incident in Regulation (EU) No 996/2010 of the European Parliament and of the Council of 20 October 2010 on the investigation and prevention of accidents and incidents in civil aviation.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the relationship between Regulation (EU) No 996/2010 of the European Parliament and of the Council of 20 October 2010 on the investigation and prevention of accidents and incidents in civil aviation and Regulation (EU) No 376/2014 of the European Parliament and of the Council of 3 April 2014 on the reporting, analysis and follow-up of occurrences in civil aviation.	X	X	X	X	X					
(03)		State the subject matter and scope of Regulation (EU) No 376/2014.	X	X	X	X	X					
(04)		Identify occurrences that must be reported.	X	X	X	X	X					
(05)		Identify occurrences that should be voluntarily reported.	X	X	X	X	X					
(06)		Describe how information from occurrences is collected, stored and analysed.	X	X	X	X	X					

SUBJECT 021 – AIRCRAFT GENERAL KNOWLEDGE – AIRFRAME, SYSTEMS AND POWER PLANT

ED Decision 2020/018/R

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
020 00 00 00		AIRCRAFT GENERAL KNOWLEDGE										
021 00 00 00		AIRCRAFT GENERAL KNOWLEDGE — AIRFRAME, SYSTEMS AND POWER PLANT										
021 01 00 00		SYSTEM DESIGN, LOADS, STRESSES, MAINTENANCE										
021 01 01 00		System design										
021 01 01 01		Design concepts										
(01)	X	Describe the following structural design philosophy: — safe life; — fail-safe (multiple load paths); — damage-tolerant.	X	X	X	X	X					
(02)		Explain the purpose of redundancy in aircraft design.	X	X	X	X	X					
021 01 01 02		Level of certification										
(01)	X	Explain why some systems are duplicated or triplicated.	X	X	X	X	X					
(02)	X	Explain that all aircraft are certified according to specifications determined by the competent authority, and that these certification specifications cover aspects such as design, material quality and build quality.	X	X	X	X	X					
(03)	X	State that the certification specifications for aeroplanes issued by EASA are: — CS-23 for Normal, Utility, Aerobatic and Commuter Aeroplanes; — CS-25 for Large Aeroplanes.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)	X	State that the certification specifications for rotorcraft issued by EASA are: — CS-27 for Small Rotorcraft; — CS-29 for Large Rotorcraft.			X	X	X					
021 01 02 00		Loads and stresses										
021 01 02 01		Stress, strain and loads										
(01)		Explain how stress and strain are always present in an aircraft structure both when parked and during manoeuvring. <i>Remark: Stress is the internal force per unit area inside a structural part as a result of external loads. Strain is the deformation caused by the action of stress on a material.</i>	X	X	X	X	X					
(02)		Describe the following types of loads that an aircraft may be subjected to, when they occur, and how a pilot may affect their magnitude: — static loads; — dynamic loads; — cyclic loads.	X	X	X	X	X					
(03)		Describe the areas typically prone to stress that should be given particular attention during a pre-flight inspection, and highlight the limited visual cues of any deformation that may be evident.	X	X	X	X	X					
021 01 03 00		Fatigue and corrosion										
021 01 03 01		Describe and explain fatigue and corrosion										
(01)		Describe the effects of corrosion and how it can be visually identified by a pilot during the pre-flight inspection.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the operating environments where the risk of corrosion is increased and how to minimise the effects of the environmental factors.	X	X	X	X	X					
(03)		Explain that aircraft have highly corrosive fluids on board as part of their systems and equipment.	X	X	X	X	X					
(04)		Explain fatigue, how it affects the useful life of an aircraft, and the effect of the following factors on the development of fatigue: — corrosion; — number of cycles; — type of flight manoeuvres; — stress level; — level and quality of maintenance.	X	X	X	X	X					
021 01 04 00		Intentionally left blank										
021 01 05 00		Maintenance										
021 01 05 01		Maintenance methods: hard-time and on-condition monitoring										
(01)		Explain the following terms: — hard-time or fixed-time maintenance; — on-condition maintenance; — condition monitoring.	X	X	X	X	X					
021 02 00 00		AIRFRAME										
021 02 01 00		Attachment methods										
021 02 01 01		Attachment methods and detecting the development of faulty attachments										
(01)		Describe the following attachment methods used for aircraft parts and components:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— riveting; — welding; — bolting; — pinning; — adhesives (bonding); — screwing.										
(02)		Explain how the development of a faulty attachment between aircraft parts or components can be detected by a pilot during the pre-flight inspection.	X	X	X	X	X					
021 02 02 00		Materials										
021 02 02 01		Composite and other materials										
(01)	X	Explain the principle of a composite material, and give examples of typical non-metallic materials used on aircraft: — carbon; — glass; — Kevlar aramid; — resin or filler.	X	X	X	X	X					
(02)	X	State the advantages and disadvantages of composite materials compared with metal alloys by considering the following: — strength-to-weight ratio; — capability to tailor the strength to the direction of the load; — stiffness;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— electrical conductivity (lightning); — resistance to fatigue and corrosion; — resistance to cost; — discovering damage during a pre-flight inspection.										
(03)		State that several types of materials are used on aircraft and that they are chosen based on type of structure or component and the required/desired material properties.	X	X	X	X	X					
021 02 03 00		Aeroplane: wings, tail surfaces and control surfaces										
021 02 03 01		Design										
(01)		Describe the following types of design and explain their advantages and disadvantages: — high-mounted wing; — low-mounted wing; — low- or mid-set tailplane; — T-tail.	X	X								
021 02 03 02		Structural components										
(01)		Describe the function of the following structural components: — spar and its components (web and girder or cap); — rib; — stringer; — skin;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— torsion box.										
021 02 03 03		Loads, stresses and aeroelastic vibrations (flutter)										
(01)		Describe the vertical and horizontal loads on the ground and during normal flight.	X	X								
(02)		Describe the vertical and horizontal loads during asymmetric flight following an engine failure for a multi-engine aeroplane, and how a pilot may potentially overstress the structure during the failure scenario.	X	X								
(03)		Explain the principle of flutter and resonance for the wing and control surfaces.	X	X								
(04)		Explain the following countermeasures used to achieve stress relief and reduce resonance: — chord-wise and span-wise position of masses (e.g. engines, fuel, balance masses for wing and control balance masses); — torsional stiffness; — bending flexibility; — fuel-balancing procedures during flight (automatic or applied by the pilot).	X	X								
021 02 04 00		Fuselage, landing gear, doors, floor, windscreen and windows										
021 02 04 01		Construction, functions, loads										
(01)	X	Describe the following types of fuselage construction: — monocoque, — semi-monocoque.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the construction and the function of the following structural components of a fuselage: — frames; — bulkhead; — pressure bulkhead; — stiffeners, stringers, longerons; — skin, doublers; — floor suspension (crossbeams); — floor panels; — firewall.	X	X	X	X	X					
(03)		Describe the loads on the fuselage due to pressurisation.	X	X								
(04)		Describe the following loads on a main landing gear: — touch-down loads (vertical and horizontal); — taxi loads on bogie gear (turns).	X	X								
(05)		Describe the structural danger of a nose-wheel landing with respect to: — fuselage loads; — nose-wheel strut loads.	X	X								
(06)		Describe the structural danger of a tail strike with respect to: — fuselage and aft bulkhead damage (pressurisation).	X	X								
(07)		Describe the door and hatch construction for pressurised and unpressurised aeroplanes including:	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— door and frame (plug type); — hinge location; — locking mechanism.										
(08)	X	Explain the advantages and disadvantages of the following fuselage cross sections: — circular; — double bubble; — oval; — rectangular.	X	X								
(09)		Explain why flight-deck windows are constructed with different layers.	X	X								
(10)		Explain the function of window heating for structural purposes.	X	X								
(11)		Explain the implication of a direct-vision window (see CS 25.773(b)(3)).	X	X								
(12)		Explain the need for an eye-reference position.	X	X								
(13)		Explain the function of floor venting (blow-out panels).	X	X								
(14)		Describe the construction and fitting of sliding doors.			X	X	X					
021 02 05 00		Helicopter: structural aspects of flight controls										
021 02 05 01		Design and construction										
(01)		List the functions of flight controls.			X	X	X					
(02)		Explain why vertical and horizontal stabilisers may have different shapes and alignments.			X	X	X					
021 02 05 02		Structural components and materials										
(01)		Describe the fatigue life and methods of checking for serviceability of the components and materials of flight and control surfaces.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 02 05 03		Loads, stresses and aeroelastic vibrations										
(01)		Describe the dangers and stresses regarding safety and serviceability in flight when the manufacturer's design envelope is exceeded.			X	X	X					
(02)		Explain that blade tracking is important both to minimise vibration and to help ensure uniformity of flow through the disc.			X	X	X					
(03)		Describe the early indications and vibrations which are likely to be experienced when the main-rotor blades and tail rotor are out of balance or tracking, including the possible early indications due to possible fatigue and overload.			X	X	X					
(04)		Explain how a vibration harmonic can be set up in other components which can lead to their early failure.			X	X	X					
(05)		State the three planes of vibration measurement, i.e. vertical, lateral, fore and aft.			X	X	X					
021 02 06 00		Structural limitations										
021 02 06 01		Maximum structural masses										
(01)		Define and explain the following maximum structural masses: — maximum ramp mass; — maximum take-off mass; — maximum zero fuel mass; — maximum landing mass. Remark: These limitations may also be found in the relevant part of Subjects 031 'Mass and balance', 032 'Performance (aeroplane)' and 034 'Performance (helicopter)'.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain that airframe life is limited by fatigue, created by alternating stress and the number of load cycles.	X	X								
(03)		Explain the maximum structural masses: — maximum take-off mass.			X	X	X					
(04)		Explain that airframe life is limited by fatigue, created by load cycles.			X	X	X					
021 03 00 00		HYDRAULICS										
021 03 01 00		Hydromechanics: basic principles										
021 03 01 01		Concepts and basic principles										
(01)	X	Explain the concept and basic principles of hydromechanics including: — hydrostatic pressure; — Pascal’s law; — the relationship between pressure, force and area; — transmission of power: multiplication of force, decrease of displacement.	X	X	X	X	X					
021 03 02 00		Hydraulic systems										
021 03 02 01		Hydraulic fluids: types, characteristics, limitations										
(01)	X	List and explain the desirable properties of a hydraulic fluid with regard to: — thermal stability; — corrosiveness; — flashpoint and flammability; — volatility; — viscosity.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	State that hydraulic fluids are irritating to skin and eyes.	X	X	X	X	X					
(03)		List the two different types of hydraulic fluids: — synthetic; — mineral.	X	X	X	X	X					
(04)		State that different types of hydraulic fluids cannot be mixed.	X	X	X	X	X					
(05)	X	State that at the pressures being considered, hydraulic fluid is considered incompressible.	X	X	X	X	X					
021 03 02 02		System components: design, operation, degraded modes of operation, indications and warnings										
(01)		Explain the working principle of a hydraulic system.	X	X	X	X	X					
(02)		Describe the difference in the principle of operation between a constant pressure system and a system pressurised only on specific demand.	X	X	X	X	X					
(03)		State the differences in the principle of operation between a passive hydraulic system (without a pressure pump) and an active hydraulic system (with a pressure pump).	X	X	X	X	X					
(04)	X	List the main advantages and disadvantages of system actuation by hydraulic or purely mechanical means with respect to: — weight; — size; — force.	X	X	X	X	X					
(05)		List the main uses of hydraulic systems.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		State that hydraulic systems can be classified as either high pressure (typically 3000 psi or higher) or low pressure (typically up to 2000 psi).	X	X	X	X	X					
(07)		State that a high-pressure hydraulic system is typically operating at 3000 psi but on some aircraft a hydraulic pressure of 4000 to 5000 psi may also be used.	X	X	X	X	X					
(08)		Explain the working principle of a low-pressure (0–2000 psi) system.	X	X	X	X	X					
(09)		Explain the advantages and disadvantages of a high-pressure system over a low-pressure system.	X	X	X	X	X					
(10)		Describe the working principle and functions of pressure pumps including: — constant pressure pump (swash plate or cam plate); — pressure pump whose output is dependent on pump revolutions per minute (rpm) (gear type).	X	X	X	X	X					
(11)		Explain the following different sources of hydraulic pressure, their typical application and potential operational limitations: — manual; — engine gearbox; — electrical; — air (pneumatic and ram-air turbine); — hydraulic (power transfer unit) or reversible motor pumps; — accessory.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(12)		Explain the following different sources of hydraulic pressure, their typical application and potential operational limitations: — manual; — engine; — gearbox; — electrical.			X	X	X					
(13)		Describe the working principle and functions of the following hydraulic system components: — reservoir (pressurised and unpressurised); — accumulators; — case drain lines and fluid cooler return lines; — piston actuators (single- and double-acting); — hydraulic motors; — filters; — non-return (check) valves; — relief valves; — restrictor valves; — selector valves (linear and basic rotary selectors, two and four ports); — bypass valves; — shuttle valves; — fire shut-off valves;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— priority valves; — fuse valves; — pressure and return pipes.										
(14)		Explain the function of the demand pump installed on many transport aeroplanes.	X	X								
(15)		Explain how redundancy is obtained by giving examples.	X	X	X	X	X					
(16)		Interpret a typical hydraulic system schematic to the level of detail as found in an aircraft flight crew operating manual (FCOM).	X	X	X	X	X					
(17)		Explain the implication of a high system demand.	X	X	X	X	X					
(18)		List and describe the instruments and alerts for monitoring a hydraulic system.	X	X	X	X	X					
(19)		State the indications and explain the implications of the following malfunctions: — system leak or low level; — low pressure; — high temperature.	X	X	X	X	X					
021 04 00 00		LANDING GEAR, WHEELS, TYRES, BRAKES										
021 04 01 00		Landing gear										
021 04 01 01		Types										
(01)	X	Name, for an aeroplane, the following different landing-gear configurations: — nose wheel; — tail wheel.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	Name, for a helicopter, the following different landing-gear configurations: — nose wheel; — tail wheel; — skids.			X	X	X					
021 04 01 02		System components, design, operation, indications and warnings, on-ground/in-flight protections, emergency extension systems										
(01)		Explain the function of the following components of a landing gear: — oleo leg/shock strut; — axles; — bogies and bogie beam; — drag struts; — side stays/struts; — torsion links; — locks (over centre); — gear doors.	X	X								
(02)		Explain the function of the following components of a landing gear: — oleo leg/shock strut; — axles; — drag struts; — side stays/struts;			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— torsion links; — locks (over centre); — gear doors.										
(03)		Name the different components of a landing gear, using a diagram.	X	X	X	X	X					
(04)		Describe the sequence of events during normal operation of the landing gear.	X	X	X	X	X					
(05)		State how landing-gear position indication and alerting is implemented.	X	X	X	X	X					
(06)		Describe the various protection devices to avoid inadvertent gear retraction on the ground and explain the implications of taking off with one or more protection devices in place: — ground lock (pins); — protection devices in the gear retraction mechanism.	X	X	X	X	X					
(07)		Explain the speed limitations for gear operation (VLO (maximum landing gear operating speed) and VLE (maximum landing gear extended speed)).	X	X	X	X	X					
(08)		Describe the sequence for emergency gear extension: — unlocking; — operating; — down-locking.	X	X	X	X	X					
(09)		Describe some methods for emergency gear extension including: — gravity/free fall;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- air or nitrogen pressure; - manually/mechanically.										
021 04 02 00		Nose-wheel steering										
021 04 02 01		Design, operation										
(01)		Explain the operating principle of nose-wheel steering.	X	X	X	X	X					
(02)		Explain, for a helicopter, the functioning of differential braking with free-castoring nose wheel.			X	X	X					
(03)		Describe, for an aeroplane, the functioning of the following systems: - differential braking with free-castoring nose wheel; - tiller or hand wheel steering; - rudder pedal nose-wheel steering.	X	X								
(04)		Explain the centring mechanism of the nose wheel.	X	X	X	X	X					
(05)		Define the term ‘shimmy’ and the possible consequences of shimmy for the nose- and the main-wheel system and explain the purpose of a shimmy damper to reduce the severity of shimmy.	X	X								
(06)		Explain the purpose of main-wheel (body) steering.	X	X								
021 04 03 00		Brakes										
021 04 03 01		Types and materials										
(01)		Describe the basic operating principle of a disc brake.	X	X	X	X	X					
(02)		State the different materials used in a disc brake (steel, carbon).	X	X	X	X	X					
(03)		Describe the characteristics, advantages and disadvantages of steel and carbon brake discs with regard to:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— weight; — temperature limits; — internal-friction coefficient; — wear.										
021 04 03 02		System components, design, operation, indications and warnings										
(01)		Explain the limitation of brake energy and describe the operational consequences.	X	X								
(02)		Explain how brakes are actuated: hydraulically, electrically.	X	X	X	X	X					
(03)		Explain the purpose of an in-flight wheel brake system.	X	X								
(04)		Describe the function of a brake accumulator.	X	X	X	X	X					
(05)		Describe the function of the parking brake.	X	X	X	X	X					
(06)		Explain the function of brake-wear indicators.	X	X								
(07)		Explain the reason for the brake-temperature indicator.	X	X								
021 04 03 03		Anti-skid										
(01)		Describe the operating principle of anti-skid where excessive brake pressure applied is automatically reduced for optimum braking performance.	X	X								
(02)		Explain that the anti-skid computer compares wheel speed to aeroplane reference speed to provide the following: — slip ratio for maximum braking performance; — locked-wheel prevention (protection against deep skid on one wheel);	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— touchdown protection (protection against brake-pressure application during touchdown); — hydroplane protection.										
(03)		Give examples of the impact of an anti-skid system on performance, and explain the implications of anti-skid system failure.	X	X								
021 04 03 04		Autobrake										
(01)		Describe the operating principle of an autobrake system.	X	X								
(02)		Explain why the anti-skid system must be available when using autobrakes.	X	X								
(03)		Explain the difference between the three modes of operation of an autobrake system: — OFF (system off or reset); — Armed (the system is ready to operate under certain conditions); — Activated/Deactivated (application of pressure on brakes).	X	X								
(04)		Describe how an autobrake system setting will either apply maximum braking (RTO or MAX) or result in a given rate of deceleration, where the amount of braking applied may be affected by: — the use of reverse thrust; — slippery runway.	X	X								
021 04 04 00		Wheels, rims and tyres										
021 04 04 01		Types, structural components and materials, operational limitations, thermal plugs										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Describe the different types of tyres such as: — tubeless; — diagonal (cross ply); — radial (circumferential bias).	X	X	X	X	X					
(02)	X	Define the following terms: — ply rating; — tyre tread; — tyre creep; — retread (cover).	X	X	X	X	X					
(03)		Explain the function of thermal/fusible plugs.	X	X								
(04)		Explain the implications of and how to identify tread separation and wear or damage with associated increased risk of tyre burst.	X	X								
(05)		Explain why the ground speed of tyres is limited.	X	X								
(06)		Describe the following tyre checks a pilot will perform during the pre-flight inspection and identify probable causes: — cuts and damages; — flat spots.	X	X								
021 04 05 00		Helicopter equipment										
021 04 05 01		Flotation devices										
(01)		Explain flotation devices, how they are operated, and their limitations.			X	X	X					
(02)		Explain why indicated airspeed (IAS) limitations before, during and after flotation-device deployment must be observed.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 05 00 00		FLIGHT CONTROLS										
021 05 01 00		Aeroplane: primary flight controls										
021 05 01 01		Definition and control surfaces										
(01)		Define a ‘primary flight control’.	X	X								
(02)		List the following primary flight control surfaces: — elevator; — aileron, roll spoilers, flaperon; — rudder.	X	X								
(03)		List the various means of control surface actuation including: — manual; — fully powered (irreversible); — partially powered (reversible).	X	X								
021 05 01 02		Manual controls										
(01)		Explain the basic principle of a fully manual control system.	X	X								
021 05 01 03		Fully powered controls (irreversible)										
(01)		Explain the basic principle of a fully powered control system.	X	X								
(02)		Explain the concept of irreversibility in a flight control system.	X	X								
(03)		Explain the need for a ‘feel system’ in a fully powered control system.	X	X								
(04)		Explain the operating principle of a stabiliser trim system in a fully powered control system.	X	X								
(05)		Explain the operating principle of rudder and aileron trim in a fully powered control system.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 05 01 04		<i>Partially powered controls (reversible)</i>										
(01)		Explain the basic principle of a partially powered control system.	X	X								
(02)		Explain why a 'feel system' is not necessary in a partially powered control system.	X	X								
021 05 01 05		<i>System components, design, operation, indications and warnings, degraded modes of operation, jamming</i>										
(01)		List and describe the function of the following components of a flight control system: — actuators; — control valves; — cables; — electrical wiring; — control surface position sensors.	X	X								
(02)		Explain how redundancy is obtained in primary flight control systems of large transport aeroplanes.	X	X								
(03)		Explain the danger of control jamming and the means of retaining sufficient control capability.	X	X								
(04)		Explain the methods of locking the controls on the ground and describe 'gust or control lock' warnings.	X	X								
(05)		Explain the concept of a rudder deflection limitation (rudder limiter) system and the various means of implementation (rudder ratio changer, variable stops, blow-back).	X	X								
021 05 02 00		<i>Aeroplane: secondary flight controls</i>										
021 05 02 01		<i>System components, design, operation, degraded modes of operation, indications and warnings</i>										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define a 'secondary flight control'.	X	X								
(02)		List the following secondary flight control surfaces: — lift-augmentation devices (flaps and slats); — speed brakes; — flight and ground spoilers; — trimming devices such as trim tabs, trimmable horizontal stabiliser.	X	X								
(03)		Describe secondary flight control actuation methods and sources of actuating power.	X	X								
(04)		Explain the function of a mechanical lock when using hydraulic motors driving a screw jack.	X	X								
(05)		Describe the requirement for limiting flight speeds for the various secondary flight control surfaces.	X	X								
(06)		For lift-augmentation devices, explain the load-limiting (relief) protection devices and the functioning of an auto-retraction system.	X	X								
(07)		Explain how a flap/slat asymmetry protection device functions, and describe the implications of a flap/slat asymmetry situation.	X	X								
(08)		Describe the function of an auto-slat system.	X	X								
(09)		Explain the concept of control surface blow-back (aerodynamic forces overruling hydraulic forces).	X	X								
021 05 03 00		Helicopter: flight controls										
021 05 03 01		Droop stops, control systems, trim systems, control stops										
(01)		Explain the methods of locking the controls on the ground.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe main-rotor droop stops and how rotor flapping is restricted.			X	X	X					
(03)		Explain the principle of phase lag and advance angle.			X	X	X					
(04)		Describe the following four axes of control operation, their operating principle and their associated cockpit controls: — collective control; — cyclic fore and aft (pitch axis); — cyclic lateral (roll axis); — yaw.			X	X	X					
(05)		Describe the swash plate or azimuth star control system including the following: — swash plate inputs; — the function of the non-rotating swash plate; — the function of the rotating swash plate; — how swash plate tilt is achieved; — swash plate pitch axis; — swash plate roll axis; — balancing of pitch/roll/collective inputs to the swash plate to equalise torsional loads on the blades.			X	X	X					
(06)		Describe the operation of the spider control system.			X	X	X					
(07)		State the need for artificial feel in a hydraulically actuated flight control system.			X	X	X					
(08)		Describe and explain the purpose of a trim system using the following terms:			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— force-trim switch; — force gradient; — parallel trim actuator; — cyclic 4-way trim switch; — interaction of trim system with an SAS/SCAS/ASS stability system; — trim-motor indicators.										
(09)		Describe the different types of control runs.			X	X	X					
(10)		Explain the use of control stops.			X	X	X					
021 05 04 00		Aeroplane: fly-by-wire (FBW) control systems										
021 05 04 01		Composition, explanation of operation, modes of operation										
(01)		Explain that an FBW flight control system is composed of the following: — pilot's input command (control column/sidestick/rudder pedals); — electrical signalling paths, including: — pilot input to computer; — computer to flight control surfaces; — feedback from aircraft response to computer; — flight control computers; — actuators; — flight control surfaces.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		State the advantages of an FBW system in comparison with a conventional flight control system including: — weight; — pilot workload; — flight-envelope protection.	X	X								
(03)		Explain why an FBW system is always irreversible.	X	X								
(04)		Explain the different modes of operation: — normal operation (e.g. normal law or normal mode); — downgraded operation (e.g. alternate law or secondary mode); — direct law.	X	X								
(05)		Describe the implications of mode degradation in relation to pilot workload and flight-envelope protection.	X	X								
(06)		— <i>Intentionally left blank</i>										
(07)		For aircraft using sidestick for manual control, describe the implications of: — dual control input made by the pilot; — the control takeover facility available to the pilot.	X	X								
(08)		<i>Intentionally left blank</i>										
(09)		Explain why several types of computers are needed and why they should be dissimilar.	X	X								
(10)		Explain why several control surfaces on every axis are needed on FBW aircraft.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(11)		Explain why several sensors are needed on critical parameters.	X	X								
021 05 05 00		Helicopter: fly-by-wire (FBW) control systems										
		To be introduced at a later date.			X	X	X					
021 06 00 00		PNEUMATICS — PRESSURISATION AND AIR-CONDITIONING SYSTEMS										
021 06 01 00		Pneumatic/bleed-air supply										
021 06 01 01		Piston-engine air supply										
(01)		Describe the following means of supplying air for the pneumatic systems for piston-engine aircraft: — compressor; — vacuum pump.	X	X	X	X	X					
(02)		State that an air supply is required for the following systems: — instrumentation; — heating; — de-icing.	X	X	X	X	X					
021 06 01 02		Gas turbine engine: bleed-air supply										
(01)		State that the possible bleed-air sources for gas turbine engine aircraft are the following: — engine; — auxiliary power unit (APU); — ground supply.	X	X	X	X	X					
(02)		State that for an aeroplane a bleed-air supply can be used for the following systems or components: — ice protection;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— engine air starter; — pressurisation of a hydraulic reservoir; — air-driven hydraulic pumps; — pressurisation and air conditioning.										
(03)		State that for a helicopter a bleed-air supply can be used for the following systems or components: — anti-icing; — engine air starter; — pressurisation of a hydraulic reservoir.			X	X	X					
(04)		State that the bleed-air supply system can comprise the following: — pneumatic ducts; — isolation valve; — pressure-regulating valve; — engine bleed valve (HP/IP valves); — fan-air pre-cooler; — temperature and pressure sensors.	X	X	X	X	X					
(05)		Interpret a basic pneumatic system schematic to the level of detail as found in an FCOM.	X	X	X	X	X					
(06)		Describe the cockpit indications for bleed-air systems.	X	X	X	X	X					
(07)		Explain how the bleed-air supply system is controlled and monitored.	X	X	X	X	X					
(08)		State the following bleed-air malfunctions: — over-temperature;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— over-pressure; — low pressure; — overheat/duct leak; and describe the potential consequences.										
021 06 02 00		Helicopter: air-conditioning systems										
021 06 02 01		Types, system components, design, operation, degraded modes of operation, indications and warnings										
(01)		Describe the purpose of an air-conditioning system.			X	X	X					
(02)		Explain how an air-conditioning system is controlled.			X	X	X					
(03)		Describe the vapour cycle air-conditioning system including system components, design, operation, degraded modes of operation and system malfunction indications.			X	X	X					
(04)		Identify the following components from a diagram of an air-conditioning system and describe the operating principle and function: — air-cycle machine (pack, bootstrap system); — pack-cooling fan; — water separator; — mixing valves; — flow-control valves; — isolation valves; — recirculation fans; — filters for recirculation;			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— temperature sensors.										
(05)		List and describe the controls, indications and warnings related to an air-conditioning system.			X	X	X					
021 06 03 00		Aeroplane: pressurisation and air-conditioning system										
021 06 03 01		System components, design, operation, degraded modes of operation, indications and warnings										
(01)		Explain that a pressurisation and an air-conditioning system of an aeroplane controls: — ventilation; — temperature; — pressure.	X	X								
(02)		Explain how humidity is controlled.	X	X								
(03)		Explain that the following components constitute a pressurisation system: — pneumatic system as the power source; — outflow valve; — outflow valve actuator; — pressure controller; — excessive differential pressure-relief valve; — negative differential pressure-relief valve.	X	X								
(04)		Explain that the following components constitute an air-conditioning system and describe their operating principles and function: — air-cycle machine (pack, bootstrap system);	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— pack-cooling fan; — water separator; — mixing valves; — flow-control valves (outflow valve); — isolation valves; — ram-air valve; — recirculation fans; — filters for recirculated air; — temperature sensors. <i>Remark: The bootstrap system is the only air-conditioning system considered for Part-FCL aeroplane examinations.</i>										
(05)		Describe the use of hot trim air.	X	X								
(06)		Define the following terms: — cabin altitude; — cabin vertical speed; — differential pressure; — ground pressurisation.	X	X								
(07)		Describe the operating principle of a pressurisation system.	X	X								
(08)		Describe the emergency operation by manual setting of the outflow valve position.	X	X								
(09)		Describe the working principle of an electronic cabin-pressure controller.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(10)		State how the maximum operating altitude is determined.	X	X								
(11)		Explain: — why the maximum allowed value of cabin altitude is limited; — a typical value of maximum differential pressure for large transport aeroplanes; — the relation between cabin altitude, the maximum differential pressure and maximum aeroplane operating altitude.	X	X								
(12)		Explain the typical warning on a transport category aircraft when cabin altitude exceeds 10 000 ft.	X	X								
(13)		List and interpret typical indications of the pressurisation system.	X	X								
(14)		Describe the main operational differences between a bleed-air-driven air-conditioning system and an electrically driven air-conditioning system as found on aircraft without engine bleed-air system.	X	X								
021 07 00 00		ANTI-ICING AND DE-ICING SYSTEMS										
021 07 01 00		Types, operation, indications										
021 07 01 01		Types, design, operation, indications and warnings, operational limitations										
(01)		Explain the concepts of anti-icing and de-icing.	X	X	X	X	X					
(02)		Name the components of an aircraft which can be protected from ice accretion.	X	X	X	X	X					
(03)		State that on some aeroplanes the tail does not have an ice-protection system.	X	X								
(04)		State the different types of anti-icing/de-icing systems and describe their operating principle:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- hot air; - electrical; - fluid.										
(05)		Describe the operating principle of the inflatable boot de-icing system.	X	X								
021 07 02 00		Ice warning systems										
021 07 02 01		Types, operation, and indications										
(01)		Describe the different operating principles of the following ice detectors: - mechanical systems using air pressure; - electromechanical systems using resonance frequencies.	X	X								
(02)		Describe the principle of operation of ice warning systems.	X	X								
021 07 03 00		Helicopter blade heating systems										
021 07 03 01		Limitations										
(01)		Explain the limitations on blade heating and the fact that on some helicopters the heating does not heat all the main-rotor blades at the same time.			X	X	X					
021 08 00 00		FUEL SYSTEM										
021 08 01 00		Piston engine										
021 08 01 01		Fuel: types, characteristics, limitations										
(01)		State the types of fuel used by a piston engine and their associated limitations: - diesel; - JET-A1 (for high-compression engines);	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— AVGAS; — MOGAS.										
(02)		State the main characteristics of these fuels and give typical values regarding their flash points, freezing points and density.	X	X	X	X	X					
021 08 01 02		Design, operation, system components, indications										
(01)		State the tasks of the fuel system.	X	X	X	X	X					
(02)		Name the following main components of a fuel system, and state their location and their function: — lines; — boost pump; — pressure valves; — filter, strainer; — tanks (wing, tip, fuselage); — vent system; — sump; — drain; — fuel-quantity sensor; — fuel-temperature sensor.	X	X	X	X	X					
(03)		Describe a gravity fuel feed system and a pressure feed fuel system.	X	X	X	X	X					
(04)		Describe the construction of the different types of fuel tanks and state their advantages and disadvantages: — drum tank;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— bladder tank; — integral tank.										
(05)		Explain the function of cross-feed.	X	X	X	X	X					
(06)		Define the term ‘unusable fuel’.	X	X	X	X	X					
(07)		List the following parameters that are monitored for the fuel system: — fuel quantity (low-level warning); — fuel temperature.	X	X	X	X	X					
021 08 02 00		Turbine engine										
021 08 02 01		Fuel: types, characteristics, limitations										
(01)		State the types of fuel used by a gas turbine engine: — JET-A; — JET-A1; — JET-B.	X	X	X	X	X					
(02)		State the main characteristics of these fuels and give typical values regarding their flash points, freezing points and density.	X	X	X	X	X					
(03)		State the existence of additives for freezing.	X	X	X	X	X					
021 08 02 02		Design, operation, system components, indications										
(01)		Explain the function of the fuel system: — lines; — centrifugal boost pump; — pressure valves; — fuel shut-off valve; — filter, strainer;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- tanks (wing, tip, fuselage, tail); - baffles/baffles; - sump; - vent system; - drain; - fuel-quantity sensor; - fuel-temperature sensor; - refuelling/defueling system; - fuel dump/jettison system.										
(02)		Name the main components of the fuel system and state their location and their function: - trim fuel tanks; - baffles; - refuelling/defueling system; - fuel dump/jettison system. <i>Remark: For completion of list, please see 021 08 01 02 (02).</i>	X	X	X	X	X					
(03)		Interpret a typical fuel system schematic to the level of detail as found in an aircraft FCOM.	X	X	X	X	X					
(04)		Explain the limitations in the event of loss of booster pump fuel pressure.	X	X	X	X	X					
(05)		Describe the use and purpose of drip sticks (manual magnetic indicators) (may also be known as dip stick or drop stick).	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the considerations for fitting a fuel dump/jettison system and, if fitted, its function.	X	X	X	X	X					
021 09 00 00		ELECTRICS										
		<i>Remark: For any reference to the direction of current flow, the conventional current flow shall be used, i.e. from positive to negative.</i>										
021 09 01 00		General, definitions, basic applications: circuit breakers, logic circuits										
021 09 01 01		Static electricity										
(01)		Explain static electricity and describe the flying conditions where aircraft are most susceptible to build-up of static electricity.	X	X	X	X	X					
(02)		Describe a static discharger and explain the following: — its purpose; — typical locations; — pilot's role of observing it during pre-flight inspection.	X	X	X	X	X					
(03)		Explain why an aircraft must first be grounded before refuelling/defueling.	X	X	X	X	X					
(04)		Explain the reason for electrical bonding.	X	X	X	X	X					
021 09 01 02		Direct current (DC)										
(01)		Explain the term 'direct current' (DC), and state that current can only flow in a closed circuit.	X	X	X	X	X					
(02)	X	Explain the basic principles of conductivity and give examples of conductors, semiconductors and insulators.	X	X	X	X	X					
(03)		Describe the difference in use of the following mechanical switches and explain the difference in	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		observing their state (e.g. ON/OFF), and why some switches are guarded: — toggle switch; — rocker switch; — pushbutton switch; — rotary switch. Explain the difference in observing their state (e.g. ON/OFF) and why some switches are guarded.										
(04)		Define voltage and current, and state their unit of measurement.	X	X	X	X	X					
(05)	X	Explain Ohm's law in qualitative terms.	X	X	X	X	X					
(06)	X	Explain the effect on total resistance when resistors are connected in series or in parallel.	X	X	X	X	X					
(07)	X	State that resistances can have a positive or a negative temperature coefficient (PTC/NTC) and state their use.	X	X	X	X	X					
(08)		Define electrical power and state the unit of measurement.	X	X	X	X	X					
021 09 01 03		Alternating current (AC)										
(01)	X	Explain the term 'alternating current' (AC), and compare its use to DC with regard to complexity.	X	X	X	X	X					
(02)		Define the term 'phase', and explain the basic principle of single-phase and three-phase AC.	X	X	X	X	X					
(03)		State that aircraft can use single-phase or three-phase AC.	X	X	X	X	X					
(04)		Define frequency and state the unit of measurement.	X	X	X	X	X					
(05)	X	Define 'phase shift' in qualitative terms.	X	X	X	X	X					
021 09 01 04		Intentionally left blank										
021 09 01 05		Intentionally left blank										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 09 01 06		Electromagnetism										
(01)		State that an electrical current produces a magnetic field.	X	X	X	X	X					
(02)		Describe how the strength of the magnetic field changes with the magnitude of the current.	X	X	X	X	X					
(03)		Explain the purpose and the working principle of a solenoid.	X	X	X	X	X					
(04)		Explain the purpose and the working principle of a relay.	X	X	X	X	X					
(05)		Explain the principle of electromagnetic induction and how two electrical components or systems may affect each other through this principle.	X	X	X	X	X					
021 09 01 07		Circuit protection										
(01)		Explain the working principle of a fuse and a circuit breaker.	X	X	X	X	X					
(02)		Explain how a fuse is rated.	X	X	X	X	X					
(03)		Describe the principal difference between the following types of circuit breakers: — thermal circuit breaker sensing magnitude of current; — magnetic circuit breaker sensing direction of current.	X	X	X	X	X					
(04)		Describe how circuit breakers may be used to reset aircraft systems/computers in the event of system failure (when part of a described procedure).	X	X	X	X	X					
(05)		Explain a short circuit in practical terms using Ohm's Law, power and energy expressions highlighting the risk of fire due to power transfer and extreme energy dissipation.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the risk of fire resulting from excessive heat in a circuit subjected to overcurrent.	X	X	X	X	X					
(07)		Explain that overcurrent situations may be transient.	X	X	X	X	X					
(08)		Explain the hazards of multiple resets of a circuit breaker or the use of incorrect fuse rating when replacing blown fuses.	X	X	X	X	X					
021 09 01 08		Semiconductors and logic circuits										
(01)		Describe the effect of temperature on semiconductors with regard to function and longevity of the component.	X	X	X	X	X					
(02)		Describe the following five basic logic functions, as used in aircraft FCOM documentation, and recognise their schematic symbols according to the ANSI/MIL standard: — AND; — OR; — NOT; — NOR; — NAND.	X	X	X	X	X					
(03)		Interpret a typical logic circuit schematic to the level of detail as found in an aircraft FCOM.	X	X	X	X	X					
021 09 02 00		Batteries										
021 09 02 01		Types, characteristics and limitations										
(01)		State the function of an aircraft battery.	X	X	X	X	X					
(02)		Name the types of rechargeable batteries used in aircraft: — lead-acid;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— nickel-cadmium; — lithium-ion; — lithium-polymer.										
(03)		Compare the different battery types with respect to: — load behaviour; — charging characteristics; — risk of thermal runaway.	X	X	X	X	X					
(04)		Explain the term ‘cell voltage’ and describe how a battery may consist of several cells that combined provide the desirable voltage and capacity.	X	X	X	X	X					
(05)		Explain the difference between battery voltage and charging voltage.	X	X	X	X	X					
(06)		Define the term ‘capacity of batteries’ and state the unit of measurement used.	X	X	X	X	X					
(07)		State the effect of temperature on battery capacity and performance.	X	X	X	X	X					
(08)		State that in the case of loss of all generated power (battery power only) the remaining electrical power is time-limited.	X	X	X	X	X					
(09)		Explain how lithium-type batteries pose a threat to aircraft safety and what affects this risk: — numbers of batteries on board an aircraft including those brought on board by passengers; — temperature, of both battery and environment; — physical condition of the battery; — battery charging.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(10)		Describe how to contain a battery thermal runaway highlighting the following: — how one cell can affect the neighbouring cells; — challenges if it happens in an aircraft during flight.	X	X	X	X	X					
021 09 03 00		Generation										
		Remark: For standardisation purposes, the following standard expressions are used: — DC generator: produces DC output; — DC alternator: produces AC, rectified by integrated rectifying unit, the output is DC; — DC alternator: producing a DC output by using a rectifier; — AC generator: produces AC output; — starter generator: integrated combination of a generator and a starter motor; — permanent magnet alternator/ generator: self-exciting AC generator.										
021 09 03 01		DC generation										
(01)		Describe the basic working principle of a simple DC generator or DC alternator.	X	X	X	X	X					
(02)		Explain the principle of voltage control and why it is required.	X	X	X	X	X					
(03)		Explain the purpose of reverse current protection from the battery/busbar to the alternator.	X	X	X	X	X					
(04)		Describe the basic operating principle of a starter generator and state its purpose.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 09 03 02		AC generation										
(01)		Describe the working principle of a brushless three-phase AC generator.	X	X	X	X	X					
(02)		State that the generator field current is used to control voltage.	X	X	X	X	X					
(03)		State the relationship between output frequency and the rpm of a three-phase AC generator.	X	X	X	X	X					
(04)		Explain the term 'frequency wild generator'.	X	X	X	X	X					
(05)		List the following different power sources that can be used for an aeroplane to drive an AC generator: — engine; — APU; — RAT; — hydraulic.	X	X								
(06)		List the following different power sources that can be used for a helicopter to drive an AC generator: — engine; — APU; — gearbox.			X	X	X					
021 09 03 03		Constant speed drive (CSD) and integrated drive generator (IDG) systems										
(01)		Describe the function of a CSD.	X	X								
(02)		Explain the parameters of a CSD that are monitored.	X	X								
(03)		Describe the function of an IDG.	X	X								
(04)		Explain the consequences of a mechanical disconnection during flight for a CSD and an IDG.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Explain that a CSD/IDG has its own, independent oil system and how a leak from this may appear as an engine oil leak.	X	X								
021 09 03 04		Transformers, transformer rectifier units (TRUs), static inverters										
(01)		State the function of a transformer.	X	X	X	X	X					
(02)		State the function of a TRU and its purpose, including type of output.	X	X	X	X	X					
(03)		State the function of a static inverter and its purpose, including type of output.	X	X	X	X	X					
021 09 04 00		Distribution										
021 09 04 01		General										
(01)		Explain the function of a busbar.	X	X	X	X	X					
(02)		Describe the function of the following buses: — AC bus; — DC bus; — emergency AC or DC bus; — essential AC or DC bus; — battery bus; — hot bus, ground servicing or maintenance bus.	X	X	X	X	X					
(03)		State that the aircraft structure can be used as a part of the electrical circuit (common earth) and explain the implications for electrical bonding.	X	X	X	X	X					
(04)		Explain the function of external power.	X	X	X	X	X					
(05)		State that a priority sequence exists between the different sources of electrical power on ground and in flight.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the term 'load sharing'.	X	X	X	X	X					
(07)		Explain the term 'load shedding'.	X	X	X	X	X					
(08)		Describe typical systems that can be shed in the event of a supply failure, such as passenger entertainment system and galley power.	X	X	X	X	X					
(09)		Interpret a typical electrical system schematic to the level of detail as found in an aircraft FCOM.	X	X	X	X	X					
(10)		Explain the difference between a supply (e.g. generator) failure and a bus failure, and the operating consequences of either.	X	X	X	X	X					
021 09 04 02		DC distribution										
(01)		Describe a simple DC electrical system of a single-engine aircraft.	X	X	X	X	X					
(02)		Describe a DC electrical system of a multi-engine aircraft (CS-23/CS-27) including the distribution consequences of loss of generator(s) or bus failure.	X	X	X	X	X					
(03)		Describe the DC part of an electrical system of a transport aircraft (CS-25/CS-29) including the distribution consequences of loss of DC supply or bus failure.	X	X	X	X	X					
(04)		Give examples of DC consumers.	X	X	X	X	X					
021 09 04 03		AC distribution										
(01)		Explain the difference in the principle of operation for a split AC electrical system and a parallel AC electrical system.	X	X	X	X	X					
(02)		Describe the following distribution consequences: — power transfer between different power supplies; — power transfer in the event of a supply failure;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— loss of all normal AC supplies.										
(03)		Give examples of AC consumers.	X	X	X	X	X					
(04)		Explain the conditions to be met for paralleling AC generators.	X	X	X	X	X					
(05)		State that volt-ampere (VA) is the unit for total power consumed in an AC system.	X	X	X	X	X					
021 09 04 04		Electrical load management and monitoring systems: automatic generators and bus switching during normal and failure operation, indications and warnings										
(01)		Give examples of system control, monitoring and annunciators using the following terms: — generator control unit (GCU) for monitoring generator output and providing network protection; — exciter contactor/breaker/relay for control of generator exciter field; — generator contactor/breaker/relay for connecting the generator to the network; — bus-tie contactor/breaker/relay for connecting busbars together; — generator switch on the flight deck for manual control of exciter contactor; — IDG/CSD disconnect switch on the flight deck for mechanical disconnection of the generator; — bus-tie switch on the flight deck with AUTO and OFF positions only.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe, for normal and degraded modes of operation, the following functions of an electrical load management system on ground and in flight using the terms in 021 09 04 04 (01): — distribution; — monitoring; — protection in the event of incorrect voltage; — protection in the event of incorrect frequency; — protection in the event of a differential fault.	X	X	X	X	X					
(03)		Describe the requirement for monitoring the aircraft batteries.	X	X	X	X	X					
(04)		Explain the importance of monitoring the temperature of nickel-cadmium and lithium-type batteries.	X	X	X	X	X					
(05)		Interpret various different ammeter indications of an ammeter which monitors the charge current of the battery.	X	X	X	X	X					
021 09 05 00		Electrical motors										
021 09 05 01		General										
(01)	X	State that the purpose of an electrical motor is to convert electrical energy into mechanical energy.	X	X	X	X	X					
(02)		State that because of the similarity in design, a generator and an electrical motor may be combined into a starter generator.	X	X	X	X	X					
(03)		Explain that the size of the engine determines how much energy is required for starting, and state the following:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- small turbine engines may be able to use the battery for a very limited number of start attempts; - large turbine engines require one or more power sources, either external or on-board.										
021 09 05 02		Operating principle										
(01)		Describe how the torque of an electrical motor is determined by the supplied voltage and current, and the resulting magnetic fields within the motor.	X	X	X	X	X					
(02)	X	State that electrical motors can be either AC or DC.	X	X	X	X	X					
(03)		Explain the consequences of the following: - rotor seizure; - rotor runaway.	X	X	X	X	X					
021 09 05 03		Components										
(01)	X	Name the following components of an electrical motor: - rotor (rotating part of an electrical motor); - stator (stationary part of an electrical motor).	X	X	X	X	X					
021 10 00 00		PISTON ENGINES										
		<i>Remark: This topic includes diesel and petrol engines.</i>										
021 10 01 00		General										
021 10 01 01		Types of internal-combustion engines: basic principles, definitions										
(01)		Define the following terms and expressions: - rpm; - torque;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— manifold absolute pressure (MAP); — power output; — specific fuel consumption; — compression ratio, clearance volume, swept (displaced) volume, total volume.										
021 10 01 02		Engine: design, operation, components										
(01)		Describe the basic operating principle of a piston engine: — crankcase; — crankshaft; — connecting rod; — piston; — piston pin; — piston rings; — cylinder; — cylinder head; — valves; — valve springs; — push rod; — camshaft; — rocker arm; — camshaft gear; — bearings.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Name and identify the various types of engine design with regard to cylinder arrangement and their advantages/disadvantages: — horizontally opposed; — in line; — radial; and — working cycle (four stroke: petrol and diesel).	X	X	X	X	X					
(03)		Describe the differences between petrol and diesel engines with respect to: — means of ignition; — maximum compression ratio; — regulating air or mixture supply to the cylinder; — pollution from the exhaust.	X	X	X	X	X					
021 10 02 00		Fuel										
021 10 02 01		Types, grades, characteristics, limitations										
(01)		Name the type of fuel used for petrol engines including its colour (AVGAS); — 100 (green); — 100LL (blue).	X	X	X	X	X					
(02)		Name the type of fuel normally used for aviation diesel engines (JET-A1).	X	X	X	X	X					
(03)	X	Define the term 'octane rating'.	X	X	X	X	X					
(04)		Define the term 'detonation' and describe the causes and effects of detonation for both petrol and diesel engines.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Define the term ‘pre-ignition’ and describe the causes and effects of pre-ignition for both petrol and diesel engines.	X	X	X	X	X					
(06)		Identify the conditions and power settings that promote detonation for petrol engines.	X	X	X	X	X					
(07)		Describe how detonation in petrol engines is recognised.	X	X	X	X	X					
(08)		Describe the method and occasions for checking the fuel for water content.	X	X	X	X	X					
(09)		State the typical value of fuel density for aviation gasoline and diesel fuel.	X	X	X	X	X					
(10)		Explain volatility, viscosity and vapour locking for petrol and diesel fuels.	X	X	X	X	X					
021 10 03 00		Engine fuel pumps										
021 10 03 01		Engine-driven fuel pump										
(01)		Explain the need for a separate engine-driven fuel pump.	X	X	X	X	X					
021 10 04 00		Carburettor/injection system										
021 10 04 01		Carburettor: design, operation, degraded modes of operation, indications and warnings										
(01)		State the purpose of a carburettor.	X	X	X	X	X					
(02)		Describe the operating principle of the simple float chamber carburettor.	X	X	X	X	X					
(03)		Describe the methods of obtaining mixture control over the whole operating engine power setting range (compensation jet, diffuser).	X	X	X	X	X					
(04)		Describe the methods of obtaining mixture control over the whole operating altitude range.	X	X	X	X	X					
(05)		Explain the purpose and the operating principle of an accelerator pump.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the purpose of power enrichment.	X	X	X	X	X					
(07)		Describe the function of the carburettor heat system.	X	X	X	X	X					
(08)		Explain the effect of carburettor heat on mixture ratio and power output.	X	X	X	X	X					
(09)		Explain the purpose and the operating principle of a primer pump.	X	X	X	X	X					
(10)		Discuss other methods for priming an engine (acceleration pumps).	X	X	X	X	X					
(11)		Explain the danger of carburettor fire, including corrective measures.	X	X	X	X	X					
021 10 04 02		<i>Injection: design, operation, degraded modes of operation, indications and warnings</i>										
(01)		Explain the advantages and difference in operation of an injection system compared with a carburettor system.	X	X	X	X	X					
021 10 04 03		<i>Icing</i>										
(01)		Describe the causes and effects of carburettor icing and the action to be taken if carburettor icing is suspected.	X	X	X	X	X					
(02)		Name the meteorological conditions under which carburettor icing may occur.	X	X	X	X	X					
(03)		Describe the indications of the presence of carburettor icing for both a fixed pitch and a constant speed propeller.	X	X								
(04)		Describe the indications of the presence of carburettor icing for a helicopter.			X	X	X					
(05)		Describe the indications that will occur upon selection of carburettor heat depending on whether ice is present or not.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the reason for the use of alternate air on fuel injection systems and describe its operating principle.	X	X	X	X	X					
(07)		State the meteorological conditions under which induction system icing may occur.	X	X	X	X	X					
021 10 05 00		Cooling systems										
021 10 05 01		Design, operation, indications and warnings										
(01)		Specify the reasons for cooling a piston engine.	X	X	X	X	X					
(02)		Describe the design features to enhance cylinder air cooling for aeroplanes.	X	X								
(03)		Describe the design features to enhance cylinder air cooling for helicopters (e.g. engine-driven impeller and scroll assembly, baffles).			X	X	X					
(04)		Compare the differences between liquid- and air-cooling systems.	X	X	X	X	X					
(05)		Identify the cylinder head temperature indication to monitor engine cooling.	X	X	X	X	X					
(06)		Describe the function and the operation of cowl flaps.	X	X								
021 10 06 00		Lubrication systems										
021 10 06 01		Lubricants: characteristics, limitations										
(01)		Describe the term 'viscosity' including the effect of temperature.	X	X	X	X	X					
(02)		Describe the viscosity grade numbering system used in aviation.	X	X	X	X	X					
021 10 06 02		Design, operation, indications and warnings										
(01)		State the functions of a piston-engine lubrication system.	X	X	X	X	X					
(02)		Describe the working principle of a dry-sump lubrication system and describe the functions of the following components:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- oil tank (reservoir) and its internal components: hot well, de-aerator, vent, expansion space; - check valve (non-return valve); - pressure pump and pressure-relief valve; - scavenge pump; - filters (suction, pressure and scavenge); - oil cooler; - oil cooler bypass valve (anti-surge and thermostatic); - pressure and temperature sensors; - lines.										
(03)		Describe a wet-sump lubrication system.	X	X	X	X	X					
(04)		State the differences between a wet- and a dry-sump lubrication system and their advantages and disadvantages.	X	X	X	X	X					
(05)		List the following factors that influence oil consumption: - oil grade; - cylinder and piston wear; - condition of piston rings.	X	X	X	X	X					
(06)		Describe the interaction between oil pressure, oil temperature and oil quantity.	X	X	X	X	X					
021 10 07 00		Ignition circuits										
021 10 07 01		Design, operation										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the working principle of a magneto-ignition system and the functions of the following components: — magneto; — contact-breaker points; — capacitor (condenser); — coils or windings; — ignition switches; — distributor; — spark plug; — high-tension (HT) cable.	X	X	X	X	X					
(02)		State why piston engines are equipped with two electrically independent ignition systems.	X	X	X	X	X					
(03)		State the function and operating principle of the following methods of spark augmentation: — starter vibrator (booster coil); — impulse-start coupling.	X	X								
(04)		State the function and operating principle of the following methods of spark augmentation: — starter vibrator (booster coil); — both magnetos live.			X	X	X					
(05)		Explain the function of the magneto check.	X	X	X	X	X					
(06)		Explain how combustion is initiated in diesel engines.	X	X	X	X	X					
021 10 08 00		Mixture										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 10 08 01		Definition, characteristic mixtures, control instruments, associated control levers, indications										
(01)		Define the following terms: — mixture; — chemically correct ratio (stoichiometric); — best power ratio; — lean (weak) mixture (lean or rich side of the exhaust gas temperature (EGT) top); — rich mixture.	X	X	X	X	X					
(02)		State the typical fuel-to-air ratio values or range of values for the above mixtures.	X	X	X	X	X					
(03)		Describe the advantages and disadvantages of weak and rich mixtures.	X	X	X	X	X					
(04)		Describe the relation between engine-specific fuel consumption and mixture ratio.	X	X	X	X	X					
(05)		Describe the use of the exhaust gas temperature as an aid to mixture-setting.	X	X	X	X	X					
(06)		Explain the relation between mixture ratio, cylinder head temperature, detonation and pre-ignition.	X	X	X	X	X					
(07)		Explain the absence of mixture control in diesel engines.	X	X	X	X	X					
021 10 09 00		Aeroplane: propellers										
021 10 09 01		Definitions, general										
		<i>Remark: Definitions and aerodynamic concepts are detailed in Subject 081 'Principles of flight (aeroplane)', Topic 07 (Propellers), but need to be appreciated for this Subject as well.</i>	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 10 09 02		Constant-speed propeller: design, operation, system components										
(01)		Describe the operating principle of a constant-speed propeller system under normal flight operations with the aid of a schematic.	X	X								
(02)		Explain the need for a MAP indicator to control the power setting with a constant-speed propeller.	X	X								
(03)		State the purpose of a torque-meter.	X	X								
(04)		State the purpose and describe the operation of a low-pitch stop (centrifugal latch).	X	X								
(05)		Describe the operating principle of a single-acting and a double-acting variable pitch propeller for single- and multi-engine aeroplanes.	X	X								
(06)		Describe the function and the basic operating principle of synchronising and synchro-phasing systems.	X	X								
(07)		Explain the purpose and the basic operating principle of an auto-feathering system and unfeathering.	X	X								
021 10 09 03		Reduction gearing: design										
(01)		State the purpose of reduction gearing.	X	X								
021 10 09 04		Propeller handling: associated control levers, degraded modes of operation, indications and warnings										
(01)		Describe the checks to be carried out on a constant-speed propeller system after engine start.	X	X								
(02)		Describe the operation of a constant-speed propeller system during flight at different true airspeeds (TAS) and rpm including an overspeeding propeller.	X	X								
(03)		Describe the operating principle of a variable pitch propeller when feathering and unfeathering, including the operation of cockpit controls.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Describe the operating principle of a variable pitch propeller when reverse pitch is selected, including the operation of cockpit controls.	X	X								
(05)		Describe the operation of the propeller levers during different phases of flight.	X	X								
021 10 10 00		Performance and engine handling										
021 10 10 01		Performance										
(01)		Describe the effect on power output of a petrol and diesel engine taking into consideration the following parameters: — ambient pressure, exhaust back pressure; — temperature; — density altitude; — humidity.	X	X	X	X	X					
(02)		Explain the term ‘normally aspirated engine’.	X	X	X	X	X					
(03)		Power-augmentation devices: explain the requirement for power augmentation (turbocharging) of a piston engine.	X	X	X	X	X					
(04)		Describe the function and the principle of operation of the following main components of a turbocharger: — turbine; — compressor; — waste gate; — waste-gate actuator.	X	X	X	X	X					
(05)		Explain the difference between an altitude-boosted turbocharger and a ground-boosted turbocharger.	X	X	X	X	X					
(06)		Explain turbo lag.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Define the term ‘critical altitude’.	X	X	X	X	X					
(08)		Explain the function of an intercooler.	X	X	X	X	X					
(09)		Define the terms ‘full-throttle height’ and ‘rated altitude’.	X	X	X	X	X					
(10)		Explain the purpose of a supercharger and the basic differences from a turbocharger.	X	X	X	X	X					
021 10 10 02		Engine handling										
(01)		State the correct procedures for setting the engine controls when increasing or decreasing power.	X	X	X	X	X					
(02)		Define the following terms: — take-off power; — maximum continuous power.	X	X	X	X	X					
(03)		Describe the start problems associated with extreme cold weather.	X	X	X	X	X					
(04)		Describe the principal difference between a full-authority digital engine control (FADEC) system-controlled engine and traditional manual engine controls.	X	X	X	X	X					
(05)		Describe the engine controls available on the flight deck for a FADEC-controlled engine.	X	X	X	X	X					
(06)		Explain that the FADEC has full authority of the control of all engine parameters ensuring efficient and correct running of the engine, including protection in the event of failure.	X	X	X	X	X					
(07)		Explain the need for FADEC redundancy with regard to power supply and data input and output.	X	X	X	X	X					
021 11 00 00		TURBINE ENGINES										
021 11 01 00		Basic principles										
021 11 01 01		Basic generation of thrust and the thrust formula										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe how thrust is produced by a basic gas turbine engine.	X	X								
(02)		Describe the simple form of the thrust formula for a basic, straight jet engine and perform simple calculations (including pressure thrust).	X	X								
(03)		State that thrust can be considered to remain approximately constant over the whole aeroplane subsonic speed range.	X	X								
021 11 01 02		Design, types and components of turbine engines										
(01)		List the main components of a basic gas turbine engine: — inlet; — compressor; — combustion chamber; — turbine; — outlet.	X	X	X	X	X					
(02)		Describe the variation of static pressure, temperature and axial velocity in a gas turbine engine under normal operating conditions and with the aid of a working cycle diagram.	X	X	X	X	X					
(03)		Describe the differences between absolute, circumferential (tangential) and axial velocity.	X	X	X	X	X					
(04)		List the different types of gas turbine engines: — straight jet; — turbofan; — turboprop.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		State that a gas turbine engine can have one or more spools.	X	X	X	X	X					
(06)		Describe how thrust is produced by turbojet and turbofan engines.	X	X								
(07)		Describe how power is produced by turboprop engines.	X	X								
(08)		Describe the term 'equivalent horsepower' (= thrust horsepower + shaft horsepower).	X	X								
(09)		Explain the principle of a free turbine or free-power turbine.	X	X	X	X	X					
(10)		Define the term 'bypass ratio' and perform simple calculations to determine it.	X	X								
(11)		Define the terms 'propulsive power', 'propulsive efficiency', 'thermal efficiency' and 'total efficiency'.	X	X								
(12)		Describe the influence of compressor-pressure ratio on thermal efficiency.	X	X	X	X	X					
(13)		Explain the variations of propulsive efficiency with forward speed for turbojet, turbofan and turboprop engines.	X	X								
(14)		Define the term 'specific fuel consumption' for turbojets and turboprops.	X	X								
021 11 01 03		<i>Coupled turbine engine: design, operation, components and materials</i>										
(01)		Name the main assembly parts of a coupled turbine engine and explain its operation.			X	X	X					
(02)		Explain the limitations of the materials used with regard to maximum turbine temperature, engine and drive train torque limits.			X	X	X					
(03)		Describe the possible effects on engine components when limits are exceeded.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain that when engine limits are exceeded, this event must be reported.			X	X	X					
021 11 01 04		Free-turbine engine: design, components and materials										
(01)		Describe the design methods to keep the engine's size small for installation in helicopters.			X	X	X					
(02)		List the main components of a free-turbine engine.			X	X	X					
(03)		Describe how the power is developed by a turboshaft/free-turbine engine.			X	X	X					
(04)		Explain how the exhaust gas temperature is used to monitor turbine stress.			X	X	X					
021 11 02 00		Main-engine components										
021 11 02 01		Aeroplane: air intake										
(01)		State the functions of the engine air inlet/air intake.	X	X								
(02)		Describe the geometry of a subsonic (pitot-type) air inlet.	X	X								
(03)		Explain the gas-parameter changes in a subsonic air inlet at different flight speeds.	X	X								
(04)		Describe the reasons for, and the dangers of, the following operational problems concerning the engine air inlet: — airflow separation; — inlet icing; — inlet damage; — foreign object damage (FOD); — heavy in-flight turbulence.	X	X								
021 11 02 02		Compressor and diffuser										
(01)		State the purpose of the compressor.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the working principle of a centrifugal and an axial flow compressor.	X	X	X	X	X					
(03)		Name the following main components of a single stage and describe their function for a centrifugal compressor: — impeller; — diffuser.	X	X	X	X	X					
(04)		Name the following main components of a single stage and describe their function for an axial compressor: — rotor vanes; — stator vanes.	X	X	X	X	X					
(05)		Describe the gas-parameter changes in a compressor stage.	X	X	X	X	X					
(06)		Define the term ‘pressure ratio’ and state a typical value for one stage of a centrifugal and an axial flow compressor and for the complete compressor.	X	X	X	X	X					
(07)		State the advantages and disadvantages of increasing the number of stages in a centrifugal compressor.	X	X	X	X	X					
(08)		Explain the difference in sensitivity for FOD of a centrifugal compressor compared with an axial flow type.	X	X	X	X	X					
(09)		Explain the convergent air annulus through an axial flow compressor.	X	X	X	X	X					
(10)		Describe the reason for twisting the compressor blades.	X	X	X	X	X					
(11)		State the tasks of inlet guide vanes (IGVs).	X	X	X	X	X					
(12)		State the reason for the clicking noise whilst the compressor slowly rotates on the ground.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(13)		State the advantages of increasing the number of spools.	X	X	X	X	X					
(14)		Explain the implications of tip losses and describe the design features to minimise the problem.	X	X	X	X	X					
(15)		Explain the problems of blade bending and flapping and describe the design features to minimise the problem.	X	X	X	X	X					
(16)		Explain the following terms: — compressor stall; — engine surge.	X	X	X	X	X					
(17)		State the conditions that are possible causes of stall and surge.	X	X	X	X	X					
(18)		Describe the indications of stall and surge.	X	X	X	X	X					
(19)		Describe the design features used to minimise the occurrence of stall and surge.	X	X	X	X	X					
(20)		Describe a compressor map (surge envelope) with rpm lines, stall limit, steady state line and acceleration line.	X	X	X	X	X					
(21)		Describe the function of the diffuser.	X	X	X	X	X					
021 11 02 03		Combustion chamber										
(01)		Define the purpose of the combustion chamber.	X	X	X	X	X					
(02)		List the requirements for combustion.	X	X	X	X	X					
(03)		Describe the working principle of a combustion chamber.	X	X	X	X	X					
(04)		Explain the reason for reducing the airflow axial velocity at the combustion chamber inlet (snout).	X	X	X	X	X					
(05)		State the function of the swirl vanes (swirler).	X	X	X	X	X					
(06)		State the function of the drain valves.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Define the terms ‘primary airflow’ and ‘secondary airflow’, and explain their purpose.	X	X	X	X	X					
(08)		Explain the following two mixture ratios: — primary airflow to fuel; — total airflow (within the combustion chamber) to fuel.	X	X	X	X	X					
(09)		Describe the gas-parameter changes in the combustion chamber.	X	X	X	X	X					
(10)		State a typical maximum value of the outlet temperature of the combustion chamber.	X	X	X	X	X					
(11)		Describe the following types of combustion chambers and state the differences between them: — can type; — can-annular, cannular or turbo-annular; — annular; — reverse-flow annular.	X	X	X	X	X					
021 11 02 04		Turbine										
(01)		Explain the purpose of a turbine in different types of gas turbine engines.	X	X	X	X	X					
(02)		Describe the principles of operation of impulse, reaction and impulse-reaction axial flow turbines.	X	X	X	X	X					
(03)		Name the main components of a turbine stage and their function.	X	X	X	X	X					
(04)		Describe the working principle of a turbine.	X	X	X	X	X					
(05)		Describe the gas-parameter changes in a turbine stage.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Describe the function and the working principle of active clearance control.	X	X								
(07)		Describe the implications of tip losses and the means to minimise them.	X	X	X	X	X					
(08)		Explain why the available engine thrust is limited by the turbine inlet temperature.	X	X								
(09)		Explain the divergent gas-flow annulus through an axial-flow turbine.	X	X	X	X	X					
(10)		Explain the high mechanical thermal stress in the turbine blades and wheels/discs.	X	X	X	X	X					
021 11 02 05		Aeroplane: exhaust										
(01)		Name the following main components of the exhaust unit and their function: — jet pipe; — propelling nozzle; — exhaust cone.	X	X								
(02)		Describe the working principle of the exhaust unit.	X	X								
(03)		Describe the gas-parameter changes in the exhaust unit.	X	X								
(04)		Define the term ‘choked exhaust nozzle’ (not applicable to turboprops).	X									
(05)		Explain how jet exhaust noise can be reduced.	X	X								
021 11 02 06		Helicopter: air intake										
(01)		Name and explain the main task of the engine air intake.			X	X	X					
(02)		Describe the use of a convergent air-intake ducting on helicopters.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Describe the reasons for and the dangers of the following operational problems concerning engine air intake: — airflow separations; — intake icing; — intake damage; — FOD; — heavy in-flight turbulence.			X	X	X					
(04)		Describe the conditions and circumstances during ground operations when FOD is most likely to occur.			X	X	X					
(05)		Describe and explain the principles of air intake filter systems that can be fitted to some helicopters for operations in icing and sand conditions.			X	X	X					
(06)		Describe the function of the heated pads on some helicopter air intakes.			X	X	X					
021 11 02 07		Helicopter: exhaust										
(01)		Describe the working principle of the exhaust unit.			X	X	X					
(02)		Describe the gas-parameter changes in the exhaust unit.			X	X	X					
021 11 03 00		Additional components and systems										
021 11 03 01		Engine fuel system										
(01)		Name the main components of the engine fuel system and state their function: — filters; — low-pressure (LP) pump; — high-pressure (HP) pump;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— fuel manifold; — fuel nozzles; — HP fuel cock; — fuel control; or — hydromechanical unit.										
(02)		Name the two types of engine-driven high-pressure pumps, such as: — gear-type; — swash plate-type.	X	X	X	X	X					
(03)		State the tasks of the fuel control unit.	X	X	X	X	X					
(04)		List the possible input parameters to a fuel control unit to achieve a given thrust/power setting.	X	X	X	X	X					
021 11 03 02		Engine control system										
(01)		State the tasks of the engine control system.	X	X	X	X	X					
(02)		List the following different types of engine control systems: — hydromechanical; — hydromechanical with a limited authority electronic supervisor; — single-channel FADEC with hydromechanical backup; — dual-channel FADEC with no backup or any other combination.	X	X	X	X	X					
(03)		Describe a FADEC as a full-authority dual-channel system including functions such as an electronic engine control unit, wiring, sensors, variable vanes,	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		active clearance control, bleed configuration, electrical signalling of thrust lever angle (TLA) (see also AMC to CS-E-50), and an EGT protection function and engine overspeed.										
(04)		Explain how redundancy is achieved by using more than one channel in a FADEC system.	X		X	X						
(05)		State the consequences of a FADEC single input data failure.	X		X	X						
(06)		State that all input and output data is checked by both channels in a FADEC system.	X		X	X						
(07)		State that a FADEC system uses its own sensors and that, in some cases, also data from aircraft systems is used.	X		X	X						
(08)		State that a FADEC must have its own source of electrical power.	X		X	X						
021 11 03 03		Engine lubrication										
(01)		State the tasks of an engine lubrication system.	X	X								
(02)		Name the following main components of a lubrication system and state their function: — oil tank and centrifugal breather; — oil pumps (pressure and scavenge pumps); — oil filters (including the bypass); — oil sumps; — chip detectors; — coolers.	X	X								
(03)		Explain that each spool is fitted with at least one ball bearing and two or more roller bearings.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the use of compressor air in oil-sealing systems (e.g. labyrinth seals).	X	X								
021 11 03 04		Engine auxiliary gearbox										
(01)		State the tasks of the auxiliary gearbox.	X	X								
(02)		Describe how the gearbox is driven and lubricated.	X	X								
021 11 03 05		Engine ignition										
(01)		State the task of the ignition system.	X	X								
(02)		Name the following main components of the ignition system and state their function: — power sources; — igniters.	X	X								
(03)		State why jet turbine engines are equipped with two electrically independent ignition systems.	X	X								
(04)		Explain the different modes of operation of the ignition system.	X	X								
021 11 03 06		Engine starter										
(01)		Name the main components of the starting system and state their function.	X	X								
(02)		Explain the principle of a turbine engine start.	X	X								
(03)		Describe the following two types of starters: — electric; — pneumatic.	X	X								
(04)		Describe a typical start sequence (on ground/in flight) for a turbofan.	X	X								
(05)		Define 'self-sustaining rpm'.	X	X								
021 11 03 07		Reverse thrust										
(01)		Name the following main components of a reverse-thrust system and state their function:	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- reverse-thrust select lever; - power source (pneumatic or hydraulic); - actuators; - doors; - annunciations.										
(02)		Explain the principle of a reverse-thrust system.	X	X								
(03)		Identify the advantages and disadvantages of using reverse thrust.	X	X								
(04)		Describe and explain the following different types of thrust-reverser systems: - hot-stream reverser; - clamshell or bucket-door system; - cold-stream reverser (only turbofan engines); - blocker doors; - cascade vanes.	X	X								
(05)		Explain the implications of reversing the cold stream (fan reverser) only on a high bypass ratio engine.	X	X								
(06)		Describe the protection features against inadvertent thrust-reverse deployment in flight as present on most transport aeroplanes.	X	X								
(07)		Describe the controls and indications provided for the thrust-reverser system.	X	X								
021 11 03 08		Helicopter specifics on design, operation and components for additional components and systems such as lubrication system, ignition circuit, starter, accessory gearbox										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		State the task of the lubrication system.			X	X	X					
(02)		List and describe the common helicopter lubrication systems.			X	X	X					
(03)		Name the following main components of a helicopter lubrication system: — reservoir; — pump assembly; — external oil filter; — magnetic chip detectors, electronic chip detectors; — thermostatic oil coolers; — breather.			X	X	X					
(04)		Identify and name the components of a helicopter lubrication system from a diagram.			X	X	X					
(05)		Identify the indications used to monitor a lubrication system including warning systems.			X	X	X					
(06)		Explain the differences and appropriate use of straight oil and compound oil, and describe the oil numbering system for aviation use.			X	X	X					
(07)		Explain and describe the ignition circuit for engine start and engine relight facility when the selection is set for both automatic and manual functions.			X	X	X					
(08)		Explain and describe the starter motor and the sequence of events when starting, and that for most helicopters the starter becomes the generator after the starting sequence is over.			X	X	X					
(09)		Explain and describe why the engine drives the accessory gearbox.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 11 04 00		Engine operation and monitoring										
021 11 04 01		General										
(01)		Explain the following aeroplane engine ratings: — take-off; — go-around; — maximum continuous thrust/power; — maximum climb thrust/power.	X	X								
(02)		Explain spool-up time.	X	X	X	X	X					
(03)		Explain the reason for the difference between ground and approach flight idle values (rpm).	X	X								
(04)		State the parameters that can be used for setting and monitoring the thrust/power.	X	X	X	X	X					
(05)		Describe the terms ‘alpha range’, ‘beta range’ and ‘reverse thrust’ as applied to a turboprop power lever.	X	X								
(06)		Explain the dangers of inadvertent beta-range selection in flight for a turboprop.	X	X								
(07)		Explain the purpose of engine trending.	X	X								
(08)		Explain how the exhaust gas temperature is used to monitor turbine stress.	X	X	X	X	X					
(09)		Describe the effect of engine acceleration and deceleration on the EGT.	X	X	X	X						
(10)		Describe the possible effects on engine components when EGT limits are exceeded.	X	X	X	X	X					
(11)		Explain why engine-limit exceedances must be reported.	X	X	X	X	X					
(12)		Explain the limitations on the use of the thrust-reverser system at low forward speed.	X	X			X					
(13)		Explain the term ‘engine seizure’.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(14)		State the possible causes of engine seizure and explain their preventative measures.	X	X	X	X	X					
(15)		Describe the potential consequences of a leak in the following two designs of fuel and oil heat exchanger: - oil pressure higher than fuel pressure with oil leaking into the fuel system, potentially affecting the combustion and running of the engine; - fuel pressure higher than oil pressure with fuel leaking into the oil system, potentially increasing the risk of a fire due to fuel entering warm parts of the engine that should be free from fuel.	X	X	X	X	X					
(16)		Explain oil-filter clogging (blockage) and the implications for the lubrication system.	X	X	X	X	X					
(17)		Give examples of monitoring instruments of an engine.	X	X	X	X	X					
(18)		Describe how to identify and assess engine damage based on instrument indications.	X	X	X	X	X					
021 11 04 02		Starting malfunctions										
(01)		Describe the indications and the possible causes of the following aeroplane starting malfunctions: - false (dry or wet) start; - tailpipe fire (torching); - hot start; - abortive (hung) start; - no N1 rotation; - no FADEC indications.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the indications and the possible causes of the following helicopter starting malfunctions: — false (dry or wet) start; — tailpipe fire (torching); — hot start; — abortive (hung) start; — no N1 rotation; — freewheel failure; — no FADEC indications.			X	X	X					
021 11 04 03		Relight envelope										
(01)		Explain the relight envelope.	X	X								
021 11 05 00		Performance aspects										
021 11 05 01		Thrust, performance aspects, and limitations										
(01)		Describe the variation of thrust and specific fuel consumption with altitude at constant TAS.	X	X								
(02)		Describe the variation of thrust and specific fuel consumption with TAS at constant altitude.	X	X								
(03)		Explain the term ‘flat-rated engine’ by describing the change of take-off thrust, turbine inlet temperature and engine rpm with outside air temperature (OAT).	X	X								
(04)		Define the term ‘engine pressure ratio’ (EPR).	X	X								
(05)		Explain the use of reduced (flexible) and derated thrust for take-off, and explain the advantages and disadvantages when compared with a full-rated take-off.	X	X								
(06)		Describe the effects of use of bleed air on rpm, EGT, thrust, and specific fuel consumption.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 11 05 02		Helicopter engine ratings, engine performance and limitations, engine handling: torque, performance aspects and limitations										
(01)		Describe engine rating torque limits for take-off, transient and maximum continuous.			X	X	X					
(02)		Describe turbine outlet temperature (TOT) limits for take-off.			X	X	X					
(03)		Explain why TOT is a limiting factor for helicopter performance.			X	X	X					
(04)		Describe and explain the relationship between maximum torque available and density altitude, which leads to decreasing torque available with the increase of density altitude.			X	X	X					
(05)		Explain that hovering downwind, on some helicopters, will noticeably increase the engine TOT.			X	X	X					
(06)		Explain the reason why the engine performance is less when aircraft accessories (i.e. anti-ice, heating, hoist, filters) are switched on.			X	X	X					
(07)		Describe the effects of use of bleed air on engine parameters.			X	X	X					
(08)		Explain that, on some helicopters, exceeding the TOT limit may cause the main rotor to droop (slow down).			X	X	X					
(09)		Describe overtorquing and explain the consequences.			X	X	X					
021 11 06 00		Auxiliary power unit (APU)										
021 11 06 01		Design, operation, functions, operational limitations										
(01)		State that an APU is a gas turbine engine and list its tasks.	X		X	X						
(02)		State the difference between the two types of APU inlets.	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Define ‘maximum operating and maximum starting altitude’.	X		X	X						
(04)		Name the typical APU control and monitoring instruments.	X		X	X						
(05)		Describe the APU’s automatic shutdown protection.	X		X	X						
021 12 00 00		PROTECTION AND DETECTION SYSTEMS										
021 12 01 00		Smoke detection										
021 12 01 01		<i>Types, design, operation, indications and warnings</i>										
(01)		Explain the operating principle of the following types of smoke detection sensors: — optical; — ionising.	X	X	X	X	X					
(02)		Give an example of warnings, indications and function tests.	X	X	X	X	X					
021 12 02 00		Fire-protection systems										
021 12 02 01		<i>Fire extinguishing (engine and cargo compartments)</i>										
(01)		Explain the operating principle of a built-in fire-extinguishing system and describe its components.	X	X	X	X	X					
(02)		State that two discharges must be provided for each engine (see CS 25.1195(c) Fire-extinguisher systems).	X	X								
021 12 02 02		<i>Fire detection</i>										
(01)		Explain the following principles of fire detection: — resistance and capacitance; — gas pressure.	X	X	X	X	X					
(02)		Explain fire-detection applications such as: — bimetallic;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— continuous loop; — gaseous loop (gas-filled detectors).										
(03)		Explain why generally double-loop systems are used.	X	X	X	X	X					
(04)		Give an example of warnings, indications and function tests of a fire-protection system.	X	X	X	X	X					
021 12 03 00		Rain-protection system										
021 12 03 01		Principle and method of operation										
(01)		Explain the principle and method of operation of the following windshield rain-protection systems for an aeroplane: — wipers; — liquids (rain-repellent); — coating.	X	X								
(02)		Explain the principle and method of operation of wipers for a helicopter.			X	X	X					
021 13 00 00		OXYGEN SYSTEMS										
021 13 01 00		Cockpit, portable and chemical oxygen systems										
021 13 01 01		Operating principles, actuation methods, comparison										
(01)		Describe the basic operating principle of a cockpit oxygen system and describe the following different modes of operation: — normal (diluter demand); — 100 %; — emergency.	X	X								
(02)		Describe the operating principle and the purposes of the following two portable oxygen systems:	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— smoke hood; — portable bottle.										
(03)		Describe the following two oxygen systems that can be used to supply oxygen to passengers: — fixed system (chemical oxygen generator or gaseous system); — portable.	X	X								
(04)		Describe the actuation methods (automatic and manual) and the functioning of a passenger oxygen mask.	X	X								
(05)		Compare chemical oxygen generators to gaseous systems with respect to: — capacity; — flow regulation.	X	X								
(06)		State the dangers of grease or oil related to the use of oxygen systems.	X	X								
021 14 00 00		HELICOPTER: MISCELLANEOUS SYSTEMS										
021 14 01 00		Variable rotor speed, active vibration suppression, night-vision goggles (NVG)										
021 14 01 01		Variable rotor speed										
(01)		Explain the system for ‘beeping’ the NR to its upper limit.			X	X	X					
021 14 01 02		Active vibration suppression										
(01)		Explain and describe how the active vibration suppression system works through high-speed actuators and accelerometer inputs.			X	X	X					
021 14 01 03		NVG										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		To be introduced at a later date.			X	X	X					
021 15 00 00		HELICOPTER: ROTOR HEADS										
021 15 01 00		Main rotor										
021 15 01 01		Types										
(01)		Describe the following rotor-head systems: — teetering (semi-articulated); — articulated; — hingeless (rigid); — bearingless (semi-articulated).			X	X	X					
(02)		Describe in basic terms the following configuration of rotor systems and their advantages and disadvantages: — tandem; — coaxial; — side by side.			X	X	X					
(03)		Explain how flapping, dragging and feathering is achieved in each rotor-head system.			X	X	X					
021 15 01 02		Structural components and materials, stresses, structural limitations										
(01)		Identify from a diagram the main structural components of the main types of rotor-head systems.			X	X	X					
(02)		List and describe the methods used to detect damage and cracks.			X	X	X					
(03)		Explain and describe the structural limitations to respective rotor systems, including the dangers of negative G inputs to certain rotor-head systems.			X	X	X					
(04)		Describe the various rotor-head lubrication methods.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 15 01 03		Design and construction										
(01)		Describe the material technology used in rotor-head design, including construction, using the following materials or mixture of materials: — composites; — fibreglass; — alloys; — elastomers.			X	X	X					
021 15 01 04		Adjustment										
(01)		Describe and explain the methods of adjustment which are possible on various helicopter rotor-head assemblies.			X	X	X					
021 15 02 00		Tail rotor										
021 15 02 01		Types										
(01)		Describe the following tail-rotor systems: — delta-3 hinge effect; — multi-bladed delta-3 effect; — Fenestron or ducted fan tail rotor; — no tail rotor (NOTAR) low-velocity air jet flows from tangential slots (the Coandă effect); — NOTAR high-velocity air jet flows from adjustable nozzles (the Coandă effect).			X	X	X					
(02)		Identify from a diagram the main structural components of the four main types of tail-rotor systems.			X	X	X					
(03)		Explain and describe the methods to detect damage and cracks on the tail rotor and assembly.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain and describe the structural limitations to the respective tail-rotor systems and possible limitations regarding the turning rate of the helicopter.			X	X	X					
(05)		Explain and describe the following methods that helicopter designers use to minimise tail-rotor drift and roll: — reducing the couple arm (tail rotor on a pylon); — offsetting the rotor mast; — use of ‘bias’ in cyclic control mechanism.			X	X	X					
(06)		Explain pitch-input mechanisms.			X	X	X					
(07)		Explain the relationship between tail-rotor thrust and engine power.			X	X	X					
(08)		Describe how the vertical fin on some types reduces the power demand of the tail rotor.			X	X	X					
021 15 02 02		Design and construction										
(01)		List and describe the various tail-rotor designs and construction methods used on helicopters currently in service.			X	X	X					
021 16 00 00		HELICOPTER: TRANSMISSION										
021 16 01 00		Main gearbox										
021 16 01 01		Different types, design, operation, limitations										
(01)		Describe the following main principles of helicopter transmission systems for single- and twin-engine helicopters: — drive for the main and tail rotor; — accessory drive for the generator(s), alternator(s), hydraulic and oil pumps, oil cooler(s) and tachometers.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the reason for limitations on multi-engine helicopter transmissions in various engine-out situations.			X	X	X					
(03)		Describe how the passive vibration control works with gearbox mountings.			X	X	X					
021 16 02 00		Rotor brake										
021 16 02 01		Types, operational considerations										
(01)		Describe the main function of the disc type of rotor brake.			X	X	X					
(02)		Describe both hydraulic- and cable-operated rotor-brake systems.			X	X	X					
(03)		Describe the different options for the location of the rotor brake.			X	X	X					
(04)		List the following operational considerations for the use of rotor brakes: — rotor speed at engagement of rotor brake; — risk of blade sailing in windy conditions; — risk of rotor-brake overheating and possible fire when brake is applied above the maximum limit, particularly when spilled hydraulic fluid is present; — avoid stopping blades over jet-pipe exhaust with engine running; — cockpit annunciation of rotor-brake operation.			X	X	X					
021 16 03 00		Auxiliary systems										
021 16 03 01		Powering the air-conditioning system										
(01)		Explain how power for the air-conditioning system is taken from the auxiliary gearbox.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 16 04 00		Driveshaft and associated installation										
021 16 04 01		Power, construction, materials, speed and torque										
(01)		Describe how power is transmitted from the engine to the main-rotor gearbox.			X	X	X					
(02)		Describe the material and construction of the driveshaft.			X	X	X					
(03)		Explain the need for alignment between the engine and the main- rotor gearbox.			X	X	X					
(04)		Identify how temporary misalignment occurs between driving and driven components.			X	X	X					
(05)		Explain the relationship between driveshaft speed and torque.			X	X	X					
(06)		Describe the methods with which power is delivered to the tail rotor.			X	X	X					
(07)		Describe and identify the construction and materials of tail-rotor/Fenestron driveshafts.			X	X	X					
021 16 05 00		Intermediate and tail gearbox										
021 16 05 01		Lubrication, gearing										
(01)		Explain and describe the various arrangements when the drive changes direction and the need for an intermediate or tail gearbox.			X	X	X					
(02)		Explain the lubrication requirements for intermediate and tail-rotor gearboxes and methods of checking levels.			X	X	X					
(03)		Explain how on most helicopters the tail-rotor gearbox contains gearing, etc., for the tail-rotor pitch-change mechanism.			X	X	X					
021 16 06 00		Clutches										
021 16 06 01		Purpose, operation, components, serviceability										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain the purpose of a clutch.			X	X	X					
(02)		Describe and explain the operation of a: — centrifugal clutch; — actuated clutch.			X	X	X					
(03)		List the typical components of the various clutches.			X	X	X					
(04)		Identify the following methods by which clutch serviceability can be ascertained: — brake-shoe dust; — vibration; — main-rotor run-down time; — engine speed at time of main-rotor engagement; — belt tensioning; — start protection in a belt-drive clutch system.			X	X	X					
021 16 07 00		Freewheels										
021 16 07 01		Purpose, operation, components, location										
(01)		Explain the purpose of a freewheel.			X	X	X					
(02)		Describe and explain the operation of a: — cam- and roller-type freewheel; — sprag-clutch-type freewheel.			X	X	X					
(03)		List the typical components of the various freewheels.			X	X	X					
(04)		Identify the various locations of freewheels in power plant and transmission systems.			X	X	X					
(05)		Explain the implications regarding the engagement and disengagement of the freewheel.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
021 17 00 00		HELICOPTER: BLADES										
021 17 01 00		Main-rotor design and blade design										
021 17 01 01		Design, construction										
(01)		Describe the different types of blade construction and the need for torsional stiffness.			X	X	X					
(02)		Describe the principles of heating systems/pads on some blades for anti-icing/de-icing.			X	X	X					
(03)		Describe the fully articulated rotor with hinges and feathering hinges.			X	X	X					
021 17 01 02		Structural components and materials										
(01)		List the materials used in the construction of main-rotor blades.			X	X	X					
(02)		List the main structural components of a main-rotor blade and their function.			X	X	X					
(03)		Describe the drag hinge of the fully articulated rotor and the lag flexure in the hingeless rotor.			X	X	X					
(04)		Explain the necessity for drag dampers.			X	X	X					
021 17 01 03		Forces and stresses										
(01)		Describe main-rotor blade-loading on the ground and in flight.			X	X	X					
(02)		Describe where the most common stress areas are on rotor blades.			X	X	X					
(03)		Show how the centrifugal forces depend on rotor rpm and blade mass and how they pull on the blade's attachment to the hub. Justify the upper limit of the rotor rpm.			X	X	X					
(04)		Assume a rigid attachment and show how thrust may cause huge oscillating bending moments which stress the attachment.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Explain why flapping hinges do not transfer such moments. Show the small flapping hinge offset on fully articulated rotors and zero offset in the case of teetering rotors.			X	X	X					
(06)		Describe the working principle of the flexible element in the hingeless rotor and describe the equivalent flapping hinge offset compared to that of the articulated rotor.			X	X	X					
021 17 01 04		Structural limitations										
(01)		Explain the structural limitations in terms of bending and rotor rpm.			X	X	X					
021 17 01 05		Adjustment										
(01)	X	Explain the use of trim tabs.			X	X	X					
021 17 01 06		Tip shape										
(01)		Describe the various blade-tip shapes used by different manufacturers and compare their advantages and disadvantages.			X	X	X					
021 17 01 07		Origins of the vertical vibrations										
(01)		Explain the lift (thrust) variations per revolution of a blade and the resulting vertical total rotor thrust (TRT) variation in the case of perfectly identical blades.			X	X	X					
(02)		Show the resulting frequencies and amplitudes as a function of the number of blades.			X	X	X					
(03)		Explain the thrust variation in the case of an out-of-track blade, causes, and frequencies (one-per-revolution).			X	X	X					
021 17 01 08		Lateral vibrations										
(01)		Explain blade imbalances, causes, and effects.			X	X	X					
021 17 02 00		Tail-rotor design and blade design										
021 17 02 01		Design, construction										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the most common design of tail-rotor blade construction, consisting of stainless steel shell reinforced by a honeycomb filler and stainless steel leading abrasive strip.			X	X	X					
(02)		Explain that ballast weights are located at the inboard trailing edge and tip of blades, and that the weights used are determined when the blades are manufactured.			X	X	X					
(03)		Describe how, for some helicopters, anti-icing/de-icing systems are designed into the blade construction.			X	X	X					
(04)		Describe the two-bladed rotor with a teetering hinge, and rotors with more than two blades.			X	X	X					
(05)		Describe the dangers to ground personnel and to the rotor blades, and how to minimise these dangers.			X	X	X					
021 17 02 02		Intentionally left blank										
021 17 02 03		Stresses, vibrations and balancing										
(01)		Describe the tail-rotor blade-loading on the ground and in flight.			X	X	X					
(02)		Explain the sources of vibration of the tail rotor and the resulting high frequencies.			X	X	X					
(03)		Explain balancing and tracking of the tail rotor.			X	X	X					
021 17 02 04		Structural limitations										
(01)		Describe the structural limitations of the tail-rotor blades.			X	X	X					
(02)		Describe the method of checking the strike indicators placed on the tip of some tail-rotor blades.			X	X	X					
021 17 02 05		Adjustment										
(01)		Describe the adjustment of yaw pedals in the cockpit to obtain full-control authority of the tail rotor.			X	X	X					
021 17 02 06		The Fenestron										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the technical layout of a Fenestron tail rotor.			X	X	X					
(02)		Explain the advantages and disadvantages of a Fenestron tail rotor.			X	X	X					
021 17 02 07		No tail rotor (NOTAR)										
(01)		Describe the technical layout of a NOTAR design.			X	X	X					
(02)		Explain the control concepts of a NOTAR.			X	X	X					
(03)		Explain the advantages and disadvantages of a NOTAR design.			X	X	X					

SUBJECT 022 – AIRCRAFT GENERAL KNOWLEDGE – INSTRUMENTATION

ED Decision 2020/018/R

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
020 00 00 00		AIRCRAFT GENERAL KNOWLEDGE										
022 00 00 00		AIRCRAFT GENERAL KNOWLEDGE — INSTRUMENTATION										
022 01 00 00		SENSORS AND INSTRUMENTS										
022 01 01 00		Pressure gauge										
022 01 01 01		Units for pressure, sensor types, measurements										
(01)	X	Define ‘pressure’, ‘absolute pressure’ and ‘differential pressure’.	X	X	X	X	X					
(02)	X	List the following units used for pressure measurement: — Pascal; — bar; — inches of mercury (in Hg); — pounds per square inch (psi).	X	X	X	X	X					
(03)	X	State the relationship between the different units.	X	X	X	X	X					
(04)		List and describe the following different types of sensors used according to the pressure to be measured: — aneroid capsules; — bellows; — diaphragms; — bourdon tube.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(05)		Identify pressure measurements that are applicable to an aircraft: — liquid-pressure measurement (fuel, oil, hydraulic); — air-pressure measurement (bleed-air systems, air-conditioning systems); — engine-pressure measurement manifold pressure (MAP), engine pressure ratio (EPR)).	X	X	X	X	X					
(06)		Identify and read pressure measurement indications both for engine indications and other systems.	X	X	X	X	X					
(07)		Explain the implications of the following pressure measurement errors both for engine indications and other systems: — loss of pressure sensing; — incorrect pressure indications.	X	X	X	X	X					
022 01 02 00		Temperature sensing										
022 01 02 01		Units for temperature, measurements										
(01)	X	Explain temperature.	X	X	X	X	X					
(02)	X	List the following units that can be used for temperature measurement: — Kelvin; — Celsius; — Fahrenheit.	X	X	X	X	X					
(03)	X	State the relationship between these units and convert between them.	X	X	X	X	X					
(04)		Identify temperature measurements that are applicable to an aircraft:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- gas temperature measurement (ambient air, bleed-air systems, air-conditioning systems, air inlet, exhaust gas, gas turbine outlets); - liquid-temperature measurement (fuel, oil, hydraulic); - component-temperature measurement (generator, transformer rectifier unit (TRU), pumps (fuel, hydraulic), power transfer unit (PTU).										
(05)		Identify and read temperature measurement indications for both engine indications and other systems.	X	X	X	X	X					
022 01 03 00		Fuel gauge										
022 01 03 01		Units for fuel, measurements, fuel gauges										
(01)		State that the quantity of fuel can be measured by volume or mass.	X	X	X	X	X					
(02)		List the following units used for fuel quantity: - kilogramme; - pound; - litres; - gallons (US and imperial).	X	X	X	X	X					
(03)		Convert between the various units.	X	X	X	X	X					
(04)		Explain the parameters that can affect the measurement of the volume or mass of the fuel in a fuel tank: - temperature; - aircraft accelerations and attitudes;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— and explain how the fuel-gauge system design compensates for these changes.										
(05)		Describe and explain the operating principles of the following types of fuel gauges: — float system; — capacitance-type of fuel-gauge system; — ultrasound-type of fuel-gauge system: to be introduced at a later date.	X	X	X	X	X					
(06)		Describe and complete a typical post-refuelling procedure for a pilot: — recording the volume that was filled; — converting to the appropriate unit used by the aircraft fuel gauge(s) to compare the actual indicated fuel content to the calculated fuel content; — assess appropriate action if the numbers does not compare.	X	X	X	X	X					
022 01 04 00		Fuel flowmeters										
022 01 04 01		Fuel flow, units for fuel flow, total fuel consumption										
(01)		Define ‘fuel flow’ and where it is measured.	X	X	X	X	X					
(02)		State that fuel flow may be measured by volume or mass per unit of time.	X	X	X	X	X					
(03)		List the following units used for fuel flow when measured by mass per hour: — kilogrammes/hour; — pounds/hour.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(04)		List the following units used for fuel flow when measured by volume per hour: — litres/hour; — imperial gallons/hour; — US gallons/hour.	X	X	X	X	X					
(05)		Explain how total fuel consumption is obtained.	X	X	X	X	X					
022 01 05 00		Tachometer										
022 01 05 01		Types, operating principles, units for engine speed										
(01)	X	List the following types of tachometers, describe their basic operating principle and give examples of use: — mechanical (rotating magnet); — electrical (three-phase tacho-generator); — electronic (impulse measurement with speed probe and phonic wheel); — and describe the operating principle of each type.	X	X	X	X	X					
(02)		Explain the typical units for engine speed: — rpm for piston-engine aircraft; — percentage for turbine-engine aircraft.	X	X	X	X	X					
(03)		Explain that some types of rpm indicators require electrical power to provide an indication.	X	X	X	X	X					
022 01 06 00		Thrust measurement										
022 01 06 01		Parameters, operating principle										
(01)		List and describe the following two parameters used to represent thrust:	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— N1; — EPR.										
(02)		Explain the operating principle of using an engine with EPR indication and explain the consequences of incorrect or missing EPR to the operation of the engine, including reverting to N1 mode.	X	X								
(03)		Give examples of display for N1 and EPR.	X	X								
022 01 07 00		Engine torquemeter										
022 01 07 01		Torque, torquemeters										
(01)		Define 'torque'.	X	X	X	X	X					
(02)		Explain the relationship between power, torque and rpm.	X	X	X	X	X					
(03)		List the following units used for torque: — Newton meters; — inch or foot pounds.	X	X	X	X	X					
(04)		State that engine torque can be displayed as a percentage.	X	X	X	X	X					
(05)	X	List and describe the following different types of torquemeters, and explain their operating principles: — mechanical; — electronic.	X	X	X	X	X					
(06)	X	Compare the two systems with regard to design and weight.	X	X	X	X	X					
(07)		Give examples of display.	X	X	X	X	X					
022 01 08 00		Synchroscope										
022 01 08 01		Purpose, operating principle, display										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(01)		State the purpose of a synchroscope.	X	X								
(02)	X	Explain the operating principle of a synchroscope.	X	X								
(03)		Give examples of display.	X	X								
022 01 09 00		Engine-vibration monitoring										
022 01 09 01		Purpose, operating principle of a vibration-monitoring system, display										
(01)		State the purpose of a vibration-monitoring system for a jet engine.	X	X								
(02)	X	Describe the operating principle of a vibration-monitoring system using the following two types of sensors: — piezoelectric crystal; — magnet.	X	X								
(03)		Explain that there is no specific unit for vibration monitoring, i.e. it is determined by specified numeric threshold values.	X	X								
(04)		Give examples of display.	X	X								
022 01 10 00		Time measurement										
022 01 10 01		On-board clock										
(01)		Explain that the on-board aircraft clock provides a time reference for several of the on-board systems including aircraft communications addressing and reporting system (ACARS) and engine and systems maintenance.	X	X	X	X	X					
022 02 00 00		MEASUREMENT OF AIR-DATA PARAMETERS										
022 02 01 00		Pressure measurement										
022 02 01 01		Definitions										
(01)		Define the following pressure measurements and state the relationship between them:	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- static pressure; - dynamic pressure; - total pressure.										
022 02 01 02		Pitot/static system: design and errors										
(01)		Describe the design and the operating principle of a: - static port/source; - pitot tube; - combined pitot/static probe.	X	X	X	X	X	X	X	1		
(02)		For each of these indicate the various locations and describe the following associated errors and how to correct, minimise the effect of or compensate for them: - position errors; - instrument errors; - errors due to a non-longitudinal axial flow (including manoeuvre-induced errors).	X	X	X	X	X	X	X	1		
(03)		Describe a typical pitot/static system and list the possible outputs.	X	X	X	X	X	X				
(04)		Explain the redundancy and the interconnections that typically exist in complex pitot/static systems found in large aircraft.	X	X	X	X	X	X				
(05)		Explain the purpose of pitot/static system heating.	X	X	X	X	X	X	X	1		
(06)		Describe alternate static sources and their effects when used, particularly in unpressurised aircraft.	X	X	X	X	X	X	X	1		
(07)		Describe a modern pitot static system using solid-state sensors near the pitot probe or static port converting the air data to numerical data	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		(electrical signals) before being sent to the air-data computer(s).										
022 02 02 00		Temperature measurement										
022 02 02 01		Definitions										
(01)		Define the following and explain the relationship between them: — outside air temperature (OAT); — total air temperature (TAT); — static air temperature (SAT).	X	X	X	X	X	X	X		1	
(02)		Explain the term ‘ram rise’ and convert TAT to SAT.	X					X				
(03)		Explain why TAT is often displayed and that TAT is the temperature input to the air-data computer.	X	X	X	X	X	X	X			
022 02 02 02		Design and operation										
(01)		Indicate typical locations for both direct-reading and remote-reading temperature probes, and describe the following errors: — position error; — instrument error.	X	X	X	X	X	X				
(02)		Explain the purpose of temperature probe heating and interpret the effect of heating on sensed temperature unless automatically compensated for.	X	X	X	X	X	X				
022 02 03 00		Angle-of-attack (AoA) measurement										
022 02 03 01		Sensor types, operating principles, ice protection, displays, incorrect indications										
(01)		Describe the following two types of AoA sensors: — null-seeking (slotted) probe; — vane detector.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(02)		For each type, explain the operating principles.	X	X								
(03)		Explain how both types are protected against ice.	X	X								
(04)		Give examples of systems that use the AoA as an input, such as: — air-data computer; — stall warning systems; — flight-envelope protection systems.	X	X								
(05)		Give examples of and interpret different types of AoA displays: — simple light arrays of green, amber and red lights; — gauges showing a numerical scale.	X	X								
(06)		Explain the implications for the pilot if the AoA indication becomes incorrect but still provides data, e.g. if the sensor is frozen in a fixed position.	X	X								
(07)		Explain how an incorrect AoA measurement can affect the controllability of an aircraft with flight-envelope protection.	X	X								
022 02 04 00		Altimeter										
022 02 04 01		Units, terms, types, operating principles, displays, errors, corrections										
(01)		List the following two units used for altimeters and state the relationship between them: — feet; — metres.	X	X	X	X	X	X				
(02)	X	Define the following terms: — height, altitude;	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— indicated altitude, true altitude; — pressure altitude, density altitude.										
(03)	X	Define the following barometric references: 'QNH', 'QFE', '1013.25'.	X	X	X	X	X	X	X		1	
(04)		Explain the operating principles of an altimeter.	X	X	X	X	X	X	X	1	2, 3	
(05)	X	Describe and compare the following three types of altimeters and reason(s) why particular designs may be required in certain airspace: — simple altimeter (single capsule); — sensitive altimeter (multi-capsule); — servo-assisted altimeter.	X	X	X	X	X	X	X		1	
(06)	X	Give examples of associated displays: pointer, multi-pointer, drum, vertical straight scale.	X	X	X	X	X	X	X		1	
(07)		Describe the following errors: — static system error; — instrument error; — barometric error; — temperature error (air column not at ISA conditions); — lag (altimeter response to change of height).	X	X	X	X	X	X	X		1	
(08)		Demonstrate the use of an altimeter correction table for the following errors: — temperature corrections; — aircraft position errors.	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(09)		Describe the effects of a blockage or a leakage on the static pressure line.	X	X	X	X	X	X	X	1		
(10)		Describe the use of GPS altitude as an alternative means of checking erroneous altimeter indications, and highlight the limitations of the GPS altitude indication.	X	X	X	X	X	X	X		1, 3	
022 02 05 00		Vertical speed indicator (VSI)										
022 02 05 01		VSI and instantaneous vertical speed indicator (IVSI)										
(01)		List the two units used for VSIs and state the relationship between them: — metres per second; — feet per minute.	X	X	X	X	X	X				
(02)		Explain the operating principles of a VSI and an IVSI.	X	X	X	X	X	X	X	1		
(03)		Describe and compare the following types of VSIs: — barometric type (VSI); — instantaneous barometric type (IVSI); — inertial type (inertial information provided by an inertial reference unit).	X	X	X	X	X	X	X	1		
(04)		Describe the following VSI errors: — static system errors; — instrument errors; — time lag.	X	X	X	X	X	X	X	1		
(05)		Describe the effects on a VSI of a blockage or a leakage on the static pressure line.	X	X	X	X	X	X	X	1		
(06)		Give examples of a VSI display.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(07)		Compare the indications of a VSI and an IVSI during flight in turbulence and appropriate pilot technique during manoeuvring using either type.	X	X	X	X	X	X				
022 02 06 00		Airspeed indicator (ASI)										
022 02 06 01		Units, errors, operating principles, displays, position errors, unreliable airspeed indications										
(01)		List the following three units used for airspeed and state the relationship between them: — nautical miles/hour (kt); — statute miles/hour (mph); — kilometres/hour (km/h).	X	X	X	X	X	X				
(02)		Describe the following ASI errors and state when they must be considered: — pitot/static system errors; — instrument errors; — position errors; — compressibility errors; — density errors.	X	X	X	X	X	X	X	1		
(03)		Explain the operating principles of an ASI (as appropriate to aeroplanes or helicopters).	X	X	X	X	X	X	X	1		
(04)		Give examples of an ASI display: pointer, vertical straight scale, and digital (HUD display).	X	X	X	X	X	X				
(05)		Demonstrate the use of an ASI correction table for position error.	X	X	X	X	X	X				
(06)		Define and explain the following colour codes that can be used on an ASI: — white arc (flap operating speed range);	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— green arc (normal operating speed range); — yellow arc (caution speed range); — red line (VNE) or barber's pole (VMO); — blue line (best rate of climb speed, one-engine-out for multi-engine piston light aeroplanes).										
(07)		Define and explain the following colour codes that can be used on an ASI: — green arc (normal operating speed range); — red line (VNE); — blue line (maximum airspeed during autorotation).			X	X	X					
(08)		Describe the effects on an ASI of a blockage or a leakage in the static or total pressure line(s).	X	X	X	X	X	X	X			
(09)		Define the term 'unreliable airspeed' and describe the means by which it can be recognised such as: — different airspeed indications between ASIs; — unexpected aircraft behaviour; — buffeting; — aircraft systems warning; — aircraft attitude.	X	X	X	X	X	X	X		1	
(10)		Describe the appropriate procedures available to the pilot in the event of unreliable airspeed indications:	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— combination of a pitch attitude and power setting; — ambient wind noise inside the aircraft; — use of GPS speed indications and the associated limitations.										
022 02 07 00		Machmeter										
022 02 07 01		Operating principle, display, CAS, TAS and Mach number										
(01)		Define 'Mach number' and 'local speed sound' (LSS). Calculate between LSS, TAS and Mach number.	X									
(02)	X	Describe the operating principle of a Machmeter.	X									
(03)	X	Explain why a Machmeter does not suffer from compressibility error.	X									
(04)		Give examples of a Machmeter display: pointer, drum, vertical straight scale, digital.	X									
(05)		Describe the effects on a Machmeter of a blockage or a leakage in the static or total pressure line(s).	X									
(06)		Explain the relationship between CAS, TAS and Mach number. Explain how CAS, TAS and Mach number vary in relation to each other during a climb, a descent, or in level flight in different temperature conditions.	X									
(07)		State the existence of maximum operating limit speed (VMO) and maximum operating Mach number (MMO).	X									
(08)		Describe typical indications of MMO and VMO on analogue and digital instruments.	X									
(09)		Describe the relationship between MMO and VMO with change in altitude and the implications of	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		climbing at constant IAS and descending at constant Mach number with respect to the margin to MMO and VMO.										
(10)		Describe the implications of climbing or descending at constant Mach number or constant IAS with respect to the margin to the stall speed or maximum speed.	X									
022 02 08 00		Air-data computer (ADC)										
022 02 08 01		Operating principle, data, errors, air-data inertial reference unit										
(01)		Explain the operating principle of an ADC.	X	X	X	X	X	X				
(02)	X	List the following possible input data: — TAT; — static pressure; — total pressure; — measured temperature; — AoA; — flaps position; — landing gear position; — stored aircraft data.	X	X	X	X	X	X				
(03)	X	List the following possible output data, as applicable to aeroplanes or helicopters: — IAS; — TAS; — SAT;	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— TAT; — Mach number; — AoA; — altitude; — vertical speed; — VMO/MMO pointer.										
(04)		Explain how position, instrument, compressibility and density errors can be compensated/corrected to achieve a TAS calculation.	X	X	X	X	X	X				
(05)		Give examples of instruments or systems which may use ADC output data.	X	X	X	X	X	X				
(06)		Explain that an air-data inertial reference unit (ADIRU) is an ADC integrated with an inertial reference unit (IRU), that there will be separate controls for the ADC part and inertial reference (IR) part, and that incorrect selection during failure scenarios may lead to unintended and potentially irreversible consequences.	X	X	X	X	X	X				
(07)	X	Explain the ADC architecture for air-data measurement including sensors, processing units and displays, as opposed to stand-alone air-data measurement instruments.	X	X	X	X	X	X				
(08)		Describe the consequences of the loss of an ADC compared to the failure of individual instruments.	X	X	X	X	X	X				
022 03 00 00		MAGNETISM — DIRECT-READING COMPASS AND FLUX VALVE										
022 03 01 00		Earth's magnetic field										
022 03 01 01		Magnetic field, variation, dip										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(01)		Describe the magnetic field of the Earth.	X	X	X	X	X	X				
(02)	X	Explain the properties of a magnet.	X	X	X	X	X	X				
(03)		Define the following terms: — magnetic variation; — magnetic dip (inclination).	X	X	X	X	X	X				
(04)		Describe that a magnetic compass will align itself to both the horizontal (azimuth) and vertical (dip) components of the Earth's magnetic field, thus will not function in the vicinity of the magnetic poles.	X	X	X	X	X	X				
(05)		Demonstrate the use of variation values (given as East/West (E/W) or +/-) to calculate: true heading to magnetic heading; magnetic heading to true heading.	X	X	X	X	X	X				
022 03 02 00		Aircraft magnetic field										
022 03 02 01		Permanent magnetism, electromagnetism, deviation										
(01)	X	Explain the following differences between permanent magnetism and electromagnetism: when they are present; what affects their magnitude.	X	X	X	X	X	X				
(02)	X	Explain the principles of and the reasons for: — compass swinging (determination of initial deviations); — compass compensation (correction of deviations found); — compass calibration (determination of residual deviations).	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(03)		Explain how permanent magnetism within the aircraft structure and electromagnetism from the aircraft systems affect the accuracy of a compass.	X	X	X	X	X	X				
(04)		Describe the purpose and the use of a deviation correction card.	X	X	X	X	X	X				
(05)		Demonstrate the use of deviation values (either given as E/W or +/-) from a compass deviation card to calculate: — compass heading to magnetic heading; — magnetic heading to compass heading.	X	X	X	X	X	X	X	1		
022 03 03 00		Direct-reading magnetic compass										
022 03 03 01		Purpose, errors, timed turns, serviceability										
(01)		Explain the purpose of a direct-reading magnetic compass.	X	X	X	X	X	X				
(02)		Describe how the direct-reading magnetic compass will only show correct indications during straight, level and unaccelerated flight, and that an error will occur during the following flight manoeuvres (no numerical examples): — acceleration and deceleration; — turning; — during pitch-up or pitch-down manoeuvres.	X	X	X	X	X	X				
(03)		Explain how the use of timed turns eliminates the problem of the turning errors of a direct-reading magnetic compass, and calculate the duration of a rate-1 turn for a given change of heading.	X	X	X	X	X	X				
(04)		Describe the serviceability check for a direct-reading magnetic compass prior to flight, such as:	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- the physical appearance of the device; - comparing the indication to another known direction such as a different compass or runway direction.										
022 03 04 00		Flux valve										
022 03 04 01		<i>Purpose, operating principle, location, errors</i>										
(01)		Explain the purpose of a flux valve.	X	X	X	X	X	X				
(02)	X	Explain its operating principle.	X	X	X	X	X	X				
(03)		Indicate typical locations of the flux valve(s).	X	X	X	X	X	X				
(04)		Give the remote-reading compass system as example of application for a flux valve.	X	X	X	X	X	X				
(05)		Explain that deviation is compensated for and, therefore, eliminates the need for a deviation correction card.	X	X	X	X	X	X				
(06)		Explain that a flux valve does not suffer from the same magnitude of errors as a direct-reading magnetic compass when turning, accelerating or decelerating and during pitch-up or pitch-down manoeuvres.	X	X	X	X	X	X				
022 04 00 00		GYROSCOPIC INSTRUMENTS										
022 04 01 00		Gyroscope: basic principles										
022 04 01 01		<i>Gyroscopic forces, degrees of freedom, gyro wander, driving gyroscopes</i>										
(01)	X	Define a 'gyro'.	X	X	X	X	X	X	X		1	
(02)	X	Explain the fundamentals of the theory of gyroscopic forces.	X	X	X	X	X	X	X		1	
(03)	X	Define the 'degrees of freedom' of a gyro.	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		<i>Remark: As a convention, the degrees of freedom of a gyroscope do not include its own axis of rotation (the spin axis).</i>										
(04)	X	Explain the following terms: — rigidity; — precession; — wander (drift/topple).	X	X	X	X	X	X				
(05)		Explain the three types of gyro wander: — real wander; — apparent wander; — transport wander.	X	X	X	X	X	X				
(06)		Describe the two ways of driving gyroscopes and any associated indications: — air/vacuum; — electrically.	X	X	X	X	X	X	X	1		
022 04 02 00		Rate-of-turn indicator — Turn coordinator — Balance (slip) indicator										
022 04 02 01		Indications, relation between bank angle, rate of turn and TAS										
(01)		Explain the purpose of a rate-of-turn and balance (slip) indicator.	X	X	X	X	X	X	X	1		
(02)		Define a 'rate-1 turn'.	X	X	X	X	X	X	X	1		
(03)		Describe the indications given by a rate-of-turn indicator.	X	X	X	X	X	X	X		1	
(04)		Explain the relation between bank angle, rate of turn and TAS, and how bank angle becomes the limiting factor at high speed (no calculations).	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(05)		Explain the purpose of a balance (slip) indicator and its principle of operation.	X	X	X	X	X	X	X	1		
(06)		Describe the indications of a rate-of-turn and balance (slip) indicator during a balanced, slip or skid turn.	X	X	X	X	X	X	X	1		
(07)		Describe the indications given by a turn coordinator (or turn-and-bank indicator).	X	X	X	X	X	X	X	1		
(08)		Compare the indications on the rate-of-turn indicator and the turn coordinator.	X	X	X	X	X	X	X	1		
022 04 03 00		Attitude indicator (artificial horizon)										
022 04 03 01		Purpose, types, effect of aircraft acceleration, display										
(01)		Explain the purpose of the attitude indicator.	X	X	X	X	X	X	X	1		
(02)		Identify the two types of attitude indicators: — attitude indicator; — attitude and director indicator (ADI).	X	X	X	X	X	X	X	1		
(03)	X	State the degrees of freedom.	X	X	X	X	X	X				
(04)		Describe the effects of the aircraft's acceleration and turns on instrument indications.	X	X	X	X	X	X				
(05)		Describe a typical attitude display and instrument markings.	X	X	X	X	X	X	X	1		
022 04 04 00		Directional gyroscope										
022 04 04 01		Purpose, types, drift, alignment to compass heading										
(01)		Explain the purpose of the directional gyroscope.	X	X	X	X	X	X	X	1		
(02)		Identify the two types of gyro-driven direction indicators: direction indicator; horizontal situation indicator (HSI).	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(03)		Explain how the directional gyroscope will drift over time due to the following: — rotation of the Earth; — aircraft manoeuvring; — aircraft movement over the Earth's surface/direction of travel.	X	X	X	X	X	X				
(04)		Describe the procedure for the pilot to align the directional gyroscope to the correct compass heading.	X	X	X	X	X	X				
022 04 05 00		Remote-reading compass systems										
022 04 05 01		Operating principles, components, comparison with a direct-reading magnetic compass										
(01)		Describe the principles of operation of a remote-reading compass system.	X	X	X	X	X	X	X		1	
(02)		Using a block diagram, list and explain the function of the following components of a remote-reading compass system: — flux detection unit; — gyro unit; — transducers, precession amplifiers, annunciator; — display unit (compass card, synchronising and set-heading knob, DG/compass/slave/free switch).	X	X	X	X	X	X	X		1	
(03)		State the advantages and disadvantages of a remote-reading compass system compared to a direct-reading magnetic compass with regard to:	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— design (power source, weight and volume); — deviation due to aircraft magnetism; — turning and acceleration errors; — attitude errors; — accuracy and stability of the information displayed; — availability of the information for several systems (compass card, RMI, automatic flight control system (AFCS)).										
022 04 06 00		Solid-state systems — attitude and heading reference system (AHRS)										
022 04 06 01		Components, indications										
(01)		Explain that the AHRS is a replacement for traditional gyros using solid-state technology with no moving parts and is a single unit consisting of: — solid-state accelerometers; — solid-state rate sensor gyroscopes; — solid-state magnetometers (measurement of the Earth's magnetic field).	X	X	X	X	X	X	X	1		
(02)		Explain that the AHRS senses rotation and acceleration for all three axes and senses the direction of the Earth's magnetic field where the indications are normally provided on electronic screens (electronic flight instrument system (EFIS)).	X	X	X	X	X	X	X			
022 05 00 00		INERTIAL NAVIGATION										
022 05 01 00		Basic principles										
022 05 01 01		Systems										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(01)		State that inertial navigation/reference systems are the main source of attitude and one of the main sources of navigational data in commercial air transport aeroplanes.	X		X	X						
(02)		State that inertial systems require no external input, except TAS, to determine aircraft attitude and navigational data.	X		X	X						
(03)		State that earlier gyro mechanically stabilised platforms are (technically incorrectly but conventionally) referred to as inertial navigation systems (INSs) and more modern fixed (strap down) platforms are conventionally referred to as inertial reference systems (IRSs). INSs can be considered to be stand-alone, whereas IRSs are integrated with the FMS.	X		X	X						
(04)		Explain the basic principles of inertial navigation (including double integration of measured acceleration and the necessity for north–south, east–west and vertical components to be measured/extracted).	X		X	X						
(05)		Explain the necessity of applying correction for transport precession, and Earth rate precession, coriolis and gravity.	X		X	X						
(06)		State that in modern aircraft fitted with inertial reference system (IRS) and flight management system (FMS), the flight management computer (FMC) position is normally derived from a mathematical analysis of IRS, global positioning system (GPS), and distance measuring equipment (DME) data, VHF omnidirectional radio range (VOR) and LOC.	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(07)		List all navigational data that can be determined by a stand-alone inertial navigation system.	X		X	X						
(08)		State that a strap-down system is fixed to the structure of the aircraft and normally consists of three laser ring gyros and three accelerometers.	X		X	X						
(09)		State the differences between a laser ring gyro and a conventional mechanical gyro.	X		X	X						
022 05 02 00		Alignment and operation										
022 05 02 01		Alignment process, incorrect data entry, and control panels										
(01)		State that during the alignment process, the inertial platform is levelled (INS) or the local vertical is determined (IRS), and true north/aircraft heading is established.	X		X	X						
(02)		Explain that the aircraft must be stationary during alignment, the aircraft position is entered during the alignment phase, and that the alignment process takes around 10 to 20 minutes at mid latitudes (longer at high latitudes).	X		X	X						
(03)		State that in-flight realignment is not possible and loss of alignment leads to loss of navigational data although attitude information may still be available.	X		X	X						
(04)		Explain that the inertial navigation system (INS) platform is maintained level and north-aligned after alignment is complete and the aircraft is in motion.	X		X	X						
(05)		State that an incorrect entry of latitude may lead to a loss of alignment and is more critical than the incorrect entry of longitude.	X		X	X						
(06)		State that the positional error of a stand-alone INS varies (a typical value can be quoted as 1–2 NM/h) and is dependent on the gyro drift rate,	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		accelerometer bias, misalignment of the platform, and computational errors.										
(07)		Explain that, on a modern aircraft, there is likely to be an air-data inertial reference unit (ADIRU), which is an inertial reference unit (IRU) integrated with an air-data computer (ADC).	X		X	X						
(08)		Identify examples of IRS control panels.	X		X	X						
(09)		Explain the following selections on the IRU mode selector: — NAV (normal operation); — ATT (attitude only).	X		X	X						
(10)		State that the majority of the IRS data can be accessed through the FMS control and display unit (CDU)/flight management and guidance system (FMGS) multifunction control and display unit (MCDU).	X		X	X						
(11)		Describe the procedure available to the pilot for assessing the performance of individual IRUs after a flight: — reviewing the residual indicated ground speed when the aircraft has parked; — reviewing the drift given as NM/h.	X		X	X						
022 06 00 00		AEROPLANE: AUTOMATIC FLIGHT CONTROL SYSTEMS										
022 06 01 00		General										
022 06 01 01		Definitions and control loops										
(01)		Describe the following purposes of an automatic flight control system (AFCS):	X	X				X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- enhancement of flight controls; - reduction of pilot workload.										
(02)		Define and explain the following two functions of an AFCS: - aircraft control: stabilise the aircraft around its centre of gravity (CG); - aircraft guidance: guidance of the aircraft's flight path.	X	X				X				
(03)		Describe the following two automatic control principles: - closed loop, where a feedback from an action or state is compared to the desired action or state; - open loop, where there is no feedback loop.	X	X								
(04)		List the following elements of a closed-loop control system and explain their basic function: - input signal; - error detector; - signal processor providing a measured output signal according to set criteria or laws; - control element such as an actuator; - feedback signal to error detector for comparison with input signal.	X	X								
(05)		Describe how a closed-loop system may enter a state of self-induced oscillation if the system	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		overcompensates for deviations from the desired state.										
(06)		Explain how a state of self-induced oscillations may be detected and describe the effects of self-induced oscillations: — aircraft controllability; — aircraft safety; — timely manual intervention as a way of mitigating loss of control; — techniques that may be used to maintain positive control of the aircraft.	X	X								
022 06 02 00		Autopilot system										
022 06 02 01		Design and operation										
(01)		Define the three basic control channels.	X	X								
(02)		Define the three different types of autopilots: — single or 1 axis (roll); — 2 axes (pitch and roll); — 3 axes (pitch, roll and yaw).	X	X								
(03)		Describe the purpose of the following components of an autopilot system: — flight control unit (FCU), mode control panel (MCP) or equivalent; — flight mode annunciator (FMA) (see Subject 022 06 04 00); — autopilot computer; — actuator.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(04)		Explain the following lateral modes: — heading (HDG)/track (TRK); — VOR (VOR)/localiser (LOC); — lateral navigation/managed navigation (LNAV or NAV).	X	X								
(05)		Describe the purpose of control laws for pitch and roll modes.	X	X								
(06)		Explain the following vertical modes: — vertical speed (V/S); — flight path angle (FPA); — level change (LVL CHG)/open climb (OP CLB) or open descent (OP DES); — speed reference system (SRS); — altitude (ALT) hold; — vertical navigation (VNAV)/managed climb (CLB) or descent (DES); — glideslope (G/S).	X	X								
(07)		Describe how the autopilot uses speed, aircraft configuration or flight phase as a measure for the magnitude of control inputs and how this may affect precision and stability.	X	X								
(08)		Explain the following mixed modes: — take-off; — go-around; — approach (APP).	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(09)		Describe the two types of autopilot configurations and explain the implications to the pilot for either and when comparing the two principles: — flight-deck controls move with the control surface when the autopilot is engaged; — flight-deck controls remain static when the autopilot is engaged.	X	X								
(10)		Describe the purpose of the following inputs and outputs for an autopilot system: — attitude information; — flight path/trajectory information; — control surface position information; — airspeed information; — aircraft configuration information; — FCU/MCP selections; — FMAs.	X	X								
(11)		Describe the purpose of the synchronisation function when engaging the autopilot and explain why the autopilot should be engaged when the aircraft is in trim.	X	X								
(12)		Define the control wheel steering (CWS) mode as manual manoeuvring of the aircraft through the autopilot computer and autopilot servos/actuators using the control column/control wheel.	X	X								
(13)		Describe the following elements of CWS: — CWS as an autopilot mode;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— flight phases where CWS cannot be used; — whether the pilot or the autopilot is controlling the flight path; — the availability of flight path/performance protections; — potential different feel and control response compared to manual flight.										
(14)		Describe touch control steering (TCS) and highlight the differences when compared to CWS: — autopilot remains engaged but autopilot servos/actuators are disconnected from the control surfaces; — manual control of the aircraft as long as TCS button is depressed; — autopilot servos/actuators reconnect when TCS button is released and the autopilot returns to previously engaged mode(s).	X	X				X				
(15)		Explain that only one autopilot may be engaged at any time except for when APP is armed in order to facilitate a fail-operational autoland.	X	X				X				
(16)		Explain the difference between an armed and an engaged mode: — not all modes have an armed state available; — a mode will only become armed if certain criteria are met;	X	X				X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— an armed mode will become engaged (replacing the previously engaged mode, if any) when certain criteria are met.										
(17)		Describe the sequence of events when a mode is engaged and the different phases: — initial phase where attitude is changed to obtain a new trajectory in order to achieve the new parameter; — the trajectory will be based on rate of closure which is again based on the difference between the original parameter and the new parameter; — capture phase where the aircraft will follow a predefined rate of change of trajectory to achieve the new parameter without overshooting/ undershooting; — tracking or hold phase where the aircraft will maintain the set parameter until a new change has been initiated.	X	X				X				
(18)		Explain automatic mode reversion and typical situations where it may occur: — no suitable data for the current mode such as flight plan discontinuity when in LNAV/managed NAV; — change of parameter during capture phase for original parameter such as change of altitude target during ALT ACQ/ALT*; — mismanagement of a mode resulting in engagement of the autopilot envelope	X	X				X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		protection, e.g. selecting excessive V/S resulting in a loss of speed control.										
(19)		Explain the dangers of mismanagement of the following modes: — use of V/S and lack of speed protection, i.e. excessive V/S or FPA may be selected with subsequent uncontrolled loss or gain of airspeed; — arming VOR/LOC or APP outside the protected area of the localiser or ILS.	X	X				X				
(20)		Describe how failure of other systems may influence the availability of the autopilot and how incorrect data from other systems may result in an undesirable aircraft state, potentially without any failure indications. Explain the importance of prompt and appropriate pilot intervention during such events.	X	X				X				
(21)		Explain an appropriate procedure for disengaging the autopilot and why both aural and visual warnings are used to indicate that the autopilot is being disengaged: — temporary warning for intended disengagement using the design method; — continuous warning for unintended disengagement or using a method other than the design method.	X	X				X				
(22)		Explain the following regarding autopilot and aircraft with manual trim:	X	X				X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- the autopilot may not engage unless the aircraft controls are in trim; - the aircraft will normally be in trim when the autopilot is disconnected; - use of manual trim when the autopilot is engaged will normally lead to autopilot disconnection and a risk of an out-of-trim situation.										
022 06 03 00		Flight director: design and operation										
022 06 03 01		Purpose, use, indications, modes, data										
(01)		Explain the purpose of a flight director system.	X	X				X				
(02)		Describe the different types of display: - pitch and roll crossbars; - V-bar.	X	X				X				
(03)		Explain the differences between a flight director and an autopilot and how the flight director provides a means of cross-checking the control/guidance commands sent to the autopilot.	X	X				X				
(04)		Explain why the flight director must be followed when engaged/shown, and describe the appropriate use of the flight director: - flight director only; - autopilot only; - flight director and autopilot; - typical job-share between pilots (pilot flying (PF)/pilot monitoring (PM)) for selecting the	X	X				X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		parameters when autopilot is engaged versus disengaged; — highlight when the flight director should not be followed or should be disengaged.										
(05)		Give examples of different scenarios and the resulting flight director indications.	X	X				X				
(06)		Explain that the flight director computes and indicates the direction and magnitude of control inputs required in order to achieve an attitude to follow a trajectory.	X	X				X				
(07)		Explain how the modes available for the flight director are the same as those available for the autopilot, and that the same panel (FCU/MCP) is normally used for selection.	X	X				X				
(08)		Explain the importance of checking the FMC data or selected autopilot modes through the FMA when using the flight directors. If the flight directors are showing incorrect guidance, they should not be followed and should be turned off.	X	X				X				
022 06 04 00		Aeroplane: flight mode annunciator (FMA)										
022 06 04 01		Purpose, modes, display scenarios										
(01)		Explain the purpose of FMAs and their importance being the only indication of the state of a system rather than a switch position.	X	X				X				
(02)		Describe where the FMAs are normally shown and how the FMAs will be divided into sections (as applicable to aircraft complexity): — vertical modes; — lateral modes;	X	X				X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— autothrust modes; — autopilot and flight director annunciators; — landing capability.										
(03)		Explain why FMAs for engaged or armed modes have different colour or different font size.	X	X				X				
(04)		Describe the following FMA display scenarios: — engagement of a mode; — mode change from armed to becoming engaged; — mode reversion.	X	X				X				
(05)		Explain the importance of monitoring the FMAs and announcing mode changes at all times (including when selecting a new mode) and why only certain mode changes will be accompanied by an aural notification or additional visual cues.	X	X				X				
(06)		Describe the consequences of not understanding what the FMAs imply or missing mode changes, and how it may lead to an undesirable aircraft state.	X	X				X				
022 06 05 00		Autoland										
022 06 05 01		Design and operation										
(01)		Explain the purpose of an autoland system.	X					X				
(02)		Explain the significance of the following components required for an autoland: — autopilot; — autothrust; — radio altimeter;	X					X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— ILS receivers.										
(03)		Explain the following terms (reference to CS-AWO 'All Weather Operations'): — fail-passive automatic landing system; — fail-operational automatic landing system; — fail-operational hybrid landing system; — alert height.	X									
(04)		Describe the autoland sequence including the following: — FMAs regarding the landing capability of the aircraft; — the significance of monitoring the FMAs to ensure the automatic arming/engagement of modes triggered by defined radio altitudes or other thresholds; — in the event of a go-around, that the aircraft performs the go-around manoeuvre both by reading the FMAs and supporting those readings by raw data; — during the landing phase, that 'FLARE' mode engages at the appropriate radio altitude, including typical time frame and actions if 'FLARE' does not engage; — after landing, that 'ROLL-OUT' mode engages and the significance of disconnecting the autopilot prior to vacating the runway.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(05)		Explain that there are operational limitations in order to legally perform an autoland beyond the technical capability of the aircraft.	X									
(06)		Explain the purpose and significance of alert height, describe the indications and implications, and consider typical pilot actions for a failure situation: — above the alert height; — below the alert height.	X									
(07)		Describe typical failures that, if occurring below the alert height, will trigger a warning: — all autopilots disengage; — loss of ILS signal or components thereof; — excessive ILS deviations; — radio-altimeter failure.	X									
(08)		Describe how the failure of various systems, including systems not directly involved in the autoland process, can influence the ability to perform an autoland or affect the minima down to which the approach may be conducted.	X									
(09)		Describe the fail-operational hybrid landing system as a primary fail-passive automatic landing system with a secondary independent guidance system such as a head-up display (HUD) to enable the pilot to complete a manual landing if the primary system fails.	X									
022 07 00 00		HELICOPTER: AUTOMATIC FLIGHT CONTROL SYSTEMS										
022 07 01 00		General principles										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
022 07 01 01		Stabilisation										
(01)		Explain the similarities and differences between SAS and AFCS (the latter can actually fly the helicopter to perform certain functions selected by the pilot). Some AFCSs just have altitude and heading hold whilst others include a vertical speed or IAS hold mode, where a constant rate of climb/decent or IAS is maintained by the AFCS.			X	X	X					
022 07 01 02		Reduction of pilot workload										
(01)		Appreciate how effective the AFCS is in reducing pilot workload by improving basic aircraft control harmony and decreasing disturbances.			X	X	X					
022 07 01 03		Enhancement of helicopter capability										
(01)		Explain how an AFCS improves helicopter flight safety during: — search and rescue (SAR) because of increased capabilities; — flight by sole reference to instruments; — underslung load operations; — white-out conditions in snow-covered landscapes; — an approach to land with lack of visual cues.			X	X	X					
(02)		Explain that the SAR modes of AFCS include the following functions: — ability to autohover; — facility for mark on target (MOT) approach to hover;			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— automatically transition from cruise down to a predetermined point or over-flown point; — ability for the rear crew to move the helicopter around in the hover; — the ability to automatically transition from the hover back to cruise flight; — the ability to fly various search patterns.										
(03)		Explain that earlier autohover systems use Doppler velocity sensors and modern systems use inertial sensors plus GPS, and normally include a two-dimensional hover-velocity indicator for the pilots.			X	X	X					
(04)		Explain why some SAR helicopters have both radio-altimeter height hold and barometric altitude hold.			X	X	X					
022 07 01 04		Failures										
(01)		Explain the various redundancies and independent systems that are built into the AFCSS.			X	X	X					
(02)		Appreciate that the pilot can override the system in the event of a failure.			X	X	X					
(03)		Explain a series actuator ‘hard over’ which equals aircraft attitude runaway.			X	X	X					
(04)		Explain the consequences of a saturation of the series actuators.			X	X	X					
022 07 02 00		Components: operation										
022 07 02 01		Basic sensors										
(01)		Explain the basic sensors in the system and their functions.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(02)		Explain that the number of sensors will be dependent on the number of coupled modes of the system.			X	X	X					
022 07 02 02		Specific sensors										
(01)		Explain the function of the microswitches and strain gauges in the system which sense pilot input to prevent excessive feedback forces from the system.			X	X	X					
022 07 02 03		Actuators										
(01)		Explain the principles of operation of the series and parallel actuators, spring-box clutches and the autotrim system.			X	X	X					
(02)		Explain the principle of operation of the electronic hydraulic actuators in the system.			X	X	X					
022 07 02 04		Pilot–system interface: control panels, system indications, warnings										
(01)		Describe the typical layout of the AFCS control panel.			X	X	X					
(02)		Describe the system indications and warnings.			X	X	X					
022 07 02 05		Operation										
(01)		Explain the functions of the redundant sensors' simplex and duplex channels (single/dual channel).			X	X	X					
022 07 03 00		Stability augmentation system (SAS)										
022 07 03 01		General principles and operation										
(01)		Explain the general principles and operation of an SAS with regard to: — rate damping; — short-term attitude hold; — effect on static stability;			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— effect on dynamic stability; — aerodynamic cross-coupling; — effect on manoeuvrability; — control response; — engagement/disengagement; — authority.										
(02)		Explain and describe the general working principles and primary use of an SAS by damping pitch, roll and yaw motions.			X	X	X					
(03)		Describe a simple SAS with force trim system which uses magnetic clutch and springs to hold cyclic control in the position where it was last released.			X	X	X					
(04)		Explain the interaction of trim with SAS/stability and control augmentation system (SCAS).			X	X	X					
(05)		Appreciate that the system can be overridden by the pilot and that individual channels can be deselected.			X	X	X					
(06)		Describe the operational limits of the system.			X	X	X					
(07)		Explain why the system should be turned off in severe turbulence or when extreme flight attitudes are reached.			X	X	X					
(08)		Explain the safety design features built into some SASs to limit the authority of the actuators to 10–20 % of the full-control throw in order to allow the pilot to override if actuators demand an unsafe control input.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(09)		Explain how cross-coupling produces an adverse effect on roll-to-yaw coupling when the helicopter is subjected to gusts.			X	X	X					
(10)		Explain the collective-to-pitch coupling, side-slip-to-pitch coupling and inter-axis coupling.			X	X	X					
022 07 04 00		Autopilot — automatic stability equipment										
022 07 04 01		General principles										
(01)		Explain the general autopilot principles with regard to: — long-term attitude hold; — fly-through; — changing the reference (beep trim, trim release).			X	X	X					
022 07 04 02		Basic modes (3/4 axes)										
(01)		Explain the AFCS operation on cyclic axes (pitch/roll), yaw axis, and on collective (fourth axis).			X	X	X					
022 07 04 03		Automatic guidance (upper modes of AFCS)										
(01)		Explain the function of the attitude-hold system in an AFCS.			X	X	X					
(02)		Explain the function of the heading-hold system in an AFCS.			X	X	X					
(03)		Explain the function of the vertical-speed hold system in an AFCS.			X	X	X					
(04)		Explain the function of the navigation-coupling system in an AFCS.			X	X	X					
(05)		Explain the function of the VOR-/ILS-coupling system in an AFCS.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(06)		Explain the function of the hover-mode system in an AFCS (including Doppler and radio-altimeter systems).			X	X	X					
(07)		Explain the function of the SAR mode (automatic transition to hover and back to cruise) in an AFCS.			X	X	X					
022 07 04 04		Flight director: design and operation										
(01)		Explain the purpose of a flight director system.			X	X	X					
(02)		Describe the different types of display: — pitch and roll crossbars; — V-bar.			X	X	X					
(03)		State the difference between the flight director system and the autopilot system. Explain how each can be used independently.			X	X	X					
(04)		List and describe the main components of the flight director system.			X	X	X					
(05)		Give examples of different situations with the respective indications of the command bars.			X	X	X					
(06)		Explain the architecture of the different flight directors fitted to helicopters and the importance to monitor other instruments as well as the flight director.			X	X	X					
(07)		Explain how some helicopter types have the collective setting as a flight director command; however, the command does not provide protection against a transmission overtorque.			X	X	X					
(08)		Describe the collective setting and yaw depiction on flight director for some helicopters.			X	X	X					
022 07 04 05		Automatic flight control panel (AFCP)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(01)		Explain the purpose and the importance of the AFCP.			X	X	X					
(02)		State that the AFCP provides: — AFCS basic and upper modes; — flight director selection, SAS and AP engagement; — failure and alert messages.			X	X	X					
022 08 00 00		TRIMS — YAW DAMPER — FLIGHT-ENVELOPE PROTECTION										
022 08 01 00		Trim systems										
022 08 01 01		Design and operation										
(01)		Explain the purpose of the trim system and describe the layout with one trim system for each control axis, depending on the complexity of the aircraft.	X	X								
(02)		Give examples of trim indicators and their function, and explain the significance of a 'green band/area' for the pitch trim.	X	X								
(03)		Describe and explain an automatic pitch-trim system for a conventional aeroplane.	X	X								
(04)		Describe and explain an automatic pitch-trim system for an FBW aeroplane and that it is also operating during manual flight; however, during certain phases it may be automatically disabled to alter the handling characteristics of the aircraft.	X	X								
(05)		Describe the consequences of manual operation on the trim wheel when the automatic pitch-trim system is engaged.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(06)		Describe and explain the engagement and disengagement conditions of the autopilot according to trim controls.	X	X								
(07)		Define 'Mach trim' and state that the Mach-trim system can be independent.	X	X								
(08)		Describe the implications for the pilot in the event of a runaway trim or significant out-of-trim state.	X	X								
022 08 02 00		Yaw damper										
022 08 02 01		Design and operation										
(01)		Explain the purpose of the yaw-damper system.	X	X								
(02)		Explain the purpose of the Dutch-roll filter (filtering of the yaw input signal).	X	X								
(03)		Explain the operation of a yaw-damper system and state the difference between a yaw-damper system and a 3-axis autopilot operation on the rudder channel.	X	X								
022 08 03 00		Flight-envelope protection (FEP)										
022 08 03 01		Purpose, input parameters, functions										
(01)		Explain the purpose of the FEP.	X	X				X				
(02)		Explain typical input parameters to the FEP: AoA; — aircraft configuration; — airspeed information.	X	X				X				
(03)		Explain the following functions of the FEP: — stall protection; — overspeed protection.	X	X				X				
(04)		Explain how the stall-protection function and the overspeed-protection function apply to both	X	X				X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		mechanical/conventional and FBW control systems, but other functions (e.g. pitch or bank limitation) can only apply to FBW control systems.										
022 09 00 00		AUTOTHRUST — AUTOMATIC THRUST CONTROL SYSTEM										
022 09 01 00		Autothrust system										
022 09 01 01		<i>Purpose, operation, overcompensation, speed control</i>										
(01)		Describe the purpose of the autothrust system and explain how the FMAs will be the only indication on active autothrust modes.	X									
(02)		Explain the operation of an autothrust system with regard to the following modes: — take-off/go-around (TOGA); — climb or maximum continuous thrust (MCT), N1 or EPR targeted (THR CLB, THR MCT, N1, THR HOLD, EPR); — speed (SPEED, MCP SPD); — idle thrust (THR IDLE, RETARD/ARM); — landing (RETARD, THR IDLE).	X									
(03)		Describe the two main variants of autothrust systems: — mode selections available on the FCU/MCP and thrust levers move with autothrust commands; — mode selections made using the thrust levers which remain static during autothrust operation.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(04)		Explain how flight in turbulence/wind shear giving fluctuating airspeed indications may lead to the autothrust overcompensating in an oscillating manner and that manual thrust may be required to settle the airspeed. Airspeed indications/trend vectors may give an indication of appropriate thrust adjustments but any reaction should not be too aggressive.	X									
(05)		Explain the threats associated with the use of autothrust resulting in the pilot losing the sense of energy awareness (e.g. speed, thrust).	X									
(06)		Explain the relationship between autopilot pitch modes and autothrust modes, and how the autopilot and autothrust will interact upon selecting modes for one of the systems.	X									
(07)		Explain the principles of speed control and how speed can be controlled: — by varying the engine thrust; — by varying the aircraft pitch.	X									
(08)		Explain the potential implications on speed control when the autothrust controls speed and the autopilot pitch channel has a fixed pitch target for the following mode combinations: — MCP SPD/SPEED and ALT HOLD/ALT; — MCP SPD/SPEED and VSP (climb); — MCP SPD/SPEED and VSP (descent).	X									
(09)		Explain the potential implications on speed control when the autothrust has a fixed thrust target and	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		the autopilot pitch channel controls speed for the following mode combinations: — N1/THR CLB and LVL CHG/OP CLB; — ARM/THR IDLE and LVL CHG/OP DES.										
022 10 00 00		COMMUNICATION SYSTEMS										
022 10 01 00		Voice communication, data-link transmission										
022 10 01 01		Definitions and transmission modes										
(01)		Describe the purpose of a data-link transmission system.	X		X	X						
(02)		Compare voice communication versus data-link transmission systems.	X		X	X						
(03)		Describe the communication links that are used in aircraft: — high-frequency (HF) communications; — very high-frequency (VHF) communications; — satellite communications (SATCOM).	X		X	X						
(04)		Consider the properties of the communication links with regard to: — signal quality; — range/area coverage; — range; — line-of-sight limitations; — quality of the signal received; — interference due to ionospheric conditions; — data transmission speed.	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(05)		Define and explain the following terms in relation to aircraft data-link communications: — message/data uplink; — message/data downlink.	X		X	X						
022 10 01 02		Systems: architecture, design and operation										
(01)		Describe the purpose of the ACARS network.	X									
(02)		Describe the systems using the ACARS network through the air traffic service unit (ATSU) suite: — aeronautical/airline operational control (AOC); — air traffic control (ATC).	X									
(03)		Explain the purpose of the following parts of the on-board equipment: — ATSU communications computer; — control and display unit (CDU)/multifunction control and display unit (MCDU); — data communication display unit (DCDU); — ATC message visual annunciator; — printer.	X									
(04)		Give examples of airline operations communications (AOC) data-link messages such as: — out of the gate, off the ground, on the ground, into the gate (OOOI); — load sheet; — passenger information (connecting flights);	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- weather reports (METAR, TAF); - maintenance reports (engine exceedances); - aircraft technical data; - free-text messages.										
(05)		Give examples of ATC data-link messages such as: - departure clearance; - oceanic clearance; - digital ATIS (D-ATIS); - controller-pilot data-link communications (CPDLC).	X									
022 10 02 00		Future air navigation systems (FANSs)										
022 10 02 01		Versions, applications, CPDLC messages, ADS contracts										
(01)		Describe the existence of the ICAO communication, navigation, surveillance/air traffic management (CNS/ATM) concept.	X									
(02)		Explain the two versions of FANSs: - FANS A/FANS 1 using the ACARS network; - FANS B/FANS 2 using the ACARS network and the aeronautical telecommunication network (ATN).	X									
(03)		List and explain the following FANS A/FANS 1 applications: - ATS facility notification (AFN); - automatic dependent surveillance (ADS);	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— CPDLC.										
(04)		Compare the ADS application with the secondary surveillance radar function, and the CPDLC application with VHF communication systems.	X									
(05)		State that an ATCU can use the ADS application only, or the CPDLC application only, or both of them (not including AFN).	X									
(06)		Describe the AFN process for logging on with an ATCU and typical data that will be included in the message.	X									
(07)		Describe typical types of CPDLC messages and the typical pilot work practices when requesting or accepting a CPDLC clearance.	X									
(08)		List and describe the different types of ADS contracts that are controlled by the ATCU and beyond the control of the pilot: — periodic: data sent at set time intervals; — on demand: data sent when requested; — on event: data sent when an event occurs (e.g. heading change, climb initiated, etc.); — emergency mode.	X									
(09)		Describe the purpose of the ADS emergency mode contract and highlight the difference to the ATCU controlled contracts.	X									
022 11 00 00		FLIGHT MANAGEMENT SYSTEM (FMS)/ FLIGHT MANAGEMENT AND GUIDANCE SYSTEM (FMGS)										
022 11 01 00		Design										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
022 11 01 01		Purpose, architecture, failures, functions										
(01)		Explain the purpose of an FMS.	X		X	X		X				
(02)		Describe a typical dual FMS architecture including the following components: — flight management computer (FMC); — CDU/MCDU; — cross-talk bus.	X		X	X						
(03)		Describe the following failures of a dual FMS architecture and explain the potential implications to the pilots: — failure of one FMC; — failure of one CDU/MCDU; — failure of the cross-talk bus.	X		X	X						
(04)		Describe how the FMS integrates with other systems and gathers data in order to provide outputs depending on its level of complexity.	X		X	X		X				
(05)		Explain how the FMS may provide the following functions: — navigation; — lateral and vertical flight planning; — performance parameters.	X	X	X	X		X				
022 11 02 00		FMC databases										
022 11 02 01		Navigation database										
(01)		Explain the purpose of, and describe typical content of, the navigation database.	X		X	X		X	X	3	1, 2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(02)		Describe the 28-day aeronautical information regulation and control (AIRAC) update cycle of the navigation database and explain the reason for having two navigation databases (one active, one standby) and the implication this has to the pilot.	X		X	X		X	X	3	1, 2	
(03)		Explain the purpose of typical user-defined waypoints such as: — latitude/longitude coordinates; — place/bearing/distance (PBD); — place/bearing place/bearing (PBX); — place/distance (PD).	X		X	X		X				
(04)		Explain that the pilot cannot change or overwrite any of the data in the navigation database and that any user-defined waypoints, routes and inputted data will be erased when a different database is activated.	X		X	X		X	X	3	1, 2	
(05)		Explain the threats and implications to the pilot of changing the database by error either on the ground or while flying.	X		X	X		X		3	1	
022 11 02 02		Aircraft performance database										
(01)		Explain the purpose of, and describe the typical content of, the aircraft performance database.	X		X	X		X				
(02)		Explain the importance of verifying that the aircraft performance database is based on the correct data, such as engine type and aircraft variant.	X		X	X		X				
(03)		Explain that the contents of the aircraft performance database cannot be modified by the pilot.	X		X	X		X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(04)		Explain the purpose of performance factor and how it influences the calculations.	X		X	X		X				
(05)		Explain the purpose of cost index (CI) and how it influences the calculations.	X									
022 11 03 00		Operations, limitations										
022 11 03 01		Data, calculations, position inputs, raw data										
(01)		Describe typical data that may be provided by the FMS: — lateral and vertical navigation guidance; — present position; — time predictions; — fuel predictions; — altitude/flight level predictions.	X		X	X		X				
(02)		Explain how the FMS will use a combination of inputted/database and measured data in order to calculate projections and provide output data.	X		X	X		X				
(03)		Explain the issues and threats using inputted/database data and give examples of consequences of inputting data incorrectly/using incorrect data.	X		X	X		X			2, 3	
(04)		Describe fuel consumption calculations during standard operations and explain typical data that will have an influence on the accuracy of the calculations.	X		X	X					2, 3	
(05)		Explain the implications on the accuracy of the calculations during flight in abnormal configurations (such as engine out, gear down, flaps extended,	X		X	X					2, 3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		spoilers extended, etc.) if the FMS is unable to detect the failure.										
(06)		Describe and explain the purpose of an FMS having dedicated radio-navigation receivers that it will tune automatically.	X		X	X						
(07)		Explain typical position inputs to an FMS: — GPS; — IRS; — DME; — VOR; — LOC; — runway threshold (RWY THR).	X		X	X			X	3		
(08)		Explain how the FMS will create its own FMS position fix and that the FMS calculations will be based on the FMS position. Depending on the type of system, the FMS position may be calculated from: — a single source of position data where the most accurate data available at a given time will be used; — multiple sources from which a position will be derived using the combined inputs.	X		X	X						
(09)		Explain the implications of a reduction in available position inputs to the FMS, especially GPS in relation to the capability of performing RNP/PBN approaches.	X		X	X						
(10)		Explain the difference between following the FMS data compared to following raw data from	X		X	X		X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		radio-navigation receivers and describe how there may be limitations for using FMS data as primary source to follow an instrument approach procedure (IAP) such as LOC, VOR or NDB.										
022 11 04 00		Human-machine interface (control and display unit (CDU)/ multifunction control and display unit (MCDU))										
022 11 04 01		Purpose, scratchpad, data input, set-up process										
(01)		Describe the purpose of a CDU/MCDU.	X		X	X		X				
(02)		Describe the typical layout of a CDU/MCDU and the general purpose of the following: — screen; — line select keys; — menu select keys; — alphanumeric keys.	X		X	X		X				
(03)		Explain the function of the ‘scratchpad’ part of the screen.	X		X	X		X				
(04)		Describe how input of some data is compulsory for the function of the FMS and other data is optional, and that different symbology is used to highlight this: — rectangular boxes = compulsory information; — dashed line = optional information.	X		X	X		X				
(05)		Describe a typical FMS pre-flight set-up process through the CDU/MCDU to cover the most basic information (with the aim to create awareness of required information as this is irrespective of aircraft type and FMS/FMGS make):	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- ident page (who am I = aircraft type/variant, engine type/rating and appropriate navigation database); - position initialisation (where am I = position for aligning the IRS and FMS position); - route initialisation (where am I going to = place of departure/destination and alternate(s)); - route programming (how will I get there = SIDs, STARS, route (company or otherwise)); - performance initialisation (when will I arrive = weights, flap setting, FLEX/assumed temperature/derate, take-off speeds).										
022 12 00 00		ALERTING SYSTEMS, PROXIMITY SYSTEMS										
022 12 01 00		General										
022 12 01 01		Alerting systems according to CS-25 and CS-29										
(01)		State definitions, category, criteria and characteristics of alerting systems according to CS-25/AMC 25.1322 for aeroplanes and CS-29 for helicopters as appropriate.	X	X	X	X	X					
022 12 02 00		Flight warning systems (FWSs)										
022 12 02 01		Annunciations, master warning, master caution, advisory										
(01)		State the annunciations given by the FWS and typical location for the annunciator(s): - master warning; - master caution;	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— advisory.										
(02)		Explain master warning: — colour of annunciator: red; — nature of aural alerts: continuous; — typical failure scenarios triggering the alert.	X	X	X	X	X	X				
(03)		Explain master caution: — colour of the annunciator: amber or yellow; — nature of aural alerts: attention-getter; — typical failure scenarios triggering the alert.	X	X	X	X	X	X				
(04)		Describe a typical procedure following a master warning or master caution alert: — acknowledging the failure; — silencing the aural warning; — initiating the appropriate response/procedure.	X	X	X	X	X	X				
(05)		Explain advisory: — colour of the annunciator: any other than red, amber, yellow or green; — absence of aural alert; — typical scenarios triggering the advisory.	X	X	X	X	X	X				
022 12 03 00		Stall warning systems (SWSs)										
022 12 03 01		Function, types, components										
(01)		Describe the function of an SWS and explain why the warning must be unique.	X	X								
(02)		Describe the different types of SWSs.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(03)		List the main components of an SWS.	X	X								
(04)		Explain the difference between the stall warning speed and the actual stalling speed of the aeroplane.	X	X								
022 12 04 00		Stall protection										
022 12 04 01		Function, types										
(01)		Describe the function of a stall protection system.	X									
(02)		Describe the different types of stall protection systems including the difference between mechanical and FBW controls.	X									
(03)		Explain the difference between an SWS and a stall protection system.	X									
022 12 05 00		Overspeed warning										
022 12 05 01		Purpose, aural warning, VMO/MMO pointer										
(01)		Explain the purpose of an overspeed warning system (V _{MO} /M _{MO} pointer).	X	X								
(02)		State that for large aeroplanes, an aural warning must be associated to the overspeed warning if an electronic display is used (see AMC 25.11, paragraph 10.b(2), p. 2-GEN-22).	X	X								
(03)		Describe and give examples of VMO/MMO pointer: barber's/barber pole pointer, barber's/barber pole vertical scale.	X	X								
022 12 06 00		Take-off warning										
022 12 06 01		Purpose										
(01)		Explain the purpose of a take-off warning system and list the typical abnormal situations which generate a warning (see AMC 25.703, paragraphs 4 and 5).	X									
022 12 07 00		Altitude alert system										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
022 12 07 01		Function, displays, alerts										
(01)		Describe the function of an altitude alert system.	X	X	X	X	X	X				
(02)		Describe different types of displays and possible alerts.	X	X	X	X	X	X				
022 12 08 00		Radio altimeter										
022 12 08 01		Purpose, range, displays, incorrect indications										
(01)		Explain the purpose of a low-altitude radio altimeter.	X	X	X	X	X	X				
(02)		Describe the principle of the distance (height) measurement.	X	X	X	X	X	X				
(03)		Describe the different types of radio-altimeter displays.	X	X	X	X	X	X				
(04)		Describe how the radio altimeter provides input to other systems and how a radio-altimeter failure may impact on the functioning of these systems.	X	X	X	X	X	X				
(05)		State the range of a radio altimeter.	X	X	X	X	X	X				
(06)		Explain the potential implications of a faulty radio-altimeter and how this in particular may affect the following systems: — autothrust (flare/retard); — ground-proximity warning systems (GPWSs).	X	X				X				
022 12 09 00		Ground-proximity warning systems (GPWSs)										
022 12 09 01		GPWSs: design, operation, indications										
(01)		Explain the purpose of GPWSs.	X		X	X						
(02)		Explain inputs and outputs of a GPWS and describe its operating principle.	X		X	X						
(03)		List and describe the different modes of operation of a GPWS.	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
022 12 09 02		<i>Terrain-avoidance warning system (TAWS); other name: enhanced GPWS (EGPWS)</i>										
(01)		Explain the purpose of a TAWS for aeroplanes and of a HTAWS for helicopters, and explain the difference from a GPWS.	X		X	X						
(02)		Explain inputs and outputs of a TAWS/HTAWS and describe its working principle.	X		X	X						
(03)		Give examples of terrain displays and list the different possible alerts.	X		X	X						
(04)		Give examples of time response left to the pilot according to look-ahead distance, speed and aircraft performances.	X		X	X						
(05)		Explain why the TAWS/HTAWS must be coupled to a precise-position sensor.	X		X	X						
(06)		Explain the possibility of triggering spurious TAWS/HTAWS warnings as a result of mismanaging the flight path in the proximity to obstacles: — high rate of descent; — high airspeed; — a combination of high rate of descent and high airspeed.	X		X	X						
022 12 09 03		<i>Intentionally left blank</i>										
022 12 10 00		ACAS/TCAS										
022 12 10 01		<i>Principles and operations</i>										
(01)		State that ACAS II is an ICAO standard for anti-collision purposes.	X	X	X	X	X	X				
(02)		Explain that ACAS II is an anti-collision system and does not guarantee any specific separation.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(03)		Describe the purpose of an ACAS II system as an anti-collision system.	X	X	X	X	X	X				
(04)		Describe the following outputs from a TCAS: — other intruders; — proximate intruders; — traffic advisory (TA); — resolution advisory (RA).	X	X	X	X	X	X				
(05)		State that ACAS II will issue commands in the vertical plane only (climb, descent or maintain), and that the commands are complied with as a manual manoeuvre.	X	X	X	X	X	X				
(06)		Explain that an RA may or may not require any active control input and the implications of reacting instinctively without awareness of actual control inputs required to comply with the RA.	X	X	X	X	X	X				
(07)		Explain that if two aircraft are fitted with ACAS II, the RA will be coordinated.	X	X	X	X	X	X				
(08)		State that ACAS II equipment can take into account several threats simultaneously.	X	X	X	X	X	X				
(09)		State that a detected aircraft without altitude-reporting can only generate a TA; describe typical type of traffic and how this can create distractions during flight in certain areas of significant air traffic activity.	X	X	X	X	X	X				
(10)		Describe the interaction between the TCAS II system and the transponder, radio altimeter and the air-data computer: — antenna used;	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— computer and links with radio altimeter, air-data computer and mode-S transponder.										
(11)		Explain the principle of TCAS II interrogations.	X	X	X	X	X	X				
(12)		State the typical standard detection range for TCAS II: — 35–40 NM horizontally; — approximately 2 000 ft above and below (any setting); — extension to approximately 10 000 ft above (ABV selected) or approximately 10 000 ft below (BLW selected).	X	X	X	X	X	X				
(13)		Explain the principle of ‘reduced surveillance’.	X	X	X	X	X	X				
(14)		Explain that in high-density traffic areas the range may automatically be decreased in order to enable detection of the threats in the proximity of the aircraft due to a limitation of the maximum number of possible intruders the system is able to process.	X	X	X	X	X	X				
(15)		Identify the equipment which an intruder must be fitted with in order to be detected by TCAS II.	X	X	X	X	X	X				
(16)		Explain in the anti-collision process: — the criteria used to trigger an alarm (TA or RA) are the time to reach the closest point of approach (CPA) (called TAU) and the difference of altitude; — an intruder will be classified as ‘proximate’ when being less than 6 NM and 1 200 ft from the TCAS-equipped aircraft;	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- the time limit to CPA is different depending on aircraft altitude, is linked to a sensitivity level (SL), and state that the value to trigger an RA is from 15 to 35 seconds; - in case of an RA, the intended vertical separation varies from 300 to 600 ft (700 ft above FL420), depending on the SL; - below 1 000 ft above ground, no RA can be generated; - below 1 450 ft (radio-altimeter value) 'increase descent' RA is inhibited; - at high altitude, performances of the type of aircraft are taken into account to inhibit 'climb' and 'increase climb' RA.										
(17)		List and interpret the following information available from TCAS: - the different possible statuses of a detected aircraft: 'other', 'proximate', 'intruder'; - the appropriate graphic symbols and their position on the horizontal display; - different aural warnings.	X	X	X	X	X	X				
(18)		Explain the indications of a TA and an RA and how an RA will generate a red area on the VSI. Some variants will also include a green area. To manoeuvre the aircraft to comply with the RA, the pilot should 'avoid the red' or 'fly the green'.	X	X	X	X	X	X				
(19)		Explain that the pilot must not interpret the horizontal track of an intruder upon the display.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
022 12 11 00		Rotor/engine overspeed alert system										
022 12 11 01		Design, operation, displays, alarms										
(01)		Describe the basic design principles, operation, displays and warning/alarm systems fitted to different helicopters.			X	X	X					
022 13 00 00		INTEGRATED INSTRUMENTS — ELECTRONIC DISPLAYS										
022 13 01 00		Electronic display units										
022 13 01 01		Design, limitations										
(01)		List the different technologies used, e.g. CRT and LCD, and the associated limitations: — cockpit temperature; — glare; — resolution.	X	X	X	X	X	X	X	1		
022 13 02 00		Mechanical integrated instruments										
022 13 02 01		Attitude and director indicator (ADI)/ horizontal situation indicator (HSI)										
(01)		Describe an ADI and an HSI.	X	X	X	X	X	X	X	1		
(02)		List all the information that can be displayed on either instrument.	X	X	X	X	X	X	X	1		
022 13 03 00		Electronic flight instrument systems (EFISs)										
022 13 03 01		Design, operation										
(01)		List the following parts of an EFIS: — control panel; — display units; — symbol generator; — remote light sensor.	X	X	X	X	X	X		1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(02)		Describe the typical layout of the EFIS display units and how there may be a facility to transfer the information from one display unit on to another if a display unit fails.	X	X	X	X	X	X				
(03)		Explain the need for standby instruments to supplement the EFIS in the event of all the display units failing and the challenge of using these standby instruments, namely their size and position on the flight deck.	X	X	X	X	X	X		1		
(04)		Explain the difference between a symbol generator failing and a display unit failing, and the implications if there are redundant symbol generators available.	X	X	X	X	X	X		1		
(05)		Describe the purpose of an EFIS control panel and typical selections that may be available: — altimeter pressure setting; — navigation display (ND) mode selector; — ND range selector; — ND data selector (waypoints, facilities, constraints, data, etc.); — radio-navigation aids selector (VOR 1/2 or ADF 1/2); — decision altitude (DA)/decision height (DH) selection.	X	X	X	X	X	X		1		
022 13 03 02		Primary flight display (PFD), electronic attitude director indicator (EADI)										
(01)		Describe that a PFD (or an EADI) presents a dynamic colour display of all the parameters necessary to control the aircraft, and that the main layout conforms with the 'basic T' principle:	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- attitude information in the centre; - airspeed information on the left; - altitude information on the right; - heading/track indication lower centre; - flight mode annunciation; - basic T; - take-off and landing reference speeds; - minimum airspeed; - lower selectable airspeed; - Mach number.										
(02)		Describe the typical design of the attitude information: - artificial horizon with aircraft symbol; - superimposed flight director command bars.	X	X	X	X	X	X	X	1		
(03)		Describe the typical design of the speed tape: - rolling speed scale with numerical read-out of current speed; - limiting airspeeds according to configuration; - speed trend vector; - bug/indication for selected airspeed.	X	X	X	X	X	X	X	1		
(04)		Explain the Mach number indications and how a selected Mach number is presented with the speed bug on a corresponding IAS on the speed tape with	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		the Mach number shown as a numerical indication outside the speed tape.										
(05)		Describe the typical design of the altitude information: — rolling altitude scale with numerical read-out of current altitude; — altimeter pressure setting; — bug/indication for selected altitude; — means of highlighting the altitude if certain criteria are met.	X	X	X	X	X	X	X	1		
(06)		Describe the typical design of the heading/track information: — rolling compass scale/rose with numerical read-out of current heading/track; — bug/indication for selected heading/track.	X	X	X	X	X	X	X	1		
(07)		Describe the typical design and location of the following information: — flight mode annunciators (FMAs); — vertical speed indicator including TCAS RA command indications; — radio altitude; — ILS localiser/glideslope and RNP/PBN, GBAS or SBAS horizontal/vertical flight path deviation indicator; — decision altitude/height (DA/H).	X	X	X	X	X	X	X	1	3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
022 13 03 03		Navigation display (ND), electronic horizontal situation indicator (EHSI)										
(01)		Describe that an ND (or an EHSI) provides a mode-selectable colour flight ND.	X	X	X	X	X	X	X	1		
(02)		List the following four modes typically available to be displayed on an ND unit: — MAP (or ARC); — VOR (or ROSE VOR); — APP (or ROSE LS); — PLAN.	X	X	X	X	X	X		1		
(03)		List and explain the following information that can be displayed with the MAP (or ARC) mode selected on an ND unit: — aircraft symbol, compass scale and range markers; — current heading and track (either one may be 'up' depending on selection), true or magnetic; — selected heading and track; — TAS/GS; — wind direction and speed (W/V); — raw data radio magnetic indicator (RMI) needles/pointers for VOR/automatic direction-finding equipment (ADF), if selected, including the frequency or ident of the selected navigation facility;	X	X	X	X	X	X		1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— route/flight plan data from the FMS; — TO/next waypoint data from the FMS; — data from the navigation database such as airports, waypoints or navigation facilities as selected; — weather radar information; — TCAS traffic information (no TCAS commands); — TAWS (EGPWS) terrain information; — failure flags and messages.										
(04)		List and explain the following information that can be displayed with the VOR or APP (or ROSE VOR or ROSE LS) mode selected on an ND unit: — aircraft symbol and compass scale; — current heading and track (either one may be 'up' depending on selection), true or magnetic; — selected heading and track; — TAS/ground speed (GS); — wind direction and speed (W/V); — VOR or ILS frequency and identification of the selected navigation aid; — VOR selected course, deviation indicator and a TO/FROM indicator in a HSI-type display format when in VOR mode;	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— localiser selected course, deviation indicator and glideslope indicator in a HSI-type display format when in APP mode. — weather radar information; — TCAS traffic information (no TCAS commands); — TAWS (EGPWS) terrain information; — failure flags and messages.										
(05)		List and explain the following information that can be displayed with the PLAN mode selected on an ND unit: — north-up compass rose and range markers; — aircraft symbol oriented according to aircraft heading; — TAS/GS; — wind direction and speed (W/V); — route/flight plan data from the FMS; — TO/next waypoint data from the FMS; — data from the navigation database such as airports, waypoints or navigation facilities as selected; — failure flags and messages.	X	X	X	X	X	X				
(06)		Explain the purpose of PLAN mode and its characteristics such as: — no compass information;	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— north is up on the display unit at all times; — the centre waypoint is the selected waypoint on the FMS CDU; — scrolling through the flight plan on the FMS CDU will shift the map view along the flight path; — the aircraft symbol will be positioned in the appropriate place along the flight path; — using PLAN mode as the primary mode during flight may lead to disorientation and loss of situational awareness.										
(07)		Distinguish the difference between the appearance of an EXPANDED or FULL/ROSE mode and how the displayed range differs between them.	X	X	X	X	X	X				
(08)		Explain the combination of mode and range selection including how selecting the appropriate range and displayed data can improve situational awareness for a given phase of flight.	X	X	X	X	X	X				
022 13 04 00		Engine parameters, crew warnings, aircraft systems, procedure and mission display systems										
022 13 04 01		Purposes of systems, display systems, checklists										
(01)		State the purpose of the following systems: — engine instruments centralised display unit; — crew alerting system/aircraft display unit; — facility for appropriate on-screen checklists;	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- that the aircraft systems display unit enables the display of normal and degraded modes of operation of the aircraft systems; - that the systems/aircraft display unit is able to show pictorial systems diagrams/schematics and associated parameters.										
(02)		Describe the similarities to EFIS with regard to basic system architecture.	X		X	X						
(03)		Give the following different names by which engine parameters, crew warnings, aircraft systems and procedures display systems are known: - multifunction display unit (MFDU); - engine indication and crew alerting systems (EICASs); - engine and warning display (EWD); - electronic centralised aircraft monitor (ECAM); - systems display (S/D).	X									
(04)		Give the names of the following different display systems and describe their main functions: - vehicle engine monitoring display (VEMD); - integrated instruments display system (IIDS).			X	X						
(05)		State the purpose of a mission display unit.			X	X						
(06)		Describe the architecture of each system and give examples of display.			X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(07)		Explain why awareness of the consequences of the actions commanded by the automatic checklist is required.	X		X	X						
(08)		Explain the limited ability of the computer to assess a situation other than using the exceedance of certain thresholds to trigger the main and subsequent events and programmed actions.	X		X	X						
(09)		Describe an appropriate procedure for following an on-screen checklist associated with a failure scenario including the following: — confirm the failure with the other flight crew member prior to performing any of the actions; — seek confirmation prior to manipulating any guarded switches or thrust levers; — follow the checklist slowly and methodically; — assess the possible implications of making certain selections, such as opening the fuel cross-feed if there is a fuel leak even though the electronic checklist may ask for the action.	X		X	X						
022 13 05 00		Engine first limit indicator										
022 13 05 01		Design, operation, information on display										
(01)		Describe the principles of design and operation, and compare the different indications and displays available.			X	X	X					
(02)		Describe what information can be displayed on the screen, when the screen is in the limited composite mode.			X	X	X					
022 13 06 00		Electronic flight bag (EFB)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
022 13 06 01		Purpose, certification, malfunctions										
(01)		Explain the purpose of the EFB and list typical equipment: — computer laptop; — tablet device; — integrated avionics suite in the aircraft.	X	X	X	X	X	X		1		
(02)		Describe the ‘class’ hardware certification: — portable: portable electronic device (PED) that can be used inside or outside the aircraft, is not part of the certified aircraft configuration and does not require tools to remove it from the flight-deck cradle, if one exists; — installed: an electronic device that is considered an aircraft part covered by the aircraft airworthiness approval, thus is a minimum equipment list (MEL) item in the event of failure.	X		X	X						
(03)		Describe the ‘type’ software certification: — type A: applications whose misuse or malfunctions have no adverse effect on flight safety; — type B: applications for which evaluation of the hazards presented by misuse or malfunctions is required.	X		X	X						
(04)		Explain implications of malfunctions with the EFB installation in a fully electronic flight-deck environment:	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— mass and balance calculations; — performance calculations; — access to charts; — access to manuals.										
022 13 07 00		Head-up display (HUD), synthetic vision system (SVS) and enhanced visual system (EVS)										
022 13 07 01		Components, benefits, modes of operation										
		State the components of a typical HUD installation: — HUD projector and stowable combiner; — HUD controls such as declutter and dimmer; — HUD computer.	X		X	X						
		Explain the reasons and benefits of having an HUD: — increased situational awareness due to reduced need to look inside to view primary flight information; — lower minima for both departure and landing; — improved accuracy of flying thus reduced susceptibility to enter a state of aircraft upset.	X		X	X						
		Describe how the HUD replicates the information on the primary flight display (PFD) by showing the following data: — altitude; — speed, including speed trend; — heading;	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— flight path vector (track and vertical flight path); — flight mode annunciator (FMA); — CAS, TAWS and wind shear command annunciations.										
		Describe the following modes of operation of an HUD: — normal display mode that may automatically adapt the information based on the phase of flight; — declutter function.	X		X	X						
		Describe the principle of SVS: — an enhanced database used as reference to provide terrain and ground features to be shown on the PFD; — limitations due to being a synthetic image not based on actual sensory information thus not lowering landing minima; — implications if aircraft position accuracy becomes reduced.	X		X	X						
		Describe the principle of EVS: — includes external sensors such as infrared cameras to generate a real-time image on the PFD or on the HUD; — limitation of the fact that an infrared camera uses temperature and temperature difference in order to produce an image;	X		X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— enables lower minima because of the real-time image, thus enhancing the visibility as experienced by the pilot.										
022 14 00 00		MAINTENANCE, MONITORING AND RECORDING SYSTEMS										
022 14 01 00		Cockpit voice recorder (CVR)										
022 14 01 01		<i>Purpose, components, parameters</i>										
(01)		Describe the purpose of a CVR, its typical location, and explain the implications of knowingly erasing or tampering with any information or equipment.	X	X	X	X	X					
(02)		List the main components of a CVR: — a shock-resistant tape recorder or digital storage associated with an underwater locating beacon (ULB); — a cockpit area microphone (CAM); — a control unit with the following controls: auto/on, test and erase, and a headset jack; — limited flight-deck controls such as erase and test switches.	X	X	X	X	X					
(023)		List the following main parameters recorded on the CVR: — voice communications transmitted from or received on the flight deck; — the aural environment of the flight deck; — voice communication of flight crew members using the aeroplane's interphone system;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		- voice or audio signals introduced into a headset or speaker; - voice communication of flight crew members using the public address system, if installed.										
022 14 02 00		Flight data recorder (FDR)										
022 14 02 01		Purpose, components, parameters										
(02)		Describe the purpose of an FDR and its typical location.	X	X								
(02)		List the main components of an FDR: - a shock-resistant data recorder associated with a ULB; - a data interface and acquisition unit; - a recording system (digital flight data recorder); - two control units (start sequence, event mark setting); - limited flight-deck controls, but includes an event switch.	X	X								
(02)		List the following main parameters recorded on the FDR: - time or relative time count; - attitude (pitch and roll); - airspeed; - pressure altitude; - heading;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— normal acceleration; — propulsive/thrust power on each engine and flight-deck thrust/power lever position, if applicable; — flaps/slats configuration or flight-deck selection; — ground spoilers or speed brake selection.										
(02)		State that additional parameters can be recorded according to FDR capacity and applicable operational requirements.	X									
022 14 03 00		Maintenance and monitoring systems										
022 14 03 01		Helicopter operations monitoring program (HOMP): design, operation, performance										
(01)		Describe the HOMP as a helicopter version of the aeroplane flight data monitoring (FDM) program.			X	X						
(02)		State that the HOMP software consists of three integrated modules: — flight data events (FDEs); — flight data measurements (FDMs); — flight data traces (FDTs).			X	X						
(03)		Describe and explain the information flow of an HOMP.			X	X						
(04)		Describe HOMP operation and management processes.			X	X						
022 14 03 02		Integrated health and usage monitoring system (IHUMS): design, operation, performance										
(01)		Describe the main features of an IHUMS:			X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
		— rotor system health; — cockpit voice recorder (CVR)/flight data recorder (FDR); — gearbox system health; — engine health; — exceedance monitoring; — usage monitoring; — transparent operation; — ground station features; — monitoring; — rotor track and balance; — engine performance trending; — quality controlled to level 2.										
(02)		Describe the ground station features of an IHUMS.			X	X						
(03)		Summarise the benefits of an IHUMS including: — reduced risk of catastrophic failure of rotor or gearbox; — improved rotor track and balance giving lower vibration levels; — accurate recording of flight exceedances; — CVR/FDR allows accurate accident/incident investigation and HOMP; — maintenance cost savings.			X	X						
(04)		State the benefits of an IHUMS and an HOMP.			X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
022 14 03 03		<i>Aeroplane condition monitoring system (ACMS): general, design, operation</i>										
(01)		State the purpose of an ACMS.	X									
(02)		Describe the structure of an ACMS including: — inputs: aircraft systems (such as air conditioning, autoflight, flight controls, fuel, landing gear, navigation, pneumatic, APU, engine), MCDU; — data management unit; — recording unit: digital recorder; — outputs: printer, ACARS or ATSU.	X									
(03)		State that maintenance messages sent by an ACMS can be transmitted without crew notification.	X									
(05)		Explain that data from the ACMS can be used as part of an FDM and safety programme.	X									
(05)		Explain that the FDM program collects data anonymously; however, grave exceedance of parameters may warrant a further investigation of the event by the operator.	X									
(06)		Explain the purpose of FDM as a system for identifying adverse safety trends and tailoring training programmes in order to enhance the overall safety of the operation.	X									
022 15 00 00		DIGITAL CIRCUITS AND COMPUTERS										
022 15 01 00		Digital circuits and computers										
022 15 01 01		<i>General, definitions and design</i>										
(01)		Define a ‘computer’ as a machine for manipulating data according to a list of instructions.	X		X	X		X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL /IR	ATPL	CPL					
(02)		Explain the term ‘bus’ being used as a term for a facility (wiring, optical fibre, etc.) transferring data between different parts of a computer, both internally and externally.	X		X	X		X	X		1	
(03)		Define the terms ‘hardware’ and ‘software’.	X		X	X		X	X		1	
(04)	X	With the help of the relevant 022 references, give examples of airborne computers and list the possible peripheral equipment for each system, such as: — ADC with pitot probe(s), static port(s) and indicators; — FMS with GPS, CDU/MCDU and ND; — GPWS with radio altimeter, ADC and ND.	X		X	X		X	X		1	

SUBJECT 031 – FLIGHT PERFORMANCE AND PLANNING: MASS AND BALANCE – AEROPLANES/HELICOPTERS

ED Decision 2020/018/R

Note that the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The professional pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

(1) DEFINITIONS OF MASSES, LOADS AND INDEXES

Allowed take-off mass

The mass taking into consideration all possible limitations for take-off including restrictions caused by regulated take-off mass and regulated landing mass.

Area load or floor load

The load (or mass) distributed over a defined area. Example units:

- SI: N/m², kg/m²;
- Non-SI: psi, lb/ft².

Basic empty mass (BEM)

The mass of an aircraft plus standard items such as: unusable fuel; full operating fluids; fire extinguishers; emergency oxygen equipment. (The lowest mass that is used in FCL exams.)

Dry operating mass (DOM)

The total mass of an aircraft ready for a specific type of operation excluding all usable fuel and traffic load. This mass includes items such as:

- crew and crew baggage;
- catering and removable passenger service equipment (food, beverages, potable water, lavatory chemicals, etc.);

- special operational equipment (e.g. stretchers, rescue hoist, cargo sling).

Dry operating index (DOI)

The aircraft index at dry operating mass.

Index

An index is a moment reduced in a numerical value by an index formula.

In-flight mass/gross mass

The mass of an aircraft in flight at a specified time.

Landing mass

The mass of an aircraft at landing.

Maximum structural in-flight mass with external loads (applicable to helicopters only)

The maximum permissible total mass of the helicopter with external loads.

Maximum structural landing mass

The maximum permissible total mass of an aircraft at landing under normal circumstances.

Maximum structural mass

The maximum permissible total mass of an aircraft at any time. It will be given only if there is no difference between maximum structural taxi mass, maximum structural take-off mass and maximum structural landing mass.

Maximum structural take-off mass

The maximum permissible total mass of an aircraft at commencement of take-off.

Maximum (structural) taxi mass or maximum (structural) ramp mass

The maximum permissible total mass of an aircraft at commencement of taxiing.

Maximum zero fuel mass

The maximum permissible mass of an aircraft with no usable fuel.

Minimum mass (applicable to helicopters only)

The minimum permissible total mass for specific helicopter operations.

Operating mass

The dry operating mass plus take-off fuel.

Payload

The total mass of passengers, baggage and cargo but excluding any non-revenue load.

Performance-limited landing mass

The mass subject to the destination airfield limitations.

Performance-limited take-off mass

The take-off mass subject to departure airfield limitations.

Ramp mass

See 'taxi mass'.

Regulated landing mass

The lower of performance-limited landing mass and maximum structural landing mass.

Regulated take-off mass

The lower of performance-limited take-off mass and maximum structural take-off mass.

Running (or linear) load

The load (or mass) distributed over a defined length of a cargo compartment irrespective of load width. Example units:

- SI: N/m, kg/m;
- Non-SI: lb/in, lb/ft.

Take-off fuel

The total amount of usable fuel at take-off.

Take-off mass

The mass of an aircraft including everything and everyone carried at the commencement of the take-off for helicopters and take-off run for aeroplanes.

Taxi mass or ramp mass

The mass of an aircraft at the commencement of taxiing.

Traffic load

The total mass of passengers, baggage and cargo, including any non-revenue load.

Zero fuel mass

The dry operating mass plus traffic load.

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
030 00 00 00		FLIGHT PERFORMANCE AND PLANNING										
031 00 00 00		MASS AND BALANCE — AEROPLANES/HELICOPTERS										
031 01 00 00		PURPOSE OF MASS-AND-BALANCE CONSIDERATIONS										
031 01 01 00		Mass limitations										
031 01 01 01		Importance with regard to structural limitations										
(01)	X	Describe the relationship between aircraft mass and structural stress. <i>Remark: See also Subject 021 01 01 00.</i>	X	X	X	X	X					
(02)	X	Describe why mass must be limited to ensure adequate margins of strength.	X	X	X	X	X					
031 01 01 02		Importance with regard to performance <i>Remark: See also Subjects 032/034 and 081/082.</i>										
(01)		Describe the relationship between aircraft mass and aircraft performance.	X	X	X	X	X					
(02)	X	Describe why aircraft mass must be limited to ensure adequate aircraft performance.	X	X	X	X	X					
031 01 02 00		Centre-of-gravity (CG) limitations										
031 01 02 01		Importance with regard to stability and controllability <i>Remark: See also Subjects 081/082.</i>										
(01)	X	Describe the relationship between CG position and stability/controllability of the aircraft.	X	X	X	X	X					
(02)		Describe the consequences if CG is in front of the forward limit.	X	X	X	X	X					
(03)		Describe the consequences if CG is behind the aft limit.	X	X	X	X	X					
031 01 02 02		Importance with regard to performance <i>Remark: See also Subjects 032/034 and 081/082.</i>										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Describe the relationship between CG position and aircraft performance.	X	X	X	X	X					
(02)		Describe the effects of CG position on performance parameters (speeds, altitude, endurance and range).	X	X	X	X	X					
031 02 00 00		LOADING										
031 02 01 00		Terminology										
031 02 01 01		Mass terms										
(01)	X	Define the following mass terms: — basic empty mass; — dry operating mass; — operating mass; — take-off mass; — landing mass; — ramp/taxi mass; — in-flight mass (gross mass); — zero fuel mass.	X	X	X	X	X					
031 02 01 02		Load terms (including fuel terms) <i>Remark: See also Subject 033.</i>										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Define the following load terms: — payload/traffic load; — block fuel; — taxi fuel; — take-off fuel; — trip fuel; — reserve fuel (contingency, alternate, final reserve and additional fuel); — extra fuel.	X	X	X	X	X					
(02)		Explain the relationship between the various load-and-mass components listed in 031 02 01 01 and 031 02 01 02.	X	X	X	X	X					
(03)		Calculate the mass of particular components from other given components.	X	X	X	X	X					
(04)		Convert fuel mass, fuel volume and fuel density given in different units used in aviation.	X	X	X	X	X					
031 02 02 00		Mass limits										
031 02 02 01		Structural limitations										
(01)	X	Define the maximum zero fuel mass.	X	X								
(02)	X	Define the maximum ramp/taxi mass.	X									
(03)	X	Define the maximum take-off mass.	X	X	X	X	X					
(04)	X	Define the maximum in-flight (gross) mass with external load.			X	X	X					
(05)	X	Define the maximum landing mass.	X	X	X	X	X					
031 02 02 02		Performance and regulated limitations										
(01)		Describe the following performance and regulated mass limitations:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— performance-limited take-off mass; — performance-limited landing mass; — regulated take-off mass; — regulated landing mass.										
031 02 02 03		Cargo compartment limitations										
(01)	X	Describe the maximum floor load (maximum load per unit of area).	X	X	X	X	X					
(02)	X	Describe the maximum running load (maximum load per unit of fuselage length).	X	X	X	X	X					
031 02 03 00		Mass calculations										
031 02 03 01		Maximum masses for take-off and landing										
(01)		Calculate the maximum mass for take-off (regulated take-off mass) given mass-and-load components and structural/ performance limits.	X	X	X	X	X					
(02)		Calculate the maximum mass for landing (regulated landing mass) given mass-and-load components and structural/ performance limits.	X	X	X	X	X					
(03)		Calculate the allowed mass for take-off.	X	X	X	X	X					
031 02 03 02		Allowed traffic load and fuel load										
(01)		Calculate the maximum allowed traffic load and fuel load in order not to exceed the given allowed take-off mass.	X	X	X	X	X					
(02)		Calculate ‘under load’/‘over load’ given the allowed mass for take-off, operating mass and actual traffic load.	X	X	X	X	X					
031 02 03 03		Use of standard masses for passengers, baggage and crew										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Extract the appropriate standard masses for passengers, baggage and crew from relevant documents or operator requirements.	X	X	X	X	X					
(02)		Calculate the traffic load by using standard masses.	X	X	X	X	X					
031 03 00 00		INTENTIONALLY LEFT BLANK										
031 04 00 00		MASS-AND-BALANCE DETAILS OF AIRCRAFT										
031 04 01 00		Contents of mass-and-balance documentation										
031 04 01 01		Datum, moment arm										
(01)	X	State where the datum and moment arms for aircraft can be found.	X	X	X	X	X					
(02)	X	Extract the appropriate data from given documents.	X	X	X	X	X					
(03)	X	Define 'datum' (reference point), 'moment arm' and 'moment'.	X	X	X	X	X					
031 04 01 02		CG position as distance from datum										
(01)	X	State where the CG position for an aircraft at basic empty mass can be found.	X	X	X	X	X					
(02)	X	State where the CG limits for an aircraft can be found.	X	X	X	X	X					
(03)		Describe the different forms in presenting CG position as distance from datum or other references.	X	X	X	X	X					
(04)		Explain the meaning of centre of gravity (CG).	X	X	X	X	X					
031 04 01 03		CG position as percentage of mean aerodynamic chord (% MAC) <i>Remark: Knowledge of the definition of MAC is covered under Subject 081 01 01 05.</i>										
(01)		Extract MAC information from aircraft documents.	X	X								
(02)		Explain the principle of using % MAC for the description of the CG position.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Calculate the CG position as % MAC.	X	X								
031 04 01 04		Longitudinal CG limits										
(01)		Extract the appropriate data from given sample documents.	X	X	X	X	X					
031 04 01 05		Lateral CG limits										
(01)		Extract the appropriate data from given sample documents.			X	X	X					
031 04 01 06		Details of passenger and cargo compartments										
(01)		Extract the appropriate data (e.g. seating schemes, compartment dimensions and limitations) from given sample documents.	X	X	X	X	X					
031 04 01 07		Details of fuel system relevant to mass-and-balance considerations										
(01)	X	Extract the appropriate data (e.g. fuel-tank capacities and fuel-tank positions) from given sample documents.	X	X	X	X	X					
(02)		Explain aircraft CG movement as flight progresses given location of fuel tank (inner wing, outer wing, central, additional aft central, horizontal stabiliser) and mass of fuel consumed from that tank and aeroplane's previous CG.	X									
(03)		Explain advantages and risks associated with fuel tanks in the aeroplane's fin or horizontal stabiliser.	X									
031 04 02 00		Determination of aircraft empty mass and CG position by weighing										
031 04 02 01		Weighing of aircraft (general aspects)										
(01)		Describe the general procedure and regulations relating to when an aircraft should be weighed, reweighed or data recalculated.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		<i>Remark: See the applicable operational requirements.</i>										
(02)	X	Extract and interpret entries from/in 'mass (weight) report' of an aircraft.	X	X	X	X	X					
031 04 02 02		Calculation of mass and CG position of an aircraft using weighing data										
(01)		Calculate the mass and CG position of an aircraft from given reaction forces on jacking points.	X	X	X	X	X					
031 04 03 00		Extraction of basic empty mass (BEM) and CG data from aircraft documentation										
031 04 03 01		BEM or dry operating mass (DOM)										
(01)	X	Extract values for BEM or DOM from given documents.	X	X	X	X	X					
031 04 03 02		CG position or moment at BEM/DOM										
(01)		Extract values for CG position and moment at BEM or DOM from given documents.	X	X	X	X	X					
031 04 03 03		Deviations from standard configuration										
(01)		Extract values from given documents for deviation from standard configuration as a result of varying crew, optional equipment, optional fuel tanks, etc.	X	X	X	X	X					
031 05 00 00		DETERMINATION OF CG POSITION										
031 05 01 00		Methods										
031 05 01 01		Arithmetic method										
(01)		Calculate the CG position of an aircraft by using the formula: — CG position = sum of moments / total mass.	X	X	X	X	X					
031 05 01 02		Graphic method										
(01)		Determine the CG position of an aircraft by using the loading graphs given in sample documents.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
031 05 01 03		<i>Index method</i>										
(01)	X	Explain the principle of the index method.	X	X	X	X	X					
(02)		Define the terms ‘index’ and ‘dry operating index’ (DOI), and calculate the DOI given the relevant formula and data.	X	X	X	X	X					
(03)		Explain the advantage(s) of the index method.	X	X	X	X	X					
031 05 02 00		<i>Load and trim sheet</i>										
031 05 02 01		<i>General considerations</i>										
(01)	X	Explain the principle and the purpose of load sheets.	X	X								
(02)	X	Explain the principle and the purpose of trim sheets.	X									
031 05 02 02		<i>Load sheet/balance schedule and CG envelope for light aeroplanes and for helicopters</i>										
(01)		Add loading data and calculate masses in a sample load sheet/balance schedule.	X	X	X	X	X					
(02)		Calculate moments and CG positions.	X	X	X	X	X					
(03)		Check CG position at zero fuel mass and take-off mass to be within the CG envelope including last-minute changes, if applicable.	X	X	X	X	X					
031 05 02 03		<i>Load sheet for large aeroplanes</i>										
(01)		Complete a sample load sheet to determine the ‘allowed mass for take-off’, ‘allowed traffic load’ and ‘under load’.	X									
(02)		Explain the purpose of each load sheet section.	X									
(03)		Explain that the purpose of boxed maximum figures in load sheet sections is to cross-check the actual and limiting mass values.	X									
(04)		Complete and cross-check a sample load sheet.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
031 05 02 04		Trim sheet for large aeroplanes										
(01)		Explain the purpose of the trim sheet and the methods to determine the CG position.	X									
(02)		Check if the zero fuel mass CG or index is within the limits.	X									
(03)		Determine the fuel index by using the 'fuel index correction table' and determine the CG position as % MAC.	X									
(04)		Check that the take-off mass CG or index are within the limits.	X									
(05)		Determine 'stabiliser trim units' for take-off.	X									
(06)		Explain the difference between certified and operational CG limits.	X									
(07)		Determine the zero fuel mass CG or index.	X									
(08)		Explain the relationship between pitch control and CG position and the operational significance.	X									
031 05 02 05		Intentionally left blank										
031 05 02 06		Other methods to present load and trim information										
(01)	X	Describe information from other methods of presenting load and balance information, e.g. aircraft communications addressing and reporting system (ACARS), electronic flight bags (EFBs), and the 'less paper in the cockpit' (LPC) software.	X									
031 05 03 00		Repositioning of CG										
031 05 03 01		Repositioning of CG by shifting the load										
(01)		Calculate the mass to be moved over a given distance, or to/from given compartments, to establish a defined CG position.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Calculate the distance to move a given mass to establish a defined CG position.	X	X	X	X	X					
(03)	X	Describe the methods to check that cargo has been loaded in correct position in relation to the loading manifest, including identifying hazard of cargo loaded in reverse order (visual inspection of one or more unit load devices (ULDs)).	X	X								
(04)		Determine whether CG remains within limits if cargo has been loaded in incorrect order or at incorrect location.	X	X								
031 05 03 02		<i>Repositioning of CG by additional load or ballast or by load or ballast removal</i>										
(01)		Calculate the amount of additional load or ballast to be loaded at or removed from a given position or compartment to establish a defined CG position.	X	X	X	X	X					
(02)		Calculate the loading position or compartment for a given amount of additional load or ballast to establish a defined CG position.	X	X	X	X	X					
031 06 00 00		CARGO HANDLING										
031 06 01 00		Types of cargo										
031 06 01 01		<i>Types of cargo (general aspects)</i>										
(01)		Describe the typical types of cargo, e.g. containerised cargo, palletised cargo, bulk cargo, and the advantages of containerised and palletised cargo.	X	X	X	X	X					
031 06 02 00		Floor-area load and running-load limitations										
031 06 02 01		<i>Floor-area load and running-load limitations in cargo compartments</i>										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Calculate the required floor-contact area for a given load to avoid exceeding the maximum permissible floor load of a cargo compartment.	X	X	X	X	X					
(02)		Calculate the maximum mass of a container with given floor-contact area to avoid exceeding the maximum permissible floor load of a cargo compartment.	X	X	X	X	X					
(03)		Calculate the linear load distribution of a container to avoid exceeding the maximum permissible running load.	X	X	X	X	X					
031 06 03 00		Securement of load										
031 06 03 01		Securement of load (reasons and methods)										
(01)		Explain the reasons to restrain or secure cargo and baggage.	X	X	X	X	X					
(02)		Describe the basic methods to restrain or secure loads (unit load devices secured by latches on roller tracks or to tie down points by straps; bulk cargo restrained by restraining nets attached to attachment points and tie-down points).	X	X	X	X	X					

SUBJECT 032 – FLIGHT PERFORMANCE AND PLANNING – PERFORMANCE – AEROPLANES

ED Decision 2020/018/R

Note that the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The professional pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

For theoretical knowledge examination purposes:

‘climb angle’ is assumed to be air-mass-related;

‘flight-path angle’ is assumed to be ground-related;

‘screen height for take-off’ is the vertical distance between the take-off surface and the take-off flight path at the end of the take-off distance;

‘screen height for landing’ is the vertical distance between the landing surface and the landing flight path from which the landing distance begins.

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
030 00 00 00		FLIGHT PERFORMANCE AND PLANNING										
032 00 00 00		PERFORMANCE — AEROPLANES										
032 01 00 00		GENERAL										
032 01 01 00		Performance legislation										
032 01 01 01		Applicability of airworthiness requirements of CS-23 and CS-25										
(01)	X	Describe the application of certification specification (CSs) with regard to the different kinds of aeroplanes.	X	X								
(02)	X	Describe the general differences between aeroplanes certified according to CS-23 and CS-25.	X									
032 01 01 02		Operational regulations and safety										

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Describe the basic concept that the applicable operational requirements differ depending on aeroplane performance.	X	X								
(02)		Describe the performance classes for commercial air transport according to the applicable operational requirements.	X	X								
032 01 01 03		Performance and safety										
(01)	X	State that aeroplane performance required for commercial air transport may limit the weight of a dispatched aeroplane in order to achieve a sufficient level of safety.	X	X								
(02)	X	Describe that the minimum level of safety required for commercial air transport is ensured through the combination of airworthiness requirements and operational limitations, i.e. the more stringent airworthiness requirements of CS-25 enable a wider range of operating conditions for these aeroplanes.	X	X								
032 01 01 04		Performance definitions and safety factors										
(01)	X	Describe measured performance and explain how it is determined.	X	X								
(02)		Describe gross performance.	X	X								
(03)		Describe net performance and safety factors.	X	X								
(04)	X	Describe that the size of a safety factor depends on the likelihood of the event and the range of the measured performance data.	X	X								
(05)		Describe the relationship between net and gross take-off and landing distances, and net and gross climb and descent gradients.	X	X								
032 01 02 00		General performance theory										
032 01 02 01		Intentionally left blank										

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
032 01 02 02		Definitions and terms										
(01)	X	Define the terms ‘climb angle’ and ‘climb gradient’.	X	X								
(02)	X	Define the terms ‘flight-path angle’ and ‘flight-path gradient’.	X	X								
(03)	X	Define the terms ‘descent angle’ and ‘descent gradient’.	X	X								
(04)	X	Explain the difference between climb/descent angle and flight-path angle.	X	X								
(05)	X	Define ‘absolute ceiling’.	X	X								
(06)		Describe ‘clearway’ and ‘stopway’ according to CS-Definitions.	X	X								
(07)		Describe: — take-off run available (TORA); — take-off distance available (TODA); — accelerate-stop distance available (ASDA); and determine each from given data or appropriate aerodrome charts.	X	X								
(08)		Describe ‘screen height’ including its various values.	X	X								
(09)	X	Define the terms ‘range’ and ‘endurance’.	X	X								
(10)		Define an aeroplane’s ‘specific range’ (SR) in terms of nautical air miles (NAM) per unit of fuel, and ‘specific range over the ground’ (SRG) in terms of nautical ground miles (NGM) per unit of fuel.	X	X								
(11)		Define the power available and power required.	X	X								
032 01 02 03		Variables influencing performance										
(01)	X	Name the following factors that affect aeroplane performance: pressure altitude and temperature, wind, aeroplane weight, aeroplane configuration, aeroplane anti-skid	X	X								

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		status, aeroplane centre of gravity (CG), aerodrome runway surface, and aerodrome runway slope.										
(02)	X	Describe how, for different density altitudes, the thrust and power available vary with speed for a propeller-driven aeroplane.	X	X								
(03)	X	Describe how, for different density altitudes, the thrust and power available vary with speed for a turbojet aeroplane.	X									
(04)		Describe how, for different density altitudes, the drag and power required vary with indicated airspeeds (IAS) and true airspeeds (TAS).	X	X								
(05)		Describe how, for different aeroplane weights and configurations, the drag and power required vary with IAS and TAS.	X	X								
032 01 03 00		Level flight, range and endurance										
032 01 03 01		Steady level flight										
(01)	X	Explain how drag (thrust) and power required vary with speed in straight and level flight.	X	X								
(02)	X	Explain the effect of excess thrust and power on speed in level flight.	X	X								
(03)		Interpret the 'thrust/power required' and 'thrust/power available' curves in straight and level flight.	X	X								
(04)		Describe how the maximum achievable straight and level flight IAS and TAS vary with altitude.	X	X								
(05)		Describe situations in which a pilot may elect to fly for 'maximum endurance' or 'maximum range'.	X	X								
032 01 03 02		Range										

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define a turbojet aeroplane's specific fuel consumption (SFC) and describe how it affects fuel flow and specific range.	X									
(02)		Define a propeller-driven aeroplane's SFC and describe how it affects fuel flow and specific range.	X	X								
(03)		Explain the optimum speed for maximum SR for a turbojet aeroplane in relation to the drag curve.	X									
(04)		Explain the optimum speed to achieve maximum SR for a propeller-driven aeroplane in relation to the power required and drag graphs.	X	X								
(05)		Explain the effect of aeroplane weight and CG position on fuel consumption, range and the optimum speed for maximum SR.	X	X								
(06)		State how a turbojet engine's SFC varies with temperature and thrust setting.	X									
(07)		Explain how SR for a turbojet aeroplane varies with altitude and under different meteorological conditions.	X									
(08)		Explain how SRG for a propeller-driven aeroplane varies with altitude and under different meteorological conditions.	X	X								
(09)		Explain the effect of weight on the optimum altitude for maximum range.	X	X								
(10)		Describe the effect of wind on SRG and the optimum speed for SRG, when compared to SR, and the optimum speed for SR.	X	X								
032 01 03 03		Maximum endurance										
(01)		Explain fuel flow in relation to TAS and thrust for a turbojet aeroplane.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		State the speed for maximum endurance for a turbojet aeroplane.	X									
(03)		Explain fuel flow in relation to TAS and thrust for a propeller-driven aeroplane.	X	X								
(04)		State the speed for maximum endurance for a propeller-driven aeroplane and the disadvantages of holding at this speed (e.g. high angle of attack (AoA) and lack of speed stability).	X	X								
(05)		Explain the effect of wind and altitude on endurance, and the maximum endurance speed for a turbojet aeroplane.	X									
(06)		Explain the effect of wind and altitude on endurance, and the maximum endurance speed for a propeller-driven aeroplane.	X	X								
(07)		Describe the benefits of managing your en-route airspeed to reduce or avoid holding time, and the operational situations when it could be used (commanded by the pilot or air traffic control (ATC), when delays at arrival airport occur).	X	X								
032 01 04 00		Climbing										
032 01 04 01		Climbing (climb performance)										
(01)		Resolve the forces during a steady climb.	X	X								
(02)		Define and explain the following terms: — critical engine; — speed for best angle of climb (V_x); — speed for best rate of climb (V_y).	X	X								
(03)		Explain climb performance in relation to the thrust available and thrust required (angle of climb), and power available and power required (rate of climb).	X	X								

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the meaning and effect of ‘excess thrust’ and ‘excess power’ in a steady climb.	X	X								
(05)		Interpret the ‘thrust/power required’ and ‘thrust/power available’ curves in a steady climb.	X	X								
(06)		State the difference between climb angle and gradient.	X	X								
(07)		Explain the effect of weight on the climb angle and rate of climb, and the speed for best angle and best rate of climb.	X	X								
(08)		Explain the effects of pressure altitude and temperature, including an inversion on climb performance (angle and rate of climb).	X	X								
(09)		Explain the effect of configuration on climb performance (angle and rate of climb, and V_X and V_Y).	X	X								
(10)		Describe the effect of engine failure on climb performance (angle and rate of climb, and V_X and V_Y).	X	X								
(11)		Calculate the all-engine and one-engine-out climb gradient from given values of engine thrust and aeroplane drag and weight.	X	X								
032 01 05 00		Descending										
032 01 05 01		Descending (descent performance)										
(01)		Resolve the forces during steady descent and in the glide.	X	X								
(02)		Explain descent performance in relation to thrust available and thrust required (drag), and power available and power required.	X	X								
(03)		Explain the meaning of ‘excess thrust required’ (excess drag) and ‘excess power required’ in a steady descent.	X	X								

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Interpret the ‘thrust/power required’ and ‘thrust/power available’ curves in a steady descent.	X	X								
(05)		Explain the effect of mass, altitude, wind, speed and configuration on the glide descent.	X	X								
(06)		Explain the effect of mass, altitude, wind, speed and configuration on the powered descent.	X	X								
032 02 00 00		CS-23/APPLICABLE OPERATIONAL REQUIREMENTS PERFORMANCE CLASS B — THEORY										
032 02 01 00		Airworthiness requirements										
032 02 01 01		Airworthiness requirements and definitions										
(01)	X	Define the following speeds: — stall speeds V_S , V_{S0} and V_{S1} ; — rotation speed V_R ; — speed at 50 ft above the take-off surface level; — reference landing speed V_{REF} .	X	X								
(02)		Describe the limitations on V_R , on the speed at 50 ft above the take-off surface and on V_{REF} , and given the appropriate stall speed, estimate the values based on these limitations for a single-engine, class B aeroplane.	X	X								
(03)		Describe the limitations on V_R , on the speed at 50 ft above the take-off surface and on V_{REF} , and given the appropriate stall speed, estimate the values based on these limitations for a multi-engine, class B aeroplane.	X	X								
(04)	X	Describe the European Union airworthiness requirements according to CS-23 relating to aeroplane performance (stall, take-off, climb, landing).	X	X								
(05)		Define and identify the critical engine of a multi-engine propeller aeroplane.	X	X								

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the effect of an engine failure on the power required, the total drag (thrust required) and climb performance of a multi-engine aeroplane.	X	X								
(07)		Explain the effect of engine failure on the minimum control speed of a multi-engine aeroplane under given conditions (temperature and pressure altitude).	X	X								
032 02 02 00		Intentionally left blank										
032 02 03 00		Take-off and landing										
032 02 03 01		Take-off and landing (definitions and effects)										
(01)	X	Define the following distances and masses: — take-off distance; — landing distance; — ground-roll distance; — maximum allowed take-off mass; — maximum allowed landing mass.	X	X								
(02)		Explain the effect of flap-setting on the take-off, landing and ground-roll distances.	X	X								
(03)		Explain the effects of the following runway (RWY) variables on take-off distances: — RWY slope; — RWY surface conditions: dry, wet and contaminated; — RWY elevation.	X	X								
(04)		For both fixed-pitch and constant-speed propeller aeroplanes, explain the effect of airspeed on thrust during the take-off run.	X	X								

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Describe the effects of brake release before take-off power is set on the TOD and ASD.	X	X								
(06)		Explain the effect of wind on take-off and landing distances, and determine the actual headwind/tailwind component given the runway direction, wind speed and direction, by use of wind component graphs, mathematical calculations, and rule of thumb.	X	X								
(07)		Explain why an aeroplane has maximum crosswind limit(s) and determine the crosswind component given the runway direction, wind speed and direction, by use of wind component graphs, mathematical calculations, and rule of thumb.	X	X								
(08)		Explain the percentage of accountability for headwind and tailwind components during take-off and landing calculations.	X	X								
(09)		Explain the effect of runway conditions on the landing distance.	X	X								
(10)		Explain the effects of pressure altitude and temperature on the take-off distance, take-off climb, landing distance and approach climb.	X	X								
(11)		Describe the landing airborne distance and ground-roll distance and estimate the effect on the landing distance when the aeroplane is too fast or too high at the screen.	X	X								
(12)		Describe the take-off flight path for a multi-engine, class B aeroplane.	X	X								
(13)		Describe the dimensions of the take off flight path accountability area (domain).	X	X								
032 02 04 00		Climb, cruise and descent										

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
032 02 04 01		Climb, cruise and descent (requirements and calculations)										
(01)		Describe the climb and en-route requirements according to the applicable operational requirements.	X	X								
(02)		For a single-engine aeroplane, calculate the expected obstacle clearance (in visual meteorological conditions (VMC)) given gross climb performance, obstacle height and distance from reference zero.	X	X								
(03)		For a single-engine aeroplane, calculate the net glide gradient and net glide distance, given aeroplane altitude, terrain elevation, gross gradient or lift/drag ratio (L/D ratio), and headwind or tailwind component.	X	X								
032 03 00 00		CS-23/APPLICABLE OPERATIONAL REQUIREMENTS PERFORMANCE CLASS B — USE OF AEROPLANE PERFORMANCE DATA FOR SINGLE- AND MULTI-ENGINE AEROPLANES										
032 03 01 00		Intentionally left blank										
032 03 02 00		Intentionally left blank										
032 03 03 00		Use of aeroplane performance data										
032 03 03 01		Take-off										
(01)		Determine the field-length-limited take-off mass and take-off speeds given defactored distance, configuration, pressure altitude, temperature and headwind/tailwind component.	X	X								
(02)		Determine the accelerate-go distance and accelerate-stop distance data.	X	X								
(03)		Determine the ground-roll distance and take-off distance from graphs.	X	X								

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Determine the all-engine-out and critical-engine-out take-off climb data.	X	X								
(05)		Determine take off flight path for a MEP aeroplane of given mass and given airfield conditions, and calculate the obstacle clearance based on the take off flight path.	X	X								
(06)		Determine the minimum headwind or maximum tailwind component required for take-off for a given mass and given airfield conditions.	X	X								
(07)		Given take-off run available (TORA), TODA and ASDA, slope and surface conditions, calculate the defactored distance to be used for commercial air transport using the appropriate take-off graphs.	X	X								
(08)		Calculate the minimum TORA or TODA for commercial air transport given the defactored take-off distance or run, runway surface and slope.	X	X								
032 03 03 02		Climb										
(01)		Determine rate of climb.	X	X								
(02)		Calculate obstacle clearance climb data.	X	X								
(03)		Determine the still-air and flight-path gradients for given IAS, altitude, temperature, aeroplane weight and, if relevant, wind component.	X	X								
032 03 03 03		Intentionally left blank										
032 03 03 04		Landing										
(01)		Determine the field-length-limited landing mass and landing speeds given defactored distance, configuration, pressure altitude, temperature and headwind or tailwind component.	X	X								
(02)		Determine landing climb data in the event of balked landing.	X	X								

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Determine landing distance and ground-roll distance for given flap position, aeroplane weight and airfield data.	X	X								
(04)		Calculate, given the landing distance available (LDA), slope and surface type and condition, the defactored distance to be used for commercial air transport using the appropriate landing graphs.	X	X								
(05)		Calculate the minimum landing distance (LD) that must be available for commercial air transport given the defactored landing distance, runway surface and slope.	X	X								
032 04 00 00		CS-25/APPLICABLE OPERATIONAL REQUIREMENTS PERFORMANCE CLASS A — THEORY										
032 04 01 00		Take-off										
032 04 01 01		<i>Take-off performance, definitions of and relationships between terms</i>										
(01)	X	Explain the forces affecting the aeroplane during the take-off run.	X									
(02)	X	State the effects of thrust-to-weight ratio and flap-setting on ground roll.	X									
(03)		Describe the European Union airworthiness requirements according to CS-25 relating to large aeroplane performance (General and Take-off).	X									
(04)		Describe the terms ‘aircraft classification number’ (ACN) and ‘pavement classification number’ (PCN), and the requirements and hazards of operating on aerodrome surfaces with PCNs smaller than the ACNs.	X									
(05)		Define and explain the following speeds in accordance with CS-25 or CS-Definitions: — reference stall speed (V_{SR});	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- reference stall speed in a specific configuration (V_{SR1}); 1-g stall speed at which the aeroplane can develop a lift force (normal to the flight path) equal to its weight (V_{S1g}); - minimum control speed with critical engine inoperative (V_{MC}); - minimum control speed on or near the ground (V_{MCG}); - minimum control speed at take-off climb (V_{MCA}); - engine failure speed (V_{EF}); - take-off decision speed (V_1); - rotation speed (V_R); - take-off safety speed (V_2); - minimum take-off safety speed (V_{2MIN}); - minimum unstick speed (V_{MU}); - lift-off speed (V_{LOF}); - maximum brake energy speed (V_{MBE}); - maximum tyre speed ($V_{Max Tyre}$).										
(06)		Explain the interdependence between the above-mentioned speeds where relevant.	X									
(07)		Define the following distances in accordance with CS-25: take-off run with all engines operating and one-engine-inoperative;	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— take-off distance with all engines operating and one-engine-inoperative; — accelerate-stop distance with all engines operating and one-engine-inoperative.										
(08)		Explain how loss of TORA due to alignment is accounted for.	X									
(09)		Explain the effect of the interdependency of relevant speeds in 032 04 01 01 (05) and the situations in which these interdependencies can cause speed and performance restrictions.	X									
032 04 01 02		Take-off distances										
(01)		Explain the effects of the following runway (RWY) variables on take-off distances: — RWY slope; — RWY surface conditions: dry, wet and contaminated; — RWY elevation.	X									
(02)		Explain the effects of the following aeroplane variables on take-off distance: — aeroplane mass; — take-off configuration; — bleed-air configurations.	X									
(03)		Explain the effects of the following meteorological variables on take-off distances: — wind; — temperature;	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— pressure altitude.										
(04)		Explain the consequence of errors in rotation technique on take-off distance: — early and late rotation; — too high and too low rotation angle; — too high and too low rotation rate.	X									
(05)		Compare the take-off distance for specified conditions and configuration for all engines operating and one-engine-inoperative.	X									
(06)		Explain the effect of using clearway on the field-length-limited take-off mass.	X									
(07)		Explain the influence of aeroplane mass, air density and flap settings on V_1 , V_2 and V_{2MIN} and thereby on take-off distance.	X									
(08)		Explain the effect of an error in V_1 on the resulting one-engine-out take-off distance.	X									
032 04 01 03		Accelerate-stop distance										
(01)		Explain how the accelerate-stop distance is affected by given conditions and configuration for all engines operating and one-engine-inoperative.	X									
(02)		Explain the effect of using a stopway on the field-length-limited take-off mass.	X									
(03)		Explain the effect of an error in V_1 on the resulting accelerate-stop distance.	X									
(04)		Explain the effect of runway slope or wind component on the accelerate-stop distance.	X									
(05)		Explain how the accelerate-stop distance is determined and discuss the deceleration procedure.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain how the accelerate-stop distance is affected by the use of brakes, anti-skid, reverse thrust, ground spoilers (lift dumpers) and by brake energy absorption limits, delayed temperature rise and brake temperature indication.	X									
(07)	X	Explain the hazards of rejecting a take-off from high ground speed or high take-off mass, and how to manage these hazards.	X									
032 04 01 04		Balanced field length concept										
(01)	X	Define the term ‘balanced field length’.	X									
(02)		Describe the relationship between take-off distance and accelerate-stop distance, and identify on a diagram the balanced field length and balanced V_1 .	X									
(03)	X	Describe the applicability of a balanced field length.	X									
032 04 01 05		Unbalanced field length concept										
(01)	X	Describe the applicability of an unbalanced field length.	X									
(02)		Explain the effect of additional stopway on the allowed take-off mass and appropriate V_1 when using an unbalanced field.	X									
(03)		Explain the effect of additional clearway on the allowed take-off mass and appropriate V_1 when using an unbalanced field.	X									
032 04 01 06		Field-length-limited take-off mass (FLLTOM)										
(01)		Explain the factors that affect the FLLTOM.	X									
(02)		Explain the concept of a ‘range of V_1 ’ and explain reasons for the placement of the designated V_1 towards the faster or slower end of the range.	X									
032 04 01 07		Contaminated runways <i>Remark: See 010 09 02 04 and 071 02 13 02</i>										

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define a 'contaminated runway', 'wet runway', and a 'dry runway'.	X	X								
(02)		Describe the different types of contamination: wet or water patches, frost-covered, dry snow, wet snow, slush, ice, compacted or rolled snow, frozen ruts or ridges.	X	X								
(03)		<i>Intentionally left blank</i>										
(04)		<i>Intentionally left blank</i>										
(05)		Define the different types of hydroplaning.	X	X								
(06)		Explain the difference between the two dynamic hydroplaning speeds and state which of them is the most limiting for an aircraft operating on a wet runway.	X	X								
(07)		State that some wind limitations may apply in case of contaminated runways. Those limitations are to be found in Part B of the Operations Manual — Limitations.	X	X								
(08)		State that the procedures associated with take-off and landing on contaminated runways are to be found in Part B of the Operations Manual — Normal procedures.	X	X								
(09)		State that the performance associated with contaminated runways is to be found in Part B of the Operations Manual — Performance.	X	X								
032 04 01 08		Take-off climb										
(01)		Explain the difference between the flat-rated and non-flat-rated part in performance charts.	X									
(02)		State the differences in climb-gradient requirements for two-, three- and four-engined aeroplanes.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the effects of aeroplane configuration and meteorological conditions on the take-off climb.	X									
(04)		Determine the climb-limited take-off mass.	X									
032 04 01 09		Obstacle-limited take-off										
(01)		Describe the operational regulations for obstacle clearance in the net take-off flight path (NTOFP).	X									
(02)		Define the actual and NTOFP with one-engine-inoperative in accordance with CS-25.	X									
(03)		Explain the effects of aeroplane configuration and meteorological conditions on the obstacle-limited take-off mass.	X									
(04)		Describe the segments of the actual take-off flight path.	X									
(05)		Describe the changes in the configuration, power, thrust and speed in the NTOFP climb segments.	X									
(06)		State the standard maximum bank angle(s) in the first and second segment, and determine the effect on the stall speed and implication on V ₂ .	X									
(07)		Explain the influence of airspeed selection, acceleration and turns on the climb gradient.	X									
(08)		Describe the European Union airworthiness requirements according to CS-25 relating to aeroplane performance take-off climb and flight path.	X									
032 04 01 10		Performance-limited take-off mass (PLTOM) and regulated take-off mass (RTOM) tables										
(01)		Define PLTOM and RTOM.	X									
(02)	X	Describe the use of RTOM tables or similar to find PLTOM and how this can also be done using an EFB.	X									
(03)		Interpret what take-off limitation (field length, obstacle, climb, structural, etc.) is restricting a	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		particular RTOM as it is presented in RTOM tables or similar.										
(04)		Describe why data from an EFB can differ from data derived from RTOM tables or similar.	X									
032 04 01 11		Take-off performance on wet and contaminated runways										
(01)		Explain the differences between the take-off performance determination on a wet or contaminated runway and on a dry runway.	X									
(02)		Describe a wet V_1 and explain the consequences of using a wet V_1 .	X									
(03)		Describe the hazards, effects and management of operating from a contaminated runway.	X									
(04)		Describe displacement drag, impingement drag, and the methods to monitor acceleration.	X									
(05)		Explain the benefits and implications of using a derated take-off on a contaminated runway.	X									
032 04 01 12		Use of reduced (flexible or flex) and derated thrust										
(01)		Explain the advantages and disadvantages of using reduced (flex) and derated thrust.	X									
(02)		Explain the difference between and principles behind reduced (flex) and derated thrust.	X									
(03)		Explain when reduced (flex) and derated thrust may and may not be used.	X									
(04)		Explain the effect of using reduced (flex) and derated thrust on take-off performance including take-off speeds, take-off distance, climb performance and obstacle clearance.	X									
(05)		Explain the assumed temperature method for determining reduced (flex) thrust performance.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
032 04 01 13		Take-off performance using different take-off flap settings										
(01)		Explain the advantages and disadvantages of using different take-off flap settings to optimise the performance-limited take-off mass (PLTOM).	X									
(02)		Determine the optimum flap position and PLTOM from given figures.	X									
032 04 01 14		Take-off performance using increased V_2 speeds ('improved climb performance')										
(01)		Explain the advantages and disadvantages of the increased V_2 procedure.	X									
(02)		Explain under what circumstances this procedure can be used.	X									
(03)		Explain the hazards of the fast V_1 and V_{LOF} speeds associated with the increased V_2 procedure and how they can be managed.	X									
032 04 01 15		Brake-energy and tyre-speed limit										
(01)		Explain the effects on take-off performance of brake-energy and tyre-speed limits.	X									
(02)		Explain under what conditions they are more likely to become limiting.	X									
032 04 02 00		Climb										
032 04 02 01		Climb techniques										
(01)		Explain the effect of climbing at constant IAS on: — TAS; — Mach number; — climb gradient; — rate of climb.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain the effect of climbing at constant Mach number on: — TAS; — IAS; — climb gradient; — rate of climb.	X									
(03)		Explain the correct sequence of climb speeds for turbojet transport aeroplanes.	X									
(04)		Determine the effect on TAS when climbing in and above the troposphere at constant Mach number.	X									
032 04 02 02		<i>Influence of variables on climb performance</i>										
(01)		Explain the effect on the operational speed limit when climbing at constant IAS and at constant Mach number.	X									
(02)		Explain the term ‘crossover altitude’ which occurs during the climb speed schedule (IAS–Mach number).	X									
032 04 03 00		Cruise										
032 04 03 01		<i>Intentionally left blank</i>										
032 04 03 02		<i>Intentionally left blank</i>										
032 04 03 03		<i>Intentionally left blank</i>										
032 04 03 04		<i>Long-range cruise</i>										
(01)		Define the term ‘long-range cruise’.	X									
(02)		Explain the differences between flying at long-range speed and maximum-range speed with regard to fuel-flow and speed stability.	X									
032 04 03 05		<i>Intentionally left blank</i>										
032 04 03 06		<i>Cruise altitudes</i>										
(01)	X	Define the term ‘optimum cruise altitude’.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain the factors that affect optimum cruise altitude.	X									
(03)		Explain the factors that can affect or limit the maximum operating cruise altitude.	X									
(04)		Explain the purpose of, and operational reasons for, a step climb and when such a climb would be initiated for optimum range.	X									
(05)		Describe the buffet onset boundary (BOB) and determine the high- and low-speed buffet (speed/Mach number only).	X									
(06)		Analyse the influence of bank angle, mass and the 1.3g buffet margin on a step climb.	X									
(07)		Describe that the high-speed buffet can occur at speeds slower or faster than M_{MO} .	X									
(08)		Explain the reasons why a step climb may not be used (e.g. for short sectors, advantageous winds, avoiding turbulence, and due to air traffic restrictions).	X									
032 04 03 07		Cost index (CI)										
(01)		Describe 'cost index'.	X									
(02)		Describe the reason for economical cruise speed.	X									
(03)		Describe the effect of cost index on climb, cruise and descent speeds.	X									
032 04 04 00		En-route one-engine-inoperative										
032 04 04 01		Drift-down										
(01)		Describe the determination of en-route flight-path data with one-engine-inoperative in accordance with the CS-25 provision on en-route flight paths.	X									
(02)		Describe the minimum obstacle-clearance height prescribed in the applicable operational requirements.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Describe the optimum speed that the pilot should select during drift-down.	X									
(04)		Explain the influence of deceleration on the drift-down profiles.	X									
032 04 04 02		<i>Influence of variables on the en-route one-engine-inoperative performance</i>										
(01)		Describe and explain the factors which affect the en-route net drift-down flight path.	X									
032 04 05 00		Descent										
032 04 05 01		<i>Descent techniques</i>										
(01)		Explain the effect of descending at constant Mach number.	X									
(02)		Explain the effect of descending at constant IAS.	X									
(03)		Explain the correct sequence of descent speeds for turbojet transport aeroplanes.	X									
(04)		Determine the effect on TAS when descending in and above the troposphere at constant Mach number.	X									
(05)		Describe the following limiting speeds for descent: — maximum operating speed (V_{MO}); — maximum Mach number (M_{MO}).	X									
(06)		Explain the effect of a descent at constant Mach number on the margin to low- and high-speed buffet.	X									
032 04 05 02		<i>Energy management in the descent</i>										
(01)		Explain the advantages and principle of a continuous descent.	X									
(02)	X	Describe energy management in terms of chemical, potential and kinetic energy.	X									
(03)		Describe the effect of increasing/decreasing headwind and tailwind on profile management.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Describe the effect of the Mach number to IAS transition (speed conversion) on profile management.	X									
(05)		Describe situations during the descent and approach in which a pilot could find that an aeroplane flies high or fast, and explain how the pilot can manage descent angle/excess energy.	X									
032 04 06 00		Approach and landing										
032 04 06 01		Approach requirements										
(01)		Describe the CS-25 requirements for the approach climb (one-engine-inoperative).	X									
(02)		Describe the CS-25 requirements for the landing climb.	X									
(03)		Explain the effect of temperature and pressure altitude on approach and landing-climb performance.	X									
032 04 06 02		Landing-field-length and landing-speed requirements										
(01)	X	Describe the landing distance determined according to CS 25 ('demonstrated' landing distance).	X									
(02)		Describe the landing-field-length requirements for dry, wet and contaminated runways and the applicable operational requirements.	X									
(03)	X	Define the 'landing distance available' (LDA).	X									
(04)		Define and explain the following speeds in accordance with CS-25 or CS-Definitions: — reference stall speed in the landing configuration (V_{SR0}); — reference landing speed (V_{REF}); — minimum control speed, approach and landing (V_{MCL}).	X									
032 04 06 03		Influence of variables on landing performance										

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain the effect of runway slope, surface conditions and wind on the maximum landing mass for a given landing distance available in accordance with the applicable operational requirements.	X									
(02)		Explain the effect on landing distance and maximum allowable landing mass of the following devices affecting deceleration: — reverse; — anti-skid; — ground spoilers or lift dumpers; — autobrakes.	X									
(03)		Explain the effect of temperature and pressure altitude on the maximum landing mass for a given landing distance available.	X									
(04)		Explain the effect of hydroplaning on landing distance required and methods of managing landing on contaminated or wet runways.	X									
032 04 06 04		Quick turnaround limit										
(01)		Describe how brake temperature limits the turnaround times.	X									
032 05 00 00		CS-25/APPLICABLE OPERATIONAL REQUIREMENTS PERFORMANCE CLASS A — USE OF AEROPLANE PERFORMANCE DATA	X									
032 05 01 00		Take-off										
032 05 01 01		Take-off (performance data)										
(01)		Determine from given graphs the field-length-limited take-off mass (FLLTOM) and describe situations in which this limitation could be most restrictive for take-off.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Determine from given graphs the climb-limited take-off mass and describe situations in which this limitation could be most restrictive for take-off.	X									
(03)		Determine from given graphs the obstacle-limited mass and describe situations in which this limitation could be most restrictive for take-off.	X									
(04)		Determine from given graphs the tyre-speed-limited take-off mass.	X									
(05)		Determine from given graphs the maximum brake-energy-limited take-off mass.	X									
(06)		Determine the take-off V speeds for the actual take-off mass.	X									
(07)		Determine the maximum take-off mass using given RTOM tables.	X									
(08)		Using RTOM tables, determine the take-off V speeds for the actual take-off weight using appropriate corrections.	X									
(09)		Determine the assumed/flex temperature and take-off V speeds using the RTOM tables.	X									
(10)		Calculate the break cooling time following a rejected take-off given appropriate data.	X									
032 05 02 00		Drift-down and stabilising altitude										
032 05 02 01		Drift-down and stabilising altitude (performance data)										
(01)		Determine the one-engine-out net stabilising altitude (level-off altitude) from given graphs/tables.	X									
(02)		Determine the maximum mass at which the net stabilising altitude with one-engine-out clears the highest relevant obstacle by the required clearance margin.	X									

Syllabus	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Determine, using drift-down graphs, fuel used, time and distance travelled in a descent from a cruise flight level to a given altitude.	X									
032 05 03 00		Landing										
032 05 03 01		Landing (performance data)										
(01)		Determine the field length required for landing with a given landing mass from the aeroplane performance data sheets.	X									
(02)		Determine the landing and approach climb-limited landing mass from the aeroplane performance data sheets.	X									
(03)		Calculate the maximum allowable landing mass as the lowest of: — approach-climb- and landing-climb-limited landing mass; — landing-field-length-limited landing mass; — structural-limited landing mass.	X									
(04)		Determine the brake cooling time for different landing masses using the aeroplane performance data sheets.	X									

SUBJECT 033 – FLIGHT PERFORMANCE AND PLANNING – FLIGHT PLANNING AND MONITORING

ED Decision 2020/018/R

General Student Pilot Route Manual (GSPRM)

This document shall be referred to as the General Student Pilot Route Manual (GSPRM) and should contain as a minimum:

1. a table of contents and a list of effective pages;
2. introduction with the instrument flight rules (IFR) charts' legends;
3. 1:500 000 visual flight rule (VFR) aeronautical chart of Germany;
4. en-route low- and high-altitude IFR charts to cover the airspace above all EU Member States plus Norway, Switzerland, Liechtenstein and the Balkans;
5. en-route high-altitude chart of the North Pole (a polar stereographic projection) to illustrate current polar routes;
6. a plotting chart of the North Atlantic (with information on extended range operations with two-engined aeroplanes (ETOPS));
7. area, aerodrome/heliport, aerodrome ground movement, standard instrument departure (SID), standard instrument arrival (STAR) and instrument approach charts (IACs) for Alicante Elche, Amsterdam Schiphol, Dubrovnik Čilipi, London Heathrow, Nantes/Atlantique, Santorini and Stuttgart for aeroplane operations, and Aberdeen, De Kooy and Tromsø for helicopter operations;
8. microwave landing system (MLS) approach chart for Galbraith Lake Alaska;
9. an example of a completed air traffic service (ATS) flight plan (with instructions on how to complete it), including the ICAO model flight plan form;
10. introduction with the VFR charts' legends, aerodrome directories for Croatia, France, Germany, Spain and United Kingdom, and area, aerodrome/heliport and visual approach charts (VACs) for Aberdeen Dyce, Alicante Elche, Dubrovnik Čilipi, Friedrichshafen, Gloucestershire and Nantes/Atlantique.

The charts should have a frozen date (e.g. 01.01.2017), and be reissued on a regular basis (e.g. every 4–5 years).

The charts listed above will form the basis for the questions in licensing examinations.

There will be no obligation for any student or approved training organisation (ATO) to buy, use or issue the GSPRM (nor will it have any other subject-matter material in it), but the content will be the basis for charts which may appear in Part-FCL exams. Any chart provider (Lido, Jeppesen, Navtech, etc.) may provide the GSPRM, but the students will not be expected to learn non-ICAO standard symbology or chart requirements.

Note that the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The professional pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
030 00 00 00		FLIGHT PERFORMANCE AND PLANNING										
033 00 00 00		FLIGHT PLANNING AND MONITORING										
033 01 00 00		FLIGHT PLANNING FOR VFR FLIGHTS <i>Remark: Using the GSPRM VFR charts.</i>										
033 01 01 00		VFR navigation plan										
033 01 01 01		Airspace, communication, visual and radio-navigation data from VFR charts										
(01)		Select routes taking the following criteria into account: — classification of airspace; — restricted areas; — VFR semicircular rules; — visually conspicuous points; — radio-navigation aids.	X	X	X	X	X					
(02)		Find the frequencies or identifiers of radio-navigation aids from charts.	X	X	X	X	X					
(03)		Find the communication frequencies and call signs for the following: — control agencies and service facilities; — flight information service (FIS); — weather information stations; — automatic terminal information service (ATIS).	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
033 01 01 02		Planning courses, distances and cruising levels with VFR charts										
(01)		Choose visual waypoints in accordance with specified criteria (large, unique, contrast, vertical extent, etc.).	X	X	X	X	X					
(02)		Measure courses and distances from a VFR chart.	X	X	X	X	X					
(03)		Find the highest obstacle within a given distance on either side of the course.	X	X	X	X	X					
(04)		Find the following data from a VFR chart and transfer them to a navigation plan: — waypoints or turning points; — distances; — true/magnetic courses.	X	X	X	X	X					
(05)		Calculate the minimum pressure altitude with a given obstacle clearance or true altitude from a given altitude or pressure altitude from minimum grid-area altitude using outside air temperature (OAT) and QNH.	X	X	X	X	X					
(06)		Calculate the vertical or horizontal distance and time to climb or descend to/from a given level or altitude with given data.	X	X	X	X	X					
(07)		Explain how to determine the position of a significant VFR point for insertion into a global navigation satellite system (GNSS) flight plan, using the distance and bearing from an existing significant point and using coordinates.	X	X	X	X	X					
033 01 01 03		Aerodrome charts and aerodrome directory										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Explain the reasons for studying the visual departure procedures and the available approach procedures.	X	X	X	X	X					
(02)		Find all visual procedures which can be expected at the departure, destination and alternate aerodromes.	X	X	X	X	X					
(03)		Find all relevant aeronautical and regulatory information required for VFR flight planning from the aerodrome charts or aerodrome directory.	X	X	X	X	X					
033 01 01 04		Intentionally left blank										
033 01 01 05		Completion of navigation plan										
(01)		Calculate the true airspeed (TAS) from given aircraft performance data, altitude and OAT.	X	X	X	X	X					
(02)		Calculate wind correction angles (WCAs), drift and ground speeds (GS).	X	X	X	X	X					
(03)		Calculate individual and accumulated times for each leg to destination and alternate aerodromes.	X	X	X	X	X					
033 02 00 00		FLIGHT PLANNING FOR IFR FLIGHTS <i>Remark: Using the GSPRM IFR charts.</i>										
033 02 01 00		IFR navigation plan										
033 02 01 01		Air traffic service (ATS) routes										
(01)		Identify suitable routings by identifying all relevant aeronautical and regulatory information (including information published in the national aeronautical information publication (AIP)) required for IFR flight planning.	X		X			X	X	2	3	
(02)		Identify and describe ATS routes (conventional, area navigation (RNAV), required navigation performance (RNP), conditional routes (CDRs), and direct routes).	X		X			X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
033 02 01 02		Courses and distances from en-route charts										
(01)		Determine courses and distances.	X		X			X	X	2	3	
(02)		Determine bearings and distances of waypoints from radio-navigation aids.	X		X			X	X	2	3	
033 02 01 03		Altitudes										
(01)		Define the following altitudes: — minimum en-route altitude (MEA); — minimum obstacle clearance altitude (MOCA); — minimum sector altitude (MSA); — minimum off-route altitude (MORA); — grid minimum off-route altitude (Grid MORA); — maximum authorised altitude (MAA); — minimum crossing altitude (MCA); — minimum holding altitude (MHA).	X		X			X	X	2	3	
(02)		Extract the following altitudes from the chart(s): — MEA; — MOCA; — MSA; — MORA; — Grid MORA; — MAA; — MCA; — MHA.	X		X			X	X	2	3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		State who is responsible for terrain separation during IFR flight inside and outside controlled airspace.	X		X			X	X			
(04)		State the minimum obstacle clearance requirements for en-route IFR flight inside and outside controlled airspace.	X		X			X	X			
(05)		State when a temperature error correction must be applied by either the pilot or ATC.	X		X			X	X			
(06)		Identify and explain the use of minimum radar vectoring altitudes.	X		X			X	X			
(07)		Calculate the minimum pressure altitude required with a given obstacle clearance, magnetic track, OAT, QNH and reduced vertical separation minimum (RVSM)/non-RVSM information.	X		X			X	X			
(08)		Calculate true altitude above a given datum using a given pressure altitude, OAT and QNH.	X		X			X	X			
033 02 01 04		Standard instrument departure (SID) and standard instrument arrival (STAR) routes										
(01)	X	State the reasons for studying SID and STAR charts.	X		X			X	X			
(02)	X	State that SID and STAR charts show procedures only in a pictorial presentation style which may not be true to scale.	X		X			X	X			
(03)		Interpret all data and information represented on SID and STAR charts, particularly: — routings; — distances; — courses; — radials;	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- altitudes/levels; - frequencies; - restrictions; - RNAV waypoints and non-RNAV intersection; - fly-over and fly-by waypoints.										
(04)		Identify SID and STAR charts which might be relevant for a planned flight.	X		X			X	X	2		
(05)		Define SID and STAR for RNAV only.	X		X			X	X			
(06)		Describe the difference between SID/STAR, RNAV SID/STAR and RNAV SID/STAR overlay.	X		X			X	X			
033 02 01 05		Instrument-approach charts										
(01)	X	State the reasons for being familiar with instrument-approach procedures (IAPs) and appropriate data for departure, destination and alternate aerodromes.	X		X			X	X			
(02)		Select IAPs appropriate for departure, destination and alternate aerodromes.	X		X			X	X	2		
(03)		Interpret all procedures, data and information represented on instrument-approach charts, particularly: - courses and radials; - distances; - altitudes/levels/heights; - restrictions; - obstructions; - frequencies;	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— speeds and times; — decision altitudes/heights (DAs/Hs); — (DA/H) and minimum descent altitudes/heights (MDAs/Hs); — visibility and runway visual ranges (RVRs); — approach-light systems.										
(04)		Explain the following IAP terms: — type A and B; — 2D and 3D; — CAT I, II and III; — precision approach (conventional and ground-based augmentation system (GBAS)); — non-precision approach (conventional and required navigation performance approach (RNP APCH) (lateral navigation (LNAV), LNAV/vertical navigation (VNAV), localiser performance (LP), localiser performance with vertical guidance (LPV), and required navigation performance authorisation required approach (RNP AR APCH)); — approach procedure with vertical guidance (APV) (APV Baro and APV satellite-based augmentation system (SBAS)).	X		X			X	X			
033 02 01 06		Communications and radio-navigation planning data										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Find the communication frequencies and call signs for aeronautical services for IFR flights from en-route charts.	X		X			X	X	2	3	
(02)		Find the frequency or identifiers of radio-navigation aids for IFR flights from en-route charts.	X		X			X	X	2	3	
033 02 01 07		Completion of a manual navigation plan										
(01)		Complete a navigation plan with the courses, distances and frequencies taken from charts.	X		X			X	X		3	
(02)		Find the SID and STAR routes to be flown or to be expected.	X		X			X	X	2	3	
(03)		Determine the position of top of climb (TOC) and top of descent (TOD) from given appropriate data.	X		X			X	X	2	3	
(04)		Determine variation and calculate magnetic/true courses.	X		X			X	X		3	
(05)		Calculate TAS from given aircraft performance data, altitude and OAT.	X		X			X	X		3	
(06)		Calculate wind correction angles (WCAs)/drift and ground speeds (GSs).	X		X			X	X		3	
(07)		Calculate individual and accumulated times for each leg to destination and alternate aerodromes.	X		X			X	X			
(08)		Describe the advantages of global navigation satellite system/flight management computer (GNSS/FMC) equipment regarding: — automatic calculation and display of tracks and leg distances; — additional route information in the database (minimum altitudes, approach procedures); — time and fuel estimates over waypoints;	X		X			X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— ability to adjust speed to arrive over a waypoint at a defined time; — time and fuel revisions based on predicted and actual wind.										
(09)		Describe the limitations of using GNSS/FMC equipment: — pilot-inputted errors (flight levels, wind, temperature, fuel); — the effect of other than predicted wind on fuel and time estimates; — the effect of aircraft's non-standard configuration on flight management system (FMS) predictions.	X		X			X	X			
033 03 00 00		FUEL PLANNING — OPERATIONAL REQUIREMENTS										
033 03 01 00		General										
033 03 01 01		<i>Fuel planning (general)</i>										
(01)		Convert to volume, mass and density given in different units which are commonly used in aviation.	X	X	X	X	X	X	X		3	
(02)		Determine relevant data, such as fuel capacity, fuel flow/ consumption at different power/thrust settings, altitudes and atmospheric conditions, from the flight manual.	X	X	X	X	X	X	X		3	
(03)		Calculate the attainable flight time/range from given average fuel flow/consumption and available amount of fuel.	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Calculate the required fuel from given average fuel flow/ consumption and required time/range to be flown.	X	X	X	X	X	X	X	3		
(05)		Calculate the required fuel for a VFR flight from given forecast meteorological conditions.	X	X	X	X	X	X	X	3		
(06)		State the minimum amount of remaining fuel required on arrival at the destination and alternate aerodromes/ heliports.	X	X	X	X	X	X	X			
(07)		Explain and describe how to calculate nautical air miles (NAM) from nautical ground miles (NGM).	X	X	X	X	X	X	X			
(08)		Calculate the required fuel for an IFR flight from given forecast meteorological conditions.	X		X			X	X	3		
033 03 02 00		Pre-flight fuel planning for commercial flights										
033 03 02 01		Taxi fuel										
(01)		Determine the fuel required for engine start and taxiing by consulting the fuel-usage tables or graphs from the flight manual taking into account all the relevant conditions.	X	X	X	X	X					
033 03 02 02		Trip fuel										
(01)		Define trip fuel and name the segments of flight for which the trip fuel is relevant.	X	X	X	X	X					
(02)		Determine the trip fuel for the flight by using data from the fuel tables or graphs from the flight manual.	X	X	X	X	X					
033 03 02 03		Reserve fuel and its components										
		Contingency fuel										
(01)		Explain the reasons for having contingency fuel.	X	X	X	X	X					
(02)		Calculate the contingency fuel according to the applicable operational requirements.	X	X	X	X	X					
		Alternate fuel										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the reasons and regulations for having alternate fuel and name the segments of flight for which the alternate fuel is relevant.	X	X	X	X	X					
(04)		Calculate the alternate fuel in accordance with the applicable operational requirements and relevant data from the navigation plan and the flight manual.	X	X	X	X	X					
		Final reserve fuel										
(05)		Explain the reasons and regulations for having final reserve fuel.	X	X	X	X	X					
(06)		Calculate the final reserve fuel for an aircraft in accordance with the applicable operational requirements and by using relevant data from the flight manual.	X	X	X	X	X					
		Additional fuel										
(07)		Explain the reasons and regulations for having additional fuel.	X	X	X	X	X					
(08)		Calculate the additional fuel for a flight in accordance with the applicable operational requirements.	X	X	X	X	X					
033 03 02 04		Extra fuel										
(01)		Explain the reasons and regulations for having extra fuel in accordance with the applicable operational requirements.	X	X	X	X	X					
(02)		Calculate the possible extra fuel under given conditions.	X	X	X	X	X					
(03)		Explain the fuel penalty incurred when loading extra fuel (i.e. the additional fuel consumption due to increased mass).	X	X	X	X	X					
033 03 02 05		Calculation of total fuel and completion of the fuel section of the navigation plan (fuel plan)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Calculate the total fuel required for a given flight.	X	X	X	X	X					
(02)		Complete the fuel plan.	X	X	X	X	X					
033 03 03 00		Specific fuel-calculation procedures										
033 03 03 01		Reduced contingency fuel procedure										
(01)	X	Explain the reasons and regulations for reduced contingency fuel as stated in the applicable operational requirements.	X									
(02)		Calculate the contingency fuel and trip fuel required in accordance with the reduced contingency fuel procedure.	X									
033 03 03 02		Isolated aerodrome or heliport procedure										
(01)	X	Explain the basic procedures for an isolated aerodrome or heliport as stated in the applicable operational requirements.	X		X	X						
(02)		Calculate the additional fuel for aeroplanes or helicopters according to the isolated aerodrome or heliport procedures.	X		X	X						
033 03 03 03		Predetermined-point procedure										
(01)	X	Explain the basic idea of the predetermined-point procedure as stated in the applicable operational requirements.	X									
033 03 03 04		Fuel-tankering										
(01)		Explain the basic idea of fuel-tankering procedures.	X									
(02)		Calculate how much fuel to tank by using given appropriate graphs, tables or data.	X									
033 03 03 05		Intentionally left blank										
033 04 00 00		PRE-FLIGHT PREPARATION										
033 04 01 00		Notice to airmen (NOTAM) briefing										
033 04 01 01		Ground- and satellite-based facilities and services										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Check that the ground- and satellite-based facilities and services required for the planned flight are available and adequate.	X	X	X	X	X	X	X		3	
033 04 01 02		Departure, destination and alternate aerodromes										
(01)		Find and analyse the latest state at the departure, destination and alternate aerodromes, in particular for: — opening hours; — work in progress (WIP); — special procedures due to WIP; — obstructions; — changes of frequencies for communications, navigation aids and facilities.	X	X	X	X	X	X	X	3		
(02)		Check that satellite-based facilities are available during the expected time of use.	X	X	X	X	X	X	X			
(03)		Check that GBAS/SBAS augmentation is available during the expected time of use.	X	X	X	X	X	X	X			
033 04 01 03		Airway routings and airspace structure										
(01)		Find and analyse the latest en-route state for: — airway(s) or route(s); — restricted, danger and prohibited areas; — changes of frequencies for communications, navigation aids and facilities.	X	X	X	X	X	X	X	3		
033 04 01 04		Pre-flight preparation of GNSS achievability										
(01)		Define why it is important to check GNSS achievability.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Define receiver autonomous integrity monitoring (RAIM), NOTAM and notice advisory to NavStar users (NANU) messages.	X									
(03)		Explain the difference in use of augmented and non-augmented GNSS in connection with the achievability check.	X									
(04)		Explain the difference in planned and unplanned outage of GNSS or SBAS.	X									
033 04 02 00		Meteorological briefing										
033 04 02 01		<i>Intentionally left blank</i>										
033 04 02 02		<i>Update of navigation plan using the latest meteorological information</i>										
(01)		Confirm the most fuel-efficient altitude from given wind, temperature and aircraft data.	X	X					X	3		
(02)		Confirm true altitudes from given atmospheric data to ensure that statutory minimum clearance is attained.	X	X	X	X	X	X				
(03)		Confirm magnetic headings and GSs.	X	X	X	X	X	X	X		3	
(04)		Confirm the individual leg times and the total time en route.	X	X	X	X	X	X	X		3	
(05)		Confirm the total time en route for the trip to the destination.	X	X	X	X	X	X	X			
(06)		Confirm the total time from destination to the alternate aerodrome.	X	X	X	X	X	X	X		3	
033 04 02 03		<i>Intentionally left blank</i>										
033 04 02 04		<i>Intentionally left blank</i>										
033 04 02 05		<i>Update of fuel plan</i>										
(01)		Calculate the revised fuel data in accordance with the changed conditions.	X	X	X	X	X	X			3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
033 04 03 00		Point of equal time (PET) and point of safe return (PSR)										
033 04 03 01		Point of equal time (PET)										
(01)		Define 'PET'.	X	X	X	X	X					
(02)		Calculate the position of a PET and the estimated time of arrival (ETA) at the PET from given relevant data.	X	X	X	X	X					
033 04 03 02		Point of safe return (PSR)										
(01)		Define 'PSR'.	X	X	X	X	X					
(02)		Calculate the position of a PSR and the ETA at the PSR from given relevant data.	X	X	X	X	X					
033 05 00 00		ICAO FLIGHT PLAN (ATS flight plan (FPL))										
033 05 01 00		Individual FPL										
033 05 01 01		Format of FPL										
(01)	X	State the reasons for a fixed format of an ICAO ATS FPL.	X	X	X	X	X	X	X			
(02)		Determine the correct entries to complete an ATS FPL plus decode and interpret the entries in a completed ATS FPL, particularly for the following: — aircraft identification (Item 7); — flight rules and type of flight (Item 8); — number and type of aircraft and wake-turbulence category (Item 9); — equipment (Item 10); — departure aerodrome and time (Item 13); — route (Item 15);	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- destination aerodrome, total estimated elapsed time and alternate aerodrome (Item 16); - other information (Item 18); - supplementary information (Item 19).										
033 05 01 02		Intentionally left blank										
033 05 02 00		Repetitive flight plan (RPL)										
033 05 02 01		Repetitive flight plan (RPL)										
(01)	X	Explain the difference between an individual FPL and an RPL.	X		X	X						
033 06 00 00		FLIGHT MONITORING AND IN-FLIGHT REPLANNING										
033 06 01 00		Flight monitoring										
033 06 01 01		Monitoring of track and time										
(01)		State the reasons for possible deviations from the planned track and planned timings.	X	X	X	X	X	X				
(02)		Calculate GS by using actual in-flight parameters.	X	X	X	X	X	X				
(03)		Calculate the expected leg times by using actual in-flight parameters.	X	X	X	X	X	X				
(04)		Enter, in the progress of flight, at the checkpoint or turning point, the 'actual time-over' and the 'estimated time-over' for the next checkpoint into the flight plan.	X	X	X	X	X					
(05)		State that it is necessary to determine the position of the aircraft accurately before commencing descent in order to ensure safe ground clearance.	X	X	X	X	X					
(06)		Calculate revised ETA based on changes to the pre-flight plan, including changes of W/V, cruise level, OAT, distances, Mach number and calibrated airspeed (CAS).	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
033 06 01 02		<i>In-flight fuel management</i>										
(01)		Explain why fuel checks must be carried out in flight at regular intervals and why relevant fuel data must be recorded.	X	X	X	X	X	X				
(02)		Assess deviations of actual fuel consumption from planned consumption.	X	X	X	X	X	X				
(03)		Calculate fuel quantity used, fuel consumption, and fuel remaining at navigation checkpoints/waypoints.	X	X	X	X	X	X				
(04)		Compare the actual with the planned fuel consumption by means of calculation.	X	X	X	X	X	X				
(05)		Determine the remaining range and endurance by means of calculation.	X	X	X	X	X	X				
(06)		Calculate the revised fuel consumption based on changes to the pre-flight plan, including changes of W/V, cruise level, OAT, distances, Mach number and CAS.	X	X	X	X	X	X				
033 06 02 00		<i>In-flight replanning</i>										
033 06 02 01		<i>Deviation from planned data</i>										
(01)		State that the commander is responsible for ensuring that, even in case of diversion, the remaining fuel is not less than the fuel required to proceed to an aerodrome where a safe landing can be made, with final reserve fuel remaining.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain that, in the case of an in-flight update, the commander has to check the following: — the suitability of the new destination or alternate aerodrome; — meteorological conditions on revised routing and at revised destination or alternate aerodrome; — the aircraft must be able to land with the prescribed final reserve fuel.	X	X	X	X	X					
(03)		Calculate the revised destination/alternate aerodrome landing mass from given latest data.	X	X	X	X	X					

SUBJECT 034 – FLIGHT PERFORMANCE AND PLANNING – PERFORMANCE – HELICOPTERS

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Note that the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The professional pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
030 00 00 00		FLIGHT PERFORMANCE AND PLANNING										
034 00 00 00		PERFORMANCE — HELICOPTERS										
034 01 00 00		GENERAL										
034 01 01 00		Performance legislation										
034 01 01 01		Airworthiness requirements										
(01)		Interpret the airworthiness requirements of CS-27 and CS-29.			X	X	X					
(02)		Name the general differences between helicopters certified according to CS-27 and CS-29.			X	X	X					
034 01 01 02		Operational regulations										
(01)		State that the person responsible for complying with operational procedures is the commander.			X	X	X					
(02)		Use and interpret diagrams and tables associated with CAT A and CAT B procedures in order to select and develop Class 1, 2 and 3 performance profiles according to available heliport size and location (surface or elevated).			X	X						
(03)		Interpret the charts showing minimum clearances associated with CAT A and CAT B procedures.			X	X						
034 01 02 00		General performance theory										
034 01 02 01		Phases of flight										
(01)		Explain the following phases of flight:			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— take-off; — climb; — level flight; — descent; — approach and landing.										
(02)		Describe the necessity for different take-off and landing procedures.			X	X	X					
034 01 02 02		Definitions and terms										
(01)		Define the following terms: — CAT A; — CAT B; — Performance Class 1, 2 and 3; — congested area; — elevated heliport; — helideck; — heliport; — hostile environment; — maximum operational passenger seating configuration (MOPSC); — non-hostile environment; — obstacle; — rotor radius (R); — take-off mass;			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— touchdown and lift-off area (TLOF); — safe forced landing; — speed for best rate of climb (V_Y); — never exceed speed (V_{NE}); — velocity landing gear extended (V_{LE}); — velocity landing gear operation (V_{LO}); — cruising speed and maximum cruising speed.										
(02)		Define the following terms: — reported headwind component; — take-off decision point (TDP); — defined point after take-off (DPATO); — take-off distance required helicopter (TODRH); — take-off distance available helicopter (TODAH); — distance required (DR); — rejected take-off distance required helicopter (RTODRH); — rotation point (RP); — committal point (CP); — defined point before landing (DPBL); — landing decision point (LDP); — landing distance available helicopter (LDAH);			X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— landing distance required helicopter (LDRH); — ditching (see operations).										
(03)		Understand the meaning and significance of the acronyms AEO and OEI.			X	X						
(04)		Define the terms ‘climb angle’ and ‘climb gradient’.			X	X						
(05)		Define the terms ‘flight-path angle’ and ‘flight-path gradient’.			X	X						
(06)		Define ‘V _{maxRange} ’ (speed for maximum range) and V _{maxEnd} (speed for maximum endurance).			X	X	X					
(07)		Define and calculate the gradient by using power, wind, and helicopter mass.			X	X						
(08)		Explain the terms ‘operational ceiling’ and ‘absolute ceiling’.			X	X	X					
(09)		Explain the term ‘service ceiling OEI’.			X	X	X					
(10)		Explain the difference between hovering in ground effect (HIGE) and hovering out of ground effect (HOGE).			X	X	X					
034 01 02 03		Power required/power available curves										
(01)		Understand and interpret the power required/power available versus TAS graphs.			X	X	X					
034 01 02 04		Height–velocity graphs										
(01)		Understand and interpret height–velocity graphs.			X	X	X					
034 01 02 05		Influencing variables on performance										
(01)		Explain how the following factors affect helicopter performance: — pressure altitude; — humidity;			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— temperature; — wind; — helicopter mass; — helicopter configuration; — helicopter centre of gravity (CG).										
034 02 00 00		PERFORMANCE CLASS 3 — SINGLE-ENGINE HELICOPTERS										
034 02 01 00		Effect of variables on single-engine (SE) helicopter performance										
034 02 01 01		Effect of variables on SE helicopter performance										
(01)		Determine the wind component, altitude and temperature for hovering, take-off and landing.			X	X	X					
(02)		Explain that operations are to be conducted only from/to heliports and over such routes, areas and diversions contained in a non-hostile environment where a safe forced landing can be carried out (point CAT.OP.MPA.137 of the EU Regulation on air operations, except when the helicopter is approved to operate in accordance with point CAT.POL.H.420). (Consider the exception: Operations may be conducted in a hostile environment. Ground level exposure — and exposure for elevated final approach and take-off areas (FATOs) or helidecks in non-hostile environments — is allowed for operations approved under CAT.POL.H.305, during the take-off and landing phases.)			X	X	X					
(03)		Explain the effect of temperature, wind and altitude on climb, cruise and descent performance.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
034 02 02 00		Take-off and landing										
034 02 02 01		Take-off and landing (including hover)										
(01)		Explain the take-off and landing requirements.			X	X	X					
(02)		Explain the maximum allowed take-off and landing mass.			X	X	X					
(03)		Explain that mass has to be restricted to HIGE.			X	X	X					
(04)		Explain that if HIGE is unlikely to be achieved (for example, blocked by an obstruction), then mass must be restricted to HOGE.			X	X	X					
034 02 03 00		Climb, cruise and descent										
034 02 03 01		Climb, cruise and descent (capabilities)										
(01)		State that the helicopter must be capable of flying its intended track without flying below the appropriate minimum flight altitude and be able to perform a safe forced landing.			X	X	X					
(02)		Explain the effect of altitude on the maximum endurance speed.			X	X	X					
034 02 04 00		Use of helicopter performance data										
034 02 04 01		Take-off (including hover)										
(01)		Find the maximum wind component.			X	X	X					
(02)		Find the maximum allowed take-off mass for certain conditions.			X	X	X					
(03)		Find the height–velocity parameters.			X	X	X					
034 02 04 02		Climb										
(01)		Find the time, distance and fuel required to climb for certain conditions.			X	X	X					
(02)		Find the rate of climb under given conditions and the best rate-of-climb speed V_Y .			X	X	X					
034 02 04 03		Cruise										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Find the cruising speed and fuel consumption for certain conditions.			X	X	X					
(02)		Calculate the range and endurance under given conditions.			X	X	X					
034 02 04 04		Landing (including hover)										
(01)		Find the maximum wind component.			X	X	X					
(02)		Find the maximum allowed landing mass for certain conditions.			X	X	X					
(03)		Find the height–velocity parameters.			X	X	X					
034 03 00 00		PERFORMANCE CLASS 2										
		<i>General remark: The Learning Objectives for Performance Class 2 are principally identical with those for Performance Class 1. (See 034 04 00 00) Additional Learning Objectives are shown below.</i>										
034 03 01 00		Operations without an assured safe forced landing capability										
034 03 01 01		Responsibility for operations without an assured safe forced landing capability										
(01)		State the responsibility of the operator for assuring safe forced landings (point CAT.POL.H.305 of the EU Regulation on air operations).			X	X						
034 03 02 00		Take-off										
034 03 02 01		Take-off requirements										
(01)		State the climb and other requirements for take-off.			X	X						
034 03 03 00		Take-off flight path										
034 03 03 01		Take-off flight path requirements										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		State the height above the take-off surface at which at least the requirements for the take-off flight path for Performance Class 1 are to be met.			X	X						
034 03 04 00		Landing										
034 03 04 01		Landing requirements										
(01)		State the requirements for the climb capability when OEI.			X	X						
(02)		State the options for a Performance Class 2 operation in the case of a critical power-unit failure at any point in the approach path.			X	X						
(03)		State the limitations for operations to/from a helideck.			X	X						
034 04 00 00		PERFORMANCE CLASS 1 — HELICOPTERS CERTIFIED ACCORDING TO CS-29 ONLY										
034 04 01 00		Take-off										
034 04 01 01		Take-off distances										
(01)		Explain the effects of the following variables on the flight-path and take-off distances: — take-off with HIGE or HOGE; — take-off procedure; — obstacle clearances both laterally and vertically; — take-off from non-elevated heliports; — take-off from elevated heliports or helidecks; — take-off from a TLOF.			X	X						
(02)		Explain the effects of the following variables on take-off distances:			X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— mass; — take-off configuration; — bleed-air configurations.										
(03)		Explain the effects of the following meteorological conditions on take-off distances: — wind; — temperature; — pressure altitude.			X	X						
(04)		Explain the take-off distances for specified conditions and configuration for AEO and OEI.			X	X						
(05)		Explain the effect of obstacles on the take-off distance required.			X	X						
(06)		State the assumed reaction time between engine failure and recognition.			X	X						
(07)		Explain that the flight must be carried out visually up to TDP.			X	X						
034 04 01 02		Rejected take-off distance required (helicopter) (RTODR(H))										
(01)		Explain RTODR(H) for specified conditions and configuration for AEO and OEI.			X	X						
(02)		Explain the time-to-decide allowance (decision time) and deceleration procedure.			X	X						
034 04 01 03		Intentionally left blank										
034 04 01 04		Take-off climb										
(01)		Define the segments of the take-off flight path.			X	X						
(02)		Explain the effect of changes in the configuration on power and speed in the segments.			X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the climb-gradient requirements for OEI.			X	X						
(04)		State the minimum altitude over the take-off path when flying at the take-off safety speed in a Category A helicopter (V_{TOSS}).			X	X						
(05)		Describe the influence of airspeed selection, acceleration and turns on the climb gradient and best rate-of-climb speed.			X	X						
034 04 01 05		Obstacle-limited take-off										
(01)		Describe the operational regulations for obstacle clearance of the take-off flight path in the departure sector with OEI.			X	X						
034 04 01 06		Use of helicopter performance data										
(01)		Determine from helicopter performance data sheets the maximum mass that satisfies the operational regulations for take-off in terms of regulated take-off mass, TODRH and minimum gradients for climb and obstacle clearance.			X	X						
034 04 02 00		Climb										
034 04 02 01		Climb techniques										
(01)		Explain the effect of climbing with best rate-of-climb speed (V_Y).			X	X						
(02)		Explain the influence of altitude on V_Y .			X	X						
034 04 02 02		Use of helicopter flight data										
(01)		Find the rate of climb and calculate the time to climb to a given altitude.			X	X						
034 04 03 00		Cruise										
034 04 03 01		Cruise techniques										
(01)		Explain the cruise procedures for 'maximum endurance' and 'maximum range'.			X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
034 04 03 02		Maximum endurance										
(01)		Explain fuel flow in relation to true airspeed (TAS).			X	X						
(02)		Explain the speed for maximum endurance.			X	X						
034 04 03 03		Maximum range										
(01)		Explain the speed for maximum range.			X	X						
034 04 03 04		Maximum cruise										
(01)		Explain the speed for maximum cruise.			X	X						
034 04 03 05		Cruise altitudes										
(01)		Explain the factors which might affect or limit the operating altitude.			X	X						
(02)		Understand the relation between power setting, fuel consumption, cruising speed and altitude.			X	X						
034 04 03 06		Use of helicopter performance data										
(01)		Determine the fuel consumption from the helicopter performance data sheets in accordance with altitude and helicopter mass.			X	X						
034 04 04 00		En-route one-engine-inoperative (OEI)										
034 04 04 01		Requirements for en-route flights with OEI										
(01)		State the flight-path clearance requirements.			X	X						
(02)		Explain drift-down techniques.			X	X						
(03)		State the reduction in the flight-path width when navigational accuracy can be achieved.			X	X						
034 04 04 02		Use of helicopter flight data										
(01)		Find the single-engine service ceiling, range and endurance from given engine-inoperative charts.			X	X						
(02)		Find OEI operating data from suitable charts.			X	X						
(03)		Find the amount of fuel to be jettisoned in order to reduce helicopter mass.			X	X						

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Calculate the relevant parameters for drift-down procedures.			X	X						
034 04 05 00		Descent										
034 04 05 01		Use of helicopter flight data										
(01)		Find the rate of descent and calculate the time to descend to a given altitude.			X	X						
034 04 06 00		Landing										
034 04 06 01		Landing requirements										
(01)		State the requirements for landing.			X	X						
034 04 06 02		Landing procedures										
(01)		Explain the procedure for critical power-unit failure before and after the landing decision point.			X	X						
(02)		Explain that the portion of flight after the landing decision point must be carried out visually.			X	X						
(03)		Explain the procedures and required obstacle clearances for landings on different heliports/helidecks.			X	X						
034 04 06 03		Use of helicopter performance data										
(01)		Determine from helicopter performance data sheets the maximum mass that satisfies the operational regulations for landing in terms of regulated landing mass, LDRH and minimum gradients for climb and obstacle clearance.			X	X						

SUBJECT 040 – HUMAN PERFORMANCE AND LIMITATIONS

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Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
040 00 00 00		HUMAN PERFORMANCE AND LIMITATIONS										
040 01 00 00		HUMAN FACTORS: BASIC CONCEPTS										
040 01 01 00		Human factors in aviation										
040 01 01 01		<i>Becoming a competent pilot</i>										
(01)		State that competence is based on knowledge, skills and attitudes of the individual pilot, and list the ICAO eight core competencies: — application of procedures; — communication; — aircraft flight path management, automation; — aircraft flight path management, manual control; — leadership and teamwork; — problem-solving and decision-making; — situation awareness; — workload management.	X	X	X	X	X	X				
040 01 02 00		Intentionally left blank										
040 01 03 00		Flight safety concepts										
040 01 03 01		<i>Threat and error management (TEM) model and SHELL model</i>										
(01)		Explain the three components of the TEM model.	X	X	X	X	X	X	X			
(02)		Explain and give examples of latent threats.	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain and give examples of environmental threats.	X	X	X	X	X	X	X	1		
(04)		Explain and give examples of organisational threats.	X	X	X	X	X	X	X	1		
(05)		Explain and give a definition of ‘error’ according to the TEM model of ICAO Doc 9683 (Part II, Chapter 2).	X	X	X	X	X	X	X	1		
(06)		Give examples of different countermeasures which may be used in order to manage threats, errors, and undesired aircraft states.	X	X	X	X	X	X	X	1		
(07)		Explain and give examples of procedural error, communication errors, and aircraft handling errors.	X	X	X	X	X	X	X	1		
(08)		Explain and give examples of ‘undesired aircraft states’.	X	X	X	X	X	X				
(09)		State the components of the SHELL model.	X	X	X	X	X	X				
(10)		State the relevance of the SHELL model to the work in the cockpit.	X	X	X	X	X	X				
040 01 04 00		Safety culture										
040 01 04 01		Safety culture and safety management										
(01)		Distinguish between ‘open cultures’ and ‘closed cultures’.	X	X	X	X	X	X	X		1	
(02)		Illustrate how safety culture is reflected in national culture.	X	X	X	X	X	X	X		1	
(03)		Discuss the established expression ‘safety first’ in a commercial entity.	X	X	X	X	X	X				
(04)		Explain James Reason’s ‘Swiss Cheese Model’.	X	X	X	X	X	X	X	1		
(05)		State the important factors that promote a good safety culture.	X	X	X	X	X	X	X		1	
(06)		Distinguish between ‘just culture’ and ‘non-punitive culture’.	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Name the five components which form safety culture (according to James Reason: informed culture, reporting culture, learning culture, just culture, flexible culture).	X	X	X	X	X	X	X		1	
(08)		Name the basic concepts of safety management system (SMS) (including hazard identification and risk management) and its relationship with safety culture in order to: — define how the organisation is set up to manage risks; — identify workplace risk and implement suitable controls; — implement effective communication across all levels of the organisation.	X	X	X	X	X	X	X			
040 02 00 00		BASICS OF AVIATION PHYSIOLOGY AND HEALTH MAINTENANCE										
040 02 01 00		Basics of flight physiology										
040 02 01 01		<i>The atmosphere</i>										
(01)		State that the volume percentage of the gases in ambient air will remain constant at all altitudes at which conventional aircraft operate.	X	X	X	X	X	X				
040 02 01 02		<i>Respiratory and circulatory system</i>										
(01)		List the main components of the respiratory system and their function.	X	X	X	X	X	X				
(02)		Identify the different volumes of air in the lungs and state the normal respiratory rate.	X	X	X	X	X	X			1	
(03)		Explain the role of carbon dioxide in the control and regulation of respiration.	X	X	X	X	X	X			1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Describe the basic processes of external respiration and internal respiration.	X	X	X	X	X	X				
(05)		List the factors that determine pulse rate.	X	X	X	X	X	X				
(06)		Name the major components of the circulatory system and describe their function.	X	X	X	X	X	X				
(07)		State the values for a normal pulse rate and the average cardiac output (heart rate × stroke volume) of an adult at rest.	X	X	X	X	X	X				
(08)		Define ‘systolic’ and ‘diastolic’ blood pressure.	X	X	X	X	X	X				
(09)		State the normal blood pressure ranges and units of measurement.	X	X	X	X	X	X				
(10)		List the main constituents of blood and describe their functions.	X	X	X	X	X	X				
(11)		Stress the function of haemoglobin in the circulatory system.	X	X	X	X	X	X				
(12)		Define ‘anaemia’ and state its common causes.	X	X	X	X	X	X				
(13)		Indicate the effect of increasing altitude on haemoglobin oxygen saturation.	X	X	X	X	X	X			1	
		Hypertension and hypotension										
(14)		Define ‘hypertension’ and ‘hypotension’.	X	X	X	X	X	X				
(15)		List the effects that high and low blood pressure will have on some normal functions of the human body.	X	X	X	X	X	X				
(16)		State that both hypotension and hypertension may disqualify a pilot from obtaining medical clearance to fly.	X	X	X	X	X	X				
(17)		List the factors which can lead to hypertension for an individual.	X	X	X	X	X	X				
(18)		State the corrective actions that may be taken to reduce high blood pressure.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(19)		Stress that hypertension is the major factor of strokes in the general population.	X	X	X	X	X	X				
		Coronary artery disease										
(20)		Differentiate between ‘angina’ and ‘heart attack’.	X	X	X	X	X	X				
(21)		Explain the major risk factors for coronary disease.	X	X	X	X	X	X				
(22)		State the role physical exercise plays in reducing the chances of developing coronary disease.	X	X	X	X	X	X				
		Hypoxia										
(23)		Define the two major forms of hypoxia (hypoxic and anaemic), and the common causes of both.	X	X	X	X	X	X				
(24)		State the symptoms of hypoxia.	X	X	X	X	X	X			1	
(25)		State that healthy people are able to compensate for altitudes up to approximately 10 000–12 000 ft.	X	X	X	X	X	X				
(26)		Name the three physiological thresholds and allocate the corresponding altitudes for each of them: — reaction threshold (7 000 ft); — disturbance threshold (10–12 000 ft); and — critical threshold (22 000 ft).	X	X	X	X	X	X				
(27)		State the altitude at which short-term memory begins to be affected by hypoxia.	X	X	X	X	X	X			1	
(28)		Define the terms ‘time of useful consciousness’ (TUC) and ‘effective performance time’ (EPT).	X	X	X	X	X	X				
(29)		State that TUC varies among individuals, but the approximate values for a person seated (at rest) are: — 20 000 ft 30 min — 30 000 ft 1–2 min	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— 35 000 ft 30–90 s — 40 000 ft 15–20 s										
(30)		List the factors that determine the severity of hypoxia.	X	X	X	X	X	X				
(31)		State the equivalent altitudes when breathing ambient air and 100 % oxygen at mean sea level (MSL) and at approximately 10 000, 30 000 and 40 000 ft.	X	X	X	X	X	X				
		Hyperventilation										
(32)		Describe the role of carbon dioxide in hyperventilation.	X	X	X	X	X	X				
(33)		Define the term ‘hyperventilation’.	X	X	X	X	X	X				
(34)		List the factors that cause hyperventilation.	X	X	X	X	X	X				
(35)		State that hyperventilation may be caused by psychological or physiological reasons.	X	X	X	X	X	X				
(36)		List the signs and symptoms of hyperventilation.	X	X	X	X	X	X				
(37)		List the measures which may be taken to counteract hyperventilation: breath slowly, close one opening of the nose, speak loudly, place a paper bag over nose and mouth.	X	X	X	X	X	X				
		Decompression sickness/illness										
(38)		State the normal range of cabin pressure altitude in pressurised commercial air transport aircraft and describe its protective function for aircrew and passengers.	X	X	X	X	X	X				
(39)		List the vital actions the crew has to perform when cabin pressurisation is lost (oxygen mask on, emergency descent, land as soon as possible, and no further flight for the next minimum 24 hours).	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		State that decompression sickness symptoms can occur up to 24 hours later.										
(40)		Identify the causes of decompression sickness in flight operation.	X	X	X	X	X	X				
(41)		State how decompression sickness can be prevented.	X	X	X	X	X	X				
(42)		List the symptoms of decompression sickness (bends, creeps, chokes, staggers).	X	X	X	X	X	X				
(43)		Indicate how decompression sickness may be treated.	X	X	X	X	X	X				
(44)		Define the hazards of diving and flying, and give the recommendations associated with these activities.	X	X	X	X	X	X			1	
		Acceleration										
(45)		Define 'linear acceleration' and 'angular acceleration'.	X	X	X	X	X	X	X	1		
(46)		Describe the effects of z-acceleration on the circulation and blood volume distribution.	X	X	X	X	X	X		1		
(47)		List magnitude, duration and onset as factors that determine the effects of acceleration on the human body.	X	X	X	X	X	X	X	1		
(48)		List the effects of positive acceleration with respect to type, sequence and corresponding G-load.	X	X	X	X	X	X		1		
		Carbon monoxide										
(49)		State how carbon monoxide is produced.	X	X	X	X	X	X		1		
(50)		State how the presence of carbon monoxide in the blood affects the distribution of oxygen.	X	X	X	X	X	X				
(51)		List the signs and symptoms of carbon-monoxide poisoning.	X	X	X	X	X	X		1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(52)		Explain immediate countermeasures on suspicion of carbon-monoxide poisoning and how poisoning can be treated later on the ground.	X	X	X	X	X	X		1		
040 02 01 03		High-altitude environment										
(01)		State how an increase in altitude may change the proportion of ozone in the atmosphere and that aircraft can be equipped with special ozone removers.	X									
		Radiation										
(02)		State the sources of radiation at high altitude.	X									
(03)		List the effects of excessive exposure to radiation.	X									
		Humidity										
(04)		List the factors that affect the relative humidity of both the atmosphere and cabin air.	X									
(05)		List the effects of low humidity on human body to be spurious thirst, dry eyes, skin and mucous membranes, and indicate measures that can be taken: drinking water, using eye drops and aqueous creams.	X									
040 02 02 00		People and the environment: the sensory system										
040 02 02 01		The different senses										
(01)		List the different senses.	X	X	X	X	X	X	X		1	
040 02 02 02		Central, peripheral and autonomic nervous system										
(01)		Define the term ‘sensory threshold’.	X	X	X	X	X	X				
(02)		Define the term ‘sensitivity’, especially in the context of vision.	X	X	X	X	X	X				
(03)		Give examples of sensory adaptation.	X	X	X	X	X	X				
(04)		Define the term ‘habituation’ and state its implication for flight safety.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
040 02 02 03		Vision										
		Functional anatomy										
(01)		Name the most important parts of the eye and the pathway to the visual cortex.	X	X	X	X	X					
(02)		State the basic functions of the parts of the eye.	X	X	X	X	X	X		1		
(03)		Define ‘accommodation’.	X	X	X	X	X	X				
(04)		Distinguish between the functions of the rod and cone cells.	X	X	X	X	X	X				
(05)		Describe the distribution of rod and cone cells in the retina and explain their relevance to vision.	X	X	X	X	X	X				
		The fovea (fovea centralis) and peripheral vision										
(06)		Explain the terms ‘visual acuity’, ‘visual field’, ‘central vision’, ‘peripheral vision’ and ‘the fovea’, and explain their function in the process of vision.	X	X	X	X	X	X				
(07)		List the factors that may degrade visual acuity and the importance of ‘lookout’.	X	X	X	X	X	X				
(08)		State the limitations of night vision and the different scanning techniques at both night and day.	X	X	X	X	X	X				
(09)		State the time necessary for the eye to adapt both to bright light and the dark.	X	X	X	X	X	X		1		
(10)		State the effect of hypoxia, smoking and altitude in excess of 5 000 ft on night vision.	X	X	X	X	X	X				
(11)		Explain the nature of colour blindness.	X	X	X	X	X	X				
		Binocular and monocular vision										
(12)		Distinguish between monocular and binocular vision.	X	X	X	X	X	X				
(13)		Explain the basis of depth perception and its relevance to flight performance.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(14)		List the possible monocular cues for depth perception.	X	X	X	X	X	X				
(15)		State that for high-energy blue light and UV rays, sunglasses can prevent damage to the retina.	X	X	X	X	X	X				
		Defective vision										
(16)		Explain long-sightedness, short-sightedness and astigmatism.	X	X	X	X	X	X				
(17)		List the causes of and the precautions that may be taken to reduce the probability of vision loss due to: — presbyopia; — cataract; — glaucoma.	X	X	X	X	X	X				
(18)		List the types of sunglasses that could cause perceptual problems in flight.	X	X	X	X	X	X				
(19)		List the measures that may be taken to protect oneself from flash blindness.	X	X	X	X	X	X				
(20)		State the possible problems associated with contact lenses.	X	X	X	X	X	X				
(21)		State the current rules/regulations governing the wearing of corrective spectacles and contact lenses when operating as a pilot.	X	X	X	X	X	X				
(22)		Explain the significance of the 'blind spot' on the retina in detecting other traffic in flight.	X	X	X	X	X	X				
040 02 02 04		Hearing										
		Descriptive and functional anatomy										
(01)		State the basic parts and functions of the outer, the middle and the inner ear.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Differentiate between the functions of the vestibular apparatus and the cochlea in the inner ear.	X	X	X	X	X	X				
		Hearing loss										
(03)		Define the main causes of the following hearing defects/loss: — ‘conductive deafness’; — ‘noise-induced hearing loss’ (NIHL); — ‘presbycusis’.	X	X	X	X	X	X				
(04)		Summarise the effects of environmental noise on hearing.	X	X	X	X	X	X				
(05)		State the decibel level of received noise that will cause NIHL.	X	X	X	X	X	X				
(06)		Identify the potential occupational risks that may cause hearing loss.	X	X	X	X	X	X				
(07)		List the main sources of hearing loss in the flying environment.	X	X	X	X	X	X				
(08)		List the precautions that may be taken to reduce the probability of onset of hearing loss.	X	X	X	X	X	X				
040 02 02 05		Equilibrium										
		Functional anatomy										
(01)		List the main elements of the vestibular apparatus.	X	X	X	X	X	X		1		
(02)		State the functions of the vestibular apparatus on the ground and in flight.	X	X	X	X	X	X		1		
(03)		Distinguish between the component parts of the vestibular apparatus in the detection of linear and angular acceleration as well as on gravity.	X	X	X	X	X	X		1		
(04)		Explain how the semicircular canals are stimulated.	X	X	X	X	X	X			1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		Motion sickness										
(05)		Describe air sickness and its accompanying symptoms.	X	X	X	X	X	X	X	1		
(06)		List the causes of air sickness.	X	X	X	X	X	X	X	1		
(07)		Describe the necessary actions to be taken to counteract the symptoms of air sickness.	X	X	X	X	X	X		1		
040 02 02 06		Integration of sensory inputs										
(01)		State the interaction between vision, equilibrium, proprioception and hearing to obtain spatial orientation in flight.	X	X	X	X	X	X	X	1		
(02)		Define the term ‘illusion’.	X	X	X	X	X	X	X	1		
(03)		Give examples of visual illusions based on shape constancy, size constancy, aerial perspective, atmospheric perspective, the absence of focal or ambient cues, autokinesis, vectional false horizons, field myopia, and surface planes.	X	X	X	X	X	X	X	1		
(04)		Relate these illusions to problems that may be experienced in flight and identify the danger attached to them.	X	X	X	X	X	X	X	1		
(05)		List approach and landing illusions for slope of the runway, black-hole approach, and terrain around runway, and state the danger involved with recommendations to avoid or counteract the problems with high or low approach or flare at the wrong time.	X	X	X	X	X	X	X	1		
(06)		State the problems associated with flickering lights (strobe lights, anti-collision lights, propellers and rotors under certain light conditions, etc.).	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Describe vestibular illusions caused by the angular accelerations (the Leans, Coriolis) and linear accelerations (somatogravic, G-effect).	X	X	X	X	X	X	X	1		
(08)		Relate the above-mentioned vestibular illusions to problems encountered in flight and state the dangers involved.	X	X	X	X	X	X	X	1		
(09)		State that the ‘seat-of-the-pants’ sense is completely unreliable when visual contact with the ground is lost or when flying in instrument meteorological conditions (IMC) or with a poor visual horizon.	X	X	X	X	X	X	X	1		
(10)		Differentiate between vertigo, Coriolis effect, and spatial disorientation.	X	X	X	X	X	X	X	1		
(11)		List the measures to prevent or overcome spatial disorientation.	X	X	X	X	X	X	X	1		
040 02 03 00		Health and hygiene										
040 02 03 01		<i>Intentionally left blank</i>										
040 02 03 02		<i>Body rhythm and sleep</i>										
(01)		Name some internal body rhythms and their relevance to sleep. Explain that the most important of which is body temperature.	X	X	X	X						
(02)		Explain the term ‘circadian rhythm’.	X	X	X	X	X					
(03)		State the approximate duration of a ‘free-running’ rhythm.	X	X	X	X	X					
(04)		Explain the significance of the ‘internal clock’ in regulating the normal circadian rhythm.	X	X	X	X	X					
(05)		State the effect of the circadian rhythm of body temperature on an individual’s performance standard and on an individual’s sleep patterns.	X	X	X	X	X					
(06)		List and describe the stages of a sleep cycle.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Differentiate between rapid eye movement (REM) and non-REM sleep.	X	X	X	X	X					
(08)		Explain the function of sleep and describe the effects of insufficient sleep on performance.	X	X	X	X	X					
(09)		Explain the simple calculations for the sleep/wake credit/debit situation.	X	X	X	X	X					
(10)		Explain how sleep debit can become cumulative.	X	X	X	X	X					
(11)		State the time formula for the adjustment of body rhythms to the new local time scale after crossing time zones.	X	X	X	X	X					
(12)		State the problems caused by circadian dysrhythmia (jet lag) with regard to an individual's performance and sleep.	X	X	X	X	X					
(13)		Differentiate between the effects of westbound and eastbound travel.	X	X	X	X	X					
(14)		Explain the interactive effects of circadian rhythm and vigilance on a pilot's performance during flight as the duty day elapses.	X	X	X	X	X					
(15)		Describe the main effects of lack of sleep on an individual's performance.	X	X	X	X	X					
(16)		List the possible strategies to cope with jet lag.	X	X	X	X	X					
040 02 03 03		Problem areas for pilots										
		Common minor ailments										
(01)		State the role of the Eustachian tube in equalising pressure between the middle ear and the environment.	X	X	X	X	X	X				
(02)		State that the in-flight environment may increase the severity of symptoms which may be minor while on the ground.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		List the negative effects of suffering from colds or flu on flight operations especially with regard to the middle ear, the sinuses, and the teeth.	X	X	X	X	X	X				
(04)		State when a pilot should seek medical advice from an aeromedical examiner (AME) or aeromedical centre (AeMC).	X	X	X	X	X	X				
(05)		Describe the measures to prevent or clear problems due to pressure changes during flight.	X	X	X	X	X	X				
		Entrapped gases and barotrauma										
(06)		Define ‘barotrauma’.	X	X	X	X	X	X				
(07)		Differentiate between otic, sinus, gastrointestinal and aerodontalgia (of the teeth) barotraumas and explain avoidance strategies.	X	X	X	X	X	X				
(08)		Explain why the effects of otic barotrauma can be worse in the descent.	X	X	X	X	X	X				
		Gastrointestinal upsets										
(09)		State the effects of gastrointestinal upsets that may occur during flight.	X	X	X	X	X	X				
(10)		List the precautions that should be observed to reduce the occurrence of gastrointestinal upsets.	X	X	X	X	X	X				
(11)		Indicate the major sources of gastrointestinal upsets.	X	X	X	X	X	X				
		Obesity										
(12)		Define ‘obesity’.	X	X	X	X	X	X				
(13)		State the following harmful effects obesity can cause: — possibility of developing coronary problems; — increased chances of developing diabetes; — reduced ability to withstand G-forces;	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— development of problems with the joints of the limbs; — general circulatory problems; — reduced ability to cope with hypoxia or decompression sickness; — sleep apnoea.										
(14)		Describe the problems associated with Type 2 (mostly adult) diabetes: — risk factors; — insulin resistance; — complications (vascular, neurological) and the consequences for the medical licence; — pilots are not protected from Type 2 diabetes more than other people.	X	X	X	X	X	X				
(15)		Describe the typical back problems (unspecific back pain, slipped disc) that pilots have. Explain also the ways of preventing and treating these problems: — good sitting posture; — lumbar support; — good physical condition; — in-flight exercise, if possible; — physiotherapy.	X	X	X	X	X	X				
		Food hygiene										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(16)		Stress the importance of and methods to be adopted by aircrew, especially when travelling abroad, to avoid contaminated food and liquids.	X	X	X	X	X	X				
(17)		List the major contaminating sources in foodstuffs.	X	X	X	X	X	X				
(18)		State the major constituents of a healthy diet.	X	X	X	X	X	X				
(19)		State the measure to avoid hypoglycaemia.	X	X	X	X	X	X				
(20)		State the importance of adequate hydration.	X	X	X	X	X	X				
		Tropical climates										
(21)		List the problems associated with operating in tropical climates.	X	X	X	X	X					
(22)		State the possible causes/sources of incapacitation in tropical countries with reference to: — standards of hygiene; — quality of water supply; — insect-borne diseases; — parasitic worms; — rabies or other diseases that may be spread through contact with animals; — sexually transmitted diseases.	X	X	X	X	X					
(23)		State the precautions to be taken to reduce the risks of developing problems in tropical areas.	X	X	X	X	X					
		Infectious diseases										
(24)		State the major infectious diseases that may severely incapacitate or kill individuals.	X	X	X	X	X	X				
(25)		State the precautions that must be taken to ensure that disease-carrying insects are not transported between areas.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
040 02 03 04		Intoxication										
		Tobacco										
(01)		State the harmful effects of tobacco on: — the respiratory system; — the cardiovascular system; — the ability to resist hypoxia; — the ability to withstand G-forces; — night vision.	X	X	X	X	X	X			1	
		Caffeine										
(02)		Indicate the level of caffeine dosage at which performance is degraded.	X	X	X	X	X	X				
(03)		Besides coffee, indicate other beverages containing caffeine.	X	X	X	X	X	X				
		Alcohol										
(04)		State the maximum acceptable limit of alcohol for flight crew according to the applicable regulations.	X	X	X	X	X	X				
(05)		State the effects of alcohol consumption on: — the ability to reason; — inhibitions and self-control; — vision; — the sense of balance and sensory illusions; — sleep patterns; — hypoxia.	X	X	X	X	X	X				
(06)		State the effects alcohol may have if consumed together with other drugs.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		List the signs and symptoms of alcoholism.	X	X	X	X	X	X				
(08)		List the factors that may be associated with the development of alcoholism.	X	X	X	X	X	X				
(09)		Define the 'unit' of alcohol and state the approximate elimination rate from the blood.	X	X	X	X	X	X				
(10)		State the maximum daily and weekly intake of units of alcohol which may be consumed without causing damage to the organs and systems of the human body.	X	X	X	X	X	X				
(11)		Discuss the actions that might be taken if a crew member is suspected of being an alcoholic.	X		X	X						
		<i>Prescription and non-prescription drugs and self-medication</i>										
(12)		State the dangers associated with the use of non-prescription drugs.	X	X	X	X	X	X			1	
(13)		State the side effects of common non-prescription drugs used to treat colds, flu, hay fever and other allergies, especially medicines containing antihistamine preparations.	X	X	X	X	X	X			1	
(14)		Interpret the rules relevant to using (prescription or non-prescription) drugs that the pilot has not used before.	X	X	X	X	X	X			1	
(15)		Interpret the general rule that 'if a pilot is so unwell that they require any medication, then they should consider themselves unfit to fly'.	X	X	X	X	X	X			1	
		<i>Toxic materials</i>										
(16)		List those materials present in an aircraft which may, when uncontained, cause severe health problems.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(17)		List those aircraft-component parts which if burnt may give off toxic fumes.	X	X	X	X	X	X				
(18)		Describe a fume event and the possible incapacitating effects on those exposed to it.	X	X	X	X	X	X				
040 02 03 05		Incapacitation in flight										
(01)		State that incapacitation is most dangerous when its onset is insidious.	X	X	X	X	X	X				
(02)		List the major causes of in-flight incapacitation.	X	X	X	X	X	X				
(03)		State the importance of crew to be able to recognise and promptly react upon incapacitation of other crew members, should it occur in flight.	X		X	X						
(04)		Explain methods and procedures to cope with incapacitation in flight.	X	X	X	X	X	X				
040 03 00 00		BASIC AVIATION PSYCHOLOGY										
040 03 01 00		Human information processing										
040 03 01 01		Attention and vigilance										
(01)		Differentiate between 'attention' and 'vigilance'.	X	X	X	X	X	X				
(02)		Differentiate between 'selective' and 'divided' attention.	X	X	X	X	X	X				
(03)		Define 'hypovigilance'.	X	X	X	X	X	X				
(04)		Identify the factors that may affect the state of vigilance.	X	X	X	X	X	X				
(05)		List the factors that may forestall hypovigilance during flight.	X	X	X	X	X	X				
(06)		Indicate the signs of reduced vigilance.	X	X	X	X	X	X				
(07)		List the factors that affect a person's level of attention.	X	X	X	X	X	X				
040 03 01 02		Perception										
(01)		Name the basis of the perceptual process.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the mechanism of perception ('bottom-up'/'top-down' process).	X	X	X	X	X	X				
(03)		Illustrate why perception is subjective and state the relevant factors that influence interpretation of perceived information.	X	X	X	X	X	X				
(04)		Describe some basic perceptual illusions.	X	X	X	X	X	X				
(05)		Illustrate some basic perceptual concepts.	X	X	X	X	X	X				
(06)		Give examples where perception plays a decisive role in flight safety.	X	X	X	X	X	X				
(07)		Stress how persuasive and believable mistaken perception can manifest itself both for an individual and a group.	X	X	X	X	X	X				
040 03 01 03		Memory										
(01)		Explain the link between the types of memory (to include sensory, working/short-term and long-term memory).	X	X	X	X	X	X				
(02)		Describe the differences between the types of memory in terms of capacity and retention time.	X	X	X	X	X	X				
(03)		Justify the importance of sensory-store memories in processing information.	X	X	X	X	X	X				
(04)		State the average maximum number of separate items that may be held in working memory (5 ± 2).	X	X	X	X	X	X				
(05)		Stress how interruption can affect short-term/working memory.	X	X	X	X	X	X				
(06)		Give examples of items that are important for pilots to hold in working memory during flight.	X	X	X	X	X	X				
(07)		Describe how the capacity of the working-memory store may be increased.	X	X	X	X	X	X				
(08)		State the subdivisions of long-term memory and give examples of their content.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(09)		Explain that skills are kept primarily in the long-term memory.	X	X	X	X	X	X				
(10)		Describe amnesia and how it affects memory.	X	X	X	X	X	X				
(11)		Name the common problems with both the long- and short-term memories and the best methods to try to counteract them.	X	X	X	X	X	X				
040 03 01 04		Response selection										
		Learning principles and techniques										
(01)		Explain and distinguish between the following basic forms of learning: — classic and operant conditioning (behaviouristic approach); — learning by insight (cognitive approach); — learning by imitating (modelling).	X	X	X	X	X	X				
(02)		Recognise pilot-related examples as behaviouristic, cognitive or modelling forms of learning.	X	X	X	X	X	X				
(03)		State the factors that are necessary for and promote the quality of learning: — intrinsic motivation; — good mental health; — rehearsals for improvement of memory; — consciousness; — vigilance; — application in practical exercises.	X	X	X	X	X	X				
(04)		Explain ways to facilitate the memorisation of information with the following learning techniques:	X	X	X	X	X	X			1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- mnemonics; - mental training.										
(05)		Describe the advantage of planning and anticipation of future actions: - define the term ‘skills’; - state the three phases of learning a skill (Anderson: cognitive, associative and autonomous phase).	X	X	X	X	X	X			1	
(06)		Explain the term ‘motor programme’ or ‘mental schema’.	X	X	X	X	X	X				
(07)		Describe the advantages and disadvantages of mental schemas.	X	X	X	X	X	X				
(08)		Explain the Rasmussen model which describes the guidance of a pilot’s behaviour in different situations.	X	X	X	X	X	X				
(09)		State the possible problems or risks associated with skill-, rule- and knowledge-based behaviour.	X	X	X	X	X	X				
		Motivation										
(10)		Define ‘motivation’.	X	X	X	X	X	X				
(11)		Explain the relationship between motivation and learning.	X	X	X	X	X	X				
(12)		Explain the problems of over-motivation, especially in the context of the extreme need to achieve.	X	X	X	X	X	X				
040 03 02 00		Human error and reliability										
040 03 02 01		Reliability of human behaviour										
(01)		Name and explain the factors that influence human reliability.	X	X	X	X	X	X				
040 03 02 02		Mental models and situation awareness										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define the term ‘situation awareness’.	X	X	X	X	X	X	X		1	
(02)		List the cues that indicate loss of situation awareness and name the steps to regain it.	X	X	X	X	X	X	X	1		
(03)		List the factors that influence one’s situation awareness both positively and negatively, and stress the importance of situation awareness in the context of flight safety.	X	X	X	X	X	X	X	1		
(04)		Define the term ‘mental model’ in relation to a surrounding complex situation.	X	X	X	X	X	X	X	1		
(05)		Describe the advantages/disadvantages of mental models.	X	X	X	X	X	X	X	1		
(06)		Explain the relationship between personal ‘mental models’ and the creation of cognitive illusions.	X	X	X	X	X	X	X	1		
040 03 02 03		Theory and model of human error										
(01)		Explain the concept of the ‘error chain’.	X	X	X	X	X	X	X	1		
(02)		Differentiate between an isolated error and an error chain.	X	X	X	X	X	X	X	1		
(03)		Distinguish between the main forms/types of errors (i.e. slips, faults, omissions and violations).	X	X	X	X	X	X	X	1		
(04)		Discuss the above errors and their relevance in flight.	X	X	X	X	X	X	X	1		
(05)		Distinguish between an active and a latent error, and give examples.	X	X	X	X	X	X	X	1		
040 03 02 04		Error generation										
(01)		Distinguish between internal and external factors in error generation.	X	X	X	X	X	X	X	1		
(02)		Identify possible sources of internal error generation.	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Define and discuss the two errors associated with motor programmes (action slip and environmental capture).	X	X	X	X	X	X	X	1		
(04)		List the three main sources of external error generation in the flight crew compartment.	X	X	X	X	X	X	X	1		
(05)		Give examples to illustrate the following factors in external error generation in the flight crew compartment: — ergonomics; — economics; — social environment.	X	X	X	X	X	X	X	1		
(06)		Name the major goals in the design of human-centred human-machine interfaces.	X	X	X	X	X	X	X	1		
(07)		Define the term 'error tolerance'.	X	X	X	X	X	X	X	1		
(08)		List and describe the strategies that are used to reduce human error.	X	X	X	X	X	X	X	1		
(09)		Describe the advantage of planning and the anticipation of future actions.	X	X	X	X	X	X	X			
040 03 03 00		Decision-making										
040 03 03 01		Decision-making concepts										
(01)		Define the terms 'deciding' and 'decision-making'.	X	X	X	X	X	X	X		1	
(02)		Describe the major factors on which decision-making should be based during the course of a flight.	X	X	X	X	X	X	X		1	
(03)		Describe the main human attributes with regard to decision-making.	X	X	X	X	X	X	X		1	
(04)		Discuss the nature of bias and its influence on the decision-making process.	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Describe the main error sources and limits in an individual's decision-making mechanism.	X	X	X	X	X	X	X		1	
(06)		State the factors upon which an individual's risk assessment is based.	X	X	X	X	X	X	X		1	
(07)		Explain the relationship between risk assessment, commitment and pressure of time in decision-making strategies.	X	X	X	X	X	X	X		1	
(08)		Explain the risks associated with dispersion or channelised attention during the application of procedures requiring a high workload within a short time frame (e.g. a go-around).	X	X	X	X	X	X			1	
(09)		Describe the positive and negative influences exerted by other group members on an individual's decision-making process (risky shift).	X	X	X	X	X	X	X		1	
(10)		Explain the general idea behind the creation of a model for decision-making based upon: — definition of the aim; — collection of information; — risk assessment; — development of options; — evaluation of options; — decision; — implementation; — consequences; — review and feedback.	X	X	X	X	X	X	X		1	
040 03 04 00		Avoiding and managing errors: cockpit management										

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
040 03 04 01		Safety awareness										
(01)		Justify the need for being aware of not only one's own performance but that of others before and during a flight and the possible consequences or risks.	X	X	X	X	X	X	X	1		
040 03 04 02		Coordination (multi-crew concepts)										
(01)		Name the objectives of the multi-crew concept.	X		X	X						
(02)		State and explain the elements of multi-crew concepts.	X		X	X						
(03)		Describe the concepts of 'standard operating procedures' (SOPs), checklists and crew briefings.	X	X	X	X	X					
(04)		Describe the purpose of and procedure for crew briefings.	X		X	X						
(05)		Describe the purpose of and procedure for checklists.	X	X	X	X	X					
(06)		Describe the function of communication in a coordinated team.	X		X	X						
(07)		Explain the advantages of SOPs.	X	X	X	X	X					
(08)		Explain how SOPs contribute to avoiding, reducing and managing threats and errors.	X	X	X	X	X					
(09)		Explain potential threats of SOPs, for example during company or type conversion (e.g. motor programmes, company culture, hazardous attitudes, developed habits).	X	X	X	X	X					
040 03 04 03		Cooperation										
(01)		Distinguish between cooperation and coercion.	X	X	X	X	X					
(02)		Define the term 'group'.	X	X	X	X	X					
(03)		Illustrate the influence of interdependence in a group.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		List the advantages and disadvantages of teamwork.	X	X	X	X	X					
(05)		Explain the term 'synergy'.	X	X	X	X	X					
(06)		Define the term 'cohesion'.	X	X	X	X	X					
(07)		Define the term 'groupthink'.	X	X	X	X	X					
(08)		State the essential conditions for good teamwork.	X	X	X	X	X					
(09)		Explain the function of role and norm in a group.	X	X	X	X	X					
(10)		Name the different role patterns which occur in a group situation.	X	X	X	X	X					
(11)		Explain how behaviour can be affected by the following factors: — persuasion; — conformity; — compliance; — obedience.	X	X	X	X	X					
(12)		Distinguish between status and role.	X	X	X	X	X					
(13)		Stress the inherent dangers of a situation where there is a mix of role and status within the flight crew compartment.	X	X	X	X	X					
(14)		Explain the terms 'leadership' and 'followership'.	X	X	X	X	X					
(15)		Describe the trans-cockpit authority gradient and its affiliated leadership styles (i.e. autocratic, laissez-faire and synergistic).	X	X	X	X	X					
(16)		Name the most important attributes of a positive leadership style.	X	X	X	X	X					
040 03 04 04		Communication										
(01)		Define the term 'communication'.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		List the most basic components of interpersonal communication.	X	X	X	X	X	X				
(03)		Explain the advantages of in-person two-way communication as opposed to one-way communication.	X	X	X	X	X	X				
(04)		<i>Intentionally left blank</i>										
(05)		Name the importance of non-verbal communication.	X	X	X	X	X	X				
(06)		Describe the general aspects of non-verbal communication.	X	X	X	X	X	X				
(07)		Describe the advantages/disadvantages of implicit and explicit communication.	X	X	X	X	X	X				
(08)		Describe the advantages and possible problems of using 'social' and 'professional' language in high-and low-workload situations.	X	X	X	X	X	X				
(09)		Name and explain the major obstacles to effective communication.	X	X	X	X	X	X				
(10)		Explain the difference between intrapersonal and interpersonal conflict.	X	X	X	X	X	X				
(11)		Describe the escalation process in human conflict.	X	X	X	X	X	X				
(12)		List the typical consequences of conflicts between crew members.	X	X	X	X	X	X				
(13)		Explain the following terms as part of the communication practice with regard to preventing or resolving conflicts: — inquiry; — active listening; — advocacy;	X	X	X	X	X	X				

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— feedback; — metacommunication; — negotiation.										
(14)		Describe the limitations of communication in situations of high workload in the flight crew compartment in view of listening, verbal, non-verbal and visual effects.	X	X	X	X	X	X				
040 03 05 00		Human behaviour										
040 03 05 01		Personality, attitude and behaviour										
(01)		Describe the factors that determine an individual's behaviour.	X	X	X	X	X	X				
(02)		Define and distinguish between 'personality', 'attitude' and 'behaviour'.	X	X	X	X	X	X				
(03)		State the origin of personality and attitude.	X	X	X	X	X	X				
(04)		State that with behaviour good and bad habits can be formed.	X	X	X	X	X	X				
(05)		Explain how behaviour is generally a product of personality, attitude and the environment to which one was exposed at significant moments (childhood, schooling and training).	X	X	X	X	X	X				
(06)		State that personality differences and selfish attitude may have effects on flight crew performance.	X	X	X	X	X	X				
040 03 05 02		Individual differences in personality and motivation										
(01)		Describe the individual differences in personality by means of a common trait model (e.g. Eysenck's personality factors) and use it to describe today's ideal pilot.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		Self-concept										
(02)		Define the term ‘self-concept’ and the role it plays in any change of personality.	X	X	X	X	X	X				
(03)		Explain how a self-concept of under-confidence may lead to an outward show of aggression and self- assertiveness.	X	X	X	X	X	X				
		Self-discipline										
(04)		Define ‘self-discipline’ and justify its importance for flight safety.	X	X	X	X	X	X				
040 03 05 03		Identification of hazardous attitudes (error proneness)										
(01)		Explain dangerous attitudes in aviation: — anti-authority; — macho; — impulsivity; — invulnerability; — complacency; — resignation.	X	X	X	X	X				1	
(02)		Describe the personality, attitude and behaviour patterns of an ideal crew member.	X	X	X	X	X					
(03)		Summarise how a person’s attitude influences their work in the flight crew compartment.	X	X	X	X	X					
040 03 06 00		Human overload and underload										
040 03 06 01		Arousal										
(01)		Explain the term ‘arousal’.	X	X	X	X	X	X				
(02)		Describe the relationship between arousal and performance.	X	X	X	X	X	X				

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			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the circumstances under which underload may occur and its possible dangers.	X	X	X	X	X	X			1	
040 03 06 02		Stress										
(01)		Explain the term ‘stress’ and why stress is a natural human reaction.	X	X	X	X	X	X				
(02)		State that the physiological response to stress is generated by the ‘fight or flight’ response.	X	X	X	X	X	X				
(03)		Describe the function of the autonomic nervous system (ANS) in stress response.	X	X	X	X	X	X				
(04)		Explain the relationship between arousal and stress.	X	X	X	X	X	X				
(05)		State the relationship between stress and performance.	X	X	X	X	X	X	X			
(06)		State the basic categories of stressors.	X	X	X	X	X	X	X			
(07)		List and discuss the major environmental sources of stress in the flight crew compartment.	X	X	X	X	X	X	X		1	
(08)		Discuss the concept of ‘break point’ with regard to stress, overload and performance.	X	X	X	X	X	X	X		1	
(09)		Name the principal causes of domestic stress.	X	X	X	X	X	X				
(10)		State that the stress experienced as a result of particular demands varies among individuals.	X	X	X	X	X	X				
(11)		Explain the factors that lead to differences in the levels of stress experienced by individuals.	X	X	X	X	X	X	X			
(12)		List the factors that influence the tolerance of stressors.	X	X	X	X	X	X			1	
(13)		State that stress is a result of perceived demands and perceived ability.	X	X	X	X	X	X				
(14)		Explain the relationship between stress and anxiety.	X	X	X	X	X	X	X			
(15)		Describe the effects of anxiety on human performance.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(16)		State the general effect of acute stress on people.	X	X	X	X	X	X	X			
(17)		Describe the relationship between stress, arousal and vigilance.	X	X	X	X	X	X				
(18)		State the general effect of chronic stress and the biological reaction by means of the three stages of the general adaptation syndrome (Selye): alarm, resistance, and exhaustion.	X	X	X	X	X	X				
(19)		Explain the differences between psychological, psychosomatic and somatic stress reactions.	X	X	X	X	X	X				
(20)		Name the typical common physiological and psychological symptoms of human overload.	X	X	X	X	X	X				
(21)		Describe the effects of stress on human behaviour.	X	X	X	X	X	X				
(22)		Explain how stress is cumulative and how stress from one situation can be transferred to a different situation.	X	X	X	X	X	X	X	1		
(23)		Explain how successful completion of a stressful task will reduce the amount of stress experienced when a similar situation arises in the future.	X	X	X	X	X	X	X	1		
(24)		Describe the effect of human underload/overload on effectiveness in the flight crew compartment.	X	X	X	X	X	X	X	1		
(25)		List sources and symptoms of human underload.	X	X	X	X	X	X	X	1		
040 03 06 03		Intentionally left blank										
040 03 06 04		Intentionally left blank										
040 03 06 05		Fatigue and stress management										
(01)		Explain the term ‘fatigue’ and differentiate between the two types of fatigue (short-term and chronic fatigue).	X	X	X	X	X	X				
(02)		Name the causes of short-term and chronic fatigue.	X	X	X	X	X	X				
(03)		Identify the symptoms and describe the effects of fatigue.	X	X	X	X	X	X			1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		List the strategies that prevent or delay the onset of fatigue and hypovigilance.	X	X	X	X	X	X				
(05)		List and describe strategies for coping with stress factors and stress reactions.	X	X	X	X	X	X				
(06)		Distinguish between short-term and long-term methods of stress management.	X	X	X	X	X	X				
(07)		Give examples of short-term methods of stress management.	X	X	X	X	X	X				
(08)		Give examples of long-term methods of coping with stress.	X	X	X	X	X	X				
(09)		Describe the fatigue risk management system (FRMS) as follows: a data-driven means of continuously monitoring and managing fatigue-related safety risks, based upon scientific principles and knowledge as well as operational experience that aims to ensure relevant personnel are performing at adequate levels of alertness.	X	X	X	X	X	X				
040 03 07 00		Advanced cockpit automation										
040 03 07 01		Advantages and disadvantages										
(01)		Compare the two basic concepts of automation: — as per Boeing, where the pilot remains the last operator; — and as per Airbus, where automated systems can correct erroneous pilot action.	X	X	X	X	X	X	X	1		
(02)		Explain the fundamental restrictions of autoflight systems to be lack of creativity in unknown situations, and lack of personal motivation with regard to safety.	X	X	X	X	X	X	X	1		
(03)		List the principal strengths and weaknesses of pilot versus autopilot systems to be creativity,	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		decision-making, prioritisation of tasks, safety attitude versus precision, reliability.										
(04)		Explain the ‘ironies of automation’: designers’ errors due to wrong interpretation of the data, leaving tasks to the pilot that are too complex to automate, loss of manual and cognitive skills of the pilot. State the necessity for regular training flights as one possible countermeasure.	X	X	X	X	X	X	X	1		
(05)		Describe methods to overcome the drawbacks of autoflight systems to be loss of manual flying capabilities, additional workload through programming, risk of slips during programming, and hypovigilance during cruise.	X	X	X	X	X	X	X	1		
040 03 07 02		Automation complacency										
(01)		State the main weaknesses in the monitoring of automatic systems to be hypovigilance during flight, and loss of flying skills.	X	X	X	X	X	X	X			
(02)		Explain some basic flight crew errors and terms that arise with the introduction of automation: — passive monitoring; — blinkered concentration; — confusion; — mode awareness.	X	X	X	X	X	X	X	1		
(03)		Explain how the method of call-outs counteracts ineffective monitoring of automatic systems.	X	X	X	X	X	X	X	1		
(04)		Define ‘complacency’.	X	X	X	X	X	X	X	1		
040 03 07 03		Working concepts										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain that the potential disadvantages of automation on crew communication are loss of awareness of input errors, flight modes, failure detection, failure comprehension, status of the aircraft and aircraft position.	X		X	X						
(02)		Explain how the negative effects of automation on pilots may be alleviated by degrading to a lower level of automation to recover comprehension of the flight status from VNAV/LNAV to ALT/HDG or even to manual flying.	X	X	X	X	X	X	X	1		
(03)		Interpret the role of automation with respect to flight safety regarding the basic principle of the use of manual versus autoflight in normal operations, frequent changes in the flight profile, and in abnormal situations.	X	X	X	X	X	X	X	1		

SUBJECT 050 – METEOROLOGY

ED Decision 2020/018/R

The operation of an aircraft is affected by the weather conditions within the atmosphere. The pilot should prove that they fulfil the following objectives in order to complete a flight safely in given meteorological conditions.

(1) Training aims

(i) Knowledge. After completion of the training, the pilot should be able to:

- understand the physical processes in the atmosphere;
- interpret the actual and forecast weather conditions in the atmosphere; and
- demonstrate understanding of the meteorological hazards and their effects on aircraft.

(ii) Skills. After completion of the training, the pilot should be able to:

- collect all the weather information which may affect a given flight;
- analyse and evaluate available weather information before flight as well as that collected in flight; and
- resolve any problems presented by the given weather conditions.

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 00 00 00		METEOROLOGY										
050 01 00 00		THE ATMOSPHERE										
050 01 01 00		Composition, extent, vertical division										
050 01 01 01		Structure of the atmosphere										
(01)		Describe the vertical division of the atmosphere up to flight level (FL) 650, based on the temperature variations with height.	X	X	X	X	X	X	X			
(02)		List the different layers and their main qualitative characteristics up to FL 650.	X	X	X	X	X	X	X			
050 01 01 02		Troposphere										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the troposphere.	X	X	X	X	X	X	X			
(02)		Describe the main characteristics of the tropopause.	X	X	X	X	X	X	X			
(03)		Describe the proportions of the most important gases in the air in the troposphere.	X	X	X	X	X	X	X			
(04)		Describe the variations of the FL and temperature of the tropopause from the poles to the equator.	X	X	X	X	X	X	X			
(05)		Describe the breaks in the tropopause along the boundaries of the main air masses.	X	X	X	X	X	X	X			
(06)		Indicate the variations of the FL of the tropopause with the seasons and the variations of atmospheric pressure.	X		X	X						
050 01 01 03		Stratosphere										
(01)		Describe the stratosphere up to FL 650.	X									
(02)		Describe that ozone can occur at jet cruise altitudes and that it constitutes a hazard.	X		X	X						
050 01 02 00		Air temperature										
050 01 02 01		Definition and units										
(01)		Define 'air temperature'.	X	X	X	X	X	X	X			
(02)	X	List the units of measurement of air temperature used in aviation meteorology (Celsius, Fahrenheit, Kelvin). (Refer to Subject 050 10 01 01)	X	X	X	X	X	X	X			
050 01 02 02		Vertical distribution of temperature										
(01)		Describe the mean vertical distribution of temperature up to FL 650.	X	X	X	X	X	X	X			
(02)		Mention the general causes of the cooling of the air in the troposphere with increasing altitude.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Calculate the temperature and temperature deviations (in relation to International Standard Atmosphere (ISA)) at specified levels.	X	X	X	X	X	X	X			
050 01 02 03		Transfer of heat										
(01)		Explain how local cooling or warming processes result in transfer of heat.	X	X	X	X	X	X	X			
(02)		Describe radiation.	X	X	X	X	X	X	X			
(03)		Describe solar radiation reaching the Earth.	X	X	X	X	X	X	X			
(04)		Describe the filtering effect of the atmosphere on solar radiation.	X	X	X	X	X	X	X			
(05)		Describe terrestrial radiation.	X	X	X	X	X	X	X			
(06)		Explain how terrestrial radiation is absorbed by some components of the atmosphere.	X	X	X	X	X	X	X			
(07)		Explain the effect of absorption and radiation in connection with clouds.	X	X	X	X	X	X	X			
(08)		Explain the process of conduction.	X	X	X	X	X	X	X			
(09)		Explain the role of conduction in the cooling and warming of the atmosphere.	X	X	X	X	X	X	X			
(10)		Explain the process of convection.	X	X	X	X	X	X	X			
(11)		Name the situations in which convection occurs.	X	X	X	X	X	X	X			
(12)		Explain the process of advection.	X	X	X	X	X	X	X			
(13)		Name the situations in which advection occurs.	X	X	X	X	X	X	X			
(14)		Describe the transfer of heat by turbulence.	X	X	X	X	X	X	X			
(15)		Describe the transfer of latent heat.	X	X	X	X	X	X	X			
050 01 02 04		Lapse rates										
(01)		Describe qualitatively and quantitatively the temperature lapse rates of the troposphere (mean value 0.65 °C/100 m or 2 °C/1 000 ft and actual values).	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 01 02 05		Development of inversions, types of inversions										
(01)		Describe the development and types of inversions.	X	X	X	X	X	X	X		1	
(02)		Explain the characteristics of inversions and of an isothermal layer concerning stability and vertical motions.	X	X	X	X	X	X	X		1	
(03)		Explain the reasons for the formation of the following inversions: — ground inversion (nocturnal radiation/advection), subsidence inversion, frontal inversion, inversion above friction layer, valley inversion.	X	X	X	X	X	X	X	1		
050 01 02 06		Temperature near the Earth's surface, insolation, surface effects, effect of clouds, effect of wind										
(01)		Explain the cooling/warming of the surface of the Earth by radiation.	X	X	X	X	X	X	X			
(02)		Explain the cooling/warming of the air by molecular or turbulent heat transfer to/from the earth or sea surfaces.	X	X	X	X	X	X	X	1		
(03)		Describe qualitatively the influence of the clouds on the cooling and warming of the surface and the air near the surface.	X	X	X	X	X	X	X	1		
(04)		Explain the influence of the wind on the cooling and warming of the air near the surfaces.	X	X	X	X	X	X	X	1		
050 01 03 00		Atmospheric pressure										
050 01 03 01		Barometric pressure, isobars										
(01)		Define 'atmospheric pressure'.	X	X	X	X	X	X	X		1	
(02)	X	List the units of measurement of the atmospheric pressure used in aviation (hPa, inches of mercury). (Refer to Subject 050 10 01 01)	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)	X	Describe the principle of the barometers (mercury barometer, aneroid barometer).	X	X	X	X	X	X				
(04)		Define isobars and identify them on surface weather charts.	X	X	X	X	X	X	X		1	
(05)		Define 'high', 'low', 'trough', 'ridge', 'col'.	X	X	X	X	X	X	X		1	
050 01 03 02		Pressure variation with height, contours (isohypses)										
(01)		Explain the pressure variation with height.	X	X	X	X	X	X	X		1	
(02)		Describe quantitatively the variation of the barometric lapse rate. <i>Remark: An approximation of the average value for the barometric lapse rate near mean sea level (MSL) is 30 ft (9 m) per 1 hPa.</i>	X	X	X	X	X	X	X		1	
(03)		State that (under conditions of ISA) pressure is approximately 50 % of MSL at 18 000 ft and density is approximately 50 % of MSL at 22 000 ft and 25 % of MSL at 40 000 ft.	X	X	X	X	X	X	X			
050 01 03 03		Reduction of pressure to QFF (MSL)										
(01)		Define 'QFF'.	X	X	X	X	X	X	X		1	
(02)		Explain the reduction of measured pressure (QFE) to QFF (MSL).	X	X	X	X	X	X	X		1	
(03)		Mention the use of QFF for surface weather charts.	X	X	X	X	X	X	X		1	
050 01 03 04		Relationship between surface pressure centres and pressure centres aloft										
(01)		Illustrate with a vertical cross section of isobaric surfaces the relationship between surface pressure systems and upper-air pressure systems.	X	X	X	X	X	X	X		1	
050 01 04 00		Air density										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 01 04 01		Relationship between pressure, temperature and density										
(01)		Describe the relationship between pressure, temperature and density.	X	X	X	X	X	X	X	1		
(02)		Describe the vertical variation of the air density in the atmosphere.	X	X	X	X	X	X	X	1		
050 01 05 00		International Standard Atmosphere (ISA)										
050 01 05 01		International Standard Atmosphere (ISA)										
(01)		Explain the use of standardised values for the atmosphere.	X	X	X	X	X	X	X	1		
(02)		List the main values of the ISA MSL pressure, MSL temperature, the vertical temperature lapse rate up to FL 650, height and temperature of the tropopause.	X	X	X	X	X	X	X	1		
050 01 06 00		Altimetry										
050 01 06 01		Terminology and definitions										
(01)		Define the following terms and explain how they are related to each other: height, altitude, pressure altitude, FL, pressure level, true altitude, true height, elevation, QNH, QFE, and standard altimeter setting.	X	X	X	X	X	X	X	1		
(02)		Describe the terms ‘transition altitude’, ‘transition level’, ‘transition layer’, ‘terrain clearance’, ‘lowest usable flight level’.	X	X	X	X	X	X	X			
050 01 06 02		Altimeter settings										
(01)		Name the altimeter settings associated to height, altitude, pressure altitude and FL.	X	X	X	X	X	X	X			
(02)		Describe the altimeter-setting procedures.	X	X	X	X	X	X	X	1		
050 01 06 03		Calculations										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Calculate the different readings on the altimeter when the pilot uses different settings (QNH, 1013.25, QFE).	X	X	X	X	X	X	X			
(02)		Illustrate with a numbered example the changes of altimeter setting and the associated changes in reading when the pilot climbs through the transition altitude or descends through the transition level.	X	X	X	X	X	X	X			
(03)		Derive the reading of the altimeter of an aircraft on the ground when the pilot uses the different settings.	X	X	X	X	X	X	X			
(04)		Explain the influence of the air temperature on the distance between the ground and the level read on the altimeter and between two FLs.	X	X	X	X	X	X	X	1		
(05)		Explain the influence of pressure areas on true altitude.	X	X	X	X	X	X	X			
(06)		Determine the true altitude/height for a given altitude/height and a given ISA temperature deviation.	X	X	X	X	X	X	X			
(07)		Calculate the terrain clearance and the lowest usable FL for given atmospheric temperature and pressure conditions.	X	X	X	X	X	X	X			
(08)		State that the 4 %-rule can be used to calculate true altitude from indicated altitude, and also indicated altitude from true altitude (not precise but sufficient due to the approximation of the 4 %-rule.)	X	X	X	X	X	X	X			
		<i>Remark: The following rules should be considered for altimetry calculations:</i>										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		a) All calculations are based on rounded pressure values to the nearest lower hPa. b) The value for the barometric lapse rate between MSL and 700 hPa to be used is 30 ft/hPa as an acceptable approximation of the barometric lapse rate. c) To determine the true altitude/height, the following rule of thumb, called the '4 %-rule', shall be used: the altitude/height changes by 4 % for each 10 °C temperature deviation from ISA. d) If no further information is given, the deviation of the outside-air temperature from ISA is considered to be constantly the same given value in the whole layer. e) The elevation of the aerodrome has to be taken into account. The temperature correction has to be considered for the layer between the ground and the position of the aircraft.										
050 01 06 04		Effect of accelerated airflow due to topography										
(01)		Describe qualitatively how the effect of accelerated airflow due to topography (the Bernoulli effect) affects altimetry.	X	X	X	X	X	X	X			
050 02 00 00		WIND										
050 02 01 00		Definition and measurement of wind										
050 02 01 01		Definition and measurement										
(01)		Define 'wind' and 'surface wind'.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		State the units of wind directions (degrees true in reports; degrees magnetic from tower) and speed (kt, m/s).	X	X	X	X	X	X	X			
(03)		Describe that the reported wind is an average wind derived from measurements with an anemometer at a height of 10 m over 2 min for local routine and special reports and ATS units, and over 10 min for aerodrome routine meteorological reports (METARs) and aerodrome special meteorological reports (SPECIs).	X	X	X	X	X	X	X			
050 02 02 00		Primary cause of wind										
050 02 02 01		Primary cause of wind, pressure gradient, Coriolis force, gradient wind										
(01)		Define the term ‘horizontal pressure gradient’.	X	X	X	X	X	X	X			
(02)		Explain how the pressure gradient force acts in relation to the pressure gradient.	X	X	X	X	X	X	X			
(03)		Explain how the Coriolis force acts in relation to the wind.	X	X	X	X	X	X	X			
(04)		Explain the development of the geostrophic wind.	X	X	X	X	X	X	X			
(05)		Indicate how the geostrophic wind flows in relation to the isobars/isohypses in the northern and in the southern hemisphere.	X	X	X	X	X	X	X			
(06)		Analyse the effect of changing latitude on the geostrophic wind speed.	X		X	X						
(07)		Explain the gradient wind effect and indicate how the gradient wind differs from the geostrophic wind in cyclonic and anticyclonic circulation.	X	X	X	X	X	X				
050 02 02 02		Variation of wind in the friction layer										
(01)		Describe why and how the wind changes direction and speed with height in the friction layer in the	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives		Aeroplane		Helicopter		IR	CB-IR(A)	BIR Exam	BIR BK	Remarks	
				ATPL	CPL	ATPL/IR	ATPL						CPL
		northern and in the southern hemisphere (rule of thumb).											
(02)		State the surface and air-mass conditions that influence the wind in the friction layer (diurnal variation).		X	X	X	X	X	X				
(03)		Name terrain, wind speed and stability as the main factors that influence the vertical extent of the friction layer.		X	X	X	X	X	X				
(04)		Explain the relationship between isobars and wind (direction and speed).		X	X	X	X	X	X	1			
		Remark: Approximate value for variation of wind in the friction layer (values to be used in examinations):											
		Type of landscape	Wind speed in friction layer in % of the geostrophic wind							The wind in the friction layer blows across the isobars towards the low pressure. Angle between wind direction and isobars.			
		over water	ca 70 %							ca 10°			
		over land	ca 50 %							ca 30°			
		WMO - No. 266											
050 02 02 03		Effects of convergence and divergence											

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe atmospheric convergence and divergence.	X	X	X	X	X	X	X	1		
(02)		Explain the relationship between convergence and divergence on the following: pressure systems at the surface and aloft; wind speed; vertical motion and cloud formation (relationship between upper-air conditions and surface pressure systems).	X	X	X	X	X	X	X	1		
050 02 03 00		General global circulation										
050 02 03 01		General circulation around the globe										
(01)		Describe the general global circulation. (Refer to Subject 050 08 01 01)	X	X	X	X	X	X	X			
(02)		Name and sketch or indicate on a map the global distribution of the surface pressure and the resulting wind pattern for all latitudes at low level in January and July.	X		X	X						
(03)		Sketch or indicate on a map the westerly and easterly tropospheric winds at high level in January and July.	X		X	X						
050 02 04 00		Local winds										
050 02 04 01		Anabatic and katabatic winds, mountain and valley winds, Venturi effects, land and sea breezes										
(01)		Describe and explain anabatic and katabatic winds.	X	X	X	X	X	X	X		1	
(02)		Describe mountain and valley winds.	X	X	X	X	X	X	X		1	
(03)		Describe the Venturi effect, convergence in valleys and mountain areas.	X	X	X	X	X	X	X		1	
(04)		Describe land and sea breezes, and sea-breeze front.	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Describe that local, low-level jet streams can develop in the evening.	X	X	X	X	X	X	X			
050 02 05 00		Mountain waves (standing waves, lee waves)										
050 02 05 01		Origin and characteristics										
(01)		Explain the origin and formation of mountain waves.	X	X	X	X	X	X	X		1	
(02)		State the conditions necessary for the formation of mountain waves.	X	X	X	X	X	X	X		1	
(03)		Describe the structure and properties of mountain waves.	X	X	X	X	X	X	X		1	
(04)		Explain how mountain waves may be identified by their associated meteorological phenomena.	X	X	X	X	X	X	X	1		
(05)		Describe that mountain wave effects can exceed the performance or structural capability of aircraft.	X	X	X	X	X	X	X			
(06)		Describe that mountain wave effects can propagate from low to high level, e.g. over Greenland and elsewhere.	X	X	X	X	X	X	X			
050 02 06 00		Turbulence										
050 02 06 01		Description and types of turbulence										
(01)		Describe turbulence and gustiness.	X	X	X	X	X	X	X		1	
(02)		List the common types of turbulence (convective, mechanical, orographic, frontal, clear-air turbulence).	X	X	X	X	X	X	X		1	
050 02 06 02		Formation and location of turbulence										
(01)		Explain the formation of convective turbulence, mechanical and orographic turbulence, and frontal turbulence.	X	X	X	X	X	X	X		1	
(02)		State where turbulence will normally be found (rough-ground surfaces, relief, inversion layers,	X	X	X	X	X	X	X		1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		cumulonimbus (CB), thunderstorm (TS) zones, unstable layers).										
050 02 06 03		Clear-air turbulence (CAT) — description, cause and location										
(01)		Describe CAT.	X	X				X	X			
(02)		Describe the formation of CAT.	X	X	X	X	X	X	X			
(03)		State where CAT is found in association with jet streams, in high-level troughs and in other disturbed high-level air flows. (Refer to Subject 050 09 02 02)	X									
(04)		State that remote sensing of CAT from satellites is not possible and that forecasting is limited.	X	X				X	X			
(05)		State that pilot reports of turbulence are a very valuable source of information as remote measurements are not available.	X	X	X	X	X	X	X			
050 02 07 00		Jet streams										
050 02 07 01		Description										
(01)		Describe jet streams.	X	X				X	X			
(02)		State the defined minimum speed of a jet stream (60 kt).	X	X				X	X			
(03)		State the typical figures for the dimensions of jet streams.	X	X				X	X			
050 02 07 02		Formation and properties of jet streams										
(01)		Explain the formation and state the heights, the speeds, the seasonal variations of speeds, the geographical positions, the seasonal occurrence and the seasonal movements of the arctic (front) jet stream, the polar (front) jet stream, the subtropical jet stream, and the tropical (easterly/equatorial) jet stream.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 02 07 03		Location of jet streams and associated CAT areas										
(01)		Sketch or describe where polar front and arctic jet streams are found in the troposphere in relation to the tropopause and to fronts.	X	X								
(02)		Describe and indicate the areas of worst wind shear and CAT.	X	X								
050 02 07 04		Intentionally left blank										
050 03 00 00		THERMODYNAMICS										
050 03 01 00		Humidity										
050 03 01 01		Water vapour in the atmosphere										
(01)		State that the density of moist air is less than the density of dry air.	X	X	X	X	X	X	X		1	
(02)		Describe the significance for meteorology of water vapour in the atmosphere.	X	X	X	X	X	X	X		1	
(03)		Indicate the sources of atmospheric humidity.	X	X	X	X	X	X	X		1	
(04)		Define ‘saturation of air by water vapour’.	X	X	X	X	X	X				
050 03 01 02		Intentionally left blank										
050 03 01 03		Temperature/dew point, relative humidity										
(01)		Define ‘dew point’.	X	X	X	X	X	X	X		1	
(02)		Define ‘relative humidity’.	X	X	X	X	X	X	X		1	
(03)		Explain the factors that influence the relative humidity at constant pressure.	X	X	X	X	X	X	X		1	
(04)		Explain the diurnal variation of the relative humidity.	X	X	X	X	X	X	X		1	
(05)		Describe the relationship between temperature and dew point.	X	X	X	X	X	X	X		1	
(06)		Estimate the relative humidity of the air from the difference between dew point and temperature.	X	X	X	X	X	X	X	1		
050 03 02 00		Change of state of water										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 03 02 01		Condensation, evaporation, sublimation, freezing and melting, latent heat										
(01)		Define ‘condensation’, ‘evaporation’, ‘sublimation’, ‘freezing and melting’ and ‘latent heat’.	X	X	X	X	X	X	X			
(02)		List the conditions for condensation/evaporation.	X	X	X	X	X	X	X			
(03)		Explain the condensation process.	X	X	X	X	X	X	X			
(04)		Explain the nature of and the need for condensation nuclei.	X	X	X	X	X	X	X			
(05)		Explain the effects of condensation on the weather.	X	X	X	X	X	X	X			
(06)		List the conditions for freezing/melting.	X	X	X	X	X	X	X	1		
(07)		Explain the process of freezing.	X	X	X	X	X	X	X			
(08)		Explain the nature of and the need for freezing nuclei.	X	X	X	X	X	X	X			
(09)		Define ‘supercooled water’. (Refer to Subject 050 09 01 01)	X	X	X	X	X	X	X	1		
(10)		List the conditions for sublimation.	X	X	X	X	X	X	X			
(11)		Explain the sublimation process.	X	X	X	X	X	X	X			
(12)		Explain the nature of and the need for sublimation nuclei.	X	X	X	X	X	X	X			
(13)		Describe the absorption or release of latent heat in each change of state of water.	X	X	X	X	X	X	X			
(14)		Illustrate all the changes of state of water with practical examples.	X	X	X	X	X	X	X	1		
050 03 03 00		Adiabatic processes										
050 03 03 01		Adiabatic processes, stability of the atmosphere										
(01)		Describe the adiabatic process in an unsaturated rising or descending air particle.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain the variation of temperature of an unsaturated rising or descending air particle.	X	X	X	X	X	X	X			
(03)		Explain the variation of humidity of an unsaturated rising or descending air particle.	X	X	X	X	X	X	X			
(04)		Describe the adiabatic process in a saturated rising or descending air particle.	X	X	X	X	X	X	X			
(05)		Explain the variation of temperature of a saturated air particle with changing altitude.	X	X	X	X	X	X	X			
(06)		Explain the static stability of the atmosphere using the actual temperature curve with reference to the adiabatic lapse rates.	X	X	X	X	X	X	X			
(07)		Define qualitatively and quantitatively the terms 'stable', 'conditionally unstable', 'unstable' and 'indifferent'.	X	X	X	X	X	X	X			
(08)		Illustrate with a schematic sketch the formation of Foehn.	X	X	X	X	X	X	X	1		
(09)		Explain the effect of the advection of air (warm or cold) on the stability of the air. <i>Remark: Dry adiabatic lapse rate = 1 °C/100 m or 3 °C/1 000 ft; average value at lower levels for saturated adiabatic lapse rate = 0.6 °C/100 m or 1.8 °C/1 000 ft (values to be used in examinations).</i>	X	X	X	X	X	X	X			
050 04 00 00		CLOUDS AND FOG										
050 04 01 00		Cloud formation and description										
050 04 01 01		Cloud formation										
(01)		Explain cloud formation by adiabatic cooling, conduction, advection and radiation.	X	X	X	X	X	X	X	1		
(02)		Describe cloud formation based on the following lifting processes:	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- unorganised lifting in thin layers and turbulent mixing; - forced lifting at fronts or over mountains; - free convection.										
(03)		List cloud types typical for stable and unstable air conditions.	X	X	X	X	X	X	X	1		
(04)		Summarise the conditions for the dissipation of clouds.	X	X	X	X	X	X	X	1		
050 04 01 02		Cloud types and cloud classification										
(01)		Describe the different cloud types and their classification.	X	X	X	X	X	X	X		1	
(02)		Identify by shape cirriform, cumuliform and stratiform clouds.	X	X	X	X	X	X	X		1	
(03)		Identify by shape and typical level the 10 cloud types (general).	X	X	X	X	X	X	X		1	
(04)		Describe and identify by shape the following species and supplementary features: castellanus, lenticularis, congestus, calvus, capillatus and virga.	X	X	X	X	X	X	X		1	
(05)		Distinguish between low-, medium- and high-level clouds according to the World Meteorological Organization's (WMO) 'cloud etage'.	X	X	X	X	X	X	X	1		
(06)		Distinguish between ice clouds, mixed clouds and pure-water clouds.	X	X	X	X	X	X	X	1		
050 04 01 03		Influence of inversions on cloud development										
(01)		Explain the influence of inversions on vertical movements in the atmosphere.	X	X	X	X	X	X	X	1		
(02)		Explain the influence of an inversion on the formation of stratus clouds.	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the influence of ground inversion on the formation of fog.	X	X	X	X	X	X	X	1		
(04)		Describe the role of the tropopause inversion with regard to the vertical development of clouds.	X	X	X	X	X					
050 04 01 04		Flying conditions in each cloud type										
(01)		Assess the 10 cloud types for icing and turbulence.	X	X	X	X	X	X	X		1	
050 04 02 00		Fog, mist, haze										
050 04 02 01		General aspects										
(01)		Define ‘fog’, ‘mist’ and ‘haze’ with reference to the WMO standards of visibility range.	X	X	X	X	X	X	X		1	
(02)		Explain briefly the formation of fog, mist and haze.	X	X	X	X	X	X	X		1	
(03)		Name the factors that generally contribute to the formation of fog and mist.	X	X	X	X	X	X	X		1	
(04)		Name the factors that contribute to the formation of haze.	X	X	X	X	X	X	X		1	
(05)		Describe freezing fog and ice fog.	X	X	X	X	X	X	X		1	
050 04 02 02		Radiation fog										
(01)		Explain the formation of radiation fog.	X	X	X	X	X	X	X	1		
(02)		Describe the significant characteristics of radiation fog, and its vertical extent.	X	X	X	X	X	X	X	1		
(03)		Summarise the conditions for the dissipation of radiation fog.	X	X	X	X	X	X	X	1		
050 04 02 03		Advection fog										
(01)		Explain the formation of advection fog.	X	X	X	X	X	X	X	1		
(02)		Describe the different possibilities of advection-fog formation (over land, sea and coastal regions).	X	X	X	X	X	X	X	1		
(03)		Describe the significant characteristics of advection fog.	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Summarise the conditions for the dissipation of advection fog.	X	X	X	X	X	X	X	1		
050 04 02 04		Sea smoke										
(01)		Explain the formation of sea smoke.	X	X	X	X	X	X	X	1		
(02)		Explain the conditions for the development of sea smoke.	X	X	X	X	X	X	X	1		
(03)		Summarise the conditions for the dissipation of sea smoke.	X	X	X	X	X	X	X	1		
050 04 02 05		Frontal fog										
(01)		Explain the formation of frontal fog.	X	X	X	X	X	X	X	1		
(02)		Describe the significant characteristics of frontal fog.	X	X	X	X	X	X	X	1		
(03)		Summarise the conditions for the dissipation of frontal fog.	X	X	X	X	X	X	X	1		
050 04 02 06		Orographic fog (hill fog)										
(01)		Summarise the features of orographic fog.	X	X	X	X	X	X	X	1		
(02)		Describe the significant characteristics of orographic fog.	X	X	X	X	X	X	X	1		
(03)		Summarise the conditions for the dissipation of orographic fog.	X	X	X	X	X	X	X	1		
050 05 00 00		PRECIPITATION										
050 05 01 00		Development of precipitation										
050 05 01 01		Process of development of precipitation										
(01)		Describe the two basic processes of forming precipitation (the Wegener–Bergeron–Findeisen process, Coalescence).	X	X	X	X	X	X	X	1		
(02)		Summarise the outlines of the ice-crystal process (the Wegener–Bergeron–Findeisen process).	X	X	X	X	X	X	X	1		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Summarise the outlines of the coalescence process.	X	X	X	X	X	X	X	1		
(04)		Explain the development of snow, rain, drizzle and hail.	X	X	X	X	X	X	X	1		
050 05 02 00		Types of precipitation										
050 05 02 01		Types of precipitation, relationship with cloud types										
(01)		List and describe the types of precipitation given in the aerodrome forecast (TAF) and METAR codes (drizzle, rain, snow, snow grains, ice pellets, hail, small hail, snow pellets, ice crystals, freezing drizzle, freezing rain).	X	X	X	X	X	X	X		1	
(02)		State the ICAO/WMO approximate diameters for cloud, drizzle and rain drops.	X	X	X	X	X	X	X	1		
(03)		State that, because of their size, hail stones can cause significant damage to aircraft.	X	X	X	X	X	X	X	1		
(04)	X	Explain the mechanism for the formation of freezing precipitation.	X	X	X	X	X	X	X		1	
(05)		Describe the weather conditions that give rise to freezing precipitation.	X	X	X	X	X	X	X	1		
(06)		Distinguish between the types of precipitation generated in convective and stratiform clouds.	X	X	X	X	X	X	X	1		
(07)		Assign typical precipitation types and intensities to different cloud types.	X	X	X	X	X	X	X	1		
(08)		Explain the relationship between moisture content and visibility during different types of winter precipitation (e.g. large vs small snowflakes).	X	X	X	X	X	X	X			
050 06 00 00		AIR MASSES AND FRONTS										
050 06 01 00		Air masses										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 06 01 01		Description, classification and source regions of air masses										
(01)		Define the term ‘air mass’.	X	X	X	X	X	X	X		3	
(02)		Describe the properties of the source regions.	X	X	X	X	X	X	X		3	
(03)		Summarise the classification of air masses by source regions.	X	X	X	X	X	X	X		3	
(04)		State the classifications of air masses by temperature and humidity at source.	X	X	X	X	X	X	X		3	
(05)		State the characteristic weather in each of the air masses.	X	X	X	X	X	X	X		3	
(06)		Name the three main air masses that affect Europe.	X	X	X	X	X	X	X		3	
(07)		Classify air masses on a surface weather chart.	X	X	X	X	X	X	X		3	
		Remark: Names and abbreviations of air masses used in examinations: — first letter: humidity — continental (c) — maritime (m) — second letter: type of air mass — arctic (A) — polar (P) — tropical (T) — equatorial (E) — third letter: temperature — cold (c) — warm (w)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 06 01 02		Modifications of air masses										
(01)		List the environmental factors that affect the final properties of an air mass.	X	X	X	X	X	X	X		3	
(02)		Explain how maritime and continental tracks modify air masses.	X	X	X	X	X	X	X		3	
(03)		Explain the effect of passage over cold or warm surfaces.	X	X	X	X	X	X	X		3	
(04)		Explain how air-mass weather is affected by the season, the air-mass track and by orographic and thermal effects over land.	X	X	X	X	X	X	X		3	
(05)		Assess the tendencies of the stability of an air mass and describe the typical resulting air-mass weather including the hazards for aviation.	X	X	X	X	X	X	X		3	
050 06 02 00		Fronts										
050 06 02 01		General aspects										
(01)		Describe the boundaries between air masses (fronts).	X	X	X	X	X	X	X		3	
(02)		Define ‘front’ and ‘frontal zone’.	X	X	X	X	X	X	X		3	
(03)		Name the global frontal systems (polar front, arctic front).	X	X	X	X	X	X				
(04)		State the approximate seasonal latitudes and geographic positions of the polar front and the arctic front.	X	X	X	X	X	X				
050 06 02 02		Warm front, associated clouds and weather										
(01)		Define a ‘warm front’.	X	X	X	X	X	X	X		3	
(02)		Describe the cloud, weather, ground visibility and aviation hazards at a warm front depending on the stability of the warm air.	X	X	X	X	X	X	X	3		
(03)		Explain the seasonal differences in the weather at warm fronts.	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Describe the structure, slope and dimensions of a warm front.	X	X	X	X	X	X	X		3	
(05)		Sketch a cross section of a warm front showing weather, cloud and aviation hazards.	X	X	X	X	X	X	X		3	
050 06 02 03		Cold front, associated clouds and weather										
(01)		Define a 'cold front'.	X	X	X	X	X	X	X		3	
(02)		Describe the cloud, weather, ground visibility and aviation hazards at a cold front depending on the stability of the warm air.	X	X	X	X	X	X	X	3		
(03)		Explain the seasonal differences in the weather at cold fronts.	X	X	X	X	X	X	X	3		
(04)		Describe the structure, slope and dimensions of a cold front.	X	X	X	X	X	X	X		3	
(05)		Sketch a cross section of a cold front showing weather, cloud and aviation hazards.	X	X	X	X	X	X	X		3	
050 06 02 04		Warm sector, associated clouds and weather										
(01)		Describe fronts and air masses associated with the warm sector.	X	X	X	X	X	X	X		3	
(02)		Describe the cloud, weather, ground visibility and aviation hazards in a warm sector.	X	X	X	X	X	X	X	3		
(03)		Explain the seasonal differences in the weather in the warm sector.	X	X	X	X	X	X	X	3		
(04)		Sketch a cross section of a warm sector showing weather, cloud and aviation hazards.	X	X	X	X	X	X	X		3	
050 06 02 05		Weather behind the cold front										
(01)		Describe the cloud, weather, ground visibility and aviation hazards behind the cold front.	X	X	X	X	X	X	X	3		
(02)		Explain the seasonal differences in the weather behind the cold front.	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 06 02 06		Occlusions, associated clouds and weather										
(01)	X	Define the term ‘occlusion’ and ‘occluded front’.	X	X	X	X	X	X	X		3	
(02)		Describe the cloud, weather, ground visibility and aviation hazards in a cold occlusion.	X	X	X	X	X	X	X	3		
(03)		Describe the cloud, weather, ground visibility and aviation hazards in a warm occlusion.	X	X	X	X	X	X	X	3		
(04)		Explain the seasonal differences in the weather at occlusions.	X	X	X	X	X	X	X	3		
(05)		Sketch a cross section of occlusions showing weather, cloud and aviation hazards.	X	X	X	X	X	X	X		3	
(06)		On a sketch illustrate the development of an occlusion and the movement of the occlusion point.	X	X	X	X	X	X	X		3	
050 06 02 07		Stationary front, associated clouds and weather										
(01)		Define a ‘stationary front’.	X	X	X	X	X	X	X		3	
(02)		Describe the cloud, weather, ground visibility and aviation hazards in a stationary front.	X	X	X	X	X	X	X	3		
050 06 02 08		Movement of fronts and pressure systems, life cycle										
(01)		Describe the movements of fronts and pressure systems and the life cycle of a mid-latitude depression.	X	X	X	X	X	X	X		3	
(02)		State the rules for predicting the direction and the speed of movement of fronts.	X	X	X	X	X	X	X		3	
(03)		State the difference in the speed of movement between cold and warm fronts.	X	X	X	X	X	X	X		3	
(04)		State the rules for predicting the direction and the speed of movement of frontal depressions.	X	X	X	X	X	X	X		3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Describe, with a sketch if required, the genesis, development and life cycle of a frontal depression with associated cloud and rain belts.	X	X	X	X	X	X	X		3	
050 06 02 09		<i>Changes of meteorological elements at a frontal wave</i>										
(01)		Sketch a plan and a cross section of a frontal wave (warm front, warm sector, and cold front) and illustrate the changes of pressure, temperature, surface wind and wind in the vertical axis.	X	X	X	X	X	X	X		3	
050 07 00 00		PRESSURE SYSTEMS										
050 07 01 00		The principal pressure areas										
050 07 01 01		<i>Location of the principal pressure areas</i>										
(01)		Identify or indicate on a map the principal global high-pressure and low-pressure areas in January and July.	X		X	X						
(02)		Explain how these pressure areas are formed.	X		X	X						
(03)		Explain how the pressure areas move with the seasons.	X		X	X						
050 07 02 00		Anticyclone										
050 07 02 01		<i>Anticyclones, types, general properties, cold and warm anticyclones, ridges and subsidence</i>										
(01)		List the different types of anticyclones.	X	X	X	X	X	X	X		3	
(02)		Describe the effect of high-level convergence in producing areas of high pressure at ground level.	X	X	X	X	X	X	X		3	
(03)		Describe air-mass subsidence, its effect on the environmental lapse rate, and the associated weather.	X	X	X	X	X	X	X		3	
(04)		Describe the formation of warm and cold anticyclones.	X	X	X	X	X	X	X	3		
(05)		Describe the formation of ridges.	X	X	X	X	X	X	X		3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Describe the properties of and the weather associated with warm and cold anticyclones.	X	X	X	X	X	X	X			
(07)		Describe the properties of and the weather associated with ridges.	X	X	X	X	X	X	X	3		
(08)		Describe the blocking anticyclone and its effects.	X	X	X	X	X	X	X		3	
050 07 03 00		Non-frontal depressions										
050 07 03 01		<i>Thermal, orographic, polar and secondary depressions; troughs</i>										
(01)		Describe the effect of high-level divergence in producing areas of low pressure at ground level.	X	X	X	X	X	X	X		3	
(02)		Describe the formation and properties of thermal, orographic (lee lows), polar and secondary depressions.	X	X	X	X	X	X	X	3		
(03)		Describe the formation, the properties and the associated weather at troughs.	X	X	X	X	X	X	X		3	
050 07 04 00		Tropical revolving storms										
050 07 04 01		<i>Characteristics of tropical revolving storms</i>										
(01)		State the conditions necessary for the formation of tropical revolving storms.	X	X	X	X	X					
(02)		State how a tropical revolving storm generally moves in its area of occurrence.	X	X	X	X	X					
(03)		Name the stages of the development of tropical revolving storms (tropical disturbance, tropical depression, tropical storm, severe tropical storm, tropical revolving storm).	X	X	X	X	X					
(04)		Describe the meteorological conditions in and near a tropical revolving storm.	X	X	X	X	X					
(05)		State the approximate dimensions of a tropical revolving storm.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		State that the movement of a tropical revolving storm can only rarely be forecast exactly, and that utmost care is necessary near a tropical revolving storm.	X	X	X	X	X					
050 07 04 02		Origin and local names, location and period of occurrence										
(01)		List the areas of origin and occurrence of tropical revolving storms, and their specified names (hurricane, typhoon, tropical cyclone).	X	X	X	X	X					
(02)		State the expected times of occurrence of tropical revolving storms in each of the source areas, and their approximate frequency.	X	X	X	X	X					
050 08 00 00		CLIMATOLOGY										
050 08 01 00		Climatic zones										
050 08 01 01		General circulation in the troposphere and lower stratosphere										
(01)	X	Describe the general tropospheric and low stratospheric circulation. (Refer to Subject 050 02 03 01)	X	X	X	X	X					
050 08 01 02		Climatic classification										
(01)		Describe the characteristics of the tropical rain climate, the dry climate, the mid-latitude climate (warm temperate rain climate), the subarctic climate (cold snow forest climate) and the snow climate (polar climate).	X	X	X	X	X					
(02)		Explain how the seasonal movement of the sun generates the transitional climate zones.	X	X	X	X	X					
(03)		State the typical locations of each major climatic zone.	X		X	X						
050 08 02 00		Tropical climatology										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 08 02 01		<i>Cause and development of tropical showers and thunderstorms: humidity, temperature, tropopause</i>										
(01)		State the conditions necessary for the formation of tropical showers and thunderstorms (mesoscale convective complex, cloud clusters).	X	X	X	X	X					
(02)		Describe the characteristics of tropical squall lines.	X	X	X	X	X					
(03)		Explain the formation of convective cloud structures caused by convergence at the boundary of the NE and SE trade winds (Intertropical Convergence Zone (ITCZ)).	X	X	X	X	X					
(04)		State the typical figures for tropical surface air temperatures and humidities, and for heights of the zero-degree isotherm.	X	X	X	X	X					
050 08 02 02		<i>Seasonal variations of weather and wind, typical synoptic situations</i>										
(01)		Indicate on a map the trade winds (tropical easterlies) and describe the associated weather.	X	X	X	X	X					
(02)		Indicate on a map the doldrums and describe the associated weather.	X	X	X	X	X					
(03)		Indicate on a sketch the latitudes of subtropical high (horse latitudes) and describe the associated weather.	X	X								
(04)		Indicate on a map the major monsoon winds.	X	X	X	X	X					
050 08 02 03		<i>Intertropical Convergence Zone (ITCZ), weather in the ITCZ, general seasonal movement</i>										
(01)		Identify or indicate on a map the positions of the ITCZ in January and July.	X	X								
(02)		Explain the seasonal movement of the ITCZ.	X	X								
(03)		Describe the weather and winds at the ITCZ.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the flight hazards associated with the ITCZ.	X	X								
050 08 02 04		Monsoon, sandstorms, cold-air outbreaks										
(01)		Define in general the term 'monsoon' and give a general overview of regions of occurrence.	X	X	X	X	X					
(02)		Describe the major monsoon conditions. (Refer to Subject 050 08 02 02)		X			X					
(03)		Explain how trade winds change character after a long track and become monsoon winds.	X	X	X	X	X					
(04)		Explain the weather and the flight hazards associated with a monsoon.	X	X	X	X	X					
(05)		Explain the formation of the SW/NE monsoon over West Africa and describe the weather, stressing the seasonal differences.	X	X	X	X	X					
(06)		Explain the formation of the SW/NE monsoon over India and describe the weather, stressing the seasonal differences.	X	X	X	X	X					
(07)		Explain the formation of the monsoon over the Far East and northern Australia and describe the weather, stressing the seasonal differences.	X	X	X	X	X					
(08)		Describe the formation and properties of sandstorms.	X	X	X	X	X					
(09)		Indicate when and where outbreaks of cold polar air can enter subtropical weather systems.	X	X	X	X	X					
(10)		Name well-known examples of polar-air outbreaks (Blizzard, Pampero).	X	X	X	X	X					
050 08 02 05		Easterly waves										
(01)		Explain the effect of easterly waves on tropical weather systems.	X		X	X						
050 08 03 00		Typical weather situations in the mid-latitudes										
050 08 03 01		Westerly situation (westerlies)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Identify on a weather chart the typical westerly situation with travelling polar front waves.	X	X				X	X		3	
050 08 03 02		High-pressure area										
(01)		Describe the high-pressure zones with the associated weather.	X	X	X	X	X	X	X		3	
(02)		Identify on a weather chart the high-pressure regions.	X	X	X	X	X	X	X		3	
050 08 03 03		Intentionally left blank										
050 08 03 04		Cold-air drop										
(01)		Define ‘cold-air drop’.	X	X	X	X	X	X	X			
(02)		Describe the formation of a cold-air drop.	X	X	X	X	X	X	X			
(03)		Identify cold-air drops on weather charts.	X	X	X	X	X	X	X			
(04)		Explain the problems and dangers of cold-air drops for aviation.	X	X	X	X	X	X	X			
050 08 04 00		Local winds and associated weather										
050 08 04 01		Foehn, Mistral, Bora										
(01)		Describe the mechanism for the development of Foehn winds (including Chinook).	X	X	X	X	X	X				
(02)		Describe the weather associated with Foehn winds.	X	X	X	X	X	X				
(03)		Describe the formation of, the characteristics of, and the weather associated with Mistral and Bora.	X	X	X	X	X	X				
050 08 04 02		Harmattan										
(01)		Describe the Harmattan wind and the associated visibility problems as an example of local winds affecting visibility.	X	X	X	X	X					
050 09 00 00		FLIGHT HAZARDS										
050 09 01 00		Icing										
050 09 01 01		Conditions for ice accretion										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Summarise the general conditions under which ice accretion occurs on aircraft (temperatures of outside air; temperature of the airframe; presence of supercooled water in clouds, fog, rain and drizzle; possibility of sublimation).	X	X	X	X	X	X	X	3		
(02)		Explain the general weather conditions under which ice accretion occurs in a venturi carburettor.	X	X	X	X	X	X	X		3	
(03)		Explain the general weather conditions under which ice accretion occurs on airframe.	X	X	X	X	X	X	X	3		
(04)		<i>Explain the formation of supercooled water in clouds, rain and drizzle. (Refer to Subject 050 03 02 01)</i>	X	X	X	X	X	X	X	3		
(05)		Explain qualitatively the relationship between the air temperature and the amount of supercooled water.	X	X	X	X	X	X	X	3		
(06)		Explain qualitatively the relationship between the type of cloud and the size and number of the droplets in cumuliform and stratiform clouds.	X	X	X	X	X	X	X	3		
(07)		Indicate in which circumstances ice can form on an aircraft on the ground: air temperature, humidity, precipitation.	X	X	X	X	X	X	X	3		
(08)		Explain in which circumstances ice can form on an aircraft in flight: inside clouds, in precipitation, and outside clouds and precipitation.	X	X	X	X	X	X	X	3		
(09)		Explain the influence of fuel temperature, radiative cooling of the aircraft surface and temperature of the aircraft surface (e.g. from previous flight) on ice formation.	X	X	X	X	X	X	X	3		
(10)		Describe the different factors that influence the intensity of icing: air temperature, amount of supercooled water in a cloud or in precipitation,	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		amount of ice crystals in the air, speed of the aircraft, shape (thickness) of the airframe parts (wings, antennas, etc.).										
(11)		Explain the effects of topography on icing.	X	X	X	X	X	X	X	3		
(12)		Explain the higher concentration of water drops in stratiform orographic clouds.	X	X	X	X	X	X	X	3		
050 09 01 02		Types of ice accretion										
(01)	X	Define 'clear ice'.	X	X	X	X	X	X	X			
(02)		Describe the conditions for the formation of clear ice.	X	X	X	X	X	X	X			
(03)		Explain the formation of the structure of clear ice with the release of latent heat during the freezing process.	X	X	X	X	X	X	X			
(04)		Describe the aspects of clear ice: appearance, weight, solidity.	X	X	X	X	X	X	X			
(05)		Define 'rime ice'.	X	X	X	X	X	X	X			
(06)		Describe the conditions for the formation of rime ice.	X	X	X	X	X	X	X			
(07)		Describe the aspects of rime ice: appearance, weight, solidity.	X	X	X	X	X	X	X	3		
(08)		Define 'mixed ice'.	X	X	X	X	X	X	X	3		
(09)		Describe the conditions for the formation of mixed ice.	X	X	X	X	X	X	X	3		
(10)		Describe the aspects of mixed ice: appearance, weight, solidity.	X	X	X	X	X	X	X	3		
(11)		Describe the possible process of ice formation in snow conditions.	X	X	X	X	X	X	X	3		
(12)		Define 'hoar frost'.	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(13)		Describe the conditions for the formation of hoar frost.	X	X	X	X	X	X	X	3		
(14)		Describe the aspects of hoar frost: appearance, solidity.	X	X	X	X	X	X	X	3		
050 09 01 03		Hazards of ice accretion, avoidance										
(01)		State the ICAO qualifying terms for the intensity of icing.	X	X	X	X	X	X	X	3		
(02)		Describe, in general, the hazards of icing.	X	X	X	X	X	X	X	3		
(03)		Assess the dangers of the different types of ice accretion.	X	X	X	X	X	X	X	3		
(04)		Describe the position of the dangerous zones of icing in fronts, in stratiform and cumuliform clouds, and in the different precipitation types.	X	X	X	X	X	X	X	3		
(05)		Indicate the possibilities of avoiding dangerous zones of icing: — in the flight planning: weather briefing, selection of track and altitude; — during flight: recognition of the dangerous zones, selection of appropriate track and altitude.	X	X	X	X	X	X	X	3		
050 09 01 04		Ice crystal icing										
(01)		Describe ice crystal icing.	X	X	X	X	X	X	X		3	
(02)		Describe the atmospheric processes leading to high ice crystal concentration. Define the variable ice water content (IWC).	X	X	X	X	X	X	X		3	
(03)		Identify weather situations and their relevant areas where high concentrations of ice crystals are likely to occur.	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Name, in general, the flight hazards associated with high concentrations of ice crystals.	X	X	X	X	X	X	X	3		
(05)		Explain how a pilot may possibly avoid areas with a high concentration of ice crystals.	X	X	X	X	X	X	X	3		
050 09 02 00		Turbulence										
050 09 02 01		Effects on flight, avoidance										
(01)		State the ICAO qualifying terms for the intensity of turbulence.	X	X	X	X	X	X	X	3		
(02)		Describe the effects of turbulence on an aircraft in flight.	X	X	X	X	X	X	X	3		
(03)		Indicate the possibilities of avoiding turbulence: — in the flight planning: weather briefing, selection of track and altitude; — during flight: selection of appropriate track and altitude.	X	X	X	X	X	X	X	3		
(04)		Describe atmospheric turbulence and distinguish between turbulence, gustiness and wind shear.	X	X	X	X	X	X	X			
(05)		Describe that forecasts of turbulence are not very reliable and state that pilot reports of turbulence are very valuable as they help others to prepare for or avoid turbulence.	X	X	X	X	X	X	X			
050 09 02 02		Clear-air turbulence (CAT): effects on flight, avoidance										
(01)		Describe the effects of CAT on flight. (Refer to Subject 050 02 06 03)	X	X	X	X	X					
(02)		Indicate the possibilities of avoiding CAT in flight: — in the flight planning: weather briefing, selection of track and altitude;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— during flight: selection of appropriate track and altitude.										
050 09 03 00		Wind shear										
050 09 03 01		Definition of wind shear										
(01)		Define ‘wind shear’ (vertical and horizontal).	X	X	X	X	X	X	X	3		
(02)		Define ‘low-level wind shear’.	X	X	X	X	X	X	X	3		
050 09 03 02		Weather conditions for wind shear										
(01)		Describe the conditions, where and how wind shear can form (e.g. thunderstorms, squall lines, fronts, inversions, land and sea breeze, friction layer, relief).	X	X	X	X	X	X	X	3		
050 09 03 03		Effects on flight, avoidance										
(01)		Describe the effects of wind shear on flight.	X	X	X	X	X	X	X	3		
(02)		Indicate the possibilities of avoiding wind shear in flight: — in the flight planning; — during flight.	X	X	X	X	X	X	X	3		
050 09 04 00		Thunderstorms										
050 09 04 01		Conditions for and process of development, forecast, location, type specification										
(01)		Name the cloud types which indicate the development of thunderstorms.	X	X	X	X	X	X	X	3		
(02)		Describe the different types of thunderstorms, their location, the conditions for and the process of development, and list their properties (air-mass thunderstorms, frontal thunderstorms, squall lines, supercell storms, orographic thunderstorms).	X	X	X	X	X	X	X	3		
050 09 04 02		Structure of thunderstorms, life cycle										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Assess the average duration of thunderstorms and their different stages.	X	X	X	X	X	X	X	3		
(02)		Describe a supercell storm: initial, supercell, tornado and dissipating stage.	X	X	X	X	X	X	X	3		
(03)		Summarise the flight hazards associated with a fully developed thunderstorm.	X	X	X	X	X	X	X	3		
(04)		Indicate on a sketch the most dangerous zones in and around a single-cell and a multi-cell thunderstorm.	X	X	X	X	X	X	X	3		
050 09 04 03		Electrical discharges										
(01)		Describe the basic outline of the electric field in the atmosphere.	X	X	X	X	X	X	X	3		
(02)		Describe types of lightning, i.e. ground stroke, intra-cloud lightning, cloud-to-cloud lightning, upward lightning.	X	X	X	X	X	X	X	3		
(03)		Describe and assess the 'St. Elmo's fire' weather phenomenon.	X	X	X	X	X	X	X		3	
(04)		Describe the development of lightning discharges.	X	X	X	X	X	X	X	3		
(05)		Describe the effect of lightning strike on aircraft and flight execution.	X	X	X	X	X	X	X	3		
050 09 04 04		Development and effects of downbursts										
(01)		Define the term 'downburst'.	X	X	X	X	X	X	X	3		
(02)		Distinguish between macroburst and microburst.	X	X	X	X	X	X	X	3		
(03)		State the weather situations leading to the formation of downbursts.	X	X	X	X	X	X	X	3		
(04)		Describe the process of development of a downburst.	X	X	X	X	X	X	X	3		
(05)		Give the typical duration of a downburst.	X	X	X	X	X	X	X	3		
(06)		Describe the effects of downbursts.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 09 04 05		Thunderstorm avoidance										
(01)		Explain how the pilot can anticipate each type of thunderstorm: through pre-flight weather briefing, observation in flight, use of specific meteorological information, use of information given by ground weather radar and by airborne weather radar. (Refer to Subject 050 10 01 04), use of a lightning detector (stormscope). (Refer to Subject 050 10 01 04), use of the stormscope (lightning detector).	X	X	X	X	X	X	X	3		
(02)		Describe practical examples of flight techniques used to avoid the hazards of thunderstorms.	X	X	X	X	X	X	X	3		
050 09 05 00		Tornadoes										
050 09 05 01		Properties and occurrence										
(01)		Define ‘tornado’.	X	X	X	X	X	X	X		3	
(02)		Describe the formation of a tornado.	X	X	X	X	X					
(03)		Describe the typical features of a tornado such as appearance, season, time of day, stage of development, speed of movement, and wind speed.	X	X	X	X	X					
(04)		Compare the occurrence of tornadoes in Europe with the occurrence in other locations, especially in the United States of America.	X	X	X	X	X					
(05)		Compare the dimensions and properties of tornadoes and dust devils.	X	X	X	X	X					
050 09 06 00		Inversions										
050 09 06 01		Influence on aircraft performance										
(01)		Compare the flight hazards during take-off and approach associated with a strong inversion alone	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		and with a strong inversion combined with marked wind shear.										
050 09 07 00		Stratospheric conditions										
050 09 07 01		<i>Influence on aircraft performance</i>										
(01)		Summarise the advantages of stratospheric flights.	X	X	X	X	X					
(02)		List the influences of the phenomena associated with the lower stratosphere (wind, temperature, air density, turbulence).	X	X	X	X	X					
050 09 08 00		Hazards in mountainous areas										
050 09 08 01		<i>Influence of terrain on clouds and precipitation, frontal passage</i>										
(01)		Describe the influence of mountainous area on a frontal passage.	X	X	X	X	X	X	X	3		
050 09 08 02		<i>Vertical movements, mountain waves, wind shear, turbulence, ice accretion</i>										
(01)		Describe the vertical movements, wind shear and turbulence that are typical of mountain areas.	X	X	X	X	X	X	X	3		
(02)		Indicate on a sketch of a chain of mountains the turbulent zones (mountain waves, rotors).	X	X	X	X	X	X	X	3		
(03)		Explain the influence of relief on ice accretion.	X	X	X	X	X	X	X	3		
050 09 08 03		<i>Development and effect of valley inversions</i>										
(01)		Describe the formation of a valley inversion due to katabatic winds.	X	X	X	X	X	X	X	3		
(02)		Describe the valley inversion formed by warm winds aloft.	X	X	X	X	X	X	X	3		
(03)		Describe the effects of a valley inversion for an aircraft in flight.	X	X	X	X	X	X	X	3		
050 09 09 00		Visibility-reducing phenomena										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 09 09 01		Reduction of visibility caused by precipitation and obscurations										
(01)		Describe the reduction of visibility caused by precipitation: drizzle, rain, snow.	X	X	X	X	X	X	X	3		
(02)		Describe the reduction of visibility caused by obscurations: — fog, mist, haze, smoke, volcanic ash.	X	X	X	X	X	X	X	3		
(03)		Describe the reduction of visibility caused by obscurations: — sand (SA), dust (DU).	X		X	X				3		
(04)		Describe the differences between ground and flight visibility, and slant and vertical visibility when an aircraft is above or within a layer of haze or fog.	X	X	X	X	X	X	X	3		
050 09 09 02		Reduction of visibility caused by other phenomena										
(01)		Describe the reduction of visibility caused by low drifting and blowing snow.	X	X	X	X	X	X	X	3		
(02)		Describe the reduction of visibility caused by low drifting and blowing dust and sand.	X	X	X	X	X					
(03)		Describe the reduction of visibility caused by dust storm (DS) and sandstorm (SS).	X	X	X	X	X					
(04)		Describe the reduction of visibility caused by icing (windshield).	X	X	X	X	X	X	X	3		
(05)		Describe the reduction of visibility caused by the position of the sun relative to the visual direction.	X	X	X	X	X	X	X	3		
(06)		Describe the reduction of visibility caused by the reflection of the sun's rays from the top of the layers of haze, fog and clouds.	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 10 00 00		METEOROLOGICAL INFORMATION										
050 10 01 00		Observation										
050 10 01 01		Surface observations										
(01)		Define ‘gusts’, as given in METARs.	X	X	X	X	X	X	X			
(02)		Distinguish wind given in METARs and wind given by the control tower for take-off and landing.	X	X	X	X	X	X	X			
(03)		Define ‘visibility’.	X	X	X	X	X	X	X		2	
(04)		Describe the meteorological measurement of visibility.	X	X	X	X	X	X	X		2	
(05)		Define ‘prevailing visibility’.	X	X	X	X	X	X	X			
(06)		Define ‘ground visibility’.	X	X	X	X	X	X	X	2		
(07)		List the units used for visibility (m, km, statute mile).	X	X	X	X	X	X	X		2	
(08)		Define ‘runway visual range’.	X	X	X	X	X	X	X	2		
(09)		Describe the meteorological measurement of runway visual range.	X	X	X	X	X	X	X	2		
(10)		Indicate where the transmissometers/ forward-scatter meters are placed on the aerodrome.	X	X	X	X	X	X	X	2		
(11)		List the units used for runway visual range (m, ft).	X	X	X	X	X	X	X		2	
(12)		List the different possibilities to transmit information to pilots about runway visual range.	X	X	X	X	X	X	X	2		
(13)		Compare ground visibility, prevailing visibility, and runway visual range.	X	X	X	X	X	X	X	2		
(14)		Indicate the means of observation of present weather.	X	X	X	X	X	X				
(15)		Indicate the means of observing clouds for the purpose of recording: type, amount, height of base (ceilometers), and top.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(16)		State the clouds which are indicated in METAR, TAF and SIGMET.	X	X	X	X	X	X	X	2		
(17)		Define 'oktas'.	X	X	X	X	X	X	X		2	
(18)		Define 'cloud base'.	X	X	X	X	X	X	X		2	
(19)		Define 'ceiling'.	X	X	X	X	X	X	X		2	
(20)		Name the unit and the reference level used for information about cloud base (ft).	X	X	X	X	X	X	X		2	
(21)		Define 'vertical visibility'.	X	X	X	X	X	X	X		2	
(22)		Explain briefly how and when vertical visibility is measured.	X	X	X	X	X	X	X	2		
(23)		Name the units used for vertical visibility (ft, m).	X	X	X	X	X	X	X		2	
(24)		Indicate the means of observation of air temperature (thermometer).	X	X	X	X	X	X	X			
(25)		Name the units of relative humidity (%) and dew-point temperature (Celsius, Fahrenheit).	X	X	X	X	X	X				
050 10 01 02		Radiosonde observations										
(01)		Describe the principle of radiosondes.	X	X	X	X	X	X				
(02)	X	Describe and interpret the sounding by radiosonde given on a simplified temperature–pressure (T–P) diagram.	X	X	X	X	X	X				
050 10 01 03		Satellite observations										
(01)		Describe the basic outlines of satellite observations.	X	X	X	X	X	X	X			
(02)		Name the main uses of satellite pictures in aviation meteorology.	X	X	X	X	X	X	X			
(03)		Describe the different types of satellite imagery.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Interpret qualitatively the satellite pictures in order to get useful information for flights: — location of clouds (distinguish between stratiform and cumuliform clouds).	X	X	X	X	X	X	X			
(05)		Interpret qualitatively the satellite pictures in order to get useful information for flights: — location of fronts.	X	X	X	X	X	X	X			
(06)		Interpret qualitatively the satellite pictures in order to get useful information for flights using atmospheric motion vector images to locate jet streams.	X									
050 10 01 04		Weather radar observations (Refer to Subject 050 09 04 05)										
(01)		Describe the basic principle and the type of information given by a ground weather radar.	X	X	X	X	X	X				
(02)		Interpret ground weather radar images.	X	X	X	X	X	X	X	2		
(03)		Describe the basic principle and the type of information given by airborne weather radar.	X	X	X	X	X	X	X	2		
(04)		Describe the limits and the errors of airborne weather radar information.	X	X	X	X	X	X	X	2		
(05)		Interpret typical airborne weather radar images.	X	X	X	X	X	X	X	2		
050 10 01 05		Aircraft observations and reporting										
(01)		Describe routine air-report and special air-report (ARS).	X	X	X	X	X	X	X			
(02)		State the obligation of a pilot to prepare air-reports.	X	X	X	X	X	X	X			
(03)		Name the weather phenomena to be stated in an ARS.	X	X	X	X	X	X	X			
050 10 02 00		Weather charts										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
050 10 02 01		Significant weather charts										
(01)		Decode and interpret significant weather charts (low, medium and high level).	X	X	X	X	X	X	X	2		
(02)		Describe from a significant weather chart the flight conditions at designated locations or along a defined flight route at a given FL.	X	X	X	X	X	X	X	2		
050 10 02 02		Surface charts										
(01)		Recognise the following weather systems on a surface weather chart (analysed and forecast): ridges, cols and troughs; fronts; frontal side, warm sector and rear side of mid-latitude frontal lows; high- and low-pressure areas.	X	X	X	X	X	X	X	2		
(02)		Determine from surface weather charts the wind direction and speed.	X	X	X	X	X	X	X			
050 10 02 03		Upper-air charts										
(01)		Define 'constant-pressure chart'.	X	X	X							
(02)		Define 'isohypse (contour line)'. (Refer to Subject 050 01 03 02)	X	X	X							
(03)		Define 'isotherm'.	X	X	X							
(04)		Define 'isotach'.	X	X	X							
(05)		Describe forecast upper-wind and temperature charts.	X	X	X							
(06)		For designated locations or routes determine from forecast upper-wind and temperature charts, if necessary by interpolation, the spot/average values for outside-air temperature, temperature deviation from ISA, wind direction, and wind speed.	X	X	X							
050 10 02 04		Gridded forecast products										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		State that numerical weather prediction uses a 3D grid of weather data, consisting of horizontal data (latitude-longitude) and vertical data (height or pressure).	X	X	X	X	X					
(02)		Explain that world area forecast centres prepare global sets of gridded forecasts for flight planning purposes (upper wind, temperature, humidity).	X	X	X	X	X					
(03)		State that the WAFCs also produce gridded datasets for Flight Level and temperature of the tropopause, direction and speed of maximum wind, cumulonimbus clouds, icing and turbulence.	X	X	X	X	X					
(04)		Explain that the data on CB and turbulence can be used in the visualization of flight hazards.	X	X	X	X	X					
(05)		Explain that the gridded forecasts can be merged in information processing systems with data relayed from aircraft or pilot reports, e.g. of turbulence, to provide improved situation awareness.	X	X	X	X	X					
050 10 03 00		Information for flight planning										
050 10 03 01		Aviation weather messages										
(01)		Describe, decode and interpret the following aviation weather messages (given in written or graphical format): METAR, aerodrome special meteorological report (SPECI), trend forecast (TREND), TAF, information concerning en-route weather phenomena which may affect the safety of aircraft operations (SIGMET), information concerning en-route weather phenomena which may affect the safety of low-level aircraft operations (AIRMET), area forecast for low-level	X	X	X	X	X	X	X	2	3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		flights (GAMET), ARS, volcanic ash advisory information.										
(02)		Describe, decode and interpret the tropical cyclone advisory information in written and graphical form.	X	X	X	X	X					
(03)		Describe the general meaning of MET REPORT and SPECIAL REPORT.	X	X	X	X	X	X	X	2	1	
(04)		List, in general, the cases when a SIGMET and an AIRMET are issued.	X	X	X	X	X	X	X	2	1	
(05)		Describe, decode (by using a code table) and interpret the following messages: runway state message (as written in a METAR). <i>Remark: For runway state message, refer to ICAO Doc 7754 'Air Navigation Plan — European Region'.</i>	X	X	X	X	X	X	X	2	1	
050 10 03 02		Meteorological broadcasts for aviation										
(01)		Describe the meteorological content of broadcasts for aviation: — meteorological information for aircraft in flight (VOLMET); — automatic terminal information service (ATIS).	X	X	X	X	X	X	X	2	1	
(02)		Describe the meteorological content of broadcasts for aviation: — HF-VOLMET.	X	X	X	X	X					
050 10 03 03		Use of meteorological documents										
(01)		Describe meteorological briefing and advice.	X	X	X	X	X	X	X	2	1	
(02)		List the information that a flight crew can receive from meteorological services for pre-flight	X	X	X	X	X	X	X	2	1	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		planning and apply the content of this information on a designated flight route.										
(03)		List the meteorological information that a flight crew can receive from flight information services during flight and apply the content of this information for the continuation of the flight.	X	X	X	X	X	X	X	2	1	
050 10 03 04		Meteorological warnings										
(01)		Describe and interpret aerodrome warnings and wind-shear warnings and alerts.	X	X	X	X	X	X	X	2	1	
050 10 04 00		Meteorological services										
050 10 04 01		World area forecast system and meteorological offices										
(01)	X	Name the world area forecast centres (WAFCs) as the provider for upper-air forecasts: WAFCs prepare upper-air gridded forecasts of upper winds; upper-air temperature and humidity; direction, speed and flight level of maximum wind; flight level and temperature of tropopause, areas of cumulonimbus clouds, icing, clear-air and in-cloud turbulence, and geopotential altitude of flight levels.	X	X	X	X	X	X	X			
(02)	X	Name the meteorological (MET) offices as the provider for aerodrome forecasts and briefing documents.	X	X	X	X	X	X	X			
(03)	X	Name the meteorological watch offices (MWOs) as the provider for SIGMET and AIRMET information.	X	X	X	X	X	X				
(04)	X	Name the aeronautical meteorological stations as the provider for METAR and MET reports.	X	X	X	X	X	X				
(05)	X	Name the volcanic ash advisory centres (VAACs) as the provider for forecasts of volcanic ash clouds.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)	X	Name the tropical cyclone advisory centres (TCACs) as the provider for forecasts of tropical cyclones.	X		X	X						
050 10 04 02		International organisations										
(01)	X	Describe briefly the following organisations and their chief activities in relation to weather for aviation: — International Civil Aviation Organization (ICAO) (Refer to Subject 010 'Air Law'); — World Meteorological Organization (WMO).	X	X	X	X	X	X				

SUBJECT 061 – NAVIGATION – GENERAL NAVIGATION

ED Decision 2020/018/R

Mental dead reckoning (MDR)

Where the term ‘mental dead reckoning’ (MDR) is used within a Learning Objective (LO), the applicable technique which will be used for the European Central Question Bank (ECQB) questions is based on the methods shown below.

Examination questions will state that an MDR technique is required to produce the solution. If other techniques (e.g. trigonometry) are used to determine the answer, then the determined answer may be incorrect.

MDR crosswind component (XWC)

The XWC can be calculated using a ‘clock code rule’, where each 15° of wind angle is represented by 1/4 of an hour — meaning 1/4 the wind strength.

The XWC can be estimated using the values from the table below:

Wind angle	15°	30°	45°	60°
% of wind speed	25	50	75	100

(Wind angle (WA) is the angle between the wind vector and the track/runway direction to the nearest 10°)

Example:

RWY 04 and surface wind from tower is 085°/20 kt. What is the XWC?

WA = 45°

XWC = (0.75) × 20

= 15 kt

MDR headwind component (HWC)/tailwind component (TWC)

The H/TWC can be estimated using the values from the following table:

90° – wind angle	10°	20°	30°	40°	50°	60°
% of wind speed	0.2	0.3	0.5	0.6	0.8	0.9

To assist recall, an aid is shown below:

90° – wind angle	10°	20°	30°	40°	50°	60°
Aid	1	1	2	2	3	3
% of wind speed	0.2	0.3	0.5	0.6	0.8	0.9

Example:

RWY 04 and surface wind from tower is 080°/20 kt. What is the HWC?

WA = 40°

90° – WA = 50°

HWC = (0.8) × 20

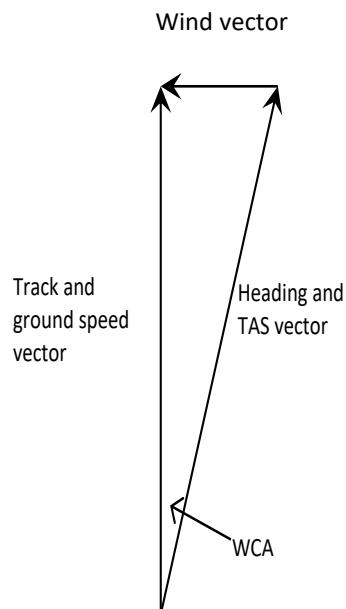
= 16 kt

Alternately, for XWC and TWC/HWC MDR calculations, the values in the following table can be used, assuming XWC = wind velocity × sine WA and TWC/HWC = wind velocity × cosine WA:

Wind angle	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°
Sine	0	0.2	0.3	0.5	0.6	0.8	0.9	0.9	1	1
Aid	0	1	1	2	2	3	3	2	2	1

MDR triangle of velocities (TOV)

Heading is determined by calculating the XWC as previously described, then applying the 1:60 rule to the TOV as follows:



This MDR technique works for the relatively small WCAs which are typical for medium to high TAS values (the ground speed (GS) therefore can be assumed to be equal to the TAS for application of the 1:60 rule).

Example 1:

Planned track = 070° (T)

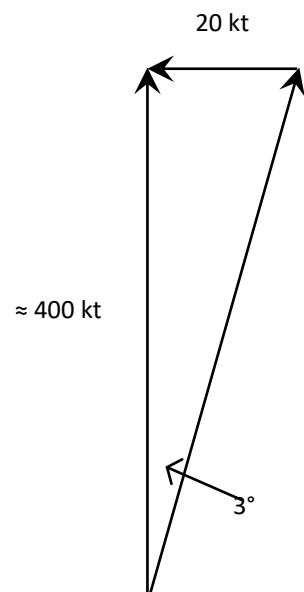
TAS = 400 kt

WV = 100° (T)/40 kt

WA = 30°

XWC = (0.5) × 40

= 20 kt



Heading required = 073° (T)

GS is determined by using the headwind/tailwind example previously explained.

$$WA = 30^\circ$$

$$90^\circ - 30^\circ = 60^\circ$$

$$\text{HWC} = (0.9) \times 40$$

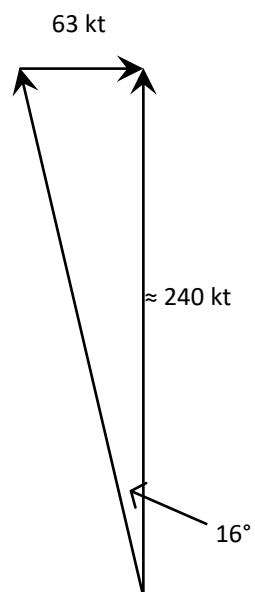
$$= 36 \text{ kt}$$

$$\text{GS} = 400 - 36 = \underline{364 \text{ kt}}$$

Example 2:

Planned track = 327° (T)

TAS = 240 kt

WV = 210° (T)/70 ktWA = 60° XWC = $(0.9) \times 7$ = 63 ktWCA = 16° Heading required = 311° (T)

GS is determined by using the headwind/tailwind example previously explained.

$$\begin{aligned} \text{WA} &= 60^\circ \\ 90^\circ - 60^\circ &= 30^\circ \\ \text{TWC} &= (0.5) \times 70 \\ &= 35 \text{ kt} \\ \text{GS} &= 240 + 35 = \underline{275 \text{ kt}} \end{aligned}$$

VFR navigation (061 02 00 00)

The techniques referred to within the LOs are based on the methods as described below.

Mental dead reckoning (MDR) off-track corrections

Based on the 1:60 rule

1 NM of cross-track error (XTE) for every 60 NM along track from waypoint = 1° of track error angle (TKE).

1 NM of XTE for every 60 NM along track to waypoint = 1° of closing angle (CA).

Change of heading required to regain track in same distance as covered from waypoint to position off track = $\underline{2 \times \text{TKE}}$.

Change of heading required to reach next waypoint from position off track = $\underline{\text{TKE} + \text{CA}}$.

Example 1:

Planned heading is 162° (T), and after 40 NM along track the aircraft position is fixed 2 NM right of planned track. What heading is required to regain track in approximately the same time as has taken to the fix position?

$$\begin{aligned} \text{TKE} &= 3^\circ \\ \text{Heading required} &= \underline{156^\circ \text{ (T)}} \end{aligned}$$

Example 2:

Planned heading is 317° (T), and after 22 NM along track the aircraft position is fixed 3.5 NM left of planned track. What heading is required to fly direct to the next waypoint which is another 45 NM down track?

TKE = 10° , CA = 5°

Heading required = 332° (T)

Mental dead reckoning (MDR) estimated time of arrival (ETA) calculations

Round the GS to the nearest NM/min, and then make the same percentage adjustment for the distance.

Example:

Distance to go = 42 NM

GS = 132 kt

GS rounded to 120 kt = 2 NM/min

Percentage change = 10 %

Distance = $42 - 10\% = 38$ NM

Time = $38 / 2 = 19$ min

Unsure-of-position procedure

As soon as the position of the aircraft is in doubt:

1. note the time;
2. communicate if in contact with an air traffic control (ATC) unit to request assistance;
3. consider using any radio-navigation aids that may be available to give position information (do not become distracted from flying the aircraft safely);
4. if short of fuel or near controlled airspace, and not in contact with ATC, set 121.5 MHz and make a PAN call;
5. if that is not necessary, check the directional indicator (DI) and compass are still synchronised and continue to fly straight and level and on route plan heading;
6. estimate the distance travelled since the last known position;

7. compare the ground with your estimated position on the map (look at the terrain for hills and valleys or line features such as a motorway, railway, river or coastline);
8. once the position has been re-established, keep checking the heading (and look out for other aircraft) and continue the flight by updating the estimated position regularly while looking for unique features such as a lake, wood, built-up area, mast, or a combination of roads, rivers and railways.

Procedure when lost

If the unsure-of-position procedure does not resolve the problem:

1. inform someone — call first on the working frequency and state the word 'LOST';
2. if there is no contact on that frequency or there is no frequency selected, change to 121.5 MHz and make a PAN call; select 7700 with ALT on the transponder if fitted.

In all cases: maintain visual meteorological conditions (VMC), note the fuel state, and try to identify an area suitable for a precautionary landing.

Consider the 'HELP ME' mnemonic:

- H. High ground/obstructions — are there any nearby?
- E. Entering controlled airspace — is that a possibility?
- L. Limited experience, low time or student pilot — let someone know.
- P. PAN call in good time — don't leave it too late.
- M. MET conditions — is the weather deteriorating?
- E. Endurance — is fuel getting low?

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
060 00 00 00		NAVIGATION										
061 00 00 00		GENERAL NAVIGATION										
061 01 00 00		BASICS OF NAVIGATION										
061 01 01 00		The Earth										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
061 01 01 01		Form										
(01)	X	State that the geoid is an irregular shape based on the surface of the oceans influenced only by gravity and centrifugal force.	X	X	X	X	X					
(02)	X	State that a number of different ellipsoids are used to describe the shape of the Earth for mapping but that WGS-84 is the reference ellipsoid required for geographical coordinates.	X	X	X	X	X					
(03)		State that the circumference of the Earth is approximately 40 000 km or approximately 21 600 NM.	X	X	X	X	X					
061 01 01 02		Earth rotation										
(01)	X	Describe the rotation of the Earth around its own spin axis and the plane of the ecliptic (including the relationship of the spin axis to the plane of the ecliptic).	X	X	X	X	X					
(02)		Explain the effect that the inclination of the Earth's spin axis has on insolation and duration of daylight.	X	X	X	X	X					
061 01 02 00		Position										
061 01 02 01		Position reference system										
(01)	X	State that geodetic latitude and longitude is used to define a position on the WGS-84 ellipsoid.	X	X	X	X	X					
(02)		Define geographic (geodetic) latitude and parallels of latitude.	X	X	X	X	X					
(03)		Calculate the difference in latitude between any two given positions.	X	X	X	X	X					
(04)		Define geographic (geodetic) longitude and meridians.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Calculate the difference in longitude between any two given positions.	X	X	X	X	X					
061 01 03 00		Direction										
061 01 03 01		Datums										
(01)	X	Define ‘true north’ (TN).	X	X	X	X	X					
(02)		Measure a true direction on any given aeronautical chart.	X	X	X	X	X					
(03)	X	Define ‘magnetic north’ (MN).	X	X	X	X	X					
(04)		Define and apply variation.	X	X	X	X	X					
(05)		Explain changes of variation with time and position.	X	X	X	X	X					
(06)	X	Define ‘compass north’ (CN).	X	X	X	X	X					
(07)		Apply deviation.	X	X	X	X	X					
061 01 03 02		Track and heading										
(01)		Calculate XWC by: – trigonometry; and – MDR.	X	X	X	X	X					
(02)		Explain and apply the concepts of drift and WCA.	X	X	X	X	X					
(03)		Calculate the actual track with appropriate data of heading and drift.	X	X	X	X	X					
(04)		Calculate TKE with appropriate data of WCA and drift.	X	X	X	X	X					
(05)		Calculate the heading change at an off-course fix to directly reach the next waypoint using the 1:60 rule.	X	X	X	X	X					
(06)		Calculate the average drift angle based upon an off-course fix observation.	X	X	X	X	X					
061 01 04 00		Distance										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
061 01 04 01		WGS-84 ellipsoid										
(01)	X	State that 1 NM is equal to 1 852 km, which is the average distance of 1' of latitude change on the WGS-84 ellipsoid.	X	X	X	X	X					
(02)		State that 1' of longitude change at the equator on the WGS-84 ellipsoid is approximately equal to 1 NM.	X	X	X	X	X					
061 01 04 02		Units										
(01)		Convert between units of distance (nautical mile (NM), kilometre (km), statute mile (SM), feet (ft), inches (in)).	X	X	X	X	X					
061 01 04 03		Graticule distances										
(01)		Calculate the distance between positions on the same meridian, on opposite (antipodal) meridians, on the same parallel of latitude, and calculate new latitude/longitude when given distances north-south and east-west.	X	X	X	X	X					
061 01 04 04		Air mile										
(01)		Evaluate the effect of wind and altitude on air distance.	X	X	X	X	X					
(02)		Convert between ground distance (NM) and air distance (NAM) using the formula: NAM = NM × TAS/GS.	X	X	X	X	X					
061 01 05 00		Speed										
061 01 05 01		True airspeed (TAS)										
(01)		Calculate TAS from CAS, and CAS from TAS by: – mechanical computer; and – rule of thumb (2 % per 1 000 ft).	X	X	X	X	X					
061 01 05 02		Mach number (M)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Calculate TAS from M, and M from TAS.	X	X								
061 01 05 03		CAS/TAS/M relationship										
(01)		Deduce the CAS, TAS and M relationship in climb/descent/cruise (flying at constant CAS or M).	X	X								
(02)		Deduce CAS and TAS in climb/descent/cruise (flying at constant CAS).			X	X	X					
061 01 05 04		Ground speed (GS)										
(01)		Calculate headwind component (HWC) and tailwind component (TWC) by: – trigonometry; and – MDR.	X	X	X	X	X					
(02)		Apply HWC and TWC to determine GS from TAS and vice versa.	X	X	X	X						
(03)	X	Explain the relationship between GS and TAS with increasing WCA.	X	X	X	X	X					
(04)		Calculate GS with: – mechanical computer (TOV solution); and – MDR (given track, TAS and WV).	X	X	X	X	X					
(05)		Perform GS, distance and time calculations.	X	X	X	X	X					
(06)		Calculate revised GS to reach a waypoint at a specific time.	X	X	X	X	X					
(07)		Calculate the average GS based on two observed fixes.	X	X	X	X	X					
061 01 05 05		Flight log										
(01)		Enter revised navigational en-route data, for the legs concerned, into the flight plan (e.g. updated	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		wind and GS and correspondingly losses or gains in time and fuel consumption).										
061 01 05 06		Gradient versus rate of climb/descent										
(01)		Estimate average climb/descent gradient (%) or glide path degrees according to the following rule of thumb: — Gradient in degrees = (vertical distance (ft) / 100) / ground distance (NM)) — Gradient in % = (vertical distance (ft) / 60) / ground distance (NM)) — Gradient in degrees = arctan (altitude difference (ft) / ground distance (ft)). <i>N.B. These rules of thumb approximate 1 NM to 6 000 ft and are based on the 1:60 rule.</i>	X	X	X	X	X					
(02)		Calculate rate of descent (ROD) on a given glide-path angle or gradient using the following rule of thumb formulae: — $ROD \text{ (ft/min)} = GP^\circ \times GS \text{ (NM/min)} \times 100$ — $ROD \text{ (ft/min)} = GP\% \times GS \text{ (kt)}$	X	X	X	X	X					
(03)		Calculate climb/descent gradient (ft/NM, % and degrees), GS or vertical speed according to the following formula: — $\text{Vertical speed (ft/min)} = (GS \text{ (kt)} \times \text{gradient (ft/NM)}) / 60.$	X	X	X	X	X					
(04)	X	State that it is necessary to determine the position of the aircraft accurately before commencing descent in order to ensure safe ground clearance.	X	X	X	X	X					
061 01 06 00		Triangle of velocities (TOV)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
061 01 06 01		Construction										
(01)		Draw and correctly label the TOV.	X	X	X	X	X					
061 01 06 02		Solutions										
(01)		Resolve the TOV for: — heading and GS (with mechanical computer and MDR); — WV (with mechanical computer); and — track and GS (with mechanical computer and MDR.	X	X	X	X	X					
061 01 07 00		Dead reckoning (DR)										
061 01 07 01		Dead reckoning (DR) technique										
(01)		Determine a DR position.	X	X	X	X	X					
(02)		Evaluate the difference between a DR and a fix position.	X	X	X	X	X					
(03)		Define ‘speed factor’ (SF). Speed divided by 60, used for mental flight-path calculations.	X	X	X	X	X					
(04)		Calculate wind correction angle (WCA) using the formula: — $WCA = XWC \text{ (crosswind component)}/SF$	X	X	X	X	X					
061 01 08 00		Navigation in climb and descent										
061 01 08 01		Average airspeed										
(01)		Average TAS used for climb problems is calculated at the altitude 2/3 of the cruising altitude.	X	X	X	X	X					
(02)		Average TAS used for descent problems is calculated at the altitude 1/2 of the descent altitude.	X	X	X	X	X					
061 01 08 02		Average wind velocity (WV)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		WV used for climb problems is the WV at the altitude 2/3 of the cruising altitude.	X	X	X	X	X					
(02)		WV used for descent problems is the WV at the altitude 1/2 of the descent altitude.	X	X	X	X	X					
(03)		Calculate the average climb/descent GS from given TAS at various altitudes, and WV at various altitudes and true track.	X	X	X	X	X					
061 01 08 03		Ground speed (GS)/distance covered during climb or descent										
(01)	X	State that most aircraft operating handbooks supply graphical material to calculate climb and descent problems.	X	X	X	X	X					
(02)		Calculate the flying time and distance during climb/descent from given average rate of climb/descent and using average GS using the following formulae valid for a 3°-glide path: — rate of descent = $(GS \times 10) / 2$ — rate of descent = speed factor (SF) $\times$ glide-path angle $\times 100$	X	X	X	X	X					
(03)		Given distance, speed and present altitude, calculate the rate of climb/descent in order to reach a certain position at a given altitude.	X	X	X	X	X					
(04)		Given speed, rate of climb/descent and altitude, calculate the distance required in order to reach a certain position at a given altitude.	X	X	X	X	X					
(05)		Given speed, distance to go and altitude to climb/descent, calculate the rate of climb/descent.	X	X	X	X	X					
061 02 00 00		VISUAL FLIGHT RULES (VFR) NAVIGATION										
061 02 01 00		Ground features										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
061 02 01 01		Ground features										
(01)		Recognise which elements would make a ground feature suitable for use for VFR navigation.	X	X	X	X	X					
061 02 01 02		Visual identification										
(01)		Describe the problems of VFR navigation at lower levels and the causes of reduced visibility.	X	X	X	X	X					
(02)		Describe the problems of VFR navigation at night.	X	X	X	X	X					
061 02 02 00		VFR navigation techniques										
061 02 02 01		Use of visual observations and application to in-flight navigation										
(01)	X	Describe what is meant by the term ‘map reading’.	X	X	X	X	X					
(02)	X	Define the term ‘visual checkpoint’.	X	X	X	X	X					
(03)		Discuss the general features of a visual checkpoint and give examples.	X	X	X	X	X					
(04)		State that the evaluation of the differences between DR positions and actual position can refine flight performance and navigation.	X	X	X	X	X					
(05)	X	Establish fixes on navigational charts by plotting visually derived intersecting lines of position.	X	X	X	X	X					
(06)	X	Describe the use of a single observed position line to check flight progress.	X	X	X	X	X					
(07)	X	Describe how to prepare and align a map/chart for use in visual navigation.	X	X	X	X	X					
(08)		Describe visual-navigation techniques including: — use of DR position to locate identifiable landmarks; — identification of charted features/landmarks;	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— factors affecting the selection of landmarks; — an understanding of seasonal and meteorological effects on the appearance and visibility of landmarks; — selection of suitable landmarks; — estimation of distance from landmarks from successive bearings; — estimation of the distance from a landmark using an approximation of the sighting angle and the flight altitude.										
(09)		Describe the action to be taken if there is no visual checkpoint available at a scheduled turning point.	X	X	X	X	X					
(10)		Understand the difficulties and limitations that may be encountered in map reading in some geographical areas due to the nature of terrain, lack of distinctive landmarks, or lack of detailed and accurate charted data.	X	X	X	X	X					
(11)	X	State the function of contour lines on a topographical chart.	X	X	X	X	X					
(12)	X	Indicate the role of ‘layer tinting’ (colour gradient) in relation to the depiction of topography on a chart.	X	X	X	X	X					
(13)		Using the contours shown on a chart, describe the appearance of a significant feature.	X	X	X	X	X					
(14)		Apply the techniques of DR, map reading, orientation, timing and revision of ETAs and headings.	X	X	X	X	X					
061 02 02 02		Unplanned events										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain what needs to be considered in case of diversion, when unsure of position and when lost.	X	X	X	X	X					
061 03 00 00		GREAT CIRCLES AND RHUMB LINES										
061 03 01 00		Great circles										
061 03 01 01		Properties										
(01)		Describe the geometric properties of a great circle (including the vertex) and a small circle.	X	X								
(02)		Describe the geometric properties of a great circle and a small circle, up to 30° difference of longitude.			X	X	X					
(03)	X	Explain why a great-circle route is the shortest distance between any two positions on the Earth.	X	X	X	X	X					
(04)		Name examples of great circles on the surface of the Earth.	X	X	X	X	X					
061 03 01 02		Convergence										
(01)	X	Explain why the track direction of a great-circle route (other than following a meridian or the equator) changes.	X	X	X	X	X					
(02)		State the formula used to approximate the value of Earth convergence as change of longitude × sine mean latitude.	X	X	X	X	X					
(03)		Calculate the approximate value of Earth convergence between any two positions, up to 30° difference of longitude.	X	X	X	X	X					
061 03 02 00		Rhumb lines										
061 03 02 01		Properties										
(01)	X	Describe the geometric properties of a rhumb line.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	State that a rhumb-line route is not the shortest distance between any two positions on the Earth (excluding meridians and equator).	X	X	X	X	X					
061 03 03 00		Relationship										
061 03 03 01		Distances										
(01)		Explain that the variation in distance of the great-circle route and rhumb-line route between any two positions increases with increasing latitude or change in longitude.	X	X	X	X	X					
061 03 03 02		Conversion angle										
(01)		Calculate and apply the conversion angle.	X	X								
061 04 00 00		CHARTS										
061 04 01 00		Chart requirements										
061 04 01 01		ICAO Annex 4 ‘Aeronautical Charts’										
(01)		State the requirement for conformality and for a straight line to approximate a great circle.	X	X	X	X	X					
061 04 01 02		Convergence										
(01)		Explain and calculate the constant of the cone (sine of parallel of origin).	X	X	X	X	X					
(02)		Explain the relationship between Earth and chart convergence with respect to the ICAO requirement for a straight line to approximate a great circle.	X	X	X	X	X					
061 04 01 03		Scale										
(01)		Recognise methods of representing scale on aeronautical charts.	X	X	X	X	X					
(02)		Perform scale calculations based on typical en-route chart scales.	X	X	X	X	X					
061 04 02 00		Projections										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
061 04 02 01		Methods of projection										
(01)	X	Identify azimuthal, cylindrical and conical projections.	X	X	X	X	X					
061 04 02 02		Polar stereographic										
(01)		State the properties of a polar stereographic projection.	X	X	X	X	X					
(02)		Calculate straight line track changes on a polar stereographic chart.	X	X	X	X	X					
061 04 02 03		Direct Mercator										
(01)		State the properties of a direct Mercator projection.	X	X	X	X	X					
(02)		Given the scale at one latitude, calculate the scale at different latitudes.	X	X	X	X	X					
(03)		Given a chart length at one latitude, show that it represents a different Earth distance at other latitudes.	X	X	X	X	X					
061 04 02 04		Lambert										
(01)		State the properties of a Lambert projection.	X	X	X	X	X					
(02)		Calculate straight line track changes on a Lambert chart.	X	X	X	X	X					
(03)		Explain the scale variation throughout the charts as follows: — the scale indicated on the chart will be correct at the standard parallels; — the scale will increase away from the parallel of origin; — the scale within the standard parallels differs by less than 1 % from the scale stated on the chart.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Given appropriate data, calculate initial, final or rhumb-line tracks between two positions (lat./long.).	X	X	X	X	X					
(05)		Given two positions (lat./long.) and information to determine convergency between the two positions, calculate the parallel of origin.	X	X	X	X	X					
(06)		Given a Lambert chart, determine the parallel of origin, or constant of cone.	X	X	X	X	X					
(07)		Given constant of cone or parallel of origin, great-circle track at one position and great-circle track at another position, calculate the difference of longitude between the two positions.	X	X	X	X	X					
061 04 03 00		Practical use										
061 04 03 01		Symbology										
(01)		Recognise ICAO Annex 4 symbology.	X	X	X	X	X					
061 04 03 02		Plotting										
(01)		Measure tracks and distances on VFR and IFR en-route charts.	X	X	X	X	X					
(02)		Fix the aircraft position on an en-route chart with information from VOR and DME equipment.	X	X	X	X	X					
(03)		Resolve bearings of an NDB station for plotting on an aeronautical chart.	X	X	X	X	X					
061 05 00 00		TIME										
061 05 01 00		Local Mean Time (LMT)										
061 05 01 01		Mean solar day										
(01)	X	Explain the concepts of a mean solar day and LMT.	X	X	X	X	X					
061 05 01 02		Local Mean Time (LMT) and Universal Time Coordinated (UTC)										
(01)		Perform LMT and UTC calculations.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
061 05 02 00		Standard time										
061 05 02 01		Standard time and daylight saving time										
(01)		Explain and apply the concept of standard time and daylight saving time, and perform standard time and daylight saving time calculations.	X	X	X	X	X					
061 05 02 02		International Date Line										
(01)		State the changes when crossing the International Date Line.	X	X	X	X	X					
061 05 03 00		Sunrise and sunset										
061 05 03 01		Sunrise and sunset times										
(01)		Define sunrise, sunset, and civil twilight, and extract times from a suitable source (e.g. an almanac).	X	X	X	X	X					
(02)		Explain the changes to sunrise, sunset, and civil twilight times with date, latitude and altitude.	X	X	X	X	X					
(03)		Explain at which time of the year the duration of daylight changes at the highest rate.	X	X	X	X	X					

SUBJECT 062 – NAVIGATION – RADIO NAVIGATION

ED Decision 2020/018/R

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
060 00 00 00		NAVIGATION										
062 00 00 00		RADIO NAVIGATION										
062 01 00 00		BASIC RADIO PROPAGATION THEORY										
062 01 01 00		Basic principles										
062 01 01 01		<i>Electromagnetic waves</i>										
(01)	X	State that radio waves travel at the speed of light, being approximately 300 000 km/s.	X	X	X	X	X	X				
(02)	X	Define a ‘cycle’: a complete series of values of a periodical process.	X	X	X	X	X	X				
062 01 01 02		<i>Frequency, wavelength, amplitude, phase angle</i>										
(01)	X	Define ‘frequency’: the number of cycles occurring in 1 second expressed in Hertz (Hz).	X	X	X	X	X	X				
(02)	X	Define ‘wavelength’: the physical distance travelled by a radio wave during one cycle of transmission.	X	X	X	X	X	X				
(03)	X	Define ‘amplitude’: the maximum deflection in an oscillation or wave.	X	X	X	X	X	X				
(04)	X	State that the relationship between wavelength and frequency is: wavelength (λ) = speed of light (c) / frequency (f).	X	X	X	X	X	X				
(05)	X	Define ‘phase angle’: the fraction of one wavelength expressed in degrees from 000° to 360°.	X	X	X	X	X	X				
(06)	X	Define ‘phase angle difference/shift’: the angular difference between the corresponding points of	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		two cycles of equal wavelength, which is measurable in degrees (°).										
062 01 01 03		Frequency bands, sidebands, single sideband										
(01)		List the bands of the frequency spectrum for electromagnetic waves: — very low frequency (VLF): 3–30 kHz; — low frequency (LF): 30–300 kHz; — medium frequency (MF): 300–3 000 kHz; — high frequency (HF): 3–30 MHz; — very high frequency (VHF): 30–300 MHz; — ultra-high frequency (UHF): 300–3 000 MHz; — super high frequency (SHF): 3–30 GHz; — extremely high frequency (EHF): 30–300 GHz.	X	X	X	X	X	X				
(02)		State that when a carrier wave is modulated, the resultant radiation consists of the carrier frequency plus additional upper and lower sidebands.	X	X	X	X	X	X				
(03)		State that HF meteorological information for aircraft in flight (VOLMET) and HF two-way communication use a single sideband.	X	X	X	X	X	X				
(04)		State that the following abbreviations (classifications according to International Telecommunication Union (ITU) regulations) are used for aviation applications:	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— NON: carrier without modulation as used by non-directional radio beacons (NDBs); — A1A: carrier with keyed Morse code modulation as used by NDBs; — A2A: carrier with amplitude modulated Morse code as used by NDBs; — A3E: carrier with amplitude modulated speech used for communication (VHF-COM).										
062 01 01 04		Pulse characteristics										
(01)		Define the following terms that are associated with a pulse string: — pulse length; — pulse power; — continuous power.	X	X	X	X	X	X				
062 01 01 05		Carrier, modulation										
(01)	X	Define ‘carrier wave’: the radio wave acting as the carrier or transporter.	X	X	X	X	X	X				
(02)	X	Define ‘modulation’: the technical term for the process of impressing and transporting information by radio waves.	X	X	X	X	X	X				
062 01 01 06		Kinds of modulation (amplitude, frequency, pulse, phase)										
(01)	X	Define ‘amplitude modulation’: the information that is impressed onto the carrier wave by altering the amplitude of the carrier.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	Define ‘frequency modulation’: the information that is impressed onto the carrier wave by altering the frequency of the carrier.	X	X	X	X	X	X				
(03)	X	Describe ‘pulse modulation’: a modulation form used in radar by transmitting short pulses followed by larger interruptions.	X	X	X	X	X	X				
(04)	X	Describe ‘phase modulation’: a modulation form used in GPS where the phase of the carrier wave is reversed.	X	X	X	X	X	X				
062 01 02 00		Antennas										
062 01 02 01		Characteristics										
(01)	X	Define ‘antenna’: an antenna or aerial is an electrical device which converts electric power into radio waves, and vice versa.	X	X	X	X	X	X				
(02)	X	State that the simplest type of antenna is a dipole, which is a wire of length equal to one half of the wavelength.	X	X	X	X	X	X				
(03)	X	State that an electromagnetic wave always consists of an oscillating electric (E) and an oscillating magnetic (H) field which propagates at the speed of light.	X	X	X	X	X	X				
(04)	X	State that the E and H fields are perpendicular to each other. The oscillations are perpendicular to the propagation direction and are in-phase.	X	X	X	X	X	X				
062 01 02 02		Polarisation										
(01)	X	State that the polarisation of an electromagnetic wave describes the orientation of the plane of oscillation of the electrical component of the wave with regard to its direction of propagation.	X	X	X	X	X	X				
062 01 02 03		Types of antennas										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Name the common different types of directional antennas: — loop antenna used in old automatic direction-finding (ADF) receivers; — parabolic antenna used in weather radars; — slotted planar array used in more modern weather radars.	X	X	X	X	X	X				
(02)		Explain ‘antenna shadowing’.	X	X	X	X	X					
(03)		Explain the importance of antenna placement on aircraft.	X	X	X	X	X					
062 01 03 00		Wave propagation										
062 01 03 01		Structure of the ionosphere and its effect on radio waves										
(01)	X	State that the ionosphere is the ionised component of the Earth’s upper atmosphere from approximately 60 to 400 km above the surface, which is vertically structured in three regions or layers.	X	X	X	X	X	X				
(02)	X	State that the layers of the ionosphere are named D, E and F layers, and their depth varies with time.	X	X	X	X	X	X				
(03)	X	State that electromagnetic waves refracted from the E and F layers of the ionosphere are called sky waves.	X	X	X	X	X	X				
(04)	X	Explain how the different layers of the ionosphere influence wave propagation.	X	X	X	X	X	X				
062 01 03 02		Ground waves										
(01)	X	Define ‘ground or surface waves’: the electromagnetic waves travelling along the surface of the Earth.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
062 01 03 03		Space waves										
(01)	X	Define ‘space waves’: the electromagnetic waves travelling through the air directly from the transmitter to the receiver.	X	X	X	X	X	X				
062 01 03 04		Propagation with the frequency bands										
(01)		State that radio waves in VHF, UHF, SHF and EHF propagate as space waves.	X	X	X	X	X	X				
(02)		State that radio waves in LF, MF and HF propagate as surface/ground waves and sky waves.	X	X	X	X	X	X				
062 01 03 05		Doppler principle										
(01)	X	State that the Doppler effect is the phenomenon where the frequency of a wave will increase or decrease if there is relative motion between the transmitter and the receiver.	X	X	X	X	X	X				
062 01 03 06		Factors affecting propagation										
(01)	X	Define ‘skip distance’: the distance between the transmitter and the point on the surface of the Earth where the first sky wave return arrives.	X	X	X	X	X	X				
(02)		State that skip zone/dead space is the distance between the limit of the surface wave and the sky wave.	X	X	X	X	X	X				
(03)		Describe ‘fading’: when a receiver picks up two signals with the same frequency, and the signals will interfere with each other causing changes in the resultant signal strength and polarisation.	X	X	X	X	X	X				
(04)		State that radio waves in the VHF band and above are limited in range as they are not reflected by the ionosphere and do not have a surface wave.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)	X	Describe the physical phenomena ‘reflection’, ‘refraction’, ‘diffraction’, ‘absorption’ and ‘interference’.	X	X	X	X	X	X				
(06)		State that multipath is when the signal arrives at the receiver via more than one path (the signal being reflected from surfaces near the receiver).	X	X	X	X	X	X				
062 02 00 00		RADIO AIDS										
062 02 01 00		Ground direction finding (DF)										
062 02 01 01		Principles										
(01)	X	Describe the use of a ground DF.	X	X	X	X	X	X				
(02)		Explain the limitation of range because of the path of the VHF signal.	X	X	X	X	X	X				
062 02 01 02		Presentation and interpretation										
(01)		Define the term ‘QDM’: the magnetic bearing to the station.	X	X	X	X	X	X	X		1	
(02)		Define the term ‘QDR’: the magnetic bearing from the station.	X	X	X	X	X	X	X		1	
(03)		Explain that by using more than one ground station, the position of an aircraft can be determined and transmitted to the pilot.	X	X	X	X	X	X				
062 02 01 03		Coverage and range										
(01)		Use the formula: $1.23 \times \sqrt{\text{transmitter height in feet}} + 1.23 \times \sqrt{\text{receiver height in feet}}$ to calculate the range in NM.	X	X	X	X	X	X	X		2	
062 02 01 04		Errors and accuracy										
(01)	X	Explain why synchronous transmissions will cause errors.	X	X	X	X	X	X				
(02)	X	Describe the effect of ‘multipath signals’.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain that VDF information is divided into the following classes according to ICAO Annex 10: – Class A: accurate to a range within $\pm 2^\circ$; – Class B: accurate to a range within $\pm 5^\circ$; – Class C: accurate to a range within $\pm 10^\circ$; – Class D: accurate to less than Class C.	X	X	X	X	X	X				
062 02 02 00		Non-directional radio beacon (NDB)/automatic direction finding (ADF)										
062 02 02 01		Principles										
(01)	X	Define the acronym 'NDB': non-directional radio beacon.	X	X	X	X	X	X	X		2	
(02)	X	Define the acronym 'ADF': automatic direction-finding equipment.	X	X	X	X	X	X	X		2	
(03)	X	State that the NDB is the ground part of the system.	X	X	X	X	X	X	X		2	
(04)	X	State that the ADF is the airborne part of the system.	X	X	X	X	X	X	X		2	
(05)		State that the NDB operates in the LF and MF frequency bands.	X	X	X	X	X	X	X	2		
(06)		State that the frequency band assigned to aeronautical NDBs according to ICAO Annex 10 is 190–1 750 kHz.	X	X	X	X	X	X	X	2		
(07)		Define a 'locator beacon': an LF/MF NDB used as an aid to final approach usually with a range of 10–25 NM.	X	X	X	X	X	X	X	2		
(08)	X	State that certain commercial radio stations transmit within the frequency band of the NDB.	X	X	X	X	X	X	X		2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(09)	X	State that according to ICAO Annex 10, an NDB station has an automatic ground monitoring system.	X	X	X	X	X	X	X			
(10)		Describe the use of NDBs for navigation.	X	X	X	X	X	X	X	2		
(11)		Describe the procedure to identify an NDB station.	X	X	X	X	X	X	X	2		
(12)	X	Interpret the term ‘cone of confusion’ in respect of an NDB.	X	X	X	X	X	X	X		2	
(13)	X	State that an NDB station emits a NON/A1A or a NON/A2A signal.	X	X	X	X	X	X	X		2	
(14)	X	State the function of the beat frequency oscillator (BFO).	X	X	X	X	X	X	X		2	
(15)	X	State that in order to identify a NON/A1A NDB, the BFO circuit of the receiver has to be activated.	X	X	X	X	X	X	X		2	
(16)	X	State that on modern aircraft, the BFO is activated automatically.	X	X	X	X	X	X	X		2	
062 02 02 02		Presentation and interpretation										
(01)	X	Name the types of indicators commonly in use: – electronic display; – radio magnetic indicator (RMI); – fixed-card ADF (radio compass); – moving-card ADF.	X	X	X	X	X	X	X		2	
(02)		Interpret the indications given on RMI, fixed-card and moving-card ADF displays.	X	X	X	X	X	X	X	2		
(03)		Given a display, interpret the relevant ADF information.	X	X	X	X	X	X	X	2		
(04)		Calculate the true bearing from the compass heading and relative bearing.	X	X	X	X	X	X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Convert the compass bearing into magnetic bearing and true bearing.	X	X	X	X	X	X	X	2		
(06)		Describe how to fly the following in-flight ADF procedures: – homing and tracking, and explain the influence of wind; – interception of inbound QDM and outbound QDR; – changing from one QDM/QDR to another; – determining station passage and the abeam point.	X	X	X	X	X	X	X	2		
062 02 02 03		Coverage and range										
(01)	X	State that the power of the transmitter limits the range of an NDB.	X	X	X	X	X	X	X		2	
(02)		Explain the relationship between power and range.	X	X	X	X	X	X	X			
(03)	X	Describe the propagation path of NDB radio waves with respect to the ionosphere and the Earth's surface.	X	X	X	X	X	X	X		2	
(04)		Explain that the interference between sky waves and ground waves leads to 'fading'.	X	X	X	X	X	X	X	2		
(05)		Define that the accuracy the pilot has to fly the required bearing in order to be considered established during approach, according to ICAO Doc 8168, has to be within $\pm 5^\circ$.	X	X	X	X	X	X	X	2		
(06)		State that there is no warning indication of NDB failure.	X	X	X	X	X	X	X	2		
062 02 02 04		Errors and accuracy										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Explain ‘coastal refraction’: as a radio wave travelling over land crosses the coast, the wave speeds up over water and the wave front bends.	X	X	X	X	X	X	X		2	
(02)	X	Define ‘night/twilight effect’: the influence of sky waves and ground waves arriving at the ADF receiver with a difference of phase and polarisation which introduce bearing errors.	X	X	X	X	X	X	X		2	
(03)		State that interference from other NDB stations on the same frequency may occur at night due to sky-wave contamination.	X	X	X	X	X	X	X	2		
062 02 02 05		Factors affecting range and accuracy										
(01)		Describe diffraction of radio waves in mountainous terrain (mountain effect).	X	X	X	X	X	X	X	2		
(02)		State that static radiation energy from a cumulonimbus cloud may interfere with the radio wave and influence the ADF bearing indication.	X	X	X	X	X	X	X	2		
(03)		Explain that the bank angle of the aircraft causes a dip error.	X	X	X	X	X	X	X			
062 02 03 00		VHF omnidirectional radio range (VOR): conventional VOR (CVOR) and Doppler VOR (DVOR)										
062 02 03 01		Principles										
(01)	X	Explain the working principle of VOR using the following general terms: — reference phase; — variable phase; — phase difference.	X	X	X	X	X	X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		State that the frequency band allocated to VOR according to ICAO Annex 10 is VHF, and the frequencies used are 108.0–117.975 MHz.	X	X	X	X	X	X	X	2		
(03)		State that frequencies within the allocated VOR range 108.0–111.975 MHz, which have an odd number in the first decimal place, are used by instrument landing system (ILS).	X	X	X	X	X	X	X	2		
(04)		State that the following types of VOR are in operation: — conventional VOR (CVOR): a first-generation VOR station emitting signals by means of a rotating antenna; — Doppler VOR (DVOR): a second-generation VOR station emitting signals by means of a combination of fixed antennas utilising the Doppler principle; — en-route VOR for use by IFR traffic; — terminal VOR (TVOR): a station with a shorter range used as part of the approach and departure structure at major aerodromes; — test VOR (VOT): a VOR station emitting a signal to test VOR indicators in an aircraft.	X	X	X	X	X	X	X	2		
(05)		State that automatic terminal information service (ATIS) information is transmitted on VOR frequencies.	X	X	X	X	X	X	X	2		
(06)	X	List the three main components of VOR airborne equipment: — the antenna;	X	X	X	X	X	X	X		2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- the receiver; - the indicator.										
(07)		Describe the identification of a VOR in terms of Morse-code letters and additional plain text.	X	X	X	X	X	X	X	2		
(08)	X	State that according to ICAO Annex 10, a VOR station has an automatic ground monitoring system.	X	X	X	X	X	X				
(09)		State that failure of the VOR station to stay within the required limits can cause the removal of identification and navigation components from the carrier or radiation to cease.	X	X	X	X	X	X	X	2		
062 02 03 02		Presentation and interpretation										
(01)		Read off the radial on an RMI.	X	X	X	X	X	X		2		
(02)		Read off the angular displacement in relation to a preselected radial on a horizontal situation indicator (HSI) or omnibearing indicator (OBI).	X	X	X	X	X	X		2		
(03)		Explain the use of the TO/FROM indicator in order to determine aircraft position relative to the VOR considering also the heading of the aircraft.	X	X	X	X	X	X		2		
(04)		Interpret VOR information as displayed on HSI, CDI and RMI.	X	X	X	X	X	X		2		
(05)		Describe the following in-flight VOR procedures: - tracking, and explain the influence of wind when tracking; - interception of a radial inbound and outbound to/from a VOR; - changing from one radial inbound/outbound to another;	X	X	X	X	X	X		2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— determining station passage and the abeam point.										
(06)		State that when converting a radial into a true bearing, the variation at the VOR station has to be taken into account.	X	X	X	X	X	X		2		
062 02 03 03		Intentionally left blank										
062 02 03 04		Errors and accuracy										
(01)		Define that the accuracy the pilot has to fly the required bearing in order to be considered established on a VOR track when flying approach procedures, according to ICAO Doc 8168, has to be within the half-full scale deflection of the required track.	X	X	X	X	X	X	X	2		
(02)		State that due to reflections from terrain, radials can be bent and lead to wrong or fluctuating indications, which is called ‘scalloping’.	X	X	X	X	X	X	X	2		
062 02 04 00		Distance-measuring equipment (DME)										
062 02 04 01		Principles										
(01)		State that DME operates in the UHF band.	X	X	X	X	X	X	X	2		
(02)	X	State that the system comprises two basic components: — the aircraft component: the interrogator; — the ground component: the transponder.	X	X	X	X	X	X	X		2	
(03)		Describe the principle of distance measurement using DME in terms of a timed transmission from the interrogator and reply from the transponder on different frequencies.	X	X	X	X	X	X				
(04)		Explain that the distance measured by DME is slant range.	X	X	X	X	X	X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Illustrate that a position line using DME is a circle with the station at its centre.	X	X	X	X	X	X	X	2		
(06)		State that the pairing of VHF and UHF frequencies (VOR/DME) enables the selection of two items of navigation information from one frequency setting.	X	X	X	X	X	X	X		2	
(07)	X	Describe, in the case of co-location with VOR and ILS, the frequency pairing and identification procedure.	X	X	X	X	X	X	X		2	
(08)		State that military UHF tactical air navigation aid (TACAN) stations may be used for DME information.	X	X	X	X	X	X	X	2		
062 02 04 02		Presentation and interpretation										
(01)	X	State that when identifying a DME station co-located with a VOR station, the identification signal with the higher-tone frequency is the DME which identifies itself approximately every 40 seconds.	X	X	X	X	X	X	X		2	
(02)		Calculate ground distance from given slant range and altitude.	X	X	X	X	X	X	X	2		
(03)		Describe the use of DME to fly a DME arc in accordance with ICAO Doc 8168 Volume 1.	X	X	X	X	X	X	X	2		
(04)	X	State that a DME system may have a ground speed (GS) and time to station read-out combined with the DME read-out.	X	X	X	X	X	X	X		2	
062 02 04 03		Coverage and range										
(01)		Explain why a ground station can generally respond to a maximum of 100 aircraft.	X	X	X	X	X	X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain which aircraft will be denied a DME range first when more than 100 interrogations are being made.	X	X	X	X	X	X	X	2		
062 02 04 04		<i>Intentionally left blank</i>										
062 02 04 05		<i>Factors affecting range and accuracy</i>										
(01)		Explain why the GS read-out from a DME can be less than the actual GS, and is zero when flying a DME arc.	X	X	X	X	X	X	X			
062 02 05 00		Instrument landing system (ILS)										
062 02 05 01		<i>Principles</i>										
(01)		Name the three main components of an ILS: — the localiser (LOC); — the glide path (GP); — range information (markers or DME).	X		X			X	X	2		
(02)	X	State the site locations of the ILS components: — the LOC antenna should be located on the extension of the runway centre line at the stop-end; — the GP antenna should be located beyond the runway threshold, laterally displaced to the side of the runway centre line.	X		X			X	X		2	
(03)		Explain that marker beacons produce radiation patterns to indicate predetermined distances from the threshold along the ILS GP.	X		X			X	X	2		
(04)		State that marker beacons are sometimes replaced by a DME paired with the LOC frequency.	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		State that in the ILS LOC frequency assigned band 108.0–111.975 MHz, only frequencies which have an odd number in the first decimal are ILS LOC frequencies.	X		X			X	X	2		
(06)		State that the GP operates in the UHF band.	X		X			X	X	2		
(07)	X	Describe the use of the 90-Hz and the 150-Hz signals in the LOC and GP transmitters/receivers, stating how the signals at the receivers vary with angular deviation.	X		X			X	X			
(08)		State that the UHF GP frequency is selected automatically by being paired with the LOC frequency.	X		X			X				
(09)		Explain that both the LOC and the GP antenna radiates side lobes (false beams) which can give rise to false centre-line and false GP indication.	X		X			X	X	2		
(10)	X	Explain that the back beam from the LOC antenna may be used as a published ‘non-precision approach’.	X		X			X	X		2	
(11)		State that the recommended GP is 3°.	X		X			X	X	2		
(12)		Name the frequency, modulation and identification assigned to all marker beacons. All marker beacons operate on 75-MHz carrier frequency. The modulation frequencies of the audio are: — outer marker: low; — middle marker: medium; — inner marker: high. The audio frequency modulation (for identification) is the continuous modulation of the audio frequency and is keyed as follows:	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- outer marker: 2 dashes per second continuously; - middle marker: a continuous series of alternate dots and dashes; - inner marker: 6 dots per second continuously. - The outer-marker cockpit indicator is coloured blue, the middle marker amber, and the inner marker white.										
(13)		State that the final-approach area contains a fix or facility that permits verification of the ILS GP–altimeter relationship. The outer marker or DME is usually used for this purpose.	X		X			X	X	2		
062 02 05 02		Presentation and interpretation										
(01)		Describe the ILS identification regarding frequency and Morse code or plain text.	X		X			X	X	2		
(02)		State that an ILS installation has an automatic ground monitoring system.	X		X			X				
(03)		State that the LOC and GP monitoring system monitors any shift in the LOC and GP mean course line or reduction in signal strength.	X		X			X				
(04)		State that warning flags will appear for both the LOC and the GP if the received signal strength is below a threshold value.	X		X			X	X	2		
(05)		Describe the circumstances in which warning flags will appear for both the LOC and the GP: - absence of the carrier frequency; - absence of the modulation simultaneously;	X		X			X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— the percentage modulation of the navigation signal reduced to 0.										
(06)		Interpret the indications on a CDI and an HSI: — full-scale deflection of the CDI needle corresponds to approximately 2.5° displacement from the ILS centre line; — full-scale deflection on the GP corresponds to approximately 0.7° from the ILS GP centre line.	X		X			X	X	2		
(07)		Interpret the aircraft's position in relation to the extended runway centre line on a back-beam approach.	X		X			X		2		
(08)		Explain the setting of the course pointer of an HSI and the course selector of an omnibearing indicator (OBI) for front-beam and back-beam approaches.	X		X			X		2		
062 02 05 03		Coverage and range										
(01)		Sketch the standard coverage area of the LOC and GP with angular sector limits in degrees and distance limits from the transmitter: LOC coverage area is 10° on either side of the centre line to a distance of 25 NM from the runway, and 35° on either side of the centre line to a distance of 17 NM from the runway; GP coverage area is 8° on either side of the centre line to a distance of minimum 10 NM from the runway.	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
062 02 05 04		Errors and accuracy										
(01)		Explain that ILS approaches are divided into facility performance categories defined in ICAO Annex 10.	X		X			X	X			
(02)		Define the following ILS operation categories: — Category I; — Category II; — Category IIIA; — Category IIIB; — Category IIIC.	X		X			X				
(03)		Explain that all Category III ILS operations guidance information is provided from the coverage limits of the facility to, and along, the surface of the runway.	X		X			X				
(04)		Explain why the accuracy requirements are progressively higher for CAT I, CAT II and CAT III ILS.	X		X			X				
(05)		Explain the following in accordance with ICAO Doc 8168: — the accuracy the pilot has to fly the ILS LOC to be considered established on an ILS track is within the half-full scale deflection of the required track; — the aircraft has to be established within the half-scale deflection of the LOC before starting descent on the GP; — the pilot has to fly the ILS GP to a maximum of half-scale fly-up deflection of	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		the GP in order to stay in protected airspace.										
(06)		State that if a pilot deviates by more than half-course deflection on the LOC or by more than half-dot deflection on the GP, an immediate go-around should be executed because obstacle clearance may no longer be guaranteed.	X		X			X	X	2		
(07)		Describe ILS beam bends as deviations from the nominal LOC and GP respectively which can be assessed by flight test.	X		X			X				
(08)		Explain that multipath interference is caused by reflections from objects within the ILS coverage area.	X		X			X				
062 02 05 05		Factors affecting range and accuracy										
(01)		Define the 'ILS-critical area': an area of defined dimensions around the LOC and GP antennas where vehicles, including aircraft, are excluded during all ILS operations.	X		X			X	X			
(02)		Define the 'ILS-sensitive area': an area extending beyond the ILS-critical area where the parking or movement of vehicles, including aircraft, is controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations.	X		X			X	X			
062 02 06 00		Microwave landing system (MLS)										
062 02 06 01		Principles										
(01)		Explain the principle of operation:	X		X			X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- horizontal course guidance during the approach; - vertical guidance during the approach; - horizontal guidance for departure and missed approach; - DME (DME/P) distance; - transmission of special information regarding the system and the approach conditions.										
(02)		State that MLS operates in the SHF band on any one of 200 channels, on assigned frequencies.	X		X			X				
(03)		Explain the reason why MLS can be installed at aerodromes where, as a result of the effects of surrounding buildings or terrain, ILS siting is difficult.	X		X			X				
062 02 06 02		Presentation and interpretation										
(01)		Interpret the display of airborne equipment designed to continuously show the position of the aircraft in relation to a preselected course and glide path, along with distance information, during approach and departure.	X		X			X				
(02)		Explain that segmented approaches can be carried out with a presentation with two cross bars directed by a computer which has been programmed with the approach to be flown.	X		X			X				
(03)		Illustrate that segmented and curved approaches can only be executed with DME/P installed.	X		X			X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain why aircraft are equipped with a multimode receiver (MMR) in order to be able to receive ILS, MLS and GPS.	X		X			X				
(05)		Explain why MLS without DME/P gives an ILS lookalike straight-line approach.	X		X			X				
062 02 06 03		Coverage and range										
(01)		Describe the coverage area for the approach direction as being within a sector of $\pm 40^\circ$ of the centre line out to a range of 20 NM from the threshold (according to ICAO Annex 10).	X		X			X				
062 03 00 00		RADAR										
062 03 01 00		Pulse techniques										
062 03 01 01		Pulse techniques and associated terms										
(01)		Name the different applications of radar with respect to air traffic control (ATC), weather observations, and airborne weather radar (AWR).	X	X	X	X	X	X	X	2		
(02)	X	Describe the pulse technique and echo principle on which primary radar systems are based.	X	X	X	X	X	X			2	
(03)	X	State that the range of a radar depends on pulse repetition frequency (PRF), pulse length, pulse power, height of aircraft, height of antenna and frequency used.	X	X	X	X	X	X				
062 03 02 00		Ground radar										
062 03 02 01		Principles										
(01)		Explain that primary radar provides bearing and distance of targets.	X		X	X		X	X	2		
(02)	X	Explain that primary ground radar is used to detect aircraft that are not equipped with a secondary radar transponder.	X		X	X		X	X		2	
062 03 02 02		Presentation and interpretation										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		State that modern ATC systems use inputs from various sensors to generate the display.	X		X	X		X	X		2	
062 03 03 00		Airborne weather radar										
062 03 03 01		Principles										
(01)		List the two main tasks of the weather radar in respect of weather and navigation.	X		X	X		X	X			
(02)		State that modern weather radars employ frequencies that give wavelengths of about 3 cm that reflect best on wet hailstones.	X		X	X		X	X			
(03)	X	State that the antenna is stabilised in the horizontal plane with signals from the aircraft's attitude reference system.	X		X	X		X	X			
(04)	X	Describe the cone-shaped pencil beam of about 3 to 5° beam width used for weather detection.	X		X	X		X	X			
062 03 03 02		Presentation and interpretation										
(01)		Explain the functions of the following different controls on the radar control panel: – off/on switch; – function switch with WX, WX+T and MAP modes; – gain-control setting (auto/manual); – tilt/autotilt switch.	X		X	X		X	X			
(02)		Name, for areas of differing reflection intensity, the colour gradations (green, yellow, red and magenta) indicating the increasing intensity of precipitation.	X		X	X		X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)	X	State the use of azimuth-marker lines and range lines in respect of the relative bearing and the distance to a thunderstorm on the screen.	X		X	X		X	X			
062 03 03 03		Coverage and range										
(01)		Explain how the radar is used for weather detection and for mapping (range, tilt and gain, if available).	X		X	X		X	X			
062 03 03 04		Errors, accuracy, limitations										
(01)		Explain why AWR should be used with extreme caution when on the ground.	X		X	X		X	X			
062 03 03 05		Factors affecting range and accuracy										
(01)		Explain the danger of the area behind heavy rain (shadow area) where no radar waves will penetrate.	X		X	X		X	X			
(02)		Describe appropriate tilt settings in relation to altitude and thunderstorms.	X		X	X		X	X			
(03)		Explain why a thunderstorm may not be detected when the tilt is set too high.	X		X	X		X	X			
062 03 03 06		Application for navigation										
(01)		Describe the navigation function of the radar in the mapping mode.	X		X	X		X	X			
(02)		Describe the use of the weather radar to avoid a thunderstorm (Cb).	X		X	X		X	X			
(03)		Explain how turbulence (not CAT) can be detected by a modern weather radar.	X		X	X		X	X			
(04)		Explain how wind shear can be detected by a modern weather radar.	X		X	X		X	X			
062 03 04 00		Secondary surveillance radar and transponder										
062 03 04 01		Principles										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		State that the ATC system is based on the replies provided by the airborne transponders in response to interrogations from the ATC secondary radar.	X	X	X	X	X	X	X	2		
(02)	X	State that the ground ATC secondary radar uses techniques which provide the ATC with information that cannot be acquired by the primary radar.	X	X	X	X	X	X	X		2	
(03)	X	State that an airborne transponder provides coded-reply signals in response to interrogation signals from the ground secondary radar and from aircraft equipped with traffic alert and collision avoidance system (TCAS).	X	X	X	X	X	X	X		2	
(04)		State the advantages of secondary surveillance radar (SSR) over a primary radar regarding range and collected information due to transponder principal information and active participation of the aircraft.	X	X	X	X	X	X	X	2		
062 03 04 02		Modes and codes										
(01)	X	State that the interrogator transmits its interrogations in the form of a series of pulse pairs.	X	X	X	X	X	X	X		2	
(02)		Name the interrogation modes: — Mode A; — Mode C; — Mode S.	X	X	X	X	X	X	X		2	
(03)		State that the interrogation frequency and the reply frequency are different.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain that the decoding of the time interval between the pulse pairs determines the operating mode of the transponder: — Mode A: transmission of aircraft transponder code; — Mode C: transmission of aircraft pressure altitude; — Mode S: selection of aircraft address and transmission of flight data for the ground surveillance.	X	X	X	X	X					
(05)		State that Mode A designation is a sequence of four digits which can be manually selected from 4 096 available codes.	X	X	X	X	X	X	X		2	
(06)		State that in Mode C reply, the pressure altitude is reported in 100-ft increments.	X	X	X	X	X	X			2	
(07)		State that in addition to the information provided, on request from ATC, a special position identification (SPI) pulse can be transmitted but only as a result of a manual selection by the pilot (IDENT button).	X	X	X	X	X	X			2	
(08)	X	State the need for compatibility of Mode S with Mode A and C.	X	X	X	X	X	X			2	
(09)		Explain that Mode S transponders receive interrogations from TCAS and SSR ground stations.	X	X	X	X	X	X			2	
(10)	X	State that Mode S interrogation contains either the aircraft address, selective call or all-call address.	X	X	X	X	X	X			2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(11)		State that every aircraft is allocated an ICAO aircraft address, which is hard-coded into the Mode S transponder (Mode S address).	X	X	X	X	X	X	X		2	
(12)		Explain that a 24-bit address is used in all Mode S transmissions, so that every interrogation can be directed to a specific aircraft.	X	X	X	X	X	X				
(13)		State that Mode S can provide enhanced vertical tracking, using a 25-ft altitude increment.	X	X	X	X	X	X				
(14)		State that SSR can be used for automatic dependent surveillance — broadcast (ADS-B).	X	X	X	X	X	X				
062 03 04 03		Presentation and interpretation										
(01)		State that an aircraft can be identified by a unique code.	X	X	X	X	X	X	X	2		
(02)		State which information can be presented on the ATC display system: — pressure altitude; — flight level; — flight number or aircraft registration number; — GS.	X	X	X	X	X	X	X	2		
(03)	X	Explain the use and function of the selector modes: OFF, Standby, ON (Mode A), ALT (Mode A, C and S), TEST, and of the reply lamp.	X	X	X	X	X	X	X		2	
062 04 00 00		INTENTIONALLY LEFT BLANK										
062 05 00 00		INTENTIONALLY LEFT BLANK										
062 06 00 00		GLOBAL NAVIGATION SATELLITE SYSTEMS (GNSSs)										
062 06 01 00		Global navigation satellite systems (GNSSs)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
062 06 01 01		General										
(01)		State that there are four main GNSSs. These are: — USA NAVigation System with Timing And Ranging Global Positioning System (NAVSTAR GPS); — Russian GLObal NAVigation Satellite System (GLONASS); — European Galileo (under construction); — Chinese BeiDou (under construction).	X	X	X	X	X	X	X	2		
(02)	X	State that all four systems (will) consist of a constellation of satellites which can be used by a suitably equipped receiver to determine position.	X	X	X	X	X	X	X		2	
062 06 01 02		Operation										
		Global navigation satellite system (GNSS)										
(01)		State that there are currently two modes of operation: standard positioning service (SPS) for civilian users, and precise positioning service (PPS) for authorised users.	X	X	X	X	X	X	X	2		
(02)		SPS was originally designed to provide civilian users with a less accurate positioning capability than PPS.	X	X	X	X	X	X	X	2		
(03)	X	Name the three GNSS segments as follows: — space segment; — control segment; — user segment.	X	X	X	X	X	X	X		2	
		Space segment (example: NAVSTAR GPS)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		State that each satellite broadcasts ranging signals on two UHF frequencies: L1 and L2.	X	X	X	X	X	X				
(05)		State that SPS is a positioning and timing service provided on frequency L1.	X	X	X	X	X	X				
(06)		State that PPS uses both frequencies L1 and L2.	X	X	X	X	X	X				
(07)	X	State that the satellites transmit a coded signal used for ranging, identification (satellite individual PRN code), timing and navigation.	X	X	X	X	X	X				
(08)	X	State that the navigation message contains: — satellite clock correction parameters; — Universal Time Coordinated (UTC) parameters; — an ionospheric model; — satellite health data.	X	X	X	X	X	X				
(09)	X	State that an ionospheric model is used to calculate the time delay of the signal travelling through the ionosphere.	X	X	X	X	X	X	X		2	
(10)	X	State that two codes are transmitted on the L1 frequency, namely a coarse acquisition (C/A) code and a precision (P) code. The P code is not used for standard positioning service (SPS).	X	X	X	X	X	X				
(11)	X	State that satellites are equipped with atomic clocks which allow the system to keep very accurate time reference.	X	X	X	X	X	X	X		2	
		Control segment										
(12)	X	State that the control segment comprises: — a master control station; — a ground antenna;	X	X	X	X	X	X	X		2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— monitoring stations.										
(13)	X	State that the control segment provides: — monitoring of the constellation status; — correction of orbital parameters; — navigation data uploading.	X	X	X	X	X	X	X			
		User segment										
(14)	X	State that GNSS supplies three-dimensional position fixes and speed data, plus a precise time reference.	X	X	X	X	X	X	X		2	
(15)	X	State that a GNSS receiver is able to determine the distance to a satellite by determining the difference between the time of transmission by the satellite and the time of reception.	X	X	X	X	X	X	X		2	
(16)	X	State that the initial distance calculated to the satellites is called pseudo-range because the difference between the GNSS receiver and the satellite time references initially creates an erroneous range.	X	X	X	X	X	X			2	
(17)	X	State that each range defines a sphere with its centre at the satellite.	X	X	X	X	X	X	X		2	
(18)	X	State that there are four unknown parameters (x, y, z and Δt) (receiver clock error) which require the measurement of ranges to four different satellites in order to get the position.	X	X	X	X	X	X	X		2	
(19)	X	State that the GNSS receiver is able to synchronise to the correct time reference when receiving four satellites.	X	X	X	X	X	X	X		2	
(20)	X	State that the receiver is able to calculate aircraft ground speed using the space vehicle (SV)	X	X	X	X	X	X			2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		Doppler frequency shift or the change in receiver position over time.										
		NAVigation System with Timing And Ranging Global Positioning System (NAVSTAR GPS) integrity										
(21)		Define ‘receiver autonomous integrity monitoring (RAIM)’ as a technique that ensures the integrity of the provided data by redundant measurements.	X	X	X	X	X	X	X	2		
(22)		State that RAIM is achieved by consistency checks among range measurements.	X	X	X	X	X	X	X	2		
(23)		State that basic RAIM requires five satellites. A sixth one is for isolating a faulty satellite from the navigation solution.	X	X	X	X	X	X	X			
(24)		State that agreements have been concluded between the appropriate agencies for the compatibility and interoperability by any approved user of NAVSTAR and GLONASS systems.	X	X	X	X	X	X				
(25)	X	State that the different GNSSs use different data with respect to reference systems, orbital data, and navigation services.	X	X	X	X	X	X				
062 06 01 03		Errors and factors affecting accuracy										
(01)		List the most significant factors that affect accuracy: — ionospheric propagation delay; — dilution of precision; — satellite clock error; — satellite orbital variations;	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— multipath.										
(02)		State that a user equivalent range error (UERE) can be computed from all these factors.	X	X	X	X	X	X				
(03)	X	State that the error from the ionospheric propagation delay (IPD) can be reduced by modelling, using a model of the ionosphere, or can almost be eliminated by using two frequencies.	X	X	X	X	X	X				
(04)	X	State that ionospheric delay is the most significant error.	X	X	X	X	X	X				
(05)		State that dilution of precision arises from the geometry and number of satellites in view. It is called geometric dilution of precision (GDOP).	X	X	X	X	X	X				
(06)		State that the UERE in combination with the geometric dilution of precision (GDOP) allows for an estimation of position accuracy.	X	X	X	X	X	X				
(07)	X	State that errors in the satellite orbits are due to: — solar winds; — gravitation of the Sun and the Moon.	X	X	X	X	X	X				
062 06 02 00		Ground-, satellite- and aircraft-based augmentation systems										
062 06 02 01		Ground-based augmentation systems (GBASs)										
(01)		Explain the principle of a GBAS: to measure on the ground the errors in the signals transmitted by GNSS satellites and relay the measured errors to the user for correction.	X	X	X	X	X	X	X			
(02)	X	State that the ICAO GBAS standard is based on this technique through the use of a data link in the VHF band of ILS–VOR systems (108–118 MHz).	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		State that for a GBAS station the coverage is about 20 NM.	X	X	X	X	X	X	X			
(04)		State that GBAS provides information for guidance in the terminal area, and for three-dimensional guidance in the final approach segment (FAS) by transmitting the FAS data block.	X	X	X	X	X	X	X			
(05)		State that one ground station can support all the aircraft subsystems within its coverage providing the aircraft with approach data, corrections and integrity information for GNSS satellites in view via a VHF data broadcast (VDB).	X	X	X	X	X	X				
(06)	X	State that the minimum software designed coverage area is 10° on either side of the final approach path to a distance between 15 and 20 NM, and 35° on either side of the final approach path up to a distance of 15 NM.	X	X	X	X	X	X				
(07)		State that outside this area the FAS data of GBAS is not used.	X	X	X	X	X	X				
(08)	X	State that GBAS based on GPS is sometimes called local area augmentation system (LAAS).	X	X	X	X	X	X				
(09)		State that a GBAS-based approach is called GLS approach (GLS-GNSS landing system).	X	X	X	X	X	X				
062 06 02 02		Satellite-based augmentation systems (SBASs)										
(01)	X	Explain the principle of an SBAS: to measure on the ground the errors in the signals received from the satellites and transmit differential corrections and integrity messages for navigation satellites.	X	X	X	X	X	X	X		2	
(02)	X	State that the frequency band of the data link is identical to that of the GPS signals.	X	X	X	X	X	X	X		2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)	X	Explain that the use of geostationary satellites enables messages to be broadcast over very wide areas.	X	X	X	X	X	X	X		2	
(04)	X	State that pseudo-range measurements to these geostationary satellites can also be made, as if they were GPS satellites.	X	X	X	X	X	X	X		2	
(05)	X	State that SBAS consists of two elements: — ground infrastructure (monitoring and processing stations); — communication satellites.	X	X	X	X	X	X	X		2	
(06)		State that SBAS allows the implementation of three-dimensional Type A and Type B approaches.	X	X	X	X	X	X	X	2		
(07)	X	State the following examples of SBAS: — European Geostationary Navigation Overlay Service (EGNOS) in western Europe and the Mediterranean; — wide area augmentation system (WAAS) in the USA; — multi-functional transport satellite (MTSAT)-based augmentation system (MSAS) in Japan; — GPS and geostationary earth orbit augmented navigation (GAGAN) in India.	X	X	X	X	X	X	X		2	
(08)	X	State that SBAS is designed to significantly improve accuracy and integrity.	X	X	X	X	X	X	X		2	
(09)		Explain that integrity and safety are improved by alerting SBAS users within 6 seconds if a GPS malfunction occurs.	X	X	X	X	X	X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
062 06 02 03		Intentionally left blank										
062 06 02 04		Aircraft-based augmentation systems (ABASs)										
(01)		Explain the principle of ABAS: to use redundant elements within the GPS constellation (e.g. multiplicity of distance measurements to various satellites) or the combination of GNSS measurements with those of other navigation sensors (such as inertial systems) in order to develop integrity control.	X	X	X	X	X	X	X	2		
(02)		State that the type of ABAS using only GNSS information is named receiver autonomous integrity monitoring (RAIM).	X	X	X	X	X	X	X	2		
(03)		State that a system using information from additional onboard sensors is named aircraft autonomous integrity monitoring (AAIM).	X	X	X	X	X	X	X	2		
(04)		Explain that the typical sensors used are barometric altimeter and inertial reference system (IRS).	X	X	X	X	X	X	X	2		
062 07 00 00		PERFORMANCE-BASED NAVIGATION (PBN)										
062 07 01 00		Performance-based navigation (PBN) concept (as described in ICAO Doc 9613)										
062 07 01 01		PBN principles										
(01)		List the factors used to define area navigation (RNAV) or required navigation performance (RNP) system performance requirements (accuracy, integrity and continuity).	X		X			X	X		3	
(02)	X	State that these RNAV and RNP systems are necessary to optimise the utilisation of available airspace.	X		X			X			2, 3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		State that it is necessary for flight crew and air traffic controllers to be aware of the on-board RNAV or RNP system capabilities in order to determine whether the performance of the RNAV or RNP system is appropriate for the specific airspace requirements.	X		X			X			2, 3	
(04)		Define accuracy as the conformance of the true position and the required position.	X		X			X			2, 3	
(05)		Define continuity as the capability of the system to perform its function without unscheduled interruptions during the intended operation.	X		X			X	X	2	3	
(06)		Define integrity as a measure of the trust that can be placed in the correctness of the information supplied by the total system. Integrity includes the ability of a system to provide timely and valid alerts to the user.	X		X			X	X	2	3	
(07)		State that, unlike conventional navigation, PBN is not sensor-specific.	X		X			X	X	2	3	
(08)		Explain the difference between raw data and computed data.	X		X			X	X	2	3	
(09)		Define availability as the percentage of time (annually) during which the system is available for use.	X		X			X	X	2	3	
062 07 01 02		PBN components										
(01)		List the components of PBN as navigational aid (NAVAID) infrastructure, navigation specification and navigation application.	X		X			X				
062 07 01 03		PBN scope										
(01)		State that in oceanic/remote, en-route and terminal phases of flight, PBN is limited to	X		X			X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		operations with linear lateral performance requirements and time constraints.										
(02)		State that in the approach phases of flight, PBN accommodates both linear and angular laterally guided operations, and explain the difference between the two.	X		X			X				
062 07 02 00		Navigation specifications										
062 07 02 01		Area navigation (RNAV) and required navigation performance (RNP)										
(01)		State the difference between RNAV and RNP in terms of the requirement for on-board performance monitoring and alerting.	X		X			X	X	2	3	
062 07 02 02		Navigation functional requirements										
(01)	X	List the basic functional requirements of the RNAV and RNP specifications (continuous indication of lateral deviation, distance/bearing to active waypoint, GS or time to active waypoint, navigation data storage and failure indication).	X		X			X				
062 07 02 03		Designation of RNP and RNAV specifications										
(01)		Interpret X in RNAV X or RNP X as the lateral navigation (LNAV) accuracy (total system error) in nautical miles, which is expected to be achieved at least 95 % of the flight time by the population of aircraft operating within the given airspace, route or procedure.	X		X			X		2	3	
(02)		State that aircraft approved to the more stringent accuracy requirements may not necessarily meet some of the functional requirements of the navigation specification that has a less stringent accuracy requirement.	X		X			X	X	2	3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		State that RNAV 10 and RNP 4 are used in the oceanic/remote phase of flight.	X		X			X				
(04)		State that RNAV 5 is used in the en-route and arrival phases of flight.	X		X			X	X	2	3	
(05)		State that RNAV 2 and RNP 2 are also used as navigation specifications.	X		X			X				
(06)		State that RNP 2 is used in the en-route and oceanic/remote phases of flight.	X		X			X				
(07)		State that RNAV 2 might be used in the en-route continental, arrival and departure phases of flight.	X		X			X				
(08)		State that RNAV 1 and RNP 1 are used in the arrival and departure phases of flight.	X		X			X	X	2		
(09)		State that required navigation performance approach (RNP APCH) is used in the approach phase of flight.	X		X			X	X	2		
(10)		State that required navigation performance authorisation required approach (RNP AR APCH) is used in the approach phase of flight.	X		X			X	X	2		
(11)		State that RNP 0.3 navigation specification is used in all phases of flight except for oceanic/remote and final approach, primarily for helicopters.	X		X			X				
(12)		State that RNAV 1, RNP 1 and RNP 0.3 may also be used in en-route phases of low-level instrument flight rule (IFR) helicopter flights.	X		X			X				
062 07 03 00		Use of performance-based navigation (PBN)										
062 07 03 01		<i>Intentionally left blank</i>										
062 07 03 02		<i>Intentionally left blank</i>										
062 07 03 03		<i>Specific RNAV and RNP system functions</i>										
(01)		Recognise the definition of radius to fix (RF) leg.	X		X			X	X	2	3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Recognise the definition of a fixed radius transition (FRT).	X		X			X	X	2	3	
(03)		State the importance of respecting the flight director guidance and the speed constraints associated with an RF procedure.	X		X			X	X	2	3	
(04)		Explain the difference between a fly-by-turn and a fly-over.	X		X			X	X	2	3	
(05)		State that the Aeronautical Radio, Incorporated (ARINC) 424 path terminators set the standards for coding the SIDs, STARs and instrument approach procedures (IAPs) from the official published government source documentation into the ARINC navigation database format.	X		X			X				
(06)		State that the path terminators define a specific type of termination of the previous flight path.	X		X			X				
(07)		Define the term ‘offset flight path’.	X		X			X	X	2	3	
062 07 03 04		<i>Intentionally left blank</i>										
062 07 04 00		Performance-based navigation (PBN) operations										
062 07 04 01		Performance-based navigation (PBN) principles										
(01)		Define ‘path definition error’ (PDE).	X		X			X	X	2	3	
(02)		Define ‘flight technical error’ (FTE) and state that the FTE is the error in following the prescribed path, either by the auto-flight system or by the pilot.	X		X			X	X	2	3	
(03)		Define ‘navigation system error’ (NSE) and state that the accuracy of a navigation system may be referred to as NSE.	X		X			X	X	2	3	
(04)		Define ‘total system error’ (TSE) and state that the geometric sum of the PDE, FTE and NSE equals the TSE.	X		X			X	X	2	3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		State that navigation accuracy depends on the TSE.	X		X			X				
062 07 04 02		On-board performance monitoring and alerting										
(01)		State that on-board performance monitoring and alerting of flight technical errors is managed by on-board systems or flight crew procedures.	X		X			X	X	2	3	
(02)		State that on-board performance monitoring and alerting of navigation system errors is a requirement of on-board equipment for RNP.	X		X			X	X	2	3	
(03)		State that, dependent on the navigation sensor, the estimated position error (EPE) is compared with the required navigation specification.	X		X			X				
(04)		Explain how a navigation system assesses the EPE.	X		X			X				
(05)		Give an example of how the loss of the ability to operate in RNP airspace may be indicated by the navigation system.	X		X			X				
(06)		State that on-board performance monitoring and alerting of path definition error is managed by gross reasonableness checks of navigation data.	X		X			X	X	2	3	
062 07 04 03		Abnormal situations										
(01)		State that abnormal and contingency procedures are to be used in case of loss of the PBN capability.	X		X			X	X	2	3	
062 07 04 04		Database management										
(01)		State that, unless otherwise specified in the operations documentation or acceptable means of compliance (AMCs), the navigational database must be valid for the current aeronautical information regulation and control (AIRAC) cycle.	X		X			X	X	2	3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
062 07 05 00		Requirements of specific RNAV and RNP specifications										
062 07 05 01		RNAV 10										
(01)		State that RNAV 10 requires that aircraft operating in oceanic and remote areas be equipped with at least two independent and serviceable long-range navigation systems (LRNSs) comprising an INS, an inertial reference system (IRS)/flight management system (FMS) or a GNSS.	X		X			X				
(02)		State that operators may extend their RNAV 10 navigation capability time by updating.	X		X			X				
062 07 05 02		RNAV 5										
(01)		State that manual data entry is acceptable for RNAV 5.	X		X			X	X	2	3	
062 07 05 03		RNAV 1/RNAV 2/RNP 1/RNP 2										
(01)		State that pilots must not fly an RNAV 1, RNAV 2, RNP 1 or RNP 2 standard instrument departure (SID) or standard instrument arrival (STAR) unless it is retrievable by route name from the on-board navigation database and conforms to the charted route.	X		X			X	X	2	3	
(02)		State that the route may subsequently be modified through the insertion (from the database) or deletion of specific waypoints in response to ATC clearances.	X		X			X	X	2	3	
(03)		State that the manual entry, or creation of new waypoints by manual entry, of either latitude and longitude or place/bearing/distance values is not permitted.	X		X			X	X	2	3	
062 07 05 04		Intentionally left blank										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
062 07 05 05		Required navigation performance approach (RNP APCH)										
(01)		State that pilots must not fly an RNP APCH unless it is retrievable by procedure name from the on-board navigation database and conforms to the charted procedure.	X		X			X	X	2		
(02)		State that an RNP APCH to LNAV minima is a non-precision IAP designed for two-dimensional approach operations.	X		X			X	X	2		
(03)		State that an RNP APCH to lateral navigation (LNAV)/vertical navigation (VNAV) minima has lateral guidance based on GNSS and vertical guidance based on either SBAS or barometric vertical navigation (Baro-VNAV).	X		X			X	X	2		
(04)		State that an RNP APCH to LNAV/VNAV minima may only be conducted with vertical guidance certified for the purpose.	X		X			X	X	2		
(05)		Explain why an RNP APCH to LNAV/VNAV minima based on Baro-VNAV may only be conducted when the aerodrome temperature is within a promulgated range if the barometric input is not automatically temperature-compensated.	X		X			X	X	2		
(06)		State that the correct altimeter setting is critical for the safe conduct of an RNP APCH using Baro-VNAV.	X		X			X	X	2		
(07)		State that an RNP APCH to LNAV/VNAV minima is a three-dimensional operation.	X		X			X	X	2		
(08)		State that an RNP APCH to localiser performance with vertical guidance (LPV) minima is a three-dimensional operation.	X		X			X	X	2		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(09)		State that RNP APCH to LPV minima requires a final approach segment (FAS) data block.	X		X			X	X	2		
(10)		State that RNP approaches to LPV minima require SBAS.	X		X			X	X	2		
(11)		State that the FAS data block is a standard data format to describe the final approach path.	X		X			X	X	2		
062 07 05 06		Required navigation performance authorisation required approach (RNP AR APCH)										
(01)		State that RNP AR APCH requires authorisation.	X		X			X	X	2		
062 07 05 07		Advanced required navigation performance (A-RNP)										
(01)		State that A-RNP incorporates the navigation specifications RNAV 5, RNAV 2, RNAV 1, RNP 2, RNP 1 and RNP APCH.	X		X			X				
062 07 05 08		PBN point-in-space (PinS) departure										
(01)		State that a PinS departure is a departure procedure designed for helicopters only.			X			X				
(02)		State that a PinS departure procedure includes either a 'proceed VFR' or a 'proceed visually' instruction from the landing location to the initial departure fix (IDF).			X			X				
(03)		Recognise the differences in the instructions 'proceed VFR' and 'proceed visually'.			X			X				
062 07 05 09		PBN point-in-space (PinS) approach										
(01)		State that a PinS approach procedure is an instrument RNP APCH procedure designed for helicopters only, and that it may be published with LNAV minima or LPV minima.			X			X				
(02)		State that a PinS approach procedure includes either a 'proceed VFR' or a 'proceed visually'			X			X				

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		instruction from the missed approach point (MAPt) to a landing location.										
(03)		Recognise the differences between 'proceed VFR' and 'proceed visually'.			X			X				

SUBJECT 070 – OPERATIONAL PROCEDURES

ED Decision 2020/018/R

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
070 00 00 00		OPERATIONAL PROCEDURES										
071 01 00 00		GENERAL REQUIREMENTS										
071 01 01 00		ICAO Annex 6										
071 01 01 01		Definitions										
(01)		Define the following: alternate aerodrome: flight time (aeroplanes); take-off alternate; en-route alternate; destination alternate.	X	X								
(02)		Define 'alternate heliport'; 'flight time (helicopters)'. Source:			X	X	X					
071 01 01 02		Applicability										
(01)		State that Part I shall be applicable to the operation of aeroplanes by operators authorised to conduct international commercial air transport (CAT) operations.	X	X								
(02)		State that Part III shall be applicable to all helicopters engaged in international CAT operations or in international general aviation operations, except helicopters engaged in aerial work.			X	X	X					
071 01 01 03		General										
(01)		Explain the compliance with laws, regulations and procedures.	X	X	X	X	X					
(02)		State the condition(s) required for the establishment of a flight data analysis programme, and state what this programme is part of.	X	X	X	X	X					
(03)		Explain what is a flight safety documents system.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain what is maintenance release.	X	X	X	X	X					
(05)		List and describe the lights to be displayed by aircraft.	X	X	X	X	X					
071 01 02 00		Operational requirements										
071 01 02 01		Applicability										
(01)	X	State the operational regulations applicable to CAT and other activities (e.g. specialised operations (SPO)).	X	X	X	X	X					
(02)		State the nature of CAT operations and exceptions.	X	X	X	X	X					
071 01 02 02		General										
(01)	X	Explain why CAT flights must meet the applicable operational requirements.	X	X	X	X	X					
(02)		Define ‘flight manual limitations — flight through the height velocity (HV) envelope’.			X	X	X					
(03)		Define ‘helicopter emergency medical service (HEMS)’.			X	X	X					
(04)		Define ‘operations over a hostile environment — applicability’. Explain that there are certain areas which should not be overflown and state possible sources of that information (e.g. governmental warnings, operator risk assessment).			X	X	X					
(05)		Define ‘local area operations — approval’.			X	X	X					
(06)		Explain the requirements about language used for crew communication and in the operations manual.	X	X	X	X	X					
(07)		Explain which are the operator requirements regarding the management system.	X	X	X	X	X					
(08)		Explain which are the operator requirements regarding accident prevention and the flight safety programme.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(09)		Explain which are the regulations concerning the carriage of persons on an aircraft.	X	X	X	X	X					
(10)		Explain the operator's and commander's responsibility concerning portable electronic devices (PEDs).	X	X	X	X	X					
(11)		Explain the operator's and commander's responsibility regarding admission in an aircraft of a person under the influence of drug or alcohol.	X	X	X	X	X					
(12)		Explain the regulations concerning the endangerment of safety.	X	X	X	X	X					
(13)		List the documents to be carried on each flight.	X	X	X	X	X					
(14)		Explain the operator's responsibility regarding manuals to be carried on board an aircraft.	X	X	X	X	X					
(15)		List the additional information and forms to be carried on board an aircraft.	X	X	X	X	X					
(16)		List the copies of items of information to be retained on the ground by the operator.	X	X	X	X	X					
(17)		Explain what responsibilities the operator and the commander have regarding the production of and access to records and documents.	X	X	X	X	X					
071 01 02 03		Operator certification and supervision										
(01)		Explain what requirement has to be satisfied for the issue of an air operator certificate (AOC).	X	X	X	X	X					
(02)		Explain what the rules applicable to air operator certification are.	X	X	X	X	X					
(03)		Explain the conditions to be met for the issue or revalidation of an AOC.	X	X	X	X	X					
(04)		Explain the contents and conditions of the AOC.	X	X	X	X	X					
071 01 02 04		Operational procedures (except preparation for long-range flight)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define the terms used for operational procedures.	X	X								
(02)		State the operator's responsibilities regarding the use of air traffic services (ATS).	X	X	X	X	X					
(03)		State the operator's responsibilities regarding authorisation of aerodromes/heliports by the operator.	X	X	X	X	X					
(04)		Explain which elements must be considered by the operator when specifying aerodrome/heliport operating minima.	X	X	X	X	X					
(05)		Explain what the operator's responsibilities are regarding departure and approach procedures.	X	X	X	X	X					
(06)		Explain which parameters should be considered in noise-abatement procedures.	X	X								
(07)		Explain which elements should be considered regarding routes and areas of operation.	X	X	X	X	X					
(08)		Explain the requirements for flights in reduced vertical separation minima (RVSM) airspace.	X	X								
(09)		List the factors to be considered when establishing minimum flight altitude.	X	X	X	X	X					
(10)		Explain the requirements for carrying persons with reduced mobility.	X	X	X	X	X					
(11)		Explain the operator's responsibilities for the carriage of inadmissible passengers, deportees or persons in custody.	X	X	X	X	X					
(12)		Explain the requirements regarding passenger seating and emergency evacuation.	X	X	X	X	X					
(13)		Detail the procedures for passenger briefing in respect of emergency equipment and exits.	X	X	X	X	X					
(14)		State the flight preparation forms to be completed before flight.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(15)		State the commander's responsibilities during flight preparation.	X	X	X	X	X					
(16)		State the rules for aerodrome/heliport selection.	X	X	X	X	X					
(17)		Explain the planning minima for instrument flight rule (IFR) flights.	X		X							
(18)		Explain the rules for refuelling/defueling with passengers on board.	X	X	X	X	X					
(19)		Explain the 'crew members at station' policy.	X	X	X	X	X					
(20)		Explain the use of seats, safety belts and harnesses.	X	X	X	X	X					
(21)		Explain the requirements for securing passenger cabin and galley.	X	X	X	X	X					
(22)		Explain the commander's responsibility regarding smoking on board.	X	X	X	X	X					
(23)		State under which conditions a commander can commence or continue a flight regarding meteorological conditions.	X	X	X	X	X					
(24)		Explain the commander's responsibility regarding ice and other contaminants.	X	X	X	X	X					
(25)		Explain the commander's responsibility regarding fuel to be carried and in-flight fuel management.	X	X	X	X	X					
(26)		Detail the rules regarding carriage and use of supplemental oxygen for passengers and aircrew.	X	X	X	X	X					
		Flight preparation										
(27)		Explain the commander's responsibility regarding approach and landing.	X	X	X	X	X					
(28)		Explain the circumstances under which a report shall be submitted.	X	X	X	X	X					
071 01 02 05		All-weather operations										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain the operator's responsibility regarding aerodrome/heliport operating minima.	X		X							
(02)		Define the following terms: 'circling', 'low-visibility procedures', 'low-visibility take-off', 'visual approach'.	X		X							
(03)		Define the following terms: 'flight control system', 'fail-passive flight control system', 'fail-operational flight control system', 'fail-operational hybrid landing system'.	X									
(04)		Define the following terms: 'final approach and take-off area'.			X							
(05)		Explain the general operating requirements for low-visibility operations.	X		X							
(06)		Define aerodrome/heliport considerations regarding low-visibility operations.	X		X							
(07)		Explain the training and qualification requirements for flight crew to conduct low-visibility operations.	X		X							
(08)		Explain the operating procedures for low-visibility operations.	X		X							
(09)		Explain the operator's and commander's responsibilities regarding minimum equipment for low-visibility operations.	X		X							
(10)		Explain the VFR operating minima.	X		X							
(11)		Aerodrome operating minima: explain under which conditions the commander can commence take-off.	X		X							
(12)		Aerodrome operating minima: explain that take-off minima are expressed as visibility or runway visual range (RVR).	X		X							
(13)		Aerodrome operating minima: explain the take-off RVR value depending on the aerodrome facilities.	X		X							

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(14)		Aerodrome operating minima: explain the system minima for non-precision approach (NPA) (minimum descent altitude/height (MDA/H) and decision altitude/height (DA/H), not RVR).	X		X							
(15)		Aerodrome operating minima: explain under which conditions a pilot can continue the approach below MDA/H or DA/H.	X		X							
(16)		Aerodrome operating minima: explain the lowest minima for precision approach category 1 (including single-pilot operations).	X		X							
(17)		Aerodrome operating minima: explain the lowest minima for precision approach category 2 operations.	X		X							
(18)		Aerodrome operating minima: explain the lowest minima for precision approach category 3 operations.	X									
(19)		Aerodrome operating minima: explain the lowest minima for circling and visual approach.	X		X							
(20)		Aerodrome operating minima: explain the RVR value and cloud ceiling depending on the aerodrome.			X							
(21)		Aerodrome operating minima: explain under which conditions an airborne radar approach can be performed and state the relevant minima.			X							
071 01 02 06		Instruments and equipment										
(01)		Explain which items do not require an equipment approval.	X	X	X	X	X					
(02)		Explain the requirements regarding availability of spare electrical fuses.	X	X								
(03)		Explain the requirements regarding windshield wipers.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		List the minimum equipment required for day and night VFR flights.	X	X	X	X	X					
(05)		List the minimum equipment required for IFR flights.	X		X							
(06)		Explain the required additional equipment for single-pilot operations under IFR.	X		X							
(07)		State the requirements for an altitude alerting system.	X	X								
(08)		State the requirements for radio altimeters.			X	X	X					
(09)		State the requirements for ground proximity warning system (GPWS)/terrain awareness and warning system (TAWS).	X	X								
(10)		State the requirements for airborne collision avoidance system (ACAS).	X	X								
(11)		State the conditions under which an aircraft must be fitted with a weather radar.	X	X	X	X	X					
(12)		State the circumstances under which a cockpit voice recorder (CVR) is compulsory (after 1998).	X	X	X	X	X					
(13)		State the rules regarding the location, construction, installation, and operation of cockpit voice recorders (CVRs) (after 1998).	X	X	X	X	X					
(14)		State the circumstances under which a flight data recorder (FDR) is compulsory (after 1998).	X	X	X	X	X					
(15)		State the rules regarding the location, construction, installation, and operation of flight data recorders (FDRs) (after 1998).	X	X	X	X	X					
(16)		Explain the requirements about seats, seat safety belts, harnesses, and child-restraint devices.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(17)		Explain the requirements about 'Fasten seat belt' and 'No smoking' signs.	X	X	X	X	X					
(18)		Explain the requirements regarding internal doors and curtains.	X	X								
		First-aid and emergency equipment										
(19)		Explain the requirements regarding first-aid kits.	X	X	X	X	X					
(20)		Explain the requirements regarding emergency medical kits and first-aid oxygen.	X	X								
(21)		Detail the rules regarding crew protective breathing equipment.	X	X								
(22)		Describe the type and location of handheld fire extinguishers.	X	X	X	X	X					
(23)		Describe the location of crash axes and crowbars.	X	X								
(24)		Specify the colours and markings used to indicate break-in points.	X	X	X	X	X					
(25)		Explain the requirements for means of emergency evacuation.	X	X								
(26)		Explain the requirements for megaphones.	X	X	X	X	X					
(27)		Explain the requirements for emergency lighting and marking.	X	X	X	X	X					
(28)		Explain the requirements for an emergency locator transmitter (ELT).	X	X	X	X	X					
(29)		Explain the requirements for life jackets, life rafts, survival kits, and ELTs.	X	X	X	X	X					
(30)		Explain the requirements for crew survival suit.			X	X	X					
(31)		Explain the requirements for survival equipment.	X	X	X	X	X					
(32)		Explain the additional requirements for helicopters operating to or from helidecks located in hostile sea areas.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(33)		Explain the requirements for emergency flotation equipment.			X	X	X					
071 01 02 07		Communication and navigation equipment										
(01)		Explain the general requirements for communication and navigation equipment.	X	X	X	X	X					
(02)		Explain why the radio-communication equipment must be able to send and receive on 121.5 MHz.	X	X	X	X	X					
(03)		Explain the requirements regarding the provision of an audio selector panel.	X	X	X	X	X					
(04)		List the requirements for radio equipment when flying under VFR by reference to visual landmarks.	X	X	X	X	X					
(05)		List the requirements for communication and navigation equipment when operating under IFR or under VFR over routes not navigated by reference to visual landmarks.	X	X	X	X	X					
(06)		Explain what equipment is required to operate in airspace with reduced vertical separation minima (RVSM).	X	X								
(07)		Explain the conditions under which a crew member interphone system and public address system are mandatory.	X	X	X	X	X					
(08)		List the equipment for operations requiring a radio communication.			X	X	X					
(09)		List the equipment for operations that require a radio navigation system.			X	X	X					
(10)		Explain the requirements regarding the provision of a transponder.	X	X	X	X	X					
(11)		Explain the requirements regarding the management of aeronautical databases.	X	X								
071 01 02 08		Intentionally left blank										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
071 01 02 09		Flight crew										
(01)		Explain the requirement regarding flight crew composition and in-flight relief.	X	X	X	X	X					
(02)		Explain the requirement for conversion training and checking.	X	X	X	X	X					
(03)		Explain the requirement for differences training and familiarisation training.	X	X	X	X	X					
(04)		Explain the conditions for upgrade from co-pilot to commander.	X	X	X	X	X					
(05)		Explain the minimum qualification requirements to operate as a commander.	X	X	X	X	X					
(06)		Explain the requirement for recurrent training and checking.	X	X	X	X	X					
(07)		Explain the requirement for a pilot to operate on either pilot's seat.	X	X	X	X	X					
(08)		Explain the minimum recent experience requirements for the commander and the co-pilot.	X	X	X	X	X					
(09)		Specify the route and aerodrome/heliport knowledge required for a PIC/commander.	X	X	X	X	X					
(10)		Explain the requirement to operate on more than one aircraft type or variant.	X	X	X	X	X					
(11)		Explain that when a flight crew member operates both helicopters and aeroplanes, the operations are limited to one of each type.	X	X	X	X	X					
(12)		Explain the requirement(s) for training records.	X	X	X	X	X					
(13)		Explain the crew members' responsibilities in the execution of their duties, and define the commander's authority.	X	X	X	X	X					
(14)		Explain the operator's and commander's responsibilities regarding persons on board,	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		admission to the flight crew compartment and carriage of unauthorised persons or cargo.										
(15)		Explain the requirements for the initial operator's crew resource management (CRM) training.	X	X	X	X	X					
071 01 02 10		Cabin crew/crew members other than flight crew										
(01)		Explain who is regarded as cabin crew member.	X	X	X	X	X					
(02)		Detail the requirements regarding the number and composition of cabin crew.	X	X	X	X	X					
(03)		Explain the conditions and the additional conditions for assignment to duties.	X	X	X	X	X					
(04)		Explain the requirements regarding senior cabin crew members.	X	X	X	X	X					
(05)		Explain the conditions for operating on more than one aircraft type or variant.	X	X	X	X	X					
(06)		Explain what is the operator's responsibility regarding the distinction between cabin crew members and additional crew members.	X	X	X	X	X					
071 01 02 11		Intentionally left blank										
071 01 02 12		Flight and duty time limitations and rest requirements										
(01)		Explain the definitions used for the regulation of flight time limitations.	X	X								
(02)		Explain the flight and duty time limitations.	X	X								
(03)		Explain the requirements regarding the maximum daily flight duty period.	X	X								
(04)		Explain the requirements regarding rest periods.	X	X								
(05)		Explain the possible extension of flight duty period due to in-flight rest.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain that it is the captain's discretion to extend flight duty in case of unforeseen circumstances in actual flight operations.	X	X								
(07)		Explain the requirement regarding standby.	X	X								
071 01 03 00		Long-range flights										
071 01 03 01		Flight management										
(01)		Minimum time routes: define and interpret minimum time route (route that gives the shortest flight time from departure to destination adhering to all ATC and airspace restrictions).	X									
(02)		State the circumstances in which a take-off alternate must be selected.	X		X							
(03)		State the maximum flight distance of a take-off alternate for: – two-engined aeroplanes; – ETOPS-approved aeroplanes; – three- or four-engined aeroplanes.	X		X							
(04)		State the factors to be considered in the selection of a take-off alternate.	X		X							
(05)		State when a destination alternate need not be selected.	X		X							
(06)		State when two destination alternates must be selected.	X		X							
(07)		State the factors to be considered in the selection of a destination alternate aerodrome.	X		X							
(08)		State the factors to be considered in the selection of an en-route alternate aerodrome.	X		X							
071 01 03 02		Transoceanic and polar flights										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		(ICAO Doc 7030 'Regional Supplementary Procedures — North Atlantic Operations and Airspace Manual')										
(01)		According to ICAO Doc 7030, explain that special rules apply to the North Atlantic (NAT) Region, and crews need to be specifically trained before flying in this area.	X									
(02)		Describe the possible indications of navigation system degradation, including any system-generated warning.	X									
(03)		Describe by what emergency means course and inertial navigation system (INS) can be cross-checked in the case of three navigation systems and two navigation systems.	X									
(04)		Describe the general ICAO procedures applicable in NAT airspace if the aircraft is unable to continue the flight in accordance with its air traffic control (ATC) clearance.	X									
(05)		Describe the ICAO procedures applicable in NAT airspace in case of radio-communication failure.	X									
(06)		Describe the recommended initial action if an aircraft is unable to obtain a revised ATC clearance.	X									
(07)		Describe the subsequent action for aircraft able to maintain assigned flight level and for aircraft unable to maintain assigned flight level.	X									
(08)		Describe determination of tracks and courses for random routes in NAT airspace.	X									
(09)		Specify the method by which planned tracks are defined (by latitude and longitude) in the NAT airspace: when operating predominately in an east–west direction south of 70°N, and when	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		operating predominately in an east–west direction north of 70°N.										
(10)		State the maximum flight time recommended between significant points on random routes.	X									
(11)		Specify the method by which planned tracks for random routes are defined for flights operating predominantly in a north–south direction.	X									
(12)		Describe how the desired random route must be specified in the ATC flight plan.	X									
(13)		Describe what precautions can be taken when operating in the area of compass unreliability as a contingency against INS failure.	X									
071 01 03 03		North Atlantic High Level Airspace (NAT HLA)										
		NAT Region										
		North Atlantic Operations and Airspace Manual										
(01)		State the lateral dimensions (in general terms) and vertical limits of the NAT HLA.	X									
(02)		Define the following acronyms: LRNS, MASPS, NAT HLA, OCA, OTS, PRM, RVSM, SLOP, and WATRS.	X									
(03)		State the NAT HLA operations.	X									
(04)		Describe the routes for aircraft with only one long-range navigation system (LRNS).	X									
(05)		Describe the routes for aircraft with short-range navigation equipment only.	X									
(06)		Explain why the horizontal (i.e. latitudinal and longitudinal) and vertical navigation performance of operators within NAT HLA is monitored on a continual basis.	X									
(07)		Describe the organised track system (OTS).	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(08)		State the OTS changeover periods.	X									
(09)		Describe the NAT track message.	X									
(10)		Illustrate routes between northern Europe and the Spain/Canaries/Lisbon flight information region (FIR) within NAT HLA.	X									
(11)		Describe the function of the North American Routes (NARs) and Shannon Oceanic Transition Area (SOTA) and Northern Oceanic Transition Area (NOTA).	X									
(12)		State that all flights should plan to operate on great-circle tracks joining successive significant waypoints.	X									
(13)		State that during the hours of validity of the OTS, operators are encouraged to plan flights: — in accordance with the OTS; — or along a route to join or leave an outer track of the OTS; — or on a random route to remain clear of the OTS, either laterally or vertically.	X									
(14)		State which flight levels are available on OTS tracks during OTS periods.	X									
(15)		State which flight levels are to be planned on random tracks or outside OTS periods.	X									
(16)		Selection of cruising altitude. Specify the appropriate cruising levels for normal long-range IFR flights and for those operating on the North Atlantic OTS.	X									
(17)		<i>Intentionally left blank</i>										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(18)		State that pilots should notify the oceanic area control centre (OAC) of the maximum acceptable flight level possible at the boundary.	X									
(19)		State that at some aerodromes which are situated close to oceanic boundaries, the oceanic clearance must be obtained before departure.	X									
(20)		State that if an aircraft encounters, whilst en-route to the NAT Oceanic Airspace, an in-flight equipment failure relevant to the airspace, then the pilot must advise ATC when requesting oceanic clearance.	X									
(21)		State that after obtaining and reading back the clearance, the pilot should monitor the forward estimate for oceanic entry, and if this changes by 3 minutes or more, unless providing position reports via automatic dependent surveillance — contract (ADS-C), the pilot must pass a revised estimate on to ATC.	X									
(22)		State that pilots should pay particular attention when the issued clearance differs from the flight plan as a significant proportion of navigation errors investigated in the NAT Region involve aircraft which have followed their flight plan rather than the differing clearance.	X									
(23)		State that if the entry point of the oceanic route for which the flight is cleared differs from that originally requested or the oceanic flight level differs from the current flight level, the pilot is responsible for requesting and obtaining the necessary domestic re-clearance.	X									
(24)		State that there are three elements to an oceanic clearance: route, speed, and flight level, and that	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		these elements serve to provide for the three basic elements of separation: lateral, longitudinal, and vertical.										
(25)		Communications and position-reporting procedures State that pilots communicate with OACCs via aeradio stations staffed by communicators who have no executive ATC authority.	X									
(26)		State that messages are relayed from the ground station to the air traffic controllers of the relevant OACC for action.	X									
(27)		State that frequencies from the lower HF bands tend to be used for communications during night-time and those from the higher bands during daytime. When initiating contact with an aeradio station, the pilot should state the HF frequency in use.	X									
(28)		State that since oceanic traffic typically communicates with ATC through aeradio facilities, a satellite communication (SATVOICE) call, made due to unforeseen inability to communicate by other means, should be made to such a facility rather than the ATC centre, unless the urgency of the communication dictates otherwise.	X									
(29)		State that an air-to-air VHF frequency has been established for worldwide use when aircraft are out of range of VHF ground stations which utilise the same or adjacent frequencies. This frequency, 123.450 MHz, is intended for pilot-to-pilot exchanges of operationally significant information.	X									
(30)		State that any pilot, who provides position reports via data link and encounters significant meteorological phenomena (such as	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		moderate/severe turbulence or icing, volcanic ash or thunderstorms), should report this information.										
(31)		State that all turbine-engined aeroplanes having a maximum certified take-off mass exceeding 5 700 kg or authorised to carry more than 19 passengers are required to carry and operate airborne collision avoidance system (ACAS) II in the NAT Region.	X									
(32)		State that even with the growing use of data-link communications, a significant volume of NAT air-ground communications are conducted using voice on single sideband (SSB) HF frequencies. To support air-ground ATC communications in the North Atlantic Region, 24 HF frequencies have been allocated, in bands ranging from 2.8 to 18 MHz.	X									
(33)		Application of the Mach number technique (NAT HLA) State that practical experience has shown that when two or more turbojet aircraft, operating along the same route at the same flight level, maintain the same Mach number, they are more likely to maintain a constant time interval between each other than when using other methods.	X									
(34)		State that after leaving oceanic airspace, pilots must maintain their assigned Mach number in domestic controlled airspace unless and until the appropriate ATC unit authorises a change.	X									
		North Atlantic High Level Airspace (NAT HLA) flight operation and navigation procedures										
(35)		NAT HLA flight operation and navigation procedures	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		State that the pre-flight procedures for any NAT HLA flight must include a Universal Time Coordinated (UTC) time check.										
(36)		Describe the function and use of the master document.	X									
(37)		State the requirements for position plotting.	X									
(38)		Describe the pre-flight procedures for: — the alignment of IRS; — the satellite navigation availability prediction programme for flights using global navigation satellite long-range navigation system (GNSS LRNS); — loading of initial waypoints; and — flight plan check.	X									
(39)		Describe the strategic lateral offset procedure (SLOP) and state that along a route or track there will be three positions that an aircraft may fly: centre line, or up to 2 NM right.	X									
(40)		State that RNAV 10 retains the RNP 10 designation, as specified in the Performance-based Navigation Manual (ICAO Doc 9613) (ICAO Doc 7030, NAT Chapter 4).	X									
(41)		State that both aircraft and operators must be RNP 10- or RNP 4-approved by the State of the Operator or the State of Registry, as appropriate.	X									
(42)		State that RNP 10 is the minimum navigation specification for the application of 93 km (50 NM) lateral separation.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(43)		Reduced vertical separation minima (RVSM) flight in NAT HLA State the altimeter cross-check to be performed before entering NAT HLA.	X									
(44)		State the altimeter cross-check to be performed when entering and flying in NAT HLA.	X									
(45)		State that pilots not using controller–pilot data-link communications (CPDLC)/ADS-C always report to ATC immediately on leaving the current cruising level and on reaching any new cruising level.	X									
(46)		State that flight crew should report when a 300-ft deviation or more occurs.	X									
(47)		Navigation planning procedures List the factors to be considered by the commander before commencing the flight.	X									
		Navigation system degradation										
(48)		For this part, consider aircraft equipped with only two operational LRNSs and state the requirements for the following situations: – one system fails before take-off; – one system fails before the OCA boundary is reached; – one system fails after the OCA boundary is crossed; and – the remaining system fails after entering NAT HLA.	X									
		Special procedures for in-flight contingencies										
(49)		State the general procedures and also state that the general concept of these NAT in-flight contingency	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		procedures is, whenever operationally feasible, to offset the assigned route by 5 NM and climb or descend to a level which differs from those normally used by 500 ft if below FL 410 or by 1 000 ft if above FL 410.										
(50)		State all the factors which may affect the direction of turn including: — direction to an alternate aerodrome; — terrain clearance; — levels allocated on adjacent routes or tracks and any known SLOP offsets adopted by other nearby traffic.	X									
(51)		State that if the deviation around severe weather is to be greater than 5 NM, the assigned flight level must be changed by ± 300 ft depending on the followed track and the direction of the deviation.	X									
071 01 03 04		Extended-range operations with two-engined aeroplanes (ETOPS)										
(01)		State that ETOPS approval is part of an AOC.	X									
(02)		State that prior to conducting an ETOPS flight, an operator shall ensure that a suitable ETOPS en-route alternate is available, within either the approved diversion time or a diversion time based on the MEL-generated serviceability status of the aeroplane, whichever is shorter.	X									
(03)		State the requirements for take-off alternate.	X									
(04)		State the planning minima for ETOPS en-route alternate.	X									
(05)		Navigation-planning procedures.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		Describe the operator's responsibilities concerning ETOPS routes.										
(06)		Selection of a route. Describe the limitations on extended-range operations with two-engined aeroplanes with and without ETOPS approval.	X									
(07)		Selection of alternate aerodrome. State the maximum flight distance of a take-off alternate for: – two-engined aeroplanes; – ETOPS-approved aeroplanes; – three- or four-engined aeroplanes.	X									
(08)		State the maximum distance from an adequate aerodrome for two-engined aeroplanes without an ETOPS approval.	X									
(09)		State the requirement for alternate aerodrome accessibility check for ETOPS operations.	X									
071 02 00 00		SPECIAL OPERATIONAL PROCEDURES AND HAZARDS — GENERAL ASPECTS										
071 02 01 00		Operations manual										
071 02 01 01		Operating procedures										
(01)		Explain the general rules for the operations manual.	X	X	X	X	X					
(02)		Explain the structure and subject headings of the operations manual.	X	X	X	X	X					
(03)		Explain the requirements for a journey log or equivalent.	X	X	X	X	X					
(04)		Describe the requirements regarding the operational flight plan.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Explain the requirements for document-storage periods.	X	X	X	X	X					
(06)		Explain that all non-type-related operational policies, instructions and procedures required for a safe operation are included in Part A of the operations manual.	X	X	X	X	X					
(07)		State that the following items are included into Part A: — de-icing and anti-icing on the ground; — adverse and potentially hazardous atmospheric conditions; — wake turbulence; — incapacitation of crew members; — use of the minimum equipment list (MEL) and configuration deviation list(s) (CDL); — security; — handling of accidents and occurrences.	X	X	X	X	X					
(08)		State that the following items are included into Part A: — altitude alerting system procedures; — ground proximity warning system procedures; — policy and procedures for the use of traffic alert and collision avoidance system (TCAS)/airborne collision avoidance system (ACAS).	X	X								
(09)		State that rotor downwash is included into Part A.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
071 02 01 02		Aeroplane/helicopter operating matters — type-related										
(01)		State that all type-related instructions and procedures required for a safe operation are included in Part B of the operations manual. They take account of any differences between types, variants or individual aircraft used by an operator.	X	X	X	X	X					
(02)		State that the following items are included into Part B: – abnormal and emergency procedures; – configuration deviation list (CDL); – minimum equipment list (MEL); – emergency evacuation procedures.	X	X								
(03)		State that the following items are included into Part B: – emergency procedures; – configuration deviation list (CDL); – minimum equipment list (MEL); – emergency evacuation procedures.			X	X	X					
071 02 01 03		Minimum equipment list (MEL) and master minimum equipment list (MMEL)										
(01)		Describe the following terms: ‘commencement of flight’, ‘inoperative’, ‘MEL’, ‘MMEL’, ‘rectification interval’.	X	X	X	X	X					
(02)		Explain the relation between MMEL and MEL.	X	X	X	X	X					
(03)		Define the ‘extent of the MEL’.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the responsibilities of the operator and the competent authority with regard to MEL and MMEL.	X	X	X	X	X					
(05)		Explain the responsibilities of the flight crew members with regard to MEL.	X	X	X	X	X					
(06)		Explain the responsibilities of the commander with regard to MEL.	X	X	X	X	X					
071 02 02 00		Icing conditions										
071 02 02 01		On-ground de-icing/anti-icing procedures, types of de-icing/anti-icing fluids										
(01)		Define the following terms: 'anti-icing', 'de-icing', 'one-step de-icing/anti-icing', 'two-step de-icing/anti-icing', 'holdover time'.	X	X								
(02)		Describe 'the clean aircraft concept' as presented in the relevant chapter of ICAO Doc 9640.	X	X								
(03)		List the types of de-icing/anti-icing fluids available.	X	X	X	X	X					
(04)		Explain the procedure to be followed when an aeroplane has exceeded the holdover time.	X	X								
(05)		Interpret the guidelines for fluid holdover times and list the factors which can reduce the fluid protection time.	X	X								
(06)		Explain how the pre-take-off check, which is the responsibility of the pilot-in-command, ensures that the critical surfaces of the aircraft are free of ice, snow, slush or frost just prior to take-off. This check shall be accomplished as close to the time of take-off as possible and is normally made from within the aeroplane by visually checking the wings.	X	X								
(07)		Explain why an aircraft has to be treated symmetrically.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(08)		Explain why an operator shall establish procedures to be followed when ground de-icing and anti-icing and related inspections of the aircraft are necessary.	X	X	X	X	X					
(09)		Explain why a commander shall not commence take-off unless the external surfaces are clear of any deposit which might adversely affect the performance or controllability of the aircraft except as permitted in the flight manual.	X	X	X	X	X					
(10)		Explain the requirements for operations in icing conditions.	X	X	X	X	X					
(11)		Explain why safety must come before commercial pressures in relation to de-icing and anti-icing of aircraft. (Consider time and financial cost versus direct and indirect effects of an incident/accident).	X	X	X	X	X					
071 02 02 02		<i>Procedure to apply in case of performance deterioration, on ground/in flight</i>										
(01)		Explain that the effects of icing are wide-ranging, unpredictable and dependent upon individual aircraft design. The magnitude of these effects is dependent upon many variables, but the effects can be both significant and dangerous.	X	X	X	X	X					
(02)		Explain that in icing conditions, for a given speed and a given angle of attack, wing lift can be reduced by as much as 30 % and drag increased by up to 40 %. State that these changes in lift and drag will significantly increase stall speed, reduce controllability, and alter flight characteristics.	X	X	X	X	X					
(03)		Explain that ice on critical surfaces and on the airframe may also break away during take-off and	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		be ingested into engines, possibly damaging fan and compressor blades.										
(04)		Explain that ice forming on pitot tubes and static ports or on angle-of-attack vanes may give false altitude, airspeed, angle-of-attack and engine-power information for air-data systems.	X	X	X	X	X					
(05)		Explain that ice, frost and snow formed on the critical surfaces on the ground can have a totally different effect on aircraft flight characteristics than ice, frost and snow formed in flight.	X	X	X	X	X					
(06)		Explain that flight in known icing conditions is subject to limitations that are contained in Part B of the operations manual.	X	X	X	X	X					
(07)		Explain where procedures and performances regarding flight in expected or actual icing conditions can be found.	X	X	X	X	X					
071 02 03 00		Bird-strike risk										
071 02 03 01		<i>Bird-strike risk and avoidance</i>										
(01)		Explain that the presence of birds that constitute a potential hazard to aircraft operations is part of the pre-flight information.	X	X	X	X	X					
(02)		Explain how information concerning the presence of birds observed by aircrews is made available to the aeronautical information service (AIS) for distribution as the circumstances dictate.	X	X	X	X	X					
(03)		Explain that the Aeronautical Information Publication (AIP) Section En-route (ENR) 5.6 contains information regarding bird migrations.	X	X	X	X	X					
(04)		Explain significant data regarding bird strikes contained in ICAO Doc 9137 'Airport Services Manual'.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Explain why birds constitute a hazard to aircraft (damage to probes, sensors, engines, windscreens, airframes, degradation in vision, etc.).	X	X	X	X	X					
(06)		Define the commander's responsibilities regarding the reporting of bird hazards and bird strikes.	X	X	X	X	X					
(07)		State that birds tend to flock to areas where food is plentiful. Such areas include: rubbish (garbage) facilities; open sewage treatment works; recently ploughed land; as well as their natural habitats.	X	X	X	X	X					
071 02 04 00		Noise abatement										
071 02 04 01		Noise-abatement procedures										
(01)		Define the operator's responsibilities regarding the establishment of noise-abatement procedures.	X	X	X	X	X					
(02)		State the main purpose of noise-abatement departure procedure (NADP) 1 and NADP 2.	X	X	X	X	X					
(03)		State that the PIC/commander has the authority to decide not to execute an NADP if conditions preclude the safe execution of the procedure.	X	X	X	X	X					
071 02 04 02		Influence of the flight procedure (departure, cruise, approach)										
(01)		List the main parameters for NADP 1 and NADP 2 (i.e. speeds, heights and configuration).	X	X								
(02)		State that a runway lead-in lighting system should be provided where it is desired to provide visual guidance along a specific approach path for noise-abatement purposes.	X	X	X	X	X					
(03)		State that detailed information about noise-abatement procedures is to be found in Part 'Aerodromes' (AD), Sections 2 and 3 of the AIP.	X	X	X	X	X					
071 02 04 03		Influence by the pilot (power setting, low drag)										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		List the adverse operating conditions under which noise-abatement procedures in the form of reduced-power take-off should not be required.	X	X								
(02)		List the adverse operating conditions under which noise-abatement procedures during approach should not be required.	X	X								
(03)		State the rule regarding the use of reverse thrust on landing.	X	X								
071 02 04 04		<i>Influence by the pilot (power setting, track of helicopter)</i>										
(01)		List the adverse operating conditions under which noise-abatement procedures in the form of reduced-power take-off should not be required.			X	X	X					
071 02 05 00		Fire and smoke										
071 02 05 01		<i>Carburettor fire</i>										
(01)		Explain that the actions to be taken in the event of a carburettor fire may be type-specific and should be known by the pilot.	X	X	X	X	X					
071 02 05 02		<i>Engine fire</i>										
(01)		Explain that the actions to be taken in the event of an engine fire may be type-specific and should be known by the pilot.	X	X	X	X	X					
071 02 05 03		<i>Fire in the cabin, in the flight crew compartment and in the cargo compartment</i>										
(01)		Identify the different types of extinguishants used in handheld fire extinguishers and the type of fire for which each one may be used.	X	X	X	X	X					
(02)		Describe the precautions to be considered when applying fire extinguishants.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Identify the appropriate handheld fire extinguishers to be used in the flight crew compartment, the passenger cabin and lavatories, and in the cargo compartments.	X	X	X	X	X					
071 02 05 04		<i>Smoke in the flight crew compartment and in the cabin</i>										
(01)		Explain which actions should be taken in the event of smoke in the flight crew compartment or in the cabin, why these actions may be type-specific, and why they should be known by the pilot.	X	X	X	X	X					
071 02 05 05		<i>Actions in case of overheated brakes</i>										
(01)		Describe the problems and safety precautions in the event that brakes overheat after a heavy-weight landing or a rejected take-off.	X	X								
(02)		Explain the difference in the way steel and carbon brakes react to energy absorption and the operational consequences.	X	X								
071 02 06 00		<i>Decompression of pressurised cabin</i>										
071 02 06 01		<i>Slow decompression</i>										
(01)		Explain what can cause, and how to detect, a slow decompression or an automatic pressurisation system failure.	X	X								
(02)		Describe the actions required following a slow decompression.	X	X								
071 02 06 02		<i>Rapid and explosive decompression</i>										
(01)		Explain what can cause, and how to detect, a rapid or an explosive decompression.	X	X								
071 02 06 03		<i>Dangers and action to be taken</i>										
(01)		Describe the actions required following a rapid or explosive decompression.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the effects on aircraft occupants of a slow decompression and of a rapid or explosive decompression.	X	X								
071 02 07 00		Wind shear and microburst										
071 02 07 01		Effects and recognition during departure and approach										
(01)		Explain how to identify low-level wind shear.	X	X	X	X	X					
071 02 07 02		Actions to avoid and actions to take when encountering wind shear										
(01)		Describe the effects of wind shear and the actions required when wind shear is encountered at take-off and approach.	X	X	X	X	X					
(02)		Describe the precautions to be taken when wind shear is suspected at take-off and approach.	X	X	X	X	X					
(03)		Describe the effects of wind shear and the actions required following entry into a strong downdraft wind shear.	X	X	X	X	X					
(04)		Describe a microburst and its effects.	X	X	X	X	X					
071 02 08 00		Wake turbulence										
071 02 08 01		Cause										
(01)		Describe the term 'wake turbulence'.	X	X	X	X	X					
(02)		Describe tip vortex circulation.	X	X	X	X	X					
(03)		State when vortex generation begins and ends.	X	X	X	X	X					
(04)		Describe vortex circulation on the ground with and without crosswind.	X	X	X	X	X					
071 02 08 02		List of relevant parameters										
(01)		List the three main factors which, when combined, give the strongest vortices (heavy, clean, slow).	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the wind conditions which are worst for wake turbulence near the ground.	X	X	X	X	X					
071 02 08 03		Actions to be taken when crossing traffic, during take-off and landing										
(01)		Describe the actions to be taken to avoid wake turbulence, specifically separations.	X	X	X	X	X					
071 02 09 00		Security (unlawful events)										
071 02 09 01		ICAO Annex 17 and Regulation (EC) No 300/2008										
(01)		Define the following terms: 'aircraft security check', 'screening', 'security', 'security-restricted area', 'unidentified baggage'.	X	X	X	X	X					
(02)		State the objectives of security.	X	X	X	X	X					
071 02 09 02		Use of secondary surveillance radar (SSR)										
(01)		Describe the commander's responsibilities concerning notifying the appropriate ATS unit.	X	X	X	X	X					
(02)		Describe the commander's responsibilities concerning operation of SSR.	X	X	X	X	X					
(03)		Describe the commander's responsibilities concerning departing from assigned track or cruising level.	X	X	X	X	X					
(04)		Describe the commander's responsibilities concerning the action required or being requested by an ATS unit to confirm SSR code and ATS interpretation response.	X	X	X	X	X					
071 02 09 03		Security (Regulation (EC) No 300/2008 and ICAO Annex 17)										
(01)		Describe the relationship between Regulation (EC) No 300/2008 and ICAO Annex 17.	X	X	X	X	X					
(02)		Explain the requirements regarding training programmes.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		State the requirements regarding reporting acts of unlawful interference.	X	X	X	X	X					
(04)		State the requirements regarding aircraft search procedures.	X	X	X	X	X					
071 02 10 00		Emergency and precautionary landing, and ditching										
071 02 10 01		Descriptions										
(01)		Describe the meaning of: ‘ditching’, ‘precautionary landing’, and ‘emergency landing’.	X	X	X	X	X					
(02)		Describe a ditching procedure.	X	X	X	X	X					
(03)		Describe a precautionary landing procedure.	X	X	X	X	X					
(04)		Describe an emergency landing procedure.	X	X	X	X	X					
(05)		Explain the factors to be considered when deciding to conduct a precautionary/emergency landing or ditching.	X	X	X	X	X					
071 02 10 02		Cause										
(01)		List some circumstances that may require a ditching, a precautionary landing or an emergency landing.	X	X	X	X	X					
071 02 10 03		Passenger information										
(01)		Describe the briefing to be given to passengers before conducting a precautionary/emergency landing or ditching (including evacuation).	X	X	X	X	X					
071 02 10 04		Action after a precautionary/emergency landing or ditching										
(01)		Describe the actions and responsibilities of crew members after landing.	X	X	X	X	X					
071 02 10 05		Evacuation										
(01)		Explain why the aircraft must be stopped and the engine(s) shut down before launching an emergency evacuation.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain the CS-25 requirements regarding evacuation procedures.	X	X								
071 02 11 00		Fuel jettisoning										
071 02 11 01		Safety aspects										
(01)		Explain why an aircraft may need to jettison fuel so as to reduce its landing mass in order to make a safe landing.	X	X								
(02)		Explain that when an aircraft that operates within controlled airspace needs to jettison fuel, the flight crew shall coordinate with ATC the following: — route to be flown which, if possible, should be clear of cities and towns, preferably over water and away from areas where thunderstorms have been reported or are expected; — the flight level to be used, which should be not less than 1 800 m (6 000 ft); and — the duration of fuel jettisoning.	X	X								
(03)		Explain how flaps and slats may adversely affect fuel jettisoning.	X	X								
071 02 11 02		Requirements										
(01)		Explain why a fuel-jettisoning system must be capable of jettisoning enough fuel within 15 minutes.	X	X								
071 02 12 00		Transport of dangerous goods by air										
071 02 12 01		ICAO Annex 18										
(01)		Define the following terms:	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		‘dangerous goods’, ‘dangerous goods accident’, ‘dangerous goods incident’, ‘exemption’, ‘incompatible’, ‘packaging’, ‘UN number’.										
(02)		Explain that detailed provisions for the transport of dangerous goods by air are contained in the Technical Instructions for the Safe Transport of Dangerous Goods by Air.	X	X	X	X	X					
(03)		State that in the event of an in-flight emergency, the pilot-in-command must inform the ATC of the transport of dangerous goods by air.	X	X	X	X	X					
071 02 12 02		Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO Doc 9284)										
(01)		Explain the principle of dangerous goods compatibility and segregation.	X	X	X	X	X					
(02)		Explain the special requirements for the loading of radioactive materials.	X	X	X	X	X					
(03)		Explain the use of the dangerous goods list.	X	X	X	X	X					
(04)		Identify the labels.	X	X	X	X	X					
071 02 12 03		Regulation (EU) No 965/2012 — Annex IV (Part-CAT) and Annex V (Part-SPA)										
(01)		Explain the terminology relevant to dangerous goods.	X	X	X	X	X					
(02)		Explain the scope of that Regulation.	X	X	X	X	X					
(03)		Explain why the transport of dangerous goods by air is subject to operator approval.	X	X	X	X	X					
(04)		Explain the limitations on the transport of dangerous goods by air.	X	X	X	X	X					
(05)		Explain the requirements for the acceptance of dangerous goods.	X	X	X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the requirements regarding inspection for damage, leakage or contamination.	X	X	X	X	X					
(07)		Explain the requirement for the provision of information to flight crew.	X	X	X	X	X					
(08)		Explain the requirements for dangerous goods incident and accident reports.	X	X	X	X	X					
(09)		State that some articles and substances, which would otherwise be classed as dangerous goods, can be exempted if they are part of the aircraft equipment, or required for use during aeromedical flights.	X	X	X	X	X					
(10)		Explain why some articles and substances may be forbidden for transport by air.	X	X	X	X	X					
(11)		Explain why packing must comply with the specifications of the Technical Instructions.	X	X	X	X	X					
(12)		Explain the need for an inspection prior to loading dangerous goods on an aircraft.	X	X	X	X	X					
(13)		Explain why some dangerous goods are designated for carriage only on cargo aircraft.	X	X	X	X	X					
(14)		Explain how misdeclared or undeclared dangerous goods found in baggage are to be reported.	X	X	X	X	X					
071 02 13 00		Contaminated runways										
071 02 13 01		Intentionally left blank										
071 02 13 02		Runway condition, braking action										
(01)		<i>Intentionally left blank</i>										
(02)		State the runway condition codes for good braking action, and when a special air report is required.	X	X								
071 02 13 03		Hydroplaning principles and effects										
(01)		Define the different types of hydroplaning.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Compute the two dynamic hydroplaning speeds using the following formulas: — spin-down speed (rotating tire) (kt) = 9 square root (pressure in PSI) — spin-up speed (non-rotating tire) (kt) = 7.7 square root (pressure in PSI).	X	X								
(03)		State that it is the spin-up speed rather than the spin-down speed which represents the actual tire situation for aircraft touchdown on flooded runways.	X	X								
071 02 13 04		Intentionally left blank										
071 02 13 05		SNOWTAM and contamination on the aerodrome										
(01)		Interpret from a SNOWTAM the contamination and braking action on a runway, taxiways and apron.	X	X								
(02)		Explain which hazards can be identified from the SNOWTAM/METAR and how to mitigate them.	X	X	X	X	X					
071 02 14 00		Rotor downwash										
071 02 14 01		Describe downwash										
(01)		Describe the downwash.			X	X	X					
071 02 14 02		Effects										
(01)		Explain its effects: soil erosion, water dispersal and spray, recirculation, damage to property, loose articles.			X	X	X					
071 02 15 00		Operation influence by meteorological conditions (helicopter)										
071 02 15 01		White-out/sand/dust										
(01)		Give the definition of 'white-out'.			X	X	X					
(02)		Describe loss of spatial orientation.			X	X	X					
(03)		Describe take-off and landing techniques.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
071 02 15 02		Strong winds										
(01)		Describe blade sailing.			X	X	X					
(02)		Describe wind operating envelopes.			X	X	X					
(03)		Describe vertical speed problems.			X	X	X					
071 02 15 03		Mountain environment										
(01)		Describe constraints associated with mountain environment.			X	X	X					
071 03 00 00		EMERGENCY PROCEDURES (HELICOPTER)										
071 03 01 00		Influence of technical problems										
071 03 01 01		Engine failure										
(01)		Describe recovery techniques in the event of engine failure during hover, climb, cruise, approach.			X	X	X					
071 03 01 02		Fire in the cabin, in the flight crew compartment and in the engine(s)										
(01)		Describe the basic actions when encountering fire in the cabin, flight deck or engine(s).			X	X	X					
071 03 01 03		Tail-rotor directional control failure										
(01)		Describe the basic actions following loss of tail rotor.			X	X	X					
(02)		Describe the basic actions following loss of directional control.			X	X	X					
071 03 01 04		Ground resonance										
(01)		Describe recovery actions.			X	X	X					
071 03 01 05		Blade stall										
(01)		Describe cause of and recovery actions when encountering retreating blade stall.			X	X	X					
071 03 01 06		Settling with power (vortex ring)										
(01)		Describe potential conditions for this event and recovery actions.			X	X	X					
071 03 01 07		Overpitch										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe recovery actions.			X	X	X					
071 03 01 08		<i>Overspeed: rotor/engine</i>										
(01)		Describe overspeed control.			X	X	X					
071 03 01 09		<i>Dynamic rollover</i>										
(01)		Describe potential conditions for this event and recovery action.			X	X	X					
071 03 01 10		<i>Mast bumping</i>										
(01)		Describe potential conditions of the ‘conductive to’ and ‘avoidance of’ effect.			X	X	X					
071 04 01 00		SPECIALISED OPERATIONS (Regulation (EU) No 965/2012 on air operations, as amended)										
071 04 01 01		<i>Additional requirements for commercial specialised operations and CAT operations (Annex III (Part-ORO), Subpart FC, Section 3)</i>										
(01)		Explain the requirements related to flight crew recurrent training and checking and operator proficiency check.	X	X	X	X	X					
071 04 01 02		<i>General requirements (Annex VIII (Part-SPO), Subpart A)</i>										
(01)		Explain the task specialist’s responsibilities.	X	X	X	X	X					
071 04 01 03		<i>Helicopter external sling load operations (HESLO) (Annex VIII (Part-SPO), Subpart E)</i>			X	X	X					
(01)		Explain the standard operating procedures and equipment requirements.			X	X	X					
071 04 01 04		<i>Human external cargo operations (HEC) (Annex VIII (Part-SPO), Subpart E)</i>			X	X	X					
(01)		Explain the standard operating procedures and equipment requirements.			X	X	X					

SUBJECT 081 – PRINCIPLES OF FLIGHT – AEROPLANES

ED Decision 2020/018/R

- (1) The following standard symbols and their corresponding meanings are used for certain mathematical operations:
 - * multiplication
 - $\geq$ greater than or equal to
 - $\leq$ less than or equal to
 - SQRT(...) square root of the function, symbol or number in round brackets
- (2) Normally, it should be assumed that the effect of a variable under review is the only variation that needs to be addressed, unless specifically stated otherwise.
- (3) Candidates are expected in simple calculations to be able to convert knots (kt) into metres/second (m/s), and know the appropriate conversion factors by heart.
- (4) In the subsonic range, as covered under Subject 081 01, compressibility effects normally are not considered, unless specifically mentioned.
- (5) For those questions related to propellers (Subject 081 07), as a simplification of the physical reality, the inflow speed into the propeller plane is taken as the aeroplane's true airspeed (TAS).
- (6) In addition, when discussing propeller rotational direction, it will always be specified as seen from behind the propeller plane.
- (7) Note that the term 'mass' is used to describe a quantity of matter, and 'weight' when describing the force. However, the term 'weight' is normally used in aviation to colloquially describe 'mass'. The professional pilot should always note the units to determine whether the term 'weight' is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
080 00 00 00		PRINCIPLES OF FLIGHT										
081 00 00 00		PRINCIPLES OF FLIGHT — AEROPLANES										
081 01 00 00		SUBSONIC AERODYNAMICS										
081 01 01 00		Basics, laws and definitions										
081 01 01 01		Laws and definitions										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		List the international system of units of measurement (SI) for mass, acceleration, weight, velocity, energy, density, temperature, pressure, force, wing loading, and power.	X	X								
(02)	X	Define ‘mass’, ‘force’, ‘acceleration’, and ‘weight’.	X	X								
(03)		State and interpret Newton’s three laws of motion.	X	X								
(04)	X	Explain air density.	X	X								
(05)	X	List the atmospheric properties that effect air density.	X	X								
(06)		Explain how temperature and pressure changes affect air density.	X	X								
(07)	X	Define ‘static pressure’.	X	X								
(08)	X	Define ‘dynamic pressure’.	X	X								
(09)	X	State the formula for ‘dynamic pressure’.	X	X								
(10)		Describe dynamic pressure in terms of an indication of the energy in the system, and how it is related to indicated airspeed (IAS) and air density for a given altitude and speed.	X	X								
(11)		State Bernoulli’s equation for incompressible flow.	X	X								
(12)		Define ‘total pressure’ and explain that the total pressure differs in different systems.	X	X								
(13)		Apply Bernoulli’s equation to flow through a venturi stream tube for incompressible flow.	X	X								
(14)		Describe how IAS is acquired from the pitot static system.	X	X								
(15)		Describe the relationship between density, temperature, and pressure for air.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(16)		Explain the equation of continuity and its application to the flow through a stream tube.	X	X								
(17)	X	Define 'IAS', 'CAS', 'EAS', and 'TAS'.	X	X								
081 01 01 02		Basics of airflow										
(01)	X	Describe steady and unsteady airflow.	X	X								
(02)	X	Explain the concept of a streamline and a stream tube.	X	X								
(03)	X	Describe and explain airflow through a stream tube.	X	X								
(04)	X	Explain the difference between two- and three-dimensional airflow.	X	X								
081 01 01 03		Aerodynamic forces on aerofoils										
(01)		Describe the originating point and direction of the resultant force caused by the pressure distribution around an aerofoil.	X	X								
(02)	X	Resolve the resultant force into the components 'lift' and 'drag'.	X	X								
(03)		Describe the direction of lift and drag.	X	X								
(04)	X	Define the 'aerodynamic moment'.	X	X								
(05)	X	List the factors that affect the aerodynamic moment.	X	X								
(06)		Describe the aerodynamic moment for a symmetrical aerofoil.	X	X								
(07)		Describe the aerodynamic moment for a positively and negatively cambered aerofoil.	X	X								
(08)	X	Define 'angle of attack' (α).	X	X								
081 01 01 04		Shape of an aerofoil section										
(01)	X	Describe the following parameter of an aerofoil section: leading edge.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	Describe the following parameter of an aerofoil section: trailing edge.	X	X								
(03)		Describe the following parameter of an aerofoil section: chord line.	X	X								
(04)		Describe the following parameter of an aerofoil section: thickness-to-chord ratio or relative thickness.	X	X								
(05)		Describe the following parameter of an aerofoil section: location of maximum thickness.	X	X								
(06)		Describe the following parameter of an aerofoil section: camber line.	X	X								
(07)		Describe the following parameter of an aerofoil section: camber.	X	X								
(08)	X	Describe the following parameter of an aerofoil section: nose radius.	X	X								
(09)	X	Describe a symmetrical and an asymmetrical aerofoil section.	X	X								
081 01 01 05		Wing shape										
(01)	X	Describe the following parameter of a wing: span.	X	X								
(02)	X	Describe the following parameter of a wing: tip and root chord.	X	X								
(03)		Describe the following parameter of a wing: taper ratio.	X	X								
(04)	X	Describe the following parameter of a wing: wing area.	X	X								
(05)		Describe the following parameter of a wing: wing planform.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)	X	Describe the following parameter of a wing: mean geometric chord.	X	X								
(07)		Describe the following parameter of a wing: mean aerodynamic chord (MAC).	X	X								
(08)		Describe the following parameter of a wing: aspect ratio.	X	X								
(09)	X	Describe the following parameter of a wing: dihedral angle.	X	X								
(10)	X	Describe the following parameter of a wing: sweep angle.	X	X								
(11)	X	Describe the following parameter of a wing: wing twist, geometric and aerodynamic.	X	X								
(12)		Describe the following parameter of a wing: angle of incidence. <i>Remark: In certain textbooks, angle of incidence is used as angle of attack (α). For Part-FCL theoretical knowledge examination purposes, this use is discontinued, and the angle of incidence is defined as the angle between the aeroplane longitudinal axis and the wing-root chord line.</i>	X	X								
081 01 02 00		Two-dimensional airflow around an aerofoil										
081 01 02 01		Streamline pattern										
(01)	X	Describe the streamline pattern around an aerofoil.	X	X								
(02)		Describe converging and diverging streamlines, and their effect on static pressure and velocity.	X	X								
(03)	X	Describe upwash and downwash.	X	X								
081 01 02 02		Stagnation point										
(01)		Describe the stagnation point.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the movement of the stagnation point as the α changes.	X	X								
081 01 02 03		Pressure distribution										
(01)		Describe pressure distribution and local speeds around an aerofoil including effects of camber and α .	X	X								
(02)		<i>Intentionally left blank</i>										
081 01 02 04		Centre of pressure (CP) and aerodynamic centre (AC)										
(01)		Explain CP and AC.	X	X								
081 01 02 05		<i>Intentionally left blank</i>										
081 01 02 06		Drag and wake										
(01)	X	List two physical phenomena that cause drag.	X	X								
(02)		Describe skin friction drag.	X	X								
(03)		Describe form (pressure) drag.	X	X								
(04)	X	Explain why drag and wake cause loss of energy (momentum).	X	X								
081 01 02 07		Influence of angle of attack (α)										
(01)		Explain the influence of α on lift.	X	X								
081 01 02 08		<i>Intentionally left blank</i>										
081 01 02 09		The lift coefficient (C_L) – angle of attack (α) graph										
(01)		Describe the C_L – α graph.	X	X								
(02)		Explain the significant points: — point where the curve crosses the horizontal axis (zero lift); — point where the curve crosses the vertical axis ($\alpha = 0$);	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— point where the curve reaches its maximum (C_{LMAX}).										
081 01 03 00		Coefficients										
081 01 03 01		General use of coefficients										
(01)	X	Explain why coefficients are used in general.	X	X								
081 01 03 02		The lift coefficient (C_L)										
(01)		Explain the lift formula, the factors that affect lift, and perform simple calculations.	X	X								
(02)		Describe the effect of camber on the C_L - α graph (symmetrical and positively/negatively cambered aerofoils).	X	X								
(03)		Describe the typical difference in the C_L - α graph for fast and slow aerofoil design.	X	X								
(04)	X	Define ' C_{LMAX} ' (maximum lift coefficient) and ' α_{CRIT} ' (stalling α) on the graph.	X	X								
(05)		Describe C_L and explain the variables that affect it in low subsonic flight.	X	X								
081 01 03 03		Drag										
(01)		Describe the two-dimensional drag formula.	X	X								
(02)		Discuss the effect of the shape of a body, cross-sectional area, and surface roughness on the drag coefficient.	X	X								
081 01 04 00		Three-dimensional airflow around an aeroplane										
081 01 04 01		Angle of attack (α)										
(01)	X	Define 'angle of attack' (α). <i>Remark: For theoretical knowledge examination purposes, the angle-of-attack definition requires a reference line. This reference line for 3D has</i>	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		<i>been chosen to be the longitudinal axis and for 2D the chord line.</i>										
(02)		Explain the difference between the α and the attitude of an aeroplane.	X	X								
081 01 04 02		Streamline pattern										
(01)		Describe the general streamline pattern around the wing, tail section, and fuselage.	X	X								
(02)		Explain and describe the causes of spanwise flow over top and bottom surfaces.	X	X								
(03)		Describe wing tip vortices and their contribution to downwash behind the wing.	X	X								
(04)		Explain why wing tip vortices vary with α .	X	X								
(05)		Describe spanwise lift distribution including the effect of wing planform.	X	X								
(06)		Describe the causes, distribution and duration of the wake turbulence behind an aeroplane.	X	X								
(07)		Describe the influence of flap deflection on the wing tip vortex.	X	X								
(08)		Describe the parameters that influence wake turbulence.	X	X								
081 01 04 03		Induced drag										
(01)		Explain the factors that cause induced drag.	X	X								
(02)		Describe the approximate formula for the induced drag coefficient (including variables but excluding constants).	X	X								
(03)		Describe the relationship between induced drag and total drag in straight and level flight with variable speed.	X	X								
(04)		Describe the effect of mass on induced drag at a given IAS.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)		Describe the means to reduce induced drag: — aspect ratio; — winglets; — tip tanks; — wing twist; — camber change.	X	X								
(06)		Describe the influence of lift distribution on induced drag.	X	X								
(07)		Describe the influence of downwash on the effective airflow.	X	X								
(08)		Explain induced and effective local α .	X	X								
(09)		Explain the influence of the induced α on the direction of the lift vector.	X	X								
(10)		Explain the relationship between induced drag and: — speed; — aspect ratio; — wing planform; — bank angle in a horizontal coordinated turn.	X	X								
(11)		Explain the induced drag coefficient and its relationship with the lift coefficient and aspect ratio.	X	X								
(12)		Explain the influence of induced drag on:	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- the C_L-α graph, and show the effect on the graph when comparing high- and low-aspect ratio wings; - the C_L-C_D (aeroplane polar), and show the effect on the graph when comparing high- and low-aspect ratio wings; - the parabolic aeroplane polar in a graph and as a formula [$C_D = C_{PD} + kC_L^2$], where C_D = coefficient of drag and C_{PD} = coefficient of parasite drag.										
(13)		Describe the C_L - C_D graph (polar).	X	X								
(14)		Indicate minimum drag on the graph.	X	X								
(15)		Explain why the C_L - C_D ratio is important as a measure of performance.	X	X								
(16)		Intentionally left blank										
081 01 05 00		Total drag										
081 01 05 01		Total drag in relation to parasite drag and induced drag										
(01)	X	State that total drag consists of parasite drag and induced drag.	X	X								
081 01 05 02		Parasite drag										
(01)		Describe the types of drag that are included in parasite drag.	X	X								
(02)		Describe form (pressure) drag and the factors which affect its magnitude.	X	X								
(03)		Describe interference drag and the factors which affect its magnitude.	X	X								
(04)		Describe friction drag and the factors which affect its magnitude.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 01 05 03		Parasite drag and speed										
(01)		Describe the relationship between parasite drag and speed.	X	X								
081 01 05 04		Induced drag and speed (Refer to 081 01 04 03)										
081 01 05 05		Total drag										
(01)		Explain the total drag–speed graph and the constituent drag components.	X	X								
(02)		Indicate the speed for minimum drag.	X	X								
081 01 05 06		Intentionally left blank										
081 01 05 07		Variables affecting the total drag–speed graph										
(01)		Describe the effect of aeroplane gross mass on the graph.	X	X								
(02)		Describe the effect of pressure altitude on: — drag–IAS graph; — drag–TAS graph.	X	X								
(03)		Describe speed stability from the graph.	X	X								
(04)		Describe non-stable, neutral, and stable IAS regions.	X	X								
(05)		Explain what happens to the IAS and drag in the non-stable region if speed suddenly decreases and why this could occur.	X	X								
081 01 06 00		Ground effect										
081 01 06 01		Influence of ground effect										
(01)		Explain the influence of ground effect on wing tip vortices, downwash, airflow pattern, lift, and drag.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe the influence of ground effect on induced α and the coefficient of induced drag (C_{Di}).	X	X								
(03)		Explain the effects of entering and leaving ground effect.	X	X								
081 01 06 02		Effect on stalling angle of attack (α_{CRIT})										
(01)		Describe the influence of ground effect on α_{CRIT} .	X	X								
081 01 06 03		Effect on lift coefficient (C_L)										
(01)		Describe the influence of ground effect on the effective α and C_L .	X	X								
081 01 06 04		Effect on take-off and landing characteristics of an aeroplane										
(01)		Describe the influence of ground effect on take-off and landing characteristics and performance of an aeroplane.	X	X								
(02)		Describe the difference in take-off and landing characteristics of high- and low-wing aeroplanes.	X	X								
081 01 07 00		The relationship between lift coefficient and speed in steady, straight, and level flight										
081 01 07 01		Represented by an equation										
(01)		Explain the effect on C_L during speed increase/decrease in steady, straight, and level flight, and perform simple calculations.	X	X								
081 01 07 02		Represented by a graph										
(01)		Explain, by using a graph, the effect on speed of C_L changes at a given weight.	X	X								
081 01 08 00		Intentionally left blank										
081 01 09 00		CLMAX augmentation										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 01 09 01		Trailing-edge flaps and the reasons for their use in take-off and landing										
(01)		From the given relevant diagrams, describe or identify the following types of trailing-edge flaps: — split flaps; — plain flaps; — slotted flaps; — Fowler flaps.	X	X								
(02)		Describe how the wing's effective camber increases the C_L and C_D , and the reasons why this can be beneficial.	X	X								
(03)		Describe their effect on: — the location of CP; — pitching moments (due to wing CP movement); — stall speed.	X	X								
(04)		Compare their influence on the C_L - α graph: — indicate the variation in C_L at any given α ; — indicate their effect on C_{LMAX} ; — indicate their effect on critical α ; — indicate their effect on the α at a given C_L .	X	X								
(05)		Compare their influence on the C_L - C_D graph: — indicate how the $(C_L/C_D)_{MAX}$ differs from that of a clean wing.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain the influence of trailing-edge flap deflection on the glide angle.	X	X								
(07)		Describe flap asymmetry: — explain the effect on aeroplane controllability.	X	X								
(08)		Describe trailing-edge flap effect on take-off and landing: — explain the advantages of lower-nose attitudes; — explain why take-off and landing speeds/distances are reduced.	X	X								
(09)		Explain the effects of flap-setting errors, such as mis-selection and premature/late extension or retraction of flaps, on: — take-off and landing distance and speeds; — climb and descent performance; — stall buffet margins.	X	X								
081 01 09 02		Leading-edge devices and the reasons for their use in take-off and landing										
(01)		From the given relevant diagrams, describe or identify the different types of leading-edge high-lift devices: — Krueger flaps; — variable camber flaps; — slats.	X	X								
(02)		Describe the function of the slot.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Describe how the wing's effective camber increases with a leading-edge flap.	X	X								
(04)		Explain the effect of leading-edge flaps on the stall speed, also in comparison with trailing-edge flaps.	X	X								
(05)		Compare their influence on the C_L - α graph, compared with trailing-edge flaps and a clean wing: — indicate the effect of leading-edge devices on C_{LMAX}; — explain how the C_L curve differs from that of a clean wing; — indicate the effect of leading-edge devices on α_{CRIT}.	X	X								
(06)		Compare their influence on the C_L - C_D graph.	X	X								
(07)		Describe slat asymmetry: — describe the effect on aeroplane controllability.	X	X								
(08)		Explain the reasons for using leading-edge high-lift devices on take-off and landing: — explain the disadvantage of increased nose-up attitudes; — explain why take-off and landing speeds/distances are reduced.	X	X								
081 01 09 03		Vortex generators										
(01)		Explain the purpose of vortex generators.	X	X								
(02)		Describe the basic operating principle of vortex generators.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		State their advantages and disadvantages.	X	X								
081 01 10 00		Means to reduce the C_L–C_D ratio										
081 01 10 01		<i>Spoilers and the reasons for their use in the different phases of flight</i>										
(01)		Describe the aerodynamic functioning of spoilers: — roll spoilers; — flight spoilers (speed brakes); — ground spoilers (lift dumpers).	X	X								
(02)		Describe the effect of spoilers on the C_L – α graph and stall speed.	X	X								
(03)		Describe the influence of spoilers on the C_L – C_D graph and lift-drag ratio.	X	X								
081 01 10 02		<i>Speed brakes and the reasons for their use in the different phases of flight</i>										
(01)		Describe speed brakes and the reasons for using them in the different phases of flight.	X	X								
(02)		State their influence on the C_L – C_D graph and lift-drag ratio.	X	X								
(03)		Explain how speed brakes increase parasite drag.	X	X								
(04)		Describe how speed brakes affect the minimum drag speed.	X	X								
(05)		Describe their effect on rate and angle of descent.	X	X								
081 01 11 00		Intentionally left blank										
081 01 12 00		Aerodynamic degradation										
081 01 12 01		<i>Ice and other contaminants</i>										
(01)		Describe the locations on an aeroplane where ice build-up will occur during flight.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain the aerodynamic effects of ice and other contaminants on: — lift (maximum C_L); — drag; — stall speed; — α_{CRIT} ; — stability and controllability.	X	X								
(03)		Explain the aerodynamic effects of icing during take-off.	X	X								
081 01 12 02		Deformation and modification of airframe, ageing aeroplanes										
(01)		Describe the effect of airframe deformation and modification of an ageing aeroplane on aeroplane performance.	X	X								
(02)		Explain the effect on boundary layer condition of an ageing aeroplane.	X	X								
081 02 00 00		HIGH-SPEED AERODYNAMICS										
081 02 01 00		Speeds										
081 02 01 01		Speed of sound										
(01)	X	Define 'speed of sound'.	X									
(02)		Explain the variation of the speed of sound with altitude.	X									
(03)		Explain the influence of temperature on the speed of sound.	X									
081 02 01 02		Mach number										
(01)		Define 'Mach number' as a function of TAS and speed of sound.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 02 01 03		<i>Influence of temperature and altitude on Mach number</i>										
(01)		Explain the absence of change of Mach number with varying temperature at constant flight level and calibrated airspeed.	X									
(02)		Explain the relationship between Mach number, TAS and IAS during climb and descent at constant Mach number or IAS, and explain variation of lift coefficient, α , pitch and flight-path angle.	X									
(03)		Explain: — risk of exceeding the maximum operation speed (VMO) when descending at constant Mach number; — risk of exceeding the maximum operating Mach number (M_{MO}) when climbing at constant IAS; — risk of a low-speed stall at high altitude when climbing at a too low Mach number.	X									
081 02 01 04		<i>Compressibility</i>										
(01)		State that compressibility means that density can change along a streamline, and that this occurs in the high subsonic (from Mach 0.4), transonic, and supersonic flow.	X									
(02)	X	State that compressibility negatively affects the pressure gradient, leading to an overall reduction of the C_L .	X									
(03)	X	State that Mach number is a measure of compressibility.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Describe that compressibility increases low-speed stall speed and decreases α_{CRIT} .	X									
081 02 01 05		Subdivision of aerodynamic flow										
(01)	X	List the subdivision of aerodynamic flow: — subsonic flow below compressibility; — subsonic flow above compressibility; — transonic flow; — supersonic flow.	X									
(02)		Describe the characteristics of the flow regimes listed above.	X									
(03)		Explain why some transport aeroplanes cruise at Mach numbers above the critical Mach number (M_{CRIT}).	X									
081 02 02 00		Shock waves										
081 02 02 01		Definition of shock wave										
(01)	X	Define a 'shock wave'.	X									
081 02 02 02		Normal shock waves										
(01)		Describe a normal shock wave with respect to changes in: — static temperature; — static and total pressure; — velocity; — local speed of sound; — Mach number; — density.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe a normal shock wave with respect to orientation relative to the wing surface.	X									
(03)		Explain the influence of increasing Mach number on a normal shock wave, at positive lift, with respect to: — strength; — position relative to the wing; — second shock wave at the lower surface.	X									
(04)		Explain the influence of α on shock-wave intensity and shock-wave location at constant Mach number.	X									
081 02 03 00		Effects of exceeding the critical Mach number (M_{CRIT})										
081 02 03 01		Critical Mach number (M_{CRIT})										
(01)		Define ' M_{CRIT} '.	X									
(02)		Explain how a change in α , aeroplane weight, manoeuvres, and centre-of-gravity (CG) position influences M_{CRIT} .	X									
081 02 03 02		Effect on lift										
(01)		Describe the behaviour of C_L versus Mach number at constant α .	X									
(02)		Explain the consequences of exceeding M_{CRIT} with respect to C_L and C_{LMAX} .	X									
(03)		Explain the change in stall indicated airspeed (IAS) with altitude.	X									
(04)		Discuss the effect on α_{CRIT} .	X									
(05)		Explain the advantages of exceeding M_{CRIT} in aeroplanes with supercritical aerofoils with respect to:	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— speed versus drag ratio; — specific range; — optimum altitude.										
081 02 03 03		Effect on drag										
(01)		Describe wave drag.	X									
(02)		Describe the behaviour of C_D versus Mach number at constant α .	X									
(03)		Explain the effect of Mach number on the C_L – C_D graph.	X									
(04)		Describe the effects and hazards of exceeding $M_{DRAG\ DIVERGENCE}$, namely: — drag rise; — instability; — Mach tuck; — shock stall.	X									
(05)		State the relation between M_{CRIT} and $M_{DRAG\ DIVERGENCE}$.	X									
081 02 03 04		Effect on pitching moment										
(01)		Discuss the effect of Mach number on the CP location.	X									
(02)		Describe the overall change in pitching moment above M_{CRIT} and explain the ‘tuck under’ or ‘Mach tuck’ effect.	X									
(03)	X	State the requirement for a Mach trim system to compensate for the effect of the CP movement and ‘tuck under’ effect.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)	X	Discuss the aerodynamic functioning of the Mach trim system.	X									
(05)		Discuss the corrective measures if the Mach trim fails.	X									
081 02 03 05		Effect on control effectiveness										
(01)		Discuss the effects on the effectiveness of control surfaces.	X									
081 02 04 00		Intentionally left blank										
081 02 05 00		Means to influence critical Mach number (M_{CRIT})										
081 02 05 01		Wing sweep										
(01)		Explain the influence of the angle of sweep on: — M_{CRIT}; — effective thickness/chord change or velocity component perpendicular to the quarter chord line.	X									
(02)		Describe the influence of the angle of sweepback at subsonic speed on: — C_{LMAX}; — efficiency of and requirement for high-lift devices; — pitch-up stall behaviour.	X									
(03)		Discuss the effect of wing sweepback on drag.	X									
081 02 05 02		Aerofoil shape										
(01)		Explain the use of thin aerofoils with reduced camber.	X									
(02)		Explain the main purpose of supercritical aerofoils.	X									
(03)		Intentionally left blank										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the advantages and disadvantages of supercritical aerofoils for wing design.	X									
081 02 05 03		Vortex generators										
(01)		Explain the use of vortex generators as a means to avoid or restrict flow separation caused by the presence of a normal shock wave.	X									
081 03 00 00		Stall, Mach tuck, and upset prevention and recovery										
081 03 01 00		The stall										
081 03 01 01		Flow separation at increasing α										
(01)	X	Define the 'boundary layer'.	X	X								
(02)	X	Describe the thickness of a typical laminar and turbulent boundary layer.	X	X								
(03)		Describe the properties, advantages and disadvantages of the laminar boundary layer.	X	X								
(04)		Describe the properties, advantages and disadvantages of the turbulent boundary layer.	X	X								
(05)		Define the 'transition point'.	X	X								
(06)		Explain why the laminar boundary layer separates easier than the turbulent boundary layer does.	X	X								
(07)		Describe why the airflow over the aft part of a wing slows down as the α increases.	X	X								
(08)		Define the 'separation point' and describe its location as a function of α .	X	X								
(09)	X	Define α_{CRIT} .	X	X								
(10)		Describe in straight and level flight the influence of increasing the α and the phenomenon that may occur regarding:	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- the forward stagnation point; - the pressure distribution; - the CP location (straight and swept-back wing); C_L; C_D and D (drag); - the pitching moment (straight and swept-back wing); - buffet onset; - deterrent buffet for a clean wing at high Mach number; - lack of pitch authority; - uncommanded pitch down; - uncommanded roll.										
(11)		Explain what causes the possible natural buffet on the aeroplane in a pre-stall condition.	X	X								
(12)		Describe the effectiveness of the flight controls in a pre-stall condition.	X	X								
(13)		Describe and explain the normal post-stall behaviour of a straight-wing aeroplane.	X	X								
(14)		Describe the effect and dangers of using the controls close to the stall.	X	X								
(15)		Describe the deterrent buffet.	X	X								
(16)		Explain the occurrence of the deterrent buffet and why this phenomenon is considered to be a stall limit.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 03 01 02		The stall speed										
(01)		Explain V_{S0} , V_{S1} , V_{SR} , and V_{S1G} .	X	X								
(02)		Solve V_{S1G} from the lift formula given varying C_L .	X	X								
(03)		Describe and explain the influence of the following parameters on stall speed: — CG; — thrust component; — slipstream; — wing loading; — mass; — wing contamination; — angle of sweep; — altitude (for compressibility effects, see 081 02 03 02).	X	X								
(04)	X	Define the 'load factor n '.	X	X								
(05)		Explain why the load factor increases in a turn.	X	X								
(06)		Explain why the load factor increases in a pull-up and decreases in a push-over manoeuvre.	X	X								
(07)		Describe and explain the influence of the 'load factor n ' on stall speed.	X	X								
(08)	X	Explain the expression 'accelerated stall'. <i>Remark: Sometimes, accelerated stall is also erroneously referred to as high-speed stall. This latter expression will not be used for Subject 081.</i>	X	X								
(09)		Calculate the change of stall speed as a function of the load factor.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(10)		Calculate the increase of stall speed in a horizontal coordinated turn as a function of bank angle.	X	X								
(11)		Calculate the change of stall speed as a function of the gross mass.	X	X								
081 03 01 03		The initial stall in spanwise direction										
(01)		Explain the initial stall sequence on the following planforms: — elliptical; — rectangular; — moderate and high taper; — sweepback or delta.	X	X								
(02)		Explain the purpose of aerodynamic and geometric twist (washout).	X	X								
(03)		<i>Intentionally left blank</i>										
(04)		Explain the influence of fences, vortilons, saw teeth, vortex generators, and strakes on engine nacelles.	X	X								
081 03 01 04		Stall warning										
(01)	X	Explain why stall warning is necessary.	X	X								
(02)	X	Explain when aerodynamic and artificial stall warnings are used.	X	X								
(03)		Explain why CS-23 and CS-25 require a margin to stall speed for take-off and landing speeds.	X	X								
(04)	X	Describe: — buffet; — stall strip;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- flapper switch (leading-edge stall-warning vane); - angle-of-attack vane; - angle-of-attack probe; - stick shaker.										
(05)		Describe the recovery after: - stall warning; - stall; - stick-pusher actuation.	X	X								
081 03 01 05		Special phenomena of stall										
(01)		<i>Intentionally left blank</i>										
(02)		Explain the difference between power-off and power-on stalls and recovery.	X	X								
(03)		Describe stall and recovery in a climbing and descending turn.	X	X								
(04)		Describe the pitch-up effect on a swept wing aeroplane and also an aeroplane with a T-tail.	X	X								
(05)		Describe super stall or deep stall.	X	X								
(06)		Describe the philosophy behind the stick-pusher system.	X	X								
(07)		Describe the factors that can lead to the absence of stall warning and explain the associated risks.	X	X								
(08)		Describe the indications and explain the consequences of premature stabiliser stall due to ice contamination (negative tail stall).	X	X								
(09)		Describe when to expect in-flight icing.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(10)		Explain how the effect is changed when retracting/extending lift-augmentation devices.	X	X								
(11)		<i>Intentionally left blank</i>										
(12)		Explain the effect of a contaminated wing on the stall speed and α_{CRIT} .	X	X								
(13)		Explain the hazards associated with airframe contamination when parked and during ground operations in winter conditions, and the aerodynamic effects when attempting a take-off.	X	X								
(14)		Explain de-icing/anti-icing holdover time and the likely hazards after it has expired.	X	X								
(15)		Describe the aerodynamic effects of heavy tropical rain on stall speed and drag, and the appropriate mitigation in such conditions.	X	X								
081 03 01 06		The spin										
(01)		Explain how to avoid spins.	X	X								
(02)		List the factors that cause a spin to develop.	X	X								
(03)		Describe an ‘incipient’ and ‘developed’ spin, recognition and recovery.	X	X								
(04)		Describe the differences in spin attitude with forward and aft CG.	X	X								
081 03 02 00		Buffet onset boundary										
081 03 02 01		Mach buffet										
(01)		Explain shock-induced separation, and describe its relationship with Mach buffet (high speed buffet) and Mach tuck.	X									
(02)		<i>Intentionally left blank</i>										
081 03 02 02		Buffet onset										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain the concept of buffet margin, and describe the influence of the following parameters on the concept of buffet margin: — α; — Mach number; — pressure altitude; — mass; — load factor; — angle of bank; — CG location.	X									
(02)		Explain how the buffet onset boundary chart can be used to determine: — manoeuvrability; — buffet margin.	X									
(03)		Describe the consequences of exceeding M_{MO} : light buffet, buffet onset.	X									
(04)		Explain ‘aerodynamic ceiling’ and ‘coffin corner’.	X									
(05)		Explain the concept of the ‘1.3g’ buffet margin altitude.	X									
(06)		Find (using an example graph): — buffet free range; — aerodynamic ceiling at a given mass; — load factor and bank angle at which buffet occurs at a given mass, Mach number, and pressure altitude.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Explain why descent increases the buffet free range.	X									
081 03 03 00		Situations in which buffet or stall could occur										
081 03 03 01		Explain why buffet or stall occurs										
(01)		Explain why buffet or stall could occur in the following pilot-induced situations, and the methods to mitigate them: — inappropriate take-off configuration, detailing the consequences of errors associated with leading-edge devices; — steep turns; — go-around using take-off/go-around (TOGA) setting (underslung engines).	X	X								
(02)		Explain why buffet or stall could occur in the following environmental conditions at low altitude, and how to mitigate them: — thunderstorms; — wind shear and microburst; — turbulence; — wake turbulence; — icing conditions.	X	X								
(03)		Explain why buffet or stall could occur in the following environmental conditions at high altitude, and how to mitigate them: — thunderstorms in the intertropical convergence zone (ITCZ); — jet streams;	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— clear-air turbulence.										
(04)		Explain why buffet or stall could occur in the following situations, and how to mitigate them: — inappropriate autopilot climb mode; — loss of, or unreliable, airspeed indication.	X	X								
081 03 04 00		Recognition of stalled condition										
081 03 04 01		Recognition and explanation of stalled condition										
(01)		Explain why a stalled condition can occur at any airspeed, or attitude or altitude.	X	X								
(02)		Explain that a stall may be recognised by continuous stall-warning activation accompanied by at least one of the following: — buffet, that can be heavy; — lack of pitch authority; — uncommanded pitch down and uncommanded roll; — inability to arrest the descent rate.	X	X								
(03)		Explain that ‘stall warning’ means a natural or synthetic indication provided when approaching the stall that may include one or more of the following indications: — aerodynamic buffeting; — reduced roll stability and aileron effectiveness; — visual or aural clues and warnings;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- reduced elevator (pitch) authority; - inability to maintain altitude or arrest a rate of descent; - stick-shaker activation.										
081 04 00 00		STABILITY										
081 04 01 00		Static and dynamic stability										
081 04 01 01		Basics and definitions										
(01)		Define 'static stability': - describe/identify a statically stable, neutral, and unstable condition (positive, neutral, and negative static stability), and - explain why aeroplanes are statically stable.	X	X								
(02)		Explain manoeuvrability.	X	X								
(03)		Explain the relationship between static stability and manoeuvrability.	X	X								
(04)		Define 'dynamic stability': - describe/identify a dynamically stable, neutral, and unstable motion (positive, neutral, and negative dynamic stability); - describe/identify periodic and aperiodic motion.	X	X								
(05)		<i>Intentionally left blank</i>										
081 04 01 02		Precondition for static stability										
(01)	X	Explain an equilibrium of forces and moments as the initial condition for static stability.	X	X								
081 04 01 03		Sum of forces										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)	X	Identify the forces considered in the equilibrium of forces.	X	X								
081 04 01 04		Sum of moments										
(01)		Identify the moments about all three axes considered in the equilibrium of moments.	X	X								
(02)		Discuss the effect of sum of moments not being zero.	X	X								
081 04 02 00		Intentionally left blank										
081 04 03 00		Static and dynamic longitudinal stability										
081 04 03 01		Methods for achieving balance										
(01)	X	Explain the stabiliser as the means to satisfy the condition of nullifying the total sum of the moments about the lateral axis.	X	X								
(02)		Explain the influence of the location of the wing CP relative to the CG on the magnitude and direction of the balancing force on the stabiliser.	X	X								
(03)		Explain the influence of the indicated airspeed on the magnitude and direction of the balancing force on the stabiliser.	X	X								
(04)		Explain the use of the elevator deflection or stabiliser angle for the generation of the balancing force and its direction.	X	X								
(05)		Explain the elevator deflection required to balance thrust change as a function of engine position.	X	X								
081 04 03 02		Static longitudinal stability										
(01)		Discuss the effect of the CG location on pitch manoeuvrability and longitudinal stability.	X	X								
081 04 03 03		Neutral point										
(01)	X	Define 'neutral point'.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	Explain why the location of the neutral point is only dependent on the aerodynamic design of the aeroplane.	X	X								
081 04 03 04		Factors affecting neutral point										
(01)		Describe the location of the neutral point relative to the locations of the aerodynamic centre of the wing and tail.	X	X								
081 04 03 05		Location of centre of gravity (CG)										
(01)		Explain the influence of the CG location on the static longitudinal stability of the aeroplane.	X	X								
(02)		Explain the CG forward and aft limits with respect to: — longitudinal control forces; — elevator effectiveness; — stability.	X	X								
(03)		Define 'static margin'.	X	X								
081 04 03 06		The $C_m-\alpha$ graph										
(01)	X	Describe the $C_m-\alpha$ graph with respect to the relationship between the slope of the graph and static stability.	X	X								
081 04 03 07		Factors affecting the $C_m-\alpha$ graph										
(01)		Explain: — the effect on the $C_m-\alpha$ graph of a shift of CG in the forward and aft direction; — the effect on the $C_m-\alpha$ graph when the elevator is moved up or down;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- the effect on the $C_m-\alpha$ graph when the trim is moved; - the effect of the wing contribution; - the tail contribution.										
081 04 03 08		Intentionally left blank										
081 04 03 09		Intentionally left blank										
081 04 03 10		The stick force versus speed graph (IAS)										
(01)		Explain how a pilot perceives stable static longitudinal stick force stability regarding changes in: - speed; - altitude; - mass distribution (CG location).	X	X								
081 04 03 11		Intentionally left blank										
081 04 03 12		The manoeuvring stability/stick force per g										
(01)	X	Define the 'stick force per g', and describe that the stick force increases linearly with increase in g.	X	X								
(02)		Explain why: - the stick force per g has a prescribed minimum and maximum value; - the stick force per g decreases with pressure altitude.	X	X								
081 04 03 13		Intentionally left blank										
081 04 03 14		Factors affecting the manoeuvring stability/stick force per g										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain the influence on stick force per g of: — CG location; — trim setting.	X	X								
081 04 03 15		Intentionally left blank										
081 04 03 16		Dynamic longitudinal stability										
(01)		Describe the phugoid and short-period motion in terms of period, damping, variations (if applicable) in speed, altitude, and α .	X	X								
(02)		Explain why the short-period motion is more hazardous than the phugoid.	X	X								
(03)		Describe ‘pilot-induced oscillations’.	X	X								
(04)		Explain the effect of high altitude on dynamic stability.	X	X								
(05)		Describe the influence of the CG location on the dynamic longitudinal stability of the aeroplane.	X	X								
081 04 04 00		Static directional stability										
081 04 04 01		Definition and effects of static directional stability										
(01)	X	Define ‘static directional stability’.	X	X								
(02)		Explain the effects of static directional stability being too weak or too strong.	X	X								
081 04 04 02		Sideslip angle										
(01)		Define ‘sideslip angle’.	X	X								
(02)		Identify β as the symbol used for the sideslip angle.	X	X								
081 04 04 03		Yaw-moment coefficient C_n										
(01)	X	Define the ‘yawing-moment coefficient C_n ’.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)	X	Define the relationship between C_n and β for an aeroplane with static directional stability.	X	X								
081 04 04 04		C_n - β graph										
(01)	X	Explain why: — C_n depends on β; — C_n equals zero for that β that provides static equilibrium about the aeroplane's normal axis; — if no asymmetric engine thrust, flight control or loading condition prevails, the equilibrium β equals zero.	X	X								
(02)	X	Identify how the slope of the C_n - β graph is a measure for static directional stability.	X	X								
(03)	X	Identify how the slope of the C_n - β graph is affected by altitude.	X	X								
081 04 04 05		Factors affecting static directional stability										
(01)		Describe how the following aeroplane components contribute to static directional stability: — wing; — fin; — dorsal fin; — ventral fin; — angle of sweep of the wing; — angle of sweep of the fin; — fuselage at high α;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— strakes.										
(02)		Explain the reduction in static directional stability when the CG moves aft.	X	X								
081 04 05 00		Static lateral stability										
081 04 05 01		Definition and effects of static lateral stability										
(01)	X	Define 'static lateral stability'.	X	X								
(02)		Explain the effects of static lateral stability being too weak or too strong.	X	X								
081 04 05 02		Bank angle ϕ										
(01)	X	Define 'bank angle ϕ '.	X	X								
081 04 05 03		The roll-moment coefficient C_l										
(01)	X	Define the 'roll-moment coefficient C_l '.	X	X								
081 04 05 04		Contribution of sideslip angle (β)										
(01)		Explain how without coordination the bank angle (ϕ) creates sideslip angle (β).	X	X								
081 04 05 05		The C_l-β graph										
(01)	X	Describe the C_l - β graph.	X	X								
(02)	X	Identify the slope of the C_l - β graph as a measure for static lateral stability.	X	X								
(03)	X	Identify how the slope of the C_l - β graph is affected by altitude.	X	X								
081 04 05 06		Factors affecting static lateral stability										
(01)		Explain the contribution to the static lateral stability of: — dihedral, anhedral; — high wing, low wing; — sweep angle of the wing;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— ventral fin; — vertical tail.										
081 04 06 00		Dynamic lateral/directional stability										
081 04 06 01		Intentionally left blank										
081 04 06 02		Tendency to spiral dive										
(01)		Explain how lateral and directional stability are coupled.	X	X								
(02)		Explain how high static directional stability and low static lateral stability may cause spiral divergence (unstable spiral dive), and under which conditions the spiral dive mode is neutral or stable.	X	X								
(03)		Describe an unstable spiral dive mode with respect to deviations in speed, bank angle, nose low-pitch attitude, and decreasing altitude.	X	X								
081 04 06 03		Dutch roll										
(01)		Describe Dutch roll.	X	X								
(02)		Explain: — why Dutch roll occurs when the static lateral stability is higher than static directional stability; — the conditions for a stable, neutral or unstable Dutch roll motion; — the function of the yaw damper; — the actions to be taken when the yaw damper is not available.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Describe how the asymmetric nature of shock waves on both wings, at high Mach numbers, can lead to Dutch roll.	X									
081 04 06 04		Effects of altitude on dynamic stability										
(01)		Explain that increased pressure altitude reduces dynamic lateral/directional stability.	X	X								
081 05 00 00		CONTROL										
081 05 01 00		General										
081 05 01 01		Basics — The three planes and three axes										
(01)	X	Define: — lateral axis; — longitudinal axis; — normal axis.	X	X								
(02)	X	— Define: — pitch angle; — bank angle (ϕ); — yaw angle.	X	X								
(03)		Describe the motion about the three axes.	X	X								
(04)		Name and describe the devices that control these motions.	X	X								
081 05 01 02		Camber change										
(01)		State that camber is changed by movement of a control surface and explain the effect.	X	X								
081 05 01 03		Angle-of-attack (α) change										
(01)	X	Explain the influence of local α change by movement of a control surface.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 05 02 00		Pitch (longitudinal) control										
081 05 02 01		Elevator/all-flying tails										
(01)		Explain the working principle of the elevator/all-flying tail and describe its function.	X	X								
081 05 02 02		Downwash effects										
(01)		Explain the effect of downwash on the tailplane α .	X	X								
(02)		<i>Intentionally left blank</i>										
081 05 02 03		<i>Intentionally left blank</i>										
081 05 02 04		Location of centre of gravity (CG)										
(01)		Explain the relationship between elevator deflection and CG location to produce a given aeroplane response.	X	X								
(02)		Explain the effect of forward CG limit on pitch control.	X	X								
081 05 02 05		Moments due to engine thrust										
(01)		Describe the effect of engine thrust on pitching moments for different engine locations.	X	X								
081 05 03 00		Yaw (directional) control										
081 05 03 01		The rudder										
(01)		Explain the working principle of the rudder and describe its function. State the relationship between rudder deflection and the moment about the normal axis. Describe the effect of sideslip on the moment about the normal axis.	X	X								
081 05 03 02		Rudder limiting										
(01)		Explain why and how rudder deflection is limited on CAT aeroplanes.	X									

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 05 04 00		Roll (lateral) control										
081 05 04 01		Ailerons										
(01)		Explain the functioning of ailerons.	X	X								
(02)		Describe the adverse effects of aileron deflection. (Refer to Subjects 081 05 04 04 and 081 06 01 02)	X	X								
(03)		Explain why some aeroplanes have inboard and outboard ailerons.	X	X								
(04)		State that the outboard ailerons are locked beyond a given speed to prevent: — over-control; — exceeding structural limitations; — aeroelastic phenomena (flutter, divergence and aileron reversal).	X	X								
(05)		Describe the use of aileron deflection in normal flight, flight with sideslip, crosswind landings, horizontal turns, flight with one-engine-inoperative.	X	X								
(06)	X	Define ‘roll rate’.	X	X								
(07)	X	List the factors that affect roll rate.	X	X								
(08)		Describe flaperons and aileron droop.	X	X								
081 05 04 02		Intentionally left blank										
081 05 04 03		Spoilers										
(01)		Explain how spoilers can be used to control the rolling movement in combination with or instead of the ailerons.	X	X								
081 05 04 04		Adverse yaw										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain why the use of ailerons induces adverse yaw.	X	X								
081 05 04 05		Means to avoid adverse yaw										
(01)		Explain how the following reduce adverse yaw: — Frise ailerons; — differential aileron deflection; — rudder aileron cross-coupling; — roll spoilers.	X	X								
081 05 05 00		Roll/yaw interaction										
081 05 05 01		Explain roll/yaw interaction										
(01)		Explain the secondary effect of roll.	X	X								
(02)		Explain the secondary effect of yaw.	X	X								
081 05 06 00		Means to reduce control forces										
081 05 06 01		Aerodynamic balance										
(01)		Describe the purpose of aerodynamic balance.	X	X								
(02)		Describe the working principle of the horn balance.	X	X								
(03)		Describe the working principle of the internal balance.	X	X								
(04)		Describe the working principle and application of: — balance tab; — anti-balance tab; — spring tab; — servo tab.	X	X								
081 05 06 02		Artificial means										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		State the differences between fully powered controls and power-assisted controls.	X	X								
(02)		Describe power-assisted controls.	X	X								
(03)		Describe the advantages of artificial feel in fully powered control.	X	X								
081 05 07 00		Fly-by-wire (FBW)										
081 05 07 01		Control laws										
(01)		Explain which parameters may be controlled in level flight with the pitch control law.	X									
(02)		Explain the advantages of using the CG position in the FBW system.	X									
(03)		Explain what type of flight-degraded control laws may be available in case of failure.	X									
(04)		Explain what are hard and soft protections.	X									
081 05 08 00		Trimming										
081 05 08 01		Reasons to trim										
(01)		State the reasons for using trimming devices.	X	X								
(02)		Explain the difference between a trim tab and the various balance tabs.	X	X								
081 05 08 02		Trim tabs										
(01)		Describe the working principle of a trim tab including cockpit indications.	X	X								
081 05 08 03		Stabiliser trim										
(01)		Describe the working principle of a stabiliser trim including the flight deck indications.	X	X								
(02)		Explain the advantages and disadvantages of a stabiliser trim compared to a trim tab.	X	X								
(03)		Explain the relationship between CG position, take-off trim setting, and stabiliser trim position.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain the effect of errors in the take-off stabiliser trim setting on the rotation characteristics and stick force during take-off rotation.	X	X								
(05)		Discuss the effects of jammed and runaway stabiliser.	X	X								
(06)		Explain the consequences of a jammed stabiliser during take-off, landing, and go-around.	X	X								
081 06 00 00		LIMITATIONS										
081 06 01 00		Operating limitations										
081 06 01 01		Flutter										
(01)		Describe the phenomenon of flutter and how IAS and mass distribution affects the likelihood of flutter occurrence.	X	X								
(02)		Describe the use of mass balance to alleviate the flutter problem by adjusting the mass distribution: — wing-mounted engines on pylons; — control surface mass balance.	X	X								
(03)		Explain what is the flight envelope free of flutter.	X	X								
081 06 01 02		Intentionally left blank										
(01)		<i>Intentionally left blank</i>										
081 06 01 03		Landing gear/flap operating										
(01)		Describe the reason for flap/landing gear limitations. Define 'V _{LO} '. Define 'V _{LE} '.	X	X								
(02)		Explain why there is a difference between V _{LO} and V _{LE} in the case of some aeroplane types.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Define 'V _{FE} ' and describe flap limiting speeds.	X	X								
(04)		Describe flap design features, procedures and warnings to prevent overload.	X	X								
081 06 01 04		V_{MO}, V_{NO}, and V_{NE}										
(01)	X	Define 'V _{MO} ', 'V _{NO} ', and 'V _{NE} '.	X	X								
(02)		Explain the significance of V _{MO} , V _{NO} and V _{NE} , and the differences between these airspeeds.	X	X								
(03)		Explain the hazards of flying at speeds above V _{NE} and V _{MO} .	X	X								
081 06 01 05		M_{MO}										
(01)		Define 'M _{MO} ' and state its limiting factors.	X									
081 06 02 00		Manoeuvring envelope										
081 06 02 01		Manoeuvring-load diagram										
(01)		Describe the manoeuvring-load diagram.	X	X								
(02)		Define limit and ultimate load factor, and explain what can happen if these values are exceeded.	X	X								
(03)		Define 'V _A ', 'V _B ', 'V _C ', and 'V _D '.	X	X								
(04)		Identify and explain the varying features on the V _N diagram: — load factor 'n'; — speed scale, equivalent airspeed; — equivalent airspeed envelope; — 1g stall speed; — stall boundary (refer to 081 03 01 02).	X	X								
(05)		Describe the relationship between V _{MO} or V _{NE} and V _C .	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		State all the manoeuvring load-factors limits applicable to CS-23 and CS-25 aeroplanes.	X	X								
(07)		Explain the relationship between V_A and V_S in a formula, and calculate the values.	X	X								
(08)		Explain the significance of V_A and the adverse consequences of applying full, abrupt nose-up elevator deflection when exceeding V_A .	X	X								
081 06 02 02		Factors affecting the manoeuvring-load diagram										
(01)		State the relationship of mass to load-factor limits and accelerated stall speed boundary limit.	X	X								
(02)		Calculate the change of V_A with changing mass.	X	X								
(03)		Explain why V_A loses significance at higher altitude.	X									
(04)	X	Define 'M _C ' and 'M _D '.	X									
081 06 03 00		Gust envelope										
081 06 03 01		Gust-load diagram										
(01)		Recognise a typical gust-load diagram, and state the minimum gust speeds in ft/s, m/s and kt that the aeroplane must be designed to withstand at V_B to V_C and V_D .	X	X								
(02)		Discuss considerations for the selection of V_{RA} .	X	X								
(03)		Explain the adverse effects on the aeroplane when flying in turbulence.	X	X								
081 06 03 02		Factors affecting the gust-load diagram										
(01)		Describe and explain the relationship between the gust-load factor and the following: lift-curve slope, aspect ratio, angle of sweep, altitude, wing loading, weight, wing area, equivalent airspeed (EAS), and speed of vertical gust.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		(Note: For examination purposes, the ECQB questions will not be calculation based.)										
081 07 00 00		PROPELLERS										
081 07 01 00		Conversion of engine torque to thrust										
081 07 01 01		<i>Explain conversion of aerodynamic force on a propeller blade</i>										
(01)		Explain the resolution of aerodynamic force on a propeller blade element into lift and drag or into thrust and torque.	X	X								
(02)		Describe how propeller thrust and aerodynamic torque vary with IAS.	X	X								
081 07 01 02		<i>Relevant propeller parameters</i>										
(01)		Describe the geometry of a typical propeller blade element at the reference section: — blade chord line; — propeller rotational velocity vector; — true airspeed vector; — blade angle of attack; — pitch or blade angle; — advance or helix angle. Define 'geometric pitch', 'effective pitch', and 'propeller slip'. <i>Remark: For theoretical knowledge examination purposes, the following definition is used for geometric pitch: the theoretical distance a propeller would advance in one revolution at zero blade angle of attack.</i>	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe how the terms 'fine pitch' and 'coarse pitch' can be used to express blade angle.	X	X								
081 07 01 03		Blade twist										
(01)	X	Define 'blade twist'.	X	X								
(02)		Explain why blade twist is necessary.	X	X								
081 07 01 04		Fixed pitch and variable pitch/constant speed										
(01)	X	List the different types of propellers: — fixed pitch; — adjustable pitch or variable pitch (non-governing); — variable pitch (governing)/constant speed.	X	X								
(02)		Discuss the advantages and disadvantages of fixed-pitch and constant-speed propellers.	X	X								
(03)		Discuss climb and cruise propellers.	X	X								
(04)		Explain the relationship between blade angle, blade angle of attack, and airspeed for fixed and variable pitch propellers.	X	X								
(05)		Describe and explain the forces that act on a rotating blade element in normal, feathered, windmilling, and reverse operation.	X	X								
(06)		Explain the effects of changing propeller pitch at constant IAS.	X	X								
081 07 01 05		Propeller efficiency versus speed										
(01)		Define 'propeller efficiency'.	X	X								
(02)		Explain and describe the relationship between propeller efficiency and speed (TAS) for different types of propellers.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the relationship between blade angle and thrust.	X	X								
081 07 01 06		Effects of ice on propeller										
(01)		Describe the effects and hazards of ice on a propeller.	X	X								
081 07 02 00		Engine failure										
081 07 02 01		Windmilling drag										
(01)		Describe the effects of an inoperative engine on the performance and controllability of an aeroplane: — thrust loss/drag increase; — influence on yaw moment during asymmetric power.	X	X								
081 07 02 02		Feathering										
(01)		Explain the reasons for feathering a propeller, including the effect on the yaw moment, performance and controllability.	X	X								
081 07 03 00		Design features for power absorption										
081 07 03 01		Propeller design characteristics that increase power absorption										
(01)	X	Name the propeller design characteristics that increase power absorption.	X	X								
081 07 03 02		Diameter of propeller										
(01)		Explain the reasons for restricting propeller diameter.	X	X								
081 07 03 03		Number of blades										
(01)	X	Define 'solidity'.	X	X								
(02)		Describe the advantages and disadvantages of increasing the number of blades.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 07 03 04		Propeller noise										
(01)	X	Describe how propeller noise can be minimised.	X	X								
081 07 04 00		Secondary effects of propellers										
081 07 04 01		Torque reaction										
(01)		Describe the effects of engine/propeller torque.	X	X								
(02)		Describe the following methods for counteracting engine/propeller torque: — counter-rotating propellers; — contra-rotating propellers.	X	X								
081 07 04 02		Gyroscopic precession										
(01)	X	Describe what causes gyroscopic precession.	X	X								
(02)	X	Describe the effect on the aeroplane due to the gyroscopic effect.	X	X								
081 07 04 03		Slipstream effect										
(01)		Describe the possible effects of the rotating propeller slipstream.	X	X								
081 07 04 04		Asymmetric blade effect										
(01)		Explain the asymmetric blade effect (also called P factor).	X	X								
(02)		Explain the influence of direction of rotation on the critical engine on twin-engine aeroplanes.	X	X								
081 07 04 05		Consideration of propeller effects										
(01)		Describe, given direction of propeller rotation, the propeller effects during take-off run, rotation and initial climb, and their consequence on controllability.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Describe, given the direction of propeller rotation, the propeller effects during a go-around and their consequence on controllability.	X	X								
(03)		Explain how propeller effects during go-around can be affected by: — high engine performance conditions and their effect on the VMC speeds; — loss of the critical engine; — crosswind; — high flap setting;	X	X								
081 08 00 00		FLIGHT MECHANICS										
081 08 01 00		Forces acting on an aeroplane										
081 08 01 01		<i>Straight, horizontal, steady flight</i>										
(01)	X	Describe the forces that act on an aeroplane in straight, horizontal, and steady flight.	X	X								
(02)	X	List the four forces and state where they act on.	X	X								
(03)		Explain how the four forces are balanced, including the function of the tailplane.	X	X								
081 08 01 02		<i>Straight, steady climb</i>										
(01)	X	Define ‘flight-path angle’ (γ).	X	X								
(02)		Describe the relationship between pitch attitude, γ and α for zero-wind and zero-bank conditions.	X	X								
(03)	X	Describe the forces that act on an aeroplane in a straight, steady climb.	X	X								
(04)		Name the forces parallel and perpendicular to the direction of flight. — Apply the formula relating to the parallel forces ($T = D + W \sin \gamma$).	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		— Apply the formula relating to the perpendicular forces ($L = W \cos \gamma$).										
(05)		Explain why thrust is greater than drag.	X	X								
(06)		Explain why lift is less than weight.	X	X								
(07)		Explain the formula (for small angles) that gives the relationship between γ , thrust, weight, and lift-drag ratio, and use this formula for simple calculations.	X	X								
(08)		Explain how IAS, α , and γ change in a climb performed with constant vertical speed and constant thrust setting.	X	X								
081 08 01 03		Straight, steady descent										
(01)	X	Describe the forces that act on an aeroplane in a straight, steady descent.	X	X								
(02)		Name the forces parallel and perpendicular to the direction of flight. — Apply the formula for forces parallel to the direction of flight ($T = D - W \sin \gamma$). — Apply the formula relating to the perpendicular forces ($L = W \cos \gamma$).	X	X								
(03)		Explain why lift is less than weight.	X	X								
(04)		Explain why thrust is less than drag.	X	X								
081 08 01 04		Straight, steady glide										
(01)	X	Describe the forces that act on an aeroplane in a straight, steady glide.	X	X								
(02)		Name the forces parallel and perpendicular to the direction of flight.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- Apply the formula for forces parallel to the direction of flight ($D = W \sin \gamma$). - Apply the formula for forces perpendicular to the direction of flight ($L = W \cos \gamma$).										
(03)		Describe the relationship between the glide gradient and the lift–drag ratio, and calculate glide range given: - initial height; - L–D ratio; - glide speed and wind speed.	X	X								
(04)		Define V_{MD} (speed for minimum drag) and explain the relationship between α , V_{MD} and the best lift–drag ratio.	X	X								
(05)		Explain the effect of wind component on glide angle, duration, and distance.	X	X								
(06)		Explain the effect of mass change on glide angle, duration, and distance, given that the aeroplane remains at either the same airspeed or at V_{MD} .	X	X								
(07)		Explain the effect of configuration change on glide angle and duration.	X	X								
(08)		Describe the relation between TAS, gradient of descent, and rate of descent.	X	X								
(09)		Define V_{MP} (speed for minimum power) and describe that the minimum rate of descent in the glide will be at V_{MP} , and explain the relationship of this speed to the optimum speed for minimum glide angle.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(10)		Discuss when a pilot could elect to fly for minimum glide rate of descent or minimum glide angle, and why speed stability or headwinds/tailwinds may favour a speed that is faster or slower than the optimum airspeed in still air.	X	X								
081 08 01 05		Steady, coordinated turn										
(01)		Describe the forces that act on an aeroplane in a steady, coordinated turn.	X	X								
(02)		Resolve the forces that act horizontally and vertically during a coordinated turn ($\tan\phi = \frac{V^2}{gR}$).	X	X								
(03)		Describe the difference between a coordinated and an uncoordinated turn, and describe how to correct an uncoordinated turn using turn and slip indicator or turn coordinator.	X	X								
(04)		Explain why the angle of bank is independent of mass, and that it only depends on TAS and radius of turn.	X	X								
(05)		Resolve the forces to show that for a given angle of bank the radius of turn is determined solely by airspeed ($\tan\phi = \frac{V^2}{gR}$).	X	X								
(06)		Calculate the turn radius of a steady turn given TAS and angle of bank.	X	X								
(07)		Explain the effects of bank angle on: – load factor ($LF = 1/\cos\phi$); – α; – thrust;	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		– drag.										
(08)	X	Define ‘angular velocity’.	X	X								
(09)	X	Define ‘rate of turn’ and ‘rate-1 turn’.	X	X								
(10)		Explain the influence of TAS on rate of turn at a given bank angle.	X	X								
(11)		Calculate the load factor and stall speed in a turn given angle of bank and 1g stall speed.	X	X								
(12)		Explain situations in which turn radius is relevant for safety, such as maximum speed limits on departure or arrival plates, or outbound speed categories on approach plates, and the implications/hazards of exceeding given speeds.	X	X								
(13)		Describe the hazards of excessive use of rudder to increase the rate of turn in a swept-wing aeroplane.	X	X								
081 08 02 00		Asymmetric thrust										
081 08 02 01		Jet-engined and propeller-driven aeroplanes										
(01)		Describe the effects on the aeroplane of asymmetric thrust during flight, for both jet-engined and propeller-driven aeroplanes.	X	X								
(02)		Explain critical engine, and explain, for a propeller-driven aeroplane, the effect of the direction of propeller rotation.	X	X								
(03)	X	Explain the effect of steady, asymmetric flight on a conventional (ball) slip indicator/turn indicator.	X	X								
(04)		Explain the effect of a crosswind on asymmetric flight.	X	X								
081 08 02 02		Balanced moments about the normal axis										
(01)		Explain the yaw moments about the CG.	X	X								

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(02)		Explain the change to the yaw moment caused by the effect of air density on thrust.	X	X								
(03)		Describe the changes to the yaw moment caused by engine distance from CG.	X	X								
(04)		Describe the methods to achieve directional balance following engine loss.	X	X								
081 08 02 03		Forces parallel to the lateral axis										
(01)		Explain: — the force on the vertical fin; — the fuselage side force due to sideslip (using wing-level method); — the use of bank angle to tilt the lift vector (in wing-down method).	X	X								
(02)		Explain the flight hazards at VMC: — α ; — side slip; — loads on the fin; — α on the fin.	X	X								
(03)		Explain the effect on fin α due to sideslip.	X	X								
081 08 02 04		Influence of aeroplane mass										
(01)		Explain why controllability with one-engine-inoperative is a typical problem arising from the low speeds associated with low aeroplane mass.	X	X								
081 08 02 05		Intentionally left blank										
081 08 02 06		Intentionally left blank										
081 08 02 07		Intentionally left blank										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
081 08 02 08		Minimum control speed (V_{MC})										
(01)		Define ' V_{MC} '.	X	X								
(02)		Describe how V_{MC} is determined.	X	X								
(03)		Explain the influence of the CG location.	X	X								
081 08 02 09		Minimum control speed during approach and landing (V_{MCL})										
(01)		Define ' V_{MCL} '.	X	X								
(02)		Describe how V_{MCL} is determined.	X	X								
(03)		Explain the influence of the CG location.	X	X								
081 08 02 10		Minimum control speed on the ground (V_{MCG})										
(01)		Define ' V_{MCG} '.	X	X								
(02)		Describe how V_{MCG} is determined.	X	X								
(03)		Explain the influence of the CG location.	X	X								
081 08 02 11		Influence of density										
(01)		Describe the influence of density on thrust during asymmetric flight.	X	X								
(02)		Explain why V_{MC} , V_{MCL} and V_{MCG} reduce with a reduction in thrust.	X	X								
081 08 03 00		Significant points on a polar curve										
081 08 03 01		Identify and explain										
(01)		Identify and explain the significant points on a polar curve.	X	X								

SUBJECT 082 – PRINCIPLES OF FLIGHT – HELICOPTERS

ED Decision 2020/018/R

(1) VOCABULARY OF MECHANICS

Speed is a scalar quantity; it has only magnitude.

Velocity is a vector quantity with magnitude and direction.

The velocity of a point on a rotor blade, when rotating around an axis, is the 'linear' or 'tangential' velocity which can be expressed in revolutions per minute (rpm).

Density is the mass of the fluid per unit volume (kg/m³) in the international system of units of measurement (SI (*Système International*)).

(2) AERONAUTICAL DEFINITIONS

A rotor blade is a high-aspect ratio aerofoil attached by its root to the rotor hub with hinges or flexible elements.

A blade element is a spanwise slice of the blade, so thin that the aerodynamic forces involved may be assumed not to vary. The forces produce lift (L), drag (D), and a pitching moment. Such a cross section has a contour, a leading and trailing edge, a chord line, a mean camber line, a maximum thickness or depth, and a thickness-to-chord ratio.

The centre of pressure (CP) is defined as the point on the chord line where the resultant of all aerodynamic forces acts.

The planform is the shape of a blade as seen from above.

The pitch angle (of a blade or an element) is the angle between the chord line and the plane of rotation.

The blade is not twisted when the pitch angle is constant from root to tip.

A blade is twisted when the pitch angle of its elements' sections varies with their distance from the root (in other words, the chord lines of the elements involved are not parallel). Washout exists when the pitch angle decreases towards the blade tip.

The vector sum of the undisturbed upstream velocity (i.e. that found in the plane of rotation of the blades) and the induced velocity is the relative airflow.

The angle between the relative airflow and the chord line of a blade element is the angle of attack (α).

Lift is the component of the aerodynamic force on a blade element that is perpendicular to the relative airflow.

Profile drag is the component of the aerodynamic force on a blade element that is parallel to the plane of rotation. Induced drag is the component of the aerodynamic force on a blade element that is parallel to the relative airflow.

Profile drag consists of pressure forces and skin friction acting on the surface of the blade element. The component of profile drag that arises from pressure forces (between the leading and trailing edges) is pressure or form drag. The component of profile drag due to shear forces over the surface is skin friction.

The total rotor thrust is the vertical upwards force from the rotor disc as a whole, as the sum of all the blade thrusts. This term has been reinstated because there is already the term 'rotor thrust' that is used to denote the thrust along the axis of rotation that acts directly opposite the weight of the helicopter in a blade element.

(3) HELICOPTER CHARACTERISTICS

Disc loading is the mass (M) of the helicopter divided by the area of the disc.

Blade loading is the mass divided by the total planform area of the blades.

The area of a rectangular blade is given by the chord multiplied by the blade tip radius. For tapered blades, the mean geometric chord is taken as an approximately equivalent chord.

Rotor solidity is the ratio of the total blade area to the disc area.

(4) PLANES, AXES AND REFERENCE SYSTEMS OF THE ROTOR

- Shaft axis: The physical axis of the rotor shaft (mast).
- Hub plane: A plane perpendicular to the shaft axis through the centre of the hub.
- Tip path plane: The plane traced out by the blade tips.
- Virtual rotation axis: The axis through the centre of the hub and perpendicular to the tip path plane.
- Rotor disc: The disc traced out by the blade tips in the tip path plane.
- Plane of rotation: The plane parallel to the tip path plane that acts through the hub centre.

(5) ANGLES OF THE BLADES, INDUCED VELOCITY

- Pitch angle of a blade element: The angle between the chord line of the element and its plane of rotation, sometimes called 'local pitch angle'.
- Blade pitch angle: Taken to be equivalent to the pitch angle of the blade element found at 75 % of the blade radius.

- Flapping angle: The angle between the longitudinal axis of the blade and the hub plane.
- Coning angle: The angle between the longitudinal axis of the blade and the tip path plane. Induced velocity is that induced by the engine power perpendicular to the plane of rotation.

Aerodynamic forces on the blades and the rotor

The thrust from a blade (blade thrust) is the sum of the thrusts from each blade element.

The sum of the thrusts from all blades is the (total) rotor thrust acting perpendicular to the tip path in the direction of the virtual rotation axis.

The result of the induced drag forces on all the blade elements of all blades is a torque on the shaft which, multiplied by the angular velocity of the blade, gives the required induced power.

The result of the profile drag forces is a torque on the shaft which, multiplied by the angular velocity of the blade, gives the required profile power.

(6) TYPES OF ROTOR HUBS

There are basically four types of rotor hubs in use:

1. Teetering rotor or seesaw rotor: The two blades are connected together; the 'hinge' is on the shaft axis, and the head is underslung. A variation is the gimballed hub; the blades and the hub are attached to the rotor shaft by means of a gimbal or universal joint (Bell 47). It is sometimes called semi-rigid because there is no movement of the blade in a drag-wise sense.
2. Fully articulated rotor: There are more than two rotor blades and each has a flapping hinge, a lead-lag (drag) hinge, and a feathering hinge or bearing.
3. Hingeless rotor: There are no flapping or dragging hinges. They are replaced by flexible elements (virtual hinges) at some part of the blade radius which allow such movements. A feathering bearing allows feathering of the blade.
4. Bearingless rotor: There are no hinges or rotating bearings. Flapping and dragging movements are obtained with flexible elements called elastomeric hinges. Feathering is obtained by twisting the element.

When referring to their equipment, Airbus call this a 'semi-articulated head' (ref.: their training material).

Two remarks:

1. Hinge offset and equivalent hinge offset

The hinge offset is the distance between the shaft axis and the axis of the hinge. Hingeless and bearingless rotors have an equivalent hinge offset.

2. Elastomeric hinges

This bearing consists of alternate layers of elastomer and metal. The flexibility of the elastomer allows flapping, dragging and feathering.

(7) DRAG AND POWERS

Induced power is that required to generate the induced velocity in the rotor disc for the production of lift. For any given thrust, induced power is minimum when the induced velocity is uniform over the rotor disc. This can be approximated by using washout and ensuring that the blades are in track (a truly uniform velocity cannot be obtained).

Rotor profile drag results from those components acting in the opposite direction to the blade velocities (i.e. the sum of all the profile drags from each blade element). The power required to overcome it is rotor profile power (the sum of the powers required to overcome the torque).

Parasite drag is the drag from the helicopter fuselage including that from the rotor hub and all external equipment such as wheels, the winch, external loads, etc. (any drag from the tail rotor is included, but not from the rotor blades, which produce profile drag). The power to overcome this drag is parasite power.

In level flight at constant speed, induced power, rotor profile power and parasite power are summed to give the total power required to drive the main rotor.

Induced power and profile power for the tail rotor are summed to give the power required to drive the tail rotor.

The power required to drive auxiliary services, such as oil pumps and electrical generators, is called accessory or ancillary power. It includes the power required to overcome mechanical friction in transmissions.

The total power required in level flight at constant speed is the sum of all the above.

When transitioning from the hover, the power required decreases as speed increases. This is called translational lift.

The term limited power means that the total power required to hover out of ground effect (HOGE) is greater than the available power.

(8) PHASE ANGLE IN FLAPPING MOVEMENT OF THE BLADE

The movement of the cyclic control tilts the rotor disc in the direction of the intended movement of the helicopter.

For teetering heads, the flapping response is 90° later than the applied cyclic control movement (less than 90° for rotors with offset hinges).

The pitch mechanism consists of the swash plate, and for each blade the pitch mechanism consists of a pitch link attached to the swash plate and a pitch horn attached to the blade.

(9) AXES THROUGH THE CENTRE OF THE HELICOPTER

Longitudinal axis or roll axis: A straight line through the centre of gravity (CG) of the helicopter from the nose to the tail about which the helicopter can roll left or right.

Lateral axis, transverse axis or pitch axis: A straight line through the CG of the helicopter about which the helicopter can pitch its nose up or down (this axis is also perpendicular to the reference plane of the aircraft, which is the plane either side of which the components that constitute the major part of the aircraft are symmetrically disposed in the port and starboard sense).

Normal axis or yaw axis: A straight line perpendicular to the plane defined by the longitudinal and lateral axes and about which the helicopter can yaw.

Note that the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The professional pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
080 00 00 00		PRINCIPLES OF FLIGHT										
082 00 00 00		PRINCIPLES OF FLIGHT — HELICOPTERS										
082 01 00 00		SUBSONIC AERODYNAMICS										
082 01 01 00		Basic concepts, laws and definitions										
082 01 01 01		<i>International system of units of measurement (SI) and conversion of SI units</i>										
(01)	X	List the fundamental quantities and units in SI, such as mass (kg), length (m), time (s).			X	X	X					
(02)	X	Be able to convert imperial units to SI units and vice versa.			X	X	X					
082 01 01 02		<i>Definitions and basic concepts of air</i>										
(01)	X	Describe air temperature and pressure as functions of height.			X	X	X					
(02)	X	Define the International Standard Atmosphere (ISA).			X	X	X					
(03)	X	Define air density, and explain the relationship between air density, pressure, and temperature.			X	X	X					
(04)	X	Explain the influence of moisture content on air density.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(05)	X	Define pressure altitude and air density altitude.			X	X	X					
082 01 01 03		Newton's laws										
(01)	X	State and interpret Newton's three laws of motion.			X	X	X					
(02)	X	Distinguish between mass and weight, and their units.			X	X	X					
082 01 01 04		Basic concepts of airflow										
(01)	X	Describe steady and unsteady airflow.			X	X	X					
(02)	X	Define 'streamline' and 'stream tube'.			X	X	X					
(03)	X	Explain the principle of the continuity equation or the conservation of mass.			X	X	X					
(04)	X	Describe the mass flow rate through a stream tube section.			X	X	X					
(05)		State Bernoulli's equation and use it to explain and define the relationship between static, dynamic and total pressure.			X	X	X					
(06)		Define the stagnation point in the flow around an aerofoil, and explain the pressure obtained at the stagnation point.			X	X	X					
(07)		Use the pitot system to explain the measurement of airspeed (no compressibility effects).			X	X	X					
(08)		Define 'TAS', 'IAS', and 'CAS'.			X	X	X					
(09)	X	Define two-dimensional airflow and its relationship to an aerofoil of infinite span (i.e. no blade tip vortices and, therefore, no induced drag). Explain the difference between two- and three-dimensional airflows.			X	X	X					
(10)	X	Explain that viscosity is a feature of any fluid (gas or liquid).			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(11)		Explain the tangential friction between air and the surface of an aerofoil, and the development of a boundary layer.			X	X	X					
(12)		Describe laminar and turbulent boundary layers and the transition from laminar to turbulent. Show the influence of the roughness of the surface on the position of the transition point.			X	X	X					
082 01 02 00		Two-dimensional airflow										
082 01 02 01		Aerofoil section geometry										
(01)	X	Define the terms: 'aerofoil section', 'aerofoil element', 'chord line', 'chord', 'thickness', 'thickness-to-chord ratio', 'camber line', 'camber', and 'leading-edge radius'.			X	X	X					
(02)		Describe symmetrical and asymmetrical aerofoil sections.			X	X	X					
082 01 02 02		Aerodynamic forces on aerofoil elements										
(01)		Define the angle of attack (α).			X	X	X					
(02)		Describe: — the resultant force from the pressure distribution and the friction at the element; — the resultant force from the boundary layers and the velocities in the wake; and — the loss of momentum due to friction forces.			X	X	X					
(03)		Resolve the aerodynamic force into the components of lift (L) and drag (D).			X	X	X					
(04)		Define the lift coefficient (C_L) and the drag coefficient (C_D).			X	X	X					
(05)		Show that the C_L is a function of the α .			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(06)		Explain how drag is caused by pressure forces on the surfaces of an aerofoil and by friction in the boundary layers. Define the term 'profile drag'.			X	X	X					
(07)		Define the L–D ratio.			X	X	X					
(08)		Use the lift and drag equations to show the influence of speed and density on lift and drag for a given α .			X	X	X					
(09)		Define the action line of the aerodynamic force and the CP.			X	X	X					
(10)		Know that symmetrical aerofoils have a CP that is approximately a quarter chord behind the leading edge.			X	X	X					
082 01 02 03		<i>Stall</i>										
(01)		Explain the boundary layer separation when α increases beyond the onset of stall and the decrease of lift and the increase of drag. Define the 'separation point'.			X	X	X					
082 01 02 04		<i>Disturbances due to profile contamination</i>										
(01)		Explain ice contamination, the modification of the section profile and surfaces due to ice and snow, the influence on L and D and the L–D ratio, the influence on α (at stall onset), and the effect of the increase in weight.			X	X	X					
(02)		Explain the effect of erosion by heavy rain on the blade and subsequent increase in profile drag.			X	X	X					
082 01 03 00		Three-dimensional airflow around a blade										
082 01 03 01		<i>The blade</i>										
(01)		Describe the various blade planforms.			X	X	X					
(02)		Define aspect ratio and blade twist.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
082 01 03 02		Airflow pattern and influence on lift (L)										
(01)		Explain the spanwise flow around a blade and the appearance of blade tip vortices which are a loss of energy.			X	X	X					
(02)		Show that the strength of the vortices increases as α and L increase.			X	X	X					
(03)		Show that downwash causes vortices.			X	X	X					
(04)		Define the relative airflow as the resultant of the undisturbed air velocity and induced velocity, and define α .			X	X	X					
(05)		Explain the spanwise L distribution and the way in which it can be modified by twist (washout).			X	X	X					
082 01 03 03		Induced drag										
(01)		Explain induced drag and the influence of α and aspect ratio.			X	X	X					
082 01 03 04		The airflow around the fuselage										
(01)		Describe the fuselage and the external components that cause (parasite) drag, the airflow around the fuselage, and the influence of the pitch angle of the fuselage. Describe fuselage shapes that minimise drag.			X	X	X					
(02)		Define profile drag as the sum of pressure (form) drag and skin friction drag.			X	X	X					
(03)		Define 'interference drag'.			X	X	X					
(04)		Know the drag formula.			X	X	X					
082 02 00 00		TRANSONIC AERODYNAMICS and COMPRESSIBILITY EFFECTS										
082 02 01 00		Airflow speeds and velocities										
082 02 01 01		Speeds and Mach number										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define the speed of sound in air.			X	X	X					
(02)		State that the speed of sound is proportional to the square root of the absolute temperature (in Kelvins).			X	X	X					
(03)		Explain the variation in the speed of sound with altitude.			X	X	X					
(04)		Define Mach number.			X	X	X					
(05)		Explain the meaning of incompressibility and compressibility of air; relate this to the value of the Mach number.			X	X	X					
(06)		Define high subsonic, transonic and supersonic flows in relation to the value of the Mach number.			X	X	X					
082 02 01 02		Shock waves										
(01)		Describe shock waves in a supersonic flow and the changes in pressure and speed.			X	X	X					
(02)		Describe the appearance of local supersonic flows on the surfaces of a blade.			X	X	X					
082 02 01 03		Influence of aerofoil section and blade planform										
(01)		Explain the different shapes that allow higher Mach numbers without generating a shock wave on the upper surface, such as: – reducing the section thickness-to-chord ratio; – a planform with a sweep angle.			X	X	X					
082 03 00 00		ROTORCRAFT TYPES										
082 03 01 00		Rotorcraft										
082 03 01 01		Rotorcraft types										
(01)		Explain the difference between an autogyro and a helicopter.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
082 03 02 00		Helicopters										
082 03 02 01		Helicopter configurations										
(01)		Describe (briefly) the single-main-rotor helicopter and other configurations: tandem, coaxial, side-by-side, synchrocopter (with intermeshing blades), the compound helicopter and tilt rotor.			X	X	X					
082 03 02 02		The helicopter, characteristics and associated terminology										
(01)		Mention the tail rotor, the Fenestron, and the no tail rotor (NOTAR).			X	X	X					
(02)		Define the rotor disc area and the blade area.			X	X	X					
(03)		Describe the teetering rotor with its hinge axis on the shaft axis, and rotors with more than two blades with offset hinge axes.			X	X	X					
(04)		Define the fuselage centre line and the three axes: roll, pitch, and normal (yaw).			X	X	X					
(05)		Define gross weight and gross mass (and the units involved), disc and blade loading.			X	X	X					
082 04 00 00		MAIN-ROTOR AERODYNAMICS										
082 04 01 00		Hover flight outside ground effect										
082 04 01 01		Airflow through the rotor disc and around the blades										
(01)	X	Based on Newton's second law (momentum), explain that the upward vertical force from the disc, i.e. the rotor thrust, is the result of vertical downward velocities inside the rotor disc.			X	X	X					
(02)		Explain why the production of the induced flow requires power applied to the shaft, i.e. induced power. Induced power is least if the induced			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		velocities have the same value on the whole disc (i.e. there is uniformity of flow over the disc).										
(03)		Explain why vertical rotor thrust must be higher than the weight of the helicopter because of the vertical drag on the fuselage.			X	X	X					
(04)		Define the pitch angle and the α of a blade element.			X	X	X					
(05)		Explain L and D relating to a blade element (including induced and profile drag).			X	X	X					
(06)		Explain the necessity for collective pitch angle changes, the influence on the α and rotor thrust, and the need for blade feathering.			X	X	X					
(07)		Describe the different blade shapes (as viewed from above).			X	X	X					
(08)		Explain how profile drag on the blade elements generates a torque on the main shaft, and define the resulting rotor profile power.			X	X	X					
(09)		Explain the influence of air density on the required powers.			X	X	X					
082 04 01 02		Anti-torque force and tail rotor										
(01)		Using Newton's third law (motion), explain the need for tail-rotor thrust, the required value being proportional to main-rotor torque. Show that tail-rotor power is proportional to tail-rotor thrust.			X	X	X					
(02)		Explain the necessity for feathering of the tail-rotor blades and their control by the yaw pedals, and the maximum and minimum values of the pitch angles of the blades.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
082 04 01 03		Total power required and hover outside ground effect (HOGE)										
(01)		Define ancillary equipment and its power requirement.			X	X	X					
(02)		Define the total power required.			X	X	X					
(03)	X	Describe the influence of ambient pressure, temperature and moisture on the required power.			X	X	X					
082 04 02 00		Vertical climb										
082 04 02 01		Relative airflow and angles of attack (α)										
(01)	X	Describe the dependence of the vertical climb speed on the opposite vertical air velocity relative to the rotor disk.			X	X	X					
(02)		Explain how α is controlled by the collective pitch angle control.			X	X	X					
082 04 02 02		Power and vertical speed										
(01)		Define total main-rotor power as the sum of parasite power, induced power, climb power, and rotor profile power.			X	X	X					
(02)		Explain why the total main-rotor power required increases when the rate of climb increases.			X	X	X					
082 04 03 00		Forward flight										
082 04 03 01		Airflow and forces in uniform inflow distribution										
(01)		Explain the assumption of a uniform inflow distribution on the rotor disc.			X	X	X					
(02)		Show the upstream air velocities relative to the blade elements and the different effects on the advancing and retreating blades. Define the area of reverse flow. Explain the influence of forward speed on the circumferential speed of the blade tip.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Assuming constant pitch angles and rigid blade attachments, explain the roll moment from the asymmetric distribution of L.			X	X	X					
(04)		Show that through cyclic feathering this imbalance could be eliminated by a low α (accomplished by a low pitch angle) on the advancing blade, and a high α (accomplished by a high pitch angle) on the retreating blade.			X	X	X					
(05)		Describe the high air velocity at the advancing blade tip and the compressibility effects which limit maximum speed.			X	X	X					
(06)		Describe the low air velocity on the retreating blade tip resulting from the difference between the circumferential speed and forward speed, the need for high α , and the onset of stall.			X	X	X					
(07)		Define the blade tip speed ratio.			X	X	X					
(08)		Explain the total rotor thrust that is perpendicular to the rotor disc and the need for tilting the thrust vector forward.			X	X	X					
(09)		Explain the conditions of equilibrium in steady straight and level flight.			X	X	X					
082 04 03 02		The flare (powered flight)										
(01)		Explain the flare in powered flight, the rearward tilt of the rotor disc and the thrust vector. Show the horizontal thrust component that is in the opposite direction to forward velocity.			X	X	X					
(02)		State the increase in thrust due to the upward inflow, and show the modifications in the α .			X	X	X					
(03)		Explain the increase in rotor rpm for a non-governed rotor.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
082 04 03 03		Non-uniform inflow distribution in relation to inflow roll										
(01)		Describe the inflow distribution which modifies α and L especially on the advancing and retreating blades.			X	X	X					
082 04 03 04		Power and maximum speed										
(01)		Explain that the induced velocities and power values decrease as the speed of the helicopter increases.			X	X	X					
(02)		Define profile drag and profile power, and the increase in their values with the speed of the helicopter.			X	X	X					
(03)		Define parasite drag and parasite power, and the increase in their values with the speed of the helicopter.			X	X	X					
(04)		Define total drag and its increase with the speed of the helicopter.			X	X	X					
(05)		Describe the power required for the tail rotor and the power required by ancillary equipment.			X	X	X					
(06)		Define the total power requirement as a sum of the above partial powers, and explain how it varies with the speed of the helicopter.			X	X	X					
(07)		Explain the influence of helicopter mass, air density, and additional external equipment on the partial powers and the total power required.			X	X	X					
(08)		Describe translational lift and show the decrease in required total power as the helicopter increases its speed from the hover.			X	X	X					
082 04 04 00		Hover and forward flight in ground effect										
082 04 04 01		Airflow in ground effect, downwash										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain how the vicinity of the ground changes the downward flow pattern and the consequences on lift (thrust) at constant rotor power. Show that ground effect depends on the height of the rotor above the ground and the rotor diameter. Show the required rotor power at constant all-up mass (AUM) as a function of height above the ground. Describe the influence of forward speed.			X	X	X					
082 04 05 00		Vertical descent										
082 04 05 01		Vertical descent, power on										
(01)		Describe the airflow around the rotor disc in a trouble-free vertical descent, power on, the airflow opposing the helicopter's velocity, the relative airflow, and α .			X	X	X					
(02)		Explain the vortex-ring state, also known as settling with power. State the approximate vertical descent speeds that allow the formation of vortex ring, related to the values of the induced velocities.			X	X	X					
(03)		Describe the airflow relative to the blades, the root stall, the loss of lift at the blade tip, and the turbulence. Show the effect of raising the lever and describe the effects on the controls.			X	X	X					
082 04 05 02		Autorotation										
(01)		State the need for early recognition and for a quick initiation of recovery. Describe the recovery actions.			X	X	X					
(02)		Explain that the collective lever must be lowered quickly enough to avoid a rapid decay of rotor rpm due to drag on the blades, and explain the influence of rotational inertia of the rotor on the rate of decay.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Show the induced flow through the rotor disc, the rotational velocity and relative airflow, the inflow and inflow angles.			X	X	X					
(04)		Show how the aerodynamic forces on the blade elements vary from root to tip and distinguish three zones: the inner stalled region, the middle driving region, and the driven region.			X	X	X					
(05)		Explain the control of the rotor rpm with collective pitch.			X	X	X					
(06)		Show the need for negative tail-rotor thrust with yaw control.			X	X	X					
(07)		Explain the final increase in rotor thrust caused by raising the collective pitch to decrease the vertical descent speed and the decay in rotor rpm.			X	X	X					
082 04 06 00		Forward flight — autorotation										
082 04 06 01		Airflow at the rotor disc										
(01)		Explain the factors that affect inflow angle and α , the autorotative power distribution, and the dissymmetry over the rotor disc in forward flight.			X	X	X					
082 04 06 02		Flight and landing										
(01)		Show the effect of forward speed on the vertical descent speed.			X	X	X					
(02)		Explain the effects of gross weight, rotor rpm, and altitude (density) on endurance and range.			X	X	X					
(03)		Explain the manoeuvres for turning and touchdown.			X	X	X					
(04)		Explain the height–velocity curves.			X	X	X					
082 05 00 00		MAIN-ROTOR MECHANICS										
082 05 01 00		Flapping of the blade in hover										
082 05 01 01		Intentionally left blank										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
082 05 01 02		Centrifugal turning moment (CTM)										
(01)		Describe the centrifugal forces on the mass elements of a blade with pitch applied and the components of those forces. Show how the forces generate a moment that tries to reduce the blade pitch angle.			X	X	X					
(02)		Explain the methods of counteracting CTM with hydraulics, bias springs, and balance masses.			X	X	X					
082 05 01 03		Coning angle in the hover										
(01)		Define the tip path plane and the coning angle.			X	X	X					
(02)		Show how the equilibrium of the moments about the flapping hinge of lift (thrust) and of the centrifugal force determine the coning angle of the blade (the blade mass being negligible).			X	X	X					
(03)		Justify the lower limit of rotor rpm.			X	X	X					
(04)		Explain the effect of the mass of a blade on the tip path and the tracking.			X	X	X					
082 05 02 00		Flapping angles of the blade in forward flight										
082 05 02 01		Forces on the blade in forward flight without cyclic feathering										
(01)		Assume rigid attachments of the blade to the hub and show the periodic lift, moment and stresses on the attachment, the ensuing metal fatigue, the roll moment on the helicopter, and justify the necessity for a flapping hinge.			X	X	X					
(02)		Assume no cyclic pitch and describe the lift on the advancing and retreating blades.			X	X	X					
(03)		State the azimuthal phase lag (90° or less) between the input (applied pitch) and the output (flapping			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		angle). Explain flapback (the rearward tilting of the tip path plane and total rotor thrust).										
082 05 02 02		<i>Cyclic pitch (feathering) in forward flight</i>										
(01)		Show that in order to assume and maintain forward flight, the total rotor thrust vector must obtain a forward component by tilting the tip path plane.			X	X	X					
(02)		Show how the applied cyclic pitch modifies the lift on the advancing and retreating blades and produces the required forward tilting of the tip path plane and the total rotor thrust.			X	X	X					
(03)		Show the cone described by the blades and define the virtual axis of rotation. Define the plane of rotation.			X	X	X					
(04)		Define the reference system in which the movements are defined: the shaft axis and the hub plane.			X	X	X					
(05)		Describe the swash plates, the pitch links and horns. Explain how the collective lever moves the non-rotating swash plate up or down the shaft axis.			X	X	X					
(06)		Describe the mechanism by which the desired cyclic blade pitch can be produced by tilting the swash plate with the cyclic stick.			X	X	X					
(07)		Explain the translational lift effect when the speed increases.			X	X	X					
(08)	X	Justify the increase of the tilt angle of the thrust vector and of the disc in order to increase the speed.			X	X	X					
082 05 03 00		Blade-lag motion in forward flight										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
082 05 03 01		Forces on the blade in the disc plane (tip path plane) in forward flight										
(01)		Explain the Coriolis force due to flapping, the resulting periodic moments in the hub plane, and the resulting periodic stresses which make lead-lag hinges necessary to avoid material fatigue.			X	X	X					
(02)		Describe the profile drag forces on the blade elements and the periodic variation of these forces.			X	X	X					
082 05 03 02		Intentionally left blank										
082 05 03 03		Ground resonance										
(01)		Explain the movement of the CG of the blades due to lead-lag movements in the multi-bladed rotor.			X	X	X					
(02)		Show the effect on the fuselage and the danger of resonance between this force and the fuselage and undercarriage when the gear touches the ground.			X	X	X					
082 05 04 00		Rotor systems										
082 05 04 01		See-saw or teetering rotor										
(01)		Explain that a teetering rotor is prone to mast bumping in low-G situations, and that it is difficult to counteract because there is no lift force to provide sideways movement.			X	X	X					
082 05 04 02		Intentionally left blank										
082 05 04 03		Hingeless rotor, bearingless rotor										
(01)		Show the forces on the flapping hinges with a large offset (virtual hinge) and the resulting moments, and compare them with other rotor systems.			X	X	X					
082 05 05 00		Blade sailing										
082 05 05 01		Blade sailing and causes										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define blade sailing, the influence of low rotor rpm and of a headwind.			X	X	X					
082 05 05 02		Minimising the danger										
(01)		Describe actions that minimise danger and the demonstrated wind envelope for engaging and disengaging rotors.			X	X	X					
082 05 05 03		Droop stops										
(01)		Explain the purpose of droop stops, and their retraction.			X	X	X					
082 05 06 00		Vibrations due to main rotor										
082 05 06 01		Intentionally left blank										
082 05 06 02		Intentionally left blank										
082 06 00 00		TAIL ROTORS										
082 06 01 00		Conventional tail rotor										
082 06 01 01		Intentionally left blank										
082 06 01 02		Tail-rotor aerodynamics										
(01)		Explain the airflow around the blades in the hover and in forward flight, and the effects of the tip speeds on noise production and compressibility.			X	X	X					
(02)		Explain the effect of wind on tail-rotor aerodynamics and thrust in the hover, and any problems.			X	X	X					
(03)		Explain tail-rotor thrust and the control through pitch alterations (feathering).			X	X	X					
(04)		Explain tail-rotor flapback, and the effects of Delta 3.			X	X	X					
(05)		Describe the roll moment and drift as side effects of the tail rotor.			X	X	X					
(06)		Explain the effects of tail-rotor failure.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(07)		Explain the loss of tail-rotor effectiveness (LTE), tail-rotor vortex-ring state, causes, crosswind, and yaw speed.			X	X	X					
082 06 01 03		Strakes on the tail boom										
(01)		Describe the strake and explain its function.			X	X	X					
082 07 00 00		EQUILIBRIUM, STABILITY AND CONTROL										
082 07 01 00		Equilibrium and helicopter attitudes										
082 07 01 01		Hover										
(01)		Explain why the vector sum of forces and moments must be zero in any acceleration-free situation.			X	X	X					
(02)		Indicate the forces and the moments about the lateral axis in a steady hover.			X	X	X					
(03)		Indicate the forces and the moments about the longitudinal axis in a steady hover.			X	X	X					
(04)		Deduce how the roll angle in a steady hover without wind results from the moments about the longitudinal axis.			X	X	X					
(05)		Explain how the cyclic is used to equalise moments about the lateral axis in a steady hover.			X	X	X					
(06)		Explain the consequence of the cyclic stick reaching its forward or aft limit during an attempt to take off to the hover.			X	X	X					
(07)		Explain the influence of density altitude on the equilibrium of forces and moments in a steady hover.			X	X	X					
082 07 01 02		Forward flight										
(01)		Explain why the vector sum of forces and of moments must be zero in unaccelerated flight.			X	X	X					
(02)		Indicate the forces and the moments about the lateral axis in steady straight and level flight.			X	X	X					

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(03)		Explain the influence of AUM on the forces and moments about the lateral axis in forward flight.			X	X	X					
(04)		Explain the influence of the CG position on the forces and moments about the lateral axis in forward flight.			X	X	X					
(05)		Explain the role of the cyclic stick position in creating equilibrium of forces and moments about the lateral axis in forward flight.			X	X	X					
(06)		Explain how forward speed influences the fuselage attitude.			X	X	X					
(07)		Describe and explain the inflow roll effect.			X	X	X					
082 07 02 00		Stability										
082 07 02 01		Static longitudinal, roll and directional stability										
(01)		Define static stability; give an example of static stability and of static instability.			X	X	X					
(02)		Explain the contribution of the main rotor to speed stability.			X	X	X					
(03)		Describe the influence of the horizontal stabiliser on static longitudinal stability.			X	X	X					
(04)		Explain the effect of hinge offset on static stability.			X	X	X					
(05)		Describe the influence of the tail rotor on static directional stability.			X	X	X					
(06)		Describe the influence of the vertical stabiliser on static directional stability.			X	X	X					
(07)		Explain the influence of the main rotor on static roll stability.			X	X	X					
(08)		Describe the influence of the longitudinal position of the CG on static longitudinal stability.			X	X	X					
082 07 02 02		Static stability in the hover										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Describe the initial movements of a hovering helicopter after the occurrence of a horizontal gust.			X	X	X					
082 07 02 03		Dynamic stability										
(01)		Define dynamic stability; give an example of dynamic stability and of dynamic instability.			X	X	X					
(02)		Explain why static stability is a precondition for dynamic stability.			X	X	X					
082 07 02 04		Longitudinal stability										
(01)		Explain the individual contributions of α and speed stability together with the stabiliser and fuselage to dynamic longitudinal stability.			X	X	X					
082 07 02 05		Roll stability and directional stability										
(01)		Know that a large static roll stability together with a small directional stability may lead to a Dutch roll.			X	X	X					
082 07 03 00		Control										
082 07 03 01		Manoeuvre stability										
(01)		Explain how helicopter control can be limited because of available stick travel.			X	X	X					
(02)		Explain how the CG position influences the remaining stick travel.			X	X	X					
082 07 03 02		Control power										
(01)		Explain the meaning of the control moment.			X	X	X					
(02)		Explain the importance of the CG position on the control moment.			X	X	X					
(03)		Explain the influence of hinge offset on controllability.			X	X	X					
082 07 03 03		Static and dynamic rollover										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Explain the mechanism which causes dynamic rollover.			X	X	X					
(02)		Explain the required pilot action when dynamic rollover is starting to develop.			X	X	X					
082 08 00 00		HELICOPTER FLIGHT MECHANICS										
082 08 01 00		Flight limits										
082 08 01 01		<i>Hover and vertical flight</i>										
(01)		Show the power required for HOGE and HIGE, and the power available.			X	X	X					
(02)		Explain the effects of AUM, ambient temperature and pressure, density altitude, and moisture.			X	X	X					
(03)		Describe the rate of climb in a vertical flight.			X	X	X					
082 08 01 02		<i>Forward flight</i>										
(01)		Compare the power required and the power available as a function of speed in straight and level flight.			X	X	X					
(02)		Define the maximum speed limited by power and the value relative to VNE and VNO.			X	X	X					
(03)		Use the power graph to determine the speeds of maximum rate of climb and the maximum angle of climb.			X	X	X					
(04)		Use the power graph to define true airspeed (TAS) for maximum range and maximum endurance, and consider the case of piston engine and turbine engine. Explain the effects of tailwind or headwind on the speed for maximum range.			X	X	X					
(05)		Explain the effects of AUM, pressure and temperature, density altitude, and humidity.			X	X	X					
082 08 01 03		<i>Manoeuvring</i>										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CB-IR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define the load factor, the radius, and the rate of turn.			X	X	X					
(02)		Explain the relationship between the angle of bank, the airspeed and the radius of turn, and between the angle of bank and the load factor.			X	X	X					
(03)		Explain the influence of AUM, pressure and temperature, density altitude, and humidity.			X	X	X					
082 08 02 00		Special conditions										
082 08 02 01		Operating with limited power										
(01)		Explain operations with limited power, use the power graph to show the limitations on vertical and level flight, and describe power checks and procedures for take-off and landing.			X	X	X					
(02)		Describe manoeuvres with limited power.			X	X	X					
082 08 02 02		Overpitch, overtorque										
(01)		Describe overpitching and show the consequences.			X	X	X					
(02)		Describe situations likely to lead to overpitching.			X	X	X					
(03)		Describe overtorquing and show the consequences.			X	X	X					
(04)		Describe situations likely to lead to overtorquing.			X	X	X					

SUBJECT 090 – COMMUNICATIONS

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Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
090 00 00 00		COMMUNICATIONS										
090 01 00 00		CONCEPTS										
090 01 01 00		Associated terms										
090 01 01 01		<i>Meanings and significance</i>										
(01)		Define commonly used air traffic services (ATS) terms for stations.	X	X	X	X	X	X	X			
(02)		Define commonly used ATS terms for communication methods.	X	X	X	X	X	X	X			
(03)		Recognise the terms used in conjunction with the approach and holding procedures.	X	X	X	X	X	X	X	3		
090 01 01 02		<i>Air traffic services (ATS) abbreviations</i>										
(01)		Define commonly used ATS abbreviations: – flight conditions; – airspace; – services; – time; – VFR-related terms; – IFR-related terms; – miscellaneous.	X	X	X	X	X	X	X	3		
090 01 01 03		<i>Q-code groups commonly used in radiotelephony (RT) air-ground communications</i>										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		Define Q-code groups commonly used in RT air-ground communications: – pressure settings; – directions and bearings.	X	X	X	X	X	X	X	3		
(02)		State the procedure for obtaining bearing information in flight.	X	X	X	X	X	X	X	3		
090 01 01 04		Categories of messages										
(01)		Identify to which category of messages a type of message belongs and identify the associated priority indicator.	X	X	X	X	X	X	X	3	2	
090 02 00 00		GENERAL OPERATING PROCEDURES										
090 02 01 00		Transmission standards										
090 02 01 01		Transmission of letters										
(01)		Know the phonetic alphabet used in RT.	X	X	X	X	X	X	X		3	
(02)		Identify the circumstances when words should be spelt out.	X	X	X	X	X	X	X		3	
090 02 01 02		Transmission of numbers										
(01)		Describe the method of transmission of numbers: – pronunciation; – single digits, whole hundreds and whole thousands; – state how numbers are transmitted in different circumstances.	X	X	X	X	X	X	X		3	
090 02 01 03		Transmission of time										
(01)		Describe the ways of transmitting time: – the standard time reference is the Coordinated Universal Time (UTC);	X	X	X	X	X	X	X		3	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		– using only minutes, or minutes and hours, when required.										
(02)		Describe the different ways in which time is to be transmitted.	X	X	X	X	X	X	X			
090 02 01 04		Transmission techniques										
(01)	X	Explain the techniques used for making good RT transmissions.	X	X	X	X	X	X	X			
090 02 01 05		Standard words and phrases										
(01)		Define the meaning of standard words and phrases.	X	X	X	X	X	X	X		3	
(02)		Recognise, describe and use the correct standard phraseology for each phase of a VFR flight (consider communication with each type of aeronautical station): – before taxi; – taxi; – departure; – en route; – circuit; – final; – landing; – after landing.	X	X	X	X	X	X	X	3		
(03)		Recognise, describe and use the correct standard phraseology for each phase of an IFR flight, including PBN operations (consider communication with each type of aeronautical station): – before pushback or taxi;	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		– pushback; – taxi; – departure; – en route; – approach; – final approach; – landing; – after landing.										
(04)		Explain phraseology for the selective calling system (SELCAL) and aircraft communications addressing and reporting system (ACARS).	X	X	X	X	X	X	X			
(05)		Explain traffic alert and collision avoidance system (TCAS) phraseology.	X	X	X	X	X	X	X			
090 02 01 06		RT call signs for aeronautical stations including use of abbreviated call signs										
(01)		Name the two parts of the call sign of an aeronautical station.	X	X	X	X	X	X	X	3		
(02)		Identify the call-sign suffixes for aeronautical stations.	X	X	X	X	X	X	X	3		
(03)		Explain when the call sign may be omitted or abbreviated to the use of suffix only.	X	X	X	X	X	X	X	3		
090 02 01 07		RT call signs for aircraft including use of abbreviated call signs										
(01)		Describe the three different ways to compose an aircraft call sign.	X	X	X	X	X	X	X			
(02)		Describe the abbreviated forms for aircraft call signs.	X	X	X	X	X	X	X			
(03)		Explain when aircraft call signs may be abbreviated.	X	X	X	X	X	X	X			

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Explain when the suffix 'HEAVY' or 'SUPER' is used with an aircraft call sign.	X	X	X	X	X	X	X	3	2	
(05)		Explain the use of the phrase 'Change your call sign to...'.	X	X	X	X	X	X	X	3		
(06)		Explain the use of the phrase 'Revert to flight plan call sign'.	X	X	X	X	X	X	X	3		
090 02 01 08		Transfer of communication										
(01)		Describe the procedure for transfer of communication: – by ground station; – by aircraft.	X	X	X	X	X	X	X	3		
090 02 01 09		Test procedures including readability scale										
(01)		Explain how to test radio transmission and reception.	X	X	X	X	X	X	X	3	2	
(02)		State the readability scale and explain its meaning.	X	X	X	X	X	X	X	3	2	
090 02 01 10		Read-back and acknowledgement requirements										
(01)		Describe the requirement to read back ATC route clearances.	X	X	X	X	X	X	X	3	2	
(02)		Describe the requirement to read back clearances related to the runway in use.	X	X	X	X	X	X	X	3	2	
(03)		Describe the requirement to read back other clearances including conditional clearances.	X	X	X	X	X	X	X	3	2	
(04)		Describe the requirement to read back other data such as runway, secondary surveillance radar (SSR) codes, etc.	X	X	X	X	X	X	X	3	2	
090 02 01 11		Radar procedural phraseology										
(01)		Use the correct phraseology for an aircraft receiving a radar service: – radar identification; – radar vectoring;	X	X	X	X	X	X	X	3	2	

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		– traffic information and avoidance; – SSR procedures.										
090 02 01 12		Level changes and reports										
(01)		Use the correct term to describe vertical position in relation to: – flight level (standard pressure setting); – altitude (metres/feet on QNH); – height (metres/feet on QFE).	X	X	X	X	X	X	X	3		
090 02 01 13		Data link messages										
(01)		List the different types of messages of the controller–pilot data link communications (CPDLC) function and give examples of data link messages.	X	X	X	X	X	X	X			
(02)		Describe a notification phase (LOG ON) and state its purpose.	X	X	X	X	X	X	X			
(03)		Explain the phrases to be used: – when voice communication is used to correct a CPDLC message; – in case of single CPDLC message failure; – when CPDLC has failed; – when reverting from CPDLC to voice communication.	X	X	X	X	X	X	X			
090 03 00 00		RELEVANT WEATHER INFORMATION										
090 03 01 00		Aerodrome weather										
090 03 01 01		Aerodrome weather terms										
(01)		List the contents of aerodrome weather reports and state units of measurement used for each item:	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
		- wind direction and speed; - variation of wind direction and speed; - visibility; - present weather; - cloud amount and type (including the definition of cloud and visibility OK (CAVOK); - air temperature and dew point; - pressure values (QNH, QFE); - supplementary information (aerodrome warnings, landing runway, runway conditions, restrictions, obstructions, wind-shear warnings, etc.).										
090 03 01 02		Weather broadcast										
(01)		List the sources (VOLMET and ATIS units) of weather information available for aircraft in flight, and describe situation(s) in which a pilot would normally obtain each.	X	X	X	X	X	X	X			
(02)	X	Explain the meaning of the acronyms 'D-ATIS', 'ATIS', and 'VOLMET'.	X	X	X	X	X	X	X			
(03)		Explain and demonstrate how to decode ATIS messages.	X	X	X	X	X	X	X	3		
(04)		Explain and demonstrate how to decode D-ATIS messages.	X	X	X	X	X	X	X			
090 04 00 00		VOICE COMMUNICATION FAILURE										
090 04 01 00		Required action										
090 04 01 01		Action required to be taken in case of communication failure										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(01)		State the action to be taken in case of communication failure on a controlled VFR flight.	X	X	X	X	X	X	X			
(02)		Identify the frequencies to be used in an attempt to establish communication.	X	X	X	X	X	X	X			
(03)		State the additional information that should be transmitted in the event of receiver failure.	X	X	X	X	X	X	X			
(04)		Identify the SSR code that may be used to indicate communication failure.	X	X	X	X	X	X	X			
(05)		Explain the action to be taken by a pilot that experiences a communication failure in the aerodrome traffic pattern at controlled aerodromes.	X	X	X	X	X	X	X			
(06)		Describe the action to be taken in case of communication failure on an IFR flight.	X	X	X	X	X	X	X	3		
(07)		Describe the action to be taken in case of communication failure on an IFR flight when flying in visual meteorological conditions (VMC) and the flight will be terminated in VMC.	X	X	X	X	X	X	X	3		
(08)		Describe the action to be taken in case of communication failure on an IFR flight when flying in instrument meteorological conditions (IMC).	X	X	X	X	X	X	X	3		
(09)		Explain the causes and possible safety impacts of a blocked frequency.	X	X	X	X	X	X	X	3		
090 05 00 00		DISTRESS AND URGENCY PROCEDURES										
090 05 01 00		Signals and procedures										
090 05 01 01		<i>Distress</i>										
(01)		State the DISTRESS signal(s) and DISTRESS procedure(s).	X	X	X	X	X	X	X		3	
(02)		Define 'DISTRESS'.	X	X	X	X	X	X	X		3	
(03)		Identify the frequencies that should be used by aircraft in DISTRESS.	X	X	X	X	X	X	X	3		

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
(04)		Specify the emergency SSR codes that may be used by aircraft, and the meaning of the codes.	X	X	X	X	X	X	X	3		
(05)		Describe the action to be taken by the station which receives a DISTRESS message.	X	X	X	X	X	X	X	3		
(06)		Describe the action to be taken by all other stations when a DISTRESS procedure is in progress.	X	X	X	X	X	X	X	3		
(07)		List the correctly sequenced elements of a DISTRESS signal/message and describe the message content.	X	X	X	X	X	X	X		3	
(08)		Describe the use of discrete frequencies (DEF) in case of distress or urgency.	X	X	X	X	X	X	X			
(09)		State that DISTRESS messages take priority over all other messages.	X	X	X	X	X	X	X	3		
090 05 01 02		Urgency										
(01)		State the URGENCY signal(s) and URGENCY procedure(s).	X	X	X	X	X	X	X		3	
(02)		Define 'URGENCY'.	X	X	X	X	X	X	X		3	
(03)		Identify the frequencies that should be used by aircraft in URGENCY.	X	X	X	X	X	X	X	3		
(04)		Describe the action to be taken by the station which receives an URGENCY message.	X	X	X	X	X	X	X	3		
(05)		Describe the action to be taken by all other stations when an URGENCY procedure is in progress.	X	X	X	X	X	X	X			
(06)		List the correctly sequenced elements of an URGENCY signal/message and describe the message content.	X	X	X	X	X	X	X		3	
(07)		State that URGENCY messages take priority over all other messages except DISTRESS.	X	X	X	X	X	X	X	3		
090 06 00 00		VHF PROPAGATION AND ALLOCATION OF FREQUENCIES										
090 06 01 00		General principles										

Syllabus reference	BK	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter			IR	CBIR(A)	BIR Exam	BIR BK	Remarks
			ATPL	CPL	ATPL/IR	ATPL	CPL					
090 06 01 01		Spectrum, bands, range										
(01)		Describe the radio-frequency spectrum with particular reference to VHF.	X	X	X	X	X	X	X		3	
(02)		Describe the radio-frequency spectrum of the bands into which the radio-frequency spectrum is divided.	X	X	X	X	X	X	X		3	
(03)		Identify the frequency range of the VHF band.	X	X	X	X	X	X	X	3		
(04)		State the band normally used for aeronautical mobile service (AMS) voice communication.	X	X	X	X	X	X	X	3		
(05)		State the frequency separation allocated between consecutive VHF frequencies.	X	X	X	X	X	X	X	3		
(06)		List the factors which reduce the effective range and quality of VHF radio transmissions.	X	X	X	X	X	X	X	3		
090 07 00 00		Other communications										
090 07 01 00		Weather observations, Morse code										
090 07 01 01		Meteorological observations										
(01)		Explain when aircraft routine meteorological observations should be made.	X	X	X	X	X	X	X	3		
(02)		Explain when aircraft special meteorological observations should be made.	X	X	X	X	X	X	X	3		
090 07 01 02		Use of Morse code										
(01)	X	Describe and list Morse code.	X	X	X	X	X	X	X			
(02)		Find the Morse code identifiers of radio navigation aids (VHF omnidirectional radio range (VOR), distance-measuring equipment (DME), non-directional radio beacon (NDB), instrument landing system (ILS)) using aeronautical charts.	X	X	X	X	X	X	X	3		

SUBJECT AREA 100 – KNOWLEDGE, SKILLS AND ATTITUDES (KSA)

ED Decision 2020/018/R

Syllabus reference	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter		
		ATPL	CPL	ATPL/IR	ATPL	CPL
100 00 00 00	KNOWLEDGE, SKILLS AND ATTITUDES (KSA)					
100 01 00 00	ICAO CORE COMPETENCIES					
(01)	Recognise the ICAO Core Competencies listed below and the associated competency descriptions (ICAO Doc 9995 'Manual of Evidence-based Training'): – Application of Procedures; – Communication; – Aircraft Flight Path Management, automation; – Aircraft Flight Path Management, manual control; – Leadership and Teamwork; – Problem Solving and Decision Making; – Situation Awareness; – Workload Management.	X	X	X	X	X
100 02 00 00	CORE COMPETENCIES LEARNING OBJECTIVES					
100 02 01 00	Communication					
(01)	Show the ability to identify whether the recipient is ready and able to receive the information.	X	X	X	X	X
(02)	Show the ability to appropriately select what, when, how and with whom to communicate.	X	X	X	X	X
(03)	Show the ability to communicate clearly, accurately and concisely.	X	X	X	X	X
(04)	Show the ability to confirm whether the recipient correctly understands important information.	X	X	X	X	X
(05)	Show the ability to listen actively and show you understand the information you receive.	X	X	X	X	X
(06)	Show the ability to ask relevant and effective questions.	X	X	X	X	X
(07)	Show the ability to adhere to standard radio-telephony phraseology.	X	X	X	X	X
(08)	Show the ability to accurately read, interpret, construct and respond to given documentation in English.	X	X	X	X	X

Syllabus reference	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter		
		ATPL	CPL	ATPL/IR	ATPL	CPL
(09)	Show the ability to correctly interpret non-verbal communication.	X	X	X	X	X
(10)	Show the ability to use appropriate eye contact, body movement and gestures that are consistent with and support verbal messages.	X	X	X	X	X
100 02 02 00	Leadership and teamwork					
(01)	Show the ability to create an atmosphere of open communication that encourages participation.	X	X	X	X	X
(02)	Show the initiative and the ability to give directions when required.	X	X	X	X	X
(03)	Show the ability to admit mistakes and take responsibility.	X	X	X	X	X
(04)	Show the ability to anticipate and respond appropriately to others' needs.	X	X	X	X	X
(05)	Show the ability to carry out instructions when directed.	X	X	X	X	X
(06)	Show the ability to communicate relevant concerns and intentions.	X	X	X	X	X
(07)	Show the ability to give and receive feedback constructively.	X	X	X	X	X
(08)	Show empathy, respect and tolerance for others.	X	X	X	X	X
(09)	Show the ability to engage others in planning and to allocate activities fairly and appropriately according to others' abilities.	X	X	X	X	X
(10)	Show the ability to address and resolve conflicts and disagreement in a constructive manner.	X	X	X	X	X
(11)	Show the ability to project self-control.	X	X	X	X	X
100 02 03 00	Problem-solving and decision-making					
(01)	Show the ability to seek accurate and adequate information from appropriate sources.	X	X	X	X	X
(02)	Show the ability to identify and verify what and why things have gone wrong.	X	X	X	X	X
(03)	Show the ability to employ proper problem-solving strategies.	X	X	X	X	X
(04)	Show the ability to persevere in working through problems.	X	X	X	X	X
(05)	Show the ability to use appropriate and timely decision-making processes.	X	X	X	X	X
(06)	Show the ability to set priorities appropriately.	X	X	X	X	X
(07)	Show the ability to identify and consider options effectively.	X	X	X	X	X
(08)	Show the ability to monitor, review and adapt decisions as required.	X	X	X	X	X
(09)	Show the ability to identify and manage risks.	X	X	X	X	X
100 02 04 00	Situation awareness					

Syllabus reference	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter		
		ATPL	CPL	ATPL/IR	ATPL	CPL
(01)	Demonstrate the ability to identify and assess accurately the general environment as it may affect the operation.	X	X	X	X	X
(02)	Demonstrate the ability to identify threats, errors and undesirable aircraft states.	X	X	X	X	X
(03)	Demonstrate the ability to manage threats, errors and undesirable aircraft states.	X	X	X	X	X
100 02 05 00	Workload management					
(01)	Show the ability to maintain self-control.	X	X	X	X	X
(02)	Show the ability to plan, prioritise and schedule tasks effectively.	X	X	X	X	X
(03)	Show the ability to manage time effectively when carrying out tasks.	X	X	X	X	X
(04)	Show the ability to offer and accept assistance, delegate when necessary and ask for help early.	X	X	X	X	X
(05)	Show the ability to manage interruptions, distractions, variations and failures effectively.	X	X	X	X	X
100 03 00 00	ADDITIONAL THREAT AND ERROR MANAGEMENT (TEM) RELATED LEARNING OBJECTIVES					
100 03 01 00	Application of knowledge					
(01)	Demonstrate the ability to complete pre-flight planning in practical exercises.	X	X	X	X	X
(02)	Demonstrate the KSA and TEM relating to phases of flight in the ground training environment.	X	X	X	X	X
100 03 02 00	Upset prevention and recovery training (UPRT) and resilience					
	<i>Note: Resilience is defined as 'the ability to recognise, absorb and adapt to disruptions'. It is supported by the pilot's core competencies and improved by experience, which can be gained by training for unexpected events or situations.</i>					
(01)	Recognise potential upset 'threats' and suggest effective 'threat management' in scenario situations.	X	X			
(02)	Recognise potential upset 'errors' and suggest effective 'error management' in scenario situations.	X	X			
(03)	Explain the causes of and contributing factors to upsets.	X	X			
(04)	Demonstrate resilience during scenario and/or other exercises.	X	X	X	X	X
(05)	Show the ability to identify the signs and discuss the effects of stress, fatigue and aviation lifestyle on situation awareness, and how to cope with them in order to maintain situation awareness.	X	X	X	X	X
100 04 00 00	MENTAL MATHS					
	<i>Note: Demonstrate, in non-calculator test scenarios or scenario exercises, the ability in a time-efficient manner to make correct mental calculation approximations for the following.</i>					

Syllabus reference	Syllabus details and associated Learning Objectives	Aeroplane		Helicopter		
		ATPL	CPL	ATPL/IR	ATPL	CPL
(01)	Convert between volumes and masses of fuel using range of units.	X	X	X	X	X
(02)	Estimate time, distance and speed.	X	X	X	X	X
(03)	Estimate the rate of climb or rate of descent, distance and time.	X	X			
(04)	Add or subtract time, distance, and fuel mass.	X	X	X	X	X
(05)	Calculate fuel burn given time and fuel flow.	X	X	X	X	X
(06)	Calculate the time available (for decision-making) given relevant fuel information.	X	X	X	X	X
(07)	Determine the top of descent using a simple method that is described by the approved training organisation (ATO).	X	X			
(08)	Determine the values that vary by a percentage, e.g. dry-to-wet landing distance and fuel burn.	X	X	X	X	X
(09)	Estimate heights at distances on a 3-degree glideslope.	X	X	X	X	X
(10)	Estimate headings using the 1-in-60 rule.	X	X	X	X	X
(11)	Estimate headwind and crosswind components given wind speed and direction and runway in use.	X	X	X	X	X

(b) Airships

SYLLABUS OF THEORETICAL KNOWLEDGE FOR CPL AND IR

The applicable items for each licence or rating are marked with 'x'. An 'x' on the main title of a subject means that all the subdivisions are applicable.

		CPL	IR
1.	AIR LAW AND ATC PROCEDURES	x	
	INTERNATIONAL LAW: CONVENTIONS, AGREEMENTS AND ORGANISATIONS		
	AIRWORTHINESS OF AIRCRAFT		
	AIRCRAFT NATIONALITY AND REGISTRATION MARKS		
	PERSONNEL LICENSING		x
	RULES OF THE AIR		x
	PROCEDURES FOR AIR NAVIGATION SERVICES: AIRCRAFT OPERATIONS		x
	AIR TRAFFIC SERVICES AND AIR TRAFFIC MANAGEMENT		x
	AERONAUTICAL INFORMATION SERVICE		x
	AERODROMES		x
	FACILITATION		
	SEARCH AND RESCUE		
	SECURITY		
	AIRCRAFT ACCIDENT AND INCIDENT INVESTIGATION		
2.	AIRSHIP GENERAL KNOWLEDGE: ENVELOPE, AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT AND EMERGENCY EQUIPMENT	x	
	DESIGN, MATERIALS, LOADS AND STRESSES		
	ENVELOPE AND AIRBAGS		
	FRAMEWORK		
	GONDOLA		
	FLIGHT CONTROLS		
	LANDING GEAR		
	HYDRAULICS AND PNEUMATICS		
	HEATING AND AIR CONDITIONING		
	FUEL SYSTEM		
	PISTON ENGINES		
	TURBINE ENGINES (BASICS)		
	ELECTRICS		
	FIRE PROTECTION AND DETECTION SYSTEMS		
	MAINTENANCE		
3.	AIRSHIP GENERAL KNOWLEDGE: INSTRUMENTATION	x	
	SENSORS AND INSTRUMENTS		
	MEASUREMENT OF AIR DATA AND GAS PARAMETERS		
	MAGNETISM: DIRECT READING COMPASS AND FLUX VALVE		
	GYROSCOPIC INSTRUMENTS		
	COMMUNICATION SYSTEMS		
	ALERTING SYSTEMS		
	INTEGRATED INSTRUMENTS: ELECTRONIC DISPLAYS		
	FLIGHT MANAGEMENT SYSTEM (GENERAL BASICS)		

		CPL	IR
	DIGITAL CIRCUITS AND COMPUTERS		
4.	FLIGHT PERFORMANCE AND PLANNING	x	
4.1.	MASS AND BALANCE: AIRSHIPS	x	
	PURPOSE OF MASS AND BALANCE CONSIDERATIONS		
	LOADING		
	FUNDAMENTALS OF CG CALCULATIONS		
	MASS AND BALANCE DETAILS OF AIRCRAFT		
	DETERMINATION OF CG POSITION		
	PASSENGER, CARGO AND BALLAST HANDLING		
4.2.	FLIGHT PLANNING AND FLIGHT MONITORING		
	FLIGHT PLANNING FOR VFR FLIGHTS	x	
	FLIGHT PLANNING FOR IFR FLIGHTS		x
	FUEL PLANNING	x	x
	PRE-FLIGHT PREPARATION	x	x
	ATS FLIGHT PLAN	x	x
	FLIGHT MONITORING AND IN-FLIGHT RE-PLANNING	x	x
4.3.	PERFORMANCE: AIRSHIPS	x	
	AIRWORTHINESS REQUIREMENTS		
	BASICS OF AIRSHIP PERFORMANCE		
	DEFINITIONS AND TERMS		
	STAGES OF FLIGHT		
	USE OF FLIGHT MANUAL		
5.	HUMAN PERFORMANCE	x	
	HUMAN FACTORS: BASIC CONCEPTS		
	BASIC AVIATION PHYSIOLOGY AND HEALTH MAINTENANCE		
	BASIC AVIATION PSYCHOLOGY		
6.	METEOROLOGY	x	
	THE ATMOSPHERE		
	WIND		
	THERMODYNAMICS		
	CLOUDS AND FOG		
	PRECIPITATION		
	AIR MASSES AND FRONTS		
	PRESSURE SYSTEMS		
	CLIMATOLOGY		
	FLIGHT HAZARDS		
	METEOROLOGICAL INFORMATION		
7.	NAVIGATION		
7.1.	GENERAL NAVIGATION	x	
	BASICS OF NAVIGATION		
	MAGNETISM AND COMPASSES		
	CHARTS		
	DR NAVIGATION		
	IN-FLIGHT NAVIGATION		
7.2.	RADIO NAVIGATION		

		CPL	IR
	BASIC RADIO PROPAGATION THEORY	x	x
	RADIO AIDS	x	x
	RADAR	x	x
	INTENTIONALLY LEFT BLANK		
	AREA NAVIGATION SYSTEMS AND RNAV/FMS		x
	GNSS	x	x
8.	OPERATIONAL PROCEDURES AIRSHIP	x	
	GENERAL REQUIREMENTS		
	SPECIAL OPERATIONAL PROCEDURES AND HAZARDS (GENERAL ASPECTS)		
	EMERGENCY PROCEDURES		
9.	PRINCIPLES OF FLIGHT	x	
9.1.	PRINCIPLES OF FLIGHT: AIRSHIPS	x	
	BASICS OF AEROSTATICS		
	BASICS OF SUBSONIC AERODYNAMICS		
	AERODYNAMICS OF AIRSHIPS		
	STABILITY		
	CONTROLLABILITY		
	LIMITATIONS		
	PROPELLERS		
	BASICS OF AIRSHIP FLIGHT MECHANICS		
10.	COMMUNICATIONS		
10.1.	VFR COMMUNICATIONS	x	
	DEFINITIONS	x	
	GENERAL OPERATING PROCEDURES	x	
	RELEVANT WEATHER INFORMATION TERMS (VFR)	x	
	ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE	x	
	DISTRESS AND URGENCY PROCEDURES	x	
	GENERAL PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES	x	
10.2.	IFR COMMUNICATIONS		
	DEFINITIONS		x
	GENERAL OPERATING PROCEDURES		x
	ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE		x
	DISTRESS AND URGENCY PROCEDURES		x
	RELEVANT WEATHER INFORMATION TERMS (IFR)		x
	GENERAL PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES		x
	MORSE CODE		x

GM1 FCL.310; FCL.515(b); FCL.615(b); FCL.835(d) Theoretical knowledge examinations

ED Decision 2020/018/R

EXPLANATION OF THE VERBS USED IN THE BENJAMIN BLOOM TAXONOMY

- (a) The depth or level of learning to be achieved during the training and the corresponding level of attainment to be examined or assessed is based on the following taxonomy. In each case, the level of knowledge or skill is signified by the learning objective (LO) verb.
- (b) The majority of the LOs relate to the cognitive domain. The taxonomy described by B. Bloom (1956) and Anderson & Krathwohl (2001) has been used as the standard.
- (c) The six sequential increasing levels of required cognitive learning are identified by the LO verb. Hence the lowest level 'remember' is signified by verbs such as 'state', 'list', 'define' and 'recall' whilst the next higher level of 'understand' is signified by verbs such as 'describe' and 'explain'. The third level of 'apply' is signified by the verbs 'calculate', 'interpret', 'relate' and 'solve'. However, the higher levels of 'analyse', which would be signified by the verbs 'plan' or 'discuss' and 'evaluate' and 'create' are less common due at least partially to questions presently possible in the ECQB examination.
- (d) The LOs used in Area 100 KSA differ in that they require a combination of knowledge and skills. However, the 'skill' level does not relate to Bloom's psychomotor taxonomy but is more closely aligned to the higher taxonomy levels required in medicine, because knowledge and skills must be combined by the student pilot in a strategy.
- (e) The verbs 'demonstrate' and 'show', with their meanings defined below, have therefore been used to supplement the cognitive LO verbs for the Area 100 KSA LOs.
 - (1) 'Demonstrate' means the selection and use of the appropriate knowledge, skills and attitudes within a strategy to achieve an effective outcome. It signifies a high taxonomy level and would normally be assessed using multiple indicators from more than one core competency.
 - (2) 'Show' means the attainment of knowledge, skill or attitude. It signifies a lower taxonomy level than 'demonstrate' and would normally be assessed by a single indicator.'

FCL.315 CPL – Training course

Regulation (EU) No 1178/2011

An applicant for a CPL shall have completed theoretical knowledge instruction and flight instruction at an ATO, in accordance with [Appendix 3](#) to this Part.

FCL.320 CPL – Skill test

Regulation (EU) No 1178/2011

An applicant for a CPL shall pass a skill test in accordance with [Appendix 4](#) to this Part to demonstrate the ability to perform, as PIC of the appropriate aircraft category, the relevant procedures and manoeuvres with the competency appropriate to the privileges granted.

SECTION 2 – SPECIFIC REQUIREMENTS FOR THE AEROPLANE CATEGORY – CPL(A)

FCL.315.A CPL – Training course

Regulation (EU) 2015/445

Theoretical knowledge and flight instruction for the issue of a CPL(A) shall include upset prevention and recovery training.

FCL.325.A CPL(A) – Specific requirements for applicants who hold an MPL

Regulation (EU) 2024/2076

- (a) Applicants for a CPL(A) who hold an MPL shall have completed in aeroplanes:
- (1) 70 hours of flight time in one of the following ways:
 - (i) as PIC;
 - (ii) made up of at least 10 hours as PIC and the additional flight time as PIC under supervision (PICUS).

Of those 70 hours, 20 hours shall be of VFR cross-country flight time as PIC, or cross-country flight time made up of at least 10 hours as PIC and 10 hours as PICUS. That shall include a VFR cross-country flight of at least 540 km (300 NM) in the course of which full-stop landings at two different aerodromes shall be flown as PIC;
 - (2) the elements of the CPL(A) modular course as specified in Chapter E, point 10(a) and point 11, of Appendix 3;
 - (3) the CPL(A) skill test, in accordance with point [FCL.320](#).
- (b) When being issued with a CPL in accordance with point (a), applicants shall be subject to the following conditions:
- (1) comply with the requirements of point [FCL.405.A\(c\)](#) in order to have the privileges of their IR(A) extended to single-pilot operations in aeroplanes;
 - (2) have the privileges of their type ratings restricted to exercise the privileges of a co-pilot only. That restriction shall be removed when applicants complete a proficiency check for the relevant type in accordance with Appendix 9 in the role of the PIC.

SUBPART E – MULTI-CREW PILOT LICENCE – MPL

FCL.400.A MPL – Minimum age

Regulation (EU) No 1178/2011

An applicant for an MPL shall be at least 18 years of age.

FCL.405.A MPL – Privileges

Regulation (EU) 2024/2076

- (a) The privileges of the holder of an MPL are to act as co-pilot in an aeroplane required to be operated with a co-pilot.
- (b) Holders of an MPL shall, upon application:
 - (1) be issued with additional privileges of the holder of a PPL(A), provided that they comply with the requirements for the PPL(A) specified in Subpart C;
 - (2) have their MPL replaced by a CPL(A), provided that they comply with the requirements specified in point [FCL.325.A](#).
- (c) The holder of an MPL shall have the privileges of his/her IR(A) limited to aeroplanes required to be operated with a co-pilot. The privileges of the IR(A) may be extended to single-pilot operations in aeroplanes, provided that the licence holder has completed the training necessary to act as PIC in single-pilot operations exercised solely by reference to instruments and passed the skill test of the IR(A) as a single-pilot.

FCL.410.A MPL – Training course and theoretical knowledge examinations

Regulation (EU) 2018/1974

- (a) Course.

Applicants for the issue of an MPL shall have completed a training course of theoretical knowledge and flight instruction at an ATO in accordance with Appendix 5 to this Annex (Part-FCL).
- (b) Examination.

Applicants for the issue of an MPL shall demonstrate a level of theoretical knowledge appropriate to the holders of an ATPL(A), in accordance with FCL.515, and to a multi-pilot type rating.

FCL.415.A MPL – Practical skill

Regulation (EU) No 1178/2011

- (a) An applicant for an MPL shall have demonstrated through continuous assessment the skills required for fulfilling all the competency units specified in [Appendix 5](#) to this Part, as pilot flying and pilot not flying, in a multi-engine turbine-powered multi-pilot aeroplane, under VFR and IFR.
- (b) On completion of the training course, the applicant shall pass a skill test in accordance with [Appendix 9](#) to this Part, to demonstrate the ability to perform the relevant procedures and manoeuvres with the competency appropriate to the privileges granted. The skill test shall be taken in the type of aeroplane used on the advanced phase of the MPL integrated training course or in an FFS representing the same type.

SUBPART F – AIRLINE TRANSPORT PILOT LICENCE – ATPL

SECTION 1 – COMMON REQUIREMENTS

FCL.500 ATPL – Minimum age

Regulation (EU) No 1178/2011

Applicants for an ATPL shall be at least 21 years of age.

FCL.505 ATPL – Privileges

Regulation (EU) No 1178/2011

- (a) The privileges of the holder of an ATPL are, within the appropriate aircraft category, to:
 - (1) exercise all the privileges of the holder of an LAPL, a PPL and a CPL;
 - (2) act as PIC of aircraft engaged in commercial air transport.
- (b) Applicants for the issue of an ATPL shall have fulfilled the requirements for the type rating of the aircraft used in the skill test.

FCL.515 ATPL – Training course and theoretical knowledge examinations

Regulation (EU) 2018/1974

- (a) Course.

Applicants for an ATPL shall have completed a training course at an ATO. The course shall be either an integrated training course or a modular course, in accordance with Appendix 3 to this Annex (Part-FCL).
- (b) Examination.

Applicants for the issue of an ATPL shall demonstrate a level of knowledge appropriate to the privileges granted in the following subjects:

 - (1) air law;
 - (2) aircraft general knowledge — airframe/systems/power plant;
 - (3) aircraft general knowledge — instrumentation;
 - (4) mass and balance;
 - (5) performance;
 - (6) flight planning and monitoring;
 - (7) human performance;
 - (8) meteorology;
 - (9) general navigation;
 - (10) radio navigation;
 - (11) operational procedures;

- (12) principles of flight; and
- (13) communications.;

AMC1 FCL.310; FCL.515(b); FCL.615(b); FCL.835(d) Theoretical knowledge examinations

ED Decision 2020/018/R

LEARNING OBJECTIVES FOR ATPL, CPL, IR, CB-IR(A) and BIR

- (a) Aeroplanes and helicopters

GENERAL

In the tables of this AMC, the applicable LOs for each ATPL, CPL, IR, CB-IR(A) are marked with an 'X', and for the BIR exam and BIR BK with the number 1, 2 or 3 (corresponding to the modules as mentioned in [FCL.835](#) 'Basic instrument rating (BIR)').

The LOs define the subject knowledge and applied knowledge, skills and attitudes that a student pilot should have assimilated during the theoretical knowledge course.

The LOs are intended to be used by an approved training organisation (ATO) when developing the Part-FCL theoretical knowledge elements of the appropriate course. It should be noted, however, that the LOs do not provide a ready-made ground training syllabus for individual ATOs, and should not be seen by organisations as a substitute for a thorough course design. Adherence to the LOs should become part of the ATO's compliance monitoring scheme as required by [ORA.GEN.200\(a\)\(6\)](#).

ATOs are required to produce a training plan for each of their courses based on the instructional systems design (ISD) methodology as specified in [AMC2 ORA.ATO.230](#).

Additional guidance on the meaning and taxonomy of the verbs used in the LOs can be found in [GM1 FCL.310; FCL.515\(b\); FCL.615\(b\); FCL.835\(d\)](#).

TRAINING AIMS

After completion of the training, a student pilot should:

- be able to understand and apply the subject knowledge in order to be able to identify and manage threats and errors effectively;
- meet at least the Area 100 KSA minimum standard.

INTERPRETATION

The abbreviations used are ICAO abbreviations listed in ICAO Doc 8400 'ICAO Abbreviations and Codes', or those listed in [GM1 FCL.010](#).

Where a LO refers to a definition, e.g. 'Define the following terms' or 'Define and understand' or 'Explain the definitions in ...', candidates are also expected to be able to recognise a given definition.

Below is a table showing the short references to applicable legislation and standards:

Reference	Legislation/Standard
The Basic Regulation	Regulation (EU) 2018/1139 of the European Parliament and of the Council of 4 July 2018
The Aircrew Regulation	Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)

Reference	Legislation/Standard
Part-FCL	Annex I to Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)
Part-MED	Annex IV to Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)
CS-23, AMC & GM to CS-23, CS-25, CS-27, CS-29, CS-E and CS-Definitions	Refer to the respective EASA Certification Specifications / AMC & GM
Single European Sky Regulations	Regulation (EC) No 549/2004 of the European Parliament and of the Council of 10 March 2004 laying down the framework for the creation of the single European sky (the framework Regulation) Regulation (EC) No 550/2004 of the European Parliament and of the Council of 10 March 2004 on the provision of air navigation services in the single European sky (the service provision Regulation) Regulation (EC) No 551/2004 of the European Parliament and of the Council of 10 March 2004 on the organisation and use of the airspace in the single European sky (the airspace Regulation) Regulation (EC) No 552/2004 of the European Parliament and of the Council of 10 March 2004 on the interoperability of the European Air Traffic Management network (the interoperability Regulation)
Passenger Rights Regulation	Regulation (EC) No 261/2004 of the European Parliament and of the Council of 11 February 2004 establishing common rules on compensation and assistance to passengers in the event of denied boarding and of cancellation or long delay of flights, and repealing Regulation (EEC) No 295/91
RTCA/EUROCAE	Refers to correspondingly numbered documents issued by the Radio Technical Commission for Aeronautics/ European Organisation for Civil Aviation Equipment
ITU Radio Regulation	International Telecommunication Union Radio Regulation
NASA TM-85652	National Aeronautics and Space Administration — Technical Memorandum 85652

‘Applicable operational requirements’ refers to, for the ATPL(A), CPL(A), ATPL(H)/IR, ATPL(H)/VFR, CPL(H), IR and CBIR, Annexes I, II, III, IV, V and VIII to Commission Regulation (EU) No 965/2012 of 5 October 2012 (as amended). For the BIR, it refers to Annexes I, II, V and VII to that Regulation.

The General Student Pilot Route Manual (GSPRM) contains planning data plus aerodrome and approach charts that may be used in theoretical knowledge training courses. The guidelines on its content can be found in this AMC, before the LO table for Subject 033 ‘Flight planning and monitoring’.

Excerpts from any aircraft manuals including but not limited to CAP 696, 697 and 698 for aeroplanes, and CAP 758 for helicopters may be used in training. Where questions refer to excerpts from aircraft manuals, the associated aircraft data will be provided in the examinations.

Some numerical data (e.g. speeds, altitudes/levels and masses) used in questions for theoretical knowledge examinations may not be representative for helicopter operations, but the data is satisfactory for the calculations required.

Note: In all subject areas, the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

DETAILED THEORETICAL KNOWLEDGE SYLLABUS AND LOs FOR ATPL, CPL, IR, CB-IR(A) and BIR

GENERAL

The detailed theoretical knowledge syllabus outlines the topics that should be taught and examined in order to meet the theoretical knowledge requirements appropriate to ATPL, MPL, CPL, IR, CB-IR(A) and BIR.

For each topic in the detailed theoretical knowledge syllabus, one or more LOs are set out in the appendices as shown below:

- [Appendix 010 AIR LAW](#)
- [Appendix 021 AIRCRAFT GENERAL KNOWLEDGE – AIRFRAME, SYSTEMS AND POWER PLANT](#)
- [Appendix 022 AIRCRAFT GENERAL KNOWLEDGE – INSTRUMENTATION](#)
- [Appendix 031 FLIGHT PERFORMANCE AND PLANNING – MASS AND BALANCE](#)
- [Appendix 032 FLIGHT PERFORMANCE AND PLANNING – PERFORMANCE – AEROPLANES](#)
- [Appendix 033 FLIGHT PERFORMANCE AND PLANNING – FLIGHT PLANNING AND MONITORING](#)
- [Appendix 034 FLIGHT PERFORMANCE AND PLANNING – PERFORMANCE – HELICOPTERS](#)
- [Appendix 040 HUMAN PERFORMANCE AND LIMITATIONS](#)
- [Appendix 050 METEOROLOGY](#)
- [Appendix 061 NAVIGATION – GENERAL NAVIGATION](#)
- [Appendix 062 NAVIGATION – RADIO NAVIGATION](#)
- [Appendix 070 OPERATIONAL PROCEDURES](#)
- [Appendix 081 PRINCIPLES OF FLIGHT – AEROPLANES](#)
- [Appendix 082 PRINCIPLES OF FLIGHT – HELICOPTERS](#)
- [Appendix 090 RADIO COMMUNICATIONS](#)
- [Appendix AREA 100 KNOWLEDGE, SKILLS AND ATTITUDES \(KSA\)](#)

(b) Airships

SYLLABUS OF THEORETICAL KNOWLEDGE FOR CPL AND IR

The applicable items for each licence or rating are marked with ‘x’. An ‘x’ on the main title of a subject means that all the subdivisions are applicable.’

		CPL	IR
1.	AIR LAW AND ATC PROCEDURES	x	
	INTERNATIONAL LAW: CONVENTIONS, AGREEMENTS AND ORGANISATIONS		
	AIRWORTHINESS OF AIRCRAFT		
	AIRCRAFT NATIONALITY AND REGISTRATION MARKS		
	PERSONNEL LICENSING		x
	RULES OF THE AIR		x
	PROCEDURES FOR AIR NAVIGATION SERVICES: AIRCRAFT OPERATIONS		x
	AIR TRAFFIC SERVICES AND AIR TRAFFIC MANAGEMENT		x
	AERONAUTICAL INFORMATION SERVICE		x
	AERODROMES		x
	FACILITATION		
	SEARCH AND RESCUE		
	SECURITY		
	AIRCRAFT ACCIDENT AND INCIDENT INVESTIGATION		
2.	AIRSHIP GENERAL KNOWLEDGE: ENVELOPE, AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT AND EMERGENCY EQUIPMENT	x	
	DESIGN, MATERIALS, LOADS AND STRESSES		
	ENVELOPE AND AIRBAGS		
	FRAMEWORK		
	GONDOLA		
	FLIGHT CONTROLS		
	LANDING GEAR		
	HYDRAULICS AND PNEUMATICS		
	HEATING AND AIR CONDITIONING		
	FUEL SYSTEM		
	PISTON ENGINES		
	TURBINE ENGINES (BASICS)		
	ELECTRICS		
	FIRE PROTECTION AND DETECTION SYSTEMS		
	MAINTENANCE		
3.	AIRSHIP GENERAL KNOWLEDGE: INSTRUMENTATION	x	

		CPL	IR
	SENSORS AND INSTRUMENTS		
	MEASUREMENT OF AIR DATA AND GAS PARAMETERS		
	MAGNETISM: DIRECT READING COMPASS AND FLUX VALVE		
	GYROSCOPIC INSTRUMENTS		
	COMMUNICATION SYSTEMS		
	ALERTING SYSTEMS		
	INTEGRATED INSTRUMENTS: ELECTRONIC DISPLAYS		
	FLIGHT MANAGEMENT SYSTEM (GENERAL BASICS)		
	DIGITAL CIRCUITS AND COMPUTERS		
4.	FLIGHT PERFORMANCE AND PLANNING	x	
4.1.	MASS AND BALANCE: AIRSHIPS	x	
	PURPOSE OF MASS AND BALANCE CONSIDERATIONS		
	LOADING		
	FUNDAMENTALS OF CG CALCULATIONS		
	MASS AND BALANCE DETAILS OF AIRCRAFT		
	DETERMINATION OF CG POSITION		
	PASSENGER, CARGO AND BALLAST HANDLING		
4.2.	FLIGHT PLANNING AND FLIGHT MONITORING		
	FLIGHT PLANNING FOR VFR FLIGHTS	x	
	FLIGHT PLANNING FOR IFR FLIGHTS		x
	FUEL PLANNING	x	x
	PRE-FLIGHT PREPARATION	x	x
	ATS FLIGHT PLAN	x	x
	FLIGHT MONITORING AND IN-FLIGHT RE-PLANNING	x	x
4.3.	PERFORMANCE: AIRSHIPS	x	
	AIRWORTHINESS REQUIREMENTS		
	BASICS OF AIRSHIP PERFORMANCE		
	DEFINITIONS AND TERMS		
	STAGES OF FLIGHT		
	USE OF FLIGHT MANUAL		

		CPL	IR
5.	HUMAN PERFORMANCE	x	
	HUMAN FACTORS: BASIC CONCEPTS		
	BASIC AVIATION PHYSIOLOGY AND HEALTH MAINTENANCE		
	BASIC AVIATION PSYCHOLOGY		
6.	METEOROLOGY	x	
	THE ATMOSPHERE		
	WIND		
	THERMODYNAMICS		
	CLOUDS AND FOG		
	PRECIPITATION		
	AIR MASSES AND FRONTS		
	PRESSURE SYSTEMS		
	CLIMATOLOGY		
	FLIGHT HAZARDS		
	METEOROLOGICAL INFORMATION		
7.	NAVIGATION		
7.1.	GENERAL NAVIGATION	x	
	BASICS OF NAVIGATION		
	MAGNETISM AND COMPASSES		
	CHARTS		
	DR NAVIGATION		
	IN-FLIGHT NAVIGATION		
7.2.	RADIO NAVIGATION		
	BASIC RADIO PROPAGATION THEORY	x	x
	RADIO AIDS	x	x
	RADAR	x	x
	INTENTIONALLY LEFT BLANK		
	AREA NAVIGATION SYSTEMS AND RNAV/FMS		x
	GNSS	x	x
8.	OPERATIONAL PROCEDURES AIRSHIP	x	
	GENERAL REQUIREMENTS		
	SPECIAL OPERATIONAL PROCEDURES AND HAZARDS (GENERAL ASPECTS)		
	EMERGENCY PROCEDURES		
9.	PRINCIPLES OF FLIGHT	x	
9.1.	PRINCIPLES OF FLIGHT: AIRSHIPS	x	
	BASICS OF AEROSTATICS		

		CPL	IR
	BASICS OF SUBSONIC AERODYNAMICS		
	AERODYNAMICS OF AIRSHIPS		
	STABILITY		
	CONTROLLABILITY		
	LIMITATIONS		
	PROPELLERS		
	BASICS OF AIRSHIP FLIGHT MECHANICS		
10.	COMMUNICATIONS		
10.1.	VFR COMMUNICATIONS	x	
	DEFINITIONS	x	
	GENERAL OPERATING PROCEDURES	x	
	RELEVANT WEATHER INFORMATION TERMS (VFR)	x	
	ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE	x	
	DISTRESS AND URGENCY PROCEDURES	x	
	GENERAL PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES	x	
10.2.	IFR COMMUNICATIONS		
	DEFINITIONS		x
	GENERAL OPERATING PROCEDURES		x
	ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE		x
	DISTRESS AND URGENCY PROCEDURES		x
	RELEVANT WEATHER INFORMATION TERMS (IFR)		x
	GENERAL PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES		x
	MORSE CODE		x

GM1 FCL.310; FCL.515(b); FCL.615(b); FCL.835(d) Theoretical knowledge examinations

ED Decision 2020/018/R

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- (a) The depth or level of learning to be achieved during the training and the corresponding level of attainment to be examined or assessed is based on the following taxonomy. In each case, the level of knowledge or skill is signified by the learning objective (LO) verb.
- (b) The majority of the LOs relate to the cognitive domain. The taxonomy described by B. Bloom (1956) and Anderson & Krathwohl (2001) has been used as the standard.
- (c) The six sequential increasing levels of required cognitive learning are identified by the LO verb. Hence the lowest level 'remember' is signified by verbs such as 'state', 'list', 'define' and 'recall' whilst the next higher level of 'understand' is signified by verbs such as 'describe' and 'explain'. The third level of 'apply' is signified by the verbs 'calculate', 'interpret', 'relate' and 'solve'. However, the higher levels of 'analyse', which would be signified by the verbs 'plan' or 'discuss' and 'evaluate' and 'create' are less common due at least partially to questions presently possible in the ECQB examination.
- (d) The LOs used in Area 100 KSA differ in that they require a combination of knowledge and skills. However, the 'skill' level does not relate to Bloom's psychomotor taxonomy but is more closely aligned to the higher taxonomy levels required in medicine, because knowledge and skills must be combined by the student pilot in a strategy.
- (e) The verbs 'demonstrate' and 'show', with their meanings defined below, have therefore been used to supplement the cognitive LO verbs for the Area 100 KSA LOs.
 - (1) 'Demonstrate' means the selection and use of the appropriate knowledge, skills and attitudes within a strategy to achieve an effective outcome. It signifies a high taxonomy level and would normally be assessed using multiple indicators from more than one core competency.
 - (2) 'Show' means the attainment of knowledge, skill or attitude. It signifies a lower taxonomy level than 'demonstrate' and would normally be assessed by a single indicator.'

SECTION 2 – SPECIFIC REQUIREMENTS FOR THE AEROPLANE CATEGORY – ATPL(A)

FCL.505.A ATPL(A) – Restriction of privileges for pilots previously holding an MPL

Regulation (EU) No 1178/2011

When the holder of an ATPL(A) has previously held only an MPL, the privileges of the licence shall be restricted to multi-pilot operations, unless the holder has complied with [FCL.405.A\(b\)\(2\) and \(c\)](#) for single-pilot operations.

FCL.510.A ATPL(A) – Prerequisites, experience and crediting

Regulation (EU) 2024/2076

- (a) Prerequisites. Applicants for an ATPL(A) shall hold:
- (1) an MPL; or
 - (2) a CPL(A) and a multi-engine IR for aeroplanes. In this case, the applicant shall also have received instruction in MCC.
- (b) Experience. Applicants for an ATPL(A) shall have completed a minimum of 1 500 hours of flight time in aeroplanes, including at least:
- (1) 500 hours in multi-pilot operations on aeroplanes;
 - (2)
 - (i) 500 hours as PIC under supervision; or
 - (ii) 250 hours as PIC; or
 - (iii) 250 hours, including at least 70 hours as PIC, and the remaining as PIC under supervision;
 - (3) 200 hours of cross-country flight time of which at least 100 hours shall be as PIC or as PIC under supervision;
 - (4) 75 hours of instrument time of which not more than 30 hours may be instrument ground time; and
 - (5) 100 hours of night flight as PIC or co-pilot.
- Of the 1500 hours of flight time, up to 100 hours of flight time may have been completed in an FFS and FNPT. Of these 100 hours, only a maximum of 25 hours may be completed in an FNPT.
- (c) Crediting.
- (1) Holders of a pilot licence for other categories of aircraft shall be credited with flight time up to a maximum of:
 - (i) for TMG or sailplanes, 30 hours flown as PIC;
 - (ii) for helicopters, 50 % of all the flight time requirements of paragraph (b).
 - (2) Holders of a flight engineer licence issued in accordance with applicable national rules shall be credited with 50 % of the flight engineer time up to a maximum credit of 250 hours. These 250 hours may be credited against the 1 500 hours requirement of

paragraph (b), and the 500 hours requirement of paragraph (b)(1), provided that the total credit given against any of these paragraphs does not exceed 250 hours.

- (3) Holders of a restricted type rating issued in accordance with point [FCL.720.A\(c\)](#) shall receive credits for their flight time completed while exercising the privileges of that type rating up to a maximum of 250 hours.
- (d) The experience required in (b) shall be completed before the skill test for the ATPL(A) is taken.

FCL.520.A ATPL(A) – Skill test

Regulation (EU) No 1178/2011

Applicants for an ATPL(A) shall pass a skill test in accordance with [Appendix 9](#) to this Part to demonstrate the ability to perform, as PIC of a multi-pilot aeroplane under IFR, the relevant procedures and manoeuvres with the competency appropriate to the privileges granted.

The skill test shall be taken in the aeroplane or an adequately qualified FFS representing the same type.

AMC1 FCL.520.A; FCL.520.H

ED Decision 2011/016/R

ATPL SKILL TEST

The ATPL skill test may serve at the same time as a skill test for the issue of the licence and a proficiency check for the revalidation of the type rating for the aircraft used in the test and may be combined with the skill test for the issue of a MP type rating.

GM1 FCL.520.A ATPL(A) — Skill test

ED Decision 2022/014/R

ATPL SKILL TEST IN AN EBT MODULE

The skill test in accordance with [Appendix 9](#) may be combined with an EBT module. It may follow the same process already described in mixed EBT for the LPC (e.g. EVAL + manoeuvres validation phase for the pilot performing the ATPL skill test). The same rationale can be applied to the regular skill test. Guidance can be found in the EASA EBT manual. The competent authority may provide additional and further guidance.

SECTION 3 – SPECIFIC REQUIREMENTS FOR THE HELICOPTER CATEGORY – ATPL(H)

FCL.510.H ATPL(H) – Prerequisites, experience and crediting

Regulation (EU) 2021/2227

Applicants for an ATPL(H) shall:

- (a) hold a CPL(H);
- (b) have received instruction in MCC in accordance with point [FCL.735.H](#);
- (c) have completed as a pilot of helicopters a minimum of 1 000 hours of flight time including at least:
 - (1) 350 hours in multi-pilot operations in helicopters;
 - (2)
 - (i) 250 hours as PIC; or
 - (ii) 100 hours as PIC and 150 hours as PIC under supervision; or
 - (iii) 250 hours as PIC under supervision in multi-pilot helicopters. In this case, the ATPL(H) privileges shall be limited to multi-pilot operations only, until 100 hours as PIC have been completed;
 - (3) 200 hours of cross-country flight time of which at least 100 hours shall be as PIC or as PIC under supervision;
 - (4) 30 hours of instrument time of which not more than 10 hours may be instrument ground time; and
 - (5) 100 hours of night flight as PIC or as co-pilot.

Of the 1 000 hours, a maximum of 100 hours may have been completed in an FSTD, of which not more than 25 hours may be completed in an FNPT;

- (d) flight time in aeroplanes shall be credited up to 50 % against the flight time requirements of point (c);
- (e) the experience required in point (c) shall be completed before the skill test for the ATPL(H) is taken;
- (f) applicants for an ATPL(H) shall receive a full credit for the requirement in point (b) when they comply with point [FCL.720.H\(a\)\(2\)\(ii\)](#) and, additionally, have received training at an ATO to meet the necessary standard for the successful completion of the course as per point [FCL.735.H](#).

AMC1 FCL.510.H(f) ATPL(H) — Prerequisites, experience and crediting

ED Decision 2022/014/R

MCC TRAINING WITH CREDITS FOR ATPL(H) — PILOTS THAT ARE EXPERIENCED IN MULTI-PILOT OPERATIONS

- (a) General

MCC training in accordance with point [FCL.510.H\(f\)](#) is intended to be completed by applicants who are undergoing integrated ATPL(H) training.

(b) Training course

The training course should include theoretical training instruction and exercises, as well as practical MCC training using one of the following helicopter simulators:

- (1) FNPT II or III qualified for MCC;
- (2) an FTD 2/3;
- (3) an FFS.

(c) Objectives

The training course should meet the objectives of [AMC1 FCL.735.A; FCL.735.H; FCL.735.As](#). The head of training of the ATO should adapt the duration of training to the individual needs of the applicant, in order to achieve these objectives.

(d) Certificate of completion

On completion of the course, once the applicant has met the objectives defined in (c), the applicant should receive a certificate of completion of the training from the ATO. The form should be based on that defined in [AMC1 FCL.735.A; FCL.735.H; FCL.735.As](#). The title of the form should read 'Training in accordance with FCL.510(f) — helicopters'.

FCL.520.H ATPL(H) – Skill test

Regulation (EU) No 1178/2011

Applicants for an ATPL(H) shall pass a skill test in accordance with [Appendix 9](#) to this Part to demonstrate the ability to perform as PIC of a multi-pilot helicopter the relevant procedures and manoeuvres with the competency appropriate to the privileges granted.

The skill test shall be taken in the helicopter or an adequately qualified FFS representing the same type.

AMC1 FCL.520.A; FCL.520.H

ED Decision 2011/016/R

ATPL SKILL TEST

The ATPL skill test may serve at the same time as a skill test for the issue of the licence and a proficiency check for the revalidation of the type rating for the aircraft used in the test and may be combined with the skill test for the issue of a MP type rating.

SUBPART G – INSTRUMENT RATING – IR

SECTION 1 – COMMON REQUIREMENTS

FCL.600 IR – General

Regulation (EU) 2020/359

Except as provided in point [FCL.835](#), operations under IFR on an aeroplane, helicopter, airship or powered-lift aircraft shall be conducted only by holders of a PPL, CPL, MPL and ATPL with an IR appropriate to the category of aircraft or, if an IR appropriate to the category of aircraft is not available, only while undergoing skill testing or dual instruction.

FCL.605 IR – Privileges and conditions

Regulation (EU) 2021/2227

(a) Privileges

The privileges of holders of an IR are to fly aircraft under IFR, including PBN operations, with a minimum decision height of:

- (1) no less than 200 ft (60 m);
- (2) less than 200 ft (60 m), provided that they are authorised to do so in accordance with Annex V (Part-SPA) to [Regulation \(EU\) No 965/2012](#).

(b) Conditions

- (1) Holders of an IR shall exercise their privileges in accordance with the conditions established in [Appendix 8](#) to this Annex.
- (2) To exercise privileges as PIC under IFR in multi-pilot operation in helicopters, holders of an IR(H) shall have at least 70 hours of instrument time, of which up to 30 hours may be instrument ground time.

FCL.610 IR – Prerequisites and crediting

Regulation (EU) No 245/2014

Applicants for an IR shall:

(a) hold:

- (1) at least a PPL in the appropriate aircraft category, and:
 - (i) the privileges to fly at night in accordance with [FCL.810](#), if the IR privileges will be used at night; or
 - (ii) an ATPL in another category of aircraft; or
- (2) a CPL, in the appropriate aircraft category;

(b) have completed at least 50 hours of cross-country flight time as PIC in aeroplanes, TMGs, helicopters or airships, of which at least 10 or, in the case of airships, 20 hours shall be in the relevant aircraft category.

(c) Helicopters only. Applicants who have completed an ATP(H)/IR, ATP(H), CPL(H)/IR or CPL(H) integrated training course shall be exempted from the requirement in (b).

FCL.615 IR – Theoretical knowledge and flight instruction

Regulation (EU) 2018/1974

(a) Course

Applicants for an IR shall have completed a course of theoretical knowledge and flight instruction at an ATO. The course shall be:

- (1) an integrated training course which includes training for the IR, in accordance with Appendix 3 to this Annex (Part-FCL); or
- (2) a modular course in accordance with Appendix 6 to this Annex (Part-FCL).

(b) Examination

Applicants shall demonstrate a level of theoretical knowledge appropriate to the privileges granted in the following subjects:

- (1) air law;
- (2) aircraft general knowledge — instrumentation;
- (3) flight planning and monitoring;
- (4) human performance;
- (5) meteorology;
- (6) radio navigation; and
- (7) communications.;

AMC1 FCL.310; FCL.515(b); FCL.615(b); FCL.835(d) Theoretical knowledge examinations

ED Decision 2020/018/R

LEARNING OBJECTIVES FOR ATPL, CPL, IR, CB-IR(A) and BIR**(a) Aeroplanes and helicopters****GENERAL**

In the tables of this AMC, the applicable LOs for each ATPL, CPL, IR, CB-IR(A) are marked with an 'X', and for the BIR exam and BIR BK with the number 1, 2 or 3 (corresponding to the modules as mentioned in [FCL.835](#) 'Basic instrument rating (BIR)').

The LOs define the subject knowledge and applied knowledge, skills and attitudes that a student pilot should have assimilated during the theoretical knowledge course.

The LOs are intended to be used by an approved training organisation (ATO) when developing the Part-FCL theoretical knowledge elements of the appropriate course. It should be noted, however, that the LOs do not provide a ready-made ground training syllabus for individual ATOs, and should not be seen by organisations as a substitute for a thorough course design. Adherence to the LOs should become part of the ATO's compliance monitoring scheme as required by [ORA.GEN.200\(a\)\(6\)](#).

ATOs are required to produce a training plan for each of their courses based on the instructional systems design (ISD) methodology as specified in [AMC2 ORA.ATO.230](#).

Additional guidance on the meaning and taxonomy of the verbs used in the LOs can be found in [GM1 FCL.310; FCL.515\(b\); FCL.615\(b\); FCL.835\(d\)](#).

TRAINING AIMS

After completion of the training, a student pilot should:

- be able to understand and apply the subject knowledge in order to be able to identify and manage threats and errors effectively;
- meet at least the Area 100 KSA minimum standard.

INTERPRETATION

The abbreviations used are ICAO abbreviations listed in ICAO Doc 8400 'ICAO Abbreviations and Codes', or those listed in [GM1 FCL.010](#).

Where a LO refers to a definition, e.g. 'Define the following terms' or 'Define and understand' or 'Explain the definitions in ...', candidates are also expected to be able to recognise a given definition.

Below is a table showing the short references to applicable legislation and standards:

Reference	Legislation/Standard
The Basic Regulation	Regulation (EU) 2018/1139 of the European Parliament and of the Council of 4 July 2018
The Aircrew Regulation	Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)
Part-FCL	Annex I to Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)
Part-MED	Annex IV to Commission Regulation (EU) No 1178/2011 of 3 November 2011 (as amended)
CS-23, AMC & GM to CS-23, CS-25, CS-27, CS-29, CS-E and CS-Definitions	Refer to the respective EASA Certification Specifications / AMC & GM
Single European Sky Regulations	Regulation (EC) No 549/2004 of the European Parliament and of the Council of 10 March 2004 laying down the framework for the creation of the single European sky (the framework Regulation) Regulation (EC) No 550/2004 of the European Parliament and of the Council of 10 March 2004 on the provision of air navigation services in the single European sky (the service provision Regulation) Regulation (EC) No 551/2004 of the European Parliament and of the Council of 10 March 2004 on the organisation and use of the airspace in the single European sky (the airspace Regulation) Regulation (EC) No 552/2004 of the European Parliament and of the Council of 10 March 2004 on the interoperability of the European Air Traffic Management network (the interoperability Regulation)
Passenger Rights Regulation	Regulation (EC) No 261/2004 of the European Parliament and of the Council of 11 February 2004 establishing common rules on compensation and assistance to passengers in the event of denied boarding and of cancellation or long delay of flights, and repealing Regulation (EEC) No 295/91
RTCA/EUROCAE	Refers to correspondingly numbered documents issued by the Radio Technical Commission for Aeronautics/ European Organisation for Civil Aviation Equipment

Reference	Legislation/Standard
ITU Radio Regulation	International Telecommunication Union Radio Regulation
NASA TM-85652	National Aeronautics and Space Administration — Technical Memorandum 85652

‘Applicable operational requirements’ refers to, for the ATPL(A), CPL(A), ATPL(H)/IR, ATPL(H)/VFR, CPL(H), IR and CBIR, Annexes I, II, III, IV, V and VIII to Commission Regulation (EU) No 965/2012 of 5 October 2012 (as amended). For the BIR, it refers to Annexes I, II, V and VII to that Regulation.

The General Student Pilot Route Manual (GSPRM) contains planning data plus aerodrome and approach charts that may be used in theoretical knowledge training courses. The guidelines on its content can be found in this AMC, before the LO table for Subject 033 ‘Flight planning and monitoring’.

Excerpts from any aircraft manuals including but not limited to CAP 696, 697 and 698 for aeroplanes, and CAP 758 for helicopters may be used in training. Where questions refer to excerpts from aircraft manuals, the associated aircraft data will be provided in the examinations.

Some numerical data (e.g. speeds, altitudes/levels and masses) used in questions for theoretical knowledge examinations may not be representative for helicopter operations, but the data is satisfactory for the calculations required.

Note: In all subject areas, the term ‘mass’ is used to describe a quantity of matter, and ‘weight’ when describing the force. However, the term ‘weight’ is normally used in aviation to colloquially describe mass. The pilot should always note the units to determine whether the term ‘weight’ is being used to describe a force (e.g. unit newton) or quantity of matter (e.g. unit kilogram).

DETAILED THEORETICAL KNOWLEDGE SYLLABUS AND LOs FOR ATPL, CPL, IR, CB-IR(A) and BIR

GENERAL

The detailed theoretical knowledge syllabus outlines the topics that should be taught and examined in order to meet the theoretical knowledge requirements appropriate to ATPL, MPL, CPL, IR, CB-IR(A) and BIR.

For each topic in the detailed theoretical knowledge syllabus, one or more LOs are set out in the appendices as shown below:

- [Appendix 010 AIR LAW](#)
- [Appendix 021 AIRCRAFT GENERAL KNOWLEDGE – AIRFRAME, SYSTEMS AND POWER PLANT](#)
- [Appendix 022 AIRCRAFT GENERAL KNOWLEDGE – INSTRUMENTATION](#)
- [Appendix 031 FLIGHT PERFORMANCE AND PLANNING – MASS AND BALANCE](#)
- [Appendix 032 FLIGHT PERFORMANCE AND PLANNING – PERFORMANCE – AEROPLANES](#)
- [Appendix 033 FLIGHT PERFORMANCE AND PLANNING – FLIGHT PLANNING AND MONITORING](#)
- [Appendix 034 FLIGHT PERFORMANCE AND PLANNING – PERFORMANCE – HELICOPTERS](#)

- [Appendix 040 HUMAN PERFORMANCE AND LIMITATIONS](#)
- [Appendix 050 METEOROLOGY](#)
- [Appendix 061 NAVIGATION – GENERAL NAVIGATION](#)
- [Appendix 062 NAVIGATION – RADIO NAVIGATION](#)
- [Appendix 070 OPERATIONAL PROCEDURES](#)
- [Appendix 081 PRINCIPLES OF FLIGHT – AEROPLANES](#)
- [Appendix 082 PRINCIPLES OF FLIGHT – HELICOPTERS](#)
- [Appendix 090 RADIO COMMUNICATIONS](#)
- [Appendix AREA 100 KNOWLEDGE, SKILLS AND ATTITUDES \(KSA\)](#)

(b) Airships

SYLLABUS OF THEORETICAL KNOWLEDGE FOR CPL AND IR

The applicable items for each licence or rating are marked with 'x'. An 'x' on the main title of a subject means that all the subdivisions are applicable.'

		CPL	IR
1.	AIR LAW AND ATC PROCEDURES	x	
	INTERNATIONAL LAW: CONVENTIONS, AGREEMENTS AND ORGANISATIONS		
	AIRWORTHINESS OF AIRCRAFT		
	AIRCRAFT NATIONALITY AND REGISTRATION MARKS		
	PERSONNEL LICENSING		x
	RULES OF THE AIR		x
	PROCEDURES FOR AIR NAVIGATION SERVICES: AIRCRAFT OPERATIONS		x
	AIR TRAFFIC SERVICES AND AIR TRAFFIC MANAGEMENT		x
	AERONAUTICAL INFORMATION SERVICE		x
	AERODROMES		x
	FACILITATION		
	SEARCH AND RESCUE		
	SECURITY		
	AIRCRAFT ACCIDENT AND INCIDENT INVESTIGATION		
2.	AIRSHIP GENERAL KNOWLEDGE: ENVELOPE, AIRFRAME AND SYSTEMS, ELECTRICS, POWERPLANT AND EMERGENCY EQUIPMENT	x	
	DESIGN, MATERIALS, LOADS AND STRESSES		

		CPL	IR
	ENVELOPE AND AIRBAGS		
	FRAMEWORK		
	GONDOLA		
	FLIGHT CONTROLS		
	LANDING GEAR		
	HYDRAULICS AND PNEUMATICS		
	HEATING AND AIR CONDITIONING		
	FUEL SYSTEM		
	PISTON ENGINES		
	TURBINE ENGINES (BASICS)		
	ELECTRICS		
	FIRE PROTECTION AND DETECTION SYSTEMS		
	MAINTENANCE		
3.	AIRSHIP GENERAL KNOWLEDGE: INSTRUMENTATION	x	
	SENSORS AND INSTRUMENTS		
	MEASUREMENT OF AIR DATA AND GAS PARAMETERS		
	MAGNETISM: DIRECT READING COMPASS AND FLUX VALVE		
	GYROSCOPIC INSTRUMENTS		
	COMMUNICATION SYSTEMS		
	ALERTING SYSTEMS		
	INTEGRATED INSTRUMENTS: ELECTRONIC DISPLAYS		
	FLIGHT MANAGEMENT SYSTEM (GENERAL BASICS)		
	DIGITAL CIRCUITS AND COMPUTERS		
4.	FLIGHT PERFORMANCE AND PLANNING	x	
4.1.	MASS AND BALANCE: AIRSHIPS	x	
	PURPOSE OF MASS AND BALANCE CONSIDERATIONS		
	LOADING		
	FUNDAMENTALS OF CG CALCULATIONS		
	MASS AND BALANCE DETAILS OF AIRCRAFT		
	DETERMINATION OF CG POSITION		
	PASSENGER, CARGO AND BALLAST HANDLING		

		CPL	IR
4.2.	FLIGHT PLANNING AND FLIGHT MONITORING		
	FLIGHT PLANNING FOR VFR FLIGHTS	x	
	FLIGHT PLANNING FOR IFR FLIGHTS		x
	FUEL PLANNING	x	x
	PRE-FLIGHT PREPARATION	x	x
	ATS FLIGHT PLAN	x	x
	FLIGHT MONITORING AND IN-FLIGHT RE-PLANNING	x	x
4.3.	PERFORMANCE: AIRSHIPS	x	
	AIRWORTHINESS REQUIREMENTS		
	BASICS OF AIRSHIP PERFORMANCE		
	DEFINITIONS AND TERMS		
	STAGES OF FLIGHT		
	USE OF FLIGHT MANUAL		
5.	HUMAN PERFORMANCE	x	
	HUMAN FACTORS: BASIC CONCEPTS		
	BASIC AVIATION PHYSIOLOGY AND HEALTH MAINTENANCE		
	BASIC AVIATION PSYCHOLOGY		
6.	METEOROLOGY	x	
	THE ATMOSPHERE		
	WIND		
	THERMODYNAMICS		
	CLOUDS AND FOG		
	PRECIPITATION		
	AIR MASSES AND FRONTS		
	PRESSURE SYSTEMS		
	CLIMATOLOGY		
	FLIGHT HAZARDS		
	METEOROLOGICAL INFORMATION		
7.	NAVIGATION		
7.1.	GENERAL NAVIGATION	x	
	BASICS OF NAVIGATION		
	MAGNETISM AND COMPASSES		
	CHARTS		
	DR NAVIGATION		
	IN-FLIGHT NAVIGATION		
7.2.	RADIO NAVIGATION		

		CPL	IR
	BASIC RADIO PROPAGATION THEORY	x	x
	RADIO AIDS	x	x
	RADAR	x	x
	INTENTIONALLY LEFT BLANK		
	AREA NAVIGATION SYSTEMS AND RNAV/FMS		x
	GNSS	x	x
8.	OPERATIONAL PROCEDURES AIRSHIP	x	
	GENERAL REQUIREMENTS		
	SPECIAL OPERATIONAL PROCEDURES AND HAZARDS (GENERAL ASPECTS)		
	EMERGENCY PROCEDURES		
9.	PRINCIPLES OF FLIGHT	x	
9.1.	PRINCIPLES OF FLIGHT: AIRSHIPS	x	
	BASICS OF AEROSTATICS		
	BASICS OF SUBSONIC AERODYNAMICS		
	AERODYNAMICS OF AIRSHIPS		
	STABILITY		
	CONTROLLABILITY		
	LIMITATIONS		
	PROPELLERS		
	BASICS OF AIRSHIP FLIGHT MECHANICS		
10.	COMMUNICATIONS		
10.1.	VFR COMMUNICATIONS	x	
	DEFINITIONS	x	
	GENERAL OPERATING PROCEDURES	x	
	RELEVANT WEATHER INFORMATION TERMS (VFR)	x	
	ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE	x	
	DISTRESS AND URGENCY PROCEDURES	x	
	GENERAL PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES	x	
10.2.	IFR COMMUNICATIONS		
	DEFINITIONS		x
	GENERAL OPERATING PROCEDURES		x

		CPL	IR
	ACTION REQUIRED TO BE TAKEN IN CASE OF COMMUNICATION FAILURE		x
	DISTRESS AND URGENCY PROCEDURES		x
	RELEVANT WEATHER INFORMATION TERMS (IFR)		x
	GENERAL PRINCIPLES OF VHF PROPAGATION AND ALLOCATION OF FREQUENCIES		x
	MORSE CODE		x

GM1 FCL.310; FCL.515(b); FCL.615(b); FCL.835(d) Theoretical knowledge examinations

ED Decision 2020/018/R

EXPLANATION OF THE VERBS USED IN THE BENJAMIN BLOOM TAXONOMY

- (a) The depth or level of learning to be achieved during the training and the corresponding level of attainment to be examined or assessed is based on the following taxonomy. In each case, the level of knowledge or skill is signified by the learning objective (LO) verb.
- (b) The majority of the LOs relate to the cognitive domain. The taxonomy described by B. Bloom (1956) and Anderson & Krathwohl (2001) has been used as the standard.
- (c) The six sequential increasing levels of required cognitive learning are identified by the LO verb. Hence the lowest level 'remember' is signified by verbs such as 'state', 'list', 'define' and 'recall' whilst the next higher level of 'understand' is signified by verbs such as 'describe' and 'explain'. The third level of 'apply' is signified by the verbs 'calculate', 'interpret', 'relate' and 'solve'. However, the higher levels of 'analyse', which would be signified by the verbs 'plan' or 'discuss' and 'evaluate' and 'create' are less common due at least partially to questions presently possible in the ECQB examination.
- (d) The LOs used in Area 100 KSA differ in that they require a combination of knowledge and skills. However, the 'skill' level does not relate to Bloom's psychomotor taxonomy but is more closely aligned to the higher taxonomy levels required in medicine, because knowledge and skills must be combined by the student pilot in a strategy.
- (e) The verbs 'demonstrate' and 'show', with their meanings defined below, have therefore been used to supplement the cognitive LO verbs for the Area 100 KSA LOs.
 - (1) 'Demonstrate' means the selection and use of the appropriate knowledge, skills and attitudes within a strategy to achieve an effective outcome. It signifies a high taxonomy level and would normally be assessed using multiple indicators from more than one core competency.
 - (2) 'Show' means the attainment of knowledge, skill or attitude. It signifies a lower taxonomy level than 'demonstrate' and would normally be assessed by a single indicator.'

AMC1 FCL.615(a) IR — Theoretical knowledge and flight instruction

ED Decision 2025/002/R

TRAINING ON PBN OPERATIONS AS PART OF IR TRAINING COURSES**GENERAL**

- (a) Since in accordance with point [FCL.605\(a\)](#) the privileges of an instrument rating (IR) include privileges to fly aircraft in PBN operations, the purpose of this AMC is to outline the training elements on PBN operations which should be part of IR training courses in accordance with point [FCL.615](#).
- (b) The training syllabi set out in this AMC should:
 - (1) include CRM training elements, where appropriate; and
 - (2) when applied, consider technological and regulatory developments.

THEORETICAL TRAINING

- (c) Theoretical knowledge instruction regarding PBN

This part should include basic principles, different classifications and description of structures as follows:

- (1) difference between conventional, RNAV and RNP (specifications and requirement);
- (2) factors used to define RNAV or RNP system performance requirements (accuracy, integrity, continuity and functionality);
- (3) overview of PBN specifications (e.g. RNAV 10, RNAV 1, RNP 1) and specific approvals (RNP-AR, point in space);
- (4) review of different existing approaches and 2D/3D concept;
- (5) linear and angular lateral guidance;
- (6) vertical guidance modes (RNP v advisory), associated limitations (AFM, temperature, QNH) and procedures (correct barometric altimeter setting);
- (7) information about receiver autonomous integrity monitoring (RAIM), airborne autonomous integrity monitoring (AAIM).

- (d) Theoretical knowledge instruction regarding requirements, procedures and equipment

- (1) This part should, while following the syllabus as set out in point (2), include theoretical knowledge instruction about:
 - (i) the operational procedures in accordance with Regulation (EU)
 - (ii) the review and study of existing means to manage the flight, including items such as charts, websites and apps.
- (2) Syllabus
 - (i) Operational and regulatory requirements:
 - (A) flight preparation:
 - (1) relevant navigation specification and system limitations (according to the AFM);
 - (2) airport/operating site selection (destination, alternate, minima);
 - (3) user waypoint management;

- (4) flight plan filing;
 - (5) NOTAM;
 - (6) RAIM prediction;
 - (B) in-flight:
 - (1) restriction in manual mode;
 - (2) radar vector restrictions (direct to IF, direct to FAF, intersection final);
 - (3) monitoring and checking;
 - (4) related phraseology;
- (ii) charting:
 - (A) representation of PBN-specific elements on routes, SID STAR, APCH (including steep angle), waypoints (fly-by and fly-over), constraint altitude/speed, navaid);
 - (B) merged intermediate fix (IF) and initial approach fix (IAF);
 - (C) waypoint naming concept;
 - (D) path terminator concept with a focus on course to fix (CF), track to fix (TF), radius to fix (RF);
 - (E) minima;
- (iii) contingency procedures:
 - (A) go-around procedure management;
 - (B) degradation or loss of the ability to operate in RNP;
 - (C) related phraseology.

PRACTICAL TRAINING**(e) Ground practical training**

- (1) Ground practical training aims at familiarisation with the aircraft on-board systems and should, while following the syllabus as set out in point (2), cover the following items:
 - (i) a theoretical overview of the on-board systems with a view to establishing links between the theory and the system to be used;
 - (ii) operating and setting a navigation system similar to the system on board of the aircraft to be operated; and
 - (iii) all functionalities provided by the navigation system likely to be used in conducting normal or abnormal procedures.

Note 1: For the purpose of ground practical training, the equipment used may be presented or managed on a computer system, on a simulation bench, on an FSTD or on an aircraft on the ground.

Note 2: The duration of this training depends on the complexity of the RNP system used and the recommendations of the manufacturer.

(2) Syllabus

- (i) aircraft on-board system:

- (A) review of basic functions (continuous indication of lateral deviation; distance/bearing to active waypoint, glideslope or time to active waypoint, navigation data storage);
 - (B) system limitations;
 - (C) failure indications and error messages;
 - (D) verification and sensor management;
 - (E) know-how:
 - (1) addressing discontinuities;
 - (2) set an 'offset flight path';
 - (3) select/delete satellite;
 - (4) manage the navigation database, retrieving a procedure from the database;
 - (5) entering data such as wind, altitude/speed constraints, vertical profile/speed;
 - (6) tactically modifying the flight plan;
 - (ii) monitoring and checking (lateral and vertical) during 2D/3D approaches;
 - (iii) contingency procedures management including aircraft failures;
 - (iv) use of automation at different stages of the procedure.
- (f) Practical training in the aircraft or FSTD
- Flight training should:
- (1) be conducted with a flight crew composition which complies with the applicable minimum flight crew composition requirements for conducting PBN operation as mentioned by the training manual and operations manual of the ATO; and
 - (2) include flight exercises that allow the applicant to do all of the following:
 - (i) apply the theoretical knowledge as specified in points (c), (d) and (e) in practice;
 - (ii) build competence in applying normal, abnormal and emergency procedures in PBN operation during both en-route and approach phases, including interruptions that are caused by the simulation of an abnormal situation (e.g. loss of RNP capacity or RAIM warning, if applicable).

AMC1 FCL.615(b) IR – Theoretical knowledge and flight instruction

ED Decision 2025/002/R

SYLLABUS OF THEORETICAL KNOWLEDGE FOR THE IR FOLLOWING THE COMPETENCY-BASED MODULAR COURSE AND BIR

- (a) The syllabus for the theoretical knowledge instruction and examination for the ATPL, MPL, CPL and IR in [AMC1 FCL.310](#); [FCL.515\(b\)](#); [FCL.615\(b\)](#), [FCL.835\(d\)](#) should be used for the CB-IR(A) and the BIR respectively.
- (b) Aspects related to threat and error management (TEM) should be included in an integrated manner, taking into account the particular risks associated to the licence and the activity.

- (c) An applicant who has completed a modular IR(A) course in accordance with [Appendix 6](#) Section A and passed the IR(A) theoretical knowledge examination should be fully credited towards the requirements of theoretical knowledge instruction and examination for a competency-based IR(A) within the validity period of the examination. An applicant wishing to transfer to a competency-based IR(A) or BIR course during a modular IR(A) course should be credited towards the requirements of theoretical knowledge instruction and examination for a competency-based IR(A) or BIR for those subjects or theory items already completed.

FCL.620 IR – Skill test

Regulation (EU) 2021/2227

Applicants for an IR shall pass a skill test in accordance with [Appendix 7](#) to this Annex to demonstrate their ability to perform the relevant procedures and manoeuvres with a degree of competency appropriate to the privileges granted.

FCL.625 IR – Validity, revalidation and renewal

Regulation (EU) 2020/2193

(a) Validity

An IR shall be valid for 1 year.

(b) Revalidation

- (1) An IR shall be revalidated within the 3 months immediately preceding its expiry date by complying with the revalidation criteria for the relevant aircraft category.
- (2) If applicants choose to fulfil the revalidation requirements earlier than prescribed in point (1), the new validity period shall commence from the date of the proficiency check.
- (3) Applicants who fail to pass the relevant section of an IR proficiency check before the expiry date of the IR shall exercise the IR privileges only if they have passed the IR proficiency check.
- (4) Applicants for the revalidation of an IR shall receive full credits for the proficiency check as required in this Subpart when they complete EBT practical assessment in accordance with [Appendix 10](#) related to the IR at an EBT operator.

(c) Renewal

If an IR has expired, in order to renew their privileges, applicants shall comply with all of the following:

- (1) in order to determine whether refresher training is necessary for the applicant to reach the level of proficiency needed to pass the instrument element of the skill test in accordance with [Appendix 9](#), they shall undergo an assessment at either of the following organisations:
 - (i) at an ATO;
 - (ii) at an EBT operator that is specifically approved for such refresher training;
- (2) if deemed necessary by the organisation providing the assessment in accordance with point (1), they shall complete refresher training at that organisation;
- (3) after complying with point (1) and, as applicable, point (2), they shall pass a proficiency check in accordance with [Appendix 9](#) or complete EBT practical assessment in accordance

with [Appendix 10](#) in the relevant aircraft category. That EBT practical assessment may be combined with the refresher training specified in point (2);

- (4) they shall hold the relevant class or type rating unless specified otherwise in this Annex.
- (d) If the IR has not been revalidated or renewed in the preceding 7 years, applicants for the IR shall pass again the IR theoretical knowledge examination and skill test.
- (e) Holders of a valid IR on a pilot licence issued by a third country in accordance with Annex 1 to the Chicago Convention shall be exempted from complying with the requirements in points (c)(1), (c)(2) and (d) when renewing the IR privileges contained in licences issued in accordance with this Annex.
- (f) The proficiency check specified in point (c)(3) may be combined with a proficiency check performed for the renewal of the relevant class or type rating.

AMC1 FCL.625(c) IR – Validity, revalidation and renewal

ED Decision 2017/022/R

RENEWAL OF INSTRUMENT RATING AT AN APPROVED TRAINING ORGANISATION (ATO): REFRESHER TRAINING

- (a) The objective of the refresher training at an ATO is to reach the level of proficiency needed to pass the instrument rating proficiency check, as described in [Appendix 9](#), or the instrument rating skill test as described in [Appendix 7](#) to Part-FCL, as applicable. The amount of refresher training needed should be determined by the ATO on a case-by-case basis, taking into account the following factors:
 - (1) the experience of the applicant;
 - (2) the amount of time elapsed since the privileges of the rating were last used;
 - (3) the complexity of the aircraft;
 - (4) whether the applicant has a current rating on another aircraft type or class; and
 - (5) where considered necessary, the performance of the applicant during a simulated proficiency check for the rating in a flight simulation training device (FSTD) or an aircraft of a relevant type or class.

The amount of training needed to reach the desired level of competency should increase with the time elapsed since the privileges of the rating were last used.

- (b) Once the ATO has determined the needs of the applicant, it should develop an individual training programme based on the ATO's approved course for the rating and focusing on those aspects where the applicant has shown the greatest needs. Theoretical-knowledge instruction should be included, as necessary. The performance of the applicant should be reviewed during the training, and additional instruction should be provided where necessary to reach the standard required for the proficiency check.
- (c) After successful completion of the training, the ATO should provide a training completion certificate to the applicant, which describes the evaluation of the factors listed under (a) above and the training received, and includes a statement that the training has been successfully completed. The training completion certificate should be presented to the examiner prior to the proficiency check. Following the successful renewal of the rating, the training completion certificate and examiner report form should be submitted to the competent authority.

- (d) Taking into account the factors listed in (a) above, an ATO may also decide that the applicant already possesses the required level of proficiency and that no refresher training is necessary. In such a case, the certificate or other documental evidence referred to in point (c) above should contain a respective statement including sufficient reasoning.

SECTION 2 – SPECIFIC REQUIREMENTS FOR THE AEROPLANE CATEGORY

FCL.620.A IR(A) – Skill test

Regulation (EU) 2021/2227

- (a) For a multi-engine IR(A), the skill test shall be taken in a multi-engine aeroplane. For a single-engine IR(A), the test shall be taken in a single-engine aeroplane. A multi-engine centreline thrust aeroplane shall be considered a single-engine aeroplane for the purposes of this point.
- (b) Applicants who have completed a skill test for a multi-engine IR(A) in a single-pilot multi-engine aeroplane for which a class rating is required shall also be issued with a single-engine IR(A) for the single-engine aeroplane class or type ratings that they hold.

FCL.625.A IR(A) – Revalidation

Regulation (EU) 2020/2193

- (a) Revalidation.
To revalidate an IR(A), applicants shall:
 - (1) hold the relevant class or type rating, unless the IR revalidation is combined with the renewal of the relevant class or type rating;
 - (2) pass a proficiency check in accordance with [Appendix 9](#), or complete EBT practical assessment in accordance with [Appendix 10](#), if the IR revalidation is combined with the revalidation of a class or type rating;
 - (3) if the IR revalidation is not combined with the revalidation of a class or type rating:
 - (i) for single-pilot aeroplanes, complete section 3b and those parts of section 1 which are relevant to the intended flight of the proficiency check in accordance with [Appendix 9](#) to this Annex;
 - (ii) for multi-engine aeroplanes, complete section 6 of the proficiency check for single-pilot aeroplanes in accordance with [Appendix 9](#) to this Annex by sole reference to instruments.
 - (4) An FNPT II or an FFS representing the relevant class or type of aeroplane may be used for the revalidation pursuant to point (3), provided that at least each alternate proficiency check for the revalidation of an IR(A) is performed in an aeroplane.
- (b) Cross-credit shall be given in accordance with [Appendix 8](#) to this Part.

SECTION 3 – SPECIFIC REQUIREMENTS FOR THE HELICOPTER CATEGORY

FCL.625.H IR(H) – Revalidation

Regulation (EU) 2019/1747

- (a) To revalidate an IR(H), applicants shall:
- (1) hold the relevant type rating, unless the IR revalidation is combined with the renewal of the relevant type rating;
 - (2) pass a proficiency check in accordance with Appendix 9 to this Annex for the relevant type of helicopter if the IR revalidation is combined with the revalidation of a type rating;
 - (3) if the IR revalidation is not combined with the revalidation of a type rating, complete Section 5 and the relevant parts of Section 1 of the proficiency check in accordance with Appendix 9 to this Annex for the relevant type of helicopter.
- (b) An FTD 2/3 or an FFS representing the relevant type of helicopter may be used for the proficiency check pursuant to point (a)(3), provided that at least each alternate proficiency check for the revalidation of an IR(H) is performed in a helicopter.
- (c) Cross-credit shall be given in accordance with [Appendix 8](#) to this Annex.

FCL.630.H IR(H) – Extension of the privileges of an IR(H) to further helicopter types

Regulation (EU) 2021/2227

Unless specified otherwise in the operational suitability data established in accordance with Annex I (Part 21) to [Commission Regulation \(EU\) No 748/2012](#), holders of an IR(H) who wish to extend their IR(H) privileges to further helicopter types shall, in addition to the relevant type rating training course, complete at an ATO 2 hours of flight training on the relevant type by sole reference to instruments according to IFR which may be conducted in an FFS or an FTD which appropriately represents the relevant type for IFR operation.

AMC1 FCL.630.H IR(H) — Extension of the privileges of an IR(H) to further helicopter types

ED Decision 2022/014/R

APPROPRIATE FFS OR FTD

The appropriate FSTD should be a FFS C/D or a FTD 2/3.

SECTION 4 – SPECIFIC REQUIREMENTS FOR THE AIRSHIP CATEGORY

FCL.620.As IR(As) – Skill test

Regulation (EU) 2021/2227

For a multi-engine IR(As), the skill test shall be taken in a multi-engine airship. For a single-engine IR(As), the test shall be taken in a single-engine airship.

FCL.625.As IR(As) – Revalidation

Regulation (EU) No 1178/2011

Applicants for the revalidation of an IR(As):

- (a) when combined with the revalidation of a type rating, shall complete a proficiency check in accordance with [Appendix 9](#) to this Part, for the relevant type of airship;
- (b) when not combined with the revalidation of a type rating, shall complete section 5 and those parts of section 1 relevant to the intended flight of the proficiency check for airships in accordance with [Appendix 9](#) of this part. In this case, an FTD 2/3 or FFS representing the relevant type may be used, but at least each alternate proficiency check for the revalidation of an IR(As) in these circumstances shall be performed in an airship.

SUBPART H – CLASS AND TYPE RATINGS

SECTION 1 – COMMON REQUIREMENTS

FCL.700 Circumstances in which class or type ratings are required

Regulation (EU) 2025/134

- (a) Holders of a pilot licence shall act as pilots of an aircraft only if they have a valid and appropriate class or type rating, unless any of the following applies:
- (1) if exercising the privileges of an LAPL;
 - (2) if they take skill tests or proficiency checks for renewal of class or type ratings;
 - (3) if they receive flight instruction;
 - (4) if they hold a flight test rating issued in accordance with point [FCL.820](#).

[applicable until 17 February 2026 - Regulation (EU) 2020/359]

- (a) Holders of a pilot licence shall act as pilots of an aircraft only if they have a valid and appropriate class or type rating, unless any of the following applies:
- (1) they exercise the privileges of an LAPL or a GPL;
 - (2) they take skill tests or proficiency checks for renewal of class or type ratings;
 - (3) they receive flight instruction;
 - (4) they hold a flight test rating issued in accordance with point [FCL.820](#).

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (b) Notwithstanding (a), in the case of flights related to the introduction or modification of aircraft types, pilots may hold a special certificate given by the competent authority, authorising them to perform the flights. This authorisation shall have its validity limited to the specific flights.

AMC1 FCL.700 Circumstances in which class or type ratings are required

ED Decision 2020/005/R

- (a) A class or type rating and license endorsement should comply with the class and type ratings that are listed in one of the following EASA publications, as applicable:
- (1) ‘List of Aeroplanes — Class and Type Ratings and Endorsement List’; and
 - (2) ‘List of Helicopters — Type Ratings List’.
- (b) Holders of Part-FCL licences should complete differences training or familiarisation training in accordance with the lists of point (a).

GM1 FCL.700 Circumstances in which class or type ratings are required

ED Decision 2025/002/R

LIST OF CLASS OR TYPE RATINGS

The following tables contain lists of aeroplanes or TMG that are included in class ratings.

(a) Class ratings (aeroplane): SP and SEP or MEP aeroplane (land or sea):

Manufacturer	Aeroplanes		Licence Endorsement
All manufacturers	SEP (land)	(D)	SEP (land)
	SEP (land) with variable pitch propellers		
	SEP (land) with retractable undercarriage		
	SEP (land) with turbocharged or supercharged engines		
	SEP (land) with cabin pressurisation		
	SEP (land) with tail wheels		
	SEP (land) with EFIS		
	SEP (land) with SLPC		
	SEP (land) with an electric engine		
	SEP (land) with multiple electric engines (Article 2(8c)(b))		
	SEP (land) with a hybrid engine (Article 2(8c)(c))		
All manufacturers	SEP (sea)	(D)	SEP (sea)
	SEP (sea) with variable pitch propellers		
	SEP (sea) with turbocharged or supercharged engines		
	SEP (sea) with cabin pressurisation		
	SEP (sea) with EFIS		
	SEP (sea) with SLPC		
	SEP (sea) with an electric engine		
	SEP (sea) with multiple electric engines (Article 2(8c)(b))		
	SEP (sea) with a hybrid engine (Article 2(8c)(c))		
	All manufacturers		
MEP (sea)		(D)	MEP (sea)

(b) Class ratings (aeroplane): SP and SEP TMG (land):

Manufacturer	Aeroplanes		Licence Endorsement
All manufacturers	All TMGs having an integrally mounted, non-retractable engine and a non-retractable propeller		TMG

(c) Additional class and type rating lists and endorsement lists are published by the Agency.

(d) Whenever (D) is indicated in one of the lists mentioned in paragraphs (a) to (c), it indicates that differences training in accordance with [FCL.710](#) is required.

FCL.705 Privileges of the holder of a class or type rating

Regulation (EU) No 1178/2011

The privileges of the holder of a class or type rating are to act as pilot on the class or type of aircraft specified in the rating.

FCL.710 Class and type ratings – variants

Regulation (EU) 2024/2076

- (a) Pilots shall complete differences training or familiarisation in order to extend their privileges to another variant of aircraft within a class or type rating, or, in the case of point FCL.710(d)(1) of this Annex, in order to maintain those privileges. In the case of variants within a class or type rating, the differences training or familiarisation shall include the relevant elements defined in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012, where applicable. When extending the privileges of an SEP aeroplane class rating to a variant with another type of engine referred to in Article 2, point (8c), of this Regulation, the differences training shall consist of dual flight instruction and theoretical knowledge instruction which shall include, with regard to that other type of engine and related aircraft systems, at least all of the following subjects:
- (1) Aircraft general knowledge;
 - (2) Operational procedures;
 - (3) Flight performance and planning.
- (b) The differences training shall be conducted at any of the following:
- (1) an ATO;
 - (2) a DTO in the case of aircraft referred to in points (a)(1)(c) and (a)(2)(c) of point DTO.GEN.110 of Annex VIII;
 - (3) an AOC holder having an approved differences training programme for the relevant class or type.
- (c) Notwithstanding the requirement in point (b), differences training for the following aircraft may be conducted by an appropriately qualified instructor unless otherwise provided in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012:
- (1) aeroplanes requiring a class rating:
 - (i) TMGs;
 - (ii) SEP aeroplanes;
 - (iii) MEP aeroplanes;
 - (iv) SET aeroplanes, except for complex or high-performance SET aeroplanes;
 - (2) single-engine helicopters with a maximum take-off mass not exceeding 3 175 kg.
- (d) Pilots who, after having extended their privileges to another variant within a class or type rating in accordance with this point, have not flown that variant within the preceding 2 years shall, before exercising their privileges in that variant, complete any of the following in that variant:
- (1) further differences training in accordance with points (a) and (b) or, if applicable, (c);
 - (2) a proficiency check;
 - (3) in the case of a variant within the SEP class rating with a particular engine type referred to in Article 2, point (8c), refresher training in accordance with point [FCL.740.A\(b\)\(1\)\(ii\)\(C\)](#).

- (da) Point FCL.710(d) shall not apply in the case of variants within the TMG class rating and variants within the SEP class rating that have the same type of engine referred to in [Article 2](#), point (8c), of this Regulation.
- (e) The differences training or the proficiency check in that variant shall be entered in the pilots' logbook or equivalent record and signed by the instructor or examiner as appropriate.

GM1 FCL.710 Class and type ratings – variants

ED Decision 2025/002/R

DIFFERENCES TRAINING AND FAMILIARISATION

- (a) Differences training requires the acquisition of additional knowledge and training on an appropriate training device or the aircraft.
- (b) Familiarisation requires the acquisition of additional knowledge.

AMC1 FCL.710(a) Class and type ratings — variants

ED Decision 2025/002/R

DIFFERENCES TRAINING FOR VARIANTS WITHIN THE SEP AEROPLANE CLASS WITH AN ELECTRIC ENGINE SYSTEM

As regards electric engines, the differences training should, on the basis of the AFM of the relevant aircraft, include at least all of the following:

(Note: The following list solely addresses the training needs with regard to electric engines and must not be understood as an exhaustive differences training syllabus.)

- (a) Theoretical knowledge
 - (1) Aircraft general knowledge:
 - (i) electric engine;
 - (ii) high-voltage batteries, including their storage;
 - (iii) electric engine and battery indication and monitoring instruments;
 - (iv) electric engine and battery limitations;
 - (v) electric battery endurance;
 - (vi) electric battery life and effects of battery age on performance.
 - (2) Operational procedures:
 - (i) electric engine and battery pre-flight inspection;
 - (ii) charging of the battery;
 - (iii) charging errors;
 - (iv) long-term battery storage procedures and limitations;
 - (v) in-flight engine/energy monitoring and management;
 - (vi) go-around with minimum energy;
 - (vii) emergency procedures:
 - (A) engine/battery fire on the ground and in the air;

- (B) engine/battery overheat;
 - (C) partial and complete loss of engine power (CLOP);
 - (D) loss of cooling function;
 - (E) failure/loss of engine/battery instruments;
 - (F) in-flight engine restart procedure.
- (3) Flight planning:
 - (i) energy consumption for all phases of the flight;
 - (ii) mission and contingency planning;
 - (iii) minimum reserve energy planning.
- (b) Flight instruction

The dual flight instruction should include the practical exercise or, in the case of emergency procedures, a simulation of all the elements listed in point (a).

FCL.725 Requirements for the issue of class and type ratings

Regulation (EU) 2024/2076

- (a) Training course. An applicant for a class or type rating shall complete a training course at an ATO. An applicant for a non-high-performance SEP aeroplane class rating, a TMG class rating or a single-engine type rating for helicopters referred to in point [DTO.GEN.110\(a\)\(2\)\(c\)](#) of Annex VIII (Part-DTO) to this Regulation may complete the training course at a DTO. The type rating training course shall include the mandatory training elements for the relevant type as defined in the operational suitability data established in accordance with Annex I (Part 21) to [Regulation \(EU\) No 748/2012](#).
- (b) Theoretical knowledge examination. The applicant for a class or type rating shall pass a theoretical knowledge examination organised by the ATO to demonstrate the level of theoretical knowledge required for the safe operation of the applicable aircraft class or type.
 - (1) For multi-pilot aircraft, the theoretical knowledge examination shall be written and comprise at least 100 multiple-choice questions distributed appropriately across the main subjects of the syllabus.
 - (2) For single-pilot multi-engine aircraft, the theoretical knowledge examination shall be written and the number of multiple-choice questions shall depend on the complexity of the aircraft.
 - (3) For single-engine aircraft, the theoretical knowledge examination shall be conducted verbally by the examiner during the skill test to determine whether or not a satisfactory level of knowledge has been achieved.
 - (4) For single-pilot aeroplanes that are classified as high performance aeroplanes, the examination shall be written and comprise at least 100 multiple-choice questions distributed appropriately across the subjects of the syllabus.
 - (5) For single-pilot single-engine and single-pilot multi-engine aeroplanes (sea), the examination shall be in a written form and shall comprise at least 30 multiple-choice questions.

- (c) Skill test. An applicant for a class or type rating shall pass a skill test in accordance with [Appendix 9](#) to this Part to demonstrate the skill required for the safe operation of the applicable class or type of aircraft.

The applicant shall pass the skill test within a period of 6 months after commencement of the class or type rating training course and within a period of 6 months preceding the application for the issue of the class or type rating.

- (d) Single- and multi-pilot operations in single-pilot aircraft – type ratings

- (1) The privileges of holders of a type rating for a single-pilot aircraft include the privileges to fly the aircraft in single-pilot operations and multi-pilot operations. However, such holders shall exercise those privileges for a particular form of operation only if they comply with all of the following:

- (i) they have successfully completed flight training for the relevant form of operation in the relevant aircraft type in accordance with Appendix 9 to this Annex, unless specified otherwise in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012. That flight training for a particular form of operation shall either be included in the initial type rating training course at an ATO or, following the initial issuance of the type rating, be completed in the form of additional training at either of the following:

- (A) an ATO;
- (B) an organisation that is subject to Annex III (Part-ORO) to Regulation (EU) No 965/2012 and that is entitled to provide such training on the basis of either an approval or a declaration;

- (ii) they have completed a skill test or a proficiency check:

- (A) for single-pilot operations in one of the following ways:

- (1) in single-pilot operations;
- (2) in multi-pilot operations with additional elements for single-pilot operations as specified in [Appendix 9](#);

- (B) for multi-pilot operations, in multi-pilot operations.

- (iii) additionally, in the case of multi-pilot operations in single-pilot aircraft:

- (A) they meet the requirements that are specified in:

- (1) point [FCL.720.A\(b\)\(4\)](#) or point [FCL.720.H\(a\)\(2\)](#), as applicable;
- (2) point [FCL.720.A\(b\)\(5\)](#), for single-pilot aeroplanes, before starting flight training for multi-pilot operations in accordance with point [FCL.725\(d\)\(1\)\(i\)](#);

- (B) they exercise their privileges only at an organisation that is subject to Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#).

- (2) When applicants for the initial issue of a type rating for a single-pilot aircraft complete the flight training and the skill test in multi-pilot operations only, the type rating shall be issued with a restriction to multi-pilot operations. That restriction shall be removed when applicants complete, in accordance with [Appendix 9](#), additional training and a proficiency check that include the necessary elements for single-pilot operations.

In all other cases, the form of operation shall not be entered onto the licence.

(da) Single- and multi-pilot operations – aeroplane class ratings

The privileges of holders of a class rating for a single-pilot aeroplane include the privileges to fly the aeroplane in multi-pilot operations, provided that such holders:

- (1) comply with the requirements set out in:
 - (i) point [FCL.720.A\(b\)\(4\)](#);
 - (ii) point [FCL.720.A\(b\)\(5\)](#), before starting flight training for multi-pilot operations in accordance with point [FCL.725\(da\)\(2\)](#);
- (2) have completed flight training and passed a proficiency check for multi-pilot operations in the relevant aeroplane class in accordance with Section B, point 5(g), of [Appendix 9](#) at an organisation that is specified in point [FCL.725\(d\)\(1\)\(i\)](#); and
- (3) exercise their privileges only at an organisation that is subject to Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#).

Additional privileges for multi-pilot operations in a single-pilot aeroplane class shall not be entered onto the licence.

(db) If a skill test or proficiency check for a single-pilot aircraft class or type rating is conducted in either of the following, the form or forms of operation in which that skill test or a proficiency check is conducted shall be entered in the logbook of the applicants and signed by the examiner:

- (1) multi-pilot operations;
- (2) single-pilot and multi-pilot operations.

(e) Notwithstanding the paragraphs above, pilots holding a flight test rating issued in accordance with [FCL.820](#) who were involved in development, certification or production flight tests for an aircraft type, and have completed either 50 hours of total flight time or 10 hours of flight time as PIC on test flights in that type, shall be entitled to apply for the issue of the relevant type rating, provided that they comply with the experience requirements and the prerequisites for the issue of that type rating, as established in this Subpart for the relevant aircraft category.

(f) Applicants for a class rating for TMGs who also hold an SPL in accordance with Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), including the privileges to fly on TMGs, shall receive full credits towards the requirements in paragraphs (a), (b) and (c).

AMC1 FCL.725(a) Requirements for the issue of class and type ratings

ED Decision 2025/002/R

SYLLABUS OF THE THEORETICAL KNOWLEDGE FOR CLASS AND TYPE RATINGS

I. TMGs, SE AND ME AEROPLANES

- (a) Detailed listing for aircraft structure and equipment (as applicable), normal operation of systems and malfunctions:
 - (1) dimensions: minimum required runway width for 180 ° turn.
 - (2) engine, including auxiliary power unit:
 - (i) type of engine or engines;
 - (ii) in general, function of the following systems or components:

- (A) engine;
- (B) auxiliary power unit;
- (C) oil system;
- (D) storage and distribution system for fuel/energy;
- (E) ignition system;
- (F) starting system;
- (G) engine/battery cooling system;
- (H) fire-warning and fire-extinguishing system;
- (I) generators and generator drives;
- (J) power indication;
- (K) reverse thrust;
- (L) water injection.
- (iii) for propeller engines, the functioning of the following systems:
 - (A) propeller system;
 - (B) feathering system.
- (iv) engine/battery controls (including starter), engine instruments and indications in the cockpit, their function, interrelation and interpretation;
- (v) engine operation, including APU, during engine start, start and engine malfunctions, procedures for normal operation in the correct sequence.
- (3) fuel/energy system:
 - (i) location of the fuel tanks, fuel pumps, fuel lines to the engines, tank capacities, valves and measuring;
 - (ia) location of the batteries and integration into the aircraft systems;
 - (ii) location of the following systems:
 - (A) filtering;
 - (B) heating;
 - (C) fuelling and defueling/charging;
 - (D) dumping;
 - (E) venting;
 - (F) cooling.
 - (iii) in the cockpit:
 - (A) monitors and indicators of the fuel/energy system;
 - (B) quantity and flow indication, and interpretation.
 - (iv) procedures:
 - (A) fuelling/charging procedures. including distribution of fuel/energy into the various tanks/batteries;

- (B) fuel supply, fuel temperature control, and fuel dumping.
- (4) pressurisation and air conditioning:
 - (i) components of the system and protection devices;
 - (ii) cockpit monitors and indicators;
 - (iii) interpretation of the operational condition;
 - (iv) normal operation of the system during start, cruise, approach and landing, air-conditioning airflow and temperature control.
- (5) ice and rain protection, windshield wipers and rain-repellent system:
 - (i) ice-protected components of the aircraft including engines, heat sources, controls and indications;
 - (ii) operation of the anti-icing or de-icing system during take-off, climb, cruise and descent, conditions that require the use of the protection systems;
 - (iii) controls and indications of the windshield wipers and rain-repellent system operation.
- (6) hydraulic system:
 - (i) components of the hydraulic system(s), quantities and system pressure, hydraulically actuated components associated to the respective hydraulic system;
 - (ii) controls, monitors and indicators in the cockpit, function and interrelation and interpretation of indications.
- (7) landing gear:
 - (i) main components of the:
 - (A) main landing gear;
 - (B) nose landing gear;
 - (C) gear steering;
 - (D) wheel brake system, including anti-skid.
 - (ii) gear retraction and extension (including changes in trim and drag caused by gear operation);
 - (iii) required tyre pressure, or location of the relevant placard;
 - (iv) controls and indicators including warning indicators in the cockpit in relation to the retraction or extension condition of the landing gear and brakes;
 - (v) components of the emergency extension system.
- (8) flight controls and high-lift devices:
 - (i) (A) aileron system;
 - (B) elevator system;
 - (C) rudder system;
 - (D) trim system;
 - (E) spoiler system;

- (F) lift devices;
 - (G) stall warning system;
 - (H) take-off configuration warning system;
 - (I) energy recuperation function.
 - (ii) components of the flight control system, including their functioning (from the cockpit controls to the flight control or surfaces);
 - (iii) controls, monitors and indicators, including warning indicators of the systems mentioned under (8)(i), interrelation and dependencies.
- (9) electrical power supply:
- (i) number, power, voltage, frequency and location of the main power system (AC or DC), auxiliary power system location and external power system;
 - (ii) location of the controls, monitors and indicators in the cockpit;
 - (iii) flight instruments, communication and navigation systems, main and backup power sources;
 - (iv) location of vital circuit breakers;
 - (v) generator/voltage level convertor operation and monitoring procedures of the electrical power supply.
- (10) flight instruments, communication, radar and navigation equipment, autoflight and flight data recorders:
- (i) visible antennae;
 - (ii) controls and instruments of the following equipment in the cockpit during normal operation:
 - (A) flight instruments;
 - (B) flight management systems;
 - (C) radar equipment, including radio altimeter;
 - (D) communication and navigation systems;
 - (E) autopilot;
 - (F) flight data recorder, cockpit voice recorder and data link communication recording function;
 - (G) TAWS;
 - (H) collision avoidance system;
 - (I) warning systems; and
 - (J) weather radar system, best practices for optimum use, interpretation of displayed information.
- (11) cockpit, cabin and cargo compartment:
- (i) operation of the exterior, cockpit, cabin and cargo compartment lighting and the emergency lighting;
 - (ii) operation of the cabin and cargo doors, stairs, windows and emergency exits;

- (iii) main components of the oxygen system and their location, oxygen masks and operation of the oxygen systems for the crew and passengers, required amount of oxygen by means of a table or diagram.
 - (12) emergency equipment operation and correct application of the following emergency equipment in the aircraft:
 - (i) handheld fire extinguisher;
 - (ii) first-aid kits;
 - (iii) portable oxygen equipment;
 - (iv) emergency ropes;
 - (v) life jacket;
 - (vi) life rafts;
 - (vii) emergency transmitters;
 - (viii) crash axes;
 - (ix) megaphones;
 - (x) emergency signals.
 - (13) pneumatic system:
 - (i) components of the pneumatic system, pressure source and actuated components;
 - (ii) controls, monitors and indicators in the cockpit and function of the pneumatic system;
 - (iii) vacuum system.
- (b) Limitations:
 - (1) general limitations:
 - (i) certification of the aircraft, category of operation, noise certification and maximum and minimum performance data for all flight profiles, conditions and aircraft systems:
 - (A) maximum tail and crosswind-components at take-off and landing;
 - (B) maximum speeds for flap extension v_{fo} ;
 - (C) at various flap settings v_{fe} ;
 - (D) for landing gear operation v_{lo} , M_{lo} ;
 - (E) for extended landing gear v_{le} , M_{le} ;
 - (F) for maximum rudder deflection v_a , M_a ;
 - (G) for tyres;
 - (H) one propeller feathered.
 - (ii)
 - (A) minimum control air speed v_{mca} ;
 - (B) minimum control ground speed v_{mcg} ;
 - (C) stall speed under various conditions v_{so} , v_{s1} ;
 - (D) maximum speed v_{ne} , M_{ne} ;

- (E) maximum speed for normal operation v_{mo} , M_{mo} ;
 - (F) altitude and temperature limitations;
 - (G) stick shaker activation.
- (iii)
 - (A) maximum airport pressure altitude, runway slope;
 - (B) maximum taxi mass;
 - (C) maximum take-off mass;
 - (D) maximum lift-off mass;
 - (E) maximum landing mass;
 - (F) maximum zero fuel mass;
 - (G) maximum dumping speed v_{dco} , M_{dco} , v_{dce} , M_{dce} ;
 - (H) maximum load factor during operation;
 - (I) certified range of centre of gravity.
- (2) engine limitations (as applicable):
 - (i) operating data of the engines:
 - (A) time limits and maximum temperatures;
 - (B) minimum RPMs and temperatures;
 - (C) time limits and maximum values for take-off and go-around on pressure altitude or flight altitude and temperature for:
 - (1) maximum torque; and/or
 - (2) maximum power;
 - (D) piston engines: certified range of mixture;
 - (E) minimum and maximum oil temperature and pressure;
 - (F) maximum starter time and required cooling;
 - (G) time between two start attempts for engines and auxiliary power unit;
 - (H) for propeller: maximum RPM of the propeller for triggering the automatic feathering device.
 - (ii) certified oil grades.
- (3) systems' limitations (as applicable):
 - (i) operating data of the following systems:
 - (A) pressurisation, air conditioning maximum pressures;
 - (B) electrical power supply, maximum load of main power system (AC or DC);
 - (C) maximum time of battery power supply in case of emergency;
 - (D) Mach trim system and yaw damper speed limits;
 - (E) autopilot limitations of various modes;
 - (F) ice protection;

- (G) speed and temperature limits of window heat;
 - (H) temperature limits of engine and wing anti-ice;
 - (I) maximum value of power recuperation.
 - (ii) fuel system: certified fuel specifications, minimum and maximum fuel pressures and fuel temperature.
 - (iii) energy system:
 - (A) minimum and maximum state of charge of the battery;
 - (B) effects of temperature on the battery and battery operating temperatures;
 - (C) minimum and maximum battery pack voltage;
 - (D) minimum state of battery health;
 - (E) maximum system power output with one or more batteries inoperative;
 - (F) maximum charging power.
 - (4) minimum equipment list.
- (c) Performance, flight planning and monitoring (as applicable):
 - (1) performance calculation about speeds, gradients, masses in all conditions for take-off, en-route, approach and landing according to the documentation available (for example for take-off v_1 , v_{mbe} , v_r , v_{lof} , v_2 , take-off distance, maximum take-off mass and the required stop distance) on the following factors:
 - (i) accelerate or stop distance;
 - (ii) take-off run and distance available (TORA, TODA);
 - (iii) ground temperature, pressure altitude, slope, wind;
 - (iv) maximum load and maximum mass (for example, ZFM);
 - (v) minimum climb gradient after engine failure or battery malfunction;
 - (vi) influence of snow, slush, moisture and standing water on the runway;
 - (vii) possible single or dual engine failure during cruise flight;
 - (viii) use of anti-icing systems;
 - (ix) failure of the water injection system or the anti-skid system;
 - (x) speeds at reduced thrust, v_1 , v_{1red} , v_{mbe} , v_{mu} , v_r , v_{lof} , v_2 ;
 - (xi) safe approach speed v_{ref} on v_{mca} and turbulent conditions;
 - (xii) effects of excessive approach speed and abnormal glideslope on the landing distance;
 - (xiii) minimum climb gradient during approach and landing;
 - (xiv) limiting values for a go-around with minimum fuel/energy;
 - (xv) maximum allowable landing mass and landing distance for the destination and alternate aerodrome on the following factors:
 - (A) landing distance available;
 - (B) ground temperature, pressure altitude, runway slope and wind;

- (C) fuel/energy consumption to destination or alternate aerodrome;
 - (D) influence of moisture, snow, slush and standing water on the runway;
 - (E) failure of the water injection system or the anti-skid system;
 - (F) influence of thrust reverser and spoilers.
- (2) flight planning for normal and abnormal conditions:
 - (i) optimum or maximum flight level;
 - (ii) minimum required flight altitude;
 - (iii) drift-down procedure after engine failure during cruise flight;
 - (iv) power setting of the engines during climb, cruise and holding under various circumstances, as well as the most economic cruising flight level;
 - (v) calculation of a short-range or long-range flight plan;
 - (vi) optimum and maximum flight level and power setting of the engines after engine failure;
 - (vii) effects of battery ageing on available energy and power.
- (3) flight monitoring.
- (d) Load and balance and servicing (as applicable):
 - (1) load and balance:
 - (i) load and trim sheet on the maximum masses for take-off and landing;
 - (ii) centre-of-gravity limits;
 - (iii) influence of fuel consumption on the centre of gravity;
 - (iv) lashing points, load clamping, maximum ground load.
 - (2) servicing on ground, servicing connections for:
 - (i) fuel/energy;
 - (ii) oil;
 - (iii) water;
 - (iv) hydraulic;
 - (v) oxygen;
 - (vi) nitrogen;
 - (vii) conditioned air;
 - (viii) electric power;
 - (ix) starter air;
 - (x) lavatory, galley and safety regulations.
- (e) Emergency procedures (as applicable):
 - (1) recognition of the situation as well as immediate memory actions in the correct sequence, and for those conditions that are recognised as emergencies by the manufacturer and the competent authority for certification:

- (i) engine failure/battery malfunction during take-off before and after v_1 , as well as in flight;
 - (ii) malfunctions of the propeller system;
 - (iii) engine/battery overheat, engine/battery fire on ground and in flight;
 - (iv) wheel-well fire;
 - (v) electrical smoke or fire;
 - (vi) rapid decompression and emergency descent;
 - (vii) air-conditioning overheat, anti-ice system overheat;
 - (viii) fuel pump/battery cooling system failure;
 - (ix) fuel freezing overheat;
 - (x) electric power failure;
 - (xi) equipment cooling failure;
 - (xii) flight instrument failure;
 - (xiii) partial or total hydraulic failure;
 - (xiv) failures at the lift devices and flight controls including boosters
 - (xv) cargo compartment smoke or fire.
 - (2) actions according to the approved checklist for abnormal and emergency procedures:
 - (i) engine restart in flight;
 - (ii) landing gear emergency extension;
 - (iii) application of the emergency brake system;
 - (iv) emergency extension of lift devices;
 - (v) fuel dumping;
 - (vi) emergency descent.
 - (f) Special requirements for ‘glass cockpit’ aircraft with EFIS additional learning objectives:
 - (1) general rules for aircraft computer hardware and software design;
 - (2) logic of all flight crew information and alerting systems and their limitations;
 - (3) interaction of the different aircraft computer systems, their limitations, the possibilities of computer fault recognition, and the actions to be performed on computer failures;
 - (4) normal procedures including all flight crew coordination duties;
 - (5) aircraft operation with different computer degradations (basic flying).
 - (g) Flight management systems.
- II. SE AND ME HELICOPTERS
- (a) Detailed listing for helicopters structure, transmissions, rotors and equipment, normal and abnormal operation of systems:
 - (1) dimensions.

- (2) engine including aux. power unit, rotor and transmissions; if an initial type rating for a turbine engine helicopter is applied for, the applicant should have received turbine engine instruction:
 - (i) type of engine or engines;
 - (ii) in general, the function of the following systems or components:
 - (A) engine;
 - (B) auxiliary power unit;
 - (C) oil system;
 - (D) fuel system;
 - (E) ignition system;
 - (F) starting system;
 - (G) fire warning and extinguishing system;
 - (H) generators and generator drive;
 - (I) power indication;
 - (J) water or methanol injection.
 - (iii) engine controls (including starter), engine instruments and indications in the cockpit, their function and interrelation and interpretation;
 - (iv) engine operation, including APU, during engine start and engine malfunctions, procedures for normal operation in the correct sequence;
 - (v) transmission system:
 - (A) lubrication;
 - (B) generators and generator drives;
 - (C) freewheeling units;
 - (D) hydraulic drives;
 - (E) indication and warning systems.
 - (vi) type of rotor systems: indication and warning systems.
- (3) fuel system:
 - (i) location of the fuel tanks, fuel pumps, fuel lines to the engines tank capacities, valves and measuring;
 - (ii) the following systems:
 - (A) filtering;
 - (B) fuelling and defuelling heatings;
 - (C) dumping;
 - (D) transferring;
 - (E) venting.
 - (iii) in the cockpit: the monitors and indicators of the fuel system, quantity and flow indication, interpretation;

- (iv) fuel procedures distribution into the various tanks fuel supply and fuel dumping.
- (4) air conditioning:
 - (i) components of the system and protection devices;
 - (ii) cockpit monitors and indicators;

Note: interpretation about the operational condition: normal operation of the system during start, cruise approach and landing, air conditioning airflow and temperature control.
- (5) ice and rain protection, windshield wipers and rain repellent:
 - (i) ice protected components of the helicopter, including engines and rotor systems, heat sources, controls and indications;
 - (ii) operation of the anti-icing or de-icing system during take-off, climb, cruise and descent, conditions requiring the use of the protection systems;
 - (iii) controls and indications of the windshield wipers and rain repellent system operation.
- (6) hydraulic system:
 - (i) components of the hydraulic system(s), quantities and system pressure, hydraulically actuated components associated to the respective hydraulic system;
 - (ii) controls, monitors and indicators in the cockpit, function and interrelation and interpretation of indications.
- (7) landing gear, skids fixed and floats:
 - (i) main components of the:
 - (A) main landing gear;
 - (B) nose gear;
 - (C) tail gear;
 - (D) gear steering;
 - (E) wheel brake system.
 - (ii) gear retraction and extension;
 - (iii) required tyre pressure, or location of the relevant placard;
 - (iv) controls and indicators including warning indicators in the cockpit in relation to the retraction or extension condition of the landing gear;
 - (v) components of the emergency extension system.
- (8) flight controls, stab- and autopilot systems: controls, monitors and indicators including warning indicators of the systems, interrelation and dependencies.
- (9) electrical power supply:
 - (i) number, power, voltage, frequency and if applicable phase and location of the main power system (AC or DC) auxiliary power system location and external power system;
 - (ii) location of the controls, monitors and indicators in the cockpit;

- (iii) main and back-up power sources flight instruments, communication and navigation systems, main and back-up power sources;
 - (iv) location of vital circuit breakers;
 - (v) generator operation and monitoring procedures of the electrical power supply.
- (10) flight instruments, communication, radar and navigation equipment, autoflight and flight data recorders:
 - (i) antennas;
 - (ii) controls and instruments of the following equipment in the cockpit:
 - (A) flight instruments (for example air speed indicator, pitot static system, compass system, flight director);
 - (B) flight management systems;
 - (C) radar equipment, including radio;
 - (D) communication and navigation system (for example HF, VHF, ADF, VOR/DME, ILS, marker beacon) and area navigation systems;
 - (E) stabilisation and autopilot system;
 - (F) flight data recorder, cockpit voice recorder, data-link communication recording function and radio altimeter;
 - (G) collision avoidance system;
 - (H) TAWS;
 - (I) HUMSS;
 - (J) weather radar system, best practices for optimum use, interpretation of displayed information.
- (11) cockpit, cabin and cargo compartment:
 - (i) operation of the exterior, cockpit, cabin and cargo compartment lighting and the emergency lighting;
 - (ii) operation of the cabin doors and emergency exits.
- (12) emergency equipment:
 - (i) operation and correct application of the following mobile emergency equipment in the helicopter:
 - (A) portable fire extinguisher;
 - (B) first-aid kits;
 - (C) portable oxygen equipment;
 - (D) emergency ropes;
 - (E) life-jacket;
 - (F) life rafts;
 - (G) emergency transmitters;
 - (H) crash axes;

- (I) megaphones;
 - (J) emergency signals;
 - (K) torches.
 - (ii) operation and correct application of the fixed emergency equipment in the helicopter: emergency floats.
- (b) Limitations:
 - (1) general limitations, according to the helicopter flight manual;
 - (2) minimum equipment list.
- (c) Performance, flight planning and monitoring:
 - (1) performance calculation about speeds, gradients, masses in all conditions for take-off, en-route, approach and landing:
 - (i) take-off:
 - (A) hover performance in and out of ground effect;
 - (B) all approved profiles, cat A and B;
 - (C) HV diagram;
 - (D) take-off and rejected take-off distance;
 - (E) take-off decision point (TDP) or (DPATO);
 - (F) calculation of first and second segment distances;
 - (G) climb performance.
 - (ii) en-route:
 - (A) air speed indicator correction;
 - (B) service ceiling;
 - (C) optimum or economic cruising altitude;
 - (D) max endurance;
 - (E) max range;
 - (F) cruise climb performance.
 - (iii) landing:
 - (A) hovering in and out of ground effect;
 - (B) landing distance;
 - (C) landing decision point (LDP) or (DPBL).
 - (iv) knowledge or calculation of: v_{lo} , v_{le} , v_{mo} , v_x , v_y , v_{toss} , v_{ne} , $v_{max\ range}$, v_{mini} .
 - (2) flight planning for normal and abnormal conditions:
 - (i) optimum or maximum flight level;
 - (ii) minimum required flight altitude;
 - (iii) drift down procedure after an engine failure during cruise flight;

- (iv) power setting of the engines during climb, cruise and holding under various circumstances as well as at the most economic cruising flight level;
 - (v) optimum and maximum flight level and power setting after an engine failure.
- (3) effect of optional equipment on performance.
- (d) Load, balance and servicing:
 - (1) load and balance:
 - (i) load and trim sheet on the maximum masses for take-off and landing;
 - (ii) centre of gravity limits;
 - (iii) influence of the fuel consumption on the centre of gravity;
 - (iv) lashing points, load clamping, max ground load.
 - (2) servicing on the ground, servicing connections for:
 - (i) fuel;
 - (ii) oil, etc.;
 - (iii) and safety regulations for servicing.
- (e) Emergency procedures.
- (f) Special requirements for helicopters with EFIS.
- (g) Optional equipment.

III. AIRSHIPS

- (a) Detailed listing for airship structure and equipment, normal operation of systems and malfunctions:
 - (1) dimensions;
 - (2) structure and envelope:
 - (i) internal structure;
 - (ii) envelope;
 - (iii) pressure system;
 - (iv) gondola;
 - (v) empennage.
 - (3) flight controls;
 - (4) systems:
 - (i) hydraulic;
 - (ii) pneumatic.
 - (5) landing gear;
 - (6) fuel system;
 - (7) fire warning and extinguishing system;
 - (8) emergency equipment;

- (9) electrical systems;
- (10) avionics, radio navigation and communication equipment;
- (11) instrumentation;
- (12) engines and propellers;
- (13) heating, ventilation and air-condition;
- (14) operational procedures during start, cruise, approach and landing:
 - (i) normal operations;
 - (ii) abnormal operations.
- (b) Limitations:
 - (1) general limitations:
 - (i) certification of the airship, category of operation, noise certification and maximum and minimum performance data for all flight profiles, conditions and aircraft systems;
 - (ii) speeds;
 - (iii) altitudes.
 - (2) engine limitations;
 - (3) systems limitations;
 - (4) minimum equipment list.
- (c) Performance and flight planning:
 - (1) performance calculation;
 - (2) flight planning.
- (d) Load and balance and servicing:
 - (1) load and balance;
 - (2) servicing.
- (e) Emergency procedures:
 - (1) recognition of emergency situations;
 - (2) actions according

AMC2 FCL.725(a) Requirements for the issue of class and type ratings

ED Decision 2025/002/R

TRAINING COURSE

FLIGHT INSTRUCTION FOR TYPE RATINGS: HELICOPTERS

- (a) The amount of flight instruction depends on:
 - (i) complexity of the helicopter type, handling characteristics, level of technology;
 - (ii) category of helicopter (SEP or SE turbine helicopter, ME turbine and MP helicopter);

- (iii) previous experience of the applicant;
- (iv) the availability of FSTDs.

(b) FSTDs

In accordance with [Appendix 9](#) to Part-FCL Section A paragraph 1 third subparagraph, the level of qualification and the complexity of the type will determine the amount of practical training that may be accomplished in FSTDs, including completion of the skill test. Before undertaking the skill test, a student should demonstrate competency in the skill test items during the practical training.

(c) Initial issue

The flight instruction (excluding skill test) should comprise training time as specified in column 2 in the table below unless otherwise foreseen in the operational suitability data established in accordance with Annex I (Part 21) to [Commission Regulation \(EU\) No 748/2012](#) (OSD). If, in accordance with [Appendix 9](#) to Part-FCL Section A paragraph 1c, training is conducted in a combination of FSTDs (other than FFS) and the helicopter, the in-aircraft training time will depend on the specificities of the FSTD used. In such a case, in-aircraft training time should comprise, as a minimum, the training time given in column 3 of the table below unless otherwise foreseen in the OSD.

The minimum training time given in column 2 may include the time of the take-off and landing training referred to in [Appendix 9](#) to Part-FCL Section A paragraph 1a.

(1)	(2)	(3)
Helicopter types	Minimum training time	Minimum training time in the helicopter, when also using FSTDs other than FFS
SEP (H)	5 hrs	4 hrs
SET(H) under 3175 kg MTOM	5 hrs	4 hrs
SET(H) at or over 3175 kg MTOM	8 hrs	4 hrs
SPH MET (H) CS and FAR 27 and 29	8 hrs	4 hrs
MPH, when the applicant has already completed: - MCC course; - 50 hours of flight time in multi-pilot operations in SPH	5 hrs	Using FTD2/3: At least 2 hrs in the helicopter
MPH	10 hrs	4 hrs

(d) Additional types

The flight instruction (excluding the skill test) should comprise training time as specified in column 2 in the table below unless otherwise foreseen in the OSD. If, in accordance with [Appendix 9](#) to Part-FCL Section A paragraph 1c, training is conducted in a combination of FSTDs (other than FFS) and the helicopter, the in-aircraft training time will depend on the specificities

of the FSTD used. In such a case, in-aircraft training time should comprise, as a minimum, the time given in column 3 of the table below unless otherwise foreseen in the OSD.

The minimum training time given in column 2 may include the time of the take-off and landing training referred to in Appendix 9 to Part-FCL Section A paragraph 1a.

(1)	(2)	(3)
Helicopter types	Minimum training time	Minimum training time in the helicopter, when also using FSTDs other than FFS
SEP(H) to SEP(H) within AMC1 FCL.740.H(b)	2 hrs	1 hr
SEP(H) to SEP(H) not included in AMC1 FCL.740.H(b)	5 hrs	2 hrs
SET(H) to SET(H)	2 hrs	1 hr
SE difference training	1 hr	N/A
MET(H) to MET(H)	3 hrs	2 hrs
ME difference training	1 hr	N/A
MPH to MPH	5 hrs	2 hrs
Extend privileges on the same type rating: — from SPH to MPH (except for initial MP issue), or — from MPH to SPH operated in single-pilot operation	2 hrs	N/A
Extend privileges on the same type rating from SPH to MPH (except for initial MP issue), when applicants have completed 50 hours of flight time in multi-pilot operations on that type	1 hr	Training in helicopter or FFS C/D

GM1 FCL.725(d); (da) Requirements for the issue of class and type ratings

ED Decision 2025/002/R

MULTI-PILOT OPERATION IN SINGLE-PILOT AEROPLANES OR HELICOPTERS IN ACCORDANCE WITH ANNEX III (PART-ORO) TO [REGULATION \(EU\) No 965/2012](#)

Points [FCL.725\(d\)\(1\)\(iii\)\(B\)](#) and [FCL.725\(d\)\(da\)\(3\)](#) require pilots to exercise their type or class rating privileges for multi-pilot operation in single-pilot aeroplanes or helicopters only in accordance with the requirements of Part-ORO, to ensure that pilots conduct multi-pilot operation in single-pilot aircraft only in accordance with approved operational procedures of an operator that is subject to Part-ORO. As a consequence, an ATO that provides training for multi-pilot operation in single-pilot aeroplanes or helicopters will need to base that training on the operational procedures of the operator for which the pilot is flying. That ATO will either be an operator itself or will have an arrangement with an operator on behalf of which the training will be carried out.

GM1 FCL.725(e) Requirements for the issue of class and type ratings

ED Decision 2017/022/R

The hours gained during the instruction flights for category 1 or 2 flight tests are not considered as flight tests related to development, certification or production.

FCL.740 Validity and renewal of class and type ratings

Regulation (EU) 2024/2076

(a) Validity

- (1) The validity period of class and type ratings shall be 1 year, except for single-pilot single-engine class ratings for which the validity period shall be 2 years, unless otherwise determined in the OSD. If pilots choose to fulfil the revalidation requirements earlier than prescribed in points [FCL.740.A](#), [FCL.740.H](#), [FCL.740.PL](#) and [FCL.740.As](#), the new validity period shall commence from the date of the proficiency check.
- (2) Applicants for the revalidation of a class or type rating shall receive full credits for the proficiency check as required in this Subpart when they complete EBT practical assessment in accordance with [Appendix 10](#) at an operator that has implemented EBT for the relevant class or type rating.

(b) Renewal

For the renewal of a class or type rating, applicants shall comply with all of the following:

- (1) in order to determine whether refresher training is necessary for the applicant to reach the level of proficiency to safely operate the aircraft, they shall undergo an assessment at one of the following:
 - (i) at an ATO;
 - (ii) at a DTO or at an ATO, if the expired rating concerned a non-high-performance SEP aeroplane class rating, a TMG class rating or a single-engine type rating for helicopters referred to in point [DTO.GEN.110\(a\)\(2\)\(c\)](#) of Annex VIII;
 - (iii) at a DTO, at an ATO or with an instructor, if the rating expired no more than 3 years ago and the rating concerned a non-high-performance SEP aeroplane class rating or a TMG class rating;
 - (iv) at an EBT operator that is specifically approved for such refresher training;
- (2) if deemed necessary by the organisation or the instructor providing the assessment as per point (1), they shall complete refresher training at that organisation or with that instructor;

- (3) after complying with point (1) and, as applicable, point (2), they shall pass a proficiency check in accordance with [Appendix 9](#) or complete EBT practical assessment in accordance with [Appendix 10](#). That EBT practical assessment may be combined with the refresher training specified in point (2).

By way of derogation from points (b)(1), (b)(2) and (b)(3), pilots holding a flight test rating issued in accordance with point [FCL.820](#) who were involved in the development, certification or production flight tests for an aircraft type and have completed either 50 hours of total flight time or 10 hours of flight time as PIC in test flights in that type during the year prior to the date of their application, shall be entitled to apply for the revalidation or renewal of the relevant type rating.

Applicants shall be exempted from the requirement in points (b)(1) and (b)(2) if they hold a valid rating for the same class or type of aircraft on a pilot licence issued by a third country in accordance with Annex 1 to the Chicago Convention and if they are entitled to exercise the privileges of that rating.

- (c) Pilots who leave an operator's EBT programme after having failed to demonstrate an acceptable level of competence in accordance with that EBT programme shall not exercise the privileges of that type rating until they have complied with one of the following:
- (1) they have completed EBT practical assessment in accordance with [Appendix 10](#);
 - (2) they have passed a proficiency check in accordance with point [FCL.625\(c\)\(3\)](#) or point [FCL.740\(b\)\(3\)](#), as applicable. In such a case, point [FCL.625\(b\)\(4\)](#) and point [FCL.740\(a\)\(2\)](#) shall not apply.

AMC1 FCL.740(b) Validity and renewal of class and type ratings

ED Decision 2025/002/R

RENEWAL OF CLASS AND TYPE RATINGS: REFRESHER TRAINING AT AN ATO, A DTO OR WITH AN INSTRUCTOR

- (a) The objective of the refresher training is for the applicant to reach the level of proficiency necessary to safely operate the relevant type or class of aircraft. The amount of refresher training needed should be determined on a case-by-case basis by the ATO, the DTO or the instructor, as applicable, taking into account the following factors:
- (1) the experience of the applicant;
 - (2) the amount of time elapsed since the privileges of the rating were last used;
 - (3) the complexity of the aircraft;
 - (4) whether the applicant has a current rating on another aircraft type or class; and
 - (5) where considered necessary, the performance of the applicant during a simulated proficiency check for the rating in an FSTD or an aircraft of the relevant type or class.

It should be expected that the amount of training needed to reach the desired level of proficiency will increase analogously to the time elapsed since the privileges of the rating were last used.

- (b) After having determined the needs of the applicant, the ATO, the DTO or the instructor, as applicable, should develop an individual training programme based on the initial training for the rating, focusing on the aspects where the applicant has shown the greatest needs.
- (c) With the exception of refresher training for ratings for aircraft referred to in point [FCL.740\(b\)\(1\)\(ii\)](#), refresher training should include theoretical knowledge instruction, as

necessary, such as for type-specific system failures in complex aircraft. The performance of the applicant should be reviewed during the training and additional instruction should be provided to the applicant, where necessary, to reach the standard required for the proficiency check.

- (d) After successful completion of the training, the ATO, the DTO or the instructor, as applicable, should issue the applicant with a training completion certificate or another document specified by the competent authority, describing the evaluation of the factors listed in (a), the training received, and a statement that the training has been successfully completed. The training completion certificate should be presented to the examiner prior to the proficiency check. Following the successful renewal of the rating, the training completion certificate or the other document specified by the competent authority and the examiner report form should be submitted to the competent authority.
- (e) Taking into account the factors listed in (a) above, the ATO, the DTO or the instructor, as applicable, may also decide that the applicant already possesses the required level of proficiency and that no refresher training is necessary. In such a case, the certificate or other documental evidence referred to in (d) above should contain a respective statement including sufficient reasoning.

GM1 FCL.740(b) Validity and renewal of class and type ratings

ED Decision 2021/002/R

RENEWAL OF CLASS AND TYPE RATINGS: REFRESHER TRAINING AT AN AOC HOLDER

It is recommended that an AOC holder approved for renewal of type ratings under Part-ORO may provide refresher training if the applicant is enrolled in the EBT programme; and if the rating has lapsed by no more than 1 year.

If the rating has lapsed by more than 1 year, it is recommended that the applicant consider to follow the training at an ATO and [AMC1 FCL.740\(b\)](#) applies.

SECTION 2 – SPECIFIC REQUIREMENTS FOR THE AEROPLANE CATEGORY

FCL.720.A Experience requirements and prerequisites for the issue of class or type ratings – aeroplanes

Regulation (EU) 2024/2076

Unless otherwise determined in the operational suitability data established in accordance with Annex I (Part-21) to Regulation (EU) No 748/2012 (OSD), applicants for the issue of a class or type rating shall comply with the following experience requirements and prerequisites for the issue of the relevant rating:

(a) Single-pilot aeroplanes

(1) Single-pilot multi-engine aeroplanes

Applicants for the issue of a first class or type rating on a single-pilot multi-engine aeroplane shall have completed at least 70 hours as PIC in aeroplanes.

(2) Single-pilot high-performance non-complex aeroplanes

Before starting flight training, applicants for the issue of a class or type rating for a single-pilot aeroplane classified as a high-performance aeroplane shall:

- (i) have at least 200 hours of total flying experience, of which 70 hours as PIC in aeroplanes; and
- (ii) comply with one of the following requirements:
 - (A) hold a certificate of satisfactory completion of a course for additional theoretical knowledge undertaken at an ATO; or
 - (B) have passed the ATPL(A) theoretical knowledge examinations in accordance with this Annex (Part-FCL); or
 - (C) hold, in addition to a licence issued in accordance with this Annex (Part-FCL), an ATPL(A) or CPL(A)/IR with theoretical knowledge credit for ATPL(A), issued in accordance with Annex 1 to the Chicago Convention.

(3) Single-pilot high-performance complex aeroplanes

Applicants for the issue of a type rating for a complex single-pilot aeroplane classified as a high-performance aeroplane shall, in addition to meeting the requirements in point (2), comply with all of the following:

- (i) they shall hold or have held a single- or multi-engine IR(A), as appropriate and as established in Subpart G;
- (ii) for the issue of the first type rating, they shall, before starting the type rating training course, meet the requirements in point (b)(5).

(b) Applicants for the issue of the first type rating for a multi-pilot aeroplane shall be student pilots currently undergoing training on an MPL training course or shall, before starting the type rating training course, comply with the following requirements:

- (1) have at least 70 hours of flight experience as PIC in aeroplanes;
- (2) hold or have held a multi-engine IR(A);

- (3) have passed the ATPL(A) theoretical knowledge examinations in accordance with this Annex (Part-FCL);
- (4) except when the type rating course is combined with an MCC course:
 - (i) hold a certificate of satisfactory completion of an MCC course in aeroplanes; or
 - (ii) hold a certificate of satisfactory completion of MCC in helicopters and have more than 100 hours of flight experience as pilots of multi-pilot helicopters; or
 - (iii) have at least 500 hours as pilots of multi-pilot helicopters; or
 - (iv) have at least 500 hours as pilots in multi-pilot operations on single-pilot multi-engine aeroplanes, in commercial air transport in accordance with the applicable air operations requirements; and
- (5) have completed the training course specified in point [FCL.745.A](#), unless they comply with any of the following:
 - (i) they completed, within the preceding 3 years, the training and checking in accordance with points ORO.FC.220 or ORO.FC.230 of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#);
 - (ii) they have completed the training specified in point [FCL.915\(e\)\(1\)\(ii\)](#).
- (c) Notwithstanding point FCL.720.A(b), applicants who comply with Section B, point 6(i), of [Appendix 9](#) shall be issued with a type rating with restricted privileges for a multi-pilot aeroplane that allows holders of such a rating to act as cruise relief co-pilots above Flight Level 200, provided that two other members of the crew have a type rating in accordance with point FCL.720.A(b).

To remove the restriction, applicants shall comply with all of the following:

 - (1) undergo an assessment at an ATO which shall determine their training needs to have the restriction removed;
 - (2) based on the assessment specified in point (1), complete, at that ATO, training which shall allow them to reach the level of proficiency to safely operate the aircraft;
 - (3) after the completion of the training specified in point (2), complete a skill test in accordance with [Appendix 9](#);
 - (4) after the completion of the skill test specified in point (3), meet one of the following conditions:
 - (i) comply with point ORO.FC.220(e) of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#), provided that they meet the experience requirements specified in point [FCL.730.A](#) of this Annex;
 - (ii) complete flight training in the aircraft which includes manoeuvres on take-off, landing and go-around.
- (d) When so determined in the OSD, the exercise of the privileges of a type rating may be initially limited to flight under the supervision of an instructor. The flight hours under supervision shall be entered in the pilots' logbook or equivalent record and signed by the instructor. The limitation shall be removed when pilots demonstrate that the hours of flight under supervision required in the OSD have been completed.

AMC1 FCL.720.A(a)(2)(ii)(A) Experience requirements and prerequisites for the issue of class or type ratings – aeroplanes

ED Decision 2025/002/R

ADDITIONAL THEORETICAL KNOWLEDGE FOR A CLASS OR TYPE RATING FOR HIGH-PERFORMANCE SINGLE-PILOT (SP) AEROPLANES

- (a) A number of aeroplanes certificated for SP operation have similar performances, systems and navigation capabilities to those more usually associated with MP types of aeroplanes, and regularly operate within the same airspace. The level of knowledge required to operate safely in this environment is not part of, or not included to the necessary depth of knowledge in the training syllabi for the PPL, CPL or IR(A) but these licence holders may fly as PIC of such aeroplanes. The additional theoretical knowledge required to operate such aeroplanes safely is obtained by completion of a course at an ATO.
- (b) The aim of the theoretical knowledge course is to provide the applicant with sufficient knowledge of those aspects of the operation of aeroplanes capable of operating at high speeds and altitudes, and the aircraft systems necessary for such operation.

COURSE SYLLABUS

- (c) The course will be divided in a VFR and an IFR part, and should cover at least the following items of the aeroplane syllabus to the ATPL(A) level:

FOR VFR OPERATIONS:

Subject ref.:	Syllabus content:
021 00 00 00	AIRCRAFT GENERAL KNOWLEDGE: AIRFRAME, SYSTEMS, AND POWER PLANT
021 09 01 03	Alternating current
021 09 03 00	Generation
021 09 03 02	AC generation
021 09 03 03	Constant speed drive (CSD) and integrated drive generator (IDG) systems
021 09 04 00	Distribution
021 09 04 01	General
021 09 04 03	AC distribution
021 09 04 04	Electrical load management and monitoring systems: automatic generators and bus switching during normal and failure operation, indications and warnings
021 06 01 01	Piston-engine air supply
021 06 01 02	Gas turbine engine: bleed-air supply
021 10 10 01	Performance
021 11 03 01	Engine fuel system
021 10 04 01	Carburettor: design, operation, degraded modes of operation, indications and warnings
021 03 01 09	Mixture
021 11 00 00 to 021 11 01 04	Turbine engines
021 13 00 00	Oxygen systems
032 03 00 00	Performance class B: ME aeroplanes
032 03 03 01	Take-off
032 03 03 02	Climb
032 03 03 04	Landing
032 01 03 00	Level flight, range and endurance
032 01 04 00	Climbing
032 01 05 00	Descending
032 02 04 00	Climb, cruise and descent

Subject ref.:	Syllabus content:
040 00 00 00	HUMAN PERFORMANCE
040 02 01 00 to 040 02 01 03	Basic human physiology and High-altitude environment
050 00 00 00	METEOROLOGY
050 02 07 00 050 02 05 00	Jet streams Standing waves
050 09 01 00 to 050 09 04 05	Flight hazards Icing and turbulence Thunderstorms
062 03 00 00	Basic radar principles
062 03 00 01 to 062 03 04 00	Basic radar principles Airborne radar SSR
081 00 00 00	PRINCIPLES OF FLIGHT: AEROPLANES
081 02 01 00 081 02 02 00 081 02 03 00	Speeds Shock waves Effects of exceeding M_{CRIT}

FOR IFR OPERATIONS

Subject ref.:	Syllabus content:
010 00 00 00	AIR LAW
010 06 07 00	Simultaneous operation on parallel or near-parallel instrument runways
010 06 08 00	Secondary surveillance radar (transponder) operating procedures
022 00 00 00	AIRCRAFT GENERAL KNOWLEDGE - INSTRUMENTATION
022 01 02 00	Temperature sensing
022 03 04 00	Flux valve
022 12 00 00	ALERTING SYSTEMS, PROXIMITY SYSTEMS
022 12 07 00	Altitude alert system
022 12 08 00	Radio-altimeter
022 12 10 00	ACAS/TCAS principles and operation
022 13 03 01	Electronic flight instrument system (EFIS) — Design, operation
050 00 00 00	METEOROLOGY
050 02 06 03	Clear-air turbulence (CAT) - Description, cause and location
050 10 02 03	Upper-air charts
062 00 00 00	RADIO NAVIGATION
062 02 05 04	ILS — Errors and accuracy

- (d) Demonstration of acquisition of this knowledge is undertaken by passing an examination set by an ATO. A successful pass of this examination results in the issue of a certificate indicating that the course and examination have been completed.
- (e) The certificate represents a 'once only' qualification and satisfies the requirement for the addition of all future high performance aeroplanes to the holder's licence. The certificate is valid indefinitely and is to be submitted with the application for the first HPA type or class rating.
- (f) A pass in any theoretical knowledge subjects as part of the HPA course will not be credited against meeting future theoretical examination requirements for issue of a CPL(A), IR(A) or ATPL(A).

- (g) The applicant who has completed a competency-based modular IR(A) course in accordance with Appendix 6 Aa or, before 8 September 2021, an EIR course in accordance with point [FCL.825](#) needs to complete both VFR and IFR parts of this course.
- (h) The applicant who has completed a modular IR(A) course according to [Appendix 6 A](#) only needs to complete the VFR part of this course.

AMC2 FCL.720.A(a)(2)(ii)(A) Experience requirements and prerequisites for the issue of class or type ratings – aeroplanes

ED Decision 2025/002/R

ADDITIONAL THEORETICAL KNOWLEDGE FOR A CLASS OR TYPE RATING FOR HIGH-PERFORMANCE SINGLE-PILOT (SP) AEROPLANES

An applicant for an additional class or type rating for a single-pilot aeroplane classified as a high-performance aeroplane (HPA), who:

- (a) has held a single-pilot HPA class or type rating prior to the application of Commission Regulation (EU) No 245/2014; and
- (b) has completed a competency-based modular IR(A) course in accordance with Appendix 6 Aa or, before 8 September 2021, an EIR course in accordance with point [FCL.825](#); and
- (c) does not fulfil the requirements of points (B) or (C) of point [FCL.720.A\(a\)\(2\)\(ii\)](#),

should pass the theoretical knowledge instruction and examination for the VFR and IFR parts of the course required in accordance with point [FCL.720.A\(a\)\(2\)\(ii\)\(A\)](#).

AMC1 FCL.720.A(c)(4)(ii) Experience requirements and prerequisites for the issue of class or type ratings — aeroplanes

ED Decision 2025/002/R

TRAINING ON TAKE-OFF, LANDING AND GO-AROUND MANOEUVRES

The ATO where the pilot is undergoing the training in accordance with point [FCL.720.A \(c\)\(4\)\(ii\)](#) should determine the number of take-off, landing and go-around manoeuvres, which should in any case comply with point (k) of [AMC2 ORA.ATO.125](#).

FCL.725.A Theoretical knowledge and flight instruction for the issue of class and type ratings – aeroplanes

Regulation (EU) 2024/2076

Unless otherwise determined in the operational suitability data established in accordance with Annex I (Part-21) to Regulation (EU) No 748/2012:

- (a) for single-pilot multi-engine aeroplanes:
 - (1) the theoretical knowledge course for a single-pilot multi-engine class rating shall include at least 7 hours of instruction in multi-engine aeroplane operations; and
 - (2) the flight training course for a single-pilot multi-engine class or type rating shall include at least 2 hours and 30 minutes of dual flight instruction under normal conditions of multi-

engine aeroplane operations, and not less than 3 hours 30 minutes of dual flight instruction in engine failure procedures and asymmetric flight techniques.

- (b) for single-pilot aeroplanes (sea):
 - (1) the training course for single-pilot aeroplane (sea) ratings shall include theoretical knowledge and flight instruction; and
 - (2) the flight training for a class or type rating (sea) for single-pilot aeroplanes (sea) shall include at least 8 hours of dual flight instruction if applicants hold the land version of the relevant class or type rating, or 10 hours if applicants do not hold such a rating; and
- (c) for single-pilot non-high-performance complex aeroplanes, single-pilot high-performance complex aeroplanes and multi-pilot aeroplanes, the training courses shall include UPRT theoretical knowledge and flight instruction related to the specificities of the relevant class or type. For the flight instruction, FSTDs may be used in accordance with Section A, point 18 of [Appendix 9](#).

AMC1 FCL.725.A(b) Theoretical knowledge and flight instruction for the issue of class and type ratings – aeroplanes

ED Decision 2011/016/R

CLASS RATING SEA

- (a) The theoretical knowledge instruction should be conducted by an instructor having appropriate experience of class rating sea.
- (b) Depending on the equipment and systems installed, the instruction should include, but not be limited to, the following content:
 - (1) theoretical knowledge:
 - (i) the aim of the training is to teach:
 - (A) the importance of preparation for flight and the safe planning taking into consideration all the factors for manoeuvring the aircraft on the wind, tidal currents, high and low water times and water movements at sea, river estuaries and lakes. In addition, icing conditions, ice covered water and broken ice flows;
 - (B) the techniques about the most critical moments at take-off, landing, taxiing and mooring the aircraft;
 - (C) the construction methods and characteristics of floats and water rudders and the importance of checking for leaks in the floats;
 - (D) the necessary requirements for the compliance of the rules for the avoidance of collisions at sea, in regard to sea charts, buoys and lights and horns.
 - (ii) after completing the training, the student should be able to:
 - (A) describe the factors that have significance for planning and decision about initiation of seaplane flying and alternative measures for completion of flight;
 - (B) describe how the water level is affected by air pressure, wind, tide, regularisations and the flight safety depending on changes in the water level;

- (C) describe the origin of different ice conditions in water areas;
 - (D) interpret nautical charts and maps about depths and shoals and risk for water currents, shifts of the wind, turbulence;
 - (E) decide what required equipment to bring during seaplane flying according to the operational requirements;
 - (F) describe the origin and extension of water waves, swells and water currents and their effect on the aeroplane;
 - (G) describe how water and air forces effect the aeroplane on water;
 - (H) describe the effect of water resistance on the aeroplanes' performance on glassy water and during different wave conditions;
 - (I) describe the consequences of taxiing with too high engine RPM;
 - (J) describe the effect of pressure and temperature on performance at take-off and climb from lakes located at higher altitude;
 - (K) describe the effect of wind, turbulence, and other meteorological conditions of special importance for flight over lakes, islands in mountain areas and other broken ground;
 - (L) describe the function of the water rudder and its handling, including the effect of lowered water rudder at take-off and landing;
 - (M) describe the parts of the float installation and their function;
 - (N) describe the effect of the floats on the aeroplanes' aerodynamics and performance in water and in air;
 - (O) describe the consequences of water in the floats and fouling of float bottoms;
 - (P) describe aviation requirements that apply specifically for the conduct of aircraft activity on water;
 - (Q) describe requirements about animal, nature and environment protection of significance for flight by seaplane, including flight in national parks;
 - (R) describe the meaning of navigation buoys;
 - (S) describe the organisation and working methods of the Sea Rescue Service;
 - (T) describe the requirements in ICAO Annex 2 as set out in paragraph 3.2.6 'Water operation', including relevant parts of the Convention on the International Regulations for Preventing Collisions at Sea.
- (2) practical training:
- (i) the aim of the practical training is to learn:
 - (A) the skills in manoeuvring aeroplanes on water and in mooring the aeroplane;
 - (B) the skills required for the reconnaissance of landing and mooring areas from the air, including the take-off area;
 - (C) the skills for assessing the effects of different water depths, shoals, wind, height of waves and swell;

- (D) the skills for flying with floats about their effect on performance and flight characteristics;
 - (E) the skills for flying in broken ground during different wind and turbulence conditions;
 - (F) the skills for take-off and landing on glassy water, different ° of swell and water current conditions.
- (ii) after the training, the student should be able to:
- (A) handle the equipment that shall be brought during seaplane flying;
 - (B) perform pre-flight daily inspection on aeroplane, float installation and special seaplane equipment, including emptying of floats;
 - (C) sail, taxi and turn the aeroplane at swell with correct handling of the water rudder;
 - (D) taxi on the step and perform turns;
 - (E) establish the wind direction with the aeroplane;
 - (F) take necessary actions if loss of steering ability and person falling overboard;
 - (G) make land and moor aeroplane at bridge, buoy and beach with the use of appropriate knots to secure the aircraft;
 - (H) maintain given rate of descent by means of variometer only;
 - (I) perform take-off and landing on glassy water with and without outer references;
 - (J) perform take-off and landing under swell;
 - (K) perform power-off landing;
 - (L) from the air, reconnaissance of landing, mooring and takeoff areas, observing;
 - (M) wind direction and strength during landing and take-off;
 - (N) surrounding terrain;
 - (O) overhead wires and other obstacles above and under water;
 - (P) congested areas;
 - (Q) determine wind direction and assess wind strength from water level and when airborne;
 - (R) state, for the aeroplane type in question;
 - (a) maximum wave height allowed;
 - (b) maximum number of ERPM allowed during taxi;
 - (S) describe how flying with floats affects the performance and flight characteristics of the aeroplane;
 - (T) take corrective action at critical moments due to wind shear and turbulence;
 - (U) navigate on the water with reference to buoys markers, obstacles and other traffic on the water.

- (c) For the initial issue of class rating sea for SP, SE and ME aeroplanes, the number of multi-choice questions in the written or computer-based examination should at least comprise thirty questions, and may be conducted by the training organisation. The pass mark should be 75 %.

FCL.730.A Specific requirements for pilots undertaking a zero flight time type rating (ZFTT) course – aeroplanes

Regulation (EU) 2024/2076

- (a) A pilot undertaking instruction at a ZFTT course shall have completed, on a multi-pilot turbo-jet aeroplane certificated to the standards of CS-25 or equivalent airworthiness code or on a multi-pilot turbo-prop aeroplane having a maximum certificated take-off mass of not less than 10 tonnes or a certificated passenger seating configuration of more than 19 passengers, at least:
- (1) if an FFS qualified to level CG, C or interim C is used during the course:
 - (i) in the case of cruise relief co-pilots, 1 500 hours of flight time;
 - (ii) in all other cases, 1 500 hours of flight time or 250 route sectors;
 - (2) if an FFS qualified to level DG or D is used during the course:
 - (i) in the case of cruise relief co-pilots, 750 hours of flight time;
 - (ii) in all other cases, 500 hours of flight time or 100 route sectors.
- (b) When a pilot is changing from a turbo-prop to a turbo-jet aeroplane or from a turbo-jet to a turbo-prop aeroplane, additional simulator training shall be required.

FCL.735.A Multi-crew cooperation training course – aeroplanes

Regulation (EU) No 1178/2011

- (a) The MCC training course shall comprise at least:
- (1) 25 hours of theoretical knowledge instruction and exercises; and
 - (2) 20 hours of practical MCC training, or 15 hours in the case of student pilots attending an ATP integrated course.
- An FNPT II MCC or an FFS shall be used. When the MCC training is combined with initial type rating training, the practical MCC training may be reduced to no less than 10 hours if the same FFS is used for both the MCC and type rating training.
- (b) The MCC training course shall be completed within 6 months at an ATO.
- (c) Unless the MCC course has been combined with a type rating course, on completion of the MCC training course the applicant shall be given a certificate of completion.
- (d) An applicant having completed MCC training for any other category of aircraft shall be exempted from the requirement in (a)(1).

AMC1 FCL.735.A; FCL.735.H; FCL.735.As Multi-crew cooperation (MCC) training course

ED Decision 2020/005/R

- (a) Competency is a combination of knowledge, skills and attitudes required to perform a task to the prescribed standard.

- (b) The objectives of MCC training are to develop the technical and non-technical components of the knowledge, skills and attitudes required to operate a multicrew aircraft.
- (c) Training should comprise both theoretical and practical elements and be designed to achieve the competencies/training objectives (see Table 1 below).

Table 1 — Competencies/training objectives			
Competency/objective	Performance indicators	Knowledge	Practical exercises
Communication	(a) Know what, how much and who to communicate to; (b) Ensure the recipient is ready and able to receive the information; (c) Pass messages and information clearly, accurately, timely and adequately; (d) Check if the other person has the correct understanding when passing important information; (e) Listen actively, patiently and demonstrate understanding when receiving information; (f) Ask relevant and effective questions, and offer suggestions; (g) Use appropriate body language, eye contact and tone; (h) Open and receptive to other people's view.	(a) Human Factors, TEM and CRM; (b) Application of TEM and CRM principles to training.	In a commercial air transport environment, apply multi-crew procedures, including principles of TEM and CRM to the following: (a) Pre-flight preparation: (1) FMS initialisation; (2) radio and navigation equipment preparation; (3) flight documentation; (4) computation of take-off performance data. (b) Take-off and climb: (1) before take-off checks; (2) normal take-offs; (3) rejected take-offs; (4) take-offs with abnormal and emergency situations included. (c) Cruise: emergency descent. (d) Descent and approach: (1) instrument flight procedures; (2) holding; (3) 3D Operations using raw data; (4) 3D Operations using flight director; (5) 3D Operations using autopilot; (6) one-engine-inoperative approach; (7) 2D Operations and circling; (8) computation of approach and landing data; (9) all engines go-around; (10) go-around with one engine inoperative;
Leadership and team working	(a) Friendly, enthusiastic, motivating and considerate of others; (b) Use initiative, give direction and take responsibility when required; (c) Open and honest about thoughts, concerns and intentions; (d) Give and receive criticism and praise well, and admit mistakes; (e) Confidently do and say what is important to him or her; (f) Demonstrate respect and tolerance towards other people; (g) Involve others in planning and share activities fairly.		

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
Situational awareness	(a) Be aware of what the aircraft and its systems are doing; (b) Be aware of where the aircraft is and its environment; (c) Keep track of time and fuel; (d) Be aware of the condition of people involved in the operation including passengers; (e) Recognise what is likely to happen, plan and stay ahead of the game; (f) Develop what-if scenarios and make pre-decisions; (g) Identify threats to the safety of the aircraft and of the people.		(11) wind shear during approach. (e) landing: transition from instrument to visual flight on reaching decision altitude or height or minimum descent altitude or height; (f) after landing and post flight procedures; (g) selected emergency and abnormal procedures.
Workload management	(a) Be calm, relaxed, careful and not impulsive; (b) Prepare, prioritise and schedule tasks effectively; (c) Use time efficiently when carrying out tasks; (d) Offer and accept assistance, delegate when necessary and ask for help early; (e) Review and monitor and cross-check actions conscientiously; (f) Follow procedures appropriately and consistently; (g) Concentrate on one thing at a time, ensure tasks are completed and does not become distracted; (h) Carry out instructions as directed.		
Problem-solving and decision-making	(a) Identify and verify why things have gone wrong and do not jump to conclusions or make assumptions; (b) Seek accurate and adequate information from appropriate resources; (c) Persevere in working through a problem; (d) Use and agree an appropriate decision making process;		

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
	(e) Agree essential and desirable criteria and prioritises; (f) Consider as many options as practicable; (g) Make decisions when they need to, reviews and changes if required; (h) Consider risks but do not take unnecessary risks.		
Monitoring and cross-checking	(a) Monitor and cross-checks all actions; (b) Monitor aircraft trajectory in critical flight phases; (c) Take appropriate actions in response to deviations from the flight path.	(a) SOPs; (b) Aircraft systems; (c) Undesired aircraft states.	
Task sharing	(a) Apply SOPs in both PF and pilot monitoring (PM) roles; (b) Makes and responds to standard call-outs.	(a) PF and PM roles; (b) SOPs.	
Use of checklists	Utilise checklists appropriately according to SOPs.	(a) SOPs; (b) Checklist philosophy.	
Briefings	Prepare and deliver appropriate briefings.	(a) SOPs; (b) Interpretation of FMS data and in-flight documentation.	
Flight management	(a) Maintain a constant awareness of the aircraft automation state; (b) Manage automation to achieve optimum trajectory and minimum workload; (c) Take effective recovery actions from automation anomalies; (d) Manage aircraft navigation, terrain clearance; (e) Manage aircraft fuel state and take appropriate actions.	(a) Understanding of aircraft performance and configuration; (b) Systems; (c) SOPs; (d) Interpretation of FMS data and in-flight documentation; (e) Minimum terrain clearance; (f) Fuel management IFR and VFR regulation.	
FMS use	Programme, manage and monitor FMS in accordance with SOPs.	(a) Systems (FMS); (b) SOPs; (c) Automation.	

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
Systems normal operations	Perform and monitor normal systems operation in accordance with SOPs.	(a) Systems; (b) SOPs.	
Systems abnormal and emergency operations	(a) Perform and monitor abnormal systems operation in accordance with SOPs; (b) Utilise electronic and paper abnormal checklists in accordance with SOPs.	(a) Systems; (b) SOPs; (c) Emergency and abnormal procedures and checklists; (d) Recall items.	
Environment, weather and ATC	(a) Communicate effectively with ATC; (b) Avoid misunderstandings by requesting clarification; (c) Adhere to ATC instructions; (d) Construct a mental model of the local ATC and weather environment.	(a) Systems; (b) SOPs; (c) ATC environment and phraseology; (d) Procedures for hazardous weather conditions.	

CERTIFICATE OF COMPLETION FORM

CERTIFICATE OF COMPLETION OF MCC			
Applicant's last name(s):		First name(s):	
Type of licence:		Number:	State:
ME/IR training completed		OR	ME/IR validity date: ME/IR skill test date:
Issued on:		passed on:	
	Signature of applicant:		

The satisfactory completion of MCC-Training according to requirements is certified below:

TRAINING			
Multi-crew co-operation training received during period:			
from:	to:	at:	ATO / operator*

Location and date:	Signature of head of ATO or authorised instructor*:
Type and number of licence and state of issue:	Name(s) in capital letters of authorised instructor:

* Delete as appropriate

AMC2 FCL.735.A Multi-crew cooperation (MCC) training course – aeroplanes

ED Decision 2017/022/R

ENHANCED MCC TRAINING TO AIRLINE PILOT STANDARDS (APS MCC) COURSE

- (a) The APS MCC training course should comprise both theoretical and practical elements and should be designed to achieve the training objectives, as set out in Table 1 below.

Table 1 — Training objectives			
Training objectives	Performance indicators	Knowledge	Practical exercises
Monitoring and cross-checking	(a) Monitor and cross-check all actions; (b) Monitor aeroplane trajectory in critical flight phases; (c) Take appropriate actions in response to deviations from the flight path.	(a) SOPs; (b) Aeroplane systems; (c) Undesired aeroplane states.	In a commercial air transport environment, apply multi-crew procedures, including principles of TEM and CRM to the following: (a) Pre-flight preparation: (1) FMS initialisation; (2) radio and navigation equipment preparation; (3) flight documentation; (4) computation of take-off performance data.
Task sharing	(a) Apply SOPs in both PF and PM roles; (b) Make and respond to standard call-outs.	(a) PF and PM roles; (b) SOPs.	(b) Take-off and climb: (1) before take-off checks; (2) normal take-offs; (3) rejected take-offs; (4) take-offs with abnormal and emergency situations included.
Use of checklists	Utilise checklists appropriately according to SOPs.	(a) SOPs; (b) Checklist philosophy.	(c) Cruise: emergency descent. (d) Descent and approach:
Briefings	Prepare and deliver appropriate briefings.	(a) SOPs; (b) Interpretation of FMS data and in-flight documentation.	
Flight management	(a) Maintain a constant awareness of the aeroplane automation state; (b) Manage automation to achieve optimum trajectory and minimum workload; (c) Take effective recovery actions from automation anomalies;	(a) Understanding of aeroplane performance and configuration; (b) Systems; (c) SOPs; (d) Interpretation of FMS data and in-flight documentation;	

Table 1 — Training objectives

Training objectives	Performance indicators	Knowledge	Practical exercises
	(d) Manage aeroplane navigation, terrain clearance; (e) Manage aeroplane fuel state and take appropriate actions.	(e) Minimum terrain clearance; (f) Fuel management IFR and VFR regulation.	(1) instrument flight procedures; (2) holding; (3) 3D Operations using raw data; (4) 3D Operations using flight director; (5) 3D Operations using autopilot; (6) one-engine-inoperative approach; (7) 2D Operations and circling; (8) computation of approach and landing data; (9) all engines go-around; (10) go-around with one engine inoperative; (11) wind shear during approach.
FMS use	Programme, manage and monitor FMS in accordance with SOPs.	(a) Systems (FMS); (b) SOPs; (c) Automation.	
Systems normal operations	Perform and monitor normal systems operation in accordance with SOPs.	(a) Systems; (b) SOPs.	
Systems abnormal and emergency operations	(a) Perform and monitor abnormal systems operation in accordance with SOPs; (b) Utilise electronic and paper abnormal checklists in accordance with SOPs.	(a) Systems; (b) SOPs; (c) Emergency and abnormal procedures and checklists; (d) Recall items.	
Environment, weather and air traffic control (ATC)	(a) Communicate effectively with ATC; (b) Avoid misunderstandings by requesting clarification; (c) Adhere to ATC instructions; (d) Construct a mental model of the local ATC and weather environment.	(a) Systems; (b) SOPs; (c) ATC environment and phraseology; (d) Procedures for hazardous weather conditions.	(e) landing: transition from instrument to visual flight on reaching decision altitude or height or minimum descent altitude or height; (f) after landing and post flight procedures; (g) selected emergency and abnormal procedures.

- (b) The APS MCC training course should include advanced swept-wing jet aeroplane training and airline operations scenario training to equip a pilot with the knowledge, skills, and attitudes required to commence initial type rating training to the standards generally required by a commercial air transport (CAT) operator certified pursuant to Regulation (EU) No 965/2012 (the 'Air OPS Regulation').
- (c) The APS MCC course should consist of the following:
- (1) the content of the MCC training course;
 - (2) advanced swept-wing jet aeroplane training;
 - (3) advanced airline operations scenario training; and
 - (4) a final assessment.
- (d) The flight simulation training device (FSTD) time per crew during practical training should be a minimum of 40 hours, or 35 for an integrated airline transport pilot licence (ATPL) holders, as set out in Table 2 below.

Table 2 — Minimum hours

Training element	Minimum FSTD time per crew
MCC TRAINING	20 hours/15 hours
ADVANCED SWEEP-WING JET AEROPLANE TRAINING	12 hours
ADVANCED AIRLINE OPERATIONS SCENARIO TRAINING	6 hours
FINAL ASSESSMENT	2 hours

The training elements may be ordered, split and combined, as determined by the approved training organisation (ATO)'s course design.

- (e) The ATO should provide generic stand-alone or CAT-operator-specific APS MCC training, advanced swept-wing jet aeroplane training and advanced airline operations scenario training. In the case of generic stand-alone training, the ATO should establish appropriate documentation and manuals representative of a CAT operator, such as manuals for aeroplane original-equipment manufacturers (OEMs), standard operating procedures (SOPs), flight documentation, as well as reporting and documentation for management systems.

FSTDs

- (f) The practical training in the APS MCC training course should be based on a multi-pilot, multi-engine aeroplane type capable of carrying at least 50 passengers or equivalent mass. The FSTD used should be type-specific and equipped with a visual system that provides at least 180° horizontal and 40° vertical field of view. However, an FNPT II MCC that has a similar visual cueing system to the above or is approved for MCC pursuant to [FCL.735.A](#) may also be acceptable provided that the device is representative of the same class of multi-pilot, multi-engine aeroplane specified in this paragraph in terms of passenger load, mass and performance, and equipped with equivalent aeroplane systems and avionics functionality.
- (g) In the case of advanced swept-wing jet aeroplane practical training, an FSTD representing a swept-wing multi-engine jet aeroplane should be used.

INSTRUCTOR QUALIFICATION

- (h) The minimum qualification level of an instructor to deliver the training course should be an MCCI(A). The ATO should ensure that:
 - (1) all the instructors, before delivering the training course content, have received training on the application of core competencies as well as competency-based training; and
 - (2) before the MCCI(A) delivers the advanced swept-wing jet handling or airline operations scenario training elements, they have satisfactorily completed relevant specific handling, systems and technical instructor training under the supervision of an SFI or TRI with the privilege to instruct for multi-pilot aeroplanes.
- (i) The final assessment should be completed by an instructor nominated by the head of training (HT) for this purpose.

COURSE DESIGN AND CORE COMPETENCIES

- (j) The course should be designed using instructional systems design (ISD) methodology.
- (k) Progress should be monitored throughout the course in accordance with the course design.
- (l) A final progress assessment should be conducted at the end of the practical training.

PROGRESS ASSESSMENTS AND COURSE COMPLETION CERTIFICATE

- (m) Practical training and progress assessments should be conducted to ensure that the student pilot has demonstrated the required level of competency (see Tables 1, 2, 3, 4 and 5 of this AMC).
- (n) During progress assessments, the student's knowledge, skills and attitudes in both pilot flying and pilot monitoring roles should be assessed; those assessments should be integrated into the training sessions.
- (o) All assessments should be graded. An example of a grading system for the APS MCC is provided in [GM3 FCL.735.A](#).
- (p) For the final assessment, the minimum standard for each competency should be at least 'satisfactory'. 'Satisfactory' is defined as demonstrating 75 % or greater of the relevant performance indicators/observable behaviours set out in the table of [GM3 FCL.735.A](#).
- (q) A student pilot who has reached a satisfactory or higher standard at the final assessment of the practical training should be awarded the APS MCC course completion certificate pursuant to [AMC2 FCL.735.A](#).
- (r) Alternatively, a student pilot who completes the APS MCC course but does not achieve the APS MCC standard should be awarded the MCC course completion certificate pursuant to [AMC1 FCL.735.A](#); [FCL.735.H](#); [FCL.735.As](#).

APS MCC TRAINING COURSE CONTENT AND PERFORMANCE INDICATORS

- (s) The elements of AMC1 FCL.735.A(c) should be enhanced as a result of the additional training in an airline context.
- (t) CRM training should be provided to an APS MCC standard.

Table 3 — APS MCC CRM TRAINING CONTENT AND PERFORMANCE INDICATORS			
Training	Performance indicators	Knowledge	Practical exercises
CRM training	(a) Display competency in the relevant CRM-related behaviours. (b) Successfully complete the final progress check.	Understand the CRM concepts set out in ORO.FC.115 of Annex III (Part-ORO) to the Air OPS Regulation.	Integrate CRM into all practical exercises of the APS MCC.

- (1) The ATO should ensure that the student pilot understands how multi-crew coordination as well as the content and intent of CRM in ORO.FC.115 is applied in an airline context.
- (2) In order to impart maximum learning to the student pilot, the ATO should ensure the following:
 - (i) CRM is integrated into all practical exercises of the APS MCC; and
 - (ii) Threat-and-error management (TEM) is central to the course instruction; the concepts of threat anticipation, threat recognition, recovery to safe flight, error management, and consequent avoidance of undesired aeroplanes states is emphasised at all times.

Table 4 — ADVANCED APS MCC FLYING TRAINING COURSE CONTENT AND PERFORMANCE INDICATORS			
Training	Performance indicators	Knowledge	Practical exercises
Advanced swept-wing flying training	(a) Understand and apply combinations of thrust and attitude that ensure a stable, safe flight in various	Elements and components of jet orientation:	(a) Take-off, approach, landing, go-around. (b) Flight deck management practices.

Table 4 — ADVANCED APS MCC FLYING TRAINING COURSE CONTENT AND PERFORMANCE INDICATORS

Training	Performance indicators	Knowledge	Practical exercises
	aeroplane configurations and altitudes. (b) Manage the (much) wider range of speed and thrust at both low level and high level. (c) Demonstrate good judgement and correct use of lift and drag devices during various phases of the flight. (d) Use displays along with all available aids to stay mentally ahead when piloting all profiles. (e) Understand and recognise the precursors of high-energy approaches. (f) Know angle-of-attack (AoA) versus attitude indications at low level as well as at high level. (g) Practice upset prevention as a priority, and clearly recognise when and how recovery is necessary, by using the required pilot skills to mitigate loss of control in-flight (LOC-I) events.	(a) glass cockpit displays; (b) propulsion; (c) aerodynamics; (d) flight controls; (e) performance; (f) jet flight planning; (g) weight and balance; (h) basic jet flying; (i) pilot techniques for jet flying, advanced-handling-skills development; (j) flight path management; (k) auto flight; (l) high-altitude operations; (m) introduction into prevention and recovery of upsets.	(c) Complex problem-solving techniques. (d) Advanced handling. (e) Manual handling skills (no autopilot, no auto thrust, and where possible, no flight director). (f) Flight at different speeds, including slow flight and altitudes within the normal flight envelope. (g) Steep turns. (h) Aeroplane stability and stall awareness. (i) Upset prevention techniques and approach-to-stall recovery events (appropriate to FSTD limitations and capabilities). (j) High-energy approach prevention. (k) Go-around management of approach and landing configurations.

Table 4 — ADVANCED APS MCC FLYING TRAINING COURSE CONTENT AND PERFORMANCE INDICATORS

Training	Performance indicators	Knowledge	Practical exercises
Advanced airline operations scenario training	(a) Execute pre-flight preparation in accordance with airline or OEM SOPs. (b) Conduct an effective crew briefing, including cabin crew managers (CCMs). (c) Display good airmanship and TEM skills in assessing aeroplane serviceability, weather planning, fuel planning, and destination facilities. (d) Conduct cockpit preparation and briefings in an effective and accurate manner. (e) Manage and execute engine start, taxi-out and pre-take-off checks safely and in accordance with airline or OEM SOPs. (f) Manage and execute runway line-up, take-off, climb, cruising, descent, approach, landing and taxi-in safely and in accordance with airline or OEM SOPs. (g) During non-normal operations, display good system knowledge, and apply non-normal procedures, communications, TEM, situational awareness (SA), decision-making and aeroplane handling.	(a) Knowledge of systems as set out in this AMC. (b) SOPs. (c) Normal-and non-normal operations' checklists and procedures.	(a) CHECK-IN PROCEDURES. (b) PRE-FLIGHT PREPARATION: (1) weather analysis; (2) flight planning; (3) fuel planning; (4) configuration deviation list (CDL), dispatch deviation procedures guide (DDPG), and minimum equipment list (MEL) analysis; and (5) cabin crew briefing. (c) NORMAL PROCEDURES: cockpit preparation, pushback, engine starting, taxiing, take-off, climb, cruising, descent, landing, shutdown, and disembarkation procedures. (d) ON TIME PERFORMANCE: (1) weather analysis; (2) flight planning; and (3) fuel planning. (e) NON-NORMAL PROCEDURES: (1) as per (c) above, in case of a technical or operational non-normal event; (2) TEM; (3) diversion decision-making; (4) communication; (5) diversion; (6) fuel SA; and (7) passenger and crew care.

Table 5 — ADVANCED APS MCC AIRLINE TRAINING CONTENT AND PERFORMANCE INDICATORS

Training	Performance Indicators	Knowledge	Practical Exercises
Airline-oriented training	(a) Understand the roles of airline departments. (b) Understand the challenges faced by airline departments. (c) Understand the relationships between airline departments. (d) Understand airline responsibilities.	Appropriate elements of the applicable Regulation (Regulation (EU) No 1178/2012 (the 'Aircrew Regulation') and the Air OPS Regulation).	The exercise should provide the student pilot with a practical understanding of airline operations. This may be achieved through a visit to an airline or alternative means.

Table 5 — ADVANCED APS MCC AIRLINE TRAINING CONTENT AND PERFORMANCE INDICATORS			
Training	Performance Indicators	Knowledge	Practical Exercises
	(e) Understand a pilot's responsibilities as a crew member.		

CERTIFICATE OF COMPLETION FORM

CERTIFICATE OF COMPLETION OF APS MCC-TRAINING			
Applicant's last name(s):		First name(s):	
Type of licence:		Number:	State:
ME/IR:		OR	ME/IR skill test:
Issued on:		passed on:	
	Signature of applicant:		

The satisfactory completion of APS MCC training according to requirements is certified below:

TRAINING			
Multi-crew cooperation training to airline pilot standards received during period:			
from:	to:	at:	ATO/operator*
Location and date:		Signature of head of ATO or authorised instructor*:	
Type and number of licence and state of issue:		Name(s) in capital letters of authorised instructor:	

* Delete as appropriate

GM1 FCL.735.A Multi-crew cooperation (MCC) training course – aeroplanes

ED Decision 2017/022/R

ENHANCED MCC TRAINING TO AIRLINE PILOT STANDARDS (APS MCC) COURSE

- (a) The ATO should be responsible for the initial course design based on the instructional systems design (ISD) methodology, as well as for the integral evaluation and further development of the course.

- (b) Technical-knowledge instruction

To maximise the benefit during the training in a flight simulation training device (FSTD), it is essential that the student pilot understands the aeroplane systems. Consequently, the approved training organisation (ATO) should provide sufficient systems training to ensure that student pilots are capable of effective situational awareness (SA) of the aeroplane systems when following normal and non-normal procedures and completing the related checklists. The standard of technical-knowledge training should be limited to this goal unless the course is part of a combined APS MCC/type rating course. ATOs providing APS MCC training in a combined APS MCC/type rating course may provide systems training up to type rating standard.

Aeroplane systems training may be delivered by any means provided that the training ensures knowledge transfer to a standard within the scope of the ATO's APS MCC training course approval. This training may be delivered either through distance learning or instructor-led classroom instruction or a combination thereof. If distance learning is utilised as an element of the course, it should be supplemented by instructor-led training.

Aeroplane systems knowledge at the required level should be confirmed by an assessment determined by the ATO's course design.

- (c) Advanced swept-wing jet flying training (see Table 4 of [AMC2 FCL.735.A](#))

The student pilot should develop a flight path management competency, including energy management, as pilot flying (PF), and associated active monitoring skills as pilot monitoring (PM). Aeroplane and airline procedures used during this training should develop the student pilot's understanding of the aeroplane flight envelope and inertia, as well as of the relationship between thrust and attitude. This phase should include an introduction to prevention and recovery of upsets, which builds confidence, skill, and resilience.

- (d) Advanced airline operations scenario training (see Table 4 of [AMC2 FCL.735.A](#))

- (1) The student pilot should be trained to apply the core competencies to conduct a safe and efficient operation in realistic airline operations scenarios.
- (2) The airline-representative scenarios should include normal and non-normal situations.
- (3) Operations should be run in real time according to a typical schedule.
- (4) The scenarios should be constructed in an airline context in order to emphasise the following:
 - (i) threat-and-error management (TEM);
 - (ii) crew resource management (CRM);
 - (iii) flight path management, including energy management; and
 - (iv) interaction with internal and external stakeholders in the resolution of scenarios.

- (e) Airline-oriented training (see Table 5 of [AMC2 FCL.735.A](#))

The training should provide an understanding of the regulatory framework that an airline must operate in. The student pilot should understand the context and operational environment that applies to airline employees. Subjects should include but are not limited to the following:

- (1) regulation of operations and aircrew;

- (2) safety management systems (SMSs) with emphasis on the pilot's reporting obligations and 'just culture';
- (3) fatigue management and fatigue risk management system (FRMS) with emphasis on the airline's and pilot's obligations;
- (4) flight time limitations (FTLs), including crew scheduling and crew control functions;
- (5) flight operations planning and flight watch reporting systems;
- (6) airline maintenance department and interaction with flight operations;
- (7) ground operations and interaction with flight operations; and
- (8) in-flight department and interaction with flight operations.

GM2 FCL.735.A Multi-crew cooperation (MCC) training course – aeroplanes

ED Decision 2017/022/R

ENHANCED MCC TRAINING TO AIRLINE PILOT STANDARDS (APS MCC) COURSE

The approved training organisation (ATO) should ensure that their course design develops the required core competencies through their training and assessment plan based on the competency framework provided in Table 1 below. An ATO may adapt this framework to include additional competencies and/or performance indicators/observable behaviours

Table 1 — COMPETENCIES		
Competency	Description	Performance indicators/observable behaviours
Application of knowledge	Relates and applies relevant knowledge in the operational environment and in scenario settings.	– Demonstrates the acquisition and retention of required aviation knowledge; – Relates knowledge between subject areas; – Applies knowledge to the operational environment; – Correctly identifies threats and errors in a timely manner; – Uses knowledge to create valid options of managing threats, errors, and undesirable aeroplane states; – Mentally resolves basic-mathematics problems relating to operational situations, both under normal circumstances and under pressure; – Shares knowledge with others openly and constructively, as and when appropriate.
Application of regulations and procedures	Identifies and applies appropriate procedures in accordance with published operating instructions and pursuant to applicable regulations.	– Identifies where to find the information; – Follows standard operating procedures (SOPs) unless a higher degree of safety dictates an appropriate deviation therefrom; – Follows all operating instructions in a timely manner; – Correctly operates aeroplane systems and associated equipment; – Monitors the status of aeroplane systems; – Complies with applicable regulations; – Applies relevant procedural knowledge.

Table 1 — COMPETENCIES

Competency	Description	Performance indicators/observable behaviours
Communication	Communicates through appropriate means in normal and non-normal situations.	– Ensures that the recipient is ready and able to receive the information; – Shares appropriate information; – Selects appropriately what, when, how, and with whom to communicate; – Conveys messages clearly, accurately, and concisely; – Confirms that the recipient correctly understands important information; – Listens actively and demonstrates understanding when receiving information; – Asks relevant and effective questions; – Communicates in order to resolve deviations identified through monitoring; – Adheres to standard radiotelephony phraseology and procedures; – Accurately reads, interprets, drafts, and responds to data link messages in English; – Correctly uses and interprets non-verbal communication.
Aeroplane flight path management — automation	Controls the aeroplane flight path through automation.	– Uses appropriate flight management and guidance systems as well as automation, as installed and as appropriate to the conditions; – Monitors and detects deviations from the desired aeroplane trajectory and takes appropriate action; – Manages the flight path to optimise the operational performance; – Maintains the desired flight path during flight using automation, whilst managing other tasks and distractions; – Effectively monitors automation, including engagement and automatic-mode transitions.
Aeroplane flight path management — manual control	Controls the aeroplane flight path through manual flight.	– Uses appropriate flight management and guidance systems and automation, as installed and appropriate to the conditions; – Manually controls the aeroplane using only the relationship between aeroplane attitude, speed and thrust, as well as navigation signals or visual information; – Monitors and detects deviations from the desired aeroplane trajectory and takes appropriate action; – Manages the flight path to optimise the operational performance; – Maintains the desired flight path during manual flight, whilst managing other tasks and distractions; – Effectively monitors flight guidance systems, including engagement and automatic-mode transitions.
Leadership and teamwork	Influences others so that they contribute to a shared purpose. Collaborates to accomplish the goals of the team.	– Creates an atmosphere of open communication and encourages team participation; – Displays initiative and gives directions when required; – Admits mistakes and takes responsibility; – Carries out instructions when directed; – Gives and receives feedback constructively;

Table 1 — COMPETENCIES

Competency	Description	Performance indicators/observable behaviours
		– Applies effective intervention strategies to resolve deviations identified whilst monitoring; – Takes into account cultural differences; – Engages others in planning; – Addresses and resolves conflicts and disagreements in a constructive manner; – Exercises decisive leadership.
Problem-solving and decision-making	Identifies problem precursors and resolves actual problems, using decision-making techniques, in a timely manner.	– Seeks accurate and appropriate information from appropriate sources; – Identifies and verifies what and why has failed; – Perseveres with resolving problems whilst prioritising safety; – Uses appropriate and timely decision-making techniques; – Sets priorities appropriately; – Identifies and considers options, as appropriate; – Monitors, reviews, and adapts decisions, as required; – Identifies, assesses, and manages risks effectively; – Adapts when faced with situations where no guidance or procedure exists.
Situational awareness (SA) and information management	Perceives, comprehends, and manages information, as well as anticipates its effect on the operation.	– Monitors, identifies, and assesses accurately the aeroplane's state and systems; – Monitors, identifies, and assesses accurately the aeroplane's energy state and anticipated flight path; – Monitors, identifies, and assesses accurately the general environment as it may affect the operation; – Validates the accuracy of information and checks for gross errors; – Maintains the awareness of the people involved in or affected by the operation as well as their capacity to perform as expected; – Anticipates what could happen, plans, and stays ahead of the situation; – Develops effective contingency plans based upon potential threats; – Recognises and effectively responds to indications of reduced SA.
Workload management	Maintains available workload capacity through prioritisation and distribution of tasks, using resources.	– Exercises self-control in all situations; – Plans, prioritises, and schedules tasks effectively; – Manages time efficiently when carrying out tasks; – Offers and gives assistance, delegates when necessary; – Seeks and accepts assistance, when necessary; – Monitors, reviews, and cross-checks taken action conscientiously; – Verifies that tasks are completed as expected; – Manages and recovers from interruptions, distractions, variations, and failures effectively, while performing tasks.

GM3 FCL.735.A Multi-crew cooperation (MCC) training course – aeroplanes

ED Decision 2017/022/R

EXAMPLE OF AN ENHANCED MCC TRAINING TO AIRLINE PILOT STANDARDS (APS MCC) GRADING SYSTEM

EXAMPLE OF AN APS MCC GRADING SYSTEM					
Competency	Unsatisfactory	Satisfactory	Good	Very Good	Exemplary
General description of each competency level.	The pilot's performance in this competency was unsatisfactory with a negative effect on safety. The pilot did not demonstrate the majority of the relevant performance indicators.	The pilot's performance in this competency was satisfactory with a slightly positive effect on safety. The pilot demonstrated most of the relevant performance indicators in this competency to at least a satisfactory standard.	The pilot's performance in this competency was effective with a significant contribution to safety. The pilot consistently demonstrated most of the relevant performance indicators in this competency to a good standard.	The pilot's performance in this competency was very effective, which significantly enhanced safety. The pilot regularly demonstrated all of the relevant performance indicators in this competency to a very good standard.	The pilot's performance in this competency was exemplary with an outstanding effect on safety. The pilot always demonstrated all of the relevant performance indicators in this competency to an exemplary standard.
Notes			— Most: 75 % or greater. — Relevant performance indicator: a performance indicator/observable behaviour that is expected to be demonstrated during the assessment.		

GM4 FCL.735.A Multi-crew cooperation (MCC) training course – aeroplanes

ED Decision 2017/022/R

ENHANCED MCC TRAINING TO AIRLINE PILOT STANDARDS (APS MCC) TRAINING — SPECIFIC ARRANGEMENT

The specific arrangement, pursuant to ORA.GEN.205, between an approved training organisation (ATO) and an operator for the APS MCC course should cover at least the following points:

- (1) pre-entry requirements (including screening and selection);
- (2) provision of the relevant documentation (operations manuals (OMs) and training manuals);
- (3) design of the training programme;
- (4) content of the course, including criteria to ensure that the operator's documentation, manuals, standard operating procedures (SOPs), reporting structures, and management system are represented throughout the training course;
- (5) training effectiveness;
- (6) performance data feedback from the ATO to the operator;

- (7) course evaluation and improvement;
- (8) alignment of the grading and assessment criteria; and
- (9) use of the operator's crew resource management (CRM) content and utilisation of a flight crew CRM trainer, standardised by the operator.

The ATO and the operator may use their OM and training manuals to identify additional areas to be covered by the specific arrangement.

FCL.740.A Revalidation of class and type ratings – aeroplanes

Regulation (EU) 2024/2076

- (a) Revalidation of multi-engine class ratings and type ratings. For revalidation of multi-engine class ratings and type ratings, the applicant shall:
 - (1) pass a proficiency check in accordance with [Appendix 9](#) or complete EBT practical assessment in accordance with [Appendix 10](#) in the relevant class or type of aeroplane or an FSTD representing that class or type, within the 3 months immediately preceding the expiry date of the rating; and
 - (2) complete during the period of validity of the rating, at least:
 - (i) 10 route sectors as pilot of the relevant class or type of aeroplane; or
 - (ii) 1 route sector as pilot of the relevant class or type of aeroplane or FFS, flown with an examiner. This route sector may be flown during the proficiency check.
 - (3) A pilot working for a commercial air transport operator approved in accordance with the applicable air operations requirements who has passed the operators proficiency check combined with the proficiency check for the revalidation of the class or type rating shall be exempted from complying with the requirement in (2).
 - (4) The revalidation of a BIR or an IR(A), if held, may be combined with a proficiency check for the revalidation of a class or type rating.
- (b) Revalidation of single-pilot single-engine class ratings.
 - (1) SEP aeroplane class ratings and TMG class ratings. For the revalidation of SEP aeroplane class ratings or TMG class ratings, applicants shall:
 - (i) within the 3 months preceding the expiry date of the rating, pass a proficiency check in the relevant class in accordance with [Appendix 9](#) to this Part with an examiner; or
 - (ii) within the 12 months preceding the expiry date of the rating, complete 12 hours of flight time in the relevant class, including the following:
 - (A) 6 hours as PIC;
 - (B) 12 take-offs and 12 landings;
 - (C) refresher training of at least 1 hour of total flight time with and to the satisfaction of a flight instructor (FI) or a class rating instructor (CRI) who shall select those flight exercises that allow the applicant to refresh their competence in safely operating the aircraft and applying normal, abnormal and emergency procedures. Applicants shall be exempted from this refresher training if they have passed any of the following, in any class or type of aeroplane:

- (1) a class or type rating proficiency check;
 - (2) a skill test;
 - (3) an EBT practical assessment;
 - (4) an assessment of competence.
 - (2) When applicants hold both an SEP aeroplane-land class rating and a TMG class rating, they may complete the requirements of point (1) in either class or a combination of those classes, and achieve revalidation of both ratings.
 - (3) Single-pilot single-engine turbo-prop aeroplanes. For revalidation of single-engine turbo-prop class ratings applicants shall pass a proficiency check on the relevant class in accordance with [Appendix 9](#) to this Part with an examiner, within the 3 months preceding the expiry date of the rating.
 - (4) When applicants hold both an SEP aeroplane-land class rating and an SEP aeroplane-sea class rating, they may complete the requirements of point (1)(ii) in either class or a combination of those classes, and achieve the fulfilment of those requirements for both ratings. At least 1 hour of the required PIC time and 6 of the required 12 take-offs and 12 landings shall be completed in each class.
 - (5) The proficiency check for the revalidation of a single-pilot single-engine aeroplane class rating may be combined with the proficiency check for the revalidation of a BIR, in accordance with point [FCL.835\(g\)\(8\)](#).
- (c) Applicants who fail to achieve a pass in all sections of a proficiency check before the expiry date of a class or type rating shall not exercise the privileges of that rating until a pass in the proficiency check has been achieved.

GM1 FCL.740.A Revalidation of class and type ratings — aeroplanes

ED Decision 2022/014/R

COMPLETE EBT PRACTICAL ASSESSMENT IN ACCORDANCE WITH [APPENDIX 10](#)

- (a) The completion of an EBT practical assessment includes:
- (1) the assessment of pilot performance either in a simulated or an operational environment; and
 - (2) the administrative action which includes the completion of the [Appendix 10](#) form.
- (b) The assessment as per point (1) usually occurs during the entire validity period of the rating as the EBT programme includes several FSTD sessions, while the administrative action as per point (2) is completed within the 3 months immediately preceding the expiry date of the rating.

AMC1 FCL.740.A(b)(1)(ii)(C) Revalidation of class and type ratings — aeroplanes

ED Decision 2025/002/R

CONTENT OF THE REFRESHER TRAINING

- (a) Before the training takes place, the instructor should hold a briefing with the candidate. That briefing should include a discussion on all of the following:
- (1) TEM with special emphasis on decision-making when encountering adverse meteorological conditions or unintentional IMCs;

- (2) navigation flight techniques;
 - (3) recovery strategies for different stall scenarios.
- (b) Flight training items should be based on the exercise items of the proficiency check, as deemed relevant by the instructor, and depending on the experience of the candidate. In any case, the flight training items should include exercises related to the recognition of and the recovery from the following scenarios:
- (1) simulated loss or partial loss of engine power during different phases of flight;
 - (2) selection of different stall scenarios (as specified in Exercise 2.3 of the table in point (5) of Section B of [Appendix 9](#)).

GM2 FCL.740.A Revalidation of type ratings — aeroplanes

ED Decision 2025/002/R

SAFETY AWARENESS BRIEFING BEFORE REFRESHER TRAINING OR PROFICIENCY CHECKS

It is recommended that the pre-flight briefing before refresher training or a proficiency check in accordance with point [FCL.740.A](#) with the pilot, the instructor or examiner, as applicable, includes elements to raise the pilot's safety awareness with regard to safely flying aeroplanes. This part of the briefing (safety awareness briefing) should have a duration of at least 15 minutes to allow discussions on several safety issues, referring to accidents and incidents in general or risks specifically related to the type of flights usually undertaken by the pilot. Threat and error management (TEM) should be promoted as effective mitigation, including the illustration of the practical application of TEM using real-life examples. There is no restriction on the subjects that could be covered. They may range from weather-related issues to personal or passenger-induced pressure. The material that can be used to support this briefing could come from accident & incident reports, mandatory or voluntary safety reporting, safety campaigns of different sources as well as from personal experience.

FCL.745.A Advanced UPRT course – aeroplanes

Regulation (EU) 2024/2076

- (a) The advanced UPRT course shall be completed at an ATO and shall comprise at least:
- (1) 5 hours of theoretical knowledge instruction;
 - (2) preflight briefings and postflight debriefings; and
 - (3) 3 hours of dual flight instruction with a flight instructor for aeroplanes FI(A) qualified in accordance with point [FCL.915](#)(e) and consisting of advanced UPRT in an aeroplane qualified for the training task. Flight time that does not include advanced UPRT but serves to go to or return from the UPRT training area shall not count towards those 3 hours.
- (b) Upon completion of the UPRT course, applicants shall be issued with a certificate of completion by the ATO.

AMC1 FCL.745.A Advanced UPRT course – aeroplanes

ED Decision 2019/005/R

COURSE OBJECTIVE AND CONTENT

COURSE OBJECTIVE

- (a) The objective of the course is for the pilot under training:

- (1) to understand how to cope with the physiological and psychological aspects of dynamic upsets in aeroplanes; and
 - (2) to develop the necessary competence and resilience to be able to apply appropriate recovery techniques during upsets.
- (b) In order to meet the objective as specified in point (a), the course should:
- (1) emphasise physiological and psychological effects of an upset and develop strategies to mitigate those effects;
 - (2) be delivered in a suitable training aircraft in order to expose trainees to conditions that cannot be replicated in an FSTD; and
 - (3) employ recovery techniques that are suitable for the aircraft used for training in order to support the training objectives. In order to minimise the risk associated with potential negative transfer of training, the recovery techniques used during the course should be compatible with techniques typically used for transport category aeroplanes.

THEORETICAL KNOWLEDGE

- (c) Theoretical knowledge instruction supports the objectives of the course and should include the following:
- (1) a review of basic aerodynamics typically applicable to aeroplane upsets in transport category aeroplanes, including case studies of incidents involving potential or actual upsets.
 - (2) aerodynamics relevant to the aeroplane and exercises used in the practical training, including differences to aerodynamics as referred to in point (1);
 - (3) possible physiological and psychological effects of an upset, including surprise and startle effect;
 - (4) strategies to develop resilience and mitigate startle effect; and
 - (5) memorising the appropriate procedures and techniques for upset recovery.

FLIGHT INSTRUCTION

- (d) Flight instruction should include:
- (1) exercises to demonstrate:
 - (i) the relationship between speed, attitude and AoA;
 - (ii) the effect of g-load on aeroplane performance, including stall events at different attitudes and airspeeds;
 - (iii) aerodynamic indications of a stall including buffeting, loss of control authority and inability to arrest a descent;
 - (iv) the physiological effects of different g-loads between -1 and 2.5G; and
 - (v) surprise and the startle effect;
 - (2) training in techniques to recover from:
 - (i) nose high at various bank angles;
 - (ii) nose low at various bank angles;
 - (iii) spiral dives;

- (iv) stall events; and
- (v) incipient spin; and
- (3) training to develop resilience and to employ strategies to mitigate the startle effect.

COURSE COMPLETION

- (e) The course is considered to have been satisfactorily completed if the trainee is able to successfully:
 - (1) apply strategies to mitigate psychological and physical effects;
 - (2) recognise upsets;
 - (3) apply correct recovery techniques from upset scenarios as specified in point (d)(2).

GM1 FCL.745.A Advanced UPRT course – aeroplanes*ED Decision 2019/005/R***UPSET RECOVERY TRAINING EXERCISES****GENERAL**

- (a) The objective of this GM is to provide instructors with further guidance on the conduct of the various upset recovery exercises, which requires instructor performance beyond that experienced in normal operations.
- (b) Instructors should:
 - (1) ensure that the risk mitigation measures determined by the ATO are strictly adhered to;
 - (2) continuously assess the performance of the student to ensure that the training objectives of the upset recovery exercises are achieved;
 - (3) understand that all-attitude/on-aeroplane upset recovery exercises serve primarily as resilience-builder. In other words, the training serves mainly human-factor training objectives and not only flying skills training;
 - (4) understand the differences between all-attitude UPRT and aerobatics training;
 - (5) have knowledge and understanding of how:
 - (i) on-aeroplane and FSTD UPRT complement each other; and
 - (ii) to ensure that negative transfer of training from small aeroplanes to heavier transport category aeroplanes is avoided. This may be achieved by observing UPRT in an FSTD, especially in a type-specific FFS; and
 - (6) have knowledge and understanding of the upset prevention theoretical knowledge and flight instruction elements taught during the CPL(A) and ATPL(A) training courses to ensure continuity and consistency in delivering UPRT.

Note: Instructors should be aware that the safety and potential human factor implications of poor upset recovery instructional technique or misleading information are *more significant* than in any other areas of pilot training.

- (c) In order to increase the applicant's resilience related to the handling of aeroplane upsets, the advanced UPRT course needs to include the development of confidence and competence in recognising and recovering safely from upsets under the presence of the real human factors. Such confidence building is specifically addressed by:

- (i) successfully overcoming natural stress response (startle and surprise); and
- (ii) performing critically important counter-intuitive actions.

Advanced UPRT therefore considers pitch attitudes, bank angles, AOA/airspeeds, sideslip and g-loads, none of which are normally experienced during routine operations.

- (d) Aeroplanes used in this course should be:
 - (1) appropriately certified and operated by the ATO in a manner that takes into account the effects of repeated training manoeuvres on airframe fatigue life; and
 - (2) provide sufficient safety margins to cater for student and instructor errors.
- (e) This course complements UPRT in FSTDs by providing exposure to psycho-physiological conditions, which cannot be delivered by the motion systems of today's qualified FSTDs. At completion of the course, the student should pilot to be able to:
 - (1) recognise and confirm the upset-situation;
 - (2) manage stress response;
 - (3) apply the correct recovery strategy timely and effectively;
 - (4) stay within the defined training envelope;
 - (5) stabilise the flight path after recovery; and
 - (6) become competent and confident in recovering from upsets.

SPECIFIC EXERCISES

- (f) Exercise 1 — Nose HIGH recovery

Exercise 1

Recovery from **Nose HIGH** upsets at various bank angles

(1) Training objectives	The student pilot should: (i) recognise and confirm the Nose HIGH situation (AOA, attitude, energy, trends); (ii) announce 'Nose High'; and (iii) apply the correct recovery strategy.
(2) Training tasks	The student pilot should: (i) regain situation awareness; (ii) recognise and analyse AOA, pitch, bank, energy state and trends; (iii) note natural and synthetic indications for AOA, attitude, and energy; (iv) manage human factors, stress response (startle and surprise, counter-intuitive actions); (v) take manual control; (vi) identify and apply the Nose HIGH recovery strategy; (vii) correct any out-of-trim condition; (viii) manage nose-down movement; (ix) manage g-load; (x) use the effects of power to assist nose-down movement; (xi) use bank to orient the lift vector as necessary; (xii) stabilise the flight path after recovery using basic pitch/power settings;
(3) Enabling objectives	The student pilot should: (i) decide if Stall Recovery or Nose HIGH recovery is applicable; (ii) perform control inputs deliberately; (iii) use up to full control deflections; (iv) avoid unnecessary low or high loads;

- (v) use secondary flight controls (trim/power) as necessary to support primary flight control inputs (i.e. nose-down movement); (vi) apply control inputs in the correct sequence (see Table 1, Nose-HIGH Recovery Strategy); (vii) apply counter-intuitive actions as necessary:
- (A) unloading;
 - (B) power-reduction in Nose-HIGH attitude (depending on engine mounting); and
 - (C) using bank to orient the lift vector downwards.

Note: Refer to GM1 to Appendix 9, Table 2: Recommended nose-high recovery strategy template.

(g) Exercise 2 — Nose LOW Recovery

Exercise 2

Recovery from **Nose LOW** upsets at various bank angles

(1) Training objectives	The student pilot should: (i) recognise and confirm the situation (AOA, attitude, energy, trends); (ii) announce 'Nose LOW'; (iii) apply the correct recovery strategy.
(2) Training tasks	The student pilot should: (i) regain situation awareness; (ii) recognise and analyse AOA, pitch, bank, energy state and trends; (iii) note natural and synthetic indications for AOA, attitude and energy; (iv) manage human factors, stress response (startle and surprise, counter-intuitive actions); (v) take manual control; (vi) identify and apply the Nose LOW recovery strategy; (vii) correct out-of-trim condition; (viii) decide if aircraft is stalled; (ix) manage g-load; (x) identify the correct direction to roll; (xi) roll to wings level to orient the lift vector upwards; (xii) manage power and drag; and (xiii) stabilise the flight path after recovery using basic pitch/power settings.
(3) Enabling objectives	The student pilot should: (i) perform control inputs deliberately; (ii) use up to full control deflections; (iii) avoid unnecessary low or high loads; (iv) apply control inputs in the correct sequence (see Table 2, Nose-LOW Recovery Strategy); and (v) apply counter-intuitive actions as necessary: (A) apply Stall Recovery in nose low attitude first if needed; (B) unloading instead of pulling; (C) unloading to increase roll rate; (D) avoid 'rolling-pull'; and (E) accept the priority of rolling to wings level first, before reducing power and before pulling.

Note: Refer to GM1 to Appendix 9, Table 3: Recommended nose-low recovery strategy template.

(h) Exercise 3 — Recovery from spiral dive

Exercise 3

Recovery from **Spiral Dive**

(1) Training objectives	The student pilot should: (i) recognise the spiral dive as a result of improper nose-up elevator input during a Nose LOW turning situation; and (i) apply the Nose LOW Recovery Strategy.
(2) Training tasks	The student pilot should: (i) maintain/regain situation awareness; (ii) recognise and analyse AOA, pitch, bank, energy state and trends; (iii) manage human factors, stress response (startle and surprise, counter-intuitive actions); (iv) take manual control; (v) identify and apply the Nose LOW recovery strategy; and (vi) stabilise the flight path after recovery using basic pitch/power settings.
(3) Enabling objectives	The student pilot should: (i) perform control inputs deliberately and in the correct sequence; (ii) use up to full control deflections, if required; and (iii) apply counter-intuitive actions as necessary: (A) unloading instead of pulling; (B) unloading to increase roll rate; (C) avoid 'rolling-pull'; and (D) accepting the priority of rolling to wings level first, before reducing power and before pulling.

(i) Exercise 4 — Stall Event Recovery

Exercise 4

Recovery from **Stall event**

(1) Training objectives	The student pilot should: (i) recognise and confirm the situation (AOA, attitude, energy, trends); (ii) announce 'Stall'; (iii) apply the Stall Event Recovery Strategy.
(2) Training tasks	The student pilot should: (i) regain situation awareness; (ii) recognise and analyse AOA, pitch, bank, energy state and trends; (iii) note natural and synthetic indications for high AOA/stall; (iv) manage human factors, stress response (startle and surprise, counter-intuitive actions); (v) recover from: (A) approach to stall (B) full stall, wings level and during turn (C) slipping stall (D) skidding stall (E) accelerated stall (F) secondary stall (vi) take manual control; (vii) identify and apply the Stall Event Recovery Template or the aircraft manufacturer Stall Recovery SOP; (viii) apply nose-down elevator input to reduce AOA; (ix) manage trim; (x) consider power reduction (if engine mounting induces a nose-up effect); (xi) accept altitude loss; (xii) identify the correct direction to roll to wings level; (xiii) manage power and drag; (xiv) manage g-load and energy to avoid secondary stall; and (xv) stabilise the flight path after recovery using basic pitch/power settings.

(3) Enabling objectives	The student pilot should: (i) perform control inputs deliberately; (ii) use up to full control deflections; (iii) apply control inputs in the correct sequence (see Table 3, Stall Event Recovery Strategy Template); and (iv) apply counter-intuitive actions as necessary: (A) unloading to reduce AOA; (B) unloading before rolling; (C) power reduction if necessary; (D) accepting altitude loss; and (E) waiting for airspeed increase before loading again.
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Note: Refer to GM1 to Appendix 9, Table 1: Recommended stall event recovery template

(j) Exercise 5 — Recovery from spin

Exercise 5 Recovery from incipient spin	
(1) Training objectives	The pilot should: (i) recognise and confirm the spin (AOA, yaw, attitude, energy, roll, trends); (ii) apply the OEM Incipient Spin Recovery procedure.
(2) Training tasks	The pilot should: (i) be aware of the aircraft response to all possible pitch and roll control inputs and to thrust/power changes during (incipient) spin; (ii) maintain/regain situation awareness; (iii) recognise and analyse AOA, attitude, energy, yaw, roll, trends); (iv) note natural and synthetic indications for high AOA, stall, spin; (v) manage human factors, stress response (startle and surprise, counter-intuitive actions); (vi) take manual control; (vii) identify and apply the OEM Incipient Spin Recovery Procedure; (viii) manage AOA, g-load and energy to avoid secondary stall; and (ix) stabilise the flight path after recovery using basic pitch/power settings.
(3) Enabling objectives	The pilot should: (i) perform control inputs deliberately and in the correct sequence; (ii) use up to full control deflections as required by the procedure; (iii) apply counter-intuitive actions as necessary; (iv) avoid unreflected control inputs; and (v) allow time for control inputs to show results.

(k) Assessment of student performance

By collecting evidence from observable behaviours, the instructor will continuously assess whether the student meets the required competency standards under the given conditions.

Pilot competencies and behavioural indicators in the context of the Advanced UPRT Course	
(1) Application of procedures	(i) Follows the recommended Nose HIGH or Nose LOW recovery strategy or the Stall Event Recovery Template / STALL RECOVERY SOP (ii) Identifies and follows operating instructions in a timely manner (iii) Correctly operates aircraft systems and equipment (iv) Applies relevant procedural knowledge
(2) Communication	(i) Adheres to callouts (ii) Verbalises the essential steps during the recoveries

- (3) Aeroplane flight path management — automation**
Disconnects autopilot and autothrust/autothrottle before initiating the recovery (to be simulated if the training aeroplane is not fitted with autothrust/autothrottle)
- (4) Aeroplane flight path management — manual control**
 - (i) Detects deviations from the desired aircraft trajectory and takes appropriate action
 - (ii) Controls the aircraft using appropriate attitude and power settings
 - (iii) Contains the aircraft within the defined flight envelope
- (5) Leadership and teamwork**
 - (i) Understands and agrees with the crew's roles and objectives
 - (ii) Uses initiative and gives directions when required
 - (iii) Admits mistakes and takes responsibility
 - (iv) Communicates relevant concerns and intentions
 - (v) Gives and receives feedback constructively
 - (vi) Projects self-control in all situations
- (6) Problem-solving and decision-making**
 - (i) Seeks accurate and adequate information from appropriate sources
 - (ii) Identifies and verifies what and why things have gone wrong
 - (iii) Perseveres in working through the event safely
 - (iv) Sets priorities appropriately
- (7) Situation awareness and information management**
 - (i) Identifies and assesses accurately the state of the aircraft and its systems
 - (ii) Identifies and assesses accurately the aircraft's vertical and lateral position, and its anticipated flight path
 - (iii) Anticipates accurately what could happen, plans and stays ahead of the situation
 - (iv) Recognises and effectively responds to indications of reduced situation awareness.
- (8) Workload management**
 - (i) Maintains self-control in all situations Manages and recovers from stress response (startle surprise), interruptions, distractions, variations and errors effectively
 - (ii) Reviews, monitors and cross-checks actions conscientiously
 - (iii) Verifies that tasks are completed to the expected outcome
 - (iv) Offers and accepts assistance, delegates when necessary, and asks for help early
 - (v) Manages and recovers from interruptions, distractions, variations and failures effectively

SECTION 3 – SPECIFIC REQUIREMENTS FOR THE HELICOPTER CATEGORY

FCL.720.H Experience requirements and prerequisites for the issue of type ratings – helicopters

Regulation (EU) 2021/2227

Unless otherwise determined in the operational suitability data established in accordance with Annex I (Part 21) to [Commission Regulation \(EU\) No 748/2012](#), an applicant for the issue of the first helicopter type rating shall comply with the following experience requirements and prerequisites for the issue of the relevant rating:

- (a) Multi-pilot helicopters. An applicant for a type rating for a multi-pilot helicopter type shall, before starting the type rating training course:
 - (1) have at least 70 hours as PIC on helicopters;
 - (2) except when the type rating course is combined with an MCC course:
 - (i) hold a certificate of satisfactory completion of an MCC course in helicopters; or
 - (ii) have completed at least 500 hours of flight time as a pilot in multi-pilot operations in any aircraft category.
 - (3) have passed the ATPL(H) theoretical knowledge examinations.
- (b) A graduate from an ATP(H)/IR, ATP(H), CPL(H)/IR or CPL(H) integrated course who does not comply with the requirement of point (a)(1), shall be entitled to undergo the type rating training course for a multi-pilot helicopter type rating and shall have the type rating issued with the privileges limited to exercising functions as co-pilot only. The limitation shall be removed once the pilot has complied with all of the following:
 - (1) completed 70 hours as PIC or pilot-in-command under supervision of helicopters;
 - (2) passed the multi-pilot skill test on the applicable helicopter type as PIC.
- (c) Multi-engine helicopters. An applicant for the issue of a first type rating for a multi-engine helicopter shall:
 - (1) before starting flight training:
 - (i) have passed the ATPL(H) theoretical knowledge examinations; or
 - (ii) hold a certificate of completion of a pre-entry course conducted by an ATO. The course shall cover the following subjects of the ATPL(H) theoretical knowledge course:
 - Aircraft General Knowledge: airframe/systems/power plant, and instrument/electronics,
 - Flight Performance and Planning: mass and balance, performance;
 - (2) in the case of applicants who have not completed an ATP(H)/IR, ATP(H), or CPL(H)/IR integrated training course, have completed at least 70 hours as PIC on helicopters.

FCL.735.H Multi-crew cooperation training course – helicopters

Regulation (EU) No 1178/2011

- (a) The MCC training course shall comprise at least:
- (1) for MCC/IR:
 - (i) 25 hours of theoretical knowledge instruction and exercises; and
 - (ii) 20 hours of practical MCC training or 15 hours, in the case of student pilots attending an ATP(H)/IR integrated course. When the MCC training is combined with the initial type rating training for a multi-pilot helicopter, the practical MCC training may be reduced to not less than 10 hours if the same FSTD is used for both MCC and type rating;
 - (2) for MCC/VFR:
 - (i) 25 hours of theoretical knowledge instruction and exercises; and
 - (ii) 15 hours of practical MCC training or 10 hours, in the case of student pilots attending an ATP(H)/IR integrated course. When the MCC training is combined with the initial type rating training for a multi-pilot helicopter, the practical MCC training may be reduced to not less than 7 hours if the same FSTD is used for both MCC and type rating.
- (b) The MCC training course shall be completed within 6 months at an ATO.
An FNPT II or III qualified for MCC, an FTD 2/3 or an FFS shall be used.
- (c) Unless the MCC course has been combined with a multi-pilot type rating course, on completion of the MCC training course the applicant shall be given a certificate of completion.
- (d) An applicant having completed MCC training for any other category of aircraft shall be exempted from the requirement in (a)(1)(i) or (a)(2)(i), as applicable.
- (e) An applicant for MCC/IR training who has completed MCC/VFR training shall be exempted from the requirement in (a)(1)(i), and shall complete 5 hours of practical MCC/IR training.

AMC1 FCL.735.A; FCL.735.H; FCL.735.As Multi-crew cooperation (MCC) training course

ED Decision 2020/005/R

- (a) Competency is a combination of knowledge, skills and attitudes required to perform a task to the prescribed standard.
- (b) The objectives of MCC training are to develop the technical and non-technical components of the knowledge, skills and attitudes required to operate a multi-crew aircraft.
- (c) Training should comprise both theoretical and practical elements and be designed to achieve the competencies/training objectives (see Table 1 below).

Table 1 — Competencies/training objectives			
Competency/objective	Performance indicators	Knowledge	Practical exercises
Communication	(a) Know what, how much and who to communicate to;	(a) Human Factors, TEM and CRM;	In a commercial air transport environment, apply multi-crew procedures, including

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
	(b) Ensure the recipient is ready and able to receive the information; (c) Pass messages and information clearly, accurately, timely and adequately; (d) Check if the other person has the correct understanding when passing important information; (e) Listen actively, patiently and demonstrate understanding when receiving information; (f) Ask relevant and effective questions, and offer suggestions; (g) Use appropriate body language, eye contact and tone; (h) Open and receptive to other people's view.	(b) Application of TEM and CRM principles to training.	principles of TEM and CRM to the following: (a) Pre-flight preparation: (1) FMS initialisation; (2) radio and navigation equipment preparation; (3) flight documentation; (4) computation of take-off performance data. (b) Take-off and climb: (1) before take-off checks; (2) normal take-offs; (3) rejected take-offs; (4) take-offs with abnormal and emergency situations included. (c) Cruise: emergency descent. (d) Descent and approach: (1) instrument flight procedures; (2) holding; (3) 3D Operations using raw data; (4) 3D Operations using flight director; (5) 3D Operations using autopilot; (6) one-engine-inoperative approach; (7) 2D Operations and circling; (8) computation of approach and landing data; (9) all engines go-around; (10) go-around with one engine inoperative; (11) wind shear during approach.
Leadership and team working	(a) Friendly, enthusiastic, motivating and considerate of others; (b) Use initiative, give direction and take responsibility when required; (c) Open and honest about thoughts, concerns and intentions; (d) Give and receive criticism and praise well, and admit mistakes; (e) Confidently do and say what is important to him or her; (f) Demonstrate respect and tolerance towards other people; (g) Involve others in planning and share activities fairly.		(e) landing: transition from instrument to visual flight on reaching decision altitude or height or minimum descent altitude or height;
Situational awareness	(a) Be aware of what the aircraft and its systems are doing; (b) Be aware of where the aircraft is and its environment; (c) Keep track of time and fuel;		

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
	(d) Be aware of the condition of people involved in the operation including passengers; (e) Recognise what is likely to happen, plan and stay ahead of the game; (f) Develop what-if scenarios and make pre-decisions; (g) Identify threats to the safety of the aircraft and of the people.		(f) after landing and post flight procedures; (g) selected emergency and abnormal procedures.
Workload management	(a) Be calm, relaxed, careful and not impulsive; (b) Prepare, prioritise and schedule tasks effectively; (c) Use time efficiently when carrying out tasks; (d) Offer and accept assistance, delegate when necessary and ask for help early; (e) Review and monitor and cross-check actions conscientiously; (f) Follow procedures appropriately and consistently; (g) Concentrate on one thing at a time, ensure tasks are completed and does not become distracted; (h) Carry out instructions as directed.		
Problem-solving and decision-making	(a) Identify and verify why things have gone wrong and do not jump to conclusions or make assumptions; (b) Seek accurate and adequate information from appropriate resources; (c) Persevere in working through a problem; (d) Use and agree an appropriate decision making process; (e) Agree essential and desirable criteria and prioritises; (f) Consider as many options as practicable;		

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
	(g) Make decisions when they need to, reviews and changes if required; (h) Consider risks but do not take unnecessary risks.		
Monitoring and cross-checking	(a) Monitor and cross-checks all actions; (b) Monitor aircraft trajectory in critical flight phases; (c) Take appropriate actions in response to deviations from the flight path.	(a) SOPs; (b) Aircraft systems; (c) Undesired aircraft states.	
Task sharing	(a) Apply SOPs in both PF and pilot monitoring (PM) roles; (b) Makes and responds to standard call-outs.	(a) PF and PM roles; (b) SOPs.	
Use of checklists	Utilise checklists appropriately according to SOPs.	(a) SOPs; (b) Checklist philosophy.	
Briefings	Prepare and deliver appropriate briefings.	(a) SOPs; (b) Interpretation of FMS data and in-flight documentation.	
Flight management	(a) Maintain a constant awareness of the aircraft automation state; (b) Manage automation to achieve optimum trajectory and minimum workload; (c) Take effective recovery actions from automation anomalies; (d) Manage aircraft navigation, terrain clearance; (e) Manage aircraft fuel state and take appropriate actions.	(a) Understanding of aircraft performance and configuration; (b) Systems; (c) SOPs; (d) Interpretation of FMS data and in-flight documentation; (e) Minimum terrain clearance; (f) Fuel management IFR and VFR regulation.	
FMS use	Programme, manage and monitor FMS in accordance with SOPs.	(a) Systems (FMS); (b) SOPs; (c) Automation.	
Systems normal operations	Perform and monitor normal systems operation in accordance with SOPs.	(a) Systems; (b) SOPs.	

Table 1 — Competencies/training objectives			
Competency/ objective	Performance indicators	Knowledge	Practical exercises
Systems abnormal and emergency operations	(a) Perform and monitor abnormal systems operation in accordance with SOPs; (b) Utilise electronic and paper abnormal checklists in accordance with SOPs.	(a) Systems; (b) SOPs; (c) Emergency and abnormal procedures and checklists; (d) Recall items.	
Environment, weather and ATC	(a) Communicate effectively with ATC; (b) Avoid misunderstandings by requesting clarification; (c) Adhere to ATC instructions; (d) Construct a mental model of the local ATC and weather environment.	(a) Systems; (b) SOPs; (c) ATC environment and phraseology; (d) Procedures for hazardous weather conditions.	

CERTIFICATE OF COMPLETION FORM

CERTIFICATE OF COMPLETION OF MCC			
Applicant's last name(s):		First name(s):	
Type of licence:		Number:	State:
ME/IR training completed		OR	ME/IR validity date: ME/IR skill test date:
Issued on:		passed on:	
	Signature of applicant:		

The satisfactory completion of MCC-Training according to requirements is certified below:

TRAINING			
Multi-crew co-operation training received during period:			
from:	to:	at:	ATO / operator*

Location and date:	Signature of head of ATO or authorised instructor*:
Type and number of licence and state of issue:	Name(s) in capital letters of authorised instructor:

* Delete as appropriate

FCL.740.H Revalidation of type ratings – helicopters

Regulation (EU) 2024/2076

- (a) Revalidation. For the revalidation of type ratings for helicopters, applicants shall comply with one of the following, as applicable:
- (1) they shall take the following steps:
 - (i) within the validity period of the type rating, complete at least 2 hours as pilot in the relevant helicopter type;
 - (ii) within the 3 months immediately preceding the expiry date of the type rating, pass a proficiency check:
 - (A) which is conducted in accordance with [Appendix 9](#) in the relevant type of helicopter or an FSTD representing that type;
 - (B) the duration of which may be counted towards the flight time specified in point (i);
 - (2) for type ratings for single-engine helicopters up to a maximum take-off mass of 3 175 kg, they shall meet one of the following conditions:
 - (i) comply with point (a)(1);
 - (ii) complete, in the relevant type of helicopter, both of the following:
 - (A) within the validity period of the type rating, at least 6 hours as PIC;
 - (B) within the 3 months immediately preceding the expiry date of the rating, a refresher training of at least 1 hour of total flight time with and to the satisfaction of an instructor who shall select those flight exercises that allow the applicant to refresh their competence in safely operating the aircraft and applying normal, abnormal and emergency procedures. Applicants may choose to complete the refresher training in the aircraft or an FSTD representing the relevant type, or in a combination of both.
- (b) When applicants hold more than one type rating for SEP helicopters, they may achieve revalidation of all the relevant type ratings by passing the proficiency check in accordance with point (a)(1)(ii) or completing the refresher training in accordance with point (a)(2)(ii)(B) in only one of the relevant types held, provided that they have completed either of the following flight times as PIC in each of the other relevant types during the validity period:
- (1) if they completed a proficiency check in accordance with point (a)(1)(ii), at least 2 hours;
 - (2) if they completed a refresher training in accordance with point (a)(2)(ii)(B), at least 6 hours.

The proficiency check or the refresher training, as applicable, shall be performed each time on a different type. The new validity period of all type ratings revalidated in accordance with this point shall commence together with the validity period of the type rating for which the proficiency check or the refresher training is performed.

- (c) When applicants hold more than one type rating for SET helicopters with a maximum certified take-off mass up to 3 175 kg, they may achieve revalidation of all the relevant type ratings by passing the proficiency check in accordance with point (a)(1)(ii) or completing the refresher training in accordance with point (a)(2)(ii)(B) in only one of the relevant types held, provided that they have completed all of the following:
- (1) 300 hours as PIC on helicopters;
 - (2) 15 hours on each of the types held;
 - (3) either of the following flight times as PIC in each of the other relevant types during the validity period:
 - (i) if they completed a proficiency check in accordance with point (a)(1)(ii), at least 2 hours;
 - (ii) if they completed a refresher training in accordance with point (a)(2)(ii)(B), at least 6 hours.

The proficiency check shall be performed each time on a different type. The new validity period of all type ratings revalidated in accordance with this point shall commence together with the validity period of the type rating for which the proficiency check or the refresher training is completed.

- (d) Applicants who successfully complete a skill test for the issue of an additional type rating or a proficiency check for the renewal of a type rating shall achieve revalidation for the relevant type ratings already held in the common groups, as specified in points (b) and (c).

The new validity period of all type ratings revalidated in accordance with this point shall commence together with the validity period of the type rating for which the skill test or proficiency check is completed.

- (e) The revalidation of an IR(H), if held, may be combined with a proficiency check for a type rating.
- (f) Applicants who fail to achieve a pass in all sections of a proficiency check before the expiry date of a type rating shall not exercise the privileges of that rating until a pass in the proficiency check has been achieved. In the case of points (b) and (c), applicants shall not exercise their privileges in any of the types.

GM1 FCL.740.H Revalidation of type ratings — helicopters

ED Decision 2025/002/R

SAFETY AWARENESS BRIEFING BEFORE REFRESHER TRAINING OR PROFICIENCY CHECKS

It is recommended that the pre-flight briefing before refresher training or a proficiency check in accordance with point [FCL.740.H](#) with the pilot, the instructor or examiner, as applicable, includes elements to raise the pilot's safety awareness with regard to safely flying helicopters. This part of the briefing (safety awareness briefing) should have a duration of at least 15 minutes to allow discussions on several safety issues, referring to accidents and incidents in general or risks specifically related to the type of flights usually undertaken by the pilot. Threat and error management (TEM) should be promoted as effective mitigation, including the illustration of the practical application of TEM using real-life examples. There is no restriction on the subjects that could be covered. They may range from

weather-related issues to personal or passenger-induced pressure. The material that can be used to support this briefing could come from accident & incident reports, mandatory or voluntary safety reporting, safety campaigns of different sources as well as from personal experience.

AMC1 FCL.740.H(a)(2)(ii)(B) Revalidation of type ratings — helicopters

ED Decision 2025/002/R

CONTENT OF THE REFRESHER TRAINING — SINGLE-ENGINE HELICOPTERS UP TO A MAXIMUM TAKE-OFF MASS OF 3 175 KG

- (a) Before the training takes place, the instructor should hold a briefing with the candidate. That briefing should include a discussion on all of the following:
 - (1) TEM with special emphasis on decision-making when encountering adverse meteorological conditions or unintentional IMC;
 - (2) navigation flight techniques;
 - (3) exercises as specified in point (b), as applicable.
- (b) Training items should be based on the exercise items of the proficiency check, as deemed relevant by the instructor, and depending on the experience of the candidate. In any case, the training flight items should include the following exercises from the PPL(H) flight training syllabus ([AMC2 FCL.210](#)):
 - (1) Exercise 18: Hover OGE — Vortex ring — unanticipated yaw (LTE);
 - (2) Exercise 21: Practice forced landings;
 - (3) Exercise 29: Confined areas.

AMC1 FCL.740.H(b) Revalidation of type ratings – helicopters

ED Decision 2025/002/R

CREDITING OF THE PROFICIENCY CHECK TOWARDS SEP HELICOPTER TYPES

Only the following SEP helicopter types can be considered for crediting of the proficiency check. Other SEP helicopters (for example, R22) should not be given credit for.

Manufacturer	SEP Helicopter type
Agusta-Bell	Bell47
Bell Helicopters	Bell47
Westland	Bell47
Brantley	Brantley B2
Enstrom	ENF28
Hélicoptères Guimbal	Cabri G2
Hiller	UH12
Robinson	R44
Hughes or Schweizer	HU269
Breda Nardi	HU269

SECTION 4 – SPECIFIC REQUIREMENTS FOR THE POWERED-LIFT AIRCRAFT CATEGORY

FCL.720.PL Experience requirements and prerequisites for the issue of type ratings – powered-lift aircraft

Regulation (EU) No 1178/2011

Unless otherwise determined in the operational suitability data established in accordance with Part-21, an applicant for the first issue of a powered-lift type rating shall comply with the following experience requirements and prerequisites:

- (a) for pilots of aeroplanes:
 - (1) hold a CPL/IR(A) with ATPL theoretical knowledge or an ATPL(A);
 - (2) hold a certificate of completion of an MCC course;
 - (3) have completed more than 100 hours as pilot on multi-pilot aeroplanes;
 - (4) have completed 40 hours of flight instruction in helicopters;
- (b) for pilots of helicopters:
 - (1) hold a CPL/IR(H) with ATPL theoretical knowledge or an ATPL/IR(H);
 - (2) hold a certificate of completion of an MCC course;
 - (3) have completed more than 100 hours as a pilot on multi-pilot helicopters;
 - (4) have completed 40 hours of flight instruction in aeroplanes;
- (c) for pilots qualified to fly both aeroplanes and helicopters:
 - (1) hold at least a CPL(H);
 - (2) hold an IR and ATPL theoretical knowledge or an ATPL in either aeroplanes or helicopters;
 - (3) hold a certificate of completion of an MCC course in either helicopters or aeroplanes;
 - (4) have completed at least 100 hours as a pilot on multi-pilot helicopters or aeroplanes;
 - (5) have completed 40 hours of flight instruction in aeroplanes or helicopters, as applicable, if the pilot has no experience as ATPL or on multi-pilot aircraft.

GM1 FCL.720.PL Experience requirements and prerequisites for the issue of type ratings – powered-lift aircraft

ED Decision 2011/016/R

The endorsement of a powered-lift type rating to an aeroplane or helicopter licence does not confer upon its holder the privileges to fly helicopters or aeroplanes, respectively.

FCL.725.PL Flight instruction for the issue of type ratings – powered-lift aircraft

Regulation (EU) No 1178/2011

The flight instruction part of the training course for a powered-lift type rating shall be completed in both the aircraft and an FSTD representing the aircraft and adequately qualified for this purpose.

FCL.740.PL Revalidation of type ratings – powered-lift aircraft

Regulation (EU) No 1178/2011

- (a) Revalidation. For revalidation of powered-lift type ratings, the applicant shall:
- (1) pass a proficiency check in accordance with [Appendix 9](#) to this Part in the relevant type of powered-lift within the 3 months immediately preceding the expiry date of the rating;
 - (2) complete during the period of validity of the rating, at least:
 - (i) 10 route sectors as pilot of the relevant type of powered-lift aircraft; or
 - (ii) 1 route sector as pilot of the relevant type of powered-lift aircraft or FFS, flown with an examiner. This route sector may be flown during the proficiency check.
 - (3) A pilot working for a commercial air transport operator approved in accordance with the applicable air operations requirements who has passed the operators proficiency check combined with the proficiency check for the revalidation of the type rating shall be exempted from complying with the requirement in (2).
- (b) An applicant who fails to achieve a pass in all sections of a proficiency check before the expiry date of a type rating shall not exercise the privileges of that rating until the a pass in the proficiency check has been achieved.

SECTION 5 – SPECIFIC REQUIREMENTS FOR THE AIRSHIP CATEGORY

FCL.720.As Prerequisites for the issue of type ratings – airships

Regulation (EU) No 1178/2011

Unless otherwise determined in the operational suitability data established in accordance with Part-21, an applicant for the first issue of an airship type rating shall comply with the following experience requirements and prerequisites:

- (a) for multi-pilot airships:
 - (1) have completed 70 hours of flight time as PIC on airships;
 - (2) hold a certificate of satisfactory completion of MCC on airships.
 - (3) An applicant who does not comply with the requirement in (2) shall have the type rating issued with the privileges limited to exercising functions as co-pilot only. The limitation shall be removed once the pilot has completed 100 hours of flight time as PIC or pilot-in-command under supervision of airships.

FCL.735.As Multi-crew cooperation training course – airships

Regulation (EU) No 245/2014

- (a) The MCC training course shall comprise at least:
 - (1) 12 hours of theoretical knowledge instruction and exercises; and
 - (2) 5 hours of practical MCC training;An FNPT II, or III qualified for MCC, an FTD 2/3 or an FFS shall be used.
- (b) The MCC training course shall be completed within 6 months at an ATO.
- (c) Unless the MCC course has been combined with a multi-pilot type rating course, on completion of the MCC training course the applicant shall be given a certificate of completion.
- (d) An applicant having completed MCC training for any other category of aircraft shall be exempted from the requirements in (a).

AMC1 FCL.735.A; FCL.735.H; FCL.735.As Multi-crew cooperation (MCC) training course

ED Decision 2020/005/R

- (a) Competency is a combination of knowledge, skills and attitudes required to perform a task to the prescribed standard.
- (b) The objectives of MCC training are to develop the technical and non-technical components of the knowledge, skills and attitudes required to operate a multicrew aircraft.
- (c) Training should comprise both theoretical and practical elements and be designed to achieve the competencies/training objectives (see Table 1 below).

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
Communication	(a) Know what, how much and who to communicate to; (b) Ensure the recipient is ready and able to receive the information; (c) Pass messages and information clearly, accurately, timely and adequately; (d) Check if the other person has the correct understanding when passing important information; (e) Listen actively, patiently and demonstrate understanding when receiving information; (f) Ask relevant and effective questions, and offer suggestions; (g) Use appropriate body language, eye contact and tone; (h) Open and receptive to other people's view.	(a) Human Factors, TEM and CRM; (b) Application of TEM and CRM principles to training.	In a commercial air transport environment, apply multi-crew procedures, including principles of TEM and CRM to the following: (a) Pre-flight preparation: (1) FMS initialisation; (2) radio and navigation equipment preparation; (3) flight documentation; (4) computation of take-off performance data. (b) Take-off and climb: (1) before take-off checks; (2) normal take-offs; (3) rejected take-offs; (4) take-offs with abnormal and emergency situations included. (c) Cruise: emergency descent. (d) Descent and approach: (1) instrument flight procedures; (2) holding; (3) 3D Operations using raw data; (4) 3D Operations using flight director; (5) 3D Operations using autopilot; (6) one-engine-inoperative approach; (7) 2D Operations and circling; (8) computation of approach and landing data; (9) all engines go-around; (10) go-around with one engine inoperative; (11) wind shear during approach. (e) landing: transition from instrument to visual flight on reaching decision altitude or height or minimum descent altitude or height;
Leadership and team working	(a) Friendly, enthusiastic, motivating and considerate of others; (b) Use initiative, give direction and take responsibility when required; (c) Open and honest about thoughts, concerns and intentions; (d) Give and receive criticism and praise well, and admit mistakes; (e) Confidently do and say what is important to him or her; (f) Demonstrate respect and tolerance towards other people; (g) Involve others in planning and share activities fairly.		
Situational awareness	(a) Be aware of what the aircraft and its systems are doing; (b) Be aware of where the aircraft is and its environment; (c) Keep track of time and fuel;		

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
	(d) Be aware of the condition of people involved in the operation including passengers; (e) Recognise what is likely to happen, plan and stay ahead of the game; (f) Develop what-if scenarios and make pre-decisions; (g) Identify threats to the safety of the aircraft and of the people.		(f) after landing and post flight procedures; (g) selected emergency and abnormal procedures.
Workload management	(a) Be calm, relaxed, careful and not impulsive; (b) Prepare, prioritise and schedule tasks effectively; (c) Use time efficiently when carrying out tasks; (d) Offer and accept assistance, delegate when necessary and ask for help early; (e) Review and monitor and cross-check actions conscientiously; (f) Follow procedures appropriately and consistently; (g) Concentrate on one thing at a time, ensure tasks are completed and does not become distracted; (h) Carry out instructions as directed.		
Problem-solving and decision-making	(a) Identify and verify why things have gone wrong and do not jump to conclusions or make assumptions; (b) Seek accurate and adequate information from appropriate resources; (c) Persevere in working through a problem; (d) Use and agree an appropriate decision making process; (e) Agree essential and desirable criteria and prioritises; (f) Consider as many options as practicable;		

Table 1 — Competencies/training objectives

Competency/ objective	Performance indicators	Knowledge	Practical exercises
	(g) Make decisions when they need to, reviews and changes if required; (h) Consider risks but do not take unnecessary risks.		
Monitoring and cross-checking	(a) Monitor and cross-checks all actions; (b) Monitor aircraft trajectory in critical flight phases; (c) Take appropriate actions in response to deviations from the flight path.	(a) SOPs; (b) Aircraft systems; (c) Undesired aircraft states.	
Task sharing	(a) Apply SOPs in both PF and pilot monitoring (PM) roles; (b) Makes and responds to standard call-outs.	(a) PF and PM roles; (b) SOPs.	
Use of checklists	Utilise checklists appropriately according to SOPs.	(a) SOPs; (b) Checklist philosophy.	
Briefings	Prepare and deliver appropriate briefings.	(a) SOPs; (b) Interpretation of FMS data and in-flight documentation.	
Flight management	(a) Maintain a constant awareness of the aircraft automation state; (b) Manage automation to achieve optimum trajectory and minimum workload; (c) Take effective recovery actions from automation anomalies; (d) Manage aircraft navigation, terrain clearance; (e) Manage aircraft fuel state and take appropriate actions.	(a) Understanding of aircraft performance and configuration; (b) Systems; (c) SOPs; (d) Interpretation of FMS data and in-flight documentation; (e) Minimum terrain clearance; (f) Fuel management IFR and VFR regulation.	
FMS use	Programme, manage and monitor FMS in accordance with SOPs.	(a) Systems (FMS); (b) SOPs; (c) Automation.	
Systems normal operations	Perform and monitor normal systems operation in accordance with SOPs.	(a) Systems; (b) SOPs.	

Table 1 — Competencies/training objectives			
Competency/ objective	Performance indicators	Knowledge	Practical exercises
Systems abnormal and emergency operations	(a) Perform and monitor abnormal systems operation in accordance with SOPs; (b) Utilise electronic and paper abnormal checklists in accordance with SOPs.	(a) Systems; (b) SOPs; (c) Emergency and abnormal procedures and checklists; (d) Recall items.	
Environment, weather and ATC	(a) Communicate effectively with ATC; (b) Avoid misunderstandings by requesting clarification; (c) Adhere to ATC instructions; (d) Construct a mental model of the local ATC and weather environment.	(a) Systems; (b) SOPs; (c) ATC environment and phraseology; (d) Procedures for hazardous weather conditions.	

CERTIFICATE OF COMPLETION FORM

CERTIFICATE OF COMPLETION OF MCC			
Applicant's last name(s):		First name(s):	
Type of licence:		Number:	State:
ME/IR training completed		OR	ME/IR validity date: ME/IR skill test date:
Issued on:		passed on:	
	Signature of applicant:		

The satisfactory completion of MCC-Training according to requirements is certified below:

TRAINING			
Multi-crew co-operation training received during period:			
from:	to:	at:	ATO / operator*

Location and date:	Signature of head of ATO or authorised instructor*:
Type and number of licence and state of issue:	Name(s) in capital letters of authorised instructor:

* Delete as appropriate

FCL.740.As Revalidation of type ratings – airships

Regulation (EU) No 1178/2011

- (a) Revalidation. For revalidation of type ratings for airships, the applicant shall:
- (1) pass a proficiency check in accordance with Appendix 9 to this Part in the relevant type of airship within the 3 months immediately preceding the expiry date of the rating; and
 - (2) complete at least 2 hours as a pilot of the relevant airship type within the validity period of the rating. The duration of the proficiency check may be counted towards the 2 hours.
 - (3) The revalidation of an IR(As), if held, may be combined with a proficiency check for the revalidation of a class or type rating.
- (b) An applicant who fails to achieve a pass in all sections of a proficiency check before the expiry date of a type rating shall not exercise the privileges of that rating until a pass in the proficiency check has been achieved.

SUBPART I – ADDITIONAL RATINGS

FCL.800 Aerobatic rating

Regulation (EU) 2020/359

- (a) Holders of a pilot licence with privileges to fly aeroplanes or TMGs shall undertake aerobatic flights only if they hold an aerobatic rating in accordance with this point.
- (b) Applicants for an aerobatic rating shall have completed:
 - (1) after the issue of the licence, at least 30 hours of flight time as PIC in aeroplanes or TMGs;
 - (2) a training course at DTO or at an ATO, including:
 - (i) theoretical knowledge instruction appropriate for the rating;
 - (ii) at least 5 hours of aerobatic instruction in aeroplanes or TMGs flown with engine power.
- (c) The privileges of the aerobatic rating shall be limited to aerobatic flight in either aeroplanes or TMGs flown with engine power, depending on which aircraft the requirements of (b)(1) and (b)(2)(ii) were complied with. This limitation shall be lifted upon application if a pilot has successfully completed at least 3 dual training flights in aeroplanes or TMGs flown with engine power, as applicable, covering the full aerobatic training syllabus.
- (d) Applicants for an aerobatic rating who also hold a TMG class rating as well as advanced aerobatic privileges for sailplanes with privileges as specified in point SFCL.200(d) of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#) shall:
 - (1) be exempted from getting their aerobatic rating limited to aeroplanes, as specified in paragraph (c) if they have complied with the requirements of (b)(1) and (b)(2)(ii) in aeroplanes, or
 - (2) receive full credit towards the requirements in paragraph (b) for the issue of an aerobatic rating restricted to TMGs flown with engine power. This limitation shall be lifted upon application if a pilot has completed the training as specified in paragraph (c).

AMC1 FCL.800 Aerobatic rating

ED Decision 2025/002/R

THEORETICAL KNOWLEDGE AND FLYING TRAINING

- (a) The aim of the aerobatic training is to qualify licence holders to perform aerobatic manoeuvres.
- (b) The DTO or the ATO should issue a certificate of satisfactory completion of the instruction to licence endorsement.
- (c) Theoretical knowledge

The theoretical knowledge syllabus should cover the revision or explanation of:

- (1) human factors and body limitation:
 - (i) spatial disorientation;
 - (ii) airsickness;
 - (iii) body stress and G-forces, positive and negative;
 - (iv) effects of grey- and blackouts.

- (2) technical subjects:
 - (i) legislation affecting aerobatic flying to include environmental and noise subjects;
 - (ii) principles of aerodynamics to include slow flight, stalls and spins, flat and inverted;
 - (iii) general airframe and engine limitations (if applicable).
- (3) limitations applicable to the specific aircraft category (and type):
 - (i) air speed limitations (aeroplane or TMG, as applicable);
 - (ii) symmetric load factors (type-related, as applicable);
 - (iii) rolling Gs (type-related, as applicable).
- (4) aerobatic manoeuvres and recovery:
 - (i) entry parameters;
 - (ii) planning systems and sequencing of manoeuvres;
 - (iii) rolling manoeuvres;
 - (iv) looping manoeuvres;
 - (v) combination manoeuvres;
 - (vi) entry and recovery from developed spins, flat, accelerated and inverted.
- (5) emergency procedures:
 - (i) recovery from unusual attitudes;
 - (ii) drills to include the use of parachutes (if worn) and aircraft abandonment.
- (d) Flying training

The exercises of the aerobatic flying training syllabus should be repeated as necessary until the applicant achieves a safe and competent standard. At the end of the flight training, the student pilot should be able to perform an aerobatic flight containing a sequence of aerobatic manoeuvres. The training flights should be tailored to the category of aircraft and limited to the permitted manoeuvres of that type of aircraft. The exercises should comprise at least the following practical training items:

- (1) confidence manoeuvres and recoveries:
 - (i) slow flights and stalls;
 - (ii) steep turns;
 - (iii) side slips;
 - (iv) engine restart in-flight (in case of unintentional engine stop during an aerobatic manoeuvre);
 - (v) spins and recovery;
 - (vi) recovery from spiral dives;
 - (vii) recovery from unusual attitudes.
- (2) aerobatic manoeuvres:
 - (i) Chandelle;
 - (ii) Lazy Eight;

- (iii) rolls;
- (iv) loops;
- (v) inverted flight;
- (vi) Hammerhead turn;
- (vii) Immelmann.

FCL.805 Sailplane towing and banner towing ratings

Regulation (EU) 2020/359

- (a) Holders of a pilot licence with privileges to fly aeroplanes or TMGs shall only tow sailplanes or banners when they hold the appropriate sailplane towing or banner towing rating.
- (b) Applicants for a sailplane towing rating shall have completed:
 - (1) at least 30 hours of flight time as PIC and 60 take-offs and landings in aeroplanes, if the activity is to be carried out in aeroplanes, or in TMGs, if the activity is to be carried out in TMGs, completed after the issue of the licence;
 - (2) a training course at a DTO or at an ATO, including:
 - (i) theoretical knowledge instruction on towing operations and procedures;
 - (ii) at least 10 instruction flights towing a sailplane, including at least 5 dual instruction flights; and
 - (iii) except for holders of an SPL in accordance with Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), 5 familiarisation flights in a sailplane which is launched by an aircraft.
- (c) Applicants for a banner towing rating shall have completed:
 - (1) at least 100 hours of flight time and 200 take-offs and landings as PIC on aeroplanes or TMG, after the issue of the licence. At least 30 of these hours shall be in aeroplanes, if the activity is to be carried out in aeroplanes, or in TMG, if the activity is to be carried out in TMGs;
 - (2) a training course at a DTO or at an ATO including:
 - (i) theoretical knowledge instruction on towing operations and procedures;
 - (ii) at least 10 instruction flights towing a banner, including at least 5 dual flights.
- (d) The privileges of the sailplane and banner towing ratings shall be limited to aeroplanes or TMGs appropriately to aircraft on which the flight instruction was completed. For banner towing, the privileges shall be limited to the towing method used for flight instruction. The privileges shall be extended if pilots have successfully completed at least three dual training flights covering the full towing training syllabus in either aircraft and towing method for banner towing.
- (e) In order to exercise the privileges of the sailplane or banner towing ratings, the holder of the rating shall have completed a minimum of 5 tows during the last 24 months.
- (f) When the pilot does not comply with the requirement in (e), before resuming the exercise of his/her privileges, the pilot shall complete the missing tows with or under the supervision of an instructor.
- (g) Applicants for a sailplane towing or banner towing rating on TMGs in accordance with this point shall receive full credit towards the requirements of paragraph (b) or (c), as applicable, if they

hold a sailplane towing or banner towing rating in accordance with point SFCL.205 of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), as applicable, or if they have fulfilled all the requirements for the issue of that rating.

AMC1 FCL.805 Sailplane towing and banner towing rating

ED Decision 2018/009/R

THEORETICAL KNOWLEDGE AND FLYING TRAINING

- (a) The aim of the towing instruction is to qualify licence holders to tow banners or sailplanes.
- (b) The DTO or the ATO should issue a certificate of satisfactory completion of the instruction to licence endorsement.

- (c) Theoretical knowledge: towing of sailplanes

The theoretical knowledge syllabus for towing of sailplanes should cover the revision or explanation of:

- (1) regulations about towing flights;
- (2) equipment for the towing activity;
- (3) sailplane towing techniques, including:
 - (i) signals and communication procedures;
 - (ii) take-off (normal and crosswind);
 - (iii) in-flight launch procedures;
 - (iv) descending on tow;
 - (v) sailplane release procedure;
 - (vi) tow rope release procedure;
 - (vii) landing with tow rope connected (if applicable);
 - (viii) emergency procedures during tow, including equipment malfunctions;
 - (ix) safety procedures;
 - (x) flight performance of the applicable aircraft type when towing sailplanes;
 - (xi) look-out and collision avoidance;
 - (xii) performance data sailplanes, including:
 - (A) suitable speeds;
 - (B) stall characteristics in turns.

- (d) Theoretical knowledge: banner towing

The theoretical knowledge syllabus for banner towing should cover the revision or explanation of:

- (1) regulations about banner towing;
- (2) equipment for the banner towing activity;
- (3) ground crew coordination;
- (4) pre-flight procedures;

-
- (5) banner towing techniques, including:
- (i) take-off launch;
 - (ii) banner pickup manoeuvres;
 - (iii) flying with a banner in tow;
 - (iv) release procedure;
 - (v) landing with a banner in tow (if applicable);
 - (vi) emergency procedures during tow, including equipment malfunctions;
 - (vii) safety procedures;
 - (viii) flight performance of the applicable aircraft type when towing a heavy or light banner;
 - (ix) prevention of stall during towing operations.
- (e) Flying training: towing of sailplanes
- The exercises of the towing training syllabus for towing sailplanes should be repeated as necessary until the student achieves a safe and competent standard and should comprise at least the following practical training items:
- (1) take-off procedures (normal and crosswind take-offs);
 - (2) 360° circles on tow with a bank of 30° and more;
 - (3) descending on tow;
 - (4) release procedure of the sailplane;
 - (5) landing with the tow rope connected (if applicable);
 - (6) tow rope release procedure in-flight;
 - (7) emergency procedures (simulation);
 - (8) signals and communication during tow.
- (f) Flying training: banner towing
- The exercises of the towing training syllabus for banner towing should be repeated as necessary until the student achieves a safe and competent standard and should comprise at least the following practical training items:
- (1) pickup manoeuvres;
 - (2) towing in-flight techniques;
 - (3) release procedures;
 - (4) flight at critically low air speeds;
 - (5) maximum performance manoeuvres;
 - (6) emergency manoeuvres to include equipment malfunctions (simulated);
 - (7) specific banner towing safety procedures;
 - (8) go-around with the banner connected;
 - (9) loss of engine power with the banner attached (simulated).

FCL.810 Night rating

Regulation (EU) 2025/134

- (a) Aeroplanes, TMGs, airships.
- (1) Applicants shall have completed a training course within a period of up to 6 months at a DTO or at an ATO to exercise the privileges of an LAPL or a PPL for aeroplanes, TMGs or airships in VFR conditions at night. The course shall comprise:
 - (i) theoretical knowledge instruction;
 - (ii) at least 5 hours of flight time in the appropriate aircraft category at night, including at least 3 hours of dual instruction, including at least 1 hour of cross-country navigation with at least one dual cross-country flight of at least 50 km (27 NM) and 5 solo take-offs and 5 solo full-stop landings.
 - (2) Before completing the training at night, LAPL holders shall have completed the basic instrument flight training required for the issue of the PPL.
 - (3) When applicants hold both an SEP aeroplane-land and a TMG class rating, they may meet the requirements in point (1) in either class or in both classes.
 - (4) Applicants for a night rating for aeroplanes or TMGs in accordance with this subparagraph shall receive full credit towards the requirements of subparagraphs (1) and (2) if they hold a TMG night rating in accordance with point SFCL.210 of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#) or if they have fulfilled all the requirements for the issue of that rating.
- (b) Helicopters. If the privileges of a PPL for helicopters are to be exercised in VFR conditions at night, the applicant shall have:
- (1) completed at least 100 hours of flight time as pilot in helicopters after the issue of the licence, including at least 60 hours as PIC on helicopters and 20 hours of cross-country flight;
 - (2) completed a training course at a DTO or at an ATO. The course shall be completed within a period of six months and comprise
 - (i) 5 hours of theoretical knowledge instruction;
 - (ii) 10 hours of helicopter dual instrument instruction time; and
 - (iii) 5 hours of flight time at night, including at least 3 hours of dual instruction, including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include a take-off and a landing.

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- 2) completed a training course at a DTO or at an ATO. The course shall be completed within a period of 6 months and comprise:
 - (i) 5 hours of theoretical knowledge instruction;
 - (ii) 10 hours of helicopter dual instrument instruction time; and
 - (iii) 5 hours of flight time at night, including at least 3 hours of dual instruction, including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include a take-off and a landing.

An applicant that holds or has held an IR in an aeroplane or TMG shall be credited with 5 hours towards the requirement in point (2)(ii) above.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (3) An applicant who holds or has held an IR in an aeroplane or TMG, shall be credited with 5 hours towards the requirement in (2)(ii) above.

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (c) Gyroplanes. If the privileges of a GPL are to be exercised in VFR conditions at night, applicants shall have:
- (1) completed at least 50 hours of flight time as pilot in gyroplanes after the issue of the licence, including at least 20 hours as PIC on gyroplanes and 20 hours of cross-country flight;
 - (2) completed a training course at a DTO or at an ATO. The course shall be completed within a period of 6 months and comprise:
 - (i) 5 hours of theoretical knowledge instruction;
 - (ii) 3 hours of gyroplane dual instrument instruction time; and
 - (iii) 5 hours of flight time at night, including at least 3 hours of dual instruction, including at least 1 hour of cross-country navigation and five solo night circuits; each circuit shall include a take-off and a landing.

An applicant that holds or has held an instrument rating in an aeroplane or a helicopter shall be credited with 2 hours towards the requirement in point (2)(ii) above. An applicant that holds a night rating in an aeroplane, helicopter or TMG shall receive a full credit for the element specified in point (2)(i).

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

GM1 FCL.810 Night rating

ED Decision 2025/002/R

SOLO FLIGHTS DURING NIGHT RATING TRAINING COURSES

Solo flights, as required by point [FCL.810](#), should take place only under the conditions established by the ATO or the DTO to ensure the safe conduct of these flights, considering, for example:

- (a) the rapid change of weather;
- (b) the reduced usability of the aerodrome;
- (c) the need for diversion to an alternate aerodrome;
- (d) other exceptional conditions.

AMC1 FCL.810(a) Night rating

ED Decision 2025/002/R

AEROPLANE NIGHT RATING COURSE

- (a) The aim of the course is to qualify applicants for or holders of Part-FCL licences with privileges to fly aeroplanes or TMGs to exercise their privileges at night.
- (b) The ATO or DTO should issue a certificate of satisfactory completion of the instruction that can be used for licence endorsement.
- (c) Theoretical knowledge

The theoretical knowledge syllabus should cover the revision or explanation of:

- (1) night VMC minima;
- (2) rules about airspace control at night and facilities available;
- (3) rules about aerodrome ground, runway, and obstruction lighting;
- (4) aircraft navigation lights and collision avoidance rules;
- (5) physiological aspects of night vision and orientation;
- (6) dangers of disorientation at night;
- (7) dangers of weather deterioration at night;
- (8) instrument systems or functions and errors;
- (9) instrument lighting and emergency cockpit lighting systems;
- (10) map marking for use under cockpit lighting;
- (11) practical navigation principles;
- (12) radio navigation principles;
- (13) planning and use of safety altitude; and
- (14) danger from icing conditions, as well as from avoidance and escape manoeuvres.

(d) Flying training

The exercises of the night rating flight training syllabus should be repeated as necessary until the student pilot has developed the necessary competence to safely operate the aircraft under visual flight rules at night.

- (1) The flight training required by point FCL.810(a)(1)(ii) should follow the syllabus set out in point (4) below.
- (2) Additional training may be completed in an FSTD(A), in order to allow for more efficient training in the aircraft.
- (3) Starred items (*) should be completed in simulated IMC and may be completed in daylight.
- (4) Night rating flight training syllabus:
 - (i) exercise 1:
 - (A) revise basic manoeuvres when flying by sole reference to instruments*;
 - (B) explain and demonstrate transition from visual flight to instrument flight*;
and
 - (C) explain and revise recovery from unusual attitudes by sole reference to instruments*;
 - (ii) exercise 2:
explain and demonstrate the use of radio navigation aids when flying by sole reference to instruments, to include position finding and tracking*;
 - (iii) exercise 3:
explain and demonstrate the use of radar assistance*;

- (iv) exercise 4:
 - (A) explain and demonstrate night take-off techniques;
 - (B) explain and demonstrate night circuit techniques;
 - (C) explain and demonstrate night approaches with or without visual approach aids; and
 - (D) practise take-offs, circuits, as well as approaches and landings;
- (v) exercise 5:

explain and demonstrate night emergency procedures including:

 - (A) simulated engine failure (to be terminated with recovery at a safe altitude);
 - (B) simulated engine failure at various phases of flight;
 - (C) simulated inadvertent entry to IMC (not on base leg or final approach);
 - (D) internal and external lighting failure; and
 - (E) other malfunctions and emergency procedures, as required by the AFM;
- (vi) exercise 6:

solo night circuits; and
- (vii) exercise 7:
 - (A) explain and demonstrate night cross-country techniques; and
 - (B) practise night cross-country dual flight and optionally supervised solo to a satisfactory standard.

AMC1 FCL.810(b) Night rating

ED Decision 2020/005/R

HELICOPTER NIGHT RATING COURSE

- (a) The aim of the course is to qualify helicopter licence holders to exercise the privileges of the licence at night.
- (b) The DTO or the ATO should issue a certificate of satisfactory completion of the instruction to licence endorsement.
- (c) Theoretical knowledge

The theoretical knowledge syllabus should cover the revision or explanation of:

 - (1) night VMC minima;
 - (2) rules about airspace control at night and facilities available;
 - (3) rules about aerodrome ground, runway, landing site and obstruction lighting;
 - (4) aircraft navigation lights and collision avoidance rules;
 - (5) physiological aspects of night vision and orientation;
 - (6) dangers of disorientation at night;
 - (7) dangers of weather deterioration at night;
 - (8) instrument systems or functions and errors;

- (9) instrument lighting and emergency cockpit lighting systems;
 - (10) map marking for use under cockpit lighting;
 - (11) practical navigation principles;
 - (12) radio navigation principles;
 - (13) planning and use of safety altitude;
 - (14) danger from icing conditions, avoidance and escape manoeuvres.
- (d) Flying training

The exercises of the night rating flight syllabus should be repeated as necessary until the student achieves a safe and competent standard:

- (1) In all cases, exercises 4 to 6 of the night rating flight syllabus should be completed in a helicopter in flight.
- (2) For exercises 1 to 3, up to 50 % of the required flight training may be completed in an FSTD(H). However, each item of exercises 1 to 3 should be completed in a helicopter in-flight.
- (3) Items marked (*) should be completed in simulated IMC and may be completed in daylight.
- (4) The flying exercises should comprise:
 - (i) Exercise 1:
 - (A) revise basic manoeuvres when flying by sole reference to instruments*;
 - (B) explain and demonstrate transition to instrument flight from visual flight*;
 - (C) explain and revise recovery from unusual attitudes by sole reference to instruments*.
 - (ii) Exercise 2:

Explain and demonstrate the use of radio navigation aids when flying by sole reference to instruments, to include position finding and tracking*.
 - (iii) Exercise 3:

Explain and demonstrate the use of radar assistance*.
 - (iv) Exercise 4:
 - (A) explain and demonstrate the use and adjustment of landing light;
 - (B) explain and demonstrate night hovering:
 - (a) higher and slower than by day;
 - (b) avoidance of unintended sideways or backwards movements.
 - (C) explain and demonstrate night take-off techniques;
 - (D) explain and demonstrate night circuit technique;
 - (E) explain and demonstrate night approaches (constant angle) with or without visual approach aids to:
 - (a) heliports;

- (b) illuminated touchdown areas.
- (F) practise take-off's, circuits and approaches;
- (G) explain and demonstrate night emergency procedures to include:
 - (a) simulated engine failure (to be terminated with power recovery at a safe altitude);
 - (b) simulated engine failure, including SE approach and landing (ME only);
 - (c) simulated inadvertent entry to IMC (not on base leg or final);
 - (d) simulated hydraulic control failure (to include landing);
 - (e) internal and external lighting failure;
 - (f) other malfunctions and emergency procedures as required by the aircraft flight manual.
- (v) Exercise 5:
Solo night circuits.
- (vi) Exercise 6:
 - (A) explain and demonstrate night cross-country techniques;
 - (B) practise night cross-country dual flight and either flight as SPIC or supervised solo to a satisfactory standard.

FCL.815 Mountain rating

Regulation (EU) 2024/2076

- (a) Privileges. The privileges of the holder of a mountain rating are to conduct flights with aeroplanes or TMGs to and from surfaces which are designated to require such a rating by the appropriate authorities designated by the Member States.

Holders of an LAPL or a PPL with privileges to fly aeroplanes or TMGs may obtain the initial mountain rating either on:
 - (1) wheels, to grant the privilege to fly to and from such surfaces when they are not covered by snow; or
 - (2) skis, to grant the privilege to fly to and from such surfaces when they are covered by snow.
 - (3) The privileges of the initial rating may be extended to either wheel or ski privileges when the pilot has undertaken an appropriate additional familiarisation course, including theoretical knowledge instruction and flight training, with a mountain flight instructor.
- (b) Training course. Applicants for a mountain rating shall have completed, within a period of 24 months, a course of theoretical knowledge instruction and flight training at a DTO or at an ATO. The content of the course shall be appropriate to the privileges of the mountain rating applied for.
- (c) Skill test. After the completion of the training, the applicant shall pass a skill test with an FE qualified for this purpose. The skill test shall contain:
 - (1) a verbal examination of theoretical knowledge;

- (2) 6 landings on at least 2 different surfaces designated as requiring a mountain rating other than the surface of departure.
- (d) In order to exercise the privileges of the mountain rating, the holder of the rating shall, during the last 2 years, have complied with either of the following conditions:
 - (1) completed at least six landings on a surface designated to require a mountain rating;
 - (2) passed a proficiency check that complies with the requirements in point (c). The completion of the proficiency check shall be entered in the logbook of the pilot and signed by the responsible examiner.

AMC1 FCL.815 Mountain rating

ED Decision 2011/016/R

THEORETICAL KNOWLEDGE AND FLYING TRAINING

THEORETICAL KNOWLEDGE	
WHEEL	SKI
1. Equipment	
W.1.1 Personal equipment for the flight	S.1.1 Personal equipment for the flight
W.1.2 Aircraft equipment for the flight	S.1.2 Aircraft equipment for the flight
2. Take-off techniques	
W.2.1 Technique for approach and landing on a mountain surface	S.2.1 Technique for approach and landing on a mountain surface S.2.2 Landing technique on skis
W.2.2 Rolling techniques of the aircraft on various runway profiles	S.2.3 Rolling techniques of the aircraft on skis about the snow nature
W.2.3 Take-off technique	S.2.4 Take-off technique on surfaces covered with snow
W.2.4 Aircraft and engine performances about altitude	S.2.5. Aircraft and engine performances about altitude
3. Rules	
W.3.1 Mountain rating	S.3.1 Mountain rating
W.3.2 Overflight rules	S.3.2 Overflight rules
W.3.3 Surfaces classification	S.3.3 Surfaces classification
W.3.4 PIC responsibilities	S.3.4 PIC responsibilities
W.3.5 Responsibilities of the surface manager	S.3.5 Responsibilities of the surface manager
W.3.6 Flight plan	S.3.6 Flight plan S.3.7 Certification of the ski mounted aeroplanes
4. Meteorology	
W.4.1 Movements of the air mass	S.4.1 Movements of the air mass
W.4.2 Flight consequences	S.4.2 Flight consequences
W.4.3 Relief effect on the movement of the air masses	S.4.3 Relief effect on the movement of the air masses
W.4.4 Altimetry	S.4.4 Altimetry
5. Human Performance and Limitations	
W.5.1 The cold	S.5.1 The cold
W.5.2 The food	S.5.2 The food
W.5.3 The hypoxia	S.5.3 The hypoxia
W.5.4 The radiance	S.5.4 The radiance

THEORETICAL KNOWLEDGE	
W.5.5 The thirst	S.5.5 The thirst
W.5.6 The tiredness	S.5.6 The tiredness
W.5.7 Turbulence effects in altitude	S.5.7 Turbulence effects in altitude
6. Navigation	
W.6.1 Progress of the flight	S.6.1 Progress of the flight
W.6.2 Dead reckoning	S.6.2 Dead reckoning
W.6.3 The path over the relief	S.6.3 The path over the relief
W.6.4 Progress in the valleys	S.6.4 Progress in the valleys
W.6.5 Detection of obstacles (high voltage lines, chairlifts, cables, etc.).	S.6.5 Detection of obstacles (high voltage lines, chairlifts, cables, etc.)
7. Specific items	
	S.7.1 Knowledge of the snow and assessment of the snow nature in-flight S.7.2 Knowledge of the glacier S.7.3 Life of the glacier S.7.4 Formation of the cracks S.7.5 Snow bridges S.7.6 Avalanches
8. Survival	
	S.8.1 Ways of survival (psychological aspects) S.8.2 Use of the equipments S.8.3 Removal of snow from the aircraft S.8.4 Building of a shelter S.8.5 How to eat and feed
FLIGHT INSTRUCTION	
WHEEL	SKI
I.- Navigation	
W.I.1 Flight techniques in the valleys	S.I.1 Flight techniques in the valleys
W.I.2 Flight over mountain passes and ridges.	S.I.2 Flight over mountain passes and ridges
W.I.3 U-turn in narrow valleys	S.I.3 U-turn in narrow valleys
W.I.4 Choice of the flight path of aerology	S.I.4 Choice of the flight path of aerology
W.I.5 Map reading	S.I.5 Map reading
II. – Arrival and reconnaissance	
W.II.1 Choice of the altitude of arrival	S.II.1 Choice of the arrival altitude
W.II.2 Choice of the arrival and overflight pattern	S.II.2 Choice of the arrival and overflight pattern
W.II.3 Choice of the landing pattern	S.II.3 Description of the circuit pattern
W.II.4 Aerology awareness	S.II.4 Aerology awareness
W.II.5 Evaluation of the length of the runway	S.II.5 Evaluation of the runway length
W.II.6 Evaluation of the runway profile (slope and banking)	S.II.6 Evaluation of the runway profile (slope and banking)
W.II.7 Collision avoidance.	S.II.7 Collision avoidance
W.II.8 Definition of the references for the landing (touchdown point)	S.II.8 Definition of the references for the landing (touchdown point)
W.II.9 Determination of the circuit pattern altitude	S.II.9 Determination of the circuit pattern altitude
W.II.10 Choice of the final speed depending on the runway profile	S.II.10 Choice of the final speed depending on the runway profile S.II.11 Choice of the take-off axis S.II.12. Choice of the landing axis

THEORETICAL KNOWLEDGE	
	S.II.13 Choice of the parking area S.II.14 Observation of the obstacles on the ground (cracks, snow bridges, avalanches) S.II.15 Estimation of the snow nature S.II.16 Observation of the way to reach a refuge from the landing area
III – Approach and landing	
W.III.1 Landing pattern altitude	S.III.1 Landing pattern altitude
W.III.2 Precision of flight along the landing path	S.III.2 Precision of flight along the landing path
W.III.3 Corrections on the landing path (accuracy and effectiveness)	S.III.3 Corrections on the landing path (accuracy and effectiveness)
W.III.4 Landing (precision of the flare and of the touchdown point)	S.III.4 Landing (precision of the flare and of the touchdown point)
W.III.5 Taxiing (use of the engine power) on various profiles	S.III.5 Taxi of the aircraft on various snows and various runway profiles
W.III.6 Parking of the aircraft (depending on the runway profile, the traffic, etc.)	S.III.6 Parking of the aircraft (depending on the snow nature and the profile of the apron) S.III.7 Turns on various snow nature and various ground profiles
IV. – Take-off	
W.IV.1 Safety checks before take-off	S. IV.1 Safety checks before take-off.
W.IV.2 Lining up on the runway	S.IV.2 Lining up on the runway
W.IV.3 Control of the runway axis during take-off	S.IV.3 Control of the runway axis during take-off
W.IV.4 Choice and use of the visual references of the take-off axis	S.IV.4 Choice and use of the visual references of the take-off axis S.IV.5 Acceleration depending on the nature of the snow S.IV.6 Short take-off S.IV.7 Take-off avoiding the skid of the skis
V. - Survival	
	S.V.1 Use of the snowshoes S.V.2 Use of the markings

AMC2 FCL.815 Mountain rating

ED Decision 2011/016/R

SKILL TEST AND PROFICIENCY CHECK

The skill test for the issue or the proficiency check for the revalidation or renewal of a mountain rating should contain the following elements:

(a) oral examination

This part should be done before the flight and should cover all the relevant parts of the theoretical knowledge. At least one question for each of the following sections should be asked:

- (1) specific equipment for a mountain flight (personal and aircraft);
- (2) rules of the mountain flight.

If the oral examination reveals a lack in theoretical knowledge, the flight test should not be done and the skill test is failed.

(b) practical skill test

During the flight test, two sites different from the departure airport should be used for recognition, approach, landing and take-off. For the mountain rating ski or the extension from wheel to ski, one of the two different sites should be a glacier.

FCL.820 Flight test rating

Regulation (EU) 2024/2076

- (a) Holders of a pilot licence for aeroplanes or helicopters shall only act as PIC in category 1 or 2 flight tests, as defined in Part-21, when they hold a flight test rating.
- (b) The obligation to hold a flight test rating established in point (a) shall only apply to flight tests conducted on:
- (1) helicopters certificated or to be certificated in accordance with the standards of CS-27 or CS-29 or equivalent airworthiness codes; or
 - (2) aeroplanes certificated or to be certificated in accordance with:
 - (i) the standards of CS-25 or equivalent airworthiness codes; or
 - (ii) the standards of CS-23 or equivalent airworthiness codes, except for aeroplanes with a maximum take-off mass of 2 000 kg or less.
- (c) The privileges of the holder of a flight test rating are to, within the relevant aircraft category:
- (1) in the case of a category 1 flight test rating, conduct all categories of flight tests, as defined in Part-21, either as PIC or co-pilot;
 - (2) in the case of a category 2 flight test rating:
 - (i) conduct category 1 flight tests, as defined in Part-21:
 - as a co-pilot, or
 - as PIC, in the case of aeroplanes referred to in (b)(2)(ii), except for those within the commuter category or having a design diving speed above 0,6 mach or a maximum ceiling above 25 000 feet;
 - (ii) conduct all other categories of flight tests, as defined in Part-21, either as PIC or co-pilot;
 - (3) conduct flights without a type or class rating as defined in Subpart H, except that the flight test rating shall not be used for commercial air transport operations.
- (d) Applicants for the first issue of a flight test rating shall:
- (1) hold at least a CPL and an IR in the appropriate aircraft category;
 - (2) have completed at least 1 000 hours of flight time in the appropriate aircraft category, of which at least 400 hours as PIC;
 - (3) have completed a training course at an ATO appropriate to the intended aircraft and category of flights. The training shall cover at least the following subjects:
 - Performance,
 - Stability and control/Handling qualities,
 - Systems,

- Test management,
 - Risk/Safety management.
- (e) The privileges of holders of a flight test rating may be extended to another category of flight test and another category of aircraft when they have completed an additional course of training at an ATO.

AMC1 FCL.820 Flight test rating

ED Decision 2020/005/R

TRAINING COURSE

GENERAL

- (a) Competency-based training:
- (1) Training courses for the flight test rating should be competency-based. The training programme should follow as much as possible the syllabus outlined below, but may be adapted taking into account the previous experience, skill and theoretical knowledge level of the applicants.
 - (2) It should also be recognised that the syllabi below assume that suitable flight test experience will be gained subsequent to attendance at the course. Should the applicant be significantly experienced already, then consideration should be made of that experience and it is possible that course content might be reduced in areas where that experience has been obtained.
 - (3) Furthermore, it should be noted that flight test ratings are specific to both a certain category of aircraft (aeroplanes or helicopters) and to a certain category of flight test (category 1 or 2). Therefore, holders of a flight test rating wishing to extend their privileges to further categories of aircraft or to further categories of flight test (this is only relevant for holders of a category 2 flight test rating since the category one flight test rating includes the privileges for category 2 test flights) should not be requested to undertake the same course as an 'ab-initio' applicant. In these cases, the ATO should develop specific 'bridge courses' taking into account the same principles mentioned above.
 - (4) To allow proper consideration of the applicant's previous experience, a pre-entry assessment of the applicant's skills should be undertaken, on the basis of which the ATO may evaluate the level of the applicant to better tailor the course. Thus, the syllabi listed below should be regarded as a list of individual demonstrable competencies and qualifications rather than a list of mandatory training objectives.
- (b) Continuous evaluation
- Training courses for the flight test rating should be built on a continuous evaluation model to guarantee that successful completion of the course ensures that the applicant has reached the level of competence (both theoretical and practical) to be issued a flight test rating.

CONTENT OF THE COURSE

- (c) In addition, the content of the course should vary taking into account whether the applicant seeks privileges for a category 1 or 2 flight test rating, as well as the relevant category of aircraft, and their level of complexity. To better take these factors into account, training courses for the flight test rating have been divided into two conditions:

- (1) condition 1 courses apply to category 1 flight test ratings on:
 - (i) helicopters certificated in accordance with the standards of CS-27 or CS-29 or equivalent airworthiness codes;
 - (ii) aeroplanes certificated in accordance with:
 - (A) the standards of CS-25 or equivalent airworthiness codes; or
 - (B) the standards of CS-23 or equivalent airworthiness codes, within the commuter category or having an M_D above 0.6 or a maximum ceiling above 25 000 ft.
- (2) condition 2 training courses apply to:
 - (i) category 2 flight test ratings for:
 - (A) helicopters certificated in accordance with the standards of CS-27 or CS-29 or equivalent airworthiness codes;
 - (B) aeroplanes certificated in accordance with:
 - (a) the standards of CS-25 or equivalent airworthiness codes; or
 - (b) the standards of CS-23 or equivalent airworthiness codes (included those mentioned in (c)(1)(ii)(B)), except for aeroplanes with a maximum take-off mass of less than 2 000 kg.
 - (ii) category 1 flight tests for aeroplanes certificated in accordance with the standards of CS-23, with a maximum take-off mass of more than 2 000kg, with the exclusion of those mentioned in (c)(1)(ii)(B) (which are subject to condition 1 courses).

AEROPLANES

- (d) Condition 1 courses for aeroplanes
 - (1) These courses should include approximately:
 - (i) 350 hours of ground training;
 - (ii) 100 hours of flight test training, during which at least 15 flights should be made without an instructor on board;
 - (iii) principles of test management and risk and safety managements should be integrated throughout the course. In addition, principles and methods applicable to the certification activity, as well as safety assessments should be taught.
 - (2) These courses should include instruction on at least 10 different aeroplane types, of which at least one should be certificated in accordance with CS-25 standards or equivalent airworthiness codes.
 - (3) During the course the student should be required to develop at least five substantial flight test reports.
 - (4) The student should be evaluated through examinations on all of the theoretical knowledge subjects, and undertake a final in-flight test upon completion of the syllabus.
 - (5) Syllabus. The following subjects should be covered in the course:

CONDITION 1 - AEROPLANES		
Theoretical knowledge	(a) aerodynamics; (b) stability and control or handling qualities; (c) engines and performance; (d) measurements and flight test instrumentation (including telemetry).	
Flight test techniques and flight training	(a) performance: (at least one flight test report should be developed)	(1) air speed calibration; (2) climb ME; (3) take-off and landing, including turboprop or turbofan OEI.
	(b) engines	Turboprop or turbofan limitations and relight envelope
	(c) handling qualities (at least two flight test reports should be developed)	(1) flight controls characteristics; (2) longitudinal handling qualities; (3) longitudinal manoeuvre stability; (4) take-off and landing MET or ME turbofan, including v_{mcg} and v_{mu} ; (5) lateral, directional handling qualities; (6) handling qualities evaluation; (7) variable stability demo flights including HOFCS; (8) stalls; (9) spins; (10) v_{mca} .
	(d) systems (at least one flight test report should be developed)	At least three different systems, for example: (1) autopilot or AFCS; (2) glass cockpit evaluation; (3) radio navigation, instruments qualification and integrated avionics; (4) TAWS; (5) ACAS.
	(e) high speed certification test	
	(f) final evaluation exercise (a flight test report should be developed)	

(e) Condition 2 courses for aeroplanes

(1) These courses should include approximately:

- (i) 150 hours of ground training;
- (ii) 50 hours of flight test training, during which at least eight flights should be made without an instructor on board.

Principles of test management and risk and safety managements should be integrated throughout the course. In addition, principles and methods applicable to the certification activity, as well as safety assessments should be taught.

- (2) These courses should include instruction on at least seven different aeroplane types, of which at least one should be certificated in accordance with CS-25 standards or equivalent airworthiness codes.
- (3) During the course the student should be required to develop at least three substantial flight test reports.
- (4) The student should be evaluated through examinations on all of the theoretical knowledge subjects, and undertake a final in-flight test upon completion of the syllabus.

- (5) Syllabus. The following subjects should be covered in the course:

CONDITION 2 - AEROPLANES			
Theoretical knowledge	(a)	aerodynamics;	
	(b)	stability and control or handling qualities;	
	(c)	engines and performance;	
	(d)	measurements and flight test instrumentation (including telemetry).	
Flight test techniques and flight training	(a)	performance: (at least one flight test report should be developed)	(1) air speed calibration; (2) climb ME; (3) take-off and landing MET or ME turbofan.
	(b)	handling qualities	(1) flight control characteristics; (2) longitudinal static, dynamic stability and control or handling qualities; (3) lateral, directional stability and control or handling qualities; (4) stalls; (5) spins.
	(c)	systems (at least one flight test report should be developed)	At least three different systems, for example: (1) autopilot or AFCS; (2) glass cockpit evaluation; (3) radio navigation, instruments qualification and integrated avionics; (4) TAWS; (5) ACAS.
	(d)	final evaluation exercise (a) flight test report should be developed)	

HELICOPTERS

- (f) Condition 1 courses for helicopters:

- (1) These courses should include approximately:
 - (i) 350 hours of ground training;
 - (ii) 100 hours of flight test training, during which at least 20 flights should be made without an instructor on board.

Principles of test management and risk and safety managements should be integrated throughout the course. In addition, principles and methods applicable to the certification activity, as well as safety assessments should be taught.

- (2) These courses should include instruction on at least eight different helicopter types, of which at least one should be certificated in accordance with CS-29 standards or equivalent airworthiness codes.
- (3) During the course the student should be required to develop at least five substantial flight test reports.
- (4) The student should be evaluated through examinations on all of the theoretical knowledge subjects, and undertake a final in-flight test upon completion of the syllabus.
- (5) Syllabus. The following subjects should be covered in the course:

CONDITION 1 - HELICOPTERS	
Theoretical knowledge	(a) aerodynamics;
	(b) stability and control or handling qualities;
	(c) engines and performance;

CONDITION 1 - HELICOPTERS		
	(d) measurements and flight test instrumentation (including telemetry).	
Flight test techniques and flight training	(a) performance: (at least one flight test report should be developed)	(1) air speed calibration; (2) level flight, climb and descent, vertical and hover performance;
	(b) engines	(1) digital engine governing; (2) turbine or piston engine evaluation.
	(c) handling qualities (at least one flight test report should be developed)	(1) flight control characteristics; (2) longitudinal static, dynamic stability and control or handling qualities; (3) lateral, directional stability and control or handling qualities; (4) ADS 33; (5) teetering rotor assessment; (6) rigid rotor assessment; (7) variable stability demo flights including HOFCS.
	(d) systems (at least one flight test report should be developed)	At least three different systems, for example: (1) navigation management systems; (2) autopilot or AFCS; (3) night vision goggles or electro-optics; (4) glass cockpit evaluation;
	(e) height and velocity envelope and EOL, including relights	
	(f) category A procedure	
	(g) vibrations and rotor adjustments	
	(h) auto rotations	
	(i) final evaluation exercise (a flight test report should be developed)	

(g) Condition 2 courses for helicopters

(1) These courses should include approximately:

- (i) 150 hours of ground training;
- (ii) 50 hours of flight test training, during which at least eight flights should be made without an instructor on board.

Principles of test management and risk and safety management should be integrated throughout the course. In addition, principles and methods applicable to the certification activity, as well as safety assessments should be taught.

- (2) These courses should include instruction on at least four different helicopters types, of which at least one should be certificated in accordance with CS-29 standards or equivalent airworthiness codes.
- (3) During the course the student should be required to develop at least three substantial flight test reports.
- (4) The student should be evaluated through examinations on all of the theoretical knowledge subjects, and undertake a final in-flight test upon completion of the syllabus.
- (5) Syllabus. The following subjects should be covered in the course:

CONDITION 2 - HELICOPTERS

Theoretical knowledge	(a) aerodynamics; (b) stability and control or handling qualities; (c) engines and performance; (d) measurements and flight test instrumentation (including telemetry).	
Flight test techniques and flight training	(a) performance: (at least one flight test report should be developed)	(1) air speed calibration; (2) level flight, climb and descent, vertical and hover performance.
	(b) engines	(1) digital engines governing; (2) turbine or piston engine evaluation.
	(c) handling qualities	(1) flight control characteristics; (2) longitudinal static, dynamic stability and control or handling qualities; (3) lateral, directional stability and control or handling qualities.
	(d) systems (at least one flight test report should be developed)	At least three different systems, for example: (1) navigation management systems; (2) autopilot or AFCS; (3) night vision goggles or electro-optics; (4) glass cockpit evaluation.
	(e) vibration and rotor adjustments	
	(f) final evaluation exercise (a flight test report should be developed)	

FCL.835 Basic instrument rating (BIR)

Regulation (EU) 2024/2076

(a) Privileges and conditions

- (1) The privileges of a BIR holder are to conduct flights under IFR on single-pilot aeroplanes for which class ratings are held, with the exception of the following:
 - (i) high-performance aeroplanes;
 - (ii) aeroplane variants if operational suitability data has determined that an IR is required.
- (2) BIR privileges shall only be exercised in accordance with point [FCL.205.A](#).
- (3) BIR privileges may be exercised at night only if the pilot holds a night rating in accordance with point [FCL.810](#).
- (4) The privileges of a multi-engine BIR shall also be valid on single-engine aeroplanes for which the pilot holds a valid single-engine class rating.
- (5) The exercise of BIR privileges shall be subject to all of the following conditions:
 - (i) the decision height (DH) or minimum descent height (MDH) used in aerodrome operating minima shall be at least 200 ft greater than what would otherwise be calculated according to point 'NCO.OP.110 Aerodrome operating minima – aeroplanes and helicopters' and point 'NCO.OP.111 Aerodrome operating minima – NPA, APV, CAT I operations' to Annex VII of [Regulation \(EU\) No 965/2012](#); and
 - (ii) the visibility used in aerodrome operating minima shall not be less than 1 500 m;

- (iii) the pilot-in-command shall not commence a flight under IFR or undertake a VFR-to-IFR transition, unless:
 - (A) at the aerodrome of departure, the visibility is at least 1 500 m and the cloud ceiling is at least 600 ft, or the published circling minimum applicable to the aeroplane category, whichever is the greater; and
 - (B) at the destination aerodrome and at any required alternate aerodrome the available current meteorological information indicates, for the period from 1 hour before until 1 hour after the estimated time of arrival, or from the actual time of departure to 1 hour after the estimated time of arrival, whichever period is shorter, a visibility of at least 1 500 m and a cloud ceiling of at least 600 ft, or the published circling minimum applicable to the aeroplane category, or the DH/MDH incremented by 200 ft in accordance with (i), whichever is the greater.
- (b) Prerequisites. Applicants for the BIR shall hold at least a PPL(A).
- (c) Training course. Applicants for the BIR shall have completed at an ATO:
 - (1) theoretical knowledge instruction in accordance with point [FCL.615\(a\)](#); and
 - (2) flight instruction that comprises the following instrument flight instruction modules:
 - (i) module 1 – the core flying training module of flight handling skills by sole reference to instruments;
 - (ii) module 2 – the applied flying training module of IFR departure, holding, 2D and 3D approach procedures;
 - (iii) module 3 – the applied flying training module of en-route IFR flight procedures; and
 - (iv) module 4 – if a multi-engine BIR is sought, the applied flying training module with one engine inoperative shall include asymmetric instrument approach and go-around procedures; and
 - (3) flight instruction that complies with the following requirements:
 - (i) The module specified in paragraph (c)(2)(i) shall be completed first. The modules specified in paragraphs (c)(2)(ii) and (c)(2)(iii), and, if applicable, (c)(2)(iv), may be completed in an order chosen by the applicant.
 - (ii) The modules specified in paragraph (c)(2) may be completed in aeroplanes, FSTDs or a combination of these. In any case, the applicant shall receive training in the aeroplane to be used for the skill test.
 - (iii) The modules specified in paragraphs (c)(2)(i), (c)(2)(ii) and (c)(2)(iv) may be commenced outside an ATO but shall be completed at an ATO. The module specified in point (c)(2)(iii) may be completed outside an ATO.
 - (iv) Prior to commencing the module specified in paragraph (c)(2)(iv), a pilot who does not hold a multi-engine aeroplane class or type rating shall have received the multi-engine training specified in Subpart H of this Annex (Part-FCL).
- (d) Theoretical knowledge. Prior to taking the skill test and through examinations in the subjects referred to in point [FCL.615\(b\)](#), the applicants shall demonstrate a level of theoretical knowledge that is appropriate to the privileges granted. The theoretical knowledge examination

shall consist of one examination paper associated with each module as specified in paragraphs (c)(2)(i), (c)(2)(ii) and (c)(2)(iii).

- (e) Skill test. After the completion of the training course specified in paragraph (c), the applicants shall pass a skill test in an aeroplane in accordance with Appendix 7 to this Annex. For a multi-engine BIR, the skill test shall be taken in a multi-engine aeroplane. For a single-engine BIR, the skill test shall be taken in a single-engine aeroplane. A multi-engine centreline thrust aeroplane shall be considered to be a single-engine aeroplane for the purposes of this paragraph.
- (f) By way of derogation from paragraph (d), holders of a single-engine BIR who also hold a multi-engine class rating and who wish to obtain a multi-engine BIR for the first time shall complete a training course at an ATO that comprises the training as specified in paragraph (c)(2)(iv) and shall pass the skill test referred to in paragraph (e).
- (g) Validity, revalidation and renewal
 - (1) A BIR shall be valid for 1 year.
 - (2) Applicants for the revalidation of a BIR shall:
 - (i) within a period of three months immediately preceding the expiry date of the rating, pass a proficiency check in accordance with [Appendix 9](#) to this Part; or
 - (ii) within the validity period, complete 6 hours as PIC under IFR including three instrument approach procedures and complete a training flight of at least one hour with an instructor who holds privileges to provide training for the BIR.
 - (3) For each alternate subsequent revalidation, the holder of the BIR shall pass a proficiency check in accordance with paragraph (2)(i) in an aeroplane.
 - (4) If a pilot chooses to fulfil the revalidation requirements specified in paragraph (g)(2)(i) earlier than what is prescribed in that paragraph, the new validity period shall commence from the date of the proficiency check.
 - (5) Applicants who fail to pass the relevant sections of a BIR proficiency check before the expiry date of the BIR shall not exercise the BIR privileges until they have passed the proficiency check.
 - (6) If a BIR has expired, in order to renew their privileges, applicants shall:
 - (i) where necessary to reach the level of proficiency needed, complete refresher training provided by an ATO or, if the BIR is expired for three years or less, by an instructor who holds privileges to provide training for the BIR; and
 - (ii) pass a proficiency check in an aeroplane.
 - (7) For a multi-engine BIR, the proficiency check for the revalidation or renewal as well as the flying training required in paragraph (g)(2)(ii) shall be completed in a multi-engine aeroplane.
 - (8) The proficiency check for the revalidation or renewal of a BIR may be combined with a proficiency check for the revalidation or renewal of a single-pilot aeroplane class rating on which BIR privileges may be exercised in accordance with point FCL.835(a)(1).
- (h) Applicants for the BIR who hold a PPL or CPL issued in accordance with Annex I (Part-FCL) and a valid IR(A) issued in accordance with the requirements of Annex 1 to the Chicago Convention by a third country may be credited in full towards the training course referred to in paragraph (c)(2). In order to be issued with the BIR, the applicants shall:

- (1) successfully complete the skill test referred to in paragraph (e);
 - (2) demonstrate orally to the examiner during the skill test that they have acquired an adequate level of theoretical knowledge of air law, meteorology, and flight planning and performance; and
 - (3) have experience of at least 25 hours of flight time under IFR as PIC on aeroplanes.
- (j) The holder of an IR shall receive full credits for the requirement in paragraph (c)(2).

AMC1 FCL.835 Basic instrument rating (BIR)

ED Decision 2025/002/R

BASIC INSTRUMENT RATING (BIR) COMPETENCIES

This AMC provides the competency criteria required for the relevant training modules of the BIR.

(a) Modules

The following modules are applicable:

- (1) Module 1: Pre-flight operations and general handling;
- (2) Module 2: Departure, precision (3D) approach procedures and non-precision (2D) approach procedures;
- (3) Module 3: En-route IFR procedures;
- (4) Module 4: Optional flight with one engine inoperative (multi-engine aeroplanes only).

Upon completion of the training, an applicant for a BIR should have received instruction on the same class of aeroplane to be used in the test.

(b) Flight tolerances

The following limits should apply and it should be borne in mind that such tolerances are expected only at the end of the training. Due consideration should be given to make allowance for turbulent conditions and the handling qualities and performance of the aircraft used:

Height

Generally	± 100 feet
Starting a go-around at decision height or altitude	+ 50 feet/– 0 feet
Minimum descent height, MAP or altitude	+ 50 feet/– 0 feet

On radio aids	±5°
For 'angular' deviations	Half-scale deflection, azimuth and glide path (e.g. LPV, ILS, MLS, GLS)
2D (LNAV) and 3D (LNAV/VNAV) 'linear' lateral deviations	Cross-track error/deviation shall normally be limited to ± ½ the RNP value associated with the procedure. Brief deviations from this standard up to a maximum of one time the RNP value are allowable.
3D linear vertical deviations (e.g. RNP APCH (LNAV/VNAV) using Baro VNAV)	Not more than – 75 feet below the vertical profile at any time, and not more than + 75 feet above the vertical profile at or below 1 000 feet above aerodrome level.

Heading

All engines operating	$\pm 5^{\circ}$
With simulated engine failure	$\pm 10^{\circ}$

Speed

All engines operating	± 5 knots
With simulated engine failure	+ 10 knots/– 5 knots

Given that the intention of the training for the BIR is to be entirely competency-based, the student and instructor need detailed guidance on these competencies. The following information is intended to provide that guidance. Each element of the training modules is described in text followed by a table which gives guidance on the competencies required and how to assess them using the key competencies model of:

OBJECTIVE (of the training item), and **SKILL — KNOWLEDGE — ATTITUDE** (to achieve the objective)

(c) Sample table

The table is separated into four rows as follows:

Training element	
Title of assessed item taken from training module	
OBJECTIVE	This cell describes the applicant's proficiency to be assessed by the training organisation or instructor.
SKILL	This cell describes the competency criteria that involve the applicant demonstrating: — manual aircraft control; — effective flight path management through proper use of flight management system guidance and automation; and — application of procedures.
KNOWLEDGE	This cell describes the knowledge needed to meet the objective's proficiency requirements.
ATTITUDE	This cell describes the competency criteria encapsulated by airmanship, crew resource management (CRM), and threat and error management (TEM), such as: — situation awareness; — effective communication; — leadership and teamwork; — effective workload management; — effective problem-solving and decision-making.
General	
In most phases of flight there are competencies that apply to a group of manoeuvres, e.g. turns, or even to the whole phase of flight. In order to avoid repetition, the common competencies are grouped under the 'General' item heading.	

(d) Content of the training

(1) Module 1: Pre-flight operations and general handling

Use of flight manual (or equivalent), especially for aircraft performance calculation, and mass and balance

Module 1: Pre-flight operations and general handling	
Use of flight manual (or equivalent), especially for aircraft performance, and mass and balance	
OBJECTIVE	(A) Proficient in the use of the flight manual (or equivalent). (B) Proficient in the mass and balance schedule. (C) Proficient in the aircraft performance calculation.
SKILL	(A) Use proficiently performance charts, tables, graphs or other data, when available, relating to items such as: (1) accelerate-stop distance available; (2) landing distance available; (3) take-off performance; (4) one engine inoperative; (5) climb performance; (6) cruise performance; (7) fuel consumption, range, and endurance; (8) go-around from rejected landing; (9) operational factors affecting aircraft performance; (10) other performance data appropriate to the test aircraft; (11) airspeeds used during specific phases of flight; (12) effects of meteorological conditions upon performance characteristics and correctly application of these factors to a specific chart, table, graph or other performance data; (13) impact of relevant NOTAMs on the conduct of the flight; (14) aircraft documentation.
KNOWLEDGE	(A) Part-NCO (non-commercial air operations) (B) Pilot operating manual (POM) or flight manual chapters dedicated to: (1) limitations; ¹ (2) performance calculation in general; (3) performance calculation and associated procedures when specific conditions exist.
ATTITUDE	(A) Situation awareness: Understand the responsibilities of proper pre-departure planning and preparations. (B) Effective communication: Ensure appropriate and clear communication with all ground service personnel (ATC, dispatch, MET). (C) Leadership and teamwork: Manage passengers and ground personnel, as applicable. (D) Effective workload management: Provide sufficient time and manage the workload for pre-flight procedures (including documentation) to be completed in an efficient manner. (E) Effective problem-solving and decision-making: (1) Make appropriate decisions on all identified threats; (2) Plan and implement suitable mitigation actions.

¹ The numbering of (15)-(17) under point (B) is an editorial error and should read (1)-(3). This will be corrected with the next update of AMC & GM to Part-FCL.

Pre-flight inspection

Module 1: Pre-flight operations and general handling	
Pre-flight inspection	
OBJECTIVE	Full initial pre-flight inspection in accordance with the approved checklist assuming the risk to IFR flights such as icing conditions, database, etc.
SKILL	(A) Perform all elements of the aeroplane pre-flight inspections. (B) Confirm that the aeroplane is in a serviceable and safe condition for IFR flight.
KNOWLEDGE	(A) Confirm the validity of database and receiver autonomous integrity monitoring (RAIM) prediction, if applicable. (B) Be aware of the possible effects of equipment defects or unserviceability.
ATTITUDE	(A) Situation awareness: (1) Note the position of the aircraft, any surrounding hazards, and location of emergency equipment, and take appropriate action to minimise potential risks; (2) Note effects of engine start on the surrounding environment; (3) Note the limitations of software and equipment such as flight director (FD), autopilot (AP), etc. (B) Effective communication: (1) Demonstrate correct communication; (2) Make a correct passenger and departure briefing. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: (1) Confirm from the checklist that all pre-flight requirements have been fulfilled; (2) Demonstrate an organised approach to performing inspection of aircraft and equipment. (E) Effective problem-solving and decision-making: (1) Identify possible defects and threats; (2) Take corrective action.

Taxiing

Module 1: Pre-flight operations and general handling	
Taxiing	
OBJECTIVE	(A) Be proficient in all recommended taxiing checks and procedures. (B) Comply with ATC instructions, airport markings and signals.
SKILL	(A) Obtain appropriate clearance before taxiing and before crossing or entering active runways. (B) Comply with instructions issued by ATC. (C) Maintain correct and positive aircraft control. (D) Take due consideration of environmental conditions (e.g. surface wind, contamination, surface condition, etc.). (E) Maintain adequate separation from other aircraft, obstructions, and persons. (F) Accomplish the applicable briefing or checklist items, and follow the recommended procedures.
KNOWLEDGE	(A) The need to correctly perform taxiing checks. (B) Understanding the following: (1) runway hold lines and stop bar lighting as applicable; (2) localiser and glide slope sensitive and critical areas; (3) beacons, as well as other surface control markings and lighting; (4) taxiing speeds; (5) rules and procedures in the event of loss of communication (priority, lighting signals); (6) rules for manoeuvring in reduced meteorological conditions.
ATTITUDE	(A) Situation awareness: (1) Maintain constant vigilance and lookout during the taxiing operation; (2) Use headings in poor visibility conditions to confirm the path; (3) Maintain awareness of taxiing speeds appropriate to the conditions and limitations. (B) Effective communication: Demonstrate correct ATC communication (where applicable). (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Divide attention properly inside and outside the cockpit. (E) Effective problem-solving and decision-making: (1) Stop the aircraft to check position when in doubt; (2) Assess major risks: collision with other aircraft, obstacles, and aircraft security.

Transition to instrument flight

Module 1: Pre-flight operations and general handling

Transition to instrument flight (must be performed by sole reference to instruments)

OBJECTIVE	Establish the climb, complete a smooth transition to instrument flight, and complete post-take-off checks and drills.
SKILL	Following the initial take-off procedure: (A) Compare the visual attitude achieved with the attitude indicator display; (B) Assess the performance instrument information to confirm that the aircraft has achieved the desired climb parameters; (C) Commence appropriate instrument scanning techniques.
KNOWLEDGE	(A) Demonstrate the required technical knowledge of the function of the instruments in order to safely fly the aircraft by sole reference to instruments. (B) Understand the need to compare the attitude indicator with the real world. (C) Understand the need to verify that the expected performance has been achieved.
ATTITUDE	(A) Situation awareness: Monitor aircraft flight path at all stages of the transition to instrument flight. (B) Effective communication: Demonstrate effective communication (as applicable). (C) Leadership and teamwork: Demonstrate effective coordination (as applicable). (D) Effective problem-solving and decision-making: (1) Correctly assess take-off and climb hazards, particularly those related to other aircraft, aerodrome infrastructure, obstacles, and weather; (2) Have a strategy to mitigate the threats.

ATC liaison — compliance, radio-telephony (RTF) procedures

Module 1: Pre-flight operations and general handling

ATC liaison — compliance, radio-telephony (RTF) procedures (must be performed by sole reference to instruments)

OBJECTIVE	(A) Ability to communicate clearly with ATC using appropriate RTF phraseology in order to perform the flight as planned in compliance with ATC instructions. (B) In the event of changes to the plan, such changes should be negotiated with ATC to ensure continued compliance.
SKILL	(A) ICAO language proficiency level 4 or greater. (B) The ability to use standard and, where applicable, non-standard RTF procedures. (C) Understand the implications of the received clearance, and be able to action the same safely and effectively. (D) Interpretation of charts and maps.
KNOWLEDGE	(A) Specific ATC phrases, e.g. ETA vs EAT. (B) Aircraft category for instrument approaches. (C) Performance of the aircraft and its ability to meet the ATC clearance. (D) ICAO standard phraseology and national differences. (E) Pilot or controller responsibilities including tower, en-route, and appropriate clearances. (F) Adequate knowledge of RTF failure procedures.
ATTITUDE	(A) Situation awareness: Establish communication with ATC on the correct frequencies and at the appropriate times. (B) Effective communication: Read back correctly, in a timely manner, the ATC clearance in the sequence received. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Copy correctly, in a timely manner, the ATC clearance as issued. (E) Effective problem-solving and decision-making: Interpret correctly the ATC clearance received and, when necessary, request clarification, verification, or change.

Control of the aeroplane by reference solely to instruments, including: level flight at various speeds, level turns at rate 1 and up to 30 degrees angle of bank, trim

**Module 1: Pre-flight operations and general handling
(must be performed by sole reference to instruments)**

Control of the aeroplane by reference solely to instruments, including: level flight at various speeds, level turns at rate 1 and up to 30 degrees angle of bank, trim

OBJECTIVE	(A) Smooth control of heading, altitude, speed, power, trim and ancillary controls. (B) Correct use of autopilot, where appropriate. (C) Demonstrate correct technique for instrument flight manoeuvring within specified limits. (D) Maintain balanced and trimmed flight.
SKILL	(A) Maintain altitude, heading and balance, by sole reference to instruments, using correct instrument confirmation, and coordinated control application. (B) Maintain altitude, heading and balance, whilst accelerating or decelerating to specific speeds, as determined by the aircraft flight manual, or as specified by the examiner. (C) Complete coordinated level turns at rate 1 and maintain entry speed onto specified headings. (D) Complete coordinated level turns at up to 30 degrees bank whilst maintaining entry speed onto specified headings. (E) Demonstrate correct procedure for pre-flight functional check of autopilot or flight director. (F) Demonstrate correct operating procedure for autopilot or flight director in all modes.
KNOWLEDGE	(A) Procedures for controlling the aircraft in accordance with the POM, aircraft flight manual and operations manual, as appropriate. (B) Autopilot system fitted to the aircraft. (C) Procedures for controlling the aircraft with automatic flight control systems, in accordance with the POM, aircraft flight manual and operations manual, as appropriate.
ATTITUDE	(A) Situation awareness: (1) Maintain awareness of the autopilot modes selected, where applicable; (2) Understand the need for trimmed, in-balance flight when manually flying the aircraft. (B) Effective communication: - As applicable to the specific situation. (C) Leadership and teamwork: - As applicable to the specific situation. (D) Effective workload management: - Use an appropriate 'division of attention' when completing flight log, etc., whilst manually controlling the aircraft. (E) Effective problem-solving and decision-making: - Prioritise activities to allow maintenance of correct instrument scan.

Climbing and descending turns with sustained rate-1 turn

**Module 1: Pre-flight operations and general handling
(must be performed by sole reference to instruments)**

Climbing and descending turns with sustained rate-1 turn

OBJECTIVE	Complete a coordinated climb or descent and turn at rate 1 using: (A) the recommended climbing speed; or (B) descent speed and nominated rates of descent for the aircraft.
SKILL	(A) Establish the recommended entry airspeed in straight and level flight. (B) Roll into a coordinated climbing or descending turn with a bank angle commensurate with the speed to produce a rate-1 turn. Maintain the bank angle in a stable, balanced turn. (C) Apply smooth, coordinated pitch, bank, and power adjustments to maintain the specified attitude and airspeed. (D) Roll out of the turn and stabilise the aircraft in straight and level flight. (E) Recover accurately onto the desired heading and at the desired airspeed for straight and level flight.
KNOWLEDGE	(A) Speed and bank angle relationship to establish a rate-1 turn. (B) Recommended climb speed and power settings. (C) Recommended speed and power settings for descent at nominated descent rates.
ATTITUDE	(A) Effective workload management: Demonstrate orientation throughout the manoeuvre. (B) Effective problem-solving and decision-making: React to departure from stabilised steep turn attitude.

Recovery from unusual attitudes, including sustained 45° bank turns and steep descending turns

**Module 1: Pre-flight operations and general handling
(must be performed by sole reference to instruments)**

Recovery from unusual attitudes, including sustained 45° bank turns and steep descending turns

OBJECTIVE	Recover from unusual attitudes, including sustained 45° bank turns and steep descending turns using the correct technique to minimise height loss.
SKILL	(A) Interpretation of the instrument displays to identify the reason behind the unusual attitude. (B) Application of the correct recovery technique. (C) Avoid any indication of an approaching stall, abnormal flight attitude, or exceeding any structural or operating limitation during any part of the manoeuvre.
KNOWLEDGE	Correct recovery technique using ‘full’ panel instruments, as appropriate.
ATTITUDE	(A) Situation awareness: (1) recognition of unusual attitude; (2) after recovery: why did the aircraft enter the unusual attitude, e.g. distraction, instrument failure, mishandling, hypoxia? (3) after recovery: is the aircraft above safety altitude? (4) which is a safe direction to fly whilst assessing the situation? (B) Effective workload management: Address the situation to recover situation awareness. (C) Effective communication: (1) Advise other crew members of the situation; (2) Advise ATC if appropriate. (D) Leadership and teamwork: Communicate and coordinate, as appropriate, during the recovery manoeuvre. (E) Effective problem-solving and decision-making: React promptly to departure from controlled flight.

Recovery from approach to stall in level flight, climbing/descending turns and in landing configuration

Module 1: Pre-flight operations and general handling
(must be performed by sole reference to instruments)

Recovery from approach to stall in level flight, climbing/descending turns and in landing configuration (may be performed in an FSTD, if approved for this procedure)

OBJECTIVE	(A) Demonstrate how to conduct appropriate safety checks before stalling. (B) Establish the required aircraft configuration and stall entry, as appropriate, from straight and level or manoeuvring flight. (C) Maintain heading (or 10–30° bank angle, as required) to stall entry. (D) Recognise the symptoms of stall or approaching stall, and initiate the correct recovery action. (E) Recover, using the correct techniques, to return to a clean configuration best rate climb, or as otherwise directed by the examiner. (F) Complete all the necessary checks and drills.
SKILL	(A) Select an entry altitude in accordance with safety requirements. When accomplished in an FSTD, the entry altitude may be at low, intermediate or high altitude as appropriate for the aircraft and the configuration. (B) Slowly establish the pitch attitude (using trim, elevator or stabiliser), bank angle, and power setting that will induce stall at the desired target airspeed. Normal trim should be used as the aircraft speed reduces, with trim at different, or as stated in the flight manual restrictions. (C) Recognise and announce the first indication of a stall appropriate to the specific aircraft design and initiate recovery. (D) Recover to a reference airspeed, altitude and heading, allowing only the acceptable altitude or airspeed loss and heading deviation using the procedures described in the aircraft flight manual or operator safety manual, as applicable. (E) Demonstrate smooth, positive control during entry, approach to a stall, and recovery.
KNOWLEDGE	(A) Academic knowledge. (B) Limitations. (C) Safety procedures before starting with stall exercises. (D) Stall recovery procedures and techniques. (E) Flight manual. (F) Operator safety manual.
ATTITUDE	(A) Situation awareness: Ensure the aircraft is in a safe area and clear of hazards prior to accomplishing an approach to a stall. (B) Effective communication: Communicate and coordinate. (C) Leadership and teamwork: Coordinate to ensure that there is adequate separation from other aircraft before initiating the stall. (D) Effective workload management: As applicable to the specific situation. (E) Effective problem-solving and decision-making: As applicable to the specific situation.

Limited panel instrument flight: stabilised climb or descent, level turns at rate 1 onto given headings, recovery from unusual attitudes

Module 1: Pre-flight operations and general handling
(must be performed by sole reference to instruments)

Limited panel instrument flight: stabilised climb or descent, level turns at rate 1 onto given headings, recovery from unusual attitudes — only applicable to aeroplanes
NB: Most modern light aircraft are now fitted with a 'standby' horizon in addition to or instead of turn rate gyros. Where this is the case, the pilot under training is to be taught these exercises using the 'standby' horizon.

OBJECTIVE	Demonstrate continued control of the aircraft by interpreting aircraft attitude from aircraft standby instruments.
SKILL	(A) Complete flight in straight and level, and climbing and descending, at nominated speeds. Fly turns at rate 1 onto nominated headings using the correct technique and demonstrating correct instrument scan and interpretation. (B) Recover from unusual attitudes including sustained 45° bank turns and steep descending and climbing turns using the correct technique to minimise height loss.
KNOWLEDGE	(A) Demonstrate the theoretical knowledge and understand the dangers of 'looping error'. (B) Variation of techniques. (C) Limitations of the use of direct-reading compass systems.
ATTITUDE	(A) Situation awareness: (1) recognition of the reason behind the unusual attitude; (2) after recovery: why did the aircraft enter the unusual attitude, e.g. distraction, instrument failure, mishandling, etc.? (3) after recovery: is the aircraft above safety altitude? (4) which is a safe direction to fly whilst assessing the situation? (B) Effective workload management: Address the situation to recover situation awareness. (C) Effective communication: Advise ATC if appropriate. (D) Leadership and teamwork: Communicate and coordinate as appropriate. (E) Effective problem-solving and decision-making: React promptly to departure from controlled flight.

(2) Module 2: Departure, precision (3D) approach procedures and non-precision (2D) approach procedures

Weather minima

Module 2: Departure and arrivals, 3D approach and 2D approach	
Weather minima	
OBJECTIVE	Confirmation of weather affecting departure, route, destination and diversion; acceptability for the flight. Determination of the expected instrument approach minimum heights/altitudes in accordance with NCO requirements.
SKILL	Ability to interpret published weather charts such as synoptic charts and coded messages (TAF, METAR, SNOWTAM, etc.).
KNOWLEDGE	(A) Air masses and local weather effects. (B) Weather codes. (C) NCO requirements.
ATTITUDE	(A) Situation awareness: (1) Be able to interpret and understand the weather factors and all the associated potential hazards likely to affect the planned flight; (2) Assess correctly whether the weather minima required at destination and diversion airfields are satisfactory for the conduct of the flight. (B) Effective communication: As applicable to the specific situation. (C) Leadership and teamwork: As applicable to the specific situation. (D) Effective workload management: As applicable to the specific situation. (E) Effective problem-solving and decision-making: Make appropriate decisions based on available weather information.

Pre-take-off briefing, take-off

Module 2: Departure and arrivals, 3D approach and 2D approach

Pre-take-off briefing, take-off

OBJECTIVE	(A)	Perform a safe take-off in compliance with ATC clearance, procedure margins and within the flight manual limits taking into account environmental conditions.
	(B)	Obtain ATC clearance for departure, flight deck preparation, confirmation of departure, and passenger emergency briefing. Actions to be taken with regard to the aeroplane if an emergency occurs during departure should be covered in the pre-flight main briefing.
SKILL	(A)	Obtain appropriate take-off clearance using standard RTF phraseology, and perform all required pre-take-off checks (including visually scanning for other aircraft).
	(B)	Position the aircraft correctly for take-off taking into account any crosswind condition.
	(C)	Apply the controls correctly to maintain longitudinal alignment on the centre line of the runway prior to initiating and during the take-off.
	(D)	Set the throttle(s) to take-off power with appropriate checks (e.g. verify the expected engine performance, monitor engine controls, settings and instruments during take-off to ensure all predetermined parameters are maintained).
	(E)	Use the correct take-off technique by applying recommended speeds for rotation, lift-off and initial climb.
	(F)	Adjust the controls to attain the desired pitch attitude at the predetermined airspeed to obtain the desired performance.
	(G)	Ensure a safe climb and departure in accordance with clearance and with due regard for other air traffic, noise abatement and wake turbulence avoidance procedures, adjusting power and aircraft configuration, and maintain desired path (or heading) as appropriate.
	(H)	Complete all necessary post-take-off checks.
	(I)	Perform or call for and verify the accomplishment of landing gear and flap retractions, power adjustments, and other required pilot-related activities at the required airspeeds within the tolerances established in the flight manual.
KNOWLEDGE	(A)	Limitations, procedure margins.
	(B)	Normal procedures (understand the different techniques dependent on varying flap settings and environmental conditions).
	(C)	Abnormal and emergency procedures.
	(D)	Performance.
	(E)	Applicable rules on wake turbulence separation.
ATTITUDE	(A)	Situation awareness: (1) Monitor engine parameters for any deviations; (2) Monitor aircraft acceleration during take-off; (3) Monitor aircraft ground and flight path at all stages of the take-off procedure.
	(B)	Effective communication: Demonstrate effective communication with ATC (as applicable).
	(C)	Leadership and teamwork: Demonstrate effective coordination with ATC (as applicable).
	(D)	Effective problem-solving and decision-making: Correctly assess take-off and climb hazards, particularly those related to other aircraft, aerodrome infrastructure, obstacles and weather, and have a strategy to mitigate the threats.

Instrument departure procedures, altimeter setting

Module 2: Departure and arrivals, 3D approach and 2D approach

Instrument departure procedures, altimeter setting (must be performed by sole reference to instruments)

OBJECTIVE	Complete the standard instrument departure (SID) procedure or follow the ATC departure instructions; use the correct altimeter-setting procedure; maintain aeroplane control, speed, heading and level.
SKILL	(A) Identify any navigation aids used. (B) Follow any noise routing or departure procedures and ATC clearances. (C) Take appropriate anti-icing/de-icing actions. (D) Use the current and appropriate navigation publications for the proposed departure. (E) Make correct use of instruments, flight director, autopilot, navigation equipment and communication equipment appropriate to the performance of the departure. (F) Intercept and follow, in a timely manner, all courses, radials and bearings (QDM/QDRs) appropriate to the departure route and ATC clearance. (G) Comply, in a timely manner, with all ATC clearances, instructions and restrictions. (H) Perform the aircraft briefing or checklist items appropriate to the departure. (I) Adhere to airspeed restrictions and adjustments required by regulations, ATC and the flight manual. (J) Maintain the appropriate airspeed, altitude, headings and accurately track radials, courses, and bearing. (K) Complete the appropriate checklist.
KNOWLEDGE	(A) Weather phenomena, particularly the conditions favouring the formation of ice on the airframe and engines. (B) Limitations of the use of ground-based navigation aids. (C) Limitations of the use of RNAV (GNSS) derived navigational information. (D) Division of airspace and altimeter-setting procedures associated with the current airspace environment. (E) The departure procedure in use and the safety implications of not adhering to the procedure. (F) Altimetry procedures in accordance with the applicable regulations.
ATTITUDE	(A) Situation awareness: (1) Understanding of any clearance limits or variations to SID/initial departure clearance instructed by ATC; (2) Awareness of the aircraft performance and the ability to conform to ATC clearances (speed, height, time limits, etc.). (B) Effective communication: - Demonstrate correct communication with ATC (where applicable). (C) Leadership and teamwork: - Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: - Prioritise attention properly between aircraft control, navigation and communication tasks. (E) Effective problem-solving and decision-making: - Make the necessary decisions to mitigate the effect of changing conditions that may affect aircraft (weather, navigation aid serviceability, ATC, etc.).

Holding procedure

Common to both 3D and 2D procedures (must be performed by sole reference to instruments)

Holding procedure

OBJECTIVE	Complete the appropriate entry procedure followed by a standard ICAO holding fix, using information in order to maintain the protected area.
SKILL	(A) Make appropriate adjustments in order to arrive over the holding fix as close as possible to the 'expected approach time', if required. (B) Recognise arrival at the clearance limit or holding fix. (C) Comply with ATC reporting requirements. (D) Change to the recommended holding airspeed appropriate for the aircraft and holding altitude, so as to cross the holding fix at or below the maximum holding airspeed. (E) Follow the appropriate entry procedures in accordance with standard operational procedures or as required by ATC. (F) Use the correct timing criteria where required by the holding procedure or ATC. (G) Use wind-drift correction techniques accurately to maintain the appropriate joining and holding pattern and to establish and maintain the correct tracks and bearings. (H) Maintain the appropriate airspeed, altitude and headings accurately to establish and maintain the correct tracks and bearings. (I) Make appropriate adjustments to the procedure timing to allow for the effects of known wind.
KNOWLEDGE	(A) Holding endurance, including but not necessarily limited to fuel on board. (B) Fuel flow while holding. (C) Fuel required to alternate, etc.
ATTITUDE	(A) Situation awareness: Establish communication with ATC on the correct frequencies and at the appropriate times. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Monitor to ensure that the flight profile complies with the cleared holding pattern. (E) Effective problem-solving and decision-making: React to navigation errors or unexpected systems malfunctions.

Setting and checking of navigation aids, identification of facilities

Module 2: 3D approach procedures

(must be performed by sole reference to instruments)

Setting and checking of navigation aids, identification of facilities

OBJECTIVE	(A) Use of navigation aids with regard to promulgated range, identification and interpretation. (B) Use the RAIM prediction, if applicable. (C) Use the correct RNP approach specifications (LPV, LNAV/VNAV).
SKILL	(A) Set and identify relevant navigation aids. (B) Confirm the availability and serviceability of selected navigation equipment.
KNOWLEDGE	(A) Systems: communication, navigation and auto-flight systems. (B) RNP approach specifications (LPV, LNAV/VNAV).
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Select radio aids appropriate to the intended approach; (3) PBN limitations; (4) Temperature limitations (LNAV/VNAV). (B) Effective workload management: Monitor to ensure safe flight profile whilst selecting and checking radio aids. (C) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (D) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (E) Effective problem-solving and decision-making: React to deviation errors or unexpected systems malfunctions.

Arrival procedures, altimeter checks

Module 2: 3D approach procedures

(must be performed by sole reference to instruments)

Arrival procedures, altimeter checks

OBJECTIVE	Descent planning and consideration of minimum sector altitude (MSA) or terminal arrival altitude (TAA). Completion of the published arrival procedure or as instructed by ATC, including altimeter setting or protected area, ATC liaison and RTF procedures.
SKILL	(A) Set and cross-check the appropriate altimeter settings. (B) Use the correct RTF procedures and terminology and comply with all ATC instructions and clearances. (C) Establish the appropriate aircraft configuration and airspeed for the phase of the approach. (D) Comply with the published arrival procedure or as required by ATC. (E) Interpretation of arrival charts.
KNOWLEDGE	(A) Altimetry procedures in accordance with the applicable regulations. (B) Knowledge of legends used in the approach charts. (C) Understanding of ATC procedures and RTF phraseology for the type of approach to be completed. (D) Knowledge of RNP arrival procedure.
ATTITUDE	(A) Situation awareness: Establish communication with ATC on the correct frequencies and at the appropriate times. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC, as appropriate. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Monitor to ensure that the flight profile complies with the approach procedure. (E) Effective problem-solving and decision-making: react to deviation errors or unexpected systems malfunctions.

Approach and landing briefing, including descent, approach, landing checks and missed approach

Module 2: 3D approach procedures

(must be performed by sole reference to instruments)

Approach and landing briefing, including descent, approach, landing checks and missed approach

OBJECTIVE	The approach briefing including weather and confirmation of instrument approach procedure minima, and applicable procedures.
SKILL	(A) Complete the checks for landing and configure the aircraft appropriately. (B) Complete a short self-briefing with regard to arrival, holding, approach, minima, weather conditions, associated performances, taxiing and missed approach procedure.
KNOWLEDGE	(A) Use of checklist as appropriate. (B) Determination of approach minima. (C) Make the necessary adjustments to the published approach minima criteria for the aircraft approach category, and with due regard for: (1) NOTAMs; (2) inoperative navigation equipment; (3) inoperative visual aids associated with the landing environment; (4) reported weather conditions; (5) aircraft status (effects of any inoperative systems).
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Aircraft technical status. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Monitor to ensure that the flight profile complies with the approach procedure. (E) Effective problem-solving and decision-making: React to deviation errors or unexpected systems malfunctions.

Compliance with published approach procedure

Module 2: 3D approach procedures

(must be performed by sole reference to instruments)

Compliance with published approach procedure

OBJECTIVE	(A) Compliance with the published 3D approach procedure. (B) Vertical and horizontal profile to the nominated minima in accordance with protected areas.
SKILL	(A) Manage the appropriate source of navigation system. (B) Complete the manoeuvring pattern as required to establish the final approach segment within the specified flight tolerances. (C) Establish a predetermined rate of descent at the point where the glide path begins, in order to follow the glide path. (D) Intercept and track within the prescribed limits. (E) Interpretation of approach chart.
KNOWLEDGE	(A) Systems: communication, navigation and auto-flight systems. (B) Correctly interpret and understand the procedure to be flown from the approach chart for runway and procedure in use. (C) Autopilot and flight director limitations. (D) Software and capacity system.
ATTITUDE	(A) Situation awareness: Establish communication with ATC on the correct frequencies and at the appropriate times; (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Monitor to ensure that the flight profile complies with the cleared procedure. (E) Effective problem-solving and decision-making: React to navigation errors or unexpected systems malfunctions.

Altitude, speed, heading control (stabilised approach)

Module 2: 3D approach procedures

(must be performed by sole reference to instruments)

Altitude, speed, heading control (stabilised approach)

OBJECTIVE	(A) Establish a stabilised approach, in trim for the aeroplane configuration and speed, using the correct techniques for attitude, heading and power control. (B) Correct assessment of track and vertical path.
SKILL	(A) Establish the final approach and maintain the approach path in horizontal and vertical profile to minima. (B) Control the aircraft as necessary to achieve a stable approach path. (C) Arrive at the minima on a stabilised approach in order to make a correct decision to perform a landing, go-around or circling approach safely. (D) Prepare backup radio aids for continued approach in the event of radio aid or display equipment failure. (E) Use correct RTF procedures and terminology and comply with all ATC instructions and clearances.
KNOWLEDGE	(A) Horizontal and vertical tolerances. (B) Actions to be taken in the event of radio aid or display equipment failure. (C) Procedure in the event of loss of communication with ATC. (D) Procedure in the event of loss of integrity.
ATTITUDE	(A) Situation awareness: Confirm that approach is stabilised. (B) Effective communication: Advise ATC if appropriate. (C) Leadership and teamwork: (1) Demonstrate correct coordination with ATC (where applicable); (2) Procedures for loss of approach capability. (D) Effective workload management: Monitor to ensure that the flight profile remains safe. (E) Effective problem-solving and decision-making: Make appropriate decision to abandon approach if required.

Setting and checking of navigation aids, identification of facilities

Module 2: 2D approach procedures

(must be performed by sole reference to instruments)

Setting and checking of navigation aids, identification of facilities

OBJECTIVE	(A) Use of navigation aids with regard to promulgated range, identification and interpretation. (B) Use the RAIM prediction, if applicable. (C) Use the correct RNP approach specifications. (D) Calculate the true altitude as required.
SKILL	(A) Set and identify relevant navigation aids. (B) Confirm the availability and serviceability of selected navigation equipment.
KNOWLEDGE	(A) Systems: communication, navigation and auto-flight systems. (B) RNP approach specifications (LNAV). (C) True altitude corrections for temperature.
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Select radio aids appropriate to the intended approach. (B) Effective workload management: Monitor to ensure safe flight profile whilst selecting and checking radio aids. (C) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (D) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (E) Effective problem-solving and decision-making: React to deviation errors or unexpected systems malfunctions.

Arrival procedures, altimeter checks

Module 2: 2D approach procedures

(must be performed by sole reference to instruments)

Arrival procedures, altimeter checks

OBJECTIVE	(A) Descent planning and consideration of MSA or TAA. (B) Completion of the published arrival procedure or as instructed by ATC, including altimeter setting or protected area, ATC liaison and RTF procedures.
SKILL	(A) Set and cross-check the appropriate altimeter settings. (B) Use the correct RTF procedures and terminology and comply with all ATC instructions and clearances. (C) Establish the appropriate aircraft configuration and airspeed for the phase of the approach. (D) Comply with the published arrival procedure or as required by ATC. (E) Interpretation of arrival charts.
KNOWLEDGE	(A) Altimetry procedures, in accordance with the applicable regulations. (B) Knowledge of the legends used in the approach charts. (C) Understanding of ATC procedures and RTF phraseology for the type of approach to be completed. (D) Knowledge of RNP arrival procedure.
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) PBN protected area. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (C) Leadership and teamwork: - Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: - Monitor to ensure that the flight profile complies with the approach procedure. (E) Effective problem-solving and decision-making: - React to deviation errors or unexpected systems malfunctions.

Approach and landing briefing, including descent, approach, landing checks and missed approach

Module 2: 2D approach procedures

(must be performed by sole reference to instruments)

Approach and landing briefing, including descent, approach, landing checks and missed approach

OBJECTIVE	The approach briefing including weather and confirmation of instrument approach procedure minima, and applicable procedures.
SKILL	(A) Complete the landing and configure the aircraft as appropriate. (B) Complete a short self-briefing with regard to arrival, holding, approach, minima, weather conditions, associated performances, taxiing and missed approach procedure.
KNOWLEDGE	(A) Use of checklist as appropriate. (B) Determination of approach minima. (C) Adjustments necessary to the published approach minima criteria for the aircraft approach category, and with due regard for: (1) NOTAMs; (2) inoperative navigation equipment; (3) inoperative visual aids associated with the landing environment; (4) reported weather conditions.
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Aircraft technical status. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Monitor to ensure that the flight profile complies with the approach procedure. (E) Effective problem-solving and decision-making: React to deviation errors or unexpected systems malfunctions.

Compliance with published approach procedure

Module 2: 2D approach procedures

(must be performed by sole reference to instruments)

Compliance with published 2D approach procedure

OBJECTIVE	(A) Compliance with the published approach procedure. (B) Vertical and horizontal profile to the nominated minima in accordance with protected areas. (C) Use of the CDFA technique where appropriate.
SKILL	(A) Manage the appropriate source of navigation system. (B) Select and comply with the appropriate 2D instrument approach procedure. (C) Complete the manoeuvring pattern as required to establish the final approach segment within the specified flight tolerances and protected area. (D) Establish a predetermined rate of descent in order to follow the published path. (E) Intercept and track the final approach track within the prescribed limits. (F) Interpretation of approach chart. (G) Ability to interpret deviation. (H) Correct selection of navigation input to the display.
KNOWLEDGE	(A) Systems: communication, navigation and auto-flight systems. (B) Correctly interpret and understand the procedure to be flown from the approach chart for runway and procedure in use. (C) CDFA technique where appropriate. (D) Autopilot and flight director limitations.
ATTITUDE	(A) Situation awareness: Establish communication with ATC on the correct frequencies and at the appropriate times; (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Monitor to ensure that the flight profile complies with the cleared procedure. (E) Effective problem-solving and decision-making; React to navigation errors or unexpected systems malfunctions.

Altitude, speed and heading control (stabilised approach)

Module 2: 2D approach procedures

(must be performed by sole reference to instruments)

Altitude, speed and heading control (stabilised approach)

OBJECTIVE	(A) Establish a stabilised approach, in trim for the aeroplane configuration and speed, using the correct techniques for attitude, heading and power control. (B) Correct assessment of track and rate of descent or vertical path angle.
SKILL	(A) Establish the final approach and maintain the approach path in horizontal and vertical profile to minima. (B) Control the aircraft as necessary to achieve a stable final approach. (C) Arrive at the minima on a stabilised approach in order to make a correct decision to perform a landing, go-around or circling approach safely. (D) Prepare backup radio aids for continued approach in the event of radio aid or display equipment failure. (E) Use correct RTF procedures and terminology, and comply with all ATC instructions and clearances.
KNOWLEDGE	(A) Horizontal and vertical tolerances. (B) Actions to be taken in the event of radio aid/display equipment failure. (C) Procedure in the event of loss of communication with ATC. (D) Procedure in the event of loss of integrity.
ATTITUDE	(A) Situation awareness: Confirm that the approach is stabilised. (B) Effective communication: Advise ATC if appropriate. (C) Leadership and teamwork: (1) Demonstrate correct coordination with ATC (where applicable); (2) Procedures for loss-of-approach capability. (D) Effective workload management: Monitor to ensure that the flight profile remains safe. (E) Effective problem-solving and decision-making: Make appropriate decision to abandon approach if required.

Approach timing

Module 2: Specificities of conventional 2D approach procedures
(must be performed by sole reference to instruments)

Approach timing

OBJECTIVE	Monitor or control the approach procedure using timing as necessary.
SKILL	Where DME information from ground-based beacons (VOR or NDB) or marker is not available, the applicant makes appropriate adjustments to the procedure timing to allow for the effects of known wind.
KNOWLEDGE	(A) Use of wind-effect correction techniques. (B) Use of wind-drift correction techniques to maintain the correct tracks, bearings and approximate distances.
ATTITUDE	(A) Situation awareness: (1) Understand when approach timing techniques are required; (2) Understand the impact required on the descent technique for the intermediate approach phase. (B) Effective workload management: Use an appropriate 'division of attention' whilst controlling the aircraft in order to apply wind-corrected timing. (C) Effective communication: As applicable to the specific situation. (D) Leadership and teamwork: As applicable to the specific situation. (E) Effective problem-solving and decision-making: As applicable to the specific situation.

Go-around and missed approach action

Module 2: Common to both 3D and 2D procedures
(must be performed by sole reference to instruments)

Go-around and missed approach action

OBJECTIVE	Make a smooth transition to a climb at the correct speed and complete the checks when: (1) reaching the minima; (2) directed by ATC; (3) being in an unstabilised approach; (4) experiencing a loss of integrity; or (5) any other reasons affecting safety approach.
SKILL	(A) Initiate go-around action in case of unstabilised approach or loss of integrity. (B) Initiate go-around action at or above minima if safe landing is not possible. (C) Control the aircraft as necessary to achieve a stable and trimmed initial climb profile. (D) Ensure a safe climb and departure in accordance with ATC clearance and with due regard for other air traffic, noise abatement and wake turbulence avoidance procedures adjusting power and aircraft configuration, and maintain desired path (or heading) as appropriate. (E) Complete all necessary procedures and checks. (F) Select the missed approach if available.
KNOWLEDGE	(A) Go-around procedure. (B) Aircraft limitations for landing gear retraction, flap retraction and power plant. (C) Necessary RTF procedures. (D) Performance limitation. (E) Climb gradient. (F) Protected areas. (G) RNP approach specifications.
ATTITUDE	(A) Situation awareness: Monitor aircraft flight path at all stages of the go-around. (B) Effective communication: (1) Demonstrate effective communication (as applicable); (2) Communicate with ATC when safe to do so. (C) Leadership and teamwork: Demonstrate effective coordination with ATC (as applicable). (D) Effective problem-solving and decision-making: Correctly assess go-around and climb hazards, particularly those related to other aircraft, aerodrome infrastructure, obstacles and weather, and have a strategy to mitigate the threats.

Landing

**Module 2: Common to both 3D and 2D procedures
(must be performed by sole reference to instruments)**

Landing

OBJECTIVE	(A) Visual landing or circle for landing, as appropriate, in a safe and controlled manner. (B) Define a strategy for track management in case of missed approach or go-around in the circle to land.
SKILL	Landing: (A) acquire the required visual references and continue to land the aircraft; (B) make a smooth transition from instrument to visual flight; (C) join smoothly, if necessary, the visual approach flight path; (D) maintain a stable (speed, power, heading) approach until the flare; (E) complete post-landing checklist.
KNOWLEDGE	(A) Flight manual. (B) Limitations. (C) Normal procedures: Demonstrate adequate judgement and knowledge of the aircraft performance and systems in order to comply with published approach procedures for the equipment used for the approach.
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Controlled flight into terrain (CFIT); (3) Balked landing. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Monitor to ensure that the flight profile complies with the approach procedure. (E) Effective problem-solving and decision-making: React to deviation errors or unexpected systems malfunctions.

ATC liaison — compliance, RTF procedures

Module 2: Common to both 3D and 2D procedures
(must be performed by sole reference to instruments)

ATC liaison — compliance, RTF procedures

OBJECTIVE	(A) Use correct and standard RTF phraseology throughout. (B) Where appropriate, obtain ATC clearances and appropriate level of service. (C) Where required, comply with ATC clearances and instructions.
SKILL	(A) Comply with all ATC instructions and clearances. (B) Use correct RTF for ILS reporting procedure.
KNOWLEDGE	(A) ICAO standard phraseology. (B) Pilot/controller responsibilities to include tower en-route control and clearance. (C) Demonstrate adequate knowledge of two-way communications failure procedures.
ATTITUDE	(A) Situation awareness: Establish communication with ATC on the correct frequencies and at the appropriate times. (B) Effective communication: Read back correctly, in a timely manner, the ATC clearance in the sequence received. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Copy correctly, in a timely manner, the ATC clearance as issued. (E) Effective problem-solving and decision-making: Interpret correctly the ATC clearance received and, when necessary, request clarification, verification, or change.

(3) Module 3: En-route IFR procedures

Use of air traffic services document and weather document

Module 3: En-route IFR procedures	
Use of air traffic services document and weather document	
OBJECTIVE	(A) Use of the correct documents, including maps. (B) Use of charts and approach procedure plates to prepare flight plan and flight log. (C) Collating and interpreting weather documents to determine the route weather.
SKILL	(A) Ensure all required paperwork is correctly completed prior to the flight. (B) Interpretation of weather charts and coded messages (TAF, METAR, etc.).
KNOWLEDGE	(A) Weather factors that may affect the safe conduct of the flight (thunderstorms, fog, strong winds, gust factor, crosswinds at departure and destination aerodromes, snow, icing, etc.). (B) Type of approach to be flown, how to calculate approach minima from charts, operational limitations of ground-based aids when planning route, ability to interpret SID and STAR charts. (C) Coordination with ATC when submitting flight plan, implications of 'calculated take-off time', etc.
ATTITUDE	(A) Situation awareness: Note potential weather hazards and act accordingly, submit flight plan in good time for planned departure. (B) Effective communication: Communicate with ATC and ground crew to ensure timely start. (C) Leadership and teamwork: Demonstrate correct crew coordination with ATC (where applicable). (D) Effective workload management: Prioritise tasks to produce a safe and effective plan for the conduct of the flight. (E) Effective problem-solving and decision-making: (1) Identify possible defects and threats; (2) Take corrective action.

Preparation of ATC flight plan and IFR flight plan or log

Module 3: En-route IFR procedures

Preparation of ATC flight plan and IFR flight plan or log

OBJECTIVE	Preparation of the ATC IFR flight plan for the route, including any off-airway sectors, and preparation of a full navigation and RTF flight log.
SKILL	(A) Prepare the flight navigation log, update maps and charts, flight plan, and fuel plan. (B) Obtain and assess all elements of the prevailing and forecast weather conditions for the route and evaluate threats (e.g. icing conditions, convection, wind conditions, potential deterioration below minima). (C) Complete an appropriate flight navigation log. (D) Complete the required ATC flight plan(s) and ensure that all required airfields are addressed. (E) Determine that the aeroplane is correctly fuelled, loaded and legal for the flight. (F) Confirm any aeroplane performance criteria and limitations applicable in relation to runway and weather conditions.
KNOWLEDGE	Demonstrate sufficient knowledge of the regulatory requirements relating to instrument flight.
ATTITUDE	(A) Situation awareness: (1) Understand the responsibilities of proper pre-departure planning and preparations; (2) Appropriate threat and error management for the flight (B) Effective communication: - Ensure appropriate and clear communication with all ground service personnel (ATC, dispatch, MET). (C) Leadership and teamwork. (D) Effective workload management: - Provide sufficient time, and manage the workload for departure procedures (including documentation) to be completed in an efficient manner. (E) Effective problem-solving and decision-making: - Make appropriate decisions on all identified threats, and plan and implement suitable mitigation actions.

Tracking, including interception, e.g. NDB, VOR, RNAV

Module 3: En-route IFR procedures

Tracking, including interception, e.g. NDB, VOR, RNAV

OBJECTIVE	(A) Intercept and maintain the route or amended route, including tracking to and from a position derived from NDB or VOR or RNAV (GNSS) using aircraft display. (B) Follow the flight-planned route or any other ATC route requirements within the specified limits. (C) Identify and use navigation systems correctly. (D) Use the correct altimeter setting procedures and show awareness of protected areas.
SKILL	(A) Use the current and appropriate navigation publications for the proposed flight. (B) Intercept, in a timely manner, all courses, radials and bearings appropriate to the procedure, route, and ATC clearance. (C) Comply, in a timely manner, with all ATC clearances, instructions and restrictions. (D) Perform the aircraft briefing or checklist items appropriate to the arrival. (E) Adhere to airspeed restrictions and adjustments required by regulations, ATC and aircraft flight manual. (F) Maintain the appropriate airspeed, altitude and heading, and accurately track radials, courses and bearing (QDM/QDRs).
KNOWLEDGE	(A) Basic instrument rating knowledge. (B) Proper ATC phraseology. (C) Demonstrate adequate knowledge of: (1) flight manual; (2) limitations; (3) instrument patterns; (4) two-way communications failure procedures. (D) Systems: communication, navigation and auto-flight systems. (E) PBN specifications.
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Awareness of aircraft position in space. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Communicate with ATC as appropriate. (C) Leadership and teamwork: - Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: - Monitor to ensure that the flight profile complies with the cleared en-route routing. (E) Effective problem-solving and decision-making: - React to navigation errors or unexpected systems malfunctions.

Use of radio aids

Module 3: En-route IFR procedures

(must be performed by sole reference to instruments)

Use of radio aids

OBJECTIVE	(A) Correct use of RNAV system and radio aids with regard to promulgated range, identification and interpretation. (B) Use of ATIS/VOLMET where available.
SKILL	(A) Use the current and appropriate navigation publications for the proposed flight. (B) Select a combination of radio aids that allow the aircraft position to be correctly determined. (C) Manage the display of such aids so that the navigational information is readily available. (D) Correctly identify the chosen radio aids using Morse code where appropriate, i.e. when there is no 'auto-ident'. (E) Correctly assess the functionality of radio aids, including RNAV, before using them for navigation. (F) Correctly check receiver autonomous integrity monitoring (RAIM) of GNSS systems, if applicable. (G) Correctly input navigation planning data into the GNSS system where appropriate.
KNOWLEDGE	(A) Demonstrate the theoretical knowledge and understanding of: (1) the limitations and errors of VOR and NDB, the limitations and errors of VOR and NDB receivers in the aircraft, and the resulting potential navigational error; (2) information pertinent to radio aids or RNAV operations contained in NOTAMs; (3) correct identification of ground-based radio aids; (4) the radio aid equipment and associated displays fitted to the aircraft. (B) Identify when a ground-based radio aid is radiating but the signal is not available for navigation.
ATTITUDE	(A) Situation awareness: (1) Monitor flight progress and select the appropriate navigation systems to enable successful completion of the planned route; (2) Awareness of aircraft position in space. (B) Effective communication: As applicable to the specific situation. (C) Leadership and teamwork: As applicable to the specific situation. (D) Effective workload management: Use an appropriate 'division of attention' appropriately whilst controlling the aircraft and reset navigation aids. (E) Effective problem-solving and decision-making: React to navigation errors or unexpected systems malfunctions.

Level flight, control of heading, attitude and airspeed, power-setting, trim technique

Module 3: En-route IFR procedures

Level flight, control of heading, attitude and airspeed, power-setting, trim technique

OBJECTIVE	(A) Smooth control of heading, attitude and airspeed, power, trim and ancillary controls. (B) Correct use of autopilot where appropriate. (C) Demonstrate correct technique for instrument flight manoeuvring within specified limits. (D) Maintain balanced and trimmed flight.
SKILL	(A) Maintain altitude, heading and balance, by sole reference to instruments, using correct instrument confirmation, and coordinated control application. (B) Maintain altitude, heading and balance, whilst accelerating or decelerating to specific speeds, as determined by the aircraft flight manual. (C) Demonstrate correct procedure for pre-flight functional check of autopilot, flight director and aircraft navigation system, as applicable. (D) Demonstrate correct operating procedure for aircraft navigation systems, autopilot or flight director in all modes.
KNOWLEDGE	(A) Procedures for controlling the aircraft in accordance with the aircraft flight manual and flight manual, as appropriate. (B) Autopilot, flight director and navigation system fitted to the aircraft.
ATTITUDE	(A) Situation awareness: (1) Maintain awareness of the autopilot modes selected, where applicable; (2) Understand the need for trimmed, in-balance flight when manually flying the aircraft; (3) Maintain adequate scan rate before, during and after execution of any manoeuvre by reference to instruments and autopilot performance. (B) Effective communication: - As applicable to the specific situation. (C) Leadership and teamwork: - As applicable to the specific situation. (D) Effective workload management: - Use an appropriate 'division of attention' when completing flight log, etc., whilst manually controlling the aircraft. (E) Effective problem-solving and decision-making: - Prioritise activities to allow maintenance of correct instrument scan.

Altimeter setting

Module 3: En-route IFR procedures	
Altimeter setting	
OBJECTIVE	Follow the altimeter-setting procedure, and cross-check and monitor en-route protected areas.
SKILL	(A) Correct use and interpretation of altimeter subscale setting. (B) Cross-check against a second altimeter.
KNOWLEDGE	(A) National procedures, if different, regarding altimeter settings for the airspace the aircraft is occupying. (B) Effects of extremely low temperatures on altimeter indications. (C) Limitations and errors in altimeters due to construction or systems installed in the aircraft, etc.
ATTITUDE	(A) Situation awareness: (1) Understand the airspace structure and make appropriate altimeter settings; (2) Be aware of minimum safe altitude, sector safe altitude, etc. (B) Effective workload management: As applicable to the specific situation. (C) Effective communication: Use appropriate RTF procedures to update pressure settings. (D) Leadership and teamwork: As applicable to the specific situation. (E) Effective problem-solving and decision-making: Where necessary, identify and make appropriate decisions when confronted with system failures.

Timing and revision of estimated time of arrival (ETA) (en-route hold, if required)

Module 3: En-route IFR procedures	
Timing and revision of estimated time of arrival (ETA) (en-route hold, if required)	
OBJECTIVE	Understand the flight plan, and that the clearance is to be completed correctly.
SKILL	(A) Use appropriate and up-to-date aeronautical charts. (B) Extract and record pertinent information from NOTAMs, the aerodrome or facility directory, and other flight publications. (C) Plot a course for the intended route of flight. (D) Select the most favourable altitudes. (E) Compute headings, flight time, and fuel requirements.
KNOWLEDGE	(A) Weather reports and forecasts. (B) Pilot and radar reports. (C) Winds and temperatures aloft. (D) ATC procedures related to timing, e.g. update of ETA if changed by ± 3 minutes, clearance limit, etc.
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Identify airspace, obstructions, and terrain features. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Demonstrate correct communication with ATC (where applicable). (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Select the appropriate navigation systems or facilities and communication frequencies. (E) Effective problem-solving and decision-making: Deal with unexpected navigation errors or systems malfunctions.

Monitoring of flight progress, flight log, fuel usage and management, systems management

Module 3: En-route IFR procedures

Monitoring of flight progress, flight log, fuel usage and management, systems management

OBJECTIVE	(A) Maintain a flight log by recording sufficient information. (B) Monitor the engine and aircraft systems throughout the flight. (C) Monitor fuel consumption versus fuel available and fuel required throughout the flight.
SKILL	(A) Follow the flight plan route in accordance with ATC. (B) Navigate by means of an appropriate navigation system for the cleared route. (C) Use the correct altimetry procedures. (D) Verify the aircraft's position in relation to the flight-planned route. (E) Correctly assess track error and make suitable adjustments to heading. (F) Correct and record the differences between pre-flight fuel, ground speed, and heading and time calculations and those determined en-route. (G) Complete all appropriate checklists. (H) Manage the flight in accordance with minimum altitude.
KNOWLEDGE	(A) Part-SERA requirements and national rules regarding use of aerodromes procedures. (B) Policy concerning IFR flights (e.g. national procedures in the AIP). (C) Services expected in different classes of airspace. (D) Danger restricted, and prohibited areas. (E) Minimum altitude and protected areas.
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Identify airspace and minimum altitudes. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Demonstrate correct communication (where applicable). (C) Leadership and teamwork: - Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: - Select appropriate navigation systems or facilities and communication frequencies. (E) Effective problem-solving and decision-making: - Deal with unexpected navigation errors or systems malfunctions.

Ice protection procedures, simulated if necessary

Module 3: En-route IFR procedures	
Ice protection procedures, simulated if necessary	
OBJECTIVE	(A) Monitoring of outside air temperature (OAT), icing risk and ice accretion rate (on FSTD if necessary); correct use of anti-icing and de-icing procedures. (B) Manage flight in icing conditions.
SKILL	(A) Assessment of ice accretion on aircraft. (B) Appropriate selection of anti-icing or de-icing systems. (C) Adapt the aircraft speed to stay within the flight manual limitations, if any. (D) Adapt the performance within the icing conditions. (E) Decision-making to avoid icing conditions.
KNOWLEDGE	(A) Weather reports and forecasts. (B) ATC, pilot and radar reports. (C) Surface analysis charts. (D) Ground radar summary charts. (E) Significant weather prognostics. (F) Forecast upper wind and temperature for aviation (WINTEN). (G) Freezing level. (H) SIGMETs. (I) ATIS and VOLMET reports. (J) Aircraft anti-icing and de-icing system limitations. (K) Significant weather chart (TEMSE).
ATTITUDE	(A) Situation awareness: (1) Understand the environmental conditions which can lead to the formation of ice on the aircraft; (2) Assess when ice accretion is outside the capability of the aircraft systems. (B) Effective communication: (1) Liaise with ATC to avoid known icing conditions; (2) Request change of route or level to avoid icing conditions. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Select appropriate navigation systems or facilities and communication frequencies. (E) Effective problem-solving and decision-making: (1) Deal with unexpected encounters with icing conditions or systems malfunctions; (2) Seek reroute or change of level in a timely manner.

ATC liaison — compliance, RTF procedures

Module 3: En-route IFR procedures

ATC liaison — compliance, RTF procedures

OBJECTIVE	ATC liaison using the correct RTF procedures and phraseology, and compliance with ATC procedures and clearances.
SKILL	(A) Follow the flight-planned route or any other ATC route requirements within the specified operating limits. (B) Identify and use navigation systems correctly. (C) Monitor whether ATC clearance is in accordance with a safe flight. (D) Use the correct RTF procedures and phraseology.
KNOWLEDGE	ICAO (language proficiency level 4, as a minimum) and national RTF procedures.
ATTITUDE	(A) Situation awareness: (1) Establish communication with ATC on the correct frequencies and at the appropriate times; (2) Identify airspace, and understand ATC clearances. (B) Effective communication: (1) Read back correctly, in a timely manner, the ATC clearance in the sequence received; (2) Demonstrate correct communication with ATC (where applicable). (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Select the appropriate navigation systems or facilities and communication frequencies. (E) Effective problem-solving and decision-making: Deal with unexpected navigation errors or systems malfunctions.

(4) Module 4: Optional flight with one engine inoperative (multi-engine aeroplanes only)

Simulated engine failure after take-off or during go-around

Module 4: Optional flight with one engine inoperative (multi-engine aeroplanes only)
(must be performed by sole reference to instruments)
(Multi-engine aeroplanes only)

Simulated engine failure after take-off or during go-around (at a safe altitude unless conducted in an adapted FSTD)

OBJECTIVE	(A) Maintain the flight path after take-off or during go-around with one engine inoperative. (B) Comply with ATC instructions.
SKILL	(A) Maintain control following engine failure with sole reference to instruments. (B) Prepare a strategy in case of engine failure or go-around. (C) Calculate one-engine-inoperative performance. (D) Adapt minima on take-off or in approach in accordance with the performance. (E) Carry out the recommended emergency procedures.
KNOWLEDGE	(A) Operating manual: (1) all systems; (2) limitations; (3) abnormal procedures; (4) Part-NCO; (5) performance; (6) CS-23. (B) Operator policy dedicated to failure during take-off: in particular, operator engine-out path during take-off.
ATTITUDE	(A) Situation awareness: (1) Recognise engine failure, and confirm correct engine; (2) Performance limitations; (3) Strategy threats. (B) Effective communication: Communicate appropriately with ATC. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: (1) Apply appropriate abnormal or emergency procedures, time permitting, to resolve reason for engine failure; (2) Management of flight path close to the ground. (E) Effective problem-solving and decision-making: Identify critical situation and make timely decision on suitable actions to carry out a safe asymmetric flight path.

Approach, go-around and procedural missed approach with one engine inoperative

Module 4: Optional flight with one engine inoperative (multi-engine aeroplanes only)
(must be performed by sole reference to instruments)
(Multi-engine aeroplanes only)

Approach, go-around and procedural missed approach with one engine inoperative

OBJECTIVE	(A) Manage IFR approach path during engine failure. (B) Maintain a stable approach in the correct configuration. (C) Make a clear decision to land or go around no later than the appropriate committal height or minima. (D) Complete asymmetric approach and go-around into visual circuit, circling approach or further instrument approach, maintaining control and correct speeds. (E) Initiate go-around action in case of destabilised approach. (F) Complete procedures and checks.
SKILL	(A) Apply the appropriate power setting for the flight condition and establish a pitch attitude necessary to achieve the desired performance. (B) Retract the wing flaps or drag devices and landing gear, if appropriate, in the correct sequence. (C) Accomplish the appropriate procedures or checklist items in a timely manner in accordance with the flight manual.
KNOWLEDGE	(A) Flight manual: (1) all systems; (2) limitations; (3) abnormal procedures; (4) patterns; (5) Part-NCO; (6) performance; (7) CS-23. (B) Operator policy dedicated to approach stabilisation criteria.
ATTITUDE	(A) Situation awareness: Recognise whether the approach profile is stabilised. (B) Effective communication: Communicate appropriately with ATC. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Apply appropriate abnormal procedures for asymmetric approach and go-around. (E) Effective problem-solving and decision-making: (1) Identify whether a critical situation is occurring due to inappropriate approach profile; (2) Make a timely decision to execute a go-around.

Approach and landing with one engine inoperative

Module 4: Optional flight with one engine inoperative (multi-engine aeroplanes only)
(must be performed by sole reference to instruments)
(Multi-engine aeroplanes only)

Approach and landing with one engine inoperative

OBJECTIVE	(A) Establish the approach and landing configuration appropriate for the selected runway and prevailing meteorological conditions, and adjust the engine controls as required. (B) Complete the applicable pre-landing checklist. (C) Maintain a stabilised approach at the desired airspeed. (D) Maintain the operating engine(s) within acceptable operating limits. (E) Accomplish a smooth, positively controlled transition from instrument reference to visual reference. (F) Join smoothly, if necessary, the visual approach flight path. (G) Complete the applicable post-landing briefing or checklist items in a timely manner, after clearing the runway, and as recommended by the manufacturer.
SKILL	(A) Consider the actual weather and wind conditions, landing surface and obstructions. (B) Maintain a stable approach in the correct configuration. (C) Plan and follow suitable approach pattern and orientation with the landing runway. (D) Establish the correct approach configuration, adjusting speed and rate of descent to maintain a stabilised approach path. (E) Make a clear decision to land or go around no later than the appropriate committal height or minima. (F) Select and achieve the appropriate touchdown area at the required speed.
KNOWLEDGE	(A) Flight manual: (1) all systems; (2) limitations; (3) abnormal procedures; (4) patterns; (5) Part-NCO; (6) performance; (7) CS-23. (B) Understand the factors affecting asymmetric committal height/altitude (ACH/A).
ATTITUDE	(A) Situation awareness: Recognise whether the approach profile is stabilised, leading to a safe asymmetric landing. (B) Effective communication: Liaise with ATC. (C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable). (D) Effective workload management: Apply appropriate abnormal procedures for asymmetric approach and landing. (E) Effective problem-solving and decision-making: Make appropriate decision at asymmetric committal height (ACH) to commit to final flap selection and landing.

ATC liaison — compliance, RTF procedures

Module 4: Optional flight with one engine inoperative (multi-engine aeroplanes only)
(must be performed by sole reference to instruments)
(Multi-engine aeroplanes only)

ATC liaison — compliance, RTF procedures

OBJECTIVE	(A) Inform ATC of abnormal flight condition and any assistance required.
	(B) Comply with ATC procedures and instructions.
SKILL	(A) Use standard RTF phraseology as far as possible and plain language as required when declaring an emergency.
	(B) Seek assistance as appropriate.
KNOWLEDGE	ICAO (language proficiency level 4 or higher) standard phraseology.
ATTITUDE	(A) Situation awareness: Communicate with ATC that an emergency has occurred.
	(B) Effective communication: Read back correctly, in a timely manner, the ATC clearance in the sequence received.
	(C) Leadership and teamwork: Demonstrate correct coordination with ATC (where applicable).
	(D) Effective workload management: Copy correctly, in a timely manner, the ATC clearance as issued.
	(E) Effective problem-solving and decision-making: Interpret correctly the ATC clearance received and ensure that it is compliant with aircraft in an asymmetric configuration.

GM1 FCL.310; FCL.515(b); FCL.615(b); FCL.835(d) Theoretical knowledge examinations

ED Decision 2020/018/R

EXPLANATION OF THE VERBS USED IN THE BENJAMIN BLOOM TAXONOMY

- The depth or level of learning to be achieved during the training and the corresponding level of attainment to be examined or assessed is based on the following taxonomy. In each case, the level of knowledge or skill is signified by the learning objective (LO) verb.
- The majority of the LOs relate to the cognitive domain. The taxonomy described by B. Bloom (1956) and Anderson & Krathwohl (2001) has been used as the standard.
- The six sequential increasing levels of required cognitive learning are identified by the LO verb. Hence the lowest level 'remember' is signified by verbs such as 'state', 'list', 'define' and 'recall' whilst the next higher level of 'understand' is signified by verbs such as 'describe' and 'explain'. The third level of 'apply' is signified by the verbs 'calculate', 'interpret', 'relate' and 'solve'. However, the higher levels of 'analyse', which would be signified by the verbs 'plan' or 'discuss'

and ‘evaluate’ and ‘create’ are less common due at least partially to questions presently possible in the ECQB examination.

- (d) The LOs used in Area 100 KSA differ in that they require a combination of knowledge and skills. However, the ‘skill’ level does not relate to Bloom’s psychomotor taxonomy but is more closely aligned to the higher taxonomy levels required in medicine, because knowledge and skills must be combined by the student pilot in a strategy.
- (e) The verbs ‘demonstrate’ and ‘show’, with their meanings defined below, have therefore been used to supplement the cognitive LO verbs for the Area 100 KSA LOs.
 - (1) ‘Demonstrate’ means the selection and use of the appropriate knowledge, skills and attitudes within a strategy to achieve an effective outcome. It signifies a high taxonomy level and would normally be assessed using multiple indicators from more than one core competency.
 - (2) ‘Show’ means the attainment of knowledge, skill or attitude. It signifies a lower taxonomy level than ‘demonstrate’ and would normally be assessed by a single indicator.’

SUBPART J – INSTRUCTORS

SECTION 1 – COMMON REQUIREMENTS

FCL.900 Instructor certificates

Regulation (EU) 2019/1747

- (a) General. A person shall only carry out:
- (1) flight instruction in aircraft when he/she holds:
 - (i) a pilot licence issued or accepted in accordance with this Regulation;
 - (ii) an instructor certificate appropriate to the instruction given, issued in accordance with this Subpart;
 - (2) synthetic flight instruction or MCC instruction when he/she holds an instructor certificate appropriate to the instruction given, issued in accordance with this Subpart.
- (b) Special conditions:
- (1) The competent authority may issue a specific certificate granting privileges for flight instruction when compliance with the requirements established in this Subpart is not possible in the case of the introduction of:
 - (i) new aircraft in the Member States or in an operator's fleet; or
 - (ii) new training courses in this Annex (Part-FCL).

Such a certificate shall be limited to the training flights necessary for the introduction of the new type of aircraft or the new training course and its validity shall not, in any case, exceed 1 year.;
 - (2) Holders of a certificate issued in accordance with (b)(1) who wish to apply for the issue of an instructor certificate shall comply with the prerequisites and revalidation requirements established for that category of instructor. Notwithstanding [FCL.905.TRI\(b\)](#), a TRI certificate issued in accordance with this (sub)paragraph will include the privilege to instruct for the issue of a TRI or SFI certificate for the relevant type.
- (c) Instruction provided outside the territory of the Member States:
- (1) By way of derogation from point (a), in the case of flight instruction provided during a training course approved in accordance with this Annex outside the territory for which Member States are responsible under the Chicago Convention, the competent authority shall issue an instructor certificate to applicants who:
 - (i) holds a pilot licence that meets all of the following criteria:
 - (A) it complies with Annex 1 to the Chicago Convention;
 - (B) in any case, it is at least a CPL in the relevant aircraft category with a relevant rating or certificate;
 - (ii) complies with the requirements established in this Subpart for the issue of the relevant instructor certificate;

- (iii) demonstrates to the competent authority an adequate level of knowledge of European aviation safety rules to be able to exercise instructional privileges in accordance with this Annex.
- (2) The certificate shall be limited to providing flight instruction during a training course approved in accordance with this Annex which meets all of the following conditions:
 - (i) it is provided outside the territory for which Member States are responsible under the Chicago Convention;
 - (ii) it is provided to student pilots who have sufficient knowledge of the language in which flight instruction is provided.

GM1 FCL.900 Instructor certificates

ED Decision 2025/002/R

GENERAL

- (a) Nine instructor categories are recognised:
 - (1) FI certificate: aeroplane (FI(A)), helicopter (FI(H)) and airship (FI(As));
 - (2) TRI certificate: aeroplane (TRI(A)), helicopter (TRI(H)), powered-lift aircraft (TRI(PL));
 - (3) CRI certificate: aeroplane (CRI(A));
 - (4) IRI certificate: aeroplane (IRI(A)), helicopter (IRI(H)) and airship (IRI(As));
 - (5) SFI certificate: aeroplane (SFI(A)), helicopter (SFI(H)) and poweredlift aircraft (SFI(PL));
 - (6) MCCI certificate: aeroplanes (MCCI(A)), helicopters (MCCI(H)), powered-lift aircraft(MCCI(PL)) and airships (MCCI(As));
 - (7) STI certificate: aeroplane (STI(A)) and helicopter (STI(H));
 - (8) MI certificate: (MI);
 - (9) FTI certificate: (FTI).
- (b) For categories (1) to (4) and for (8) and (9) the applicant needs to hold a pilot licence. For categories (5) to (7) no licence is needed, only an instructor certificate.
- (c) A person may hold more than one instructor certificate.

SPECIAL CONDITIONS

- (a) When new aircraft are introduced, requirements such as to hold a licence and rating equivalent to the one for which instruction is being given, or to have adequate flight experience, may not be possible to comply with. In this case, to allow for the first instruction courses to be given to applicants for licences or ratings for these aircraft, competent authorities need the possibility to issue a specific certificate that does not have to comply with the requirements established in this Subpart.
- (b) The competent authority should only give these certificates to holders of other instruction qualifications. As far as possible, preference should be given to persons with at least 100 hours of experience in similar types or classes of aircraft.
- (c) When the new aircraft type introduced in an operator's fleet already existed in a Member State, the competent authority should only give the specific certificate to an applicant that is qualified as PIC on that aircraft.

- (d) The certificate should ideally be limited in validity to the time needed to qualify the first instructors for the new aircraft in accordance with this Subpart, but in any case it should not exceed the 1 year established in the rule.

GM1 FCL.900(c); FCL.1000(c) Instruction or examination outside the territory of the Member States

ED Decision 2020/005/R

Instruction or examination outside the territory of the Member States is possible within the scope of:

- ATOs that have their principal place of business outside the territory of the Member States; or
- ATOs that have their principal place of business in a Member State and one or more additional training sites outside the territory of the Member States.

GM1 FCL.900(c)(1) Instructor certificates

ED Decision 2020/005/R

INSTRUCTION OUTSIDE THE TERRITORY OF THE MEMBER STATES

The competent authority may issue an unrestricted flight instructor (FI) certificate (FI(A) for aeroplanes or FI(H) for helicopters) to an applicant that has at least 100 hours of experience in flight instruction and 25 hours in solo-flight supervision.

FCL.915 General prerequisites and requirements for instructors

Regulation (EU) 2024/2076

- (a) General.

Applicants for the issue of an instructor certificate shall be at least 18 years of age.

- (b) Additional requirements for instructors that provide flight instruction in aircraft.

Applicants for the issue of or holders of an instructor certificate with privileges to conduct flight instruction in an aircraft shall:

- (1) for licence training, hold at least the licence or, in the case of point [FCL.900\(c\)](#), the equivalent licence, for which flight instruction is to be given;
- (2) for a rating training, hold the relevant rating or, in the case of point [FCL.900\(c\)](#), the equivalent rating, for which flight instruction is to be given;
- (3) except in the case of flight test instructors (FTIs), have:
 - (i) completed at least 15 hours of flight time as pilots of the class or type of aircraft on which flight instruction is to be given, of which a maximum of 7 hours may be in an FSTD representing the class or type of aircraft, if applicable; or
 - (ii) passed an assessment of competence for the relevant category of instructor on that class or type of aircraft;
- (4) be entitled to act as PIC in the aircraft during such flight instruction; and
- (5) when providing flight instruction in a variant of the SEP aeroplane class with a particular type of engine referred to in Article 2, point (8c), of this Regulation, have complied with one of the following conditions:
 - (i) completed at least 5 hours of flight time as pilots in that variant;

- (ii) passed an assessment of competence for the relevant category of instructor in that variant.
- (c) Credit towards further instructor certificates and for the purpose of revalidation:
 - (1) Full credit towards the teaching and learning skills may be granted to:
 - (i) holders of an instructor certificate who apply for further instructor certificates; and
 - (ii) applicants for an instructor certificate who already hold an instructor certificate issued in accordance with Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) or with Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#).
 - (2) Hours flown as an examiner during skill tests or proficiency checks shall be credited in full towards revalidation requirements for all instructor certificates held.
- (d) Credit for extension to further types shall take into account the relevant elements as defined in the operational suitability data established in accordance with Annex I (Part-21) to Regulation (EU) No 748/2012 (OSD).
- (e) Additional requirements for instructing in a training course in accordance with [FCL.745.A](#):
 - (1) In addition to (b), before acting as instructors for a training course according to [FCL.745.A](#), holders of an instructor certificate shall:
 - (i) have at least 500 hours of flight time as pilots of aeroplanes, including 200 hours of flight instruction;
 - (ii) after complying with the experience requirements in point (e)(1)(i), have completed a UPRT instructor training course at an ATO, during which the competence of applicants shall have been assessed continuously; and
 - (iii) upon completion of the course, have been issued with a certificate of course completion by the ATO, whose Head of Training (HT) shall have entered the privileges specified in point (e)(1) in the logbook of the applicants.
 - (2) The privileges referred to in point (e)(1) shall only be exercised if instructors have, during the last year, received refresher training at an ATO during which the competence required to instruct on a course in accordance with point [FCL.745.A](#) is assessed to the satisfaction of the HT.
 - (3) Instructors holding the privileges specified in point (e)(1) may act as instructors for a course as specified in point (e)(1)(ii), provided that they:
 - (i) have 25 hours of flight instruction experience during training according to [FCL.745.A](#);
 - (ii) have completed an assessment of competence for this privilege; and
 - (iii) comply with the recency requirements in point (e)(2).
 - (4) These privileges shall be entered in the logbook of the instructors and signed by the examiner.

AMC1 FCL.915(e) General prerequisites and requirements for instructors

ED Decision 2019/005/R

ADDITIONAL REQUIREMENTS FOR INSTRUCTING IN A TRAINING COURSE IN ACCORDANCE WITH FCL.745.A – GENERAL

- (a) The objective of the course required by point [FCL.915\(e\)\(1\)](#) is to train instructors to deliver training on the advanced UPRT course according to point [FCL.745.A](#) using the train-to-proficiency concept.
- (b) Training should be both theoretical and practical. Practical elements should include the development of specific instructor skills, particularly in the area of teaching upset recovery techniques and strategies, whilst exploring the associated physiological and psychological aspects.
- (c) Within 6 months preceding the start of the course, the instructor should have completed a pre-course assessment with an instructor holding the privilege in accordance with [FCL.915\(e\)\(1\)](#) to assess their ability to undertake the course.
- (d) The training course should comprise:
 - (1) theoretical knowledge instruction on the theoretical knowledge elements presented in the advanced UPRT course and the additional elements required for an instructor to deliver effective training;
 - (2) flight instruction on the exercises used in the advanced UPRT course; and
 - (3) flight instruction on recovery from upsets that could result from students mis-handling the aircraft during the advanced UPRT course including spin recovery.
- (e) The content of the theoretical knowledge and flight instruction should be tailored to the competence of the applicant as demonstrated during both pre-course and continuous assessment.
- (f) Successful completion of the course requires that the instructor:
 - (1) demonstrates the resilience to be able to recover from any feasible upset in the aircraft to be used for training;
 - (2) demonstrates the ability to provide instruction to achieve the objectives of the advanced UPRT course to a wide range of trainees; and
 - (3) manages the physiological and psychological well-being of students during training.
- (g) The instructor should be issued with a certificate following successful completion of the course.

AMC2 FCL.915(e) General prerequisites and requirements for instructors

ED Decision 2019/005/R

ADDITIONAL REQUIREMENTS FOR INSTRUCTING IN A TRAINING COURSE IN ACCORDANCE WITH FCL.745.A – SYLLABUS

The following tables contain theoretical knowledge (Table 1) and practical training exercises (Table 2) that should be taught in the context of the advanced UPRT course as per point [FCL.745.A](#).

TABLE 1: THEORETICAL KNOWLEDGE

1.	Completion of a flight risk assessment
2.	Resilience-building strategies, managing startle and surprise
3	The limitations and type-specific characteristics of the aeroplane used for training
4	The importance of adhering to the scenarios that have been validated by the training programme developer
5.	Instructor techniques to induce and manage startle and surprise
6.	Upset recognition and recovery strategies
7.	Disorientation
8.	Distraction
9.	Immediate recognition of student pilot errors
10.	Intervention strategies
11.	Delivery of the theoretical knowledge instruction of the advanced UPRT course

TABLE 2: PRACTICAL TRAINING EXERCISES

SECTION 1 — PRE-FLIGHT PREPARATION

- | | |
|-----|---|
| 1.1 | Correct completion of a flight risk assessment (such as weather, terrain, traffic density, student's experience level and capabilities) |
| 1.2 | Safety briefing |

SECTION 2 — FLIGHT

- | | |
|------|---|
| 2.1 | Selection of suitable airspace for the conduct of recovery exercises |
| 2.2 | Accurate execution of all of the manoeuvres required for the advanced UPRT course |
| 2.3. | Recovery from upsets that could result from the student or instructor mishandling the aeroplane including: <ul style="list-style-type: none"> – timely and appropriate intervention; – accelerated stall; – secondary stall; – incipient spin; – fully developed spin; and – Spiral dive. |
| 2.4 | Delivery of all of the training exercises in the advanced UPRT course |
| 2.5 | Anticipating and immediately recognising incorrect student inputs which might exceed aeroplane limitations and acting swiftly and appropriately to maintain the necessary margins of safety |
| 2.6 | Exercises to surprise the student |
| 2.7 | Adapt the training programme to take account of the physiological and psychological state of the student |
| 2.8 | Ensure the safety of the operation during training by maintaining awareness of the operating environment |
| 2.9 | Assess the competence of the student |

SECTION 3 — POST-FLIGHT

- | | |
|-----|---|
| 3.1 | Provide effective instructor feedback to the student and plan subsequent training details |
| 3.2 | Avoid negative transfer of training |

GM1 FCL.915(e) General prerequisites and requirements for instructors

ED Decision 2019/005/R

TRAINING ON SPIN AVOIDANCE AND SPIN RECOVERY

- (a) While the purpose of advanced UPRT course is to expose students to psychological and physiological effects, students' responses and actions on controls may take any conceivable variations, including some which can initiate spin entry or, most importantly, can highly aggravate the upset or loss-of-control they are supposed to recover from.
- (b) The advanced UPRT course in accordance with point [FCL.745.A](#) is not aerobatic training and only requires training for the incipient spin as well as uncoordinated side slipped stalls which are prone to initiating spins. Full spin training or the development of spin recovery proficiency is reserved for the training course in accordance with point [FCL.915\(e\)](#).
- (c) Even though most flights will go exactly as planned without an unanticipated departure from controlled flight, the instructor is responsible for the safety of flight despite anomalies or unexpected student inputs.
- (d) Even in a case where an aeroplane is not certified for intentional flat or aggravated or inverted spins, it does not mean that mishandled student recovery avoids placing the aeroplane in such a situation. Some student inputs will take the aeroplane uncontrolled far beyond the normal scope of the aerobatic rating as defined in point [FCL.800](#). Those situations might also have the potential to draw the aeroplane outside its certified flight envelope (e.g. overloads, snap-roll departures above limit speed, spin or inverted spin when not certified for, flat spins, etc.). Most importantly, those resulting situations could startle the instructor.
- (e) For the reasons specified in point (d), instructors should:
 - (1) be trained to the extent of proficiency on the specific type of aircraft they use to deliver the course;
 - (2) have academic understanding of the factors assisting or deterring spin recoveries (upright and inverted spins), altitude requirements for safe recovery margins, and other operational considerations;
 - (3) demonstrate that they have the ability to early recognise abnormal situations, timely take action, and safely recover from all the conditions that they may encounter in the delivery of training; and
 - (4) demonstrate their ability to recover from all spin types, not only from spins entered intentionally, but from spins of unannounced direction of autorotation, and from all potential spin variations, including:
 - (i) normal (non-aggravated) spins;
 - (ii) flat spins;
 - (iii) accelerated spins; and
 - (iv) transition spins (incorrect recovery resulting in reversal of rotation).
- (f) In the context of points (d) and (e), it is recommended that candidates either hold an aerobatic rating for aeroplanes or have equivalent experience.

AMC1 FCL.915(e)(2) General prerequisites and requirements for instructors

ED Decision 2019/005/R

CONTENT OF THE REFRESHER TRAINING FOR UPRT INSTRUCTIONAL PRIVILEGES

- (a) The objective of the refresher training is for the instructor to maintain or to re-obtain, as applicable, the level of competence required for instructing on a training course as per point [FCL.745.A](#).
- (b) The content of the refresher training should:
- (1) consist of elements from the initial UPRT instructor training course as per point [FCL.915\(e\)\(1\)\(ii\)](#); and
 - (2) be determined by the ATO on a case-by-case basis, considering the needs of the individual instructor and taking into account the following factors:
 - (i) the experience of the instructor;
 - (ii) the amount of time elapsed since the instructor provided instruction on a training course as per point [FCL.745.A](#) for the last time; and
 - (iii) the performance of the instructor during a simulated UPRT training session comprising exercises from the advanced UPRT course as per point [FCL.745.A](#). During this simulated training session, another instructor qualified in accordance with point [FCL.915\(e\)](#) should play the role of the student on the advanced UPRT course.
- (c) Taking into account the factors listed in (b)(2) above, the ATO may also count the simulated training session as per point (b)(2)(iii) as refresher training without the need for further refresher training sessions, provided that the instructor demonstrates that he or she already possesses the required level of competence.
- (d) The completion of the refresher training should be entered in the logbook of the instructor and should be signed by the head of training of the ATO.

FCL.920 Instructor competencies and assessment

Regulation (EU) No 1178/2011

All instructors shall be trained to achieve the following competences:

- Prepare resources,
- Create a climate conducive to learning,
- Present knowledge,
- Integrate Threat and Error Management (TEM) and crew resource management,
- Manage time to achieve training objectives,
- Facilitate learning,
- Assess trainee performance,
- Monitor and review progress,
- Evaluate training sessions,
- Report outcome.

AMC1 FCL.920 Instructor competencies and assessment

ED Decision 2019/005/R

- (a) Training should be both theoretical and practical. Practical elements should include the development of specific instructor skills, particularly in the area of teaching and assessing threat and error management and CRM.
- (b) The training and assessment of instructors should be made against the following performance standards:

Competence	Performance	Knowledge
Prepare resources	(a) ensures adequate facilities; (b) prepares briefing material; (c) manages available tools; (d) plans training within the training envelope of the training platform, as determined by the ATO (Note: See GM1 ORA.ATO.125 point (f)).	(a) understand objectives; (b) available tools; (c) competency-based training methods; (d) understands the training envelope of the training platform, as determined by the ATO (Note: See GM1 ORA.ATO.125 point (f)) and avoids training beyond the boundaries of this envelope.
Create a climate conducive to learning	(a) establishes credentials, role models appropriate behaviour; (b) clarifies roles; (c) states objectives; (d) ascertains and supports student pilot's needs.	(a) barriers to learning; (b) learning styles.
Present knowledge	(a) communicates clearly; (b) creates and sustains realism; (c) looks for training opportunities.	teaching methods
Integrate TEM and CRM	(a) makes TEM and CRM links with technical training; (b) for aeroplanes: makes upset prevention links with technical training.	(a) TEM and CRM; (b) Causes and countermeasures against undesired aircraft states
Manage time to achieve training objectives	Allocates the appropriate time to achieve competency objective.	syllabus time allocation
Facilitate learning	(a) encourages trainee participation; (b) shows motivating, patient, confident and assertive manner; (c) conducts one-to-one coaching; (d) encourages mutual support.	(a) facilitation; (b) how to give constructive feedback; (c) how to encourage trainees to ask questions and seek advice.
Assesses trainee performance	(a) assesses and encourages trainee self-assessment of performance against competency standards; (b) makes assessment decision and provides clear feedback; (c) observes CRM behaviour.	(a) observation techniques; (b) methods for recording observations.
Monitor and review progress	(a) compares individual outcomes to defined objectives; (b) identifies individual differences in learning rates; (c) applies appropriate corrective action.	(a) learning styles; (b) strategies for training adaptation to meet individual needs.

Competence	Performance	Knowledge
Evaluate training sessions	(a) elicits feedback from student pilots; (b) tracks training session processes against competence criteria; (c) keeps appropriate records.	(a) competency unit and associated elements; (b) performance criteria.
Report outcome	Reports accurately using only observed actions and events.	(a) phase training objectives; (b) individual versus systemic weaknesses.

FCL.925 Additional requirements for instructors for the MPL

Regulation (EU) No 1178/2011

- (a) Instructors conducting training for the MPL shall:
- (1) have successfully completed an MPL instructor training course at an ATO; and
 - (2) additionally, for the basic, intermediate and advanced phases of the MPL integrated training course:
 - (i) be experienced in multi-pilot operations; and
 - (ii) have completed initial crew resource management training with a commercial air transport operator approved in accordance with the applicable air operations requirements.
- (b) MPL instructors training course
- (1) The MPL instructor training course shall comprise at least 14 hours of training.
Upon completion of the training course, the applicant shall undertake an assessment of instructor competencies and of knowledge of the competency-based approach to training.
 - (2) The assessment shall consist of a practical demonstration of flight instruction in the appropriate phase of the MPL training course. This assessment shall be conducted by an examiner qualified in accordance with Subpart K.
 - (3) Upon successful completion of the MPL training course, the ATO shall issue an MPL instructor qualification certificate to the applicant.
- (c) In order to maintain the privileges, the instructor shall have, within the preceding 12 months, conducted within an MPL training course:
- (1) 1 simulator session of at least 3 hours; or
 - (2) 1 air exercise of at least 1 hour comprising at least 2 take-offs and landings.
- (d) If the instructor has not fulfilled the requirements of (c), before exercising the privileges to conduct flight instruction for the MPL he/she shall:
- (1) receive refresher training at an ATO to reach the level of competence necessary to pass the assessment of instructor competencies; and
 - (2) pass the assessment of instructor competencies as set out in (b)(2).

AMC1 FCL.925 Additional requirements for instructors for the MPL

ED Decision 2011/016/R

MPL INSTRUCTOR COURSE

- (a) The objectives of the MPL instructors training course are to train applicants to deliver training in accordance with the features of a competency-based approach to training and assessment.
- (b) Training should be both theoretical and practical. Practical elements should include the development of specific instructor skills, particularly in the area of teaching and assessing threat and error management and CRM in the multicrew environment.
- (c) The course is intended to adapt instructors to conduct competency-based MPL training. It should cover the items specified below:

THEORETICAL KNOWLEDGE

- (d) Integration of operators and organisations providing MPL training:
 - (1) reasons for development of the MPL;
 - (2) MPL training course objective;
 - (3) adoption of harmonised training and procedures;
 - (4) feedback process.
- (e) The philosophy of a competency-based approach to training: principles of competency-based training.
- (f) Regulatory framework, instructor qualifications and competencies:
 - (1) source documentation;
 - (2) instructor qualifications;
 - (3) syllabus structure.
- (g) Introduction to Instructional systems design methodologies (see ICAO PANSTRG Doc):
 - (1) analysis;
 - (2) design and production;
 - (3) evaluation and revision.
- (h) Introduction to the MPL training scheme:
 - (1) training phases and content;
 - (2) training media;
 - (3) competency units, elements and performance criteria.
- (i) Introduction to human performance limitations, including the principles of threat and error management and appropriate countermeasures developed in CRM:
 - (1) definitions;
 - (2) appropriate behaviours categories;
 - (3) assessment system.
- (j) Application of the principles of threat and error management and CRM principles to training:
 - (1) application and practical uses;

- (2) assessment methods;
 - (3) individual corrective actions;
 - (4) debriefing techniques.
- (k) The purpose and conduct of assessments and evaluations:
- (1) basis for continuous assessment against a defined competency standard;
 - (2) individual assessment;
 - (3) collection and analysis of data;
 - (4) training system evaluation.

PRACTICAL TRAINING

- (l) Practical training may be conducted by interactive group classroom modules, or by the use of training devices. The objective is to enable instructors to:
- (1) identify behaviours based on observable actions in the following areas:
 - (i) communications;
 - (ii) team working;
 - (iii) situation awareness;
 - (iv) workload management;
 - (v) problem solving and decision making.
 - (2) analyse the root causes of undesirable behaviours;
 - (3) debrief students using appropriate techniques, in particular:
 - (i) use of facilitative techniques;
 - (ii) encouragement of student self-analysis.
 - (4) agree corrective actions with the students;
 - (5) determine achievement of the required competency.

AMC2 FCL.925(d)(1) Additional requirements for instructors for the MPL

ED Decision 2011/016/R

RENEWAL OF PRIVILEGES: REFRESHER TRAINING

- (a) Paragraph (d) of [FCL.925](#) determines that if the applicant has not complied with the requirements to maintain his/her privileges to conduct competency-based approach training, he or she shall receive refresher training at an ATO to reach the level of competence necessary to pass the assessment of instructor competencies. The amount of refresher training needed should be determined on a case-by-case basis by the ATO, taking into account the following factors:
- (1) the experience of the applicant;
 - (2) the amount of time lapsed since the last time the applicant has conducted training in an MPL course. The amount of training needed to reach the desired level of competence should increase with the time lapsed. In some cases, after evaluating the instructor, and

when the time lapsed is very limited, the ATO may even determine that no further refresher training is necessary.

- (b) Once the ATO has determined the needs of the applicant, it should develop an individual training programme, which should be based on the MPL instructor course and focus on the aspects where the applicant has shown the greatest needs.

GM1 FCL.925 Additional requirements for instructors for the MPL

ED Decision 2011/016/R

MPL INSTRUCTORS

The following table summarises the instructor qualifications for each phase of MPL integrated training course:

Phase of training	Qualification
Line flying under supervision according to operational requirements	Line training captain or TRI(A)
Phase 4: Advanced base training	TRI(A)
Phase 4: Advanced skill test	TRE(A)
Phase 4: Advanced	SFI(A) or TRI(A)
Phase 3: Intermediate	SFI(A) or TRI(A)
Phase 2: Basic	(a) FI(A) or IRI(A) and IR(A)/ME/MCC and 1500 hours multi-crew environment and IR(A) instructional privileges, or (b) FI(A) and MCCI(A), or (c) FI(A) and SFI(A), or (d) FI(A) and TRI(A)
Phase 1: Core flying skills	FI(A) and 500 hours, including 200 hours of instruction Instructor qualifications and privileges should be in accordance with the training items within the phase. STI for appropriate exercises conducted in an FNPT or BITD.

FCL.930 Training course

Regulation (EU) 2024/2076

- (a) An applicant for an instructor certificate shall have completed a course of theoretical knowledge and flight instruction at an ATO.
- (b) In addition to the specific elements set out in this Annex (Part-FCL) for each category of instructor, the training course shall contain the elements required in point [FCL.920](#).

FCL.935 Assessment of competence

Regulation (EU) No 1178/2011

- (a) Except for the multi-crew cooperation instructor (MCCI), the synthetic training instructor (STI), the mountain rating instructor (MI) and the flight test instructor (FTI), an applicant for an instructor certificate shall pass an assessment of competence in the appropriate aircraft category to demonstrate to an examiner qualified in accordance with Subpart K the ability to instruct a student pilot to the level required for the issue of the relevant licence, rating or certificate.

- (b) This assessment shall include:
 - (1) the demonstration of the competencies described in [FCL.920](#), during pre-flight, post-flight and theoretical knowledge instruction;
 - (2) oral theoretical examinations on the ground, pre-flight and post-flight briefings and in-flight demonstrations in the appropriate aircraft class, type or FSTD;
 - (3) exercises adequate to evaluate the instructor's competencies.
- (c) The assessment shall be performed on the same class or type of aircraft or FSTD used for the flight instruction.
- (d) When an assessment of competence is required for revalidation of an instructor certificate, an applicant who fails to achieve a pass in the assessment before the expiry date of an instructor certificate shall not exercise the privileges of that certificate until the assessment has successfully been completed.

AMC1 FCL.935 Assessment of competence

ED Decision 2025/002/R

GENERAL

- (a) The format and application form for the assessment of competence are determined by the competent authority.
- (b) When an aircraft is used for the assessment, it should meet the requirements for training aircraft.
- (c) If an aircraft is used for the test or check, the examiner acts as the PIC, except in circumstances agreed upon by the examiner when another instructor is designated as PIC for the flight.
- (d) During the assessment of competence, the applicant occupies the seat that is normally occupied by the instructor (instructor seat if in an FSTD, or pilot seat if in an aircraft). The examiner, another instructor or, for MPA in an FFS, a real crew member under instruction, functions as the 'student'. The applicant is required to explain the relevant exercises and to demonstrate their conduct to the 'student', where appropriate. Following that, the 'student' executes the same manoeuvres (if the 'student' is the examiner or another instructor, this can include typical mistakes inexperienced students make). The applicant is expected to correct the mistakes orally or, if necessary, by intervening physically.
- (e) The assessment of competence should also include additional demonstration exercises, as decided by the examiner and agreed upon with the applicant before the assessment. These additional exercises should be related to the training requirements for the applicable instructor certificate.
- (f) All relevant exercises should be completed within a period of 6 months. However, all exercises should, where possible, be completed on the same day. In principle, failure in any exercise requires a retest covering all exercises, with the exception of those that may be retaken separately. The examiner may terminate the assessment at any stage if they consider that a retest is required.

AMC2 FCL.935 Assessment of competence

ED Decision 2011/016/R

MCCI, STI AND MI

In the case of the MCCI, STI and MI, the instructor competencies are assessed continuously during the training course.

AMC3 FCL.935 Assessment of competence

ED Decision 2011/016/R

CONTENT OF THE ASSESSMENT FOR THE FI

(a) In the case of the FI, the content of the assessment of competence should be the following:

SECTION 1 THEORETICAL KNOWLEDGE ORAL	
1.1	Air law
1.2	Aircraft general knowledge
1.3	Flight performance and planning
1.4	Human performance and limitations
1.5	Meteorology
1.6	Navigation
1.7	Operational procedures
1.8	Principles of flight
1.9	Training administration

Sections 2 and 3 selected main exercises:

SECTION 2 PRE-FLIGHT BRIEFING	
2.1	Visual presentation
2.3	Technical accuracy
2.4	Clarity of explanation
2.5	Clarity of speech
2.6	Instructional technique
2.7	Use of models and aids
2.8	Student participation

SECTION 3 FLIGHT	
3.1	Arrangement of demo
3.2	Synchronisation of speech with demo
3.3	Correction of faults
3.4	Aircraft handling
3.5	Instructional technique
3.6	General airmanship and safety
3.7	Positioning and use of airspace

SECTION 4 ME EXERCISES	
4.1	Actions following an engine failure shortly after take-off ¹
4.2	SE approach and go-around ¹
4.3	SE approach and landing ¹

SECTION 5 POST-FLIGHT DE-BRIEFING	
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¹ These exercises are to be demonstrated at the assessment of competence for FI for ME aircraft.

5.1	Visual presentation
5.2	Technical accuracy
5.3	Clarity of explanation
5.4	Clarity of speech
5.5	Instructional technique
5.6	Use of models and aids
5.7	Student participation

- (b) Section 1, the oral theoretical knowledge examination part of the assessment of competence, is for all FI and is subdivided into two parts:
- (1) The applicant is required to give a lecture under test conditions to other 'student(s)', one of whom will be the examiner. The test lecture is to be selected from items of section 1. The amount of time for preparation of the test lecture is agreed upon beforehand with the examiner. Appropriate literature may be used by the applicant. The test lecture should not exceed 45 minutes;
 - (2) The applicant is tested orally by an examiner for knowledge of items of section 1 and the 'core instructor competencies: teaching and learning' content given in the instructor courses.
- (c) Sections 2, 3 and 5 are for all FIs. These sections comprise exercises to demonstrate the ability to be an FI (for example instructor demonstration exercises) chosen by the examiner from the flight syllabus of the FI training courses. The applicant is required to demonstrate FI abilities, including briefing, flight instruction and de-briefing.
- (d) Section 4 comprises additional instructor demonstration exercises for an FI for ME aircraft. This section, if applicable, is done in an ME aircraft, or an FFS or FNPT II simulating an ME aircraft. This section is completed in addition to sections 2, 3 and 5.

AMC4 FCL.935 Assessment of competence

ED Decision 2011/016/R

CONTENT OF THE ASSESSMENT FOR THE SFI

The assessment should consist of at least 3 hours of flight instruction related to the duties of an SFI on the applicable FFS or FTD 2/3.

AMC5 FCL.935 Assessment of competence

ED Decision 2018/009/R

REPORT FORMS FOR THE INSTRUCTOR CERTIFICATES

- (a) Assessment of competence form for the FI, IRI and CRI certificates:

APPLICATION AND REPORT FORM FOR THE INSTRUCTOR ASSESSMENT OF COMPETENCE			
1 Applicant's personal particulars:			
Applicant's last name(s):		First name(s):	
Date of birth:		Tel (home):	Tel (work):

Address:				Country:					
2 Licence details									
Licence type:					Number:				
Class ratings included in the licence:					Exp. Date:				
Type ratings included in the licence:			1.						
			2.						
			3.						
			4.						
			5.						
Other ratings included in the licence:			1.						
			2.						
			3.						
			4.						
			5.						
3 Pre-course flying experience									
Total flying hours		PIC hours in instructor training course aircraft		Hours in instructor training course aircraft in the preceding 6 months		Instrument flight instruction		Cross-country hours	
4 Pre-entry flight test									
I recommendfor the FI course.									

Name of ATO:		Date of flight test:	
Name(s) of FI conducting the test (capital letters):			
Licence number:			
Signature:			
5 Declaration by the applicant			
<i>I have received a course of training in accordance with the syllabus for the: (tick as applicable)</i>			
FI certificate FI(A)/(H)/(As)	IRI certificate IRI(A)/(H)/(As)	CRI certificate CRI(A)	
Applicant's name(s): (capital letters)		Signature:	
6 Declaration by the CFI			
<i>I certify that has satisfactorily completed an approved course of training for the</i>			
FI certificate FI(A)/(H)/(As)	IRI certificate IRI(A)/(H)/(As)	CRI certificate CRI(A)	
<i>in accordance with the relevant syllabus.</i>			
Flying hours during the course:			
Aircraft or FSTDs used :			
Name(s) of CFI:			
Signature:			
Name of ATO:			
7 Flight instructor examiner's certificate			
<i>I have tested the applicant according to to Part-FCL</i>			
A. FLIGHT INSTRUCTOR EXAMINER'S ASSESSMENT (in case of partial pass):			
Theoretical oral examination:		Skill test:	
Passed	Failed	Passed	Failed
<i>I recommend further flight or ground training with an instructor before re-test</i>			
<i>I do not consider further flight or theoretical instruction necessary before re-test (tick as applicable)</i>			
B. FLIGHT INSTRUCTOR EXAMINER'S ASSESSMENT:			
FI certificate			
IRI certificate			
CRI certificate (tick as applicable)			
Name(s) of FIE (capital letters):			
Signature:			
Licence number:		Date:	

FCL.940 Validity of instructor certificates

Regulation (EU) 2019/1747

With the exception of the MI, and without prejudice to [FCL.900\(b\)\(1\)](#) and FCL.915 (e)(2), instructor certificates shall be valid for a period of 3 years.

FCL.945 Obligations for instructors

Regulation (EU) 2024/2076

Upon completion of the training flight for the revalidation of an SEP aeroplane or TMG class rating in accordance with point [FCL.740.A\(b\)\(1\)\(ii\)\(C\)](#) or for the revalidation of a single-engine helicopter type rating in accordance with point [FCL.740.H\(a\)\(2\)\(ii\)\(B\)](#), and only in the event of fulfilment of all the other revalidation criteria required by point [FCL.740.A\(b\)\(1\)\(ii\)](#) or point [FCL.740.H\(a\)\(2\)\(ii\)](#), as applicable, the instructor shall endorse the applicant's licence with the new expiry date of the rating, if specifically authorised for that purpose by the competent authority responsible for the applicant's licence.

SECTION 2 – SPECIFIC REQUIREMENTS FOR THE FLIGHT INSTRUCTOR – FI

FCL.905.FI FI – Privileges and conditions

Regulation (EU) 2025/134

The privileges of FIs are to conduct flight instruction for the issue, revalidation or renewal of:

- (a) a PPL and LAPL in the appropriate aircraft category;

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (a) a PPL, LAPL and GPL in the appropriate aircraft category;

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (b) class and type ratings for single-pilot aircraft, except for single-pilot high-performance complex aeroplanes;
- (c) class and type ratings for single-pilot aeroplanes, except for single-pilot high-performance complex aeroplanes, in multi-pilot operations, provided that FIs meet any of the following conditions:
- (1) hold or have held a TRI certificate for multi-pilot aeroplanes;
 - (2) have completed all of the following:
 - (i) at least 500 hours as pilots in multi-pilot operations on aeroplanes;
 - (ii) the training course for an MCCI in accordance with point FCL.930.MCCI;
- (d) type ratings for single or multi-pilot airships;
- (e) a CPL in the appropriate aircraft category, provided that FIs have completed at least 200 hours of flight instruction in that aircraft category;
- (f) the night rating, provided that FIs meet all of the following conditions:
- (1) are qualified to fly at night in the appropriate aircraft category;
 - (2) have demonstrated the ability to instruct at night to an FI qualified in accordance with point (j);
 - (3) comply with the night experience requirement laid down in point FCL.060(b)(2);
- (g) a towing or aerobatic rating, provided that such privileges are held and the FI has demonstrated the ability to instruct for that rating to an FI who is qualified in accordance with point (j);
- (h) a BIR or an IR in the appropriate aircraft category, provided that FI meets the following conditions:
- (1) they have completed as student pilots the IRI training course and have passed an assessment of competence for the IRI certificate;
 - (2) they comply with points [FCL.915.CRI\(a\)](#), [FCL.930.CRI](#) and [FCL.935](#) in the case of multi-engine aeroplanes and with points [FCL.910.TRI\(c\)\(1\)](#) and [FCL.915.TRI\(d\)\(2\)](#) in the case of multi-engine helicopters;

In addition to conditions (1) and (2):

- (3) if during an approved training course at an ATO, the FI is providing training in FSTDs or supervising SPIC training flights that take place under IFR, the FI shall have completed at

- least 50 hours of flight time under IFR after the issuance of the BIR or the IR, of which a maximum of 10 hours may be instrument ground time in an FFS, an FTD 2/3 or an FNPT II,;
- (4) if the FI is providing training in an aircraft, the FI shall have completed at least 200 hours of flight time under IFR, of which up to 50 hours may be instrument ground time in an FFS, an FTD 2/3 or an FNPT II.'
- (i) single-pilot multi-engine class or type ratings, except for single-pilot high-performance complex aeroplanes, provided that they meet the following conditions:
- (1) in the case of aeroplanes, comply with points FCL.915.CRI(a), FCL.930.CRI and FCL.935;
- (2) in the case of helicopters, comply with points FCL.910.TRI(c)(1) and FCL.915.TRI(d)(2);
- (j) an FI, an IRI, a CRI, an STI or an MI certificate provided that they meet all of the following conditions:
- (1) they have completed at least 500 hours of flight instruction in the appropriate aircraft category;
- (2) they have passed an assessment of competence in accordance with point FCL.935 in the appropriate aircraft category to demonstrate to a flight instructor examiner (FIE) the ability to instruct for the relevant certificate;
- (k) an MPL, provided that the FIs meet all of the following conditions:
- (1) for the core flying phase of the training, have completed at least 500 hours of flight time as a pilot of aeroplanes, including at least 200 hours of flight instruction;
- (2) for the basic phase of the training:
- (i) hold a multi-engine aeroplane IR and the privilege to instruct for an IR;
- (ii) have completed at least 1 500 hours of flight time in multi-crew operations;
- (3) in the case of FIs already qualified to instruct on ATP(A) or CPL(A)/IR integrated courses, the requirement in point (2)(ii) may be replaced by the completion of a structured course of training consisting of:
- (i) MCC qualification;
- (ii) observation of five sessions of flight instruction in Phase 3 of an MPL course;
- (iii) observation of five sessions of flight instruction in Phase 4 of an MPL course;
- (iv) observation of five operator recurrent line-oriented flight training sessions;
- (v) the content of the MCCI course.

In this case, FIs shall conduct their first five instructor sessions under the supervision of a TRI(A), an MCCI(A) or an SFI(A) qualified for MPL flight instruction.

GM1 FCL.905.FI(h)(2) Privileges and conditions

ED Decision 2020/005/R

FSTDs should not be used to pass an assessment of competence on the class or type of aircraft.

GM1 FCL.905.FI(h)(3) FI — privileges and conditions

ED Decision 2025/002/R

CONSEQUENCES OF AN INTERVENTION BY AN INSTRUCTOR QUALIFIED IN ACCORDANCE WITH POINT [FCL.905.FI\(h\)\(3\)](#) DURING AN SPIC TRAINING FLIGHT

Point [FCL.905.FI\(h\)](#) sets out the conditions for FI certificates to include privileges for providing instruction towards an IR. Compared to the regular case (point (h)(4), requiring 200 hours of IFR experience), point (h)(3) allows holders of an FI certificate with reduced IFR experience (50 hours) to already provide elements of IR instruction at an ATO (FSTD instruction, supervision of SPIC training flights). If however, during such an SPIC training flight, an instructor qualified in accordance with point [FCL.905.FI\(h\)\(3\)](#), needs to intervene for safety reasons, the instructor needs to either take over controls and end that flight as PIC or, if possible, continue with VFR instruction, since such an instructor is not entitled to provide IFR instruction in an aeroplane in flight.

FCL.910.FI FI – Restricted privileges

Regulation (EU) 2025/134

- (a) An FI shall have his or her privileges limited to conducting flight instruction under the supervision of an FI for the same category of aircraft nominated by the DTO or the ATO for this purpose, in the following cases:
 - (1) for the issue of the PPL and LAPL;
 - (2) in all integrated courses at PPL level, in case of aeroplanes and helicopters;
 - (3) for class and type ratings for single-pilot, single-engine aircraft, except for single-pilot high-performance complex aeroplanes;
 - (4) for the night, towing or aerobatic ratings.
- (b) While conducting training under supervision, in accordance with (a), the FI shall not have the privilege to authorise student pilots to conduct first solo flights and first solo cross-country flights.
- (c) The limitations in (a) and (b) shall be removed from the FI certificate when the FI has completed at least:
 - (1) for the FI(A), 100 hours of flight instruction in aeroplanes or TMGs and, in addition has supervised at least 25 student solo flights;
 - (2) for the FI(H) 100 hours of flight instruction in helicopters and, in addition has supervised at least 25 student solo flight air exercises;
 - (3) for the FI(As), 15 hours or 50 take-offs of flight instruction covering the full training syllabus for the issue of a PPL(As).
 - 4) for the FI(G):
 - (i) 100 hours of flight instruction in gyroplanes and, in addition, has supervised at least 25 solo flights;
 - (ii) in the case of an FI(G) who also holds an FI(A) or FI(H) certificate and complies with points (1) or (2), as applicable, 25 hours of flight instruction in gyroplanes and, in addition, has supervised at least 10 student solo flights.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

FCL.915.FI FI – Prerequisites

Regulation (EU) 2025/134

An applicant for an FI certificate shall:

(a) in the case of the FI(A) and FI(H):

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

(a) in the case of the FI(A), FI(H) and FI(G):

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (1) have received at least 10 hours of instrument flight instruction on the appropriate aircraft category, of which not more than 5 hours may be instrument ground time in an FSTD;
 - (2) have completed 20 hours of VFR cross-country flight on the appropriate aircraft category as PIC; and
- (b) additionally, for the FI(A):
- (1) hold at least a CPL(A); or
 - (2) hold at least a PPL(A), or hold an MPL with additional PPL privileges in accordance with point [FCL.405.A\(b\)\(1\)](#), and have complied with the following conditions:
 - (i) except for an FI(A) providing training for the LAPL(A) only, passed the CPL theoretical knowledge examination, which may be taken without completing a CPL theoretical knowledge training course and which in that case shall not be valid for any of the following:
 - (A) the issue of a CPL;
 - (B) receiving credits for CPL theoretical knowledge in accordance with point [FCL.035](#) and [Appendix 1](#);
 - (ii) completed at least 200 hours of flight time on aeroplanes or TMGs, of which 150 hours as PIC;
 - (3) have completed at least 30 hours in the class or type of aeroplane used for the FI(A) training course, of which at least 5 hours shall have been completed during the 6 months preceding the pre-entry flight assessment set out in point [FCL.930.FI\(a\)](#);
 - (4) have completed a VFR cross-country flight as PIC, including a flight of at least 540 km (300 NM) in the course of which full stop landings at 2 different aerodromes shall be made;
- (c) additionally, for the FI(H), have completed 250 hours total flight time as pilot on helicopters of which:
- (1) at least 100 hours shall be as PIC, if the applicant holds at least a CPL(H); or
 - (2) at least 200 hours as PIC if the applicant holds at least a PPL(H) and has passed the CPL theoretical knowledge examination, which may be taken without completing a CPL theoretical knowledge training course and which shall not be valid for the issue of a CPL;
- (ca) additionally, for the FI(G), have completed at least 150 hours of flight time on gyroplanes, of which at least 100 hours as PIC.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (d) for an FI(As), have completed 500 hours of flight time on airships as PIC, of which 400 hours shall be as PIC holding a CPL(As).

FCL.930.FI FI – Training course

Regulation (EU) 2025/134

- (a) Applicants for the FI certificate shall, at an ATO, have passed a specific pre-entry flight assessment with an FI qualified in accordance with point [FCL.905.FI\(j\)](#) within the 6 months preceding the start of the course, to assess their ability to undertake the course. That pre-entry flight assessment shall be based on the proficiency check for the class and type ratings as set out in [Appendix 9](#).

- (b) The FI training course shall include:

- (1) 25 hours of teaching and learning;
- (2) at least 100 hours of theoretical knowledge instruction, including progress tests;
- (3) (i) in the case of an FI(A) and (H), at least 30 hours of flight instruction, of which 25 hours shall be dual flight instruction, of which 5 hours may be conducted in an FFS, an FNPT I or II or an FTD 2/3;

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (i) in the case of an FI(A), (H) and (G), at least 30 hours of flight instruction, of which 25 hours shall

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (ii) in the case of an FI(As), at least 20 hours of flight instruction, of which 15 hours shall be dual flight instruction.

- (4) Applicants for an FI certificate in another category of aircraft who are holding or have held an FI(A), (H) or (As) shall be credited with 55 hours towards the requirement in point (b)(2).

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (c) Applicants for the FI certificate who hold or have held any other instructor certificate issued in accordance with this Annex shall be deemed to meet the requirements in point (b)(1).

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (c) Applicants for an FI certificate in another category of aircraft who are holding or have held an FI(A), (H), (As) or (G) shall be credited with 55 hours towards the requirement in point (b)(2).';
- (d) Applicants for an FI(G) certificate that hold or have held an FI(A), (H) or (As) certificate shall be credited with 15 hours towards the requirement in point (b)(3)(i), of which a maximum of 10 hours shall be a credit towards the dual instruction time.
- (e) Applicants for the FI certificate who hold or have held any other instructor certificate issued in accordance with this Annex shall be deemed to meet the requirements in point (b)(1).

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

AMC1 FCL.930.FI FI Training course

ED Decision 2020/005/R

FI(A), FI(H) AND FI(AS) TRAINING COURSE

GENERAL

- (a) The aim of the FI training course is to train aircraft licence holders to the level of competence defined in [FCL.920](#).
- (b) The training course should develop safety awareness throughout by teaching the knowledge, skills and attitudes relevant to the FI task including at least the following:
 - (1) refresh the technical knowledge of the student instructor;
 - (2) train the student instructor to teach the ground subjects and air exercises;
 - (3) ensure that the student instructor's flying is of a sufficiently high standard;
 - (4) teach the student instructor the principles of basic instruction and to apply them at the PPL level.

FLIGHT INSTRUCTION

- (c) The remaining 5 hours in [FCL.930.FI\(b\)\(3\)](#) may be mutual flying (that is, two applicants flying together to practice flight demonstrations).
- (d) The tassessment of competence is additional to the course training time.

CONTENT

- (e) The training course consists of two parts:
 - (1) Part 1, theoretical knowledge, including the teaching and learning instruction that should comply with [AMC1 FCL.920](#);
 - (2) Part 2, flight instruction.

Part 1**TEACHING AND LEARNING**

- (a) The course should include at least 125 hours of theoretical knowledge instruction, including at least 25 hours teaching and learning instruction.

CONTENT OF THE TEACHING AND LEARNING INSTRUCTIONS (INSTRUCTIONAL TECHNIQUES):

- (b) The learning process:
 - (1) motivation;
 - (2) perception and understanding;
 - (3) memory and its application;
 - (4) habits and transfer;
 - (5) obstacles to learning;
 - (6) incentives to learning;
 - (7) learning methods;
 - (8) rates of learning.
- (c) The teaching process:
 - (1) elements of effective teaching;
 - (2) planning of instructional activity;

- (3) teaching methods;
- (4) teaching from the 'known' to the 'unknown';
- (5) use of 'lesson plans'.
- (d) Training philosophies:
 - (1) value of a structured (approved) course of training;
 - (2) importance of a planned syllabus;
 - (3) integration of theoretical knowledge and flight instruction.
- (e) Techniques of applied instruction:
 - (1) theoretical knowledge: classroom instruction techniques:
 - (i) use of training aids;
 - (ii) group lectures;
 - (iii) individual briefings;
 - (iv) student participation or discussion.
 - (2) flight: airborne instruction techniques:
 - (i) the flight or cockpit environment;
 - (ii) techniques of applied instruction;
 - (iii) post-flight and in-flight judgement and decision making.
- (f) Student evaluation and testing:
 - (1) assessment of student performance:
 - (i) the function of progress tests;
 - (ii) recall of knowledge;
 - (iii) translation of knowledge into understanding;
 - (iv) development of understanding into actions;
 - (v) the need to evaluate rate of progress.
 - (2) analysis of student errors:
 - (i) establish the reason for errors;
 - (ii) tackle major faults first, minor faults second;
 - (iii) avoidance of over criticism;
 - (iv) the need for clear concise communication.
- (g) Training programme development:
 - (1) lesson planning;
 - (2) preparation;
 - (3) explanation and demonstration;
 - (4) student participation and practice;
 - (5) evaluation.

- (h) Human performance and limitations relevant to flight instruction:
 - (1) physiological factors:
 - (i) psychological factors;
 - (ii) human information processing;
 - (iii) behavioural attitudes;
 - (iv) development of judgement and decision making.
 - (2) threat and error management.
- (i) Specific hazards involved in simulating systems failures and malfunctions in the aircraft during flight:
 - (i) importance of ‘touch drills’;
 - (ii) situational awareness;
 - (iii) adherence to correct procedures.
- (j) Training administration:
 - (1) flight or theoretical knowledge instruction records;
 - (2) pilot’s personal flying logbook;
 - (3) the flight or ground curriculum;
 - (4) study material;
 - (5) official forms;
 - (6) flight manual or equivalent document (for example owner’s manual or pilot’s operating handbook);
 - (7) flight authorisation papers;
 - (8) aircraft documents;
 - (9) the private pilot’s licence regulations.

A. Aeroplanes

Part 2

AIR EXERCISES

- (a) The air exercises are similar to those used for the training of PPL(A) but with additional items designed to cover the needs of an FI.
- (b) The numbering of exercises should be used primarily as an exercise reference list and as a broad instructional sequencing guide: therefore the demonstrations and practices need not necessarily be given in the order listed. The actual order and content will depend upon the following interrelated factors:
 - (1) the applicant’s progress and ability;
 - (2) the weather conditions affecting the flight;
 - (3) the flight time available;

- (4) instructional technique considerations;
- (5) the local operating environment.
- (c) It follows that student instructors will eventually be faced with similar interrelated factors. They should be shown and taught how to construct flight lesson plans, taking these factors into account, so as to make the best use of each flight lesson, combining parts of the set exercises as necessary.

GENERAL

- (d) The briefing normally includes a statement of the aim and a brief allusion to principles of flight only if relevant. An explanation is to be given of exactly what air exercises are to be taught by the instructor and practised by the student during the flight. It should include information on how the flight will be conducted, who is to fly the aeroplane and what airmanship, weather and flight safety aspects currently apply. The nature of the lesson will govern the order in which the constituent parts are to be taught.
- (e) The four basic components of the briefing will be:
 - (1) the aim;
 - (2) principles of flight (briefest reference only);
 - (3) the air exercise(s) (what, and how and by whom);
 - (4) airmanship (weather, flight safety etc.).

PLANNING OF FLIGHT LESSONS

- (f) The preparation of lesson plans is an essential prerequisite of good instruction and the student instructor is to be given supervised practice in the planning and practical application of flight lesson plans.

GENERAL CONSIDERATIONS

- (g) The student instructor should complete flight training to practise the principles of basic instruction at the PPL(A) level.
- (h) During this training, except when acting as a student pilot for mutual flights, the student instructor occupies the seat normally occupied by the FI(A).
- (i) It is to be noted that airmanship and look-out is a vital ingredient of all flight operations. Therefore, in the following air exercises the relevant aspects of airmanship are to be stressed at all times.
- (j) If the privileges of the FI(A) certificate are to include instruction for night flying, exercises 19 and 20 of the flight instruction syllabus should be undertaken at night in addition to by day either as part of the course or subsequent to certification issue.
- (k) The student instructor should learn how to identify common errors and how to correct them properly, which should be emphasised at all times.

SYLLABUS OF FLIGHT INSTRUCTION CONTENTS**LONG BRIEFINGS AND AIR EXERCISES**

Note: though exercise 11b is not required for the PPL(A) course, it is a requirement for the FI course.

EXERCISE 1: FAMILIARISATION WITH THE AEROPLANE

- (a) Long briefing objectives:
 - (1) introduction to the aeroplane;
 - (2) explanation of the cockpit layout;
 - (3) aeroplane and engine systems;
 - (4) checklists, drills and controls;
 - (5) propeller safety;
 - (i) precautions general;
 - (ii) precautions before and during hand turning;
 - (iii) hand swinging technique for starting (if applicable to type).
 - (6) differences when occupying the instructor's seat;
 - (7) emergency drills:
 - (i) action if fire in the air and on the ground: engine, cock or cabin and electrical fire;
 - (ii) system failure as applicable to type;
 - (iii) escape drills: location and use of emergency equipment and exits.
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 2: PREPARATION FOR AND ACTION AFTER FLIGHT

- (a) Long briefing objectives:
 - (1) flight authorisation and aeroplane acceptance, including technical log (if applicable) and certificate of maintenance;
 - (2) equipment required for flight (maps, etc.);
 - (3) external checks;
 - (4) internal checks;
 - (5) student comfort, harness, seat or rudder pedal adjustment;
 - (6) starting and warming up checks;
 - (7) power checks;
 - (8) running down, system checks and switching off the engine;
 - (9) leaving the aeroplane, parking, security and picketing;
 - (10) completion of authorisation sheet and aeroplane serviceability documents.
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 3: AIR EXPERIENCE

- (a) Long briefing objectives:

Note: there is no requirement for a long briefing for this exercise.

- (b) Air exercise:
 - (1) air experience;
 - (2) cockpit layout, ergonomics and controls;
 - (3) cockpit procedures: stability and control.

EXERCISE 4: EFFECTS OF CONTROLS

- (a) Long briefing objectives:
 - (1) function of primary flying controls: when laterally level and banked;
 - (2) further effect of ailerons and rudder;
 - (3) effect of inertia;
 - (4) effect of air speed;
 - (5) effect of slipstream;
 - (6) effect of power;
 - (7) effect of trimming controls;
 - (8) effect of flaps;
 - (9) operation of mixture control;
 - (10) operation of carburettor heat control;
 - (11) operation of cabin heat or ventilation systems;
- (b) Air exercise:
 - (1) primary effects of flying controls: when laterally level and banked;
 - (2) further effects of ailerons and rudder;
 - (3) effect of air speed;
 - (4) effect of slipstream;
 - (5) effect of power;
 - (6) effect of trimming controls;
 - (7) effect of flaps;
 - (8) operation of mixture control;
 - (9) operation of carburettor heat control;
 - (10) operation of cabin heat or ventilation systems;
 - (11) effect of other controls as applicable.

EXERCISE 5: TAXIING

- (a) Long briefing objectives:
 - (1) pre-taxiing checks;
 - (2) starting, control of speed and stopping;

- (3) engine handling;
 - (4) control of direction and turning (including manoeuvring in confined spaces);
 - (5) parking area procedures and precautions;
 - (6) effect of wind and use of flying controls;
 - (7) effect of ground surface;
 - (8) freedom of Rudder movement;
 - (9) marshalling signals;
 - (10) instrument checks;
 - (11) ATC procedures;
 - (12) emergencies: steering failure and brake failure.
- (b) Air exercise:
- (1) pre-taxiing checks;
 - (2) starting, control of speed and stopping;
 - (3) engine handling;
 - (4) control of direction and turning;
 - (5) turning in confined spaces;
 - (6) parking area procedures and precautions;
 - (7) effect of wind and use of flying control;
 - (8) effect of ground surface;
 - (9) freedom of Rudder movement;
 - (10) marshalling signals;
 - (11) instrument checks;
 - (12) ATC procedures;
 - (13) emergencies: steering failure and brake failure.

EXERCISE 6: STRAIGHT AND LEVEL FLIGHT

- (a) Long briefing objectives:
- (1) the forces;
 - (2) longitudinal stability and control in pitch;
 - (3) relationship of CG to control in pitch;
 - (4) lateral and directional stability (control of lateral level and balance);
 - (5) attitude and balance control;
 - (6) trimming;
 - (7) power settings and air speeds;
 - (8) drag and power curves;

- (9) range and endurance.
- (b) Air exercise:
 - (1) at normal cruising power;
 - (2) attaining and maintaining straight and level flight;
 - (3) demonstration of inherent stability;
 - (4) control in pitch, including use of elevator trim control;
 - (5) lateral level, direction and balance, use of rudder trim controls as applicable at selected air speeds (use of power):
 - (i) effect of drag and use of power (two air speeds for one power setting);
 - (ii) straight and level in different aeroplane configurations (flaps and landing gear);
 - (iii) use of instruments to achieve precision flight.

EXERCISE 7: CLIMBING

- (a) Long briefing objectives:
 - (1) the forces;
 - (2) relationship between power or air speed and rate of climb (power curves maximum rate of climb (v_y));
 - (3) effect of mass;
 - (4) effect of flaps;
 - (5) engine considerations;
 - (6) effect of density altitude;
 - (7) the cruise climb;
 - (8) maximum angle of climb (v_x).
- (b) Air exercise:
 - (1) entry and maintaining the normal maximum rate climb;
 - (2) levelling off;
 - (3) levelling off at selected altitudes;
 - (4) climbing with flaps down;
 - (5) recovery to normal climb;
 - (6) en-route climb (cruise climb);
 - (7) maximum angle of climb;
 - (8) use of instruments to achieve precision flight.

EXERCISE 8: DESCENDING

- (a) Long briefing objectives:
 - (1) the forces;

- (2) glide descent: angle, air speed and rate of descent;
 - (3) effect of flaps;
 - (4) effect of wind;
 - (5) effect of mass;
 - (6) engine considerations;
 - (7) power assisted descent: power or air speed and rate of descent;
 - (8) cruise descent;
 - (9) sideslip.
- (b) Air exercise:
- (1) entry and maintaining the glide;
 - (2) levelling off;
 - (3) levelling off at selected altitudes;
 - (4) descending with flaps down;
 - (5) powered descent: cruise descent (including effect of power and air speed);
 - (6) side-slipping (on suitable types);
 - (7) use of instrument to achieve precision flight.

EXERCISE 9: TURNING

- (a) Long briefing objectives:
- (1) the forces;
 - (2) use of controls;
 - (3) use of power;
 - (4) maintenance of attitude and balance;
 - (5) medium level turns;
 - (6) climbing and descending turns;
 - (7) slipping turns;
 - (8) turning onto selected headings: use of gyro heading indicator and magnetic compass.
- (b) Air exercise:
- (1) entry and maintaining medium level turns;
 - (2) resuming straight flight;
 - (3) faults in the turn (incorrect pitch, bank and balance);
 - (4) climbing turns;
 - (5) descending turns;
 - (6) slipping turns (on suitable types);
 - (7) turns to selected headings: use of gyro heading indicator and magnetic compass

- (8) use of instruments to achieve precision flight;

Note: stall or spin awareness and avoidance training consists of exercises 10a, 10b and 11a.

EXERCISE 10a: SLOW FLIGHT

- (a) Long briefing objectives:

- (1) aeroplane handling characteristics during slow flight at:
 - (i) v_{s1} & $v_{so} + 10$ knots;
 - (ii) v_{s1} & $v_{so} + 5$ knots.
- (2) slow flight during instructor induced distractions;
- (3) effect of overshooting in configurations where application of engine power causes a strong 'nose-up' trim change.

- (b) Air exercise:

- (1) safety checks;
- (2) introduction to slow flight;
- (3) controlled slow flight in the clean configuration at:
 - (i) $v_{s1} + 10$ knots and with flaps down;
 - (ii) $v_{so} + 10$ knots;
 - (iii) straight and level flight;
 - (iv) level turns;
 - (v) climbing and descending;
 - (vi) climbing and descending turns.
- (4) controlled slow flight in the clean configuration at:
 - (i) $v_{s1} + 5$ knots and with flaps down;
 - (ii) $v_{so} + 5$ knots;
 - (iii) straight and level flight;
 - (iv) level turns;
 - (v) climbing and descending;
 - (vi) climbing and descending turns;
 - (vii) descending 'unbalanced' turns at low air speed: the need to maintain balanced flight.
- (5) 'instructor induced distractions' during flight at low air speed: the need to maintain balanced flight and a safe air speed;
- (6) effect of going around in configurations where application of engine power causes a strong 'nose up' trim change.

EXERCISE 10b: STALLING

- (a) Long briefing objectives:

- (1) characteristics of the stall;
- (2) angle of attack;
- (3) effectiveness of the controls at the stall;
- (4) factors affecting the stalling speed:
 - (i) effect of flaps, slats and slots;
 - (ii) effect of power, mass, CG and load factor.
- (5) effects of unbalance at the stall;
- (6) symptoms of the stall;
- (7) stall recognition and recovery;
- (8) stalling and recovery:
 - (i) without power;
 - (ii) with power on;
 - (iii) with flaps down;
 - (iv) maximum power climb (straight and turning flight to the point of stall with uncompensated yaw);
 - (v) stalling and recovery during manoeuvres involving more than 1 G (accelerated stalls, including secondary stalls and recoveries);
 - (vi) recovering from incipient stalls in the landing and other configurations and conditions;
 - (vii) recovering at the incipient stage during change of configuration;
 - (viii) stalling and recovery at the incipient stage with 'instructor induced' distractions.

Note: consideration is to be given to manoeuvre limitations and references to the flight manual or equivalent document (for example owner's manual or pilot's operating handbook) in relation to mass and balance limitations. The safety checks should take into account the minimum safe altitude for initiating such exercises in order to ensure an adequate margin of safety for the recovery. If specific procedures for stalling or spinning exercises and for the recovery techniques are provided by the flight manual or equivalent document (for example owner's manual or pilot's operating handbook), they have to be taken into consideration. These factors are also covered in the next exercise spinning.

- (b) Air exercise:
- (1) safety checks;
 - (2) symptoms of the stall;
 - (3) stall recognition and recovery:
 - (i) without power;
 - (ii) with power on;
 - (iii) recovery when a wing drops at the stall;
 - (iv) stalling with power 'on' and recovery;
 - (v) stalling with flap 'down' and recovery;

- (vi) maximum power climb (straight and turning flight) to the point of stall with uncompensated yaw: effect of unbalance at the stall when climbing power is being used;
- (vii) stalling and recovery during manoeuvres involving more than 1 G (accelerated stalls, including secondary stalls and recoveries);
- (viii) recoveries from incipient stalls in the landing and other configurations and conditions;
- (ix) recoveries at the incipient stage during change of configuration;
- (x) instructor induced distractions during stalling.

Note: consideration of manoeuvre limitations and the need to refer to the aeroplane manual and weight (mass) and balance calculations. The safety checks should take into account the minimum safe altitude for initiating such exercises in order to ensure an adequate margin of safety for the recovery. If specific procedures for stalling or spinning exercises and for the recovery techniques are provided by the flight manual or equivalent document (for example owner's manual or pilot's operating handbook), they have to be taken into consideration. These factors are to be covered in the next exercise: spinning.

EXERCISE 11a: SPIN RECOVERY AT THE INCIPIENT STAGE

- (a) Long briefing objectives:
 - (1) causes, stages, autorotation and characteristics of the spin;
 - (2) recognition and recovery at the incipient stage: entered from various flight attitudes;
 - (3) aeroplane limitations.
- (b) Air exercise:
 - (1) aeroplane limitations;
 - (2) safety checks;
 - (3) recognition at the incipient stage of a spin;
 - (4) recoveries from incipient spins entered from various attitudes with the aeroplane in the clean configuration, including instructor induced distractions.

EXERCISE 11b: SPIN RECOVERY AT THE DEVELOPED STAGE

- (a) Long briefing objectives:
 - (1) spin entry;
 - (2) recognition and identification of spin direction;
 - (3) spin recovery;
 - (4) use of controls;
 - (5) effects of power or flaps (flap restriction applicable to type);
 - (6) effect of the CG upon spinning characteristics;
 - (7) spinning from various flight attitudes;
 - (8) aeroplane limitation;

- (9) safety checks.
- (b) Air exercise:
 - (1) aeroplane limitations;
 - (2) safety checks;
 - (3) spin entry;
 - (4) recognition and identification of the spin direction;
 - (5) spin recovery (reference to flight manual);
 - (6) use of controls;
 - (7) effects of power or flaps (restrictions applicable to aeroplane type);
 - (8) spinning and recovery from various flight attitudes.

EXERCISE 12: TAKE-OFF AND CLIMB TO DOWNWIND POSITION

- (a) Long briefing objectives:
 - (1) handling: factors affecting the length of take-off run and initial climb;
 - (2) correct lift off speed, use of elevators (safeguarding the nose wheel), rudder and power;
 - (3) effect of wind (including crosswind component);
 - (4) effect of flaps (including the decision to use and the amount permitted);
 - (5) effect of ground surface and gradient upon the take-off run;
 - (6) effect of mass, altitude and temperature on take-off and climb performance;
 - (7) pre take-off checks;
 - (8) ATC procedure before take-off;
 - (9) drills, during and after take-off;
 - (10) noise abatement procedures;
 - (11) tail wheel considerations (as applicable);
 - (12) short or soft field take-off considerations or procedures;
 - (13) emergencies:
 - (i) aborted take-off;
 - (ii) engine failure after take-off.
 - (14) ATC procedures.
- (b) Air exercise:
 - (1) take-off and climb to downwind position;
 - (2) pre take-off checks;
 - (3) into wind take-off;
 - (4) safeguarding the nose wheel;
 - (5) crosswind take-off;

- (6) drills during and after take-off;
- (7) short take-off and soft field procedure or techniques (including performance calculations);
- (8) noise abatement procedures.

EXERCISE 13: CIRCUIT, APPROACH AND LANDING

- (a) Long briefing objectives:
 - (1) downwind leg, base leg and approach: position and drills;
 - (2) factors affecting the final approach and the landing run;
 - (3) effect of mass;
 - (4) effects of altitude and temperature;
 - (5) effect of wind;
 - (6) effect of flap;
 - (7) landing;
 - (8) effect of ground surface and gradient upon the landing run;
 - (9) types of approach and landing:
 - (i) powered;
 - (ii) crosswind;
 - (iii) flapless (at an appropriate stage of the course);
 - (iv) glide;
 - (v) short field;
 - (vi) soft field.
 - (10) tail wheel aeroplane considerations (as applicable);
 - (11) missed approach;
 - (12) engine handling;
 - (13) wake turbulence awareness;
 - (14) windshear awareness;
 - (15) ATC procedures;
 - (16) mislanding and go-around;
 - (17) special emphasis on look-out.
- (b) Air exercise:
 - (1) circuit approach and landing;
 - (2) circuit procedures: downwind and base leg;
 - (3) powered approach and landing;
 - (4) safeguarding the nose wheel;

- (5) effect of wind on approach and touchdown speeds and use of flaps;
- (6) crosswind approach and landing;
- (7) glide approach and landing;
- (8) flapless approach and landing (short and soft field);
- (9) short field and soft field procedures;
- (10) wheel landing (tail wheel aircraft);
- (11) missed approach and go-around;
- (12) mislanding and go-around;
- (13) noise abatement procedures.

EXERCISE 14: FIRST SOLO AND CONSOLIDATION

Note: a summary of points to be covered before sending the student on first solo.

(a) Long briefing objectives:

During the flights immediately following the solo circuit consolidation period the following should be covered:

- (1) procedures for leaving and rejoining the circuit;
 - (2) local area (restrictions, controlled airspace, etc.);
 - (3) compass turns;
 - (4) QDM meaning and use.
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 15: ADVANCED TURNING

(a) Long briefing objectives:

- (1) the forces;
- (2) use of power;
- (3) effect of load factor:
 - (i) structural considerations
 - (ii) increased stalling speed.
- (4) physiological effects;
- (5) rate and radius of turn;
- (6) steep, level, descending and climbing turns;
- (7) stalling in the turn and how to avoid it;
- (8) spinning from the turn: recovery at the incipient stage;
- (9) spiral dive;
- (10) unusual attitudes and recoveries.

Note: considerations are to be given to manoeuvre limitations and reference to the flight manual or equivalent document (for example owner's manual or pilot's operating handbook) in relation to mass and balance, and any other restrictions for practice entries to the spin.

- (b) Air exercise:
 - (1) level, descending and climbing steep turns;
 - (2) stalling in the turn;
 - (3) spiral dive;
 - (4) spinning from the turn;
 - (5) recovery from unusual attitudes;
 - (6) maximum rate turns.

EXERCISE 16: FORCED LANDING WITHOUT POWER

- (a) Long briefing objectives:
 - (1) selection of forced landing areas;
 - (2) provision for change of plan;
 - (3) gliding distance: consideration;
 - (4) planning the descent;
 - (5) key positions;
 - (6) engine failure checks;
 - (7) use of radio: R/T 'distress' procedure;
 - (8) base leg;
 - (9) final approach;
 - (10) go-around;
 - (11) landing considerations;
 - (12) actions after landing: aeroplane security;
 - (13) causes of engine failure.
- (b) Air exercise:
 - (1) forced landing procedures;
 - (2) selection of landing area:
 - (i) provision for change of plan;
 - (ii) gliding distance considerations.
 - (3) planning the descent;
 - (4) key positions;
 - (5) engine failure checks;
 - (6) engine cooling precautions;
 - (7) use of radio;

- (8) base leg;
- (9) final approach;
- (10) landing;
- (11) actions after landing: when the exercise is conducted at an aerodrome;
- (12) aeroplane security.

EXERCISE 17: PRECAUTIONARY LANDING

- (a) Long briefing objectives:
 - (1) occasions when necessary (in-flight conditions);
 - (2) landing area selection and communication (R/T procedure);
 - (3) overhead inspection;
 - (4) simulated approach;
 - (5) climb away;
 - (6) landing area selection:
 - (i) normal aerodrome;
 - (ii) disused aerodrome;
 - (iii) ordinary field;
 - (7) circuit and approach;
 - (8) actions after landing; aeroplane security.
- (b) Air exercise:
 - (1) occasions when necessary (in-flight conditions);
 - (2) landing area selection
 - (3) overhead inspection
 - (4) simulated approach
 - (5) climb away
 - (6) landing area selection:
 - (i) normal aerodrome;
 - (ii) disused aerodrome;
 - (iii) ordinary field;
 - (7) circuit and approach;
 - (8) actions after landing; aeroplane security;

EXERCISE 18a: NAVIGATION

- (a) Long briefing objectives:
 - (1) flight planning;
 - (i) weather forecast and actual(s);

- (ii) map selection, orientation, preparation and use:
 - (A) choice of route;
 - (B) regulated or controlled airspace;
 - (C) danger, prohibited and restricted areas;
 - (D) safety altitude.
- (iii) calculations:
 - (A) magnetic heading(s) and time(s) en-route;
 - (B) fuel consumption;
 - (C) mass and balance;
 - (D) mass and performance.
- (iv) flight information:
 - (A) NOTAMs etc.;
 - (B) noting of required radio frequencies;
 - (C) selection of alternate aerodrome(s).
 - (v) aeroplane documentation.
- (vi) notification of the flight:
 - (A) pre-flight administration procedures;
 - (B) flight plan form (where appropriate).
- (2) departure;
 - (i) organisation of cockpit workload;
 - (ii) departure procedures:
 - (A) altimeter settings;
 - (B) setting heading procedures;
 - (C) noting of ETA(s).
 - (iii) en-route map reading: identification of ground features;
 - (iv) maintenance of altitudes and headings;
 - (v) revisions to ETA and heading, wind effect, drift angle and groundspeed checks;
 - (vi) log keeping;
 - (vii) use of radio (including VDF if applicable);
 - (viii) minimum weather conditions for continuance of flight;
 - (ix) 'in-flight' decisions;
 - (x) diversion procedures;
 - (xi) operations in regulated or controlled airspace;
 - (xii) procedures for entry, transit and departure;
 - (xiii) navigation at minimum level;

- (xiv) uncertainty of position procedure, including R/T procedure;
- (xv) lost procedure;
- (xvi) use of radio navaids.
- (3) arrival procedures and aerodrome circuit joining procedures:
 - (i) ATC liaison, R/T procedure, etc.;
 - (ii) altimeter setting,
 - (iii) entering the traffic pattern (controlled or uncontrolled aerodromes);
 - (iv) circuit procedures;
 - (v) parking procedures;
 - (vi) security of aircraft;
 - (vii) refuelling;
 - (viii) booking in.
- (b) Air exercise:
 - (1) flight planning:
 - (i) weather forecast and actual(s);
 - (ii) map selection and preparation:
 - (A) choice of route;
 - (B) regulated or controlled airspace;
 - (C) danger, prohibited and restricted areas;
 - (D) safety altitude.
 - (iii) calculations:
 - (A) magnetic heading(s) and time(s) en-route;
 - (B) fuel consumption;
 - (C) mass and balance;
 - (D) mass and performance.
 - (iv) flight information:
 - (A) NOTAMs etc.;
 - (B) noting of required radio frequencies;
 - (C) selection of alternate aerodromes.
 - (v) aircraft documentation;
 - (vi) notification of the flight:
 - (A) flight clearance procedures (as applicable);
 - (B) flight plans.
 - (2) aerodrome departure;
 - (i) organisation of cockpit workload;

- (ii) departure procedures:
 - (A) altimeter settings;
 - (B) en-route;
 - (C) noting of ETA(s).
 - (iii) wind effect, drift angle and ground speed checks;
 - (iv) maintenance of altitudes and headings;
 - (v) revisions to ETA and heading;
 - (vi) log keeping;
 - (vii) use of radio (including VDF if applicable);
 - (viii) minimum weather conditions for continuance of flight;
 - (ix) 'in-flight' decisions;
 - (x) diversion procedure;
 - (xi) operations in regulated or controlled airspace;
 - (xii) procedures for entry, transit and departure;
 - (xiii) uncertainty of position procedure;
 - (xiv) lost procedure;
 - (xv) use of radio nav aids.
- (3) arrival procedures and aerodrome joining procedures:
- (i) ATC liaison, R/T procedure etc.;
 - (ii) altimeter setting,
 - (iii) entering the traffic pattern;
 - (iv) circuit procedures;
 - (v) parking procedures
 - (vi) security of aircraft;
 - (vii) refuelling;
 - (viii) booking in.

EXERCISE 18b: NAVIGATION AT LOWER LEVELS AND IN REDUCED VISIBILITY

- (a) Long briefing objectives:
- (1) general considerations:
 - (i) planning requirements before flight in entry or exit lanes;
 - (ii) ATC rules, pilot qualifications and aircraft equipment;
 - (iii) entry or exit lanes and areas where specific local rules apply.
 - (2) low level familiarisation:
 - (i) actions before descending;

- (ii) visual impressions and height keeping at low altitude;
 - (iii) effects of speed and inertia during turns;
 - (iv) effects of wind and turbulence;
- (3) low level operation:
 - (i) weather considerations;
 - (ii) low cloud and good visibility;
 - (iii) low cloud and poor visibility;
 - (iv) avoidance of moderate to heavy rain showers;
 - (v) effects of precipitation;
 - (vi) joining a circuit;
 - (vii) bad weather circuit, approach and landing.
- (b) Air exercise:
 - (1) general considerations: entry or exit lanes and areas where specific local rules apply;
 - (2) low level familiarisation:
 - (i) actions before descending;
 - (ii) visual impressions and height keeping at low altitude;
 - (iii) effects of speed and inertia during turns;
 - (iv) effects of wind and turbulence;
 - (v) hazards of operating at low levels;
 - (3) low level operation:
 - (i) weather considerations;
 - (ii) low cloud and good visibility;
 - (iii) low cloud and poor visibility;
 - (iv) avoidance of moderate to heavy rain showers;
 - (v) effects of precipitation (forward visibility);
 - (vi) joining a circuit;
 - (vii) bad weather circuit, approach and landing.

EXERCISE 18c: USE OF RADIO NAVIGATION AIDS UNDER VFR

- (a) Long briefing objectives:
 - (1) use of VOR:
 - (i) availability, AIP and frequencies;
 - (ii) signal reception range;
 - (iii) selection and identification;
 - (iv) radials and method of numbering;

- (v) use of OBS;
 - (vi) to or from indication and station passage;
 - (vii) selection, interception and maintaining a radial;
 - (viii) use of two stations to determine position.
- (2) use of ADF equipment:
 - (i) availability of NDB stations, AIP and frequencies;
 - (ii) signal reception range;
 - (iii) selection and identification;
 - (iv) orientation in relation to NDP;
 - (v) homing to an NDP.
- (3) use of VHF/DF:
 - (i) availability. AIP and frequencies;
 - (ii) R/T procedures;
 - (iii) obtaining QDMs and QTEs.
- (4) use of radar facilities:
 - (i) availability and provision of service and AIS;
 - (ii) types of service;
 - (iii) R/T procedures and use of transponder:
 - (A) mode selection;
 - (B) emergency codes.
- (5) use of distance DME:
 - (i) availability and AIP;
 - (ii) operating modes;
 - (iii) slant range.
- (6) use of GNSS (RNAV – SATNAV):
 - (i) availability;
 - (ii) operating modes;
 - (iii) limitations.
- (b) Air exercise:
 - (1) use of VOR:
 - (i) availability, AIP and frequencies;
 - (ii) selection and identification;
 - (iii) use of OBS;
 - (iv) to or from indications: orientation;
 - (v) use of CDI;

- (vi) determination of radial;
 - (vii) intercepting and maintaining a radial;
 - (viii) VOR passage;
 - (ix) obtaining a fix from two VORs.
- (2) use of ADF equipment;
 - (i) availability of NDB stations, AIP and frequencies;
 - (ii) selection and identification;
 - (iii) orientation relative to the beacon;
 - (iv) homing.
- (3) use of VHF/DF:
 - (i) availability, AIP and frequencies;
 - (ii) R/T procedures and ATC liaison;
 - (iii) obtaining a QDM and homing.
- (4) use of en-route or terminal radar:
 - (i) availability and AIP;
 - (ii) procedures and ATC liaison;
 - (iii) pilot's responsibilities;
 - (iv) secondary surveillance radar;
 - (v) transponders;
 - (vi) code selection;
 - (vii) interrogation and reply.
- (5) use of DME:
 - (i) station selection and identification;
 - (ii) modes of operation.
- (6) use of GNSS (RNAV – SATNAV):
 - (i) setting up;
 - (ii) operation;
 - (iii) interpretation.

EXERCISE 19: BASIC INSTRUMENT FLIGHT

- (a) Long briefing objectives:
 - (1) flight instruments;
 - (i) physiological sensations;
 - (ii) instrument appreciation;
 - (iii) attitude instrument flight;

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- (iv) pitch indications;
 - (v) bank indications;
 - (vi) different dial presentations;
 - (vii) introduction to the use of the attitude indicator;
 - (viii) pitch attitude;
 - (ix) bank attitude;
 - (x) maintenance of heading and balanced flight;
 - (xi) instrument limitations (inclusive system failures).
 - (2) attitude, power and performance;
 - (i) attitude instrument flight:
 - (ii) control instruments;
 - (iii) performance instruments;
 - (iv) effect of changing power and configuration;
 - (v) cross-checking the instrument indications;
 - (vi) instrument interpretation;
 - (vii) direct and indirect indications (performance instruments);
 - (viii) instrument lag;
 - (ix) selective radial scan;
 - (3) basic flight manoeuvres (full panel);
 - (i) straight and level flight at various air speeds and aeroplane configurations;
 - (ii) climbing;
 - (iii) descending;
 - (iv) standard rate turns onto pre-selected headings:
 - (A) level;
 - (B) climbing;
 - (C) Descending.
- (b) Air exercise:
- (1) Introduction to instrument flying
 - (i) flight instruments;
 - (ii) physiological sensations;
 - (iii) instrument appreciation;
 - (iv) attitude instrument flight;
 - (v) pitch attitude;
 - (vi) bank attitude;
 - (vii) maintenance of heading and balanced flight;

- (2) attitude, power and performance;
 - (i) attitude instrument flight;
 - (ii) effect of changing power and configuration;
 - (iii) cross-checking the instruments;
 - (iv) selective radial scan;
- (3) basic flight manoeuvres (full panel);
 - (i) straight and level flight at various air speeds and aeroplane configurations;
 - (ii) climbing;
 - (iii) descending;
 - (iv) standard rate turns onto pre-selected headings:
 - (A) level;
 - (B) climbing;
 - (C) Descending

EXERCISE 20: NIGHT FLYING (if night instructional qualification required)

- (a) Long briefing objectives:
 - (1) start up procedures;
 - (2) local procedures: including ATC liaison;
 - (3) taxiing:
 - (i) parking area and taxiway lighting;
 - (ii) judgement of speed and distances;
 - (iii) use of taxiway lights;
 - (iv) avoidance of hazards: obstruction lighting;
 - (v) instrument checks;
 - (vi) holding point: lighting procedure;
 - (vii) initial familiarisation at night;
 - (viii) local area orientation;
 - (ix) significance of lights on other aircraft;
 - (x) ground obstruction lights;
 - (xi) division of piloting effort: external or instrument reference;
 - (xii) rejoining procedure;
 - (xiii) aerodrome lighting: approach and runway lighting (including VASI and PAPI):
 - (A) threshold lights;
 - (B) approach lighting;
 - (C) visual approach slope indicator systems.

- (4) night circuits;
 - (i) take-off and climb:
 - (A) line up;
 - (B) visual references during the take-off run;
 - (C) transfer to instruments;
 - (D) establishing the initial climb;
 - (E) use of flight instruments;
 - (F) instrument climb and initial turn.
 - (ii) circuit:
 - (A) aeroplane positioning: reference to runway lighting;
 - (B) the traffic pattern and look-out;
 - (C) initial approach and runway lighting demonstration;
 - (D) aeroplane positioning;
 - (E) changing aspect of runway lights and VASI (or PAPI);
 - (F) intercepting the correct approach path;
 - (G) the climb away.
 - (iii) approach and landing:
 - (A) positioning, base leg and final approach;
 - (B) diurnal wind effect;
 - (C) use of landing lights;
 - (D) the flare and touchdown;
 - (E) the roll out;
 - (F) turning off the runway: control of speed.
 - (iv) missed approach:
 - (A) use of instruments;
 - (B) re-positioning in the circuit pattern;
- (5) night navigation:
 - (i) particular emphasis on flight planning;
 - (ii) selection of ground features visible at night:
 - (A) air light beacons;
 - (B) effect of cockpit lighting on map colours;
 - (C) use of radio aids;
 - (D) effect of moonlight upon visibility at night;
 - (iii) emphasis on maintaining a 'minimum safe altitude';
 - (iv) alternate aerodromes: restricted availability;

- (v) restricted recognition of weather deterioration;
 - (vi) lost procedures;
- (6) night emergencies;
 - (i) radio failure;
 - (ii) failure of runway lighting;
 - (iii) failure of aeroplane landing lights;
 - (iv) failure of aeroplane internal lighting;
 - (v) failure of aeroplane navigation lights;
 - (vi) total electrical failure;
 - (vii) abandoned take-off;
 - (viii) engine failure;
 - (ix) obstructed runway procedure.
- (b) Air exercise: during the air exercise all long briefing objectives mentioned above should also be trained on site and the student instructor should demonstrate the following items:
 - (1) how to plan and to perform a flight at night;
 - (2) how to advise the student pilot to plan and prepare a flight at night;
 - (3) how to advise the student pilot to perform a flight at night;
 - (4) how to analyse and correct errors as necessary.

B. Helicopters

GROUND INSTRUCTION

Note: During ground instruction the student instructor should pay specific attention to the teaching of enhanced ground instruction in weather interpretation, planning and route assessment, decision making on encountering DVE including reversing course or conducting a precautionary landing.

Part 2

AIR EXERCISES

- (a) The air exercises are similar to those used for the training of PPL(H) but with additional items designed to cover the needs of an FI.
- (b) The numbering of exercises should be used primarily as an exercise reference list and as a broad instructional sequencing guide: therefore the demonstrations and practices need not necessarily be given in the order listed. The actual order and content will depend upon the following interrelated factors:
 - (1) the applicant's progress and ability;
 - (2) the weather conditions affecting the flight;
 - (3) the flight time available;
 - (4) instructional technique considerations;

- (5) the local operating environment;
- (6) applicability of the exercises to the helicopter type.
- (c) It follows that student instructors will eventually be faced with similar interrelated factors. They should be shown and taught how to construct flight lesson plans, taking these factors into account, so as to make the best use of each flight lesson, combining parts of the set exercises as necessary.

GENERAL

- (d) The briefing normally includes a statement of the objectives and a brief reference to principles of flight only if relevant. An explanation is to be given of exactly what air exercises are to be taught by the instructor and practised by the student during the flight. It should include how the flight will be conducted about who is to fly the helicopter and what airmanship, weather and flight safety aspects currently apply. The nature of the lesson will govern the order in which the constituent parts are to be taught.
- (e) The four basic components of the briefing will be:
 - (1) the aim;
 - (2) principles of flight (briefest reference only);
 - (3) the air exercise(s) (what, and how and by whom);
 - (4) airmanship (weather, flight safety etc.).

PLANNING OF FLIGHT LESSONS

- (f) The preparation of lesson plans is an essential prerequisite of good instruction and the student instructor is to be given supervised practice in the planning and practical application of flight lesson plans.

GENERAL CONSIDERATIONS

- (g) The student instructor should complete flight training to practise the principles of basic instruction at the PPL(H) level.
- (h) During this training, except when acting as a student pilot for mutual flights, the student instructor occupies the seat normally occupied by the FI(H).
- (i) It is to be noted that airmanship and look-out is a vital ingredient of all flight operations. Therefore, in the following air exercises the relevant aspects of airmanship are to be stressed at all times.
- (j) If the privileges of the FI(H) certificate are to include instruction for night flying, exercise 28 should be undertaken either as part of the course or subsequent to certificate issue.
- (k) The student instructor should learn how to identify common errors and how to correct them properly, which should be emphasised at all times.
- (l) The student instructor should be trained to keep in mind that wherever possible, flight simulation should be used to demonstrate to student pilots the effects of flight into DVE and to enhance their understanding and need for avoidance of this potentially fatal flight regime.

SYLLABUS OF FLIGHT INSTRUCTION CONTENTS

LONG BRIEFINGS AND AIR EXERCISES

EXERCISE 1: FAMILIARISATION WITH THE HELICOPTER

- (a) Long briefing objectives:
 - (1) introduction to the helicopter;
 - (2) explanation of the cockpit layout;
 - (3) helicopter and engine systems;
 - (4) checklist(s) and procedures;
 - (5) familiarisation with the helicopter controls;
 - (6) differences when occupying the instructor's seat;
 - (7) emergency drills:
 - (i) action if fire in the air and on the ground: engine, cockpit or cabin and electrical fire;
 - (ii) system failure drills as applicable to type;
 - (iii) escape drills: location and use of emergency equipment and exits.
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 2: PREPARATION FOR AND ACTION AFTER FLIGHT

- (a) Long briefing objectives:
 - (1) flight authorisation and helicopter acceptance, including technical log (if applicable) and certificate of maintenance;
 - (2) equipment required for flight (maps, etc.);
 - (3) external checks;
 - (4) internal checks;
 - (5) student comfort, harness, seat and rudder pedal adjustment;
 - (6) starting and after starting checks;
 - (7) system, power or serviceability checks (as applicable);
 - (8) closing down or shutting down the helicopter (including system checks).
 - (9) parking and leaving the helicopter (including safety or security as applicable);
 - (10) completion of authorisation sheet and helicopter serviceability documents.
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 3: AIR EXPERIENCE

- (a) Long briefing objectives:

Note: there is no requirement for a long briefing for this exercise.
- (b) Air exercise:

- (1) air experience;
- (2) cockpit layout, ergonomics and controls;
- (3) cockpit procedures: stability and control.

EXERCISE 4: EFFECTS OF CONTROLS

- (a) Long briefing objectives:
 - (1) function of the flying controls (primary and secondary effect);
 - (2) effect of air speed;
 - (3) effect of power changes (torque);
 - (4) effect of yaw (sideslip);
 - (5) effect of disc loading (bank and flare);
 - (6) effect on controls of selecting hydraulics on/off;
 - (7) effect of control friction;
 - (8) use of instruments;
 - (9) operation of carburettor heat or anti-icing control.
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 5: POWER AND ATTITUDE CHANGES

- (a) Long briefing objectives:
 - (1) relationship between cyclic control position, disc attitude, fuselage attitude and air speed flap back;
 - (2) power required diagram in relation to air speed;
 - (3) power and air speed changes in level flight;
 - (4) use of the instruments for precision;
 - (5) engine and air speed limitations;
- (b) Air exercise:
 - (1) relationship between cyclic control position, disc attitude, fuselage attitude and air speed flap back;
 - (2) power and air speed changes in level flight;
 - (3) use of instruments for precision (including instrument scan and lookout).

EXERCISE 6: LEVEL FLIGHT, CLIMBING, DESCENDING AND TURNING

Note: for ease of training this exercise is divided into four separate parts in the PPL(H) syllabus but may be taught complete or in convenient parts.

- (a) Long briefing objectives:
 - (1) basic factors involved in level flight;

- (2) normal power settings;
 - (3) use of control friction or trim;
 - (4) importance of maintaining direction and balance;
 - (5) power required or power available diagram;
 - (6) optimum climb and descent speeds, angles or rates;
 - (7) importance of balance, attitude and co-ordination in the turn;
 - (8) effects of turning on rate of climb or descent;
 - (9) use of the gyro direction or heading indicator and compass;
 - (10) use of instruments for precision.
- (b) Air exercises:
- (1) maintaining straight and level flight at normal cruise power;
 - (2) control in pitch, including use of control friction or trim;
 - (3) use of the ball or yaw string to maintain direction and balance;
 - (4) setting and use of power for selected air speeds and speed changes;
 - (5) entry to climb;
 - (6) normal and maximum rate of climb;
 - (7) levelling off from climb at selected altitudes or heights;
 - (8) entry to descent;
 - (9) effect of power and air speed on rate of descent;
 - (10) levelling off from descent at selected altitudes or heights;
 - (11) entry to medium rate turns;
 - (12) importance of balance, attitude and co-ordination to maintain level turn;
 - (13) resuming straight and level flight;
 - (14) turns onto selected headings, use of direction indicator and compass;
 - (15) turns whilst climbing and descending;
 - (16) effect of turn on rate of climb or descent;
 - (17) use of instruments for precision (including instrument scan and lookout).

EXERCISE 7: AUTOROTATION

- (a) Long briefing objectives:
- (1) characteristics of autorotation;
 - (2) safety checks (including look-out and verbal warning);
 - (3) entry and development of autorotation;
 - (4) effect of AUM, IAS, disc loading, G forces and density altitude on RRPM and rate of descent;

- (5) rotor and engine limitations;
 - (6) control of air speed and RRPM;
 - (7) recovery to powered flight;
 - (8) throttle override and control of ERPM or RRPM during re-engagement (as applicable);
 - (9) danger of vortex condition during recovery.
- (b) Air exercise:
- (1) safety checks (including verbal warning and look-out);
 - (2) entry to and establishing in autorotation;
 - (3) effect of IAS and disc loading on RRPM and rate of descent;
 - (4) control of air speed and RRPM;
 - (5) recovery to powered flight;
 - (6) medium turns in autorotation;
 - (7) simulated engine off landing (as appropriate).

EXERCISE 8: HOVERING AND HOVER TAXIING

- (a) Long briefing objectives:
- (1) ground effect and power required;
 - (2) effect of wind, attitude and surface;
 - (3) stability in hover and effects of over controlling;
 - (4) effect of control in hover;
 - (5) control and co-ordination during spot turns;
 - (6) requirement for slow hover speed to maintain ground effect;
 - (7) effect of hydraulic failure in hover;
 - (8) specific hazards, for example snow, dust, etc.
- (b) Air exercise:
- (1) ground effect and power or height relationship;
 - (2) effect of wind, attitude and surface;
 - (3) stability in hover and effects of over controlling;
 - (4) effect of control and hover technique;
 - (5) gentle forward running touchdown;
 - (6) control and co-ordination during spot (90 ° clearing) turns;
 - (7) control and co-ordination during hover taxi;
 - (8) dangers of mishandling and over pitching;
 - (9) (where applicable) effect of hydraulics failure in hover;
 - (10) simulated engine failure in the hover and hover taxi.

EXERCISE 9: TAKE-OFF AND LANDING

- (a) Long briefing objectives:
 - (1) pre take-off checks or drills;
 - (2) importance of good look-out;
 - (3) technique for lifting to hover;
 - (4) after take-off checks;
 - (5) danger of horizontal movement near ground;
 - (6) dangers of mishandling and over pitching;
 - (7) technique for landing;
 - (8) after landing checks;
 - (9) take-off and landing crosswind and downwind.
- (b) Air exercise:
 - (1) pre take-off checks or drills;
 - (2) pre take-off look-out technique;
 - (3) lifting to hover;
 - (4) after take-off checks;
 - (5) landing;
 - (6) after landing checks or drills;
 - (7) take-off and landing crosswind and downwind.

EXERCISE 10: TRANSITIONS FROM HOVER TO CLIMB AND APPROACH TO HOVER

- (a) Long briefing objectives:
 - (1) revision of ground effect;
 - (2) translational lift and its effects;
 - (3) inflow roll and its effects;
 - (4) revision of flap back and its effects;
 - (5) avoidance of curve diagram and associated dangers;
 - (6) effect or dangers of wind speed and direction during transitions;
 - (7) transition to climb technique;
 - (8) constant angle approach;
 - (9) transition to hover technique.
- (b) Air exercise:
 - (1) revision of take-off and landing;
 - (2) transition from hover to climb;
 - (3) effect of translational lift, inflow roll and flap back;

- (4) constant angle approach;
- (5) technique for transition from descent to hover;
- (6) a variable flare simulated engine off landing.

EXERCISE 11: CIRCUIT, APPROACH AND LANDING

- (a) Long briefing objectives:
 - (1) circuit and associated procedures;
 - (2) take-off and climb (including checks or speeds);
 - (3) crosswind leg (including checks, speeds or angles of bank in turns);
 - (4) downwind leg (including pre-landing checks);
 - (5) base leg (including checks, speeds or angles of bank in turns);
 - (6) final approach (including checks or speeds);
 - (7) effect of wind on approach and hover IGE;
 - (8) crosswind approach and landing technique;
 - (9) missed approach and go-around technique (as applicable);
 - (10) steep approach technique (including danger of high sink rate);
 - (11) limited power approach technique (including danger of high speed at touchdown);
 - (12) use of the ground effect;
 - (13) abandoned take-off technique;
 - (14) hydraulic failure drills and hydraulics off landing technique (where applicable);
 - (15) drills or technique for tail rotor control or tail rotor drive failure;
 - (16) engine failure drills in the circuit to include;
 - (17) engine failure
 - (18) on take-off:
 - (i) crosswind;
 - (ii) downwind;
 - (iii) base leg;
 - (iv) on final approach.
 - (19) noise abatement procedures (as applicable).
- (b) Air exercise:
 - (1) revision of transitions and constant angle approach;
 - (2) basic training circuit, including checks;
 - (3) crosswind approach and landing technique;
 - (4) missed approach and go-around technique (as applicable);
 - (5) steep approach technique;

- (6) basic limited power approach or run on technique;
- (7) use of ground effect;
- (8) hydraulic failure and approach to touchdown with hydraulics off and to recover at safe height (as applicable);
- (9) simulated engine failure on take-off, crosswind, downwind, base leg and finals;
- (10) variable flare simulated engine off landing.

EXERCISE 12: FIRST SOLO

- (a) Long briefing objectives:
 - (1) warning of change of attitude due to reduced and laterally displaced weight;
 - (2) low tail, low skid or wheel during hover or landing;
 - (3) dangers of loss of RRPM and over pitching;
 - (4) pre take-off checks;
 - (5) into wind take-off;
 - (6) drills during and after take-off;
 - (7) normal circuit, approach and landing;
 - (8) action if an emergency.
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 13: SIDEWAYS AND BACKWARDS HOVER MANOEUVRING

- (a) Long briefing objectives:
 - (1) revision of hovering;
 - (2) directional stability and weather cocking effect;
 - (3) danger of pitching nose down on recovery from backwards manoeuvring;
 - (4) helicopter limitations for sideways and backwards manoeuvring;
 - (5) effect of CG position.
- (b) Air exercise:
 - (1) revision of hovering and 90 ° clearing turns;
 - (2) manoeuvring sideways heading into wind;
 - (3) manoeuvring backwards heading into wind;
 - (4) manoeuvring sideways and backwards heading out of wind;
 - (5) manoeuvring backwards too fast and recovery action.

EXERCISE 14: SPOT TURNS

- (a) Long briefing objectives:
 - (1) revision of ground effect and effect of wind;

- (2) weather cocking and control actions;
 - (3) control of RRPM;
 - (4) torque effect;
 - (5) cyclic limiting stops due to CG position (where applicable);
 - (6) rate of turn limitations;
 - (7) spot turn about pilot position;
 - (8) spot turn about tail rotor position;
 - (9) spot turn about helicopter geometric centre;
 - (10) square (safe visibility) and clearing turn.
- (b) Air exercise:
- (1) weather cocking, torque effect and control actions;
 - (2) rate of turn;
 - (3) spot turn about pilot position;
 - (4) spot turn about tail rotor position;
 - (5) spot turn about helicopter geometric centre;
 - (6) square and clearing turn.

EXERCISE 15: HOVER OUT OF GROUND EFFECT AND VORTEX RING

- (a) Long briefing objectives:
- (1) revision of ground effect and power required diagram;
 - (2) drift, height and power control, look-out or scan;
 - (3) vortex ring, (including dangers, recognition and recovery actions);
 - (4) loss of tail rotor effectiveness.
- (b) Air exercise:
- (1) to demonstrate hover OGE;
 - (2) drift, height, power control and look-out, and instrument scan technique;
 - (3) recognition of incipient stage of vortex ring and settling with power;
 - (4) recovery action from incipient stage of vortex ring;
 - (5) recognition of loss of tail rotor effectiveness and recovery actions.

EXERCISE 16: SIMULATED ENGINE OFF LANDINGS

- (a) Long briefing objectives:
- (1) revision of basic autorotation;
 - (2) effect of AUM, disc loading, density altitude and RRPM decay;
 - (3) use of cyclic and collective to control speed or RRPM;
 - (4) torque effect;

- (5) use of flare or turn to restore RRPM;
 - (6) technique for variable flare simulated EOL;
 - (7) technique for constant attitude simulated EOL;
 - (8) revision of technique for hover or hover taxi simulated EOL;
 - (9) emergency technique for engine failure during transition;
 - (10) technique for low level simulated EOL.
- (b) Air exercise
- (1) revision of entry to and control in autorotation;
 - (2) variable flare simulated EOL
 - (3) constant attitude simulated EOL;
 - (4) hover simulated EOL;
 - (5) hover taxi simulated EOL;
 - (6) low level simulated EOL.

EXERCISE 17: ADVANCED AUTOROTATIONS

- (a) Long briefing objectives:
- (1) effect of air speed or AUM on angles or rates of descent
 - (2) effect of RRPM setting on angle or rate of descent;
 - (3) reason and technique for range autorotation;
 - (4) reason and technique for constant attitude autorotation;
 - (5) reason and technique for low speed and 'S' turns in autorotation;
 - (6) speed or bank limitations in turns in autorotation;
 - (7) revision of re-engagement or go-around procedures.
- (b) Air exercise:
- (1) selection of ground marker and standard datum height to determine distance covered during various autorotation techniques;
 - (2) revision of basic autorotation;
 - (3) technique for range autorotation;
 - (4) technique for constant attitude autorotation;
 - (5) technique for low speed autorotation, including need for timely speed recovery;
 - (6) technique for 'S' turn in autorotation;
 - (7) 180 and 360 ° turns in autorotation;
 - (8) revision of re-engagement and go-around technique.

EXERCISE 18: PRACTICE FORCED LANDINGS

- (a) Long briefing objectives:

- (1) types of terrain or surface options for choice of best landing area;
 - (2) practice forced landing procedure;
 - (3) forced landing checks and crash actions;
 - (4) rules or height for recovery and go-around.
- (b) Air exercise:
- (1) recognition of types of terrain from normal cruise height or altitude;
 - (2) practice forced landing technique;
 - (3) revision of recovery or go-around technique.

EXERCISE 19: STEEP TURNS

- (a) Long briefing objectives:
- (1) air speed or angle of bank limitations;
 - (2) technique for co-ordination to hold bank or attitude;
 - (3) revision of speed or bank limitations in autorotation including RRPM control;
 - (4) significance of disc loading, vibration and control feedback;
 - (5) effect of wind in turns at low level.
- (b) Air exercise:
- (1) technique for turning at 30 ° of bank;
 - (2) technique for turning at 45 ° of bank (where possible);
 - (3) steep autorotative turns;
 - (4) explanation of faults in the turn: balance, attitude, bank and coordination;
 - (5) effect of wind at low level.

EXERCISE 20: TRANSITIONS

- (a) Long briefing objectives:
- (1) revision of effect of ground cushion, translational lift and flap back;
 - (2) training requirement for precision exercise;
 - (3) technique for transition to forward flight and back to hover as precision exercise;
 - (4) effect of wind.
- (b) Air exercise:
- (1) transition from hover to minimum 50 knots IAS and back to hover;
Note: select constant height (20 - 30 ft) and maintain.
 - (2) effect of wind.

EXERCISE 21: QUICK STOPS

- (a) Long briefing objectives:

- (1) power control co-ordination;
- (2) revision of effect of wind;
- (3) technique for quick stop into wind;
- (4) technique for quick stop from crosswind;
- (5) revision of air speed and angles of bank limitations;
- (6) technique for emergency turn from downwind;
- (7) technique for quick stop from downwind from high speed: flare and turn;
- (8) technique for quick stop from downwind from low speed: turn and flare;
Note: use reasonable datum speed for example high speed, low speed.
- (9) danger of holding flare when downwind, (vortex ring) - (minimum speed 70 knots);
- (10) to revise danger of high disc loading.

(b) Air exercise:

- (1) technique for quick stop into wind;
- (2) technique for quick stop from crosswind;
- (3) danger of vortex ring and disc loading;
- (4) technique for quick stop from downwind with low speed;
- (5) technique for quick stop from downwind with high speed;
- (6) emergency turns from downwind.

EXERCISE 22: NAVIGATION

(a) Long briefing objectives:

Note: to be broken down into manageable parts at discretion of instructor.

- (1) flight planning:
 - (i) weather forecasts and actuals;
 - (ii) map selection, orientation, preparation and use:
 - (A) choice of route;
 - (B) regulated or controlled airspace;
 - (C) danger, prohibited and restricted areas;
 - (D) safety altitude.
 - (iii) calculations:
 - (A) magnetic heading(s), time(s) en route;
 - (B) fuel consumption;
 - (C) mass and balance.
 - (iv) flight information:
 - (A) NOTAMs etc;

- (B) noting of required radio frequencies;
 - (C) selection of alternate landing sites.
- (v) helicopter documentation;
- (vi) notification of the flight:
 - (A) pre-flight administration procedures;
 - (B) flight plan form (where appropriate).
- (2) departure:
 - (i) organisation of cockpit workload;
 - (ii) departure procedures:
 - (A) altimeter settings;
 - (B) ATC liaison in controlled or regulated airspace;
 - (C) setting heading procedure;
 - (D) noting of ETA(s);
 - (E) maintenance of height or altitude and heading.
 - (iii) procedure for revisions of ETA and headings to include:
 - (A) 10° line, double track, track error and closing angle;
 - (B) 1 in 60 rule;
 - (iv) amending an ETA;
 - (v) log keeping;
 - (vi) use of radio;
 - (vii) use of nav aids;
 - (viii) weather monitoring and minimum weather conditions for continuation of flight;
 - (ix) significance of in-flight decision making;
 - (x) technique for transiting controlled or regulated airspace;
 - (xi) uncertainty of position procedure;
 - (xii) lost procedure.
- (3) arrival:
 - (i) aerodrome joining procedure, in particular ATC liaison in controlled or regulated airspace:
 - (A) altimeter setting;
 - (B) entering traffic pattern;
 - (C) circuit procedures.
 - (ii) parking procedures, in particular:
 - (A) security of helicopter;
 - (B) refuelling;

- (C) closing of flight plan, (if appropriate);
 - (D) post flight administrative procedures.
- (4) navigation problems at low heights and reduced visibility:
 - (i) actions before descending;
 - (ii) significance of hazards, (for example obstacles and other traffic);
 - (iii) difficulties of map reading;
 - (iv) effects of wind and turbulence;
 - (v) significance of avoiding noise sensitive areas;
 - (vi) procedures for joining a circuit from low level;
 - (vii) procedures for a bad weather circuit and landing;
 - (viii) actions in the event of encountering DVE;
 - (ix) appropriate procedures and choice of landing area for precautionary landings;
 - (x) decision to divert or conduct precautionary landing;
 - (xi) precautionary landing.
- (5) radio navigation:
 - (i) use of VOR:
 - (A) availability, AIP and frequencies;
 - (B) selection and identification;
 - (C) use of OBS;
 - (D) to or from indications: orientation;
 - (E) use of CDI;
 - (F) determination of radial;
 - (G) intercepting and maintaining a radial;
 - (H) VOR passage;
 - (I) obtaining a fix from two VORs.
 - (ii) use of ADF equipment:
 - (A) availability of NDB stations, AIP and frequencies;
 - (B) selection and identification;
 - (C) orientation relative to beacon;
 - (D) homing.
 - (iii) use of VHF/DF
 - (A) availability, AIP and frequencies;
 - (B) R/T procedures and ATC liaison;
 - (C) obtaining a QDM and homing.
 - (iv) use of en-route or terminal radar:

- (A) availability and AIP;
 - (B) procedures and ATC liaison;
 - (C) pilots responsibilities;
 - (D) secondary surveillance radar:
 - (a) transponders;
 - (b) code selection;
 - (E) interrogation and reply.
 - (v) use of DME:
 - (A) station selection and identification;
 - (B) modes of operation: distance, groundspeed and time to run.
 - (vi) use of GNSS:
 - (A) selection of waypoints;
 - (B) to or from indications and orientation;
 - (C) error messages;
 - (D) hazards of over-reliance in the continuation of flight in DVE.
- (b) Air exercise:
- (1) navigation procedures as necessary;
 - (2) to advise student and correct errors as necessary;
 - (3) map reading techniques;
 - (4) the significance of calculations;
 - (5) revision of headings and ETA's;
 - (6) use of radio;
 - (7) use of nav aids: ADF/NDB, VOR, VHF/DF, DME and transponder;
 - (8) cross-country flying by using visual reference, DR, GNNS and, where available, radio navigation aids; simulation of deteriorating weather conditions and actions to divert or conduct precautionary landing;
 - (8) log keeping;
 - (9) importance of decision making;
 - (10) procedure to deal with uncertainty of position;
 - (11) lost procedure;
 - (12) appropriate procedures and choice of landing area for precautionary landings;
 - (13) aerodrome joining procedure;
 - (14) parking and shut-down procedures;
 - (15) post-flight administration procedures.

EXERCISE 23: ADVANCED TAKE-OFF, LANDINGS AND TRANSITIONS

- (a) Long briefing objectives:
 - (1) revision of landing and take-off out of wind (performance reduction);
 - (2) revision of wind limitations;
 - (3) revision of directional stability variation when out of wind;
 - (4) revision of power required diagram;
 - (5) technique for downwind transitions;
 - (6) technique for vertical take-off over obstacles;
 - (7) reconnaissance technique for landing site;
 - (8) power checks;
 - (9) technique for running landing;
 - (10) technique for zero speed landing;
 - (11) technique for crosswind and downwind landings;
 - (12) steep approach, including dangers;
 - (13) revision of go-around procedures.
- (b) Air exercise
 - (1) technique for downwind transition;
 - (2) technique for vertical take-off over obstacles;
 - (3) reconnaissance technique for landing site;
 - (4) power check and assessment;
 - (5) technique for running landing;
 - (6) technique for zero speed landing;
 - (7) technique for crosswind and downwind landings;
 - (8) technique for steep approach;
 - (9) go-around procedures.

EXERCISE 24: SLOPING GROUND

- (a) Long briefing objectives:
 - (1) limitations;
 - (2) wind and slope relationship, including blade and control stops;
 - (3) effect of CG when on slope;
 - (4) ground effect and power required when on slope;
 - (5) landing technique when on slope, left, right and nose-up;
 - (6) avoidance of dynamic rollover, dangers of soft ground and sideways movement;
 - (7) dangers of over controlling near ground on slope;
 - (8) danger of striking main or tail rotor on up slope.

- (b) Air exercise
 - (1) technique for assessing slope angle;
 - (2) technique for landing and take-off left skid up slope;
 - (3) technique for landing and take-off right skid up slope;
 - (4) technique for landing nose up slope;
 - (5) dangers of over controlling near ground.

EXERCISE 25: LIMITED POWER

- (a) Long briefing objectives:
 - (1) use of appropriate helicopter performance graphs;
 - (2) selection of technique according to available power;
 - (3) effect of wind on available power.
- (b) Air exercise: to revise and refine techniques demonstrated in exercise 23.

EXERCISE 26: CONFINED AREAS

- (a) Long briefing objectives:
 - (1) revision of use of helicopter performance graphs;
 - (2) procedure for locating landing site and selecting site marker;
 - (3) procedures for assessing wind speed and direction;
 - (4) landing site reconnaissance techniques;
 - (5) reason for selecting landing markers;
 - (6) procedure for selecting direction and type of approach;
 - (7) dangers of out of wind approach;
 - (8) circuit procedures;
 - (9) reason for approach to committal point and go-around, (practice approach);
 - (10) approach technique;
 - (11) revision of clearing turn and landing (sloping ground technique);
 - (12) hover power check or performance assessment IGE and OGE (if necessary);
 - (13) take-off procedures.
- (b) Air exercise
 - (1) procedures for locating landing site and selecting site marker;
 - (2) procedures for assessing wind speed and direction;
 - (3) landing site reconnaissance techniques;
 - (4) selecting landing markers, direction and type of approach;
 - (5) circuit procedure;
 - (6) practice approach, go-around and approach technique;

- (7) revision of clearing turn and landing (sloping ground technique);
- (8) hover power check or performance assessment IGE and OGE (if necessary);
- (9) take-off procedures.

EXERCISE 27: BASIC INSTRUMENT FLIGHT

- (a) Long briefing objectives:
 - (1) physiological sensations;
 - (2) instrument appreciation;
 - (3) attitude instrument flight;
 - (4) instrument scan;
 - (5) instrument limitations;
 - (6) basic manoeuvres by sole reference to instruments:
 - (i) straight and level flight at various air speeds and configurations;
 - (ii) climbing and descending;
 - (iii) standard rate turns, climbing and descending, onto selected headings;
 - (iv) recoveries from climbing and descending turns (unusual attitudes).
- (b) Air exercise:
 - (1) attitude instrument flight and instrument scan;
 - (2) basic manoeuvres by sole reference to instruments:
 - (i) straight and level flight at various air speeds and configurations;
 - (ii) climbing and descending;
 - (iii) standard rate turns, climbing and descending, onto selected headings;
 - (iv) recoveries from climbing and descending turns (unusual attitudes).

EXERCISE 28: NIGHT FLYING (if night instructional qualification required)

- (a) Long briefing objectives:
 - (1) medical or physiological aspects of night vision;
 - (2) requirement for torch to be carried (pre-flight inspection, etc.);
 - (3) use of the landing light;
 - (4) take-off and hover taxi procedures at night;
 - (5) night take-off procedure;
 - (6) cockpit procedures at night;
 - (7) approach techniques;
 - (8) night landing techniques;
 - (9) night autorotation techniques (power recovery at safe height);
 - (10) technique for practice forced landing at night (using appropriate illumination);

- (11) emergency procedures at night;
- (12) navigation principles at night;
- (13) map marking for night use (highlighting built up or lit areas with thicker lines, etc.).
- (b) Air exercise:
 - (1) use of torch for pre-flight inspection;
 - (2) use of landing light;
 - (3) night take-off to hover (no sideways or backwards movement);
 - (4) night hover taxi (higher and slower than by day);
 - (5) night transition procedure;
 - (6) night circuit;
 - (7) night approach and landing (including use of landing light);
 - (8) night autorotation (power recovery at safe height);
 - (9) practice forced landing at night (using appropriate illumination);
 - (10) night emergency procedures;
 - (11) night cross country techniques, as appropriate.

C. Airships

Part 2

AIR EXERCISES

- (a) The air exercises are similar to those used for the training of PPL(As) but with additional items designed to cover the needs of an FI.
- (b) The numbering of exercises should be used primarily as an exercise reference list and as a broad instructional sequencing guide: therefore the demonstrations and practices need not necessarily be given in the order listed. The actual order and content will depend upon the following interrelated factors:
 - (1) the applicant's progress and ability;
 - (2) the weather conditions affecting the flight;
 - (3) the flight time available;
 - (4) instructional technique considerations;
 - (5) the local operating environment.
- (c) It follows that student instructors will eventually be faced with similar interrelated factors. They should be shown and taught how to construct flight lesson plans, taking these factors into account, so as to make the best use of each flight lesson, combining parts of the set exercises as necessary.

GENERAL

- (d) The briefing normally includes a statement of the aim and a brief allusion to principles of flight only if relevant. An explanation is to be given of exactly what air exercises are to be taught by

the instructor and practised by the student during the flight. It should include how the flight will be conducted about who is to fly the airship and what airmanship, weather and flight safety aspects currently apply. The nature of the lesson will govern the order in which the constituent parts are to be taught.

- (e) The four basic components of the briefing will be:
 - (1) the aim;
 - (2) principles of flight (briefest reference only);
 - (3) the air exercise(s) (what, and how and by whom);
 - (4) airmanship (weather, flight safety etc.).

PLANNING OF FLIGHT LESSONS

- (f) The preparation of lesson plans is an essential prerequisite of good instruction and the student instructor is to be given supervised practice in the planning and practical application of flight lesson plans.

GENERAL CONSIDERATIONS

- (g) The student instructor should complete flight training to practise the principles of basic instruction at the PPL(As) level.
- (h) During this training, except when acting as a student pilot for mutual flights, the student instructor occupies the seat normally occupied by the FI(As).
- (i) It is to be noted that airmanship and look-out is a vital ingredient of all flight operations. Therefore, in the following air exercises the relevant aspects of airmanship are to be stressed at all times.
- (j) The exercises 15 and 16 of the flight instruction syllabus should be undertaken at night in addition to by day as part of the course.
- (k) The student instructor should learn how to identify common errors and how to correct them properly, which should be emphasised at all times.

SYLLABUS OF FLIGHT INSTRUCTION CONTENTS

LONG BRIEFINGS AND AIR EXERCISES

Note: although exercise 16 is not required for the PPL(As) course it is a requirement for the FI(As) course.

EXERCISE 1: FAMILIARISATION WITH THE AIRSHIP

- (a) Long briefing objectives:
 - (1) introduction to the airship;
 - (2) characteristics of the airship;
 - (3) cockpit layout;
 - (4) airship and engine systems;
 - (5) use of the checklist(s) and procedures;

- (6) to familiarise the student with the airship controls;
- (7) differences when occupying the instructor's seat;
- (8) emergency drills:
 - (i) action if fire in the air or on the ground: engine, cockpit or cabin and electrical fire;
 - (ii) system failure drills as applicable to type;
 - (iii) escape drills: location and use of emergency equipment and exits.
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 2: PREPARATION FOR AND ACTION AFTER FLIGHT

- (a) Long briefing objectives:
 - (1) flight authorisation and airship acceptance including tech log (if applicable) and certificate of maintenance;
 - (2) equipment required for flight (maps, etc.);
 - (3) external checks;
 - (4) internal checks;
 - (5) student comfort, harness, seat and rudder pedal adjustment;
 - (6) starting and after starting checks;
 - (7) system, power or serviceability checks (as applicable);
 - (8) closing down or shutting down the airship (including system checks);
 - (9) parking, masting and unmasting, leaving the airship (including safety or security as applicable);
 - (10) completion of the authorisation sheet and airship serviceability documents;
- (b) Air exercise: all long briefing objectives mentioned above should also be trained on site during the air exercise.

EXERCISE 3: AIR EXPERIENCE

- (a) Long briefing objectives:

Note: there is no requirement for a long briefing for this exercise.
- (b) Air exercise:
 - (1) air experience;
 - (2) cockpit layout, ergonomics and controls;
 - (3) cockpit procedures: stability and control.

EXERCISE 4: EFFECTS OF CONTROLS

- (a) Long briefing objectives:
 - (1) function of the flying controls (primary and secondary effect);

- (2) effect of air speed;
 - (3) effect of power changes;
 - (4) effect of trimming and other controls;
 - (5) use of instruments;
 - (6) use of carburettor heat.
- (b) Air exercise:
- (1) function of the flying controls;
 - (2) effect of air speed;
 - (3) effect of power changes;
 - (4) effect of trimming and other controls;
 - (5) use of instruments (including instrument scan);
 - (6) use of carburettor heat.

EXERCISE 5: GROUND MANOEUVERING

- (a) Long briefing objectives:
- (1) pre-taxi checks;
 - (2) starting, control of speed and stopping;
 - (3) engine handling;
 - (4) mastering procedures;
 - (5) control of direction and turning;
 - (6) effects of wind;
 - (7) effects of ground surface;
 - (8) marshalling signals;
 - (9) instrument checks;
 - (10) ATC procedures;
 - (11) emergencies.
- (b) Air exercise:
- (1) starting, control of speed and stopping;
 - (2) engine handling;
 - (3) mastering procedures;
 - (4) control of direction and turning;
 - (5) effect of wind.

EXERCISE 6: TAKE-OFF PROCEDURES

- (a) Long briefing objectives:
- (1) pre take-off checks;

- (2) take-off with different static heaviness;
 - (3) drills during and after take-off;
 - (4) noise abatement procedures.
- (b) Air exercise:
- (1) take-off with different static heaviness;
 - (2) drills during and after take-off.

EXERCISE 6e: EMERGENCIES

- (a) Long briefing objectives:
- (1) abandoned take-off;
 - (2) engine failures and actions after take-off;
 - (3) malfunctions of thrust vector control;
 - (4) aerodynamic control failures;
 - (5) electrical and system failures.
- (b) Air exercise:
- (1) how to abandon a take-off;
 - (2) engine failure and suitable action;
 - (3) malfunctions of thrust vector control;
 - (4) aerodynamic control failures.

EXERCISE 7: CLIMBING

- (a) Long briefing objectives:
- (1) entry and how to maintain the normal and max rate of climb;
 - (2) levelling off procedure;
 - (3) how to level off at selected altitudes;
 - (4) maximum angle of climb;
 - (5) maximum rate of climb.
- (b) Air exercise:
- (1) how to level off at selected altitudes;
 - (2) maximum angle of climb.

EXERCISE 8: STRAIGHT AND LEVEL FLIGHT

- (a) Long briefing objectives:
- (1) how to attain and maintain straight and level flight;
 - (2) flight at or close to pressure height;
 - (3) control in pitch, including use of trim;

- (4) at selected air speeds (use of power);
 - (5) during speed changes;
 - (6) use of instruments for precision.
- (b) Air exercise:
 - (1) how to attain and maintain straight and level flight;
 - (2) flight at or close to pressure height;
 - (3) control in pitch, including use of trim;
 - (4) at selected air speeds (use of power);
 - (5) during speed changes.

EXERCISE 9: DESCENDING

- (a) Long briefing objectives:
 - (1) entry, maintaining and levelling off techniques;
 - (2) levelling off at selected altitudes;
 - (3) maximum rate of descent;
 - (4) maximum angle of descent;
 - (5) use of instruments for precision flight.
- (b) Air exercise:
 - (1) levelling off at selected altitudes;
 - (2) maximum rate of descent;
 - (3) maximum angle of descent.

EXERCISE 10: TURNING

- (a) Long briefing objectives:
 - (1) entry and maintaining level turns;
 - (2) resuming straight flight;
 - (3) faults in the turn;
 - (4) climbing turns;
 - (5) descending turns;
 - (6) turns to selected headings: use of gyro heading indicator and compass;
 - (7) use of instruments for precision.
- (b) Air exercise
 - (1) faults in the turn and correction techniques;
 - (2) climbing turns;
 - (3) descending turns.

EXERCISE 11: HOVERING

- (a) Long briefing objectives: hovering manoeuvres (as applicable).
- (b) Air exercise: hovering manoeuvres (as applicable).

EXERCISE 12: APPROACH AND LANDING

- (a) Long briefing objectives:
 - (1) effect of wind on approach and touchdown speeds;
 - (2) landing with different static heaviness;
 - (3) missed approach and go-around procedures;
 - (4) noise abatement procedures.
- (b) Air exercise
 - (1) a landing with different static heaviness;
 - (2) missed approach and go-around procedures.

EXERCISE 12e: EMERGENCIES

- (a) Long briefing objectives:
 - (1) aborted approach or go-around;
 - (2) malfunction of thrust vector control;
 - (3) envelope emergencies;
 - (4) fire emergencies;
 - (5) aerodynamic control failures;
 - (6) electrical and system failures.
- (b) Air exercise: emergency drills and actions.

EXERCISE 13: PRECAUTIONARY LANDING

- (a) Long briefing objectives:
 - (1) occasions necessitating a precautionary landing;
 - (2) in-flight conditions;
 - (3) landing area selection;
 - (4) circuit and approach.
- (b) Air exercise:
 - (1) how to perform the landing area selection;
 - (2) circuit and approach.

EXERCISE 14a: NAVIGATION

- (a) Long briefing objectives:

- (1) how to do the flight planning;
- (2) departure for a navigation flight;
- (3) in-flight navigational techniques;
- (4) arrival and aerodrome joining procedures;
- (b) Air exercise:
 - (1) complete flight planning of a navigation flight;
 - (2) departure for a navigation flight;
 - (3) in-flight navigational techniques;
 - (4) arrival and aerodrome joining procedures.

EXERCISE 14b: NAVIGATION AT LOWER LEVELS AND IN REDUCED VISIBILITY

- (a) Long briefing objectives:
 - (1) actions before descending;
 - (2) possible hazards (for example obstacles and terrain) and actions;
 - (3) student difficulties of map reading;
 - (4) effects of winds, turbulence and precipitation;
 - (5) vertical situational awareness;
 - (6) avoidance of noise sensitive areas;
 - (7) joining the circuit;
 - (8) bad weather circuit and landing.
- (b) Air exercise:
 - (1) actions before descending;
 - (2) map reading techniques;
 - (3) vertical situational awareness;
 - (4) avoidance of noise sensitive areas;
 - (5) joining the circuit;
 - (6) bad weather circuit and landing.

EXERCISE 14c: RADIO NAVIGATION

- (a) Long briefing objectives:
 - (1) use of VOR;
 - (2) use of ADF equipment;
 - (3) use of NDB stations;
 - (4) use of VHF/DF;
 - (5) use of en-route or terminal radar;
 - (6) use of DME equipment.

- (b) Air exercise
 - (1) use of nav aids;
 - (2) procedure to deal with uncertainty of position.

EXERCISE 15: BASIC INSTRUMENT FLIGHT

- (a) Long briefing objectives:
 - (1) physiological sensations;
 - (2) instrument appreciation;
 - (3) attitude instrument flight;
 - (4) instrument scan;
 - (5) instrument limitations;
 - (6) basic manoeuvres by sole reference to the instruments:
 - (i) straight and level;
 - (ii) climbing and descending;
 - (iii) turns, climbing and descending, onto selected headings;
 - (iv) recoveries from climbing and descending turns.
- (b) Air exercise:
 - (1) attitude instrument flight and instrument scan;
 - (2) the basic manoeuvres:
 - (i) straight and level;
 - (ii) climbing and descending;
 - (iii) turns, climbing and descending, onto selected headings;
 - (iv) recoveries from climbing and descending turns.

EXERCISE 16: NIGHT FLYING (if night instructional qualification required)

- (a) Long briefing objectives:
 - (1) medical and physiological aspects of night vision;
 - (2) requirement for torch to be carried (pre-flight inspection, etc.);
 - (3) use of the landing light;
 - (4) ground manoeuvring procedures at night;
 - (5) night take-off procedure;
 - (6) cockpit procedures at night;
 - (7) approach techniques;
 - (8) night landing techniques
 - (9) emergency procedures at night;
 - (10) navigation principles at night.

(b) Air exercise:

- (1) use of landing light;
- (2) night ground manoeuvring;
- (3) night take-off, circuit or approach and landing (including use of landing light).

FCL.940.FI FI – Revalidation and renewal*Regulation (EU) 2025/134***(a) Revalidation**

- (1) To revalidate an FI certificate, holders shall fulfil at least two out of the three following requirements before the expiry date of the FI certificate:

- (i) they have completed

- (A) in the case of an FI(A) and an FI(H), at least 50 hours of flight instruction in the appropriate aircraft category as FIs, TRIs, CRIs, IRIs, MIs or examiners. If the privileges to instruct for the BIR and the IR are to be revalidated, 10 of those 50 hours shall be flight instruction for a BIR or an IR and shall have been completed within the last 12 months immediately preceding the expiry date of the FI certificate;
 - (B) in the case of an FI(As), at least 20 hours of flight instruction in airships as FIs, IRIs or as examiners. If the privileges to instruct for the IR are to be revalidated, 10 of those hours shall be flight instruction for an IR and shall have been completed in the period of 12 months immediately preceding the expiry date of the FI certificate;
 - (C) in the case of an FI(G), at least 50 hours of flight instruction in gyroplanes as FIs or as examiners;

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- (ii) they have completed instructor refresher training as an FI at an ATO or at the competent authority;
 - (iii) they have passed an assessment of competence in accordance with point [FCL.935](#) in the period of 12 months immediately preceding the expiry date of the FI certificate.

- (2) For at least each alternate revalidation, in the case of FI(A) or FI(H), or each third revalidation, in the case of FI (As), holders of the relevant FI certificate shall pass an assessment of competence in accordance with point [FCL.935](#).

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (2) For at least each alternate revalidation, in the case of FI(A), FI(H) or FI(G), or each third revalidation, in the case of FI(As), holders of the relevant FI certificate shall pass an assessment of competence in accordance with point [FCL.935](#).

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

(b) Renewal.

If the FI certificate has expired, applicants shall, within a period of 12 months before the application date for the renewal, complete instructor refresher training as an FI at an ATO or at

a competent authority and complete an assessment of competence in accordance with point [FCL.935](#).

AMC1 FCL.940.FI; FCL.940.IRI Revalidation and renewal

ED Decision 2025/002/R

- (a) The instructor refresher training for the revalidation of the FI and IRI certificates should be provided as a seminar by either an ATO, DTO, or competent authority.
- (1) FI or IRI refresher seminars made available in Member States should have due regard to geographical location, numbers attending, and periodicity throughout the territory of the Member State concerned.
 - (2) Such seminars should run for at least 2 days (1 day = 6 hours), and attendance from participants will be required for the whole duration of the seminar including breakout groups and workshops. Different aspects, such as inclusion of participants holding certificates in other categories of aircraft, should be considered.
 - (3) Appropriately experienced FIs or IRIs currently involved with flying training and with a practical understanding of the revalidation requirements and current instructional techniques should be included as speakers at these seminars.
 - (4) The attendance form will be completed and signed by the organiser of the seminar as approved by the competent authority, following attendance and satisfactory participation by the FI or IRI.
 - (5) The content of the FI or IRI refresher seminar should be selected from the following:
 - (i) new or current rules or regulations, with emphasis on knowledge of Part-FCL and operational requirements;
 - (ii) teaching and learning;
 - (iii) instructional techniques;
 - (iv) the role of the instructor;
 - (v) national regulations (as applicable);
 - (vi) human factors;
 - (vii) flight safety, incident and accident prevention;
 - (viii) airmanship;
 - (ix) legal aspects and enforcement procedures;
 - (x) navigational skills including new or current radio navigation aids;
 - (xi) teaching instrument flying;
 - (xii) weather-related topics including methods of distribution;
 - (xiii) any additional topic selected by the competent authority.
 - (6) Formal sessions should allow time for presentations and related questions. The use of visual aids is recommended, with interactive videos and other teaching aids (where available) for breakout groups and workshops.

- (b) If the instructor certificate lapsed, the ATO, DTO, or competent authority, whichever is appropriate to the category of aircraft, should consider all the above as well as the following, when assessing the refresher training programme:
- (1) the ATO, DTO, or competent authority should determine on a case-by-case basis the amount of refresher training needed, following an assessment of the candidate taking into account the following factors:
 - (i) the experience of the applicant;
 - (ii) the amount of time elapsed since the expiry of the FI or IRI certificate; and
 - (iii) the technical elements of the FI or IRI training course, as determined by the assessment of the candidate by the ATO, DTO, or competent authority; and
 - (2) the individual training programme should be based on the content of the FI or IRI training course and focus on the aspects where the applicant showed the greatest needs.
- (c) After successful completion of the seminar or refresher training, as applicable, the ATO, DTO, or competent authority should:
- (1) in case of a seminar, in accordance with point (a), issue the applicant with a seminar completion certificate or another document specified by the competent authority, which describes the content of the seminar as in point (a), as well as a statement that the seminar was successfully completed; and
 - (2) in case of refresher training, in accordance with point (b), issue the applicant with a training completion certificate or another document specified by the competent authority, which describes the evaluation of the factors listed in point (b)(1) and the training received, as well as a statement that the training was successfully completed; the training completion certificate should be presented to the examiner prior to the assessment of competence.
- Upon successful completion of the refresher seminar or refresher training, as applicable, the ATO or DTO should submit the seminar or training completion certificate, or the other document specified by the competent authority, to the competent authority.
- (d) Taking into account the factors listed in point (b)(1), the ATO, DTO, or competent authority, as applicable, may also decide that it is sufficient for the candidate to complete a seminar in accordance with point (a). In such a case, the completion certificate or the other document that is referred to in point (c) should contain a related statement with sufficient reasoning.

FI — Revalidation and renewal

FI CERTIFICATE: REVALIDATION AND RENEWAL FORM

A. AEROPLANES

INSTRUCTIONAL FLYING EXPERIENCE				
Instructors applying for revalidation of the FI certificate should enter the instructional hours flown during the preceding 36 months.				
SINGLE-ENGINE		MULTI-ENGINE		INSTRUMENT
DAY	NIGHT	DAY	NIGHT	
Total instructional hours (preceding 36 months):				
Total instructional hours (preceding 12 months):				

FI REFRESHER SEMINAR					
1	This is to certify that the undersigned attended an FI seminar				
2	Attendee's personal particulars:				
<table border="1"> <tr> <td>Name(s):</td> <td>Address:</td> </tr> <tr> <td>Licence number:</td> <td>Expiration date of FI(A) certificate</td> </tr> </table>		Name(s):	Address:	Licence number:	Expiration date of FI(A) certificate
Name(s):	Address:				
Licence number:	Expiration date of FI(A) certificate				
3	Seminar particulars:				
<table border="1"> <tr> <td>Date(s) of seminar:</td> <td>Place:</td> </tr> </table>		Date(s) of seminar:	Place:		
Date(s) of seminar:	Place:				
4	Declaration by the responsible organiser:				
I certify that the above data are correct and that the FI seminar was carried out.					
<table border="1"> <tr> <td>Date of approval:</td> <td>Name(s) of organiser: (capital letters)</td> </tr> <tr> <td>Date and place:</td> <td>Signature:</td> </tr> </table>		Date of approval:	Name(s) of organiser: (capital letters)	Date and place:	Signature:
Date of approval:	Name(s) of organiser: (capital letters)				
Date and place:	Signature:				
5	Declaration by the attendee:				
I confirm the data under 1 through 3					
Attendee's signature:					
PROFICIENCY CHECK					
(Name(s) of applicant) has given proof of flying instructional ability during a proficiency check flight. This was done to the required standard.					
<table border="1"> <tr> <td>Flying time:</td> <td>Aeroplane or FFS used:</td> </tr> </table>		Flying time:	Aeroplane or FFS used:		
Flying time:	Aeroplane or FFS used:				
Main exercise:					
<table border="1"> <tr> <td>Name(s) of FIE:</td> <td>Licence number:</td> </tr> <tr> <td>Date and place:</td> <td>Signature:</td> </tr> </table>		Name(s) of FIE:	Licence number:	Date and place:	Signature:
Name(s) of FIE:	Licence number:				
Date and place:	Signature:				

B. HELICOPTERS

INSTRUCTIONAL FLYING EXPERIENCE
Instructors applying for revalidation of the FI certificate should enter the instructional hours flown during the preceding 36 months.
Instrument:

Total instructional hours (preceding 36 months):			
Total instructional hours (preceding 12 months):			
FI REFRESHER SEMINAR			
1	This is to certify that the undersigned attended an FI seminar		
2	Attendees personal particulars:		
Name(s):		Address:	
Licence number:		Expiration date of FI(H) certificate:	
3	Seminar particulars:		
Date(s) of seminar:		Place:	
4	Declaration by the responsible organiser:		
I certify that the above data are correct and that the FI seminar was carried out.			
Date of approval:		Name(s) of organiser: (capital letters)	
Date and place:		Signature:	
5	Declaration by the attendee:		
I confirm the data under 1 through 3			
Attendee's signature:			
PROFICIENCY CHECK			
(Name(s) of applicant) has given proof of flying instructional ability during a proficiency check flight. This was done to the required standard.			
Flying time:		Helicopter or FFS used:	
Main exercise:			
Name(s) of FIE:		Licence number:	
Date and place:			
Signature:			

C. AIRSHIPS

INSTRUCTIONAL FLYING EXPERIENCE				
Instructors applying for revalidation of the FI certificate should enter the instructional hours flown during the preceding 36 months.				
SINGLE-ENGINE		MULTI-ENGINE		INSTRUMENT
DAY	NIGHT	DAY	NIGHT	
Total instructional hours (preceding 36 months):				
Total instructional hours (preceding 12 months):				
FLIGHT INSTRUCTOR REFRESHER SEMINAR				
1	This is to certify that the undersigned attended an FI seminar			
2	Attendee's personal particulars:			
Name(s):			Address:	
Licence number:			Expiration date of FI(As) certificate:	
3	Seminar particulars:			
Date(s) of seminar:			Place:	
4	Declaration by the responsible organiser:			
I certify that the above data are correct and that the FI seminar was carried out.				
Date of approval:			Name(s) of organiser: (capital letters)	
Date and place:			Signature:	
5	Declaration by the attendee:			
I confirm the data under 1 through 3				
Attendee's signature:				
PROFICIENCY CHECK				
(Name(s) of applicant) has given proof of flying instructional ability during a proficiency check flight. This was done to the required standard.				
Flying time:			Airship or FFS used:	
Main exercise:				
Name(s) of FIE:			Licence number:	

Date and place:	Signature:
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SECTION 4 – SPECIFIC REQUIREMENTS FOR THE TYPE RATING INSTRUCTOR – TRI

FCL.905.TRI TRI – Privileges and conditions

Regulation (EU) 2024/2076

- (a) The privileges of a TRI are to instruct for:
- (1) the revalidation and renewal of an IR, provided the TRI holds a valid IR;
 - (2) the issue of a TRI or SFI certificate, provided that the holder meets all of the following conditions:
 - (i) has at least 50 hours of instructional experience as a TRI or SFI in accordance with this Regulation or [Regulation \(EU\) No 965/2012](#);
 - (ii) has conducted the flight instruction syllabus of the relevant part of the TRI training course in accordance with point [FCL.930.TRI\(ab\)\(3\)](#) to the satisfaction of the head of training of an ATO;
 - (3) in the case of the TRI for single-pilot aeroplanes:
 - (i) the issue, revalidation and renewal of type ratings for single-pilot high-performance complex aeroplanes provided that the applicant seeks privileges to operate in single-pilot operations.

The privileges of the TRI(SPA) may be extended to flight instruction for single-pilot high-performance complex aeroplane type ratings in multi-pilot operations, provided that the TRI meets any of the following conditions:

 - (A) holds or has held a TRI certificate for multi-pilot aeroplanes;
 - (B) has at least 500 hours on aeroplanes in multi-pilot operations and completed an MCCI training course in accordance with point [FCL.930.MCCI](#);
 - (ii) the MPL course on the basic phase, provided that he or she has the privileges extended to multi-pilot operations and holds or has held an FI(A) or an IRI(A) certificate;
 - (4) in the case of the TRI for multi-pilot aeroplanes:
 - (i) the issue, revalidation and renewal of type ratings for:
 - (A) multi-pilot aeroplanes;
 - (B) single-pilot high-performance complex aeroplanes when the applicant seeks privileges to operate in multi-pilot operations;
 - (ii) MCC training;
 - (iii) the MPL course on the basic, intermediate and advanced phases, provided that, for the basic phase, he or she holds or has held an FI(A) or IRI(A) certificate;
 - (5) in the case of the TRI for helicopters:
 - (i) the issue, revalidation and renewal of helicopter type ratings;
 - (ii) MCC training, provided that he or she has completed 350 hours of flight time as a pilot in multi-pilot operations in any aircraft category.
 - (6) in the case of the TRI for powered-lift aircraft:

- (i) the issue, revalidation and renewal of powered-lift type ratings;
 - (ii) MCC training.
- (b) The privileges of a TRI include privileges to conduct EBT practical assessment at an EBT operator, provided that the instructor complies with the requirements of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#) for EBT instructor standardisation at that EBT operator.

GM1 FCL.905.TRI(b) Privileges and conditions

ED Decision 2019/005/R

INSTRUCTORS INSTRUCTING FOR THE ISSUE OF A TRI OR SFI CERTIFICATE

Training in an aeroplane is not a requirement for the issue of an SFI or a TRI certificate. In order to deliver effective UPRT, it is beneficial for the instructor to have first-hand experience of the critical psychological and physiological human factors, which might be present during recoveries from developed upsets. These human factors (effects of unusual acceleration, such as variations from normal 1G flight, the difficulty to perform counter-intuitive actions, and the management of the associated stress response) can only be experienced during training in an aeroplane because FFSs are not capable of reproducing sustained accelerations. Completion of the advanced UPRT course in accordance with [FCL.745.A](#) would provide such experience and is therefore useful for instructors providing instruction for the issue of a TRI or an SFI certificate.

FCL.910.TRI TRI – Restricted privileges

Regulation (EU) 2024/2076

- (a) General. If the TRI training is carried out in FSTDs only, the privileges of TRIs shall be restricted to training in FSTDs. This restriction shall however include the following privileges for conducting, in the aircraft:
 - (1) LIFUS, provided that the TRI training course has included the relevant training specified in point [FCL.930.TRI\(ab\)\(4\)](#)
 - (2) landing training, provided that the TRI training course has included the relevant training specified in point [FCL.930.TRI\(ab\)\(4\)](#); or
 - (3) the training flight specified in point FCL.060(c)(2), provided that the TRI training course has included the training referred to in points (a)(1) or (a)(2).

The restriction to FSTD shall be removed if TRIs have completed an assessment of competence in the aircraft.

- (b) TRIs for aeroplanes and for powered-lift aircraft — TRI(A) and TRI(PL). The privileges of TRIs are restricted to the type of aeroplane or powered-lift aircraft in which the training and the assessment of competence were conducted. Unless otherwise determined in the OSD, to extend the privileges of TRIs to further types, TRIs shall have:
 - (1) completed within the 12 months preceding the application, at least 15 route sectors, including take-offs and landings on the applicable aircraft type, of which of maximum of 7 sectors may be completed in an FSTD;
 - (2) completed the relevant parts of the technical training and the flight instruction parts of the applicable TRI course;
 - (3) passed the relevant sections of the assessment of competence in accordance with point [FCL.935](#) in order to demonstrate to an FIE or a TRE qualified in accordance with Subpart

K to this Annex their ability to instruct a pilot to the level required for the issue of a type rating, including pre-flight, post-flight and theoretical knowledge instruction.

The privileges of TRIs shall be extended to further variants, provided that they comply with point [FCL.710](#) in those other variants and, where applicable, the elements set out in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012 for those variants

(c) TRIs for helicopters — TRI(H).

(1) The privileges of TRIs(H) are restricted to the type of helicopter in which the assessment of competence for the issue of the TRI certificate was taken. Unless otherwise determined in the OSD, the privileges of the TRIs shall be extended to further types if TRIs have:

- (i) completed the relevant parts of the technical training and flight instruction parts of the TRI course;
- (ii) completed within the 12 months preceding the date of application, at least 10 hours on the applicable helicopter type, of which a maximum of 5 hours may be completed in an FFS or FTD 2/3; and
- (iii) passed the relevant sections of the assessment of competence in accordance with point [FCL.935](#) in order to demonstrate to an FIE or a TRE qualified in accordance with Subpart K of this Annex their ability to instruct a pilot to the level required for the issue of a type rating, including pre-flight, post-flight and theoretical knowledge instruction.

The privileges of TRIs shall be extended to further variants, provided that they comply with point [FCL.710](#) in those other variants and, where applicable, the elements set out in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012 for those variants.

- (2) In order to extend the privileges of a TRI(H) to multi-pilot operations in the same type of single-pilot helicopters, the holder shall have at least 350 hours of flight time as a pilot in multi-pilot operations in any aircraft category or have at least 100 hours of flight time as a pilot in multi-pilot operations on the specific type within the last 2 years.
- (3) Before the privileges of a TRI(H) are extended from single-pilot helicopters to multi-pilot helicopters, the holder shall comply with point [FCL.915.TRI\(d\)\(3\)](#).

(d) Notwithstanding the points above, holders of a TRI certificate who received a type rating in accordance with point [FCL.725\(e\)](#) shall be entitled to have their TRI privileges extended to that new type of aircraft.

GM1 FCL.910.TRI TRI Restricted privileges

ED Decision 2020/005/R

(a) The restrictions of the TRI privileges are annotated on the license under 'Remarks and Restrictions' against the appropriate TRI certificate, along with the following endorsements:

- (1) if the training is carried out in an FSTD: 'TRI/r' (r=restricted);
- (2) if the TRI training, as specified in point [FCL.910.TRI\(a\)\(1\)](#), includes the LIFUS training: endorsement as per point (a) and 'LIFUS'; and

- (3) if the landing training, as specified in point FCL.910.TRI(a)(2), is included in the TRI training course: endorsement as per point (a) and 'LT' (LT = landing training).
- (b) For example a TRI restricted with LIFUS and landing training privileges will have on their license the following endorsement: 'TRI/r LIFUS LT'.

GM1 FCL.910.TRI(b)(2) TRI training for type extension

ED Decision 2020/005/R

'Relevant parts of the technical training and the flight instruction parts of the applicable TRI training course' means that the training should be relevant to its purpose, taking into consideration the experience of the individual TRI on other aircraft types that are similar to the one for which the extension of TRI privileges is applied for.

FCL.915.TRI TRI – Prerequisites

Regulation (EU) 2021/2227

An applicant for a TRI certificate shall:

- (a) hold a CPL, MPL or ATPL pilot licence on the applicable aircraft category;
- (b) for a TRI(MPA) certificate:
 - (1) have completed 1 500 hours flight time as a pilot on multi-pilot aeroplanes; and
 - (2) have completed, within the 12 months preceding the date of application, 30 route sectors, including take-offs and landings, as PIC or co-pilot on the applicable aeroplane type, of which 15 sectors may be completed in an FFS representing that type;
- (c) for a TRI(SPA) certificate:
 - (1) have completed, within the 12 months preceding the date of the application, at least 30 route sectors, including take-offs and landings, as PIC on the applicable aeroplane type, of which a maximum of 15 sectors may be completed in an FSTD representing that type; and
 - (2) (i) have competed at least 500 hours flight time as pilot on aeroplanes, including 30 hours as PIC on the applicable type of aeroplane; or
 - (ii) hold or have held an FI certificate for multi-engine aeroplanes with IR(A) privileges;
- (d) for TRI(H):
 - (1) for a TRI(H) certificate for single-pilot single-engine helicopters, either:
 - (i) have completed 250 hours as a pilot on helicopters; or
 - (ii) hold an FI(H) certificate.
 - (2) for a TRI(H) certificate for single-pilot multi-engine helicopters, either:
 - (i) have completed 500 hours as pilot of helicopters, including 100 hours as PIC in single-pilot multi-engine helicopters; or
 - (ii) hold an FI(H) certificate and have completed 100 hours of flight time as a pilot in multi-engine helicopters.
 - (3) for a TRI(H) certificate for multi-pilot helicopters, have completed 1 000 hours of flight time as a pilot on helicopters, and have either 350 hours in multi-pilot operations on any

aircraft category or 100 hours of flight time as a pilot in multi-pilot operations on the type for which the TRI(H) certificate is sought;

- (e) for TRI(PL):
- (1) have completed 1 500 hours flight time as a pilot on multi-pilot aeroplanes, powered-lift, or multi-pilot helicopters; and
 - (2) have completed, within the 12 months preceding the application, 30 route sectors, including take-offs and landings, as PIC or co-pilot on the applicable powered-lift type, of which 15 sectors may be completed in an FFS representing that type.

FCL.930.TRI TRI – Training course

Regulation (EU) 2024/2076

- (a) The TRI training course shall be conducted:
- (1) in the case of MPA, in an FFS or a combination of FSTD(s) and FFS;
 - (2) in the case of type ratings for SPA and helicopters, in either of the following:
 - (i) an available and accessible FFS, or in a combination of such FFS and FSTD(s);
 - (ii) a combination of FSTD(s) and the aircraft if an FFS is not available or accessible;
 - (iii) the aircraft if no FSTD is available or accessible.
- (aa) By way of derogation from point (a), the TRI training course:
- (1) for non-complex helicopters may be conducted in a combination of FSTD(s) and the aircraft even if an FFS is available and accessible;
 - (2) for aircraft referred to in Section A, point 1e, of Appendix 9 may be conducted in accordance with points FCL.930.TRI(a)(2)(i), (ii) or (iii), irrespective of the availability and accessibility of FFS or FSTD.
- (ab) The TRI training course shall include:
- (1) 25 hours of teaching and learning;
 - (2) 10 hours of technical training, including revision of technical knowledge, the preparation of lesson plans and the development of classroom or simulator instructional skills;
 - (3) 5 hours of flight instruction on the appropriate aircraft or an FSTD representing that aircraft for single-pilot aircraft and 10 hours for multi-pilot aircraft or an FSTD representing that aircraft;
 - (4) if privileges for conducting LIFUS or take-off and landing training are sought, the following training, as applicable:
 - (i) in an FSTD and related to the relevant privileges, additional specific training which, in the case of privileges for conducting take-off and landing training, shall include training for emergency procedures related to the aircraft;
 - (ii) consolidation of the specific training in accordance with point (4)(i), where the applicant shall, in the aircraft, provide LIFUS or take-off and landing training, as applicable, under the supervision of a TRI that is nominated for that purpose by the ATO.
- (b) Applicants holding or having held an instructor certificate shall be fully credited towards the requirement of (a)(1).

- (c) An applicant for a TRI certificate who holds an SFI certificate for the relevant type shall be fully credited towards the requirements of this paragraph for the issue of a TRI certificate restricted to flight instruction in simulators.

AMC1 FCL.930.TRI TRI Training course

ED Decision 2025/002/R

TRI TRAINING COURSE — AEROPLANES

(a) General

- (1) The training course should develop safety awareness throughout by imparting knowledge, skills, and attitudes relevant to the TRI task, and should be designed to adequately train the candidate instructor in theoretical-knowledge instruction, flight instruction, and FSTD instruction to enable the candidate instructor to instruct others on an aeroplane type rating for which the candidate instructor is qualified.
- (2) The TRI(A) training course should place particular emphasis on the role of the individual, human factors in the man–machine environment, and CRM.
- (3) Special attention should be given to the candidate instructor’s maturity and judgment including their understanding of adults, behavioural attitudes, and variable levels of learning ability. During the training course, the candidate instructor should be made aware of their own attitude towards the importance of flight safety.
- (4) For a TRI(A), the amount of time for flight training should vary depending on the complexity of the aeroplane type. A similar number of hours should be allotted to the instruction on, and practice of, both preflight and postflight briefing for each exercise.
- (5) The flight instruction should ensure that the candidate instructor is able to teach the air exercises safely and efficiently and should be related to the type of aeroplane on which the candidate instructor wishes to instruct. The content of the training programme should cover training exercises applicable to the aeroplane type, which are set out in the applicable type rating training courses.
- (6) Airmanship is a vital element of all flight operations. Therefore, in the following exercises, the relevant aspects of airmanship should be stressed at the appropriate times during each flight.
- (7) The candidate instructor should learn how to identify common errors and how to correct them properly, which should be emphasised at all times.

(b) Content

The training course consists of three parts:

- Part 1: teaching and learning instruction in accordance with [AMC1 FCL.920](#);
 - Part 2: technical theoretical-knowledge instruction (technical training); and
 - Part 3: flight instruction.
- (1) Part 1 — Teaching and learning The content of the teaching and learning part of the FI training course as described in [AMC1 FCL.930.FI](#) should be used as guidance to develop the course syllabus.
 - (2) Part 2 — Technical theoretical-knowledge instruction syllabus

- (i) If a TRI(A) certificate for MP aeroplanes is sought, particular attention should be given to MCC. If a TRI(A) certificate for SP aeroplanes is sought, particular attention should be given to the duties in SP operations.
 - (ii) The technical theoretical-knowledge instruction should comprise at least 10 hours of training to refresh Part-1 theoretical topics, as necessary, and aircraft technical knowledge. It should include preparation of lesson plans and development of briefing-room instructional skills. A proportion of the allotted 10 hours could be integrated into the practical flight instruction lessons of Part 3, using expanded preflight and postflight briefing sessions. Consequently, for practical purposes, Part 2 and Part 3 could be considered complementary to each other.
 - (iii) The type rating theoretical syllabus should be used to develop the TRI(A)'s teaching skills in relation to the type technical course syllabus. The course instructor should deliver example lectures from the applicable type technical syllabus and the candidate instructor should prepare and deliver lectures on topics that are selected by the course instructor from the type rating course.
- (3) Part 3 — Flight instruction
- (i) General
 - (A) The course should be related to the type of aeroplane on which the applicant wishes to instruct. It should consist of at least 5 hours of flight instruction for SP aeroplanes that are operated in SP operations, and at least 10 hours for MP aeroplanes or SP-certified aeroplanes that are operated in MP operations, per candidate instructor.
 - (B) TEM, CRM, and the appropriate use of behavioural markers should be integrated throughout.
 - (C) Training courses should be developed to help the candidate instructor gain experience in the training of a variety of exercises, covering both normal and abnormal operations.
 - (D) The syllabus should be tailored and appropriate to the aeroplane type, and the exercises used should be more demanding for each individual student.
 - (E) The course should cover the whole range of instructor skills to enable the candidate instructor to plan sessions, brief, train and debrief using all relevant training techniques that are appropriate to pilot training.
 - (ii) Use of FSTDs
 - (A) The applicant for a TRI(A) certificate should be instructed in using the device and made familiar with its limitations, capabilities, and safety features, including emergency evacuation.
 - (B) The applicant for a TRI(A) certificate should be instructed in providing and evaluating training from the instructor station and from all pilot operating positions, including demonstrations of handling exercises.
 - (C) The syllabus should include engine-out handling and engine-out operations in addition to representative exercises from the type rating course.
 - (D) Where no FSTD exists for the type of aeroplane for which the certificate is sought, or if the FSTD is not suitable to complete all the elements of the training programme for the TRI certificate, the entire course or a part of it

should be conducted in the applicable aeroplane type, and the synthetic-device elements should be replaced with appropriate exercises in the aeroplane.

The assessment of competence should be performed:

- when no FSTD exists, in the aeroplane; and
- when not all elements of the training are completed in the FSTD, in both the aeroplane and the FSTD; this combined use of aeroplane and FSTD in the assessment of competence should reflect and be similar to the combined use of aeroplane and FSTD during the training course.

(F) In general, TRI training is designed to develop the competencies of a pilot to become an instructor. From this perspective, the training may be provided in several arrangements:

- the candidate instructor is seating in either pilot seat;
- the candidate instructor is seating at the IOS; or
- the candidate instructor is observing (seating as an observer).

The combination of the above-mentioned training arrangements and the allocation of time to each one of them depends on an analysis of several elements, including but not limited to the following:

- previous experience and curriculum of each candidate (e.g. previous instructor experience, experience on aeroplane type, total flight experience, etc.) in isolation and as part of the course group(s);
- specific requirements for aeroplane type and related training exercises;
- overall maturity and experience of the ATO in providing TRI training courses; and
- type, fidelity level, and reliability of the available devices.

Subject to particular training arrangements that are determined by the ATO and approved by the competent authority, a TRI may instruct in parallel two TRI candidate instructors under the following scenarios:

- one candidate is sitting at the controls (supported by a suitable pilot), while the second candidate is sitting at the IOS; this scenario may be used for demonstration of flight manoeuvres or engine out exercises; or
- both candidates receive instruction (general introduction and handling) at the IOS.

In this way, both candidates can independently develop specific competencies.

Additional TRI candidate instructors may be present as observers during such an instruction given in parallel, with no credit of hours for their TRI training.

For an initial TRI training course, such ‘in parallel’ instruction should be given only for a reasonable part of the overall TRI training course duration. For a

TRI type extension, the amount of hours required for such an instruction may be increased.

In any case, the way of instruction largely depends on the experience of the TRI trainer in the various training arrangements and on the general experience of the candidate instructor.

(iii) SP MET aeroplane training for asymmetric power flight

During this part of the training, particular emphasis should be placed on:

- (A) the circumstances under which the actual feathering and unfeathering is practised, e.g. safe altitude, compliance with regulations regarding minimum altitude or height for feathering, weather conditions, distance from the nearest available aerodrome;
- (B) the procedure that should be used for cooperation between instructor and student, e.g. the correct use of touch drills and the prevention of misunderstandings, especially during feathering and unfeathering and when zero thrust is used for asymmetric circuits; this procedure should include a positive agreement on which engine should be shut down or restarted or set at zero thrust, as well as on identifying each control and the engine it will affect;
- (C) avoiding overworking the operating engine and preventing degraded performance when operating the aeroplane in asymmetric flight; and
- (D) the need to use the specific checklist for the given aeroplane type.

(iv) Long briefings

Note: Hereunder are listed the subjects of the long briefings for the SP MET aeroplanes. Those long briefings should be adapted, if applicable, to the type of aeroplane for which the privileges are sought.

Long briefings provide an essential link between academic principles and air exercises. They introduce aeronautical theory and the practical application of aeronautical principles to the student.

The instructor should ensure that the candidate instructor is able to teach long briefings with regard to all the following subjects:

- (A) Asymmetric power flight:
 - (a) introduction to asymmetric flight;
 - (b) feathering the propeller: method of operation;
 - (c) effects on aeroplane handling at cruising speed;
 - (d) introduction to the effects upon aeroplane performance;
 - (e) identification of the foot load to maintain a constant heading (no rudder trim);
 - (f) feathering the propeller: regaining normal flight;
 - (g) finding the zero-thrust setting: comparison of foot load when the propeller is feathered and thrust is set to zero;
 - (h) effects and recognition of engine failure in level flight;

- (i) forces and effects of yaw;
 - (j) types of failure:
 - (1) sudden or gradual, and
 - (2) complete or partial;
 - (k) yaw direction and further effects of yaw;
 - (l) flight instrument indications;
 - (m) identification of failed engine;
 - (n) couples and residual out-of-balance forces: resultant flight attitude;
 - (o) use of rudder to counteract yaw;
 - (p) use of aileron: dangers of misuse;
 - (q) use of elevator to maintain level flight;
 - (r) use of power to maintain safe airspeed and altitude;
 - (s) supplementary recovery to straight and level flight: simultaneous increase in speed and reduction in power;
 - (t) identification of failed engine: idle engine;
 - (u) use of engine instruments for identification:
 - (1) fuel pressure or flow;
 - (2) RPM gauge response effect of constant speed unit (CSU) action at lower and higher airspeed; and
 - (3) engine temperature gauges;
 - (v) confirmation of identification: closing the throttle of the identified failed engine;
 - (w) effects and recognition of engine failure in turns;
 - (x) identification and control; and
 - (y) side forces and effects of yaw.
- (B) Turning flight:
- (a) effect of 'inside' engine failure: sudden and pronounced effect;
 - (b) effect of 'outside' engine failure: less sudden and pronounced effect;
 - (c) possible confusion in identification (particularly at low power):
 - (1) correct use of rudder; and
 - (2) possible need to return to lateral level flight to confirm correct identification;
 - (d) visual and flight instrument indications;
 - (e) effect of varying speed and power;
 - (f) speed and thrust relationship;

- (g) at normal cruising speed and cruising power: engine failure clearly recognised;
 - (h) at low safe speed and climb power: engine failure most likely recognised; and
 - (i) at high-speed descent and low power: asymmetry (engine failure) possibly not recognised.
- (C) Minimum control speeds:
- (a) Air speed indicator (ASI) colour coding: red radial line. Note: this exercise is intended to explore the ultimate boundaries of controllability of the aeroplane aircraft in an asymmetric state in various conditions with a steady power setting. A steady power setting is achieved by using a fixed power setting and adjusting the aircraft attitude to obtain a gradual speed reduction. The failure exercise should not be performed as a sudden and complete failure at the VMCA given in the AFM. The purpose of the exercise is to continue the gradual introduction of a student to the control of an aeroplane in asymmetric power flight in extreme or critical situations, and not to demonstrate VMCA.
 - (b) Techniques for assessing critical speeds at wings level, and recovery from those speeds; dangers involved when minimum control speed and stalling speed are very close: use of safe single-engine speed (V_{sse}).
 - (c) Establishing a minimum control speed for each asymmetrically disposed engine: establishing the critical engine (if applicable).
 - (d) Effects on minimum control speeds of:
 - (i) bank;
 - (ii) zero-thrust setting; and
 - (iii) take-off configuration:
 - (A) landing gear down and take-off flap set; and
 - (B) landing gear up and take-off flap set.

Note: the use of 5 ° of bank towards the operating engine results in a better climb performance than that obtained with wings level held. Manufacturers may use these conditions when determining the asymmetric climb performance of the aircraft.

Thus, the VMCA quoted in the AFM may be different from the speeds that are determined during this exercise.
- (D) Feathering and unfeathering:
- (a) minimum heights for practising feathering and unfeathering drills; and
 - (b) engine-handling precautions (overheating, icing conditions, priming, warm-up, method of simulating an engine failure: refer to the aircraft engine manual, service instructions, and bulletins).

- (E) Engine failure procedure:
 - (a) once control is maintained, the phase of operation and the aircraft type determine in which order the procedures should be followed; and
 - (b) the flight phase should be:
 - (1) in cruising flight; or
 - (2) a critical phase, e.g. immediately after take-off or during approach to landing or during a go-around.

- (F) Aircraft type:

Variations in the order of certain drills and checks inevitably occur due to differences between aeroplane types and perhaps between models of the same aeroplane type. The AFM should be consulted to establish the exact order of the related procedures.

For example, one AFM may call for the raising of flaps and landing gear before feathering, whereas another AFM may recommend feathering as a first step. The reason for this latter procedure may be that some engines cannot be feathered if RPM drop below a certain figure.

However, in some aeroplanes, the raising of the landing gear may create more drag during retraction due to the transient position of the landing gear doors, and as a result, retraction should be avoided until feathering is completed and propeller drag reduced.

Therefore, the order in which the drills and checks are presented under immediate and subsequent actions in this syllabus should be considered as general guidance only; the exact order of precedence is determined by reference to the AFM for the specific aeroplane type used in the course.

- (G) In-flight engine failure during cruising or other flight phase not including take-off or landing:
 - (a) immediate actions:
 - (1) control of the aircraft;
 - (2) recognition of asymmetric condition;
 - (3) identification and confirmation of failed engine:
 - (i) idle leg = idle engine; and
 - (ii) closing of throttle or pulling back of power lever, as appropriate, for confirmation;
 - (4) identification of failure cause and fire check:
 - (i) typical reasons for failure; and
 - (ii) methods of rectification; and
 - (5) feathering decision and procedure:
 - (i) reduction of other drag;
 - (ii) need for speed but not haste; and

- (iii) use of rudder trim;
- (b) subsequent actions:
 - (1) operating engine:
 - (i) temperature, pressure, and power;
 - (ii) remaining services;
 - (iii) electrical load: assess and reduce, as necessary;
 - (iv) effect on power source for air-driven instruments;
 - (v) landing gear; and (vi) flaps and other services;
 - (2) replanning of the flight:
 - (i) ATC and weather;
 - (ii) terrain clearance, SE cruising speed; and
 - (iii) decision to divert or continue;
 - (3) fuel management: best use of remaining fuel;
 - (4) dangers of restarting damaged engine;
 - (5) action if unable to maintain altitude: effect of altitude on available power;
 - (6) effects on performance;
 - (7) effects on available power and required power;
 - (8) effects on various airframe configurations and propeller settings;
 - (9) use of AFM:
 - (i) cruising;
 - (ii) climbing: ASI colour coding (blue line);
 - (iii) descending; and (iv) turning;
 - (10) limitations and handling of operating engine; and
 - (11) control and performance of take-off and approach.
- (H) Significant factors:
 - (a) significance of take-off safety speed:
 - (1) effect on aeroplane performance of landing gear, flap, feathering, take-off, trim setting, and systems for operating landing gear and flaps; and
 - (2) effect on aeroplane performance of mass, altitude, and temperature;
 - (b) significance of best SE climb speed (Vyse):
 - (1) accelerating to Vyse and establishing a positive climb;
 - (2) relationship between Vyse and normal climb speed; and

- (3) action, if unable to climb; and
 - (c) significance of asymmetric committal height and speed: action, if baulked below asymmetric committal height.
- (I) Engine failure during take-off:
 - (a) below VMCA or unstick speed:
 - (1) use AFM data, if available ; and
 - (2) accelerate or stop distance considerations;
 - (b) above VMCA or unstick speed and below safety speed;
 - (c) immediate relanding or use of remaining power for forced landing; and
 - (d) considerations:
 - (1) degree of engine failure;
 - (2) speed at the time;
 - (3) mass, altitude, temperature performance;
 - (4) configuration;
 - (5) length of remaining runway; and
 - (6) position of any obstacles ahead.
- (J) Engine failure after take-off:
 - (a) simulated at a safe height and at or above take-off safety speed;
 - (b) considerations:
 - (1) need to maintain control;
 - (2) use of bank technique towards operating engine;
 - (3) use of available power to reach Vyse;
 - (4) mass, altitude, temperature performance; and
 - (5) effect of prevailing conditions and circumstances;
 - (c) immediate actions:
 - (1) maintaining control, including airspeed and use of power;
 - (2) recognition of asymmetric condition;
 - (3) identification and confirmation of failed engine;
 - (4) feathering and removal of drag (procedure for specific type); and
 - (5) reaching and maintaining Vyse; and
 - (d) subsequent actions, whilst carrying out an asymmetric power climb to the downwind position at Vyse:
 - (1) identification of failure and fire check;
 - (2) handling considerations for operating engine;

- (3) remaining services;
- (4) liaison with ATC; and
- (5) fuel management.

Note: these procedures are dependent upon the aeroplane type concerned and actual flight situation.

(K) Asymmetric committal height

- (a) Asymmetric committal height is the minimum height needed to put the aircraft into a positive climb, whilst maintaining an adequate speed to control the aircraft and reduce drag during an approach to landing.
- (b) Due to the significantly reduced performance of many CS-23 aeroplanes when operating with one engine, a minimum height should be considered from which it would be safe to attempt a go-around procedure during an approach when the aeroplane must change from descent to climb in a high-drag configuration.
- (c) Due to the height loss that occurs when the operating engine is turned to full power, with landing gear and flap retracted, and the aeroplane is put into a climb at Vyse, a minimum height (often referred to as 'asymmetric committal height') should be selected below which the pilot should not attempt to fly another circuit. This height should be compatible with the aeroplane type, all-up weight, altitude of the aerodrome used, air temperature, wind, height of obstructions along the climb-out path, and the pilot's competence.
- (d) Circuit approach and landing with asymmetric power:
 - (1) definition and use of asymmetric committal height;
 - (2) use of standard pattern and normal procedures;
 - (3) action, if unable to maintain circuit height;
 - (4) speed and power settings required; and
 - (5) decision to land or execute a go-around at asymmetric committal height: factors to be considered.
- (e) Undershooting: importance of maintaining an appropriate airspeed.

(L) Speed and heading control:

- (a) relationship between height, speed, and power: need for minimum possible drag; and
- (b) reaching a positive climb at Vyse:
 - (1) effect of availability of systems, and power for the flap and landing gear; and
 - (2) operation and rapid clean-up.

Note 1: the airspeed at which the decision is made to make a landing or execute a go-around should normally be Vyse and not lower than the safety speed.

Note 2: instrument approach ‘decision height’ and its associated procedures should not be confused with the selection of minimum height for initiating a go-around in asymmetric power flight.

- (M) Engine failure during an all-engine approach or missed approach:
 - (a) use of asymmetric committal height, and speed considerations; and
 - (b) speed and heading control: decision to attempt a landing, go-around or forced landing depending on circumstances. Note: at least one demonstration and practice of engine failure in this situation should be performed during the course.
- (N) Instrument flying with asymmetric power:
 - (a) considerations relating to aircraft performance during:
 - (1) straight and level flight;
 - (2) climb and descent;
 - (3) standard rate turns; and
 - (4) level, climbing, and descending turns including turns to preselected headings;
 - (b) availability of vacuum-operated instruments; and
 - (c) electrical power source.

- (v) Specific trainings: LIFUS training and landing training

The applicant for a TRI(A) certificate should receive instruction in an FSTD in accordance with point [FCL.930.TRI\(ab\)\(4\)](#).

- (A) LIFUS training: content
 - (a) Training in an FSTD:
 - (1) familiarisation as PF on both seats, as applicable, which should include at least the following:
 - (i) pre-flight preparation and use of checklists;
 - (ii) taxiing;
 - (iii) take-off;
 - (iv) rejected take-off;
 - (v) engine failure during take-off, after take-off decision speed (V1);
 - (vi) one-engine-inoperative approach and go-around;
 - (vii) one-engine-inoperative (critical, simulated) landing;
 - (viii) other emergency and abnormal operating procedures (as necessary);
 - (ix) emergency evacuations; and
 - (x) task sharing and decision-making; and

- (2) aeroplane training techniques:
 - (i) methods of providing appropriate commentary; and
 - (ii) intervention strategies developed from situations that are role-played by a TRI training course instructor, taken from but not limited to:
 - (A) take-off:
 - tail strike awareness and avoidance,
 - rejected take-off,
 - actual engine failure,
 - take-off configuration warning, and
 - overcontrolling;
 - (B) approach and landing:
 - normal approach,
 - high flare, long float, no flare,
 - immediate go-around after touchdown,
 - baulked landing,
 - rejected landing,
 - crosswind, and
 - overcontrolling; and
 - (C) flight management:
 - task sharing and handover of controls,
 - effect of ATC-delaying actions on endurance,
 - alternate management and diversion, and
 - traffic awareness when flying in pattern.

(b) Consolidation of FSTD training in an aeroplane

Upon completion of the FSTD training in accordance with point (a), the candidate instructor should complete at least one route sector where he or she:

- (1) either observes a TRI(A) who conducts line flying under supervision, or
- (2) conducts role play line flying under supervision for a TRI(A) who is qualified for line flying under supervision.

Upon completion of the above-mentioned tasks under supervision, the candidate instructor should complete a route sector in the role of a TRI under the supervision and to the satisfaction of a TRI(A) who is nominated for that purpose by the ATO.

(B) Landing training: content

(a) Training in an FSTD

The training in an FSTD should be tailored and appropriate to the aeroplane type, and the exercises should be more demanding for each candidate instructor. In addition to the LIFUS training items in the FSTD (listed under (a)(1) and (a)(2) above), the landing training should comprise a variety of exercises that cover both normal and abnormal operations including the following:

- (1) consideration of threats during touch-and-go:
 - operating at low altitude;
 - General Aviation (GA) traffic;
 - increased fuel consumption;
 - bird strikes;
 - decision to continue touch-and-go or make a full-stop landing; and
 - aspects of performance and associated risks;
- (2) incorrect rudder inputs;
- (3) failure of a critical engine;
- (4) approach and full-stop landing in simulated engine-out flight; and
- (5) go-around in simulated engine-out flight. The applicant needs to be additionally trained in other abnormal items during the training course, if required.

(b) Consolidation of FSTD training in an aeroplane

- (1) Upon completion of the FSTD training in accordance with point (a), the applicant should perform role-play flying for landing training under the supervision and to the satisfaction of a TRI(A) who is nominated for that purpose by the ATO.

The training should cover at least the following elements:

- take-off,
 - traffic pattern,
 - touch-and-go,
 - go-around, and
 - full-stop landing with different flap settings.
- (2) In exceptional circumstances, it may be necessary to perform simulated engine-out handling and engine-out operations in an aeroplane in addition to representative exercises from the type rating course.

(vi) UPRT

Instructors should have the specific competence to provide UPRT during the type rating training course, including the ability to demonstrate knowledge and

understanding of the type-specific upset recovery procedures and of the recommendations that are developed by the original equipment manufacturers (OEMs). Therefore, during the TRI training course, the student instructor should:

- (A) be able to apply the correct upset recovery techniques for the specific aeroplane type;
- (B) understand the importance of applying type-specific OEM procedures for recovery manoeuvres;
- (C) be able to distinguish between the applicable SOPs and OEM recommendations (if available);
- (D) understand the capabilities and limitations of the FSTDs that are used for UPRT;
- (E) ensure that the training remains within the FSTD training envelope to avoid the risk of negative transfer of training;
- (F) understand and be able to use the IOS of the FSTD in the context of providing effective UPRT;
- (G) understand and be able to use the available FSTD instructor tools to provide accurate feedback on pilot performance;
- (H) understand the importance of adhering to the FSTD UPRT scenarios that are validated by the training programme developer; and
- (I) understand the missing critical human factor aspects due to the limitations of the FSTD, and convey this to the student pilot(s) receiving the training.

AMC2 FCL.930.TRI TRI — Training course

ED Decision 2025/002/R

HELICOPTERS

GENERAL

- (a) The aim of the TRI(H) course is to train helicopter licence holders to the level of competence defined in [FCL.920](#) and adequate for a TRI.
- (b) The training course should develop safety awareness throughout by teaching the knowledge, skills and attitudes relevant to the TRI(H) task, and should be designed to give adequate training to the applicant in theoretical knowledge instruction, flight instruction and FSTD instruction to instruct for a helicopter type rating for which the applicant is qualified.
- (c) The TRI(H) training course should give particular emphasis to the role of the individual in relation to the importance of human factors in the man-machine environment and the role of CRM.
- (d) Special attention should be given to the applicant's maturity and judgment including an understanding of adults, their behavioural attitudes and variable levels of learning ability. During the training course the applicants should be made aware of their own attitudes to the importance of flight safety. It will be important during the course of training to aim at giving the applicant the knowledge, skills and attitudes relevant to the role of the TRI.
- (e) For a TRI(H) certificate the amount of flight training will vary depending on the complexity of the helicopter type.

- (f) A similar number of hours should be used for the instruction and practice of pre-flight and post flight briefing for each exercise. The flight instruction should aim to ensure that the applicant is able to teach the air exercises safely and efficiently and should be related to the type of helicopter on which the applicant wishes to instruct. The content of the training program should cover training exercises applicable to the helicopter type as set out in the applicable type rating course syllabus.
- (g) A TRI(H) may instruct in a TRI(H) course once he or she has conducted a minimum of four type rating instruction courses.

CONTENT

- (h) The training course consists of three parts:
 - (1) Part 1: teaching and learning, that should comply with [AMC1 FCL.920](#);
 - (2) Part 2: technical theoretical knowledge instruction (technical training);
 - (3) Part 3: flight instruction.

Part 1

The content of the teaching and learning part of the FI training course, as established in [AMC1 FCL.930.FI](#), should be used as guidance to develop the course syllabus.

Part 2

TECHNICAL THEORETICAL KNOWLEDGE INSTRUCTION SYLLABUS

- (a) The technical theoretical knowledge instruction should comprise of not less than 10 hours training to include the revision of technical knowledge, the preparation of lesson plans and the development of classroom instructional skills to enable the TRI(H) to instruct the technical theoretical knowledge syllabus.
- (b) If a TRI(H) certificate for MP helicopters is sought, particular attention should be given to multi-crew cooperation.
- (c) The type rating theoretical syllabus should be used to develop the TRI(H)'s teaching skills in relation to the type technical course syllabus. The course instructor should deliver example lectures from the applicable type technical syllabus and the candidate instructor should prepare and deliver lectures on topics selected by the course instructor from the subject list below:
 - (1) helicopter structure, transmissions, rotor and equipment, normal and abnormal operation of systems:
 - (i) dimensions;
 - (ii) engine including aux. power unit, rotors and transmissions;
 - (iii) fuel system;
 - (iv) air-conditioning;
 - (v) ice protection, windshield wipers and rain repellent;
 - (vi) hydraulic system;
 - (vii) landing gear;
 - (viii) flight controls, stability augmentation and autopilot systems;

- (ix) electrical power supply;
 - (x) flight instruments, communication, radar and navigation equipment;
 - (xi) cockpit, cabin and cargo compartment;
 - (xii) emergency equipment.
- (2) limitations:
 - (i) general limitations, according to the helicopter flight manual;
 - (ii) minimum equipment list.
- (3) performance, flight planning and monitoring:
 - (i) performance;
 - (ii) flight planning.
- (4) load and balance and servicing:
 - (i) load and balance;
 - (ii) servicing on ground;
- (5) emergency procedures;
- (6) special requirements for helicopters with EFIS;
- (7) optional equipment.

Part 3

FLIGHT INSTRUCTION SYLLABUS

- (a) The amount of flight training will vary depending on the complexity of the helicopter type. At least 5 hours flight instruction for a SP helicopter and at least 10 hours for a MP ME helicopter should be counted. A similar number of hours should be used for the instruction and practice of pre-flight and post flight briefing for each exercise. The flight instruction should aim to ensure that the applicant is able to teach the air exercises safely and efficiently and related to the type of helicopter on which the applicant wishes to instruct. The content of the training programme should only cover training exercises applicable to the helicopter type as set out in [Appendix 9](#) to Part-FCL.
- (b) If a TRI(H) certificate for MP helicopters is sought, particular attention should be given to MCC.
- (c) If a TRI(H) certificate for revalidation of instrument ratings is sought, then the applicant should hold a valid instrument rating.

FLIGHT OR FSTD TRAINING

- (d) The training course should be related to the type of helicopter on which the applicant wishes to instruct.
- (e) For MP helicopter type ratings MCC, CRM and the appropriate use of behavioural markers should be integrated throughout.
- (f) The content of the training programme should cover identified and significant exercises applicable to the helicopter type.

FSTD TRAINING

- (g) The applicant for a TRI(H) certificate should be taught and made familiar with the device, its limitations, capabilities and safety features, and the instructor station.
- (h) The applicant for a TRI(H) certificate should be taught and made familiar with giving instruction from the instructor station seat as well as the pilot's seats, including demonstrations of appropriate handling exercises.
- (i) Training courses should be developed to give the applicant experience in training a variety of exercises, covering both normal and abnormal operations. The syllabus should be tailored appropriate to the helicopter type, using exercises considered more demanding for the student. This should include engine-out handling and engine-out operations in addition to representative exercises from the type transition course.
- (j) The applicant should be required to plan, brief, train and debrief sessions using all relevant training techniques.
- (ja) In general, TRI training is designed to develop the competencies of a pilot to become an instructor. From this perspective, the training may be provided in several arrangements:
 - the candidate instructor is seating in either pilot seat;
 - the candidate instructor is seating at the IOS; or
 - the candidate instructor is observing (seating as an observer).

The combination of the above-mentioned training arrangements and the allocation of time to each one of them depends on an analysis of several elements, including but not limited to the following:

- previous experience and curriculum of each candidate (e.g. previous instructor experience, experience on aeroplane type, total flight experience, etc.) in isolation and as part of the course group(s);
- specific requirements for aeroplane type and related training exercises;
- overall maturity and experience of the ATO in providing TRI training courses; and
- type, fidelity level and reliability of the available devices.

Subject to particular training arrangements that are determined by the ATO and approved by the competent authority, a TRI may instruct in parallel two TRI candidate instructors under the following scenarios:

- one candidate is sitting at the controls (supported by a suitable pilot), while the second candidate is sitting at the IOS; this scenario may be used for demonstration of flight manoeuvres or engine-out exercises; or
- both candidates receive instruction (general introduction and handling) at the IOS.

In this way, both candidates can independently develop specific competencies.

Additional TRI candidate instructors may be present as observers during such an instruction given in parallel, with no credit of hours for their TRI training.

For an initial TRI training course, such 'in parallel' instruction should be given only for a reasonable part of the overall TRI training course duration. For a TRI type extension, the number of hours required for such an instruction may be increased.

In any case, the way of instruction largely depends on the experience of the TRI trainer in the various training arrangements and on the general experience of the candidate instructor.

HELICOPTER TRAINING

- (k) The applicant for a TRI(H) certificate should receive instruction in an FSTD to a satisfactory level in:
- (1) left hand seat familiarisation, and in addition right hand seat familiarisation where instruction is to be given to co-pilots operating in the left hand seat, which should include at least the following as pilot flying:
 - (i) pre-flight preparation and use of checklists;
 - (ii) taxiing: ground and air;
 - (iii) take-off and landings;
 - (iv) engine failure during take-off, before DPATO;
 - (v) engine failure during take-off, after DPATO;
 - (vi) engine inoperative approach and go-around;
 - (vii) one engine simulated inoperative landing;
 - (viii) autorotation to landing or power recovery;
 - (ix) other emergency and abnormal operating procedures (as necessary);
 - (x) instrument departure, approach and go-around with one engine simulated inoperative should be covered where TRI(H) privileges include giving instrument instruction for the extension of an IR(H) to additional types.
 - (2) helicopter training techniques:
 - (i) methods for giving appropriate commentary;
 - (ii) instructor demonstrations of critical manoeuvres with commentary;
 - (iii) particularities and safety considerations associated with handling the helicopter in critical manoeuvres such as one-engine-inoperative and autorotation exercises;
 - (iv) where relevant, the conduct of instrument training with particular emphasis on weather restrictions, dangers of icing and limitations on the conduct of critical manoeuvres in instrument meteorological conditions;
 - (v) intervention strategies developed from situations role-played by a TRI(H) course instructor, taken from but not limited to:
 - (A) incorrect helicopter configuration;
 - (B) over controlling;
 - (C) incorrect control inputs;
 - (D) excessive flare close to the ground;
 - (E) one-engine-inoperative take-off and landings;
 - (F) incorrect handling of autorotation;
 - (G) static or dynamic rollover on take-off or landing;
 - (H) too high on approach with associated danger of vortex ring or settling with power;
 - (I) incapacitation;

- (L) abnormal and emergency procedures and appropriate methods and minimum altitudes for simulating failures in the helicopter;
- (M) failure of the driving engine during OEI manoeuvres.

TRAINING WHERE NO FSTD EXISTS

- (m) Where no FSTD exists for the type for which the TRI(H) certificate is sought, a similar course of training should be conducted in the applicable helicopter type. This includes all elements listed under sub paragraphs (k)(1) and (2) of this AMC, the FSTD elements being replaced with appropriate exercises in a helicopter of the applicable type, subject to any restrictions placed on the conduct of critical exercises associated with helicopter flight manual limitations and safety considerations.

SPECIFIC TRAINING: LANDING TRAINING

- (n) If applicants for, or holders of, a TRI(H) certificate wish to include privileges for landing training in their certificate, the specific training for that purpose should include at least all the following:
 - (1) Considerations of threats related to:
 - (i) taxi, take-off, approach and landing;
 - (ii) operating at low altitude;
 - (iii) general aviation (GA) traffic;
 - (iv) increased fuel consumption;
 - (v) bird strikes;
 - (vi) aspects of performance and associated risks.
 - (2) FSTD training:
 - (i) taxiing on the ground and in the air;
 - (ii) aircraft handling manoeuvres in hover;
 - (iii) take-off, including different take-off profiles, as applicable;
 - (iv) traffic pattern;
 - (v) approach, go-around and landing, including different approach and landing profiles, as applicable;
 - (vi) engine failure during take-off and landing, including different take-off and landing profiles, as applicable;
 - (vii) engine inoperative during hover, take-off, approach and landing;
 - (viii) slope landing;
 - (ix) take-off, approach and go-around with one engine inoperative (multi-engine helicopters only);
 - (x) any further exercises, if specified in the applicable OSD.
 - (3) Consolidation of FSTD training in a helicopter

After completion of the specific training in the FSTD, applicants should demonstrate their ability to conduct landing training in accordance with the above syllabus as TRIs during role-play flying in the aircraft under the supervision and to the satisfaction of a TRI(H) who is nominated for that purpose by the ATO.

FCL.935.TRI TRI – Assessment of competence

Regulation (EU) 2019/1747

- (a) The assessment of competence for a TRI for MPA and PL shall be conducted in an FFS. If no FFS is available or accessible, an aircraft shall be used.
- (b) The assessment of competence for a TRI for single-pilot high-performance complex aeroplanes and helicopters shall be conducted in any of the following:
 - (1) an available and accessible FFS;
 - (2) if no FFS is available or accessible, in a combination of FSTD(s) and an aircraft;
 - (3) if no FSTD is available or accessible, in an aircraft.

FCL.940.TRI TRI – Revalidation and renewal

Regulation (EU) 2019/1747

(a) Revalidation

(1) Aeroplanes

To revalidate a TRI(A) certificate, applicants shall, within the 12 months immediately preceding the expiry date of the certificate fulfil at least two out of the three following requirements:

- (i) conduct one of the following parts of a complete type rating or recurrent training course: simulator session of at least 3 hours or one air exercise of at least 1 hour comprising a minimum of two take-offs and landings;
- (ii) complete instructor refresher training as a TRI(A) at an ATO;
- (iii) pass the assessment of competence in accordance with point [FCL.935](#). Applicants who have complied with point [FCL.910.TRI\(b\)\(3\)](#) shall be deemed to comply with this requirement.

(2) Helicopters and powered lift

To revalidate a TRI (H) or TRI(PL) certificate, applicants shall, within the validity period of the TRI certificate fulfil at least two out of the three following requirements:

- (i) completed at least 50 hours of flight instruction in each of the types of aircraft for which instructional privileges are held or in an FSTD representing those types, of which at least 15 hours shall be completed in the period of 12 months immediately preceding the expiry date of the TRI certificate. In the case of a TRI(PL), those hours shall be completed as a TRI or a type rating examiner (TRE), or as an SFI or a synthetic flight examiner (SFE). In the case of a TRI(H), the time flown as FIs, instrument rating instructors (IRIs), synthetic training instructors (STIs) or as any kind of examiners shall be accounted for this purpose;
- (ii) complete instructor refresher training as a TRI(H) or TRI(PL), as relevant, at an ATO;
- (iii) in the period of 12 months immediately preceding the expiry date of the certificate, passed an assessment of competence in accordance with points [FCL.935](#), [FCL.910.TRI\(b\)\(3\)](#) or [FCL.910.TRI\(c\)\(3\)](#), as applicable.

- (3) For at least each alternate revalidation of a TRI certificate, holders shall pass the assessment of competence in accordance with point [FCL.935](#).

- (4) If TRIs hold a certificate for more than one type of aircraft within the same category, the assessment of competence taken on one of those types of aircraft shall revalidate the TRI certificate for the other types held within the same category of aircraft, unless it is otherwise determined in the OSD.
 - (5) Specific requirements for the revalidation of a TRI(H) certificate
TRIs(H) holding an FI(H) certificate in the relevant type shall be deemed to comply with the requirements in point (a). In that case, the TRI(H) certificate shall be valid until the expiry date of the FI(H) certificate.
- (b) **Renewal**
- To renew a TRI certificate, applicants shall, within the 12 months immediately preceding the date of the application, have passed the assessment of competence in accordance with point [FCL.935](#) and shall have completed the following:
- (1) for aeroplanes:
 - (i) at least 30 route sectors, including take-offs and landings on the applicable aeroplane type, of which maximum 15 sectors may be completed in an FFS;
 - (ii) instructor refresher training as a TRI at an ATO which shall cover the relevant elements of the TRI training course;
 - (2) for helicopters and powered lift:
 - (i) at least 10 hours of flight time, including take-offs and landings on the applicable aircraft type, of which maximum 5 hours may be completed in an FFS or FTD 2/3;
 - (ii) instructor refresher training as a TRI at an ATO, which shall cover the relevant elements of the TRI training course.
 - (3) If applicants held a certificate for more than one type of aircraft within the same category, the assessment of competence taken on one of those types of aircraft shall renew the TRI certificate for the other types held within the same category of aircraft, unless it is otherwise determined in the OSD.

**AMC1 FCL.940.TRI(a)(1)(ii), (a)(2)(ii), (b)(1)(ii), (b)(2)(ii);
FCL.940.SFI(a)(2), (e)(1)**

ED Decision 2020/005/R

- (a) The refresher training for revalidation of the TRI and SFI certificates should be provided as a seminar. The seminar should consist of 6 hours of learning and may be held in the form of either one or more of the following: e-learning, two-way online meetings, face-to-face seminars. The content of the refresher seminar for revalidation should be selected from the following items:
- (1) relevant changes to national or EU regulations;
 - (2) the role of the instructor;
 - (3) teaching and learning styles;
 - (4) observational skills;
 - (5) instructional techniques;
 - (6) briefing and debriefing skills;
 - (7) TEM;

- (8) human performance and limitations;
 - (9) flight safety, prevention of incidents and accidents, including those specific to the ATO;
 - (10) significant changes in the content of the relevant part of the aviation system;
 - (11) legal aspects and enforcement procedures;
 - (12) developments in competency-based instruction;
 - (13) report writing; and
 - (14) any additional topics proposed by the competent authority.
- (b) For the refresher training for renewal of the TRI and SFI certificates:
- (1) the ATO should determine on a case-by-case basis the amount of refresher training needed, through an assessment of the candidate, taking into account the following factors:
 - (i) the experience of the applicant;
 - (ii) the amount of time elapsed since the expiry of the TRI or SFI certificate; and
 - (iii) the technical elements of the TRI or SFI training course, as determined by the assessment of the candidate by the ATO;
 - (2) the ATO should also consider the elements defined in point (a) above to determine the refresher training needed; and
 - (3) once the ATO has determined the needs of the applicant, it should develop an individual training programme that should be based on the content of the TRI or SFI training course and focus on the aspects where the applicant has the greatest needs.
- (c) After successful completion of the seminar or refresher training, as applicable, the ATO should:
- (1) in case of a seminar, in accordance with point (a), issue the applicant with a seminar completion certificate or another document specified by the competent authority, which describes the content of the seminar as in point (a), as well as a statement that the seminar was successfully completed; and
 - (2) in case of refresher training, in accordance with point (b), issue the applicant with a training completion certificate or another document specified by the competent authority, which describes the evaluation of the factors listed in point (b)(1) and the training received, as well as a statement that the training was successfully completed; the training completion certificate should be presented to the examiner prior to the assessment of competence.
- (d) Upon successful completion of the seminar or refresher training, as applicable, the ATO should submit the seminar or training completion certificate, or the other document specified by the competent authority, to the competent authority.

SECTION 5 – SPECIFIC REQUIREMENTS FOR THE CLASS RATING INSTRUCTOR – CRI

FCL.905.CRI CRI – Privileges and conditions

Regulation (EU) 2024/2076

- (a) The privileges of a CRI are to instruct for:
- (1) the issue, revalidation or renewal of a class or type rating for single-pilot aeroplanes, except for single-pilot high performance complex aeroplanes, when the privileges sought by the applicant are to fly in single-pilot operations;
 - (2) a towing or aerobatic rating for the aeroplane category, provided that the CRI holds the relevant rating and has demonstrated the ability to instruct for that rating to an FI qualified in accordance with point [FCL.905.FI\(j\)](#);
 - (3) extension of LAPL(A) privileges to another class or variant of aeroplane.
- (b) The privileges of a CRI are restricted to the class or type of aeroplane in which the instructor assessment of competence was taken. The privileges of the CRI shall be extended to further classes or types when the CRI has completed, within the last 12 months:
- (1) 15 hours flight time as PIC on aeroplanes of the applicable class or type of aeroplane;
 - (2) one training flight from the right hand seat under the supervision of another CRI or FI qualified for that class or type occupying the other pilot's seat.
- (ba) The privileges of CRIs are to instruct for class and type ratings for single-pilot aeroplanes, except for single-pilot high-performance complex aeroplanes, in multi-pilot operations, provided that CRIs meet any of the following conditions:
- (1) hold or have held a TRI certificate for multi-pilot aeroplanes;
 - (2) have at least 500 hours on aeroplanes in multi-pilot operations and completed an MCCI training course in accordance with point [FCL.930.MCCI](#).
- (c) Applicants for a CRI for multi-engine aeroplanes holding a CRI certificate for single-engine aeroplanes shall have fulfilled the prerequisites for a CRI established in [FCL.915.CRI\(a\)](#) and the requirements of [FCL.930.CRI\(a\)\(3\)](#) and [FCL.935](#).

FCL.915.CRI CRI – Prerequisites

Regulation (EU) No 1178/2011

An applicant for a CRI certificate shall have completed at least:

- (a) for multi-engine aeroplanes:
- (1) 500 hours flight time as a pilot on aeroplanes;
 - (2) 30 hours as PIC on the applicable class or type of aeroplane;
- (b) for single-engine aeroplanes:
- (1) 300 hours flight time as a pilot on aeroplanes;
 - (2) 30 hours as PIC on the applicable class or type of aeroplane.

FCL.930.CRI CRI – Training course

Regulation (EU) 2019/1747

- (a) The training course for the CRI shall include, at least:
- (1) 25 hours of teaching and learning instruction;
 - (2) 10 hours of technical training, including revision of technical knowledge, the preparation of lesson plans and the development of classroom/simulator instructional skills;
 - (3) 5 hours of flight instruction on multi-engine aeroplanes or an FSTD representing that class or type of aeroplane, including at least 3 hours on the aeroplane, or at least 3 hours of flight instruction on single-engine aeroplanes, given by an FI(A) qualified in accordance with point [FCL.905.FI\(j\)](#).
- (b) Applicants holding or having held an instructor certificate shall be fully credited towards the requirement of (a)(1).

AMC1 FCL.930.CRI CRI — training course

ED Decision 2025/002/R

GENERAL

- (a) The aim of the CRI training course is to train aircraft licence holders to the level of competence defined in [FCL.920](#) and adequate to a CRI.
- (b) The training course should be designed to give adequate training to the applicant in theoretical knowledge instruction, flight instruction and FSTD instruction to instruct for any class or type rating, except for single-pilot high-performance complex aeroplanes, for which the applicant is qualified.
- (c) The flight training should be aimed at ensuring that the applicant is able to teach the air exercises safely and efficiently to students undergoing a course of training for the issue of a class or type rating, except for single-pilot high-performance complex aeroplanes.
- (d) It is to be noted that airmanship is a vital ingredient of all flight operations. Therefore, in the following air exercises the relevant aspects of airmanship are to be stressed at the appropriate times during each flight.
- (e) The student instructor should learn how to identify common errors and how to correct them properly, which should be emphasised at all times.

CONTENT

- (f) The training course consists of three parts:
- (1) Part 1: teaching and learning that should be in accordance with [AMC1 FCL.920](#);
 - (2) Part 2: technical theoretical knowledge instruction (technical training);
 - (3) Part 3: flight instruction.

Part 1

The content of the teaching and learning part of the FI training course, as established in [AMC1 FCL.930.FI](#), should be used as guidance to develop the course syllabus.

Part 2

- (a) The technical theoretical-knowledge instruction should comprise at least 10 hours of training to include the revision of technical knowledge, preparation of lesson plans, and development of classroom instructional skills to enable the CRI to teach the technical theoretical-knowledge syllabus.
- (b) The type or class rating theoretical syllabus should be used to develop the CRI teaching skills in relation to the type or class technical course syllabus. The course instructor should deliver example lectures from the applicable type or class technical syllabus. The candidate instructor should prepare and deliver lectures on topics that are selected by the course instructor from the type/class rating course and the generic topics listed further below.
- (c) The 10 hours of technical theoretical-knowledge instruction should develop the applicant's ability to teach a student the knowledge and understanding that are required for the relevant air exercises for either SE or ME aeroplanes, depending on the privileges sought by the candidate.
- (d) If CRI privileges for both SE and ME aeroplanes are sought, the applicant should complete 10 hours of technical theoretical-knowledge instruction related to SE and ME aeroplanes each.
- (e) This following syllabus of general subjects concerns training only on ME aeroplanes.

GENERAL SUBJECTS

- (a) Air legislation:
 - (1) aeroplane performance group definitions;
 - (2) methods of factoring gross performance.
- (b) Asymmetric power flight;
- (c) Principles of flight;
- (d) The problems:
 - (1) asymmetry;
 - (2) control;
 - (3) performance;
- (e) The forces and couples:
 - (1) offset thrust line;
 - (2) asymmetric blade effect;
 - (3) offset drag line;
 - (4) failed engine propeller drag;
 - (5) total drag increase;
 - (6) asymmetry of lift;
 - (7) uneven propeller slipstream effect;
 - (8) effect of yaw in level and turning flight;
 - (9) thrust and rudder side force couples;
 - (10) effect on moment arms.
- (f) Control in asymmetric power flight:

- (1) use, misuse and limits of:
 - (i) rudder;
 - (ii) aileron;
 - (iii) elevators.
- (2) effect of bank or sideslip and balance;
- (3) decrease of aileron and rudder effectiveness;
- (4) fin stall possibility;
- (5) effect of IAS and thrust relationship;
- (6) effect of residual unbalanced forces;
- (7) foot loads and trimming.
- (g) Minimum control and safety speeds:
 - (1) minimum control speed (v_{mc});
 - (2) definition;
 - (3) origin;
 - (4) factors affecting (v_{mc}):
 - (i) thrust;
 - (ii) mass and centre of gravity position;
 - (iii) altitude;
 - (iv) landing gear;
 - (v) flaps;
 - (vi) cowl flaps or cooling gills;
 - (vii) turbulence or gusts;
 - (viii) pilot reaction or competence;
 - (ix) banking towards the operating engine;
 - (x) drag;
 - (xi) feathering;
 - (xii) critical engine.
 - (5) take-off safety speed;
 - (6) definition or origin of v_2 ;
 - (7) other relevant v codes;
- (h) Aeroplane performance: one engine inoperative:
 - (1) effect on excess power available;
 - (2) SE ceiling;
 - (3) cruising, range and endurance;
 - (4) acceleration and deceleration;

- (5) zero thrust, definition and purpose;
- (i) Propellers:
 - (1) variable pitch: general principles;
 - (2) feathering and un-feathering mechanism and limitations (for example minimum RPM);
- (j) Specific aeroplane type;
- (k) Aeroplane and engine systems:
 - (1) operation normal;
 - (2) operation abnormal;
 - (3) emergency procedures.
- (l) Limitations: airframe:
 - (1) load factors;
 - (2) landing gear and flap limiting speeds (v_{lo} and v_{fe});
 - (3) rough air speed (v_{ra});
 - (4) maximum speeds (v_{no} and v_{ne}).
- (m) Limitations: engine:
 - (1) RPM and manifold pressure;
 - (2) oil temperature and pressure;
 - (3) emergency procedures.
- (n) Mass and balance:

(to be covered in conjunction with the flight manual or equivalent document (for example owner's manual or pilot's operating handbook))

 - (1) mass and balance documentation for aeroplane type;
 - (2) revision of basic principles;
 - (3) calculations for specific aeroplane type.
- (o) Mass and performance:

(to be covered in conjunction with the flight manual or equivalent document (for example owner's manual or pilot's operating handbook))

 - (1) calculations for specific aeroplane type (all engines operating);
 - (2) take-off run;
 - (3) take-off distance;
 - (4) accelerate and stop distance;
 - (5) landing distance;
 - (6) landing run;
 - (7) take-off or climb out flight path;
 - (8) calculations for specific aeroplane type (one engine operating);

- (9) climb out flight path;
- (10) landing distance;
- (11) landing run.

Part 3

FLIGHT INSTRUCTION SYLLABUS: NORMAL FLIGHT

- (a) This part is similar to the air exercise sections of the SE FI course, including ‘Introduction to instrument flying’ except that the objectives, airmanship considerations and common errors are related to the operation of an ME aeroplane.
- (b) The purpose of this part is to acquaint the applicant with the teaching aspects of the operational procedures and handling of an ME aeroplane with all engines functioning.
- (c) The following items should be covered:
 - (1) aeroplane familiarisation;
 - (2) pre-flight preparation and aeroplane inspection;
 - (3) engine starting procedures;
 - (4) taxiing;
 - (5) pre take-off procedures;
 - (6) the take-off and initial climb:
 - (i) into wind;
 - (ii) crosswind;
 - (iii) short field.
 - (7) climbing;
 - (8) straight and level flight;
 - (9) descending (including emergency descent procedures);
 - (10) turning;
 - (11) slow flight;
 - (12) stalling and recoveries;
 - (13) instrument flight: basic;
 - (14) emergency drills (not including engine failure);
 - (15) circuit, approach and landing:
 - (i) into wind;
 - (ii) crosswind;
 - (iii) short field;
 - (16) mislanding and going round again;
 - (17) actions after flight.

AIR EXERCISES

- (d) The syllabus for CRI SE and ME training courses should comprise air exercises 1 to 4 and should not last less than 3 hours. In addition, the syllabus for a CRI ME training course should also include air exercise 5 to address asymmetric power flight and should not last less than 2 hours.

EXERCISE 1: FAMILIARISATION WITH THE AEROPLANE

- (a) Long briefing objectives:
- (1) introduction to the aeroplane;
 - (2) explanation of the cockpit layout;
 - (3) systems and controls;
 - (4) aeroplane power plant;
 - (5) checklists and drills;
 - (6) differences when occupying the instructor's seat;
 - (7) emergency drills:
 - (i) action in event of fire in the air and on the ground;
 - (ii) escape drills: location of exits and use of emergency equipment (for example fire extinguishers, etc.).
 - (8) pre-flight preparation and aeroplane inspection:
 - (i) aeroplane documentation;
 - (ii) external checks;
 - (iii) internal checks;
 - (iv) harness, seat or rudder pedal adjustment;
 - (9) engine starting procedures:
 - (i) use of checklists;
 - (ii) checks before starting;
 - (iii) checks after starting.
- (b) Air exercise:
- (1) external features;
 - (2) cockpit layout;
 - (3) aeroplane systems;
 - (4) checklists and drills;
 - (5) action if fire in the air and on the ground;
 - (i) engine;
 - (ii) cabin;
 - (iii) electrical.
 - (6) systems failure (as applicable to type);
 - (7) escape drills (location and use of emergency equipment and exits);

- (8) preparation for and action after flight:
 - (i) flight authorisation and aeroplane acceptance;
 - (ii) technical log or certificate of maintenance release;
 - (iii) mass and balance and performance considerations;
 - (iv) external checks;
 - (v) internal checks, adjustment of harness or rudder pedals;
 - (vi) starting and warming up engines;
 - (vii) checks after starting;
 - (viii) radio navigation and communication checks;
 - (ix) altimeter checks and setting procedures;
 - (x) power checks;
 - (xi) running down and switching off engines;
 - (xii) completion of authorisation sheet and aeroplane serviceability documents.

EXERCISE 2: TAXIING

- (a) Long briefing objectives:
 - (1) pre-taxiing area precautions (greater mass: greater inertia);
 - (2) effect of differential power;
 - (3) precautions on narrow taxiways;
 - (4) pre take-off procedures:
 - (i) use of checklist;
 - (ii) engine power checks;
 - (iii) pre take-off checks;
 - (iv) instructor's briefing to cover the procedure to be followed should an emergency occur during take-off, for example engine failure.
 - (5) the take-off and initial climb:
 - (i) ATC considerations;
 - (ii) factors affecting the length of the take-off run or distance;
 - (iii) correct lift-off speed;
 - (iv) importance of safety speed;
 - (v) crosswind take-off, considerations and procedures;
 - (vi) short field take-off, considerations and procedures;
 - (vii) engine handling after take-off: throttle, pitch and engine synchronisation.
 - (6) climbing:
 - (i) pre-climbing checks;

- (ii) engine considerations (use of throttle or pitch controls);
 - (iii) maximum rate of climb speed;
 - (iv) maximum angle of climb speed;
 - (v) synchronising the engines.
- (b) Air exercise
 - (1) pre-taxing checks;
 - (2) starting, control of speed and stopping;
 - (3) control of direction and turning;
 - (4) turning in confined spaces;
 - (5) leaving the parking area;
 - (6) freedom of rudder movement (importance of pilot ability to use full rudder travel);
 - (7) instrument checks;
 - (8) emergencies (brake or steering failure);
 - (9) pre take-off procedures:
 - (i) use of checklist;
 - (ii) engine power and system checks;
 - (iii) pre take-off checks;
 - (iv) instructor's briefing if emergencies during take-off.
 - (10) the take-off and initial climb:
 - (i) ATC considerations;
 - (ii) directional control and use of power;
 - (iii) lift-off speed;
 - (iv) crosswind effects and procedure;
 - (v) short field take-off and procedure.
 - (vi) procedures after take-off (at an appropriate stage of the course):
 - (A) landing gear retraction;
 - (B) flap retraction (as applicable);
 - (C) selection of manifold pressure and RPM;
 - (D) engine synchronisation;
 - (E) other procedures (as applicable).
 - (11) climbing:
 - (i) pre-climbing checks;
 - (ii) power selection for normal and maximum rate climb;
 - (iii) engine and RPM limitations;
 - (iv) effect of altitude on manifold pressure, full throttle;

- (v) levelling off: power selection;
- (vi) climbing with flaps down;
- (vii) recovery to normal climb;
- (viii) en-route climb (cruise climb);
- (ix) maximum angle of climb;
- (x) altimeter setting procedures;
- (xi) prolonged climb and use of cowl flaps or cooling gills;
- (xii) instrument appreciation.

EXERCISE 3: STRAIGHT AND LEVEL FLIGHT

- (a) Long briefing objectives:
 - (1) selection of power: throttle or pitch controls;
 - (2) engine synchronisation;
 - (3) fuel consumption aspects;
 - (4) use of trimming controls: elevator and rudder (aileron as applicable);
 - (5) operation of flaps:
 - (i) effect on pitch attitude;
 - (ii) effect on air speed.
 - (6) operation of landing gear:
 - (i) effect on pitch attitude;
 - (ii) effect on air speed.
 - (7) use of mixture controls;
 - (8) use of alternate air or carburettor heat controls;
 - (9) operation of cowl flaps or cooling gills;
 - (10) use of cabin ventilation and heating systems;
 - (11) operation and use of the other systems (as applicable to type);
 - (12) descending:
 - (i) pre-descent checks;
 - (ii) normal descent;
 - (iii) selection of throttle or pitch controls;
 - (iv) engine cooling considerations;
 - (v) emergency descent procedure.
 - (13) turning:
 - (i) medium turns;
 - (ii) climbing and descending turns;

- (iii) steep turns (45 ° of bank or more).
- (b) Air exercise:
 - (1) at normal cruising power:
 - (i) selection of cruise power;
 - (ii) manifold pressure or RPM;
 - (iii) engine synchronisation;
 - (iv) use of trimming controls;
 - (v) performance considerations: range or endurance.
 - (2) instrument appreciation;
 - (3) operation of flaps (in stages):
 - (i) air speed below v_{fe} ;
 - (ii) effect on pitch attitude;
 - (iii) effect on air speed.
 - (4) operation of landing gear:
 - (i) air speed below v_{lo} / v_{le} ;
 - (ii) effect on pitch attitude;
 - (iii) effect on air speed.
 - (5) use of mixture controls;
 - (6) use of alternate air or carburettor control;
 - (7) operation of cowl flaps or cooling gills;
 - (8) operation of cabin ventilation or heating systems;
 - (9) operation and use of other systems (as applicable to type);
 - (10) descending;
 - (i) pre-descent checks;
 - (ii) power selection: manifold pressure or RPM;
 - (iii) powered descent (cruise descent);
 - (iv) engine cooling considerations: use of cowl flaps or cooling gills;
 - (v) levelling off;
 - (vi) descending with flaps down;
 - (vii) descending with landing gear down;
 - (viii) altimeter setting procedure;
 - (ix) instrument appreciation;
 - (x) emergency descent:
 - (A) as applicable to type;
 - (B) limitations in turbulence v_{no} .

- (11) turning:
 - (i) medium turns;
 - (ii) climbing and descending turns;
 - (iii) steep turns: 45 ° of bank;
 - (iv) instrument appreciation.

EXERCISE 4: SLOW FLIGHT

- (a) Long briefing objectives:
 - (1) aeroplane handling characteristics during slow flight: flight at v_{s1} and $v_{so} + 5$ knots;
 - (2) simulated go-around from slow flight:
 - (i) at V_{sse} with flaps down;
 - (ii) note pitch trim change.
 - (3) stalling:
 - (i) power selection;
 - (ii) symptoms approaching the stall;
 - (iii) full stall characteristics;
 - (iv) recovery from the full stall;
 - (v) recovery at the incipient stall;
 - (vi) stalling and recovery in the landing configuration;
 - (vii) recovery at the incipient stage in the landing configuration.
 - (4) instrument flight (basic):
 - (i) straight and level;
 - (ii) climbing;
 - (iii) turning;
 - (iv) descending.
 - (5) emergency drills (not including engine failure), as applicable to type;
 - (6) circuit approach and landing:
 - (i) downwind leg:
 - (A) air speed below v_{fe} ;
 - (B) use of flaps (as applicable);
 - (C) pre-landing checks;
 - (D) position to turn onto base leg.
 - (ii) base leg:
 - (A) selection of power (throttle or pitch), flaps and trimming controls;
 - (B) maintenance of correct air speed.

- (iii) final approach:
 - (A) power adjustments (early reaction to undershooting);
 - (B) use of additional flaps (as required);
 - (C) confirmation of landing gear down;
 - (D) selection 'touch down' point;
 - (E) air speed reduction to V_{at} ;
 - (F) maintenance of approach path.
 - (iv) landing:
 - (A) greater sink rate;
 - (B) longer landing distance and run;
 - (C) crosswind approach and landing;
 - (D) crosswind considerations;
 - (E) short field approach and landing;
 - (F) short field procedure: considerations.
- (b) Air exercise
 - (1) safety checks;
 - (2) setting up and maintaining (flaps up);
 - (i) $v_{s1} + 5$ knots;
 - (ii) note aeroplane handling characteristics.
 - (3) setting up and maintaining (flaps down):
 - (i) $v_{s0} + 5$ knots;
 - (ii) note aeroplane handling characteristics.
 - (4) simulated go-around from a slow flight with flaps:
 - (i) down and air speed not below V_{sse} , for example air speed at V_{sse} or $v_{mca} + 10$ knots;
 - (ii) increase to full power and enter a climb;
 - (iii) note pitch change.
 - (5) resume normal flight.
 - (6) stalling;
 - (i) selection of RPM;
 - (ii) stall symptoms;
 - (iii) full stall characteristics;
 - (iv) recovery from the full stall: care in application of power;
 - (v) recovery at the incipient stage;
 - (vi) stalling and recovery in landing configuration;
 - (vii) stall recovery at the incipient stage in the landing configuration.

- (7) instrument flight (basic):
 - (i) straight and level;
 - (ii) climbing;
 - (iii) turning;
 - (iv) descending.
- (8) emergency drills (not including engine failure), as applicable to type;
- (9) circuit, approach and landing:
 - (i) downwind leg:
 - (A) control of speed (below v_{fe});
 - (B) flaps as applicable;
 - (C) pre-landing checks;
 - (D) control of speed and height;
 - (E) base leg turn.
 - (ii) base leg:
 - (A) power selection;
 - (B) use of flap and trimming controls;
 - (C) maintenance of correct air speed.
 - (iii) final approach:
 - (A) use of additional flap (as required);
 - (B) confirmation of landing gear down;
 - (C) selection of touchdown point;
 - (D) air speed reduction to V_{at} ;
 - (E) maintaining correct approach path: use of power.
 - (iv) landing:
 - (A) control of sink rate during flare;
 - (B) crosswind considerations;
 - (C) longer landing roll;
 - (D) short or soft field approach and landing;
 - (E) considerations and precautions.
- (10) Asymmetric power flight.

During this part, special emphasis is to be placed on the:

 - (i) circumstances in which actual feathering and un-feathering practice will be done, for example safe altitude; compliance with regulations about minimum altitude or height for feathering practice, weather conditions, distance from nearest available aerodrome;

- (ii) procedure to use for instructor and student co-operation, for example the correct use of touch drills and the prevention of misunderstandings, especially during feathering and unfeathering practice and when zero thrust is being used for asymmetric circuits. This procedure is to include positive agreement as to which engine is being shut down or re-started or set at zero thrust and identifying each control and naming the engine it is going to affect;
- (iii) consideration to be given to avoid over-working the operating engine, and the degraded performance when operating the aeroplane during asymmetric flight;
- (iv) need to use the specific checklist for the aeroplane type.

EXERCISE 5: FLIGHT ON ASYMMETRIC POWER

- (a) Long briefing objectives:
 - (1) introduction to asymmetric flight;
 - (2) feathering the propeller: method of operation;
 - (3) effects on aeroplane handling at cruising speed;
 - (4) introduction to effects upon aeroplane performance;
 - (5) note foot load to maintain a constant heading (no rudder trim);
 - (6) un-feathering the propeller;
 - (7) return to normal flight finding the zero thrust setting;
 - (8) comparison of foot load when feathered and with zero thrust set.
 - (9) effects and recognition of engine failure in level flight;
 - (10) forces and the effects of yaw;
 - (11) types of failure:
 - (i) sudden or gradual;
 - (ii) complete or partial.
 - (12) yaw, direction and further effects of yaw;
 - (13) flight instrument indications;
 - (14) identification of failed engine;
 - (15) the couples and residual out of balance forces: resultant flight attitude;
 - (16) use of rudder to counteract yaw;
 - (17) use of aileron: dangers of misuse;
 - (18) use of elevator to maintain level flight;
 - (19) use of power to maintain a safe air speed and altitude;
 - (20) supplementary recovery to straight and level flight: simultaneous increase of speed and reduction in power;
 - (21) identification of failed engine: idle leg = idle engine;
 - (22) use of engine instruments for identification:

- (i) fuel pressure or flow;
 - (ii) RPM gauge response effect of CSU action at lower and higher air speed;
 - (iii) engine temperature gauges.
- (23) confirmation of identification: close the throttle of identified failed engine;
- (24) effects and recognition of engine failure in turns;
- (25) identification and control;
- (26) side forces and effects of yaw.
- (27) During turning flight:
- (i) effect of 'inside' engine failure: effect sudden and pronounced;
 - (ii) effect of 'outside' engine failure: effect less sudden and pronounced;
 - (iii) the possibility of confusion in identification (particularly at low power):
 - (A) correct use of rudder;
 - (B) possible need to return to lateral level flight to confirm correct identification.
 - (iv) visual and flight instrument indications;
 - (v) effect of varying speed and power;
 - (vi) speed and thrust relationship;
 - (vii) at normal cruising speed and cruising power: engine failure clearly recognised;
 - (viii) at low safe speed and climb power: engine failure most positively recognised;
 - (ix) high speed descent and low power: possible failure to notice asymmetry (engine failure).
- (28) Minimum control speeds:
- (i) ASI colour coding: red radial line.

Note: this exercise is concerned with the ultimate boundaries of controllability in various conditions that a student can reach in a steady asymmetric power state, approached by a gradual speed reduction. Sudden and complete failure should not be given at the Flight Manual v_{mca} . The purpose of the exercise is to continue the gradual introduction of a student to control an aeroplane in asymmetric power flight during extreme or critical situations. It is not a demonstration of v_{mca} .
 - (ii) Techniques for assessing critical speeds with wings level and recovery: dangers involved when minimum control speed and the stalling speed are very close: use of V_{sse} ;
 - (iii) Establish a minimum control speed for each asymmetrically disposed engine to establish critical engine (if applicable);
 - (iv) Effects on minimum control speeds of:
 - (A) bank;
 - (B) zero thrust setting;
 - (C) take-off configuration:
 - (a) landing gear down and take-off flap set;

- (b) landing gear up and take-off flap set.

Note: it is important to appreciate that the use of 5 ° of bank towards the operating engine produces a lower v_{mca} and also a better performance than that obtained with the wings held level. It is now normal for manufacturers to use 5 ° of bank in this manner when determining the v_{mca} for the specific type. Thus, the v_{mca} quoted in the aeroplane manual will have been obtained using the technique.

- (29) Feathering and un-feathering:

- (i) minimum heights for practising feathering or un-feathering drills;
- (ii) engine handling: precautions (overheating, icing conditions, priming, warm-up, method of simulating engine failure: reference to aircraft engine manual and service instructions and bulletins).

- (30) Engine failure procedure:

- (i) once the maintenance of control has been achieved, the order in which the procedures are carried out will be determined by the phase of operation and the aircraft type.
- (ii) flight phase:
 - (A) in cruising flight;
 - (B) critical phase such as immediately after take-off or during the approach to landing or during a go-around.

- (31) Aircraft type:

Variations will inevitably occur in the order of certain drills and checks due to differences between aeroplane types and perhaps between models of the same type, and the flight manual or equivalent document (for example owner's manual or pilot's operating handbook) is to be consulted to establish the exact order of these procedures.

For example, one flight manual or equivalent document (for example owner's manual or pilot's operating handbook) may call for the raising of flaps and landing gear before feathering, whilst another may recommend feathering as a first step. The reason for this latter procedure could be due to the fact that some engines cannot be feathered if the RPM drops below a certain figure.

Again, in some aeroplanes, the raising of the landing gear may create more drag during retraction due to the transient position of the landing gear doors and as a result of this retraction would best be left until feathering has been accomplished and propeller drag reduced.

Therefore, the order in which the drills and checks are shown in this syllabus under 'immediate actions' and 'subsequent actions' are to be used as a general guide only and the exact order of precedence is determined by reference to the flight manual or equivalent document (for example owner's manual or pilot's operating handbook) for the specific aeroplane type being used on the course.

- (32) In-flight engine failure in cruise or other flight phase not including take-off or landing:

- (i) immediate actions:
 - (A) recognition of asymmetric condition and control of the aircraft;

- (B) identification and confirmation of failed engine:
 - (a) idle leg = idle engine;
 - (b) closing of throttle for confirmation.
- (C) cause and fire check:
 - (a) typical reasons for failure;
 - (b) methods of rectification.
- (D) feathering decision and procedure:
 - (a) reduction of other drag;
 - (b) need for speed but not haste;
 - (c) use of rudder trim.
- (ii) subsequent actions;
 - (A) live engine:
 - (a) temperature, pressures and power;
 - (b) remaining services;
 - (c) electrical load: assess and reduce as necessary;
 - (d) effect on power source for air driven instruments;
 - (e) landing gear;
 - (f) flaps and other services.
 - (B) re-plan flight:
 - (a) ATC and weather;
 - (b) terrain clearance, SE cruise speed;
 - (c) decision to divert or continue.
 - (C) fuel management: best use of remaining fuel;
 - (D) dangers of re-starting damaged engine;
 - (E) action if unable to maintain altitude: effect of altitude on power available;
 - (F) effects on performance;
 - (G) effects on power available and power required;
 - (H) effects on various airframe configuration and propeller settings;
 - (I) use of flight manual or equivalent document (for example owner's manual or pilot's operating handbook):
 - (a) cruising;
 - (b) climbing: ASI colour coding (blue line);
 - (c) descending;
 - (d) turning.
 - (J) 'live' engine limitations and handling;

- (K) take-off and approach: control and performance.
- (33) Significant factors:
- (i) significance of take-off safety speed:
 - (A) effect of landing gear, flap, feathering, take-off, trim setting, systems for operating landing gear and flaps;
 - (B) effect on mass, altitude and temperature (performance).
 - (ii) significance of best SE climb speed (V_{yse}):
 - (A) acceleration to best engine climb speed and establishing a positive climb;
 - (B) relationship of SE climb speed to normal climb speed;
 - (C) action if unable to climb.
 - (iii) significance of asymmetric committal height and speed: action if baulked below asymmetric committal height.
- (34) Engine failure during take-off:
- (i) below v_{mca} or unstick speed:
 - (A) accelerate or stop distance considerations;
 - (B) prior use of flight manual data if available.
 - (ii) above v_{mca} or unstick speed and below safety speed;
 - (iii) immediate re-landing or use of remaining power to achieve forced landing;
 - (iv) considerations:
 - (A) degree of engine failure;
 - (B) speed at the time;
 - (C) mass, altitude and temperature (performance);
 - (D) configuration;
 - (E) length of runway remaining;
 - (F) position of any obstacles ahead.
- (35) Engine failure after take-off:
- (i) simulated at a safe height and at or above take-off safety speed;
 - (ii) considerations:
 - (A) need to maintain control;
 - (B) use of bank towards operating engine;
 - (C) use of available power achieving best SE climb speed;
 - (D) mass, altitude, temperature (performance);
 - (E) effect of prevailing conditions and circumstances.
- (36) Immediate actions: maintenance of control, including air speed and use of power:
- (i) recognition of asymmetric condition;

- (ii) identification and confirmation of failed engine;
 - (iii) feathering and removal of drag (procedure for type);
 - (iv) establishing best SE climb speed.
- (37) Subsequent actions: whilst carrying out an asymmetric power climb to the downwind position at SE best rate of climb speed:
- (i) cause and fire check;
 - (ii) live engine, handling considerations;
 - (iii) remaining services;
 - (iv) ATC liaison;
 - (v) fuel management.

Note: these procedures are applicable to aeroplane type and flight situation.

- (38) Significance of asymmetric committal height:
- (i) Asymmetric committal height is the minimum height needed to establish a positive climb whilst maintaining adequate speed for control and removal of drag during an approach to a landing.

Because of the significantly reduced performance of many CS/JAR/FAR 23 aeroplanes when operating on one engine, consideration is to be given to a minimum height from which it would be safely possible to attempt a go-around procedure, during an approach when the flight path will have to be changed from a descent to a climb with the aeroplane in a high drag configuration.

Due to the height loss which will occur during the time that the operating engine is brought up to full power, landing gear and flap retracted, and the aeroplane established in a climb at v_{yse} a minimum height (often referred to as 'Asymmetric committal height') is to be selected, below which the pilot should not attempt to take the aeroplane round again for another circuit. This height will be compatible with the aeroplane type, all up weight, altitude of the aerodrome being used, air temperature, wind, the height of obstructions along the climb out path, and pilot competence.
 - (ii) circuit approach and landing on asymmetric power:
 - (A) definition and use of asymmetric committal height;
 - (B) use of standard pattern and normal procedures;
 - (C) action if unable to maintain circuit height;
 - (D) speed and power settings required;
 - (E) decision to land or go-around at asymmetric committal height: factors to be considered.
 - (iii) undershooting importance of maintaining correct air speed (not below v_{yse}).
- (39) Speed and heading control:
- (i) height, speed and power relationship: need for minimum possible drag;
 - (ii) establishing positive climb at best SE rate of climb speed:

- (A) effect of availability of systems, power for flap and landing gear;
- (B) operation and rapid clean up.

Note 1: The air speed at which the decision is made to commit the aeroplane to a landing or to go-around should normally be the best SE rate of climb speed and in any case not less than the safety speed.

Note 2: On no account should instrument approach 'decision height' and its associated procedures be confused with the selection of minimum height for initiating a go-around in asymmetric power flight.

(40) Engine failure during an all engines approach or missed approach:

- (i) use of asymmetric committal height and speed considerations;
- (ii) speed and heading control;
- (iii) decision to attempt a landing, go-around or force land as circumstances dictate.

Note: at least one demonstration and practice of engine failure in this situation should be performed during the course.

(41) Instrument flying on asymmetric power:

- (i) considerations relating to aircraft performance during:
 - (A) straight and level flight;
 - (B) climbing and descending;
 - (C) standard rate turns;
 - (D) level, climbing and descending turns including turns onto preselected headings.
- (ii) availability of vacuum operated instruments;
- (iii) availability of electrical power source.

(b) Air exercise

This section covers the operation of a SP ME aeroplane when one engine has failed and it is applicable to all such light piston aeroplanes. Checklists should be used as applicable.

- (1) introduction to asymmetric flight;
- (2) close the throttle of one engine;
- (3) feather its propeller;
- (4) effects on aeroplane handling at cruising speed;
- (5) effects on aeroplane performance for example cruising speed and rate of climb;
- (6) note foot load to maintain a constant heading;
- (7) un-feather the propeller;
- (8) return to normal flight finding the zero thrust throttle setting;
- (9) comparison of foot load when feathered and with zero thrust set.
- (10) effects and recognition of engine failure in level flight with the aeroplane straight and level at cruise speed:

- (i) slowly close the throttle of one engine;
 - (ii) note yaw, roll and spiral descent.
- (11) return to normal flight:
 - (i) close throttle of other engine;
 - (ii) note same effects in opposite direction.
- (12) methods of control and identification of failed engine close one throttle and maintain heading and level flight by use of:
 - (i) rudder to control yaw;
 - (ii) aileron to hold wings level;
 - (iii) elevators to maintain level flight;
 - (iv) power (as required) to maintain air speed and altitude.
- (13) alternative or supplementary method of control:
 - (i) simultaneously;
 - (ii) lower aeroplane nose to increase air speed;
 - (iii) reduce power;
 - (iv) loss of altitude: inevitable.
- (14) identification of failed engine: idle foot = idle engine;
- (15) use of instruments for identification:
 - (i) fuel pressure or fuel flow;
 - (ii) RPM gauge or CSU action may mask identification;
 - (iii) engine temperature gauges.
- (16) confirmation of identification: close the throttle of the identified failed engine;
- (17) effects and recognition of engine failure in turns and effects of 'inside' engine failure:
 - (i) more pronounced yaw;
 - (ii) more pronounced roll;
 - (iii) more pronounced pitch down.
- (18) effects of 'outside' engine failure:
 - (i) less pronounced yaw;
 - (ii) less pronounced roll;
 - (iii) less pronounced pitch down.
- (19) possibility of confusion in identification:
 - (i) use of correct rudder application;
 - (ii) return to lateral level flight if necessary.
- (20) flight instrument indications;
- (21) effect of varying speed and power;

- (22) failure of one engine at cruise speed and power: engine failure clearly recognised;
- (23) failure of one engine at low speed and high power (not below v_{sse}): engine failure most positively recognised;
- (24) failure of one engine at higher speeds and low power: possible failure to recognise engine failure;
- (25) minimum control speeds;
- (26) establish the v_{yse} :
 - (i) select maximum permitted manifold pressure and RPM;
 - (ii) close the throttle on one engine;
 - (iii) raise the aeroplane nose and reduce the air speed;
 - (iv) note the air speed when maximum rudder deflection is being applied and when directional control can no longer be maintained;
 - (v) lower the aeroplane nose and reduce power until full directional control is regained;
 - (vi) the lowest air speed achieved before the loss of directional control will be the V_{mc} for the flight condition;
 - (vii) repeat the procedure closing the throttle of the other engine;
 - (viii) the higher of these two air speeds will identify the most critical engine to fail.

Note: warning - in the above situations the recovery is to be initiated immediately before directional control is lost with full rudder applied, or when a safe margin above the stall remains, for example when the stall warning device operates, for the particular aeroplane configuration and flight conditions. On no account should the aeroplane be allowed to decelerate to a lower air speed.

- (27) establish the effect of using 5° of bank at v_{mc} :
 - (i) close the throttle of one engine;
 - (ii) increase to full power on the operating engine;
 - (iii) using 5° of bank towards the operating engine reduce speed to the V_{mc} ;
 - (iv) note lower V_{mc} when 5° of bank is used.
- (28) 'in-flight' engine failure procedure;
- (29) in cruise and other flight circumstances not including take-off and landing.
- (30) Immediate actions: maintenance of control including air speed and use of power:
 - (i) identification and confirmation of failed engine;
 - (ii) failure cause and fire check;
 - (iii) feathering decision and implementation;
 - (iv) reduction of any other drag, for example flaps, cowl flaps etc.;
 - (v) retrim and maintain altitude.
- (31) Subsequent actions:
 - (i) live engine:

- (A) oil temperature, pressure, fuel flow and power;
 - (B) remaining services;
 - (C) electrical load: assess and reduce as necessary;
 - (D) effect on power source for air driven instruments;
 - (E) landing gear;
 - (F) flaps and other services.
- (ii) re-plan flight:
- (A) ATC and weather;
 - (B) terrain clearance;
 - (C) SE cruise speed;
 - (D) decision to divert or continue;
- (iii) fuel management: best use of
- (iv) dangers of re-starting damaged engine;
- (v) action if unable to maintain altitude:
- (A) adopt V_{yse} ;
 - (B) effect of altitude on power available.
- (vi) effects on performance;
- (vii) effects on power available and power required;
- (viii) effects on various airframe configurations and propeller settings;
- (ix) use of flight manual or equivalent document (for example owner's manual or pilot's operating handbook):
- (A) cruising;
 - (B) climbing: ASI colour coding (blue line);
 - (C) descending;
 - (D) turning.
- (x) 'live' engine limitations and handling;
- (xi) take-off and approach: control and handling;
- Note: to be done at a safe height away from the circuit;
- (xii) take-off case with landing gear down and take-off flap set (if applicable);
- (xiii) significance of take-off at or above safety speed (at safety speed. The ability to maintain control and to accelerate to SE climb speed with aeroplane clean and zero thrust set. Thereafter to achieve a positive climb);
- (xiv) significance of flight below safety speed (below safety speed and above v_{mca} . A greater difficulty to maintain control, a possible loss of height whilst maintaining speed, cleaning up, accelerating to SE climb speed and establishing a positive climb);

- (xv) significance of best SE climb speed (the ability to achieve the best rate of climb on one engine with minimum delay).
- (32) Significance of asymmetric committal height:
 - (i) the ability to maintain or accelerate to the best SE rate of climb speed and to maintain heading whilst cleaning up with perhaps a slight height loss before climbing away;
 - (ii) below this height, the aeroplane is committed to continue the approach to a landing.
- (33) Engine failure during take-off run and below safety speed briefing only;
- (34) Engine failure after take-off;

Note: to be initiated at a safe height and at not less than take-off safety speed with due regard to the problems of a prolonged SE climb in the prevailing conditions.

 - (i) immediate actions:
 - (A) control of direction and use of bank;
 - (B) control of air speed and use of power;
 - (C) recognition of asymmetric condition;
 - (D) identification and confirmation of failed engine feathering and reduction of drag (procedure for type);
 - (E) re-trim;
 - (ii) subsequent actions: whilst carrying out an asymmetric power climb to the downwind position at SE best rate of climb speed:
 - (A) cause and fire check;
 - (B) live engine, handling considerations;
 - (C) drills and procedures applicable to aeroplane type and flight situation;
 - (D) ATC liaison;
 - (E) fuel management.
- (35) Asymmetric circuit, approach and landing;
 - (i) downwind and base legs:
 - (A) use of standard pattern;
 - (B) normal procedures;
 - (C) landing gear and flap lowering considerations;
 - (D) position for base leg;
 - (E) live engine handling;
 - (F) air speed and power settings;
 - (G) maintenance of height.
 - (ii) final approach:
 - (A) asymmetric committal height drill;

- (B) control of air speed and descent rate;
- (C) flap considerations.
- (iii) going round again on asymmetric power (missed approach):
 - (A) not below asymmetric committal height;
 - (B) speed and heading control;
 - (C) reduction of drag, landing gear retraction;
 - (D) maintaining V_{yse} ;
 - (E) establish positive rate of climb.
- (36) Engine failure during all engines approach or missed approach:

Note: to be started at not less than asymmetric committal height and speed and not more than part flap set:

 - (i) speed and heading control;
 - (ii) reduction of drag flap;
 - (iii) decision to attempt landing or go-around;
 - (iv) control of descent rate if approach is continued;
 - (v) if go-around is initiated, maintain v_{yse} , flaps and landing gear retracted and establish positive rate of climb.

Note: at least one demonstration and practice of engine failure in this situation should be performed during the course.
- (37) Instrument flying on asymmetric power;
- (38) Flight instrument checks and services available:
 - (i) straight and level flight;
 - (ii) climbing and descending;
 - (iii) standard rate turns;
 - (iv) level, climbing and descending turns including turns onto preselected headings.

EXERCISE 6: UPRT

Instructors should have the specific competence to provide UPRT during the type rating course, including the ability to demonstrate knowledge and understanding of the type-specific upset recovery procedures and of the recommendations that are developed by the OEMs. Therefore, during the CRI training course, the student instructor should:

- (a) be able to apply the correct upset recovery techniques for the specific aeroplane type;
- (b) understand the importance of applying type-specific OEM procedures for recovery manoeuvres;
- (c) be able to distinguish between the applicable SOPs and OEM recommendations (if available);
- (d) understand the capabilities and limitations of the FSTDs that are used for UPRT;
- (e) ensure that the training remains within the FSTD training envelope to avoid the risk of negative transfer of training;

- (f) understand and be able to use the IOS of the FSTD in the context of providing effective UPRT;
- (g) understand and be able to use the available FSTD instructor tools to provide accurate feedback on pilot performance;
- (h) understand the importance of adhering to the FSTD UPRT scenarios that are validated by the training programme developer; and
- (i) understand the missing critical human factor aspects due to the limitations of the FSTD, and convey this to the student pilot(s) receiving the training.

FCL.940.CRI CRI – Revalidation and renewal

Regulation (EU) 2019/1747

- (a) To revalidate a CRI certificate, applicants shall fulfil, within the validity period of the CRI certificate, at least two out of the following three requirements:
 - (1) conduct at least 10 hours of flight instruction as a CRI. If applicants have CRI privileges on both single-engine and multi-engine aeroplanes, those hours of flight instruction shall be equally divided between single-engine and multi-engine aeroplanes;
 - (2) complete a refresher training as a CRI at an ATO or at a competent authority;
 - (3) pass the assessment of competence in accordance with point [FCL.935](#) for multi-engine or single-engine aeroplanes, as relevant.
- (b) For at least each alternate revalidation of a CRI certificate, holders shall have complied with the requirement in point (a)(3).
- (c) **Renewal**

If the CRI certificate has expired, it shall be renewed if the applicants in the period of 12 months before the application for the renewal:

 - (1) have completed a refresher training as a CRI at an ATO or at a competent authority;
 - (2) have completed the assessment of competence as required by point [FCL.935](#).

AMC1 FCL.940.CRI CRI Revalidation and renewal

ED Decision 2020/005/R

REFRESHER TRAINING

- (a) Paragraph (c)(1) of [FCL.940.CRI](#) determine that an applicant for renewal of a CRI certificate shall complete refresher training as a CRI at an ATO or competent authority. Paragraph (a)(2) also establishes that an applicant for revalidation of the CRI certificate that has not completed a minimum amount of instruction hours (established in paragraph (a)(1)) during the validity period of the certificate shall undertake refresher training at an ATO or competent authority for the revalidation of the certificate. The amount of refresher training needed should be determined on a case by case basis by the ATO or competent authority, taking into account the following factors:
 - (1) the experience of the applicant;
 - (2) whether the training is for revalidation or renewal;
 - (3) the amount of time elapsed since the last time the applicant has conducted training, in the case of revalidation, or since the certificate has lapsed, in the case of renewal. The

amount of training needed to reach the desired level of competence should increase with the time elapsed.

- (b) Once the ATO or competent authority has determined the needs of the applicant, it should develop an individual training programme that should be based on the CRI training course and focus on the aspects where the applicant has shown the greatest needs.
- (c) After successful completion of the refresher training, as applicable, the ATO or competent authority, should, in accordance with point (b), issue the applicant with a training completion certificate or another document specified by the competent authority, which describes the evaluation of the factors listed in point (a)(1) (the experience of the applicant) and the training received, as well as a statement that the training was successfully completed. The training completion certificate should be presented to the examiner prior to the assessment of competence.

Upon successful completion of the refresher training, as applicable, the ATO should submit the training completion certificate, or the other document specified by the competent authority, to the competent authority.

SECTION 6 – SPECIFIC REQUIREMENTS FOR THE INSTRUMENT RATING

INSTRUCTOR – IRI

FCL.905.IRI IRI – Privileges and conditions

Regulation (EU) 2024/2076

- (a) The privileges of IRIs are to instruct for the issue, revalidation and renewal of a BIR and an IR in the appropriate aircraft category.
- (b) Specific requirements for the MPL course. To instruct for the basic phase of training on an MPL course, the IRI(A) shall:
 - (1) hold an IR for multi-engine aeroplanes; and
 - (2) have completed at least 1 500 hours of flight time in multi-crew operations.
 - (3) In the case of IRI already qualified to instruct on ATP(A) or CPL(A)/IR integrated courses, the requirement of point FCL.905.IRI(b)(2) may be replaced by the completion of the course provided for in point [FCL.905.FI\(k\)\(3\)](#).

FCL.915.IRI IRI – Prerequisites

Regulation (EU) 2021/2227

Applicants for an IRI certificate shall:

- (a) in case of an IRI(A):
 - (1) to provide training in FSTDs during an approved training course at an ATO, have completed at least 200 hours of flight time under IFR after the issuance of the BIR or the IR, of which at least 50 hours shall be in aeroplanes;
 - (2) to provide training in an aeroplane, have completed at least 800 hours of flight time under IFR, of which at least 400 hours shall be in aeroplanes;
 - (3) to apply for an IRI(A) for multi-engine aeroplanes, meet the requirements of points [FCL.915.CRI\(a\)](#), [FCL.930.CRI](#) and [FCL.935](#);
- (b) for an IRI(H):
 - (1) to provide training in FSTDs during an approved training course at an ATO, have completed at least 125 hours of flight time under IFR after the issuance of the IR, of which at least 65 hours shall be instrument flight time in helicopters;
 - (2) to provide training in a helicopter, have completed at least 500 hours of flight time under IFR, of which at least 250 hours shall be instrument flight time in helicopters; and
 - (3) when seeking privileges to provide training in multi-engine helicopters, meet the requirements of points [FCL.910.TRI\(c\)\(1\)](#) and [FCL.915.TRI\(d\)\(2\)](#);
- (c) Applicants for an IRI(As) certificate shall have completed at least 300 hours of flight time under IFR, of which at least 100 hours shall be instrument flight time in airships.

FCL.930.IRI IRI – Training course

Regulation (EU) 2024/2076

- (a) The training course for the IRI shall include, at least:

- (1) 25 hours of teaching and learning instruction;
- (2) 10 hours of technical training, including revision of instrument theoretical knowledge, the preparation of lesson plans and the development of classroom instructional skills;
- (3)
 - (i) for the IRI(A), at least 10 hours of flight instruction on an aeroplane, FFS, FTD 2/3 or FPNT II. In the case of applicants holding an FI(A) certificate, these hours are reduced to 5;
 - (ii) for the IRI(H), at least 10 hours of flight instruction on a helicopter, FFS, FTD 2/3 or FNPT II/III. In the case of applicants holding an FI(H) certificate, those hours are reduced to at least 5;
 - (iii) for the IRI(As), at least 10 hours of flight instruction on an airship, FFS, FTD 2/3 or FNPT II.
- (b) Flight instruction shall be given by an FI qualified in accordance with point [FCL.905.FI\(j\)](#).
- (c) Applicants holding or having held an instructor certificate shall be fully credited towards the requirement of (a)(1).

AMC1 FCL.930.IRI IRI – Training course

ED Decision 2025/002/R

GENERAL

- (a) The aim of the IRI training course is to train aircraft licence holders to the level of competence defined in [FCL.920](#), and adequate for an IRI.
- (b) The IRI training course should give particular stress to the role of the individual in relation to the importance of human factors in the manmachine environment.
- (c) Special attention should be paid to the applicant's levels of maturity and judgement including an understanding of adults, their behavioural attitudes and variable levels of education.
- (d) With the exception of the section on 'teaching and learning', all the subject detail contained in the theoretical and flight training syllabus is complementary to the instrument rating pilot course syllabus which should already be known by the applicant. Therefore, the objective of the course is to:
 - (1) refresh and bring up to date the technical knowledge of the student instructor;
 - (2) train pilots in accordance with the requirements of the modular instrument flying training course;
 - (3) enable the applicant to develop the necessary instructional techniques required for teaching of instrument flying, radio navigation and instrument procedures to the level required for the issue of an instrument rating;
 - (4) ensure that the student instrument rating instructor's flying is of a sufficiently high standard.
- (e) In part 3 some of the air exercises of the flight instruction syllabus of this AMC may be combined in the same flight.
- (f) During the training course the applicants should be made aware of their own attitudes to the important aspects of flight safety. Improving safety awareness should be a fundamental objective throughout the training course. It will be of major importance for the training course to aim at giving applicants the knowledge, skills and attitudes relevant to an instructor's task.

To achieve this, the course curriculum, in terms of objectives, should comprise at least the following areas.

- (g) It is to be noted that airmanship is a vital ingredient of all flight operations. Therefore, in the following air exercises the relevant aspects of airmanship are to be stressed at the appropriate times during each flight.
- (h) The student instructor should learn how to identify common errors and how to correct them properly, which should be emphasised at all times.

CONTENT

- (i) The training course consists of three parts:
 - (1) Part 1: teaching and learning that should follow the content of [AMC1 FCL.920](#).
 - (2) Part 2: instrument technical theoretical knowledge instruction (technical training).
 - (3) Part 3: flight instruction.

Part 1

The content of the teaching and learning part of the FI training course, as established in [AMC1 FCL.930.FI](#), should be used as guidance to develop the course syllabus.

Part 2

THEORETICAL KNOWLEDGE INSTRUCTION SYLLABUS

- (a) The instrument theoretical knowledge instruction should comprise not less than 10 hours training to include the revision of instrument theoretical knowledge, the preparation of lesson plans and the development of classroom instructional skills to enable the IRI to instruct the instrument theoretical knowledge syllabus.
- (b) All the subject detail contained in the instrument theoretical knowledge instruction syllabus and flight instruction syllabus is complementary to the instrument rating pilot course syllabus which should already be known by the applicant. Therefore, the objective of the course is to:
 - (1) refresh and bring up to date the technical knowledge of the student instructor;
 - (2) train pilots in accordance with the requirements of the modular instrument flying training course;
 - (3) enable the applicant to develop the necessary instructional techniques required for teaching of instrument flying, radio navigation and instrument procedures to the level required for the issue of an instrument rating; and
 - (4) ensure that the student instrument rating instructor's flying is of a sufficiently high standard.
- (c) The theoretical subjects covered below should be used to develop the instructor's teaching skills. The items selected should relate to the student's background and should be applied to training for an IR.

GENERAL SUBJECTS

- (d) Physiological and psychological factors:
 - (1) the senses;

- (2) spatial disorientation;
- (3) sensory illusions;
- (4) stress.
- (e) Flight instruments:
 - (1) air speed indicator;
 - (2) altimeter;
 - (3) vertical speed indicator;
 - (4) attitude indicator;
 - (5) heading indicator;
 - (6) turn and slip indicator;
 - (7) magnetic compass;
 - (8) in relation to the above instruments the following items should be covered:
 - (i) principles of operation;
 - (ii) errors and in-flight serviceability checks;
 - (iii) system failures.
- (f) Radio navigation aids:
 - (1) basic radio principles;
 - (2) use of VHF RTF channels;
 - (3) the Morse code;
 - (4) basic principles of radio aids;
 - (5) use of VOR;
 - (6) ground and aeroplane equipment;
 - (7) use of NDB/ADF;
 - (8) ground and aeroplane equipment;
 - (9) use of VHF/DF;
 - (10) radio detection and ranging (radar);
 - (11) ground equipment;
 - (12) primary radar;
 - (13) secondary surveillance radar;
 - (14) aeroplane equipment;
 - (15) transponders;
 - (16) precision approach system;
 - (17) other navigational systems (as applicable) in current operational use;
 - (18) ground and aeroplane equipment;
 - (19) use of DME;

- (20) ground and aeroplane equipment;
 - (21) marker beacons;
 - (22) ground and aeroplane equipment;
 - (23) pre-flight serviceability checks;
 - (24) range, accuracy and limitations of equipment.
- (g) Flight planning considerations;
- (h) Aeronautical information publications:
- (1) the training course should cover the items listed below, but the applicant's aptitude and previous aviation experience should be taken into account when determining the amount of instructional time allotted. Although a number of items contained under this heading are complementary to those contained in the PPL/CPL/IR syllabi, the instructor should ensure that they have been covered during the applicant's training and due allowance should be made for the time needed to revise these items as necessary.
 - (2) AIP
 - (3) NOTAM class 1 and 2;
 - (4) AIC;
 - (5) information of an operational nature;
 - (6) the rules of the air and ATS;
 - (7) visual flight rules and instrument flight rules;
 - (8) flight plans and ATS messages;
 - (9) use of radar in ATS;
 - (10) radio failure;
 - (11) classification of airspace;
 - (12) airspace restrictions and hazards;
 - (13) holding and approach to land procedures;
 - (14) precision approaches and non precision approaches;
 - (15) radar approach procedures;
 - (16) missed approach procedures;
 - (17) visual manoeuvring after an instrument approach;
 - (18) conflict hazards in uncontrolled airspace;
 - (19) communications;
 - (20) types of services;
 - (21) extraction of AIP data relating to radio aids;
 - (22) charts available;
 - (23) en-route;
 - (24) departure and arrival;

- (25) instrument approach and landing;
- (26) amendments, corrections and revision service.
- (i) flight planning general:
 - (1) the objectives of flight planning;
 - (2) factors affecting aeroplane and engine performance;
 - (3) selection of alternate(s);
 - (4) obtaining meteorological information;
 - (5) services available;
 - (6) meteorology briefing;
 - (7) telephone or electronic data processing;
 - (8) actual weather reports (TAFs, METARs and SIGMET messages);
 - (9) the route forecast;
 - (10) the operational significance of the meteorological information obtained (including icing, turbulence and visibility);
 - (11) altimeter considerations;
 - (12) definitions of:
 - (i) transition altitude;
 - (ii) transition level;
 - (iii) flight level;
 - (iv) QNH;
 - (v) regional QNH;
 - (vi) standard pressure setting;
 - (vii) QFE.
 - (13) altimeter setting procedures;
 - (14) pre-flight altimeter checks;
 - (15) take-off and climb;
 - (16) en-route;
 - (17) approach and landing;
 - (18) missed approach;
 - (19) terrain clearance;
 - (20) selection of a minimum safe en-route altitude;
 - (21) IFR;
 - (22) preparation of charts;
 - (23) choice of routes and flight levels;
 - (24) compilation of flight plan or log sheet;

- (25) log sheet entries;
 - (26) navigation ground aids to be used;
 - (27) frequencies and identification;
 - (28) radials and bearings;
 - (29) tracks and fixes;
 - (30) safety altitude(s);
 - (31) fuel calculations;
 - (32) ATC frequencies (VHF);
 - (33) tower, approach, en-route, radar, FIS, ATIS, and weather reports;
 - (34) minimum sector altitudes at destination and alternate aerodromes;
 - (35) determination of minimum safe descent heights or altitudes (decision heights) at destination and alternate aerodromes.
- (j) The privileges of the instrument rating:
- (1) outside controlled airspace;
 - (2) within controlled airspace;
 - (3) period of validity and renewal procedures.

Part 3

FLIGHT INSTRUCTION SYLLABUS

- (a) When conducting the 10 hours of flight instruction, a similar number of hours should be used for the instruction and practice of preflight and post-flight briefing for each exercise.
- (b) The flight instruction should aim to ensure that the applicant is able to teach the air exercises safely and efficiently.

A. AEROPLANES

LONG BRIEFINGS AND AIR EXERCISES

EXERCISE 1: INSTRUMENT FLYING (Basic)

(for revision, as deemed necessary by the instructor)

- (a) Long briefing objectives:
 - (1) flight instruments;
 - (2) physiological considerations;
 - (3) instrument appreciation:
 - (i) attitude instrument flight;
 - (ii) pitch indications;
 - (iii) bank indications;

- (iv) different instrument presentations;
 - (v) introduction to the use of the attitude indicator;
 - (vi) pitch attitude;
 - (vii) bank attitude;
 - (viii) maintenance of heading and balanced flight;
 - (ix) instrument limitations (inclusive system failures).
- (4) attitude, power and performance:
 - (i) attitude instrument flight;
 - (ii) control instruments;
 - (iii) performance instruments;
 - (iii) effect of changing power and configuration;
 - (iv) cross-checking the instrument indications;
 - (v) instrument interpretation;
 - (vi) direct and indirect indications (performance instruments);
 - (vii) instrument lag;
 - (viii) selective radial scan.
- (5) the basic flight manoeuvres (full panel):
 - (i) straight and level flight at various air speeds and aeroplane configurations;
 - (ii) climbing;
 - (iii) descending;
 - (iv) standard rate turns;
 - (v) level, climbing and descending on to pre-selected headings.
- (b) Air exercise:
 - (1) instrument flying (basic);
 - (i) physiological sensations;
 - (ii) instrument appreciation;
 - (iii) attitude instrument flight;
 - (iv) pitch attitude;
 - (v) bank attitude;
 - (vi) maintenance of heading and balanced flight;
 - (vii) attitude instrument flight;
 - (viii) effect of changing power and configuration;
 - (ix) cross-checking the instruments;
 - (x) selective radial scan;
 - (2) the basic flight manoeuvres (full panel):

- (i) straight and level flight at various air speeds and aeroplane configurations;
- (ii) climbing;
- (iii) descending;
- (iv) standard rate turns;
- (v) level, climbing and descending on to pre-selected headings.

EXERCISE 2: INSTRUMENT FLYING (Advanced)

- (a) Long briefing objectives:
 - (1) full panel;
 - (2) 30 ° level turns;
 - (3) unusual attitudes: recoveries;
 - (4) transference to instruments after take-off;
 - (5) limited panel;
 - (6) basic flight manoeuvres;
 - (7) unusual attitudes: recoveries.
- (b) Air exercise:
 - (1) full panel;
 - (2) 30 ° level turns;
 - (3) unusual attitudes: recoveries;
 - (4) limited panel;
 - (5) repeat of the above exercises.

EXERCISE 3: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF VOR

- (a) Long briefing objectives:
 - (1) availability of VOR stations en-route;
 - (2) station frequencies and identification;
 - (3) signal reception range;
 - (4) effect of altitude;
 - (5) VOR radials;
 - (6) use of OBS;
 - (7) to or from indicator;
 - (8) orientation;
 - (9) selecting radials;
 - (10) intercepting a pre-selected radial;
 - (11) assessment of distance to interception;
 - (12) effects of wind;

- (13) maintaining a radial;
 - (14) tracking to and from a VOR station;
 - (15) procedure turns;
 - (16) station passage;
 - (17) use of two stations for obtaining a fix;
 - (18) pre-selecting fixes along a track;
 - (19) assessment of ground speed and timing;
 - (20) holding procedures;
 - (21) various entries;
 - (22) communication (R/T procedures and ATC liaison).
- (b) Air exercise:
- (1) station selection and identification;
 - (2) orientation;
 - (3) intercepting a pre-selected radial;
 - (4) R/T procedures and ATC liaison;
 - (5) maintaining a radial inbound;
 - (6) recognition of station passage;
 - (7) maintaining a radial outbound;
 - (8) procedure turn;
 - (9) use of two stations to obtain a fix along the track;
 - (10) assessment of ground speed and timing;
 - (11) holding procedures and entries;
 - (12) holding at a pre-selected fix;
 - (13) holding at a VOR station.

EXERCISE 4: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF NDB

- (a) Long briefing objectives:
- (1) availability of an NDB facilities en-route;
 - (2) location, frequencies, tuning (as applicable) and identification codes;
 - (3) signal reception range;
 - (4) static interference;
 - (5) night effect;
 - (6) station interference;
 - (7) mountain effect;
 - (8) coastal refraction;

- (9) orientation in relation to an NDB;
 - (10) homing;
 - (11) intercepting a pre-selected magnetic bearing and tracking inbound;
 - (12) station passage;
 - (13) tracking outbound;
 - (14) time and distance checks;
 - (15) use of two NDBs to obtain a fix or alternatively use of one NDB and one other navaid;
 - (16) holding procedures and various approved entries;
 - (17) communication (R/T procedures and ATC liaison).
- (b) Air exercise:
- (1) selecting, tuning and identifying an NDB;
 - (2) ADF orientation;
 - (3) communication (R/T procedures and ATC liaison);
 - (4) homing;
 - (5) tracking inbound;
 - (6) station passage;
 - (7) tracking outbound;
 - (8) time and distance checks;
 - (9) intercepting a pre-selected magnetic bearing;
 - (10) determining the aeroplane's position from two NDBs or alternatively from one NDB and one other navaid;
 - (11) ADF holding procedures and various approved entries.

EXERCISE 5: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF VHF/DF

- (a) Long briefing objectives:
- (1) availability of VHF/DF facilities en-route;
 - (2) location, frequencies, station call signs and hours of operation;
 - (3) signal and reception range;
 - (4) effect of altitude;
 - (5) communication (R/T procedures and ATC liaison);
 - (6) obtaining and using types of bearings, for example QTE, QDM and QDR;
 - (7) homing to a station;
 - (8) effect of wind;
 - (9) use of two VHF/DF stations to obtain a fix (or alternatively one VHF/DF station and one other navaid);
 - (10) assessment of groundspeed and timing.

- (b) Air exercise:
- (1) establishing contact with a VHF/DF station;
 - (2) R/T Procedures and ATC liaison;
 - (3) obtaining and using a QDR and QTE;
 - (4) homing to a station;
 - (5) effect of wind;
 - (6) use of two VHF/DF stations to obtain a fix (or alternatively one VHF/DF station and one other navaid);
 - (7) assessment of groundspeed and timing.

EXERCISE 6: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF DME

- (a) Long briefing objectives:
- (1) availability of DME facilities;
 - (2) location, frequencies and identification codes;
 - (3) signal reception range;
 - (4) slant range;
 - (5) use of DME to obtain distance, groundspeed and timing;
 - (6) use of DME to obtain a fix.
- (b) Air exercise:
- (1) station selection and identification;
 - (2) use of equipment functions;
 - (3) distance;
 - (4) groundspeed;
 - (5) timing;
 - (6) DME arc approach;
 - (7) DME holding.

EXERCISE 7: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF TRANSPONDERS (SSR)

- (a) Long briefing objectives:
- (1) operation of transponders;
 - (2) code selection procedure;
 - (3) emergency codes;
 - (4) precautions when using airborne equipment.
- (b) Air exercise:
- (1) operation of transponders;
 - (2) types of transponders;

- (3) code selection procedure;
- (4) emergency codes;
- (5) precautions when selecting the required code.

EXERCISE 8: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF ENROUTE RADAR

- (a) Long briefing objectives:
 - (1) availability of radar services;
 - (2) location, station frequencies, call signs and hours of operation;
 - (3) AIP and NOTAMs;
 - (4) provision of service;
 - (5) communication (R/T, procedures and ATC liaison);
 - (6) airspace radar advisory service;
 - (7) emergency service;
 - (8) aircraft separation standards.
- (b) Air exercise:
 - (1) communication (R/T procedures and ATC liaison);
 - (2) establishing the service required and position reporting;
 - (3) method of reporting conflicting traffic;
 - (4) terrain clearance.

EXERCISE 9: PRE-FLIGHT AND AERODROME DEPARTURE AND ARRIVAL PROCEDURES

- (a) Long briefing objectives:
 - (1) determining the serviceability of the aeroplane radio;
 - (2) navigation equipment;
 - (3) obtaining the departure clearance;
 - (4) setting up radio nav aids before take-off for example VOR frequencies, required radials, etc.;
 - (5) aerodrome departure procedures, frequency changes;
 - (6) altitude and position reporting as required;
 - (7) SID procedures;
 - (8) obstacle clearance considerations.
- (b) Air exercise:
 - (1) radio equipment serviceability checks;
 - (2) departure clearance;
 - (3) nav aid selection;
 - (4) frequencies, radials, etc.;

- (5) aerodrome departure checks, frequency changes, altitude and position reports;
- (6) SID procedures.

EXERCISE 10: INSTRUMENT APPROACH: ILS APPROACHES TO SPECIFIED MINIMA AND MISSED APPROACH PROCEDURE

- (a) Long briefing objectives:
 - (1) precision approach charts;
 - (2) approach to the initial approach fix and minimum sector altitude;
 - (3) navaid requirements, for example radar, ADF, etc.;
 - (4) communication (ATC liaison and R/T phraseology);
 - (5) holding procedure;
 - (6) the final approach track;
 - (7) forming a mental picture of the approach;
 - (8) completion of aerodrome approach checks;
 - (9) initial approach procedure;
 - (10) selection of the ILS frequency and identification;
 - (11) obstacle clearance altitude or height;
 - (12) operating minima;
 - (13) achieving the horizontal and vertical patterns;
 - (14) assessment of distance, groundspeed time, and rate of descent from the final approach fix to the aerodrome;
 - (15) use of DME (as applicable);
 - (16) go-around and missed approach procedure;
 - (17) review of the published instructions;
 - (18) transition from instrument to visual flight (sensory illusions);
 - (19) visual manoeuvring after an instrument approach:
 - (i) circling approach;
 - (ii) visual approach to landing.
- (b) Air exercise:
 - (1) initial approach to the ILS;
 - (2) completion of approach planning;
 - (3) holding procedure;
 - (4) frequency selection and identification of ILS;
 - (5) review of the published procedure and minimum sector altitude;
 - (6) communication (ATC liaison and R/T phraseology);
 - (7) determination of operating minima and altimeter setting;

- (8) weather consideration, for example cloud base and visibility;
- (9) availability of runway lighting;
- (10) ILS entry methods;
- (11) radar vectors;
- (12) procedural method;
- (13) assessment of approach time from the final approach fix to the aerodrome;
- (14) determination of:
 - (i) the descent rate on final approach;
 - (ii) the wind velocity at the surface and the length of the landing runway;
 - (iii) the obstruction heights to be borne in mind during visual manoeuvring after an instrument approach;
- (15) circling approach;
- (16) the approach:
 - (i) at the final approach fix;
 - (ii) use of DME (as applicable);
 - (iii) ATC liaison;
 - (iv) note time and establish air speed and descent rate;
 - (v) maintaining the localiser and glide path;
 - (vi) anticipation in change of wind velocity and its effect on drift;
 - (vii) decision height;
- (17) runway direction;
- (18) overshoot and missed approach procedure;
- (19) transition from instrument to visual flight;
- (20) circling approach;
- (21) visual approach to landing.

EXERCISE 11: INSTRUMENTS APPROACH: NDB APPROACHES TO SPECIFIED MINIMA AND MISSED APPROACH PROCEDURES

- (a) Long briefing objectives:
 - (1) non-precision approach charts;
 - (2) initial approach to the initial approach fix and minimum sector altitude;
 - (3) ATC liaison;
 - (4) communication (ATC procedures and R/T phraseology);
 - (5) approach planning;
 - (6) holding procedure;
 - (7) the approach track;

- (8) forming a mental picture of the approach;
 - (9) initial approach procedure;
 - (10) operating minima;
 - (11) completion of approach planning;
 - (12) achieving the horizontal and vertical patterns;
 - (13) assessment of distance, groundspeed time, and rate of descent from the final approach fix to the aerodrome;
 - (14) use of DME (as applicable);
 - (15) go-around and missed approach procedure;
 - (16) review of the published instructions;
 - (17) transition from instrument to visual flight (sensory illusions);
 - (18) visual manoeuvring after an instrument approach;
 - (19) circling approach
 - (20) visual approach to landing.
- (b) Air exercise:
- (1) completion of approach planning including determination of:
 - (i) descent rate from the final approach fix;
 - (ii) the wind velocity at the surface and length of the landing runway;
 - (iii) the obstruction heights to be borne in mind during visual manoeuvring after an instrument approach;
 - (2) circling approach;
 - (3) go-around and missed approach procedure;
 - (4) initial approach;
 - (5) frequency selection and identification;
 - (6) review of the published procedure and minimum safe sector altitude;
 - (7) ATC liaison and R/T phraseology;
 - (8) determination of decision height and altimeter setting;
 - (9) weather considerations, for example cloud base and visibility;
 - (10) availability of runway lighting;
 - (11) determination of inbound track;
 - (12) assessment of time from final approach fix to the missed approach point;
 - (13) ATC liaison;
 - (14) the outbound procedure (inclusive completion of pre-landing checks);
 - (15) the inbound procedure;
 - (16) re-check of identification code;

- (17) altimeter setting re-checked;
- (18) the final approach;
- (19) note time and establish air speed and descent rate
- (20) maintaining the final approach track;
- (21) anticipation of change in wind velocity and its effect on the drift;
- (22) minimum descent altitude or height;
- (23) runway direction;
- (24) go-around and missed approach procedure;
- (25) transition from instrument to visual flight (sensory illusions);
- (26) visual approach.

EXERCISE 12: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF GNSS (to be developed)

- (a) Long briefing objectives: use of GNSS.
- (b) Air exercise: use of GNSS.

B. HELICOPTERS**LONG BRIEFINGS AND AIR EXERCISES****EXERCISE 1: INSTRUMENT FLYING (Basic)**

(for revision as deemed necessary by the instructor)

- (a) Long briefing objectives:
 - (1) flight instruments;
 - (2) physiological considerations;
 - (3) instrument appreciation:
 - (i) attitude instrument flight;
 - (ii) pitch indications;
 - (iii) bank indications;
 - (iv) different instrument presentations;
 - (v) introduction to the use of the attitude indicator;
 - (vi) pitch attitude;
 - (vii) bank attitude;
 - (viii) maintenance of heading and balanced flight;
 - (ix) instrument limitations (inc. system failures);
 - (4) attitude, power and performance:
 - (i) attitude instrument flight;
 - (ii) control instruments;

- (iii) performance instruments;
 - (iv) effect of changing power;
 - (v) cross-checking the instrument indications;
 - (vi) instrument interpretation;
 - (vii) direct and indirect indications (performance instruments);
 - (viii) instrument lag;
 - (ix) selective radial scan;
- (5) the basic flight manoeuvres (full panel):
 - (i) straight and level flight at various air speeds;
 - (ii) climbing;
 - (iii) descending;
 - (iv) standard rate turns;
 - (v) level, climbing and descending on to pre-selected headings.
- (b) Air exercise:
 - (1) physiological sensations;
 - (2) instrument appreciation;
 - (3) attitude instrument flight;
 - (4) pitch attitude;
 - (5) bank attitude;
 - (6) maintenance of heading and balanced flight;
 - (7) attitude instrument flight;
 - (8) effect of changing power;
 - (9) cross-checking the instruments;
 - (10) selective radial scan;
 - (11) the basic flight manoeuvres (full panel):
 - (i) straight and level flight at various air speeds and helicopter configurations;
 - (ii) climbing;
 - (iii) descending;
 - (iv) standard rate turns;
 - (v) level, climbing and descending on to pre-selected headings; (vi) manoeuvring at minimum and maximum IMC speed.

EXERCISE 2: INSTRUMENT FLYING (Advanced)

- (a) Long briefing objectives:
 - (1) full panel;

- (2) 30° level turns;
 - (3) unusual attitudes: recoveries;
 - (4) transition to instruments after take-off;
 - (5) limited panel;
 - (6) basic flight manoeuvres;
 - (7) unusual attitudes: recoveries.
- (b) Air exercise:
- (1) full panel;
 - (2) 30° level turns;
 - (3) unusual attitudes: recoveries;
 - (4) identification and recovery from low pitch steep bank and high pitch steep bank attitudes (at low and high power settings);
 - (5) limited panel;
 - (6) repeat of the above exercises.

EXERCISE 3: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF VOR

- (a) Long briefing objectives:
- (1) availability of VOR stations en-route;
 - (2) station frequencies and identification;
 - (3) signal reception range;
 - (4) effect of altitude;
 - (5) VOR radials;
 - (6) use of OBS;
 - (7) to and from indicator;
 - (8) orientation;
 - (9) selecting radials;
 - (10) intercepting a pre-selected radial;
 - (11) assessment of distance to interception;
 - (12) effects of wind;
 - (13) maintaining a radial;
 - (14) tracking to and from a VOR station;
 - (15) procedure turns;
 - (16) station passage;
 - (17) use of two stations for obtaining a fix;
 - (18) pre-selecting fixes along a track;

- (19) assessment of ground speed and timing;
 - (20) holding procedures;
 - (21) various entries;
 - (22) communication (R/T procedures and ATC liaison).
- (b) Air exercise:
- (1) station selection and identification;
 - (2) orientation;
 - (3) intercepting a pre-selected radial;
 - (4) R/T procedures and ATC liaison;
 - (5) maintaining a radial inbound;
 - (6) recognition of station passage;
 - (7) maintaining a radial outbound;
 - (8) procedure turns;
 - (9) use of two stations to obtain a fix along the track;
 - (10) assessment of ground speed and timing;
 - (11) holding procedures and entries;
 - (12) holding at a pre-selected fix;
 - (13) holding at a VOR station.

EXERCISE 4: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF NDB

- (a) Long briefing objectives:
- (1) availability of NDB facilities en-route;
 - (2) location, frequencies, tuning (as applicable) and identification codes;
 - (3) signal reception range;
 - (4) static interference;
 - (5) night effect;
 - (6) station interference;
 - (7) mountain effect;
 - (8) coastal refraction;
 - (9) orientation in relation to an NDB;
 - (10) homing;
 - (11) intercepting a pre-selected magnetic bearing and tracking inbound;
 - (12) station passage;
 - (13) tracking outbound;
 - (14) time and distance checks;

- (15) use of two NDBs to obtain a fix or alternatively use of one NDB and one other navaid;
- (16) holding procedures;
- (17) communication (R/T procedures and ATC liaison).
- (b) Air exercise:
 - (1) selecting, tuning and identifying an NDB;
 - (2) ADF orientation;
 - (3) communication (R/T procedures and ATC liaison);
 - (4) homing;
 - (5) tracking inbound;
 - (6) station passage;
 - (7) tracking outbound;
 - (8) time and distance checks;
 - (9) intercepting a pre-selected magnetic bearing;
 - (10) determining the helicopter's position from two NDBs or alternatively from one NDB and one other navaid;
 - (11) ADF holding procedures.

EXERCISE 5: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF VHF/DF

- (a) Long briefing objectives:
 - (1) availability of VHF/DF facilities en-route;
 - (2) location, frequencies, station call signs and hours of operation;
 - (3) signal and reception range;
 - (4) effect of altitude;
 - (5) communication (R/T procedures and ATC liaison);
 - (6) obtaining and using types of bearings, for example QTE, QDM, QDR;
 - (7) homing to a station;
 - (8) effect of wind;
 - (9) use of two VHF/DF stations to obtain a fix (or alternatively one VHF/DF station and one other navaid);
 - (10) assessment of groundspeed and timing.
- (b) Air exercise:
 - (1) establishing contact with a VHF/DF station;
 - (2) R/T procedures and ATC liaison;
 - (3) obtaining and using a QDR and QTE;
 - (4) homing to a station;
 - (5) effect of wind;

- (6) use of two VHF/DF stations to obtain a fix (or alternatively one VHF/DF station and one other navaid);
- (7) assessment of groundspeed and timing.

EXERCISE 6: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF DME

- (a) Long briefing objectives:
 - (1) availability of DME facilities;
 - (2) location, frequencies and identification codes;
 - (3) signal reception range;
 - (4) slant range;
 - (5) use of DME to obtain distance, groundspeed and timing;
 - (6) use of DME to obtain a fix;
- (b) Air exercise:
 - (1) station selection and identification;
 - (2) use of equipment functions;
 - (3) distance;
 - (4) groundspeed;
 - (5) timing;
 - (6) DME arc approach;
 - (7) DME holding.

EXERCISE 7: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF TRANSPONDERS

- (a) Long briefing objectives:
 - (1) operation of transponders;
 - (2) code selection procedure;
 - (3) emergency codes;
 - (4) precautions when using airborne equipment.
- (b) Air exercise:
 - (1) operation of transponders;
 - (2) types of transponders;
 - (3) code selection procedure;
 - (4) emergency codes;
 - (5) precautions when selecting the required code.

EXERCISE 8: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF ENROUTE RADAR SERVICES

- (a) Long briefing objectives:

- (1) availability of radar services;
 - (2) location, station frequencies, call signs and hours of operation;
 - (3) AIP and NOTAMS;
 - (4) provision of service;
 - (5) communication (R/T procedures and ATC liaison);
 - (6) airspace radar advisory service;
 - (7) emergency service
 - (8) aircraft separation standards.
- (b) Air exercise:
- (1) communication (R/T procedures and ATC liaison);
 - (2) establishing the service required and position reporting;
 - (3) method of reporting conflicting traffic;
 - (4) terrain clearance.

EXERCISE 9: PRE-FLIGHT AND AERODROME DEPARTURE AND ARRIVAL PROCEDURES

- (a) Long briefing objectives:
- (1) determining the serviceability of the radio equipment;
 - (2) navigation equipment;
 - (3) obtaining the departure clearance;
 - (4) setting up radio nav aids before take-off for example VOR frequencies, required radials, etc.;
 - (5) aerodrome departure procedures, frequency changes;
 - (6) altitude and position reporting as required;
 - (7) SID procedures;
 - (8) obstacle clearance considerations.
- (b) Air exercise:
- (1) radio equipment serviceability checks;
 - (2) departure clearance;
 - (3) nav aid selection;
 - (4) frequencies, radials, etc.;
 - (5) aerodrome departure checks, frequency changes, altitude and position reports;
 - (6) SID procedures.

EXERCISE 10: INSTRUMENT APPROACH: PRECISION APPROACH AID TO SPECIFIED MINIMA AND MISSED APPROACH PROCEDURES

- (a) Long briefing objectives:

- (1) precision approach charts;
 - (2) approach to the initial approach fix and minimum sector altitude;
 - (3) navaid requirements, for example radar, ADF, etc.;
 - (4) communication (ATC liaison and R/T phraseology);
 - (5) holding procedure;
 - (6) the final approach track;
 - (7) forming a mental picture of the approach;
 - (8) completion of aerodrome approach checks;
 - (9) initial approach procedure;
 - (10) selection of the ILS frequency and identification;
 - (11) obstacle clearance altitude or height;
 - (12) operating minima;
 - (13) achieving the horizontal and vertical patterns;
 - (14) assessment of distance, groundspeed time, and rate of descent from the final approach fix to the aerodrome;
 - (15) use of DME (as applicable);
 - (16) go-around and missed approach procedure;
 - (17) review of the published instructions;
 - (18) transition from instrument to visual flight (sensory illusions);
 - (19) visual manoeuvring after an instrument approach;
 - (i) circling approach;
 - (ii) visual approach to landing.
- (b) Air exercise:
- (1) initial approach to the ILS;
 - (2) completion of approach planning;
 - (3) holding procedure;
 - (4) frequency selection and identification of ILS;
 - (5) review of the published procedure and minimum sector altitude;
 - (6) communication (ATC liaison and R/T phraseology);
 - (7) determination of operating minima and altimeter setting;
 - (8) weather consideration, for example cloud base and visibility;
 - (9) availability of landing site lighting;
 - (10) ILS entry methods;
 - (11) radar vectors;
 - (12) procedural method;

- (13) assessment of approach time from the final approach fix to the aerodrome;
- (14) determination of:
 - (i) the descent rate on final approach;
 - (ii) the wind velocity at the surface and the length of the landing site;
 - (iii) the obstruction heights to be borne in mind during visual manoeuvring after an instrument approach;
- (15) circling approach;
- (16) the approach:
 - (i) at the final approach fix;
 - (ii) use of DME (as applicable);
 - (iii) ATC liaison;
 - (iv) note time and establish air speed and descent rate;
 - (v) maintaining the localizer and glide path;
 - (vi) anticipation in change of wind velocity and its effect on drift;
 - (vii) decision height.
- (17) landing direction;
- (18) go-around and missed approach procedure;
- (19) transition from instrument to visual flight;
- (20) circling approach;
- (21) visual approach to landing.

EXERCISE 11: INSTRUMENT APPROACH: NON-PRECISION APPROACH TO SPECIFIED MINIMA AND MISSED APPROACH PROCEDURES

- (a) Long briefing objectives:
 - (1) non-precision approach charts;
 - (2) initial approach to the initial approach fix and minimum sector altitude;
 - (3) ATC liaison;
 - (4) communication (ATC procedures and R/T phraseology);
 - (5) approach planning;
 - (6) holding procedure;
 - (7) the approach track;
 - (8) forming a mental picture of the approach;
 - (9) initial approach procedure;
 - (10) operating minima;
 - (11) completion of approach planning;
 - (12) achieving the horizontal and vertical patterns;

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- (13) assessment of distance, groundspeed time, and rate of descent from the final approach fix to the aerodrome;
 - (14) use of DME (as applicable);
 - (15) go-around and missed approach procedure;
 - (16) review of the published instructions;
 - (17) transition from instrument to visual flight (sensory illusions);
 - (18) visual manoeuvring after an instrument approach;
 - (19) circling approach;
 - (20) visual approach to landing.
- (b) Air exercise:
- (1) completion of approach planning, including determination of:
 - (i) descent rate from the final approach fix;
 - (ii) the wind velocity at the surface and length of the landing site;
 - (iii) the obstruction heights to be borne in mind during visual manoeuvring after an instrument approach.
 - (2) circling approach;
 - (3) go-around and missed approach procedure;
 - (4) initial approach;
 - (5) frequency selection and identification;
 - (6) review of the published procedure and minimum safe sector altitude;
 - (7) ATC liaison and R/T phraseology;
 - (8) determination of decision height and altimeter setting;
 - (9) weather considerations, for example cloud base and visibility;
 - (10) availability of landing site lighting;
 - (11) determination of inbound track;
 - (12) assessment of time from final approach fix to the missed approach point;
 - (13) ATC liaison;
 - (14) the outbound procedure (incl. completion of pre-landing checks);
 - (15) the inbound procedure;
 - (16) re-check of identification code;
 - (17) altimeter setting re-checked;
 - (18) the final approach;
 - (19) note time and establish air speed and descent rate;
 - (20) maintaining the final approach track;
 - (21) anticipation of change in wind velocity and its effect on the drift;

- (22) minimum descent altitude or height;
- (23) landing site direction;
- (24) go-around and missed approach procedure;
- (25) transition from instrument to visual flight (sensory illusions);
- (26) visual approach.

EXERCISE 12: USE OF GNSS (to be developed)

- (a) Long briefing objectives: use of GNSS.
- (b) Air exercise: use of GNSS.

C. AIRSHIPS**LONG BRIEFINGS AND AIR EXERCISES****EXERCISE 1: INSTRUMENT FLYING (Basic)**

(for revision as deemed necessary by the instructor)

- (a) Long briefing objectives:
 - (1) flight instruments;
 - (2) physiological considerations;
 - (3) instrument appreciation:
 - (i) attitude instrument flight;
 - (ii) pitch indications;
 - (iii) different instrument presentations;
 - (iv) introduction to the use of the attitude indicator;
 - (v) pitch attitude;
 - (vi) maintenance of heading and balanced flight;
 - (vii) instrument limitations (inclusive system failures).
 - (4) attitude, power and performance:
 - (i) attitude instrument flight;
 - (ii) control instruments;
 - (iii) performance instruments;
 - (iii) effect of changing power, trim and configuration;
 - (iv) cross-checking the instrument indications;
 - (v) instrument interpretation;
 - (vi) direct and indirect indications (performance instruments);
 - (vii) instrument lag;
 - (viii) selective radial scan.

- (5) the basic flight manoeuvres (full panel):
 - (i) straight and level flight at various air speeds and airship configurations;
 - (ii) climbing;
 - (iii) descending;
 - (iv) standard rate turns;
 - (v) level, climbing and descending on to pre-selected headings.
- (b) Air exercise:
 - (1) physiological sensations;
 - (2) instrument appreciation;
 - (3) attitude instrument flight;
 - (4) pitch attitude;
 - (5) bank attitude;
 - (6) maintenance of heading and balanced flight;
 - (7) attitude instrument flight;
 - (8) effect of changing power and configuration;
 - (9) cross-checking the instruments;
 - (10) selective radial scan;
 - (11) the basic flight manoeuvres (full panel):
 - (i) straight and level flight at various air speeds and airship configurations;
 - (ii) climbing;
 - (iii) descending;
 - (iv) standard rate turns;
 - (v) level, climbing and descending on to pre-selected headings.

EXERCISE 2: INSTRUMENT FLYING (Advanced)

- (a) Long briefing objectives:
 - (1) full panel;
 - (2) unusual attitudes: recoveries;
 - (3) transference to instruments after take-off;
 - (4) limited panel;
 - (5) basic flight manoeuvres;
 - (6) unusual attitudes: recoveries.
- (b) Air exercise:
 - (1) full panel;
 - (2) unusual attitudes: recoveries;

- (3) limited panel;
- (4) repeat of the above exercises.

EXERCISE 3: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF VOR

- (a) Long briefing objectives:
 - (1) availability of VOR stations en-route;
 - (2) station frequencies and identification;
 - (3) signal reception range;
 - (4) effect of altitude;
 - (5) VOR radials;
 - (6) use of OBS;
 - (7) to or from indicator;
 - (8) orientation;
 - (9) selecting radials;
 - (10) intercepting a pre-selected radial;
 - (11) assessment of distance to interception;
 - (12) effects of wind;
 - (13) maintaining a radial;
 - (14) tracking to and from a VOR station;
 - (15) procedure turns;
 - (16) station passage;
 - (17) use of two stations for obtaining a fix;
 - (18) pre-selecting fixes along a track;
 - (19) assessment of ground speed and timing;
 - (20) holding procedures;
 - (21) various entries;
 - (22) communication (R/T procedures and ATC liaison).
- (b) Air exercise:
 - (1) station selection and identification;
 - (2) orientation;
 - (3) intercepting a pre-selected radial;
 - (4) R/T procedures and ATC liaison;
 - (5) maintaining a radial inbound;
 - (6) recognition of station passage;
 - (7) maintaining a radial outbound;

- (8) procedure turns;
- (9) use of two stations to obtain a fix along the track;
- (10) assessment of ground speed and timing;
- (11) holding procedures and entries;
- (12) holding at a pre-selected fix;
- (13) holding at a VOR station.

EXERCISE 4: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF ADF

(Automatic DF equipment)

- (a) Long briefing objectives:
 - (1) availability of NDB facilities en-route;
 - (2) location, frequencies, tuning (as applicable) and identification codes;
 - (3) signal reception range;
 - (4) static interference;
 - (5) night effect;
 - (6) station interference;
 - (7) mountain effect;
 - (8) coastal refraction;
 - (9) orientation in relation to an NDB;
 - (10) homing;
 - (11) intercepting a pre-selected magnetic bearing and tracking inbound;
 - (12) station passage;
 - (13) tracking outbound;
 - (14) time and distance checks;
 - (15) use of two NDBs to obtain a fix or alternatively use of one NDB and one other navaid;
 - (16) holding procedures and various approved entries;
 - (17) communication (R/T procedures and ATC liaison).
- (b) Air exercise:
 - (1) selecting, tuning and identifying an NDB;
 - (2) ADF orientation;
 - (3) communication (R/T procedures and ATC liaison);
 - (4) homing;
 - (5) tracking inbound;
 - (6) station passage;
 - (7) tracking outbound;

- (8) time and distance checks;
- (9) intercepting a pre-selected magnetic bearing;
- (10) determining the airship's position from two NDBs or alternatively from one NDB and one other navaid;
- (11) ADF holding procedures and various approved entries.

EXERCISE 5: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF VHF/DF

- (a) Long briefing objectives:
 - (1) availability of VHF/DF facilities en-route;
 - (2) location, frequencies, station call signs and hours of operation;
 - (3) signal and reception range;
 - (4) effect of altitude;
 - (5) communication (R/T procedures and ATC liaison);
 - (6) obtaining and using types of bearings, for example QTE, QDM, QDR;
 - (7) homing to a station;
 - (8) effect of wind;
 - (9) use of two VHF/DF stations to obtain a fix (or alternatively one VHF/DF station and one other navaid);
 - (10) assessment of groundspeed and timing.
- (b) Air exercise:
 - (1) establishing contact with a VHF/DF station;
 - (2) R/T procedures and ATC liaison;
 - (3) obtaining and using a QDR and QTE;
 - (4) homing to a station;
 - (5) effect of wind;
 - (6) use of two VHF/DF stations to obtain a fix (or alternatively one VHF/DF station and one other navaid);
 - (7) assessment of groundspeed and timing.

EXERCISE 6: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF DME

- (a) Long briefing objectives:
 - (1) availability of DME facilities;
 - (2) location, frequencies and identification codes;
 - (3) signal reception range;
 - (4) slant range;
 - (5) use of DME to obtain distance, groundspeed and timing;
 - (6) use of DME to obtain a fix.

- (b) Air exercise:
- (1) station selection and identification;
 - (2) use of equipment functions;
 - (3) distance;
 - (4) groundspeed;
 - (5) timing;
 - (6) DME arc approach;
 - (7) DME holding.

EXERCISE 7: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF TRANSPONDERS

- (a) Long briefing objectives:
- (1) operation of transponders;
 - (2) code selection procedure;
 - (3) emergency codes;
 - (4) precautions when using airborne equipment.
- (b) Air exercise:
- (1) operation of transponders;
 - (2) types of transponders;
 - (3) code selection procedure;
 - (4) emergency codes;
 - (5) precautions when selecting the required code.

EXERCISE 8: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF ENROUTE RADAR SERVICES

- (a) Long briefing objectives:
- (1) availability of radar services;
 - (2) location, station frequencies, call signs and hours of operation;
 - (3) AIP and NOTAMS;
 - (4) provision of service;
 - (5) communication (R/T, procedures and ATC liaison);
 - (6) airspace radar advisory service;
 - (7) emergency service;
 - (8) aircraft separation standards.
- (b) Air exercise:
- (1) communication (R/T procedures and ATC liaison);
 - (2) establishing the service required and position reporting;
 - (3) method of reporting conflicting traffic;

- (4) terrain clearance.

EXERCISE 9: PRE-FLIGHT AND AERODROME DEPARTURE AND ARRIVAL PROCEDURES

- (a) Long briefing objectives:
 - (1) determining the serviceability of the airship radio;
 - (2) navigation equipment;
 - (3) obtaining the departure clearance;
 - (4) setting up radio nav aids before take-off for example VOR frequencies, required radials, etc.;
 - (5) aerodrome departure procedures, frequency changes;
 - (6) altitude and position reporting as required;
 - (7) SID procedures;
 - (8) obstacle clearance considerations.
- (b) Air exercise:
 - (1) radio equipment serviceability checks;
 - (2) departure clearance;
 - (3) nav aid selection;
 - (4) frequencies, radials, etc.;
 - (5) aerodrome departure checks, frequency changes, altitude and position reports;
 - (6) SID procedures.

EXERCISE 10: INSTRUMENT APPROACHES: ILS APPROACHES TO SPECIFIED MINIMA AND MISSED APPROACHES PROCEDURES

- (a) Long briefing objectives:
 - (1) precision approach charts;
 - (2) approach to the initial approach fix and minimum sector altitude;
 - (3) nav aid requirements, for example radar, ADF, etc.;
 - (4) communication (ATC liaison and R/T phraseology);
 - (5) review;
 - (6) holding procedure;
 - (7) the final approach track;
 - (8) forming a mental picture of the approach;
 - (9) completion of aerodrome approach checks;
 - (10) initial approach procedure;
 - (11) selection of the ILS frequency and identification;
 - (12) obstacle clearance altitude or height;

- (13) operating minima;
 - (14) achieving the horizontal and vertical patterns;
 - (15) assessment of distance, groundspeed time, and rate of descent from the final approach fix to the aerodrome;
 - (16) use of DME (as applicable);
 - (17) go-around and missed approach procedure;
 - (18) review of the published instructions;
 - (19) transition from instrument to visual flight (sensory illusions);
 - (20) visual manoeuvring after an instrument approach;
 - (i) circling approach;
 - (ii) visual approach to landing.
- (b) Air exercise:
- (1) initial approach to the ILS;
 - (2) completion of approach planning;
 - (3) holding procedure;
 - (4) frequency selection and identification of ILS;
 - (5) review of the published procedure and minimum sector altitude;
 - (6) communication (ATC liaison and R/T phraseology);
 - (7) determination of operating minima and altimeter setting;
 - (8) weather consideration, for example cloud base and visibility;
 - (9) availability of runway lighting;
 - (10) ILS entry methods;
 - (11) radar vectors;
 - (12) procedural method;
 - (13) assessment of approach time from the final approach fix to the aerodrome;
 - (14) determination of:
 - (i) the descent rate on final approach;
 - (ii) the wind velocity at the surface (and the length of the landing runway);
 - (iii) the obstruction heights to be borne in mind during visual manoeuvring after an instrument approach;
 - (15) circling approach;
 - (16) the approach:
 - (i) at the final approach fix;
 - (ii) use of DME (as applicable);
 - (iii) ATC liaison;

- (iv) note time and establish air speed and descent rate;
- (v) maintaining the localiser and glide path;
- (vi) anticipation in change of wind velocity and its effect on drift;
- (vii) decision height;
- (viii) runway direction.
- (17) missed approach procedure;
- (18) transition from instrument to visual flight;
- (19) circling approach;
- (20) visual approach to landing.

EXERCISE 11: INSTRUMENT APPROACHES: NDB APPROACHES TO SPECIFIED MINIMA AND MISSED APPROACHES PROCEDURE

- (a) Long briefing objectives:
 - (1) non-precision approach charts;
 - (2) initial approach to the initial approach fix and minimum sector altitude;
 - (3) ATC liaison;
 - (4) communication (ATC procedures and R/T phraseology);
 - (5) approach planning:
 - (i) holding procedure;
 - (ii) the approach track;
 - (iii) forming a mental picture of the approach;
 - (iv) initial approach procedure;
 - (v) operating minima;
 - (vi) completion of approach planning.
 - (6) achieving the horizontal and vertical patterns;
 - (7) assessment of distance, groundspeed time, and rate of descent from the final approach fix to the aerodrome;
 - (8) use of DME (as applicable);
 - (9) go-around and missed approach procedure;
 - (10) review of the published instructions;
 - (11) transition from instrument to visual flight (sensory illusions);
 - (12) visual manoeuvring after an instrument approach;
 - (13) circling approach;
 - (14) visual approach to landing.
- (b) Air exercise:
 - (1) completion of approach planning including;

- (2) determination of:
 - (i) descent rate from the final approach fix;
 - (ii) the wind velocity at the surface and length of the landing runway;
 - (iii) the obstruction heights to be borne in mind during visual manoeuvring after an instrument approach.
- (3) circling approach;
- (4) go-around and missed approach procedure;
- (5) initial approach;
- (6) frequency selection and identification;
- (7) review of the published procedure and minimum safe sector altitude;
- (8) ATC liaison and R/T phraseology;
- (9) determination of decision height and altimeter setting;
- (10) weather considerations, for example cloud base and visibility;
- (11) availability of runway lighting;
- (12) determination of inbound track;
- (13) assessment of time from final approach fix to the missed approach point;
- (14) ATC liaison;
- (15) the outbound procedure (inclusive completion of pre-landing checks);
- (16) the inbound procedure;
- (17) re-check of identification code;
- (18) altimeter setting re-checked;
- (19) the final approach;
- (20) note time and descent rate;
- (21) maintaining the final approach track;
- (22) anticipation of change in wind velocity and its effect on the drift;
- (23) minimum descent altitude or height;
- (24) runway direction;
- (25) go-around and missed approach procedure;
- (26) transition from instrument to visual flight (sensory illusions);
- (27) visual approach.

EXERCISE 12: RADIO NAVIGATION (APPLIED PROCEDURES): USE OF GNSS (to be developed)

- (a) Long briefing objectives: use of GNSS.
- (b) Air exercise: use of GNSS.

FCL.940.IRI IRI – Revalidation and renewal

Regulation (EU) No 1178/2011

For revalidation and renewal of an IRI certificate, the holder shall meet the requirements for revalidation and renewal of an FI certificate, in accordance with [FCL.940.FI](#).

SECTION 7 – SPECIFIC REQUIREMENTS FOR THE SYNTHETIC FLIGHT INSTRUCTOR – SFI

FCL.905.SFI SFI – Privileges and conditions

Regulation (EU) 2021/2227

- (a) The privileges of SFIs are to carry out synthetic flight instruction, within the relevant aircraft category, for:
- (1) the revalidation and renewal of an IR, provided that they hold or have held an IR in the relevant aircraft category;
 - (2) the issue of an IR, provided that they hold or have held an IR in the relevant aircraft category and have completed an IRI training course.
- (b) The privileges of SFIs for single-pilot aeroplanes are to carry out synthetic flight instruction for:
- (1) the issue, revalidation and renewal of type ratings for single-pilot high performance complex aeroplanes, if applicants seek privileges to operate in single-pilot operations.

The privileges of SFIs for single-pilot aeroplanes may be extended to flight instruction for single-pilot high performance complex aeroplanes type ratings in multi-pilot operations, provided that they meet any of the following conditions:
 - (i) hold or have held a TRI certificate for multi-pilot aeroplanes;
 - (ii) have at least 500 hours on aeroplanes in multi-pilot operations and have completed an MCCI training course in accordance with point [FCL.930.MCCI](#);
 - (2) the MCC and the MPL training courses on the basic phase, provided that the privileges of SFIs(SPA) have been extended to multi-pilot operations in accordance with point (1).
- (c) The privileges of SFIs for multi-pilot aeroplanes are to carry out synthetic flight instruction for:
- (1) the issue, revalidation and renewal of type ratings for multi-pilot aeroplanes and if applicants seek privileges to operate in multi-pilot operations, for single-pilot high-performance complex aeroplanes;
 - (2) the MCC training course;
 - (3) the MPL course on the basic, intermediate and advanced phases, provided that, for the basic phase, they hold or have held an FI(A) or an IRI(A) certificate;
- (d) The privileges of SFIs for helicopters are to carry out synthetic flight instruction for:
- (1) the issue, revalidation and renewal of helicopter type ratings;
 - (2) MCC training, provided that they have at least 350 hours as a pilot in multi-pilot operations in any aircraft category.
- (e) The privileges of an SFI include privileges to conduct EBT practical assessment at an EBT operator, provided that the instructor complies with the requirements of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#) for EBT instructor standardisation at that EBT operator.

FCL.910.SFI SFI – Restricted privileges

Regulation (EU) 2024/2076

The privileges of SFIs shall be restricted to the FTD 2/3 or FFS of the aircraft type in which the SFI training course was taken.

Unless otherwise specified in the operational suitability data established in accordance with Annex I (Part 21) to Regulation (EU) No 748/2012, to extend the privileges of SFIs to other FSTDs representing further types of the same category of aircraft, SFIs shall have:

- (a) completed the simulator content of the relevant type rating course;
- (b) completed the relevant parts of the technical training and the FSTD content of the flight instruction syllabus of the applicable TRI course;
- (c) conducted on a complete type rating course at least 3 hours of flight instruction related to the duties of an SFI on the applicable type under the supervision and to the satisfaction of a TRE or an SFE qualified for this purpose.

The privileges of the SFI shall be extended to further variants in accordance with the OSD if the SFI has completed the type relevant parts of the technical training and the FSTD content of the flight instruction syllabus of the applicable TRI course.

FCL.915.SFI SFI – Prerequisites

Regulation (EU) 2021/2227

An applicant for an SFI certificate shall:

- (a) hold or have held a CPL, MPL or ATPL in the appropriate aircraft category;
- (b) have completed the proficiency check for the issue of the specific aircraft type rating in an FFS representing the applicable type, within the 12 months preceding the application; and
- (c) additionally, for an SFI(A) for multi-pilot aeroplanes or SFI(PL), have:
 - (1) at least 1 500 hours flight time as a pilot on multi-pilot aeroplanes or powered-lift, as applicable;
 - (2) completed, as a pilot or as an observer, within the 12 months preceding the application, at least:
 - (i) 3 route sectors on the flight deck of the applicable aircraft type; or
 - (ii) 2 line-orientated flight training-based simulator sessions conducted by qualified flight crew on the flight deck of the applicable type. These simulator sessions shall include 2 flights of at least 2 hours each between 2 different aerodromes, and the associated pre-flight planning and de-briefing;
- (d) additionally, for an SFI(A) for single-pilot high performance complex aeroplanes:
 - (1) have completed at least 500 hours of flight time as PIC on single-pilot aeroplanes;
 - (2) hold or have held a multi-engine IR(A) rating; and
 - (3) have met the requirements in (c)(2);
- (e) additionally, for an SFI(H), have:
 - (1) completed, as a pilot or as an observer, at least 1 hour of flight time on the flight deck of the applicable type, within the 12 months preceding the application; and

- (2) in the case of multi-pilot helicopters, at least 1 000 hours of flying experience as a pilot in helicopters, including at least 350 hours in multi-pilot operations in any aircraft category;
- (3) in the case of single-pilot multi-engine helicopters, completed 500 hours as pilot of helicopters, including 100 hours as PIC on single-pilot multi-engine helicopters;
- (4) in the case of single-pilot single-engine helicopters, completed 250 hours as a pilot on helicopters;
- (5) in the case of single-pilot helicopters in multi-pilot operations, completed at least 350 hours in multi-pilot operations in any aircraft category.

FCL.930.SFI SFI – Training course

Regulation (EU) 2024/2076

- (a) The training course for the SFI shall include:
 - (1) the FSTD content of the applicable type rating course;
 - (2) the relevant parts of the technical training and the FSTD content of the flight instruction syllabus of the applicable TRI training course;
 - (3) 25 hours of teaching and learning instruction.
- (b) An applicant for an SFI certificate who:
 - (1) holds a TRI certificate for the relevant type shall be fully credited towards the requirements of point (a);
 - (2) holds or has held an instructor certificate shall be fully credited towards the requirement of point (a)(3).

FCL.940.SFI SFI – Revalidation and renewal

Regulation (EU) 2019/1747

- (a) Revalidation

To revalidate an SFI certificate, applicants shall fulfil, before the expiry date of the SFI certificate, at least two out of the following three requirements:

 - (1) have completed at least 50 hours as instructors or examiners in FSTDs, of which at least 15 hours in the period of 12 months immediately preceding the expiry date of the SFI certificate;
 - (2) have completed instructor refresher training as an SFI at an ATO;
 - (3) have passed the relevant sections of the assessment of competence in accordance with point [FCL.935](#).
- (b) Additionally, applicants shall have completed, on an FFS, the proficiency checks for the issue of the specific aircraft type ratings representing the types for which privileges are held.
- (c) For at least each alternate revalidation of an SFI certificate, holders shall comply with the requirement in point (a)(3).
- (d) If an SFI holds a certificate in more than one type of aircraft within the same category, the assessment of competence taken on one of those types shall revalidate the SFI certificate for the other types held within the same category of aircraft, unless otherwise is determined in the OSD.

(e) Renewal

To renew the SFI certificate, applicants shall, within the period of 12 months immediately preceding the application for the renewal, comply with all of the following conditions:

- (1) have completed instructor refresher training as an SFI at an ATO;
- (2) have passed the assessment of competence in accordance with point [FCL.935](#);
- (3) have completed, on an FSTD, the skill test for the issue of the specific aircraft type ratings representing the types for which privileges are to be renewed.

SECTION 8 – SPECIFIC REQUIREMENTS FOR THE MULTI-CREW COOPERATION

INSTRUCTOR – MCCI

FCL.905.MCCI MCCI – Privileges and conditions

Regulation (EU) No 1178/2011

- (a) The privileges of an MCCI are to carry out flight instruction during:
- (1) the practical part of MCC courses when not combined with type rating training; and
 - (2) in the case of MCCI(A), the basic phase of the MPL integrated training course, provided he/she holds or has held an FI(A) or an IRI(A) certificate.

FCL.910.MCCI MCCI – Restricted privileges

Regulation (EU) No 1178/2011

The privileges of the holder of an MCCI certificate shall be restricted to the FNPT II/III MCC, FTD 2/3 or FFS in which the MCCI training course was taken.

The privileges may be extended to other FSTDs representing further types of aircraft when the holder has completed the practical training of the MCCI course on that type of FNPT II/III MCC, FTD 2/3 or FFS.

FCL.915.MCCI MCCI – Prerequisites

Regulation (EU) 2021/2227

An applicant for an MCCI certificate shall:

- (a) hold or have held a CPL, MPL or ATPL in the appropriate aircraft category;
- (b) have at least:
- (1) in the case of aeroplanes, airships and powered-lift aircraft, 1 500 hours of flying experience as a pilot in multi-pilot operations, of which at least 350 hours in the appropriate aircraft category;
 - (2) in the case of helicopters, 1 000 hours of flying experience as a pilot in multi-pilot operations, of which at least 350 hours in helicopters.

FCL.930.MCCI MCCI – Training course

Regulation (EU) No 1178/2011

- (a) The training course for the MCCI shall include, at least:
- (1) 25 hours of teaching and learning instruction;
 - (2) technical training related to the type of FSTD where the applicant wishes to instruct;
 - (3) 3 hours of practical instruction, which may be flight instruction or MCC instruction on the relevant FNPT II/III MCC, FTD 2/3 or FFS, under the supervision of a TRI, SFI or MCCI nominated by the ATO for that purpose. These hours of flight instruction under supervision shall include the assessment of the applicant's competence as described in [FCL.920](#).

- (b) Applicants holding or having held an FI, TRI, CRI, IRI or SFI certificate shall be fully credited towards the requirement of (a)(1).

AMC1 FCL.930.MCCI MCCI — Training course

ED Decision 2011/016/R

AEROPLANES

GENERAL

- (a) The objective of the technical training is to apply the core instructor competencies acquired during the teaching and learning training to MCC training.
- (b) During the practical training the applicant should demonstrate the ability to instruct a pilot in MCC.
- (c) To supervise applicants for MCCI certificates, the adequate experience should include at least three type rating or MCC courses.
- (d) It is to be noted that airmanship is a vital ingredient of all flight operations. Therefore, in the following air exercises the relevant aspects of airmanship are to be stressed at the appropriate times during each flight.
- (e) The student instructor should learn how to identify common errors and how to correct them properly, which should be emphasised at all times.

COURSE OBJECTIVE

- (f) The course should be designed to give adequate training to the applicant in theoretical knowledge instruction and FSTD instruction to instruct those aspects of MCC required by an applicant for a type rating on a first MP aeroplane.
- (g) Confirmation of competency of the applicant to be authorised as an MCCI(A) will be determined by the applicant conducting at least 3 hours MCC instruction to a satisfactory standard on the relevant FNPT or FFS under the supervision of a TRI(A), SFI(A) or MCCI(A) nominated by the ATO for this purpose.
- (h) The course consists of three parts:
 - (1) Part 1: teaching and learning that should follow the content of [AMC1 FCL.920](#);
 - (2) Part 2: technical theoretical knowledge instruction (technical training);
 - (3) Part 3: flight instruction.

Part 1

The content of the teaching and learning part of the FI training course, as established in [AMC1 FCL.930.FI](#), should be used as guidance to develop the course syllabus.

Part 2

TECHNICAL THEORETICAL KNOWLEDGE INSTRUCTION SYLLABUS

- (a) The FSTD training consists of the application of core instructor competencies to MCC training in a commercial air transport environment, including principles of threat and error management and CRM.

The content of the training programme should cover MCC course exercises in sufficient depth to meet the standard required for issue of the MCCI(A) certificate.

- (b) The course should be related to the type of FSTD on which the applicant wishes to instruct. A training programme should give details of all theoretical knowledge instruction.
- (c) Identification and application of human factors (as set in the ATPL syllabus 040) related to MCC aspects of the training.

Part 3

FLIGHT INSTRUCTION SYLLABUS

- (a) The content of the instruction programme should cover training exercises as applicable to the MCC requirements of an applicant for a MP type rating.
- (b) Training exercises:

The exercises should be accomplished as far as possible in a simulated commercial air transport environment. The instruction should cover the following areas:

- (1) pre-flight preparation, including documentation, and computation of take-off performance data;
- (2) pre-flight checks, including radio and navigation equipment checks and setting;
- (3) before take-off checks, including powerplant checks, and take-off briefing by the PF;
- (4) normal take-offs with different flap settings, tasks of PF and PNF, callouts;
- (5) rejected take-offs; crosswind take-offs; take-offs at maximum takeoff mass; engine failure after v1;
- (6) normal and abnormal operation of aircraft systems, use of checklists;
- (7) selected emergency procedures to include engine failure and fire, smoke control and removal, windshear during take-off and landing, emergency descent, incapacitation of a flight crew member;
- (8) early recognition of and reaction on approaching stall in differing aircraft configurations;
- (9) instrument flight procedures, including holding procedures; precision approaches using raw navigation data, flight director and automatic pilot, one engine simulated inoperative approaches, non-precision and circling approaches, approach briefing by the PF, setting of navigation equipment, call-out procedures during approaches; computation of approach and landing data;
- (10) go-arounds; normal and with one engine simulated inoperative, transition from instrument to visual flight on reaching decision height or minimum descent height or altitude;
- (11) landings, normal, crosswind and with one engine simulated inoperative, transition from instrument to visual flight on reaching decision height or minimum descent height or altitude.

FCL.940.MCCI MCCI – Revalidation and renewal

Regulation (EU) No 1178/2011

- (a) For revalidation of an MCCI certificate the applicant shall have completed the requirements of [FCL.930.MCCI\(a\)\(3\)](#) on the relevant type of FNPT II/III, FTD 2/3 or FFS, within the last 12 months of the validity period of the MCCI certificate.
- (b) Renewal. If the MCCI certificate has lapsed, the applicant shall complete the requirements of [FCL.930.MCCI\(a\)\(2\) and \(3\)](#) on the relevant type of FNPT II/III MCC, FTD 2/3 or FFS.

SECTION 9 – SPECIFIC REQUIREMENTS FOR THE SYNTHETIC TRAINING INSTRUCTOR – STI

FCL.905.STI STI – Privileges and conditions

Regulation (EU) 2020/359

- (a) The privileges of an STI are to carry out synthetic flight instruction in the appropriate aircraft category for:
 - (1) the issue of a licence;
 - (2) the issue, revalidation or renewal of a BIR and an IR and a class or type rating for single-pilot aircraft, except for single-pilot high-performance complex aeroplanes.
- (b) Additional privileges for the STI(A). The privileges of an STI(A) shall include synthetic flight instruction during the core flying skills training of the MPL integrated training course.

FCL.910.STI STI – Restricted privileges

Regulation (EU) 2019/1747

The privileges of STIs shall be restricted to the FSTD in which the STI training course was taken.

The privileges may be extended to other FSTDs representing further types of aircraft if in the period of 12 months immediately preceding the application the holders have:

- (a) completed the FSTD content of the CRI or TRI course on the class or type of aircraft for which instructional privileges are sought;
- (b) passed in the FSTD on which flight instruction is to be conducted, the applicable section of the proficiency check in accordance with Appendix 9 to this Annex for the appropriate class or type of aircraft.

For STIs(A) instructing on BITD only, the proficiency check shall include only the exercises appropriate for the skill test for the issue of a PPL(A);

- (c) conducted, on a CPL, an IR, a PPL or a class or type rating course, at least 3 hours of flight instruction under the supervision of an FI, a CRI(A), an IRI or a TRI nominated by the ATO for this purpose, including at least 1 hour of flight instruction that is supervised by an FIE in the appropriate aircraft category.

FCL.915.STI STI – Prerequisites

Regulation (EU) 2019/1747

- (a) Applicants for the issue of an STI certificate shall:
 - (1) hold, or have held within the 3 years prior to the application, a pilot licence and instructional privileges appropriate to the courses on which instruction is intended;
 - (2) have completed in an FSTD the relevant proficiency check for the class or type rating, in the period of 12 months immediately preceding the application.

Applicants for the issue of an STI(A) wishing to instruct on BITDs only, shall complete the exercises appropriate for a skill test for the issue of a PPL(A) only;

- (b) Additionally to the requirements laid down in point (a), applicants for the issue of an STI(H) certificate shall have completed at least 1 hour of flight time as an observer on the flight deck of the applicable type of helicopter, in the period of 12 months immediately preceding the application.

FCL.930.STI STI – Training course

Regulation (EU) No 1178/2011

- (a) The training course for the STI shall comprise at least 3 hours of flight instruction related to the duties of an STI in an FFS, FTD 2/3 or FNPT II/III, under the supervision of an FIE. These hours of flight instruction under supervision shall include the assessment of the applicant's competence as described in [FCL.920](#).

Applicants for an STI(A) wishing to instruct on a BITD only, shall complete the flight instruction on a BITD.

- (b) For applicants for an STI(H), the course shall also include the FFS content of the applicable TRI course.

FCL.940.STI Revalidation and renewal of the STI certificate

Regulation (EU) 2019/1747

- (a) Revalidation

To revalidate an STI certificate, applicants shall, within the period of 12 months immediately preceding the expiry date of the STI certificate, comply with all of the following conditions:

- (1) have conducted at least 3 hours of flight instruction in an FSTD, as part of a complete CPL, IR, PPL or class or type rating course;
- (2) have passed in the FSTD on which flight instruction is conducted, the applicable sections of the proficiency check in accordance with Appendix 9 to this Annex for the appropriate class or type of aircraft.

For STIs(A) instructing on BITDs only, the proficiency check shall include the exercises appropriate for a skill test for the issue of a PPL(A) only.

- (b) Renewal

To renew STI certificate, the applicants shall within the period of 12 months immediately preceding the application for the renewal:

- (1) complete a refresher training as an STI at an ATO;
- (2) pass in the FSTD on which flight instruction is conducted, the applicable sections of the proficiency check in accordance with Appendix 9 to this Annex for the appropriate class or type of aircraft.

For an STI(A) instructing on BITDs only, the proficiency check shall include the exercises appropriate for a skill test for the issue of a PPL(A) only;

- (3) conduct, in the relevant aircraft category, on a complete CPL, IR, PPL or class or type rating course, at least 3 hours of flight instruction under the supervision of an FI, a CRI, an IRI or a TRI nominated by the ATO for this purpose, including at least 1 hour of flight instruction supervised by a flight instructor examiner (FIE).

SECTION 10 – MOUNTAIN RATING INSTRUCTOR – MI

FCL.905.MI MI – Privileges and conditions

Regulation (EU) No 1178/2011

The privileges of an MI are to carry out flight instruction for the issue of a mountain rating.

FCL.915.MI MI – Prerequisites

Regulation (EU) No 1178/2011

An applicant for an MI certificate shall:

- (a) hold a, FI, CRI, or TRI certificate, with privileges for single-pilot aeroplanes;
- (b) hold a mountain rating.

FCL.930.MI MI – Training course

Regulation (EU) No 1178/2011

- (a) The training course for the MI shall include the assessment of the applicant's competence as described in [FCL.920](#).
- (b) Before attending the course, applicants shall have passed a pre-entry flight test with an MI holding an FI certificate to assess their experience and ability to undertake the training course.

FCL.940.MI Validity of the MI certificate

Regulation (EU) No 245/2014

The MI certificate is valid as long as the, FI, TRI or CRI certificate is valid.

SECTION 11 – SPECIFIC REQUIREMENTS FOR THE FLIGHT TEST INSTRUCTOR – FTI

FCL.905.FTI FTI – Privileges and conditions

Regulation (EU) No 1178/2011

- (a) The privileges of a flight test instructor (FTI) are to instruct, within the appropriate aircraft category, for:
 - (1) the issue of category 1 or 2 flight test ratings, provided he/she holds the relevant category of flight test rating;
 - (2) the issue of an FTI certificate, within the relevant category of flight test rating, provided that the instructor has at least 2 years of experience instructing for the issue of flight test ratings.
- (b) The privileges of an FTI holding a category 1 flight test rating include the provision of flight instruction also in relation to category 2 flight test ratings.

FCL.915.FTI FTI – Prerequisites

Regulation (EU) No 1178/2011

An applicant for an FTI certificate shall:

- (a) hold a flight test rating issued in accordance with [FCL.820](#);
- (b) have completed at least 200 hours of category 1 or 2 flight tests.

FCL.930.FTI FTI – Training course

Regulation (EU) No 1178/2011

- (a) The training course for the FTI shall include, at least:
 - (1) 25 hours of teaching and learning;
 - (2) 10 hours of technical training, including revision of technical knowledge, the preparation of lesson plans and the development of classroom/simulator instructional skills;
 - (3) 5 hours of practical flight instruction under the supervision of an FTI qualified in accordance with [FCL.905.FTI\(b\)](#). These hours of flight instruction shall include the assessment of the applicant's competence as described in [FCL.920](#).
- (b) Crediting:
 - (1) Applicants holding or having held an instructor certificate shall be fully credited towards the requirement of (a)(1).
 - (2) In addition, applicants holding or having held an FI or TRI certificate in the relevant aircraft category shall be fully credited towards the requirements of (a)(2).

FCL.940.FTI FTI – Revalidation and renewal

Regulation (EU) No 1178/2011

- (a) Revalidation. For revalidation of an FTI certificate, the applicant shall, within the validity period of the FTI certificate, fulfil one of the following requirements:

- (1) complete at least:
 - (i) 50 hours of flight tests, of which at least 15 hours shall be within the 12 months preceding the expiry date of the FTI certificate; and
 - (ii) 5 hours of flight test flight instruction within the 12 months preceding the expiry date of the FTI certificate; or
 - (2) receive refresher training as an FTI at an ATO. The refresher training shall be based on the practical flight instruction element of the FTI training course, in accordance with [FCL.930.FTI\(a\)\(3\)](#), and include at least 1 instruction flight under the supervision of an FTI qualified in accordance with [FCL.905.FTI\(b\)](#).
- (b) Renewal. If the FTI certificate has lapsed, the applicant shall receive refresher training as an FTI at an ATO. The refresher training shall comply at least with the requirements of [FCL.930.FTI\(a\)\(3\)](#).

SUBPART K – EXAMINERS

SECTION 1 – COMMON REQUIREMENTS

FCL.1000 Examiner certificates

Regulation (EU) 2019/1747

(a) General

Holders of an examiner certificate shall:

- (1) hold, unless otherwise determined in this Annex, an equivalent licence, rating or certificate to the ones for which they are authorised to conduct skill tests, proficiency checks or assessments of competence and the privilege to instruct for them;
- (2) be qualified to act as PIC in the aircraft during a skill test, proficiency check or assessment of competence if conducted on the aircraft.

(b) Special conditions:

- (1) The competent authority may issue a specific certificate granting privileges for the conduct of skill tests, proficiency checks and assessments of competence if compliance with the requirements established in this Subpart is not possible because of the introduction of any of the following:

- (i) new aircraft in the Member States or in an operator's fleet;
- (ii) new training courses in this Annex.

Such a certificate shall be limited to the skill tests, proficiency checks and assessments of competence necessary for the introduction of the new type of aircraft or the new training course and its validity shall not, in any case, exceed 1 year.

- (2) Holders of a certificate issued in accordance with point (b)(1) who wish to apply for an examiner certificate shall comply with the prerequisites and revalidation requirements for that category of examiner certificate.
- (3) Where no qualified examiner is available, competent authorities may, on a case-by-case basis, authorise inspectors or examiners who do not meet the relevant instructor, type or class rating requirements as specified in (a), to perform skill tests, proficiency checks and assessments of competence.

(c) Examination provided outside the territory of the Member States:

- (1) By way of derogation from point (a), in the case of skill tests and proficiency checks provided outside the territory for which Member States are responsible under the Chicago Convention, the competent authority shall issue an examiner certificate to applicants holding a pilot licence that is compliant with Annex 1 to the Chicago Convention, provided that those applicants:
 - (i) hold at least an equivalent licence, rating, or certificate to the one for which they are authorised to conduct skill tests, proficiency checks or assessments of competence, and in any case at least a CPL;
 - (ii) are qualified to act as PIC in the aircraft during a skill test or proficiency check that is conducted in the aircraft;

- (iii) comply with the requirements established in this Subpart for the issue of the relevant examiner certificate; and
 - (iv) demonstrate to the competent authority an adequate level of knowledge of European aviation safety rules to be able to exercise examiner privileges in accordance with this Annex.
- (2) The certificate referred to in point (1) shall be limited to performing skill tests and proficiency checks:
- (i) outside the territories for which the Member States are responsible under the Chicago Convention; and
 - (ii) to pilots who have sufficient knowledge of the language in which the test/check is given.

GM1 FCL.1000 Examiner certificates

ED Decision 2020/005/R

SPECIAL CONDITIONS

When new aircraft are introduced, requirements such as to hold a licence and rating equivalent to the one for which the skill test is being conducted, or to have adequate flight experience, may not be possible to comply with. In this case, to allow for the first ratings for these aircraft to be issued to applicants, competent authorities need the possibility to issue a specific certificate that does not have to comply with the requirements established in this Subpart.

The competent authority should only give these certificates to holders of other examiner certificates. As far as possible, preference should be given to persons with experience in similar types or classes of aircraft, for example, in aircraft having the same kind and number of engines or rotors and of the same order of mass or technology.

The certificate should be limited in validity to the time needed to qualify the first examiners for the new aircraft in accordance with this Subpart, but in any case, it should not exceed the 1 year established in the rule.

GM2 FCL.1000 Examiner certificates

ED Decision 2020/005/R

When examiners conduct a skill test, proficiency check or assessment of competence, in addition to a licence for the relevant aircraft category, they are required to hold the rating or certificate equivalent to the one for which they conduct the skill test, proficiency check or assessment of competence.

For example, a candidate who holds a CPL(A) may make a class rating proficiency check on an SE piston aeroplane with an examiner who holds a PPL(A) with an SE piston class rating and related examiner privileges.

FCL.1005 Limitation of privileges in case of vested interests

Regulation (EU) 2019/1747

Examiners shall not conduct:

- (a) skill tests or assessments of competence of applicants for the issue of a licence, rating or certificate to whom they have provided more than 25 % of the required flight instruction for the licence, rating or certificate for which the skill test or assessment of competence is being taken; and

- (b) skill tests, proficiency checks or assessments of competence whenever they feel that their objectivity may be affected.

GM1 FCL.1005(b) Limitation of privileges in case of vested interests

ED Decision 2020/005/R

Examples of a situation where the examiner should consider if their objectivity is affected are when the applicant is a relative or a friend of the examiner, or when they are linked by economic interests or political affiliations, etc.

FCL.1010 Prerequisites for examiners

Regulation (EU) No 1178/2011

Applicants for an examiner certificate shall demonstrate:

- (a) relevant knowledge, background and appropriate experience related to the privileges of an examiner;
- (b) that they have not been subject to any sanctions, including the suspension, limitation or revocation of any of their licences, ratings or certificates issued in accordance with this Part, for non-compliance with the Basic Regulation and its Implementing Rules during the last 3 years.

AMC1 FCL.1010 Prerequisites for examiners

ED Decision 2011/016/R

When evaluating the applicant's background, the competent authority should evaluate the personality and character of the applicant, and his/her cooperation with the competent authority.

The competent authority may also take into account whether the applicant has been convicted of any relevant criminal or other offenses, taking into account national law and principles of non-discrimination.

FCL.1015 Examiner standardisation

Regulation (EU) 2020/2193

- (a) An applicant for an examiner certificate shall undertake a standardisation course which is provided by the competent authority or which is provided by an ATO and approved by the competent authority.
- (b) The standardisation course shall consist of theoretical and practical instruction and shall include, at least:
 - (1) the conduct of 2 skill tests, proficiency checks or assessments of competences for the licences, ratings or certificates for which the applicant seeks the privilege to conduct tests and checks;
 - (2) instruction on the applicable requirements in this part and the applicable air operations requirements, the conduct of skill tests, proficiency checks and assessments of competence, and their documentation and reporting;
 - (3) a briefing on the national administrative procedures, requirements for protection of personal data, liability, accident insurance and fees.

- (4) a briefing on the need to review and apply the items in (3) when conducting skill tests, proficiency checks or assessments of competence of an applicant for which the competent authority is not the same that issued the examiner's certificate; and
 - (5) an instruction on how to get access to these national procedures and requirements of other competent authorities when needed;
- (c) Holders of an examiners certificate shall not conduct skill tests, proficiency checks or assessments of competence of an applicant for which the competent authority is not the same that issued the examiner's certificate, unless they have reviewed the latest available information containing the relevant national procedures of the applicant's competent authority.

AMC1 FCL.1015 Examiner standardisation

ED Decision 2025/002/R

GENERAL

- (a) The competent authority should provide the course itself or through an arrangement with an ATO.
- This arrangement should clearly state that the ATO acts under the management system of the competent authority.
- (b) The course should last:
- (1) for the FE and FIE, at least 1 day, divided into theoretical and practical training;
 - (2) for other examiners, at least 3 days, divided into theoretical training (1 day) and practical training conducting real or role-played proficiency checks, skill tests or assessments of competence (at least 2 days).
- (c) The competent authority or the ATO should determine any further training required before presenting the candidate for the examiner assessment of competence.

CONTENT

- (d) The training should comprise:
- (1) Theoretical training covering at least:
 - (i) the contents of [AMC2 FCL.1015](#) and the FEM;
 - (ii) Part-FCL and related AMCs and GM relevant to their duties;
 - (iii) operational requirements and related AMCs and GM relevant to their duties;
 - (iv) national requirements relevant to their examination duties;
 - (v) fundamentals of human performance and limitations relevant to flight examination;
 - (vi) fundamentals of evaluation relevant to applicant's performance;
 - (vii) the management system of ATOs and the organisational structure of DTOs;
 - (viii) MCC, human performance and limitations, if applicable.
 - (2) Examiners should also be briefed on the protection requirements for personal data, liability, accident insurance and fees, as applicable in the member state concerned.

- (3) All items above are the core knowledge requirements for an examiner and are recommended as the core course material. This core course may be studied before recommended examiner training is commenced. The core course may utilise any suitable training format.
- (4) Practical training consisting of at least:
 - (i) knowledge and management of the test for which the certificate is to be sought. These are described in the relevant modules in the FEM;
 - (ii) knowledge of the administrative procedures pertaining to that test or check.
- (5) For an initial examiner certificate, practical training should include the examination of the test profile sought, consisting of the conduct of at least two test or check profiles in the role of examiner (these two tests or checks profiles can be performed in the same simulator session), including briefing, conduct of the skill test and proficiency check, assessment of the applicant to whom the test or check is given, debriefing and recording or documentation under the supervision of an examiner of the appropriate category on the applicable type. This training is conducted in the aircraft if approval for testing or checking in the aircraft is required. If examiner privileges in FSTD's are required, practical instruction in the use of FSTD(s) for testing or checking should also be completed.
- (6) If examiner privileges are to include the conduct of proficiency checks for the revalidation or renewal of an instrument rating, practical instruction should include the conduct of at least four instrument check profiles in the role of examiner, including briefing, conduct of the skill test and proficiency check, assessment of the applicant to whom the test or check is given, debriefing and recording or documentation under the supervision of an examiner of the appropriate category on the applicable type. This training is conducted in the aircraft if approval for testing or checking in the aircraft is required. If examiner privileges in both FSTD and aircraft are required, at least one of the instrument check profiles should be conducted in an FSTD.
- (7) For extension of an examiner certificate to further types (as required for TRE), further practical training on the new type may be required, consisting of the conduct of at least one test or check profile in the role of examiner on the new type, including briefing, conduct of the skill test and proficiency check, assessment of the applicant to whom the test or check is given, debriefing and recording or documentation under the supervision of an examiner of the appropriate category on the applicable type. A further examiner check on the new type may be required, which may be supervised by an inspector of the competent authority or a suitably authorised senior examiner.

AMC2 FCL.1015 Examiner standardisation

ED Decision 2020/005/R

STANDARDISATION ARRANGEMENTS FOR EXAMINERS

LIMITATIONS

- (a) An examiner should allow an applicant adequate time to prepare for a test or check, normally not more than 1 hour.
- (b) An examiner should plan a test or check flight so that all required exercises can be performed while allowing sufficient time for each of the exercises and with due regard to the weather conditions, traffic situation, ATC requirements and local procedures.

PURPOSE OF A TEST OR CHECK

- (c) Determine through practical demonstration during a test or check that an applicant has acquired or maintained the required level of knowledge and skill or proficiency.
- (d) Improve training and flight instruction in ATOs or DTOs by feedback of information from examiners about items or sections of tests or checks that are most frequently failed.
- (e) Assist in maintaining and, where possible, improving air safety standards by having examiners display good airmanship and flight discipline during tests or checks.

CONDUCT OF TEST OR CHECK

- (f) An examiner will ensure that an applicant completes a test or check in accordance with Part-FCL requirements and is assessed against the required test or check standards.
- (g) Each item within a test or check section should be completed and assessed separately. The test or check schedule, as briefed, should not normally be altered by an examiner. A failed item is not always a failed section, for example type rating skill test where a failure of an item in a section does not fail the entire section, only the failed item is taken again.
- (h) Marginal or questionable performance of a test or check item should not influence an examiner's assessment of any subsequent items.
- (i) An examiner should verify the requirements and limitations of a test or check with an applicant during the pre-flight briefing.
- (j) When a test or check is completed or discontinued, an examiner should debrief the applicant and give reasons for items or sections failed. In case of a failed or discontinued skill test and proficiency check, the examiner should provide appropriate advice to assist the applicant in re-tests or re-checks.
- (k) Any comment on, or disagreement with, an examiner's test or check evaluation or assessment made during a debriefing will be recorded by the examiner on the test or check report, and will be signed by the examiner and countersigned by the applicant.

EXAMINER PREPARATION

- (l) An examiner should supervise all aspects of the test or check flight preparation, including, where necessary, obtaining or assuring an ATC 'slot' time.
- (m) An examiner will plan a test or check in accordance with Part-FCL requirements. Only the manoeuvres and procedures set out in the appropriate test or check form will be undertaken. The same examiner should not reexamine a failed applicant without the agreement of the applicant.

EXAMINER APPROACH

- (n) An examiner should encourage a friendly and relaxed atmosphere to develop both before and during a test or check flight. A negative or hostile approach should not be used. During the test or check flight, the examiner should avoid negative comments or criticisms and all assessments should be reserved for the debriefing.

ASSESSMENT SYSTEM

- (o) Although test or checks may specify flight test tolerances, an applicant should not be expected to achieve these at the expense of smoothness or stable flight. An examiner should make due allowance for unavoidable deviations due to turbulence, ATC instructions, etc. An examiner should terminate a test or check only when it is clear that the applicant has not been able to demonstrate the required level of knowledge, skill or proficiency and that a full re-test will be necessary or for safety reasons. An examiner will use one of the following terms for assessment:

- (1) a 'pass', provided that the applicant demonstrates the required level of knowledge, skill or proficiency and, where applicable, remains within the flight test tolerances for the licence or rating;
- (2) a 'fail' provided that any of the following apply:
 - (i) the flight test tolerances have been exceeded after the examiner has made due allowance for turbulence or ATC instructions;
 - (ii) the aim of the test or check is not completed;
 - (iii) the aim of exercise is completed but at the expense of safe flight, violation of a rule or regulation, poor airmanship or rough handling;
 - (iv) an acceptable level of knowledge is not demonstrated;
 - (v) an acceptable level of flight management is not demonstrated;
 - (vi) the intervention of the examiner or safety pilot is required in the interest of safety.
- (3) a 'partial pass' in accordance with the criteria shown in the relevant skill test appendix of Part-FCL.

METHOD AND CONTENTS OF THE TEST OR CHECK

- (p) Before undertaking a test or check, an examiner will verify that the aircraft or FSTD intended to be used is suitable and appropriately equipped for the test or check. Aircraft that fall under points (a), (b), (c), or (d) of Annex I to the Basic Regulation can be used provided that they are subject to an authorisation as per point ORA.ATO.135 or point DTO.GEN.240.
- (q) A test or check flight will be conducted in accordance with the AFM and, if applicable, the AOM.
- (r) A test or check flight will be conducted within the limitations contained in the operations manual of an ATO or the operator for which the applicant is flying, as applicable, or, if available, within the limitations placed by the DTO.
- (s) Contents:
 - (1) a test or check is comprised of:
 - (i) oral examination on the ground (where applicable);
 - (ii) pre-flight briefing;
 - (iii) in-flight exercises;
 - (iv) post-flight debriefing.
 - (2) oral examination on the ground should include:
 - (i) aircraft general knowledge and performance;
 - (ii) planning and operational procedures;
 - (iii) other relevant items or sections of the test or check.
 - (3) pre-flight briefing should include:
 - (i) test or check sequence;
 - (ii) power setting, speeds and approach minima, if applicable;
 - (iii) safety considerations.
 - (4) in-flight exercises will include each relevant item or section of the test or check;

- (5) post-flight debriefing should include:
 - (i) assessment or evaluation of the applicant;
 - (ii) documentation of the test or check with the applicant's FI present, if possible.
- (t) A test or check is intended to simulate a practical flight. Thus, an examiner may set practical scenarios for an applicant while ensuring that the applicant is not confused and air safety is not compromised.
- (u) When manoeuvres are to be flown by sole reference to instruments, the examiner should ensure that a suitable method of screening is used to simulate IMC.
- (v) An examiner should maintain a flight log and assessment record during the test or check for reference during the post or flight debriefing.
- (w) An examiner should be flexible to the possibility of changes arising to preflight briefings due to ATC instructions, or other circumstances affecting the test or check.
- (x) Where changes arise to a planned test or check an examiner should be satisfied that the applicant understands and accepts the changes. Otherwise, the test or check flight should be terminated.
- (y) Should an applicant choose not to continue a test or check for reasons considered inadequate by an examiner, the applicant will be assessed as having failed those items or sections not attempted. If the test or check is terminated for reasons considered adequate by the examiner, only these items or sections not completed will be tested during a subsequent test or check.
- (z) An examiner may terminate a test or check at any stage, if it is considered that the applicant's competency requires a complete re-test or re-check.

GM1 FCL.1015 Examiner standardisation

ED Decision 2025/002/R

- (a) An examiner should plan per day not more than:
 - (1) three tests or checks relating to PPL, CPL, IR or class ratings;
 - (2) four tests or checks relating to the LAPL;
 - (3) two tests or checks related to MPL or ATPL;
 - (4) two assessments of competence related to instructor certificates;
 - (5) four tests or checks relating to SP type ratings.
- (b) An examiner should plan at least 2 hours for a LAPL, 3 hours for a PPL, CPL, IR or class rating test or checks, and at least 4 hours for instructor certificates, MPL, ATPL or MP type rating tests or checks, including preflight briefing and preparation, conduct of the test, check or assessment of competence, de-briefing, evaluation of the applicant and documentation.
- (c) For the conduct of the test, check or assessment of competence, without the additional activities specified in point (b), the following values may be used as guidance:
 - (1) 45 minutes for SP class ratings VFR only;
 - (2) 90 minutes for LAPL(A) or LAPL(H), PPL(A) or PPL(H), and CPL(A) or CPL(H), including the navigation section;
 - (3) 60 minutes for PPL(As) and CPL(As);

- (4) 60 minutes for IR, BIR, instructor certificates, and SP type or class ratings; and
- (5) 120 minutes for MPL, ATPL, and MP type ratings.

FCL.1020 Examiners assessment of competence

Regulation (EU) No 1178/2011

Applicants for an examiner certificate shall demonstrate their competence to an inspector from the competent authority or a senior examiner specifically authorised to do so by the competent authority responsible for the examiner's certificate through the conduct of a skill test, proficiency check or assessment of competence in the examiner role for which privileges are sought, including briefing, conduct of the skill test, proficiency check or assessment of competence, and assessment of the person to whom the test, check or assessment is given, debriefing and recording documentation.

AMC1 FCL.1020 Examiners assessment of competence

ED Decision 2011/016/R

GENERAL

- (a) The competent authority may nominate either one of its inspectors or a senior examiner to assess the competence of applicants for an examiner certificate.

DEFINITIONS

- (b) Definitions:
 - (1) 'Inspector': the inspector of the competent authority conducting the examiner competence assessment;
 - (2) 'Examiner applicant': the person seeking certification as an examiner;
 - (3) 'Candidate': the person being tested or checked by the examiner applicant. This person may be a pilot for whom the test or check would be required, or the inspector of the competent authority who is conducting the examiner certification acceptance test.

CONDUCT OF THE ASSESSMENT

- (c) An inspector of the competent authority or a senior examiner will observe all examiner applicants conducting a test on a 'candidate' in an aircraft for which examiner certificate is sought. Items from the related training course and test or check schedule will be selected by the inspector for examination of the 'candidate' by the examiner applicant. Having agreed with the inspector the content of the test, the examiner applicant will be expected to manage the entire test. This will include briefing, the conduct of the flight, assessment and debriefing of the 'candidate'. The inspector will discuss the assessment with the examiner applicant before the 'candidate' is debriefed and informed of the result.

BRIEFING THE 'CANDIDATE'

- (d) The 'candidate' should be given time and facilities to prepare for the test flight. The briefing should cover the following:
 - (1) the objective of the flight;
 - (2) licensing checks, as necessary;
 - (3) freedom for the 'candidate' to ask questions;
 - (4) operating procedures to be followed (for example operators manual);
 - (5) weather assessment;

- (6) operating capacity of 'candidate' and examiner;
 - (7) aims to be identified by 'candidate';
 - (8) simulated weather assumptions (for example icing and cloud base);
 - (9) use of screens (if applicable);
 - (10) contents of exercise to be performed;
 - (11) agreed speed and handling parameters (for example V-speeds, bank angle, approach minima);
 - (12) use of R/T;
 - (13) respective roles of 'candidate' and examiner (for example during emergency);
 - (14) administrative procedures (for example submission of flight plan).
- (e) The examiner applicant should maintain the necessary level of communication with the 'candidate'. The following check details should be followed by the examiner applicant:
- (1) involvement of examiner in a MP operating environment;
 - (2) the need to give the 'candidate' precise instructions;
 - (3) responsibility for safe conduct of the flight;
 - (4) intervention by examiner, when necessary;
 - (5) use of screens;
 - (6) liaison with ATC and the need for concise, easily understood intentions;
 - (7) prompting the 'candidate' about required sequence of events (for example following a go-around);
 - (8) keeping brief, factual and unobtrusive notes.

ASSESSMENT

- (f) The examiner applicant should refer to the flight test tolerances given in the relevant skill test. Attention should be paid to the following points:
- (1) questions from the 'candidate';
 - (2) give results of the test and any sections failed;
 - (3) give reasons for failure.

DEBRIEFING

- (g) The examiner applicant should demonstrate to the inspector the ability to conduct a fair, unbiased debriefing of the 'candidate' based on identifiable factual items. A balance between friendliness and firmness should be evident. The following points should be discussed with the 'candidate', at the applicant's discretion:
- (1) advise the candidate on how to avoid or correct mistakes;
 - (2) mention any other points of criticism noted;
 - (3) give any advice considered helpful.

RECORDING OR DOCUMENTATION

- (h) The examiner applicant should demonstrate to the inspector the ability to complete the relevant records correctly. These records may be:
- (1) the relevant test or check form;
 - (2) licence entry;
 - (3) notification of failure form;
 - (4) relevant company forms where the examiner has privileges of conducting operator proficiency checks.

DEMONSTRATION OF THEORETICAL KNOWLEDGE

- (i) The examiner applicant should demonstrate to the inspector a satisfactory knowledge of the regulatory requirements associated with the function of an examiner.

FCL.1025 Validity, revalidation and renewal of examiner certificates*Regulation (EU) 2020/2193*

(a) Validity

An examiner certificate shall be valid for 3 years.

(b) Revalidation

To revalidate an examiner certificate, holders shall comply with all of the following conditions:

- (1) before the expiry date of the certificate, have conducted at least six skill tests, proficiency checks, assessments of competence, or EBT evaluation phases during an EBT module referred to in point ORO.FC.231 of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#);
- (2) in the period of 12 months immediately preceding the expiry date of the certificate, have completed an examiner refresher course which is provided by the competent authority or which is provided by an ATO and approved by the competent authority;
- (3) one of the skill tests, proficiency checks, assessments of competence or EBT evaluation phases conducted in accordance with point (1) shall take place in the period of 12 months immediately preceding the expiry date of the examiner certificate and shall:
 - (i) have been assessed by an inspector from the competent authority or by a senior examiner specifically authorised to do so by the competent authority responsible for the examiner certificate; or
 - (ii) comply with the requirements in point [FCL.1020](#).

If applicants for the revalidation hold privileges for more than one category of examiner, all examiner privileges may be revalidated if applicants comply with the requirements laid down in points (b)(1) and (2) and point [FCL.1020](#) for one of the categories of examiner certificates held, in agreement with the competent authority.

(c) Renewal

If the certificate has expired, before resuming the exercise of the privileges, the applicants shall comply with the requirements in point (b)(2) and point [FCL.1020](#) in the period of 12 months immediately preceding the application for the renewal.

- (d) An examiner certificate shall only be revalidated or renewed if applicants demonstrate continued compliance with the requirements laid down in points [FCL.1010](#) and [FCL.1030](#).

AMC1 FCL.1020; FCL.1025

ED Decision 2011/016/R

QUALIFICATION OF SENIOR EXAMINERS

- (a) A senior examiner specifically tasked by the competent authority to observe skill tests or proficiency checks for the revalidation of examiner certificates should:
 - (1) hold a valid or current examiner certificate appropriate to the privileges being given;
 - (2) have examiner experience level acceptable to the competent authority;
 - (3) have conducted a number of skill tests or proficiency checks as a Part-FCL examiner.
- (b) The competent authority may conduct a pre-assessment of the applicant or candidate carrying out a skill test and proficiency check under supervision of an inspector of the competent authority.
- (c) Applicants should be required to attend a senior examiner briefing, course or seminar arranged by the competent authority. Content and duration will be determined by the competent authority and should include:
 - (1) pre-course self-study;
 - (2) legislation;
 - (3) the role of the senior examiner;
 - (4) an examiner assessment;
 - (5) national administrative requirements.
- (d) The validity of the authorisation should not exceed the validity of the examiners certificate, and in any case should not exceed 3 years. The authorisation may be revalidated in accordance with procedures established by the competent authority.

AMC1 FCL.1025 Validity, revalidation and renewal of examiner certificates

ED Decision 2018/009/R

EXAMINER REFRESHER COURSE

The examiner refresher course should follow the content of the examiner standardisation course, included in [AMC1 FCL.1015](#), and take into account specific contents adequate to the category of examiner affected.

GM1 FCL.1015(a); FCL.1025(b)(2)

ED Decision 2018/009/R

EXAMINER STANDARDISATION COURSES AT AN ATO OR A DTO

In point [FCL.1015\(a\)](#) (second sentence) and in point [FCL.1025\(b\)\(2\)](#) (second sentence), the word ‘may’ is used to indicate that completing an examiner standardisation course or an examiner refresher course at a DTO is an option which can be used by examiners for sailplanes and balloons as an alternative to completing such courses provided by the competent authority or an ATO (first sentence in both point [FCL.1015\(a\)](#) and point [FCL.1025\(b\)\(2\)](#)).

FCL.1030 Conduct of skill tests, proficiency checks and assessments of competence

Regulation (EU) No 245/2014

- (a) When conducting skill tests, proficiency checks and assessments of competence, examiners shall:
 - (1) ensure that communication with the applicant can be established without language barriers;
 - (2) verify that the applicant complies with all the qualification, training and experience requirements in this Part for the issue, revalidation or renewal of the licence, rating or certificate for which the skill test, proficiency check or assessment of competence is taken;
 - (3) make the applicant aware of the consequences of providing incomplete, inaccurate or false information related to their training and flight experience.
- (b) After completion of the skill test or proficiency check, the examiner shall:
 - (1) inform the applicant of the result of the test. In the event of a partial pass or fail, the examiner shall inform the applicant that he/she may not exercise the privileges of the rating until a full pass has been obtained. The examiner shall detail any further training requirement and explain the applicant's right of appeal;
 - (2) in the event of a pass in a proficiency check or assessment of competence for revalidation or renewal, endorse the applicant's licence or certificate with the new expiry date of the rating or certificate, if specifically authorised for that purpose by the competent authority responsible for the applicant's licence;
 - (3) provide the applicant with a signed report of the skill test or proficiency check and submit without delay copies of the report to the competent authority responsible for the applicant's licence, and to the competent authority that issued the examiner certificate. The report shall include:
 - (i) a declaration that the examiner has received information from the applicant regarding his/her experience and instruction, and found that experience and instruction complying with the applicable requirements in this Part;
 - (ii) confirmation that all the required manoeuvres and exercises have been completed, as well as information on the verbal theoretical knowledge examination, when applicable. If an item has been failed, the examiner shall record the reasons for this assessment;
 - (iii) the result of the test, check or assessment of competence;
 - (iv) a declaration that the examiner has reviewed and applied the national procedures and requirements of the applicant's competent authority if the competent authority responsible for the applicant's licence is not the same that issued the examiner's certificate;
 - (v) a copy of the examiner certificate containing the scope of his/her privileges as examiner in the case of skill tests, proficiency checks or assessments of competence of an applicant for which the competent authority is not the same that issued the examiner's certificate.

- (c) Examiners shall maintain records for 5 years with details of all skill tests, proficiency checks and assessments of competence performed and their results.
- (d) Upon request by the competent authority responsible for the examiner certificate, or the competent authority responsible for the applicant's licence, examiners shall submit all records and reports, and any other information, as required for oversight activities.

AMC1 FCL.1030(b)(3) Conduct of skill tests, proficiency checks and assessments of competence

ED Decision 2025/002/R

OBLIGATIONS FOR EXAMINERS' APPLICATION AND REPORT FORMS

Common application and report forms for examiners can be found:

- (a) For skill tests or proficiency checks for the issue, revalidation or renewal of LAPL, PPL, CPL and IR, in [AMC1 to Appendix 7](#);
- (b) For training, skill tests or proficiency checks for ATPL, MPL or class and type ratings, in [AMC1 to Appendix 9](#);
- (c) For EBT practical assessment, in [AMC1 to Appendix 10](#);
- (d) For assessments of competence for instructors, in [AMC5 FCL.935](#).

GM1 FCL.1030(b)(3)(ii) Conduct of skill tests, proficiency checks and assessments of competence

ED Decision 2021/002/R

REVALIDATION OF CLASS AND TYPE RATINGS — AEROPLANES — REQUIRED MANOEUVRES AND EXERCISES IN THE CONTEXT OF APPENDIX 10 (EBT PRACTICAL ASSESSMENT)

The confirmation that all the required manoeuvres and exercises have been completed means that during the period of validity of the type rating, the applicant has completed the operator's EBT programme applicable to that period.

SECTION 2 – SPECIFIC REQUIREMENTS FOR FLIGHT EXAMINERS – FE

FCL.1005.FE FE – Privileges and conditions

Regulation (EU) 2025/134

- (a) FE(A). The privileges of an FE for aeroplanes are to conduct:
- (1) skill tests for the issue of the PPL(A) and skill tests and proficiency checks for associated/single-pilot class and type ratings, except for single-pilot high performance complex aeroplanes, provided that the examiner has completed at least 1 000 hours of flight time as a pilot on aeroplanes or TMGs, including at least 250 hours of flight instruction;
 - (2) skill tests for the issue of the CPL(A) and skill tests and proficiency checks for the associated single-pilot class and type ratings, except for single-pilot high performance complex aeroplanes, provided that the examiner has completed at least 2 000 hours of flight time as a pilot on aeroplanes or TMGs, including at least 250 hours of flight instruction;
 - (3) skill tests and proficiency checks for the LAPL(A), provided that the examiner has completed at least 500 hours of flight time as a pilot on aeroplanes or TMGs, including at least 100 hours of flight instruction;
 - (4) skill tests for the issue of a mountain rating, provided that the examiner has completed at least 500 hours of flight time as a pilot on aeroplanes or TMGs, including at least 500 take-offs and landings of flight instruction for the mountain rating.
 - (5) proficiency checks for the revalidation and renewal of EIRs, provided that the FE has completed at least 1 500 hours as a pilot on aeroplanes and complies with the requirements in [FCL.1010.IRE\(a\)\(2\)](#).
- (b) FE(H). The privileges of an FE for helicopters are to conduct:
- (1) skill tests for the issue of the PPL(H) and skill tests and proficiency checks for single-pilot single-engine helicopter type ratings entered in a PPL(H), provided that the examiner has completed 1 000 hours of flight time as a pilot on helicopters, including at least 250 hours of flight instruction;
 - (2) skill tests for the issue of the CPL(H) and skill tests and proficiency checks for single-pilot single-engine helicopter type ratings entered in a CPL(H), provided the examiner has completed 2 000 hours of flight time as pilot on helicopters, including at least 250 hours of flight instruction;
 - (3) skill tests and proficiency checks for single-pilot multi-engine helicopter type ratings entered in a PPL(H) or a CPL(H), provided the examiner has completed the requirements in (1) or (2), as applicable, and holds a CPL(H) or ATPL(H) and, when applicable, an IR(H);
 - (4) skill tests and proficiency checks for the LAPL(H), provided that the examiner has completed at least 500 hours of flight time as a pilot on helicopters, including at least 150 hours of flight instruction.
- (c) FE(As). The privileges of an FE for airships are to conduct skill tests for the issue of the PPL(As) and CPL(As) and skill tests and proficiency checks for the associated airship type ratings, provided that the examiner has completed 500 hours of flight time as a pilot on airships, including 100 hours of flight instruction.

- (d) FE(G). The privileges of an FE for gyroplanes are to conduct skill tests and proficiency checks for the GPL and for associated privileges for gyroplane classes and types, provided that the FE has completed at least 500 hours of flight time as a pilot on gyroplanes, including at least 100 hours of flight instruction, of which up to 40 hours may be flight instruction conducted as FI(A) or FI(H).

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

FCL.1010.FE FE – Prerequisites

Regulation (EU) No 1178/2011

An applicant for an FE certificate shall hold:
an FI certificate in the appropriate aircraft category.

SECTION 3 – SPECIFIC REQUIREMENTS FOR TYPE RATING EXAMINERS – TRE

FCL.1005.TRE TRE – Privileges and conditions

Regulation (EU) 2021/2227

- (a) TRE(A) and TRE(PL). The privileges of a TRE for aeroplanes or powered-lift aircraft are to conduct:
- (1) skill tests for the initial issue of type ratings for aeroplanes or powered-lift aircraft, as applicable;
 - (2) proficiency checks for the revalidation or renewal of type ratings and IRs;
 - (3) skill tests for ATPL(A) issue;
 - (4) skill tests for MPL issue, provided that the examiner has complied with the requirements in [FCL.925](#);
 - (5) assessments of competence for the issue, revalidation or renewal of a TRI or SFI certificates in the applicable aircraft category, provided that they have completed at least 3 years as a TRE and have undergone specific training for the assessment of competence in accordance with point FCL.1015 (b).
- (b) TRE(H). The privileges of a TRE(H) are to conduct:
- (1) skill tests and proficiency checks for the issue, revalidation or renewal of helicopter type ratings;
 - (2) proficiency checks for the revalidation or renewal of IRs, provided the TRE(H) holds a valid IR(H);
 - (3) skill tests for ATPL(H) issue;
 - (4) assessments of competence for the issue, revalidation or renewal of a TRI(H) or SFI(H) certificates, provided that they have completed at least 3 years as a TRE and have undergone specific training for the assessment of competence in accordance with point FCL.1015 (b).

FCL.1010.TRE TRE – Prerequisites

Regulation (EU) 2021/2227

- (a) TRE(A) and TRE(PL). Applicants for a TRE certificate for aeroplanes and powered-lift aircraft shall:
- (1) in the case of multi-pilot aeroplanes or powered-lift aircraft, have completed 1500 hours of flight time as a pilot of multi-pilot aeroplanes or powered-lift aircraft, as applicable, of which at least 500 hours shall be as PIC;
 - (2) in the case of single-pilot high performance complex aeroplanes, have completed 500 hours of flight time as a pilot of single-pilot aeroplanes, of which at least 200 hours shall be as PIC;
 - (3) hold a CPL or ATPL and a TRI certificate for the applicable type;
 - (4) for the initial issue of an TRE certificate, have completed at least 50 hours of flight instruction as a TRI, FI or SFI in the applicable type or an FSTD representing that type.

- (b) TRE(H). Applicants for a TRE (H) certificate for helicopters shall:
- (1) hold a TRI(H) certificate or, in the case of single-pilot single-engine helicopters, a valid FI(H) certificate, for the applicable type;
 - (2) for the initial issue of a TRE certificate, have completed 50 hours of flight instruction as a TRI, FI or SFI in the applicable type or an FSTD representing that type;
 - (3) in the case of multi-pilot helicopters, hold a CPL(H) or ATPL(H) and have completed 1500 hours of flight as a pilot on multi-pilot helicopters, of which at least 500 hours shall be as PIC;
 - (4) in the case of single-pilot multi-engine helicopters:
 - (i) have completed 1000 hours of flight as pilot on helicopters, of which at least 500 hours shall be as PIC;
 - (ii) hold a CPL(H) or ATPL(H) and, when applicable, a valid IR(H);
 - (5) in the case of single-pilot single-engine helicopters:
 - (i) have completed 750 hours of flight as a pilot on helicopters, of which at least 500 hours shall be as PIC;
 - (ii) hold a CPL(H) or ATPL(H).
 - (6) Before the privileges of a TRE(H) are extended from single-pilot operations to multi-pilot operations on the same type of helicopter, the holder shall have either:
 - (i) at least 100 hours in multi-pilot operations on this type; or
 - (ii) at least 350 hours in multi-pilot operations in any aircraft category.
 - (7) In the case of applicants for the first multi-pilot multi-engine TRE certificate, the 1500 hours of flight experience on multi-pilot helicopters required in (b)(3) may be considered to have been met if they have completed the 500 hours of flight time as PIC on a multi-pilot helicopter of the same type.

SECTION 4 – SPECIFIC REQUIREMENTS FOR CLASS RATING EXAMINER – CRE

FCL.1005.CRE CRE – Privileges

Regulation (EU) 2020/359

The privileges of a CRE are to conduct, for single-pilot aeroplanes, except for single-pilot high performance complex aeroplanes:

- (a) skill tests for the issue of class and type ratings;
- (b) proficiency checks for:
 - (1) revalidation or renewal of class and type ratings;
 - (2) revalidation of IRs, provided that they have completed at least 1500 hours as pilots of aeroplanes and have completed at least 450 hours of flight time under IFR;
 - (3) renewal of IRs, provided that they comply with the requirements laid down in point [FCL.1010.IRE\(a\)](#); and (4) revalidation and renewal of EIRs, provided that they have completed at least 1 500 hours as a pilot on aeroplanes and comply with the requirements laid down in point [FCL.1010.IRE\(a\)\(2\)](#).
 - (4) revalidation and renewal of BIRs, provided that the CRE has completed:
 - (i) 1 500 hours of flight time as a pilot of aeroplanes; and
 - (ii) 450 hours of flight time under IFR; and
- (c) skill tests for the extension of LAPL(A) privileges to another class or variant of aeroplane.

FCL.1010.CRE CRE – Prerequisites

Regulation (EU) 2019/1747

Applicants for a CRE certificate shall:

- (a) hold a CPL(A), MPL(A) or ATPL(A) with single-pilot privileges or have held it and hold a PPL(A);
- (b) hold a CRI or FI certificate with instructional privileges for the applicable class or type;
- (c) have completed 500 hours of flight time as a pilot on aeroplanes.

SECTION 5 – SPECIFIC REQUIREMENTS FOR INSTRUMENT RATING EXAMINER – IRE

FCL.1005.IRE IRE – Privileges

Regulation (EU) 2020/359

The privileges of holders of an instrument rating examiner (IRE) certificate are to conduct skill tests for the issue and proficiency checks for the revalidation or renewal of BIRs and IRs.

FCL.1010.IRE IRE – Prerequisites

Regulation (EU) 2019/1747

(a) IRE(A)

Applicants for an IRE certificate for aeroplanes shall hold an IRI(A) or an FI(A) certificate with the privilege to instruct for the IR(A) and shall have completed:

- (1) 2000 hours of flight time as pilots of aeroplanes; and
- (2) 450 hours of flight time under IFR, of which 250 hours shall be as an instructor.

(b) IRE(H)

Applicants for an IRE certificate for helicopters shall hold an IRI(H) or an FI(H) certificate with the privilege to instruct for the IR(H) and shall have completed:

- (1) 2 000 hours of flight time as a pilot on helicopters; and
- (2) 300 hours of instrument flight time on helicopters, of which 200 hours shall be as an instructor.

(c) IRE(As)

Applicants for an IRE certificate for airships shall hold an IRI(As) or an FI(As) certificate with the privilege to instruct for the IR(As) and shall have completed:

- (1) 500 hours of flight time as a pilot on airships; and
- (2) 100 hours of instrument flight time on airships, of which 50 hours shall be as an instructor.

SECTION 6 – SPECIFIC REQUIREMENTS FOR SYNTHETIC FLIGHT EXAMINER – SFE

FCL.1005.SFE SFE – Privileges and conditions

Regulation (EU) 2019/1747

(a) SFE for aeroplanes (SFE(A)) and SFE for powered-lift aircraft (SFE(PL))

The privileges of SFEs for aeroplanes or powered-lift aircraft are to conduct in an FFS, or for the assessments in point (5) on the applicable FSTD:

- (1) skill tests and proficiency checks for the issue, revalidation or renewal of type ratings for aeroplanes or powered-lift aircraft, as applicable;
- (2) proficiency checks for the revalidation or renewal of IRs if combined with the revalidation or renewal of a type rating, provided that they have passed a proficiency check for the aircraft type including the instrument rating within the last year;
- (3) skill tests for ATPL(A) issue;
- (4) skill tests for MPL issue, provided that they have complied with the requirements laid down in point [FCL.925](#); and
- (5) assessments of competence for the issue, revalidation or renewal of an SFI certificate in the relevant aircraft category, provided that they have completed at least 3 years as an SFE(A) and have undergone specific training for the assessment of competence in accordance with point [FCL.1015\(b\)](#).

(b) SFE for helicopters (SFE(H))

The privileges of an SFEs(H) are to conduct in an FFS or for the assessments in point (4) on the applicable FSTD:

- (1) skill tests and proficiency checks for the issue, revalidation and renewal of type ratings;
- (2) proficiency checks for the revalidation and renewal of IRs if those checks are combined with the revalidation or renewal of a type rating, provided that the SFEs have passed a proficiency check for the aircraft type including the instrument rating within the last year preceding the proficiency check;
- (3) skill tests for ATPL(H) issue; and
- (4) assessments of competence for the issue, revalidation or renewal of an SFI(H) certificate, provided that they have completed at least 3 years as an SFE(H) and have undergone specific training for the assessment of competence in accordance with point [FCL.1015\(b\)](#).

FCL.1010.SFE SFE – Prerequisites

Regulation (EU) 2021/2227

(a) SFE(A)

Applicants for an SFE(A) certificate shall comply with all of the following conditions:

- (1) in the case of multi-pilot aeroplanes:
 - (i) hold or have held an ATPL(A) and a type rating;
 - (ii) hold an SFI(A) certificate for the applicable type of aeroplane; and

- (iii) have at least 1 500 hours of flight time as pilots of multi-pilot aeroplanes;
 - (2) in the case of single-pilot high-performance complex aeroplanes:
 - (i) hold or have held a CPL(A) or an ATPL(A) and a type rating;
 - (ii) hold an SFI(A) certificate for the applicable class or type of aeroplane; and
 - (iii) have at least 500 hours of flight time as pilots of single-pilot aeroplanes;
 - (3) for the initial issue of an SFE certificate, have completed at least 50 hours of synthetic flight instruction as a TRI(A) or an SFI(A) on the applicable type.
- (b) SFE(H)

Applicants for an SFE(H) certificate shall comply with all of the following conditions:

 - (1) hold or have held an ATPL(H), and a type rating for the applicable type of helicopter;
 - (2) hold an SFI(H) certificate for the applicable type of helicopter;
 - (3) in the case of multi-pilot helicopters, have at least 1 000 hours of flight time as pilots of multi-pilot helicopters;
 - (4) in the case of single-pilot helicopters in multi-pilot operations, have completed at least 350 hours in multi-pilot operations in any aircraft category;
 - (5) for the initial issue of an SFE certificate, have completed at least 50 hours of synthetic flight instruction as a TRI(H) or an SFI(H) on the applicable type.

GM1 FCL.1010.SFE(a)(1)(i) SFE — Prerequisites

ED Decision 2025/002/R

PREREQUISITE FOR AN SFE(A) TO HOLD A TYPE RATING

Point [FCL.1010.SFE\(a\)\(1\)\(i\)](#) requires an applicant for an SFE(A) certificate to hold or have held a type rating. This general reference to ‘a type rating’ allows pilots to seek SFE privileges for a type that they do not or did not fly during their active career. Still, the intention of this requirement is that, even if that type rating referred to in point [FCL.1010.SFE\(a\)\(1\)\(i\)](#) is not related to the type for which SFE privileges are sought, it should at least be related to an aeroplane type with similar type of propulsion (including number of engines) and MTOM.

SECTION 7 – SPECIFIC REQUIREMENTS FOR THE FLIGHT INSTRUCTOR EXAMINER – FIE

FCL.1005.FIE FIE – Privileges and conditions

Regulation (EU) 2025/134

- (a) FIE(A). The privileges of an FIE on aeroplanes are to conduct assessments of competence for the issue, revalidation or renewal of certificates for FI(A), CRI(A), IRI(A) and TRI(A) on single-pilot aeroplanes, provided that the relevant instructor certificate is held.
- (b) FIE(H). The privileges of an FIE on helicopters are to conduct assessments of competence for the issue, revalidation or renewal of certificates for FI(H), IRI(H) and TRI(H) on single-pilot helicopters, provided that the relevant instructor certificate is held.
- (c) FIE(As). The privileges of an FIE on airships are to conduct assessments of competence for the issue, revalidation or renewal of instructor certificates of airships, provided that the relevant instructor certificate is held.

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

- (c) FIE(As) and FIE(G). The privileges of an FIE on airships and gyroplanes are to conduct assessments of competence for the issue, revalidation, or renewal of instructor certificates in the applicable aircraft category, provided that the relevant instructor certificate is held.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

FCL.1010.FIE FIE – Prerequisites

Regulation (EU) 2025/134

- (a) FIE(A). Applicants for an FIE certificate for aeroplanes shall:
in case of applicants wishing to conduct assessments of competence:
 - (1) hold the relevant instructor certificate, as applicable;
 - (2) have completed 2000 hours of flight time as a pilot on aeroplanes or TMGs; and
 - (3) have at least 100 hours of flight time instructing applicants for an instructor certificate.
- (b) FIE(H). Applicants for an FIE certificate for helicopters shall:
 - (1) hold the relevant instructor certificate, as applicable;
 - (2) have completed 2000 hours of flight time as pilot on helicopters;
 - (3) have at least 100 hours of flight time instructing applicants for an instructor certificate.
- (c) FIE(As). Applicants for an FIE certificate for airships shall:
 - (1) have completed 500 hours of flight time as a pilot on airships;
 - (2) have at least 20 hours of flight time instructing applicants for an FI(AS) certificate;
 - (3) hold the relevant instructor certificate.
- (d) FIE(G). Applicants for an FIE certificate for gyroplanes shall:
 - (1) hold an FI(G) instructor certificate;
 - (2) have completed 1 000 hours of flight time as a pilot on gyroplanes; and

- (3) have completed at least 100 hours of flight time instructing applicants for an FI(G) certificate. In the case of applicants holding a FIE(A) or FIE(H) certificate, this requirement is reduced to 50 hours

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

APPENDICES TO ANNEX I

Appendix 1 – Crediting of theoretical knowledge

Regulation (EU) 2025/134

A. CREDITING OF THEORETICAL KNOWLEDGE IN THE SAME OR ANOTHER CATEGORY OF AIRCRAFT — BRIDGE INSTRUCTION AND EXAMINATION REQUIREMENTS

1. LAPL and PPL

[applicable until 17 February 2026 - Regulation (EU) 2024/2076]

1. LAPL, PPL and GPL

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

- 1.1. For the issue of an LAPL, the holder of an LAPL in another category of aircraft shall be fully credited towards requirements of theoretical knowledge on the common subjects established in point [FCL.120\(a\)](#).
- 1.2. For the issue of an LAPL or a PPL, holders of a PPL, CPL or ATPL in another category of aircraft shall be credited towards requirements of theoretical knowledge on the common subjects established in point [FCL.215\(a\)](#). This credit shall also apply to applicants for an LAPL or a PPL who hold a BPL issued in accordance with Annex III (Part-BFCL) to [Regulation \(EU\) 2018/395](#) or an SPL issued in accordance with Annex III (Part-SFCL) to [Implementing Regulation \(EU\) 2018/1976](#), except that the subject 'navigation' shall not be credited.
- 1.3. For the issue of a PPL, the holder of an LAPL in the same category of aircraft shall be fully credited towards the requirements of theoretical knowledge instruction and examination. That credit shall also apply to applicants for a PPL who have already successfully completed the theoretical knowledge examination for the LAPL in the same aircraft category, as long as it is within the validity period specified in point [FCL.025\(c\)](#).
- 1.4. By way of derogation from point 1.2, for the issue of an LAPL(A), the holder of an SPL issued in accordance with Annex III (Part-SFCL) to [Implementing Regulation \(EU\) 2018/1976](#) with privileges to fly TMGs shall demonstrate an adequate level of theoretical knowledge for the SEP aeroplane-land class in accordance with point [FCL.135.A\(a\)\(2\)](#).
- 1.5. For the issue of a GPL, the holder of a pilot licence for aeroplanes or helicopters in accordance with this Regulation shall be credited in full towards the theoretical knowledge instruction and examination requirements in the following subjects:
 - (a) Air law and ATC procedures,
 - (b) Human performance,
 - (c) Meteorology,
 - (d) Communications,
 - (e) Navigation.

[applicable from 18 February 2026 - Regulation (EU) 2025/134]

2. CPL

-
- 2.1. Applicants for the issue of a CPL holding a CPL in another category of aircraft shall have received theoretical knowledge bridge instruction at an ATO on an approved course according to the differences identified between the CPL syllabi for different aircraft categories.
 - 2.2. Applicants shall pass theoretical knowledge examinations as set out in this Annex (Part-FCL) for the following subjects in the appropriate aircraft category:
 - 021 — Aircraft general knowledge: airframe and systems, electrics, power plant and emergency equipment;
 - 022 — Aircraft general knowledge: instrumentation;
 - 032/034 — Performance aeroplanes or helicopters, as applicable;
 - 070 — Operational procedures;
 - 081/082 — Principles of flight for aeroplanes or helicopters, as applicable.
 - 2.3. Applicants for the issue of a CPL having passed the relevant theoretical knowledge examinations for an IR in the same category of aircraft are credited towards the theoretical knowledge requirements in human performance and meteorology unless they have completed the IR training course in accordance with Appendix 6, Section Aa, to this Annex (Part-FCL).
 - 2.4. Applicants for a CPL that have passed the relevant theoretical knowledge examinations for an IR in the same category of aircraft are credited towards the theoretical knowledge requirements in the communications subject.
- 3. ATPL**
- 3.1. Applicants for the issue of an ATPL holding an ATPL in another category of aircraft shall have received theoretical knowledge bridge instruction at an ATO on an approved course according to the differences identified between the ATPL syllabi for different aircraft categories.
 - 3.2. Applicants shall pass theoretical knowledge examinations as defined in this Annex (Part-FCL) for the following subjects in the appropriate aircraft category:
 - 021 — Aircraft general knowledge: airframe and systems, electrics, power plant and emergency equipment;
 - 022 — Aircraft general knowledge: instrumentation;
 - 032/034 — Performance aeroplanes or helicopters, as applicable;
 - 070 — Operational procedures;
 - 081/082 — Principles of flight for aeroplanes or helicopters, as applicable.
 - 3.3. Applicants for the issue of an ATPL(A) having passed the relevant theoretical examination for a CPL(A) are credited towards the theoretical knowledge requirements in the subject 'communications'.
 - 3.4. Applicants for the issue of an ATPL(H) having passed the relevant theoretical examinations for a CPL(H) are credited towards the theoretical knowledge requirements in the following subjects:
 - air law;
 - principles of flight (helicopter); and

- communications.
- 3.5. Applicants for the issue of an ATPL(A) having passed the relevant theoretical examination for an IR(A) are credited towards the theoretical knowledge requirements in the subject ‘communications’.
- 3.6. Applicants for the issue of an ATPL(H) with an IR(H) having passed the relevant theoretical examinations for a CPL(H) are credited towards the theoretical knowledge requirements in the following subjects:
 - principles of flight (helicopter); and
 - communications.
- 4. **IR**
 - 4.1. Applicants for an IR, or for a BIR, who have passed the relevant theoretical examinations for a CPL in the same aircraft category, shall be credited towards the theoretical knowledge requirements in the following subjects:
 - Human Performance,
 - Meteorology,
 - Communication.
 - 4.2. Applicants for the issue of an IR(H) having passed the relevant theoretical examinations for an ATPL(H) VFR are required to pass the following examination subjects:
 - air law;
 - flight planning and flight monitoring; and
 - radio navigation.

AMC1 to Appendix 1 Crediting of theoretical knowledge

ED Decision 2025/002/R

CROSS-CREDITS FOR THE SUBJECT AREA 100 KSA

- (a) Applicants who already hold another licence for the issuance of which they already completed the relevant Area 100 KSA may receive credits for the Area 100 KSA foreseen for the licence sought.
- (b) In order to receive such credits for Area 100 KSA for the licence sought, applicants specified in point (a) should do all of the following:
 - (1) undergo a pre-entry assessment at an ATO, as regards Area 100 KSA, in order to determine areas where applicants require bridge instruction and assessment(s);
 - (2) complete bridge instruction and assessment(s) in Area 100 KSA, as determined by the ATO on the basis of the assessment specified in point (1).
- (c) The ATO should record credits granted in line with this AMC in the applicants’ training records and may decide that applicants do not need to demonstrate, during the licence training course, those knowledge, skills and attitudes which they already demonstrated during the pre-entry assessment in accordance with point (1).
- (d) Before recommending applicants for theoretical knowledge examinations in accordance with point [FCL.025\(a\)\(2\)](#), the ATO should ensure that applicants’ instruction and assessments in total (credits and bridge instructions and assessments) have covered all aspects of Area 100 KSA for

the licence sought. In the case of applicants for an ATPL theoretical knowledge examination holding a CPL in the same aircraft category, the ATO should specifically ensure that all aspects of Area 100 KSA related to multi-pilot operations have been covered.

Appendix 2 – Language Proficiency Rating Scale – Expert, extended and operational level

Regulation (EU) No 1178/2011

LEVEL	PRONUNCIATION	STRUCTURE	VOCABULARY	FLUENCY	COMPREHENSION	INTERACTIONS
Expert (Level 6)	Pronunciation, stress, rhythm, and intonation, though possibly influenced by the first language or regional variation, almost never interfere with ease of understanding.	Both basic and complex grammatical structures and sentence patterns are consistently well controlled.	Vocabulary range and accuracy are sufficient to communicate effectively on a wide variety of familiar and unfamiliar topics. Vocabulary is idiomatic, nuanced and sensitive to register.	Able to speak at length with a natural, effortless flow. Varies speech flow for stylistic effect, e.g. to emphasize a point. Uses appropriate discourse markers and connectors spontaneously.	Comprehension is consistently accurate in nearly all contexts and includes comprehension of linguistic and cultural subtleties.	Interacts with ease in nearly all situations. Is sensitive to verbal and non-verbal cues, and responds to them appropriately.
Extended (Level 5)	Pronunciation, stress, rhythm, and intonation, though influenced by the first language or regional variation, rarely interfere with ease of understanding.	Basic grammatical structures and sentence patterns are consistently well controlled. Complex structures are attempted but with errors which sometimes interfere with meaning.	Vocabulary range and accuracy are sufficient to communicate effectively on common, concrete, and work-related topics. Paraphrases consistently and successfully. Vocabulary is sometimes idiomatic.	Able to speak at length with relative ease on familiar topics, but may not vary speech flow as a stylistic device. Can make use of appropriate discourse markers or connectors.	Comprehension is accurate on common, concrete, and work-related topics and mostly accurate when the speaker is confronted with a linguistic or situational complication or an unexpected turn of events. Is able to comprehend a range of speech varieties (dialect and/or accent) or registers.	Responses are immediate, appropriate, and informative. Manages the speaker/listener relationship effectively.
Operational (Level 4)	Pronunciation, stress, rhythm, and intonation are influenced by the first language or regional variation but only sometimes	Basic grammatical structures and sentence patterns are used creatively and are usually well controlled. Errors may occur, particularly in	Vocabulary range and accuracy are usually sufficient to communicate effectively on common, concrete, and work-related topics.	Produces stretches of language at an appropriate tempo. There may be occasional loss of fluency on transition from rehearsed or formulaic speech to spontaneous	Comprehension is mostly accurate on common, concrete, and work-related topics when the accent or variety used is sufficiently intelligible for an international community of users.	Responses are usually immediate, appropriate, and informative. Initiates and maintains exchanges even when dealing with an unexpected

LEVEL	PRONUNCIATION	STRUCTURE	VOCABULARY	FLUENCY	COMPREHENSION	INTERACTIONS
	interfere with ease of understanding.	unusual or unexpected circumstances, but rarely interfere with meaning.	Can often paraphrase successfully when lacking vocabulary particularly in unusual or unexpected circumstances.	interaction, but this does not prevent effective communication. Can make limited use of discourse markers and connectors. Fillers are not distracting.	When the speaker is confronted with a linguistic or situational complication or an unexpected turn of events, comprehension may be slower or require clarification strategies.	turn of events. Deals adequately with apparent misunderstandings by checking, confirming, or clarifying.

Note: The initial text of [Appendix 2](#) has been transferred to AMC, see also the Explanatory Note.

Appendix 3 – Training courses for the issue of a CPL and an ATPL

Regulation (EU) 2021/2227

1. This appendix describes the requirements for the different types of training courses for the issue of a CPL and an ATPL, with and without an IR.
2. An applicant wishing to transfer to another ATO during a training course shall apply to the competent authority for a formal assessment of the further hours of training required.

A. ATP integrated course – Aeroplanes

GENERAL

1. The aim of the ATP(A) integrated course is to train pilots to the level of proficiency necessary to enable them to operate as co-pilot on multi-pilot multi-engine aeroplanes in commercial air transport and to obtain the CPL(A)/IR.
2. An applicant wishing to undertake an ATP(A) integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(A) or PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of a PPL(A) or PPL(H) entrant, 50% of the hours flown prior to the course shall be credited, up to a maximum of 40 hours flying experience, or 45 hours if an aeroplane night rating has been obtained, of which up to 20 hours may count towards the requirement for dual instruction flight time.
4. The course shall comprise:
 - (a) theoretical knowledge instruction to the ATPL(A) knowledge level;
 - (b) visual and instrument flying training;
 - (c) training in MCC for the operation of multi-pilot aeroplanes; and
 - (d) UPRT in accordance with FCL.745.A unless applicants have already completed this training course before starting the ATP integrated course.’;
5. Applicants failing or being unable to complete the entire ATP(A) course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges and an IR if the applicable requirements are met.

THEORETICAL KNOWLEDGE

6. An ATP(A) theoretical knowledge course shall comprise at least 750 hours of instruction.
- 7.1. The MCC course shall comprise at least 25 hours of theoretical knowledge instruction and exercises.
- 7.2. The theoretical knowledge instruction in UPRT shall be conducted in accordance with FCL.745.A.

THEORETICAL KNOWLEDGE EXAMINATION

8. An applicant shall demonstrate the level of knowledge appropriate to the privileges granted to the holder of an ATPL(A).

FLYING TRAINING

9. The flying training, not including type rating training, shall comprise a total of at least 195 hours, including all progress tests, of which up to 55 hours for the entire course may be instrument ground time. Within the total of 195 hours, applicants shall complete at least:
 - (a) 95 hours of dual instruction, of which up to 55 hours may be instrument ground time;

- (b) 70 hours as PIC, of which up to 55 hours may be SPIC. The instrument flight time as SPIC shall only be counted as PIC flight time up to a maximum of 20 hours;
 - (c) 50 hours of cross-country flight as PIC, including one VFR cross-country flight of at least 540 km (300 NM), in the course of which full-stop landings at two aerodromes different from the aerodrome of departure shall be made; and
 - (d) 5 hours of flight time at night, comprising 3 hours of dual instruction, which shall include at least:
 - (1) 1 hour of cross-country navigation;
 - (2) five solo take-offs; and
 - (3) five solo full-stop landings;
 - (e) UPRT flight instruction in accordance with FCL.745.A;
 - (f) 115 hours of instrument time comprising, at least:
 - (1) 20 hours as SPIC;
 - (2) 15 hours of MCC, for which an FFS or an FNPT II may be used;
 - (3) 50 hours of instrument flight instruction, of which up to:
 - (i) 25 hours may be instrument ground time in an FNPT I; or
 - (ii) 40 hours may be instrument ground time in an FNPT II, an FTD 2 or an FFS, of which up to 10 hours may be conducted in an FNPT I.
- Applicants holding a BIR or a course completion certificate for the Basic Instrument Flight Module shall be credited with up to 10 hours towards the required instrument instruction time. Hours done in a BITD shall not be credited.
- (g) 5 hours in an aeroplane which:
 - (1) is certificated for the carriage of at least 4 persons; and
 - (2) has a variable pitch propeller and retractable landing gear.

SKILL TEST

10. Upon completion of the related flying training, the applicant shall take the CPL(A) skill test on either a single-engine or a multi-engine aeroplane and the IR skill test on a multi-engine aeroplane.

B. ATP modular course – Aeroplanes

1. Applicants for an ATPL(A) who complete their theoretical knowledge instruction at a modular course shall:
- (a) hold at least a PPL(A) issued in accordance with Annex 1 to the Chicago Convention; and complete at least the following hours of theoretical knowledge instruction:
 - (1) for applicants holding a PPL(A): 650 hours;
 - (2) for applicants holding a CPL(A): 400 hours;
 - (3) for applicants holding an IR(A): 500 hours;
 - (4) for applicants holding a CPL(A) and an IR(A): 250 hours.

The theoretical knowledge instruction shall be completed before the skill test for the ATPL(A) is taken.

C. CPL/IR integrated course – Aeroplanes

GENERAL

1. The aim of the CPL(A) and IR(A) integrated course is to train pilots to the level of proficiency necessary to operate single-pilot single-engine or multi-engine aeroplanes in commercial air transport and to obtain the CPL(A)/IR.
2. An applicant wishing to undertake a CPL(A)/IR integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(A) or PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of a PPL(A) or PPL(H) entrant, 50% of the hours flown prior to the course shall be credited, up to a maximum of 40 hours flying experience, or 45 hours if an aeroplane night rating has been obtained, of which up to 20 hours may count towards the requirement for dual instruction flight time.
4. The course shall comprise:
 - (a) theoretical knowledge instruction to CPL(A) and IR knowledge level; and
 - (b) visual and instrument flying training.
5. An applicant failing or unable to complete the entire CPL/IR(A) course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges and an IR if the applicable requirements are met.

THEORETICAL KNOWLEDGE

6. A CPL(A)/IR theoretical knowledge course shall comprise at least 500 hours of instruction.

THEORETICAL KNOWLEDGE EXAMINATION

7. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(A) and an IR.

FLYING TRAINING

8. The flying training, not including type rating training, shall comprise a total of at least 180 hours, to include all progress tests, of which up to 40 hours for the entire course may be instrument ground time. Within the total of 180 hours, applicants shall complete at least:
 - (a) 80 hours of dual instruction, of which up to 40 hours may be instrument ground time;
 - (b) 70 hours as PIC, of which up to 55 hours may be SPIC. The instrument flight time as SPIC shall only be counted as PIC flight time up to a maximum of 20 hours;
 - (c) 50 hours of cross-country flight as PIC, including a VFR cross-country flight of at least 540 km (300 NM), in the course of which full stop landings at two aerodromes different from the aerodrome of departure shall be made;
 - (d) 5 hours flight time shall be completed at night, comprising 3 hours of dual instruction, which shall include at least 1 hour of cross-country navigation and 5 solo take-offs and 5 solo full stop landings; and
 - (e) 100 hours of instrument time comprising, at least:
 - (1) 20 hours as SPIC; and
 - (2) 50 hours of instrument flight instruction, of which up to:
 - (i) 25 hours may be instrument ground time in an FNPT I, or

- (ii) 40 hours may be instrument ground time in an FNPT II, FTD 2 or FFS, of which up to 10 hours may be conducted in an FNPT I.

Applicants holding a BIR or a course completion certificate for the Basic Instrument Flight Module shall be credited with up to 10 hours towards the required instrument instruction time. Hours done in a BITD shall not be credited; and

- (f) 5 hours to be carried out in an aeroplane certificated for the carriage of at least 4 persons that has a variable pitch propeller and retractable landing gear.

SKILL TESTS

- 10. Upon completion of the related flying training the applicant shall take the CPL(A) skill test and the IR skill test on either a multi-engine aeroplane or a single-engine aeroplane.

D. CPL integrated course – Aeroplanes

GENERAL

- 1. The aim of the CPL(A) integrated course is to train pilots to the level of proficiency necessary for the issue of a CPL(A).
- 2. An applicant wishing to undertake a CPL(A) integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
- 3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(A) or PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of a PPL(A) or PPL(H) entrant, 50% of the hours flown prior to the course shall be credited, up to a maximum of 40 hours flying experience, or 45 hours if an aeroplane night rating has been obtained, of which up to 20 hours may count towards the requirement for dual instruction flight time.
- 4. The course shall comprise:
 - (a) theoretical knowledge instruction to CPL(A) knowledge level; and
 - (b) visual and instrument flying training.
- 5. An applicant failing or unable to complete the entire CPL(A) course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges, if the applicable requirements are met.

THEORETICAL KNOWLEDGE

- 6. A CPL(A) theoretical knowledge course shall comprise at least 350 hours of instruction.

THEORETICAL KNOWLEDGE EXAMINATION

- 7. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(A).

FLYING TRAINING

- 8. The flying training, not including type rating training, shall comprise a total of at least 150 hours, to include all progress tests, of which up to 5 hours for the entire course may be instrument ground time. Within the total of 150 hours, applicants shall complete at least:
 - (a) 80 hours of dual instruction, of which up to 5 hours may be instrument ground time;
 - (b) 70 hours as PIC, of which up to 55 hours may be as SPIC;
 - (c) 20 hours of cross-country flight as PIC, including a VFR cross-country flight of at least 540 km (300 NM), in the course of which full stop landings at two aerodromes different from the aerodrome of departure shall be made;

- (d) 5 hours flight time shall be completed at night, comprising 3 hours of dual instruction, which shall include at least 1 hour of cross-country navigation and 5 solo take-offs and 5 solo full stop landings;
- (e) 10 hours of instrument flight instruction, of which up to 5 hours may be instrument ground time in an FNPT I, FTD 2, FNPT II or FFS. Applicants holding a BIR or a course completion certificate for the Basic Instrument Flight Module shall be credited with up to 10 hours towards the required instrument instruction time. Hours done in a BITD shall not be credited;
- (f) 5 hours to be carried out in an aeroplane certificated for the carriage of at least four persons that has a variable pitch propeller and retractable landing gear.

SKILL TEST

- 9. Upon completion of the flying training the applicant shall take the CPL(A) skill test on a single-engine or a multi-engine aeroplane.

E. CPL modular course – Aeroplanes**GENERAL**

- 1. The aim of the CPL(A) modular course is to train PPL(A) holders to the level of proficiency necessary for the issue of a CPL(A).
- 2. Before commencing a CPL(A) modular course an applicant shall be the holder of a PPL(A) issued in accordance with Annex 1 to the Chicago Convention.
- 3. Before commencing the flight training the applicant shall:
 - (a) have completed 150 hours flight time; including 50 hours as PIC on aeroplanes of which 10 hours shall be cross-country.

Except for the requirement of 50 hours as PIC in aeroplanes, hours as PIC in other categories of aircraft may account for the 150 hours of aeroplane flight time in any of the following cases:

 - (1) 20 hours in helicopters, if applicants hold a PPL(H);
 - (2) 50 hours in helicopters, if applicants hold a CPL(H);
 - (3) 10 hours in TMGs or sailplanes;
 - (4) 20 hours in airships, if applicants hold a PPL(As);
 - (5) 50 hours in airships, if applicants hold a CPL(As).
 - (b) have complied with the prerequisites for the issue of a class or type rating for multi-engine aeroplanes in accordance with Subpart H, if a multi-engine aeroplane is to be used on the skill test.
- 4. An applicant wishing to undertake a modular CPL(A) course shall complete all the flight instructional stages in one continuous course of training as arranged by an ATO. The theoretical knowledge instruction may be given at an ATO conducting theoretical knowledge instruction only.
- 5. The course shall comprise:
 - (a) theoretical knowledge instruction to CPL(A) knowledge level; and
 - (b) visual and instrument flying training.

THEORETICAL KNOWLEDGE

6. An approved CPL(A) theoretical knowledge course shall comprise at least 250 hours of instruction.

THEORETICAL KNOWLEDGE EXAMINATION

7. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(A).

FLYING TRAINING

8. Applicants without an IR shall be given at least 25 hours dual flight instruction, including 10 hours of instrument instruction of which up to 5 hours may be instrument ground time in a BITD, an FNPT I or II, an FTD 2 or an FFS.
9. Applicants holding a valid IR(A) shall be credited towards the dual instrument instruction time. Applicants holding a valid IR(H) shall be credited up to 5 hours of the dual instrument instruction time, in which case at least 5 hours dual instrument instruction time shall be given in an aeroplane. Applicants holding a BIR or a Course Completion Certificate for the Basic Instrument Flight Module shall be credited with up to 10 hours towards the required instrument instruction time.
10.
 - (a) Applicants with a valid IR shall be given at least 15 hours dual visual flight instruction.
 - (b) Applicants without a night rating aeroplane shall be given additionally at least 5 hours night flight instruction, comprising 3 hours of dual instruction, which shall include at least 1 hour of cross-country navigation and 5 solo take-offs and 5 solo full stop landings.
11. At least 5 hours of the flight instruction shall be carried out in an aeroplane certificated for the carriage of at least 4 persons and have a variable pitch propeller and retractable landing gear.

EXPERIENCE

12. The applicant for a CPL(A) shall have completed at least 200 hours flight time, including at least:
 - (a) 100 hours as PIC, of which 20 hours of cross-country flight as PIC, which shall include a VFR cross-country flight of at least 540 km (300 NM), in the course of which full stop landings at two aerodromes different from the aerodrome of departure shall be made;
 - (b) 5 hours of flight time shall be completed at night, comprising 3 hours of dual instruction, which shall include at least 1 hour of cross-country navigation and 5 solo take-offs and 5 solo full stop landings; and
 - (c) 10 hours of instrument flight instruction, of which up to 5 hours may be instrument ground time in an FNPT I, or FNPT II or FFS. Applicants holding a BIR or a course completion certificate for the Basic Instrument Flight Module shall be credited with up to 10 hours towards the required instrument instruction time. Hours done in a BITD shall not be credited;
 - (d) 6 hours of flight time shall be completed in a multi-engine aeroplane, if a multi-engine aeroplane is used for the skill test.
 - (e) Hours as PIC of other categories of aircraft may count towards the 200 hours flight time, in the following cases:
 - (i) 30 hours in helicopter, if the applicant holds a PPL(H); or
 - (ii) 100 hours in helicopters, if the applicant holds a CPL(H); or
 - (iii) 30 hours in TMGs or sailplanes; or
 - (iv) 30 hours in airships, if the applicant holds a PPL(As); or

- (v) 60 hours in airships, if the applicant holds a CPL(As).

SKILL TEST

13. Upon completion of the flying training and relevant experience requirements the applicant shall take the CPL(A) skill test on either a single-engine or a multi-engine aeroplane.

F. ATP/IR integrated course – Helicopters**GENERAL**

1. The aim of the ATP(H)/IR integrated course is to train pilots to the level of proficiency necessary to enable them to operate as co-pilot on multi-pilot multi-engine helicopters in commercial air transport and to obtain the CPL(H)/IR.
2. An applicant wishing to undertake an ATP(H)/IR integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of a PPL(H) entrant, 50% of the relevant experience shall be credited, up to a maximum of:
 - (a) 40 hours, of which up to 20 hours may be dual instruction; or
 - (b) 50 hours, of which up to 25 hours may be dual instruction, if a helicopter night rating has been obtained.
4. The course shall comprise:
 - (a) theoretical knowledge instruction to the ATPL(H) and IR knowledge level;
 - (b) visual and instrument flying training; and
 - (c) training in MCC for the operation of multi-pilot helicopters.
5. An applicant failing or unable to complete the entire ATP(H) /IR course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges and an IR, if the applicable requirements are met.

THEORETICAL KNOWLEDGE

6. An ATP(H)/IR theoretical knowledge course shall comprise at least 750 hours of instruction.
7. The MCC course shall comprise at least 25 hours of theoretical knowledge instruction exercises.

THEORETICAL KNOWLEDGE EXAMINATION

8. An applicant shall demonstrate the level of knowledge appropriate to the privileges granted to the holder of an ATPL(H) and an IR.

FLYING TRAINING

9. The flying training shall comprise a total of at least 195 hours, to include all progress tests. Within the total of 195 hours, applicants shall complete at least:
 - (a) 140 hours of dual instruction, of which:
 - (1) 75 hours visual instruction may include:
 - (i) 30 hours in a helicopter FFS, level C/D, or
 - (ii) 25 hours in a FTD 2,3, or
 - (iii) 20 hours in a helicopter FNPT II/III, or

- (iv) 20 hours in an aeroplane or TMG;
- (2) 50 hours instrument instruction may include:
 - (i) up to 20 hours in a helicopter FFS or FTD 2,3 or FNPT II/III, or
 - (ii) 10 hours in at least a helicopter FNPT 1 or an aeroplane;
- (3) 15 hours MCC, for which a helicopter FFS or helicopter FTD 2,3(MCC) or FNPT II/III(MCC) may be used.

If the helicopter used for the flying training is of a different type from the helicopter FFS used for the visual training, the maximum credit shall be limited to that allocated for the helicopter FNPT II/III.
- (b) 55 hours as PIC, of which 40 hours may be as SPIC. At least 14 hours solo day and 1 hour solo night shall be made.
- (c) 50 hours of cross-country flight, including at least 10 hours of cross-country flight as SPIC including a VFR cross-country flight of at least 185 km (100 NM) in the course of which landings at two different aerodromes from the aerodrome of departure shall be made;
- (d) 5 hours flight time in helicopters shall be completed at night comprising 3 hours of dual instruction including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include a take-off and a landing;
- (e) 50 hours of dual instrument time comprising:
 - (i) 10 hours basic instrument instruction time, and
 - (ii) 40 hours IR Training, which shall include at least 10 hours in a multi-engine IFR-certificated helicopter.

SKILL TESTS

10. Upon completion of the related flying training, the applicant shall take the CPL(H) skill test on a multi-engine helicopter and the IR skill test on an IFR certificated multi-engine helicopter and shall comply with the requirements for MCC training.

G. ATP integrated course – Helicopters**GENERAL**

1. The aim of the ATP(H) integrated course is to train pilots to the level of proficiency necessary to enable them to operate as co-pilot on multi-pilot multi-engine helicopters limited to VFR privileges in commercial air transport and to obtain the CPL(H).
2. An applicant wishing to undertake an ATP(H) integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of a PPL(H) entrant, 50% of the relevant experience shall be credited, up to a maximum of:
 - (a) 40 hours, of which up to 20 hours may be dual instruction; or
 - (b) 50 hours, of which up to 25 hours may be dual instruction, if a helicopter night rating has been obtained.
4. The course shall comprise:
 - (a) theoretical knowledge instruction to the ATPL(H) knowledge level;

- (b) visual and basic instrument flying training; and
 - (c) training in MCC for the operation of multi-pilot helicopters.
5. An applicant failing or unable to complete the entire ATP(H) course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges, if the applicable requirements are met.

THEORETICAL KNOWLEDGE

6. An ATP(H) theoretical knowledge course shall comprise at least 650 hours of instruction.
7. The MCC course shall comprise at least 20 hours of theoretical knowledge instruction exercises.

THEORETICAL KNOWLEDGE EXAMINATION

8. An applicant shall demonstrate the level of knowledge appropriate to the privileges granted to the holder of an ATPL (H).

FLYING TRAINING

9. The flying training shall comprise a total of at least 150 hours, to include all progress tests. Within the total of 150 hours, applicants shall complete at least:

- (a) 95 hours of dual instruction, of which:
 - (i) 75 hours visual instruction may include:
 - (1) 30 hours in a helicopter FFS level C/D, or
 - (2) 25 hours in a helicopter FTD 2,3, or
 - (3) 20 hours in a helicopter FNPT II/III, or
 - (4) 20 hours in an aeroplane or TMG;
 - (ii) 10 hours basic instrument instruction may include 5 hours in at least a helicopter FNPT I or an aeroplane;
 - (iii) 10 hours MCC, for which a helicopter: helicopter FFS or FTD 2,3(MCC) or FNPT II/III(MCC) may be used.

If the helicopter used for the flying training is of a different type from the helicopter FFS used for the visual training, the maximum credit shall be limited to that allocated for the helicopter FNPT II/III.

- (b) 55 hours as PIC, of which 40 hours may be as SPIC. At least 14 hours solo day and 1 hour solo night shall be made;
- (c) 50 hours of cross-country flight, including at least 10 hours of cross-country flight as SPIC, including a VFR cross-country flight of at least 185 km (100 NM) in the course of which landings at two different aerodromes from the aerodrome of departure shall be made;
- (d) 5 hours flight time in helicopters shall be completed at night comprising 3 hours of dual instruction including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include a take-off and a landing.

SKILL TESTS

10. Upon completion of the related flying training the applicant shall take the CPL(H) skill test on a multi-engine helicopter and comply with MCC requirements.

H. ATP modular course – Helicopters

1. Applicants for an ATPL(H) who complete their theoretical knowledge instruction at a modular course shall hold at least a PPL(H) and complete at least the following hours of instruction within a period of 18 months:
 - (a) for applicants holding a PPL(H) issued in accordance with Annex 1 to the Chicago Convention: 550 hours;
 - (b) for applicants holding a CPL(H): 300 hours.
2. Applicants for an ATPL(H)/IR who complete their theoretical knowledge instruction at a modular course shall hold at least a PPL(H) and complete at least the following hours of instruction:
 - (a) for applicants holding a PPL(H): 650 hours;
 - (b) for applicants holding a CPL(H): 400 hours;
 - (c) for applicants holding an IR(H): 500 hours;
 - (d) for applicants holding a CPL(H) and an IR(H): 250 hours.

I. CPL/IR integrated course – Helicopters**GENERAL**

1. The aim of the CPL(H)/IR integrated course is to train pilots to the level of proficiency necessary to operate single-pilot multi-engine helicopters and to obtain the CPL(H)/IR multi-engine helicopter.
2. An applicant wishing to undertake a CPL(H)/IR integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of an entrant holding a PPL(H), 50% of the relevant experience shall be credited, up to a maximum of:
 - (a) 40 hours, of which up to 20 hours may be dual instruction; or
 - (b) 50 hours, of which up to 25 hours may be dual instruction, if a helicopter night rating has been obtained.
4. The course shall comprise:
 - (a) theoretical knowledge instruction to CPL(H) and IR knowledge level, and the initial multi-engine helicopter type rating; and
 - (b) visual and instrument flying training.
5. An applicant failing or unable to complete the entire CPL(H)/IR course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges and an IR, if the applicable requirements are met.

THEORETICAL KNOWLEDGE

6. A CPL(H)/IR theoretical knowledge course shall comprise at least 500 hours of instruction.

THEORETICAL KNOWLEDGE EXAMINATION

7. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(H) and an IR.

FLYING TRAINING

8. The flying training shall comprise a total of at least 180 hours including all progress tests. Within the 180 hours, applicants shall complete at least:
- (a) 125 hours of dual instruction, of which:
 - (i) 75 hours visual instruction, which may include:
 - (1) 30 hours in a helicopter FFS level C/D, or
 - (2) 25 hours in a helicopter FTD 2,3, or
 - (3) 20 hours in a helicopter FNPT II/III, or
 - (4) 20 hours in an aeroplane or TMG;
 - (ii) 50 hours instrument instruction which may include:
 - (1) up to 20 hours in a helicopter FFS or FTD 2,3, or FNPT II,III, or
 - (2) 10 hours in at least a helicopter FNPT I or an aeroplane.

If the helicopter used for the flying training is of a different type from the FFS used for the visual training, the maximum credit shall be limited to that allocated for the FNPT II/III.
 - (b) 55 hours as PIC, of which 40 hours may be as SPIC. At least 14 hours solo day and 1 hour solo night shall be made;
 - (c) 10 hours dual cross-country flying;
 - (d) 10 hours of cross-country flight as PIC, including a VFR cross-country flight of at least 185 km (100 NM) in the course of which full stop landings at two different aerodromes from the aerodrome of departure shall be made;
 - (e) 5 hours of flight time in helicopters shall be completed at night comprising 3 hours of dual instruction including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include a take-off and a landing;
 - (f) 50 hours of dual instrument time comprising:
 - (i) 10 hours basic instrument instruction time; and
 - (ii) 40 hours IR Training, which shall include at least 10 hours in a multi-engine IFR-certificated helicopter.

SKILL TEST

9. Upon completion of the related flying training, the applicant shall take the CPL(H) skill test on either a multi-engine or a single-engine helicopter and the IR skill test on an IFR-certificated helicopter.

J. CPL integrated course – Helicopters**GENERAL**

1. The aim of the CPL(H) integrated course is to train pilots to the level of proficiency necessary for the issue of a CPL(H).
2. An applicant wishing to undertake a CPL(H) integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of an entrant holding a PPL(H), 50% of the relevant experience shall be credited, up to a maximum of:

- (a) 40 hours, of which up to 20 hours may be dual instruction; or
 - (b) 50 hours, of which up to 25 hours may be dual instruction if a helicopter night rating has been obtained.
4. The course shall comprise:
- (a) theoretical knowledge instruction to CPL(H) knowledge level; and
 - (b) visual and instrument flying training.
5. An applicant failing or unable to complete the entire CPL(H) course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges, if the applicable requirements are met.

THEORETICAL KNOWLEDGE

6. An approved CPL(H) theoretical knowledge course shall comprise at least 350 hours of instruction or 200 hours if the applicant is the holder of a PPL.

THEORETICAL KNOWLEDGE EXAMINATION

7. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(H).

FLYING TRAINING

8. The flying training shall comprise a total of at least 135 hours, to include all progress tests, of which up to 5 hours may be instrument ground time. Within the 135 hours total, applicants shall complete at least:
- (a) 85 hours of dual instruction, of which:
 - (i) up to 75 hours may be visual instruction, and may include:
 - (1) 30 hours in a helicopter FFS level C/D, or
 - (2) 25 hours in a helicopter FTD 2,3, or
 - (3) 20 hours in a helicopter FNPT II/III, or
 - (4) 20 hours in an aeroplane or TMG.
 - (ii) up to 10 hours may be instrument instruction, and may include 5 hours in at least a helicopter FNPT I or an aeroplane.

If the helicopter used for the flying training is of a different type from the FFS used for the visual training, the maximum credit shall be limited to that allocated for the FNPT II/III.
 - (b) 50 hours as PIC, of which 35 hours may be as SPIC. At least 14 hours solo day and 1 hour solo night shall be made;
 - (c) 10 hours dual cross-country flying;
 - (d) 10 hours of cross-country flight as PIC, including a VFR cross-country flight of at least 185 km (100 NM) in the course of which full stop landings at two different aerodromes from the aerodrome of departure shall be made;
 - (e) 5 hours flight time in helicopters shall be completed at night comprising 3 hours of dual instruction including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include a take-off and a landing;
 - (f) 10 hours of instrument dual instruction time, including at least 5 hours in a helicopter.

SKILL TEST

9. Upon completion of the related flying training, the applicant shall take the CPL(H) skill test.

K. CPL modular course – Helicopters**GENERAL**

1. The aim of the CPL(H) modular course is to train PPL(H) holders to the level of proficiency necessary for the issue of a CPL(H).
2. Before commencing a CPL(H) modular course an applicant shall be the holder of a PPL(H) issued in accordance with Annex 1 to the Chicago Convention.
3. Before commencing the flight training the applicant shall:

- (a) have completed 155 hours flight time, including 50 hours as PIC in helicopters of which 10 hours shall be cross- country.

Except for the requirement of 50 hours as PIC in helicopters, hours as PIC in other categories of aircraft may account for the 155 hours of helicopter flight time in any of the following cases:

- (1) 20 hours in aeroplanes if applicants hold a PPL(A);
 - (2) 50 hours in aeroplanes if applicants hold a CPL(A);
 - (3) 10 hours in TMGs or sailplanes;
 - (4) 20 hours in airships if applicants hold a PPL(As);
 - (5) 50 hours in airships if applicants hold a CPL(As);
- (b) have complied with [FCL.725](#) and [FCL.720.H](#) if a multi-engine helicopter is to be used on the skill test.
4. An applicant wishing to undertake a modular CPL(H) course shall complete all the flight instructional stages in one continuous course of training as arranged by an ATO. The theoretical knowledge instruction may be given at an ATO that conducts theoretical knowledge instruction only.
 5. The course shall comprise:
 - (a) theoretical knowledge instruction to CPL(H) knowledge level; and
 - (b) visual and instrument flying training.

THEORETICAL KNOWLEDGE

6. An approved CPL(H) theoretical knowledge course shall comprise at least 250 hours of instruction.

THEORETICAL KNOWLEDGE EXAMINATION

7. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(H).

FLYING TRAINING

8. Applicants without an IR shall be given at least 30 hours dual flight instruction, of which:
 - (a) 20 hours visual instruction, which may include 5 hours in a helicopter FFS or FTD 2,3 or FNPT II,III; and

- (b) 10 hours instrument instruction, which may include 5 hours in at least a helicopter FTD 1 or FNPT I or aeroplane.
- 9. Applicants holding a valid IR(H) shall be fully credited towards the dual instrument instruction time. Applicants holding a valid IR(A) shall complete at least 5 hours of the dual instrument instruction time in a helicopter.
- 10. Applicants without a night rating helicopter shall be given additionally at least 5 hours night flight instruction comprising 3 hours of dual instruction including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include a take-off and a landing.

EXPERIENCE

- 11. The applicant for a CPL(H) shall have completed at least 185 hours flight time, including 50 hours as PIC, of which 10 hours of cross-country flight as PIC, including a VFR cross-country flight of at least 185 km (100 NM), in the course of which full stop landings at two aerodromes different from the aerodrome of departure shall be made.

Hours as pilot-in-command of other categories of aircraft may count towards the 185 hours flight time, in the following cases:

- (a) 20 hours in aeroplanes, if the applicant holds a PPL(A); or
- (b) 50 hours in aeroplanes, if the applicant holds a CPL(A); or
- (c) 10 hours in TMGs or sailplanes; or
- (d) 20 hours in airships, if the applicant holds a PPL(As); or
- (e) 50 hours in airships, if the applicant holds a CPL(As).

SKILL TEST

- 12. Upon completion of the related flying training and relevant experience, the applicant shall take the CPL(H) skill test.

L. CPL/IR integrated course – Airships

- 1. The aim of the CPL(As)/IR integrated course is to train pilots to the level of proficiency necessary to operate airships and to obtain the CPL(As)/IR.
- 2. An applicant wishing to undertake a CPL(As)/IR integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
- 3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(As), PPL(A) or PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of an entrant holding a PPL(As), PPL(A) or PPL(H) shall be credited up to a maximum of:
 - (a) 10 hours, of which up to 5 hours may be dual instruction; or
 - (b) 15 hours, of which up to 7 hours may be dual instruction, if an airship night rating has been obtained.
- 4. The course shall comprise:
 - (a) theoretical knowledge instruction to CPL(As) and IR knowledge level, and the initial airship type rating; and
 - (b) visual and instrument flying training.

5. An applicant failing or unable to complete the entire CPL/IR(As) course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges and an IR, if the applicable requirements are met.

THEORETICAL KNOWLEDGE

6. A CPL(As)/IR theoretical knowledge course shall comprise at least 500 hours of instruction.

THEORETICAL KNOWLEDGE EXAMINATION

7. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(As) and an IR.

FLYING TRAINING

8. The flying training shall comprise a total of at least 80 hours including all progress tests. Within the 80 hours, applicants shall complete at least:

- (a) 60 hours of dual instruction, of which:
 - (i) 30 hours visual instruction, which may include:
 - (1) 12 hours in an airship FFS, or
 - (2) 10 hours in an airship FTD, or
 - (3) 8 hours in an airship FNPT II/III, or
 - (4) 8 hours in an aeroplane, helicopter or TMG;
 - (ii) 30 hours instrument instruction which may include:
 - (1) up to 12 hours in an airship FFS or FTD or FNPT II,III, or
 - (2) 6 hours in at least a airship FTD 1 or FNPT I or aeroplane.
- If the airship used for the flying training is of a different type from the FFS used for the visual training, the maximum credit shall be limited to 8 hours.
- (b) 20 hours as PIC, of which 5 hours may be as SPIC. At least 14 hours solo day and 1 hour solo night shall be made;
 - (c) 5 hours of cross-country flight as PIC, including a VFR cross-country flight of at least 90 km (50 NM) in the course of which two full stop landings at the destination aerodrome shall be made;
 - (d) 5 hours flight time in airships shall be completed at night comprising 3 hours of dual instruction including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include take-off and landing;
 - (e) 30 hours of dual instrument time comprising:
 - (i) 10 hours basic instrument instruction time; and
 - (ii) 20 hours IR Training, which shall include at least 10 hours in a multi-engine IFR-certificated airship.

SKILL TEST

9. Upon completion of the related flying training, the applicant shall take the CPL(As) skill test on either a multi-engine or a single-engine airship and the IR skill test on an IFR-certificated multi-engine airship.

M. CPL integrated course – Airships**GENERAL**

1. The aim of the CPL(As) integrated course is to train pilots to the level of proficiency necessary for the issue of a CPL(AS).
2. An applicant wishing to undertake a CPL(As) integrated course shall complete all the instructional stages in one continuous course of training as arranged by an ATO.
3. An applicant may be admitted to training either as an ab-initio entrant, or as a holder of a PPL(As), PPL(A) or PPL(H) issued in accordance with Annex 1 to the Chicago Convention. In the case of an entrant holding a PPL(As), PPL(A) or PPL(H) shall be credited up to a maximum of:
 - (a) 10 hours, of which up to 5 hours may be dual instruction; or
 - (b) 15 hours, of which up to 7 hours may be dual instruction if a airship night rating has been obtained.
4. The course shall comprise:
 - (a) theoretical knowledge instruction to CPL(As) knowledge level; and
 - (b) visual and instrument flying training.
5. An applicant failing or unable to complete the entire CPL(As) course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges, if the applicable requirements are met.

THEORETICAL KNOWLEDGE

6. An approved CPL(As) theoretical knowledge course shall comprise at least 350 hours of instruction or 200 hours if the applicant is a PPL holder.

THEORETICAL KNOWLEDGE EXAMINATION

7. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(As).

FLYING TRAINING

8. The flying training shall comprise a total of at least 50 hours, to include all progress tests, of which up to 5 hours may be instrument ground time. Within the 50 hours total, applicants shall complete at least:
 - (a) 30 hours of dual instruction, of which up to 5 hours may be instrument ground time;
 - (b) 20 hours as PIC;
 - (c) 5 hours dual cross-country flying;
 - (d) 5 hours of cross-country flight as PIC, including a VFR cross-country flight of at least 90 km (50 NM) in the course of which two full stop landings at the destination aerodrome shall be made;
 - (e) 5 hours flight time in airships shall be completed at night comprising 3 hours of dual instruction including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include take-off and landing;
 - (f) 10 hours of instrument dual instruction time, including at least 5 hours in an airship.

SKILL TEST

9. Upon completion of the related flying training, the applicant shall take the CPL(As) skill test.

N. CPL modular course – Airships**GENERAL**

1. The aim of the CPL(As) modular course is to train PPL(As) holders to the level of proficiency necessary for the issue of a CPL(As).
2. Before commencing a CPL(As) modular course an applicant shall:
 - (a) hold a PPL(As) issued in accordance with Annex 1 to the Chicago Convention;
 - (b) have completed 200 hours flight time as a pilot on airships, including 100 hours as PIC, of which 50 hours shall be cross-country.
3. An applicant wishing to undertake a modular CPL(As) course shall complete all the flight instructional stages in one continuous course of training as arranged by an ATO. The theoretical knowledge instruction may be given at an ATO that conducts theoretical knowledge instruction only.
4. The course shall comprise:
 - (a) theoretical knowledge instruction to CPL(As) knowledge level; and
 - (b) visual and instrument flying training.

THEORETICAL KNOWLEDGE

5. An approved CPL(As) theoretical knowledge course shall comprise at least 250 hours of instruction.

THEORETICAL KNOWLEDGE EXAMINATION

6. An applicant shall demonstrate a level of knowledge appropriate to the privileges granted to the holder of a CPL(As).

FLYING TRAINING

7. Applicants without an IR shall be given at least 20 hours dual flight instruction, of which:
 - 10 hours visual instruction, which may include 5 hours in an airship FFS or FTD 2,3 or FNPT II,III; and
 - 10 hours instrument instruction, which may include 5 hours in at least an airship FTD 1 or FNPT I or aeroplane.
8. Applicants holding a valid IR(As) shall be fully credited towards the dual instrument instruction time. Applicants holding a valid IR in another category of aircraft shall complete at least 5 hours of the dual instrument instruction time in an airship.
9. Applicants without a night rating airship shall be given additionally at least 5 hours night flight instruction comprising 3 hours of dual instruction including at least 1 hour of cross-country navigation and 5 solo night circuits. Each circuit shall include a take-off and a landing.

EXPERIENCE

10. The applicant for a CPL(As) shall have completed at least 250 hours flight time in airships, including 125 hours as PIC, of which 50 hours of cross-country flight as PIC, including a VFR cross-country flight of at least 90 km (50 NM), in the course of which a full stop landing at destination aerodrome.

Hours as PIC of other categories of aircraft may count towards the 185 hours flight time, in the following cases;

- (a) 30 hours in aeroplanes or helicopters, if the applicant holds a PPL(A) or PPL(H) respectively; or
- (b) 60 hours in aeroplanes or helicopters, if the applicant holds a CPL(A) or CPL(H) respectively; or
- (c) 10 hours in TMGs or sailplanes; or
- (d) 10 hours in balloons.

SKILL TEST

11. Upon completion of the related flying training and relevant experience, the applicant shall take the CPL(As) skill test.

AMC1 to Appendix 3 Training courses for the issue of a CPL and an ATPL

*ED Decision 2025/002/R***GENERAL**

- (a) When ensuring that the applicant complies with the prerequisites for the course, in accordance with ORA.ATO.145, the ATO should check that the applicant has enough knowledge of mathematics, physics and English to facilitate the understanding of the theoretical knowledge instruction content of the course.
- (b) Whenever reference is made to a certain amount of hours of training, this means a full hour. Time not directly assigned to training (such as breaks, etc.) is not to be counted towards the total amount of time that is required.
- (c) The UPRT elements and components specified in AMC2 to Appendix 3; AMC1 to Appendix 5 point (a) should be integrated into the flying training phases or modules.
- (d) The flight instruction syllabus should take into account the principles of TEM.

A. ATP integrated course: aeroplanes

- (a) The ATP integrated course should last between 12 and 36 months. This period may be extended where additional flying training or ground instruction is provided by the ATO.

CREDITING

- (b) Credit for previous experience given to an applicant who already holds a PPL should be entered into the applicant's training record. In the case of a student pilot who does not hold a pilot licence and with the approval of the competent authority, an ATO may designate certain dual exercises to be flown in a helicopter or a TMG up to a maximum of 20 hours.

THEORETICAL KNOWLEDGE

- (c) The 750 hours of instruction, which also cover the Area 100 KSA, may include in suitable proportions:
 - (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;

- (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
- (6) exercises that use demonstration equipment or training devices;
- (7) directed study including workbook exercises or assignments;
- (8) aerodrome or aviation industry field trips;
- (9) computer-based training and e-learning elements;
- (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
- (11) other training methods, media and tools approved by the competent authority.

The 750 hours of instruction should be divided in such a way that in each subject the minimum hours are:

- | | |
|---------------------------------------|-----------|
| (1) Air law | 35 hours |
| (2) Aircraft general knowledge | 100 hours |
| (3) Flight performance and planning | 120 hours |
| (4) Human performance and limitations | 35 hours |
| (5) Meteorology | 60 hours |
| (6) Navigation | 90 hours |
| (7) Operational procedures | 25 hours |
| (8) Principles of flight | 55 hours |
| (9) Communications | 20 hours |

Other subdivisions of hours may be agreed upon between the competent authority and the ATO.

FLYING TRAINING

- (d) The flying instruction is divided into six phases:

- (1) Phase 1:

Exercises up to the first solo flight comprise a total of at least 10 hours dual flight instruction on an SE aeroplane including:

- (i) pre-flight operations, mass and balance determination, aeroplane inspection and servicing;
- (ii) aerodrome and traffic pattern operations, collision avoidance and precautions;
- (iii) control of the aeroplane by external visual references;
- (iv) normal take-offs and landings;
- (v) the basic UPRT exercises as specified in point (b) of AMC2 to Appendix 3; AMC1 to Appendix 5;
- (vi) simulated engine failure.

- (2) Phase 2:

Exercises up to the first solo cross-country flight comprise a total of at least 10 hours of dual flight instruction and at least 10 hours solo flight including:

- (i) maximum performance (short field and obstacle clearance) takeoffs and short-field landings;
 - (ii) flight by reference solely to instruments, including the completion of a 180 ° turn;
 - (iii) dual cross-country flying using external visual references, DR and radio navigation aids, diversion procedures;
 - (iv) aerodrome and traffic pattern operations at different aerodromes;
 - (v) crosswind take-offs and landings;
 - (vi) abnormal and emergency procedures and manoeuvres, including simulated aeroplane equipment malfunctions;
 - (vii) operations to, from and transiting controlled aerodromes, compliance with ATS procedures, R/T procedures and phraseology;
 - (viii) knowledge of meteorological briefing arrangements, evaluation of weather conditions for flight and use of AIS.
- (3) Phase 3:
- Exercises up to the VFR navigation progress test comprise a total of at least 5 hours of dual instruction and at least 40 hours as PIC.
- The dual instruction and testing up to the VFR navigation progress test should comprise:
- (i) repetition of exercises of phases 1 and 2;
 - (ii) VFR navigation progress test conducted by an FI not connected with the applicant's training;
 - (iii) dual night flight instruction.
- (4) Phase 4:
- Exercises up to the instrument rating skill test comprise:
- (i) at least 55 hours instrument flight, which may contain up to 25 hours of instrument ground time in an FNPT I or up to 40 hours in an FNPT II or FFS which should be conducted by an FI or an authorised SFI;
 - (ii) 20 hours instrument time flown as SPIC;
 - (iii) pre-flight procedures for IFR flights, including the use of the flight manual and appropriate ATS documents in the preparation of an IFR flight plan;
 - (iv) procedures and manoeuvres for IFR operation under normal, abnormal and emergency conditions covering at least:
 - (A) transition from visual to instrument flight on take-off;
 - (B) SIDs and arrivals;
 - (C) en-route IFR procedures;
 - (D) holding procedures;
 - (E) instrument approaches to specified minima;
 - (F) missed approach procedures;
 - (G) landings from instrument approaches, including circling;

- (v) in-flight manoeuvres and specific flight characteristics and the basic UPRT exercises as specified in Sections A, B, C and D of Table 2 in point (b) of AMC2 to Appendix 3; AMC1 to Appendix 5;
 - (vi) operation of an ME aeroplane in the exercises of (iv), including operation of the aeroplane solely by reference to instruments with one engine simulated inoperative, and engine shut-down and restart (the latter training should be conducted at a safe altitude unless carried out in an FSTD);
 - (vii) after completion of instrument training that is equivalent to the basic instrument flight module set out in AMC2 to Appendix 6, take-offs and landings as PIC at night.
- (5) Phase 5: Advanced UPRT in accordance with point FCL.745.A;
- (6) Phase 6:
- (i) instruction and testing in MCC comprising the relevant training requirements;
 - (ii) if a type rating for single-pilot aeroplanes in multi-pilot operations, or multi-pilot aeroplanes is not required on completion of this phase, the applicant should be issued with a certificate of course completion for MCC training.

B. ATP modular theoretical knowledge course: aeroplanes

- (a) The aim of this course is to train pilots who have not received the theoretical knowledge instruction during an integrated course to the level of theoretical knowledge required for the ATPL.
- (b) An approved course may contain in suitable proportions:
- (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;
 - (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
 - (6) exercises that use demonstration equipment or training devices;
 - (7) directed study including workbook exercises or assignments;
 - (8) aerodrome or aviation industry field trips;
 - (9) computer-based training and e-learning elements;
 - (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
 - (11) other training methods, media and tools approved by the competent authority.
- Approved distance-learning (correspondence) courses may also be offered as part of the course. The minimum amount of classroom instruction, as required by ORA.ATO.305, may include all of the above except item (b)(9).
- (c) The ATP modular course should be completed within 18 months. This period may be extended where additional training is provided by the ATO. The flight instruction and skill test need to be completed within the period of validity of the pass in the theoretical examinations.

C. CPL/IR integrated course: aeroplanes

- (a) The CPL/IR integrated course should last between 9 and 30 months. This period may be extended where additional flying training or ground instruction is provided by the ATO.

CREDITING

- (b) Credit for previous experience given to an applicant who already holds a PPL should be entered into the applicant's training record. In the case of a student pilot who does not hold a pilot licence and with the approval of the competent authority, an ATO may designate certain dual exercises to be flown in a helicopter or a TMG up to a maximum of 20 hours.

THEORETICAL KNOWLEDGE

- (c) The 500 hours of instruction, which also cover the Area 100 KSA, may include in suitable proportions:
- (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;
 - (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
 - (6) exercises that use demonstration equipment or training devices;
 - (7) directed study including workbook exercises or assignments;
 - (8) aerodrome or aviation industry field trips;
 - (9) computer-based training and e-learning elements;
 - (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
 - (11) other training methods, media and tools approved by the competent authority.

The 500 hours of instruction should be divided in such a way that in each subject the minimum hours are:

(1) Air law	25 hours
(2) Aircraft general knowledge	75 hours
(3) Flight performance and planning	80 hours
(4) Human performance and limitations	20 hours
(5) Meteorology	40 hours
(6) Navigation	55 hours
(7) Operational procedures	15 hours
(8) Principles of flight	35 hours
(9) Communications	15 hours

Other subdivisions of hours may be agreed upon between the competent authority and the ATO.

FLYING TRAINING

(d) The flying instruction is divided into four phases:

(1) Phase 1:

Exercises up to the first solo flight comprise a total of at least 10 hours dual flight instruction on an SE aeroplane, including:

- (i) pre-flight operations, mass and balance determination, aeroplane inspection and servicing;
- (ii) aerodrome and traffic pattern operations, collision avoidance and precautions;
- (iii) control of the aeroplane by external visual references;
- (iv) normal take-offs and landings;
- (v) the basic UPRT exercises as specified in point (b) of AMC2 to Appendix 3; AMC1 to Appendix 5;
- (vi) simulated engine failure.

(2) Phase 2:

Exercises up to the first solo cross-country flight comprise a total of at least 10 hours of dual flight instruction and at least 10 hours solo flight including:

- (i) maximum performance (short field and obstacle clearance) take-offs and short-field landings;
- (ii) flight by reference solely to instruments, including the completion of a 180° turn;
- (iii) dual cross-country flying using external visual references, DR and radio navigation aids, diversion procedures;
- (iv) aerodrome and traffic pattern operations at different aerodromes;
- (v) crosswind take-offs and landings;
- (vi) abnormal and emergency operations and manoeuvres, including simulated aeroplane equipment malfunctions;
- (vii) operations to, from and transiting controlled aerodromes, compliance with ATS procedures, R/T procedures and phraseology;
- (viii) knowledge of meteorological briefing arrangements, evaluation of weather conditions for flight and use of AIS.

(3) Phase 3:

Exercises up to the VFR navigation progress test comprise a total of at least 5 hours of instruction and at least 40 hours as PIC.

The dual instruction and testing up to the VFR navigation progress test and the skill test should contain the following:

- (i) repetition of exercises of phases 1 and 2;
- (ii) VFR navigation progress test conducted by an FI not connected with the applicant's training;
- (iii) dual night flight instruction.

(4) Phase 4:

Exercises up to the instrument rating skill test comprise:

- (i) at least 55 hours instrument time, which may contain up to 25 hours of instrument ground time in an FNPT I or up to 40 hours in an FNPT II or FFS which should be conducted by an FI or SFI;
- (ii) 20 hours instrument time flown as SPIC;
- (iii) pre-flight procedures for IFR flights, including the use of the flight manual and appropriate ATS documents in the preparation of an IFR flight plan;
- (iv) procedures and manoeuvres for IFR operation under normal, abnormal and emergency conditions covering at least:
 - (A) transition from visual to instrument flight on take-off;
 - (B) SIDs and arrivals;
 - (C) en-route IFR procedures;
 - (D) holding procedures;
 - (E) instrument approaches to specified minima;
 - (F) missed approach procedures;
 - (G) landings from instrument approaches, including circling.
- (v) in-flight manoeuvres and particular flight characteristics and the basic UPRT exercises as specified in Sections A, B, C and D of Table 2 in paragraph (b) of AMC2 to Appendix 3; AMC1 to Appendix 5;
- (vi) operation of either an SE or an ME aeroplane in the exercises of (iv), including in the case of an ME aeroplane operation of the aeroplane solely by reference to instruments with one engine simulated inoperative and engine shut-down and restart. The latter exercise is to be conducted at a safe altitude unless carried out in an FSTD;
- (vii) after completion of instrument training that is equivalent to the basic instrument flight module set out in AMC2 to Appendix 6, take-offs and landings as PIC at night.

D. CPL integrated course: aeroplanes

- (a) The CPL integrated course should last between 9 and 24 months. This period may be extended where additional flying training or ground instruction is provided by the ATO.

CREDITING

- (b) Credit for the hours flown should be entered into the applicant's training record. In the case of a student pilot who does not hold a pilot licence and with the approval of the competent authority, an ATO may designate certain dual exercises to be flown in a helicopter or a TMG up to a maximum of 20 hours.

THEORETICAL KNOWLEDGE

- (c) The 350 hours of instruction, which also cover the Area 100 KSA, may include in suitable proportions:
 - (1) classroom work;
 - (2) lessons;

- (3) tutorials;
- (4) demonstrations, including those supported by demonstration equipment;
- (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
- (6) exercises that use demonstration equipment or training devices;
- (7) directed study including workbook exercises or assignments;
- (8) aerodrome or aviation industry field trips;
- (9) computer-based training and e-learning elements;
- (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
- (11) other training methods, media and tools approved by the competent authority.

FLYING TRAINING

- (d) The flying instruction is divided into four phases:

- (1) Phase 1:

Exercises up to the first solo flight comprise a total of at least 10 hours dual flight instruction on an SE aeroplane, including:

- (i) pre-flight operations, mass and balance determination, aeroplane inspection and servicing;
- (ii) aerodrome and traffic pattern operations, collision avoidance and precautions;
- (iii) control of the aeroplane by external visual references;
- (iv) normal take-offs and landings;
- (v) the basic UPRT exercises as specified in point (b) of AMC2 to Appendix 3; AMC1 to Appendix 5;
- (vi) simulated engine failure.

- (2) Phase 2:

Exercises up to the first solo cross-country flight comprise a total of at least 10 hours of dual flight instruction and at least 10 hours solo flight including:

- (i) maximum performance (short field and obstacle clearance) take-offs and short-field landings;
- (ii) flight by reference solely to instruments, including the completion of a 180° turn;
- (iii) dual cross-country flying using external visual references, DR and radio navigation aids, diversion procedures;
- (iv) aerodrome and traffic pattern operations at different aerodromes;
- (v) crosswind take-offs and landings;
- (vi) abnormal and emergency procedures and manoeuvres, including simulated aeroplane equipment malfunctions;
- (vii) operations to, from and transiting controlled aerodromes, compliance with ATS procedures, R/T procedures and phraseology;

- (viii) knowledge of meteorological briefing arrangements, evaluation of weather conditions for flight and use of AIS.

(3) Phase 3:

Exercises up to the VFR navigation progress test comprise a total of at least 30 hours instruction and at least 58 hours as PIC, including:

- (i) at least 10 hours instrument time, which may contain 5 hours of instrument ground time in an FNPT or an FFS and should be conducted by an FI or SFI;
- (ii) repetition of exercises of phases 1 and 2, which should include at least 5 hours in an aeroplane certificated for the carriage of at least four persons and have a variable pitch propeller and retractable landing gear;
- (iii) night flight time including, g, after completion of instrument training that is equivalent to the basic instrument flight module set out in AMC2 to Appendix 6, take-offs and landings as PIC.

(4) Phase 4:

The dual instruction and testing up to the CPL(A) skill test contain the following:

- (i) up to 30 hours instruction which may be allocated to specialised aerial work training;
- (ii) repetition of exercises in Phase 3, as required;
- (iii) in-flight manoeuvres and particular flight characteristics including the basic UPRT exercises as specified in point (b) of AMC2 to Appendix 3; AMC1 to Appendix 5;
- (iv) ME training.

If required, operation of an ME aeroplane including operation of the aeroplane with one engine simulated inoperative, and engine shutdown and restart (the latter exercise at a safe altitude unless carried out in an FSTD).

E. CPL modular course: aeroplanes

- (a) The CPL modular course should be completed within 18 months. This period may be extended where additional training is provided by the ATO. The flight instruction and skill test need to be completed within the period of validity of the pass in the theoretical examinations.

CREDITING

Applicants with prior experience as PIC may be credited with an amount of hours to meet the requirement of 150 hours of flight time of Appendix 3, Part E, point (3)(a). The amount of credited hours should be decided by the ATO where the applicant takes the training course on the basis of a pre-entry flight test, but in any case, should have been completed only in one aircraft category other than aeroplane.

THEORETICAL KNOWLEDGE

- (b) The 250 hours of instruction, which also cover the Area 100 KSA, may include in suitable proportions:
 - (1) classroom work;
 - (2) lessons;

- (3) tutorials;
- (4) demonstrations, including those supported by demonstration equipment;
- (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
- (6) exercises that use demonstration equipment or training devices;
- (7) directed study including workbook exercises or assignments;
- (8) aerodromes or aviation industry field trips;
- (9) computer-based training and e-learning elements;
- (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
- (11) other training methods, media and tools approved by the competent authority.

Approved distance-learning (correspondence) courses may also be offered as part of the course. The minimum amount of classroom instruction, as required by ORA.ATO.305, may include all of the above except item (b)(9).

FLYING TRAINING

(c) The following flight time is suggested for the flying training:

- | (1) visual flight training: | suggested flight time |
|--|-----------------------|
| (i) Exercise 1: | |
| pre-flight operations: mass and balance determination, aeroplane inspection and servicing. | |
| (ii) Exercise 2: | |
| take-off, traffic pattern, | 0:45 hours |
| approach and landing, | |
| use of checklist, collision avoidance and checking procedures. | |
| (iii) Exercise 3: | |
| traffic patterns: simulated | 0:45 hours |
| engine failure during and after take-off. | |
| (iv) Exercise 4: | |
| maximum performance | 1:00 hours |
| (short field and obstacle clearance) | |
| take-offs and short-field landings. | |
| (v) Exercise 5: | |
| crosswind take-offs, | 1:00 hours |
| landings and go-arounds. | |
| (vi) Exercise 6: | |

Arresting divergence of the aeroplane 0:45 hours
from intended flight path, Preventing flight
at airspeeds inappropriate for the (intended flight)
conditions, High airspeed (including flight at
relatively high airspeed), Steep turns Nose-low
attitudes at various bank angles (including spiral dive).

(vii) Exercise 7:

Arresting divergence of the aeroplane 0:45 hours
from intended flight path, Preventing flight
at airspeeds inappropriate for the (intended flight)
conditions, slow flight, nose-high attitudes
at various bank angles, spin avoidance, stall events
in the following configurations:
— take-off configuration,
— clean configuration, and
— landing configuration.

(viii) Exercise 8:

cross-country flying 10:00 hours
using DR and radio navigation aids; flight
planning by the applicant; filing of ATC flight plan;
evaluation of weather briefing documentation,
NOTAM, etc.; R/T procedures and phraseology;
positioning by radio navigation aids;
operation to, from and transiting controlled
aerodromes, compliance with ATS procedures
for VFR flights, simulated radio communication
failure, weather deterioration, diversion procedures;
simulated engine failure during cruise flight;
selection of an emergency landing strip.

(2) instrument flight training:

- (i) This module's content is identical to that of the 10-hour basic instrument flight module as set out in [AMC2 to Appendix 6](#). This module is focused on the basics of flying by sole reference to instruments, including limited panel and basic UPRT exercises as specified in Sections A, B and C of Table 2 in point (b) of AMC2 Appendix 3; AMC1 Appendix 5.

- (ii) All exercises may be performed in an FNPT I or II or an FFS. If instrument flight training is in VMC, a suitable means of simulating IMC for the student should be used.
- (iii) A BITD may be used for the following exercises: (9), (10), (11) and (14).
- (iv) The use of the BITD is subject to the following:
 - (A) the training is complemented by exercises in an aeroplane;
 - (B) the record of the parameters of the flight is available;
 - (C) an FI(A) or IRI(A) conducts the instruction.
- (v) Exercise 9:
Basic instrument flying without 0:30 hours
external visual cues; horizontal flight;
power changes for acceleration or
deceleration, maintaining straight and
level flight; turns in level flight with 15°
and 25° bank, left and right; roll-out
onto predetermined headings.
- (vi) Exercise 10:
Repetition of exercise 9; 0:45 hours
additionally climbing and descending,
maintaining heading and speed, transition to
horizontal flight; climbing and descending turns.
- (vii) Exercise 11:
Instrument pattern: 0:45 hours
 - (1) start exercise, decelerate to approach speed, flaps into approach configuration;
 - (2) initiate standard turn (left or right);
 - (3) roll out on opposite heading, maintain new heading for 1 minute;
 - (4) standard turn, gear down, descend 500 ft/min;
 - (5) roll out on initial heading, maintain descent (500 ft/min) and new heading for 1 minute;
 - (6) transition to horizontal flight, 1.000 ft below initial flight level;
 - (7) initiate go-around;
 - (8) climb at best rate of climb speed.
- (viii) Exercise 12:
Repetition of exercise 9 and steep 0:45 hours
turns with 45° bank; recovery from unusual attitudes.

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- | | | |
|--------|---|------------|
| (ix) | Exercise 13:
Repetition of exercise 12 | 0:45 hours |
| (x) | Exercise 14:
Radio navigation using VOR,
NDB or, if available, VDF; interception of
predetermined QDM and QDR. | 0:45 hours |
| (xi) | Exercise 15:
Repetition of exercise 9 and
recovery from nose-high attitudes at various
bank angles, recovery from nose-low
attitudes at various bank angles | 0:45 hours |
| (xii) | Exercise 16:
Repetition of exercise 9, turns and
level change and recovery from nose-high
attitudes at various bank angles, recovery from
nose-low attitudes at various bank angles with
simulated failure of the artificial horizon
or directional gyro. | 0:45 hours |
| (xiii) | Exercise 17:
Basic UPRT exercises as specified
in point (b) of AMC2 to Appendix 3;
AMC1 to Appendix 5, excluding those
manoeuvres which have already been
completed during exercises 15 and 16 | 0:45 hours |
| (xiv) | Exercise 18:
Repetition of exercises (14), (16) and (17). | 3:00 hours |
- (3) ME training
- If required, operation of an ME aeroplane Exercises 1 through 17, including operation of the aeroplane with one engine simulated inoperative, and engine shutdown and restart (the latter exercise at a safe altitude unless carried out in an FSTD). Before commencing training, the applicant should have complied with the type and class ratings prerequisites as appropriate to the aeroplane used for the test.
- (4) Applicants who need to complete night training in accordance with point 10(b) of Section E of Appendix 3 to Part-FCL should perform take-offs and landings as PIC at night only after having completed the instrument flight training specified in point (2)(i) of 'FLYING TRAINING' of Section E of this AMC.

F. ATP/IR integrated course: helicopters

- (a) The ATP/IR integrated course should last between 12 and 36 months. This period may be extended where additional flying training or ground instruction is provided by the ATO.

CREDITING

- (b) Credit for the hours flown should be entered into the applicant's training record. In case of a student pilot who does not hold a pilot licence and with the approval of the competent authority, an ATO may designate certain dual exercises to be flown in an aeroplane or a TMG up to a maximum of 20 hours.

THEORETICAL KNOWLEDGE

- (c) The 750 hours of instruction, which also cover the Area 100 KSA, may include in suitable proportions:
- (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;
 - (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
 - (6) exercises that use demonstration equipment or training devices;
 - (7) directed study including workbook exercises or assignments;
 - (8) aerodrome or aviation industry field trips;
 - (9) computer-based training and e-learning elements;
 - (10) progress test, Area 100 KSA assessments and mental maths test(s); and
 - (11) other training methods, media and tools approved by the competent authority.

The 750 hours of instruction should be divided in such a way that in each subject the minimum hours are:

(1) Air law	35 hours
(2) Aircraft general knowledge	100 hours
(3) Flight performance and planning	120 hours
(4) Human performance and limitations	35 hours
(5) Meteorology	60 hours
(6) Navigation	90 hours
(7) Operational procedures	25 hours
(8) Principles of flight	55 hours
(9) Communications	20 hours

Other subdivisions of hours may be agreed upon between the competent authority and the ATO.

- (d) The flight instruction is divided into four phases:

(1) phase 1:

Flight exercises up to the first solo flight comprise a total of not less than 12 hours dual flight instruction on a helicopter, including:

- (1) pre-flight operations, mass and balance determination, helicopter inspection and servicing;
- (2) aerodrome and traffic pattern operations, collision avoidance and procedures;
- (3) control of the helicopter by external visual reference;
- (4) take-offs, landings, hovering, look-out turns and normal transitions from and to the hover;
- (5) emergency procedures, basic auto-rotations, simulated engine failure, ground resonance recovery if relevant to type.

(2) phase 2:

Flight exercises until general handling and day VFR navigation progress check, and basic instrument flying progress check. This phase comprises a total flight time of not less than 128 hours including 73 hours of dual flight instruction flight time and including at least 5 hours VFR conversion training on an ME helicopter, 15 hours of solo flight and 40 hours flown as student PIC. The instruction and testing contain the following:

- (i) sideways and backwards flight, turns on the spot;
- (ii) recognition of and recovery from the incipient vortex ring state;
- (iii) advanced/touchdown auto-rotations, simulated engine-off landings, practice forced landings. Simulated equipment malfunctions and emergency procedures relating to malfunctions of engines, controls, electrical and hydraulic circuits;
- (iv) steep turns;
- (v) transitions, quick stops, out of wind manoeuvres, sloping ground landings and take-offs;
- (vi) limited power and confined area operations, including low level operations to and from unprepared sites;
- (vii) flight by sole reference to basic flight instruments, including completion of a 180 ° turn and recovery from unusual attitudes to simulate inadvertent entry into cloud;
- (viii) cross-country flying by external visual reference, DR and radio navigation aids, diversion procedures;
- (ix) aerodrome and traffic pattern operations at different aerodromes;
- (x) operations to, from and transiting controlled aerodromes; compliance with ATS procedures, R/T procedures and phraseology;
- (xi) application of meteorological briefing arrangements, evaluation of weather conditions for flight and use of AIS;
- (xii) night flight, including take-offs and landings as PIC;
- (xiii) general handling, day VFR navigation and basic instrument flying progress checks in accordance with [Appendix 4](#) to Part-FCL, conducted by an FI not connected with the applicant's training.

(3) phase 3:

Flight exercises up to IR skill test. This part comprises a total of 40 hours dual instrument flight time, including 10 hours of an ME IFR certificated helicopter.

The instruction and testing should contain the following:

- (i) pre-flight procedures for IFR flights, including the use of the flight manual and appropriate ATS documents in the preparation of an IFR flight plan;
- (ii) procedures and manoeuvres for IFR operation under normal, abnormal and emergency conditions covering at least:
 - (A) transition from visual to instrument flight on take-off;
 - (B) SIDs and arrivals;
 - (C) en-route IFR procedures;
 - (D) holding procedures;
 - (E) instrument approaches to specified minima;
 - (F) missed approach procedure;
 - (G) landings from instrument approaches;
 - (H) in-flight manoeuvres and particular flight characteristics;
 - (I) instrument exercises with one engine simulated inoperative.

(4) phase 4:

Instruction in MCC should comprise the relevant training set out in [FCL.735.H](#) and [AMC1 FCL.735.A, FCL.735.H and FCL.735.As](#).

If a type rating for MP helicopter is not required on completion of this part, the applicant should be provided with a certificate of course completion for MCC training.

G. ATP integrated course: helicopters

- (a) The ATP integrated course should last between 12 and 36 months. This period may be extended where additional flying training or ground instruction is provided by the ATO.

CREDITING

- (b) Credit for the hours flown should be entered into the applicant's training record. In case of a student pilot who does not hold a pilot licence and with the approval of the competent authority, an ATO may designate certain dual exercises to be flown in an aeroplane or a TMG up to a maximum of 20 hours.

THEORETICAL KNOWLEDGE

- (c) The 650 hours of instruction, which also cover the area 100 KSA, may include in suitable proportions:
- (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;

- (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
- (6) exercises that use demonstration equipment or training devices;
- (7) directed study including workbook exercises or assignments;
- (8) aerodrome or aviation industry field trips;
- (9) computer-based training and e-learning elements;
- (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
- (11) other training methods, media and tools approved by the competent authority.

The 650 hours of instruction should be divided in such a way that in each subject the minimum hours are:

- | | |
|---------------------------------------|----------|
| (1) Air law | 30 hours |
| (2) Aircraft general knowledge | 90 hours |
| (3) Flight performance and planning | 90 hours |
| (4) Human performance and limitations | 30 hours |
| (5) Meteorology | 50 hours |
| (6) Navigation | 70 hours |
| (7) Operational procedures | 20 hours |
| (8) Principles of flight | 45 hours |
| (9) Communications | 15 hours |

Other subdivisions of hours may be agreed upon between the competent authority and the ATO.

- (d) The flight instruction is divided into three phases:

- (1) phase 1:

Flight exercises up to the first solo flight comprise a total of not less than 12 hours dual flight instruction on a helicopter, including:

- (i) pre-flight operations, mass and balance determination, helicopter inspection and servicing;
- (ii) aerodrome and traffic pattern operations, collision avoidance and procedures;
- (iii) control of the helicopter by external visual reference;
- (iv) take-offs, landings, hovering, look-out turns and normal transitions from and to the hover;
- (v) emergency procedures, basic auto-rotations, simulated engine failure, ground resonance recovery if relevant to type.

- (2) phase 2:

Flight exercises until general handling and day VFR navigation progress and basic instrument flying progress check conducted by an FI not connected with the applicant's training. This phase comprises a total flight time of not less than 128 hours, including 73 hours of dual instruction flight time and including at least 5 hours VFR conversion training

on an ME helicopter, 15 hours of solo flight and 40 hours flown as student PIC. The instruction and testing contain the following:

- (i) sideways and backwards flight, turns on the spot;
- (ii) recognition of and recovery from the incipient vortex ring state;
- (iii) touchdown or advanced auto-rotations, simulated engine-off landings, practice forced landings. Simulated equipment malfunctions and emergency procedures relating to malfunctions of engines, controls, electrical and hydraulic circuits;
- (iv) steep turns;
- (v) transitions, quick stops, out of wind manoeuvres, sloping ground landings and take-offs;
- (vi) limited power and confined area operations, including low level operations to and from unprepared sites;
- (vii) 10 hours flight by sole reference to basic flight instruments, including completion of a 180 ° turn and recovery from unusual attitudes to simulate inadvertent entry into cloud;
- (viii) cross-country flying by external visual reference, DR and radio navigation aids, diversion procedures;
- (ix) aerodrome and traffic pattern operations at different aerodromes;
- (x) operations to, from and transiting controlled aerodromes, compliance with ATS procedures, R/T procedures and phraseology;
- (xi) application of meteorological briefing arrangements, evaluation of weather conditions for flight and use of AIS;
- (xii) night flight, including take-offs and landings as PIC;
- (xiii) general handling, day VFR navigation and basic instrument flying progress checks in accordance with [Appendix 4](#) to Part-FCL, conducted by an FI not connected with the applicant's training.

(3) phase 3:

Instruction in MCC comprises the relevant training set out in [FCL.735.H](#) and [AMC1 FCL.735.A, FCL.735.H and FCL.735.As](#).

If a type rating for MP helicopter is not required on completion of this part, the applicant should be provided with a certificate of course completion for MCC training.

H. ATP modular theoretical knowledge course: helicopters

- (a) The aim of this course is to train pilots who have not received the theoretical knowledge instruction during an integrated course to the level of theoretical knowledge required for the ATPL.
- (b) An approved course, which also covers the area 100 KSA, may contain in suitable proportions:
 - (1) classroom work;
 - (2) lessons;
 - (3) tutorials;

- (4) demonstrations, including those supported by demonstration equipment;
- (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
- (6) exercises that use demonstration equipment or training devices;
- (7) directed study including workbook exercises or assignments;
- (8) aerodrome or aviation industry field trips;
- (9) computer-based training and e-learning elements;
- (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
- (11) other training methods, media and tools approved by the competent authority.

Approved distance-learning (correspondence) courses may also be offered as part of the course. The minimum amount of classroom instruction, as required by ORA.ATO.305, may include all of the above except item (b)(9).

- (c) The ATP modular course should be completed within 18 months. This period may be extended where additional training is provided by the ATO. The flight instruction and skill test need to be completed within the period of validity of the pass in the theoretical examinations.

I. CPL/IR integrated course: helicopters

- (a) The CPL/IR integrated course should last between 9 and 30 months. This period may be extended where additional flying training or ground instruction is provided by the ATO.

CREDITING

- (b) Credit for the hours flown should be entered into the applicant's training record. In case of a student pilot who does not hold a pilot licence and with the approval of the competent authority, an ATO may designate certain dual exercises to be flown in an aeroplane or a TMG up to a maximum of 20 hours.

THEORETICAL KNOWLEDGE

- (c) The 500 hours of instruction, which also cover the Area 100 KSA, may include in suitable proportions:
 - (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;
 - (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
 - (6) exercises that use demonstration equipment or training devices;
 - (7) directed study including workbook exercises or assignments;
 - (8) aerodrome or aviation industry field trips;
 - (9) computer-based training and e-learning elements;
 - (10) progress tests, Area 100 KSA assessments and mental maths test(s); and

- (11) other training methods, media and tools approved by the competent authority.

The 500 hours of instruction should be divided in such a way that in each subject the minimum hours are:

(1)	Air law	25 hours
(2)	Aircraft general knowledge	75 hours
(3)	Flight performance and planning	80 hours
(4)	Human performance and limitations	20 hours
(5)	Meteorology	40 hours
(6)	Navigation	55 hours
(7)	Operational procedures	15 hours
(8)	Principles of flight	35 hours
(9)	Communications	15 hours

Other subdivision of hours may be agreed upon between the competent authority and the ATO.

FLYING TRAINING

- (d) The flight instruction is divided into three phases:

- (1) phase 1:

Flight exercises up to the first solo flight. This part comprises a total of at least 12 hours dual flight instruction on a helicopter including:

- (i) pre-flight operations: mass and balance determination, helicopter inspection and servicing;
- (ii) aerodrome and traffic pattern operations, collision avoidance and procedures;
- (iii) control of the helicopter by external visual reference;
- (iv) take-offs, landings, hovering, look-out turns and normal transitions from and to the hover;
- (v) emergency procedures, basic auto-rotation, simulated engine failure, ground resonance recovery if relevant to type.

- (2) phase 2:

Flight exercises until general handling and day VFR navigation progress check conducted by an FI not connected with the applicant's training, and basic instrument progress check. This part comprises a total flight time of not less than 128 hours, including 73 hours of dual instruction flight time and including at least 5 hours VFR conversion training on an ME helicopter, 15 hours of solo flight and 40 hours flown as SPIC. The instruction and testing contain the following:

- (i) sideways and backwards flight, turns on the spot;
- (ii) recognition of and recovery from the incipient vortex ring state;
- (iii) touchdown or advanced auto-rotation and simulated engine-off landings, practice forced landings. Simulated equipment malfunctions and emergency procedures relating to malfunctions of engines, controls, electrical and hydraulic circuits;
- (iv) steep turns;

- (v) transitions, quick stops, out of wind manoeuvres, sloping ground landings and take-offs;
- (vi) limited power and confined area operations, including selection of and low level operations to and from unprepared sites;
- (vii) flight by sole reference to basic flight instruments, including completion of 180 degree turn and recovery from unusual attitudes to simulate inadvertent entry into cloud;
- (viii) cross-country flying by external visual reference, DR and radio navigation aids and diversion procedures;
- (ix) aerodrome and traffic pattern operations at different aerodromes;
- (x) operations to, from and transiting controlled aerodromes, compliance with ATS procedures, R/T procedures and phraseology;
- (xi) application of meteorological briefing arrangements, evaluation of weather conditions for flight and use of AIS;
- (xii) night flight, including take-offs and landings as PIC;
- (xiii) general handling, day VFR navigation and basic instrument flying progress checks in accordance with [Appendix 4](#) to Part-FCL, conducted by an FI not connected with the applicant's training.

(3) phase 3:

Flight exercises up to IR skill test. This part comprises a total of 40 hours dual instrument flight time, including 10 hours of an ME IFR certificated helicopter.

The instruction and testing should contain the following:

- (i) pre-flight procedures for IFR flights, including the use of the flight manual and appropriate ATS documents in the preparation of an IFR flight plan;
- (ii) procedures and manoeuvres for IFR operation under normal, abnormal and emergency conditions covering at least:
 - (A) transition from visual to instrument flight on take-off;
 - (B) SIDs and arrivals;
 - (C) en-route IFR procedures;
 - (D) holding procedures;
 - (E) instrument approaches to specified minima;
 - (F) missed approach procedure;
 - (G) landings from instrument approaches;
 - (H) in-flight manoeuvres and particular flight characteristics;
 - (I) instrument exercises with one engine simulated inoperative.

J. CPL integrated course: helicopters

- (a) The CPL integrated course should last between 9 and 24 months. This period may be extended where additional flying training or ground instruction is provided by the ATO.

CREDITING

- (b) Credit for the hours flown should be entered into the applicant's training record. In case of a student pilot who does not hold a pilot licence and with the approval of the competent authority, an ATO may designate certain dual exercises to be flown in an aeroplane or a TMG up to a maximum of 20 hours.

THEORETICAL KNOWLEDGE

- (c) The 350 hours of instruction, which also cover the Area 100 KSA, may include in suitable proportions:
- (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;
 - (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
 - (6) exercises that use demonstration equipment or training devices;
 - (7) directed study including workbook exercises or assignments;
 - (8) aerodrome or aviation industry field trips;
 - (9) computer-based training and e-learning elements;
 - (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
 - (11) other training methods, media and tools approved by the competent authority.

The 350 hours of instruction should be divided in such a way that in each subject the minimum hours are:

(1)	Air law	15 hours
(2)	Aircraft general knowledge	40 hours
(3)	Flight performance and planning	35 hours
(4)	Human performance and limitations	10 hours
(5)	Meteorology	30 hours
(6)	Navigation	35 hours
(7)	Operational procedures	10 hours
(8)	Principles of flight	30 hours
(9)	Communications	10 hours

Other subdivisions of hours may be agreed upon between the competent authority and the ATO.

FLYING TRAINING

- (d) The flight instruction is divided into two phases:
- (1) phase 1:

Flight exercises up to the first solo flight. This part comprises a total of not less than 12 hours dual flight instruction on a helicopter, including:

- (1) pre-flight operations: mass and balance determination, helicopter inspection and servicing;
- (2) aerodrome and traffic pattern operations, collision avoidance and procedures;
- (3) control of the helicopter by external visual reference;
- (4) take-offs, landings, hovering, look-out turns and normal transitions from and to the hover;
- (5) emergency procedures, basic auto-rotations, simulated engine failure, ground resonance recovery if relevant to type.

(2) phase 2:

Flight exercises until general handling and day VFR navigation progress check conducted by an FI not connected with the applicant's training, and basic instrument progress check. This part comprises a total flight time of not less than 123 hours, including 73 hours of dual instruction flight time, 15 hours of solo flight and 35 hours flown as SPIC. The instruction and testing contain the following:

- (i) sideways and backwards flight, turns on the spot;
- (ii) recognition of and recovery from the incipient vortex ring state;
- (iii) touchdown or advanced auto-rotations and simulated engine-off landings, practice forced landings. Simulated equipment malfunctions and emergency procedures relating to malfunctions of engines, controls, electrical and hydraulic circuits;
- (iv) steep turns;
- (v) transitions, quick stops, out of wind manoeuvres, sloping ground landings and take-offs;
- (vi) limited power and confined area operations, including selection of and low level operations to and from unprepared sites;
- (vii) flight by sole reference to basic flight instruments, including completion of a 180° turn and recovery from unusual attitudes to simulate inadvertent entry into cloud;
- (viii) cross-country flying by external visual reference, DR and radio navigation aids, diversion procedures;
- (ix) aerodrome and traffic pattern operations at different aerodromes;
- (x) operations to, from and transiting controlled aerodromes, compliance with ATS procedures, R/T procedures and phraseology;
- (xi) application of meteorological briefing arrangements, evaluation of weather conditions for flight and use of AIS;
- (xii) night flight, including take-offs and landings as PIC;
- (xiii) general handling, day VFR navigation and basic instrument flying progress checks in accordance with Appendix 4 to Part-FCL, conducted by an FI not connected with the applicant's training.

K. CPL modular course: helicopters

- (a) The CPL modular course should be completed within 18 months. This period may be extended where additional training is provided by the ATO. The flight instruction and skill test need to be completed within the period of validity of the pass in the theoretical examinations.

CREDITING

Applicants with prior experience as PIC may be credited with an amount of hours to meet the requirement of 150 hours of flight time of Appendix 3, Part K, point (3)(a).

The amount of credited hours should be decided by the ATO where the applicant takes the training course on the basis of a pre-entry flight test, but in any case, should have been completed only in one aircraft category other than helicopter, and not be a combination of hours in more than two different aircraft categories.

THEORETICAL KNOWLEDGE

- (b) The 250 hours of instruction, which also covers the Area 100 KSA may include in suitable proportions:
- (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;
 - (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
 - (6) exercises that use demonstration equipment or training devices;
 - (7) directed study including workbook exercises or assignments;
 - (8) aerodrome or aviation industry field trips;
 - (9) computer-based training and e-learning elements;
 - (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
 - (11) other training methods, media and tools approved by the competent authority.

Approved distance-learning (correspondence) courses may also be offered as part of the course. The minimum amount of classroom instruction, as required by ORA.ATO.305, may include all of the above except item (b)(9).

FLYING TRAINING

- (c) The flying instruction comprises the following items. The flight time allocated to each exercise is at the discretion of the FI, provided that at least 5 hours flight time is allocated to cross-country flying.

VISUAL INSTRUCTION

- (d) Within the total of dual flight instruction time, the applicant may have completed during the visual phase up to 5 hours in a helicopter FFS or FTD 2, 3 or FNPT II, III.
- (1) pre-flight operations: mass and balance calculations, helicopter inspection and servicing;
 - (2) level flight speed changes, climbing, descending, turns, basic autorotations, use of checklist, collision avoidance and checking procedures;

- (3) take-offs and landings, traffic pattern, approach, simulated engine failures in the traffic pattern. Sideways and backwards flight and spot turns in the hover;
- (4) recognition of and recovery from the incipient vortex ring state;
- (5) advanced auto-rotations covering the speed range from low speed to maximum range and manoeuvre in auto-rotations (180°, 360° and 'S' turns) and simulated engine-off landings;
- (6) selection of emergency landing areas, auto-rotations following simulated emergencies to given areas and steep turns at 30° and 45° bank;
- (7) manoeuvres at low level and quick-stops;
- (8) landings, take-offs and transitions to and from the hover when heading out of wind;
- (9) landings and take-offs from sloping or uneven ground;
- (10) landings and take-offs with limited power;
- (11) low level operations into and out of confined landing sites;
- (12) cross-country flying using dead reckoning and radio navigation aids, flight planning by the applicant, filing of ATC flight plan, evaluation of weather briefing documentation, NOTAM, etc., R/T procedures and phraseology, positioning by radio navigation aids; operation to, from and transiting controlled aerodromes, compliance with ATS procedures for VFR flights, simulated radio communication failure, weather deterioration, diversion procedures; location of an off airfield landing site and simulated approach.

BASIC INSTRUMENT INSTRUCTION

- (e) A maximum of 5 hours of the following exercises may be performed in an FFS or FTD or FNPT. Flight training should be carried out in VMC using a suitable means of simulating IMC for the student.
- (1) Exercise 1:
Instrument flying without external visual cues. Level flight performing speed changes, maintaining flight altitude (level, heading) turns in level flight at rate 1 and 30° bank, left and right; roll-out on predetermined headings;
 - (2) Exercise 2:
repetition of exercise 1; additionally climbing and descending, maintaining heading and speed, transition to horizontal flight; climbing and descending turns;
 - (3) Exercise 3:
repetition of exercise 1; and recovery from unusual attitudes;
 - (4) Exercise 4:
radio navigation;
 - (5) Exercise 5:
repetition of exercise 1; and turns using standby magnetic compass and standby artificial horizon (if fitted).

AMC2 to Appendix 3; AMC1 to Appendix 5

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BASIC UPRT FOR AEROPLANE ATP INTEGRATED, CPL/IR INTEGRATED, CPL INTEGRATED AND CPL MODULAR COURSES AS WELL AS MPL COURSE PHASES 1 TO 3

(a) BASIC UPRT ELEMENTS AND COMPONENTS

In order for student pilots to develop the competencies to prevent and recover from aeroplane upsets, the basic UPRT elements and respective components in the following Table 1 should be integrated into the flying training modules and phases, such that all the elements are covered.

Table 1: Basic UPRT elements and components		Pre-flight briefing	Flying training
A.	Aerodynamics		
1.	General aerodynamic characteristics	•	•
2.	Aeroplane certification and limitations	•	•
4.	Aerodynamics (high and low altitude)	•	
5.	Aeroplane performance (high and low altitude)	•	
6.	AoA and stall awareness	•	•
7.	Aeroplane stability	•	•
8.	Control surface fundamentals	•	•
9.	Use of trim	•	•
10.	Icing and contamination effects	•	•
11.	Propeller slipstream (as applicable)	•	•
B.	Causes of and contributing factors to upsets		
1.	Environmental	•	
2.	Pilot-induced	•	
3.	Mechanical (aeroplane systems)	•	
C.	Safety review of accidents and incidents relating to aeroplane upsets		
1.	Safety review of accidents and incidents relating to aeroplane upsets	•	
D.	G-load awareness and management		
1.	Positive/negative/increasing/decreasing G-loads	•	•
2.	Lateral G awareness (sideslip)	•	•
3.	G-load management	•	•
E.	Energy management		
1.	Kinetic energy vs potential energy vs chemical energy (power)	•	•
F.	Flight path management		
1.	Relationship between pitch, power and performance	•	•
2.	Performance and effects of differing power plants	•	•
3.	Manual and automation inputs for guidance and control (if applicable)	•	•
4.	Class-specific characteristics of flight path management	•	•
5.	Management of go-arounds from various stages during the approach	•	•
6.	Automation management (if applicable)	•	•
7.	Proper use of rudder	•	•
G.	Recognition		

Table 1: Basic UPRT elements and components		Pre-flight briefing	Flying training
1.	Class-specific examples of physiological, visual and instrument clues during developing and developed upset	•	•
2.	Pitch/power/roll/yaw	•	•
3.	Effective scanning (effective monitoring)	•	•
4.	Stall protection systems and cues	•	•
5.	Criteria for identifying stalls and upsets	•	•
H.	System malfunction (including immediate handling and subsequent operational considerations, as applicable)		
1.	Flight control defects	•	•
2.	Engine failure (partial or full)	•	•
3.	Instrument failures	•	•
4.	Loss of reliable airspeed (training elements as per point (Ib) of AMC2 ORA.ATO.125 ¹).	•	•
5.	Automation failures	•	•
6.	Stall protection system failures, including icing alerting systems	•	•

(b) MANOEUVRE-BASED UPRT EXERCISES

The following Table 2 contains manoeuvre-based basic UPRT exercises.

Table 2: Manoeuvre-based basic UPRT exercises		Pre-flight briefing	Flying training
A.	Timely and appropriate intervention		
1.	Arresting divergence of the aeroplane from intended flight path	•	•
2.	Preventing flight at airspeeds inappropriate for the (intended flight) condition	•	•
3.	Avoiding spins	•	•
B.	Flight path management		
1.	Steep turns	•	•
2.	Slow flight (including flight at critically low airspeed)	•	•
3.	High airspeed (including flight at relatively high airspeed)	•	•
C.	Application of OEM recommendations (if applicable) during developing upsets		
1.	Nose-high attitudes at various bank angles	•	•
2.	Nose-low attitudes at various bank angles (including spiral dive)	•	•
D.	Stall events in the following configurations		
1.	Take-off configuration	•	•
2.	Clean configuration	•	•
3.	Landing configuration	•	•

(c) INTEGRATION OF TEM, PILOT CORE COMPETENCIES, AND HUMAN FACTORS

¹ Please refer to ED Decision 2012/007/R.

Threat and Error Management (TEM), pilot competencies and human factors, as shown in the following Table 3 below, should be integrated into the flying training modules and phases as appropriate.

Table 3: Core elements and components of TEM, pilot competencies and human factors		Pre-flight briefing	Flying training
A.	TEM		
1.	TEM framework	•	•
2.	Recognition of threats and errors	•	•
3.	Management of threats and errors	•	•
4.	Countermeasures against threats and errors to prevent undesired aircraft states, including early intervention and, when necessary to prevent upsets, timely application of countermeasures to manage undesired aircraft states	•	•
B.	Pilot Competencies, including CRM		
1.	All elements listed in Table 1 of GM2 FCL.735.A	•	•
C.	Human factors		
1.	Instrument interpretation, active monitoring, checking	•	•
2.	Distraction, inattention, fixation, fatigue	•	•
3.	Human information processing, cognitive effects	•	•
4.	Perceptual illusions (visual or physiological) and spatial disorientation, effects of G-loads	•	•
5.	Stress, startle and surprise effect	•	•
6.	Intuitive and counter-intuitive behaviour	•	•

GM1 to Appendix 3; Appendix 5

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BASIC UPRT EXERCISES

(a) GENERAL

The training objective of the basic UPRT exercises is for the student to achieve competence in applying prevention and recovery techniques. In order to meet the training objectives, some UPRT exercises will involve operation at altitudes, speeds and g-loadings that are not required for other parts of the training course. When designing training courses, ATOs should ensure that the aircraft used for these exercises will allow the training objectives to be achieved while maintaining a margin of safety to aircraft limitations in accordance with the training envelope, as determined by the ATO (see GM1 ORA.ATO.125 point (f)).

(b) UPRT WITH REFERENCE TO INSTRUMENTS

Basic UPRT exercises completed by reference to instruments (i.e. in simulated instrument meteorological conditions (IMC)) should involve only moderate excursions from the speeds and attitudes used in normal instrument flight. Exercises conducted in IMC should not be planned to involve 'unusual attitudes'.

(c) INSTRUCTORS DELIVERING BASIC UPRT

Instructors conducting basic UPRT training during the CPL or ATP course do not require any additional qualifications. It is the responsibility of the ATO to ensure that instructors are competent to deliver effective training on all parts of the course and also that they are

competent to recover the aircraft in the event that a student erroneously conducts any UPRT exercise.

(d) **APPLICATION OF OEM RECOMMENDATIONS DURING DEVELOPING UPSETS**

Stall recovery training exercises as well as nose-high and nose-low prevention training exercises use the recovery strategies recommended by the OEMs contained in Tables 1, 2 and 3 below.

Note: As OEM procedures always take precedence over the general strategies as recommended by the OEMs, ATOs should consult the OEM on whether any approved specific procedures are available prior to using the templates.

Refer to revision 3 of the Airplane Upset Prevention and Recovery Training Aid (AUPRTA) for a detailed explanation and rationale of nose-high and nose-low recovery strategies as recommended by the OEMs.

Table 1: Stall event recovery template

Pilot Flying (PF)	
Immediately do the following at first indication of a stall (aerodynamic buffeting, reduced roll stability and aileron effectiveness, visual or aural cues and warnings, reduced elevator (pitch) authority, inability to maintain altitude or arrest rate of descent, stick shaker activation (if installed)) during any flight phases except at lift-off.	
1.	AUTOPILOT — DISCONNECT (IF APPLICABLE) (A large out-of-trim condition could be encountered when the autopilot is disconnected)
2.	AUTOTHROTTLE — OFF (IF APPLICABLE)
3.	(a) NOSE-DOWN PITCH CONTROL apply until stall warning is eliminated (b) NOSE-DOWN PITCH TRIM (as needed) (Reduce the AoA whilst accepting the resulting altitude loss.)
4.	BANK — WINGS LEVEL
5.	POWER — ADJUST (as needed) (Thrust reduction for aeroplanes with underwing-mounted engines may be needed)
6.	SPEEDBRAKES/SPOILERS — RETRACT
7.	When airspeed is sufficiently increasing — RECOVER to level flight (Avoid the secondary

Table 2: Nose-high recovery strategy template

Recognise and confirm the developing situation by announcing ‘nose high’	
Pilot Flying (PF)	
1.	AUTOPILOT — DISCONNECT (if applicable) (A large out-of-trim condition could be encountered when the autopilot is disconnected)
2.	AUTOTHROTTLE — OFF (if applicable)
3.	APPLY as much nose-down control input as required to obtain a nose-down pitch rate
4.	POWER — ADJUST (if required)
5.	ROLL — ADJUST (if required) (Avoid exceeding 60-degree bank)
6.	When airspeed is sufficiently increasing — RECOVER to level flight (Avoid the secondary stall due to premature recovery or excessive G-loading)
NOTE: (1) Recovery to level flight may require use of pitch trim.	

- (2) **WARNING:** Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.

Table 3: Nose-low recovery strategy template

Recognise and confirm the developing situation by announcing **‘nose low’**
 (If the autopilot or autothrottle is responding correctly, it may not be appropriate to decrease the level of automation while assessing if the divergence is being stopped)

Pilot Flying (PF)

- | | |
|----|--|
| 1. | AUTOPILOT — DISCONNECT (if applicable)
(A large out-of-trim condition could be encountered when the autopilot is disconnected) |
| 2. | AUTOTHROTTLE — OFF (if applicable) |
| 3. | RECOVERY from stall (if required) |
| 4. | ROLL in the shortest direction to wings level
(It may be necessary to reduce the G-loading by applying forward control pressure to improve roll effectiveness) |
| 5. | POWER and DRAG — ADJUST (if required) |
| 6. | RECOVER to level flight
(Avoid the secondary stall due to premature recovery or excessive G-loading) |

NOTE:

- (1) Recovery to level flight may require use of pitch trim.
 (2) **WARNING:** Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.

ADDITIONAL GUIDANCE

- (e) Specific guidance on UPRT is available in the latest revision of ICAO Doc 10011 ‘Manual on Aeroplane Upset Prevention and Recovery Training’.

GM1 to Appendix 3; Appendix 6; FCL.735.H
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OVERVIEW OF FSTD TRAINING CREDITS FOR DUAL INSTRUCTION IN HELICOPTER FLYING TRAINING COURSES

		ATPL(H)/IR integrated			FSTD credits
		Solo	SPIC	Total	
Visual, including ME T/R training	Dual 75 hrs	Solo 15 hrs	40 hrs	130 hrs	FFS; FTD; FNPT 30 hrs FFS C/D level or 25 hrs FTD 2, 3 or 20 hrs FNPT II/III
Basic instrument	10 hrs	-	-	10 hrs	20 hrs FFS or FTD 2, 3 or FNPT II/III or 10 hrs in at least an FNPT I
Instrument rating training	40 hrs	-		40 hrs	
MCC	15 hrs	-	-	15 hrs	15 hrs FFS or FTD 2, 3 (MCC) or FNPT II/III (MCC)
Total	140 hrs	55 hrs		195 hrs	Note 2

		ATPL(H)/VFR integrated			FSTD credits
		Solo	SPIC	Total	
Visual including ME T/R training	Dual 75 hrs	Solo 15 hrs	40 hrs	130 hrs	FFS; FTD; FNPT 30 hrs FFS C/D level or 25 hrs FTD 2, 3 or 20 hrs FNPT II/III

Basic instrument	10 hrs	-	-	10 hrs	5 hrs in at least an FNPT I
MCC / VFR	10 hrs	-	-	10 hrs	10 hrs FFS or FTD 2, 3 (MCC) or FNPT II/III (MCC)
Total	95 hrs	55 hrs		150 hrs	Note 2

		CPL(H)/IR integrated			
	Dual	Solo	SPIC	Total	FFS; FTD; FNPT
Visual including ME T/R training	75 hrs	15 hrs	40hrs	130 hrs	30 hrs FFS C/D level or 25 hrs FTD 2, 3 or 20 hrs FNPT II/III
Basic instrument	10 hrs	-	-	10 hrs	20 hrs FFS or FTD 2, 3 or FNPT II/III
Instrument rating training	40 hrs	-		40 hrs	or 10 hrs in at least an FNPT I
Total	125 hrs	55 hrs		180 hrs	Note 2

		CPL(H) Integrated			
	Dual	Solo	SPIC	Total	FFS; FTD; FNPT
Visual	75 hrs	15 hrs	35 hrs	125 hrs	30 hrs FFS C/D level or 25 hrs FTD 2, 3 or 20 hrs FNPT II/III
Basic instrument	10 hrs	-	-	10 hrs	5 hrs in at least an FNPT I
Total	85 hrs	50 hrs		135 hrs	Note 2

		CPL(H) modular			
	Dual	Solo	SPIC	Total	FFS; FTD; FNPT
Visual	20 hrs	-	-	20 hrs	5 hrs FFS or FTD 2, 3 or FNPT II/III
Basic instrument	10 hrs	-	-	10 hrs	5 hrs in at least an FNPT I
Total	30 hrs	-	-	30 hrs	Note 2

		IR(H) modular			
	Dual	Solo	SPIC	Total	FFS; FTD; FNPT
SE	50 hrs	-	-	50 hrs	35 hrs FFS or FTD 2, 3 or FNPT II/III or 20 hrs FNPT I (H) or (A)
ME	55 hrs	-	-	55 hrs	40 hrs FFS; FTD 2, 3 FNPT II/III or 20 hrs FNPT I (H) or (A)

		MCC(H)			
	Dual	Solo	SPIC	Total	FFS; FTD; FNPT
MCC / IR	20 hrs	-	-	20 hrs	20 hrs FFS or FTD 2, 3 (MCC) or FNPT II/III (MCC)
MCC / VFR	15 hrs	-	-	15 hrs	15 hrs FFS or FTD 2, 3 (MCC) or FNPT II/III (MCC)
MCC / IR for MCC/VFR holders	5 hrs	-	-	5 hrs	5 hrs FFS or FTD 2, 3 (MCC) or FNPT II/III (MCC)

Note 1: In this matrix, FSTD credits refer to helicopter FSTDs, if not mentioned otherwise.

Note 2: Total credits for the FSTDs used in the course are not provided in the tables as the FSTDs may be used in various combinations. The FSTD credits provided in the tables for the separate phases of the course are the maximum FSTD credits available for each phase.

GM1 to Appendix 3 Example of a grading system for practical flight training during ATP, CPL and MPL courses grading system

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An ATPL/CPL/MPL grading system may be developed by using the grading system in [GM3 FCL.735.A](#).

Appendix 4 – Skill test for the issue of a CPL

Regulation (EU) 2024/2076

A. General

1. An applicant for a skill test for the CPL shall have received instruction on the same class or type of aircraft to be used in the test.
2. An applicant shall pass all the relevant sections of the skill test. If any item in a section is failed, that section is failed. Failure in more than one section will require the applicant to take the entire test again. An applicant failing only in one section shall only repeat the failed section. Failure in any section of the retest, including those sections that have been passed on a previous attempt, will require the applicant to take the entire test again. All relevant sections of the skill test shall be completed within 6 months. Failure to achieve a pass in all relevant sections of the test in two attempts will require further training.
3. Further training may be required following any failed skill test. There is no limit to the number of skill tests that may be attempted.

CONDUCT OF THE TEST

4. Should the applicant choose to terminate a skill test for reasons considered inadequate by the Flight Examiner (FE), the applicant shall retake the entire skill test. If the test is terminated for reasons considered adequate by the FE, only those sections not completed shall be tested in a further flight.
5. At the discretion of the FE, any manoeuvre or procedure of the test may be repeated once by the applicant. The FE may stop the test at any stage if it is considered that the applicant's demonstration of flying skills requires a complete re-test.
6. An applicant shall be required to fly the aircraft from a position where the PIC functions can be performed and to carry out the test as if no other crew member is present. Responsibility for the flight shall be allocated in accordance with national regulations.
7. An applicant shall indicate to the FE the checks and duties carried out, including the identification of radio facilities. Checks shall be completed in accordance with the checklist for the aircraft on which the test is being taken. During pre-flight preparation for the test, the applicant is required to determine power settings and speeds. Performance data for take-off, approach and landing shall be calculated by the applicant in compliance with the operations manual or flight manual for the aircraft used.
8. The FE shall take no part in the operation of the aircraft except where intervention is necessary in the interests of safety or to avoid unacceptable delay to other traffic.

B. Content of the skill test for the issue of a CPL – Aeroplanes

1. The aeroplane used for the skill test shall meet the requirements for training aeroplanes, and shall be certificated for the carriage of at least four persons, have a variable pitch propeller and retractable landing gear.
2. The route to be flown shall be chosen by the FE and the destination shall be a controlled aerodrome. The applicant shall be responsible for the flight planning and shall ensure that all equipment and documentation for the execution of the flight are on board. The duration of the flight shall be at least 90 minutes.
3. The applicant shall demonstrate the ability to:

- (a) operate the aeroplane within its limitations,
- (b) complete all manoeuvres with smoothness and accuracy,
- (c) exercise good judgement and airmanship;
- (d) apply aeronautical knowledge; and
- (e) maintain control of the aeroplane at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.

FLIGHT TEST TOLERANCES

4. The following limits shall apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the aeroplane used.
- Height
 - normal flight ± 100 feet
 - with simulated engine failure ± 150 feet
 - Tracking on radio aids $\pm 5^\circ$
 - Heading
 - normal flight $\pm 10^\circ$
 - with simulated engine failure $\pm 15^\circ$
 - Speed
 - take-off and approach ± 5 knots
 - all other flight regimes ± 10 knots

CONTENT OF THE TEST

5. Items in Section 2(c) and (e)(iv), Section 3(g), and the whole of Sections 5 and 6 may be performed in an FNPT II or an FFS. The examiner may decide not to perform items in Section 6(d).

Use of the aeroplane checklists, airmanship, control of the aeroplane by external visual reference, anti-icing/de-icing procedures and principles of threat and error management apply in all sections.

SECTION 1 — PRE-FLIGHT OPERATIONS AND DEPARTURE

a	Pre-flight, including: Flight planning, Documentation, Mass and balance determination, Weather brief, NOTAMS
b	Aeroplane inspection and servicing
c	Taxiing and take-off
d	Performance considerations and trim
e	Aerodrome and traffic pattern operations
f	Departure procedure, altimeter setting, collision avoidance (lookout)
g	ATC liaison – compliance, R/T procedures

SECTION 2 GENERAL AIRWORK

a	Control of the aeroplane by external visual reference, including straight and level, climb, descent, lookout
b	Flight at critically low airspeeds including recognition of and recovery from incipient and full stalls
c	Turns, including turns in landing configuration. Steep turns 45°

d	Flight at critically high airspeeds, including recognition of and recovery from spiral dives
e	Flight by reference solely to instruments, including: (i) level flight, cruise configuration, control of heading, altitude and airspeed (ii) climbing and descending turns with 10°–30° bank (iii) recoveries from unusual attitudes (iv) limited panel instruments
f	ATC liaison – compliance, R/T procedures
SECTION 3 — EN-ROUTE PROCEDURES	
a	Control of aeroplane by external visual reference, including cruise configuration Range/Endurance considerations
b	Orientation, map reading
c	Altitude, speed, heading control, lookout
d	Altimeter setting. ATC liaison – compliance, R/T procedures
e	Monitoring of flight progress, flight log, fuel usage, assessment of track error and re-establishment of correct tracking
f	Observation of weather conditions, assessment of trends, diversion planning
g	Tracking, positioning (NDB, VOR, or track between waypoints), identification of facilities (instrument flight).
h	Implementation of diversion plan to alternate aerodrome (visual flight)
SECTION 4 — APPROACH AND LANDING PROCEDURES	
a	Arrival procedures, altimeter setting, checks, lookout
b	ATC liaison - compliance, R/T procedures
c	Go-around action from low height
d	Normal landing, crosswind landing (if suitable conditions)
e	Short field landing
f	Approach and landing with idle power (single-engine only)
g	Landing without use of flaps
h	Post flight actions
SECTION 5 — ABNORMAL AND EMERGENCY PROCEDURES	
This section may be combined with sections 1 through 4	
a	Simulated engine failure after take-off (at a safe altitude), fire drill
b	Equipment malfunctions including alternative landing gear extension, electrical and brake failure
c	Forced landing (simulated)
d	ATC liaison - compliance, R/T procedures
e	Oral questions
SECTION 6 — SIMULATED ASYMMETRIC FLIGHT AND RELEVANT CLASS OR TYPE ITEMS	
This section may be combined with sections 1 through 5	
a	Simulated engine failure during take-off (at a safe altitude unless carried out in an FFS)
b	Asymmetric approach and go-around
c	Asymmetric approach and full stop landing
d	Engine shut-down and restart (if selected by the examiner)
e	ATC liaison – compliance, R/T procedures, Airmanship
f	As determined by the FE — any relevant items of the class or type rating skill test to include, if applicable: (i) aeroplane systems including handling of autopilot (ii) operation of pressurisation system (iii) use of de-icing and anti-icing system

g Oral questions**C. Content of the skill test for the issue of the CPL – Helicopters**

1. The helicopter used for the skill test shall meet the requirements for training helicopters.
2. The area and route to be flown shall be chosen by the FE and all low level and hover work shall be at an approved aerodrome/site. Routes used for section 3 may end at the aerodrome of departure or at another aerodrome and one destination shall be a controlled aerodrome. The skill test may be conducted in 2 flights. The total duration of the flight(s) shall be at least 90 minutes.
3. The applicant shall demonstrate the ability to:
 - (a) operate the helicopter within its limitations;
 - (b) complete all manoeuvres with smoothness and accuracy;
 - (c) exercise good judgement and airmanship;
 - (d) apply aeronautical knowledge; and
 - (e) maintain control of the helicopter at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.

FLIGHT TEST TOLERANCES

4. The following limits shall apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the helicopter used.
 - Height
 - normal flight ± 100 feet
 - simulated major emergency ± 150 feet
 - Tracking on radio aids $\pm 10^\circ$
 - Heading
 - normal flight $\pm 10^\circ$
 - simulated major emergency $\pm 15^\circ$
 - Speed
 - take-off and approach multi-engine ± 5 knots
 - all other flight regimes ± 10 knots
 - Ground drift
 - T.O. hover I.G.E. ± 3 feet
 - landing no sideways or backwards movement

CONTENT OF THE TEST

5. Items in section 4 may be performed in a helicopter FNPT or a helicopter FFS. Use of helicopter checklists, airmanship, control of helicopter by external visual reference, anti-icing procedures, and principles of threat and error management apply in all sections.

SECTION 1 — PRE-FLIGHT/POST-FLIGHT CHECKS AND PROCEDURES

a	Helicopter knowledge (e.g. technical log, fuel, mass and balance, performance), flight planning, documentation, NOTAMS, weather
b	Pre-flight inspection/action, location of parts and purpose
c	Cockpit inspection, starting procedure
d	Communication and navigation equipment checks, selecting and setting frequencies
e	Pre-take-off procedure, R/T procedure, ATC liaison-compliance
f	Parking, shutdown and post-flight procedure
SECTION 2 — Hover manoeuvres, advanced handling and confined areas	
a	Take-off and landing (lift-off and touchdown)
b	Taxi, hover taxi
c	Stationary hover with head/cross/tail wind
d	Stationary hover turns, 360° left and right (spot turns)
e	Forward, sideways and backwards hover manoeuvring
f	Simulated engine failure from the hover
g	Quick stops into and downwind
h	Sloping ground/unprepared sites landings and take-offs
i	Take-offs (various profiles)
j	Crosswind, downwind take-off (if practicable)
k	Take-off at maximum take-off mass (actual or simulated)
l	Approaches (various profiles)
m	Limited power take-off and landing
n	Autorotations (FE to select two items from — Basic, range, low speed, and 360° turns)
o	Autorotative landing
p	Practice forced landing with power recovery
q	Power checks, reconnaissance technique, approach and departure technique
SECTION 3 — NAVIGATION — EN-ROUTE PROCEDURES	
a	Navigation and orientation at various altitudes/heights, map reading
b	Altitude/height, speed, heading control, observation of airspace, altimeter setting
c	Monitoring of flight progress, flight log, fuel usage, endurance, ETA, assessment of track error and re-establishment of correct track, instrument monitoring
d	Observation of weather conditions, diversion planning
e	Tracking, positioning (NDB and/or VOR), identification of facilities
f	ATC liaison and observance of regulations, etc.
SECTION 4 — FLIGHT PROCEDURES AND MANOEUVRES BY SOLE REFERENCE TO INSTRUMENTS	
a	Level flight, control of heading, altitude/height and speed
b	Rate 1 level turns onto specified headings, 180° to 360° left and right
c	Climbing and descending, including turns at rate 1 onto specified headings
d	Recovery from unusual attitudes
e	Turns with 30° bank, turning up to 90° left and right
SECTION 5 — Abnormal and Emergency procedures (simulated where appropriate)	
Note (1): Where the test is conducted on a multi-engine helicopter a simulated engine failure drill, including a single-engine approach and landing, shall be included in the test.	
Note (2): The FE shall select 4 items from the following:	
a	Engine malfunctions, including governor failure, carburettor/engine icing, oil system, as appropriate
b	Fuel system malfunction
c	Electrical system malfunction

d	Hydraulic system malfunction, including approach and landing without hydraulics, as applicable
e	Main rotor and/or anti-torque system malfunction (FFS or discussion only)
f	Fire drills, including smoke control and removal, as applicable
g	Other abnormal and emergency procedures as outlined in appropriate flight manual, including for multi-engine helicopters: Simulated engine failure at take-off: rejected take-off at or before TDP or safe forced landing at or before DPATO, shortly after TDP or DPATO. Landing with simulated engine failure: landing or go-around following engine failure before LDP or DPBL, following engine failure after LDP or safe forced landing after DPBL.

D. Content of the skill test for the issue of a CPL — Airships

1. The airship used for the skill test shall meet the requirements for training airships.
2. The area and route to be flown shall be chosen by the FE. Routes used for section 3 may end at the aerodrome of departure or at another aerodrome and one destination shall be a controlled aerodrome. The skill test may be conducted in 2 flights. The total duration of the flight(s) shall be at least 60 minutes.
3. The applicant shall demonstrate the ability to:
 - (a) operate the airship within its limitations;
 - (b) complete all manoeuvres with smoothness and accuracy;
 - (c) exercise good judgement and airmanship;
 - (d) apply aeronautical knowledge; and
 - (e) maintain control of the airship at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.

FLIGHT TEST TOLERANCES

4. The following limits shall apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the airship used.
 - Height
 - normal flight ±100 feet
 - simulated major emergency ±150 feet
 - Tracking on radio aids ±10°
 - Heading
 - normal flight ±10°
 - simulated major emergency ±15°

CONTENT OF THE TEST

5. Items in sections 5 and 6 may be performed in an Airship FNPT or an airship FFS. Use of airship checklists, airmanship, control of airship by external visual reference, anti-icing procedures, and principles of threat and error management apply in all sections.

SECTION 1 — PRE-FLIGHT OPERATIONS AND DEPARTURE

- | | |
|---|------------------------|
| a | Pre-flight, including: |
|---|------------------------|

	Flight planning, Documentation, Mass and Balance determination, Weather brief, NOTAMS
b	Airship inspection and servicing
c	Off-mast procedure, ground manoeuvring and take-off
d	Performance considerations and trim
e	Aerodrome and traffic pattern operations
f	Departure procedure, altimeter setting, collision avoidance (lookout)
g	ATC liaison – compliance, R/T procedures
SECTION 2 — GENERAL AIRWORK	
a	Control of the airship by external visual reference, including straight and level, climb, descent, lookout
b	Flight at pressure height
c	Turns
d	Steep descents and climbs
e	Flight by reference solely to instruments, including: (i) level flight, control of heading, altitude and airspeed (ii) climbing and descending turns (iii) recoveries from unusual attitudes (iv) limited panel instruments
f	ATC liaison – compliance, R/T procedures
SECTION 3 — EN-ROUTE PROCEDURES	
a	Control of airship by external visual reference, Range/Endurance considerations
b	Orientation, map reading
c	Altitude, speed, heading control, lookout
d	Altimeter setting, ATC liaison – compliance, R/T procedures
e	Monitoring of flight progress, flight log, fuel usage, assessment of track error and re-establishment of correct tracking
f	Observation of weather conditions, assessment of trends, diversion planning
g	Tracking, positioning (NDB or VOR), identification of facilities (instrument flight). Implementation of diversion plan to alternate aerodrome (visual flight)
SECTION 4 — APPROACH AND LANDING PROCEDURES	
a	Arrival procedures, altimeter setting, checks, lookout
b	ATC liaison – compliance, R/T procedures
c	Go-around action from low height
d	Normal landing
e	Short field landing
f	Approach and landing with idle power (single-engine only)
g	Landing without use of flaps
h	Post-flight actions
SECTION 5 — ABNORMAL AND EMERGENCY PROCEDURES	
This section may be combined with sections 1 through 4	
a	Simulated engine failure after take-off (at a safe altitude), fire drill
b	Equipment malfunctions
c	Forced landing (simulated)
d	ATC liaison – compliance, R/T procedures
e	Oral questions
SECTION 6 — RELEVANT CLASS OR TYPE ITEMS	
This section may be combined with sections 1 through 5	

a	Simulated engine failure during take-off (at a safe altitude unless carried out in an FFS)
b	Approach and go-around with failed engine(s)
c	Approach and full stop landing with failed engine(s)
d	Malfunctions in the envelope pressure system
e	ATC liaison – compliance, R/T procedures, Airmanship
f	As determined by the FE – any relevant items of the class or type rating skill test to include, if applicable: (i) airship systems (ii) operation of envelope pressure system
g	Oral questions

Appendix 5 – Integrated MPL training course

Regulation (EU) 2018/1974

GENERAL

1. The aim of the MPL integrated course is to train pilots to the level of proficiency necessary to enable them to operate as co-pilot of a multi-engine multi-pilot turbine-powered air transport aeroplane under VFR and IFR and to obtain an MPL.
2. Approval for an MPL training course shall only be given to an ATO that is part of a commercial air transport operator certificated in accordance with Part-ORO or having a specific arrangement with such an operator.
3. An applicant wishing to undertake an MPL integrated course shall complete all the instructional stages in one continuous course of training at an ATO. The training shall be competency based and conducted in a multi-crew operational environment.
4. Only ab-initio applicants shall be admitted to the course.
5. The course shall comprise:
 - (a) theoretical knowledge instruction to the ATPL(A) knowledge level;
 - (b) visual and instrument flying training;
 - (c) training in MCC for the operation of multi-pilot aeroplanes; and
 - (d) type rating training.
6. An applicant failing or unable to complete the entire MPL course may apply to the competent authority for the theoretical knowledge examination and skill test for a licence with lower privileges and an IR, if the applicable requirements are met.

THEORETICAL KNOWLEDGE

7. An approved MPL theoretical knowledge course shall comprise at least 750 hours of instruction for the ATPL(A) knowledge level, as well as the hours required for:
 - (a) theoretical knowledge instruction for the relevant type rating, in accordance with Subpart H; and
 - (b) UPRT theoretical knowledge instruction in accordance with FCL.745.A.

FLYING TRAINING

8. The flying training shall comprise a total of at least 240 hours, composed of hours as PF and PM, in actual and simulated flight, and covering the following four phases of training:
 - (a) Phase 1 — Core flying skills
Specific basic single-pilot training in an aeroplane
 - (b) Phase 2 — Basic
Introduction of multi-crew operations and instrument flight
 - (c) Phase 3 — Intermediate
Application of multi-crew operations to a multi-engine turbine aeroplane certified as a high-performance aeroplane in accordance with Annex I (Part-21) to Regulation (EU) No 748/2012

- (d) Phase 4 — Advanced
- Type rating training within an airline-oriented environment.
- MCC requirements shall be incorporated into the relevant phases above.
- Training in asymmetric flight shall be given either in an aeroplane or an FFS.
- 8a. Flight experience in actual flight shall include:
- (a) all the experience requirements of Subpart H;
 - (b) UPRT flight instruction in accordance with FCL.745.A;
 - (c) aeroplane UPRT exercises related to the specificities of the relevant type in accordance with FCL.725.A(c);
 - (d) night flying;
 - (e) flight solely by reference to instruments; and
 - (f) the experience required to achieve the relevant airmanship.
9. Each phase of training in the flight instruction syllabus shall be composed of both instruction in the underpinning knowledge and in practical training segments.
10. The training course shall include a continuous evaluation process of the training syllabus and a continuous assessment of the students following the syllabus. Evaluation shall ensure that:
- (a) the competencies and related assessment are relevant to the task of a co-pilot of a multi-pilot aeroplane; and
 - (b) the students acquire the necessary competencies in a progressive and satisfactory manner.
11. The training course shall include at least 12 take-offs and landings to ensure competency. Those take-offs and landings may be reduced to at least six, provided that prior to delivering the training, the ATO and the operator ensure that:
- (a) a procedure is in place to assess the required level of competency of the student pilot; and
 - (b) a process is in place to ensure that corrective action is taken if in-training evaluation indicates the need to do so.
- Those take-offs and landings shall be performed under the supervision of an instructor in an aeroplane for which the type rating shall be issued.

ASSESSMENT LEVEL

12. The applicant for the MPL shall have demonstrated performance in all 9 competency units specified in paragraph 13 below, at the advanced level of competency required to operate and interact as a co-pilot in a turbine-powered multi-pilot aeroplane, under visual and instrument conditions. Assessment shall confirm that control of the aeroplane or situation is maintained at all times, to ensure the successful outcome of a procedure or manoeuvre. The applicant shall consistently demonstrate the knowledge, skills and attitudes required for the safe operation of the applicable aeroplane type, in accordance with the MPL performance criteria.

COMPETENCY UNITS

13. The applicant shall demonstrate competency in the following 9 competency units:

- (1) apply human performance principles, including principles of threat and error management;
- (2) perform aeroplane ground operations;
- (3) perform take-off;
- (4) perform climb;
- (5) perform cruise;
- (6) perform descent;
- (7) perform approach;
- (8) perform landing; and
- (9) perform after landing and aeroplane post-flight operations.

SIMULATED FLIGHT**14. Minimum requirements for FSTDs:****(a) Phase 1 — Core flying skills**

E-training and part tasking devices approved by the competent authority that have the following characteristics:

- involve accessories beyond those normally associated with desktop computers, such as functional replicas of a throttle quadrant, a side-stick controller, or an FMS keypad; and
- involve psychomotor activity with appropriate application of force and timing of responses.

(b) Phase 2 — Basic

An FNPT II MCC that represents a generic multi-engine turbine-powered aeroplane.

(c) Phase 3 — Intermediate

An FSTD that represents a multi-engine turbine-powered aeroplane required to be operated with a co-pilot and qualified to an equivalent standard to level B, additionally including:

- a daylight/twilight/night visual system continuous cross-cockpit minimum collimated visual field of view providing each pilot with 180° horizontal and 40° vertical field of view, and
- ATC environment simulation.

(d) Phase 4 — Advanced

An FFS which is fully equivalent to level D or level C with an enhanced daylight visual system, including ATC environment simulation.

GM1 to Appendix 5 Integrated MPL training course*ED Decision 2019/005/R***GENERAL**

- (a) In broad terms, the MPL holder is expected to be able to complete the airline operators' conversion course with a high probability of success and within the time frame normally allowed

for this phase. The standard is equivalent to what is currently expected from graduates of the ATP(A) integrated course who have completed type rating training.

- (b) The general approach is to use the existing ATP(A) integrated training course as a reference and to implement progressively the MPL integrated training course and specifically the transfer from actual flight to simulated flight.
- (c) This transfer should be organised in a way that is similar to the approach used for ETOPS. Successive evolutions of the training syllabus introduce progressively a higher level of simulated flight and a reduction of actual flight. Change from one version to the next should only take place after enough experience has been gained and once its results, including those of airline operator conversion courses, have been analysed and taken into account.

MPL TRAINING SCHEME

- (d) The specific arrangement, pursuant to ORA.GEN.205, between an approved training organisation (ATO) and an operator for the multi-pilot licence (MPL) training should cover at least the following points:
 - (1) pre-entry requirements (including screening and selection);
 - (2) provision of the relevant documentation (operations manuals (OMs) and training manuals);
 - (3) design of the training programme;
 - (4) content of the operator conversion course;
 - (5) training effectiveness (e.g. continuous monitoring system, progress checks, etc.);
 - (6) provision of base training;
 - (7) graduate performance data feedback from the operator to the ATO;
 - (8) course evaluation and improvement; and
 - (9) alignment of the grading and assessment criteria.

The ATO and operator may use their OMs and training manuals to identify additional areas to be covered by the specific arrangement.

The following scheme should be applied:

MPL Training Scheme

Adapted competency model

Phases of training	Training items	Flight and simulated flight training media - Minimum level requirement -		Ground training media
Phase 4 — advanced Type rating training covering the training content of Appendix 9 to Part-FCL within an airline-oriented environment	• TEM and CRM • Landing training • All weather • LOFT • Abnormal procedures • Normal procedures • Type-specific UPRT	Aeroplane: ME Multi-crew certified FSTD FS level D or C + ATC simulation	6 to 12 take-offs and landings as PF (in accordance with point 11 of Appendix 5) One go-around with all engines operating (refer to GM1 to Appendix 9 (d)). PF / PM	• E-learning • Part-task trainer • Class-room
Phase 3 — intermediate Application of multi-crew operations in a high-performance ME turbine aeroplane	• TEM and CRM • LOFT • Abnormal procedures • Normal procedures • Multi-crew • Instrument flight • Non-type-specific UPRT	FSTD: <i>representing an ME turbine-powered aeroplane to be operated with a co-pilot and qualified to an equivalent standard to level B + ATC simulation</i>	PF / PM	
Phase 2 — basic Introduction of multi-crew operations and instrument flight	• TEM and CRM • PF / PM complement • IFR cross-country • Instrument flight • Night flight	Aeroplane: SE or ME FSTD: FNPT II + MCC	PF / PM	
Phase 1 — core flying skills Specific basic SP training	• TEM and CRM • VFR cross-country • Solo flight • Basic instrument flight • Principles of flight • Cockpit procedures • Upset recovery in an aeroplane • Night flight	Aeroplane: SE or ME FSTD: FNPT I / BITD	PF	

THEORETICAL KNOWLEDGE INSTRUCTION

(e) The 750 hours of instruction, which also cover the Area 100 KSA, may include in suitable proportions:

- (1) classroom work;
- (2) lessons;
- (3) tutorials;
- (4) demonstrations, including those supported by demonstration equipment;
- (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
- (6) exercises that use demonstration equipment or training devices;
- (7) directed study including workbook exercises or assignments;
- (8) aerodrome or aviation industry field trips;
- (9) computer-based training and e-learning elements;
- (10) progress tests, Area 100 KSA assessments and mental maths test(s); and
- (11) other training methods, media and tools approved by the competent authority.

COMPETENCY UNITS, COMPETENCY ELEMENTS AND PERFORMANCE CRITERIA

(f) Apply human performance principles, including principles of threat and error management:

- (1) cooperation;
- (2) leadership and managerial skills;
- (3) situation awareness;
- (4) decision making.

These behaviour categories are intended to help in the effective utilisation of all available resources to achieve safe and efficient operations.

These behaviour categories may be adapted and extended to incorporate issues like communication and use of automation if it is considered to be relevant to the development of the curriculum.

(g) Perform Aircraft Ground and Pre-Flight Operations

List of competency elements and performance criteria:

- | | | |
|-----|--|---------------------------------|
| (1) | demonstrate attitudes and behaviours appropriate to the safe conduct of flight, including recognising and managing potential threats and errors; | Duty Observation and assessment |
| | | Satisfactory (S) |
| | | Unsatisfactory (U) |
| (2) | perform dispatch duties: | (S) or (U) |
| | (i) verifies technical condition of the a/c, including adequate use of MEL; | PF/PNF |
| | (ii) checks technical bulletins and notices; | PF/PNF |
| | (iii) determines operational environment and pertinent weather; | PF/PNF |
| | (iv) determines impact of weather on aircraft performance; | PF/PNF |
| | (v) applies flight planning and load procedures; | PF/PNF |
| | (vi) determines fuel requirement; | PF/PNF |
| | (vii) files an ATS flight plan (if required) | PF/PNF |
| (3) | provide flight crew and cabin crew briefings; | (S) or (U) |
| | (i) briefed flight crew in all relevant matters; | PF |
| | (ii) briefed cabin crew in all relevant matters. | PF |
| (4) | perform pre-flight checks and cockpit preparation: | (S) or (U) |
| | (i) ensures the airworthiness of the aircraft; | PF |
| | (ii) performs the cockpit preparation and briefings; | PF/PNF |
| | (iii) performs FMS initialisation, data insertion and confirmation; | PF/PNF |
| | (iv) optimises and checks take-off performance and take-off data calculation. | PF/PNF |
| (5) | perform engine start: | (S) or (U) |
| | (i) asks for, receives acknowledges and checks ATC clearance; | PNF |
| | (ii) performs engine start procedure; | PF/PNF |
| | (iii) uses standard communication procedures with ground crew and ATC. | PF/PNF |

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- | | | |
|--------|---|------------|
| (6) | perform taxi out: | (S) or (U) |
| (i) | receives, checks and adheres to taxi clearance; | PNF |
| (ii) | taxis the aircraft, including use of exterior lighting; | PF |
| (iii) | complies to taxi clearance; | PF/PNF |
| (iv) | maintains look-out for conflicting traffic and obstacles; | PF/PNF |
| (v) | operates thrust, brakes and steering; | PF |
| (vi) | conducts relevant briefings; | PF |
| (vii) | uses standard communication procedures with crew and ATC; | PNF |
| (viii) | completes standard operating procedures and checklists; | PF/PNF |
| (ix) | updates and confirms FMS data; | PF/PNF |
| (x) | manages changes in performance and departure route; | PF/PNF |
| (xi) | completes de or anti-ice procedures. | PF/PNF |
| (7) | manage abnormal and emergency situations: | (S) or (U) |
| (i) | identifies the abnormal condition; | PF/PNF |
| (ii) | interprets the abnormal condition; | PF/PNF |
| (iii) | performs the procedure for the abnormal condition. | PF/PNF |
| (8) | communicate with cabin crew, passengers and company: | (S) or (U) |
| (i) | communicates relevant information with cabin crew; | PF |
| (ii) | communicates relevant information with company; | PF/PNF |
| (iii) | makes passenger announcements when appropriate. | PF/PNF |
- (h) Perform take-off
- List of competency elements and performance criteria:
- | | | |
|-------|--|------------|
| (1) | demonstrate attitudes and behaviours appropriate to the safe conduct of including recognising flight, and managing potential | |
| (2) | perform pre threats and errors. -take-off and predeparture preparation: | (S) or (U) |
| (i) | checks and acknowledges line up clearance; | PF/PNF |
| (ii) | checks correct runway selection; | PF/PNF |
| (iii) | confirms validity of performance data; | PF/PNF |
| (iv) | checks approach sector and runway are clear; | PF/PNF |
| (v) | confirms all checklists and take-off preparations completed; | PF/PNF |
| (vi) | lines up the aircraft on centreline without losing distance; | PF |
| (vii) | checks weather on departure sector; | PF/PNF |
-

	(viii) checks runway status and wind.	PF/PNF	
(3)	perform take-off roll:		(S) or (U)
	(i) applies take-off thrust;	PF	
	(ii) checks engine parameters;	PNF	
	(iii) checks air speed indicators;	PF/PNF	
	(iv) stays on runway centreline.	PF	
(4)	perform transition to instrument flight rules:		(S) or (U)
	(i) applies v1 procedures;	PF/PNF	
	(ii) rotates at vr to initial pitch attitude;	PF	
	(iii) establishes initial wings level attitude;	PF	
	(iv) retracts landing gear;	PNF	
	(v) maintains climb out speed.	PF	
(5)	perform initial climb to flap retraction altitude:		(S) or (U)
	(i) sets climb power;	PF	
	(ii) adjusts attitude for acceleration;	PF	
	(iii) selects flaps according flap speed schedule;	PF/PNF	
	(iv) observes speed restrictions;	PF	
	(v) completes relevant checklists.	PF/PNF	
(6)	perform rejected take-off:		(S) or (U)
	(i) recognises the requirement to abort the take-off;	PF	
	(ii) applies the rejected take-off procedure;	PF	
	(iii) assesses the need to evacuate the aircraft.	PF/PNF	
(7)	perform navigation:		(S) or (U)
	(i) complies to departure clearance;	PF	
	(ii) complies with published departure procedures, for example speeds;	PF	
	(iii) monitors navigation accuracy;	PF/PNF	
	(iv) communicates and coordinates with ATC.	PNF	
(8)	manage abnormal and emergency situations:		(S) or (U)
	(i) identifies the abnormal condition;	PF/PNF	
	(ii) interprets the abnormal condition;	PF/PNF	
	(iii) performs the procedure for the abnormal condition.	PF/PNF	
(i)	Perform climb		
	List of competency elements and performance criteria:		

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- | | | |
|-----|--|------------|
| (1) | demonstrate attitudes and behaviours appropriate to the safe conduct of flight, including recognising and managing potential threats and errors; | |
| (2) | perform SID or en-route navigation: | (S) or (U) |
| | (i) complies with departure clearance and procedures; | PF |
| | (ii) demonstrates terrain awareness; | PF/PNF |
| | (iii) monitors navigation accuracy; | PF/PNF |
| | (iv) adjusts flight to weather and traffic conditions; | PF |
| | (v) communicates and coordinates with ATC; | PNF |
| | (vi) observes minimum altitudes; | PF/PNF |
| | (vii) selects appropriate level of automation; | PF |
| | (viii) complies with altimeter setting procedures. | PF/PNF |
| (3) | complete climb procedures and checklists: | (S) or (U) |
| | (i) performs the after take-off items; | PF/PNF |
| | (ii) confirms and checks according checklists. | PF/PNF |
| (4) | modify climb speeds, rate of climb and cruise altitude: | (S) or (U) |
| | (i) recognises the need to change speed, Rate of climb or cruise altitude; | PF |
| | (ii) selects and maintains the appropriate climb speed or rate of climb; | PF |
| | (iii) selects optimum cruise flight level. | PF/PNF |
| (5) | perform systems operations and procedures: | (S) or (U) |
| | (i) monitors operation of all systems; | PF/PNF |
| | (ii) operates systems as required. | PF/PNF |
| (6) | manage abnormal and emergency situations: | (S) or (U) |
| | (i) identifies the abnormal condition; | PF/PNF |
| | (ii) interprets the abnormal condition; | PF/PNF |
| | (iii) performs the procedure for the abnormal condition. | PF/PNF |
| (7) | communicate with cabin crew, passengers and company: | (S) or (U) |
| | (i) communicates relevant information with cabin crew; | PF |
| | (ii) communicates relevant information with company; | PF/PNF |
| | (iii) makes passenger announcements when appropriate. | PF |
- (j) Perform cruise
- List of competency elements and performance criteria.
- | | | |
|-----|--|------------|
| (1) | demonstrate attitudes and behaviours appropriate to the safe conduct of flight, including recognising and managing potential threats and errors; | |
| (2) | monitor navigation accuracy: | (S) or (U) |

	(i)	demonstrates adequate area knowledge;	PF/PNF	
	(ii)	demonstrates adequate route knowledge;	PF/PNF	
	(iii)	navigates according to flight plan and clearance;	PF	
	(iv)	adjusts flight to weather and traffic conditions;	PF	
	(v)	communicates and coordinates with ATC;	PNF	
	(vi)	observes minimum altitudes;	PF/PNF	
	(vii)	uses all means of automation.	PF	
(3)		monitor flight progress:		(S) or (U)
	(i)	selects optimum speed;	PF	
	(ii)	selects optimum cruise flight level;	PF	
	(iii)	monitors and controls fuel status;	PF/PNF	
	(iv)	recognises the need for a possible diversion;	PF/PNF	
	(v)	creates a diversion contingency plan if required.	PF/PNF	
(4)		perform descent and approach planning:		(S) or (U)
	(i)	checks weather of destination and alternate airport;	PF/PNF	
	(ii)	checks runway in use and approach procedure;	PF/PNF	
	(iii)	sets the FMS accordingly;	PNF	
	(iv)	checks landing weight and landing distance required;	PNF	
	(v)	checks MEA, MGA and MSA;	PF/PNF	
	(vi)	identifies top of descent point.	PF	
(5)		perform systems operations and procedures:		(S) or (U)
	(i)	monitors operation of all systems;	PF/PNF	
	(ii)	operates systems as required.	PNF	
(6)		manage abnormal and emergency situations:		(S) or (U)
	(i)	identifies the abnormal condition;	PF/PNF	
	(ii)	interprets the abnormal condition;	PF/PNF	
	(iii)	performs the procedure for the abnormal condition.	PF/PNF	
(7)		communicate with cabin crew, passengers and company:		(S) or (U)
	(i)	communicates relevant information with cabin crew;	PF	
	(ii)	communicates relevant information with company;	PF/PNF	
	(iii)	makes passenger announcements when appropriate.	PF	
(k)		Perform descent		
		List of competency elements and performance criteria:		
(1)		Demonstrate attitudes and behaviours appropriate to the safe conduct of flight, including recognising and managing potential threats and errors;		

- | | | |
|-----|---|------------|
| (2) | initiate and manage descent: | (S) or (U) |
| | (i) starts descent according to ATC clearance or optimum descent point; | PF |
| | (ii) selects optimum speed and descent rate; | PF |
| | (iii) adjusts speed to existing environmental conditions; | PF |
| | (iv) recognises the need to adjust the descent path; | PF |
| | (v) adjusts the flight path as required; | PF |
| | (vi) utilises all means of FMS descent information. | PF |
| (3) | monitor and perform en route and descent navigation: | (S) or (U) |
| | (i) complies with arrival clearance and procedures; | PF |
| | (ii) demonstrates terrain awareness; | PF/PNF |
| | (iii) monitors navigation accuracy; | PF/PNF |
| | (iv) adjusts flight to weather and traffic conditions; | PF |
| | (v) communicates and coordinates with ATC; | PNF |
| | (vi) observes minimum altitudes; | PF/PNF |
| | (vii) selects appropriate level or mode of automation; | PF |
| | (viii) complies with altimeter setting procedures. | PF/PNF |
| (4) | re-planning and update of approach briefing: | (S) or (U) |
| | (i) re-checks destination weather and runway in use; | PNF |
| | (ii) briefs or re-briefs about instrument approach and landing as required; | PF |
| | (iii) reprograms the FMS as required; | PNF |
| | (iv) re-checks fuel status. | PF/PNF |
| (5) | perform holding: | (S) or (U) |
| | (i) identifies holding requirement; | PF/PNF |
| | (ii) programs FMS for holding pattern; | PNF |
| | (iii) enters and monitors holding pattern; | PF |
| | (iv) assesses fuel requirements and determines max holding time; | PF/PNF |
| | (v) reviews the need for a diversion; | PF/PNF |
| | (vi) initiates diversion. | PF |
| (6) | perform systems operations and procedures: | (S) or (U) |
| | (i) monitors operation of all systems; | PF/PNF |
| | (ii) operates systems as required. | PF/PNF |
| (7) | manage abnormal and emergency situations: | |
| | (i) identifies the abnormal condition; | PF/PNF |

	(ii)	interprets the abnormal condition;	PF/PNF	
	(iii)	performs the procedure for the abnormal condition.	PF/PNF	
(8)		communicate with cabin crew, passengers and company:		(S) or (U)
	(i)	communicates relevant information with cabin crew;	PF	
	(ii)	communicates relevant information with company;	PF/PNF	
	(iii)	makes passenger announcements when appropriate;	PF	
(I)		Perform approach		
		List of competency elements and performance criteria:		
(1)		demonstrate attitudes and behaviours appropriate to the safe conduct of flight, including recognising and managing potential threats and errors;		
(2)		perform approach in general:		(S) or (U)
	(i)	executes approach according to procedures and situation;	PF	
	(ii)	selects appropriate level or mode of automation;	PF	
	(iii)	selects optimum approach path;	PF	
	(iv)	operates controls smooth and coordinated;	PF	
	(v)	performs speed reduction and flap extension;	PF/PNF	
	(vi)	performs relevant checklists;	PF/PNF	
	(vii)	initiates final descent;	PF	
	(viii)	achieves stabilised approach criteria;	PF	
	(ix)	ensures adherence to minima;	PF/PNF	
	(x)	initiates go-around if required;	PF	
	(xi)	masters transition to visual segment.	PF	
(3)		perform precision approach:		(S) or (U)
	(i)	performs ILS approach;	PF	
	(ii)	performs MLS approach.	PF	
(4)		perform non-precision approach:		(S) or (U)
	(i)	performs VOR approach;	PF	
	(ii)	performs NDB approach;	PF	
	(iii)	performs SRE approach;	PF	
	(iv)	performs GNSS approach;	PF	
	(v)	performs ILS loc approach;	PF	
	(vi)	performs ILS back beam approach.	PF	
(5)		perform approach with visual reference to ground:		(S) or (U)
	(i)	performs standard visual approach;	PF	

	(ii) performs circling approach.	PF	
(6)	monitor the flight progress:		(S) or (U)
	(i) insures navigation accuracy;	PF/PNF	
	(ii) communicates with ATC and crew members;	PNF	
	(iii) monitors fuel status.	PF/PNF	
(7)	perform systems operations and procedures:		
	(i) monitors operation of all systems;	PF	
	(ii) operates systems as required.	PF	
(8)	manage abnormal and emergency situations:		(S) or (U)
	(i) identifies the abnormal condition;	PF/PNF	
	(ii) interprets the abnormal condition;	PF/PNF	
	(iii) performs the procedure for the abnormal condition.	PF/PNF	
(9)	perform missed approach and goaround:		(S) or (U)
	(i) initiates go-around procedure;	PF	
	(ii) navigates according to missed approach procedure;	PF	
	(iii) completes the relevant checklists;	PF/PNF	
	(iv) initiates approach or diversion after the go-around;	PF	
	(v) communicates with ATC and crew members.	PNF	
(10)	communicate with cabin crew, passengers and company:		(S) or (U)
	(i) communicates relevant information with cabin crew;	PF	
	(ii) communicates relevant information with company;	PF/PNF	
	(iii) makes passenger announcements when appropriate;	PF	
	(iv) initiates go-around procedure.	PF	
(m)	Perform landing		
	List of competency elements and performance criteria:		
(1)	demonstrate attitudes and behaviours appropriate to the safe conduct of flight, including recognising and managing potential threats and errors;		
(2)	land the aircraft;		(S) or (U)
	(i) maintains a stabilised approach path during visual segment;	PF	
	(ii) recognises and acts on changing conditions for windshift or wind shear segment;	PF	
	(iii) initiates flare;	PF	
	(iv) controls thrust;	PF	
	(v) achieves touchdown in touchdown zone on centreline;	PF	
	(vi) lowers nose wheel;	PF	

	(vii) maintains centreline;	PF	
	(viii) performs after-touchdown procedures;	PF	
	(ix) makes use of appropriate braking and reverse thrust;	PF	
	(x) vacates runway with taxi speed.	PF	
(3)	perform systems operations and procedures:		(S) or (U)
	(i) monitors operation of all systems;	PF	
	(ii) operates systems as required.	PF	
(4)	manage abnormal and emergency situations:		(S) or (U)
	(i) identifies the abnormal condition;	PF/PNF	
	(ii) interprets the abnormal condition;	PF/PNF	
	(iii) performs the procedure for the abnormal condition.	PF/PNF	
(n)	Perform after landing and post flight operations		
	List of competency elements and performance criteria:		
(1)	demonstrate attitudes and behaviours appropriate to the safe conduct of flight, including recognising and managing potential threats and errors;		
(2)	perform taxiing and parking:		(S) or (U)
	(i) receives, checks and adheres to taxi clearance;	PNF	
	(ii) taxies the aircraft including use of exterior lighting;	PF	
	(iii) controls taxi speed;	PF/PNF	
	(iv) maintains centreline;	PF	
	(v) maintains look-out for conflicting traffic and obstacles;	PF	
	(vi) identifies parking position;	PF/PNF	
	(vii) complies with marshalling or stand guidance;	PF/PNF	
	(viii) applies parking and engine shut down procedures;	PF	
	(ix) completes with relevant checklists.	PF/PNF	
(3)	perform aircraft post-flight operations:		(S) or (U)
	(i) communicates to ground personnel and crew;	PF	
	(ii) completes all required flight documentation;	PF/PNF	
	(iii) ensures securing of the aircraft;	PF	
	(iv) conducts the debriefings.	PF	
(4)	perform systems operations and procedures:		(S) or (U)
	(i) monitors operation of all systems;	PF/PNF	
	(ii) operates systems as required.	PF/PNF	
(5)	manage abnormal and emergency situations:		(S) or (U)
	(i) identifies the abnormal condition;	PF/PNF	

- (ii) interprets the abnormal condition; PF/PNF
- (iii) performs the procedure for the abnormal condition. PF/PNF
- (6) communicate with cabin crew, passengers and company: (S) or (U)
 - (i) communicates relevant information with cabin crew; PF
 - (ii) communicates relevant information with company; PF/PNF
 - (iii) makes passenger announcements when appropriate. PF

PRINCIPLES OF THREAT AND ERROR MANAGEMENT

- (o) One model that explains the principles of threat and error management is the TEM model.

- (1) The components of the TEM model:

There are three basic components in the TEM model, from the perspective of flight crews: threats, errors and undesired aircraft states. The model proposes that threats and errors are part of everyday aviation operations that must be managed by flight crews, since both threats and errors carry the potential to generate undesired aircraft states. Flight crews must also manage undesired aircraft states, since they carry the potential for unsafe outcomes. Undesired state management is an essential component of the TEM model, as important as threat and error management. Undesired aircraft state management largely represents the last opportunity to avoid an unsafe outcome and thus maintain safety margins in flight operations.

- (2) Threats:

- (i) Threats are defined as events or errors that occur beyond the influence of the flight crew, increase operational complexity, and which must be managed to maintain the margins of safety. During typical flight operations, flight crews have to manage various contextual complexities. Such complexities would include, for example, dealing with adverse meteorological conditions, airports surrounded by high mountains, congested airspace, aircraft malfunctions, errors committed by other people outside of the cockpit, such as air traffic controllers, flight attendants or maintenance workers, and so forth. The TEM model considers these complexities as threats because they all have the potential to negatively affect flight operations by reducing margins of safety;
- (ii) Some threats can be anticipated, since they are expected or known to the flight crew. For example, flight crews can anticipate the consequences of a thunderstorm by briefing their response in advance, or prepare for a congested airport by making sure they keep a watchful eye on other aircraft as they execute the approach;
- (iii) Some threats can occur unexpectedly, such as an in-flight aircraft malfunction that happens suddenly and without warning. In this case, flight crews must apply skills and knowledge acquired through training and operational experience;
- (iv) Lastly, some threats may not be directly obvious to, or observable by, flight crews immersed in the operational context, and may need to be uncovered by safety analysis. These are considered latent threats. Examples of latent threats include equipment design issues, optical illusions, or shortened turnaround schedules;
- (v) Regardless of whether threats are expected, unexpected, or latent, one measure of the effectiveness of a flight crew's ability to manage threats is whether threats

are detected with the necessary anticipation to enable the flight crew to respond to them through deployment of appropriate countermeasures;

- (vi) Threat management is a building block to error management and undesired aircraft state management. Although the threat-error linkage is not necessarily straightforward, and although it may not be always possible to establish a linear relationship, or one-to-one mapping between threats, errors and undesired states, archival data demonstrates that mismanaged threats are normally linked to flight crew errors, which in turn are often linked to undesired aircraft states. Threat management provides the most proactive option to maintain margins of safety in flight operations, by voiding safety-compromising situations at their roots. As threat managers, flight crews are the last line of defence to keep threats from impacting flight operations;
- (vii) Table 1 presents examples of threats, grouped under two basic categories derived from the TEM Model. Environmental threats occur due to the environment in which flight operations take place. Some environmental threats can be planned for and some will arise spontaneously, but they all have to be managed by flight crews in real time. Organisational threats, on the other hand, can be controlled (for example removed or, at least, minimised) at source by aviation organisations. Organisational threats are usually latent in nature. Flight crews still remain the last line of defence, but there are earlier opportunities for these threats to be mitigated by aviation organisations themselves.

Environmental threats		Organisational threats	
(A)	weather: thunderstorms, turbulence, icing, wind shear, cross or tailwind, very low or high temperatures;	(A)	operational pressure: delays, late arrivals or equipment changes;
(B)	ATC: traffic congestion, ACAS RA/TA, ATC command, ATC error, ATC language difficulty, ATC non-standard phraseology, ATC runway change, ATIS communication or units of measurement (QFE/meters);	(B)	aircraft: aircraft malfunction, automation event or anomaly, MEL/CDL;
(C)	airport: contaminated or short runway; contaminated taxiway, lack of, confusing, faded signage, markings, birds, aids unserviceable, complex surface navigation procedures or airport constructions;	(C)	cabin: flight attendant error, cabin event distraction, interruption, cabin door security;
(D)	terrain: high ground, slope, lack of references or 'black hole';	(D)	maintenance: maintenance event or error;
(E)	other: similar call-signs.	(E)	ground: ground-handling event, de-icing or ground crew error;
		(F)	dispatch: dispatch paperwork event or error;
		(G)	documentation: manual error or chart error;
		(H)	other: crew scheduling event.

Table 1. Examples of threats (list is not exhaustive)

(3) Errors:

- (i) Errors are defined actions or inactions by the flight crew that lead to deviations from organisational or flight crew intentions or expectations. Unmanaged or mismanaged errors frequently lead to undesired aircraft states. Errors in the operational context thus tend to reduce the margins of safety and increase the probability of adverse events;
- (ii) Errors can be spontaneous (for example without direct linkage to specific, obvious threats), linked to threats, or part of an error chain. Examples of errors would include the

inability to maintain stabilised approach parameters, executing a wrong automation mode, failing to give a required callout, or misinterpreting an ATC clearance;

- (iii) Regardless of the type of error, an error's effect on safety depends on whether the flight crew detects and responds to the error before it leads to an undesired aircraft state and to a potential unsafe outcome. This is why one of the objectives of TEM is to understand error management (for example detection and response), rather than to solely focus on error causality (for example causation and commission). From the safety perspective, operational errors that are timely detected and promptly responded to (for example properly managed), errors that do not lead to undesired aircraft states, do not reduce margins of safety in flight operations, and thus become operationally inconsequential. In addition to its safety value, proper error management represents an example of successful human performance, presenting both learning and training value;
- (iv) Capturing how errors are managed is then as important, if not more, as capturing the prevalence of different types of error. It is of interest to capture if and when errors are detected and by whom, the response(s) upon detecting errors, and the outcome of errors. Some errors are quickly detected and resolved, thus becoming operationally inconsequential, while others go undetected or are mismanaged. A mismanaged error is defined as an error that is linked to or induces an additional error or undesired aircraft state;
- (v) Table 2 presents examples of errors, grouped under three basic categories derived from the TEM model. In the TEM concept, errors have to be 'observable' and therefore, the TEM model uses the 'primary interaction' as the point of reference for defining the error categories;
- (vi) The TEM model classifies errors based upon the primary interaction of the pilot or flight crew at the moment the error is committed. Thus, in order to be classified as aircraft handling error, the pilot or flight crew must be interacting with the aircraft (for example through its controls, automation or systems). In order to be classified as procedural error, the pilot or flight crew must be interacting with a procedure (for example checklists; SOPs; etc.). In order to be classified as communication error, the pilot or flight crew must be interacting with people (ATC, ground crew, other crewmembers, etc.);
- (vii) Aircraft handling errors, procedural errors and communication errors may be unintentional or involve intentional non-compliance. Similarly, proficiency considerations (for example skill or knowledge deficiencies, training system deficiencies) may underlie all three categories of error. In order to keep the approach simple and avoid confusion, the TEM model does not consider intentional noncompliance and proficiency as separate categories of error, but rather as sub-sets of the three major categories of error.

Aircraft handling errors	(A)	manual handling, flight controls: vertical, lateral or speed deviations, incorrect flaps or speed brakes, thrust reverser or power settings;
	(B)	automation: incorrect altitude, speed, heading, auto throttle settings, incorrect mode executed or incorrect entries;
	(C)	systems, radio, instruments: incorrect packs, incorrect anti-icing, incorrect altimeter, incorrect fuel switches settings, incorrect speed bug or incorrect radio frequency dialled;
	(D)	ground navigation: attempting to turn down wrong taxiway or runway, taxi too fast, failure to hold short or missed taxiway or runway.

Procedural errors	(A) SOPs: failure to cross-verify automation inputs; (B) checklists: wrong challenge and response; items missed, checklist performed late or at the wrong time; (C) callouts: omitted or incorrect callouts; (D) briefings: omitted briefings; items missed; (E) documentation: wrong weight and balance, fuel information, ATIS, or clearance information recorded, misinterpreted items on paperwork; incorrect logbook entries or incorrect application of MEL procedures.
Communication errors	(A) crew to external: missed calls, misinterpretations of instructions, incorrect read-back, wrong clearance, taxiway, gate or runway communicated; (B) pilot to pilot: within crew miscommunication or mis-interpretation.

Table 2. Examples of errors (list is not exhaustive)

(4) Undesired aircraft states:

- (i) Undesired aircraft states are flight crew-induced aircraft position or speed deviations, misapplication of flight controls, or incorrect systems configuration, associated with a reduction in margins of safety. Undesired aircraft states that result from ineffective threat or error management may lead to compromising situations and reduce margins of safety in flight operations. Often considered at the cusp of becoming an incident or accident, undesired aircraft states must be managed by flight crews;
- (ii) Examples of undesired aircraft states would include lining up for the incorrect runway during approach to landing, exceeding ATC speed restrictions during an approach, or landing long on a short runway requiring maximum braking. Events such as equipment malfunctions or ATC controller errors can also reduce margins of safety in flight operations, but these would be considered threats;
- (iii) Undesired states can be managed effectively, restoring margins of safety, or flight crew response(s) can induce an additional error, incident, or accident;
- (iv) Table 3 presents examples of undesired aircraft states, grouped under three basic categories derived from the TEM model;

Aircraft handling	(A) aircraft control (attitude); (B) vertical, lateral or speed deviations; (C) unnecessary weather penetration; (D) unauthorised airspace penetration; (E) operation outside aircraft limitations; (F) unstable approach; (G) continued landing after unstable approach; (H) long, floated, firm or off-centreline landing.
Ground navigation	(A) proceeding towards wrong taxiway or runway; (B) Wrong taxiway, ramp, gate or hold spot.
Incorrect aircraft configurations	(A) incorrect systems configuration; (B) incorrect flight controls configuration; (C) incorrect automation configuration; (D) incorrect engine configuration; (E) incorrect weight and balance configuration.

Table 3. Examples of undesired aircraft states (list is not exhaustive)

- (v) An important learning and training point for flight crews is the timely switching from error management to undesired aircraft state management. An example would be as follows: a flight crew selects a wrong approach in the FMC. The flight crew subsequently identifies the error during a cross-check prior to the FAF. However, instead of using a basic mode (for example heading) or manually flying the desired track, both flight crew members become involved in attempting to reprogram the correct approach prior to reaching the FAF. As a result, the aircraft 'stitches' through the localiser, descends late, and goes into an unstable approach. This would be an example of the flight crew getting 'locked in' to error management, rather than switching to undesired aircraft state management. The use of the TEM model assists in educating flight crews that, when the aircraft is in an undesired state, the basic task of the flight crew is undesired aircraft state management instead of error management. It also illustrates how easy it is to get locked in to the error management phase;
 - (vi) Also from a learning and training perspective, it is important to establish a clear differentiation between undesired aircraft states and outcomes. Undesired aircraft states are transitional states between a normal operational state (for example a stabilised approach) and an outcome. Outcomes, on the other hand, are end states, most notably, reportable occurrences (for example incidents and accidents). An example would be as follows: a stabilised approach (normal operational state) turns into an unstabilised approach (undesired aircraft state) that results in a runway excursion (outcome);
 - (vii) The training and remedial implications of this differentiation are of significance. While at the undesired aircraft state stage, the flight crew has the possibility, through appropriate TEM, of recovering the situation, returning to a normal operational state, thus restoring margins of safety. Once the undesired aircraft state becomes an outcome, recovery of the situation, return to a normal operational state, and restoration of margins of safety is not possible.
- (5) Countermeasures:
- (i) Flight crews must, as part of the normal discharge of their operational duties, employ countermeasures to keep threats, errors and undesired aircraft states from reducing margins of safety in flight operations. Examples of countermeasures would include checklists, briefings, call-outs and SOPs, as well as personal strategies and tactics. Flight crews dedicate significant amounts of time and energies to the application of countermeasures to ensure margins of safety during flight operations. Empirical observations during training and checking suggest that as much as 70 % of flight crew activities may be countermeasures-related activities.
 - (ii) All countermeasures are necessarily flight crew actions. However, some countermeasures to threats, errors and undesired aircraft states that flight crews employ build upon 'hard' resources provided by the aviation system. These resources are already in place in the system before flight crews report for duty, and are therefore considered as systemic-based countermeasures. The following would be examples of 'hard' resources that flight crews employ as systemic-based countermeasures:
 - (A) ACAS;
 - (B) TAWS;
 - (C) SOPs;
 - (D) checklists;
 - (E) briefings;

- (F) training;
- (G) etc.
- (iii) Other countermeasures are more directly related to the human contribution to the safety of flight operations. These are personal strategies and tactics, individual and team countermeasures that typically include canvassed skills, knowledge and attitudes developed by human performance training, most notably, by CRM training. There are basically three categories of individual and team countermeasures:
 - (A) planning countermeasures: essential for managing anticipated and unexpected threats;
 - (B) execution countermeasures: essential for error detection and error response;
 - (C) review countermeasures: essential for managing the changing conditions of a flight.
- (iv) Enhanced TEM is the product of the combined use of systemic based and individual and team countermeasures. Table 4 presents detailed examples of individual and team countermeasures. Further guidance on countermeasures can be found in the sample assessment guides for terminal training objectives (PANS-TRG, Chapter 3, Attachment B) as well as in the ICAO manual, Line Operations Safety Audit (LOSA) (Doc 9803).

Planning countermeasures		
SOP briefing	The required briefing was interactive and operationally thorough	(A) Concise, not rushed, and met SOP requirements; (B) Bottom lines were established
Plans stated	Operational plans and decisions were communicated and acknowledged	Shared understanding about plans: 'Everybody on the same page'
Workload assignment	Roles and responsibilities were defined for normal and non-normal situations	Workload assignments were communicated and acknowledged
Contingency management	Crew members developed effective strategies to manage threats to safety	(A) Threats and their consequences were anticipated; (B) Used all available resources to manage threats
Execution countermeasures		
Monitor and cross-check	Crew members actively monitored and cross-checked systems and other crew members	Aircraft position, settings, and crew actions were verified
Workload management	Operational tasks were prioritised and properly managed to handle primary flight duties	(A) Avoided task fixation; (B) Did not allow work overload
Automation management	Automation was properly managed to balance situational and workload requirements	(A) Automation setup was briefed to other members (B) Effective recovery techniques from automation anomalies
Review countermeasures		
Evaluation and modification of plans	Existing plans were reviewed and modified when necessary	Crew decisions and actions were openly analysed to make sure the existing plan was the best plan
Inquiry	Crew members asked questions to investigate and/or clarify current plans of action	Crew members not afraid to express a lack of knowledge: 'Nothing taken for granted' attitude

Assertiveness	Crew members stated critical information or solutions with appropriate persistence	Crew members spoke up without hesitation
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Table 4. Examples of individual and team countermeasures

GM2 to Appendix 5 Assessment of student competency during take-off and landing training

ED Decision 2018/011/R

The required level of competency of a student pilot is assessed by observing the following:

- (a) application of knowledge;
- (b) application of regulations and procedures;
- (c) communication;
- (d) aeroplane flight path management – automation;
- (e) aeroplane flight path management – manual control;
- (f) leadership and teamwork;
- (g) problem-solving and decision-making;
- (h) situational awareness (SA) and information management; and
- (i) workload management.

The competencies referred to in points (b) and (e) are particularly relevant during the training. This means that the focus is on observing the student pilot performing take-offs and landings in accordance with the standard operating procedures (SOPs) and recommended techniques of the original equipment manufacturer (OEM).

The competency elements and sub-elements stipulated in [GM1 to Appendix 5](#) for take-off and landing provide additional guidance for instructors and student pilots.

Consistency and repeatability of all the competencies above is achieved if the student pilot is able to perform at least three successive take-offs and landings demonstrating the required observable behaviours.

The take-off and landing training in an aeroplane should include at least one go-around. Due consideration should be given to environmental conditions when evaluating competency.

Appendix 6 – Modular training courses for the IR

Regulation (EU) 2021/2227

A. IR(A) — Modular flying training course

GENERAL

1. The aim of the IR(A) modular flying training course is to train pilots to the level of proficiency necessary to operate aeroplanes under IFR and in IMC. The course consists of two modules, which may be taken separately or combined:
 - (a) **Basic Instrument Flight Module**

This comprises 10 hours of instrument time under instruction, of which up to 5 hours can be instrument ground time in a BITD, FNPT I or II, or an FFS. Upon completion of the Basic Instrument Flight Module, the candidate shall be issued a Course Completion Certificate.
 - (b) **Procedural Instrument Flight Module**

This comprises the remainder of the training syllabus for the IR(A), 40 hours single-engine or 45 hours multi-engine instrument time under instruction, and the theoretical knowledge course for the IR(A).
2. Applicants for a modular IR(A) course shall be the holder of a PPL(A) or a CPL(A). Applicants for the Procedural Instrument Flight Module, who do not hold a CPL(A), shall be a holder of a BIR or of a Course Completion Certificate for the Basic Instrument Flight Module.

The ATO shall ensure that the applicant for a multi-engine IR(A) course who has not held a multi-engine aeroplane class or type rating has received the multi-engine training specified in Subpart H prior to commencing the flight training for the IR(A) course.
3. An applicant wishing to undertake the Procedural Instrument Flight Module of a modular IR(A) course shall be required to complete all the instructional stages in one continuous approved course of training. Prior to commencing the Procedural Instrument Flight Module, the ATO shall ensure the competence of the applicant in basic instrument flying skills. Refresher training shall be given as required.
4. The course of theoretical instruction shall be completed within 18 months. The Procedural Instrument Flight Module and the skill test shall be completed within the period of validity of the pass in theoretical examinations.
5. The course shall comprise:
 - (a) theoretical knowledge instruction to the IR knowledge level;
 - (b) instrument flight instruction.

THEORETICAL KNOWLEDGE

6. An approved modular IR(A) course shall comprise at least 150 hours of theoretical knowledge instruction.

FLYING TRAINING

7. A single-engine IR(A) course shall comprise at least 50 hours instrument time under instruction of which up to 20 hours may be instrument ground time in an FNPT I, or up to 35 hours in an FFS or FNPT II. A maximum of 10 hours of FNPT II or an FFS instrument ground time may be conducted in an FNPT I.

8. A multi-engine IR(A) course shall comprise at least 55 hours instrument time under instruction, of which up to 25 hours may be instrument ground time in an FNPT I, or up to 40 hours in an FFS or FNPT II. A maximum of 10 hours of FNPT II or an FFS instrument ground time may be conducted in an FNPT I. The remaining instrument flight instruction shall include at least 15 hours in multi-engine aeroplanes.
9. The holder of a single-engine IR(A) who also holds a multi-engine class or type rating wishing to obtain a multi-engine IR(A) for the first time shall complete a course at an ATO comprising at least 5 hours instruction in instrument flying in multi-engine aeroplanes, of which 3 hours may be in an FFS or FNPT II.
- 10.1 Holders of a CPL(A), of a BIR, or of a Course Completion Certificate for the Basic Instrument Flight Module shall be credited with up to 10 hours in regard of the total amount of training required in paragraphs 7 or 8 above.
- 10.2 The holder of an IR(H) may have the total amount of training required in paragraphs 7 or 8 above reduced to 10 hours.
- 10.3 The total instrument flight instruction in aeroplane shall comply with paragraph 7 or 8, as appropriate.
11. The flying exercises up to the IR(A) skill test shall comprise:
 - (a) Basic Instrument Flight Module: Procedure and manoeuvre for basic instrument flight covering at least:
 - basic instrument flight without external visual cues:
 - horizontal flight,
 - climbing,
 - descent,
 - turns in level flight, climbing, descent;
 - instrument pattern;
 - steep turn;
 - radio navigation;
 - recovery from unusual attitudes;
 - limited panel;
 - recognition and recovery from incipient and full stalls;
 - (b) Procedural Instrument Flight Module:
 - (i) pre-flight procedures for IFR flights, including the use of the flight manual and appropriate air traffic services documents in the preparation of an IFR flight plan;
 - (ii) procedure and manoeuvres for IFR operation under normal, abnormal and emergency conditions covering at least:
 - transition from visual to instrument flight on take-off,
 - standard instrument departures and arrivals,
 - en-route IFR procedures,
 - holding procedures,

- instrument approaches to specified minima,
 - missed approach procedures,
 - landings from instrument approaches, including circling;
- (iii) in-flight manoeuvres and particular flight characteristics;
- (iv) if required, operation of a multi-engine aeroplane in the above exercises, including operation of the aeroplane solely by reference to instruments with one engine simulated inoperative and engine shutdown and restart (the latter exercise to be carried out at a safe altitude unless carried out in an FFS or FNPT II).

Aa. IR(A) — Competency-based modular flying training course**GENERAL**

1. The aim of the competency-based modular flying training course is to train PPL or CPL holders for the instrument rating, taking into account prior instrument flight instruction and experience. It is designed to provide the level of proficiency needed to operate aeroplanes under IFR and in IMC. The course shall be taken within an ATO or consist of a combination of instrument flight instruction provided by an IRI(A) or an FI(A) holding the privilege to provide training for the IR and flight instruction within an ATO.
2. An applicant for such a competency-based modular IR(A) shall be the holder of a PPL(A) or CPL(A).
3. The course of theoretical instruction shall be completed within 18 months. The instrument flight instruction and the skill test shall be completed within the period of validity of the pass of the theoretical knowledge examinations.
4. The course shall comprise:
 - (a) theoretical knowledge instruction to the IR(A) knowledge level;
 - (b) instrument flight instruction.

THEORETICAL KNOWLEDGE

5. An approved competency-based modular IR(A) course shall comprise at least 80 hours of theoretical knowledge instruction. The theoretical knowledge course may contain computer-based training and e-learning elements. A minimum amount of classroom teaching as required by ORA.ATO.305 has to be provided.

FLYING TRAINING

6. The method of attaining an IR(A) following this modular course is competency-based. However, the minimum requirements below shall be completed by the applicant. Additional training may be required to reach required competencies.
 - (a) A single-engine competency-based modular IR(A) course shall include at least 40 hours of instrument time under instruction, of which up to 10 hours may be instrument ground time in an FNPT I, or up to 25 hours in an FFS or FNPT II. A maximum of 5 hours of FNPT II or FFS instrument ground time may be conducted in an FNPT I.
 - (i) When the applicant has:
 - (A) completed instrument flight instruction provided by an IRI(A) or an FI(A) holding the privilege to provide training for the IR; or

- (B) prior experience of instrument flight time as PIC on aeroplanes, under a rating providing the privileges to fly under IFR and in IMC,
these hours may be credited towards the 40 hours above up to maximum of 30 hours,
 - (ii) When the applicant has prior instrument flight time under instruction other than specified in point (a)(i), these hours may be credited towards the required 40 hours up to a maximum of 15 hours.
 - (iii) In any case, the flying training shall include at least 10 hours of instrument flight time under instruction in an aeroplane at an ATO.
 - (iv) The total amount of dual instrument instruction shall not be less than 25 hours.
 - (b) A multi-engine competency-based modular IR(A) course shall include at least 45 hours instrument time under instruction, of which up to 10 hours may be instrument ground time in an FNPT I, or up to 30 hours in an FFS or FNPT II. A maximum of 5 hours of FNPT II or FFS instrument ground time may be conducted in an FNPT I.
 - (i) When the applicant has:
 - (A) completed instrument flight instruction provided by an IRI(A) or an FI(A) holding the privilege to provide training for the IR; or
 - (B) prior experience of instrument flight time as PIC on aeroplanes, under a rating giving the privileges to fly under IFR and in IMC
these hours may be credited towards the 45 hours above up to a maximum of 35 hours.
 - (ii) When the applicant has prior instrument flight time under instruction other than specified in point (b)(i), these hours may be credited towards the required 45 hours up to a maximum of 15 hours.
 - (iii) In any case, the flying training shall include at least 10 hours of instrument flight time under instruction in a multi-engine aeroplane at an ATO.
 - (iv) The total amount of dual instrument instruction shall not be less than 25 hours, of which at least 15 hours shall be completed in a multi-engine aeroplane.
 - (c) To determine the amount of hours credited and to establish the training needs, the applicant shall complete a pre-entry assessment at an ATO.
 - (d) The completion of the instrument flight instruction provided by an IRI(A) or FI(A) in accordance with point (a)(i) or (b)(i) shall be documented in a specific training record and signed by the instructor.
- 7. The flight instruction for the competency-based modular IR(A) shall comprise:
 - (a) procedures and manoeuvres for basic instrument flight covering at least:
 - (i) basic instrument flight without external visual cues;
 - (ii) horizontal flight;
 - (iii) climbing;
 - (iv) descent;
 - (v) turns in level flight, climbing and descent;

- (vi) instrument pattern;
 - (vii) steep turn;
 - (viii) radio navigation;
 - (ix) recovery from unusual attitudes;
 - (x) limited panel; and
 - (xi) recognition and recovery from incipient and full stall;
 - (b) pre-flight procedures for IFR flights, including the use of the flight manual and appropriate air traffic services documents for the preparation of an IFR flight plan;
 - (c) procedure and manoeuvres for IFR operation under normal, abnormal, and emergency conditions covering at least:
 - (i) transition from visual to instrument flight on take-off;
 - (ii) standard instrument departures and arrivals;
 - (iii) en route IFR procedures;
 - (iv) holding procedures;
 - (v) instrument approaches to specified minima;
 - (vi) missed approach procedures; and
 - (vii) landings from instrument approaches, including circling;
 - (d) in-flight manoeuvres and particular flight characteristics;
 - (e) if required, operation of a multi-engine aeroplane in the above exercises, including:
 - (i) operation of the aeroplane solely by reference to instruments with one engine simulated inoperative;
 - (ii) engine shutdown and restart (to be carried out at a safe altitude unless carried out in an FFS or FNPT II).
8. Applicants for the competency-based modular IR(A) holding a Part-FCL PPL or CPL and a valid IR(A) issued in compliance with the requirements of Annex 1 to the Chicago Convention by a third country may be credited in full towards the training course mentioned in paragraph 4. In order to be issued the IR(A), the applicant shall:
- (a) successfully complete the skill test for the IR(A) in accordance with Appendix 7;
 - (b) demonstrate to the examiner during the skill test that he/she has acquired an adequate level of theoretical knowledge of air law, meteorology and flight planning and performance (IR); and
 - (c) have a minimum experience of at least 50 hours of flight time under IFR as PIC on aeroplanes.
9. Applicants for the competency-based modular IR(A) who hold a BIR in accordance with point [FCL.835](#) and who have received at least 10 hours of instrument flight time under instruction at an ATO may be credited towards the training course referred to in paragraph 4, provided that all competency-based instrument rating topics have been included in that BIR training, and assessed by the ATO that provides the competency-based modular flying training course.

10. Applicants for the competency-based modular IR(A) who hold a BIR and have experience of at least 50 hours of flight time under IFR as PIC on aeroplanes, shall:
 - (a) at an ATO:
 - (i) be assessed as having an acceptable standard of competency-based instrument rating theoretical knowledge;
 - (ii) receive appropriate flight training to extend IFR privileges in accordance with [FCL.605.IR\(a\)](#);
 - (b) after completion of (a);
 - (i) successfully complete the skill test for the IR(A) in accordance with [Appendix 7](#);
 - (ii) demonstrate orally to the examiner during the skill test that they have acquired an adequate level of theoretical knowledge of air law, meteorology, and flight planning and performance.

PRE-ENTRY ASSESSMENT

11. The content and duration of the pre-entry assessment shall be determined by the ATO based on the prior instrument experience of the applicant.

MULTI-ENGINE

12. The holder of a single-engine IR(A) who also holds a multi-engine class or type rating wishing to obtain a multi-engine IR(A) for the first time shall complete a course at an ATO comprising at least 5 hours instrument time under instruction in multi-engine aeroplanes, of which 3 hours may be in an FFS or FNPT II and shall pass a skill test.

B. IR(H) — Modular flying training course

1. The aim of the IR(H) modular flying training course is to train pilots to the level of proficiency necessary to operate helicopters under IFR and in IMC.
2. An applicant for a modular IR(H) course shall be the holder of a PPL(H), or a CPL(H) or an ATPL(H). Prior to commencing the aircraft instruction phase of the IR(H) course, the applicant shall be the holder of the helicopter type rating used for the IR(H) skill test, or have completed approved type rating training on that type. The applicant shall hold a certificate of satisfactory completion of MCC if the skill test is to be conducted in Multi- Pilot conditions.
3. An applicant wishing to undertake a modular IR(H) course shall be required to complete all the instructional stages in one continuous approved course of training.
4. The course of theoretical instruction shall be completed within 18 months. The flight instruction and the skill test shall be completed within the period of validity of the pass in the theoretical examinations.
5. The course shall comprise:
 - (a) theoretical knowledge instruction to the IR knowledge level;
 - (b) instrument flight instruction.

THEORETICAL KNOWLEDGE

6. An approved modular IR(H) course shall comprise at least 150 hours of instruction.

FLYING TRAINING

7. An IR(H) course shall comprise at least 55 hours instrument time under instruction, of which:
 - (a) up to 20 hours may be instrument ground time in an FNPT I (H) or (A). These 20 hours instruction time in FNPT I (H) or (A) may be substituted by 20 hours instruction time for IR(H) in an aeroplane, approved for this course; or
 - (b) up to 40 hours may be instrument ground time in a helicopter FTD 2/3, FNPT II/III or FFS.The instrument flight instruction shall include at least 10 hours in an IFR-certificated helicopter.
- 8.1 Holders of an ATPL(H) shall have the theoretical knowledge instruction hours reduced by 50 hours.
- 8.2 The holder of an IR(A) may have the amount of training required reduced to 10 hours.
- 8.3 The holder of a PPL(H) with a helicopter night rating or a CPL(H) may have the total amount of instrument time under instruction required reduced by 5 hours.
9. The flying exercises up to the IR(H) skill test shall comprise:
 - (a) pre-flight procedures for IFR flights, including the use of the flight manual and appropriate air traffic services documents in the preparation of an IFR flight plan;
 - (b) procedure and manoeuvres for IFR operation under normal, abnormal and emergency conditions covering at least:
 - transition from visual to instrument flight on take-off,
 - standard instrument departures and arrivals,
 - en-route IFR procedures,
 - holding procedures,
 - instrument approaches to specified minima,
 - missed approach procedures,
 - landings from instrument approaches, including circling;
 - (c) in-flight manoeuvres and particular flight characteristics;
 - (d) if required, operation of a multi-engine helicopter in the above exercises, including operation of the helicopter solely by reference to instruments with one engine simulated inoperative and engine shutdown and restart (the latter exercise to be carried out in an FFS or FNPT II or FTD 2/3).

C. IR(As) — Modular flying training course

GENERAL

1. The aim of the IR(As) modular flying training course is to train pilots to the level of proficiency necessary to operate airships under IFR and in IMC. The course consists of two modules, which may be taken separately or combined:
 - (a) **Basic Instrument Flight Module**

This comprises 10 hours of instrument time under instruction, of which up to 5 hours can be instrument ground time in a BITD, FNPT I or II, or an FFS. Upon completion of the Basic Instrument Flight Module, the candidate shall be issued a Course Completion Certificate.
 - (b) **Procedural Instrument Flight Module**

This comprises the remainder of the training syllabus for the IR(As), 25 hours instrument time under instruction, and the theoretical knowledge course for the IR(As).

2. An applicant for a modular IR(As) course shall be the holder of a PPL(As) including the privileges to fly at night or a CPL(As). An applicant for the Procedural Instrument Flight Module, who does not hold a CPL(As), shall be holder of a Course Completion Certificate for the Basic Instrument Flight Module.
3. An applicant wishing to undertake the Procedural Instrument Flight Module of a modular IR(As) course shall be required to complete all the instructional stages in one continuous approved course of training. Prior to commencing the Procedural Instrument Flight Module, the ATO shall ensure the competence of the applicant in basic instrument flying skills. Refresher training shall be given as required.
4. The course of theoretical instruction shall be completed within 18 months. The Procedural Instrument Flight Module and the skill test shall be completed within the period of validity of the pass in theoretical examinations.
5. The course shall comprise:
 - (a) theoretical knowledge instruction to the IR knowledge level;
 - (b) instrument flight instruction.

THEORETICAL KNOWLEDGE

6. An approved modular IR(As) course shall comprise at least 150 hours of theoretical knowledge instruction.

FLYING TRAINING

7. An IR(As) course shall comprise at least 35 hours instrument time under instruction of which up to 15 hours may be instrument ground time in an FNPT I, or up to 20 hours in an FFS or FNPT II. A maximum of 5 hours of FNPT II or FFS instrument ground time may be conducted in an FNPT I.
8. The holder of a CPL(As) or of a Course Completion Certificate for the Basic Instrument Flight Module may have the total amount of training required in paragraph 7 reduced by 10 hours. The total instrument flight instruction in airship shall comply with paragraph 7.
9. If the applicant is the holder of an IR in another category of aircraft the total amount of flight instruction required may be reduced to 10 hours on airships.
10. The flying exercises up to the IR(As) skill test shall comprise:
 - (a) Basic Instrument Flight Module:

Procedure and manoeuvre for basic instrument flight covering at least:

basic instrument flight without external visual cues:

- horizontal flight,
- climbing,
- descent,
- turns in level flight, climbing, descent;

instrument pattern;

radionavigation;

recovery from unusual attitudes;

limited panel;

(b) Procedural Instrument Flight Module:

- (i) pre-flight procedures for IFR flights, including the use of the flight manual and appropriate air traffic services documents in the preparation of an IFR flight plan;
- (ii) procedure and manoeuvres for IFR operation under normal, abnormal and emergency conditions covering at least:
 - transition from visual to instrument flight on take-off,
 - standard instrument departures and arrivals,
 - en-route IFR procedures,
 - holding procedures,
 - instrument approaches to specified minima,
 - missed approach procedures,
 - landings from instrument approaches, including circling;
- (iii) inflight manoeuvres and particular flight characteristics;
- (iv) operation of airship in the above exercises, including operation of the airship solely by reference to instruments with one engine simulated inoperative and engine shut-down and restart (the latter exercise to be carried out at a safe altitude unless carried out in an FFS or FNPT II).

AMC1 to Appendix 6 ‘Modular training course for the IR’

ED Decision 2019/017/R

ALL MODULAR FLYING TRAINING COURSES FOR THE IR, EXCEPT COMPETENCYBASED MODULAR FLYING TRAINING COURSE

- (a) The theoretical knowledge instruction may be given at an ATO conducting theoretical knowledge instruction only, in which case the head of training (HT) of that organisation should supervise that part of the course.
- (b) The 150 hours of instruction, which include the application of threat and error management (TEM), may include in suitable proportions:
 - (1) classroom work;
 - (2) lessons;
 - (3) tutorials;
 - (4) demonstrations, including those supported by demonstration equipment;
 - (5) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
 - (6) exercises that use demonstration equipment or training devices;
 - (7) directed study including workbook exercises or assignments;
 - (8) aerodrome or aviation industry field trips;
 - (9) computer-based training and e-learning elements;
 - (10) progress tests; and

- (11) other training methods, media and tools approved by the competent authority.

Approved distance-learning (correspondence) courses may also be offered as part of the course. The minimum amount of classroom instruction, as required by ORA.ATO.305, may include all of the above except item (b)(9).

AMC2 to Appendix 6 Modular training course for the IR

ED Decision 2014/022/R

SECTION A IR(A) - MODULAR FLYING TRAINING COURSE

Basic Instrument Flight Module Training Course

- (a) This 10 hours module is focused on the basics of flying by sole reference to instruments, including limited panel and unusual attitude recovery.
- (b) All exercises may be performed in an FNPT I or II or an FFS, for a maximum of 5 hours. If instrument flight training is in VMC, a suitable means of simulating IMC for the student should be used.
- (c) A BITD may be used for the exercises 1, 2, 3, 4, 6, and 8.
- (d) The use of the BITD is subject to the following:
 - (1) the training should be complemented by exercises on an aeroplane;
 - (2) the record of the parameters of the flight must be available;
 - (3) an FI(A) or IRI(A) should conduct the instruction.

EXERCISES

- (e) Exercise 1:
 - (1) basic instrument flying without external visual cues; 0:30 hours
 - (2) horizontal flight; power changes for acceleration or deceleration;
 - (3) maintaining straight and level flight;
 - (4) turns in level flight with 15 ° and 25 ° bank, left and right;
 - (5) roll-out onto predetermined headings.
- (f) Exercise 2:
 - (1) repetition of exercise 1; 0:45 hours
 - (2) additionally climbing, descending, maintaining heading and speed, transition to horizontal flight;
 - (3) climbing and descending turns.
- (g) Exercise 3:
 - Instrument pattern: 0:45 hours
 - (1) start exercise, decelerate to approach speed, flaps into approach configuration;
 - (2) initiate standard turn (left or right);
 - (3) roll out on opposite heading, maintain new heading for 1 minute
 - (4) standard turn, gear down, descend 500 ft/min;

- (5) roll out on initial heading, maintain descent (500 ft/min) and new heading for 1 minute;
- (6) transition to horizontal flight, 1000 ft below initial flight level;
- (7) initiate go-around;
- (8) climb at best rate of climb speed.
- (h) Exercise 4:
 - Repetition of exercise 1 and steep turns with 45° bank; recovery from unusual attitudes. 0:45 hours
- (i) Exercise 5:
 - Repetition of exercise 4. 0:45 hours
- (j) Exercise 6:
 - (1) radio navigation using VOR, NDB or, if available, VDF; 0:45 hours
 - (2) interception of predetermined QDM, QDR.
- (k) Exercise 7:
 - Repetition of exercise 1 and recovery from unusual attitudes. 0:45hours
- (l) Exercise 8:
 - (1) Repetition of exercise 1; 0:45 hours
 - (2) turns, level change and recovery from unusual attitudes with simulated failure of the artificial horizon or directional gyro.
- (m) Exercise 9:
 - Recognition of, and recovery from, incipient and full stalls. 0:45 hours
- (n) Exercise 10: Repetition of exercises 6, 8 3:30 hours and 9.

CERTIFICATE OF COMPLETION OF BASIC INSTRUMENT FLIGHT MODULE

CERTIFICATE OF COMPLETION OF BASIC INSTRUMENT FLIGHT MODULE			
Pilot's last name(s):		First name(s):	
Type of licence:		Number:	State:
Flight training hours performed on SE aeroplane:		OR	Flight training hours performed on ME aeroplane:

Flight training hours performed in an FSTD (maximum 5 hours):		
	Signature of applicant:	

The satisfactory completion of basic instrument flight module according to requirements is certified below:

TRAINING			
Basic instrument flight module training received during period:			
from:	to:	at:	ATO
Location and date:		Signature of head of training:	
Type and number of licence and state of issue:		Name(s) in capital letters of authorised instructor:	

AMC3 to Appendix 6 ‘Modular training courses for the IR’

ED Decision 2019/017/R

SECTION Aa IR(A) – COMPETENCY-BASED MODULAR FLYING TRAINING COURSE

(a) THEORETICAL KNOWLEDGE INSTRUCTION

- (1) The theoretical knowledge instruction may be given at an ATO conducting theoretical knowledge instruction only, in which case the HT of that ATO should supervise that part of the course.
- (2) The hours required for the theoretical knowledge instruction for the IR following the competency-based training route should be divided between the subjects and include the application of threat and error management (TEM) as based on the ATO’s systems course design and agreed upon between the competent authority and the ATO.

An approved course, which includes the application of threat and error management, may contain in suitable proportions:

- (i) classroom work;
- (ii) lessons;
- (iii) tutorials;
- (iv) demonstrations, including those supported by demonstration equipment;

- (v) exercises carried out as groups or individuals and based on pre-flight and en-route planning, communications, presentations and projects;
- (vi) exercises that use demonstration equipment or training devices;
- (vii) directed study including workbook exercises or assignments;
- (viii) aerodrome or aviation industry field trips;
- (ix) computer-based training and e-learning elements;
- (x) progress tests; and
- (xi) other training methods, media and tools approved by the competent authority.

Approved distance-learning (correspondence) courses may also be offered as part of the course. The minimum amount of classroom instruction, as required by ORA.ATO.305, may include all of the above except item (a)(2)(ix).'

(b) THEORETICAL KNOWLEDGE EXAMINATION

The applicant for the IR following the competency-based training route should pass an examination to demonstrate a level of theoretical knowledge appropriate to the privileges granted in the subjects further detailed in [FCL.615\(b\)](#). The number of questions per subject, the distribution of questions and the time allocated to each subject is detailed in AMC1 ARA.FCL.300(b).

AMC4 to Appendix 6 Modular training courses for the IR

ED Decision 2014/022/R

SECTION Aa IR(A) – COMPETENCY-BASED MODULAR FLYING TRAINING COURSE FLYING TRAINING

- (a) The instrument flight instruction outside an ATO provided by an IRI(A) or an FI(A) holding the privilege to provide training for the IR in accordance with Appendix 6 Section Aa (6)(a)(i)(A) may consist of instrument flight time under instruction or instrument ground time or a combination thereof.

TRAINING AIRCRAFT

- (b) The aeroplane used for the instrument flight training provided outside an ATO by an IRI(A) or FI(A) should be:
- (1) fitted with primary flight controls that are instantly accessible by both the student and the instructor (for example dual flight controls or a centre control stick). Swing-over flight controls should not be used; and
 - (2) suitably equipped to simulate instrument meteorological conditions (IMC) and for the instrument flight training required.
- (c) The FSTD used for the instrument flight instruction provided outside an ATO by an IRI(A) or FI(A) should be suitably equipped to simulate instrument meteorological conditions (IMC) and for the instrument flight training required

AMC5 to Appendix 6 Modular training courses for the IR

ED Decision 2025/002/R

SECTION Aa IR(A) – COMPETENCY-BASED MODULAR FLYING TRAINING COURSE (6)(a)(i)(B); (6)(b)(i)(B)

PRIOR EXPERIENCE OF FLIGHT TIME UNDER IFR AS PIC

A rating giving privileges to fly under IFR and in IMC referred to in points (6)(a)(i)(B) and (6)(b)(i)(B) may be any of the following:

- (a) an BIR rating issued by a competent authority of a Member State; or
- (b) a national instrument rating issued by a Member State prior to the application of Commission Regulation (EU) No 1178/2011; or
- (c) an instrument rating issued in compliance with the requirements of Annex 1 to the Chicago Convention by a third country; or
- (d) an authorisation issued by a Member State under Article 4(8) of Commission Regulation (EU) No 1178/2011.

The amount of credit given should not exceed the amount of hours completed as instrument flight time.

AMC6 to Appendix 6 Modular training courses for the IR

ED Decision 2014/022/R

SECTION Aa IR(A) – COMPETENCY-BASED MODULAR FLYING TRAINING COURSE

(6)(a)(ii); (6)(b)(ii)

PRIOR INSTRUMENT FLIGHT TIME UNDER INSTRUCTION

Prior instrument flight time under instruction on aeroplanes, as referred in (6)(a)(ii) and (6)(b)(ii), may be instrument flight time completed for the issue of:

- (a) an EIR rating issued by a competent authority of a Member State; or
- (b) a national instrument rating prior to the application of Commission Regulation (EU) No 1178/2011; or
- (c) an instrument rating in compliance with the requirements of Annex 1 to the Chicago Convention by a third country; or
- (d) an authorisation issued by a Member State under Article 4(8) of Commission Regulation (EU) No 1178/2011.

AMC7 to Appendix 6 Modular training courses for the IR

ED Decision 2014/022/R

SECTION Aa IR(A) – COMPETENCY-BASED MODULAR FLYING TRAINING COURSE

(6)(c); (6)(d)

PRE-ENTRY ASSESSMENT AND TRAINING RECORD

(a) PRE-ENTRY ASSESSMENT

The assessment to establish the amount of training to be credited and to identify the training needs should be based on the training syllabus established in [Appendix 6 Aa](#).

(b) TRAINING RECORD

- (1) Before initiating the assessment the applicant should provide to an ATO a training record containing the details of the previous flight instruction provided by the IRI(A) or the FI(A). This training record should at least specify the aircraft type and registration used for the training, the number of flights and the total amount of instrument time under instruction. It should also specify all the exercises completed during the training by using the syllabus contained in [Appendix 6 Aa](#).

- (2) The instructor having provided the training should keep the training records containing all the details of the flight training given for a period of at least 5 years after the completion of the training.

AMC8 to Appendix 6 Modular training courses for the IR

ED Decision 2014/022/R

SECTION Aa IR(A) – COMPETENCY-BASED MODULAR FLYING TRAINING COURSE

(8)

In order to be credited in full towards the multi-engine IR(A) training course requirements, the applicant should

- (a) hold a multi-engine IR(A), issued in accordance with the requirements of Annex 1 to the Chicago Convention by a third country;
- (b) have the minimum experience required in [Appendix 6](#) Aa paragraph 8(c), of which at least 15 hours should be completed in a multi-engine aeroplane.

AMC9 to Appendix 6 Modular training courses for the IR

ED Decision 2014/022/R

AIRSHIPS

Basic Instrument Flight Module Training Course

- (a) This 10 hours module is focused on the basics of flying by sole reference to instruments, including limited panel and unusual attitude recovery.
- (b) All exercises may be performed in an FNPT I or II or an FFS, for a maximum of 5 hours. If instrument flight training is in VMC, a suitable means of simulating IMC for the student should be used.
- (c) A BITD may be used for the exercises 1, 2, 3, 4, 6 and 8.
- (d) The use of the BITD is subject to the following:
 - (1) the training should be complemented by exercises on an airship;
 - (2) the record of the parameters of the flight must be available;
 - (3) an FI(As) or IRI(As) should conduct the instruction.

EXERCISES

- (e) Exercise 1:
 - (1) basic instrument flying without external visual cues; 0:30 hours
 - (2) horizontal flight;
 - (3) maintaining straight and level flight;
 - (4) turns in level flight, left and right;
 - (5) rollout onto predetermined headings.
- (f) Exercise 2:
 - (1) Repetition of exercise 1; 0:45 hours

- additionally climbing and descending
- (2) maintaining heading and speed;
 - (3) transition to horizontal flight;
 - (4) climbing and descending turns.
- (g) Exercise 3:
- Instrument pattern: 0:45 hours
- (1) start exercise, decelerate to approach speed, approach configuration;
 - (2) initiate standard turn (left or right);
 - (3) rollout on opposite heading, maintain new heading for 1 minute;
 - (4) standard turn, descend with given rate (for example 500 ft/min);
 - (5) rollout on initial heading, maintain descent (for example 500 ft/min) and new heading for 1 minute;
 - (6) transition to horizontal flight (for example 1 000 ft below initial level);
 - (7) initiate go-around;
 - (8) climb at best rate of climb speed.
- (h) Exercise 4:
- (1) repetition of exercise 1; 0:45 hours
 - (2) recovery from unusual attitudes.
- (i) Exercise 5
- Repetition of exercise 4. 0:45 hours
- (j) Exercise 6
- (1) radio navigation using VOR, NDB 0:45 hours or, if available, VDF;
 - (2) interception of predetermined QDM, QDR.
- (k) Exercise 7
- (1) repetition of exercise 1; 0:45 hours
 - (2) recovery from unusual attitudes.
- (l) Exercise 8
- (1) repetition of exercise 1; 0:45 hours
 - (2) turns, level change and recovery from unusual attitudes with simulated failure of the artificial horizon or directional gyro.
- (m) Exercise 9
- Repetition of exercises (6) and (8). 4:15 hours

CERTIFICATE OF COMPLETION OF BASIC INSTRUMENT FLIGHT MODULE

CERTIFICATE OF COMPLETION OF BASIC INSTRUMENT FLIGHT MODULE

Pilot's last name(s):		First name(s):	
Type of licence:		Number:	State:
Flight training hours performed on airship:			
Flight training hours performed in an FSTD (maximum 5 hours):			
	Signature of applicant:		

The satisfactory completion of basic instrument flight module according to requirements is certified below:

TRAINING			
Basic instrument flight module training received during period:			
from:	to:	at:	ATO
Location and date:		Signature of head of training:	
Type and number of licence and state of issue:		Name(s) in capital letters of authorised instructor:	

GM1 to Appendix 6 Modular training courses for the IR

ED Decision 2017/022/R

Aa. IR(A)(8)

The following elements may be used by the examiner for the applicant's verbal demonstration of knowledge:

(a) AIR LAW:

- (1) explain the requirements for plus validity and privileges of instrument ratings;
- (2) explain why a time check has to be completed before flight;
- (3) describe the necessary action when an aircraft experiences a failure in communications;

- (4) state the responsibility of the operator when unable to utilise the published departure procedures;
- (5) explain when the omnidirectional method is used for departure;
- (6) describe the solutions when omnidirectional procedures are not possible;
- (7) justify the establishment of aircraft categories for the approach;
- (8) state the minimum obstacle clearance provided by the minimum sector altitudes (MSAs) established for an aerodrome;
- (9) describe the point of origin, shape, size, and subdivisions of the area used for MSAs;
- (10) explain why a pilot should not descend below obstacle clearance altitude/height (OCA/H) without visual reference, which is established for precision approach procedures, non-precision approach procedures and visual (circling) procedures;
- (11) translate the following acronyms into plain language: decision altitude (DA), decision height (DH), obstacle clearance altitude (OCA), obstacle clearance height (OCH), minimum decision altitude (MDA), minimum decision height (MDH), minimum obstacle clearance (MOC), decision altitude/height (DA/H), obstacle clearance altitude/height (OCA/H) and minimum decision altitude/height (MDA/H);
- (12) explain the relationship between the following: DA, DH, OCA, OCH, MDA, MDH, MOC, DA/H, OCA/H and MDA/H;
- (13) define the following terms: initial approach fix (IAF), intermediate fix (IF), final approach fix (FAF), missed approach point (MAPt) and turning point;
- (14) state the accuracy of facilities providing track (omnidirectional radio range (VOR), instrument landing system (ILS), non-directional beacon (NDB));
- (15) state the optimum descent gradient (preferred for a precision approach) in degrees and per cent;
- (16) name the five standard segments of an instrument approach procedure and state the beginning and end for each of them;
- (17) describe where an arrival (ARR) route normally ends;
- (18) state whether or not omnidirectional or sector ARRs are possible to be made;
- (19) explain the main task of the initial approach segment;
- (20) describe the main task of the intermediate approach segment;
- (21) state the main task of the final approach segment;
- (22) name the two possible aims of a final approach;
- (23) explain the term 'final approach point' in case of an ILS approach;
- (24) state what happens if an ILS glide path (GP) becomes inoperative during approach;
- (25) describe the main task of a missed approach procedure;
- (26) define 'MAPt';
- (27) state the pilot's reaction if upon reaching the MAPt, the required visual reference is not established;

- (28) describe what a pilot is expected to do in the event that a missed approach is initiated prior to arriving at the MAPt (a missed approach, after an approach flown as CDFA, should be made when reaching the MAPt or DA/H, whichever occurs first);
- (29) state whether the pilot is obliged to cross the MAPt at the A/H required by the procedure or whether they are allowed to cross the MAPt at an A/H greater than that required by the procedure;
- (30) describe what is meant by 'visual manoeuvring (circling)';
- (31) state the conditions to be fulfilled before descending below MDA/H in a visual manoeuvring (circling) approach;
- (32) state how the pilot is expected to behave after initial visual contact during a visual manoeuvring (circling);
- (33) describe what the pilot is expected to do if visual reference is lost while circling to land from an instrument approach;
- (34) describe the shape and terminology associated with the holding pattern;
- (35) state the bank angle and rate of turn to be used whilst flying in a holding pattern;
- (36) explain why pilots in a holding pattern should attempt to maintain tracks and how this is achieved;
- (37) describe where outbound timing begins in a holding pattern;
- (38) state where the outbound leg in a holding pattern terminates if the outbound leg is based on distance-measuring equipment (DME);
- (39) describe the three entry headings for entries into a holding pattern;
- (40) define the terms 'parallel entry', 'offset entry', and 'direct entry';
- (41) determine the correct entry procedure for a given holding pattern;
- (42) state the still-air time for flying on the outbound entry heading with or without DME;
- (43) define the following Q codes: 'QNH' and 'QFE';
- (44) define 'flight level' (FL);
- (45) state the intervals by which consecutive FLs should be separated;
- (46) describe how FLs are numbered;
- (47) define the term 'transition altitude';
- (48) define the term 'transition level';
- (49) state how the vertical position of the aircraft should be expressed at or below the transition altitude and transition level;
- (50) define the term 'transition layer';
- (51) state when the QNH altimeter setting should be made available to departing aircraft;
- (52) state how a QNH altimeter setting should be made available to aircraft approaching a controlled aerodrome for landing;
- (53) state where during the climb, the altimeter setting should be changed from QNH to 1013.2 hPa;

- (54) describe when a pilot of an aircraft intending to land at an aerodrome should obtain the transition level;
- (55) describe when a pilot of an aircraft intending to land at an aerodrome should obtain the actual QNH altimeter setting;
- (56) state where the altimeter settings should be changed from 1013.2 hPa to QNH during descent for landing;
- (57) state the modes and codes that the pilot should operate in the absence of any air traffic control (ATC) directions or regional air navigation agreements;
- (58) state when the pilot should 'squawk ident';
- (59) state the transponder mode and code to indicate: a state of emergency, a failure in communications, an unlawful interference;
- (60) describe the consequences of an in-flight transponder failure;
- (61) state the primary action of the pilot in the case of an unserviceable transponder before departure when no repair or replacement at that aerodrome is possible;
- (62) understand the various rules and services that apply to the various classes of airspace;
- (63) describe the aim of clearances issued by the ATC with regard to instrument flight rules (IFR), visual flight rules (VFR) or special VFR flights, and refer to the different airspaces;
- (64) explain what is meant by the expression 'clearance limit';
- (65) explain the meaning of the phrases 'cleared via flight planned route', 'cleared via (designation) departure' and 'cleared via (designation) ARR' in an ATC clearance;
- (66) list which items of an ATC clearance should always be read back by the flight crew;
- (67) justify the speed control by the ATC;
- (68) explain how the change from IFR to VFR may be initiated by the pilot in command (PIC);
- (69) define the following terms: 'transition level', 'transition layer', and 'transition altitude';
- (70) indicate how the vertical position of an aircraft in the vicinity of an aerodrome should be expressed at or below the transition altitude, at or above the transition level, and while climbing or descending through the transition layer;
- (71) list the six items that are normally included in a voice position report;
- (72) name the item of a position report which must be forwarded to the ATC with the initial call after changing to a new frequency;
- (73) understand the difference among the types of separation within the various classes of airspace and among the various types of flight;
- (74) state who is responsible for the avoidance of collision with other aircraft when operating in visual meteorological conditions (VMC);
- (75) explain the term 'expected approach time' and the procedures for its use;
- (76) state the reasons which may probably lead to the decision to use another take-off or landing direction than the one into the wind;
- (77) define the term 'radar vectoring';
- (78) explain the procedures for the conduct of surveillance radar approaches (SRAs);

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- (79) state the mode and code of secondary surveillance radar (SSR) equipment that a pilot may operate in a (general) state of emergency, or (specifically) in case the aircraft is subject to unlawful interference;
 - (80) describe the expected action of the aircraft after receiving a broadcast from air traffic services (ATS) concerning the emergency descent of another aircraft;
 - (81) name the colours used for the various markings (runway (RWY), taxiway (TWY), aircraft stands, apron safety lines);
 - (82) describe the application and characteristics of RWY centre line markings and threshold markings;
 - (83) describe the wing bars of a precision approach path indicator (PAPI) and an abbreviated precision approach path indicator (A-PAPI); and
 - (84) interpret what the pilot sees during approach, using a PAPI, an APAPI, a T visual approach slope indicating system (TVASIS), and an abbreviated T visual approach slope indicator system (ATVASIS);
- (b) **FLIGHT PLANNING AND FLIGHT MONITORING:**
- (1) select the preferred airway(s) or route(s) considering:
 - (i) altitudes and FLs,
 - (ii) standard routes,
 - (iii) ATC restrictions,
 - (iv) the shortest distance,
 - (v) obstacles, and
 - (vi) any other relevant data;
 - (2) determine courses and distances from en route charts;
 - (3) determine bearings and distances of waypoints based on radio navigation aids on en route charts;
 - (4) define the following altitudes:
 - (i) minimum en route altitude (MEA),
 - (ii) minimum obstacle clearance altitude (MOCA),
 - (iii) minimum off-route altitude (MORA),
 - (iv) grid minimum off-route altitude (Grid MORA),
 - (v) maximum authorised altitude (MAA),
 - (vi) minimum crossing altitude (MCA), and
 - (vii) minimum holding altitude (MHA);
 - (5) extract the following altitudes from the chart(s):
 - (i) MEA,
 - (ii) MOCA,
 - (iii) MORA,
 - (iv) Grid MORA,

- (v) MAA,
- (vi) MCA, and
- (vii) MHA;
- (6) explain the reasons for studying standard instrument departure (SID) and standard ARR (STAR) charts;
- (7) state the reasons why the SID and STAR charts show procedures only in a pictorial presentation style which is not to scale;
- (8) interpret all data and information represented on SID and STAR charts, particularly:
 - (i) routings,
 - (ii) distances,
 - (iii) courses,
 - (iv) radials,
 - (v) altitudes/levels,
 - (vi) frequencies, and
 - (vii) restrictions;
- (9) identify SIDs and STARs which may be relevant to a planned flight;
- (10) state the reasons why it is imperative to be familiar with instrument approach procedures and appropriate data for departure, destination, and alternate airfields prior to departure;
- (11) select instrument approach procedures appropriate for departure, destination, and alternate airfields;
- (12) interpret all procedures, data and information represented on instrument approach charts, particularly:
 - (i) courses and radials,
 - (ii) distances,
 - (iii) altitudes, levels or heights,
 - (iv) restrictions,
 - (v) obstructions,
 - (vi) frequencies,
 - (vii) speeds and times,
 - (viii) DA/Hs and MDA/H,
 - (ix) visibility and runway visual ranges (RVRs), and
 - (x) approach light systems;
- (13) find communications (COM) frequencies and call signs for the following:
 - (i) control agencies, service facilities, and flight information services (FISs),
 - (ii) weather information stations, and
 - (iii) automatic terminal information service (ATIS);

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- (14) find the frequency and/or identifiers of radio navigation aids;
 - (15) complete the navigation plan with the courses, distances, and frequencies taken from charts;
 - (16) find standard instrument departure and ARR routes to be flown or to be expected;
 - (17) determine the position of top of climb (TOC) and top of descent (TOD), considering appropriate data;
 - (18) determine variation and calculate magnetic/true courses;
 - (19) calculate true airspeed (TAS) according to given aircraft performance data, altitude, and outside air temperature (OAT);
 - (20) calculate wind correction angles (WCA)/drift and ground speeds (GSs);
 - (21) determine all relevant altitudes/levels, particularly MEA, MOCA, MORA, MAA, MCA, MRA, and MSA;
 - (22) calculate individual and accumulated times for each leg until destination and alternate airfields;
 - (23) convert between volume, mass, and density given in different units commonly used in aviation;
 - (24) determine relevant data from the flight manual, such as fuel capacity, fuel flow/consumption at different power/thrust settings, altitudes, and atmospheric conditions;
 - (25) calculate attainable flight time/range considering fuel flow/consumption and available amount of fuel;
 - (26) calculate the required fuel considering fuel flow/consumption and required time/range to be flown;
 - (27) calculate the required fuel for an IFR flight considering expected meteorological conditions and expected delays under defined conditions;
 - (28) find and analyse the latest state at the departure, destination, and alternate aerodromes, in particular with regard to:
 - (i) opening hours,
 - (ii) work in progress (WIP),
 - (iii) special procedures due to WIP,
 - (iv) obstructions, and
 - (v) changes of frequencies for COM, navigation aids, and facilities;
 - (29) find and analyse the latest en route state with regard to:
 - (i) airway(s) or route(s),
 - (ii) restricted, dangerous, and prohibited areas, and
 - (iii) changes of frequencies for COM, navigation aids, and facilities;
 - (30) state the reasons for a fixed format of an International Civil Aviation Organization (ICAO) air traffic services flight plan (ATS FPL);

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- (31) determine the correct entries to complete an FPL, as well as decode and interpret the entries in a completed FPL, particularly as regards the following:
- (i) aircraft identification (Item 7),
 - (ii) flight rules and type of flight (Item 8),
 - (iii) number and type of aircraft and wake turbulence category (Item 9),
 - (iv) equipment (Item 10),
 - (v) departure aerodrome and time (Item 13),
 - (vi) route (Item 15),
 - (vii) destination aerodrome, total estimated elapsed time, and alternate aerodrome (Item 16),
 - (viii) other information (Item 18), and
 - (ix) supplementary information (Item 19);
- (32) complete the FPL using information from the following:
- (i) navigation plan,
 - (ii) fuel plan, and
 - (iii) operator's records on basic aircraft information;
- (33) explain the requirements for the submission of an ATS FPL;
- (34) explain the action to be taken in case of FPL changes;
- (35) state the action to be taken in case of inadvertent changes to track, TAS, and time estimate, affecting the current FPL; and
- (36) explain the procedures for closing an FPL;
- (c) METEOROLOGY:
- (1) describe qualitatively and quantitatively the temperature lapse rates of the troposphere (mean value of 0.65 °C/100 m or 2 °C/1 000 ft and actual values);
 - (2) explain the characteristics of inversions and of an isothermal layer;
 - (3) explain the cooling and warming of the air on the earth or sea surfaces;
 - (4) describe qualitatively the influence of the clouds on the cooling and warming of the earth or sea surfaces as well as of the air near those surfaces;
 - (5) explain the influence of the wind on the cooling and warming of the air near the earth or sea surfaces;
 - (6) define 'atmospheric pressure';
 - (7) list the units of measurement of atmospheric pressure used in aviation (hPa, in.);
 - (8) describe isobars on the surface weather charts;
 - (9) explain the pressure variation with height;
 - (10) describe qualitatively the variation of the barometric lapse rate (note: the average value for the barometric lapse rate near mean sea level is 27 ft (8 m) per 1 hPa, whereas at about 5 500 m above mean sea level (AMSL) is 50 ft (15 m) per 1 hPa;

- (11) describe and interpret contour lines (isohypses) on a constant pressure chart;
- (12) describe the relationship between pressure, temperature, and density;
- (13) describe the vertical variation of the air density in the atmosphere;
- (14) describe the effect of humidity changes on the air density;
- (15) explain the use of standardised values for the international standard atmosphere (ISA);
- (16) list the main values of ISA (mean sea level pressure, mean sea level temperature, a vertical temperature lapse rate up to 20 km, as well as height and temperature of the tropopause);
- (17) calculate the standard temperature in Celsius degrees for a given FL;
- (18) determine a standard temperature deviation based on the difference between the given OAT and the standard temperature;
- (19) define the following terms and acronyms and explain how they are related to each other: H, A, pressure A, FL, pressure level, true A, true H, elevation, QNH, QFE, and standard altimeter setting;
- (20) describe the following terms: transition A, transition level, transition layer, terrain clearance, and lowest usable FL;
- (21) calculate the different readings on the altimeter when the pilot changes the altimeter setting;
- (22) illustrate with a numbered example the changes of the altimeter setting and the associated changes in reading when the pilot climbs through the transition altitude or descends through the transition level;
- (23) derive the reading of the altimeter of an aircraft on the ground when the pilot uses different settings;
- (24) explain the influence of the air temperature on the distance between the ground and the level reading on the altimeter as well as between two FLs;
- (25) explain the influence of pressure areas on the true altitude;
- (26) determine the true A/H for a given A/H and a given ISA temperature deviation;
- (27) describe why and how the wind changes direction and speed with H in the friction layer in the northern and southern hemisphere (rule of thumb);
- (28) describe and explain the origin and formation of mountain waves;
- (29) explain how mountain waves may be identified through their associated meteorological phenomena;
- (30) describe turbulence and gustiness;
- (31) list common types of turbulence (convective, mechanical, orographic, frontal, and clear-air turbulence);
- (32) indicate the sources of atmospheric humidity;
- (33) define 'dew point';
- (34) define 'relative humidity';
- (35) describe the relationship between temperature and dew point;

- (36) estimate the relative humidity of the air based on the difference between dew point and temperature;
- (37) explain the influence of relative humidity on the H of the cloud base;
- (38) list cloud types typical for stable and unstable air conditions;
- (39) identify by shape cirriform, cumuliform, and stratiform clouds;
- (40) explain the influence of inversions on vertical movements in the atmosphere;
- (41) name the factors contributing in general to the formation of fog and mist;
- (42) name the factors contributing to the formation of haze;
- (43) describe significant characteristics of orographic fog;
- (44) summarise the conditions for the dissipation of orographic fog;
- (45) list and describe the types of precipitation given in the aerodrome forecast (TAF) and aerodrome routine meteorological report (METAR) codes (drizzle, rain, snow, snow grains, ice pellets, hail, small hail, snow pellets, ice crystals, freezing drizzle, and freezing rain);
- (46) assign typical precipitation types and intensities to different clouds;
- (47) describe the boundaries between air masses (fronts);
- (48) define 'front' and 'frontal surface' ('frontal zone');
- (49) define 'warm front';
- (50) describe the cloud, weather, ground visibility, and aviation hazards at a warm front depending on the stability of the warm air;
- (51) explain the seasonal differences in the weather at warm fronts;
- (52) describe the structure, slope, and dimensions of a warm front;
- (53) define 'cold front';
- (54) explain the seasonal differences in the weather at cold fronts;
- (55) describe the structure, slope, and dimensions of a cold front;
- (56) describe the cloud, weather, ground visibility, and aviation hazards in a warm sector;
- (57) describe the cloud, weather, ground visibility, and aviation hazards behind the cold front;
- (58) define the term 'occlusion';
- (59) identify the typical flat pressure pattern on a surface weather chart;
- (60) describe the weather associated with a flat pressure pattern;
- (61) explain the general weather conditions under which ice accretion on airframe occurs;
- (62) indicate in which circumstances ice may form on an aircraft on the ground: air temperature, humidity, precipitation;
- (63) explain in which circumstances ice may form on an aircraft in flight: inside clouds, in precipitation, outside clouds, and in the absence of precipitation;
- (64) describe the different factors influencing the intensity of icing: air temperature, amount of supercooled water in a cloud or in precipitation, amount of ice crystals in the air, speed of the aircraft, shape (thickness) of the airframe parts (wings, antennas, etc.);

- (65) define 'clear ice';
- (66) define 'rime ice';
- (67) define 'hoar frost';
- (68) state the ICAO qualifying terms for the intensity of icing;
- (69) describe in general the hazards of icing;
- (70) assess the dangers of the different types of ice accretion;
- (71) state the ICAO qualifying terms for the intensity of turbulence;
- (72) describe the effects of turbulence on an aircraft in flight;
- (73) indicate the possibilities of avoiding turbulence
 - (i) in the flight planning: weather briefing, choice of track, and altitude, and
 - (ii) during flight: choice of appropriate track and altitude;
- (74) define 'wind shear' (vertical and horizontal);
- (75) describe the conditions in which wind shear forms and how it forms (e.g. thunderstorms, squall lines, fronts, inversions, land and sea breeze, friction layer, and relief);
- (76) describe the effects of wind shear on flight;
- (77) indicate the possibilities of avoiding wind shear in flight:
 - (i) in the flight planning, and
 - (ii) during flight;
- (78) name the cloud types which indicate the development of thunderstorms;
- (79) describe the different types of thunderstorms, their location, the conditions for and the process of their development, and list their properties (air mass thunderstorms, frontal thunderstorms, squall lines, supercell storms, orographic thunderstorms);
- (80) assess the average duration of thunderstorms and their different stages;
- (81) summarise the flight hazards of a fully developed thunderstorm;
- (82) describe and assess 'St. Elmo's fire';
- (83) describe the effect of lightning strike on aircraft and flight execution;
- (84) describe practical examples of flight techniques used to avoid the hazards of thunderstorms;
- (85) describe the influence of a mountainous terrain on cloud and precipitation;
- (86) describe the effects of the foehn;
- (87) describe the influence of a mountainous area on a frontal passage;
- (88) indicate the turbulent zones (mountain waves, rotors) on a sketch of a mountain chain;
- (89) describe the reduction of visibility caused by precipitation (drizzle, rain, and snow);
- (90) describe the differences between ground visibility, flight visibility, slant visibility, and vertical visibility when an aircraft is above or within a layer of haze or fog;
- (91) define 'ground visibility';

- (92) list the units used for visibility (m, km);
- (93) define 'RVR';
- (94) list the units used for RVR (m);
- (95) compare visibility and RVR;
- (96) define 'ceiling';
- (97) name the unit and the reference level used for information about the cloud base (ft);
- (98) define 'vertical visibility';
- (99) name the unit used for vertical visibility (ft);
- (100) interpret ground-weather radar images;
- (101) describe the basic principle of airborne weather radars as well as the type of information they provide;
- (102) describe the limits and errors of airborne weather radar information;
- (103) interpret typical airborne weather radar images;
- (104) decode and interpret significant weather charts (low-, medium-, and high-level charts);
- (105) describe the flight conditions at designated locations or along a defined flight route at a given FL, based on a significant weather chart;
- (106) describe, decode (by using a code table), and interpret the following aviation weather messages (given in written or graphical format):
 - (i) METAR;
 - (ii) aerodrome special meteorological reports (SPECI);
 - (iii) trend forecast (TREND);
 - (iv) TAF;
 - (v) information concerning en route weather phenomena which may affect the safety of aircraft operations (SIGMET);
 - (vi) information concerning en route weather phenomena which may affect the safety of low-level aircraft operations (AIRMET);
 - (vii) area forecast for low-level flights (GAMET);
 - (viii) automatic terminal information service (ATIS);
 - (ix) meteorological information for aircraft in flight (VOLMET);
 - (x) special air-report, and
 - (xi) volcanic-ash advisory information;
- (107) list in general the cases where a SIGMET and an AIRMET are issued; and
- (108) describe, decode (by using a code table), and interpret the following messages: runway state message (as written in a METAR) and general aviation forecast (GAFOR).

Appendix 7 – BIR and IR skill test

Regulation (EU) 2020/359

1. Applicants shall have received instruction on the same class or type of aircraft to be used in the test which shall be appropriately equipped for the training and testing purposes.
2. An applicant shall pass all the relevant sections of the skill test. If any item in a section is failed, that section is failed. Failure in more than one section will require the applicant to take the entire test again. An applicant failing only one section shall only repeat the failed section. Failure in any section of the retest, including those sections that have been passed on a previous attempt, will require the applicant to take the entire test again. All relevant sections of the skill test shall be completed within 6 months. Failure to achieve a pass in all relevant sections of the test in two attempts will require further training.
3. Further training may be required following a failed skill test. There is no limit to the number of skill tests that may be attempted.

CONDUCT OF THE TEST

4. The test is intended to simulate a practical flight. The route to be flown shall be chosen by the examiner. An essential element is the ability of the applicant to plan and conduct the flight from routine briefing material. The applicant shall undertake the flight planning and shall ensure that all equipment and documentation for the execution of the flight are on board. The duration of the flight shall be at least 1 hour.
5. Should the applicant choose to terminate a skill test for reasons considered inadequate by the examiner, the applicant shall retake the entire skill test. If the test is terminated for reasons considered adequate by the examiner, only those sections not completed shall be tested in a further flight.
6. At the discretion of the examiner, any manoeuvre or procedure of the test may be repeated once by the applicant. The examiner may stop the test at any stage if it is considered that the applicant's demonstration of flying skill requires a complete retest.
7. An applicant shall fly the aircraft from a position where the PIC functions can be performed and to carry out the test as if there is no other crew member. The examiner shall take no part in the operation of the aircraft, except when intervention is necessary in the interests of safety or to avoid unacceptable delay to other traffic. Responsibility for the flight shall be allocated in accordance with national regulations.
8. Decision heights/altitude, minimum descent heights/altitudes and missed approach point shall be determined by the applicant and agreed by the examiner.
9. Applicants shall indicate to the examiner the checks and duties carried out, including the identification of radio facilities. Checks shall be completed in accordance with the authorised checklist for the aircraft on which the test is being taken. During pre-flight preparation for the test applicants shall determine power settings and speeds. The applicants shall calculate performance data for take-off, approach and landing in compliance with the operations manual or flight manual for the aircraft used.

FLIGHT TEST TOLERANCES

10. The applicant shall demonstrate the ability to:
 - operate the aircraft within its limitations;
 - complete all manoeuvres with smoothness and accuracy;

exercise good judgment and airmanship;

apply aeronautical knowledge; and

maintain control of the aircraft at all times in such a manner that the successful outcome of a procedure or manoeuvre is never seriously in doubt.

11. The following limits shall apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the aircraft used.

Height

Generally	±100 feet
Starting a go-around at decision height/altitude	+50 feet/–0 feet
Minimum descent height/MAP/altitude	+50 feet/–0 feet

Tracking

On radio aids	±5°
For angular deviations	Half scale deflection, azimuth and glide path (e.g. LPV, ILS, MLS, GLS)
2D (LNAV) and 3D (LNAV/VNAV) “linear” lateral deviations	cross-track error/deviation shall normally be limited to ± ½ the RNP value associated with the procedure. Brief deviations from this standard up to a maximum of 1 time the RNP value are allowable.
3D linear vertical deviations (e.g. RNP APCH (LNAV/VNAV) using BaroVNAV)	not more than – 75 feet below the vertical profile at any time, and not more than + 75 feet above the vertical profile at or below 1 000 feet above aerodrome level.

Heading

all engines operating	±5°
with simulated engine failure	±10°

Speed

all engines operating	±5 knots
with simulated engine failure	+10 knots/–5 knots

CONTENT OF THE TEST

Aeroplanes

SECTION 1 — PRE-FLIGHT OPERATIONS AND DEPARTURE

Use of checklist, airmanship, anti-icing/de-icing procedures, etc., apply in all sections

- | | |
|---|---|
| a | Use of flight manual (or equivalent) especially a/c performance calculation, mass and balance |
| b | Use of Air Traffic Services document, weather document |
| c | Preparation of ATC flight plan, IFR flight plan/log |
| d | Identification of the required nav aids for departure, arrival and approach procedures |
| e | Pre-flight inspection |

f	Weather Minima
g	Taxiing
h	PBN departure (if applicable): — Check that the correct procedure has been loaded in the navigation system; and — Cross-check between the navigation system display and the departure chart.
i	Pre-take-off briefing, Take-off
j ^(*)	Transition to instrument flight
k ^(*)	Instrument departure procedures, including PBN departures, and altimeter setting
l ^(*)	ATC liaison — compliance, R/T procedures
SECTION 2 — GENERAL HANDLING ^(*)	
a	Control of the aeroplane by reference solely to instruments, including: level flight at various speeds, trim
b	Climbing and descending turns with sustained Rate 1 turn
c	Recoveries from unusual attitudes, including sustained 45° bank turns and steep descending turns
d ^(*)	Recovery from approach to stall in level flight, climbing/descending turns and in landing configuration
e	Limited panel: stabilised climb or descent, level turns at Rate 1 onto given headings, recovery from unusual attitudes
SECTION 3 — EN-ROUTE IFR PROCEDURES ^(*)	
a	Tracking, including interception, e.g. NDB, VOR, or track between waypoints
b	Use of navigation system and radio aids
c	Level flight, control of heading, altitude and airspeed, power setting, trim technique
d	Altimeter settings
e	Timing and revision of ETAs (en-route hold, if required)
f	Monitoring of flight progress, flight log, fuel usage, systems' management
g	Ice protection procedures, simulated if necessary
h	ATC liaison - compliance, R/T procedures
SECTION 3a — ARRIVAL PROCEDURES	
a	Setting and checking of navigational aids, and identification of facilities, if applicable
b	Arrival procedures, altimeter checks
c	Altitude and speed constraints, if applicable
d	PBN arrival (if applicable): — Check that the correct procedure has been loaded in the navigation system; and — Cross-check between the navigation system display and the arrival chart.
SECTION 4 ^(*) — 3D Operations ⁽⁺⁾	
a	Setting and checking of navigational aids Check Vertical Path angle For RNP APCH: — Check that the correct procedure has been loaded in the navigation system; and — Cross-check between the navigation system display and the approach chart.
b	Approach and landing briefing, including descent/approach/landing checks, including identification of facilities
c ⁽⁺⁾	Holding procedure
d	Compliance with published approach procedure
e	Approach timing
f	Altitude, speed heading control (stabilised approach)

g ⁽⁺⁾	Go-around action
h ⁽⁺⁾	Missed approach procedure/landing
i	ATC liaison – compliance, R/T procedures
SECTION 5^(°) – 2D OPERATIONS⁽⁺⁺⁾	
a	Setting and checking of navigational aids For RNP APCH: — Check that the correct procedure has been loaded in the navigation system; and — Cross-check between the navigation system display and the approach chart.
b	Approach and landing briefing, including descent/approach/landing checks, including identification of facilities
c ⁽⁺⁾	Holding procedure
d	Compliance with published approach procedure
e	Approach timing
f	Altitude/Distance to MAPT, speed, heading control (stabilised approach), Stop Down Fixes (SDF(s)), if applicable
g ⁽⁺⁾	Go-around action
h ⁽⁺⁾	Missed approach procedure/landing
i	ATC liaison – compliance, R/T procedures
SECTION 6 — FLIGHT WITH ONE ENGINE INOPERATIVE (multi-engine aeroplanes only) (°)	
a	Simulated engine failure after take-off or on go-around
b	Approach, go-around and procedural missed approach with one engine inoperative
c	Approach and landing with one engine inoperative
d	ATC liaison – compliance, R/T procedures

(°) Must be performed by sole reference to instruments.

(*) May be performed in an FFS, FTD 2/3 or FNPT II.

(+) May be performed in either Section 4 or Section 5.

(++) To establish PBN privileges, one approach in either Section 4 or Section 5 shall be an RNP APCH. Where an RNP APCH is not practicable, it shall be performed in an appropriately equipped FSTD.

Helicopters

SECTION 1 — DEPARTURE	
Use of checklist, airmanship, anti-icing/de-icing procedures, etc., apply in all sections	
a	Use of flight manual (or equivalent) especially aircraft performance calculation; mass and balance
b	Use of Air Traffic Services document, weather document
c	Preparation of ATC flight plan, IFR flight plan/log
d	Identification of the required navaids for departure, arrival and approach procedures
e	Pre-flight inspection
f	Weather minima
g	Taxiing/Air taxi in compliance with ATC or instructions of instructor
h	PBN departure (if applicable): — Check that the correct procedure has been loaded in the navigation system; and — Cross-check between the navigation system display and the departure chart.
i	Pre-take-off briefing, procedures and checks
j	Transition to instrument flight

k	Instrument departure procedures, including PBN procedures
SECTION 2 — GENERAL HANDLING	
a	Control of the helicopter by reference solely to instruments, including:
b	Climbing and descending turns with sustained Rate 1 turn
c	Recoveries from unusual attitudes, including sustained 30° bank turns and steep descending turns
SECTION 3 — EN-ROUTE IFR PROCEDURES	
a	Tracking, including interception, e.g. NDB, VOR, RNAV
b	Use of radio aids
c	Level flight, control of heading, altitude and airspeed, power setting
d	Altimeter settings
e	Timing and revision of ETAs
f	Monitoring of flight progress, flight log, fuel usage, systems management
g	Ice protection procedures, simulated if necessary and if applicable
h	ATC liaison – compliance, R/T procedures
SECTION 3a — ARRIVAL PROCEDURES	
a	Setting and checking of navigational aids, if applicable
b	Arrival procedures, altimeter checks
c	Altitude and speed constraints, if applicable
d	PBN arrival (if applicable) — Check that the correct procedure has been loaded in the navigation system; and — Cross-check between the navigation system display and the arrival chart.
SECTION 4 — 3D OPERATIONS⁽⁺⁾	
a	Setting and checking of navigational aids Check Vertical Path angle For RNP APCH: (a) Check that the correct procedure has been loaded in the navigation system; and (b) Cross-check between the navigation system display and the approach chart.
b	Approach and landing briefing, including descent/approach/landing checks
c ^(*)	Holding procedure
d	Compliance with published approach procedure
e	Approach timing
f	Altitude, speed, heading control (stabilised approach)
g ^(*)	Go-around action
h ^(*)	Missed approach procedure/landing
i	ATC liaison – compliance, R/T procedures
SECTION 5 — 2D OPERATIONS⁽⁺⁾	
a	Setting and checking of navigational aids For RNP APCH: — Check that the correct procedure has been loaded in the navigation system; and — Cross-check between the navigation system display and the approach chart.
b	Approach and landing briefing, including descent/approach/landing checks and identification of facilities
c ^(*)	Holding procedure
d	Compliance with published approach procedure
e	Approach timing
f	Altitude, speed, heading control (stabilised approach)
g ^(*)	Go-around action
h ^(*)	Missed approach procedure ^(*) /landing
i	ATC liaison – compliance, R/T procedures

SECTION 6 — ABNORMAL AND EMERGENCY PROCEDURES

This section may be combined with sections 1 through 5. The test shall have regard to control of the helicopter, identification of the failed engine, immediate actions (touch drills), follow-up actions and checks and flying accuracy, in the following situations:

a	Simulated engine failure after take-off and on/during approach ^(**) (at a safe altitude unless carried out in an FFS or FNPT II/III, FTD 2,3)
b	Failure of stability augmentation devices/hydraulic system (if applicable)
c	Limited panel
d	Autorotation and recovery to a pre-set altitude
e	3D operations manually without flight director ^(***) 3D operations manually with flight director ^(***)

(+) To establish PBN privileges, one approach in either Section 4 or Section 5 shall be an RNP APCH. Where an RNP APCH is not practicable, it shall be performed in an appropriately equipped FSTD.

(*) To be performed in Section 4 or Section 5.

(**) Multi-engine helicopter only.

(***) Only one item to be tested

Airships

SECTION 1 — PRE-FLIGHT OPERATIONS AND DEPARTURE

Use of checklist, airmanship, ATC liaison compliance, R/T procedures, apply in all sections

a	Use of flight manual (or equivalent) especially a/c performance calculation, mass and balance
b	Use of Air Traffic Services document, weather document
c	Preparation of ATC flight plan, IFR flight plan/log
d	Pre-flight inspection
e	Weather minima
f	Pre-take-off briefing, off mast procedure, manoeuvring on ground
g	Take-off
h	Transition to instrument flight
i	Instrument departure procedures, altimeter setting
j	ATC liaison - compliance, R/T procedures

SECTION 2 — GENERAL HANDLING

a	Control of the airship by reference solely to instruments
b	Climbing and descending turns with sustained rate of turn
c	Recoveries from unusual attitudes
d	Limited panel

SECTION 3 — EN-ROUTE IFR PROCEDURES

a	Tracking, including interception, e.g. NDB, VOR, RNAV
b	Use of radio aids
c	Level flight, control of heading, altitude and airspeed, power setting, trim technique
d	Altimeter settings
e	Timing and revision of ETAs
f	Monitoring of flight progress, flight log, fuel usage, systems' management
g	ATC liaison – compliance, R/T procedures

SECTION 4 — PRECISION APPROACH PROCEDURES

a	Setting and checking of navigational aids, identification of facilities
b	Arrival procedures, altimeter checks
c	Approach and landing briefing, including descent/approach/landing checks
d ⁽⁺⁾	Holding procedure
e	Compliance with published approach procedure
f	Approach timing
g	Stabilised approach (altitude, speed and heading control)
h ⁽⁺⁾	Go-around action
i ⁽⁺⁾	Missed approach procedure/landing
j	ATC liaison – compliance, R/T procedures
SECTION 5 — NON-PRECISION APPROACH PROCEDURES	
a	Setting and checking of navigational aids, identification of facilities
b	Arrival procedures, altimeter settings
c	Approach and landing briefing, including descent/approach/landing checks
d ⁽⁺⁾	Holding procedure
e	Compliance with published approach procedure
f	Approach timing
g	Stabilised approach (altitude, speed and heading control)
h ⁽⁺⁾	Go-around action
i ⁽⁺⁾	Missed approach procedure/landing
j	ATC liaison – compliance, R/T procedures
SECTION 6 — FLIGHT WITH ONE ENGINE INOPERATIVE	
This section may be combined with sections 1 through 5. The test shall have regard to control of the airship, identification of the failed engine, immediate actions, follow-up actions, checks and flying accuracy in the following situations:	
a	Simulated engine failure after take-off or on go-around
b	Approach and procedural go-around with one engine inoperative
c	Approach and landing, missed approach procedure, with one engine inoperative
d	ATC liaison – compliance, R/T procedures

⁽⁺⁾ May be performed in either section 4 or section 5.

AMC1 to Appendix 7 IR skill test

ED Decision 2025/002/R

LAPL, PPL, CPL, IR SKILL TEST AND PROFICIENCY CHECK APPLICATION AND REPORT FORM

APPLICATION AND REPORT FORM LAPL, PPL, CPL, IR SKILL TEST AND PROFICIENCY CHECK	
Applicant's last name(s):	
Applicant's first name(s):	LAPL: A ☐ H ☐
Signature of applicant:	
Type of licence*:	PPL: A ☒ ☐ As ☐
Licence number*:	CPL: A ☒ ☐ As ☐
State:	IR: A ☒ ☐ As ☐

1	Details of the flight			
Class, type of aircraft:		Registration:		
Aerodrome or site:	Take-off time:	Landing time:	Flight time:	
			Total flight time:	
2	Result of the test			
Skill test details:				
Pass ☐	Fail ☐	Partial pass ☐		
3	Remarks			
Location and date:				
Examiner's certificate number *:		Type and number of licence:		
Signature of examiner:		Name(s) in capital letters:		

** if applicable*

Appendix 8 – Cross-crediting of the IR part of a class or type rating proficiency check or EBT practical assessment

Regulation (EU) 2024/2076

A. Aeroplanes

Credits shall be granted only if holders are revalidating or renewing IR privileges for single-pilot single-engine and single-pilot multi-engine aeroplanes, as appropriate.

If a skill test, proficiency check or EBT practical assessment including IR is performed, and holders have a valid:	Credit is valid towards the IR part in proficiency check for:
MPA type rating; single-pilot high-performance complex aeroplane type rating	SE class rating (*), and SE type rating (*), and SP ME class or type rating except for high-performance complex type ratings, only credits for Section 3B of the proficiency check in point B.5 of Appendix 9 (*)
SP ME aeroplane class or type rating except for high- performance complex aeroplane type ratings, operated as single-pilot	SE class rating, and SE type rating, and SP ME class or type rating except for high-performance complex aeroplane type ratings
SP ME aeroplane class or type rating except for high- performance complex aeroplane type ratings, restricted to MP operations	SE class rating (*), and SE type rating (*), and SP ME class or type rating except for high-performance complex aeroplane type ratings (*)
SP SE aeroplane class or type rating	SE class rating, and SE type rating

* Provided that within the preceding 12 months the applicants have flown at least three IFR departures and approaches exercising PBN privileges, including at least one RNP APCH approach on an SP class or type of aeroplane in SP operations, or, for multi-engine, other than HP complex aeroplanes, the applicants have passed Section 6 of the skill test for SP, other than HP complex aeroplanes flown solely by reference to instruments in SP operations.

B. Helicopters

Credits shall be granted only if holders are revalidating or renewing IR privileges for single-pilot helicopters, as appropriate.

If a skill test or a proficiency check, including IR, is performed and the holders have a valid:	Credit is valid towards the IR part in a proficiency check for:
Multi-pilot helicopter (MPH) type rating	Single-pilot helicopter (SPH) of the same type, including privileges for single-pilot operations (* ¹)
Single-pilot helicopter (SPH) type rating, in multi-pilot operations	Privileges for single-pilot operations in the same type (* ¹)

*¹ Provided that within the preceding 12 months at least three IFR departures and approaches exercising PBN privileges, including one RNP APCH approach (which may be a Point in Space (PinS) approach), have been performed on a SP type of helicopter in SP operations.

Appendix 9 – Training, skill test and proficiency check for the MPL, and the ATPL, and for type and class ratings, and proficiency checks for the BIR and the IR

Regulation (EU) 2024/2076

A. General

1. Applicants for a skill test shall have received instruction in the same class or type of aircraft to be used in the test.
- 1a. Training in FFS in accordance with points 1b and 1c of this Section shall be complemented with take-off and landing training in a single-pilot aircraft operated in single-pilot or multi-pilot operations, or in a multi-pilot aircraft, as applicable, in accordance with point 17 of this Section, unless the training is completed in accordance with point [FCL.730.A](#) or constitutes training for cruise relief co-pilots in accordance with Section B, point 6(i), of this Appendix.
- 1b. The training for MPA and PL type ratings shall be conducted in an FFS or in a combination of FSTD(s) and FFS. The skill test or proficiency check for MPA and PL type ratings and the issue of an ATPL and an MPL, shall be conducted in an FFS, if available.
- 1c. The training, skill test or proficiency check for class or type ratings for SPA and helicopters shall be conducted in either of the following:
 - (a) an available and accessible FFS, or in a combination of such FFS and FSTD(s);
 - (b) a combination of FSTD(s) and the aircraft if an FFS is not available or accessible;
 - (c) the aircraft if no FSTD is available or accessible.
- 1d. By way of derogation from point 1c, the training, skill test or proficiency check for class or type ratings for non-complex SPA and for non-complex helicopters may be conducted in a combination of FSTD(s) and the aircraft even if an FFS is available and accessible.
- 1e. By way of derogation from point 1c, the training, skill test or proficiency check for any of the following may be conducted in accordance with points 1c(a), (b) or (c), irrespective of the availability and accessibility of FFS or FSTD:
 - (a) non-complex non-high-performance single-pilot aeroplanes;
 - (b) TMGs;
 - (c) non-complex helicopters for which the maximum certified seat configuration does not exceed five seats.

- 1f. If FSTDs are used during training, testing or checking, the suitability of the FSTDs used shall be verified against the applicable 'Table of functions and subjective tests' and the applicable 'Table of FSTD validation tests' contained in the primary reference document applicable for the device used. All restrictions and limitations indicated on the device's qualification certificate shall be considered.
2. Failure to achieve a pass in all sections of the test in two attempts will require further training.
3. There is no limit to the number of skill tests that may be attempted.

CONTENT OF THE TRAINING/SKILL TEST/PROFICIENCY CHECK

4. Unless otherwise determined in the operational suitability data established in accordance with Annex I (Part-21) to Regulation (EU) No 748/2012 (OSD), the syllabus of flight instruction, the skill test and the proficiency check shall comply with this Appendix. The syllabus, skill test and proficiency check may be reduced to give credit for previous experience on similar aircraft types, as determined in the OSD.
5. Except in the case of skill tests for the issue of an ATPL, when so defined in the OSD for the specific aircraft, credit may be given for skill test items common to other types or variants where the pilots are qualified.

CONDUCT OF THE TEST/CHECK

6. The examiner may choose between different skill test or proficiency check scenarios containing simulated relevant operations. Full-flight simulators and other training devices shall be used, as established in this Annex (Part-FCL).
7. During the proficiency check, the examiner shall verify that holders of the class or type rating maintain an adequate level of theoretical knowledge.
8. Should applicants choose to terminate a skill test for reasons considered inadequate by the examiner, they shall retake the entire skill test. If the test is terminated for reasons considered adequate by the examiner, only those sections not completed shall be tested in a further flight.
9. At the discretion of the examiner, any manoeuvre or procedure of the test may be repeated once by the applicants. The examiner may stop the test at any stage if it is considered that the applicants' demonstration of flying skill requires a complete retest.
10. Applicants shall be required to fly the aircraft from a position where the PIC or co-pilot functions, as relevant, can be performed. Under single-pilot conditions, the test shall be performed as if there was no other crew member present.
11. During preflight preparation for the test, applicants are required to determine power settings and speeds. Applicants shall indicate to the examiner the checks and duties carried out, including the identification of radio facilities. Checks shall be completed in accordance with the checklist for the aircraft on which the test is being taken and, if applicable, with the MCC concept. Performance data for take-off, approach and landing shall be calculated by applicants in compliance with the operations manual or flight manual for the aircraft used. Decision heights/altitudes, minimum descent heights/altitudes and missed approach point shall be agreed upon with the examiner.

12. The examiner shall take no part in the operation of the aircraft except where intervention is necessary in the interests of safety or to avoid unacceptable delay to other traffic.

SPECIFIC REQUIREMENTS FOR THE TRAINING, SKILL TEST AND PROFICIENCY CHECK FOR TYPE RATINGS FOR MULTI-PILOT AIRCRAFT, FOR SINGLE-PILOT AIRCRAFT WHEN OPERATED IN MULTI- PILOT OPERATIONS, FOR THE MPL AND FOR THE ATPL

13. The skill test for a multi-pilot aircraft or a single-pilot aircraft when operated in multi-pilot operations shall be performed in a multi-crew environment. Another applicant or another type rated qualified pilot may function as the second pilot. If an aircraft is used, the second pilot shall be the examiner or an instructor.
14. Applicants shall operate as PF during all sections of the skill test, except for abnormal and emergency procedures, which may be conducted as PF or PM in accordance with MCC. Applicants for the initial issue of a multi-pilot aircraft type rating or ATPL shall also demonstrate the ability to act as PM. Applicants may choose either the left-hand or the right-hand seat for the skill test if all items can be executed from the selected seat.
15. The following matters shall be specifically checked by the examiner for applicants for the ATPL or a type rating for multi-pilot aircraft or for multi-pilot operations in a single-pilot aircraft extending to the duties of a PIC, irrespective of whether the applicants act as PF or PM:
- (a) managing crew cooperation;
 - (b) maintaining a general survey of the aircraft operation by appropriate supervision; and
 - (c) setting priorities and making decisions in accordance with safety aspects and relevant rules and regulations appropriate to the operational situation, including emergencies.
16. The test or check should be accomplished under IFR, if the IR rating is included, and as far as possible be accomplished in a simulated commercial air transport environment. An essential element to be checked is the ability to plan and conduct the flight from routine briefing material.
17. When their type rating course has included less than 2 hours of flight training in the aircraft, applicants shall, before or after the skill test, complete flight training in the aircraft.
- Such approved flight training shall include take-off and landing manoeuvres and shall be performed by a qualified instructor under the responsibility of:
- (a) an ATO; or
 - (b) an organisation holding an AOC issued in accordance with Annex III (Part-ORO) to Regulation (EU) No 965/2012 and specifically approved for such training; or

- (c) the instructor, in cases where no aircraft flight training for SP aircraft at an ATO or AOC holder is approved, and the aircraft flight training was approved by the applicants' competent authority.

A certificate of completion of the type rating course including the flight training in the aircraft shall be forwarded to the competent authority before the new type rating is entered in the applicants' licence.

18. For the upset recovery training, 'stall event' means either an approach-to-stall or a stall. An FFS can be used by the ATO to either train recovery from a stall or demonstrate the type-specific characteristics of a stall, or both, provided that:
- (a) the FFS has been qualified in accordance with the special evaluation requirements in CS-FSTD(A); and
 - (b) the ATO has successfully demonstrated to the competent authority that any negative transfer of training is mitigated.

B. Specific requirements for the aeroplane category

PASS MARKS

1. In the case of single-pilot aeroplanes, with the exception of single-pilot high-performance complex aeroplanes, applicants shall pass all sections of the skill test or proficiency check. Failure in any item of a section will cause applicants to fail the entire section. If they fail only one section, they shall repeat only that section. Failure in more than one section will require applicants to repeat the entire test or check. Failure in any section in the case of a retest or recheck, including those sections that have been passed on a previous attempt, will require applicants to repeat the entire test or check again. For single-pilot multi-engine aeroplanes, Section 6 of the relevant test or check, addressing asymmetric flight, shall be passed.
2. In the case of multi-pilot and single-pilot high-performance complex aeroplanes, applicants shall pass all sections of the skill test or proficiency check. Failure in more than five items will require applicants to take the entire test or check again. Applicants failing five or fewer items shall take the failed items again. Failure in any item on the retest or recheck, including those items that have been passed on a previous attempt, will require applicants to repeat the entire check or test again.

FLIGHT TEST TOLERANCE

3. Applicants shall demonstrate the ability to:
 - (a) operate the aeroplane within its limitations;
 - (b) complete all manoeuvres with smoothness and accuracy;
 - (c) exercise good judgement and airmanship;
 - (d) apply aeronautical knowledge;

- (e) maintain control of the aeroplane at all times in such a manner that the successful outcome of a procedure or manoeuvre is never in doubt;
- (f) understand and apply crew coordination and incapacitation procedures, if applicable; and
- (g) communicate effectively with the other crew members, if applicable.

4. The following limits shall apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the aeroplane used:

Height

Generally	± 100 ft
Starting a go-around at decision height/altitude	+ 50 ft/– 0 ft
Minimum descent height/MAPt/altitude	+ 50 ft/– 0 ft

Tracking

On radio aids	±5°
For ‘angular’ deviations	Half-scale deflection, azimuth and glide path (e.g. LPV, ILS, MLS, GLS)

2D (LNAV) and 3D (LNAV/VNAV) ‘linear’ lateral deviations	cross-track error/deviation shall normally be limited to ± ½ of the RNP value associated with the procedure. Brief deviations from this standard up to a maximum of one time the RNP value are allowable.
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3D linear vertical deviations (e.g. RNP APCH (LNAV/VNAV) using BaroVNAV)	not more than – 75 ft below the vertical profile at any time, and not more than + 75 ft above the vertical profile at or below 1 000 ft above aerodrome level.
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Heading

all engines operating	± 5°
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with simulated engine failure	$\pm 10^\circ$
Speed	
all engines operating	± 5 knots
with simulated engine failure	+ 10 knots/– 5 knots

CONTENT OF THE TRAINING/SKILL TEST/PROFICIENCY CHECK

5. Single-pilot aeroplanes, except for high performance complex aeroplanes

(a) The following symbols mean:

P = Trained as PIC or co-pilot and as PF and PM

OTD = Other training devices may be used for this exercise

X = An FFS shall be used for this exercise; otherwise, an aeroplane shall be used if appropriate for the manoeuvre or procedure

P# = The training shall be complemented by supervised aeroplane inspection

(b) The practical training shall be conducted at least at the training equipment level shown as (P), or may be conducted on any higher level of equipment shown by the arrow (---->).

The following abbreviations are used to indicate the training equipment used:

A = aeroplane

FFS = full-flight simulator

FSTD = flight simulation training device

(c) The starred (*) items of Section 3B and, for multi-engine, Section 6, shall be flown solely by reference to instruments if revalidation/renewal of an IR is included in the skill test or proficiency check. If the starred (*) items are not flown solely by reference to instruments during the skill test or proficiency check, and when there is no crediting of IR privileges, the class or type rating will be restricted to VFR only.

(d) Section 3A shall be completed to revalidate a type or multi-engine class rating, VFR only, where the required experience of 10 route sectors within the previous 12 months has not been completed. Section 3A is not required if Section 3B is completed.

- (e) Where the letter 'M' appears in the skill test or proficiency check column, this will indicate a mandatory exercise or a choice where more than one exercise appears.
- (f) An FSTD shall be used for practical training for type or ME class ratings if they form part of an approved class or type rating course. The following considerations will apply to the approval of the course:
 - (i) the qualification of the FSTD as set out in the relevant requirements of Annex VI (Part-ARA) and Annex VII (Part-ORA);
 - (ii) the qualifications of the instructors;
 - (iii) the amount of FSTD training provided on the course; and
 - (iv) the qualifications and previous experience on similar types of the pilots under training.
- (g) To exercise the privileges of a class or type rating in multi-pilot operations in accordance with point [FCL.725\(d\)](#) or (da), pilots that are already entitled to operate the relevant class or type of aeroplane in single-pilot operations shall meet the following conditions:
 - (1) complete a bridge course containing manoeuvres and procedures including MCC as well as the exercises of Section 7 using threat and error management (TEM), CRM and human factors at an organisation that is specified in point [FCL.725\(d\)\(1\)\(i\)](#);
 - (2) pass a proficiency check in multi-pilot operations.
- (h) To exercise the privileges of a type rating in single-pilot operations in accordance with point [FCL.725\(d\)](#), pilots that are already entitled to operate the relevant type of aeroplane in multi-pilot operations shall be trained at an organisation that is specified in point [FCL.725\(d\)\(1\)\(i\)](#) and checked for the following additional manoeuvres and procedures in single-pilot operations:
 - (i) for SE aeroplanes, 1.6, 4.5, 4.6, 5.2 and, if applicable, one approach from Section 3.B;
 - (ii) for ME aeroplanes, 1.6, Section 6 and, if applicable, one approach from Section 3.B.
- (i) Pilots that are entitled to exercise the privileges of a class or type rating in both single-pilot and multi-pilot operations in accordance with points (g) and (h) may maintain privileges for both forms of operation by completing either of the following:
 - (i) two proficiency checks, one in single-pilot operations and one in multi-pilot operations;
 - (ii) a proficiency check in multi-pilot operations in addition to the exercises referred to in points (h)(i) or (h)(ii), as applicable, in single-pilot operations.

Proficiency checks for the revalidation or renewal of class ratings shall always include the exercises referred to in points (h)(i) or (h)(ii), as applicable, in single-pilot operations.

- (j) To remove a restriction to multi-pilot operations from a single-pilot aeroplane type rating in accordance with point [FCL.725\(d\)\(2\)](#), pilots shall comply with point (h).
- (k) The training, testing and checking shall follow the table mentioned below.
- (1) Training at an ATO, testing and checking requirements for single-pilot privileges
 - (2) Training at an ATO, testing and checking requirements for multi-pilot privileges
 - (3) Training at an ATO, testing and checking requirements for pilots holding single-pilot privileges seeking multi-pilot privileges for the first time (bridge course)
 - (4) Training at an ATO, testing and checking requirements for pilots holding multi-pilot privileges seeking single-pilot privileges for the first time (bridge course)
 - (5) Training at an ATO and checking requirements for combined revalidation and renewal of single and multi-pilot privileges

	(1)		(2)		(3)		(4)		(5)	
	Type of operation									
Type of aircraft	SP		MP		SP → MP (initial)		MP → SP (initial)		SP + MP	
	Training	Testing/ checking	Training	Testing/ checking	Training	Testing/ checking	Training, testing and checking (SE aeroplanes)	Training, testing and checking (ME aeroplanes)	SE aeroplanes	ME aeroplanes
Initial issue										
All (except SP complex)	Sections 1-6	Sections 1-6	MCC CRM Human factors TEM Sections 1-7	Sections 1-6	MCC CRM Human factors TEM Section 7	Sections 1-6	1.6, 4.5, 4.6, 5.2 and, if applicable, one approach	1.6, Section 6 and, if applicable, one approach from Section 3.B		
SP complex	1-7	1-6								

							from Section 3.B			
Revalidation										
All	n/a	Sections 1-6	n/a	Sections 1-6	n/a	n/a	n/a	n/a	MPO: Sections 1-7 (training) Section 1-6 (checking) SPO: 1.6, 4.5, 4.6, 5.2 and, if applicable, one approach from Section 3.B	MPO: Sections 1-7 (training) Sections 1-6 (checking) SPO: 1.6, Section 6 and, if applicable, one approach from Section 3.B
Renewal										
All	FCL.740	Sections 1-6	FCL.740	Sections 1-6	n/a	n/a	n/a	n/a	Training: FCL.740 Check: as for the revalidation	Training: FCL.740 Check: as for the revalidation

- (I) To establish or maintain PBN privileges, one approach shall be an RNP APCH. Where an RNP APCH is not practicable, it shall be performed in an appropriately equipped FSTD.

By way of derogation from the first paragraph, in cases where a proficiency check for revalidation of PBN privileges is performed in an aircraft or an FSTD representing that aircraft, which are not equipped for RNP APCH manoeuvres, the proficiency check may not include RNP APCH exercises. In such

cases, the PBN privileges of the pilot shall not include RNP APCH. The restriction shall be lifted if the pilot has completed a proficiency check including an RNP APCH exercise for the relevant class or type.

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
SECTION 1						
1	Departure	OTD				
1.1	Preflight including: – documentation; – mass and balance; – weather briefing; and – NOTAM.					
1.2	Pre-start checks					
1.2.1	External	OTD P#	P		M	
1.2.2	Internal	OTD P#	P		M	
1.3	Engine starting: normal malfunctions	P---->	---->		M	
1.4	Taxiing	P---->	---->		M	
1.5	Pre-departure checks: engine run-up (if applicable)	P---->	---->		M	
1.6	Take-off procedure: – normal with flight manual flap settings; and – crosswind (if conditions are available).	P---->	---->		M	

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
1.7	Climbing: – V _x /V _y – turns onto headings; and – level off.	P---->	---->		M	
1.8	ATC liaison – compliance, R/T procedures	P---->			M	
SECTION 2						
2	Airwork (visual meteorological conditions (VMC))	P---->	---->			
2.1	Straight and level flight at various airspeeds including flight at critically low airspeed with and without flaps (including approach to V V _{mca} when applicable)					
2.2	Steep turns (360° left and right at 45° bank)	P---->	---->		M	
2.3	Stalls and recovery: (i) clean stall; (ii) approach to stall in descending turn with bank with approach configuration and power; (iii) approach to stall in landing configuration and power; and	P---->	---->		M	

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
	(iv) approach to stall, climbing turn with take-off flap and climb power (single-engine aeroplanes only)					
2.4	Handling using autopilot and flight director (may be conducted in Section 3), if applicable	P---->	---->		M	
2.5	ATC liaison – Compliance, R/T procedures	P---->	---->		M	
SECTION 3A						
3A 3A.1	En route procedures VFR (see B.5 (c) and (d)) Flight plan, dead reckoning and map reading	P---->	---->			
3A.2	Maintenance of altitude, heading and speed	P---->	---->			
3A.3	Orientation, timing and revision of ETAs	P---->	---->			
3A.4	Use of radio navigation aids (if applicable)	P---->	---->			
3A.5	Flight management (flight log, routine checks including fuel, systems and icing)	P---->	---->			
3A.6	ATC liaison – compliance, R/T procedure	P---->	---->			
SECTION 3B						

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
3B	Instrument flight	P---->	---->		M	
3B.1*	Departure IFR	P---->	---->		M	
3B.2*	En route IFR	P---->	---->		M	
3B.3*	Holding procedures	P---->	---->		M	
3B.4*	3D operations to decision height/altitude (DH/A) of 200 ft (60 m) or to higher minima if required by the approach procedure (autopilot may be used to the final approach segment vertical path intercept)	P---->	---->		M	
3B.5*	2D operations to minimum descent height/altitude (MDH/A)	P---->	---->		M	
3B.6*	Flight exercises including simulated failure of the compass and attitude indicator: – rate 1 turns; and – recoveries from unusual attitudes.	P---->	---->		M	
3B.7*	Failure of localiser or glideslope	P---->	---->			
3B.8*	ATC liaison – compliance, R/T procedures	P---->	---->		M	
	Intentionally left blank					
SECTION 4						
4	Arrival and landings	P---->	---->		M	
4.1	Aerodrome arrival procedure					

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
4.2	Normal landing	P---->	---->		M	
4.3	Flapless landing	P---->	---->		M	
4.4	Crosswind landing (if suitable conditions)	P---->	---->			
4.5	Approach and landing with idle power from up to 2 000 ft above the runway (single-engine aeroplanes only)	P---->	---->			
4.6	Go-around from minimum height	P---->	---->		M	
4.7	Night go-around and landing (if applicable)	P---->	---->			
4.8	ATC liaison – compliance, R/T procedures	P---->	---->		M	
SECTION 5						
5	Abnormal and emergency procedures (This section may be combined with Sections 1 through 4.)					
5.1	Rejected take-off at a reasonable speed	P---->	---->		M	
5.2	Simulated engine failure after take-off (single-engine aeroplanes only)		P		M	

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
5.3	Simulated forced landing without power (single-engine aeroplanes only)		P		M	
5.4	Simulated emergencies: (i) fire or smoke in flight; and (ii) systems' malfunctions as appropriate	P---->	---->			
5.5	ME aeroplanes and TMG training only: engine shutdown and restart (at a safe altitude if performed in the aircraft)	P---->	---->			
5.6	ATC liaison – compliance, R/T procedure					
SECTION 6						
6 6.1*	Simulated asymmetric flight (This section may be combined with Sections 1 through 5.) Simulated engine failure during take-off (at a safe altitude unless carried out in an FFS or an FNPT II)	P---->	---->X		M	
6.2*	Asymmetric approach and go-around	P---->	---->		M	
6.3*	Asymmetric approach and full-stop landing	P---->	---->		M	

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
6.4	ATC liaison – compliance, R/T procedures	P---->	---->		M	
SECTION 7						
7	UPRT					
7.1	Flight manoeuvres and procedures					
7.1.1	Manual flight with and without flight directors (no autopilot, no autothrust/autothrottle, and at different control laws, where applicable)	P---->	---->			
7.1.1.1	At different speeds (including slow flight) and altitudes within the FSTD training envelope.	P---->	---->			
7.1.1.2	Steep turns using 45° bank, 180° to 360° left and right	P---->	---->			
7.1.1.3	Turns with and without spoilers	P---->	---->			
7.1.1.4	Procedural instrument flying and manoeuvring including instrument departure and arrival, and visual approach	P---->	---->			
7.2	Upset recovery training	P---->	---->			
7.2.1	Recovery from stall events in: – take-off configuration; – clean configuration at low altitude;					

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
	– clean configuration near maximum operating altitude; and – landing configuration					
7.2.2	The following upset exercises: – recovery from nose-high at various bank angles; and – recovery from nose-low at various bank angles.	P	X An aeroplane shall not be used for this exercise			
7.3	Go-around with all engines operating* from various stages during an instrument approach	P---->	---->			
7.4	Rejected landing with all engines operating: – from various heights below DH/MDH 15 m (50 ft) above the runway threshold – after touchdown (balked landing) – In aeroplanes which are not certificated as transport category aeroplanes (JAR/FAR 25) or as commuter category aeroplanes (SFAR 23), the rejected landing with all engines operating shall	P---->	---->			

TMGs AND SINGLE-PILOT AEROPLANES, EXCEPT FOR HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			CLASS OR TYPE RATING SKILL TEST/PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
	be initiated below MDH/A or after touchdown.					

6. Multi-pilot aeroplanes and single-pilot high-performance complex aeroplanes

(a) The following symbols mean:

P = Trained as PIC or co-pilot and as PF and PM for the issue of a type rating as applicable.

OTD = Other training devices may be used for this exercise.

X = An FFS shall be used for this exercise; otherwise an aeroplane shall be used if appropriate for the manoeuvre or procedure.

P# = The training shall be complemented by supervised aeroplane inspection.

(b) The practical training shall be conducted at least at the training equipment level shown as (P), or may be conducted up to any higher equipment level shown by the arrow (---->).

The following abbreviations are used to indicate the training equipment used:

A = aeroplane

FFS = full-flight simulator

FSTD = flight simulator training device

(c) The starred items (*) shall be flown solely by reference to instruments.

- (d) Where the letter 'M' appears in the skill test or proficiency check column, this will indicate a mandatory exercise or a choice where more than one exercise appears.
- (e) An FFS shall be used for practical training and testing if the FFS forms part of an approved type rating course. The following considerations will apply to the approval of the course:
 - (i) the qualifications of the instructors;
 - (ii) the qualification and the amount of training provided on the course in an FSTD; and
 - (iii) the qualifications and previous experience on similar types of the pilots under training.
- (f) Manoeuvres and procedures shall include MCC for multi-pilot aeroplane and for single-pilot high-performance complex aeroplanes in multi-pilot operations.
- (g) Manoeuvres and procedures shall be conducted in single-pilot role for single-pilot high-performance complex aeroplanes in single-pilot operations.
- (h) To remove a restriction to multi-pilot operations in accordance with point [FCL.725\(d\)\(2\)](#) from a single-pilot high-performance complex aeroplane type rating, pilots shall complete the manoeuvres/procedures in 2.5, 3.8.3.4, 4.4, 5.5 and at least one manoeuvre/procedure from Section 3.4 in single-pilot operation.
- (i) Applicants for and holders of a restricted type rating issued in accordance with point [FCL.720.A\(c\)](#) shall complete training, skill tests and proficiency checks in accordance with this Appendix. However, unless they undergo a skill test in accordance with point [FCL.720.A\(c\)\(3\)](#), they shall, during a skill test or a proficiency check, perform at least the landing manoeuvres in the role of the pilot monitoring but shall not be required to perform the following:
 - (i) take-off manoeuvres;
 - (ii) landing manoeuvres in the role of the pilot flying.
- (j) To establish or maintain PBN privileges, one approach shall be an RNP APCH. Where an RNP APCH is not practicable, it shall be performed in an appropriately equipped FSTD.

By way of derogation from the first paragraph, in cases where a proficiency check for revalidation of PBN privileges is performed in an aircraft or an FSTD representing that aircraft, which are not equipped for RNP APCH manoeuvres, the proficiency check may not include RNP APCH exercises. In such cases, the PBN privileges of the pilot shall not include RNP APCH. The restriction shall be lifted if the pilot has completed a proficiency check including an RNP APCH exercise for the relevant class or type.

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
SECTION 1						
1	Flight preparation	OTD				
1.1	Performance calculation	P				
1.2	Aeroplane external visual inspection; location of each item and purpose of inspection	OTD P#	P			
1.3	Cockpit inspection	P---->	---->			
1.4	Use of checklist prior to starting engines, starting procedures, radio and navigation equipment check, selection and setting of navigation and communication frequencies	P---->	---->		M	
1.5	Taxiing in compliance with ATC instructions or instructions of instructor	P---->	---->			
1.6	Before take-off checks	P---->	---->		M	
SECTION 2						
2	Take-offs	P---->	---->			
2.1	Normal take-offs with different flap settings, including expedited take-off					
2.2*	Instrument take-off; transition to instrument flight is required	P---->	---->			

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
	during rotation or immediately after becoming airborne					
2.3	Crosswind take-off	P---->	---->			
2.4	Take-off at maximum take-off mass (actual or simulated maximum take-off mass)	P---->	---->			
2.5	Take-offs with simulated engine failure:	P---->	---->			
2.5.1*	shortly after reaching V2					
	(In aeroplanes which are not certificated as transport category or commuter category aeroplanes, the engine failure shall not be simulated until reaching a minimum height of 500 ft above the runway end. In aeroplanes having the same performance as a transport category aeroplane regarding take-off mass and density altitude, the instructor may simulate the engine failure shortly after reaching V2)					
2.5.2*	between V1 and V2	P	X		M FFS only	

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
2.6	Rejected take-off at a reasonable speed before reaching V1	P---->	---->		M	
SECTION 3						
3	Flight manoeuvres and procedures	P---->	---->			
3.1	Manual flight with and without flight directors (no autopilot, no autothrust/autothrottle, and at different control laws, where applicable)					
3.1.1	At different speeds (including slow flight) and altitudes within the FSTD training envelope	P---->	---->			
3.1.2	Steep turns using 45° bank, 180° to 360° left and right	P---->	---->			
3.1.3	Turns with and without spoilers	P---->	---->			
3.1.4	Procedural instrument flying and manoeuvring including instrument departure and arrival, and visual approach	P---->	---->			

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
3.2	Tuck under and Mach buffets (if applicable), and other specific flight characteristics of the aeroplane (e.g. Dutch Roll)	P---->	---->X An aeroplane shall not be used for this exercise		FFS only	
3.3	Normal operation of systems and controls engineer's panel (if applicable)	OTD P---->	---->			
3.4	Normal and abnormal operations of following systems:				M	A mandatory minimum of 3 abnormal items shall be selected from 3.4.0 to 3.4.14 inclusive
3.4.0	Engine (if necessary propeller)	OTD P---->	---->			
3.4.1	Pressurisation and air conditioning	OTD P---->	---->			
3.4.2	Pitot/static system	OTD P---->	---->			

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
3.4.3	Fuel system	OTD P---->	---->			
3.4.4	Electrical system	OTD P---->	---->			
3.4.5	Hydraulic system	OTD P---->	---->			
3.4.6	Flight control and trim system	OTD P---->	---->			
3.4.7	Anti-icing/de-icing system, glare shield heating	OTD P---->				
3.4.8	Autopilot/flight director	OTD P---->			M (single pilot only)	
3.4.9	Stall warning devices or stall avoidance devices, and stability augmentation devices	OTD P---->				
3.4.10	Ground proximity warning system, weather radar, radio altimeter, transponder	P---->				
3.4.11	Radios, navigation equipment, instruments, FMS	OTD P---->				
3.4.12	Landing gear and brake	OTD P---->	---->			
3.4.13	Slat and flap system	OTD	---->			
3.4.14	Auxiliary power unit (APU)	OTD P---->	---->			

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
	Intentionally left blank					
3.6	Abnormal and emergency procedures:				M	A mandatory minimum of 3 items shall be selected from 3.6.1 to 3.6.9 inclusive
3.6.1	Fire drills, e.g. engine, APU, cabin, cargo compartment, flight deck, wing and electrical fires including evacuation	P---->	---->			
3.6.2	Smoke control and removal	P---->	---->			
3.6.3	Engine failures, shutdown and restart at a safe height	P---->	---->			
3.6.4	Fuel dumping (simulated)	P---->	---->			
3.6.5	Wind shear at take-off/landing	P	X		FFS only	
3.6.6	Simulated cabin pressure failure/emergency descent	P---->	---->			
3.6.7	Incapacitation of flight crew member	P---->	---->			
3.6.8	Other emergency procedures as outlined in the appropriate aeroplane flight manual (AFM)	P---->	---->			

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
3.6.9	TCAS event	OTD P---->	An aero-plane shall not be used		FFS only	
3.7 3.7.1	Upset recovery training Recovery from stall events in: – take-off configuration; – clean configuration at low altitude; – clean configuration near maximum operating altitude; and – landing configuration.	P FFS qualified for the training task only	X An aero-plane shall not be used for this exercise			
3.7.2	The following upset exercises: – recovery from nose-high at various bank angles; and – recovery from nose-low at various bank angles	P FFS qualified for the training task only	X An aero-plane shall not be used for this exercise		FFS only	
3.8	Instrument flight procedures					

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
3.8.1*	Adherence to departure and arrival routes and ATC instructions	P---->	---->		M	
3.8.2*	Holding procedures	P---->	---->			
3.8.3*	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach procedure					
Note: According to the AFM, RNP APCH procedures may require the use of autopilot or flight director. The procedure to be flown manually shall be chosen taking into account such limitations (for example, choose an ILS for 3.8.3.1 in the case of such AFM limitation).						
3.8.3.1*	Manually, without flight director	P---->	---->		M (skill test only)	
3.8.3.2*	Manually, with flight director	P---->	---->			
3.8.3.3*	With autopilot	P---->	---->			
3.8.3.4*	Manually, with one engine simulated inoperative during final approach, either until touchdown or through the complete missed approach procedure (as applicable), starting: (i) before passing 1 000 ft above aerodrome level; and	P---->	---->		M	

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
	(ii) after passing 1 000 ft above aerodrome level. In aeroplanes which are not certificated as transport category aeroplanes (JAR/FAR 25) or as commuter category aeroplanes (SFAR 23), the approach with simulated engine failure and the ensuing go-around shall be initiated in conjunction with the 2D approach in accordance with 3.8.4. The go-around shall be initiated when reaching the published obstacle clearance height/altitude (OCH/A); however, not later than reaching an MDH/A of 500 ft above the runway threshold elevation. In aeroplanes having the same performance as a transport category aeroplane regarding take-off mass and density altitude, the instructor may simulate the engine failure in accordance with exercise 3.8.3.4.					

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
3.8.4*	2D operations down to the MDH/A	P*---->	---->		M	
3.8.5	Circling approach under the following conditions: (a)*approach to the authorised minimum circling approach altitude at the aerodrome in question in accordance with the local instrument approach facilities in simulated instrument flight conditions; followed by: (b) circling approach to another runway at least 90° off centreline from the final approach used in item (a), at the authorised minimum circling approach altitude. Remark: If (a) and (b) are not possible due to ATC reasons, a simulated low visibility pattern may be performed.	P*---->	---->			
3.8.6	Visual approaches	P---->	---->			
SECTION 4						
4	Missed approach procedures	P*---->	---->			

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
4.1.	Go-around with all engines operating* during a 3D operation on reaching decision height	P*---->	---->			
4.2.	Go-around with all engines operating* from various stages during an instrument approach	P*---->	---->			
4.3.	Other missed approach procedures	P*---->	---->			
4.4*	Manual go-around with the critical engine simulated inoperative after an instrument approach on reaching DH, MDH or MAPt	P*---->	---->		M	
4.5.	Rejected landing with all engines operating: – from various heights below DH/MDH; – after touchdown (balked landing) In aeroplanes which are not certificated as transport category aeroplanes (JAR/FAR 25) or as commuter category aeroplanes (SFAR 23), the rejected landing with all engines operating shall be	P---->	---->			

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
	initiated below MDH/A or after touchdown.					
SECTION 5						
5	Landings	P				
5.1.	Normal landings* with visual reference established when reaching DA/H following an instrument approach operation					
5.2.	Landing with simulated jammed horizontal stabiliser in any out-of-trim position	P---->	An aeroplane shall not be used for this exercise		FFS only	
5.3.	Crosswind landings (aircraft, if practicable)	P---->	---->			
5.4.	Traffic pattern and landing without extended or with partly extended flaps and slats	P---->	---->			
5.5.	Landing with critical engine simulated inoperative	P---->	---->		M	
5.6.	Landing with two engines inoperative:	P	X		M	

MULTI-PILOT AEROPLANES AND SINGLE-PILOT HIGH-PERFORMANCE COMPLEX AEROPLANES		PRACTICAL TRAINING			ATPL/MPL/TYPE RATING SKILL TEST OR PROF. CHECK	
Manoeuvres/Procedures		FSTD	A	Instructor initials when training completed	Tested or checked in FSTD or A	Examiner initials when test or check completed
	– aeroplanes with three engines: the centre engine and one outboard engine as far as practicable according to data of the AFM; and – aeroplanes with four engines: two engines at one side				FFS only (skill test only)	

7. Class ratings – sea

Section 6 shall be completed to revalidate a multi-engine class rating sea, VFR only, where the required experience of 10 route sectors within the previous 12 months has not been completed.

CLASS RATING SEA	PRACTICAL TRAINING	CLASS RATING SKILL TEST OR PROFICIENCY CHECK
Manoeuvres/Procedures	Instructor's initials when training completed	Examiner's initials when test completed
SECTION 1		
1 Departure Preflight including: – documentation; – mass and balance; – weather briefing; and – NOTAM.		

CLASS RATING SEA		PRACTICAL TRAINING	CLASS RATING SKILL TEST OR PROFICIENCY CHECK
Manoeuvres/Procedures		Instructor's initials when training completed	Examiner's initials when test completed
1.2	Pre-start checks External/internal		
1.3	Engine start-up and shutdown Normal malfunctions		
1.4	Taxiing		
1.5	Step taxiing		
1.6	Mooring: Beach Jetty pier Buoy		
1.7	Engine-off sailing		
1.8	Pre-departure checks: Engine run-up (if applicable)		
1.9	Take-off procedure: – normal with flight manual flap settings; and – crosswind (if conditions are available).		
1.10	Climbing: – turns onto headings – level off		
1.11	ATC liaison – Compliance, R/T procedures		
SECTION 2			
2	Airwork (VFR)		
2.1	Straight and level flight at various airspeeds including flight at critically low airspeed with and without flaps (including approach to VMCA when applicable)		
2.2	Steep turns (360° left and right at 45° bank)		
2.3.	Stalls and recovery: (i) clean stall;		

CLASS RATING SEA	PRACTICAL TRAINING	CLASS RATING SKILL TEST OR PROFICIENCY CHECK
Manoeuvres/Procedures	Instructor's initials when training completed	Examiner's initials when test completed
(ii) approach to stall in descending turn with bank with approach configuration and power; (iii) approach to stall in landing configuration and power; and (iv) approach to stall, climbing turn with take-off flap and climb power (single-engine aeroplanes only).		
2.4 ATC liaison – Compliance, R/T procedures		
SECTION 3		
3 En-route procedures VFR		
3.1 Flight plan, dead reckoning and map reading		
3.2 Maintenance of altitude, heading and speed		
3.3 Orientation, timing and revision of ETAs		
3.4 Use of radio navigation aids (if applicable)		
3.5 Flight management (flight log, routine checks including fuel, systems and icing)		
3.6 ATC liaison – Compliance, R/T procedures		
SECTION 4		
4 Arrivals and landings		
4.1 Aerodrome arrival procedure (amphibians only)		
4.2 Normal landing		
4.3 Flapless landing		
4.4 Crosswind landing (if suitable conditions)		
4.5 Approach and landing with idle power from up to 2000' above the water (single-engine aeroplane only)		
4.6 Go-around from minimum height		

CLASS RATING SEA	PRACTICAL TRAINING	CLASS RATING SKILL TEST OR PROFICIENCY CHECK
Manoeuvres/Procedures	Instructor's initials when training completed	Examiner's initials when test completed
Glassy water landing Rough water landing		
4.8 ATC liaison – Compliance, R/T procedure,		
SECTION 5		
5 Abnormal and emergency procedures (This section may be combined with sections 1 through 4)		
5.1 Rejected take-off at a reasonable speed		
5.2 Simulated engine failure after take-off (single-engine aeroplane only)		
5.3 Simulated forced landing without power (single-engine aeroplane only)		
5.4 Simulated emergencies: (i) fire or smoke in flight (ii) systems' malfunctions as appropriate		
5.5 ATC liaison – Compliance, R/T procedures		
SECTION 6		
6 Simulated asymmetric flight (This section may be combined with sections 1 through 5)		
6.1 Simulated engine failure during take-off (at a safe altitude unless carried out in FFS and FNPT II)		
6.2 Engine shutdown and restart (ME skill test only)		
6.3 Asymmetric approach and go-around		
6.4 Asymmetric approach and full-stop landing		
6.5 ATC liaison – Compliance, R/T procedures		

C. Specific requirements for the helicopter category

1. In the case of skill test or proficiency check for type ratings and the ATPL, applicants shall pass Sections 1 to 4 and 6 (as applicable) of the skill test or proficiency check. Failure in more than five items will require applicants to repeat the entire test or check. Applicants failing not more than five items shall repeat the failed items. Failure in any item in the case of a retest or a recheck or failure in any other items already passed will require the applicants to repeat the entire test or check again. All sections of the skill test or proficiency check shall be completed within 6 months.
2. In the case of proficiency check for an IR, applicants shall pass Section 5 of the proficiency check. Failure in more than 3 items will require applicants to repeat the entire Section 5. Applicants failing not more than 3 items shall repeat the failed items. Failure in any item in the case of a recheck or failure in any other items of Section 5 already passed will require applicants to repeat the entire check.

FLIGHT TEST TOLERANCE

3. The applicant shall demonstrate the ability to:
 - (a) operate the helicopter within its limitations;
 - (b) complete all manoeuvres with smoothness and accuracy;
 - (c) exercise good judgement and airmanship;
 - (d) apply aeronautical knowledge;
 - (e) maintain control of the helicopter at all times in such a manner that the successful outcome of a procedure or manoeuvre is never in doubt;
 - (f) understand and apply crew coordination and incapacitation procedures, if applicable; and
 - (g) communicate effectively with the other crew members, if applicable.
4. The following limits shall apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the helicopter used.
 - (a) IFR flight limits

Height	
Generally	±100 ft
Starting a go-around at decision height/altitude	+50 ft/–0 ft

Minimum descent height/MAP/altitude	+50 ft/–0 ft
Tracking	
On radio aids	±5°
For “angular” deviations	Half-scale deflection, azimuth and glide path (e.g. LPV, ILS, MLS, GLS)
2D (LNAV) and 3D (LNAV/VNAV) “linear” lateral deviations	cross-track error/deviation shall normally be limited to ± ½ of the RNP value associated with the procedure. Brief deviations from this standard up to a maximum of one time the RNP value are allowable.
3D linear vertical deviations (e.g. RNP APCH (LNAV/VNAV) using BaroVNAV)	not more than – 75 ft below the vertical profile at any time, and not more than + 75 ft above the vertical profile at or below 1 000 ft above aerodrome level.
Heading	
all engines operating	±5°
with simulated engine failure	±10°
Speed	
all engines operating	±5 knots
with simulated engine failure	+10 knots/–5 knots

(b) VFR flight limits

Height:	
Generally	±100 ft
Heading:	
Normal operations	±5°
Abnormal operations/emergencies	±10°
Speed:	
Generally	±10 knots
With simulated engine failure	+10 knots/-5 knots
Ground drift:	
T.O. hover I.G.E.	±3 ft
Landing	±2 ft (with 0 ft rearward or lateral flight)

CONTENT OF THE TRAINING/SKILL TEST/PROFICIENCY CHECK**GENERAL**

5. The following symbols mean:

P = Trained as PIC for the issue of a type rating for single-pilot helicopters (SPH) or trained as PIC or co-pilot and as PF and PM for the issue of a type rating for multi pilot helicopters (MPH).

6. The practical training shall be conducted at least at the training equipment level shown as (P), or may be conducted up to any higher equipment level shown by the arrow (--->).

The following abbreviations are used to indicate the training equipment used:

FFS = full-flight simulator

FTD = flight training device

H = helicopter

7. The starred items (*) shall be flown in actual or simulated IMC, only by applicants wishing to renew or revalidate an IR(H) or extend the privileges of that rating to another type.
8. Instrument flight procedures (Section 5) shall be performed only by applicants wishing to renew or revalidate an IR(H) or extend the privileges of that rating to another type. An FFS or an FTD 2/3 may be used for this purpose.
- 8a. To establish or maintain PBN privileges, one approach shall be an RNP APCH. Where an RNP APCH is not practicable, it shall be performed in an appropriately equipped FSTD.

By way of derogation from subparagraph above, in cases where a proficiency check for revalidation of PBN privileges does not include an RNP APCH exercise, the PBN privileges of the pilot shall not include RNP APCH. The restriction shall be lifted if the pilot has completed a proficiency check including an RNP APCH exercise.
9. Where the letter 'M' appears in the skill test or proficiency check column, this will indicate a mandatory exercise.
10. An FSTD shall be used for practical training and testing if the FSTD forms part of a type rating course. The following considerations will apply to the course:
 - (a) the qualification of the FSTD as set out in the relevant requirements of Annex VI (Part-ARA) and Annex VII (Part-ORA);
 - (b) the qualifications of the instructor and examiner;
 - (c) the amount of FSTD training provided on the course;
 - (d) the qualifications and previous experience in similar types of the pilots under training; and
 - (e) the amount of supervised flying experience provided after the issue of the new type rating.

MULTI-PILOT HELICOPTERS

11. Applicants for the skill test for the issue of the multi-pilot helicopter type rating and ATPL(H) shall pass only Sections 1 to 4 and, if applicable, Section 6.
12. Applicants for the revalidation or renewal of the multi-pilot helicopter type rating proficiency check shall pass only Sections 1 to 4 and, if applicable, Section 6.

SINGLE/MULTI-PILOT HELICOPTERS		PRACTICAL TRAINING			SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		FSTD	H	Instructor initials when training completed	Checked in FSTD or H	Examiner initials when test completed
SECTION 1 – Preflight preparations and checks						
1.1	Helicopter exterior visual inspection; location of each item and purpose of inspection		P		M (if performed in the helicopter)	
1.2	Cockpit inspection	P	---->		M	
1.3	Starting procedures, radio and navigation equipment check, selection and setting of navigation and communication frequencies	P	---->		M	
1.4	Taxiing/air taxiing in compliance with ATC instructions or with instructions of an instructor	P	---->		M	
1.5	Pre-take-off procedures and checks	P	---->		M	
SECTION 2 – Flight manoeuvres and procedures						
2.1	Take-offs (various profiles)	P	---->		M	
2.2	Sloping ground or crosswind take-offs & landings	P	---->			
2.3	Take-off at maximum take-off mass (actual or simulated maximum take-off mass)	P	---->			
2.4	Take-off with simulated engine failure shortly before reaching TDP or DPATO	P	---->		M	

SINGLE/MULTI-PILOT HELICOPTERS		PRACTICAL TRAINING			SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		FSTD	H	Instructor initials when training completed	Checked in FSTD or H	Examiner initials when test completed
2.4.1	Take-off with simulated engine failure shortly after reaching TDP or DPATO	P	---->		M	
2.5	Climbing and descending turns to specified headings	P	---->		M	
2.5.1	Turns with 30° bank, 180° to 360° left and right, by sole reference to instruments	P	---->		M	
2.6	Autorotative descent	P	---->		M	
2.6.1	For single-engine helicopters (SEH): — autorotative landing; or — power recovery, provided that applicants, in the preceding year, completed training that included an autorotative landing and that training was entered and signed in the applicants' logbook by the instructor. For multi-engine helicopters (MEH): power recovery.	P	---->		M	
2.7	Landings, various profiles	P	---->		M	
2.7.1	Go-around or landing following simulated engine failure before LDP or DPBL	P	---->		M	
2.7.2	Landing following simulated engine failure after LDP or DPBL	P	---->		M	

SINGLE/MULTI-PILOT HELICOPTERS		PRACTICAL TRAINING			SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		FSTD	H	Instructor initials when training completed	Checked in FSTD or H	Examiner initials when test completed
SECTION 3 – Normal and abnormal operations of the following systems and procedures						
3	Normal and abnormal operations of the following systems and procedures:				M	A mandatory minimum of 3 items shall be selected from this section
3.1	Engine	P	---->			
3.2	Air conditioning (heating, ventilation)	P	---->			
3.3	Pitot/static system	P	---->			
3.4	Fuel System	P	---->			
3.5	Electrical system	P	---->			
3.6	Hydraulic system	P	---->			
3.7	Flight control and trim system	P	---->			
3.8	Anti-icing and de-icing system	P	---->			
3.9	Autopilot/Flight director	P	--->			
3.10	Stability augmentation devices	P	---->			
3.11	Weather radar, radio altimeter, transponder	P	---->			
3.12	Area navigation system	P	---->			
3.13	Landing gear system	P	---->			
3.14	APU	P	---->			

SINGLE/MULTI-PILOT HELICOPTERS		PRACTICAL TRAINING			SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		FSTD	H	Instructor initials when training completed	Checked in FSTD or H	Examiner initials when test completed
3.15	Radio, navigation equipment, instruments and FMS	P	---->			
SECTION 4 – Abnormal and emergency procedures						
4	Abnormal and emergency procedures				M	A mandatory minimum of 3 items shall be selected from this section
4.1	Fire drills (including evacuation if applicable)	P	---->			
4.2	Smoke control and removal	P	---->			
4.3	Engine failures, shutdown and restart at a safe height	P	---->			
4.4	Fuel dumping (simulated)	P	---->			
4.5	Tail rotor control failure (if applicable)	P	---->			
4.5.1	Tail rotor loss (if applicable)	P	A helicopter shall not be used for this exercise			
4.6	Incapacitation of crew member – MPH only	P	---->			

SINGLE/MULTI-PILOT HELICOPTERS		PRACTICAL TRAINING			SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		FSTD	H	Instructor initials when training completed	Checked in FSTD or H	Examiner initials when test completed
4.7	Transmission malfunctions	P	---->			
4.8	Other emergency procedures as outlined in the appropriate flight manual	P	---->			
SECTION 5 – Instrument flight procedures (to be performed in IMC or simulated IMC)						
5.1	Instrument take-off: transition to instrument flight is required as soon as possible after becoming airborne	P*	---->*			
5.1.1	Simulated engine failure during departure	P*	---->*		M*	
5.2	Adherence to departure and arrival routes and ATC instructions	P*	---->*		M*	
5.3	Holding procedures	P*	---->*			
5.4	3D operations to DH/A of 200 ft (60 m) or to higher minima if required by the approach procedure	P*	---->*			
5.4.1	Manually, without flight director. Note: According to the AFM, RNP APCH procedures may require the use of autopilot or flight director. The procedure to be flown manually shall be chosen taken into account such limitations (for example, choose	P*	---->*		M* (unless Exercise 5.4.2 is completed)	

SINGLE/MULTI-PILOT HELICOPTERS		PRACTICAL TRAINING			SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		FSTD	H	Instructor initials when training completed	Checked in FSTD or H	Examiner initials when test completed
	an ILS for 5.4.1 in the case of such AFM limitation).					
5.4.2	Manually, with flight director	P*	---->*		M* (unless Exercise 5.4.1 is completed)	
5.4.3	With coupled autopilot	P*	---->*			
5.4.4	Manually, with one engine simulated inoperative; engine failure has to be simulated during final approach before passing 1 000 ft above aerodrome level until touchdown or until completion of the missed approach procedure	P*	---->*		M*	
5.5	2D operations down to the MDA/H	P*	---->*		M*	
5.6	Go-around with all engines operating on reaching DA/H or MDA/MDH	P*	---->*			
5.6.1	Other missed approach procedures	P*	---->*			
5.6.2	Go-around with one engine simulated inoperative on reaching DA/H or MDA/MDH	P*	---->*		M*	

SINGLE/MULTI-PILOT HELICOPTERS		PRACTICAL TRAINING			SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		FSTD	H	Instructor initials when training completed	Checked in FSTD or H	Examiner initials when test completed
5.7	IMC autorotation with power recovery	P*	---->*		M*	
5.8	Recovery from unusual attitudes	P*	---->*		M*	
SECTION 6 — Use of optional equipment						
6	Use of optional equipment	P	---->			

SINGLE-PILOT HELICOPTERS

13. To exercise the privileges of the type rating in accordance with point [FCL.725\(d\)](#), applicants for the issue, revalidation or renewal of a single-pilot helicopter type rating shall:
- (a) for single-pilot operations, complete the skill test or proficiency check in single-pilot operations;
 - (b) in multi-pilot operations, complete the skill test or proficiency check in multi-pilot operations;
 - (c) for both single-pilot and multi-pilot operations, complete the skill test or proficiency check in multi-pilot operations and, additionally, the following manoeuvres and procedures in single-pilot operations:
 - (1) for single-engine helicopters: 2.1 take-off and 2.6 and 2.6.1 autorotative descent and autorotative landing;
 - (2) for multi-engine helicopters: 2.1 take-off and 2.4 and 2.4.1 engine failures shortly before and shortly after reaching TDP;
 - (3) for IR privileges, in addition to point (1) or (2), as applicable, one approach of Section 5, unless the criteria of [Appendix 8](#) are met;
 - (d) in order to remove a restriction to multi-pilot operations from a single-pilot helicopter type rating in accordance with point [FCL.725\(d\)\(2\)](#), complete a proficiency check that includes the manoeuvres and procedures referred to in points (c)(1) or (c)(2), as applicable.

D. Specific requirements for the powered-lift aircraft category

1. In the case of skill tests or proficiency checks for powered-lift aircraft type ratings, applicants shall pass Sections 1 to 5 and 6 (as applicable) of the skill test or proficiency check. Failure in more than five items will require applicants to repeat the entire test or check. Applicants failing not more than five items shall repeat the failed items. Failure in any item in the case of a retest or a recheck or failure in any other items already passed will require applicants to repeat the entire test or check. All sections of the skill test or proficiency check shall be completed within 6 months.

FLIGHT TEST TOLERANCE

2. Applicants shall demonstrate the ability to:
 - (a) operate the powered-lift aircraft within its limitations;
 - (b) complete all manoeuvres with smoothness and accuracy;
 - (c) exercise good judgement and airmanship;
 - (d) apply aeronautical knowledge;
 - (e) maintain control of the powered-lift aircraft at all times in such a manner that the successful outcome of a procedure or manoeuvre is never in doubt;
 - (f) understand and apply crew coordination and incapacitation procedures; and
 - (g) communicate effectively with the other crew members.
3. The following limits shall apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the powered-lift aircraft used.
 - (a) **IFR flight limits:**

Height	
Generally	±100 ft
Starting a go-around at decision height/altitude	+50 ft/-0 ft
Minimum descent height/altitude	+50 ft/-0 ft
Tracking	
On radio aids	±5°

Precision approach	half scale deflection, azimuth and glide path
Heading	
Normal operations	$\pm 5^\circ$
Abnormal operations/emergencies	$\pm 10^\circ$
Speed	
Generally	± 10 knots
With simulated engine failure	+10 knots/-5 knots
(b) VFR flight limits:	
Height	
Generally	± 100 ft
Heading:	
Normal operations	$\pm 5^\circ$
Abnormal operations/emergencies	$\pm 10^\circ$
Speed	
Generally	± 10 knots
With simulated engine failure	+10 knots/-5 knots
Ground drift	
T.O. hover I.G.E.	± 3 ft
Landing	± 2 ft (with 0 ft rearward or lateral flight)

CONTENT OF THE TRAINING/SKILL TEST/PROFICIENCY CHECK**4. The following symbol means:**

P= Trained as PIC or co-pilot and as PF and PM for the issue of a type rating as applicable

5. The practical training shall be conducted at least at the training equipment level shown as (P), or may be conducted up to any higher equipment level shown by the arrow (---->).
6. The following abbreviations are used to indicate the training equipment used:
 - FFS = full-flight simulator
 - FTD = flight training device
 - OTD = other training device
 - PL = powered-lift aircraft
- 6a. The starred items (*) shall be flown solely by reference to instruments. If this condition is not met during the skill test or proficiency check, the type rating will be restricted to VFR only.
7. Where the letter 'M' appears in the skill test or proficiency check column, this will indicate a mandatory exercise.
8. FSTDs shall be used for practical training and testing if they form part of an approved type rating course. The following considerations will apply to the approval of the course:
 - (a) the qualification of the FSTDs as set out in the relevant requirements of Annex VI (Part-ARA) and Annex VII (Part-ORA); and
 - (b) the qualifications of the instructor.

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
SECTION 1 – Preflight preparations and checks								
1.1	Powered-lift aircraft exterior visual inspection; location of each item and purpose of inspection				P			
1.2	Cockpit inspection	P	---->	---->	---->			

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
1.3	Starting procedures, radio and navigation equipment check, selection and setting of navigation and communication frequencies	P	---->	---->	---->		M	
1.4	Taxiing in compliance with ATC instructions or with instructions of an instructor		P	---->	---->			
1.5	Pre-take-off procedures and checks including power check	P	---->	---->	---->		M	
SECTION 2 – Flight manoeuvres and procedures								
2.1	Normal VFR take-off profiles: Runway operations (short take-off and landing (STOL) and vertical take-off and landing (VTOL)) including crosswind Elevated heliports Ground level heliports		P	---->	---->		M	
2.2	Take-off at maximum take-off mass (actual or		P	---->				

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
	simulated maximum take-off mass)							
2.3.1	Rejected take-off: – during runway operations; – during elevated heliport operations; and – during ground level operations.		P	---->			M	
2.3.2	Take-off with simulated engine failure after passing decision point: during runway operations; during elevated heliport operations; and during ground level operations.		P	---->			M	
2.4	Autorotative descent in helicopter mode to ground (an aircraft shall not be used for this exercise)	P	---->	---->			M FFS only	
2.4.1	Windmill descent in aeroplane mode (an aircraft shall not be used for this exercise)		P	---->			M FFS only	

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
2.5	Normal VFR landing profiles: runway operations (STOL and VTOL) elevated heliports ground level heliports		P	---->	---->		M	
2.5.1	Landing with simulated engine failure after reaching decision point: – during runway operations; – during elevated heliport operations; and – during ground level operations.							
2.6	Go-around or landing following simulated engine failure before decision point		P	---->			M	
SECTION 3 – Normal and abnormal operations of the following systems and procedures:								
3	Normal and abnormal operations of the following systems and procedures (may be completed in an FSTD if qualified for the exercise):						M	A mandatory minimum of 3 items shall be selected from this section

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
3.1	Engine	P	---->	---->				
3.2	Pressurisation and air conditioning (heating, ventilation)	P	---->	---->				
3.3	Pitot/static system	P	---->	---->				
3.4	Fuel System	P	---->	---->				
3.5	Electrical system	P	---->	---->				
3.6	Hydraulic system	P	---->	---->				
3.7	Flight control and trim system	P	---->	---->				
3.8	Anti-icing and de-icing system, glare shield heating (if fitted)	P	---->	---->				
3.9	Autopilot/Flight director	P	--->	--->				
3.10	Stall warning devices or stall avoidance devices and stability augmentation devices	P	---->	---->				
3.11	Weather radar, radio altimeter, transponder, ground proximity warning system (if fitted)	P	---->	---->				
3.12	Landing gear system	P	---->	---->				
3.13	APU	P	---->	---->				

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
3.14	Radio, navigation equipment, instruments and FMS	P	---->	---->				
3.15	Flap system	P	---->	---->				
SECTION 4 – Abnormal and emergency procedures								
4	Abnormal and emergency procedures (may be completed in an FSTD if qualified for the exercise)						M	A mandatory minimum of 3 items shall be selected from this section
4.1	Fire drills, engine, APU, cargo compartment, flight deck and electrical fires including evacuation if applicable	P	---->	---->				
4.2	Smoke control and removal	P	---->	---->				
4.3	Engine failures, shutdown and restart (an aircraft shall not be used for this exercise) including one engine inoperative conversion from helicopter to	P	---->	---->			FFS only	

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
	aeroplane modes and vice versa							
4.4	Fuel dumping (simulated, if fitted)	P	---->	---->				
4.5	Wind shear at take-off and landing (an aircraft shall not be used for this exercise)			P			FFS only	
4.6	Simulated cabin pressure failure/emergency descent (an aircraft shall not be used for this exercise)	P	---->	---->			FFS only	
4.7	ACAS event (an aircraft shall not be used for this exercise)	P	---->	---->			FFS only	
4.8	Incapacitation of crew member	P	---->	---->				
4.9	Transmission malfunctions	P	---->	---->			FFS only	

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
4.10	Recovery from a full stall (power on and off) or after activation of stall warning devices in climb, cruise and approach configurations (an aircraft shall not be used for this exercise)	P	---->	---->			FFS only	
4.11	Other emergency procedures as detailed in the appropriate flight manual	P	---->	---->				
SECTION 5 — Instrument flight procedures (to be performed in IMC or simulated IMC)								
5.1	Instrument take-off: transition to instrument flight is required as soon as possible after becoming airborne	P*	---->*	---->*				
5.1.1	Simulated engine failure during departure after decision point	P*	---->*	---->*			M*	
5.2	Adherence to departure and arrival routes and ATC instructions	P*	---->*	---->*			M*	
5.3	Holding procedures	P*	---->*	---->*				

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
5.4	Precision approach down to a decision height not less than 60 m (200 ft)	P*	---->*	---->*				
5.4.1	Manually, without flight director	P*	---->*	---->*			M* (Skill test only)	
5.4.2	Manually, with flight director	P*	---->*	---->*				
5.4.3	With use of autopilot	P*	---->*	---->*				
5.4.4	Manually, with one engine simulated inoperative; engine failure has to be simulated during final approach before passing the OM and continued either to touchdown or until completion of the missed approach procedure	P*	---->*	---->*			M*	
5.5	Non-precision approach down to the MDA/H	P*	---->*	---->*			M*	
5.6	Go-around with all engines operating on reaching DA/H or MDA/MDH	P*	---->*	---->*				

POWERED-LIFT AIRCRAFT CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	PL	Instructor's initials when training completed	Checked in FFS PL	Examiner's initials when test completed
5.6.1	Other missed approach procedures	P*	---->*	---->*				
5.6.2	Go-around with one engine simulated inoperative on reaching DA/H or MDA/MDH	P*					M*	
5.7	IMC autorotation with power recovery to land on runway in helicopter mode only (an aircraft shall not be used for this exercise)	P*	---->*	---->*			M* FFS only	
5.8	Recovery from unusual attitudes (this one depends on the quality of the FFS)	P*	---->*	---->*			M*	
SECTION 6 — Optional equipment								
6	Use of optional equipment		P	---->	---->			

E. Specific requirements for the airship category

1. In the case of skill tests or proficiency checks for airship type ratings, applicants shall pass Sections 1 to 5 and 6 (as applicable) of the skill test or proficiency check. Failure in more than five items will require applicants to repeat the entire test or check. Applicants failing not more than five items shall take the failed items again. Failure in any item in the case of a retest or a recheck, or failure in any other items already passed will require applicants to repeat the entire test or check again. All sections of the skill test or proficiency check shall be completed within 6 months.

FLIGHT TEST TOLERANCE

2. Applicants shall demonstrate the ability to:
- (a) operate the airship within its limitations;
 - (b) complete all manoeuvres with smoothness and accuracy;
 - (c) exercise good judgement and airmanship;
 - (d) apply aeronautical knowledge;
 - (e) maintain control of the airship at all times in such a manner that the successful outcome of a procedure or manoeuvre is never in doubt;
 - (f) understand and apply crew coordination and incapacitation procedures; and
 - (g) communicate effectively with the other crew members.
3. The following limits shall apply, corrected to make allowance for turbulent conditions and the handling qualities and performance of the airship used.
- (a) **IFR flight limits:**
 - Height
 - Generally ± 100 ft
 - Starting a go-around at decision height/altitude $+50$ ft/ -0 ft
 - Minimum descent height/altitude $+50$ ft/ -0 ft
 - Tracking
 - On radio aids $\pm 5^\circ$
 - Precision approach half-scale deflection, azimuth and glide path
 - Heading
 - Normal operations $\pm 5^\circ$
 - Abnormal operations/emergencies $\pm 10^\circ$
 - (b) **VFR flight limits:**

Height	
Generally	±100 ft
Heading	
Normal operations	±5°
Abnormal operations/emergencies	±10°

CONTENT OF THE TRAINING/SKILL TEST/PROFICIENCY CHECK

4. The following symbol means:
P = Trained as PIC or co-pilot and as PF and PM for the issue of a type rating as applicable.
5. The practical training shall be conducted at least at the training equipment level shown as (P), or may be conducted up to any higher equipment level shown by the arrow (---->).
6. The following abbreviations are used to indicate the training equipment used:
FFS = full-flight simulator
FTD = flight training device
OTD = other training device
As = airship
- 6a. The starred items (*) shall be flown solely by reference to instruments. If this condition is not met during the skill test or proficiency check, the type rating will be restricted to VFR only.
7. Where the letter 'M' appears in the skill test or proficiency check column, this will indicate a mandatory exercise.
8. FSTDs shall be used for practical training and testing if they form part of a type rating course. The following considerations will apply to the course:
 - (a) the qualification of the FSTDs as set out in the relevant requirements of Annex VI (Part-ARA) and Annex VII (Part-ORA); and
 - (b) the qualifications of the instructor.

AIRSHIP CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	As	Instructor's initials when training completed	Checked in FFS As	Examiner's initials when test completed
SECTION 1 – Preflight preparations and checks								
1.1	Preflight inspection				P			
1.2	Cockpit inspection	P	---->	---->	---->			
1.3	Starting procedures, radio and navigation equipment check, selection and setting of navigation and communication frequencies		P	---->	---->		M	
1.4	Off-mast procedure and ground manoeuvring			P	---->		M	
1.5	Pre-take-off procedures and checks	P	---->	---->	---->		M	
SECTION 2 – Flight manoeuvres and procedures								
2.1	Normal VFR take-off profile			P	---->		M	
2.2	Take-off with simulated engine failure			P	---->		M	
2.3	Take-off with heaviness > 0 (Heavy T/O)			P	---->			
2.4	Take-off with heaviness < 0 (Light/T/O)			P	---->			
2.5	Normal climb procedure			P	---->			
2.6	Climb to Pressure Height			P	---->			

AIRSHIP CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	As	Instructor's initials when training completed	Checked in FFS As	Examiner's initials when test completed
2.7	Recognising of pressure height			P	---->			
2.8	Flight at or close to pressure height			P	---->		M	
2.9	Normal descent and approach			P	---->			
2.10	Normal VFR landing profile			P	---->		M	
2.11	Landing with heaviness > 0 (Heavy Ldg.)			P	---->		M	
2.12	Landing with heaviness < 0 (Light Ldg.)			P	---->		M	
	Intentionally left blank							
SECTION 3 – Normal and abnormal operations of the following systems and procedures								
3	Normal and abnormal operations of the following systems and procedures (may be completed in an FSTD if qualified for the exercise):						M	A mandatory minimum of 3 items shall be selected from this section
3.1	Engine	P	---->	---->	---->			
3.2	Envelope pressurisation	P	---->	---->	---->			
3.3	Pitot/static system	P	---->	---->	---->			
3.4	Fuel system	P	---->	---->	---->			

AIRSHIP CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	As	Instructor's initials when training completed	Checked in FFS As	Examiner's initials when test completed
3.5	Electrical system	P	---->	---->	---->			
3.6	Hydraulic system	P	---->	---->	---->			
3.7	Flight control and trim system	P	---->	---->	---->			
3.8	Ballonet system	P	---->	---->	---->			
3.9	Autopilot/flight director	P	--->	--->	---->			
3.10	Stability augmentation devices	P	---->	---->	---->			
3.11	Weather radar, radio altimeter, transponder, ground proximity warning system (if fitted)	P	---->	---->	---->			
3.12	Landing gear system	P	---->	---->	---->			
3.13	APU	P	---->	---->	---->			
3.14	Radio, navigation equipment, instruments and FMS	P	---->	---->	---->			
	Intentionally left blank							
SECTION 4 – Abnormal and emergency procedures								
4	Abnormal and emergency procedures (may be completed in an FSTD if qualified for the exercise)						M	A mandatory minimum of three items shall be selected

AIRSHIP CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	As	Instructor's initials when training completed	Checked in FFS As	Examiner's initials when test completed
								from this section
4.1	Fire drills, engine, APU, cargo compartment, flight deck and electrical fires, including evacuation if applicable	P	---->	---->	---->			
4.2	Smoke control and removal	P	---->	---->	---->			
4.3	Engine failures, shutdown and restart: in particular phases of flight, inclusive multiple engine failure	P	---->	---->	---->			
4.4	Incapacitation of crew member	P	---->	---->	---->			
4.5	Transmission/gearbox malfunctions	P	---->	---->	---->		FFS only	
4.6	Other emergency procedures as outlined in the appropriate flight manual	P	---->	---->	---->			
SECTION 5 – Instrument Flight Procedures (to be performed in IMC or simulated IMC)								

AIRSHIP CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	As	Instructor's initials when training completed	Checked in FFS As	Examiner's initials when test completed
5.1	Instrument take-off: transition to instrument flight is required as soon as possible after becoming airborne	P*	---->*	---->*	---->*			
5.1.1	Simulated engine failure during departure	P*	---->*	---->*	---->*		M*	
5.2	Adherence to departure and arrival routes and ATC instructions	P*	---->*	---->*	---->*		M*	
5.3	Holding procedures	P*	---->*	---->*	---->*			
5.4	Precision approach down to a decision height not less than 60 m (200 ft)	P*	---->*	---->*	---->*			
5.4.1	Manually, without flight director	P*	---->*	---->*	---->*		M* (Skill test only)	
5.4.2	Manually, with flight director	P*	---->*	---->*	---->*			
5.4.3	With use of autopilot	P*	---->*	---->*	---->*			
5.4.4	Manually, with one engine simulated inoperative; engine failure has to be simulated during final	P*	---->*	---->*	---->*		M*	

AIRSHIP CATEGORY		PRACTICAL TRAINING					SKILL TEST OR PROFICIENCY CHECK	
Manoeuvres/Procedures		OTD	FTD	FFS	As	Instructor's initials when training completed	Checked in FFS As	Examiner's initials when test completed
	approach before passing the OM and continued to touchdown or until completion of the missed approach procedure							
5.5	Non-precision approach down to the MDA/H	P*	---->*	---->*	---->*		M*	
5.6	Go-around with all engines operating on reaching DA/H or MDA/MDH	P*	---->*	---->*	---->*			
5.6.1	Other missed approach procedures	P*	---->*	---->*	---->*			
5.6.2	Go-around with one engine simulated inoperative on reaching DA/H or MDA/MDH	P*					M*	
5.7	Recovery from unusual attitudes (this one depends on the quality of the FFS)	P*	---->*	---->*	---->*		M*	
SECTION 6 – Optional equipment								
6	Use of optional equipment		P	---->				

AMC1 to Appendix 9 Training, skill test and proficiency check for MPL, ATPL, type and class ratings, and proficiency check for IRs

ED Decision 2011/016/R

APPLICATION AND REPORT FORM

If applicable, this form is also the certificate of completion of the type rating course for ZFTT.

APPLICATION AND REPORT FORM				
ATPL, MPL, TYPE RATING, TRAINING, SKILL TEST AND PROFICIENCY CHECK				
AEROPLANES (A) AND HELICOPTERS (H)				
Applicant's last name(s):	Aircraft:	SE-SP: A ☐ H ☐	ME-SP: A ☐ H ☐	
Applicant's first name(s):		SE-MP: A ☐ H ☐	ME-MP: A ☐ H ☐	
Signature of applicant:	Operations:	SP ☐ MP ☐		
Type of licence held:	Checklist:	Training record: ☐	Type rating: ☐	
Licence number:		Skill test: ☐	Class rating: ☐	
		IR: ☐		
State of licence issue:		Proficiency check: ☐	ATPL: ☐	MPL: ☐

1	Theoretical training for the issue of a type or class rating performed during period			
From:		To:		At:
Mark obtained:		% (Pass mark 75%):		Type and number of licence:
Signature of HT:			Name(s) in capital letters:	
2	FSTD			
FSTD (aircraft type):		Three or more axes: Yes ☐ No ☐	Ready for service and used: ☐	
FSTD manufacturer:		Motion or system:	Visual aid: Yes ☐ No ☐	
FSTD operator:			FSTD ID code:	
Total training time at the controls:			Instrument approaches at aerodromes to a decision altitude or height of:	
Location, date and time:			Type and number of licence:	
Type rating instructor ☐ Class rating instructor ☐ instructor ☐				
Signature of instructor:			Name(s) in capital letters:	
3	Flight training: in the aircraft		in the FSTD (for ZFTT)	
Type of aircraft:	Registration:	Flight time at the controls:		

Take-offs:		Landings:		Training aerodromes or sites (take-offs, approaches and landings):	
Take-off time:				Landing time:	
Location and date:				Type and number of licence held:	
Type rating instructor ☐ Class rating instructor ☐					
Signature of instructor:				Name(s) in capital letters:	
4	Skill test		Proficiency check		
Skill test and proficiency check details:					
Aerodrome or site:				Total flight time:	
Take-off time:				Landing time:	
Pass ☐		Fail ☐		Reason(s) why, if failed:	
Location and date:				SIM or aircraft registration:	
Examiner's certificate number (if applicable):				Type and number of licence:	
Signature of examiner:				Name(s) in capital letters:	

AMC2 to Appendix 9 Training, skill test and proficiency check for MPL, ATPL, type and class ratings, and proficiency check for IRs

ED Decision 2011/016/R

TRAINING, SKILL TEST AND PROFICIENCY CHECK: SP AEROPLANES

Section 3.B of the training and skill test and proficiency check content for SP aeroplanes included in Appendix 9.B should include training on a circling approach, after an IFR approach.

GM1 to Appendix 9 Training, skill test and proficiency check for MPL, ATPL, type and class ratings, and proficiency check for IRs

ED Decision 2019/005/R

TYPE SPECIFIC UPRT AND GO-AROUND TRAINING IN FSTD

(a) General

- (1) The upset recovery training exercises should be mainly manoeuvre-based but may include some scenario-based training elements. The manoeuvre-based training enables type rating applicants to apply their handling skills and recovery strategy whilst leveraging CRM principles to return the aeroplane from an upset condition to a stabilised flight path.
- (2) If training is conducted in an FSTD, it is important that applicants understand the limitations of the FSTD in replicating the physiological and psychological aspects of upset recovery exercises.

Note: In order to avoid negative training and negative transfer of training, the ATO should ensure that the selected upset recovery exercises take into consideration the limitations of the FFS.

(b) Stall event recovery in FSTD (Appendix 9, Section B(5) exercise 7.2.1; Section B(6) exercise 3.7.1)

- (1) It is of utmost importance that stall event recovery training takes into account the capabilities of the FFS used. To deliver stall event recovery training, the FFS should be qualified against the relevant UPRT elements of CS-FSTD Issue 2. Stall event recovery training should include training up to the stall (approach-to-stall). Post-stall training may be delivered provided the device has been qualified against the relevant optional elements of CS-FSTD Issue 2 and the operator demonstrates that negative training or negative transfer of training is avoided. A 'stall event' is defined as an occurrence whereby the aeroplane experiences one or more conditions associated with an approach-to-stall or a post stall.
- (2) Stall event recovery training should emphasise the requirement to reduce the AoA whilst accepting the resulting altitude loss. High-altitude stall event training should be included so that flight crew experience the aeroplane control response, the significant altitude loss during the recovery, and the increased time required to recover. The training should also emphasise the risk of triggering a secondary stall event during the recovery.
- (3) Recovery from a stall event should always be conducted in accordance with the stall event recovery procedures of the OEMs.

Note: If an OEM-approved recovery procedure does not exist, ATOs should develop and train the aeroplane-specific stall recovery procedure based on the template in Table 1 below. Refer to Revision 3 of the Airplane Upset Prevention and Recovery Training Aid (AUPRTA) for a detailed explanation and rationale of the stall event recovery template as recommended by the OEMs.

Table 1: Recommended stall event recovery template

Stall event recovery template		
Pilot Flying (PF)		Pilot Monitoring (PM)
Immediately do the following at first indication of a stall (aerodynamic buffeting, reduced roll stability and aileron effectiveness, visual or aural cues and warnings, reduced elevator (pitch) authority, inability to maintain altitude or arrest rate of descent, stick shaker activation (if installed)) during any flight phases <i>except at lift-off</i> .		
1.	AUTOPILOT — DISCONNECT (A large out-of-trim condition could be encountered when the autopilot is disconnected)	MONITOR airspeed and attitude throughout the recovery and ANNOUNCE any continued divergence
2.	AUTOTHRUST/AUTOTHROTTLE — OFF	
3.	(a) NOSE-DOWN PITCH CONTROL apply until stall warning is eliminated (b) NOSE-DOWN PITCH TRIM (as needed) (Reduce the AoA whilst accepting the resulting altitude loss.)	
4.	BANK — WINGS LEVEL	
5.	THRUST — ADJUST (as needed) (Thrust reduction for aeroplanes with underwing-mounted engines may be needed)	
6.	SPEEDBRAKES/SPOILERS — RETRACT	
7.	When airspeed is sufficiently increasing — RECOVER to level flight (Avoid the secondary stall due to premature recovery or excessive G-loading)	

- (c) Nose-high and nose-low recovery exercises (Appendix 9, Section B(5) exercise 7.2.2; B(6) exercise 3.7.2)

Nose-high and nose-low recovery exercises should be conducted in accordance with the strategies recommended by the OEMs contained in Tables 2 and 3 below.

Note: As the OEM procedures always take precedence over the recommendations, ATOs should consult the OEM on whether any approved type-specific recovery procedures are available prior to using the templates.

Refer to Revision 3 of the Airplane Upset Prevention and Recovery Training Aid (AUPRTA) for a detailed explanation and rationale of nose-high and nose-low recovery strategies as recommended by the OEMs.

Table 2: Recommended nose-high recovery strategy template

Nose-high recovery strategy template		
Either pilot — Recognise and confirm the developing situation by announcing ‘nose high’		
PF		PM
1.	AUTOPILOT — DISCONNECT (A large out-of-trim condition could be encountered when the autopilot is disconnected)	MONITOR airspeed and attitude throughout the recovery and ANNOUNCE any continued divergence
2.	AUTOTHRUST/AUTOTHROTTLE — OFF	
3.	APPLY as much nose-down control input as required to obtain a nose-down pitch rate	
4.	THRUST — ADJUST (if required) (Thrust reduction for aeroplanes with underwing-mounted engines may be needed)	
5.	ROLL — ADJUST (if required) (Avoid exceeding 60-degree bank)	
6.	When airspeed is sufficiently increasing — RECOVER to level flight (Avoid the secondary stall due to premature recovery or excessive G-loading)	
NOTE:		
(1) Recovery to level flight may require use of pitch trim.		
(2) If necessary, consider reducing thrust in aeroplanes with underwing-mounted engines to aid in achieving nose-down pitch rate.		
(3) WARNING: Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.		

Table 3: Recommended nose-low recovery strategy template

Nose-low recovery strategy template		
Either pilot — Recognise and confirm the developing situation by announcing ‘nose low’ (If the autopilot or autothrust/autothrottle is responding correctly, it may not be appropriate to decrease the level of automation while assessing if the divergence is being stopped)		
	PF	PM
1.	AUTOPILOT — DISCONNECT (A large out-of-trim condition could be encountered when the autopilot is disconnected)	MONITOR airspeed and attitude throughout the recovery and ANNOUNCE any continued divergence
2.	AUTOTHRUST/AUTOTHROTTLE — OFF	
3.	RECOVERY from stall if required	
4.	ROLL in the shortest direction to wings level (It may be necessary to reduce the G-loading by applying forward control pressure to improve roll effectiveness)	
5.	THRUST and DRAG — ADJUST (if required)	
6.	RECOVER to level flight (Avoid the secondary stall due to premature recovery or excessive G-loading.)	

NOTE:

- (1) Recovery to level flight may require use of pitch trim.
- (2) **WARNING:** Excessive use of pitch trim or rudder may aggravate the upset situation or may result in high structural loads.

- (d) Go-around with all engines operating from various stages during an instrument approach (Appendix 9, Section B(5) exercise 7.3; B(6) exercise 4.1.)
 - (1) The objective of the go-around exercises is to expose the student pilot to the physiological effects caused by a go-around. The instructor should ensure that student pilots understand the objective of the exercises and provide students with appropriate coping strategies, including TEM. Due consideration should be given to environmental conditions when evaluating the demonstration of task proficiency and related criteria.
 - (2) A go-around may be commenced at any time during an approach, including before the aeroplane is in the landing configuration. Historically, most go-around training has been conducted when the aeroplane is in the landing configuration prior to commencing the go-around. Students must be prepared to adapt the go-around manoeuvre if the go-around is commenced prior to the point where the aeroplane is fully configured for landing. Situation awareness in relation to flap and gear configuration, aeroplane speed and missed approach altitude is important.
 - (3) Unanticipated go-arounds may startle the students (e.g. unexpected ATC constraints, automation malfunction, adverse weather, etc.). Students may find themselves faced with a situation where they have to perform a large number of critical actions under a high workload (e.g. setting thrust, landing gear retraction, flight path management). The instructor should explain that there is also a possibility of disorientation during a go-around because of the somatogravic effect produced by large longitudinal acceleration felt by the inner-ear as the aeroplane speed increases. This effect cannot be reproduced in an FSTD.
 - (4) It is vital that the correct pitch attitude is selected and maintained, while the aeroplane is kept in trim as it accelerates (depending on the aeroplane type). On some aeroplane types with under-slung engines the pitch response with all engines functioning may be amplified due to the relatively low gross weight towards the end of a flight and the high thrust available from modern aeroplane engines. It is particularly important that trim changes are anticipated on such aeroplanes.
 - (5) ATOs should develop scenarios for go-around training containing different take-off and approach stall situations that also involve surprise and startle effects and include:
 - (i) a go-around from the non-landing configuration;
 - (ii) a go-around at low gross weight using maximum go-around thrust;
 - (iii) a go-around from the outer marker or equivalent point;
 - (iv) a go-around below 500 ft using, as applicable/permitted, reduced go-around thrust;
 - (v) a go-around initiated above the published missed approach altitude; and
 - (vi) a normal go-around from the landing configuration using reduced go-around thrust (if available / type-specific).

- (6) Training should also incorporate topics such as flight path management (manual and automatic), application of procedures, startle factors, communication, workload management and situation awareness. The objective of this training is to highlight:
- (i) differences to procedures when the aircraft is in the non-landing configuration;
 - (ii) differences in handling characteristics at low gross weights and high thrust settings;
 - (iii) the threat associated with go-arounds close to the published missed approach altitudes;
 - (iv) startle and surprise associated with an unplanned go-around (ATC, blocked runway, etc.);
 - (v) the importance of effective communication between flight crew;
 - (vi) the requirement to be aware of the aircraft energy state during a go-around; and
 - (vii) the importance of engaging the autopilot or flight director in the correct modes during a go-around.
- (7) Go-around training should not be limited to addressing the somatogravic effects caused by a go-around. Training should also cover topics such as flight path management (manual and automatic), application of procedures, startle factor, communication, workload management and situation awareness. Flight path management training should address:
- (i) the handling differences of a lighter than normal aircraft which may differ to handling experienced during take-off when the aircraft is much heavier;
 - (ii) the different reaction of the aeroplane (pitch and vertical speed) comparing a go-around performed with reduced G/A thrust (if the function is available) and a go-around performed with full G/A thrust (a different weight).
- (8) The importance of correct selection of TO/GA modes by the PF should also be emphasised (pushing TO/GA, selected the correct thrust lever detent, etc.)
- (9) The importance of the PM role in the go-around manoeuvre should also be highlighted. The PM usually has higher workload as they need to reconfigure the aircraft, engage FMA modes, communicate with ATC and monitor the actions of the PF. This excessive workload for the PM may lead him or her to prioritise actions to the detriment of monitoring activities. The phenomenon of attentional tunnelling may also need to be addressed. This happens when one pilot, or both, focus exclusively on a problem at the expense of general monitoring of the flight parameters.

GM1 to Appendix 9 Section A point 1c Training, skill test and proficiency check for the MPL, and the ATPL, and for type and class ratings, and proficiency checks for the BIR and the IR

ED Decision 2025/002/R

FRAMEWORK FOR THE MANDATORY USE OF A FULL-FLIGHT SIMULATOR IN THE CASE OF SINGLE-PILOT AEROPLANES AND HELICOPTERS

- (a) Partial availability of FFS and additional in-aircraft training

In the case of single-pilot aeroplanes (SPAs) and helicopters, [Appendix 9](#), Section A, point 1c requires training to be conducted in a full-flight simulator (FFS), if available and accessible, or in a combination of such FFS and FSTD(s). In that context, an FFS can be deemed 'available' only to the extent to which that FFS can, based on its technical capabilities, serve as training platform for class rating or type rating training. For example, if the FFS that exists for a particular type of aircraft can only serve as a training platform for 80 % of the type rating training syllabus (e.g. in the case of helicopters, near-ground manoeuvres cannot be simulated appropriately), that FFS is 'available' only to 80 % of the training syllabus.

The remaining 20 % of the training syllabus need to be completed in an FSTD that is suitable for the relevant exercises, or in the aircraft.

Independently, [Appendix 9](#) Section A point 1a contains a requirement for applicants to complete additional take-off and landing training in the aircraft (unless undergoing ZFTT, or undergoing type rating training for cruise relief co-pilots).

(b) In-aircraft training arrangements in older operational suitability data (OSD) for flight crew

The framework and conditions for the mandatory use of FFSs and other flight simulation training devices (FSTDs) in training, skill tests, and proficiency checks for class ratings and type ratings were introduced into Part-FCL, [Appendix 9](#), Section A through Regulation (EU) 2018/1974 and became applicable as from 20 December 2019. From that day, training, skill tests, and proficiency checks for class ratings and type ratings for SPAs and helicopters needed to be conducted in FFSs, or, in cases where an FFS is not available and accessible, in other FSTDs in combination with in-aircraft training. The subsequent Regulation (EU) 2024/2076 complemented this framework with the option to combine training in FFS, if available and accessible, and FSTDs. Completing training courses solely in the aircraft is only possible in the case of SPAs or helicopters for which no FSTD exists.

Training considerations (not mandatory training elements) in OSD for flight crew which was established prior to 20 December 2019 may refer to in-aircraft training although an FFS exists for the respective type of aircraft. Today, that OSD for flight crew needs to be understood in the light of the requirements of the currently applicable Part-FCL, [Appendix 9](#) Section A as those requirements do not allow OSD to determine alternative training platform arrangements (OSD cannot overrule Part-FCL, [Appendix 9](#) as regards the requirements of that Appendix for the use of FSTDs). This means that the in-aircraft training content that is referred to in training considerations for flight crew in such older OSD after 20 December 2019, needs to be delivered in an FFS in accordance with Part-FCL [Appendix 9](#) Section A and considering the content of point (a) of this GM.

GM1 to Appendix 9 Section B point (5)(l); Section B point (6)(j) Training, skill test and proficiency check for the MPL, and the ATPL, and for type and class ratings, and proficiency checks for the BIR and the IR

ED Decision 2025/002/R

PBN PRIVILEGES WITHOUT RNP APCH PRIVILEGES

- (a) RNP APCH means PBN procedures for final approaches until touchdown. PBN privileges without RNP APCH therefore include all of the following:
- (1) standard instrument departures (SIDs);

- (2) standard arrivals (STARs);
 - (3) en-route flight.
- (b) The exclusion of RNP APCH privileges can be documented through an appropriate remark (e.g. 'No RNP APCH') on the licence (instrument rating).

Appendix 10 – Revalidation and renewal of type ratings, and revalidation and renewal of IRs when combined with the revalidation or renewal of type ratings – EBT practical assessment

Regulation (EU) 2020/2193

A – General

1. The revalidation and renewal of type ratings as well as the revalidation and renewal of IRs when combined with the revalidation or renewal of type ratings in accordance with this Appendix shall be completed only at EBT operators which comply with all of the following:
 - (a) they have established an EBT programme relevant for the applicable type rating or the IR in accordance with point ORO.FC.231 of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#);
 - (b) they have an experience of at least 3 years in conducting a mixed EBT programme;
 - (c) for each type rating within the EBT programme, the organisation has appointed an EBT manager. EBT managers shall comply with all of the following:
 - (i) they shall hold examiner privileges for the relevant type rating;
 - (ii) they shall have extensive experience in training as an instructor for the relevant type rating;
 - (iii) they shall either be the person nominated in accordance with point ORO.AOC.135(a)(2) of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#) or a deputy of that person.
2. The EBT manager responsible for the relevant type rating shall ensure that the applicant complies with all qualification, training and experience requirements of this Annex for the revalidation or the renewal of the relevant rating.
3. Applicants who wish to revalidate or renew a rating in accordance with this Appendix shall comply with all of the following:
 - (a) they shall be enrolled in the operator's EBT programme;
 - (b) in the case of revalidation of a rating, they shall complete the operator's EBT programme within the period of validity of the relevant rating;
 - (c) in the case of renewal of a rating, they shall comply with procedures developed by the EBT operator in accordance with point ORO.FC.231(a)(5) of Annex III (Part-ORO) to [Regulation \(EU\) No 965/2012](#).
4. The revalidation or renewal of a rating in accordance with this Appendix shall comprise all of the following:
 - (a) continuous EBT practical assessment within an EBT programme;
 - (b) demonstration of an acceptable level of performance in all competencies;
 - (c) the administrative action of licence revalidation or renewal for which the EBT manager responsible for the relevant type rating shall do all of the following:
 - (1) ensure that the requirements of point [FCL.1030](#) are complied with;
 - (2) when acting in accordance with point [FCL.1030\(b\)\(2\)](#), endorse the applicant's licence with the new expiry date of the rating. That endorsement may be

completed by another person on behalf of the EBT manager, if that person received a delegation from the EBT manager to do so in accordance with the procedures established in the EBT programme.

B – Conduct of the EBT practical assessment

The EBT practical assessment shall be conducted in accordance with the operator's EBT programme.

AMC1 to Appendix 10 Revalidation and renewal of type ratings, and revalidation and renewal of IRs when combined with the revalidation or renewal of type ratings – EBT practical assessment

ED Decision 2021/002/R

APPLICATION AND REPORT FORM — ADMINISTRATIVE PROCEDURES RELATED TO TYPE RATINGS

(a) Minimum information provided in the form for Appendix 10.

Applicant's last name(s):		Applicant's first name(s):	
Signature of applicant:		State of licence issue:	
Type of licence held:		Licence number:	
Type rating:		FSTD (aircraft type):	
EBT module 1	Session 1.....Name of the instructor: _____		
	Type and number of licence: _____		
	Location, date and time: _____ FSTD ID code: _____		
	Session 2.....Name of the instructor: _____		
	Type and number of licence: _____		
EBT module 2	Location, date and time: _____ FSTD ID code: _____		
	Session X.....Name of the instructor: _____		
	Type and number of licence: _____		
	Location, date and time: _____ FSTD ID code: _____		
	Completion of the module: ☐		
date / signature (EBT manager) _____			
EBT module 2	Session 1.....Name of the instructor: _____		
	Type and number of licence: _____		
	Location, date and time: _____ FSTD ID code: _____		
	Session 2.....Name of the instructor: _____		
	Type and number of licence: _____		
EBT module X	Location, date and time: _____ FSTD ID code: _____		
	Session X.....Name of the instructor: _____		
	Type and number of licence: _____		
	Location, date and time: _____ FSTD ID code: _____		
	Completion of the module: ☐		
date / signature (EBT manager) _____			
(...)			

Location, date and time: _____ FSTD ID code: _____	
Completion of the module: ☐	_____ date / signature (EBT manager)
Completion of the operator's EBT programme from ____ (date) to ____ (date) ☐	_____ date / signature (EBT manager)
Name(s) in capital letters: Type and number of licence: Examiner certificate number:	Signature of examiner (EBT manager) _____ Date of applicant's licence endorsement: _____
Delegation of signature for licence endorsement (instructor)	
Name: Position in the operator: Date:	Signature

- (b) AOC declaration for revalidation and renewal under the EBT programme for the purpose of [AMC1 ARA.GEN.315\(a\)](#) point (d) and for the purpose of point 1(a) of [Appendix 10](#).

I confirm all of the following:

The EBT manager holds a current type rating examiner certificate in the type rating filled in in Appendix 10 (copy to be attached);	YES ☐
The instructor(s) that conducted the training to the applicant has (have) been standardised.	YES ☐
The EBT operator has performed a verification of the grading system at least once in the last 3 years.	YES ☐
The integrity of the applicant training data is ensured.	YES ☐

Signature of the training manager or EBT manager _____

- (c) In order for the EBT manager to delegate their signature in accordance with point 4(c)(2) of [Appendix 10](#) to another person to endorse the licence of the applicant, the following should apply:
- (1) the person signing the licence should be nominated,
 - (2) the person signing the licence should hold or have held an instructor certificate,
 - (3) the approved procedure for delegation of signature should include procedures to prevent the person who received the delegation from signing the licence when the EBT programme applicable to the validity period has not been completed.
- (d) The authority may customise the form above by requesting additional information or changing the order of the elements of the form.

GM1 to Appendix 10 — Revalidation and renewal of type ratings, and revalidation and renewal of IRs when combined with the revalidation or renewal of type ratings — EBT practical assessment

ED Decision 2022/014/R

REVALIDATION AND RENEWAL OF TYPE RATING — ADMINISTRATIVE PROCEDURES

- (a) The operator may nominate several deputy EBT managers to ensure the availability of at least one examiner for each fleet, in the context of workload to manage the EBT programme, several locations of the training facilities, or bases, etc.
- (b) For the first revalidation of type rating after the transition from mixed EBT, the examiner may use mixed EBT module(s) in addition to the other EBT module(s) as a means to revalidate the type rating.
- (c) In accordance with the approved procedure in [Appendix 10](#), 4. (c)(2), and as provided in [AMC1 to Appendix 10](#) point (c), the EBT manager may nominate the EBT instructor who completed the EBT module as the person to whom the signature of the examiner is delegated. A stamp or electronic signature may exclusively be given from the EBT manager to the EBT instructor, in order to document the delegation in a transparent and secure manner. Following that process, EBT instructors on behalf of the EBT manager can endorse an applicant's licence.
- (d) The EBT programme is included in the operations manual and, in accordance with ORO.FC.145 of [Regulation \(EU\) No 965/2012](#), for CAT it is subject to prior approval including any changes, its syllabi and the use of individual FSTD.
- (e) The requirements for completion of the operator's EBT programme (see the form set out in [AMC1 to Appendix 10](#)) can be found in point ORO.FC.231(a)(3) of Annex III (Part-ORO) to [Regulation \(EU\) 965/2012](#) and include, as a minimum, the completion of:
 - (1) a minimum of two EBT modules;
 - (2) line evaluation(s) of competence. In the case of renewal of a type rating, the line evaluation of competence can be performed only when the pilot has their type rating renewed; and
 - (3) ground training.
- (f) The EBT programme includes equivalencies of approaches (point ORO.FC.231(g)) and determines the necessary training frequency for particular approaches. Therefore, pilots can maintain their PBN privileges when completing an EBT programme. Additionally, in the case of specific approaches related to PBN which require a specific approval in accordance with Subpart B of Annex V (Part-SPA) to [Regulation \(EU\) No 965/2012](#), the requirements of that Subpart (SPA.PBN) apply.

GM2 to Appendix 10 — Revalidation and renewal of type ratings, and revalidation and renewal of IRs when combined with the revalidation or renewal of type ratings – EBT practical assessment

ED Decision 2022/014/R

EBT PRACTICAL ASSESSMENT — PROFICIENCY CHECK

EBT practical assessment (or Practical assessment) is defined in [FCL.010](#). More information can be found in ICAO Doc 9868 'PANS-TRG'.

- (a) The demonstration of skills to revalidate or renew referred to in the definition of 'proficiency check' in point [FCL.010](#) is equivalent to the EBT practical assessments conducted in the EBT programme and the final review of the examiner. In fact, one single EBT practical assessment demonstrates the necessary skills performed in legacy training; however, EBT goes one step further — to revalidate or renew, the pilot performs at least two demonstrations, corresponding to at least two EBT modules within the validity period of the type rating.

- (b) A proficiency check is equivalent to the combination of the evaluation phase (EVAL) and manoeuvres training phase (MT). However, EBT also requires a scenario-based training phase (SBT) following the EVAL to complete the module.

ANNEX II – CONDITIONS FOR THE CONVERSION OF EXISTING NATIONAL LICENCES AND RATINGS FOR AEROPLANES AND HELICOPTERS

Regulation (EU) 2015/445

A. AEROPLANES

1. Pilot licences

A pilot licence issued by a Member State in accordance with the national requirements shall be converted into a Part-FCL licence provided that the applicant complies with the following requirements:

- (a) for ATPL(A) and CPL(A), complete as a proficiency check the revalidation requirements of Part-FCL for type/class and instrument rating, relevant to the privileges of the licence held;
- (b) demonstrate knowledge of the relevant parts of the operational requirements and Part-FCL;
- (c) demonstrate language proficiency in accordance with FCL.055;
- (d) comply with the requirements set out in the following table:

National licence held	Total flying hours experience	Any further requirements	Replacement Part-FCL licence and conditions (where applicable)	Removal of conditions	
(1)	(2)	(3)	(4)	(5)	
ATPL(A)	> 1 500 as PIC on multi-pilot aeroplanes	None	ATPL(A)	Not applicable	(a)
ATPL(A)	> 1 500 on multi-pilot aeroplanes	None	as in (c)(4)	as in (c)(5)	(b)
ATPL(A)	> 500 on multi-pilot aeroplanes	Demonstrate knowledge of flight planning and performance as required by FCL.515	ATPL(A), with type rating restricted to co-pilot	Demonstrate ability to act as PIC as required by Appendix 9 to Part-FCL	(c)
CPL/IR(A) and passed an ICAO ATPL theory test in the Member State of licence issue		(i) demonstrate knowledge of flight planning and performance as required by FCL.310 and FCL.615(b) (ii) meet remaining requirements of FCL.720.A(c)	CPL/IR(A) with ATPL theory credit	Not applicable	(d)
CPL/IR(A)	> 500 on multi-pilot aeroplanes,	(i) pass an examination for ATPL(A)	CPL/IR(A) with ATPL theory credit	Not applicable	(e)

	or in multi-pilot operations on single-pilot aeroplanes CS-23 commuter category or equivalent in accordance with the relevant requirements of Part-CAT and Part-ORO for commercial air transport	knowledge in the Member State of licence issue ¹ (ii) meet remaining requirements of FCL.720.A(c)			
CPL/IR(A)	> 500 as PIC on single-pilot aeroplanes	None	CPL/IR(A) with class ratings and type ratings restricted to single-pilot aeroplanes	Obtain multi-pilot type rating in accordance with Part-FCL	(f)
CPL/IR(A)	< 500 as PIC on single-pilot aeroplanes	Demonstrate knowledge of flight planning and flight performance for CPL/IR level	As (4)(f)	As (5)(f)	(g)
CPL(A)	> 500 as PIC on single-pilot aeroplanes	Night rating, if applicable	CPL(A), with type/class ratings restricted to single-pilot aeroplanes		(h)
CPL(A)	< 500 as PIC on single-pilot aeroplanes	(i) Night rating, if applicable; (ii) demonstrate knowledge of flight performance and planning as required by FCL.310	as (4)(h)		(i)
PPL/IR(A)	≥ 75 in accordance with IFR		PPL/IR(A) (the IR restricted to PPL)	Demonstrate knowledge of flight performance and planning as required by FCL.615(b)	(j)
PPL(A)	≥ 70 on aeroplanes	Demonstrate the use of radio navigation aids	PPL(A)		(k)

2. Instructor certificates

¹ CPL holders already holding a type rating for a multi-pilot aeroplane are not required to have passed an examination for ATPL(A) theoretical knowledge whilst they continue to operate that same aeroplane type, but will not be given ATPL(A) theory credit for a Part-FCL licence. If they require another type rating for a different multi-pilot aeroplane, they must comply with column (3), row (e)(i) of the above table.

An instructor certificate issued by a Member State in accordance with the national requirements shall be converted into a Part-FCL certificate provided that the applicant complies with the following requirements:

National certificate or privileges held	Experience	Any further requirements	Replacement Part-FCL certificate
(1)	(2)	(3)	(4)
FI(A)/IRI(A)/TRI(A)/CRI(A)	as required under Part-FCL for the relevant certificate	N/A	FI(A)/IRI(A)/TRI(A)/CRI(A)

3. SFI certificate

A SFI certificate issued by a Member State in accordance with the national requirements shall be converted into a Part-FCL certificate provided that the holder complies with the following requirements:

National certificate held	Experience	Any further requirements	Replacement Part-FCL certificate
(1)	(2)	(3)	(4)
SFI(A)	>1500 hours as pilot of MPA	(i) hold or have held a CPL, MPL or ATPL for aeroplanes issued by a Member State; (ii) have completed the flight simulator content of the applicable type rating course including MCC.	SFI(A)
SFI(A)	3 years recent experience as a SFI	have completed the flight simulator content of the applicable type rating course including MCC	SFI(A)

The conversion shall be valid for a maximum period of 3 years. Revalidation shall be subject to the completion of the relevant requirements set out in Part-FCL.

4. STI certificate

An STI certificate issued by a Member State in accordance with the national requirements of that State may be converted into a Part-FCL certificate provided that the holder complies with the requirements set out in the table below:

National certificate held	Experience	Any further requirements	Replacement certificate
(1)	(2)	(3)	(4)
STI(A)	> 500 hours as pilot on SPA	(i) hold or have held a pilot licence issued by a Member State; (ii) have completed a proficiency check in accordance with Appendix 9 to Part-FCL in an FSTD appropriate to the instruction intended	STI(A)
STI(A)	3 years recent experience as a STI	have completed a proficiency check in accordance with Appendix 9 to Part-FCL in an FSTD appropriate to the instruction intended	STI(A)

Revalidation of the certificate shall be subject to the completion of the relevant requirements set out in Part-FCL.

B. HELICOPTERS

1. Pilot licences

A pilot licence issued by a Member State in accordance with the national requirements shall be converted into a Part-FCL licence provided that the applicant complies with the following requirements:

- (a) complete as a proficiency check the revalidation requirements of Part-FCL for type and instrument rating, relevant to the privileges of the licence held;
- (b) demonstrate knowledge of the relevant parts of the operational requirements and Part-FCL;
- (c) demonstrate language proficiency in accordance with FCL.055;
- (d) comply with the requirements set out in the following table:

National licence held	Total flying hours experience	Any further requirements	Replacement Part-FCL licence and conditions (where applicable)	Removal of conditions	
(1)	(2)	(3)	(4)	(5)	
ATPL(H) valid IR(H)	>1000 as PIC on multi-pilot helicopters	none	ATPL(H) and IR	Not applicable	(a)
ATPL(H) no IR(H) privileges	>1000 as PIC on multi-pilot helicopters	none	ATPL(H)		(b)
ATPL(H) valid IR(H)	>1000 on multi-pilot helicopters	None	ATPL(H), and IR with type rating restricted to co-pilot	demonstrate ability to act as PIC as required by Appendix 9 to Part-FCL	(c)
ATPL(H) no IR(H) privileges	>1000 on multi-pilot helicopters	None	ATPL(H) type rating restricted to co-pilot	demonstrate ability to act as PIC as required by Appendix 9 to Part-FCL	(d)
ATPL(H) valid IR(H)	>500 on multi-pilot helicopters	demonstrate knowledge of flight planning and flight performance as required by FCL.515 and FCL.615(b)	as (4)(c)	as (5)(c)	(e)
ATPL(H) no IR(H) privileges	>500 on multi-pilot helicopters	as (3)(e)	as (4)(d)	as (5)(d)	(f)
CPL/IR(H) and passed an ICAO ATPL(H) theory test in the Member State of licence issue		(i) demonstrate knowledge of flight planning and flight performance as required by FCL.310 and FCL.615(b);	CPL/IR(H) with ATPL(H) theory credit, provided that the ICAO ATPL(H) theory test is assessed as being at Part-FCL ATPL level	Not applicable	(g)

B. HELICOPTERS

National licence held	Total flying hours experience	Any further requirements	Replacement Part-FCL licence and conditions (where applicable)	Removal of conditions	
		(ii) meet remaining requirements of FCL.720.H(b)			
CPL/IR(H)	>500 hrs on multi-pilot helicopters	(i) to pass an examination for Part-FCL ATPL(H) theoretical knowledge in the Member State of licence issue ¹ (ii) to meet remaining requirements of FCL.720.H(b)	CPL/IR(H) with Part-FCL ATPL(H) theory credit	Not applicable	(h)
CPL/IR(H)	>500 as PIC on single-pilot helicopters	None	CPL/IR(H) with type ratings restricted to single-pilot helicopters	obtain multi-pilot type rating as required by Part-FCL	(i)
CPL/IR(H)	<500 as PIC on single-pilot helicopters	demonstrate knowledge of flight planning and flight performance as required by FCL.310 and FCL.615(b)	as (4)(i)		(j)
CPL(H)	>500 as PIC on single-pilot helicopters	night rating	CPL(H), with type ratings restricted to single-pilot helicopters		(k)
CPL(H)	<500 as PIC on single-pilot helicopters	night rating demonstrate knowledge of flight performance and planning as required by FCL.310	as (4) (k)		(l)
CPL(H) Without night rating	>500 as PIC on single-pilot helicopters		As (4)(k) and restricted to day VFR operations	Obtain multi-pilot type rating as required by Part-FCL and a night rating.	(m)
CPL(H) Without night rating	<500 as PIC on single-pilot helicopters	demonstrate knowledge of flight planning and flight performance as required by FCL.310	As (4)(k) and restricted to day VFR operations		(n)

¹ CPL holders already holding a type rating for a multi-pilot helicopter are not required to have passed an examination for ATPL(H) theoretical knowledge whilst they continue to operate that same helicopter type, but will not be given ATPL(H) theory credit for a Part-FCL licence. If they require another type rating for a different multi-pilot helicopter, they must comply with column (3), row (h)(i) of the table.

National licence held	Total flying hours experience	Any further requirements	Replacement Part-FCL licence and conditions (where applicable)	Removal of conditions	
PPL/IR(H)	≥75 in accordance with IFR		PPL/IR(H) (the IR restricted to PPL)	demonstrate knowledge of flight performance and planning as required by FCL.615(b)	(o)
PPL(H)	≥75 on helicopters	demonstrate the use of radio navigation aids	PPL (H)		(p)

2. Instructor certificates

An instructor certificate issued by a Member State in accordance with the national requirements shall be converted into a Part-FCL certificate provided that the applicant complies with the following requirements:

National certificate or privileges held	Experience	Any further requirements	Replacement certificate
(1)	(2)	(3)	(4)
FI(H)/IRI(H)/TRI(H)	as required under Part-FCL for the relevant certificate		FI(H)/IRI(H)/TRI(H) ⁽¹⁾

Revalidation of the certificate shall be subject to the completion of the relevant requirements set out in Part-FCL.

3. SFI certificate

An SFI certificate issued by a Member State in accordance with the national requirements shall be converted into a Part-FCL certificate provided that the holder complies with the following requirements:

National certificate held	Experience	Any further requirements	Replacement certificate
(1)	(2)	(3)	(4)
SFI(H)	>1.000 hours as pilot of MPH	(i) hold or have held a CPL, MPL or ATPL issued by a Member State; (ii) have completed the flight simulator content of the applicable type rating course including MCC	SFI(H)
SFI(H)	3 years recent experience as an SFI	have completed the simulator content of the applicable type rating course including MCC	SFI(H)

¹ CPL holders already holding a type rating for a multi-pilot helicopter are not required to have passed an examination for ATPL(H) theoretical knowledge whilst they continue to operate that same helicopter type, but will not be given ATPL(H) theory credit for a Part-FCL licence. If they require another type rating for a different multi-pilot helicopter, they must comply with column (3), row (h)(i) of the table.

Revalidation of the certificate shall be subject to the completion of the relevant requirements set out in Part-FCL.

4. STI certificate

An STI certificate issued by a Member State in accordance with the national requirements of that State may be converted into a Part-FCL certificate provided that the holder complies with the requirements set out in the table below:

National certificate held	Experience	Any further requirements	Replacement certificate
(1)	(2)	(3)	(4)
STI(H)	>500 hours as pilot on SPH	(i) hold or have held a pilot licence issued by a Member State; (ii) have completed a proficiency check in accordance with Appendix 9 to Part-FCL in an FSTD appropriate to the instruction intended	STI(H)
STI(H)	3 years recent experience as an STI	have completed a proficiency check in accordance with Appendix 9 to Part-FCL in an FSTD appropriate to the instruction intended	STI(H)

Revalidation of the certificate shall be subject to the completion of the relevant requirements set out in Part-FCL.

ANNEX III

Regulation (EU) 2020/723

[Deleted by [Commission Delegated Regulation \(EU\) 2020/723](#)]

ANNEX IV (PART-MED)

SUBPART A – GENERAL REQUIREMENTS

SECTION 1 – GENERAL

MED.A.001 Competent authority

Regulation (EU) 2019/27

For the purpose of this Annex (Part-MED), the competent authority shall be:

- (a) for aero-medical centres (AeMCs):
 - (1) the authority designated by the Member State, where the AeMC has its principal place of business;
 - (2) the Agency, if the AeMC is located in a third country;
- (b) for aero-medical examiners (AMEs):
 - (1) the authority designated by the Member State where the AME has its principal place of practice;
 - (2) if the principal place of practice of an AME is located in a third country, the authority designated by the Member State to which the AME applies for the issue of the AME certificate;
- (c) for general medical practitioners (GMPs), the authority designated by the Member State to which the GMP notify their activity;
- (d) for occupational health medical practitioners (OHMPs) assessing the medical fitness of cabin crew, the authority designated by the Member State to which the OHMP notify their activity.

MED.A.005 Scope

Regulation (EU) 2019/27

This Annex (Part-MED) establishes the requirements for:

- (a) the issuance, validity, revalidation and renewal of the medical certificate required for exercising the privileges of a pilot licence or of a student pilot;
- (b) the medical fitness of cabin crew;
- (c) the certification of AMEs;
- (d) the qualification of GMPs and OHMPs.

MED.A.010 Definitions

Regulation (EU) 2024/2076

For the purpose of this Annex (Part-MED), the following definitions shall apply:

- ‘limitation’ means a condition placed on the medical certificate or cabin crew medical report that shall be complied with whilst exercising the privileges of the licence or cabin crew attestation;

- ‘aero-medical examination’ means an inspection, palpation, percussion, auscultation or any other means of investigation for determining the medical fitness to exercise the privileges of the licence, or to carry out cabin crew safety duties;
- ‘aero-medical assessment’ means the conclusion on the medical fitness of an applicant based on the evaluation of the applicant as required in this Annex (Part-MED) and further examinations and medical tests as clinically indicated;
- ‘significant’ means a degree of a medical condition, the effect of which would prevent the safe exercise of the privileges of the licence or of the cabin crew safety duties;
- ‘applicant’ means a person applying for, or being the holder of, a medical certificate who undergoes an aero-medical assessment of fitness to exercise the privileges of the licence, or to carry out cabin crew safety duties;
- ‘medical history’ means a narrative or record of past diseases, injuries, treatments or other medical facts, including unfit assessment(s) or limitation of a medical certificate, that are or may be relevant to an applicant’s current state of health and aero-medical fitness;
- ‘licensing authority’ means the competent authority of the Member State that issued the licence, or to which a person applies for the issuance of a licence, or, when a person has not yet applied for a licence, the competent authority as determined in accordance with FCL.001 of Annex I (Part-FCL);
- ‘colour safe’ means the ability of an applicant to readily distinguish the colours used in air navigation and to correctly identify aviation coloured lights;
- ‘helicopter emergency medical services (HEMS) operation’ means a “HEMS flight” as defined in point 61 of Annex I to Regulation (EU) No 965/2012;
- ‘investigation’ means the assessment of a suspected pathological condition of an applicant by means of examinations and tests in order to verify the presence or absence of a medical condition;
- ‘accredited medical conclusion’ means the conclusion reached by one or more medical experts acceptable to the licensing authority, on the basis of objective and non-discriminatory criteria, for the purposes of the case concerned, in consultation with flight operations or other experts as necessary, for which an operational risk assessment may be appropriate;
- ‘misuse of substances’ means the use of one or more psychoactive substances by aircrew in a way that, alternatively or jointly:
 - (a) constitutes a direct hazard to the user or endangers the lives, health or welfare of others;
 - (b) causes or worsens an occupational, social, mental or physical problem or disorder;
- ‘psychoactive substances’ means alcohol, opioids, cannabinoids, sedatives and hypnotics, cocaine, other psychostimulants, hallucinogens, and volatile solvents, with the exception of caffeine and tobacco;;
- ‘refractive error’ means the deviation from emmetropia measured in dioptres in the most ametropic meridian, measured by standard methods.

MED.A.015 Medical confidentiality

Regulation (EU) 2019/27

All persons involved in aero-medical examinations, assessments and certification shall ensure that medical confidentiality is respected at all times.

AMC1 MED.A.015 Medical confidentiality

ED Decision 2019/002/R

To ensure medical confidentiality, all medical reports and records should be securely held with accessibility restricted to personnel authorised by the medical assessor or, where applicable, by the head of the aero-medical centre (AEMC), the aero-medical examiner (AME), general medical practitioner (GMP) or occupational health medical practitioner (OHMP).

MED.A.020 Decrease in medical fitness

Regulation (EU) 2019/27

- (a) Licence holders shall not exercise the privileges of their licence and related ratings or certificates, and student pilots shall not fly solo, at any time when they:
 - (1) are aware of any decrease in their medical fitness which might render them unable to safely exercise those privileges;
 - (2) take or use any prescribed or non-prescribed medication which is likely to interfere with the safe exercise of the privileges of the applicable licence;
 - (3) receive any medical, surgical or other treatment that is likely to interfere with the safe exercise of the privileges of the applicable licence.
- (b) In addition, holders of a medical certificate shall, without undue delay and before exercising the privileges of their licence, seek aero-medical advice from the AeMC, AME or GMP, as applicable, when they:
 - (1) have undergone a surgical operation or invasive procedure;
 - (2) have commenced the regular use of any medication;
 - (3) have suffered any significant personal injury involving incapacity to function as a member of the flight crew;
 - (4) have been suffering from any significant illness involving incapacity to function as a member of the flight crew;
 - (5) are pregnant;
 - (6) have been admitted to hospital or medical clinic;
 - (7) first require correcting lenses.
- (c) In the cases referred to in point (b):
 - (1) holders of class 1 and class 2 medical certificates shall seek the aero-medical advice of an AeMC or AME. In that case, the AeMC or AME shall assess their medical fitness and decide whether they are fit to resume the exercise of their privileges;
 - (2) holders of light aircraft pilot licence medical certificates shall seek the aero-medical advice of an AeMC, an AME or the GMP who signed the medical certificate. In that case, the AeMC, AME or GMP shall assess their medical fitness and decide whether they are fit to resume the exercise of their privileges.
- (d) Cabin crew members shall not perform duties on an aircraft and, where applicable, shall not exercise the privileges of their cabin crew attestation when they are aware of any decrease in their medical fitness, to the extent that this medical condition might render them unable to discharge their safety duties and responsibilities.

- (e) In addition, if any of the medical conditions specified in points (1) to (5) of point (b) apply, cabin crew members shall, without undue delay, seek the advice of an AME, AeMC or OHMP, as applicable. In that case, the AME, AeMC or OHMP shall assess the medical fitness of the cabin crew members and decide whether they are fit to resume their safety duties.

GM1 MED.A.020 Decrease in medical fitness

ED Decision 2019/002/R

MEDICATION – GUIDANCE FOR PILOTS AND CABIN CREW MEMBERS

- (a) Any medication can cause side effects, some of which may impair the safe performance of flying duties. Equally, symptoms of colds, sore throats, diarrhoea and other abdominal upsets may cause little or no problem whilst on the ground but may distract the pilot or cabin crew member and degrade their performance whilst on duty. The in-flight environment may also increase the severity of symptoms which may only be minor whilst on the ground. Therefore, one issue with medication and flying is the underlying condition and, in addition, the symptoms may be compounded by the side effects of the medication prescribed or bought over the counter for treatment. This guidance material provides some help to pilots and cabin crew in deciding whether expert aero-medical advice by an AME, AeMC, GMP, OHMP or medical assessor is needed.
- (b) Before taking any medication and acting as a pilot or cabin crew member, the following three basic questions should be satisfactorily answered:
- (1) Do I feel fit to fly?
 - (2) Do I really need to take medication at all?
 - (3) Have I given this particular medication a personal trial on the ground to ensure that it will not have any adverse effects on my ability to fly?
- (c) Confirming the absence of adverse effects may well need expert aero-medical advice.
- (d) The following are some widely used medicines with a description of their compatibility with flying duties:
- (1) Antibiotics. Antibiotics may have short-term or delayed side effects which can affect pilot or cabin crew performance. More significantly, however, their use usually indicates that an infection is present and, thus, the effects of this infection may mean that a pilot or cabin crew member is not fit to fly and should obtain expert aero-medical advice.
 - (2) Anti-malaria drugs. The decision on the need for anti-malaria drugs depends on the geographical areas to be visited, and the risk that the pilot or cabin crew member has of being exposed to mosquitoes and of developing malaria. An expert medical opinion should be obtained to establish whether anti-malaria drugs are needed and what kind of drugs should be used. Most of the anti-malaria drugs (atovaquone plus proguanil, chloroquine, doxycycline) are compatible with flying duties. However, adverse effects associated with mefloquine include insomnia, strange dreams, mood changes, nausea, diarrhoea and headaches. In addition, mefloquine may cause spatial disorientation and lack of fine coordination and is, therefore, not compatible with flying duties.
 - (3) Antihistamines. Antihistamines can cause drowsiness. They are widely used in ‘cold cures’ and in treatment of hay fever, asthma and allergic rashes. They may be in tablet form or a constituent of nose drops or sprays. In many cases, the condition itself may preclude flying, so that, if treatment is necessary, expert aero-medical advice should be sought so

that so-called non-sedative antihistamines, which do not degrade human performance, can be prescribed.

- (4) Cough medicines. Antitussives often contain codeine, dextromethorfan or pseudoephedrine which are not compatible with flying duties. However, mucolytic agents (e.g. carbocysteine) are well-tolerated and are compatible with flying duties.
- (5) Decongestants. Nasal decongestants with no effect on alertness may be compatible with flying duties. However, as the underlying condition requiring the use of decongestants may be incompatible with flying duties, expert aero-medical advice should be sought. For example, oedema of the mucosal membranes causes difficulties in equalising the pressure in the ears or sinuses.
- (6) Nasal corticosteroids are commonly used to treat hay fever, and they are compatible with flying duties.
- (7)
 - (i) Common pain killers and antifebrile drugs. Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) and paracetamol, commonly used to treat pain, fever or headaches, may be compatible with flying duties. However, the pilot or cabin crew member should give affirmative answers to the three basic questions listed in (b) before using the medication and carrying out flying duties.
 - (ii) Strong analgesics. The more potent analgesics including codeine are opiate derivatives, and may produce a significant decrement in human performance and, therefore, are not compatible with flying duties.
- (8) Anti-ulcer medicines. Gastric secretion inhibitors such as H2 antagonists (e.g. ranitidine, cimetidine) or proton pump inhibitors (e.g. omeprazole) may be acceptable after diagnosis of the pathological condition. It is important to seek for the medical diagnosis and not to only treat the dyspeptic symptoms.
- (9) Anti-diarrhoeal drugs. Loperamide is one of the more common anti-diarrhoeal drugs and is usually safe to take whilst flying. However, the diarrhoea itself often makes the pilot and cabin crew member unfit for flying duties.
- (10) Hormonal contraceptives and hormone replacement therapy usually have no adverse effects and are compatible with flying duties.
- (11) Erectile dysfunction medication. This medication may cause disturbances in colour vision and dizziness. There should be at least 6 hours between taking sildenafil and flying duty; and 36 hours between taking vardenafil or tadalafil and flying duty.
- (12) Smoking cessation. Nicotine replacement therapy may be acceptable. However, other medication affecting the central nervous system (bupropion, varenicline) is not acceptable for pilots.
- (13) High blood pressure medication. Most anti-hypertensive drugs are compatible with flying duties. However, if the level of blood pressure is such that drug therapy is required, the pilot or cabin crew member should be monitored for any side effects before carrying out flying duties. Therefore, consultation with the AME, AeMC, GMP, OHMP or medical assessor as applicable, is needed.
- (14) Asthma medication. Asthma has to be clinically stable before a pilot or cabin crew member can return to flying duties. The use of respiratory aerosols or powders, such as corticosteroids, beta-2-agonists or chromoglycic acid may be compatible with flying duties. However, the use of oral steroids or theophylline derivatives is incompatible with

flying duty. Pilots or cabin crew members using medication for asthma should consult the AME, AeMC, GMP, OHMP or medical assessor, as applicable.

- (15) Tranquillisers and sedatives. The inability to react, due to the use of this group of medicines, has been a contributory cause to fatal aircraft accidents. In addition, the underlying condition for which these medications have been prescribed will almost certainly mean that the mental state of a pilot or cabin crew member is not compatible with flying duties.
 - (16) Sleeping tablets. Sleeping tablets dull the senses, may cause confusion and slow reaction times. The duration of effect may vary from individual to individual and may be unduly prolonged. Expert aero-medical advice should be obtained before using sleeping tablets.
 - (17) Melatonin. Melatonin is a hormone that is involved with the regulation of the circadian rhythm. In some countries it is a prescription medicine, whereas in most other countries it is regarded as a 'dietary supplement' and can be bought without any prescription. The results from the efficiency of melatonin in treatment of jet lag or sleep disorders have been contradictory. Expert aero-medical advice should be obtained.
 - (18) Coffee and other caffeinated drinks may be acceptable, but excessive coffee drinking may have harmful effects, including disturbance of the heart's rhythm. Other stimulants including caffeine pills, amphetamines, etc. (often known as 'pep' pills) used to maintain wakefulness or suppress appetite can be habit forming. Susceptibility to different stimulants varies from one individual to another, and all may cause dangerous overconfidence. Overdosage causes headaches, dizziness and mental disturbance. These other stimulants should not be used.
 - (19) Anaesthetics. Following local, general, dental and other anaesthetics, a period of time should elapse before returning to flying. The period will vary considerably from individual to individual, but a pilot or cabin crew member should not fly for at least 12 hours after a local anaesthetic, and for at least 48 hours after a general, spinal or epidural anaesthetic (see [MED.A.020](#)).
- (e) Many preparations on the market nowadays contain a combination of medicines. It is, therefore, essential that if there is any new medication or dosage, however slight, the effect should be observed by the pilot or the cabin crew member on the ground prior to flying. It should be noted that medication which would not normally affect pilot or cabin crew performance may do so in individuals who are 'oversensitive' to a particular preparation. Individuals are, therefore, advised not to take any medicines before or during flight unless they are completely familiar with their effects on their own bodies. In cases of doubt, pilots and cabin crew members should consult an AME, AeMC, GMP, OHMP or medical assessor, as applicable.
- (f) Other treatments
- Alternative or complementary medicine, such as acupuncture, homeopathy, hypnotherapy and several other disciplines, is developing and gaining greater credibility. Such treatments are more acceptable in some States than others. There is a need to ensure that 'other treatments', as well as the underlying condition, are declared and considered by the AME, AeMC, GMP, OHMP or medical assessor, as applicable, for assessing fitness.

MED.A.025 Obligations of the AeMC, AME, GMP and OHMP

Regulation (EU) 2019/27

- (a) When conducting aero-medical examinations and aero-medical assessments as required in this Annex (Part-MED), the AeMC, AME, GMP and OHMP shall:

- (1) ensure that communication with the applicant can be established without language barriers;
 - (2) make the applicant aware of the consequences of providing incomplete, inaccurate or false statements on their medical history;
 - (3) notify the licensing authority, or, in the case of cabin crew attestation holders, notify the competent authority, if the applicant provides incomplete, inaccurate or false statements on their medical history;
 - (4) notify the licensing authority if an applicant withdraws the application for a medical certificate at any stage of the process.
- (b) After completion of the aero-medical examinations and assessments, the AeMC, AME, GMP and OHMP shall:
- (1) inform the applicant whether he or she is fit, unfit or referred to the medical assessor of the licensing authority, AeMC or AME, as applicable;
 - (2) inform the applicant of any limitation that may restrict flight training or the privileges of his or her licence or cabin crew attestation, as applicable;
 - (3) if the applicant has been assessed as unfit, inform him or her of his or her right to have the decision reviewed in accordance with the procedures of the competent authority;
 - (4) in the case of applicants for a medical certificate, submit without delay to the medical assessor of the licensing authority a signed, or electronically authenticated, report containing the detailed results of the aero-medical examinations and assessments as required for the class of medical certificate—and a copy of the application form, the examination form, and the medical certificate;
 - (5) inform the applicant of his or her responsibilities in the case of decrease in medical fitness, as specified in point [MED.A.020](#).
- (c) Where consultation with the medical assessor of the licensing authority is required in accordance with this Annex (Part-MED), the AeMC and AME shall follow the procedure established by the competent authority.
- (d) AeMCs, AMEs, GMPs and OHMPs shall maintain records with details of aero-medical examinations and assessments performed in accordance with this Annex (Part-MED) and their results for a minimum of 10 years, or for a longer period if so determined by national legislation.
- (e) AeMCs, AMEs, GMPs and OHMPs shall submit to the medical assessor of the competent authority, upon request, all aero-medical records and reports, and any other relevant information, when required for:
- (1) medical certification;
 - (2) oversight functions.
- (f) AeMCs and AMEs shall enter or update the data included in the European Aero-Medical Repository in accordance with point (c) of point ARA.MED.160.

AMC1 MED.A.025 Obligations of the AeMC, AME, GMP and OHMP

ED Decision 2019/002/R

- (a) If the medical examination is carried out by two or more AMEs or GMPs, only one of them should be responsible for coordinating the results of the examination, evaluating the findings with regard to medical fitness, and signing the report.

- (b) The applicant should be made aware that the associated medical certificate or cabin crew report may be suspended or revoked if the applicant provides incomplete, inaccurate or false statements on their medical history to the AeMC, AME, GMP or OHMP.
- (c) In cases where the AeMC or AME is required to assess the fitness of an applicant for a class 2 medical certificate in consultation with the medical assessor of the licensing authority, they should document the consultation in accordance with the procedure established by the competent authority.
- (d) The AeMC, AME, GMP or OHMP should give advice to the applicant on treatment and preventive measures if, during the course of the examination, medical conditions or risk factors are identified which may endanger the medical fitness of the applicant in the future.
- (e) When data is not being properly recorded in the European aero-medical data repository (EAMR) due to unserviceability of the system, the AeMCs and AMEs should enter, or correct the existing data, in the EAMR without undue delay when the system recovers.
- (f) In case of denial or referral to the licensing authority, the AeMC, AME, GMP or OHMP should inform the applicant in writing regarding the result of the assessment in a form and manner established by the competent authority.

GM1 MED.A.025 Obligations of the AeMC, AME, GMP and OHMP

ED Decision 2019/002/R

GUIDELINES FOR THE AeMC, AME OR GMP CONDUCTING THE MEDICAL EXAMINATIONS AND ASSESSMENTS FOR MEDICAL CERTIFICATION OF PILOTS

- (a) Before performing the medical examination, the AeMC, AME or GMP should:
 - (1) verify the applicant's identity by checking their identity card, passport, driving licence or other official document containing a photograph of the applicant;
 - (2) obtain details of the applicant's flight crew licence from the applicant's licensing authority if they do not have their licence with them;
 - (3) except for initial applicants, obtain details of the applicant's most recent medical certificate from the medical assessor of the applicant's licensing authority if they do not have their certificate with them;
 - (4) in the case of a specific medical examination(s) (SIC) limitation on the existing medical certificate, obtain details of the specific medical condition and any associated instructions from the medical assessor of the applicant's licensing authority. This could include, for example, a requirement to undergo a specific examination or test;
 - (5) except for initial applicants, ascertain, from the previous medical certificate, which routine medical test(s) should be conducted, for example electrocardiography (ECG);
 - (6) provide the applicant with the application form for a medical certificate and the instructions for completion and ask the applicant to complete the form but not to sign it yet;
 - (7) go through the form with the applicant and give information to help the applicant understand the significance of the entries and ask any questions which might help the applicant to recall important historical medical data;
 - (8) verify that the form is complete and legible, ask the applicant to sign and date the form and then sign it as well. If the applicant declines to complete the application form fully,

inform the applicant that it may not be possible to issue a medical certificate regardless of the outcome of the clinical examination and assessment.

- (b) Once all the items in (a) have been addressed, the AeMC, AME or GMP should:
 - (1) perform the medical examination of the applicant in accordance with the applicable rules;
 - (2) arrange for additional specialist medical examinations, such as otorhinolaryngology (ENT) or ophthalmology, to be conducted as applicable and obtain the associated report forms or reports;
 - (3) complete the medical examination report form in accordance with the associated instructions for completion;
 - (4) ensure that all of the report forms are complete, accurate and legible.
- (c) Once all the actions in (b) have been carried out, the AeMC, AME or GMP should review the report forms and:
 - (1) if satisfied that the applicant meets the applicable medical requirements as set out in Part-MED, issue a medical certificate for the appropriate class, with limitations if necessary. The applicant should sign the certificate once signed by the AeMC, AME or GMP; or
 - (2) if the applicant does not meet the applicable medical requirements, or if the fitness of the applicant for the class of medical certificate applied for is in doubt:
 - (i) refer the decision on medical fitness to, or consult the decision on medical fitness with, the medical assessor of the licensing authority or AME in compliance with [MED.B.001](#); or
 - (ii) deny issuance of a medical certificate, explain the reason(s) for denial to the applicant and inform them of their right of a review according to the procedures of the competent authority.
- (d) The AeMC, AME or GMP should send the documents as required by [MED.A.025\(b\)](#) to the medical assessor of the applicant's licensing authority within 5 days from the date of the medical examination. If a medical certificate has been denied or the decision has been referred, the documents should be sent to the medical assessor of the licensing authority on the same day that the denial or referral decision is reached.

SECTION 2 - REQUIREMENTS FOR MEDICAL CERTIFICATES

MED.A.030 Medical certificates

Regulation (EU) 2025/134

- (a) A student pilot shall not fly solo unless that student pilot holds a medical certificate, as required for the relevant licence.
- (b) An applicant for a licence, in accordance with Annex I (Part-FCL), shall hold a medical certificate issued in accordance with this Annex (Part-MED) and appropriate to the licence privileges applied for.
- (c) When exercising the privileges of a:
 - (1) light aircraft pilot licence (LAPL), a balloon pilot licence (BPL) issued in accordance with Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#), or a sailplane pilot licence (SPL) issued in accordance with Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), the pilot shall hold at least a valid LAPL medical certificate;
 - (2) private pilot licence (PPL), the pilot shall hold at least a valid class 2 medical certificate;
[applicable until 17 February 2026 - Regulation (EU) 2024/2076]
 - (2) private pilot licence (PPL) or gyroplane pilot licence (GPL), the pilot shall hold at least a valid class 2 medical certificate;
[applicable from 18 February 2026 - Regulation (EU) 2025/134]
 - (3) BPL for the purpose of:
 - (i) commercial passenger ballooning, the pilot shall hold at least a valid class 2 medical certificate;
 - (ii) commercial operation other than commercial passenger ballooning, with more than 4 persons on board the aircraft, the pilot shall hold at least a valid class 2 medical certificate;
 - (4) SPL for the purpose of commercial sailplane operations other than those specified in Article 3(2) of [Commission Implementing Regulation \(EU\) 2018/1976](#), the pilot shall hold at least a valid class 2 medical certificate;
 - (5) a commercial pilot licence (CPL), a multi-crew pilot licence (MPL) or an airline transport pilot licence (ATPL), the pilot shall hold a valid class 1 medical certificate.
- (d) If a night rating is added to a PPL or LAPL, the licence holder shall be colour safe.
- (e) If an instrument rating or basic instrument rating is added to a PPL, the licence holder shall undergo pure tone audiometry examinations in accordance with the periodicity and the standard required for class 1 medical certificate holders.
- (f) A licence holder shall not at any time hold more than one medical certificate issued in accordance with this Annex (Part-MED).

AMC1 MED.A.030 Medical certificates

ED Decision 2019/002/R

- (a) A class 1 medical certificate includes the privileges and validities of class 2 and LAPL medical certificates.
- (b) A class 2 medical certificate includes the privileges and validities of a LAPL medical certificate.

MED.A.035 Application for a medical certificate

Regulation (EU) 2019/27

- (a) Applications for a medical certificate shall be made in a form and manner established by the competent authority.
- (b) Applicants for a medical certificate shall provide the AeMC, AME or GMP, as applicable, with:
 - (1) proof of their identity;
 - (2) a signed declaration:
 - (i) of medical facts concerning their medical history;
 - (ii) as to whether they have previously applied for a medical certificate or have undergone an aero-medical examination for a medical certificate and, if so, by whom and with what result;
 - (iii) as to whether they have ever been assessed as unfit or had a medical certificate suspended or revoked.
- (c) When applying for a revalidation or renewal of the medical certificate, applicants shall present the most recent medical certificate to the AeMC, AME or GMP, as applicable, prior to the relevant aero-medical examinations.

AMC1 MED.A.035 Application for a medical certificate

ED Decision 2019/002/R

Except for initial applicants, the AeMC, AME or GMP should not start the aero-medical examination for the issue of the medical certificate where applicants do not present the most recent medical certificate, unless relevant information is received from the medical assessor of the licensing authority.

MED.A.040 Issuance, revalidation and renewal of medical certificates

Regulation (EU) 2024/2076

- (a) A medical certificate shall only be issued, revalidated or renewed once the required aero-medical examinations and assessments, as applicable, have been completed and the applicant has been assessed as fit.
- (b) *Initial issuance*
 - (1) Class 1 medical certificates shall be issued by an AeMC.
 - (2) Class 2 medical certificates shall be issued by an AeMC or an AME.

- (3) LAPL medical certificates shall be issued by an AeMC or an AME. They may also be issued by a GMP if so permitted under the national law of the Member State of the licensing authority to which the application for the medical certificate has been made.
- (c) *Revalidation and renewal*
 - (1) Class 1 and class 2 medical certificates shall be revalidated and renewed by an AeMC or an AME. Specifically, class 1 medical certificates for applicants who have reached the age of 60 and are involved in single-pilot HEMS operations shall be revalidated and renewed primarily by an AeMC or, at the discretion of the competent authority, by an experienced AME designated by the competent authority.
 - (2) LAPL medical certificates shall be revalidated and renewed by an AeMC or an AME. They may also be revalidated or renewed by a GMP if so permitted under the national law of the Member State of the licensing authority to which the application for the medical certificate has been made.
- (d) The AeMC, AME or GMP shall only issue, revalidate or renew a medical certificate if both of the following conditions have been met:
 - (1) the applicant has provided them with a complete medical history and, if required by the AeMC, AME or GMP, with results of medical examinations and tests conducted by the applicant's physician or any medical specialists;
 - (2) the AeMC, AME or GMP has conducted the aero-medical assessment based on the medical examinations and tests as required for the relevant medical certificate to verify that the applicant complies with all the relevant requirements of this Annex (Part-MED).
- (e) The AME, AeMC or, in the case of referral, the medical assessor of the licensing authority may require the applicant to undergo additional medical examinations and investigations when there is a clinical or epidemiological indication before the medical certificate is issued, revalidated or renewed.
- (f) The medical assessor of the licensing authority may issue or reissue a medical certificate.

MED.A.045 Validity, revalidation and renewal of medical certificates

Regulation (EU) 2019/27

- (a) *Validity*
 - (1) Class 1 medical certificates shall be valid for a period of 12 months.
 - (2) By derogation from point (1), the period of validity of class 1 medical certificates shall be 6 months for licence holders who:
 - (i) are engaged in single-pilot commercial air transport operations carrying passengers and have reached the age of 40;
 - (ii) have reached the age of 60.
 - (3) Class 2 medical certificates shall be valid for a period of:
 - (i) 60 months, until the licence holder reaches the age of 40. A medical certificate issued prior to the licence holder reaching the age of 40 shall cease to be valid after the licence holder reaches the age of 42;

- (ii) 24 months, for licence holders aged between 40 and 50. A medical certificate issued prior to the licence holder reaching the age of 50 shall cease to be valid after the licence holder reaches the age of 51;
 - (iii) 12 months, for licence holders aged above 50.
 - (4) LAPL medical certificates shall be valid for a period of:
 - (i) 60 months, until the licence holder reaches the age of 40. A medical certificate issued prior to the licence holder reaching the age of 40 shall cease to be valid after the licence holder reaches the age of 42;
 - (ii) 24 months, for licence holders aged above 40.
 - (5) The validity period of a medical certificate, including any associated examination or special investigation, shall be calculated from the date of the aero-medical examination in the case of initial issue and renewal, and from the expiry date of the previous medical certificate in the case of revalidation.
- (b) *Revalidation*
- Aero-medical examinations and assessments, as applicable, for the revalidation of a medical certificate may be undertaken up to 45 days prior to the expiry date of the medical certificate.
- (c) *Renewal*
- (1) If the holder of a medical certificate does not comply with point (b), a renewal examination and assessment, as applicable, shall be required.
 - (2) In the case of class 1 and class 2 medical certificates:
 - (i) if the medical certificate has expired for less than 2 years, a routine revalidation aero-medical examination shall be performed;
 - (ii) if the medical certificate has expired for more than 2 years but less than 5 years, the AeMC or AME shall only conduct the renewal aero-medical examination after assessment of the aero-medical records of the applicant;
 - (iii) if the medical certificate has expired for more than 5 years, the aero-medical examination requirements for initial issue shall apply and the assessment shall be based on the revalidation requirements.
 - (3) In the case of LAPL medical certificates, the AeMC, AME or GMP shall assess the medical history of the applicant and perform the aero-medical examinations and assessments, as applicable, in accordance with points [MED.B.005](#) and [MED.B.095](#).

MED.A.046 Suspension or revocation of medical certificates

Regulation (EU) 2019/27

- (a) A medical certificate may be suspended or revoked by the licensing authority.
- (b) Upon suspension of the medical certificate, the holder shall return the medical certificate to the licensing authority on request of that authority.
- (c) Upon revocation of the medical certificate, the holder shall immediately return the medical certificate to the licensing authority.

MED.A.050 Referral

Regulation (EU) 2019/27

- (a) If an applicant for a class 1 or class 2 medical certificate is referred to the medical assessor of the licensing authority in accordance with point [MED.B.001](#), the AeMC or AME shall transfer the relevant medical documentation to the licensing authority.
- (b) If an applicant for a LAPL medical certificate is referred to an AME or AeMC in accordance with point [MED.B.001](#), the GMP shall transfer the relevant medical documentation to the AeMC or AME.

SUBPART B – REQUIREMENTS FOR PILOT MEDICAL CERTIFICATES

SECTION 1 – GENERAL

MED.B.001 Limitations to medical certificates

Regulation (EU) 2019/27

(a) *Limitations to class 1 and class 2 medical certificates*

- (1) If the applicant does not fully comply with the requirements for the relevant class of medical certificate but is considered to be not likely to jeopardise the safe exercise of the privileges of the applicable licence, the AeMC or AME shall:
 - (i) in the case of applicants for a class 1 medical certificate, refer the decision on fitness of the applicant to the medical assessor of the licensing authority as indicated in this Subpart;
 - (ii) in cases where a referral to the medical assessor of the licensing authority is not indicated in this Subpart, evaluate whether the applicant is able to perform his/her duties safely when complying with one or more limitations endorsed on the medical certificate and issue the medical certificate with limitation(s) as necessary;
 - (iii) in the case of applicants for a class 2 medical certificate, evaluate, in consultation with the medical assessor of the licensing authority as indicated in this Subpart, whether the applicant is able to perform his/her duties safely when complying with one or more limitations endorsed on the medical certificate and issue the medical certificate, with limitation(s) as necessary.
- (2) The AeMC or AME may revalidate or renew a medical certificate with the same limitation(s) without referring to or consulting with the medical assessor of the licensing authority.

(b) *Limitations to LAPL medical certificates*

- (1) If a GMP, after due consideration of the applicant's medical history, concludes that the applicant for a LAPL medical certificate does not fully meet the requirements for medical fitness, the GMP shall refer the applicant to an AeMC or AME, unless the applicant requires only limitation(s) related to the use of corrective lenses or to the period of validity of the medical certificate.
- (2) If an applicant for a LAPL medical certificate has been referred in accordance with point (1), the AeMC or AME shall give due consideration to points [MED.B.005](#) and [MED.B.095](#), evaluate whether the applicant is able to perform his or her duties safely when complying with one or more limitations endorsed on the medical certificate and issue the medical certificate with limitation(s) as necessary. The AeMC or AME shall always consider the need to restrict the applicant from carrying passengers (operational passenger limitation, OPL).
- (3) The GMP may revalidate or renew a LAPL medical certificate with the same limitation without referring the applicant to an AeMC or AME.

(c) When assessing whether a limitation is necessary, particular consideration shall be given to:

- (1) whether accredited medical conclusion indicates that in special circumstances the applicant's failure to meet any requirement, whether numerical or otherwise, is such that the exercise of the privileges of the licence applied for is not likely to jeopardise flight safety;
 - (2) the applicant's ability, skill and experience relevant to the operation to be performed.
- (d) *Operational limitation codes*
- (1) Operational multi-pilot limitation (OML – class 1 only)
 - (i) When the holder of a CPL, ATPL or MPL does not fully meet the requirements for a class 1 medical certificate and has been referred to a medical assessor of the licensing authority, that medical assessor shall assess whether the medical certificate may be issued with an OML 'valid only as or with qualified co-pilot'.
 - (ii) The holder of a medical certificate with an OML shall only operate an aircraft in multi-pilot operations when the other pilot is fully qualified on the relevant class and type of aircraft, is not subject to an OML and has not attained the age of 60 years.
 - (iii) The OML for class 1 medical certificates shall be initially imposed and only removed by the medical assessor of the licensing authority.
 - (2) Operational safety pilot limitation (OSL – class 2 and LAPL privileges)
 - (i) The holder of a medical certificate with an OSL shall only operate an aircraft if another pilot fully qualified to act as pilot-in-command on the relevant class and type of aircraft is carried on board, the aircraft is fitted with dual controls and the other pilot occupies a seat at the controls.
 - (ii) The OSL for class 2 medical certificates may be imposed and removed either by the medical assessor of the licensing authority, or by an AeMC or an AME in consultation with the medical assessor of the licensing authority.
 - (iii) The OSL for LAPL medical certificates may be imposed and removed by the medical assessor of the licensing authority, an AeMC or an AME.
 - (3) Operational passenger limitation (OPL – class 2 and LAPL privileges)
 - (i) The holder of a medical certificate with an OPL shall only operate an aircraft without passengers on board.
 - (ii) The OPL for class 2 medical certificates may be imposed and removed either by the medical assessor of the licensing authority, or by an AeMC or an AME in consultation with the medical assessor of the licensing authority.
 - (iii) The OPL for LAPL medical certificates may be imposed and removed by the medical assessor of the licensing authority, an AeMC or an AME.
 - (4) Operational pilot restriction limitation (ORL – class 2 and LAPL privileges)
 - (i) The holder of a medical certificate with an ORL shall only operate an aircraft if one of the two following conditions have been met:
 - (A) another pilot fully qualified to act as pilot-in-command on the relevant class and type of aircraft is on board the aircraft, the aircraft is fitted with dual controls and the other pilot occupies a seat at the controls;
 - (B) there are no passengers on board the aircraft.

- (ii) The ORL for class 2 medical certificates may be imposed and removed either by the medical assessor of the licensing authority, or by an AeMC or AME in consultation with the medical assessor of the licensing authority.
- (iii) The ORL for LAPL medical certificates may be imposed and removed by the medical assessor of the licensing authority, an AeMC or an AME.
- (5) Special restriction as specified (SSL)
 - The SSL on a medical certificate shall be followed by a description of the limitation.
- (e) Any other limitation may be imposed on the holder of a medical certificate by the medical assessor of the licensing authority, AeMC, AME or GMP, as applicable, if required to ensure flight safety.
- (f) Any limitation imposed on the holder of a medical certificate shall be specified therein.

AMC1 MED.B.001 Limitations to medical certificates

ED Decision 2019/002/R

GENERAL

- (a) An AeMC or AME may refer the decision on fitness of an applicant to the medical assessor of the licensing authority in borderline cases or where fitness is in doubt.
- (b) In cases where a fit assessment may only be considered with a limitation, the AeMC, AME, GMP or the medical assessor of the licensing authority should evaluate the medical condition of the applicant in consultation with flight operations and other experts, if necessary.
- (c) Initial application of limitations
 - (1) The limitations TML, VDL, VML, VNL and VCL, as listed in [AMC2 MED.B.001\(a\)](#), may be imposed by an AME or an AeMC for class 1, class 2, and LAPL medical certificates, or a GMP for LAPL medical certificates.
 - (2) All other limitations listed in [AMC2 MED.B.001\(a\)](#) should only be imposed:
 - (i) for class 1 medical certificates, by the medical assessor of the licensing authority where a referral is required according to [MED.B.001](#);
 - (ii) for class 2 medical certificates, by the AME or AeMC in consultation with the medical assessor of the licensing authority where consultation is required according to [MED.B.001](#);
 - (iii) for LAPL medical certificates, by an AME or AeMC.
- (d) Removal of limitations
 - (1) For class 1 medical certificates, all limitations should only be removed by the medical assessor of the licensing authority.
 - (2) For class 2 medical certificates, limitations may be removed by the medical assessor of the licensing authority or by an AeMC or AME in consultation with the medical assessor of the licensing authority.
 - (3) For LAPL medical certificates, limitations may be removed by an AeMC or AME.

AMC2 MED.B.001 Limitations to medical certificates

ED Decision 2019/002/R

LIMITATION CODES

- (a) The following abbreviations for limitations codes should be used on the medical certificates as applicable:

Code	Limitation
TML	Limited period of validity of the medical certificate
VDL	Valid only with correction for defective distant vision
VML	Valid only with correction for defective distant, intermediate and near vision
VNL	Valid only with correction for defective near vision
CCL	Correction by means of contact lenses
VCL	Valid by day only
RXO	Specialist ophthalmological examination(s)
SIC	Specific medical examination(s)
HAL	Valid only when hearing aids are worn
APL	Valid only with approved prosthesis
AHL	Valid only with approved hand controls
OML	Valid only as, or with, a qualified co-pilot
OCL	Valid only as a qualified co-pilot
OSL	Valid only with a safety pilot and in aircraft with dual controls
OPL	Valid only without passengers
ORL	Valid only with a safety pilot if passengers are carried
OAL	Restricted to demonstrated aircraft type
SSL	Special restriction(s) as specified

- (b) The abbreviations for the limitation codes should be explained to the holder of a medical certificate as follows:

(1) TML Time limitation

The period of validity of the medical certificate is limited to the duration as shown on the medical certificate. This period of validity commences on the date of the medical examination. Any period of validity remaining on the previous medical certificate is no longer valid. The holder of the medical certificate should present themselves for re-examination when advised and should follow any medical recommendations.

(2) VDL Wear corrective lenses and carry a spare set of spectacles

Correction for defective distant vision: whilst exercising the privileges of the licence, the holder of the medical certificate should wear spectacles or contact lenses that correct for defective distant vision as examined and approved by the AeMC, AME or GMP. Contact lenses may not be worn until cleared to do so by the AeMC, AME or GMP. A spare set of spectacles, approved by the AeMC, AME or GMP, should be readily available.

(3) VML Wear multifocal spectacles and carry a spare set of spectacles

Correction for defective distant, intermediate and near vision: whilst exercising the privileges of the licence, the holder of the medical certificate should wear spectacles that correct for defective distant, intermediate and near vision as examined and approved by the AeMC, AME or GMP. Contact lenses or full frame spectacles, when either correct for

near vision only, may not be worn. A spare set of spectacles, approved by the AeMC, AME or GMP, should be readily available.

- (4) VNL Have available corrective spectacles and carry a spare set of spectacles

Correction for defective near vision: whilst exercising the privileges of the licence, the holder of the medical certificate should have readily available spectacles that correct for defective near vision as examined and approved by the AeMC, AME or GMP. Contact lenses or full frame spectacles, when either correct for near vision only, may not be worn. A spare set of spectacles, approved by the AeMC, AME or GMP, should be readily available.

- (5) CCL Wear contact lenses that correct for defective distant vision

Correction for defective distant vision: whilst exercising the privileges of the licence, the holder of a medical certificate should wear contact lenses that correct for defective distant vision, as examined and approved by the AeMC, AME or GMP. A spare set of similarly correcting spectacles, approved by the AeMC, AME or GMP, should be readily available for immediate use whilst exercising the privileges of the licence.

- (6) VCL Valid by day only

This limitation allows holders of a class 2 or LAPL medical certificate with varying degrees of colour deficiency, to exercise the privileges of their licence by daytime only.

- (7) RXO Specialist ophthalmological examination(s)

Specialist ophthalmological examination(s), other than the examinations stipulated in Part-MED, are required for a significant reason.

- (8) SIC Specific regular medical examination(s) contact the medical assessor of the licensing authority

This limitation requires the AeMC, or AME to contact the medical assessor of the licensing authority before embarking upon a revalidation or renewal aero-medical assessment. The limitation is likely to concern a medical history or additional examination(s) which the AeMC or AME should be aware of prior to undertaking the assessment.

- (9) HAL Wear hearing aid(s)

Whilst exercising the privileges of the licence, the holder of the medical certificate should use hearing aid(s) that compensate for defective hearing as examined and approved by the AeMC or AME. A spare set of batteries should be readily available.

- (10) APL Valid only with approved prosthesis

This limitation applies to the holder of a medical certificate with a musculoskeletal condition when a medical flight test or a flight simulator test has shown that the use of a prosthesis is required to safely exercise the privileges of the licence. The prosthesis to be used should be approved.

- (11) AHL Valid only with approved hand controls

This limitation applies to the holder of a medical certificate who has a limb deficiency or other anatomical problem which had been shown by a medical flight test or flight simulator testing to be acceptable but to require the aircraft to be equipped with suitable, approved hand controls.

- (12) OML Valid only as or with a qualified co-pilot

This limitation applies to holders of a class 1 medical certificate who do not fully meet the aero-medical requirements for single-pilot operations, but are fit for multi-pilot operations. Refer to MED.B.001(d)(1).

- (13) OCL Valid only as a qualified co-pilot

This limitation is an extension of the OML and are restricted to the role of co-pilot.

- (14) OSL Valid only with a safety pilot and in aircraft with dual controls

This limitation applies to holders of a class 2 or a LAPL medical certificate only. The safety pilot should be made aware of the type(s) of possible incapacity that the pilot whose medical certificate has been issued with this limitation may suffer and should be prepared to take over the aircraft controls during flight. Refer to MED.B.001(d)(2).

- (15) OPL Valid only without passengers

This limitation applies to holders of a class 2 or LAPL medical certificate with a medical condition that may lead to an increased level of risk to flight safety when exercising the privileges of the licence. This limitation is to be applied when this risk is not acceptable for the carriage of passengers. Refer to MED.B.001(d)(3).

- (16) ORL Valid only with a safety pilot if passengers are carried and in aircraft with dual controls

This limitation applies to holders of a class 2 or LAPL medical certificate with a medical condition that may lead to an increased level of risk to flight safety when exercising the privileges of the licence. The safety pilot, if carried, should be made aware of the type(s) of possible incapacity that the pilot whose medical certificate has been issued with this limitation may suffer and should be prepared to take over the aircraft controls during flight. Refer to MED.B.001(d)(4).

- (17) OAL Restricted to demonstrated aircraft type

This limitation applies to a the holder of a medical certificate who has a limb deficiency or other medical problem which had been shown by a medical flight test or flight simulator testing to be acceptable but to require a restriction to a specific class and type of aircraft.

- (18) SSL Special restriction(s) as specified

This limitation may be considered when an individually specified limitation, not defined in this AMC, is appropriate to mitigate an increased level of risk to flight safety. The description of the SSL should be entered on the medical certificate or in a separate document to be carried with the medical certificate.

MED.B.005 General medical requirements

Regulation (EU) 2024/2076

Applicants for a medical certificate shall be assessed in accordance with the detailed medical requirements set out in Sections 2 and 3.

They shall, in addition, be assessed as unfit where they have any of the following medical conditions which entails a degree of functional incapacity which is likely to interfere with the safe exercise of the privileges of the licence applied for or could render the applicant likely to become suddenly unable to exercise those privileges:

- (a) abnormality, either congenital or acquired;

- (b) active, latent, acute or chronic disease or disability;
- (c) wound, injury or sequelae from operation;
- (d) effect or side effect of any prescribed or non-prescribed therapeutic, diagnostic or preventive medication taken.

In their examination AMEs shall give proper consideration to the degenerative effects of ageing on the body systems.

SECTION 2 – MEDICAL REQUIREMENTS FOR CLASS 1 AND CLASS 2 MEDICAL CERTIFICATES

MED.B.010 Cardiovascular System

Regulation (EU) 2024/2076

(a) *Examination*

- (1) A standard 12-lead resting electrocardiogram (ECG) and report shall be completed when clinically indicated and at the following moments:
 - (i) for a class 1 medical certificate, at the initial examination, then every 5 years until age 30, every 2 years until age 40, annually until age 50, and at all revalidation or renewal examinations thereafter;
 - (ii) for a class 2 medical certificate, at the initial examination, at the first examination after age 40 and then at the first examination after age 50, and every 2 years thereafter.
- (2) An extended cardiovascular assessment shall be required when clinically indicated.
- (3) For a class 1 medical certificate, an extended cardiovascular assessment shall be completed at the first revalidation or renewal examination after the age of 65 and every 4 years thereafter. For applicants involved in single-pilot HEMS operations, an extended cardiovascular assessment shall be completed at the first revalidation or renewal examination after the age of 60 and subject to a cardiovascular risk factor assessment thereafter.
- (4) For a class 1 medical certificate, estimation of serum lipids, including cholesterol fractions, shall be required at the initial examination, and at the first examination after having reached the age of 40.

(b) *Cardiovascular System – General*

- (1) Applicants for a class 1 medical certificate with any of the following medical conditions shall be assessed as unfit:
 - (i) aneurysm of the thoracic or supra-renal abdominal aorta, before surgery;
 - (ii) significant functional or symptomatic abnormality of any of the heart valves;
 - (iii) heart or heart/lung transplantation;
 - (iv) symptomatic hypertrophic cardiomyopathy.
- (2) Before further consideration is given to their application, applicants for a class 1 medical certificate with a documented medical history or diagnosis of any of the following medical conditions shall be referred to the medical assessor of the licensing authority:
 - (i) peripheral arterial disease before or after surgery;
 - (ii) aneurysm of the thoracic or supra-renal abdominal aorta after surgery;
 - (iii) aneurysm of the infra-renal abdominal aorta before or after surgery;
 - (iv) functionally insignificant cardiac valvular abnormalities;
 - (v) after cardiac valve surgery;

- (vi) abnormality of the pericardium, myocardium or endocardium;
 - (vii) congenital abnormality of the heart, before or after corrective surgery;
 - (viii) vasovagal syncope of uncertain cause;
 - (ix) arterial or venous thrombosis;
 - (x) pulmonary embolism;
 - (xi) cardiovascular condition requiring systemic anticoagulant therapy.
- (3) Applicants for a class 2 medical certificate with an established diagnosis of one of the conditions specified in points (1) and (2) shall be evaluated by a cardiologist before they may be assessed as fit, in consultation with the medical assessor of the licensing authority.
- (4) Applicants with cardiac disorders other than those specified in points (1) and (2) may be assessed as fit subject to satisfactory cardiological evaluation.
- (5) A cardiovascular risk factor assessment shall form part of examinations for class 1 and class 2 medical certificates at the first examination after reaching the age of 40 and at regular intervals thereafter.
- (c) *Blood Pressure*
- (1) Applicants' blood pressure shall be recorded at each examination.
- (2) Applicants whose's blood pressure is not within normal limits shall be further assessed with regard to their cardiovascular condition and medication with a view to determining whether they are to be assessed as unfit in accordance with points (3) and (4).
- (3) Applicants for a class 1 medical certificate with any of the following medical conditions shall be assessed as unfit:
- (i) symptomatic hypotension;
 - (ii) blood pressure at examination consistently exceeding 160 mmHg systolic or 95 mmHg diastolic, with or without treatment.
- (4) Applicants who have commenced the use of medication for the control of blood pressure shall be assessed as unfit until the absence of significant side effects has been established.
- (d) *Coronary Artery Disease*
- (1) Before further consideration is given to their application, applicants for a class 1 medical certificate with any of the following medical conditions shall be referred to the medical assessor of the licensing authority and undergo cardiological evaluation to exclude myocardial ischaemia:
- (i) suspected myocardial ischaemia;
 - (ii) asymptomatic minor coronary artery disease requiring no anti-anginal treatment.
- (2) Before further consideration is given to their application, applicants for a class 2 medical certificate with any of the medical conditions set out in point (1) shall undergo satisfactory cardiological evaluation.
- (3) Applicants with any of the following medical conditions shall be assessed as unfit:
- (i) myocardial ischaemia;
 - (ii) symptomatic coronary artery disease;

- (iii) symptoms of coronary artery disease controlled by medication.
 - (4) Applicants for the initial issue of a class 1 medical certificate with a medical history or diagnosis of any of the following medical conditions shall be assessed as unfit:
 - (i) myocardial ischaemia;
 - (ii) myocardial infarction;
 - (iii) revascularisation or stenting for coronary artery disease.
 - (5) Before further consideration is given to their application, applicants for a class 2 medical certificate who are asymptomatic following myocardial infarction or surgery for coronary artery disease shall undergo satisfactory cardiological evaluation, in consultation with the medical assessor of the licensing authority. Such applicants for the revalidation of a class 1 medical certificate shall be referred to the medical assessor of the licensing authority.
- (e) *Rhythm/Conduction Disturbances*
- (1) Applicants with any of the following medical conditions shall be assessed as unfit:
 - (i) symptomatic sinoatrial disease;
 - (ii) complete atrioventricular block;
 - (iii) symptomatic QT prolongation;
 - (iv) an automatic implantable defibrillating system;
 - (v) a ventricular anti-tachycardia pacemaker.
 - (2) Before further consideration is given to their application, applicants for a class 1 medical certificate having any significant disturbance of cardiac conduction or rhythm, including any of the following, shall be referred to the medical assessor of the licensing authority:
 - (i) disturbance of supraventricular rhythm, including intermittent or established sinoatrial dysfunction, atrial fibrillation and/or flutter and asymptomatic sinus pauses;
 - (ii) complete left bundle branch block;
 - (iii) Mobitz type 2 atrioventricular block;
 - (iv) broad and/or narrow complex tachycardia;
 - (v) ventricular pre-excitation;
 - (vi) asymptomatic QT prolongation;
 - (vii) Brugada pattern on electrocardiography.
 - (3) Before further consideration is given to their application, applicants for a class 2 medical certificate with any of the medical conditions specified in point (2) shall undergo satisfactory cardiological evaluation, in consultation with the medical assessor of the licensing authority.
 - (4) Applicants with any of the following medical conditions may be assessed as fit subject to satisfactory cardiological evaluation and in the absence of any other abnormality:
 - (i) incomplete bundle branch block;
 - (ii) complete right bundle branch block;
 - (iii) stable left axis deviation;

- (iv) asymptomatic sinus bradycardia;
 - (v) asymptomatic sinus tachycardia;
 - (vi) asymptomatic isolated uniform supra-ventricular or ventricular ectopic complexes;
 - (vii) first degree atrioventricular block;
 - (viii) Mobitz type 1 atrioventricular block.
- (5) Applicants with a medical history of any of the following medical conditions shall undergo satisfactory cardiovascular evaluation before they may be assessed as fit:
- (i) ablation therapy;
 - (ii) pacemaker implantation.

Such applicants for a class 1 medical certificate shall be referred to the medical assessor of the licensing authority. Such applicants for a class 2 medical certificate shall be assessed in consultation with the medical assessor of the licensing authority.

AMC1 MED.B.010 Cardiovascular system

ED Decision 2025/002/R

(a) Examination

Exercise electrocardiography

An exercise ECG when required as part of a cardiovascular assessment should be symptom limited and completed to a minimum of Bruce Stage IV or equivalent.

(b) General

(1) Cardiovascular risk factor assessment

- (i) Serum lipid estimation is case finding and significant abnormalities should be reviewed, investigated and supervised by the AeMC or AME in consultation with the medical assessor of the licensing authority.
- (ii) Applicants with an accumulation of two or more risk factors should undergo a cardiovascular evaluation by the AeMC or AME, if necessary in consultation with the medical assessor of the licensing authority.
- (iii) Cardiovascular risk factor assessment should be performed using risk calculators relevant for the target population and taking into consideration the latest guidelines on cardiovascular disease prevention.
- (iv) Cardiovascular risk factor assessment should take place at least once every 5 years for applicants 40 to 49 years old, once every 3 years for applicants 50 to 59 years old and once every 2 years thereafter. A more frequent assessment of the cardiovascular risk factors may be considered when risk factors have been identified.

(2) Cardiovascular assessment

- (i) Reporting of resting and exercise electrocardiograms should be by the AME or an accredited specialist.
- (ii) The extended cardiovascular assessment should be undertaken at an AeMC or may be delegated to a cardiologist.

- (iii) For applicants involved in single-pilot HEMS operations who have reached the age of 60, the extended cardiovascular assessment should include at least the following elements:
 - (A) resting ECG;
 - (B) exercise ECG;
 - (C) serum lipids;
 - (D) glycosylated haemoglobin test (HbA1c);
 - (E) echocardiography;
 - (F) arterial doppler ultrasound carotid arteries, and at clinical indication thoracic or abdominal aorta could be considered.
- (c) Peripheral arterial disease

If there is no significant functional impairment, a fit assessment may be considered provided:

 - (1) applicants without symptoms of coronary artery disease have reduced any vascular risk factors to an appropriate level;
 - (2) applicants should be on appropriate secondary prevention treatment;
 - (3) exercise electrocardiography is satisfactory. Further tests may be required which should show no evidence of myocardial ischaemia or significant coronary artery stenosis.
- (d) Aortic aneurysm
 - (1) Applicants with an aneurysm of the infra-renal abdominal aorta of less than 5 cm in diameter may be assessed as fit before surgery, with an OML subject to satisfactory evaluation by a cardiologist. Follow-up by ultra-sound scans or other imaging techniques, as necessary, should be determined by the medical assessor of the licensing authority.
 - (2) Applicants may be assessed as fit with an OML after surgery for an aneurysm of the thoracic or abdominal aorta if the blood pressure and cardiovascular evaluation is satisfactory. Regular evaluations by a cardiologist should be carried out.
- (e) Cardiac valvular abnormalities
 - (1) Applicants with previously unrecognised cardiac murmurs should undergo evaluation by a cardiologist and assessment by the medical assessor of the licensing authority. If considered significant, further investigation should include at least 2D Doppler echocardiography or equivalent imaging.
 - (2) Applicants with minor cardiac valvular abnormalities may be assessed as fit. Applicants with significant abnormality of any of the heart valves should be assessed as unfit.
 - (3) Aortic valve disease
 - (i) Applicants with a bicuspid aortic valve may be assessed as fit if no other cardiac or aortic abnormality is demonstrated. Follow-up with echocardiography, as necessary, should be determined by the medical assessor of the licensing authority.
 - (ii) Applicants with aortic stenosis may be assessed as fit provided the left ventricular function is intact and the mean pressure gradient is less than 20 mmHg. Applicants with an aortic valve orifice with indexation on the body surface of more than 0.6 cm²/m² and a mean pressure gradient above 20 mmHg, but not greater than

50 mmHg, may be assessed as fit with an OML. Follow-up with 2D Doppler echocardiography, as necessary, should be determined by the medical assessor of the licensing authority in all cases. Alternative measurement techniques with equivalent ranges may be used. Regular evaluation by a cardiologist should be considered. Applicants with a history of systemic embolism or significant dilatation of the thoracic aorta should be assessed as unfit.

- (iii) Applicants with trivial aortic regurgitation may be assessed as fit. A greater degree of aortic regurgitation should require an OML. There should be no demonstrable abnormality of the ascending aorta on 2D Doppler echocardiography. Follow-up, as necessary, should be determined by the medical assessor of the licensing authority.

(4) Mitral valve disease

- (i) Asymptomatic applicants with an isolated mid-systolic click due to mitral leaflet prolapse may be assessed as fit.
- (ii) Applicants with rheumatic mitral stenosis should normally be assessed as unfit.
- (iii) Applicants with minor regurgitation may be assessed as fit. Periodic cardiological review should be determined by the medical assessor of the licensing authority.
- (iv) Applicants with moderate mitral regurgitation may be considered as fit with an OML if the 2D Doppler echocardiogram demonstrates satisfactory left ventricular dimensions and satisfactory myocardial function is confirmed by exercise electrocardiography. Periodic cardiological review should be required, as determined by the medical assessor of the licensing authority.
- (v) Applicants with evidence of volume overloading of the left ventricle demonstrated by increased left ventricular end-diastolic diameter or evidence of systolic impairment should be assessed as unfit.

(f) Valvular surgery

Applicants who have undergone cardiac valve replacement or repair should be assessed as unfit. A fit assessment may be considered in the following cases:

- (1) Mitral leaflet repair for prolapse is compatible with a fit assessment, provided post-operative investigations reveal satisfactory left ventricular function without systolic or diastolic dilation and no more than minor mitral regurgitation.
- (2) Asymptomatic applicants with a tissue valve or with a mechanical valve who, at least 6 months following surgery, are taking no cardioactive medication may be considered for a fit assessment with an OML. Investigations which demonstrate normal valvular and ventricular configuration and function should have been completed as demonstrated by:
 - (i) a satisfactory symptom limited exercise ECG. Myocardial perfusion imaging/stress echocardiography should be required if the exercise ECG is abnormal or any coronary artery disease is suspected;
 - (ii) a 2D Doppler echocardiogram showing no significant selective chamber enlargement, a tissue valve with minimal structural alteration and a normal Doppler blood flow, and no structural or functional abnormality of the other heart valves. Left ventricular fractional shortening should be normal.

Follow-up with exercise ECG and 2D echocardiography, as necessary, should be determined by the medical assessor of the licensing authority.

- (3) Where anticoagulation is needed after valvular surgery, a fit assessment with an OML may be considered if the haemorrhagic risk is acceptable and the anticoagulation is stable. Anticoagulation should be considered stable if, within the last 6 months, at least 5 international normalised ratio (INR) values are documented, of which at least 4 are within the INR target range. The INR target range should be determined by the type of surgery performed.
- (g) Thromboembolic disorders
- Applicants with arterial or venous thrombosis or pulmonary embolism should be assessed as unfit. A fit assessment with an OML may be considered after a period of stable anticoagulation as prophylaxis, after review by the medical assessor of the licensing authority. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range and the haemorrhagic risk is acceptable. In cases of anticoagulation medication not requiring INR monitoring, a fit assessment with an OML may be considered after review by the medical assessor of the licensing authority after a stabilisation period of 3 months. Applicants with pulmonary embolism should also be evaluated by a cardiologist. Following cessation of anticoagulant therapy, for any indication, applicants should undergo a re-assessment by the medical assessor of the licensing authority.
- (h) Other cardiac disorders
- (1) Applicants with a primary or secondary abnormality of the pericardium, myocardium or endocardium should be assessed as unfit. A fit assessment may be considered following complete resolution and satisfactory cardiological evaluation which may include 2D Doppler echocardiography, exercise ECG and/or myocardial perfusion imaging/stress echocardiography and 24-hour ambulatory ECG. Coronary angiography may be indicated. Frequent review and an OML may be required after fit assessment.
- (2) Applicants with a congenital abnormality of the heart should be assessed as unfit. Applicants following surgical correction or with minor abnormalities that are functionally unimportant may be assessed as fit following cardiological evaluation. No cardioactive medication is acceptable. Investigations may include 2D Doppler echocardiography, exercise ECG and 24-hour ambulatory ECG. The potential hazard of any medication should be considered as part of the assessment. Particular attention should be paid to the potential for the medication to mask the effects of the congenital abnormality before or after surgery. Regular cardiological evaluations should be carried out.
- (i) Syncope
- (1) In the case of a single episode of vasovagal syncope which can be explained and is compatible with flight safety, a fit assessment may be considered.
- (2) Applicants with a history of recurrent vasovagal syncope should be assessed as unfit. A fit assessment may be considered after a 6-month period without recurrence, provided cardiological evaluation is satisfactory. Such evaluation should include:
- (i) a satisfactory symptom limited 12 lead exercise ECG to Bruce Stage IV, or equivalent. If the exercise ECG is abnormal, myocardial perfusion imaging/stress echocardiography or equivalent test should be carried out;
- (ii) a 2D Doppler echocardiogram showing neither significant selective chamber enlargement nor structural or functional abnormality of the heart, valves or myocardium;

- (iii) a 24-hour ambulatory ECG recording showing no conduction disturbance, complex or sustained rhythm disturbance or evidence of myocardial ischaemia.
 - (3) A tilt test, or equivalent, carried out to a standard protocol showing no evidence of vasomotor instability may be required.
 - (4) Neurological review should be required.
 - (5) An OML should be required until a period of 5 years has elapsed without recurrence. The medical assessor of the licensing authority may determine a shorter or longer period of OML according to the individual circumstances of the case.
 - (6) Applicants who experienced loss of consciousness without significant warning should be assessed as unfit.
- (j) Blood pressure
- (1) The diagnosis of hypertension should require cardiovascular evaluation to include potential vascular risk factors.
 - (2) Anti-hypertensive treatment should be agreed by the medical assessor of the licensing authority. Acceptable medication may include:
 - (i) non-loop diuretic agents;
 - (ii) ACE inhibitors;
 - (iii) angiotensin II receptor blocking agents (sartans);
 - (iv) channel calcium blocking agents;
 - (v) certain (generally hydrophilic) beta-blocking agents.
 - (3) Following initiation of medication for the control of blood pressure, applicants should be re-assessed to verify that satisfactory control has been achieved and the treatment is compatible with the safe exercise of the privileges of the applicable licence(s).
- (k) Coronary artery disease
- (1) Chest pain of uncertain cause should require full investigation. Applicants with angina pectoris should be assessed as unfit, whether or not it is alleviated by medication.
 - (2) In suspected asymptomatic coronary artery disease, exercise electrocardiography should be required. Further tests may be required, which should show no evidence of myocardial ischaemia or significant coronary artery stenosis.
 - (3) Applicants with evidence of exercise-induced myocardial ischaemia should be assessed as unfit.
 - (4) After an ischaemic cardiac event or revascularisation procedure, applicants should have reduced cardiovascular risk factors to an appropriate level. Medication, when used to control cardiac symptoms, is not acceptable. All applicants should be on appropriate secondary prevention treatment.
 - (i) A coronary angiogram obtained around the time of, or during, the ischaemic myocardial event or revascularisation procedure and a complete, detailed clinical report of the ischaemic event and of any operative procedures should be made available to the medical assessor of the licensing authority:

- (A) there should be no stenosis more than 50 % in any major untreated vessel, in any vein or artery graft or at the site of an angioplasty/stent, except in a vessel subtending a myocardial infarction;
 - (B) the whole coronary vascular tree should be assessed as satisfactory by a cardiologist, and particular attention should be paid to multiple stenoses and/or multiple revascularisations;
 - (C) Applicants with an untreated stenosis greater than 30 % in the left main or proximal left anterior descending coronary artery should be assessed as unfit.
 - (ii) At least 6 months from the ischaemic myocardial event or revascularisation procedure, the following investigations should be completed (equivalent tests may be substituted):
 - (A) an exercise ECG showing neither evidence of myocardial ischaemia nor rhythm or conduction disturbance;
 - (B) an echocardiogram showing satisfactory left ventricular function with no important abnormality of wall motion (such as dyskinesia or akinesia) and a left ventricular ejection fraction of 50 % or more;
 - (C) in cases of angioplasty/stenting, a myocardial perfusion scan or stress echocardiogram, or equivalent test, which should show no evidence of reversible myocardial ischaemia. If there is any doubt about myocardial perfusion in other cases (infarction or bypass grafting) a perfusion scan, or equivalent test, should also be carried out;
 - (D) further investigations, such as a 24-hour ECG, may be necessary to assess the risk of any significant rhythm disturbance.
 - (iii) Follow-up should be annual (or more frequently, if necessary) to ensure that there is no deterioration of the cardiovascular status. It should include a review by a cardiologist, exercise ECG and cardiovascular risk assessment. Additional investigations may be required by the medical assessor of the licensing authority.
 - (A) After coronary artery bypass grafting, a myocardial perfusion scan, or equivalent test, should be performed if there is any indication, and in all cases within 5 years from the procedure.
 - (B) In all cases, coronary angiography should be considered at any time if symptoms, signs or non-invasive tests indicate myocardial ischaemia.
 - (iv) Successful completion of the 6-month or subsequent review will allow a fit assessment with an OML.
- (I) Rhythm and conduction disturbances
- (1) Applicants with significant rhythm or conduction disturbance should undergo evaluation by a cardiologist before a fit assessment with an OML, as necessary, may be considered. Appropriate follow-up should be carried out at regular intervals. Such evaluation should include:
 - (i) exercise ECG to the Bruce protocol or equivalent. Bruce stage 4 should be achieved and no significant abnormality of rhythm or conduction, or evidence of myocardial ischaemia should be demonstrated. Withdrawal of cardioactive medication prior to the test should normally be required;

- (ii) 24-hour ambulatory ECG which should demonstrate no significant rhythm or conduction disturbance;
- (iii) 2D Doppler echocardiogram which should show no significant selective chamber enlargement or significant structural or functional abnormality, and a left ventricular ejection fraction of at least 50 %.

Further evaluation may include (equivalent tests may be substituted):

- (iv) 24-hour ECG recording repeated as necessary;
 - (v) electrophysiological study;
 - (vi) myocardial perfusion imaging;
 - (vii) cardiac magnetic resonance imaging (MRI);
 - (viii) coronary angiogram.
- (2) Applicants with frequent or complex forms of supra ventricular or ventricular ectopic complexes require full cardiological evaluation.
 - (3) Where anticoagulation is needed for a rhythm disturbance, a fit assessment with an OML may be considered if the haemorrhagic risk is acceptable and the anticoagulation is stable. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range. In cases of anticoagulation medication not requiring INR monitoring, a fit assessment with an OML may be considered after review by the medical assessor of the licensing authority after a stabilisation period of 3 months.
 - (4) Ablation

Applicants who have undergone ablation therapy should be assessed as unfit. A fit assessment may be considered following successful catheter ablation and should require an OML for at least one year, unless an electrophysiological study, undertaken at a minimum of 2 months after the ablation, demonstrates satisfactory results. For those whose long-term outcome cannot be assured by invasive or non-invasive testing, an additional period with an OML and/or observation may be necessary.
 - (5) Supraventricular arrhythmias

Applicants with significant disturbance of supraventricular rhythm, including sinoatrial dysfunction, whether intermittent or established, should be assessed as unfit. A fit assessment may be considered if cardiological evaluation is satisfactory.

- (i) Atrial fibrillation/flutter
 - (A) For initial applicants, a fit assessment should be limited to those with a single episode of arrhythmia which is considered by the medical assessor of the licensing authority to be unlikely to recur.
 - (B) For revalidation, applicants may be assessed as fit if cardiological evaluation is satisfactory and the stroke risk is sufficiently low. A fit assessment with an OML may be considered after a period of stable anticoagulation as prophylaxis, after review by the medical assessor of the licensing authority. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range. In cases of anticoagulation medication not requiring INR monitoring, a fit assessment with an OML may be considered after review

by the medical assessor of the licensing authority after a stabilisation period of 3 months.

- (ii) Applicants with asymptomatic sinus pauses up to 2.5 seconds on resting electrocardiography may be assessed as fit if exercise electrocardiography, echocardiography and 24-hour ambulatory ECG are satisfactory.
- (iii) Applicants with symptomatic sino-atrial disease should be assessed as unfit.
- (6) Mobitz type 2 atrio-ventricular block
Applicants with Mobitz type 2 AV block should require full cardiological evaluation and may be assessed as fit in the absence of distal conducting tissue disease.
- (7) Complete right bundle branch block
 - (i) Applicants with complete right bundle branch block should undergo a cardiological evaluation on first presentation. A fit assessment may be considered if there is no underlying pathology.
 - (ii) Applicants with bifascicular block may be assessed as fit with an OML after a satisfactory cardiological evaluation. The OML may be considered for removal if an electrophysiological study demonstrates no infra-Hissian block, or a 3-year period of satisfactory surveillance has been completed.
- (8) Complete left bundle branch block
 - (i) A fit assessment may be considered subject to satisfactory cardiological evaluation and a 3-year period with an OML, and without an OML after 3 years of surveillance and satisfactory cardiological evaluation.
 - (ii) Investigation of the coronary arteries is necessary for applicants over age 40.
- (9) Ventricular pre-excitation
 - (i) Asymptomatic initial applicants with pre-excitation may be assessed as fit if an electrophysiological study, including adequate drug-induced autonomic stimulation reveals no inducible re-entry tachycardia and the existence of multiple pathways is excluded.
 - (ii) Asymptomatic applicants with pre-excitation may be assessed as fit at revalidation with limitation(s) as appropriate. Limitations may not be necessary if an electrophysiological study, including adequate drug-induced autonomic stimulation, reveals no inducible re-entry tachycardia and the existence of multiple accessory pathways is excluded.
- (10) Pacemaker
Applicants with a subendocardial pacemaker should be assessed as unfit. A fit assessment with an OML may be considered at revalidation no sooner than 3 months after insertion provided:
 - (i) there is no other disqualifying condition;
 - (ii) a bipolar lead system, programmed in bipolar mode without automatic mode change has been used;
 - (iii) the applicant is not pacemaker dependent; and

- (iv) the applicant has a follow-up at least every 12 months, including a pacemaker check.
- (11) QT prolongation

Applicants with asymptomatic QT prolongation may be assessed as fit with an OML subject to satisfactory cardiological evaluation.
- (12) Brugada pattern on electrocardiography

Applicants with a Brugada pattern Type 1 should be assessed as unfit. Applicants with Type 2 or Type 3 may be assessed as fit, with limitations as appropriate, subject to satisfactory cardiological evaluation.

AMC2 MED.B.010 Cardiovascular system

ED Decision 2025/002/R

- (a) Examination

Exercise electrocardiography

An exercise ECG when required as part of a cardiovascular assessment should be symptom-limited and completed to a minimum of Bruce Stage IV or equivalent.
- (b) General
 - (1) Cardiovascular risk factor assessment

Cardiovascular risk factor assessment should take place at least once every 5 years for applicants 40 to 59 years old, and once every 2 years thereafter. A more frequent assessment of the cardiovascular risk factors may be considered when risk factors have been identified.

Applicants with an accumulation of two or more risk factors should undergo a cardiovascular evaluation by the AeMC or AME.
 - (2) Cardiovascular assessment

Reporting of resting and exercise electrocardiograms should be by the AME or an accredited specialist.
- (c) Peripheral arterial disease

A fit assessment may be considered for an applicant with peripheral arterial disease, or after surgery for peripheral arterial disease, provided there is no significant functional impairment, any vascular risk factors have been reduced to an appropriate level, the applicant is receiving acceptable secondary prevention treatment, and there is no evidence of myocardial ischaemia.
- (d) Aortic aneurysm
 - (1) Applicants with an aneurysm of the infra-renal abdominal aorta of less than 5 cm in diameter may be assessed as fit, subject to satisfactory cardiological evaluation. Regular cardiological evaluations should be carried out.
 - (2) Applicants with an aneurysm of the thoracic or supra-renal abdominal aorta of less than 5 cm in diameter may be assessed as fit with an ORL or OSL, subject to satisfactory cardiological evaluation. Regular follow-up should be carried out.

- (3) Applicants may be assessed as fit after surgery for an infra-renal abdominal aortic aneurysm, subject to satisfactory cardiological evaluation. Regular cardiological evaluations should be carried out.
- (4) Applicants may be assessed as fit with an ORL or OSL after surgery for a thoracic or supra-renal abdominal aortic aneurysm, subject to satisfactory cardiological evaluation. Regular cardiological evaluations should be carried out.
- (e) Cardiac valvular abnormalities
 - (1) Applicants with previously unrecognised cardiac murmurs should undergo further cardiological evaluation.
 - (2) Applicants with minor cardiac valvular abnormalities may be assessed as fit.
 - (3) Aortic valve disease
 - (i) Applicants with a bicuspid aortic valve may be assessed as fit if no other cardiac or aortic abnormality is demonstrated. Follow-up with echocardiography, as necessary, should be determined in consultation with the medical assessor of the licensing authority.
 - (ii) Applicants with aortic stenosis may be assessed as fit provided the left ventricular function is intact and the mean pressure gradient is less than 20 mmHg. Applicants with an aortic valve orifice of more than 1 cm² and a mean pressure gradient above 20 mmHg, but not greater than 50 mmHg, may be assessed as fit with an ORL or OSL. Follow-up with 2D Doppler echocardiography, as necessary, should be determined in consultation with the medical assessor of the licensing authority in all cases. Alternative measurement techniques with equivalent ranges may be used. Regular cardiological evaluation should be considered. Applicants with a history of systemic embolism or significant dilatation of the thoracic aorta should be assessed as unfit.
 - (iii) Applicants with trivial aortic regurgitation may be assessed as fit. Applicants with a greater degree of aortic regurgitation may be assessed as fit with an OSL. There should be no demonstrable abnormality of the ascending aorta on 2D Doppler echocardiography. Follow-up, as necessary, should be determined in consultation with the medical assessor of the licensing authority.
 - (4) Mitral valve disease
 - (i) Asymptomatic applicants with an isolated mid-systolic click due to mitral leaflet prolapse may be assessed as fit.
 - (ii) Applicants with rheumatic mitral stenosis should be assessed as unfit.
 - (iii) Applicants with minor regurgitation may be assessed as fit. Periodic cardiological review should be determined in consultation with the medical assessor of the licensing authority.
 - (iv) Applicants with moderate mitral regurgitation may be considered as fit with an ORL or OSL if the 2D Doppler echocardiogram demonstrates satisfactory left ventricular dimensions and satisfactory myocardial function is confirmed by exercise electrocardiography. Periodic cardiological review should be determined in consultation with the medical assessor of the licensing authority.

- (v) Applicants with evidence of volume overloading of the left ventricle demonstrated by increased left ventricular end-diastolic diameter or evidence of systolic impairment should be assessed as unfit.
- (f) Valvular surgery
 - (1) Applicants who have undergone cardiac valve replacement or repair may be assessed as fit without limitations subject to satisfactory post-operative cardiological evaluation and if no anticoagulants are needed.
 - (2) Where anticoagulation is needed after valvular surgery, a fit assessment with an ORL or OSL may be considered after cardiological evaluation if the haemorrhagic risk is acceptable. The review should show that the anticoagulation is stable. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range. The INR target range should be determined by the type of surgery performed. Applicants who measure their INR on a 'near patient' testing system within 12 hours prior to flight and only exercise the privileges of their licence(s) if the INR is within the target range, may be assessed as fit without the above-mentioned limitation. The INR results should be recorded and the results should be reviewed at each aero-medical assessment. Applicants taking anticoagulation medication not requiring INR monitoring, may be assessed as fit without the above-mentioned limitation in consultation with the medical assessor of the licensing authority after a stabilisation period of 3 months.
- (g) Thromboembolic disorders

Applicants with arterial or venous thrombosis or pulmonary embolism should be assessed as unfit. A fit assessment with an ORL or OSL may be considered after a period of stable anticoagulation as prophylaxis in consultation with the medical assessor of the licensing authority. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range and the haemorrhagic risk is acceptable. Applicants who measure their INR on a 'near patient' testing system within 12 hours prior to flight and only exercise the privileges of their licence(s) if the INR is within the target range may be assessed as fit without the above-mentioned limitation. The INR results should be recorded and the results should be reviewed at each aero-medical assessment. Applicants taking anticoagulation medication not requiring INR monitoring, may be assessed as fit without the above-mentioned limitation in consultation with the medical assessor of the licensing authority after a stabilisation period of 3 months. Applicants with pulmonary embolism should also undergo a cardiological evaluation. Following cessation of anticoagulant therapy for any indication, applicants should undergo a re-assessment in consultation with the medical assessor of the licensing authority.
- (h) Other cardiac disorders
 - (1) Applicants with a primary or secondary abnormality of the pericardium, myocardium or endocardium may be assessed as fit subject to satisfactory cardiological evaluation.
 - (2) Applicants with a congenital abnormality of the heart, including those who have undergone surgical correction, may be assessed as fit subject to satisfactory cardiological evaluation. Cardiological follow-up may be necessary and should be determined in consultation with the medical assessor of the licensing authority.
- (i) Syncope

- (1) In the case of a single episode of vasovagal syncope which can be explained and is compatible with flight safety, a fit assessment may be considered.
 - (2) Applicants with a history of recurrent vasovagal syncope should be assessed as unfit. A fit assessment may be considered after a 6-month period without recurrence, providing cardiological evaluation is satisfactory. Neurological review may be indicated.
- (j) Blood pressure
- (1) When the blood pressure at examination consistently exceeds 160 mmHg systolic and/or 95 mmHg diastolic, with or without treatment, the applicant should be assessed as unfit.
 - (2) The diagnosis of hypertension requires review of other potential vascular risk factors.
 - (3) Applicants with symptomatic hypotension should be assessed as unfit.
 - (4) Anti-hypertensive treatment should be compatible with flight safety.
 - (5) Following initiation of medication for the control of blood pressure, applicants should be re-assessed to verify that satisfactory control has been achieved and that the treatment is compatible with the safe exercise of the privileges of the applicable licence(s).
- (k) Coronary artery disease
- (1) Chest pain of uncertain cause requires full investigation.
 - (2) Applicants with suspected asymptomatic coronary artery disease should undergo cardiological evaluation which should show no evidence of myocardial ischaemia or significant coronary artery stenosis.
 - (3) Applicants with evidence of exercise-induced myocardial ischaemia should be assessed as unfit.
 - (4) After an ischaemic cardiac event, or revascularisation, applicants without symptoms should have reduced cardiovascular risk factors to an appropriate level. Medication, when used to control angina pectoris, is not acceptable. All applicants should be on appropriate secondary prevention treatment.
 - (i) A coronary angiogram obtained around the time of, or during, the ischaemic myocardial event and a complete, detailed clinical report of the ischaemic event and of any operative procedures should be available to the AME.
 - (A) There should be no stenosis more than 50 % in any major untreated vessel, in any vein or artery graft or at the site of an angioplasty/stent, except in a vessel subtending a myocardial infarction.
 - (B) The whole coronary vascular tree should be assessed as satisfactory by a cardiologist and particular attention should be paid to multiple stenoses and/or multiple revascularisations.
 - (C) Applicants with an untreated stenosis greater than 30 % in the left main or proximal left anterior descending coronary artery should be assessed as unfit.
 - (ii) At least 6 months from the ischaemic myocardial event, including revascularisation, the following investigations should be completed (equivalent tests may be substituted):
 - (A) an exercise ECG showing neither evidence of myocardial ischaemia nor rhythm disturbance;

- (B) an echocardiogram showing satisfactory left ventricular function with no important abnormality of wall motion and a satisfactory left ventricular ejection fraction of 50 % or more;
 - (C) in cases of angioplasty/stenting, a myocardial perfusion scan or stress echocardiogram, or equivalent test, which should show no evidence of reversible myocardial ischaemia. If there is doubt about revascularisation in myocardial infarction or bypass grafting, a perfusion scan, or equivalent test, should also be carried out;
 - (D) further investigations, such as a 24-hour ECG, may be necessary to assess the risk of any significant rhythm disturbance.
- (iii) Periodic follow-up should include a cardiological evaluation.
 - (A) After coronary artery bypass grafting, a myocardial perfusion scan (or equivalent test) should be performed if there is any indication, and in all cases within five years from the procedure for a fit assessment without an OSL, OPL or ORL.
 - (B) In all cases, coronary angiography should be considered at any time if symptoms, signs or non-invasive tests indicate myocardial ischaemia.
- (iv) Successful completion of the six-month or subsequent review will allow a fit assessment. Applicants may be assessed as fit with an ORL or OSL having successfully completed only an exercise ECG.
- (5) Applicants with angina pectoris should be assessed as unfit, whether or not it is alleviated by medication.
- (I) Rhythm and conduction disturbances
 - (1) Applicants with significant rhythm or conduction disturbance should undergo cardiological evaluation before a fit assessment may be considered with an ORL or OSL, as appropriate. Such evaluation should include:
 - (i) exercise ECG to the Bruce protocol or equivalent. Bruce stage 4 should be achieved and no significant abnormality of rhythm or conduction, or evidence of myocardial ischaemia should be demonstrated. Withdrawal of cardioactive medication prior to the test should normally be required;
 - (ii) 24-hour ambulatory ECG which should demonstrate no significant rhythm or conduction disturbance;
 - (iii) 2D Doppler echocardiogram which should show no significant selective chamber enlargement or significant structural or functional abnormality, and a left ventricular ejection fraction of at least 50 %.Further evaluation may include (equivalent tests may be substituted):
 - (iv) 24-hour ECG recording repeated as necessary;
 - (v) electrophysiological study;
 - (vi) myocardial perfusion imaging;
 - (vii) cardiac magnetic resonance imaging (MRI);
 - (viii) coronary angiogram.

- (2) Where anticoagulation is needed for a rhythm disturbance, a fit assessment with an ORL or OSL may be considered, if the haemorrhagic risk is acceptable and the anticoagulation is stable. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range. Applicants who measure their INR on a 'near patient' testing system within 12 hours prior to flight and only exercise the privileges of their licence(s) if the INR is within the target range may be assessed as fit without the above-mentioned limitation. The INR results should be recorded and the results should be reviewed at each aero-medical assessment. Applicants taking anticoagulation medication not requiring INR monitoring, may be assessed as fit without the above-mentioned limitation in consultation with the medical assessor of the licensing authority after a stabilisation period of 3 months.
- (3) Ablation

A fit assessment may be considered following successful catheter ablation subject to satisfactory cardiological review undertaken at a minimum of 2 months after the ablation.
- (4) Supraventricular arrhythmias
 - (i) Applicants with significant disturbance of supraventricular rhythm, including sinoatrial dysfunction, whether intermittent or established, may be assessed as fit if cardiological evaluation is satisfactory.
 - (ii) Applicants with atrial fibrillation/flutter may be assessed as fit if cardiological evaluation is satisfactory and the stroke risk is sufficiently low. Where anticoagulation is needed, a fit assessment with an ORL or OSL may be considered after a period of stable anticoagulation as prophylaxis, in consultation with the medical assessor of the licensing authority. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range. Applicants who measure their INR on a 'near patient' testing system within 12 hours prior to flight and only exercise the privileges of their licence(s) if the INR is within the target range may be assessed as fit without the above-mentioned limitation. The INR results should be recorded and the results should be reviewed at each aero-medical assessment. Applicants taking anticoagulation medication not requiring INR monitoring, may be assessed as fit without the above-mentioned limitation in consultation with the medical assessor of the licensing authority after a stabilisation period of 3 months.
 - (iii) Applicants with asymptomatic sinus pauses up to 2.5 seconds on resting electrocardiography may be assessed as fit if cardiological evaluation is satisfactory.
- (5) Heart block
 - (i) Applicants with first degree and Mobitz type 1 AV block may be assessed as fit.
 - (ii) Applicants with Mobitz type 2 AV block may be assessed as fit in the absence of distal conducting tissue disease.
- (6) Complete right bundle branch block

Applicants with complete right bundle branch block should undergo a cardiological evaluation on first presentation. A fit assessment may be considered if there is no underlying pathology.
- (7) Complete left bundle branch block

Applicants with complete left bundle branch block may be assessed as fit with appropriate limitations, such as an ORL, and subject to satisfactory cardiological evaluation.

(8) Ventricular pre-excitation

Asymptomatic applicants with ventricular pre-excitation may be assessed as fit with limitation(s) as appropriate, subject to satisfactory cardiological evaluation. Limitations may not be necessary if an electrophysiological study is conducted and the results are satisfactory.

(9) Pacemaker

Applicants with a subendocardial pacemaker should be assessed as unfit. A fit assessment may be considered no sooner than 3 months after insertion, providing:

- (i) there is no other disqualifying condition;
- (ii) a bipolar lead system, programmed in bipolar mode without automatic mode change, has been used;
- (iii) the applicant is not pacemaker dependent; and
- (iv) the applicant has a follow-up at least every 12 months, including a pacemaker check.

(10) QT prolongation

Applicants with asymptomatic QT prolongation may be assessed as fit with an ORL or OS� subject to satisfactory cardiological evaluation.

(11) Brugada pattern on electrocardiography

Applicants with a Brugada pattern Type 1 should be assessed as unfit. Applicants with Type 2 or Type 3 may be assessed as fit, with limitation(s) as appropriate, subject to satisfactory cardiological evaluation.

(m) Heart or heart/lung transplantation

(1) Applicants who have undergone heart or heart/lung transplantation may be assessed as fit, with appropriate limitation(s) such as an ORL, no sooner than 12 months after transplantation, provided that cardiological evaluation is satisfactory with:

- (i) no rejection in the first year following transplantation;
- (ii) no significant arrhythmias;
- (iii) a left ventricular ejection fraction $\geq 50\%$;
- (iv) a symptom limited exercise ECG; and
- (v) a coronary angiogram if indicated;

(2) Regular cardiological evaluations should be carried out.

GM1 MED.B.010 Cardiovascular system

ED Decision 2019/002/R

MITRAL VALVE DISEASE

- (a) Minor regurgitation should have evidence of no thickened leaflets or flail chordae and left atrial internal diameter of less than or equal to 4.0 cm.

- (b) The following may indicate severe regurgitation:
 - (1) LV internal diameter (diastole) > 6.0 cm; or
 - (2) LV internal diameter (systole) > 4.1 cm; or
 - (3) Left atrial internal diameter > 4.5 cm.
- (c) Doppler indices, such as width of jet, backwards extension and whether there is flow reversal in the pulmonary veins may be helpful in assessing severity of regurgitation.

GM2 MED.B.010 Cardiovascular system

ED Decision 2019/002/R

VENTRICULAR PRE-EXCITATION

Asymptomatic applicants with pre-excitation may be assessed as fit if they meet the following criteria, which may also indicate a satisfactory electrophysiological evaluation:

- (a) refractory period > 300 ms;
- (b) no induced atrial fibrillation.

GM3 MED.B.010 Cardiovascular system

ED Decision 2019/002/R

ANTICOAGULATION

Applicants taking anticoagulant medication which requires monitoring with INR testing, should measure their INR on a 'near patient' testing system within 12 hours prior to flight and the privileges of the applicable licence(s) should only be exercised if the INR is within the target range. The INR result should be recorded and the results should be reviewed at each aero-medical assessment.

GM4 MED.B.010 Cardiovascular system

ED Decision 2019/002/R

MITRAL VALVE DISEASE

- (a) Minor regurgitation should have evidence of no thickened leaflets or flail chordae and left atrial internal diameter of less than or equal to 4.0 cm.
- (b) The following may indicate severe regurgitation:
 - (1) LV internal diameter (diastole) > 6.0 cm; or
 - (2) LV internal diameter (systole) > 4.1 cm; or
 - (3) Left atrial internal diameter > 4.5 cm.
- (c) Doppler indices, such as width of jet, backwards extension and whether there is flow reversal in the pulmonary veins may be helpful in assessing severity of regurgitation.

GM5 MED.B.010 Cardiovascular system

ED Decision 2019/002/R

VENTRICULAR PRE-EXCITATION

Asymptomatic applicants with pre-excitation may be assessed as fit if they meet the following criteria:

- (a) no inducible re-entry tachycardia;

- (b) refractory period > 300 ms;
- (c) no induced atrial fibrillation;
- (d) no evidence of multiple accessory pathways.

GM1 MED.B.010(b) Cardiovascular system

ED Decision 2025/002/R

(a) Cardiovascular risk factor assessment

A risk calculator is constructed as an equation with regression coefficients for each included risk factor, based on a statistical analysis of data from a population of a certain region to provide a crude risk estimate. A risk calculator to be used for screening of CAT pilots should be relevant for the ethnicity of the pilots being screened and should predict the 5–10-year risk for non-fatal events such as acute coronary syndromes or stroke, as well as fatal cardiac events, as both may lead to total in-flight incapacitation.

It is recommended to use a risk estimation tool that is based on populations similar to your most common target population. No risk calculator is perfect, and an assessment of advantages and disadvantages should be made when deciding on which tools should be used. For example, the most common tools that are based on European population are: SCORE 2, PROCAM, AGLA and QRISK 3.

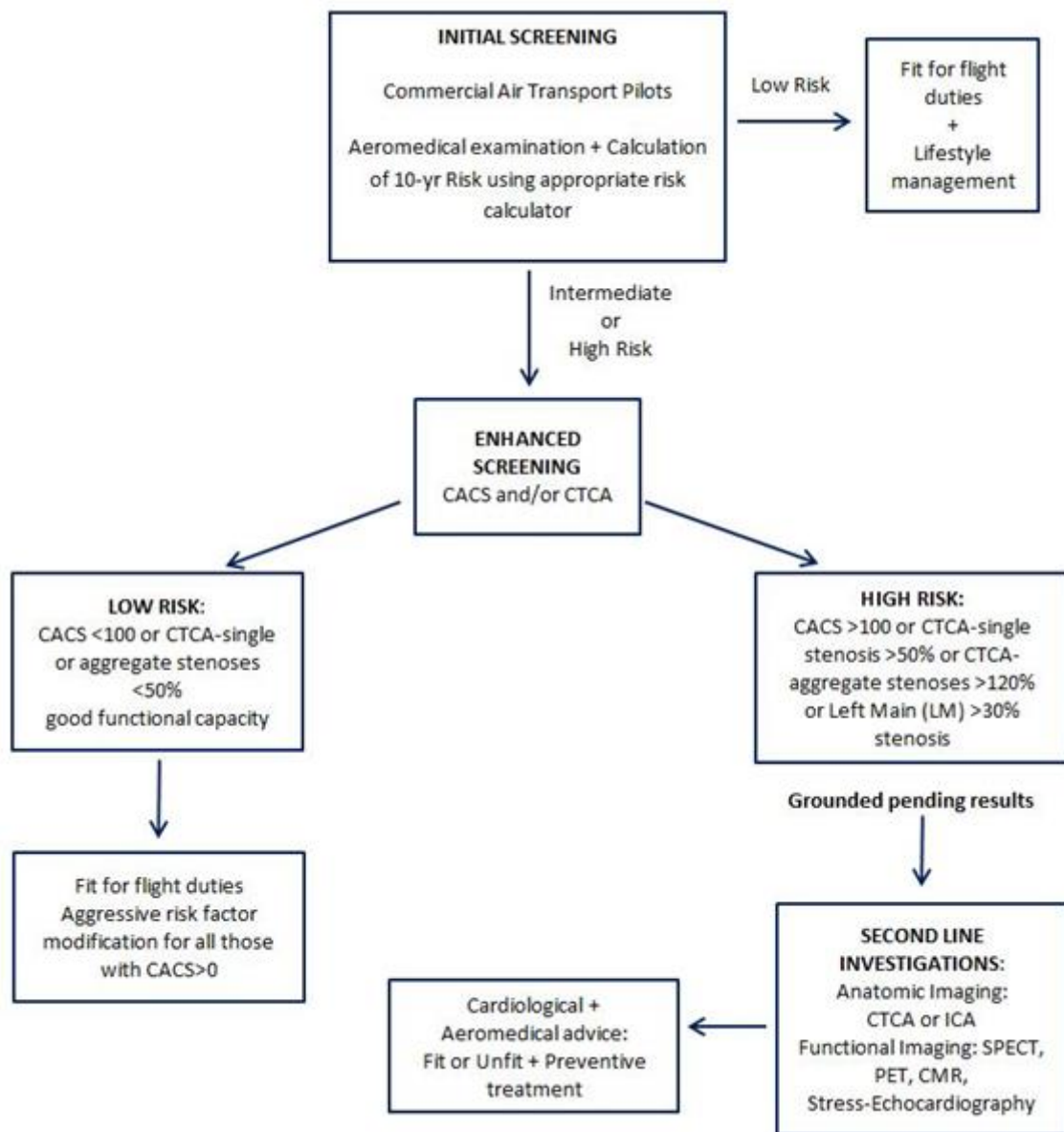
In the risk assessment AMEs should give proper consideration to the latest published guidance of the European Society of Cardiology. At the time of the drafting, the most recent guidelines are '[2021 ESC Guidelines on cardiovascular disease prevention in clinical practice](#)'.

(b) Cardiovascular assessment

It is recommended that for applicants involved in single-pilot HEMS operations who have reached the age of 60, the cardiovascular assessment considers the risk level when deciding on enhanced screening investigations.

In this regard the following flow chart algorithm adapted by Simons et al. (2019) from Gray et al.(2019) is aimed at supporting AMEs and medical assessors. The classification of low, intermediate or high risk is given by the cardiovascular score being used. The enhanced screening investigations are in the realm of the consultant cardiologist.

Abbreviations used in the flowchart: CACS=Coronary Artery Calcium Score; CTCA=Computed Tomography Coronary Angiography; SPECT=Single-Positron Emission Tomography; PET=Positron Emission Tomography; CMR=Cardiac Magnetic Resonance; ICA= Invasive Coronary Angiography. [Simons et al. (2019)]



MED.B.015 Respiratory System

Regulation (EU) 2024/2076

- (a) Applicants with significant impairment of pulmonary function shall be assessed as unfit. However, they may be assessed as fit once pulmonary function has recovered and is satisfactory.
- (b) Applicants for a class 1 medical certificate shall undertake pulmonary functional tests at the initial examination and when clinically indicated.
- (ba) For class 1 medical certificate holders involved in single-pilot HEMS operations, pulmonary functional tests and obstructive sleep apnoea (OSA) screening shall be completed at the first revalidation or renewal examination after the age of 60.
- (c) Applicants for a class 2 medical certificate shall undertake pulmonary morphological and functional tests when clinically or epidemiologically indicated.

- (d) Applicants with a medical history or diagnosis of any of the following medical conditions shall undertake respiratory evaluation with a satisfactory result before they may be assessed as fit:
- (1) asthma requiring medication;
 - (2) active inflammatory disease of the respiratory system;
 - (3) active sarcoidosis;
 - (4) pneumothorax;
 - (5) sleep apnoea syndrome;
 - (6) major thoracic surgery;
 - (7) pneumonectomy;
 - (8) chronic obstructive pulmonary disease.
- Before further consideration is given to their application, applicants with an established diagnosis of any of the medical conditions specified in points (3) and (5) shall undergo satisfactory cardiological evaluation.
- (e) Aero-medical assessment
- (1) Applicants for a class 1 medical certificate with any of the medical conditions specified in point (d) shall be referred to the medical assessor of the licensing authority.
 - (2) Applicants for a class 2 medical certificate with any of the medical conditions specified in point (d) shall be assessed in consultation with the medical assessor of the licensing authority.
- (f) Applicants for a class 1 medical certificate who have undergone a pneumonectomy shall be assessed as unfit.

AMC1 MED.B.015 Respiratory system

ED Decision 2025/002/R

- (a) Examination
- (1) Spirometry

A spirometric examination is required by points [MED.B.015](#) (b) and (c) for applicants for a class 1 medical certificate in specific situations. Applicants with an FEV1/FVC ratio of less than 70 % should be evaluated by a specialist in respiratory disease.
 - (2) Chest radiography

Posterior/anterior chest radiography may be required at initial, revalidation or renewal examinations if clinically or epidemiologically indicated
- (b) Chronic obstructive pulmonary disease
- Applicants with chronic obstructive pulmonary disease should be assessed as unfit. Applicants with only minor impairment of pulmonary function may be assessed as fit.
- (c) Asthma
- Applicants with asthma requiring medication or experiencing recurrent attacks of asthma may be assessed as fit if the asthma is considered stable with satisfactory pulmonary function tests and medication is compatible with flight safety. Applicants requiring systemic steroids should be assessed as unfit.

(d) Inflammatory disease

For applicants with active inflammatory disease of the respiratory system a fit assessment may be considered when the condition has resolved without sequelae and no medication is required.

(e) Sarcoidosis

(1) Applicants with active sarcoidosis should be assessed as unfit. Investigation should be undertaken with respect to the possibility of systemic, particularly cardiac, involvement. A fit assessment may be considered if no medication is required, and the disease is investigated and shown to be limited to hilar lymphadenopathy and inactive.

(2) Applicants with cardiac or neurological sarcoid should be assessed as unfit.

(f) Pneumothorax

(1) Applicants with a spontaneous pneumothorax should be assessed as unfit. A fit assessment may be considered if respiratory evaluation is satisfactory:

- (i) 1 year following full recovery from a single spontaneous pneumothorax;
- (ii) at revalidation, 6 weeks following full recovery from a single spontaneous pneumothorax, with an OML for at least a year after full recovery;
- (iii) following surgical intervention in the case of a recurrent pneumothorax provided there is satisfactory recovery.

(2) Applicants with a recurrent spontaneous pneumothorax that has not been surgically should be assessed as unfit.

(3) A fit assessment following full recovery from a traumatic pneumothorax as a result of an accident or injury may be acceptable once full absorption of the pneumothorax is demonstrated.

(g) Thoracic surgery

(1) Applicants requiring major thoracic surgery should be assessed as unfit until recovery is complete, the applicant is asymptomatic, and the risk of secondary complication is minimal.

(2) A fit assessment following lesser chest surgery may be considered after satisfactory recovery and full respiratory evaluation.

(h) Sleep apnoea syndrome/sleep disorder

Applicants with unsatisfactorily treated sleep apnoea syndrome should be assessed as unfit.

Obstructive sleep apnoea (OSA) screening should include an anamnestic interview and assessment of OSA risk factors such as increased BMI, and a history of cardiovascular, cerebrovascular, metabolic and ENT pathology.

AMC2 MED.B.015 Respiratory system

ED Decision 2019/002/R

(a) Examination

(1) A spirometric examination should be performed on clinical indication. Applicants with a forced expiratory volume in the first one second (FEV1)/forced vital capacity(FVC)ratio of less than 70 % should be evaluated by a specialist in respiratory disease.

- (2) Posterior/anterior chest radiography may be required if clinically or epidemiologically indicated.
- (b) Chronic obstructive pulmonary disease
Applicants with only minor impairment of pulmonary function may be assessed as fit.
- (c) Asthma
Applicants with asthma may be assessed as fit if the asthma is considered stable with satisfactory pulmonary function tests and medication is compatible with flight safety. Applicants requiring systemic steroids should be assessed as unfit.
- (d) Inflammatory disease
Applicants with active inflammatory disease of the respiratory system should be assessed as unfit pending resolution of the condition.
- (e) Sarcoidosis
 - (1) Applicants with active sarcoidosis should be assessed as unfit. Investigation should be undertaken with respect to the possibility of systemic involvement. A fit assessment may be considered once the disease is inactive.
 - (2) Applicants with cardiac sarcoid should be assessed as unfit.
- (f) Pneumothorax
 - (1) Applicants with spontaneous pneumothorax should be assessed as unfit. A fit assessment may be considered if respiratory evaluation is satisfactory:
 - (i) six weeks following full recovery from a single spontaneous pneumothorax;
 - (ii) following surgical intervention in the case of a recurrent pneumothorax, provided there is satisfactory recovery.
 - (2) A fit assessment following full recovery from a traumatic pneumothorax as a result of an accident or injury may be acceptable once full absorption of the pneumothorax is demonstrated.
- (g) Thoracic surgery
Applicants requiring major thoracic surgery should be assessed as unfit until recovery is complete, the applicant is asymptomatic, and the risk of secondary complication is minimal.
- (h) Sleep apnoea syndrome
Applicants with unsatisfactorily treated sleep apnoea syndrome should be assessed as unfit.

GM1 MED.B.015 Respiratory system

ED Decision 2025/002/R

SCREENING OF THE OBSTRUCTIVE SLEEP APNOEA (OSA) SYNDROME

- (a) AMEs may consider the following algorithm when screening their applicants regarding the OSA syndrome:
Assessment of OSA risk may be considered at every medical examination of pilots through scores that combine history questions with physical findings such as the STOP-BANG score.
- (b) AME guidance:

Indicators to initiate OSA evaluation

- History interview including at least the following:
- Daytime sleepiness (i.e. Epworth Sleepiness Scale)?
- Snoring (what does spouse/partner say?)
- Psychosocial issues due to sleepiness, heavy snoring
- Observable apnoea episodes
- Contributing factors:
 - BMI >30
 - Previous bariatric history
- Neck circumference: ≥ 40 cm
- Diagnosed arterial hypertension
- Heart troubles
 - Arrhythmia
 - Congestive heart failure
 - CHD
- Previous TIA, stroke
- Diabetes Type 2
- ENT
 - Nasal obstruction
 - Orthodontic/Retrognathia
 - Oropharyngeal examination –e.g. modified Mallampati Score or Friedman tongue position

Methodology (if indicated):

- Nocturnal oximetry
- Respiratory polygraphy
- Polysomnography in certified sleep laboratories
- Eventually evaluation of vigilance:
 - Maintenance of Wakefulness Test (MWT)
 - Multiple Sleep Latency Test (MSLT)

MED.B.020 Digestive System*Regulation (EU) 2019/27*

- (a) Applicants with any sequelae of disease or surgical intervention in any part of the digestive tract or its adnexa likely to cause incapacitation in flight, in particular any obstruction due to stricture or compression, shall be assessed as unfit.
- (b) Applicants who have herniae that might give rise to incapacitating symptoms shall be assessed as unfit.

- (c) Applicants with any of the following disorders of the gastrointestinal system may be assessed as fit subject to satisfactory gastrointestinal evaluation after successful treatment or full recovery after surgery:
 - (1) recurrent dyspeptic disorder requiring medication;
 - (2) pancreatitis;
 - (3) symptomatic gallstones;
 - (4) a clinical diagnosis or documented medical history of chronic inflammatory bowel disease;
 - (5) after surgical operation on the digestive tract or its adnexa, including surgery involving total or partial excision or a diversion of any of these organs.
- (d) Aero-medical assessment
 - (1) Applicants for a class 1 medical certificate with the diagnosis of any of the medical conditions specified in points (2), (4) and (5) of point (c) shall be referred to the medical assessor of the licensing authority.
 - (2) The fitness of applicants for a class 2 medical certificate with the diagnosis of the medical condition specified in point (2) of point (c) shall be assessed in consultation with the medical assessor of the licensing authority.

AMC1 MED.B.020 Digestive system

ED Decision 2019/002/R

- (a) Oesophageal varices

Applicants with oesophageal varices should be assessed as unfit.
- (b) Pancreatitis

Applicants with pancreatitis should be assessed as unfit pending assessment. A fit assessment may be considered if the cause is removed.
- (c) Gallstones
 - (1) Applicants with a single asymptomatic large gallstone discovered incidentally may be assessed as fit if not likely to cause incapacitation in flight.
 - (2) Applicants with asymptomatic multiple gallstones may be assessed as fit with an OML.
- (d) Inflammatory bowel disease

Applicants with an established diagnosis or history of chronic inflammatory bowel disease should be assessed as fit if the inflammatory bowel disease is in established remission and stable and if systemic steroids are not required for its control.
- (e) Peptic ulceration

Applicants with peptic ulceration should be assessed as unfit pending full recovery and demonstrated healing.
- (f) Digestive tract and abdominal surgery

Applicants who have undergone a surgical operation for medical conditions of the digestive tract or its adnexa, including a total or partial excision or a diversion of any of these organs or herniae should be assessed as unfit. A fit assessment may be considered if recovery is complete,

the applicant is asymptomatic, and there is only a minimal risk of secondary complication or recurrence.

(g) Liver disease

Applicants with morphological or functional liver disease, or after surgery, including liver transplantation, may be assessed as fit subject to satisfactory gastroenterological evaluation.

AMC2 MED.B.020 Digestive system

ED Decision 2019/002/R

(a) Oesophageal varices

Applicants with oesophageal varices should be assessed as unfit.

(b) Pancreatitis

Applicants with pancreatitis should be assessed as unfit pending satisfactory recovery.

(c) Gallstones

(1) Applicants with a single asymptomatic large gallstone or asymptomatic multiple gallstones may be assessed as fit.

(2) Applicants with symptomatic single or multiple gallstones should be assessed as unfit. A fit assessment may be considered following gallstone removal.

(d) Inflammatory bowel disease

Applicants with an established diagnosis or history of chronic inflammatory bowel disease may be assessed as fit provided that the disease is stable and not likely to interfere with the safe exercise of the privileges of the applicable licence(s).

(e) Peptic ulceration

Applicants with peptic ulceration should be assessed as unfit pending full recovery.

(f) Digestive tract and abdominal surgery

Applicants who have undergone a surgical operation:

(1) for herniae; or

(2) on the digestive tract or its adnexa, including a total or partial excision or diversion of any of these organs

should be assessed as unfit. A fit assessment may be considered if recovery is complete, the applicant is asymptomatic, and there is only a minimal risk of secondary complication or recurrence.

(g) Liver disease

Applicants with morphological or functional liver disease, or after surgery, including liver transplantation, may be assessed as fit subject to satisfactory gastroenterological evaluation.

MED.B.025 Metabolic and Endocrine Systems

Regulation (EU) 2019/27

(a) Applicants with metabolic, nutritional or endocrine dysfunction may be assessed as fit subject to demonstrated stability of the medical condition and satisfactory aero-medical evaluation.

- (b) *Diabetes mellitus*
 - (1) Applicants with diabetes mellitus requiring insulin shall be assessed as unfit.
 - (2) Applicants with diabetes mellitus not requiring insulin shall be assessed as unfit unless it can be demonstrated that blood sugar control has been achieved and is stable.
- (c) *Aero-medical assessment*
 - (1) Applicants for a class 1 medical certificate requiring medication other than insulin for blood sugar control shall be referred to the medical assessor of the licensing authority.
 - (2) The fitness of applicants for a class 2 medical certificate requiring medication other than insulin for blood sugar control shall be assessed in consultation with the medical assessor of the licensing authority.

AMC1 MED.B.025 Metabolic and endocrine systems

ED Decision 2019/002/R

- (a) *Metabolic, nutritional or endocrine dysfunction*

Applicants with metabolic, nutritional or endocrine dysfunction may be assessed as fit if the condition is asymptomatic, clinically compensated and stable with or without replacement therapy, and regularly reviewed by an appropriate specialist.
- (b) *Obesity*

Applicants with a Body Mass Index ≥ 35 may be assessed as fit only if the excess weight is not likely to interfere with the safe exercise of the applicable licence(s) and the results of a risk assessment, including evaluation of the cardiovascular system and evaluation of the possibility of sleep apnoea, are satisfactory.
- (c) *Addison's disease*

Applicants with Addison's disease should be assessed as unfit. A fit assessment with an OML may be considered, provided that cortisone is carried and available for use whilst exercising the privileges of the applicable licence(s).
- (d) *Gout*

Applicants with acute gout should be assessed as unfit. A fit assessment may be considered once asymptomatic, after cessation of treatment or the condition is stabilised on anti-hyperuricaemic therapy.
- (e) *Thyroid dysfunction*

Applicants with hyperthyroidism or hypothyroidism should be assessed as unfit. A fit assessment may be considered when a stable euthyroid state is attained.
- (f) *Abnormal glucose metabolism*

Glycosuria and abnormal blood glucose levels require investigation. A fit assessment may be considered if normal glucose tolerance is demonstrated (low renal threshold) or impaired glucose tolerance without diabetic pathology is fully controlled by diet and regularly reviewed.
- (g) *Diabetes mellitus*

Subject to good control of blood sugar with no hypoglycaemic episodes:

 - (1) applicants with diabetes mellitus not requiring medication may be assessed as fit;

- (2) the use of antidiabetic medications that are not likely to cause hypoglycaemia may be acceptable for a fit assessment with an OML.

AMC2 MED.B.025 Metabolic and endocrine systems

ED Decision 2019/002/R

- (a) Metabolic, nutritional or endocrine dysfunction

Applicants with metabolic, nutritional or endocrine dysfunction should be assessed as unfit. A fit assessment may be considered if the condition is asymptomatic, clinically compensated and stable.

- (b) Obesity

Applicants with a Body Mass Index ≥ 35 may be assessed as fit only if the excess weight is not likely to interfere with the safe exercise of the applicable licence(s) and the results of a risk assessment, including evaluation of the cardiovascular system and evaluation of the possibility of sleep apnoea, are satisfactory.

- (c) Addison's disease

Applicants with Addison's disease may be assessed as fit provided that cortisone is carried and available for use whilst exercising the privileges of the applicable licence(s).

- (d) Gout

Applicants with acute gout should be assessed as unfit until asymptomatic.

- (e) Thyroid dysfunction

Applicants with thyroid disease may be assessed as fit once a stable euthyroid state is attained.

- (f) Abnormal glucose metabolism

Glycosuria and abnormal blood glucose levels require investigation. A fit assessment may be considered if normal glucose tolerance is demonstrated (low renal threshold) or impaired glucose tolerance is fully controlled by diet and regularly reviewed.

- (g) Diabetes mellitus

Applicants with diabetes mellitus may be assessed as fit. The use of antidiabetic medications that are not likely to cause hypoglycaemia may be acceptable.

MED.B.030 Haematology

Regulation (EU) 2019/27

- (a) Applicants for a class 1 medical certificate shall be subjected to an haemoglobin test at each aero-medical examination.
- (b) Applicants with a haematological condition may be assessed as fit subject to satisfactory aero-medical evaluation.
- (c) Applicants for a class 1 medical certificate with any of the following haematological conditions shall be referred to the medical assessor of the licensing authority:
- (1) abnormal haemoglobin, including, but not limited to anaemia, erythrocytosis or haemoglobinopathy;
 - (2) significant lymphatic enlargement;

- (3) enlargement of the spleen;
 - (4) coagulation, haemorrhagic or thrombotic disorder;
 - (5) leukaemia.
- (d) The fitness of applicants for a class 2 medical certificate with any of the haematological conditions specified in points (4) and (5) of point (c) shall be assessed in consultation with the medical assessor of the licensing authority.

AMC1 MED.B.030 Haematology

ED Decision 2019/002/R

- (a) **Abnormal haemoglobin**
Applicants with abnormal haemoglobin should be investigated.
- (b) **Anaemia**
 - (1) Applicants with anaemia demonstrated by a reduced haemoglobin level require investigation. Applicants with an haematocrit of less than 32 % should be assessed as unfit. A fit assessment may be considered in cases where the primary cause, such as iron or B12 deficiency, has been treated and the haemoglobin or haematocrit has stabilised at a satisfactory level.
 - (2) Applicants with anaemia which is unamenable to treatment should be assessed as unfit.
- (c) **Erythrocytosis**
Applicants with erythrocytosis should be assessed as unfit. A fit assessment with an OML may be considered if investigation establishes that the condition is stable and no associated pathology is demonstrated.
- (d) **Haemoglobinopathy**
 - (1) Applicants with a haemoglobinopathy should be assessed as unfit. A fit assessment may be considered where minor thalassaemia or other haemoglobinopathy is diagnosed without a history of crises and where full functional capability is demonstrated. The haemoglobin level should be satisfactory.
 - (2) Applicants with sickle cell disease (homozygote) should be assessed as unfit.
- (e) **Coagulation disorders**
 - (1) Applicants with a coagulation disorder should be assessed as unfit. A fit assessment may be considered if there is no history of significant bleeding episodes.
 - (2) Applicants with thrombocytopenia with a platelet count less than $75 \times 10^9/L$ should be assessed as unfit. A fit assessment may be considered once the platelet count is above $75 \times 10^9/L$ and stable.
- (f) **Haemorrhagic disorders**
Applicants with a haemorrhagic disorder require investigation. A fit assessment with an OML may be considered if there is no history of significant bleeding.
- (g) **Thromboembolic disorders**
 - (1) Applicants with a thrombotic disorder require investigation. A fit assessment may be considered when the applicant is asymptomatic and there is only minimal risk of secondary complication or recurrence.

- (2) If anticoagulation is used as treatment, refer to [AMC1 MED.B.010\(g\)](#).
- (3) Applicants with arterial embolus should be assessed as unfit. A fit assessment may be considered once recovery is complete, the applicant is asymptomatic, and there is only minimal risk of secondary complication or recurrence.
- (h) Disorders of the lymphatic system

Applicants with significant localised and generalised enlargement of the lymphatic glands or haematological disease should be assessed as unfit and require investigation. A fit assessment may be considered in cases of an acute infectious process which is fully recovered or Hodgkin's lymphoma or other lymphoid malignancy which has been treated and is in full remission.
- (i) Leukaemia
 - (1) Applicants with acute leukaemia should be assessed as unfit. Once in established remission, applicants may be assessed as fit.
 - (2) Applicants with chronic leukaemia should be assessed as unfit. After a period of demonstrated stability a fit assessment may be considered.
 - (3) Applicants with a history of leukaemia should have no history of central nervous system involvement and no continuing side-effects from treatment of flight safety importance. Haemoglobin and platelet levels should be satisfactory. Regular follow-up is required.
- (j) Splenomegaly

Applicants with splenomegaly should be assessed as unfit and require investigation. A fit assessment may be considered when the enlargement is minimal, stable and no associated pathology is demonstrated, or if the enlargement is minimal and associated with another acceptable condition.

AMC2 MED.B.030 Haematology

ED Decision 2019/002/R

- (a) Abnormal haemoglobin

Haemoglobin should be tested when clinically indicated.
- (b) Anaemia

Applicants with anaemia demonstrated by a reduced haemoglobin level or low haematocrit may be assessed as fit once the primary cause has been treated and the haemoglobin or haematocrit has stabilised at a satisfactory level.
- (c) Erythrocytosis

Applicants with erythrocytosis may be assessed as fit if the condition is stable and no associated pathology is demonstrated.
- (d) Haemoglobinopathy

Applicants with a haemoglobinopathy may be assessed as fit if minor thalassaemia or other haemoglobinopathy is diagnosed without a history of crises and where full functional capability is demonstrated.
- (e) Coagulation and haemorrhagic disorders

Applicants with a coagulation or haemorrhagic disorder may be assessed as fit if there is no likelihood of significant bleeding.

(f) Thromboembolic disorders

Applicants with a thrombotic disorder may be assessed as fit if there is minimal likelihood of significant clotting episodes. If anticoagulation is used as treatment, refer to [AMC2 MED.B.010\(g\)](#).

(g) Disorders of the lymphatic system

Applicants with significant enlargement of the lymphatic glands or haematological disease may be assessed as fit if the condition is unlikely to interfere with the safe exercise of the privileges of the applicable licence(s). Applicants may be assessed as fit in cases of acute infectious process which is fully recovered or Hodgkin's lymphoma or other lymphoid malignancy which has been treated and is in full remission.

(h) Leukaemia

- (1) Applicants with acute leukaemia may be assessed as fit once in established remission.
- (2) Applicants with chronic leukaemia may be assessed as fit after a period of demonstrated stability.
- (3) In cases (h)(1) and (h)(2), there should be no history of central nervous system involvement and no continuing side effects from treatment of flight safety importance. Haemoglobin and platelet levels should be satisfactory. Regular follow-up is required.

(i) Splenomegaly

Applicants with splenomegaly may be assessed as fit if the enlargement is minimal, stable and no associated pathology is demonstrated, or if the enlargement is minimal and associated with another acceptable condition.

MED.B.035 Genitourinary System

Regulation (EU) 2019/27

- (a) Urinalysis shall form part of each aero-medical examination. Applicants shall be assessed as unfit where their urine contains abnormal elements considered to be of pathological significance that could entail a degree of functional incapacity which is likely to jeopardise the safe exercise of the privileges of the license or could render the applicant likely to become suddenly unable to exercise those privileges.
- (b) Applicants with any sequelae of disease or surgical procedures on the genitourinary system or its adnexa likely to cause incapacitation, in particular any obstruction due to stricture or compression, shall be assessed as unfit.
- (c) Applicants with a diagnosis or medical history of the following may be assessed as fit subject to satisfactory genitourinary evaluation, as applicable:
 - (1) renal disease;
 - (2) one or more urinary calculi, or a medical history of renal colic.
- (d) Applicants who have undergone a major surgical operation in the genitourinary system or its adnexa involving a total or partial excision or a diversion of their organs shall be assessed as unfit. However, after full recovery, they may be assessed as fit.
- (e) The applicants for a class 1 medical certificate referred to in points (c) and (d) shall be referred to the medical assessor of the licensing authority.

AMC1 MED.B.035 Genitourinary system

ED Decision 2019/002/R

- (a) Abnormal urinalysis
Investigation is required if there is any abnormal finding on urinalysis.
- (b) Renal disease
 - (1) Applicants presenting with any signs of renal disease should be assessed as unfit. A fit assessment may be considered if blood pressure is satisfactory and renal function is acceptable.
 - (2) Applicants requiring dialysis should be assessed as unfit.
- (c) Urinary calculi
 - (1) Applicants with an asymptomatic calculus or a history of renal colic require investigation.
 - (2) Applicants presenting with one or more urinary calculi should be assessed as unfit and require investigation.
 - (3) Whilst awaiting assessment or treatment, a fit assessment with an OML may be considered.
 - (4) After successful treatment for a calculus a fit assessment without an OML may be considered.
 - (5) Applicants with parenchymal residual calculi may be considered for a fit assessment with an OML.
- (d) Renal and urological surgery
 - (1) Applicants who have undergone a major surgical operation on the genitourinary system or its adnexa involving a total or partial excision or a diversion of any of its organs, should be assessed as unfit until recovery is complete, the applicant is asymptomatic, and the risk of secondary complication is minimal.
 - (2) After other urological surgery, a fit assessment may be considered when the applicant is completely asymptomatic and there is only minimal risk of secondary complication or recurrence.
 - (3) Applicants with compensated nephrectomy without hypertension or uraemia may be considered for a fit assessment.
 - (4) Applicants who have undergone renal transplantation may be considered for a fit assessment with an OML if it is fully compensated and tolerated with only minimal immuno-suppressive therapy after at least 12 months.
 - (5) Applicants who have undergone total cystectomy may be considered for a fit assessment with an OML if there is satisfactory urinary function, no infection and no recurrence of primary pathology.

AMC2 MED.B.035 Genitourinary system

ED Decision 2019/002/R

- (a) Renal disease
Applicants presenting with renal disease may be assessed as fit if blood pressure is satisfactory and renal function is acceptable. Applicants requiring dialysis should be assessed as unfit.

- (b) Urinary calculi
 - (1) Applicants presenting with one or more urinary calculi should be assessed as unfit.
 - (2) Applicants with an asymptomatic calculus or a history of renal colic require investigation.
 - (3) While awaiting assessment or treatment, a fit assessment with an OSL may be considered.
 - (4) After successful treatment the applicant may be assessed as fit.
 - (5) Applicants with parenchymal residual calculi may be assessed as fit.
- (c) Renal and urological surgery
 - (1) Applicants who have undergone a major surgical operation on the genitourinary system or its adnexa involving a total or partial excision or a diversion of any of its organs, should be assessed as unfit until recovery is complete, the applicant is asymptomatic, and the risk of secondary complication is minimal.
 - (2) After other urological surgery, a fit assessment may be considered when the applicant is completely asymptomatic and there is only minimal risk of secondary complication or recurrence.
 - (3) Applicants with compensated nephrectomy without hypertension or uraemia may be assessed as fit.
 - (4) Applicants who have undergone renal transplantation may be considered for a fit assessment if it is fully compensated and with only minimal immuno-suppressive therapy.
 - (5) Applicants who have undergone total cystectomy may be considered for a fit assessment if there is satisfactory urinary function, no infection and no recurrence of primary pathology.

MED.B.040 Infectious Disease

Regulation (EU) 2019/27

- (a) Applicants shall be assessed as unfit where they have a clinical diagnosis or medical history of any infectious disease which is likely to jeopardise the safe exercise of the privileges of the licence.
- (b) Applicants who are HIV positive may be assessed as fit subject to satisfactory aero-medical evaluation. Such applicants for a class 1 medical certificate shall be referred to the medical assessor of the licensing authority.

AMC1 MED.B.040 Infectious disease

ED Decision 2019/002/R

- (a) Infectious disease General

In cases of infectious disease, consideration should be given to a history of, or clinical signs indicating, underlying impairment of the immune system.
- (b) Tuberculosis
 - (1) Applicants with active tuberculosis should be assessed as unfit. A fit assessment may be considered following completion of therapy.

- (2) Applicants with quiescent or healed lesions may be assessed as fit. Specialist evaluation should consider the extent of the disease, the treatment required and possible side effects of medication.
- (c) Syphilis

Applicants with acute syphilis should be assessed as unfit. A fit assessment may be considered in the case of those fully treated and recovered from the primary and secondary stages.
- (d) HIV positivity
 - (1) Applicants who are HIV positive may be assessed as fit with an OML if a full investigation provides no evidence of HIV associated diseases that might give rise to incapacitating symptoms. Frequent review of the immunological status and neurological evaluation by an appropriate specialist should be carried out. A cardiological evaluation may also be required, depending on the medication.
 - (2) Applicants with signs or symptoms of an AIDS-defining condition should be assessed as unfit.
- (e) Infectious hepatitis

Applicants with infectious hepatitis should be assessed as unfit. A fit assessment may be considered once the applicant has become asymptomatic. Regular review of the liver function should be carried out.

AMC2 MED.B.040 Infectious disease

ED Decision 2019/002/R

- (a) Tuberculosis
 - (1) Applicants with active tuberculosis should be assessed as unfit. A fit assessment may be considered following completion of therapy.
 - (2) Applicants with quiescent or healed lesions may be assessed as fit. Specialist evaluation should consider the extent of the disease, the treatment required and possible side effects of medication.
- (b) HIV positivity
 - (1) Applicants who are HIV positive may be assessed as fit if a full investigation provides no evidence of HIV associated diseases that might give rise to incapacitating symptoms. Frequent review of the immunological status and neurological evaluation by an appropriate specialist should be carried out. A cardiological evaluation may be required, depending on the medication.
 - (2) Applicants with signs or symptoms of an AIDS-defining condition should be assessed as unfit.

MED.B.045 Obstetrics and Gynaecology

Regulation (EU) 2019/27

- (a) Applicants who have undergone a major gynaecological operation shall be assessed as unfit. However, they may be assessed as fit after full recovery.
- (b) Pregnancy

- (1) In the event of pregnancy, an applicant may continue to exercise her privileges until the end of the 26th week of gestation only if the AeMC or AME considers that she is fit to do so.
- (2) For holders of a class 1 medical certificate who are pregnant, an OML shall apply. Notwithstanding point [MED.B.001](#), in that case, the OML may be imposed and removed by the AeMC or AME.
- (3) An applicant may resume exercising her privileges after recovery following the end of the pregnancy.

AMC1 MED.B.045 Obstetrics and gynaecology

ED Decision 2019/002/R

(a) Gynaecological surgery

Applicants who have undergone a major gynaecological operation should be assessed as unfit. A fit assessment may be considered if recovery is complete, the applicant is asymptomatic, and the risk of secondary complication or recurrence is minimal.

(b) Pregnancy

- (1) A pregnant licence holder may be assessed as fit with an OML during the first 26 weeks of gestation following review of the obstetric evaluation by the AeMC or AME who should inform the medical assessor of the licensing authority.
- (2) The AeMC or AME should provide written advice to the applicant and the supervising physician regarding potentially significant complications of pregnancy.

AMC2 MED.B.045 Obstetrics and gynaecology

ED Decision 2019/002/R

(a) Gynaecological surgery

Applicants who have undergone a major gynaecological operation should be assessed as unfit until recovery is complete, the applicant is asymptomatic, and the risk of secondary complication or recurrence is minimal.

(b) Pregnancy

- (1) A pregnant licence holder may be assessed as fit during the first 26 weeks of gestation following satisfactory obstetric evaluation.
- (2) Licence privileges may be resumed upon satisfactory confirmation of full recovery following confinement or termination of pregnancy.

MED.B.050 Musculoskeletal System

Regulation (EU) 2019/27

- (a) Applicants who do not have sufficient sitting height, arm and leg length and muscular strength for the safe exercise of the privileges of the licence shall be assessed as unfit. However, where their sitting height, arm and leg length and muscular strength is sufficient for the safe exercise of the privileges in respect of a certain aircraft type, which can be demonstrated where necessary through a medical flight or a simulator flight test, the applicant may be assessed as fit and their privileges shall be limited accordingly.

- (b) Applicants who do not have satisfactory functional use of the musculoskeletal system to enable them to safely exercise the privileges of the licence shall be assessed as unfit. However, where their functional use of the musculoskeletal system is satisfactory for the safe exercise the privileges in respect of a certain aircraft type, which may be demonstrated where necessary through a medical flight or a simulator flight test, the applicant may be assessed as fit and their privileges shall be limited accordingly.
- (c) In case of doubt arising in the context of the assessments referred to in points (a) and (b), applicants for a class 1 medical certificate shall be referred to the medical assessor of the licensing authority and applicants for a class 2 medical certificate shall be assessed in consultation with the medical assessor of the licensing authority.

AMC1 MED.B.050 Musculoskeletal system

ED Decision 2019/002/R

- (a) Applicants with any significant sequelae from disease, injury or congenital abnormality affecting the bones, joints, muscles or tendons with or without surgery require full evaluation prior to a fit assessment.
- (b) Applicants with inflammatory, infiltrative, traumatic or degenerative disease of the musculoskeletal system may be assessed as fit, provided the condition is in remission or is stable and the applicant is taking no disqualifying medication and has satisfactorily completed a medical flight or simulator flight test. Appropriate limitation(s) apply.
- (c) Applicants with abnormal musculoskeletal system, including obesity, undertaking medical flight or flight simulator testing should satisfactorily perform all tasks required for the type of flight intended, including the emergency and evacuation procedures.

AMC2 MED.B.050 Musculoskeletal system

ED Decision 2019/002/R

- (a) Applicants with any significant sequelae from disease, injury or congenital abnormality affecting the bones, joints, muscles or tendons with or without surgery should require full evaluation prior to a fit assessment.
- (b) Applicants with inflammatory, infiltrative, traumatic or degenerative disease of the musculoskeletal system may be assessed as fit provided the condition is in remission or is stable and the applicant is taking no disqualifying medication and has satisfactorily completed a medical flight test. Appropriate limitation(s) may apply.
- (c) Applicants with abnormal musculoskeletal system, including obesity, undertaking a medical flight test should satisfactorily perform all tasks required for the type of flight intended, including the emergency and evacuation procedures.

MED.B.055 Mental Health

Regulation (EU) 2019/27

- (a) Comprehensive mental health assessment shall form part of the initial class 1 aero-medical examination.
- (b) Drugs and alcohol screening shall form part of the initial class 1 aero-medical examination.
- (c) Applicants with a mental or behavioural disorder due to use or misuse of alcohol or other psychoactive substances shall be assessed as unfit pending recovery and freedom from

- psychoactive substance use or misuse and subject to satisfactory psychiatric evaluation after successful treatment.
- (d) Applicants with a clinical diagnosis or documented medical history of any of the following psychiatric conditions shall undergo satisfactory psychiatric evaluation before they may be assessed as fit:
 - (1) mood disorder;
 - (2) neurotic disorder;
 - (3) personality disorder;
 - (4) mental or behavioural disorder;
 - (5) misuse of a psychoactive substance.
 - (e) Applicants with a documented medical history of a single or repeated acts of deliberate self-harm or suicide attempt shall be assessed as unfit. However, they may be assessed as fit after satisfactory psychiatric evaluation.
 - (f) Aero-medical assessment
 - (1) Applicants for a class 1 medical certificate with any of the conditions specified in point (c), (d) or (e) shall be referred to the medical assessor of the licensing authority.
 - (2) The fitness of applicants for a class 2 medical certificate with any of the conditions specified in point (c), (d) or (e) shall be assessed in consultation with the medical assessor of the licensing authority.
 - (g) Applicants with a documented medical history or clinical diagnosis of schizophrenia, schizotypal or delusional disorder shall be assessed as unfit.

AMC1 MED.B.055 Mental health

ED Decision 2025/002/R

- (a) Mental health assessment as part of the initial class 1 aero-medical examination
 - (1) A comprehensive mental health assessment should be conducted and recorded taking into account social, environmental and cultural contexts.
 - (2) The applicant's history and symptoms of disorders that might pose a threat to flight safety should be identified and recorded.
 - (3) The mental health assessment should include assessment and documentation of:
 - (i) general attitudes to mental health, including understanding possible indications of reduced mental health in themselves and others;
 - (ii) coping strategies under periods of psychological stress or pressure in the past, including seeking advice from others;
 - (iii) childhood behavioural problems;
 - (iv) interpersonal and relationship issues;
 - (v) current work and life stressors; and
 - (vi) overt personality disorders.

-
- (4) Where there are signs or is established evidence that an applicant may have a psychiatric or psychological disorder, the applicant should be referred for specialist opinion and advice.
 - (b) Mental health assessment as part of revalidation or renewal class 1 medical examination
 - (1) The assessment should include review and documentation of:
 - (i) current work and life stressors;
 - (ii) coping strategies under periods of psychological stress or pressure in the past, including seeking advice from others;
 - (iii) any difficulties with operational crew resource management (CRM);
 - (iv) any difficulties with employer and/or other colleagues and managers; and
 - (v) interpersonal and relationship issues, including difficulties with relatives, friends, and work colleagues.
 - (2) Where there are signs or is established evidence that an applicant may have a psychiatric or psychological disorder, the applicant should be referred for specialist opinion and advice.
 - (3) Established evidence should be verifiable information from an identifiable source related to the mental fitness or personality of a particular individual. Sources for this information can be accidents or incidents, problems in training or proficiency checks, behaviour or knowledge relevant to the safe exercise of the privileges of the applicable licence(s).
 - (c) Assessment of holders of a class 1 medical certificate referenced in MED.B.055(d)

Assessment of holders of a class 1 medical certificate referenced in MED.B.055(d) may require psychiatric and psychological evaluation as determined by the medical assessor of the licensing authority. A SIC limitation should be imposed in case of a fit assessment. Follow-up and removal of SIC limitation, as necessary, should be determined by the medical assessor of the licensing authority.
 - (d) Psychoactive substance testing
 - (1) Drug tests should screen for opioids, cannabinoids, amphetamines, cocaine, hallucinogens and sedative hypnotics. Following a risk assessment performed by the competent authority on the target population, screening tests may include additional drugs.
 - (2) For renewal/revalidation, random psychoactive substance screening test may be performed based on the risk assessment by the competent authority on the target population. If random psychoactive substance screening test is considered, it should be performed and reported in accordance with the procedures developed by the competent authority.
 - (3) In the case of a positive psychoactive substance screening result, confirmation should be required in accordance with national standards and procedures for psychoactive substance testing.
 - (4) In case of a positive confirmation test, a psychiatric evaluation should be undertaken before a fit assessment may be considered by the medical assessor of the licensing authority.
 - (e) Assessment and referral decisions

(1) Psychotic disorder

Applicants with a history, or the occurrence, of a functional psychotic disorder should be assessed as unfit. A fit assessment may be considered if a cause can be unequivocally identified as one which is transient, has ceased and the risk of recurrence is minimal.

(2) Organic mental disorder

Applicants with an organic mental disorder should be assessed as unfit. Once the cause has been treated, an applicant may be assessed as fit following satisfactory psychiatric evaluation.

(3) Psychoactive medication

Applicants who use psychoactive medication likely to affect flight safety should be assessed as unfit. If stability on maintenance psychoactive medication is confirmed, a fit assessment with an OML may be considered. If the dosage or type of medication is changed, a further period of unfit assessment should be required until stability is confirmed.

(4) Schizophrenia, schizotypal or delusional disorder

Applicants with an established history or clinical diagnosis of schizophrenia, schizotypal or delusional disorder may only be considered for a fit assessment if the medical assessor of the licensing authority concludes that the original diagnosis was inappropriate or inaccurate as confirmed by psychiatric evaluation, or, in the case of a single episode of delirium of which the cause was clear, provided that the applicant has suffered no permanent mental impairment.

(5) Mood disorder

Applicants with an established mood disorder should be assessed as unfit. After full recovery and after full consideration of the individual case, a fit assessment may be considered, depending on the characteristics and severity of the mood disorder.

(6) Neurotic, stress-related or somatoform disorder

Where there are signs or is established evidence that an applicant may have a neurotic, stress-related or somatoform disorder, the applicant should be referred for psychiatric or psychological opinion and advice.

(7) Personality or behavioural disorders

Where there are signs or is established evidence that an applicant may have a personality or behavioural disorder, the applicant should be referred for psychiatric or psychological opinion and advice.

(8) Disorders due to alcohol or other psychoactive substance(s) use or misuse

(i) Applicants with mental or behavioural disorders due to alcohol or other psychoactive substance(s) use or misuse, with or without dependency, should be assessed as unfit.

(ii) A fit assessment may be considered after a period of two years of documented sobriety or freedom from psychoactive substance use or misuse. At revalidation or renewal, a fit assessment may be considered earlier with an OML. Depending on the individual case, treatment and evaluation may include in-patient treatment of some weeks and inclusion into a support programme followed by ongoing checks,

including drug and alcohol testing and reports resulting from the support programme, which may be required indefinitely.

(9) Deliberate self-harm and suicide attempt

Applicants who have carried out a single self-destructive action or repeated acts of deliberate self-harm or suicide attempt should be assessed as unfit. A fit assessment may be considered after full consideration of an individual case and may require psychiatric or psychological evaluation. Neuropsychological evaluation may also be required.

(10) Cognitive disorders

(i) Applicants who exhibit signs of cognitive disorders should undergo a satisfactory neuropsychiatric evaluation to assess the severity of the cognitive impairment before a fit assessment may be considered. Applicants with mild cognitive impairment may be assessed as fit with an OML limitation and regular monitoring of the cognitive decline.

(ii) For applicants above the age of 60 performing single-pilot HEMS operations, AMEs should pay particular attention to early signs of cognitive decline. A comprehensive specialist evaluation should be considered where the medical assessors received information from the personnel performing regular training and checking of these applicants in accordance with ORO.FC.230 or AMEs performing the recurrent aero-medical examination documenting a potential cognitive decline of such pilots.

(11) Assessment

The assessment should take into consideration whether the indication for the treatment, side effects and addiction risks of such treatment and the characteristics of the psychiatric disorder are compatible with flight safety.

(f) Specialist opinion and advice

- (1) In case a specialist evaluation is needed, following the evaluation, the specialist should submit a written report to the AME, AeMC or medical assessor of the licensing authority as appropriate, detailing their opinion and recommendation.
- (2) Psychiatric evaluations should be conducted by a qualified psychiatrist having adequate knowledge and experience in aviation medicine.
- (3) The psychological opinion and advice should be based on a clinical psychological assessment conducted by a suitably qualified and accredited clinical psychologist with expertise and experience in aviation psychology.
- (4) The psychological evaluation may include a collection of biographical data, the administration of aptitude as well as personality tests and clinical interview.

AMC2 MED.B.055 Mental health

ED Decision 2019/002/R

(a) Mental health assessment as part of class 2 aero-medical examination

- (1) A mental health assessment should be conducted and recorded taking into account social, environmental and cultural contexts.
- (2) The applicant's history and symptoms of disorders that might pose a threat to flight safety should be identified and recorded.

- (3) Where there are signs or is established evidence that an applicant may have a psychiatric or psychological disorder, the applicant should be referred for specialist opinion and advice.
 - (4) Established evidence should be verifiable information from an identifiable source related to the mental fitness or personality of a particular individual. Sources for this information can be accidents or incidents, problems in training or proficiency checks, behaviour or knowledge relevant to the safe exercise of the privileges of the applicable licence(s).
- (b) Assessment of holders of a class 2 medical certificate referenced in MED.B.055(d)
- Assessment of holders of a class 2 medical certificate referenced in MED.B.055(d) may require psychiatric and psychological evaluation as determined by the AME, AeMC or medical assessor of the licensing authority. Follow-up, as necessary, should be determined in consultation with the medical assessor of the licensing authority.
- (c) Assessment and referral decisions
- (1) Psychotic disorder

Applicants with a history, or the occurrence, of a functional psychotic disorder should be assessed as unfit. A fit assessment may be considered if a cause can be unequivocally identified as one which is transient, has ceased and the risk of recurrence is minimal.
 - (2) Organic mental disorder

Applicants with an organic mental disorder should be assessed as unfit. Once the cause has been treated, an applicant may be assessed as fit following satisfactory psychiatric evaluation.
 - (3) Schizophrenia, schizotypal or delusional disorder

Applicants with an established history or clinical diagnosis of schizophrenia, schizotypal or delusional disorder may only be considered for a fit assessment in consultation with the medical assessor of the licensing authority if the original diagnosis was inappropriate or inaccurate as confirmed by psychiatric evaluation, or, in the case of a single episode of delirium of which the cause was clear, provided that the applicant has suffered no permanent mental impairment.
 - (4) Mood disorder

Applicants with an established mood disorder should be assessed as unfit. After full recovery and after full consideration of the individual case, a fit assessment may be considered, depending on the characteristics and severity of the mood disorder.
 - (5) Neurotic, stress-related or somatoform disorder

Where there are signs or is established evidence that an applicant may have a neurotic, stress-related or somatoform disorder, the applicant should be referred for psychiatric opinion and advice.
 - (6) Personality or behavioural disorders

Where there are signs or is established evidence that an applicant may have a personality or behavioural disorder, the applicant should be referred for psychiatric opinion and advice.
 - (7) Psychoactive medication

Applicants who use psychoactive medication likely to affect flight safety should be assessed as unfit. If stability on maintenance psychoactive medication is confirmed, a fit assessment with an OSL or OPL may be considered. If the dosage or type of medication is changed, a further period of unfit assessment should be required until stability is confirmed.

- (8) Disorders due to alcohol or other psychoactive substance(s) use or misuse
- (i) Applicants with mental or behavioural disorders due to alcohol or other psychoactive substance(s) use or misuse, with or without dependency, should be assessed as unfit.
 - (ii) Drug and alcohol tests
 - (A) In the case of a positive drug or alcohol result, confirmation should be required in accordance with national procedures for drugs and alcohol testing.
 - (B) In case of a positive confirmation test, a psychiatric evaluation should be undertaken before a fit assessment may be considered.
 - (iii) A fit assessment may be considered after a period of two years of documented sobriety or freedom from psychoactive substance use or misuse. At revalidation or renewal, a fit assessment may be considered earlier with an OSL or OPL. Depending on the individual case, treatment and evaluation may include in-patient treatment of some weeks and inclusion into a support programme followed by ongoing checks, including drug and alcohol testing and reports resulting from the support programme, which may be required indefinitely.
- (9) Deliberate self-harm
- Applicants who have carried out a single self-destructive action or repeated acts of deliberate self-harm or suicide attempt should be assessed as unfit. A fit assessment may be considered after full consideration of an individual case and may require psychiatric or psychological evaluation. Neuropsychological evaluation may also be required.
- (e) Specialist opinion and advice
- (1) In case a specialist evaluation is needed, following the evaluation, the specialist should submit a written report to the AME, AeMC or medical assessor of the licensing authority as appropriate, detailing their opinion and recommendation.
 - (2) Psychiatric evaluations should be conducted by a qualified psychiatrist having adequate knowledge and experience in aviation medicine.
 - (3) The psychological opinion and advice should be based on a clinical psychological assessment conducted by a suitably qualified and accredited clinical psychologist with expertise and experience in aviation psychology.
 - (4) The psychological evaluation may include a collection of biographical data, the administration of aptitude as well as personality tests and clinical interview.

GM1 MED.B.055 Mental health

ED Decision 2019/002/R

- (a) Symptoms of concern may include but are not limited to:
- (1) use of alcohol or other psychoactive substances;

- (2) loss of interest/energy;
 - (3) eating and weight changes;
 - (4) sleeping problems;
 - (5) low mood and, if present, any suicidal thoughts;
 - (6) family history of psychiatric disorders, particularly suicide;
 - (7) anger, agitation or high mood; and
 - (8) depersonalisation or loss of control.
- (b) The following aspects should be taken into consideration when conducting the mental health examination:
- (1) Appearance;
 - (2) Attitude;
 - (3) Behaviour;
 - (4) Mood;
 - (5) Speech;
 - (6) Thoughts process and content;
 - (7) Perception;
 - (8) Cognition;
 - (9) Insight; and
 - (10) Judgement.

GM2 MED.B.055 Mental health

ED Decision 2019/002/R

- (a) Drugs and alcohol screening tests used should:
- (1) provide information regarding medium-term consumption;
 - (2) be accepted on national level by the competent authority based on the availability and suitability for the scope mentioned in point(a)(1) above.
- (b) Statistical data of the screening campaign mentioned in [AMC1 MED.B.055\(d\)\(1\)](#) should be made available to the Agency on a yearly basis.

GM3 MED.B.055 Mental health

ED Decision 2019/002/R

- (a) The mental health assessment for class 2 applicants should include assessment and documentation of:
- (1) general attitudes to mental health, including understanding possible indications of reduced mental health in themselves and others;
 - (2) coping strategies under periods of psychological stress or pressure in the past, including seeking advice from others;
 - (3) childhood behavioural problems;

- (4) interpersonal and relationship issues, including difficulties with relatives, friends, and work colleagues;
 - (5) current work and life stressors, including difficulties with aviation operational environment; and
 - (6) overt personality disorders.
- (b) In regard to symptoms of concern and aspects to be taken into consideration when conducting mental health examination for class 2 applicants, guidance presented in [GM1 MED.B.055](#) should be used.

GM4 MED.B.055 Mental health

ED Decision 2019/002/R

Drugs and alcohol screening tests used should:

- (a) provide information regarding medium-term consumption;
- (b) be accepted on national level by the competent authority based on the availability and suitability with the scope mentioned in [GM2 MED.B.055\(a\)](#) above.

MED.B.065 Neurology

Regulation (EU) 2019/27

- (a) Applicants with clinical diagnosis or a documented medical history of any of the following medical conditions shall be assessed as unfit:
 - (1) epilepsy, except in the cases referred to in points (1) and (2) of point (b);
 - (2) recurring episodes of disturbance of consciousness of uncertain cause.
- (b) Applicants with clinical diagnosis or a documented medical history of any of the following medical conditions shall undergo further evaluation before they may be assessed as fit:
 - (1) epilepsy without recurrence after age 5;
 - (2) epilepsy without recurrence and off all treatment for more than 10 years;
 - (3) epileptiform EEG abnormalities and focal slow waves;
 - (4) progressive or non-progressive disease of the nervous system;
 - (5) inflammatory disease of the central or peripheral nervous system;
 - (6) migraine;
 - (7) a single episode of disturbance of consciousness of uncertain cause;
 - (8) loss of consciousness after head injury;
 - (9) penetrating brain injury;
 - (10) spinal or peripheral nerve injury;
 - (11) disorders of the nervous system due to vascular deficiencies including haemorrhagic and ischaemic events.

Applicants for a class 1 medical certificate shall be referred to the medical assessor of the licensing authority. The fitness of applicants for a class 2 medical certificate shall be assessed in consultation with the medical assessor of the licensing authority.

AMC1 MED.B.065 Neurology

ED Decision 2019/002/R

(a) Epilepsy

- (1) Applicants with a diagnosis of epilepsy should be assessed as unfit unless there is unequivocal evidence of a syndrome of benign childhood epilepsy associated with a very low risk of recurrence, and unless the applicant has been free of recurrence and off treatment for more than 10 years. One or more convulsive episode after the age of 5 should lead to unfitness. In the case of an acute symptomatic seizure, which is considered to have a very low risk of recurrence, a fit assessment may be considered after neurological evaluation.
- (2) Applicants may be assessed as fit with an OML if:
 - (i) there is a history of a single afebrile epileptiform seizure;
 - (ii) there has been no recurrence after at least 10 years off treatment;
 - (iii) there is no evidence of continuing predisposition to epilepsy.

(b) EEG

- (1) Electroencephalography is required when indicated by the applicant's history or on clinical grounds.
- (2) Applicants with epileptiform paroxysmal EEG abnormalities and focal slow waves should be assessed as unfit.

(c) Neurological disease

Applicants with any disease of the nervous system which is likely to cause a hazard to flight safety should be assessed as unfit. However, in certain cases, including cases of minor functional losses associated with stable disease, a fit assessment may be considered after full evaluation which should include a medical flight test which may be conducted in a flight simulation training device.

(d) Migraine

Applicants with an established diagnosis of migraine or other severe periodic headaches likely to cause a hazard to flight safety should be assessed as unfit. A fit assessment may be considered after full evaluation. The evaluation should take into account at least the following: auras, visual field loss, frequency, severity, therapy. Appropriate limitation(s) may apply.

(e) Episode of disturbance of consciousness

In the case of a single episode of disturbance of consciousness, which can be satisfactorily explained, a fit assessment may be considered, but applicants experiencing a recurrence should be assessed as unfit.

(f) Head injury

Applicants with a head injury which was severe enough to cause loss of consciousness or is associated with penetrating brain injury should be evaluated by a neurologist. A fit assessment may be considered if there has been a full recovery and the risk of epilepsy is sufficiently low.

(g) Spinal or peripheral nerve injury

Applicants with a history or diagnosis of spinal or peripheral nerve injury or a disorder of the nervous system due to a traumatic injury should be assessed as unfit. A fit assessment may be

considered if neurological evaluation is satisfactory and the conditions of [AMC1 MED.B.050](#) are satisfied.

(h) Vascular deficiencies

Applicants with a disorder of the nervous system due to vascular deficiencies including haemorrhagic and ischaemic events should be assessed as unfit. A fit assessment may be considered if neurological evaluation is satisfactory and the conditions of [AMC1 MED.B.050](#) are satisfied. A cardiological evaluation and medical flight test should be undertaken for applicants with residual deficiencies.

AMC2 MED.B.065 Neurology

ED Decision 2019/002/R

(a) Epilepsy

Applicants may be assessed as fit if:

- (1) there is a history of a single afebrile epileptiform seizure, considered to have a very low risk of recurrence;
- (2) there has been no recurrence after at least 10 years off treatment; and
- (3) there is no evidence of continuing predisposition to epilepsy.

(b) Neurological disease

Applicants with any disease of the nervous system which is likely to cause a hazard to flight safety should be assessed as unfit. However, in certain cases, including cases of functional loss associated with stable disease, a fit assessment may be considered after full evaluation which should include a medical flight test which may be conducted in a flight simulation training device.

(c) Migraine

Applicants with an established diagnosis of migraine or other severe periodic headaches likely to cause a hazard to flight safety should be assessed as unfit. A fit assessment may be considered after full evaluation. The evaluation should take into account at least the following: auras, visual field loss, frequency, severity, and therapy. Appropriate limitation(s) may apply.

(d) Head injury

Applicants with a head injury which was severe enough to cause loss of consciousness or is associated with penetrating brain injury may be assessed as fit if there has been a full recovery and the risk of epilepsy is sufficiently low. An evaluation by a neurologist may be required depending on the staging of the original injury.

(e) Spinal or peripheral nerve injury

Applicants with a history or diagnosis of spinal or peripheral nerve injury or a disorder of the nervous system due to a traumatic injury should be assessed as unfit. A fit assessment may be considered if neurological evaluation is satisfactory and the conditions of [AMC2 MED.B.050](#) are satisfied.

(f) Vascular deficiencies

Applicants with a disorder of the nervous system due to vascular deficiencies including haemorrhagic and ischaemic events should be assessed as unfit. A fit assessment may be considered if neurological evaluation is satisfactory and the provisions of [AMC2 MED.B.050](#) are

met. A cardiological evaluation and medical flight test should be undertaken for applicants with residual deficiencies.

MED.B.070 Visual System

Regulation (EU) 2024/2076

(a) Examination

(1) For a class 1 medical certificate:

- (i)** a comprehensive eye examination shall form part of the initial examination and shall be undertaken when clinically indicated and periodically, depending on the refraction and the functional performance of the eye.
- (ii)** a routine eye examination shall form part of all revalidation and renewal examinations.
- (iii)** when holders are involved in single-pilot HEMS operations, a comprehensive eye examination shall be completed at the first revalidation or renewal examination after the age of 60 and every year thereafter.

(2) For a class 2 medical certificate:

- (i)** a routine eye examination shall form part of the initial and all revalidation and renewal examinations.
- (ii)** a comprehensive eye examination shall be undertaken when clinically indicated.

(b) Visual acuity

(1) For a class 1 medical certificate:

- (i)** Distant visual acuity, with or without correction, shall be 6/9 (0,7) or better in each eye separately and visual acuity with both eyes shall be 6/6 (1,0) or better.
- (ii)** At the initial examination, applicants with substandard vision in one eye shall be assessed as unfit.
- (iii)** At revalidation and renewal examinations, notwithstanding point (b)(1)(i), applicants with acquired substandard vision in one eye or acquired monocular vision shall be referred to the medical assessor of the licensing authority and may be assessed as fit subject to a satisfactory ophthalmological evaluation.

(2) For a class 2 medical certificate:

- (i)** Distant visual acuity, with or without correction, shall be 6/12 (0,5) or better in each eye separately and visual acuity with both eyes shall be 6/9 (0,7) or better.
- (ii)** Notwithstanding point (b)(2)(i), applicants with substandard vision in one eye or monocular vision may be assessed as fit, in consultation with the medical assessor of the licensing authority and subject to a satisfactory ophthalmological evaluation.

(3) Applicants shall be able to read an N5 chart or equivalent at 30-50 cm and an N14 chart or equivalent at 100 cm, if necessary with correction.

(c) Refractive error and anisometropia

- (1)** Applicants with refractive errors or anisometropia may be assessed as fit subject to satisfactory ophthalmic evaluation.

- (2) Notwithstanding point (c)(1), applicants for a class 1 medical certificate with any of the following medical conditions shall be referred to the medical assessor of the licensing authority and may be assessed as fit subject to a satisfactory ophthalmological evaluation:
 - (i) myopia exceeding –6.0 dioptres;
 - (ii) astigmatism exceeding 2.0 dioptres;
 - (iii) anisometropia exceeding 2.0 dioptres.
- (3) Notwithstanding point (c)(1), applicants for a class 1 medical certificate with hypermetropia exceeding +5.0 dioptres shall be referred to the medical assessor of the licensing authority and may be assessed as fit subject to a satisfactory ophthalmological evaluation, provided that there are adequate fusional reserves, normal intraocular pressures and anterior angles and no significant pathology has been demonstrated. Notwithstanding point (b)(1)(i), corrected visual acuity in each eye shall be 6/6 or better.
- (4) Applicants with a clinical diagnosis of keratoconus may be assessed as fit subject to a satisfactory examination by an ophthalmologist. Such applicants for a class 1 medical certificate shall be referred to the medical assessor of the licensing authority.
- (d) **Binocular function**
 - (1) Applicants for a class 1 medical certificate shall be assessed as unfit, where they do not have normal binocular function and that medical condition is likely to jeopardise the safe exercise of the privileges of the license, taking account of any appropriate corrective measures where relevant.
 - (2) Applicants with diplopia shall be assessed as unfit.
- (e) **Visual fields**

Applicants for a class 1 medical certificate shall be assessed as unfit, where they do not have normal fields of vision and that medical condition is likely to jeopardise the safe exercise of the privileges of the license, taking account of any appropriate corrective measures where relevant.
- (f) **Eye surgery**

Applicants who have undergone eye surgery shall be assessed as unfit. However, they may be assessed as fit after full recovery of their visual function and subject to satisfactory ophthalmological evaluation.
- (g) **Spectacles and contact lenses**
 - (1) If satisfactory visual function is achieved only with the use of correction, the spectacles or contact lenses shall provide optimal visual function, be well-tolerated and suitable for aviation purposes.
 - (2) No more than one pair of spectacles shall be used to meet the visual requirements when exercising the privileges of the applicable licence(s).
 - (3) For distant vision, spectacles or contact lenses shall be worn when exercising the privileges of the applicable licence(s).
 - (4) For near vision, a pair of spectacles shall be kept available when exercising the privileges of the applicable licence(s).

- (5) A spare set of similarly correcting spectacles, for distant or near vision as applicable, shall be readily available for immediate use when exercising the privileges of the applicable licence(s).
- (6) If contact lenses are worn when exercising the privileges of the applicable licence(s), they shall be for distant vision, monofocal, and non-tinted and well-tolerated.
- (7) Applicants with a large refractive error shall use contact lenses or high-index spectacle lenses.
- (8) Orthokeratological lenses shall not be used.

AMC1 MED.B.070 Visual system

ED Decision 2025/002/R

(a) Eye examination

- (1) At each aero-medical examination, an assessment of the visual fitness should be undertaken and the eyes should be examined with regard to possible pathology.
- (2) All abnormal and doubtful cases should be referred to an ophthalmologist. Conditions which indicate ophthalmological examination include but are not limited to a substantial decrease in the uncorrected visual acuity, any decrease in best corrected visual acuity and/or the occurrence of eye disease, eye injury, or eye surgery.
- (3) Where specialist ophthalmological examinations are required for any significant reason, this should be imposed as a limitation on the medical certificate.
- (4) The possible cumulative effect of more than one eye condition should be evaluated by an ophthalmologist.
- (5) In their examination AMEs should give proper consideration to the degenerative effects of ageing on the visual system.

(b) Comprehensive eye examination

A comprehensive eye examination by an eye specialist is required at the initial examination. All abnormal and doubtful cases should be referred to an ophthalmologist. The examination should include:

- (1) history;
- (2) visual acuities near, intermediate and distant vision (uncorrected and with best optical correction if needed). Where a degeneration of the visual acuity is observed, additional specialised examinations could be considered, subject to the suspected pathology;
- (3) examination of the external eye, anatomy, media (slit lamp) and fundoscopy;
- (4) ocular motility;
- (5) binocular vision;
- (6) visual fields;
- (7) tonometry on clinical indication and for all cases where a comprehensive eye examination is required for applicants over the age of 45;
- (8) objective refraction: hyperopic initial applicants with a hyperopia of more than +2 dioptres and under the age of 25 should undergo objective refraction in cycloplegia;
- (9) assessment of mesopic contrast sensitivity; and

- (10) colour vision.
- (c) Routine eye examination

A routine eye examination may be performed by an AME and should include:

 - (1) history;
 - (2) visual acuities - near, intermediate and distant vision (uncorrected and with best optical correction if needed);
 - (3) examination of the external eye, anatomy, media and funduscopy;
 - (4) contrast sensitivity assessment test for applicants above the age of 60; and
 - (5) further examination on clinical indication.
- (d) Refractive error and anisometropia
 - (1) Applicants with the following conditions may be assessed as fit subject to satisfactory ophthalmic evaluation and provided that optimal correction has been considered and no significant pathology is demonstrated:
 - (i) hypermetropia not exceeding +5.0 dioptres;
 - (ii) myopia not exceeding -6.0 dioptres;
 - (iii) astigmatism not exceeding 2.0 dioptres;
 - (iv) anisometropia not exceeding 2.0 dioptres.
 - (2) Applicants should wear contact lenses if:
 - (i) hypermetropia exceeds +5.0 dioptres;
 - (ii) anisometropia exceeds 3.0 dioptres.
 - (3) An evaluation by an eye specialist should be undertaken 5-yearly if:
 - (i) the refractive error is between -3.0 and -6.0 dioptres or +3 and +5 dioptres;
 - (ii) astigmatism or anisometropia is between 2.0 and 3.0 dioptres.
 - (4) An evaluation by an eye specialist should be undertaken 2-yearly if:
 - (i) the refractive error is greater than -6.0 dioptres or +5.0 dioptres;
 - (ii) astigmatism or anisometropia exceeds 3.0 dioptres.
- (e) Uncorrected visual acuity

No limits apply to uncorrected visual acuity.
- (f) Visual acuity
 - (1) Reduced vision in one eye or monocularly: Applicants for revalidation or renewal with reduced central vision or acquired loss of vision in one eye may be assessed as fit with an OML if:
 - (i) the binocular visual field or, in the case of monocularly, the monocular visual field is acceptable;
 - (ii) in the case of monocularly, a period of adaptation time has passed from the known point of visual loss, during which the applicant should be assessed as unfit;

- (iii) the unaffected eye achieves distant visual acuity of 6/6 (1,0) corrected or uncorrected;
- (iv) the unaffected eye achieves intermediate visual acuity of N14 and N5 for near;
- (v) the underlying pathology is acceptable according to ophthalmological assessment and there is no significant ocular pathology in the unaffected eye; and
- (vi) a medical flight test is satisfactory.

(2) Visual fields

Applicants with a visual field defect, who do not have reduced central vision or acquired loss of vision in one eye, may be assessed as fit if the binocular visual field is normal.

(g) Keratoconus

Applicants with keratoconus may be assessed as fit if the visual requirements are met with the use of corrective lenses and periodic evaluation is undertaken by an ophthalmologist.

(h) Binocular function

Applicants with heterophoria (imbalance of the ocular muscles) exceeding:

(1) at 6 metres:

- 2.0 prism dioptres in hyperphoria,
 - 10.0 prism dioptres in esophoria,
 - 8.0 prism dioptres in exophoria
- and

(2) at 33 centimetres:

- 1.0 prism dioptre in hyperphoria,
- 8.0 prism dioptres in esophoria,
- 12.0 prism dioptres in exophoria

should be assessed as unfit. A fit assessment may be considered if an orthoptic evaluation demonstrates that the fusional reserves are sufficient to prevent asthenopia and diplopia.

(i) Eye surgery

The assessment after eye surgery should include an ophthalmological examination.

(1) After refractive surgery, a fit assessment may be considered, provided that:

- (i) stability of refraction of less than 0.75 dioptres variation diurnally has been achieved;
- (ii) examination of the eye shows no post-operative complications;
- (iii) glare sensitivity is within normal standards;
- (iv) mesopic contrast sensitivity is not impaired;
- (v) an evaluation is undertaken by an eye specialist.

(2) Following intraocular lens surgery, including cataract surgery, a fit assessment may be considered once recovery is complete and the visual requirements are met with or

without correction. Intraocular lenses should be monofocal and should not impair colour vision and night vision.

- (3) Retinal surgery entails unfitness. A fit assessment may be considered 6 months after surgery, or earlier if recovery is complete. A fit assessment may also be considered earlier after retinal laser therapy. Regular follow-up by an ophthalmologist should be carried out.
- (4) Glaucoma surgery entails unfitness. A fit assessment may be considered 6 months after surgery or earlier if recovery is complete. Regular follow-up by an ophthalmologist should be carried out.

(j) Visual correction

Correcting lenses should permit the licence holder to meet the visual requirements at all distances.

AMC2 MED.B.070 Visual system

ED Decision 2025/002/R

(a) Eye examination

- (1) At each aero-medical revalidation examination an assessment of the visual fitness of the applicant should be undertaken and the eyes should be examined with regard to possible pathology. Conditions which indicate further ophthalmological examination include but are not limited to a substantial decrease in the uncorrected visual acuity, any decrease in best corrected visual acuity and/or the occurrence of eye disease, eye injury, or eye surgery.
- (2) At the initial assessment, the examination should include:
 - (i) history;
 - (ii) visual acuities near, intermediate and distant vision (uncorrected and with best optical correction if needed);
 - (iii) examination of the external eye, anatomy, media and funduscopy;
 - (iv) ocular motility;
 - (v) binocular vision;
 - (vi) visual fields;
 - (vii) colour vision;
 - (viii) further examination on clinical indication.
- (3) At the initial assessment the applicant should submit a copy of the recent spectacle prescription if visual correction is required to meet the visual requirements.
- (4) In their examination AMEs should give proper consideration to the degenerative effects of ageing on the visual system

(b) Routine eye examination

A routine eye examination should include:

- (1) history;

- (2) visual acuities - near, intermediate and distant vision (uncorrected and with best optical correction if needed);
 - (3) examination of the external eye, anatomy, media and funduscopy;
 - (4) contrast sensitivity assessment test for applicants above the age of 60; and
 - (5) further examination on clinical indication.
- (c) Visual acuity
- Reduced vision in one eye or monocularly: Applicants with reduced vision or loss of vision in one eye may be assessed as fit if:
- (1) the binocular visual field or, in the case of monocularly, the monocular visual field is acceptable;
 - (2) in the case of monocularly, a period of adaptation time has passed from the known point of visual loss, during which the applicant should be assessed as unfit;
 - (3) the unaffected eye achieves distant visual acuity of 6/6 (1,0), corrected or uncorrected;
 - (4) the unaffected eye achieves intermediate visual acuity of N14 or equivalent and N5 or equivalent for near (Refer to [GM1 MED.B.070](#));
 - (5) there is no significant ocular pathology in the unaffected eye; and
 - (6) a medical flight test is satisfactory.
- (d) Binocular function
- Reduced stereopsis, abnormal convergence not interfering with near vision and ocular misalignment where the fusional reserves are sufficient to prevent asthenopia and diplopia may be acceptable.
- (e) Eye surgery
- (1) The assessment after eye surgery should include an ophthalmological examination.
 - (2) After refractive surgery a fit assessment may be considered provided that there is satisfactory stability of refraction, there are no post-operative complications and no increase in glare sensitivity.
 - (3) After cataract, retinal or glaucoma surgery a fit assessment may be considered once recovery is complete and the visual requirements are met with or without correction.
- (f) Visual correction
- Correcting lenses should permit the licence holder to meet the visual requirements at all distances.

GM1 MED.B.070 Visual system

ED Decision 2019/002/R

COMPARISON OF DIFFERENT READING CHARTS (APPROXIMATE FIGURES)

- (a) Test distance: 40 cm

Decimal	Nieden	Jäger	Snellen	N	Parinaud
1,0	1	2	1,5	3	2
0,8	2	3	2	4	3

0,7	3	4	2,5		
0,6	4	5	3	5	4
0,5	5	5		6	5
0,4	7	9	4	8	6
0,35	8	10	4,5		8
0,32	9	12	5,5	10	10
0,3	9	12		12	
0,25	9	12		14	
0,2	10	14	7,5	16	14
0,16	11	14	12	20	

(b) Test distance: 80 cm

Decimal	Nieden	Jäger	Snellen	N	Parinaud
1,2	4	5	3	5	4
1,0	5	5		6	5
0,8	7	9	4	8	6
0,7	8	10	4,5		8
0,63	9	12	5,5	10	10
0,6	9	12		12	10
0,5	9	12		14	10
0,4	10	14	7,5	16	14
0,32	11	14	12	20	14

GM2 MED.B.070 Visual system

ED Decision 2019/002/R

EYE SPECIALIST

The term 'eye specialist' refers to an ophthalmologist or a vision care specialist qualified in optometry and trained to recognise pathological conditions.

MED.B.075 Colour vision

Regulation (EU) 2024/2076

- (a) Applicants shall be assessed as unfit, where they cannot demonstrate their ability to readily perceive the colours that are necessary for the safe exercise of the privileges of the licence.
- (b) Examination and assessment
 - (1) Applicants shall be subjected to the Ishihara test for the initial issue of a medical certificate. For class 1 medical certificate holders involved in single-pilot HEMS operations, a colour vision assessment shall be completed at the first revalidation or renewal examination after the age of 60 and every year thereafter. Applicants who pass that test may be assessed as fit.
 - (2) For a class 1 medical certificate:

- (i) Applicants who do not pass the Ishihara test shall be referred to the medical assessor of the licensing authority and shall undergo further colour perception testing to establish whether they are colour safe.
- (ii) Applicants shall be normal trichromats or shall be colour safe.
- (iii) Applicants who fail further colour perception testing shall be assessed as unfit.
- (3) For a class 2 medical certificate:
 - (i) Applicants who do not pass the Ishihara test shall undergo further colour perception testing to establish whether they are colour safe.
 - (ii) Applicants who do not have satisfactory perception of colours shall be limited to exercising the privileges of the applicable licence in daytime only.

AMC1 MED.B.075 Colour vision

ED Decision 2019/002/R

- (a) At revalidation and renewal examinations, colour vision should be tested on clinical indication.
- (b) The Ishihara test (24 plate version) is considered passed if the first 15 plates, presented in a random order, are identified without error.
- (c) Those failing the Ishihara test should be examined either by:
 - (1) anomaloscopy (Nagel or equivalent). This test is considered passed if the colour match is trichromatic and the matching range is 4 scale units or less, or if the anomalous quotient is acceptable; or by
 - (2) lantern testing with a Spectrolux, Beynes or Holmes-Wright lantern. This test is considered passed if the applicant passes without error a test with accepted lanterns.
 - (3) Colour Assessment and Diagnosis (CAD) test. This test is considered passed if the threshold is less than 6 standard normal (SN) units for deutan deficiency, or less than 12 SN units for protan deficiency. A threshold greater than 2 SN units for tritan deficiency indicates an acquired cause which should be investigated.

AMC2 MED.B.075 Colour vision

ED Decision 2019/002/R

- (a) Colour vision should be tested on clinical indication at revalidation and renewal examinations.
- (b) The Ishihara test (24 plate version) is considered passed if the first 15 plates, presented in a random order, are identified without error.
- (c) Those failing the Ishihara test should be examined either by:
 - (1) anomaloscopy (Nagel or equivalent). This test is considered passed if the colour match is trichromatic and the matching range is 4 scale units or less, or if the anomalous quotient is acceptable; or by
 - (2) lantern testing with a Spectrolux, Beynes or Holmes-Wright lantern. This test is considered passed if the applicant passes without error a test with accepted lanterns.
 - (3) Colour Assessment and Diagnosis (CAD) test. This test is considered passed if the threshold is less than 6 standard normal (SN) units for deutan deficiency, or less than 12

SN units for protan deficiency. A threshold greater than 2 SN units for tritan deficiency indicates an acquired cause which should be investigated.

MED.B.080 Otorhinolaryngology (ENT)

Regulation (EU) 2024/2076

(a) Examination

- (1) Applicants' hearing shall be tested at all examinations.
 - (i) For a class 1 medical certificate, and for a class 2 medical certificate when an instrument rating or a basic instrument rating is to be added to the licence, hearing shall be tested with pure-tone audiometry at the initial examination, then every 5 years until the licence holder reaches the age of 40, and then every 2 years until the licence holder reaches the age of 60 and every year thereafter.
 - (ii) When tested on a pure-tone audiometer, initial applicants shall not have a hearing loss of more than 35 dB at any of the frequencies 500, 1 000 or 2 000 Hz, or more than 50 dB at 3 000 Hz, in either ear separately. Applicants for revalidation or renewal with greater hearing loss shall demonstrate satisfactory functional hearing ability.
- (2) A comprehensive ear, nose and throat examination shall be undertaken for the initial issue of a class 1 medical certificate and periodically thereafter when clinically indicated.
- (3) For class 1 medical certificate holders involved in single-pilot HEMS operations, a comprehensive ear, nose and throat examination shall be completed at the first revalidation or renewal examination after the age of 60.

(b) Applicants with any of the following medical conditions shall undergo further examination to establish that the medical condition does not interfere with the safe exercise of the privileges of the applicable licence(s):

- (1) hypoacusis;
- (2) an active pathological process of the internal or middle ear;
- (3) unhealed perforation or dysfunction of the tympanic membrane(s);
- (4) dysfunction of the Eustachian tube(s);
- (5) disturbance of vestibular function;
- (6) significant restriction of the nasal passages;
- (7) sinus dysfunction;
- (8) significant malformation or significant infection of the oral cavity or upper respiratory tract;
- (9) significant disorder of speech or voice;
- (10) any sequelae of surgery of the internal or middle ear.

(c) Aero-medical assessment

- (1) Applicants for a class 1 medical certificate with any of the medical conditions specified in points (1), (4) and (5) of point (b) shall be referred to the medical assessor of the licensing authority.

- (2) The fitness of applicants for a class 2 medical certificate with any of the medical conditions specified in point (4) and (5) of point (b) shall be assessed in consultation with the medical assessor of the licensing authority.
- (3) The fitness of applicants for a class 2 medical certificate for an instrument rating or en route instrument rating to be added to the licence with the medical condition specified in point (1) of point (b) shall be assessed in consultation with the medical assessor of the licensing authority.

AMC1 MED.B.080 Otorhinolaryngology (ENT)

ED Decision 2019/002/R

(a) Hearing

- (1) Applicants should understand correctly conversational speech when tested with each ear at a distance of 2 metres from and with the applicant's back turned towards the AME.
- (2) Applicants with hypoacusis may be assessed as fit if a speech discrimination test or functional flight deck hearing test demonstrates satisfactory hearing ability. A vestibular function test may be appropriate.
- (3) If the hearing requirements can only be met with the use of hearing aids, the hearing aids should provide optimal hearing function, be well tolerated and suitable for aviation purposes.

(b) Comprehensive ENT examination

A comprehensive ENT examination should include:

- (1) history;
- (2) clinical examination including otoscopy, rhinoscopy, and examination of the mouth and throat;
- (3) tympanometry or equivalent;
- (4) clinical examination of the vestibular system.

(c) Ear conditions

- (1) Applicants with an active pathological process of the internal or middle ear should be assessed as unfit. A fit assessment may be considered once the condition has stabilised or there has been a full recovery.
- (2) Applicants with an unhealed perforation or dysfunction of the tympanic membranes should be assessed as unfit. An applicant with a single dry perforation of non-infectious origin and which does not interfere with the normal function of the ear may be considered for a fit assessment.

(d) Vestibular disturbance

Applicants with disturbance of vestibular function should be assessed as unfit. A fit assessment may be considered after full recovery. The presence of spontaneous or positional nystagmus requires complete vestibular evaluation by specialist. Applicants with significant abnormal caloric or rotational vestibular responses should be assessed as unfit. Abnormal vestibular responses should be assessed in their clinical context.

(e) Sinus dysfunction

Applicants with any dysfunction of the sinuses should be assessed as unfit until there has been full recovery.

(f) Oral/upper respiratory tract infections

Applicants with a significant infection of the oral cavity or upper respiratory tract should be assessed as unfit. A fit assessment may be considered after full recovery.

(g) Speech disorder

Applicants with a significant disorder of speech or voice should be assessed as unfit.

(h) Air passage restrictions

Applicants with significant restriction of the nasal air passage on either side, or significant malformation of the oral cavity or upper respiratory tract may be assessed as fit if ENT evaluation is satisfactory.

(i) Eustachian tube(s)

Applicants with permanent dysfunction of the Eustachian tube(s) may be assessed as fit if ENT evaluation is satisfactory.

(j) Sequelae of surgery of the internal or middle ear

Applicants with sequelae of surgery of the internal or middle ear should be assessed as unfit until recovery is complete, the applicant is asymptomatic, and the risk of secondary complication is minimal.

AMC2 MED.B.080 Otorhinolaryngology (ENT)

ED Decision 2019/002/R

(a) Hearing

- (1) Applicants should understand correctly conversational speech when tested with each ear at a distance of 2 metres from and with the applicant's back turned towards the AME.
- (2) Applicants with hypoacusis may be assessed as fit if a speech discrimination test or functional cockpit hearing test demonstrates satisfactory hearing ability.
- (3) If the hearing requirements can be met only with the use of hearing aids, the hearing aids should provide optimal hearing function, be well tolerated and suitable for aviation purposes.
- (4) Applicants with profound deafness or major disorder of speech, or both, may be assessed as fit with an SSL, such as 'limited to areas and operations where the use of radio is not mandatory'. The aircraft should be equipped with appropriate alternative warning devices in lieu of sound warnings.

(b) Examination

An ENT examination should form part of all initial, revalidation and renewal examinations.

(c) Ear conditions

- (1) Applicants with an active pathological process of the internal or middle ear should be assessed as unfit until the condition has stabilised or there has been a full recovery.
- (2) Applicants with an unhealed perforation or dysfunction of the tympanic membranes should be assessed as unfit. An applicant with a single dry perforation of non-infectious

origin which does not interfere with the normal function of the ear may be considered for a fit assessment.

(d) Vestibular disturbance

Applicants with disturbance of vestibular function should be assessed as unfit pending full recovery.

(e) Sinus dysfunction

Applicants with any dysfunction of the sinuses should be assessed as unfit pending full recovery.

(f) Oral/upper respiratory tract infections

Applicants with a significant infection of the oral cavity or upper respiratory tract should be assessed as unfit. A fit assessment may be considered after full recovery.

(g) Speech disorder

Applicants with a significant disorder of speech or voice should be assessed as unfit.

(h) Air passage restrictions

Applicants with significant restriction of the nasal air passage on either side, or significant malformation of the oral cavity or upper respiratory tract may be assessed as fit if ENT evaluation is satisfactory.

(i) Eustachian tube dysfunction

Applicants with permanent dysfunction of the Eustachian tube(s) may be assessed as fit if ENT evaluation is satisfactory.

(j) Sequelae of surgery of the internal or middle ear

Applicants with sequelae of surgery of the internal or middle ear should be assessed as unfit until recovery is complete, the applicant is asymptomatic, and the risk of secondary complication is minimal.

GM1 MED.B.080 Otorhinolaryngology (ENT)

ED Decision 2019/002/R

PURE TONE AUDIOGRAM

The pure tone audiogram may also cover the 4 000 Hz frequency for early detection of decrease in hearing.

GM2 MED.B.080 Otorhinolaryngology (ENT)

ED Decision 2019/002/R

PURE TONE AUDIOGRAM

The pure tone audiogram may also cover the 4 000 Hz frequency for early detection of decrease in hearing.

MED.B.085 Dermatology

Regulation (EU) 2019/27

Applicants shall be assessed as unfit, where they have an established dermatological condition which is likely to jeopardise the safe exercise of the privileges of the licence.

AMC1 MED.B.085 Dermatology

ED Decision 2019/002/R

- (a) If doubt exists about the fitness of applicants with eczema (exogenous and endogenous), severe psoriasis, bacterial infections, drug induced or bullous eruptions or urticaria, the AME should refer the case to the medical assessor of the licensing authority.
- (b) Systemic effects of radiant or pharmacological treatment for a dermatological condition should be reviewed before a fit assessment may be considered.
- (c) In cases where a dermatological condition is associated with a systemic illness, full consideration should be given to the underlying illness before a fit assessment may be considered.

AMC2 MED.B.085 Dermatology

ED Decision 2019/002/R

In cases where a dermatological condition is associated with a systemic illness, full consideration should be given to the underlying illness before a fit assessment may be considered.

MED.B.090 Oncology

Regulation (EU) 2019/27

- (a) Before further consideration is given to their application, applicants with primary or secondary malignant disease shall undergo satisfactory oncological evaluation. Such applicants for a class 1 medical certificate shall be referred to the medical assessor of the licensing authority. Such applicants for a class 2 medical certificate shall be assessed in consultation with the medical assessor of the licensing authority.
- (b) Applicants with a documented medical history or clinical diagnosis of an intracerebral malignant tumour shall be assessed as unfit.

AMC1 MED.B.090 Oncology

ED Decision 2019/002/R

- (a) Applicants who have been diagnosed with a malignant disease may be assessed as fit provided that:
 - (1) after primary treatment, there is no evidence of residual malignant disease likely to jeopardise flight safety;
 - (2) time appropriate to the type of tumour and primary treatment has elapsed;
 - (3) the risk of inflight incapacitation from a recurrence or metastasis is sufficiently low;
 - (4) there is no evidence of short or long-term sequelae from treatment. Special attention should be paid to applicants who have received anthracycline chemotherapy;
 - (5) satisfactory oncology follow-up reports are provided to the medical assessor of the licensing authority.
- (b) An OML should be applied as appropriate.
- (c) Applicants receiving ongoing chemotherapy or radiation treatment should be assessed as unfit.
- (d) Applicants with pre-malignant conditions of the skin may be assessed as fit if treated or excised as necessary and there is regular follow-up.

AMC2 MED.B.090 Oncology

ED Decision 2019/002/R

- (a) Applicants who have been diagnosed with a malignant disease may be considered for a fit assessment provided that:
 - (1) after primary treatment, there is no evidence of residual malignant disease likely to jeopardise flight safety;
 - (2) time appropriate to the type of tumour and primary treatment has elapsed;
 - (3) the risk of in-flight incapacitation from a recurrence or metastasis is sufficiently low;
 - (4) there is no evidence of short or long-term sequelae from treatment that may jeopardise flight safety;
 - (5) arrangements for an oncological follow-up have been made for an appropriate period of time.
- (b) Applicants receiving ongoing chemotherapy or radiation treatment should be assessed as unfit.
- (c) Applicants with pre-malignant conditions of the skin may be assessed as fit if treated or excised as necessary and there is a regular follow-up.

SECTION 3 – SPECIFIC REQUIREMENTS FOR LAPL MEDICAL CERTIFICATES

MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

Regulation (EU) 2019/27

- (a) An applicant for a LAPL medical certificate shall be assessed based on aero-medical best practice.
- (b) Special attention shall be given to the applicant's complete medical history.
- (c) The initial assessment, all subsequent re-assessments after the licence holder reaches the age of 50 and any assessments in cases where the medical history of the applicant is not available to the examiner shall include at least all of the following:
 - (1) clinical examination;
 - (2) blood pressure;
 - (3) urine test;
 - (4) vision;
 - (5) hearing ability.
- (d) After the initial assessment, subsequent re-assessments until the licence holder reaches the age of 50 shall include at least both of the following:
 - (1) an assessment of the LAPL holder's medical history;
 - (2) the items specified in point(c) as deemed necessary by the AeMC, AME or GMP in accordance with aero-medical best practice.

AMC1 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

When a specialist evaluation is required under this section, the aero-medical assessment of the applicant should be performed by an AeMC, an AME or, in the case of [AMC5 MED.B.095\(d\)](#), by the medical assessor of the licensing authority.

AMC2 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

CARDIOVASCULAR SYSTEM

- (a) Examination
 - Pulse and blood pressure should be recorded at each examination.
- (b) General
 - (1) Cardiovascular risk factor assessment

An accumulation of risk factors (smoking, family history, lipid abnormalities, hypertension, etc.) requires cardiovascular evaluation.

(2) Aortic aneurysm

Applicants with an aortic aneurysm may be assessed as fit subject to satisfactory cardiological evaluation and a regular follow-up.

(3) Cardiac valvular abnormalities

(i) Applicants with a cardiac murmur may be assessed as fit if the murmur is assessed as being of no pathological significance.

(ii) Applicants with a cardiac valvular abnormality may be assessed as fit subject to satisfactory cardiological evaluation.

(4) Valvular surgery

After cardiac valve replacement or repair, a fit assessment may be considered, with an ORL if anticoagulation is needed, subject to satisfactory post-operative cardiological evaluation. Anticoagulation should be stable and the haemorrhagic risk should be acceptable. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range. The INR target range should be determined by the type of surgery performed. Applicants who measure their INR on a 'near patient' testing system within 12 hours prior to flight and only exercise the privileges of their licence if the INR is within the target range, may be assessed as fit without the above-mentioned limitation. The INR results should be recorded and the results should be reviewed at each aero-medical assessment. Applicants taking anticoagulation medication not requiring INR monitoring, may be assessed as fit without the above-mentioned limitation in consultation with the medical assessor of the licensing authority after a stabilisation period of 3 months.

(5) Other cardiac disorders

(i) Applicants with other cardiac disorders may be assessed as fit subject to satisfactory cardiological evaluation. A fit assessment may be considered, with an ORL if anticoagulation is needed. Anticoagulation should be stable and the haemorrhagic risk should be acceptable. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range. The INR target range should be determined by the type of surgery performed. Applicants who measure their INR on a 'near patient' testing system within 12 hours prior to flight and only exercise the privileges of their licence if the INR is within the target range, may be assessed as fit without the above-mentioned limitation. The INR results should be recorded and the results should be reviewed at each aero-medical assessment. Applicants taking anticoagulation medication not requiring INR monitoring, may be assessed as fit without the above-mentioned limitation in consultation with the medical assessor of the licensing authority after a stabilisation period of 3 months.

(ii) Applicants with symptomatic hypertrophic cardiomyopathy should be assessed as unfit.

(c) Blood pressure

(1) When the blood pressure consistently exceeds 160 mmHg systolic and/or 95 mmHg diastolic, with or without treatment, the applicant should be assessed as unfit.

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- (2) Applicants initiating medication for the control of blood pressure should be assessed as unfit until the absence of significant side effects has been established.
- (d) Coronary artery disease
- (1) Applicants with suspected myocardial ischaemia should undergo a cardiological evaluation before a fit assessment may be considered.
- (2) Applicants with angina pectoris requiring medication for cardiac symptoms should be assessed as unfit.
- (3) After an ischaemic cardiac event, including myocardial infarction or revascularisation, applicants without symptoms should have reduced cardiovascular risk factors to an appropriate level. Medication, when used to control cardiac symptoms, is not acceptable. All applicants should be on appropriate secondary prevention treatment.
- (4) In cases (d)(1), (d)(2) and (d)(3), applicants who have had a satisfactory cardiological evaluation to include an exercise test or equivalent that is negative for ischaemia may be assessed as fit.
- (e) Rhythm and conduction disturbances
- (1) Applicants with a significant disturbance of cardiac rhythm or conduction should be assessed as unfit unless a cardiological evaluation concludes that the disturbance is not likely to interfere with the safe exercise of the privileges of the licence. A fit assessment may be considered, with an ORL if anticoagulation is needed. Anticoagulation should be stable and the haemorrhagic risk should be acceptable. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range. The INR target range should be determined by the type of surgery performed. Applicants who measure their INR on a 'near patient' testing system within 12 hours prior to flight and only exercise the privileges of their licence if the INR is within the target range, may be assessed as fit without the above-mentioned limitation. The INR results should be recorded and the results should be reviewed at each aero-medical assessment. Applicants taking anticoagulation medication not requiring INR monitoring, may be assessed as fit without the above-mentioned limitation in consultation with the medical assessor of the licensing authority after a stabilisation period of 3 months.
- (2) Pre-excitation
- Applicants with ventricular pre-excitation may be assessed as fit subject to satisfactory cardiological evaluation. Applicants with ventricular pre-excitation associated with a significant arrhythmia should be assessed as unfit.
- (3) Automatic implantable defibrillating system
- Applicants with an automatic implantable defibrillating system should be assessed as unfit.
- (4) Pacemaker
- A fit assessment may be considered subject to satisfactory cardiological evaluation.

AMC3 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

RESPIRATORY SYSTEM

- (a) Applicants should undergo pulmonary morphological or functional tests when clinically indicated.
- (b) Asthma and chronic obstructive pulmonary disease
Applicants with asthma or impairment of pulmonary function may be assessed as fit provided that the condition is considered stable with satisfactory pulmonary function and medication is compatible with flight safety. Systemic steroids may be acceptable provided that the dosage required is acceptable and there are no adverse side effects.
- (c) Sarcoidosis
 - (1) Applicants with active sarcoidosis should be assessed as unfit. Investigation should be undertaken with respect to the possibility of systemic involvement. A fit assessment may be considered once the disease is inactive.
 - (2) Applicants with cardiac sarcoidosis should be assessed as unfit.
- (d) Pneumothorax
 - (1) Applicants with spontaneous pneumothorax may be assessed as fit subject to satisfactory respiratory evaluation following recovery from a single spontaneous pneumothorax or following recovery from surgical intervention for a recurrent pneumothorax.
 - (2) Applicants with traumatic pneumothorax may be assessed as fit following recovery.
- (e) Thoracic surgery
Applicants who have undergone thoracic surgery may be assessed as fit following recovery.
- (f) Sleep apnoea syndrome/sleep disorder
Applicants with unsatisfactorily treated sleep apnoea syndrome should be assessed as unfit.

AMC4 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

DIGESTIVE SYSTEM

- (a) Gallstones
Applicants with symptomatic gallstones should be assessed as unfit. A fit assessment may be considered following gallstone removal.
- (b) Inflammatory bowel disease
Applicants with an established diagnosis or history of chronic inflammatory bowel disease may be assessed as fit provided that the disease is stable and not likely to interfere with the safe exercise of the privileges of the licence.
- (c) Peptic ulceration

Applicants with peptic ulceration may be assessed as fit subject to satisfactory gastroenterological evaluation.

(d) Digestive tract and abdominal surgery

Applicants who have undergone a surgical operation:

- (1) for herniae; or
- (2) on the digestive tract or its adnexa, including a total or partial excision or diversion of any of these organs,

should be assessed as unfit. A fit assessment may be considered if recovery is complete, the applicant is asymptomatic, and there is only a minimal risk of secondary complication or recurrence.

(e) Pancreatitis

Applicants with pancreatitis may be assessed as fit after satisfactory recovery.

(f) Liver disease

Applicants with morphological or functional liver disease or after surgery, including liver transplantation, may be assessed as fit subject to satisfactory gastroenterological evaluation.

AMC5 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

METABOLIC AND ENDOCRINE SYSTEMS

(a) Metabolic, nutritional or endocrine dysfunction

Applicants with metabolic, nutritional or endocrine dysfunction may be assessed as fit subject to demonstrated stability of the condition and satisfactory aero-medical evaluation.

(b) Obesity

Obese applicants may be assessed as fit if the excess weight is not likely to interfere with the safe exercise of the licence.

(c) Thyroid dysfunction

Applicants with thyroid disease may be assessed as fit once a stable euthyroid state is attained.

(d) Diabetes mellitus

- (1) Applicants using antidiabetic medications that are not likely to cause hypoglycaemia may be assessed as fit.
- (2) Applicants with diabetes mellitus Type 1 should be assessed as unfit.
- (3) Applicants with diabetes mellitus Type 2 treated with insulin may be assessed as fit with limitations for revalidation if blood sugar control has been achieved and the process under (e) and (f) is followed. An ORL is required. A TML for 12 months may be needed to ensure compliance with the follow-up requirements below. Licence privileges should not include rotary aircraft flying.

(e) Aero-medical assessment by, or under the guidance of, the medical assessor of the licensing authority:

- (1) A diabetology review at yearly intervals, including:
 - (i) symptom review;
 - (ii) review of data logging of blood sugar;
 - (iii) cardiovascular status. Exercise ECG at age 40, at 5-yearly intervals thereafter and on clinical indication, including an accumulation of risk factors;
 - (iv) nephropathy status.
- (2) Ophthalmological review at yearly intervals, including:
 - (i) visual fields — Humphrey-perimeter;
 - (ii) retinae — full dilatation slit lamp examination;
 - (iii) cataract — clinical screening.

The development of retinopathy requires a full ophthalmological review.

- (3) Blood testing at 6-monthly intervals:
 - (i) HbA1c;
 - (ii) renal profile;
 - (iii) liver profile;
 - (iv) lipid profile.
- (4) Applicants should be assessed as temporarily unfit after:
 - (i) changes of medication/insulin leading to a change to the testing regime until stable blood sugar control can be demonstrated;
 - (ii) a single unexplained episode of severe hypoglycaemia until stable blood sugar control can be demonstrated.
- (5) Applicants should be assessed as unfit in the following cases:
 - (i) loss of hypoglycaemic awareness;
 - (ii) development of retinopathy with any visual field loss;
 - (iii) significant nephropathy;
 - (iv) any other complication of the disease where flight safety may be jeopardised.

(f) Pilot responsibility

Blood sugar testing is carried out during non-operational and operational periods. A whole blood glucose measuring device with memory should be carried and used. Equipment for continuous glucose monitoring (CGMS) should not be used. Pilots should prove to the AME or AeMC or medical assessor of the licensing authority that testing has been performed as indicated below and with which results.

- (1) Testing during non-operational periods: normally 3–4 times/day or as recommended by the treating physician, and on any awareness of hypoglycaemia.
- (2) Testing frequency during operational periods:
 - (i) 120 minutes before departure;
 - (ii) <30 minutes before departure;

- (iii) 60 minutes during flight;
- (iv) 30 minutes before landing.
- (3) Actions following glucose testing:
 - (i) 120 minutes before departure: if the test result is >15 mmol/l, piloting should not be commenced.
 - (ii) 10–15g of carbohydrate should be ingested and a re-test performed within 30 minutes if:
 - (A) any test result is <4,5 mmol/l;
 - (B) the pre-landing test measurement is missed or a subsequent go-around/diversion is performed.

AMC6 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

HAEMATOLOGY

Applicants with a haematological condition, such as:

- (a) abnormal haemoglobin including, but not limited to, anaemia, erythrocytosis or haemoglobinopathy;
- (b) coagulation, haemorrhagic or thrombotic disorder;
- (c) significant lymphatic enlargement;
- (d) acute or chronic leukaemia;
- (e) splenomegaly;

may be assessed as fit subject to satisfactory aero-medical evaluation. If anticoagulation is being used as treatment, refer to [AMC2 MED.B.095\(b\)\(4\)](#).

AMC7 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

GENITOURINARY SYSTEM

- (a) Applicants with a genitourinary disorder, such as:
 - (1) renal disease; or
 - (2) one or more urinary calculi, or a history of renal colicmay be assessed as fit subject to satisfactory renal and urological evaluation, as applicable.
- (b) Applicants who have undergone a major surgical operation on the genitourinary system or its adnexa may be assessed as fit following recovery.
- (c) Applicants who have undergone renal transplantation may be assessed as fit subject to satisfactory renal evaluation.

AMC8 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

INFECTIOUS DISEASE

- (a) Applicants who are HIV positive may be assessed as fit subject to satisfactory aero-medical evaluation.
- (b) Applicants with other chronic infections may be assessed as fit provided the infections are not likely to interfere with the safe exercise of the privileges of the licence.

AMC9 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

OBSTETRICS AND GYNAECOLOGY

- (a) Pregnancy
Holders of a LAPL medical certificate should only exercise the privileges of their licences until the end of the 26th week of gestation under routine antenatal care.
- (b) Applicants who have undergone a major gynaecological operation may be assessed as fit after recovery.

AMC10 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

MUSCULOSKELETAL SYSTEM

Applicants should have satisfactory functional use of the musculoskeletal system to enable the safe exercise of the privileges of the licence.

AMC11 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

MENTAL HEALTH

- (a) Applicants with a mental or behavioural disorder due to use or misuse of alcohol or other psychoactive substances, with or without dependency, should be assessed as unfit. A fit assessment may be considered after a period of two years of documented sobriety or freedom from psychoactive substance use or misuse, subject to satisfactory psychiatric evaluation after successful treatment. At revalidation or renewal, a fit assessment may be considered earlier. Depending on the individual case, treatment and evaluation may include in-patient treatment of some weeks followed by ongoing checks, including blood testing and peer reports, which may be required indefinitely.
- (b) Applicants with a history of, or the occurrence of, a functional psychotic disorder should be assessed as unfit. A fit assessment may be considered if a cause can be unequivocally identified as one which is transient, has ceased, and the risk of recurrence is minimal.

- (c) Applicants with an established history or clinical diagnosis of schizophrenia, schizotypal or delusional disorder should be assessed as unfit. A fit assessment may only be considered if the original diagnosis was inappropriate or inaccurate as confirmed by psychiatric evaluation or, in the case of a single episode of delirium, provided that the applicant has suffered no permanent impairment.
- (d) Psychoactive substances
- Applicants who use or misuse psychoactive substances or psychoactive medication likely to affect flight safety should be assessed as unfit. If stability on maintenance psychoactive medication is confirmed, a fit assessment with appropriate limitation(s) may be considered. If the dosage or type of medication is changed, a further period of unfit assessment should be required until stability is confirmed.
- (e) Applicants with a psychiatric condition, such as:
- (1) mood disorder;
 - (2) neurotic disorder;
 - (3) personality disorder;
 - (4) mental or behavioural disorder
- should undergo satisfactory psychiatric evaluation before a fit assessment may be considered.
- (f) Applicants with a history of significant or repeated acts of deliberate self-harm should undergo satisfactory psychiatric or psychological evaluation or both before a fit assessment may be considered.
- (g) Psychiatric evaluations and reviews may include reports from the applicant's flight instructor.
- (h) Applicants with a psychological disorder may need to be referred for psychological opinion and advice.
- (i) In case a specialist evaluation is needed, following the evaluation, the specialist should submit a written report to the AME, AeMC, GMP or medical assessor of the licensing authority as appropriate, detailing their opinion and recommendation.

AMC12 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

NEUROLOGY

- (a) Epilepsy and seizures
- (1) Applicants with an established diagnosis of and under treatment for epilepsy should be assessed as unfit. A re-assessment after all treatment has been stopped for at least 5 years should include a review of neurological reports.
 - (2) Applicants may be assessed as fit if:
 - (i) there is a history of a single afebrile epileptiform seizure considered to have a very low risk of recurrence;
 - (ii) there has been no recurrence after at least 5 years off treatment;
 - (iii) a cause has been identified and treated and there is no evidence of continuing predisposition to epilepsy.

(b) Neurological disease

Applicants with any disease of the nervous system which is likely to cause a hazard to flight safety should be assessed as unfit. However, in certain cases, including cases of functional loss associated with stable disease, a fit assessment may be considered after full evaluation including, if necessary, a medical flight test.

(c) Migraine

Applicants with an established diagnosis of migraine or other severe periodic headaches likely to cause a hazard to flight safety should be assessed as unfit. A fit assessment may be considered after full evaluation. The evaluation should take into account at least the following: auras, visual field loss, frequency, severity, therapy. Appropriate limitation(s) may apply.

(d) Head injury

Applicants with a head injury which was severe enough to cause loss of consciousness or is associated with penetrating brain injury may be assessed as fit if there has been a full recovery and the risk of epilepsy is sufficiently low. An evaluation by a neurologist may be required depending on the staging of the original injury.

(e) Spinal or peripheral nerve injury

Applicants with a history or diagnosis of spinal or peripheral nerve injury or a disorder of the nervous system due to a traumatic injury may be assessed as fit if neurological evaluation is satisfactory and the conditions of [AMC10 MED.B.095](#) are satisfied.

(f) Vascular deficiencies

Applicants with a disorder of the nervous system due to vascular deficiencies including haemorrhagic and ischaemic events should be assessed as unfit. A fit assessment may be considered if neurological evaluation is satisfactory and the conditions of [AMC10 MED.B.095](#) are satisfied. A cardiological evaluation and medical flight test should be undertaken for applicants with residual deficiencies.

AMC13 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

VISUAL SYSTEM

(a) Applicants should not possess any abnormality of the function of the eyes or their adnexa or any active pathological condition, congenital or acquired, acute or chronic, or any sequelae of eye surgery or trauma, which is likely to interfere with the safe exercise of the privileges of the applicable licence.

(b) Eye examination

The examination should include visual acuities (near, intermediate and distant vision) and visual field.

(c) Visual acuity

- (1) Visual acuity with or without corrective lenses should be 6/9 (0,7) binocularly and 6/12 (0,5) in each eye.
- (2) Applicants who do not meet the required visual acuity should be assessed by an AME or AeMC, taking into account the privileges of the licence held and the risk involved.

- (3) Applicants should be able to read, binocularly, an N5 chart (or equivalent) at 30-50 cm and an N14 chart (or equivalent) at 100 cm, with correction if prescribed (Refer to [GM1 MED.B.070](#)).
- (d) Visual acuity
- Applicants with substandard vision in one eye may be assessed as fit if the better eye:
- (1) achieves distant visual acuity of 6/6 (1,0), corrected or uncorrected;
 - (2) achieves distant visual acuity less than 6/6 (1,0) but not less than 6/9 (0,7), after ophthalmological evaluation.
- (e) Visual field defects
- Applicants with a visual field defect may be assessed as fit if the binocular visual field or, in the case of monocularly, the monocular visual field is acceptable.
- (f) Eye surgery
- (1) After refractive surgery, a fit assessment may be considered, provided that there is satisfactory stability of refraction, there are no post-operative complications and no significant increase in glare sensitivity.
 - (2) After cataract, retinal or glaucoma surgery a fit assessment may be considered once recovery is complete.
- (g) Visual correction
- Correcting lenses should permit the licence holder to meet the visual requirements at all distances.

AMC14 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

COLOUR VISION

Applicants for a night rating should correctly identify 9 of the first 15 plates of the 24-plate edition of Ishihara pseudoisochromatic plates or should be colour safe.

AMC15 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

OTORHINOLARYNGOLOGY (ENT)

- (a) Hearing
- (1) Applicants should understand correctly conversational speech when tested with or without hearing aids at a distance of 2 metres from and with the applicant's back turned towards the examiner.
 - (2) If the hearing requirements can only be met with the use of hearing aid(s), the hearing aid(s) should provide optimal hearing function, be well-tolerated, and be suitable for aviation purposes.
 - (3) Applicants with hypoacusis should demonstrate satisfactory functional hearing ability.

- (4) Applicants with profound deafness or major disorder of speech, or both, may be assessed as fit with an SSL such as 'limited to areas and operations where the use of radio is not mandatory'. The aircraft should be equipped with appropriate alternative warning devices in lieu of sound warnings.
- (b) Ear conditions
- Applicants with:
- (1) an active pathological process of the internal or middle ear;
 - (2) unhealed perforation or dysfunction of the tympanic membrane(s);
 - (3) disturbance of vestibular function;
 - (4) significant restriction of the nasal passages;
 - (5) sinus dysfunction;
 - (6) significant malformation or significant infection of the oral cavity or upper respiratory tract; or
 - (7) significant disorder of speech or voice
- should undergo further examination to establish that the condition does not interfere with the safe exercise of the privileges of the licence.

AMC16 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

DERMATOLOGY

In cases where a dermatological condition is associated with a systemic illness, full consideration should be given to the underlying illness before a fit assessment may be considered.

AMC17 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

ONCOLOGY

- (a) In the case of malignant disease, applicants may be considered for a fit assessment if:
- (1) there is no evidence of residual malignant disease likely to jeopardise flight safety;
 - (2) time appropriate to the type of tumour has elapsed since the end of primary treatment;
 - (3) the risk of in-flight incapacitation from a recurrence or metastasis is sufficiently low;
 - (4) there is no evidence of short or long-term sequelae from treatment that may jeopardise flight safety.
- (b) Arrangements for an oncological follow-up should be made for an appropriate period of time.
- (c) Applicants with an established history or clinical diagnosis of intracerebral malignant tumour should be assessed as unfit.

GM1 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

DIABETES MELLITUS TYPE 2 TREATED WITH INSULIN – GENERAL

- (a) Pilots and their treating physician should be aware that if the HbA1c target level was set to normal (non-diabetic) levels, this will significantly increase the chance of hypoglycaemia. For safety reasons the target level of HbA1c is therefore set to 7,5–8,5 % even though there is evidence that lower HbA1c levels are correlated with fewer diabetic complications.
- (b) The safety pilot should be briefed pre-flight on the potential condition of the pilot. The results of blood sugar testing before and during flight should be shared with the safety pilot for the acceptability of the values obtained.

GM2 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

DIABETES MELLITUS TYPE 2 TREATED WITH INSULIN – CONVERSION TABLE FOR HbA1c IN % AND MMOL/MOL

HbA1c	in %	HbA1c	in mmol/mol
	4,7		28
	5,0		31
	5,3		34
	5,6		38
	5,9		41
	6,2		44
	6,5		48
	6,8		51
	7,4		57
	8,0		64
	8,6		70
	9,2		77
	9,8		84
	10,4		90
	11,6		103

GM3 MED.B.095 Medical examination and assessment of applicants for LAPL medical certificates

ED Decision 2019/002/R

MOOD DISORDER

After full recovery from a mood disorder and after full consideration of the individual case, a fit assessment may be considered, depending on the characteristics and gravity of the mood disorder. If stability on maintenance psychoactive medication is confirmed, a fit assessment may be considered. If the dosage or type of medication is changed, a further evaluation may be required until stability is confirmed.

SUBPART C – REQUIREMENTS FOR MEDICAL FITNESS OF CABIN CREW

SECTION 1 – GENERAL REQUIREMENTS

MED.C.001 General

Regulation (EU) No 1178/2011

Cabin crew members shall only perform the duties and responsibilities required by aviation safety rules on an aircraft if they comply with the applicable requirements of this Part.

MED.C.005 Aero-medical assessments

Regulation (EU) No 1178/2011

- (a) Cabin crew members shall undergo aero-medical assessments to verify that they are free from any physical or mental illness which might lead to incapacitation or an inability to perform their assigned safety duties and responsibilities.
- (b) Each cabin crew member shall undergo an aero-medical assessment before being first assigned to duties on an aircraft, and after that at intervals of maximum 60 months.
- (c) Aero-medical assessments shall be conducted by an AME, AeMC, or by an OHMP if the requirements of [MED.D.040](#) are complied with.

AMC1 MED.C.005 Aero-medical assessments

ED Decision 2019/002/R

- (a) When conducting aero-medical examinations and assessments of cabin crew members, as applicable, their medical fitness should be assessed with particular regard to their physical and mental ability to:
 - (1) undergo the training required for cabin crew to acquire and maintain competence, e.g. actual fire-fighting, slide descending, using Protective Breathing Equipment (PBE) in a simulated smoke-filled environment, providing first aid;
 - (2) manipulate the aircraft systems and emergency equipment to be used by cabin crew, e.g. cabin management systems, doors/exits, escape devices, fire extinguishers, taking also into account the class and type of aircraft operated, e.g. narrow-bodied or wide-bodied, single/multi-deck, single/multi-cabin crew operation;
 - (3) continuously tolerate the aircraft environment whilst performing duties, e.g. altitude, pressure, re-circulated air, noise; and the type of operations such as short/medium/long/ultra long haul; and
 - (4) perform the required duties and responsibilities efficiently during normal and abnormal operations, and in emergency situations and psychologically demanding circumstances, e.g. assistance to crew members and passengers in case of decompression; stress management, decision-making, crowd control and effective crew coordination, management of disruptive passengers and of security threats. When relevant, operating as single cabin crew should also be taken into account when assessing the medical fitness of cabin crew.

(b) Intervals

- (1) The interval between aero-medical assessments should be determined by the competent authority. The intervals established by the competent authority apply to cabin crew members who:
 - (i) undergo aero-medical assessments by an AME, AeMC or OHMP under the oversight of that competent authority; or
 - (ii) are employed by an operator under the oversight of that competent authority.
- (2) The interval between aero-medical assessments may be reduced by the AME, AeMC or OHMP for medical reasons and in accordance with MED.C.035.
- (3) Aero-medical assessments for the revalidation of a cabin crew medical report may be undertaken up to 45 days prior to the expiry date of the previous medical report. The validity period of the aero-medical assessment should be calculated from the expiry date of the previous aero-medical assessment.

SECTION 2 – REQUIREMENTS FOR AERO-MEDICAL ASSESSMENT OF CABIN CREW

MED.C.020 General

Regulation (EU) No 1178/2011

Cabin crew members shall be free from any:

- (a) abnormality, congenital or acquired;
- (b) active, latent, acute or chronic disease or disability;
- (c) wound, injury or sequelae from operation; and
- (d) effect or side effect of any prescribed or non-prescribed therapeutic, diagnostic or preventive medication taken that would entail a degree of functional incapacity which might lead to incapacitation or an inability to discharge their safety duties and responsibilities.

MED.C.025 Content of aero-medical assessments

Regulation (EU) No 1178/2011

- (a) An initial aero-medical assessment shall include at least:
 - (1) an assessment of the applicant cabin crew member's medical history; and
 - (2) a clinical examination of the following:
 - (i) cardiovascular system;
 - (ii) respiratory system;
 - (iii) musculoskeletal system;
 - (iv) otorhino-laryngology;
 - (v) visual system; and
 - (vi) colour vision.
- (b) Each subsequent aero-medical re-assessment shall include:
 - (1) an assessment of the cabin crew member's medical history; and
 - (2) a clinical examination if deemed necessary in accordance with aero-medical best practice.
- (c) For the purpose of (a) and (b), in case of any doubt or if clinically indicated, a cabin crew member's aero-medical assessment shall also include any additional medical examination, test or investigation that are considered necessary by the AME, AeMC or OHMP.

AMC1 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

Aero-medical examinations and assessments of cabin crew members should be conducted in accordance with AMC2 to AMC18 MED.C.025.

AMC2 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

CARDIOVASCULAR SYSTEM

(a) Examination

- (1) A standard 12-lead resting electrocardiogram (ECG) and report should be completed on clinical indication, at the first examination after the age of 40 and then at least every five years after the age of 50. If cardiovascular risk factors such as smoking, abnormal cholesterol levels or obesity are present, the intervals of resting ECGs should be reduced to two years.
- (2) Extended cardiovascular assessment should be required when clinically indicated.

(b) Cardiovascular system - general

- (1) Cabin crew members with any of the following conditions:
 - (i) aneurysm of the thoracic or supra-renal abdominal aorta, before surgery;
 - (ii) significant functional abnormality of any of the heart valves; or
 - (iii) heart or heart/lung transplantationshould be assessed as unfit.
- (2) Cabin crew members with an established diagnosis of one of the following conditions:
 - (i) peripheral arterial disease before or after surgery;
 - (ii) aneurysm of the abdominal aorta, before or after surgery;
 - (iii) minor cardiac valvular abnormalities;
 - (iv) after cardiac valve surgery;
 - (v) abnormality of the pericardium, myocardium or endocardium;
 - (vi) congenital abnormality of the heart, before or after corrective surgery;
 - (vii) a cardiovascular condition requiring systemic anticoagulation;
 - (viii) vasovagal syncope of uncertain cause;
 - (ix) arterial or venous thrombosis; or
 - (x) pulmonary embolismshould be evaluated by a cardiologist before a fit assessment may be considered.

(c) Thromboembolic disorders

Whilst anticoagulation therapy is initiated, cabin crew members should be assessed as unfit. After a period of stable anticoagulation, a fit assessment may be considered with limitation(s), as appropriate. Anticoagulation should be considered stable if, within the last 6 months, at least 5 INR values are documented, of which at least 4 are within the INR target range and the haemorrhagic risk is acceptable. In cases of anticoagulation medication not requiring INR monitoring, a fit assessment may be considered after a stabilisation period of 3 months. Cabin crew members with pulmonary embolism should also be evaluated by a cardiologist. Following cessation of anticoagulant therapy, for any indication, cabin crew members should undergo a re-assessment.

(d) Syncope

- (1) In the case of a single episode of vasovagal syncope which can be satisfactorily explained, a fit assessment may be considered.

- (2) Cabin crew members with a history of recurrent vasovagal syncope should be assessed as unfit. A fit assessment may be considered after a 6-month period without recurrence, provided cardiological evaluation is satisfactory. Neurological review may be indicated.
- (e) Blood pressure

Blood pressure should be recorded at each examination.

 - (1) The blood pressure should be within normal limits and should not consistently exceed 160 mmHg systolic and/or 95 mmHg diastolic, with or without treatment, taking into account other risk factors.
 - (2) Cabin crew members initiating medication for the control of blood pressure should be assessed as unfit until the absence of any significant side effects has been established and verification that the treatment is compatible with the safe exercise of cabin crew duties has been achieved.
- (f) Coronary artery disease
 - (1) Cabin crew members with:
 - (i) cardiac ischaemia;
 - (ii) symptomatic coronary artery disease; or
 - (iii) symptoms of coronary artery disease controlled by medicationshould be assessed as unfit.
 - (2) Cabin crew members who are asymptomatic after myocardial infarction or surgery for coronary artery disease should have fully recovered before a fit assessment may be considered. The affected cabin crew members should be on appropriate secondary prevention treatment.
- (g) Rhythm/conduction disturbances
 - (1) Cabin crew members with any significant disturbance of cardiac conduction or rhythm should undergo cardiological evaluation before a fit assessment may be considered.
 - (2) Cabin crew members with a history of:
 - (i) ablation therapy; or
 - (ii) pacemaker implantationshould undergo satisfactory cardiovascular evaluation before a fit assessment may be made.
 - (3) Cabin crew members with:
 - (i) symptomatic sinoatrial disease;
 - (ii) symptomatic hypertrophic cardiomyopathy
 - (iii) complete atrioventricular block;
 - (iv) symptomatic QT prolongation;
 - (v) an automatic implantable defibrillating system; or
 - (vi) a ventricular anti-tachycardia pacemakershould be assessed as unfit.

AMC3 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

RESPIRATORY SYSTEM

- (a) Cabin crew members with significant impairment of pulmonary function should be assessed as unfit. A fit assessment may be considered once pulmonary function has recovered and is satisfactory.
- (b) Cabin crew members should undergo pulmonary morphological or functional tests on when clinically indicated.
- (c) Cabin crew members with a history or established diagnosis of:
 - (1) asthma;
 - (2) active inflammatory disease of the respiratory system;
 - (3) active sarcoidosis;
 - (4) pneumothorax;
 - (5) sleep apnoea syndrome/sleep disorder; or
 - (6) major thoracic surgeryshould undergo respiratory evaluation with a satisfactory result before a fit assessment may be considered.
- (d) Cabin crew members who have undergone a pneumonectomy should be assessed as unfit.

AMC4 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

DIGESTIVE SYSTEM

- (a) Cabin crew members with any disease or sequelae of surgical intervention in any part of the digestive tract or its adnexa likely to cause incapacitation in flight, in particular any obstruction due to stricture or compression, should be assessed as unfit.
- (b) Cabin crew members should be free from herniae that might give rise to incapacitating symptoms.
- (c) Cabin crew members with disorders of the gastro-intestinal system, including:
 - (1) recurrent severe dyspeptic disorder requiring medication;
 - (2) peptic ulceration;
 - (3) pancreatitis;
 - (4) symptomatic gallstones;
 - (5) an established diagnosis or history of chronic inflammatory bowel disease;
 - (6) after surgical operation on the digestive tract or its adnexa, including surgery involving total or partial excision or a diversion of any of these organs;
 - (7) morphological or functional liver disease; or
 - (8) after surgery, including liver transplantationmay be assessed as fit subject to satisfactory gastroenterological evaluation.

AMC5 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

METABOLIC AND ENDOCRINE SYSTEMS

- (a) Cabin crew members should not possess any functional or structural metabolic, nutritional or endocrine disorder which is likely to interfere with the safe exercise of their duties and responsibilities.
- (b) Cabin crew members with metabolic, nutritional or endocrine dysfunction may be assessed as fit, subject to demonstrated stability of the condition and satisfactory aero-medical evaluation.
- (c) Diabetes mellitus
 - (1) Cabin crew members with diabetes mellitus requiring insulin may be assessed as fit:
 - (i) if it can be demonstrated that adequate blood sugar control has been achieved and hypoglycaemia awareness is established and maintained; and
 - (ii) in the absence, within the preceding 12 months, of any;
 - (A) hospitalisation related to diabetes; or
 - (B) hypoglycaemia that resulted in a seizure, loss of consciousness, impaired cognitive function or that required the intervention by another party; or
 - (C) episode of hypoglycaemia unawareness.
 - (2) Limitations should be imposed as appropriate. A limitation to undergo specific medical examinations (SIC) and a restriction to operate only in multi-cabin crew operations (MCL) should be placed as a minimum.
 - (3) Cabin crew members with diabetes mellitus not requiring insulin may be assessed as fit if it can be demonstrated that adequate blood sugar control has been achieved and hypoglycaemia awareness, if applicable considering the medication, is achieved.

AMC6 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

HAEMATOLOGY

Cabin crew members with a haematological condition, such as:

- (a) abnormal haemoglobin including, but not limited to, anaemia, erythrocytosis or haemoglobinopathy;
- (b) coagulation, haemorrhagic or thrombotic disorder;
- (c) significant lymphatic enlargement;
- (d) acute or chronic leukaemia; or
- (e) splenomegaly

may be assessed as fit subject to satisfactory aero-medical evaluation. If anticoagulation is being used as treatment, refer to [AMC2 MED.C.025\(c\)](#).

AMC7 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

GENITOURINARY SYSTEM

- (a) Urine analysis should form part of every aero-medical examination and assessment. The urine should not contain any abnormal element(s) considered to be of pathological significance.
- (b) Cabin crew members with any disease or sequelae of surgical procedures on the kidneys or the urinary tract, in particular any obstruction due to stricture or compression likely to cause incapacitation should be assessed as unfit.
- (c) Cabin crew members with a genitourinary disorder, such as:
 - (1) renal disease; or
 - (2) a history of renal colic due to one or more urinary calculimay be assessed as fit subject to satisfactory renal/urological evaluation.
- (d) Cabin crew members who have undergone a major surgical operation in the genitourinary apparatus involving a total or partial excision or a diversion of its organs should be assessed as unfit and be re-assessed after recovery before a fit assessment may be made.
- (e) Cabin crew members who have undergone renal transplantation may be considered for a fit assessment if it is fully compensated and tolerated with only minimal immuno-suppressive therapy after at least 12 months. A requirement to undergo specific medical examinations (SIC) and a restriction to operate only in multi-cabin crew operations (MCL) should be considered.
- (f) Cabin crew members requiring dialysis should be assessed as unfit.

AMC8 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

INFECTIOUS DISEASE

Cabin crew members who are HIV positive may be assessed as fit if investigation provides no evidence of clinical disease and subject to satisfactory aero-medical evaluation.

AMC9 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

OBSTETRICS AND GYNAECOLOGY

- (a) Cabin crew members who have undergone a major gynaecological operation should be assessed as unfit until after recovery.
- (b) Pregnancy
 - (1) A pregnant cabin crew member may be assessed as fit only during the first 16 weeks of gestation following review of the obstetric evaluation by the AME or OHMP.
 - (2) A limitation not to perform duties as single cabin crew member should be considered.
 - (3) The AME or OHMP should provide written advice to the cabin crew member and supervising physician regarding potentially significant complications of pregnancy resulting from flying duties.

AMC10 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

MUSCULOSKELETAL SYSTEM

- (a) Cabin crew members should have sufficient standing height, arm and leg length and muscular strength for the safe exercise of their duties and responsibilities.
- (b) Cabin crew members should have satisfactory functional use of the musculoskeletal system. Particular attention should be paid to emergency procedures and evacuation, and related training.
- (c) Cabin crew members with any significant sequelae from disease, injury or congenital abnormality affecting the bones, joints, muscles or tendons with or without surgery require full evaluation prior to a fit assessment.
- (d) Cabin crew members with inflammatory, infiltrative, traumatic or degenerative disease of the musculoskeletal system may be assessed as fit provided the condition is in remission or is stable and the affected cabin crew member is not taking any medication that may lead to unfitness.

AMC11 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

MENTAL HEALTH

- (a) Cabin crew members with a mental or behavioural disorder due to use or misuse of alcohol or other psychoactive substances should be assessed as unfit pending recovery and freedom from psychoactive substance use or misuse and subject to satisfactory psychiatric evaluation after successful treatment.
- (b) Cabin crew members with an established history or clinical diagnosis of schizophrenia, schizotypal or delusional disorder should be assessed as unfit.
- (c) Cabin crew members with a psychiatric condition such as:
 - (1) mood disorder;
 - (2) neurotic disorder;
 - (3) personality disorder; or
 - (4) mental or behavioural disordershould undergo satisfactory psychiatric evaluation before a fit assessment may be considered.
- (d) Cabin crew members with a history of a single or repeated acts of deliberate self-harm should be assessed as unfit. Cabin crew members should undergo satisfactory psychiatric evaluation before a fit assessment may be considered.
- (e) Where there is established evidence that a cabin crew member has a psychological disorder, he/she should be referred for psychological opinion and advice.
- (f) The psychological evaluation may include a collection of biographical data, the review of aptitudes, and personality tests and psychological interview.
- (g) The psychologist should submit a report to the AME or OHMP, detailing the results and recommendation.

AMC12 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

NEUROLOGY

- (a) Cabin crew members with an established history or clinical diagnosis of:

- (1) epilepsy; or
- (2) recurring episodes of disturbance of consciousness of uncertain cause
should be assessed as unfit.
- (b) Cabin crew members with an established history or clinical diagnosis of:
 - (1) epilepsy without recurrence after 5 years of age and without treatment for more than 10 years;
 - (2) epileptiform EEG abnormalities and focal slow waves;
 - (3) progressive or non-progressive disease of the nervous system;
 - (4) inflammatory disease of the central or peripheral nervous system;
 - (5) migraine;
 - (6) a single episode of disturbance of consciousness of uncertain cause;
 - (7) loss of consciousness after head injury;
 - (8) penetrating brain injury; or
 - (9) spinal or peripheral nerve injuryshould undergo further evaluation before a fit assessment may be considered.
- (c) Cabin crew members with a disorder of the nervous system due to vascular deficiencies including haemorrhagic and ischaemic events should be assessed as unfit. A fit assessment may be considered if neurological review and musculoskeletal assessments are satisfactory.

AMC13 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

VISUAL SYSTEM

- (a) Examination
 - (1) a routine eye examination should form part of the initial and all further examinations and assessments; and
 - (2) an extended eye examination should be undertaken by an eye specialist when clinically indicated. (Refer to [GM2 MED.B.070](#))
- (b) Distant visual acuity, with or without correction, should be with both eyes 6/9 (0,7) or better.
- (c) Cabin crew members should be able to read an N5 chart (or equivalent) at 30–50 cm, with correction if prescribed (Refer to [GM1 MED.B.070](#)).
- (d) The binocular visual field or, in the case of monocularity, the monocular visual field should be acceptable.
- (e) Cabin crew members who have undergone refractive surgery may be assessed as fit subject to satisfactory ophthalmic evaluation.
- (f) Cabin crew members with diplopia should be assessed as unfit.
- (g) Spectacles and contact lenses:
If satisfactory visual function is achieved only with the use of correction:

- (1) in the case of myopia or hyperopia or both, spectacles or contact lenses should be worn whilst on duty;
- (2) in the case of presbyopia, spectacles should be readily available for immediate use;
- (3) the correction should provide optimal visual function and be well-tolerated;
- (4) a spare set of similarly correcting spectacles should be readily available for immediate use whilst on duty;
- (5) orthokeratologic lenses should not be used.

AMC14 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

COLOUR VISION

Cabin crew members should be able to correctly identify 9 of the first 15 plates of the 24-plate edition of Ishihara pseudoisochromatic plates. Alternatively, cabin crew members should demonstrate the ability to readily perceive those colours of which the perception is required for the safe performance of their duties.

AMC15 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

OTORHINOLARYNGOLOGY (ENT)

- (a) Hearing should be satisfactory for the safe exercise of cabin crew duties and responsibilities. Cabin crew with hypoacusis should demonstrate satisfactory functional hearing abilities.
- (b) Examination
 - (1) An ear, nose and throat (ENT) examination should form part of all examinations and assessments. A tympanometry or equivalent should be performed at the initial examination and when clinically indicated.
 - (2) Hearing should be tested at all examinations and assessments:
 - (i) the cabin crew member should understand correctly conversational speech when tested with each ear at a distance of 2 metres from and with the cabin crew member's back turned towards the examiner;
 - (ii) notwithstanding (b)(2)(i), hearing should be tested with pure tone audiometry at the initial examination and when clinically indicated;
 - (iii) at initial examination the cabin crew member should not have a hearing loss of more than 35 dB at any of the frequencies 500 Hz, 1 000 Hz or 2 000 Hz, or more than 50 dB at 3 000 Hz, in either ear separately.
 - (3) If the hearing requirements can be met only with the use of hearing aid(s), the hearing aid(s) should provide optimal hearing function, be well-tolerated, and suitable for aviation purposes.
- (c) Cabin crew members with:
 - (1) an active pathological process of the internal or middle ear;
 - (2) unhealed perforation or dysfunction of the tympanic membrane(s);
 - (3) disturbance of vestibular function;

- (4) significant restriction of the nasal passages;
- (5) sinus dysfunction;
- (6) significant malformation or significant infection of the oral cavity or upper respiratory tract;
- (7) significant disorder of speech or voice

should undergo further examination to establish that the condition does not interfere with the safe exercise of their duties and responsibilities.

AMC16 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

DERMATOLOGY

In cases where a dermatological condition is associated with a systemic illness, full consideration should be given to the underlying illness before a fit assessment may be made.

AMC17 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

ONCOLOGY

- (a) After treatment for malignant disease, cabin crew members should undergo satisfactory oncological and aero-medical evaluation before a fit assessment may be considered.
- (b) Cabin crew members with an established history or clinical diagnosis of intracerebral malignant tumour should be assessed as unfit. Considering the histology of the tumour, a fit assessment may be considered after successful treatment and recovery.

GM1 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

- (a) When conducting aero-medical examinations and assessments, typical cabin crew duties as listed in (b) and (c), particularly those to be performed during abnormal operations and emergency situations, and cabin crew responsibilities to the travelling public should be considered in order to identify:
 - (1) any physical and/or mental conditions that could be detrimental to the performance of the duties required from cabin crew; and
 - (2) which examination(s), test(s) or investigation(s) should be undergone to complete an appropriate aero-medical assessment.
- (b) Main cabin crew duties and responsibilities during day-to-day normal operations
 - (1) During pre/post-flight ground operations with/without passengers on board:
 - (i) monitoring of situation inside the aircraft cabin and awareness of conditions outside the aircraft including observation of visible aircraft surfaces and information to flight crew of any surface contamination such as ice or snow;
 - (ii) assistance to special categories of passengers (SCPs) such as infants and children (accompanied or unaccompanied), persons with disabilities or reduced mobility, medical cases with or without medical escort, and inadmissible persons, deportees and passengers in custody;

- (iii) observation of passengers (any suspicious behaviour, passengers under the influence of alcohol and/or drugs, mentally disturbed), observation of potential able-bodied persons, crowd control during boarding and disembarkation;
 - (iv) safe stowage of cabin luggage, safety demonstrations and cabin secured checks, management of passengers and ground services during re-fuelling, observation of use of portable electronic devices;
 - (v) preparedness to carry out safety and emergency duties at any time, and security alertness.
- (2) During flight:
 - (i) operation and monitoring of aircraft systems, surveillance of the cabin, lavatories, galleys, crew areas and flight crew compartment;
 - (ii) coordination with flight crew on situation in the cabin and turbulence events/effects;
 - (iii) management and observation of passengers (consumption of alcohol, behaviour, potential medical issues), observation of use of portable electronic devices;
 - (iv) safety and security awareness and preparedness to carry out safety and emergency duties at any time, and cabin secured checks prior to landing.
- (c) Main cabin crew duties and responsibilities during abnormal and emergency operations
 - (1) In case of planned or unplanned emergency evacuation: briefing and/or commands to passengers including SCPs and selection and briefing to able-bodied persons; crowd control monitoring and evacuation conduct including in the absence of command from the flight crew; post-evacuation duties including assistance, first aid and management of survivors and survival in particular environments; activation of applicable communication means towards search and rescue services.
 - (2) In case of decompression: checking of crew members, passengers, cabin, lavatories, galleys, crew rest areas and flight crew compartment, and administering oxygen to crew members and passengers as necessary.
 - (3) In case of pilot incapacitation: secure pilot in his/her seat or remove from flight crew compartment; administer first aid and assist operating pilot as required.
 - (4) In case of fire or smoke: identify source/cause/type of fire/smoke to perform the necessary required actions; coordinate with other cabin crew members and flight crew; select appropriate extinguisher/agent and fight the fire using portable breathing equipment (PBE), gloves, and protective clothing as required; management of necessary passengers' movement if possible; instructions to passengers to prevent smoke inhalation/suffocation; give first aid as necessary; monitor the affected area until landing; preparation for possible emergency landing.
 - (5) In case of first aid and medical emergencies: assistance to crew members and/or passengers; correct assessment and correct use of therapeutic oxygen, defibrillator, first-aid kits/emergency medical kit contents as required; management of events, of incapacitated person(s) and of other passengers; coordination and effective communication with other crew members, in particular when medical advice is transmitted by frequency to flight crew or by a telecommunication connection.
 - (6) In case of disruptive passenger behaviour: passenger management as appropriate including use of restraint technique as considered required.

- (7) In case of security threats (bomb threat on ground or in-flight and/or hijack): control of cabin areas and passengers' management as required by the type of threat, management of suspicious device, protection of flight crew compartment door.
- (8) In case of handling of dangerous goods: observing safety procedures when handling the affected device, in particular when handling chemical substances that are leaking; protection and management of self and passengers and effective coordination and communication with other crew members.

GM2 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

DIABETES MELLITUS TREATED WITH INSULIN

When considering a fit assessment for cabin crew with diabetes mellitus requiring insulin, account should be taken of the IATA Guidelines on Insulin-Treated Diabetes (Cabin Crew), as last amended.

GM3 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

COLOUR VISION – GENERAL

Examples of colours of which the perception is required for the safe performance of cabin crew members' duties are: cabin crew indication panels, pressure gauges of emergency equipment (e.g. fire extinguishers) and cabin door status.

GM4 MED.C.025 Content of aero-medical assessments

ED Decision 2019/002/R

OTORHINOLARYNGOLOGY (ENT) – PURE TONE AUDIOGRAM

The pure tone audiogram may also cover the 4 000 Hz frequency for early detection of decrease in hearing.

SECTION 3 – ADDITIONAL REQUIREMENTS FOR APPLICANTS FOR, OR HOLDERS OF, A CABIN CREW ATTESTATION

MED.C.030 Cabin crew medical report

Regulation (EU) No 1178/2011

- (a) After completion of each aero-medical assessment, applicants for, and holders of, a cabin crew attestation:
 - (1) shall be provided with a cabin crew medical report by the AME, AeMC or OHMP; and
 - (2) shall provide the related information, or a copy of their cabin crew medical report to the operator(s) employing their services.

- (b) Cabin crew medical report

A cabin crew medical report shall indicate the date of the aero-medical assessment, whether the cabin crew member has been assessed fit or unfit, the date of the next required aero-medical assessment and, if applicable, any limitation(s). Any other elements shall be subject to medical confidentiality in accordance with [MED.A.015](#).

AMC1 MED.C.030 Cabin crew medical report

ED Decision 2019/002/R

The cabin crew medical report to be provided in writing to the applicants for, and holders of, a cabin crew attestation:

- (a) should be issued in the national language(s) and/or in English; and
- (b) should include the following elements:
 - (1) The State where the aero-medical assessment of the Cabin Crew Attestation (CCA) applicant/holder was conducted (I);
 - (2) Last and first name of the CCA applicant/holder (IV);
 - (3) Date of birth of the CCA applicant/holder (dd/mm/yyyy) (XIV);
 - (4) Nationality of the CCA applicant/holder (VI);
 - (5) Signature of the CCA applicant/holder (VII);
 - (6) Aero-medical assessment result (fit or unfit) (II);
 - (7) Expiry date of the previous cabin crew medical report (dd/mm/yyyy);
 - (8) Date of issue (dd/mm/yyyy) and signature of the AeMC, AME, or OHMP (X);
 - (9) Date of the aero-medical assessment (dd/mm/yyyy);
 - (10) Seal or stamp of the AeMC, AME or OHMP (XI);
 - (11) Limitation(s), if applicable (XII);
 - (12) Expiry date of medical report (dd/mm/yyyy) (IX).

GM1 MED.C.030(b) Cabin crew medical report

ED Decision 2019/002/R

GENERAL

The format of the cabin crew medical report may be as shown in the example below, with the size of each sheet being 1/8 of A4.

State of issue CABIN CREW MEDICAL REPORT FOR CABIN CREW ATTESTATION (CCA) APPLICANT OR HOLDER	
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I The State where the aero-medical assessment is conducted: III Cabin crew attestation reference number: IV Last and first name: XIV Date of birth (dd/mm/yyyy): VI Nationality: VII Signature of CCA applicant/holder:	II Aero-medical assessment result (fit/unfit): Expiry date of the previous cabin crew medical report (dd/mm/yyyy): Date of aero-medical assessment (dd/mm/yyyy): X Date of issue* (dd/mm/yyyy): X Signature of the AeMC, AME or OHMP: XI Seal or stamp of the AeMC, AME or OHMP:
2	3

* Date of issue is the date the Cabin Crew Medical Report is issued and signed.

MED.C.035 Limitations

Regulation (EU) No 1178/2011

- ## AMC1 MED.C.035 Limitations

ED Decision 2019/002/R

- (a) a restriction to operate only in multi-cabin crew operations (MCL);
- (b) a restriction to specified aircraft type(s) (OAL) or to a specified type of operation (OOL);

- (c) a requirement to undergo the next aero-medical examination and assessment at an earlier date than required by [MED.C.005\(b\)](#) (TML);
- (d) a requirement to undergo specific medical examination(s) (SIC);
- (e) a requirement for visual correction (CVL), or by means of contact lenses that correct for defective vision (CCL);
- (f) a requirement to use hearing aids (HAL); and
- (g) special restriction as specified (SSL).

SUBPART D – AERO-MEDICAL EXAMINERS (AME), GENERAL MEDICAL PRACTITIONERS (GMP), OCCUPATIONAL HEALTH MEDICAL PRACTITIONERS (OHMP)

SECTION 1 – AERO-MEDICAL EXAMINERS

MED.D.001 Privileges

Regulation (EU) 2019/27

- (a) The privileges of holders of an aero-medical examiner (AME) certificate are to issue, revalidate and renew class 2 medical certificates and LAPL medical certificates and to conduct the relevant medical examinations and assessments.
- (b) Holders of an AME certificate may apply for an extension of their privileges to include medical examinations for the revalidation and renewal of class 1 medical certificates, if they comply with the requirements set out in point [MED.D.015](#).
- (c) The privileges of a holder of an AME certificate referred to in points (a) and (b) shall include the privileges to conduct cabin crew members' aero-medical examinations and assessments and to provide the related cabin crew members' medical reports, as applicable, in accordance with this Annex (Part-MED).
- (d) The scope of the privileges of the holder of an AME certificate, and any condition thereof, shall be specified in that certificate.
- (e) A holder of an AME certificate shall not at any time hold more than one AME certificate issued in accordance with this Regulation.
- (f) Holders of an AME certificate shall not undertake aero-medical examinations and assessments in a Member State other than the Member State that issued their AME certificate, unless they have completed all the following steps:
 - (1) they have been granted access by the other Member State concerned to exercise their professional activities as a specialised doctor;
 - (2) they have informed the competent authority of that other Member State of their intention to conduct aero-medical examinations and assessments and to issue medical certificates within the scope of their privileges as AME;
 - (3) they have received a briefing from the competent authority of that other Member State.

MED.D.005 Application

Regulation (EU) 2019/27

- (a) An application for an AME certificate or for an extension of the privileges of an AME certificate shall be made in a form and manner specified by the competent authority.
- (b) Applicants for an AME certificate shall provide the competent authority with:
 - (1) their personal details and professional address;

- (2) documentation demonstrating that they comply with the requirements of point [MED.D.010](#), including evidence of successful completion of the training course in aviation medicine appropriate to the privileges they apply for;
 - (3) a written declaration that, once the AME certificate has been issued, the AME will issue medical certificates on the basis of the requirements of this Regulation.
- (c) When AMEs undertake aero-medical examinations in more than one location, they shall provide the competent authority with relevant information regarding all practice locations and practice facilities.

MED.D.010 Requirements for the issue of an AME certificate

Regulation (EU) 2019/27

Applicants shall be issued an AME certificate, where they meet all of the following conditions:

- (a) they are fully qualified and licensed for the practice of medicine and have evidence of completion of specialist medical training;
- (b) they have successfully completed a basic training course in aviation medicine, including practical training in the examination methods and aero-medical assessments;
- (c) they have demonstrated to the competent authority that they:
 - (1) have adequate facilities, procedures, documentation and functioning equipment suitable for aero-medical examinations;
 - (2) have in place the necessary procedures and conditions to ensure medical confidentiality.

MED.D.011 Privileges of an AME certificate holder

Regulation (EU) 2019/27

Through the issuance of an AME certificate, the holder shall be granted the privileges to initially issue, revalidate and renew all of the following:

- (a) class 2 medical certificates;
- (b) LAPL medical certificates;
- (c) cabin crew members' medical reports.

MED.D.015 Requirements for the extension of privileges

Regulation (EU) 2019/27

Applicants shall be issued an AME certificate extending their privileges to the revalidation and renewal of class 1 medical certificates where they meet all of the following conditions:

- (a) they hold a valid AME certificate;
- (b) they conducted at least 30 examinations for the issue, revalidation or renewal of class 2 medical certificates or equivalent over a period of no more than 3 years preceding the application;
- (c) they successfully completed an advanced training course in aviation medicine, including practical training in the examination methods and aero-medical assessments;

- (d) they have successfully completed practical training of a duration of at least 2 days, either at an AeMC or under the supervision of the competent authority.

MED.D.020 Training courses in aviation medicine

Regulation (EU) 2024/2076

- (a) Training courses in aviation medicine referred to in [MED.D.010\(b\)](#) and [MED.D.015\(c\)](#) shall only be provided after the prior approval of the course by the competent authority of the Member State where the training organisation has its principal place of business. In order to obtain such approval, the training organisation shall demonstrate that the course syllabus contains the learning objectives to acquire the necessary competencies and that the persons in charge of providing the training have adequate knowledge and experience.
- (aa) For demonstrating compliance with points [MED.D.010\(b\)](#) and [MED.D.015\(c\)](#), an aviation medicine training course completed by an applicant outside the territories for which Member States are responsible under the Chicago Convention may be accepted by the competent authority, provided that the following conditions are met:
- (i) the competent authority has assessed and verified the course syllabus in accordance with point [ARA.MED.200\(c\)\(1\)](#) of Annex VI;
 - (ii) the applicant has completed a specific training module on the aero-medical requirements detailed in this Annex (Part-MED) as provided by the competent authority.
- (b) Except in the case of refresher training, the courses shall be concluded by a written examination on the subjects included in the course content.
- (c) The training organisation shall issue a certificate of successful completion to participants when they have obtained a pass in the examination.

AMC1 MED.D.020 Training courses in aviation medicine

ED Decision 2019/002/R

BASIC TRAINING COURSE

- (a) Basic training course for AMEs
- The basic training course for AMEs should consist of 60 hours of theoretical and practical training, including specific examination techniques.
- (b) The learning objectives to acquire the necessary competencies should include theoretical knowledge, risk management, and decision-making principles in the following subjects. Demonstrations and practical skills should also be included, where appropriate.
- (1) Introduction to aviation medicine;
 - (2) Basic aeronautical knowledge;
 - (3) Aviation physiology;
 - (4) Cardiovascular system;
 - (5) Respiratory system;
 - (6) Digestive system;

- (7) Metabolic and endocrine systems;
- (8) Haematology;
- (9) Genitourinary system;
- (10) Obstetrics and gynaecology;
- (11) Musculoskeletal system;
- (12) Psychiatry;
- (13) Psychology;
- (14) Neurology;
- (15) Visual system and colour vision;
- (16) Otorhinolaryngology;
- (17) Oncology;
- (18) Incidents and accidents escape and survival;
- (19) Medication and flying;
- (20) Legislation, rules and regulations;
- (21) Cabin crew working environment;
- (22) In-flight environment; and
- (23) Space medicine.

AMC2 MED.D.020 Training courses in aviation medicine

ED Decision 2019/002/R

ADVANCED TRAINING COURSE

- (a) Advanced training course for AMEs

The advanced training course for AMEs should consist of 66 hours of theoretical and practical training, including specific examination techniques.

- (b) The learning objectives to acquire the necessary competencies should include theoretical knowledge, risk management, and decision-making principles in the following subjects. Demonstrations and practical skills should also be included, where appropriate.

- (1) Pilot working environment;
- (2) Aerospace physiology;
- (3) Clinical medicine;
- (4) Cardiovascular system;
- (5) Neurology;
- (6) Psychiatry/psychology;
- (7) Visual system and colour vision;
- (8) Otorhinolaryngology;

- (9) Dentistry;
 - (10) Human factors in aviation;
 - (11) Incidents and accidents, escape and survival; and
 - (12) Tropical medicine.
- (c) Practical training in an AeMC should be under the guidance and supervision of the head of the AeMC.
- (d) After the successful completion of the practical training, a report of demonstrated competency should be issued.

GM1 MED.D.020 Training courses in aviation medicine

ED Decision 2019/002/R

BASIC TRAINING COURSE

- | | | |
|-----|--|----------|
| (a) | Basic training course in aviation medicine | 60 hours |
| (1) | Introduction to aviation medicine | 2 hours |
| | (i) History of aviation medicine | |
| | (ii) Specific aspects of civil aviation medicine | |
| | (iii) Different types of recreational flying | |
| | (iv) AME and pilots relationship | |
| | (v) Responsibility of the AME in aviation safety | |
| | (vi) Communication and interview techniques | |
| (2) | Basic aeronautical knowledge | 2 hours |
| | (i) Flight mechanisms | |
| | (ii) Man-machine interface, informational processing | |
| | (iii) Propulsion | |
| | (iv) Conventional instruments, 'glass cockpit' | |
| | (v) Recreational flying | |
| | (vi) Simulator/aircraft experience | |
| (3) | Aviation physiology | 9 hours |
| | (i) Atmosphere | |
| | (A) Functional limits for humans in flight | |
| | (B) Divisions of the atmosphere | |
| | (C) Gas laws — physiological significance | |
| | (D) Physiological effects of decompression | |
| | (ii) Respiration | |

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- (A) Blood gas exchange
 - (B) Oxygen saturation
 - (iii) Hypoxia signs and symptoms
 - (A) Average time of useful consciousness (TUC)
 - (B) Hyperventilation signs and symptoms
 - (C) Barotrauma
 - (D) Decompression sickness
 - (iv) Acceleration
 - (A) G-Vector orientation
 - (B) Effects and limits of G-load
 - (C) Methods to increase Gz-tolerance
 - (D) Positive/negative acceleration
 - (E) Acceleration and the vestibular system
 - (v) Visual disorientation
 - (A) Sloping cloud deck
 - (B) Ground lights and stars confusion
 - (C) Visual autokinesis
 - (vi) Vestibular disorientation
 - (A) Anatomy of the inner ear
 - (B) Function of the semicircular canals
 - (C) Function of the otolith organs
 - (D) The oculogyral and coriolis illusion
 - (E) 'Leans'
 - (F) Forward acceleration illusion of 'nose up'
 - (G) Deceleration illusion of 'nose down'
 - (H) Motion sickness — causes and management
 - (vii) Noise and vibration
 - (A) Preventive measures
- (4) Cardiovascular system 3 hours
- (i) Relation to aviation; risk of incapacitation
 - (ii) Examination procedures: ECG, laboratory testing and other special examinations
 - (iii) Cardiovascular diseases:
 - (A) Hypertension, treatment and assessment

- (B) Ischaemic heart disease
- (C) ECG findings
- (D) Assessment of satisfactory recovery from myocardial infarction, interventional procedures and surgery
- (E) Cardiomyopathies; pericarditis; rheumatic heart disease; valvular diseases
- (F) Rhythm and conduction disturbances, treatment and assessment
- (G) Congenital heart disease: surgical treatment, assessment
- (H) Cardiovascular syncope: single and repeated episodes

Topics (5) to (11) inclusive, and (17)

10 hours

(5) Respiratory system

- (i) Relation to aviation, risk of incapacitation
- (ii) Examination procedures: spirometry, peak flow, x-ray, other examinations
- (iii) Pulmonary diseases: asthma, chronic obstructive pulmonary diseases
- (iv) Infections, tuberculosis
- (v) Bullae, pneumothorax
- (vi) Obstructive sleep apnoea
- (vii) Treatment and assessment

(6) Digestive system

- (i) Relation to aviation, risk of incapacitation
- (ii) Examination of the system
- (iii) Gastro-intestinal disorders: gastritis, ulcer disease
- (iv) Biliary tract disorders
- (v) Hepatitis and pancreatitis
- (vi) Inflammatory bowel disease, irritable colon/irritable bowel disease
- (vii) Herniae
- (viii) Treatment and assessment including post-abdominal surgery

(7) Metabolic and endocrine systems

- (i) Relation to aviation, risk of incapacitation
- (ii) Endocrine disorders
- (iii) Diabetes mellitus Type 1 & 2
 - (A) Diagnostic tests and criteria
 - (B) Anti-diabetic therapy
 - (C) Operational aspects in aviation

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- (D) Satisfactory control criteria for aviation
 - (iv) Hyper/hypothyroidism
 - (v) Pituitary and adrenal glands disorders
 - (vi) Treatment and assessment
 - (8) Haematology
 - (i) Relation to aviation, risk of incapacitation
 - (ii) Blood donation aspects
 - (iii) Erythrocytosis; anaemia; leukaemia; lymphoma
 - (iv) Sickle cell disorders
 - (v) Platelet disorders
 - (vi) Haemoglobinopathies; geographical distribution; classification
 - (vii) Treatment and assessment
 - (9) Genitourinary system
 - (i) Relation to aviation, risk of incapacitation
 - (ii) Action to be taken after discovery of abnormalities in routine dipstick urinalysis, e.g. haematuria; albuminuria
 - (iii) Urinary system disorders:
 - (A) Nephritis; pyelonephritis; obstructive uropathies
 - (B) Tuberculosis
 - (C) Lithiasis: single episode; recurrence
 - (D) Nephrectomy, transplantation, other treatment and assessment
 - (10) Obstetrics and gynaecology
 - (i) Relation to aviation, risk of incapacitation
 - (ii) Pregnancy and aviation
 - (iii) Disorders, treatment and assessment
 - (11) Musculoskeletal system
 - (i) Vertebral column diseases
 - (ii) Arthropathies and arthroprosthesis
 - (iii) Pilots with a physical impairment
 - (iv) Treatment of musculoskeletal system, assessment for flying
 - (12) Psychiatry 2 hours
 - (i) Relation to aviation, risk of incapacitation
 - (ii) Psychiatric examination

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- (iii) Psychiatric disorders: neurosis; personality disorders; psychosis; organic mental illness
 - (iv) Alcohol and other psychoactive substance(s) use
 - (v) Treatment, rehabilitation and assessment
- (13) Psychology 2 hours
- (i) Introduction to psychology in aviation as a supplement to psychiatric assessment
 - (ii) Methods of psychological examination
 - (iii) Behaviour and personality
 - (iv) Workload management and situational awareness
 - (v) Flight motivation and suitability
 - (vi) Group social factors
 - (vii) Psychological stress, stress coping, fatigue
 - (viii) Psychomotor functions and age
 - (ix) Mental fitness and training
- (14) Neurology 3 hours
- (i) Relation to aviation, risk of incapacitation
 - (ii) Examination procedures
 - (iii) Neurological disorders
 - (A) Seizures — assessment of single episode
 - (B) Epilepsy
 - (C) Multiple sclerosis
 - (D) Head trauma
 - (E) Post-traumatic states
 - (F) Vascular diseases
 - (G) Tumours
 - (H) Disturbance of consciousness — assessment of single and repeated episodes
 - (iv) Degenerative diseases
 - (v) Sleep disorders
 - (vi) Treatment and assessment
- (15) Visual system and colour vision 4 hours
- (i) Anatomy of the eye
 - (ii) Relation to aviation duties
 - (iii) Examination techniques

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- (A) Visual acuity assessment
 - (B) Visual aids
 - (C) Visual fields — acceptable limits for certification
 - (D) Ocular muscle balance
 - (E) Assessment of pathological eye conditions
 - (F) Glaucoma
 - (iv) Monocularity and medical flight tests
 - (v) Colour vision
 - (vi) Methods of testing: pseudoisochromatic plates, lantern tests, anomaloscopy
 - (vii) Importance of standardisation of tests and of test protocols
 - (viii) Assessment after eye surgery
 - (16) Otorhinolaryngology 3 hours
 - (i) Anatomy of the systems
 - (ii) Clinical examination in ORL
 - (iii) Functional hearing tests
 - (iv) Vestibular system; vertigo, examination techniques
 - (v) Assessment after ENT surgery
 - (vi) Barotrauma ears and sinuses
 - (vii) Aeronautical ENT pathology
 - (viii) ENT requirements
 - (17) Oncology
 - (i) Relation to aviation, risk of metastasis and incapacitation
 - (ii) Risk management
 - (iii) Different methods of treatment and assessment
 - (18) Incidents and accidents, escape and survival 1 hour
 - (i) Accident statistics
 - (ii) Injuries
 - (iii) Aviation pathology, post-mortem examination, identification
 - (iv) Aircraft evacuation
 - (A) Fire
 - (B) Ditching
 - (C) By parachute
 - (19) Medication and flying 2 hours

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- (i) Hazards of medications
 - (ii) Common side effects; prescription medications; over-the-counter medications; herbal medications; ‘alternative’ therapies
 - (iii) Medication for sleep disturbance
 - (20) Legislation, rules and regulations 4 hours
 - (i) ICAO Standards and Recommended Practices, European provisions (e.g. Implementing Rules, AMC and GM)
 - (ii) Incapacitation: acceptable aero-medical risk of incapacitation; types of incapacitation; operational aspects
 - (iii) Basic principles in assessment of fitness for aviation
 - (iv) Operational and environmental conditions
 - (v) Use of medical literature in assessing medical fitness; differences between scientific study populations and licensed populations
 - (vi) Flexibility
 - (vii) Annex 1 to the Chicago Convention, paragraph 1.2.4.9
 - (viii) Accredited Medical Conclusion; consideration of knowledge, skill and experience
 - (ix) Trained versus untrained crews; incapacitation training
 - (x) Medical flight tests
 - (21) Cabin crew working environment 1 hour
 - (i) Cabin environment, workload, duty and rest time, fatigue risk management
 - (ii) Cabin crew safety duties and associated training
 - (iii) Types of aircraft and types of operations
 - (iv) Single-cabin crew and multi-cabin crew operations
 - (22) In-flight environment 1 hour
 - (i) Hygiene aboard aircraft: water supply, oxygen supply, disposal of waste, cleaning, disinfection and disinsection
 - (ii) Catering
 - (iii) Crew nutrition
 - (iv) Aircraft and transmission of diseases
 - (23) Space medicine 1 hour
 - (i) Microgravity and metabolism, life sciences
 - (24) Practical demonstrations of basic aeronautical knowledge 8 hours
 - (25) Concluding items 2 hours
 - (i) Final examination
 - (ii) De-briefing and critique
-

GM2 MED.D.020 Training courses in aviation medicine

ED Decision 2019/002/R

ADVANCED TRAINING COURSE

- | | | |
|-------|--|----------|
| (a) | Advanced training course in aviation medicine | 66 hours |
| (1) | Pilot working environment | 6 hours |
| (i) | Commercial aircraft flight crew compartment | |
| (ii) | Business jets, commuter flights, cargo flights | |
| (iii) | Professional airline operations | |
| (iv) | Fixed wing and helicopter, specialised operations including aerial work | |
| (v) | Air traffic control | |
| (vi) | Single-pilot/multi-pilot | |
| (vii) | Exposure to radiation and other harmful agents | |
| (2) | Aerospace physiology | 4 hours |
| (i) | Brief review of basics in physiology (hypoxia, rapid/slow decompression, hyperventilation, acceleration, ejection, spatial disorientation) | |
| (ii) | Simulator sickness | |
| (3) | Clinical medicine | 5 hours |
| (i) | Complete physical examination | |
| (ii) | Review of basics with relationship to commercial flight operations | |
| (iii) | Class 1 requirements | |
| (iv) | Clinical cases | |
| (v) | Communication and interview techniques | |
| (4) | Cardiovascular system | 4 hours |
| (i) | Cardiovascular examination and review of basics | |
| (ii) | Class 1 requirements | |
| (iii) | Diagnostic steps in cardiovascular system | |
| (iv) | Clinical cases | |
| (5) | Neurology | 3 hours |
| (i) | Brief review of basics (neurological and psychiatric examination) | |
| (ii) | Alcohol and other psychoactive substance(s) use | |
| (iii) | Class 1 requirements | |
| (iv) | Clinical cases | |
| (6) | Psychiatry/psychology | 5 hours |
| (i) | Brief review of basics (psychiatric/psychological evaluation techniques) | |

-
- (ii) Alcohol and other psychoactive substance(s) use
 - (iii) Class 1 requirements
 - (iv) Clinical cases
 - (7) Visual system and colour vision 5 hours
 - (i) Brief review of basics (visual acuity, refraction, colour vision, visual fields, night vision, stereopsis, monocularly)
 - (ii) Class 1 visual requirements
 - (iii) Implications of refractive and other eye surgery
 - (iv) Clinical cases
 - (8) Otorhinolaryngology 4 hours
 - (i) Brief review of basics (barotrauma — ears and sinuses, functional hearing tests)
 - (ii) Noise and its prevention
 - (iii) Vibration, kinetosis
 - (iv) Class 1 hearing requirements
 - (v) Clinical cases
 - (9) Dentistry 2 hours
 - (i) Oral examination including dental formula
 - (ii) Oral cavity, dental disorders and treatment, including implants, fillings, prosthesis, etc.
 - (iii) Barodontalgia
 - (iv) Clinical cases
 - (10) Human factors in aviation, including 8 hours demonstration and practical experience 22 hours
 - (i) Long-haul flight operations
 - (A) Flight time limitations
 - (B) Sleep disturbance
 - (C) Extended/expanded crew
 - (D) Jet lag/time zones
 - (ii) Human information processing and system design
 - (A) Flight Management System (FMS), Primary Flight Display (PFD), datalink, fly by wire
 - (B) Adaptation to the glass cockpit
 - (C) Crew Coordination Concept (CCC), Crew Resource Management (CRM), Line Oriented Flight Training (LOFT) etc.
 - (D) Practical simulator training

- (E) Ergonomics
- (iii) Crew commonality
 - (A) Flying under the same type rating, e.g. A-318, A-319, A-320, A-321
- (iv) Human factors in aircraft incidents and accidents
- (v) Flight safety strategies in commercial aviation
- (vi) Fear and refusal of flying
- (vii) Psychological selection criteria
- (viii) Operational requirements (flight time limitation, fatigue risk management, etc.)
- (11) Incidents and accidents, escape and survival 2 hours
 - (i) Accident statistics
 - (ii) Types of injuries
 - (iii) Aviation pathology, post-mortem examination related to aircraft accidents, identification
 - (iv) Rescue and emergency evacuation
- (12) Tropical medicine 2 hours
 - (i) Endemicity of tropical disease
 - (ii) Infectious diseases (communicable diseases, sexually transmitted diseases, HIV etc.)
 - (iii) Vaccination of flight crew and passengers
 - (iv) Diseases transmitted by vectors
 - (v) Food and water-borne diseases
 - (vi) Parasitic diseases
 - (vii) International health regulations
 - (viii) Personal hygiene of aviation personnel
- (13) Concluding items 2 hours
 - (i) Final examination
 - (ii) De-briefing and critique

GM3 MED.D.020 Training courses in aviation medicine

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GENERAL

(a) Principles of training:

To acquire knowledge and skills for the aero-medical examination and assessment, the training should be:

- (1) based on regulations;

- (2) based on general clinical skills and knowledge necessary to conduct relevant examinations for the different medical certificates;
 - (3) based on knowledge of the different risk assessments required for various types of medical certification;
 - (4) based on an understanding of the limits of the decision-making competences of an AME in assessing safety-critical medical conditions for when to defer and when to deny;
 - (5) based on knowledge of the aviation environment; and
 - (6) exemplified by clinical cases and practical demonstrations.
- (b) Training outcomes:
- The trainee should demonstrate a thorough understanding of:
- (1) the aero-medical examination and assessment process:
 - (i) principles, requirements and methods;
 - (ii) ability to investigate all clinical aspects that present aero-medical risks, the reasonable use of additional investigations;
 - (iii) the role in the assessment of the ability of the pilot or cabin crew member to safely perform their duties in special cases, such as the medical flight test;
 - (iv) aero-medical decision-making based on risk management;
 - (v) medical confidentiality; and
 - (vi) correct use of appropriate forms, and the reporting and storing of information;
 - (2) the conditions under which the pilots and cabin crew carry out their duties; and
 - (3) principles of preventive medicine, including aero-medical advice in order to help prevent future limitations.
- (c) The principles and training outcomes stated at (a) and (b) should also be taken into consideration for refresher training programmes

MED.D.025 Changes to the AME certificate

Regulation (EU) 2019/27

- (a) Holders of an AME certificate shall, without undue delay, notify the competent authority of the following circumstances which could affect their AME certificate:
- (1) the AME is subject to disciplinary proceedings or investigation by a medical regulatory body;
 - (2) there are changes to the conditions under which the certificate was granted, including the content of the statements provided with the application;
 - (3) the requirements for the issuance of the AME certificate are no longer met;
 - (4) there is a change to the aero-medical examiner's practice location(s) or correspondence address.

- (b) Failure to notify the competent authority in accordance with point (a) shall result in the suspension or revocation of the AME certificate in accordance with point ARA.MED.250 of Annex II (Part-ARA).

MED.D.030 Validity of AME certificates

Regulation (EU) 2019/27

An AME certificate shall be valid for a period of 3 years, unless the competent authority decides to reduce that period for duly justified reasons related to the individual case.

Upon application by the holder, the certificate shall be:

- (a) revalidated, provided that the holder:
- (1) continues to fulfil the general conditions required for medical practice and maintains his or her licence for the practice of medicine;
 - (2) has undertaken refresher training in aviation medicine within the last 3 years;
 - (3) has performed at least 10 aero-medical examinations or equivalent every year;
 - (4) remains in compliance with the terms of the certificate;
 - (5) exercises the privileges in accordance with the requirements of this Annex (Part-MED);
 - (6) has demonstrated that he or she maintains his or her aero-medical competency in accordance with the procedure established by the competent authority.
- (b) renewed, provided that the holder complies with either the requirements for revalidation set out in point (a) or with all of the following requirements:
- (1) continues to fulfil the general conditions required for medical practice and maintains his or her licence for the practice of medicine;
 - (2) has undertaken refresher training in aviation medicine within the previous year;
 - (3) has successfully completed practical training within the previous year, either at an AeMC or under the supervision of the competent authority;
 - (4) remains in compliance with the requirements of point [MED.D.010](#);
 - (5) has demonstrated that he or she maintains his or her aero-medical competency in accordance with the procedure established by the competent authority.

AMC1 MED.D.030 Validity of AME certificates

ED Decision 2019/002/R

REFRESHER TRAINING

- (a) It is the responsibility of the AME to continuously maintain and improve their competencies.
- (b) During the period of validity of the AME certificate, an AME should attend a minimum of 20 hours of refresher training.
- (c) An AME exercising class 1 privileges should attend at least 10 hours of refresher training per year.

- (d) A proportionate number of refresher training hours should be provided by, or conducted under the direct supervision of, the competent authority or the medical assessor.
- (e) The curricula of refresher training hours referred to in (c) should be decided by the competent authority following a risk-based assessment.
- (f) Attendance at scientific meetings and congresses, and flight deck experience may be credited by the competent authority for a specified number of hours against the training obligations of the AME, provided the competent authority has assessed it in advance as being relevant for crediting purposes.
- (g) In case of renewal of an AME certificate, the practical training should include at least 10 aero-medical assessments, in accordance with the type of the requested AME certificate.

GM1 MED.D.030 Validity of AME certificates

ED Decision 2019/002/R

REFRESHER TRAINING

- (a) The curricula for the refresher training hours that should be provided by, or conducted under the direct supervision of, the competent authority or the medical assessor may include but are not limited to subjects such as:
 - (1) Psychiatry
 - (i) Relation to aviation, risk of incapacitation;
 - (ii) Psychiatric examination;
 - (iii) Psychiatric disorders: neurosis, personality disorders, psychosis, organic mental illness;
 - (iv) Alcohol and other psychoactive substance(s) use; and
 - (v) Treatment, rehabilitation and assessment.
 - (2) Psychology
 - (i) Introduction to psychology in aviation as a supplement to psychiatric assessment;
 - (ii) Methods of psychological examination;
 - (iii) Behaviour and personality;
 - (iv) Workload management and situational awareness;
 - (v) Flight motivation and suitability;
 - (vi) Group social factors;
 - (vii) Psychological stress, stress coping, fatigue;
 - (viii) Psychomotor functions and age; and
 - (ix) Mental fitness and training.
 - (3) Communication and interview techniques
- (b) Scientific meetings, congresses or flight deck experience that may be credited by the competent authority:

International Academy of Aviation and Space Medicine Annual Congresses (ICASM)	10 hours credit
European Conference of Aerospace Medicine (ECAM)	10 hours credit
Aerospace Medical Association Annual Scientific Meetings (AsMA)	10 hours credit
Other scientific meetings (A minimum of 6 hours to be under the direct supervision of the medical assessor of the competent authority)	10 hours credit
Flight crew compartment experience (a maximum of 5 hours credit per 3 years):	
(i) Jump seat	5 sectors — 1 hour credit
(ii) Simulator	4 hours — 1 hour credit
(iii) Aircraft piloting	4 hours — 1 hour credit
(c) An AME exercising class 1 revalidation/renewal privileges should attend international aviation medicine scientific meetings or congresses at regular intervals.	
(d) Aero-medical examinations of military pilots may be considered as equivalent in accordance with MED.D.030(a)(3) , subject to approval by the medical assessor of the competent authority.	

GM2 MED.D.030 Validity of AME certificates

ED Decision 2019/002/R

AME PEER SUPPORT GROUPS

- (a) The competent authority should promote better performance of AMEs by supporting the establishment of AME peer support groups that could provide both professional support and educational enhancement.
- (b) Attendance to AME peer support group meetings may be credited by the competent authority as refresher training. The competent authority should determine a maximum of hours that can be credited as refresher training during the period of authorisation.
- (c) AME peer support groups may be established as part of, or complementary to, national associations of aerospace medicine.

SECTION 2 – GENERAL MEDICAL PRACTITIONERS

MED.D.035 Requirements for general medical practitioners

Regulation (EU) 2019/27

General medical practitioners (GMPs) may act as AMEs for issuing LAPL medical certificates, where they meet all of the following conditions:

- (a) they exercise their activity in a Member State where GMPs have access to the full medical records of applicants;
- (b) they exercise their activity in accordance with any additional requirements established in the national law of the Member State of their competent authority;
- (c) they are fully qualified and licensed for the practice of medicine in accordance with national law of the Member State of their competent authority;
- (d) they have notified the competent authority before starting such activity.

SECTION 3 – OCCUPATIONAL HEALTH MEDICAL PRACTITIONERS

MED.D.040 Requirements for occupational health medical practitioners

Regulation (EU) 2019/27

In Member States where the competent authority is satisfied that the requirements of the national health system applicable to occupational health medical practitioners (OHMPs) are such as to ensure compliance with the requirements of this Annex (Part-MED) applicable to OHMPs, OHMPs may conduct aero-medical assessments of cabin crew, provided that:

- (a) they are fully qualified and licensed in the practice of medicine and qualified in occupational medicine;
- (b) the in-flight working environment and safety duties of the cabin crew were included in their occupational medicine qualification syllabus or other training or operational experience;
- (c) they have notified the competent authority before starting such activity.

ANNEX V (PART-CC)

SUBPART GEN – GENERAL REQUIREMENTS

CC.GEN.001 Competent authority

Regulation (EU) No 1178/2011

For the purpose of this Part, the competent authority shall be the authority designated by the Member State where a person applies for the issue of a cabin crew attestation.

CC.GEN.005 Scope

Regulation (EU) No 1178/2011

This Part establishes the requirements for the issue of cabin crew attestations and the conditions for their validity and use by their holders.

CC.GEN.015 Application for a cabin crew attestation

Regulation (EU) No 1178/2011

The application for a cabin crew attestation shall be made in a form and manner established by the competent authority.

CC.GEN.020 Minimum age

Regulation (EU) No 1178/2011

The applicant for a cabin crew attestation shall be at least 18 years of age.

CC.GEN.025 Privileges and conditions

Regulation (EU) No 1178/2011

- (a) The privileges of holders of a cabin crew attestation are to act as cabin crew members in commercial air transport operation of aircraft referred to in Article 4(1)(b) and (c) of Regulation (EC) No 216/2008.
- (b) Cabin crew members may exercise the privileges specified in (a) only if they:
 - (1) hold a valid cabin crew attestation as specified in [CC.CCA.105](#); and
 - (2) comply with [CC.GEN.030](#), [CC.TRA.225](#) and the applicable requirements of Part-MED.

CC.GEN.030 Documents and record-keeping

Regulation (EU) No 1178/2011

To show compliance with the applicable requirements as specified in [CC.GEN.025\(b\)](#), each holder shall keep, and provide upon request, the cabin crew attestation, the list and the training and checking records of his/her aircraft type or variant qualification(s), unless the operator employing his/her services keeps such records and can make them readily available upon request by a competent authority or by the holder.

SUBPART CCA – SPECIFIC REQUIREMENTS FOR THE CABIN CREW ATTESTATION

CC.CCA.100 Issue of the cabin crew attestation

Regulation (EU) No 1178/2011

- (a) Cabin crew attestations shall only be issued to applicants who have passed the examination following completion of the initial training course in accordance with this Part.
- (b) Cabin crew attestations shall be issued:
 - (1) by the competent authority; and/or
 - (2) by an organisation approved to do so by the competent authority.

CC.CCA.105 Validity of the cabin crew attestation

Regulation (EU) No 1178/2011

The cabin crew attestation shall be issued with unlimited duration and shall remain valid unless:

- (a) it is suspended or revoked by the competent authority; or
- (b) its holder has not exercised the associated privileges during the preceding 60 months on at least one aircraft type.

CC.CCA.110 Suspension and revocation of the cabin crew attestation

Regulation (EU) No 1178/2011

- (a) If holders do not comply with this Part, their cabin crew attestation may be suspended or revoked by the competent authority.
- (b) In case of suspension or revocation of their cabin crew attestation by the competent authority, holders shall:
 - (1) be informed in writing of this decision, and of their right of appeal in accordance with national law;
 - (2) not exercise the privileges granted by their cabin crew attestation;
 - (3) inform, without undue delay, the operator(s) employing their services; and
 - (4) return their attestation in accordance with the applicable procedure established by the competent authority.

SUBPART TRA – TRAINING REQUIREMENTS FOR CABIN CREW ATTESTATION APPLICANTS AND HOLDERS

CC.TRA.215 Provision of training

Regulation (EU) No 1178/2011

Training required in this Part shall be:

- (a) provided by training organisations or commercial air transport operators approved to do so by the competent authority;
- (b) performed by personnel suitably experienced and qualified for the training elements to be covered; and
- (c) conducted according to a training programme and syllabus documented in the organisation's approval.

CC.TRA.220 Initial training course and examination

Regulation (EU) No 1178/2011

- (a) Applicants for a cabin crew attestation shall complete an initial training course to familiarise themselves with the aviation environment and to acquire sufficient general knowledge and basic proficiency required to perform the duties and discharge the responsibilities related to the safety of passengers and flight during normal, abnormal and emergency operations.
- (b) The programme of the initial training course shall cover at least the elements specified in [Appendix 1](#) to this Part. It shall include theoretical and practical training.
- (c) Applicants for a cabin crew attestation shall undergo an examination covering all elements of the training programme specified in (b), except CRM training, to demonstrate that they have attained the level of knowledge and proficiency required in (a).

CC.TRA.225 Aircraft type or variant qualification(s)

Regulation (EU) No 1178/2011

- (a) Holders of a valid cabin crew attestation shall only operate on an aircraft if they are qualified in accordance with the applicable requirements of Part-ORO.
- (b) To be qualified for an aircraft type or a variant, the holder:
 - (1) shall comply with the applicable training, checking and validity requirements, covering as relevant to the aircraft to be operated:
 - (i) aircraft-type specific training, operator conversion training and familiarisation;
 - (ii) differences training;
 - (iii) recurrent training; and
 - (2) shall have operated within the preceding 6 months on the aircraft type, or shall have completed the relevant refresher training and checking before operating again on that aircraft type.

APPENDIX TO ANNEX V

Appendix 1 to Part-CC Initial training course and examination

Regulation (EU) No 290/2012

TRAINING PROGRAMME

The training programme of the initial training course shall include at least the following:

1. General theoretical knowledge of aviation and aviation regulations covering all elements relevant to the duties and responsibilities required from cabin crew:

- 1.1. aviation terminology, theory of flight, passenger distribution, areas of operation, meteorology and effects of aircraft surface contamination;
- 1.2. aviation regulations relevant to cabin crew and the role of the competent authority;
- 1.3. duties and responsibilities of cabin crew during operations and the need to respond promptly and effectively to emergency situations;
- 1.4. continuing competence and fitness to operate as a cabin crew member, including as regards flight and duty time limitations and rest requirements;
- 1.5. the importance of ensuring that relevant documents and manuals are kept up-to-date, with amendments provided by the operator as applicable;
- 1.6. the importance of cabin crew performing their duties in accordance with the operations manual of the operator;
- 1.7. the importance of the cabin crew's pre-flight briefing and the provision of necessary safety information with regards to their specific duties; and
- 1.8. the importance of identifying when cabin crew members have the authority and responsibility to initiate an evacuation and other emergency procedures.

2. Communication:

During training, emphasis shall be placed on the importance of effective communication between cabin crew and flight crew, including communication techniques, common language and terminology.

3. Introductory course on human factors (HF) in aviation and crew resource management (CRM)

This course shall be conducted by at least one cabin crew CRM instructor. The training elements shall be covered in depth and shall include at least the following:

- 3.1. General: human factors in aviation, general instructions on CRM principles and objectives, human performance and limitations;
- 3.2. Relevant to the individual cabin crew member: personality awareness, human error and reliability, attitudes and behaviours, self-assessment; stress and stress management; fatigue and vigilance; assertiveness; situation awareness, information acquisition and processing.

4. Passenger handling and cabin surveillance:

- 4.1. the importance of correct seat allocation with reference to aeroplane mass and balance, special categories of passengers and the necessity of seating able-bodied passengers adjacent to unsupervised exits;

- 4.2. rules covering the safe stowage of cabin baggage and cabin service items and the risk of it becoming a hazard to occupants of the passenger compartment or otherwise obstruction or damaging emergency equipment or exits;
- 4.3. advice on the recognition and management of passengers who are, or become, intoxicated with alcohol or are under the influence of drugs or are aggressive;
- 4.4. precautions to be taken when live animals are carried in the passenger compartment;
- 4.5. duties to be undertaken in the event of turbulence, including securing the passenger compartment; and
- 4.6. methods used to motivate passengers and the crowd control necessary to expedite an emergency evacuation.

5. Aero-medical aspects and first-aid:

- 5.1. general instruction on aero-medical aspects and survival;
- 5.2. the physiological effects of flying with particular emphasis on hypoxia, oxygen requirements, Eustachian tubal function and barotraumas;
- 5.3. basic first-aid, including care of:
 - (a) air sickness;
 - (b) gastro-intestinal disturbances;
 - (c) hyperventilation;
 - (d) burns;
 - (e) wounds;
 - (f) the unconscious; and
 - (g) fractures and soft tissue injuries;
- 5.4. in-flight medical emergencies and associated first-aid covering at least:
 - (a) asthma;
 - (b) stress and allergic reactions;
 - (c) shock;
 - (d) diabetes;
 - (e) choking;
 - (f) epilepsy;
 - (g) childbirth;
 - (h) stroke; and
 - (i) heart attack;
- 5.5. the use of appropriate equipment including first-aid oxygen, first-aid kits and emergency medical kits and their contents;
- 5.6. practical cardio-pulmonary resuscitation training by each cabin crew member using a specifically designed dummy and taking account of the characteristics of an aircraft environment; and
- 5.7. travel health and hygiene, including:

- (a) hygiene on board;
- (b) risk of contact with infectious diseases and means to reduce such risks;
- (c) handling of clinical waste;
- (d) aircraft disinsection;
- (e) handling of death on board; and
- (f) alertness management, physiological effects of fatigue, sleep physiology, circadian rhythm and time zone changes.

6. Dangerous goods in accordance with the applicable ICAO Technical Instructions.

7. General security aspects in aviation, including awareness of the provisions laid down in Regulation (EC) No 300/2008.

8. Fire and smoke training:

- 8.1. emphasis on the responsibility of cabin crew to deal promptly with emergencies involving fire and smoke and, in particular, emphasis on the importance of identifying the actual source of the fire;
- 8.2. the importance of informing the flight crew immediately, as well as the specific actions necessary for coordination and assistance, when fire or smoke is discovered;
- 8.3. the necessity for frequent checking of potential fire-risk areas including toilets, and the associated smoke detectors;
- 8.4. the classification of fires and the appropriate type of extinguishing agents and procedures for particular fire situations;
- 8.5. the techniques of application of extinguishing agents, the consequences of misapplication, and of use in a confined space including practical training in fire-fighting and in the donning and use of smoke protection equipment used in aviation; and
- 8.6. the general procedures of ground-based emergency services at aerodromes.

9. Survival training:

- 9.1. principles of survival in hostile environments (e.g. polar, desert, jungle, sea) ; and
- 9.2. water survival training which shall include the actual donning and use of personal flotation equipment in water and the use of slide-rafts or similar equipment, as well as actual practice in water.

AMC1 Appendix 1 to Part-CC(3) Initial training course and examination

ED Decision 2015/023/R

CREW RESOURCE MANAGEMENT TRAINING TABLE

The CRM training table recapitulates all elements relevant to CRM training for cabin crew, specifying the following:

- The elements of the introductory course on CRM required for the cabin crew initial training course, where 'in-depth' means a training that should be instructional or interactive in style taking full advantage of group discussions, team task analysis, team task simulation, etc., for the acquisition of knowledge, skills and attitudes.
- The elements identified as 'not required' for the cabin crew initial training are listed for information as they are covered during other training in accordance with the applicable requirements of Annex III (Part-ORO) to Commission Regulation (EU) No 965/2012.

CRM TRAINING TABLE	
Training elements	Introductory course on CRM
General Principles	
Human factors in aviation; General instructions on CRM principles and objectives; Human performance and limitations; Threat and error management.	In-depth
Relevant to the individual cabin crew member	
Personality awareness, human error and reliability, attitudes and behaviours, selfassessment and self-critique; Stress and stress management; Fatigue and vigilance; Assertiveness; situation awareness, information acquisition and processing.	In-depth
Relevant to the entire aircraft crew	
Shared situation awareness, shared information acquisition and processing; Workload management; Effective communication and coordination between all crew members including the flight crew as well as inexperienced cabin crew members; Leadership, cooperation, synergy, delegation, decision-making, actions; Resilience development; Surprise and startle effect; Cultural differences; Identification and management of passenger human factors: crowd control, passenger stress, conflict management, medical factors.	Not required (covered under CRM training required by Part-ORO)
Specifics related to aircraft types (narrow-/wide-bodied, single-/multi-deck), flight crew and cabin crew composition and number of passengers	
Relevant to the operator and the organisation (principles)	
Operator’s safety culture and company culture, standard operating procedures (SOPs), organisational factors, factors linked to the type of operations; Effective communication and coordination with other operational personnel and ground services; Participation in cabin safety incident and accident reporting.	Not required (covered under CRM training required by Part-ORO)
Case studies	

ANNEX VI (PART-ARA)

List of acronyms used throughout this Annex

ED Decision 2018/009/R

The following provides a list of acronyms used throughout this Annex:

(A)	aeroplane
(H)	helicopter
A/C	aircraft
ACAS	airborne collision avoidance system
AD	airworthiness directive
AIS	aeronautical information services
AM	accountable manager
AeMC	aero-medical centre
AMC	acceptable means of compliance
AME	aero-medical examiner
APP	approach
APU	auxiliary power unit
ARA	authority requirements for aircrew
ATC	air traffic control
ATO	approved training organisation
ATPL	airline transport pilot licence
BITD	basic instrument training device
BPL	balloon pilot licence
bpm	beats per minute
CAT	category
CBT	computer-based training
CC	cabin crew
CFI	chief flying instructor
cm	centimetres
CM	compliance monitoring
CMP	compliance-monitoring programme
CMS	compliance-monitoring system
COP	code of practice
CPL	commercial pilot licence
CRM	crew resource management
CS	certification specifications
CS-FSTD(A)	Certification Specifications for aeroplane flight simulation training devices
CS-FSTD(H)	Certification Specifications for helicopter flight simulation training devices
CTKI	chief theoretical-knowledge instructor
dB	decibel
DG	dangerous goods
DH	decision height
DPATO	defined point after take-off
DPBL	decision point before landing
EC	European Community
ECG	electrocardiogram
ENT	ear, nose and throat
EOG	electro-oculography
ERP	emergency response plan
ETOPS	extended-range operations with twin-engined aeroplanes
FANS	future air navigation system

FATO	final approach and take-off area
FD	flight director
FEV ₁	forced expiratory volume in 1 second
FFS	full flight simulator
FMGC	flight management and guidance computer
FMS	flight management system
FNPT	flight navigation and procedures trainer
FSTD	flight simulation training device
ft	feet
FTD	flight training device
FTI	flight test instructor
FVC	forced vital capacity
GM	guidance material
GMP	general medical practitioner
GPS	global positioning system
HEMS	helicopter emergency medical service
HF	human factors
Hg	mercury
HHO	helicopter hoist operation
HT	head of training
Hz	Hertz
IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
IFR	instrument flight rules
IGE	in-ground effect
ILS	instrument landing system
IMC	instrument meteorological conditions
IOS	instructor operating station
IR	instrument rating
kg	kilogram
LAPL	light aircraft pilot licence
LDP	landing decision point
LIFUS	line flying under supervision
LVO	low-visibility operation
LVTO	low visibility take-off
MCC	multi-crew cooperation
MMEL	master minimum equipment list
MPA	multi-pilot aeroplane
MPL	multi-crew pilot licence
NVIS	night vision imaging system
m	metre
mm	millimetre
OGE	out-of-ground effect
OPC	operator proficiency check
ORA	organisation requirements for aircrew
ORO	organisation requirements for air operations
OSD	operational suitability data
OTD	other training device
PBN	performance-based navigation
PF	pilot flying
PIC	pilot-in-command
PM	pilot monitoring
POM	proof of match
PPL	private pilot licence
QTG	qualification test guide

ROD	rate of descent
RVR	runway visual range
RWY	runway
SMM	safety management manual
SOP	standard operating procedure
SPL	sailplane pilot licence
TAWS	terrain avoidance and warning system
TDP	take-off decision point
TRE	type rating examiner
TRI	type rating instructor
TWY	taxiway
VDR	validation data road map
VFR	visual flight rules
ZFTT	zero-flight-time training

SUBPART GEN – GENERAL REQUIREMENTS

SECTION I – GENERAL

ARA.GEN.115 Oversight documentation

Regulation (EU) No 1178/2011

The competent authority shall provide all legislative acts, standards, rules, technical publications and related documents to relevant personnel in order to allow them to perform their tasks and to discharge their responsibilities.

ARA.GEN.120 Means of compliance

Regulation (EU) No 290/2012

- (a) The Agency shall develop Acceptable Means of Compliance (AMC) that may be used to establish compliance with Regulation (EC) No 216/2008 and its Implementing Rules. When the AMC are complied with, the related requirements of the Implementing Rules are met.
- (b) Alternative means of compliance may be used to establish compliance with the Implementing Rules.
- (c) The competent authority shall establish a system to consistently evaluate that all alternative means of compliance used by itself or by organisations and persons under its oversight allow the establishment of compliance with Regulation (EC) No 216/2008 and its Implementing Rules.
- (d) The competent authority shall evaluate all alternative means of compliance proposed by an organisation in accordance with ORA.GEN.120 by analysing the documentation provided and, if considered necessary, conducting an inspection of the organisation.

When the competent authority finds that the alternative means of compliance are in accordance with the Implementing Rules, it shall without undue delay:

- (1) notify the applicant that the alternative means of compliance may be implemented and, if applicable, amend the approval or certificate of the applicant accordingly; and
 - (2) notify the Agency of their content, including copies of all relevant documentation;
 - (3) inform other MS about alternative means of compliance that were accepted.
- (e) When the competent authority itself uses alternative means of compliance to achieve compliance with Regulation (EC) No 216/2008 and its Implementing Rules it shall:
 - (1) make them available to all organisations and persons under its oversight; and
 - (2) without undue delay notify the Agency.

The competent authority shall provide the Agency with a full description of the alternative means of compliance, including any revisions to procedures that may be relevant, as well as an assessment demonstrating that the Implementing Rules are met.

AMC1 ARA.GEN.120(d)(3) Means of compliance

ED Decision 2012/006/R

GENERAL

The information to be provided to other Member States following approval of an alternative means of compliance should contain a reference to the Acceptable Means of Compliance (AMC) to which such means of compliance provides an alternative, as well as a reference to the corresponding Implementing Rule, indicating as applicable the subparagraph(s) covered by the alternative means of compliance.

GM1 ARA.GEN.120 Means of compliance

ED Decision 2012/006/R

GENERAL

Alternative means of compliance used by a competent authority or by organisations under its oversight may be used by other competent authorities or organisations only if processed again in accordance with [ARA.GEN.120\(d\) and \(e\)](#).

ARA.GEN.125 Information to the Agency

Regulation (EU) 2023/203

- (a) The competent authority shall notify the Agency in case of any significant problems with the implementation of [Regulation \(EU\) 2018/1139](#) and the delegated and implementing acts adopted on the basis thereof within 30 days from the time the authority became aware of the problems.
- (b) Without prejudice to [Regulation \(EU\) No 376/2014](#) of the European Parliament and of the Council¹ and the delegated and implementing acts adopted on the basis thereof, the competent authority shall provide the Agency with safety-significant information stemming from the occurrence reports stored in the national database, as soon as possible.
- (c) The competent authority of the Member State shall provide the Agency as soon as possible with safety-significant information stemming from the information security reports it has received pursuant to point IS.I.OR.230 of Annex II (Part-IS.I.OR) to [Implementing Regulation \(EU\) 2023/203](#).

[applicable from 22 February 2026 — Implementing Regulation (EU) 2023/203]

ARA.GEN.135 Immediate reaction to a safety problem

Regulation (EU) No 1178/2011

- (a) Without prejudice to [Regulation \(EU\) No 376/2014](#) and the delegated and implementing acts adopted on the basis thereof, the competent authority shall implement a system to appropriately collect, analyse and disseminate safety information.
- (b) The Agency shall implement a system to appropriately analyse any relevant safety information received and without undue delay provide to Member States and the Commission any information, including recommendations or corrective actions to be taken, necessary for them to react in a timely manner to a safety problem involving products, parts, non-installed equipment, persons or organisations subject to [Regulation \(EU\) 2018/1139](#) and the delegated and implementing acts adopted on the basis thereof.

¹ Regulation (EU) No 376/2014 of the European Parliament and of the Council of 3 April 2014 on the reporting, analysis and follow-up of occurrences in civil aviation, amending Regulation (EU) No 996/2010 of the European Parliament and of the Council and repealing Directive 2003/42/EC of the European Parliament and of the Council and Commission Regulations (EC) No 1321/2007 and (EC) No 1330/2007 ([OJ L 122, 24.4.2014, p. 18](#)).

- (c) Upon receiving the information referred to in (a) and (b), the competent authority shall take adequate measures to address the safety problem.
- (d) Measures taken under point (c) shall immediately be notified to all persons or organisations that need to comply with them under [Regulation \(EU\) 2018/1139](#) and the delegated and implementing acts adopted on the basis thereof. The competent authority shall also notify those measures to the Agency and, when combined action is required, the other Member States concerned.

ARA.GEN.135A Immediate reaction to an information security incident or vulnerability with an impact on aviation safety

Regulation (EU) 2023/203

- (a) The competent authority shall implement a system to appropriately collect, analyse, and disseminate information related to information security incidents and vulnerabilities with a potential impact on aviation safety that are reported by organisations. This shall be done in coordination with any other relevant authorities responsible for information security or cybersecurity within the Member State to increase the coordination and compatibility of reporting schemes.
- (b) The Agency shall implement a system to appropriately analyse any relevant safety-significant information received in accordance with point [ARA.GEN.125\(c\)](#), and without undue delay provide the Member States and the Commission with any information, including recommendations or corrective actions to be taken, necessary for them to react in a timely manner to an information security incident or vulnerability with a potential impact on aviation safety involving products, parts, non-installed equipment, persons or organisations subject to [Regulation \(EU\) 2018/1139](#) and its delegated and implementing acts.
- (c) Upon receiving the information referred to in points (a) and (b), the competent authority shall take adequate measures to address the potential impact on aviation safety of the information security incident or vulnerability.
- (d) Measures taken in accordance with point (c) shall immediately be notified to all persons or organisations that shall comply with them under [Regulation \(EU\) 2018/1139](#) and its delegated and implementing acts. The competent authority of the Member State shall also notify those measures to the Agency and, when combined action is required, the competent authorities of the other Member States concerned.

[applicable from 22 February 2026 — Implementing Regulation (EU) 2023/203]

AMC1 ARA.GEN.135A Immediate reaction to an information security incident or vulnerability with an impact on aviation safety

ED Decision 2023/010/R

- (a) To appropriately collect and analyse information related to information security incidents and vulnerabilities with a potential impact on aviation safety, the competent authority should implement means that ensure the necessary confidentiality.
- (b) When disseminating information related to information security incidents and vulnerabilities with a potential impact on aviation safety, the competent authority should properly select the appropriate recipient(s) to prevent the content of a report from being exploited to the detriment of aviation safety, by revealing, for instance, uncorrected vulnerabilities.

[applicable from 22 February 2026 — ED Decision 2023/010/R]

GM1 ARA.GEN.135A Immediate reaction to an information security incident or vulnerability with an impact on aviation safety

ED Decision 2023/010/R

When deemed necessary, a two-step mechanism could be used: a report alerting about the information security event or incident and the availability of additional data that would require controlled and confidential distribution. This report should only alert recipients of the urgency and the necessity for organisations and competent authorities to establish further communication through secure means.

Therefore, the report should consist of two parts: one limited to mostly public information and one containing the sensitive data that should be restricted to the recipients who need to know. Wherever possible, reports should be based on an agreed taxonomy.

[applicable from 22 February 2026 — ED Decision 2023/010/R]

SECTION II – MANAGEMENT

ARA.GEN.200 Management system

Regulation (EU) 2023/203

- (a) The competent authority shall establish and maintain a management system, including as a minimum:
- (1) documented policies and procedures to describe its organisation, means and methods to achieve compliance with [Regulation \(EU\) 2018/1139](#) and the delegated and implementing acts adopted on the basis thereof. The procedures shall be kept up to date and serve as the basic working documents within that competent authority for all related tasks;
 - (2) a sufficient number of personnel to perform its tasks and discharge its responsibilities. Such personnel shall be qualified to perform their allocated tasks and have the necessary knowledge, experience, initial and recurrent training to ensure continuing competence. A system shall be in place to plan the availability of personnel, in order to ensure the proper completion of all tasks;
 - (3) adequate facilities and office accommodation to perform the allocated tasks;
 - (4) a function to monitor compliance of the management system with the relevant requirements and adequacy of the procedures including the establishment of an internal audit process and a safety risk management process. Compliance monitoring shall include a feedback system of audit findings to the senior management of the competent authority to ensure implementation of corrective actions as necessary; and
 - (5) a person or group of persons, ultimately responsible to the senior management of the competent authority for the compliance monitoring function.
- (b) The competent authority shall, for each field of activity including management system, appoint one or more persons with the overall responsibility for the management of the relevant task(s).
- (c) The competent authority shall establish procedures for participation in a mutual exchange of all necessary information and assistance with other competent authorities concerned, whether from within the Member State or in other Member States, including the following information:
- (1) on all findings raised, corrective follow-up actions taken pursuant to such findings and enforcement measures taken as a result of oversight of persons and organisations exercising activities in the territory of a Member State but certified by or having made declarations to the competent authority of another Member State or the Agency;
 - (2) stemming from mandatory and voluntary occurrence reporting as required by point [ORA.GEN.160](#) of Annex VII.
- (d) A copy of the procedures related to the management system and their amendments shall be made available to the Agency for the purpose of standardisation.
- (e) In addition to the requirements contained in point (a), the management system established and maintained by the competent authority shall comply with Annex I (Part-IS.AR) to [Implementing Regulation \(EU\) 2023/203](#) in order to ensure the proper management of information security risks which may have an impact on aviation safety.

[applicable from 22 February 2026 — Implementing Regulation (EU) 2023/203]

AMC1 ARA.GEN.200(a) Management system

ED Decision 2018/009/R

GENERAL

- (a) All of the following should be considered when deciding upon the required organisational structure:
- (1) the number of certificates, attestations, authorisations and approvals to be issued;
 - (2) the number of declared training organisations;
 - (3) the number of certified persons and organisations exercising an activity within that Member State, including persons or organisations certified by, or having made a declaration to, other competent authorities;
 - (4) the possible use of qualified entities and of resources of other competent authorities to fulfil the continuing oversight obligations;
 - (5) the level of civil aviation activity in terms of:
 - (i) number and complexity of aircraft operated;
 - (ii) size and complexity of the Member State's aviation industry;
 - (6) the potential growth of activities in the field of civil aviation.
- (b) The set-up of the organisational structure should ensure that the various tasks and obligations of the competent authority do not rely solely on individuals. A continuous and undisturbed fulfilment of these tasks and obligations of the competent authority should also be guaranteed in case of illness, accident or leave of individual employees.

GM1 ARA.GEN.200(a) Management system

ED Decision 2012/006/R

GENERAL

- (a) The competent authority designated by each Member State should be organised in such a way that:
- (1) there is specific and effective management authority in the conduct of all relevant activities;
 - (2) the functions and processes described in the applicable requirements of Regulation (EC) No 216/2008¹ and its Implementing Rules and AMCs, Certification Specifications (CSs) and Guidance Material (GM) may be properly implemented;
 - (3) the competent authority's organisation and operating procedures for the implementation of the applicable requirements of Regulation (EC) No 216/2008 and its Implementing Rules are properly documented and applied;
 - (4) all competent authority personnel involved in the related activities are provided with training where necessary;

¹ Regulation (EC) No 216/2008 of the European Parliament and of the Council of 20 February 2008 on common rules in the field of civil aviation and establishing a European Aviation Safety Agency, and repealing Council Directive 91/670/EEC, Regulation (EC) No 1592/2002 and Directive 2004/36/EC. OJ L 79, 19.3.2008, p. 1.

- (5) specific and effective provision is made for the communication and interface as necessary with the Agency and the competent authorities of other Member States; and
- (6) all functions related to implementing the applicable requirements are adequately described.
- (b) A general policy in respect of activities related to the applicable requirements of Regulation (EC) No 216/2008 and its Implementing Rules should be developed, promoted and implemented by the manager at the highest appropriate level; for example the manager at the top of the functional area of the competent authority that is responsible for such activities.
- (c) Appropriate steps should be taken to ensure that the policy is known and understood by all personnel involved, and all necessary steps should be taken to implement and maintain the policy.
- (d) The general policy, whilst also satisfying additional national regulatory responsibilities, should in particular take into account:
 - (1) the provisions of Regulation (EC) No 216/2008;
 - (2) the provisions of the applicable Implementing Rules and their AMCs, CSs and GM;
 - (3) the needs of industry; and
 - (4) the needs of the Agency and of the competent authority.
- (e) The policy should define specific objectives for key elements of the organisation and processes for implementing related activities, including the corresponding control procedures and the measurement of the achieved standard.

AMC1 ARA.GEN.200(a)(1) Management system

ED Decision 2012/006/R

DOCUMENTED POLICIES AND PROCEDURES

- (a) The various elements of the organisation involved with the activities related to Regulation (EC) No 216/2008 and its Implementing Rules should be documented in order to establish a reference source for the establishment and maintenance of this organisation.
- (b) The documented procedures should be established in a way that facilitates their use. They should be clearly identified, kept up-to-date and made readily available to all personnel involved in the related activities.
- (c) The documented procedures should cover, as a minimum, all of the following aspects:
 - (1) policy and objectives;
 - (2) organisational structure;
 - (3) responsibilities and associated authority;
 - (4) procedures and processes;
 - (5) internal and external interfaces;
 - (6) internal control procedures;
 - (7) training of personnel;
 - (8) cross-references to associated documents;
 - (9) assistance from other competent authorities or the Agency (where required).

- (d) It is likely that the information is held in more than one document or series of documents, and suitable cross-referencing should be provided. For example, organisational structure and job descriptions are not usually in the same documentation as the detailed working procedures. In such cases it is recommended that the documented procedures include an index of cross-references to all such other related information, and the related documentation should be readily available when required.

AMC1 ARA.GEN.200(a)(2) Management system

ED Decision 2012/006/R

QUALIFICATION AND TRAINING - GENERAL

- (a) The competent authority should ensure appropriate and adequate training of its personnel to meet the standard that is considered necessary to perform the work. To ensure personnel remain qualified, arrangements should be made for initial and recurrent training as required.
- (b) The basic capability of the competent authority's personnel is a matter of recruitment and normal management functions in selection of personnel for particular duties. Moreover, the competent authority should provide training in the basic skills as required for those duties. However, to avoid differences in understanding and interpretation, all personnel should be provided with further training specifically related to Regulation (EC) No 216/2008, its Implementing Rules and related AMCs, CSs and GM, as well as related to the assessment of alternative means of compliance.
- (c) The competent authority may provide training through its own training organisation with qualified trainers or through another qualified training source.
- (d) When training is not provided through an internal training organisation, adequately experienced and qualified persons may act as trainers, provided their training skills have been assessed. If required, an individual training plan should be established covering specific training skills. Records should be kept of such training and of the assessment, as appropriate.

AMC2 ARA.GEN.200(a)(2) Management system

ED Decision 2017/022/R

QUALIFICATION AND TRAINING - INSPECTORS

- (a) Qualification
- (1) All inspectors should receive, as appropriate to their role, training in the following areas:
- (i) auditing techniques, as relevant to the particular duties and responsibilities of the inspector;
 - (ii) safety management systems (SMSs);
 - (iii) compliance monitoring system (CMSs);
 - (iv) the requirements of Regulation (EU) No 1178/2011 related to their duties, in particular of Annex VII (Part-ORA) and Annex VI (Part ARA) thereto; and
 - (v) ICAO Annexes and guidance material relevant to their duties.
- (2) Additional qualification criteria:
- (i) inspectors conducting sampling of training flights in aircraft or FSTD sessions should hold or have held a pilot licence and relevant ratings and certificates appropriate to the level of the training conducted;

- (ii) inspectors conducting sampling of training flights in aircraft as a member of the flight crew should hold a pilot licence and relevant ratings and certificates appropriate to the level of the training conducted;
- (iii) inspectors conducting sampling of theoretical-knowledge instruction should have a practical background in aviation in the areas relevant to the training provided as well as practical experience in instructional techniques;
- (iv) inspectors approving training programmes should have relevant experience in the same area; and
- (v) inspectors not involved in activities referred to in (i)-(iv) above should have a relevant background in aviation related to their duties.

(b) Initial training programme

The initial training programme for inspectors should include, as appropriate to their role, current knowledge of, as well as experience and skills in, at least the following:

- (1) air law – organisation and structure;
- (2) Regulation (EC) No 216/2008, as well as its implementing regulations and related AMC/GM;
- (3) the Chicago Convention, as well as relevant ICAO Annexes and guidance;
- (4) relevant national aviation and administrative legislation;
- (5) the applicable requirements and procedures (including the correct formulation of findings);
- (6) management systems, including assessment of SMSs and CMSs, as well as auditing, risk assessment, and reporting techniques;
- (7) competency-based training, including approval of training organisations;
- (8) criteria for the qualification of FSTDs;
- (9) evidence-based training;
- (10) HF training (including ‘just culture’ in aviation and conflict management);
- (11) performance-based oversight;
- (12) rights and obligations of the competent authority’s inspecting personnel;
- (13) ‘on-the-job training’;
- (14) the relevant Annexes to Regulation (EU) No 965/2012; and
- (15) suitable technical training appropriate to the role and tasks of the inspector, in particular for those areas requiring approvals.

(c) Recurrent training programme

The recurrent training programme should reflect, at least, changes in aviation legislation and industry. It should also cover the specific needs of the inspectors and of the competent authority, and include at least the following:

- (1) an inspection on behalf of the competent authority, supervised by another inspector;
- (2) licence proficiency check(LPC)/OPC on an appropriate aircraft type/class (if applicable);

- (3) instructor refresher seminar (if applicable);
- (4) audit techniques course for regulators (refresher course); and
- (5) SMS refresher course.

GM1 ARA.GEN.200(a)(2) Management system

ED Decision 2018/009/R

SUFFICIENT PERSONNEL

- (a) This GM on the determination of the required personnel is limited to the performance of certification and oversight tasks, excluding personnel required to perform tasks subject to any national regulatory requirements.
- (b) The elements to be considered when determining required personnel and planning their availability may be divided into quantitative and qualitative elements:
 - (1) Quantitative elements:
 - (i) the estimated number of initial certificates to be issued and declarations to be received;
 - (ii) the number of:
 - (A) organisations certified by the competent authority; and
 - (B) organisations having declared their activity to the competent authority;
 - (iii) the number of persons to whom the competent authority has issued a licence, certificate, rating, authorisation or attestation;
 - (iv) the estimated number of persons and organisations exercising their activity within the territory of the Member State and established or residing in another Member State.
 - (2) Qualitative elements:
 - (i) the size, nature and complexity of activities of certified and declared organisations as well as FSTD qualification certificate holders (cf. AMC1 ORA.GEN.200(b)), taking into account:
 - (A) privileges of the organisation;
 - (B) type and scope of approval or declared activities, multiple certification or declaration;
 - (C) possible certification or declaration to industry standards;
 - (D) types of aircraft / flight simulation training devices (FSTDs) operated;
 - (E) number of personnel; and
 - (F) organisational structure, existence of subsidiaries;
 - (ii) the safety priorities identified;
 - (iii) the results of past oversight activities, including audits, inspections and reviews, in terms of risks and regulatory compliance, taking into account:
 - (A) number and level of findings;
 - (B) timeframe for implementation of corrective actions; and

- (C) maturity of management systems implemented by organisations and their ability to effectively manage safety risks, taking into account also information provided by other competent authorities related to activities in the territory of the Member States concerned; and
 - (iv) the size and complexity of the Member State's aviation industry and the potential growth of activities in the field of civil aviation, which may be an indication of the number of new applications and declarations as well as changes to existing certificates and declarations to be expected.
- (c) Based on existing data from previous oversight planning cycles and taking into account the situation within the Member State's aviation industry, the competent authority may estimate:
 - (1) the standard working time required for processing:
 - (i) applications for new certificates (for persons, organisations and FSTD qualification);
 - (ii) new declarations;
 - (2) for each planning period, the number of:
 - (i) new certificates to be issued;
 - (ii) declarations to be received; and
 - (iii) changes to existing certificates and declarations to be processed;
 - (3) the number of changes to existing certificates to be processed for each planning period.
- (d) In line with the competent authority's oversight policy, the following planning data should be determined specifically for each type of organisation certified by the competent authority ining the AMC & GM to the implementing rules of Commission Regulation (EU) (approved training organisations (ATOs) and aero-medical centres (AeMCs)) and for FSTD qualification certificate holders as well as for declared training organisations:
 - (1) standard number of audits to be performed per oversight planning cycle;
 - (2) standard duration of each audit;
 - (3) standard working time for audit preparation, on-site audit, reporting and follow-up, per inspector;
 - (4) standard number of ramp and unannounced inspections to be performed;
 - (5) standard duration of inspections, including preparation, reporting and follow-up, per inspector;
 - (6) minimum number and required qualification of inspectors for each audit/inspection.
- (e) Standard working time could be expressed either in working hours per inspector or in working days per inspector. All planning calculations should then be based on the same unit (hours or working days).
- (f) It is recommended to use a spreadsheet application to process data defined under (c) and (d), to assist in determining the total number of working hours / days per oversight planning cycle required for certification, oversight and enforcement activities. This application could also serve as a basis for implementing a system for planning the availability of personnel.
- (g) For each type of organisation certified by the competent authority, FSTD qualification certificate holders and declared training organisations, the number of working hours/days per planning

period for each qualified inspector that may be allocated for certification, oversight and enforcement activities should be determined, taking into account:

- (1) purely administrative tasks not directly related to oversight and certification;
 - (2) training;
 - (3) participation in other projects;
 - (4) planned absence; and
 - (5) the need to include a reserve for unplanned tasks or unforeseeable events.
- (h) The determination of working time available for certification, oversight and enforcement activities should also consider:
- (1) the possible use of qualified entities; and
 - (2) possible cooperation with other competent authorities for approvals and declarations involving more than one Member State.
- (i) Based on the elements listed above, the competent authority should be able to:
- (1) monitor dates when audits and inspections are due and when they have been carried out;
 - (2) implement a system to plan the availability of personnel; and
 - (3) identify possible gaps between the number and qualification of personnel and the required volume of certification and oversight.

Care should be taken to keep planning data up-to-date in line with changes in the underlying planning assumptions, with particular focus on risk-based oversight principles.

GM2 ARA.GEN.200(a)(2) Management system

ED Decision 2017/022/R

- (a) The content of the initial training programme for inspectors referred to in [AMC2 ARA.GEN.200\(a\)\(2\)](#) may be selected from the following documents, as relevant to the particular duties and responsibilities of the inspector:
- (1) ICAO Annex 1 'Personnel Licensing';
 - (2) ICAO Annex 19 'Safety Management';
 - (3) ICAO Doc 9841 'Manual on the Approval of Flight Crew Training Organisations';
 - (4) ICAO Doc 9868 'Procedures for Air Navigation Services – Training';
 - (5) ICAO Doc 9859 'Safety Management Manual';
 - (6) ICAO Doc 9379 'Manual of Procedures for Establishment and Management of a States Personnel Licensing System';
 - (7) ICAO Doc 9625 'Manual of Criteria for the Qualification of Flight Simulation Training Devices';
 - (8) ICAO Doc 9995 'Manual of Evidence-based Training';
 - (9) ICAO Doc 10011 'Manual on Aeroplane Upset Prevention and Recovery Training';
 - (10) 'Airplane Upset Prevention and Recovery Training Aid' (AUPRTA), Revision 3.

- (b) A minimum of activities should be performed according to the initial training programme:
- (1) observations; and
 - (2) inspections as a team member.

GM3 ARA.GEN.200(a)(2) Management system

ED Decision 2017/022/R

The meaning of ‘relevant ratings and certificates appropriate to the level of the training conducted’, as used in [AMC2 ARA.GEN.200\(a\)\(2\)](#), is explained below:

- the range of activities in an ATO may vary from instructions for the simple single-engined aircraft to type training for CS-25-certified multi-pilot aircraft;
- in the context of the general approval of the ATO, experience in similar types or classes of aircraft is acceptable;
- the inspector has the instructional experience in the same or similar types or the same class of aircraft intended to be flown within the ATO (e.g. a type rating to assess the type training programmes); and
- the experience in CS-25-certified multi-pilot aircraft will not, for example, equip the inspector to assess the training programme in an ATO operating only single-engine piston (SEP) (land) aircraft; similarly, experience as a PPL instructor will not necessarily equip the inspector to assess a type training course for a CS-25 aircraft; in both cases, additional appropriate training in the applicable environment is necessary.

AMC1 ARA.GEN.200(d) Management system

ED Decision 2018/009/R

PROCEDURES AVAILABLE TO THE AGENCY

- (a) Copies of the procedures related to the competent authority’s management system and their amendments to be made available to the Agency for the purpose of standardisation should provide at least the following information:
- (1) Regarding continuing oversight functions undertaken by the competent authority, the competent authority’s organisational structure with description of the main processes. This information should demonstrate the allocation of responsibilities within the competent authority, and that the competent authority is capable of carrying out the full range of tasks regarding the size and complexity of the Member State’s aviation industry. It should also consider overall proficiency and authorisation scope of competent authority personnel.
 - (2) For personnel involved in oversight activities, the minimum professional qualification requirements and experience and principles guiding appointment (e.g. assessment).
 - (3) How the following are carried out: assessing applications and evaluating compliance of applications and declarations, issue of certificates, performance of continuing oversight, follow-up of findings, enforcement measures and resolution of safety concerns.
 - (4) Principles of managing exemptions and derogations.
 - (5) Processes in place to disseminate applicable safety information for timely reaction to a safety problem.

- (6) Criteria for planning continuing oversight (oversight programme), including adequate management of interfaces when conducting continuing oversight (air operations, flight crew licensing, continuing airworthiness management for example).
- (7) Outline of the initial training of newly recruited oversight personnel (taking future activities into account), and the basic framework for continuation training of oversight personnel.
- (b) As part of the continuous monitoring of a competent authority, the Agency may request details of the working methods used, in addition to the copy of the procedures of the competent authority's management system (and amendments). These additional details are the procedures and related guidance material describing working methods for competent authority personnel conducting oversight.
- (c) Information related to the competent authority's management system may be submitted in electronic format.

ARA.GEN.205 Allocation of tasks to qualified entities

Regulation (EU) 2023/203

[applicable until 21 February 2026 — Regulation (EU) No 290/2012]

ARA.GEN.205 Allocation of tasks

[applicable from 22 February 2026 — Implementing Regulation (EU) 2023/203]

- (a) Tasks related to the initial certification or continuing oversight of persons or organisations subject to Regulation (EC) No 216/2008 and its Implementing Rules shall be allocated by Member States only to qualified entities. When allocating tasks, the competent authority shall ensure that it has:
 - (1) a system in place to initially and continuously assess that the qualified entity complies with Annex V to Regulation (EC) No 216/2008.

This system and the results of the assessments shall be documented;
 - (2) established a documented agreement with a the qualified entity, approved by both parties at the appropriate management level, which clearly defines:
 - (i) the tasks to be performed;
 - (ii) the declarations, reports and records to be provided;
 - (iii) the technical conditions to be met in performing such tasks;
 - (iv) the related liability coverage; and
 - (v) the protection given to information acquired in carrying out such tasks.
- (b) The competent authority shall ensure that the internal audit process and a safety risk management process required by [ARA.GEN.200\(a\)\(4\)](#) cover all certification or continuing oversight tasks performed on its behalf.
- (c) *With regard to the certification and oversight of the organisation's compliance with point [ORA.GEN.200A](#), the competent authority may allocate tasks to qualified entities in accordance with point (a), or to any relevant authority responsible for information security or cybersecurity within the Member State. When allocating tasks, the competent authority shall ensure that:*

- (1) all aspects related to aviation safety are coordinated and taken into account by the qualified entity or relevant authority;
- (2) the results of the certification and oversight activities performed by the qualified entity or relevant authority are integrated in the overall certification and oversight files of the organisation;
- (3) its own information security management system established in accordance with point [ARA.GEN.200\(e\)](#) covers all the certification and continuing oversight tasks performed on its behalf.

[applicable from 22 February 2026 — Implementing Regulation (EU) 2023/203]

GM1 ARA.GEN.205 Allocation of tasks to qualified entities

ED Decision 2023/010/R

[applicable until 21 February 2026 — ED Decision 2012/006/R]

GM1 ARA.GEN.205 Allocation of tasks

[applicable from 22 February 2026 — ED Decision 2023/010/R]

CERTIFICATION TASKS

The tasks that may be performed by a qualified entity on behalf of the competent authority include those related to the initial certification and continuing oversight of persons and organisations as defined in this Regulation, with the exclusion of the issuance of certificates, licences, ratings or approvals.

ARA.GEN.210 Changes in the management system

Regulation (EU) No 1178/2011

- (a) The competent authority shall have a system in place to identify changes that affect its capability to perform its tasks and discharge its responsibilities as defined in [Regulation \(EU\) 2018/1139](#) and the delegated and implementing acts adopted on the basis thereof. That system shall enable it to take action as appropriate to ensure that its management system remains adequate and effective.
- (b) The competent authority shall update its management system to reflect any change to [Regulation \(EU\) 2018/1139](#) and the delegated and implementing acts adopted on the basis thereof in a timely manner, so as to ensure effective implementation.
- (c) The competent authority shall notify the Agency of changes affecting its capability to perform its tasks and discharge its responsibilities as defined in [Regulation \(EU\) 2018/1139](#) and the delegated and implementing acts adopted on the basis thereof.

ARA.GEN.220 Record-keeping

Regulation (EU) 2020/359

- (a) The competent authority shall establish a system of record-keeping providing for adequate storage, accessibility and reliable traceability of:
 - (1) the management system's documented policies and procedures;
 - (2) training, qualification and authorisation of its personnel;

- (3) the allocation of tasks, covering the elements required by [ARA.GEN.205](#) as well as the details of tasks allocated;
 - (4) certification and declaration processes as well as oversight of certified and declared organisations;
 - (5) processes for issuing personnel licences, ratings, certificates and attestations and for the continuing oversight of the holders of those licences, ratings, certificates and attestations;
 - (6) processes for issuing FSTD qualification certificates and for the continuing oversight of the FSTD and of the organisation operating it;
 - (7) oversight of persons and organisations exercising activities within the territory of the Member State, but overseen or certified by the competent authority of another Member State or the Agency, as agreed between these authorities;
 - (8) the evaluation and notification to the Agency of alternative means of compliance proposed by organisations and the assessment of alternative means of compliance used by the competent authority itself;
 - (9) findings, corrective actions and date of action closure;
 - (10) enforcement measures taken;
 - (11) safety information and follow-up measures;
 - (12) the use of flexibility provisions in accordance with Article 71 of Regulation (EU) 2018/1139; and
 - (13) the evaluation and authorisation process of aircraft laid down in points ORA.ATO.135 (a) and DTO.GEN.240 (a).
- (b) The competent authority shall establish and keep up to date a list of all organisation certificates, FSTD qualification certificates and personnel licences, certificates and attestations it issued, DTO declarations it received, and the DTO training programmes it verified or approved for compliance with Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#), or Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#).
- (c) All records shall be kept for the minimum period specified in this Regulation. In the absence of such indication, records shall be kept for a minimum period of 5 years subject to applicable data protection law.

AMC1 ARA.GEN.220(a) Record-keeping

ED Decision 2012/006/R

GENERAL

- (a) The record-keeping system should ensure that all records are accessible whenever needed within a reasonable time. These records should be organised in a way that ensures traceability and retrievability throughout the required retention period.
- (b) Records should be kept in paper form or in electronic format or a combination of both media. Records stored on microfilm or optical disc form are also acceptable. The records should remain legible and accessible throughout the required retention period. The retention period starts when the record has been created.

- (c) Paper systems should use robust material, which can withstand normal handling and filing. Computer systems should have at least one backup system, which should be updated within 24 hours of any new entry. Computer systems should include safeguards against unauthorised alteration of data.
- (d) All computer hardware used to ensure data backup should be stored in a different location from that containing the working data and in an environment that ensures they remain in good condition. When hardware- or software-changes take place, special care should be taken that all necessary data continue to be accessible at least through the full period specified in the relevant Subpart or by default in [ARA.GEN.220\(c\)](#).

AMC1 ARA.GEN.220(a)(1);(2);(3) Record-keeping

ED Decision 2012/006/R

COMPETENT AUTHORITY MANAGEMENT SYSTEM

Records related to the competent authority's management system should include, as a minimum and as applicable:

- (a) the documented policies and procedures;
- (b) the personnel files of competent authority personnel, with supporting documents related to training and qualifications;
- (c) the results of the competent authority's internal audit and safety risk management processes, including audit findings and corrective actions; and
- (d) the contract(s) established with qualified entities performing certification or oversight tasks on behalf of the competent authority.

AMC1 ARA.GEN.220(a)(4) Record-keeping

ED Decision 2018/009/R

ORGANISATIONS

Records related to an organisation certified by, or having declared its activity to, the competent authority should include, as appropriate to the type of organisation:

- (a) the application for an organisation approval or the declaration received;
- (b) the documentation based on which the approval has been granted and any amendments to that documentation or, in the case of declared training organisations, the documentation required to be submitted with the declaration and any amendments thereto;
- (c) the organisation approval certificate or any approval, including any changes;
- (d) a copy of the continuing oversight programme listing the dates when audits or inspections are due and when such audits or inspections were carried out;
- (e) continuing oversight records including all audit and inspection records;
- (f) copies of all relevant correspondence;
- (g) details of any exemption and enforcement actions;
- (h) any report from other competent authorities relating to the oversight of the organisation; and
- (i) a copy of any other document approved by the competent authority.

GM1 ARA.GEN.220(a)(4) Record-keeping

ED Decision 2018/009/R

CERTIFIED ORGANISATIONS - DOCUMENTATION

Documentation to be kept as records in support of the approval include the management system documentation, including any technical manuals, such as the operations manual, and training manual, that have been submitted with the initial application, and any amendments to these documents.

GM2 ARA.GEN.220(a)(4) Record-keeping

ED Decision 2018/009/R

DECLARED TRAINING ORGANISATIONS - DOCUMENTATION

Documents to be kept as records in support of the declaration process include the declaration form and all required attachments to it (training programmes) as well as any amendments to these documents.

AMC1 ARA.GEN.220(a)(5) Record-keeping

ED Decision 2018/011/R

PERSONS

Records related to personnel licences, certificates, ratings, authorisations or attestations issued by the competent authority should include, as a minimum:

- (a) the application for a licence, certificate, rating, authorisation or attestation or change to a licence, certificate, rating, authorisation or attestation;
- (b) documentation in support of the application for a licence, certificate, rating, authorisation or attestation or change to a licence, certificate, rating, authorisation or attestation, covering as applicable:
 - (1) the course Area 100 KSA assessment;
 - (2) theoretical examination(s);
 - (3) skill test(s);
 - (4) proficiency check(s); and
 - (5) certificates attesting required experience;
- (c) a copy of the licence or certificate including any changes;
- (d) all relevant correspondence or copies thereof;
- (e) details of any exemption;
- (f) details of any enforcement action(s); and
- (g) any report from other competent authorities relating to personnel licences, certificates, ratings, authorisations or attestations issued by the competent authority.

AMC1 ARA.GEN.220(a)(7) Record-keeping

ED Decision 2018/009/R

ACTIVITIES PERFORMED IN THE TERRITORY OF A MEMBER STATE BY PERSONS OR ORGANISATIONS ESTABLISHED OR RESIDING IN ANOTHER MEMBER STATE

- (a) Records related to the oversight of activities performed in the territory of a Member State by persons or organisations established or residing in another Member State should include, as a minimum:
 - (1) oversight records including all audit and inspection records and related correspondence;
 - (2) copies of all relevant correspondence to exchange information with other competent authorities relating to the oversight of such persons/organisations;
 - (3) details of any enforcement measures and penalties; and
 - (4) any report from other competent authorities relating to the oversight of these persons/organisations, including any notification of evidence showing non-compliance with the applicable requirements.
- (b) Records should be kept by the competent authority having performed the audit or inspection and should be made available to other competent authorities at least in the following cases:
 - (1) serious incidents or accidents;
 - (2) findings through the oversight programme where organisations certified by, or having declared its activities to, another competent authority are involved to determine the root cause;
 - (3) an organisation being certified by, having approvals issued by, or having declared its activities to, competent authorities in several Member States.
- (c) When records are requested by another competent authority, the reason for the request should be clearly stated.
- (d) The records can be made available by sending a copy or by allowing access to them for consultation.

GM1 ARA.GEN.220 Record-keeping

ED Decision 2012/006/R

GENERAL

Records are required to document results achieved or to provide evidence of activities performed. Records become factual when recorded. Therefore, they are not subject to version control. Even when a new record is produced covering the same issue, the previous record remains valid.

SECTION III – OVERSIGHT, CERTIFICATION AND ENFORCEMENT

ARA.GEN.300 Oversight

Regulation (EU) 2025/2293

- (a) The competent authority shall verify:
- (1) compliance with the requirements applicable to organisations or persons prior to the issue of an organisation certificate, approval, FSTD qualification certificate or personnel licence, certificate, rating, or attestation, as applicable;
 - (2) continued compliance with the requirements applicable to the persons holding licences, ratings and certificates, the organisations it has certified, the holders of a FSTD qualification and the organisations from which it received a declaration;
 - (3) implementation of appropriate safety measures mandated by the competent authority as defined in [ARA.GEN.135\(c\) and \(d\)](#).
- (b) This verification shall:
- (1) be supported by documentation specifically intended to provide personnel responsible for safety oversight with guidance to perform their functions;
 - (2) provide the persons and organisations concerned with the results of safety oversight activity;
 - (3) be based on audits and inspections, including ramp and unannounced inspections; and
 - (4) provide the competent authority with the evidence needed in case further action is required, including the measures foreseen by [ARA.GEN.350](#) and [ARA.GEN.355](#).
- (c) The scope of oversight defined in (a) and (b) shall take into account the results of past oversight activities and the safety priorities.
- (d) Without prejudice to the competences of the Member States and to their obligations as set out in ARO.RAMP, the scope of the oversight of activities performed in the territory of a Member State by persons or organisations established or residing in another Member State shall be determined on the basis of the safety priorities, as well as of past oversight activities.
- (e) Where the activity of a person or organisation involves more than one Member State or the Agency, the competent authority responsible for the oversight under (a) may agree to have oversight tasks performed by the competent authority(ies) of the Member State(s) where the activity takes place or by the Agency. Any person or organisation subject to such agreement shall be informed of its existence and of its scope.
- (f) The competent authority shall collect and process any information deemed useful for oversight, including for ramp and unannounced inspections.
- (g) With regard to the certification and oversight of the organisation's compliance with point [ORA.GEN.200A](#) of this Annex, in addition to complying with points (a) to (f) of this point, the competent authority shall review any approval granted under point IS.I.OR.200(e) of Annex II to Commission Implementing Regulation (EU) 2023/203¹ following the applicable

¹ Commission Implementing Regulation (EU) 2023/203 of 27 October 2022 laying down rules for the application of Regulation (EU) 2018/1139 of the European Parliament and of the Council, as regards requirements for the management of information security risks with a potential impact on aviation safety for organisations covered by Commission Regulations (EU) No 1321/2014, (EU) No 965/2012, (EU) No 1178/2011, (EU) 2015/340, Commission Implementing Regulations (EU) 2017/373 and (EU) 2021/664, and for competent authorities

oversight audit cycle and whenever changes are implemented in the scope of work of the organization.

[applicable from 22 February 2026 — Regulation (EU) 2025/2293 (enter into force date 1 December 2025)]

AMC1 ARA.GEN.300(a);(b);(c) Oversight

ED Decision 2020/018/R

EVALUATION OF APPROVED TRAINING ORGANISATIONS' OPERATIONAL SAFETY RISK ASSESSMENT

As part of the initial certification or the continuing oversight of an ATO, the competent authority should normally evaluate its safety risk assessment processes related to hazards identified by the ATO as having an interface with its operations. These safety risk assessments should be identifiable processes of the ATO's management system. As part of its continuing oversight, the competent authority should also remain satisfied as to the effectiveness of these safety risk assessments.

- (a) General methodology for operational hazards
 - (1) The competent authority should establish a methodology for evaluating the safety risk assessment processes of the ATO's management system.
 - (2) When related to operational hazards, the competent authority's evaluation under its normal oversight process should be considered satisfactory if the ATO demonstrates its competence and capability to:
 - (i) understand the hazards identified and their consequences on its operations;
 - (ii) be clear on where these hazards may exceed acceptable safety risk limits;
 - (iii) identify and implement mitigations including suspension of operations where mitigation cannot reduce the risk to within safety risk limits;
 - (iv) develop and execute effectively, robust procedures for the preparation and the safe operation of the flights subject to the hazards identified;
 - (v) assess the competence and currency of its staff in relation to the duties for the intended operations and implement any necessary training; and
 - (vi) ensure sufficient numbers of qualified and competent staff for such duties.
 - (3) The competent authority should take into account:
 - (i) the ATO's recorded mitigations for each unacceptable risk identified are in place;
 - (ii) the operational procedures specified by the ATO with the most significance to safety appear to be robust; and
 - (iii) that the staff on which the ATO depends in respect of those duties necessary for the intended operations are trained and assessed as competent in the relevant procedures.

EVALUATION OF APPROVED TRAINING ORGANISATIONS' VOLCANIC ASH SAFETY RISK ASSESSMENT

covered by Commission Regulations (EU) No 748/2012, (EU) No 1321/2014, (EU) No 965/2012, (EU) No 1178/2011, (EU) 2015/340 and (EU) No 139/2014, Commission Implementing Regulations (EU) 2017/373 and (EU) 2021/664 and amending Commission Regulations (EU) No 1178/2011, (EU) No 748/2012, (EU) No 965/2012, (EU) No 139/2014, (EU) No 1321/2014, (EU) 2015/340, and Commission Implementing Regulations (EU) 2017/373 and (EU) 2021/664 (OJ L 31, 2.2.2023, p. 1, ELI: http://data.europa.eu/eli/reg_impl/2023/203/oj).

- (b) In addition to the general methodology for operational hazards, the competent authority's evaluation under its normal oversight process should also assess the ATO's competence and capability to:
- (1) choose the correct information sources to use to interpret the information related to volcanic ash contamination forecast and to resolve correctly any conflicts among such sources; and
 - (2) take account of all information from its type certificate holders (TCHs) concerning volcanic ash-related airworthiness aspects of the aircraft it operates, and the related preflight, in-flight and post flight precautions to be observed.

GM1 ARA.GEN.300(a);(b);(c) Oversight

ED Decision 2013/006/R

VOLCANIC ASH SAFETY RISK ASSESSMENT - ADDITIONAL GUIDANCE

Further guidance on the assessment of an ATO volcanic ash safety risk assessment is given in ICAO Doc. 9974 (Flight safety and volcanic ash – Risk management of flight operations with known or forecast volcanic ash contamination).

GM1 ARA.GEN.300(d) Oversight

ED Decision 2018/009/R

ACTIVITIES WITHIN THE TERRITORY OF THE MEMBER STATE

- (a) Activities performed in the territory of the Member State by persons or organisations established or residing in another Member State include:
- (1) activities of organisations certified by the competent authority of any other Member State or the Agency as well as activities of organisations having declared their activities to the competent authority of any other Member State;
 - (2) activities of persons holding a licence, certificate, rating, or attestation issued by the competent authority of any other Member State; and
 - (3) activities of persons making declarations to the competent authority of any other Member State.
- (b) Audits and inspections of such activities, including ramp and unannounced inspections, should be prioritised towards those areas of greater safety concern, as identified through the analysis of data on safety hazards and their consequences in operations.

ARA.GEN.305 Oversight programme

Regulation (EU) 2024/2076

- (a) The competent authority shall establish and maintain an oversight programme covering the oversight activities required by [ARA.GEN.300](#) and by ARO.RAMP.
- (b) For organisations certified by the competent authority and FSTD qualification certificate holders, the oversight programme shall be developed taking into account the specific nature of the organisation, the complexity of its activities, the results of past certification and/or oversight activities and shall be based on the assessment of associated risks. It shall include within each oversight planning cycle:
- (1) audits and inspections, including ramp and unannounced inspections as appropriate; and

- (2) meetings convened between the accountable manager and the competent authority to ensure both remain informed of significant issues.
- (c) For organisations certified by the competent authority and FSTD qualification certificate holders an oversight planning cycle not exceeding 24 months shall be applied.

The oversight planning cycle may be reduced if there is evidence that the safety performance of the organisation or the FTSD qualification certificate holder has decreased.

The oversight planning cycle may be extended to a maximum of 36 months if the competent authority has established that, during the previous 24 months:

- (1) the organisation has demonstrated an effective identification of aviation safety hazards and management of associated risks;
- (2) the organisation has continuously demonstrated under ORA.GEN.130 that it has full control over all changes;
- (3) no level 1 findings have been issued; and
- (4) all corrective actions have been implemented within the time period accepted or extended by the competent authority as defined in [ARA.GEN.350\(d\)\(2\)](#).

The oversight planning cycle may be further extended to a maximum of 48 months if, in addition to the above, the organisation has established, and the competent authority has approved, an effective continuous reporting system to the competent authority on the safety performance and regulatory compliance of the organisation itself.

- (ca) Notwithstanding (c), for organisations only providing training towards the LAPL, PPL, SPL or BPL and associated ratings and certificates, an oversight planning cycle not exceeding 48 months shall be applied. The oversight planning cycle shall be reduced if there is evidence that the safety performance of the organisation holder has decreased.

The oversight planning cycle may be extended to a maximum of 72 months, if the competent authority has established that, during the previous 48 months:

- (1) the organisation has demonstrated an effective identification of aviation safety hazards and management of associated risks, as demonstrated by the results of the annual review in accordance with ORA.GEN.200(c);
 - (2) the organisation has continuously maintained control over all changes in accordance with ORA.GEN.130 as demonstrated by the results of the annual review in accordance with ORA.GEN.200(c);
 - (3) no level 1 findings have been issued; and
 - (4) all corrective actions have been implemented within the time period accepted or extended by the competent authority as defined in [ARA.GEN.350\(d\)\(2\)](#).
- (d) For persons that hold a licence, certificate, rating, or attestation issued by the competent authority, the oversight programme shall include an appropriate volume of inspections and unannounced inspections.
- (e) The oversight programme shall include records of the dates when audits, inspections and meetings are due and when such audits, inspections and meetings have been carried out.
- (f) Notwithstanding points (b), (c), and (ca), the oversight programme of DTOs shall be developed taking into account the specific nature of the organisation, the complexity of its activities and the results of past oversight activities and shall be based on the assessment of risks associated

with the type of training provided. The oversight activities shall include inspections, including unannounced inspections, and may, as deemed necessary by the competent authority, include audits.

AMC1 ARA.GEN.305(b) Oversight programme

ED Decision 2012/006/R

SPECIFIC NATURE AND COMPLEXITY OF THE ORGANISATION, RESULTS OF PAST OVERSIGHT

- (a) When determining the oversight programme for an organisation the competent authority should consider in particular the following elements, as applicable:
 - (1) the implementation by the organisation of industry standards, directly relevant to the organisation's activity subject to this Regulation;
 - (2) the procedure applied for and scope of changes not requiring prior approval;
 - (3) specific approvals held by the organisation;
 - (4) specific procedures implemented by the organisation related to any alternative means of compliance used.
- (b) For the purpose of assessing the complexity of an organisation's management system, AMC1 ORA.GEN.200(b) should be used.
- (c) Regarding results of past oversight, the competent authority should also take into account relevant results of ramp inspections of organisations it has certified that were performed in other Member States in accordance with ARO.RAMP.

AMC1 ARA.GEN.305(b)(1) Oversight programme

ED Decision 2012/006/R

AUDIT

- (a) The oversight programme should indicate which aspects of the approval will be covered with each audit.
- (b) Part of an audit should concentrate on the organisation's compliance monitoring reports produced by the compliance monitoring personnel to determine if the organisation is identifying and correcting its problems.
- (c) At the conclusion of the audit, an audit report should be completed by the auditing inspector, including all findings raised.

AMC2 ARA.GEN.305(b)(1) Oversight programme

ED Decision 2012/006/R

RAMP INSPECTIONS

When conducting a ramp inspection of aircraft used by organisations under its regulatory oversight the competent authority should, in as far as possible, comply with the requirements defined in ARO.RAMP.

AMC1 ARA.GEN.305(b);(c) Oversight programme

ED Decision 2012/006/R

INDUSTRY STANDARDS

- (a) For organisations having demonstrated compliance with industry standards, the competent authority may adapt its oversight programme, in order to avoid duplication of specific audit items.
- (b) Demonstrated compliance with industry standards should not be considered in isolation from the other elements to be considered for the competent authority's risk-based oversight.
- (c) In order to be able to credit any audits performed as part of certification in accordance with industry standards, the following should be considered:
 - (1) the demonstration of compliance is based on certification auditing schemes providing for independent and systematic verification;
 - (2) the existence of an accreditation scheme and accreditation body for certification in accordance with the industry standards has been verified;
 - (3) certification audits are relevant to the requirements defined in Annex VII (Part-ORA) and other Annexes to this Regulation as applicable;
 - (4) the scope of such certification audits can easily be mapped against the scope of oversight in accordance with Part-ORA;
 - (5) audit results are accessible to the competent authority and open to exchange of information in accordance with Article 15(1) of Regulation (EC) No 216/2008; and
 - (6) the audit planning intervals of certification audits i.a.w. industry standards are compatible with the oversight planning cycle.

AMC1 ARA.GEN.305(c) Oversight programme

ED Decision 2012/006/R

OVERSIGHT PLANNING CYCLE

- (a) When determining the oversight planning cycle and defining the oversight programme, the competent authority should assess the risks related to the activity of each organisation and adapt the oversight to the level of risk identified and to the organisation's ability to effectively manage safety risks.
- (b) The competent authority should establish a schedule of audits and inspections appropriate to each organisation. The planning of audits and inspections should take into account the results of the hazard identification and risk assessment conducted and maintained by the organisation as part of the organisation's management system. Inspectors should work in accordance with the schedule provided to them.
- (c) When the competent authority, having regard to an organisation's safety performance, varies the frequency of an audit or inspection it should ensure that all aspects of the organisation's activity are audited and inspected within the applicable oversight planning cycle.
- (d) The section(s) of the oversight programme dealing with ramp inspections should be developed based on geographical locations, taking into account aerodrome activity, and focusing on key issues that can be inspected in the time available without unnecessarily delaying the operations.

AMC2 ARA.GEN.305(c) Oversight programme

ED Decision 2012/006/R

OVERSIGHT PLANNING CYCLE

- (a) For each organisation certified by the competent authority and each FSTD qualification certificate holder all processes should be completely audited at periods not exceeding the applicable oversight planning cycle. The beginning of the first oversight planning cycle is normally determined by the date of issue of the first certificate. If the competent authority wishes to align the oversight planning cycle with the calendar year, it should shorten the first oversight planning cycle accordingly.
- (b) The interval between two audits for a particular process should not exceed the interval of the applicable oversight planning cycle.
- (c) Audits should include at least one on-site audit within each oversight planning cycle. For organisations exercising their regular activity at more than one site, the determination of the sites to be audited should consider the results of past oversight, the volume of activity at each site, as well as main risk areas identified.
- (d) For organisations holding more than one certificate, the competent authority may define an integrated oversight schedule to include all applicable audit items. In order to avoid duplication of audits, credit may be granted for specific audit items already completed during the current oversight planning cycle, subject to four conditions:
 - (1) the specific audit item should be the same for all certificates under consideration;
 - (2) there should be satisfactory evidence on record that such specific audit items were carried out and that all corrective actions have been implemented to the satisfaction of the competent authority;
 - (3) the competent authority should be satisfied that there is no reason to believe standards have deteriorated in respect of those specific audit items being granted a credit;
 - (4) the interval between two audits for the specific item being granted a credit should not exceed the applicable oversight planning cycle.

AMC1 ARA.GEN.305(d) Oversight programme

ED Decision 2012/006/R

PERSONS HOLDING A LICENCE, CERTIFICATE, RATING OR ATTESTATION

The oversight of persons holding a licence, certificate, rating or attestation should normally be ensured as part of the oversight of organisations. Additionally, the competent authority should verify compliance with applicable requirements when endorsing or renewing ratings.

To properly discharge its oversight responsibilities, the competent authority should perform a certain number of unannounced verifications.

AMC1 ARA.GEN.305(f) Oversight programme

ED Decision 2018/009/R

- (a) When determining the oversight programme for organisations that have declared their activities, the competent authority should make a selection of the DTOs to be inspected based on the elements specified in point [ARA.GEN.305\(f\)](#).
- (b) For each selected DTO, an inspection is a sample inspection of the predefined inspection criteria on the basis of key risk elements and the applicable requirements.
- (c) The results of past oversight activities should include information from the DTO's annual internal review and the DTO's annual activity reports as well as information from the verification

of the DTO's training programme for Part-FCL compliance and occurrence reports linked to the activity of the DTO, if applicable.

- (d) The oversight programme should follow a risk-based approach and should be developed on a yearly basis. At least one inspection should be performed for each DTO not later than 72 months starting from the date on which the declaration was received or, subsequently, the last inspection, as applicable.
- (e) Additional inspections or unannounced inspections to specific DTOs may be included in the oversight programme on the basis of the elements specified in point [ARA.GEN.305\(f\)](#).

AMC2 ARA.GEN.305(f) Oversight programme

ED Decision 2018/009/R

An inspection of a DTO should at least focus on:

- (a) the existence of a safety policy statement and its adequacy regarding the DTO activities;
- (b) the existence of appropriate measures aiming to achieve the objectives of the safety policy including risk mitigation measures, results of annual reviews and respective corrective actions, if applicable;
- (c) flight training in accordance with the DTO training programme, its conduct and standards as well as training records;
- (d) training aircraft in use, including their registration, associated documents and maintenance records;
- (e) use of FSTDs;
- (f) operating sites and associated facilities as appropriate; and
- (g) information on flight instructors and on the validity of their licences, certificates, ratings and logbooks.

ARA.GEN.310 Initial certification procedure – organisations

Regulation (EU) No 1178/2011

- (a) Upon receiving an application for the initial issue of a certificate for an organisation, the competent authority shall verify the organisation's compliance with the applicable requirements.
- (b) When satisfied that the organisation is in compliance with the applicable requirements, the competent authority shall issue the certificate(s), as established in Appendixes III and V to this Part. The certificate(s) shall be issued for an unlimited duration. The privileges and scope of the activities that the organisation is approved to conduct shall be specified in the terms of approval attached to the certificate(s).
- (c) To enable an organisation to implement changes without prior competent authority approval in accordance with ORA.GEN.130, the competent authority shall approve the procedure submitted by the organisation defining the scope of such changes and describing how such changes will be managed and notified.

AMC1 ARA.GEN.310(a) Initial certification procedure – organisations

ED Decision 2012/006/R

VERIFICATION OF COMPLIANCE

- (a) In order to verify the organisation's compliance with the applicable requirements, the competent authority should conduct an audit of the organisation, including interviews of personnel and inspections carried out at the organisation's facilities.
- (b) The competent authority should only conduct such audit after being satisfied that the application shows compliance with the applicable requirements.
- (c) The audit should focus on the following areas:
 - (1) detailed management structure, including names and qualifications of personnel required by ORA.GEN.210 and adequacy of the organisation and management structure;
 - (2) personnel:
 - (i) adequacy of number and qualifications with regard to the intended terms of approval and associated privileges;
 - (ii) validity of licences, ratings, certificates or attestations as applicable;
 - (3) processes for safety risk management and compliance monitoring;
 - (4) facilities – adequacy with regard to the organisation's scope of work;
 - (5) documentation based on which the certificate should be granted (organisation documentation as required by Part-ORA, including technical manuals, such as operations manual or training manual).
- (d) In case of non-compliance, the applicant should be informed in writing of the corrections that are required.
- (e) In cases where an application for an organisation certificate is refused, the applicant should be informed of the right of appeal as exists under national law.

ARA.GEN.315 Procedure for issue, revalidation, renewal or change of licences, ratings, certificates or attestations – persons

Regulation (EU) No 1178/2011

- (a) Upon receiving an application for the issue, revalidation, renewal or change of a personal licence, rating, certificate or attestation and any supporting documentation, the competent authority shall verify whether the applicant meets the applicable requirements.
- (b) When satisfied that the applicant meets the applicable requirements, the competent authority shall issue, revalidate, renew or change the licence, certificate, rating, or attestation.

AMC1 ARA.GEN.315(a) Procedure for issue, revalidation, renewal or change of licences, ratings or certificates – persons

ED Decision 2021/002/R

VERIFICATION OF COMPLIANCE

- (a) In order to verify that the applicant meets the requirements, the competent authority should review the application and any supporting documents submitted, for completeness and compliance with applicable requirements.
- (b) As part of the verification that the applicant meets the requirements, the competent authority should check that he/she:
 - (1) was not holding any personnel licence, certificate, rating, authorisation or attestation with the same scope and in the same category issued in another Member State;
 - (2) has not applied for any personnel licence, certificate, rating, authorisation or attestation with the same scope and in the same category in another Member State; and
 - (3) has never held any personnel licence, certificate, rating, authorisation or attestation with the same scope and in the same category issued in another Member State which was revoked or suspended in any other Member State.
- (c) The competent authority should request the applicant to make a declaration covering items (b)(1) to (b)(3). Such declaration should include a statement that any incorrect information could disqualify the applicant from being granted a personnel licence, certificate, rating, authorisation or attestation. In case of doubts, the competent authority should contact the competent authority of the Member State where the applicant may have previously held any personnel licence, certificate, rating, authorisation or attestation.
- (d) In addition to the requirements in points (a), (b) and (c) above, in order to verify that the applicant meets the requirements for revalidation or renewal under the EBT programme, the competent authority should request the applicant to provide the application and report form of [Appendix 10](#), including the declaration of the operator as per point (b) of [AMC1 to Appendix 10](#) — ‘Revalidation and renewal of type ratings, and revalidation and renewal of IRs when combined with the revalidation or renewal of type ratings – EBT practical assessment’.

ARA.GEN.330 Changes – organisations

Regulation (EU) 2018/1119

- (a) Upon receiving an application for a change that requires prior approval, the competent authority shall verify the organisation’s compliance with the applicable requirements before issuing the approval.

The competent authority shall prescribe the conditions under which the organisation may operate during the change, unless the competent authority determines that the organisation’s certificate needs to be suspended.

When satisfied that the organisation is in compliance with the applicable requirements, the competent authority shall approve the change.
- (b) Without prejudice to any additional enforcement measures, when the organisation implements changes requiring prior approval without having received competent authority approval as defined in (a), the competent authority shall suspend, limit or revoke the organisation’s certificate.
- (c) For changes not requiring prior approval, the competent authority shall assess the information provided in the notification sent by the organisation in accordance with ORA.GEN.130 to verify compliance with the applicable requirements. In case of any non-compliance, the competent authority shall:
 - (1) notify the organisation about the non-compliance and request further changes; and

- (2) in case of level 1 or level 2 findings, act in accordance with [ARA.GEN.350](#).
- (d) Notwithstanding points (a), (b) and (c), in the case of changes to the information contained in the declarations received from a DTO or to the training programme used by the DTO, notified to it in accordance with point DTO.GEN.116 of Annex VIII (Part-DTO), the competent authority shall act in accordance with the requirements of points [ARA.DTO.105](#) and [ARA.DTO.110](#), as applicable.

AMC1 ARA.GEN.330 Changes – organisations

ED Decision 2012/006/R

GENERAL

- (a) Changes in nominated persons:
- The competent authority should be informed of any changes to personnel specified in Part-ORA that may affect the certificate or terms of approval/approval schedule attached to it. When an organisation submits the name of a new nominee for any of the persons nominated as per ORA.GEN.210(b), the competent authority should require the organisation to produce a written résumé of the proposed person's qualifications. The competent authority should reserve the right to interview the nominee or call for additional evidence of his/her suitability before deciding upon his/her acceptability.
- (b) A simple management system documentation status sheet should be maintained, which contains information on when an amendment was received by the competent authority and when it was approved.
- (c) The organisation should provide each management system documentation amendment to the competent authority, including for the amendments that do not require prior approval by the competent authority. Where the amendment requires competent authority approval, the competent authority, when satisfied, should indicate its approval in writing. Where the amendment does not require prior approval, the competent authority should acknowledge receipt in writing within 10 working days.
- (d) For changes requiring prior approval, in order to verify the organisation's compliance with the applicable requirements, the competent authority should conduct an audit of the organisation, limited to the extent of the changes. If required for verification, the audit should include interviews and inspections carried out at the organisation's facilities.

GM1 ARA.GEN.330 Changes – organisations

ED Decision 2012/006/R

CHANGE OF NAME OF THE ORGANISATION

- (a) On receipt of the application and the relevant parts of the organisation's documentation as required by Part-ORA, the competent authority should re-issue the certificate.
- (b) A name change alone does not require the competent authority to audit the organisation, unless there is evidence that other aspects of the organisation have changed.

ARA.GEN.330A Changes to the information security management system

Regulation (EU) 2023/203

- (a) With regard to changes managed and notified to the competent authority in accordance with the procedure set out in point IS.I.OR.255(a) of Annex II (Part-IS.I.OR) to [Implementing Regulation \(EU\) 2023/203](#), the competent authority shall include the review of such changes in its continuing oversight in accordance with the principles laid down in point [ARA.GEN.300](#). If any non-compliance is found, the competent authority shall notify the organisation thereof, request further changes and act in accordance with point [ARA.GEN.350](#).
- (b) With regard to other changes requiring an application for approval in accordance with point IS.I.OR.255(b) of Annex II (Part-IS.I.OR) to [Implementing Regulation \(EU\) 2023/203](#):
 - (1) upon receiving the application for the change, the competent authority shall check the organisation's compliance with the applicable requirements before issuing the approval;
 - (2) the competent authority shall establish the conditions under which the organisation may operate during the implementation of the change;
 - (3) if it is satisfied that the organisation complies with the applicable requirements, the competent authority shall approve the change.

[applicable from 22 February 2026 — Implementing Regulation (EU) 2023/203]

ARA.GEN.350 Findings and corrective actions – organisations

Regulation (EU) 2020/359

- (a) The competent authority for oversight in accordance with [ARA.GEN.300\(a\)](#) shall have a system to analyse findings for their safety significance.
- (b) A level 1 finding shall be issued by the competent authority when any significant non-compliance is detected with the applicable requirements of Regulation (EC) No 216/2008 and its Implementing Rules, with the organisation's procedures and manuals or with the terms of an approval or certificate which lowers safety or seriously hazards flight safety.

The level 1 findings shall include:

- (1) failure to give the competent authority access to the organisation's facilities as defined in [ORA.GEN.140](#) during normal operating hours and after two written requests;
 - (2) obtaining or maintaining the validity of the organisation certificate by falsification of submitted documentary evidence;
 - (3) evidence of malpractice or fraudulent use of the organisation certificate; and
 - (4) the lack of an accountable manager.
- (c) A level 2 finding shall be issued by the competent authority when any non-compliance is detected with the applicable requirements of Regulation (EC) No 216/2008 and its Implementing Rules, with the organisation's procedures and manuals or with the terms of an approval or certificate which could lower safety or hazard flight safety.
- (d) When a finding is detected during oversight or by any other means, the competent authority shall, without prejudice to any additional action required by Regulation (EC) No 216/2008 and its Implementing Rules, communicate the finding to the organisation in writing and request

corrective action to address the non-compliance(s) identified. Where relevant, the competent authority shall inform the State in which the aircraft is registered.

- (1) In the case of level 1 findings the competent authority shall take immediate and appropriate action to prohibit or limit activities and, if appropriate, it shall take action to revoke the certificate or specific approval or to limit or suspend it in whole or in part, depending upon the extent of the level 1 finding, until successful corrective action has been taken by the organisation.
 - (2) In the case of level 2 findings, the competent authority shall:
 - (i) grant the organisation a corrective action implementation period appropriate to the nature of the finding that in any case initially shall not be more than 3 months. At the end of this period, and subject to the nature of the finding, the competent authority may extend the 3-month period subject to a satisfactory corrective action plan agreed by the competent authority; and
 - (ii) assess the corrective action and implementation plan proposed by the organisation and, if the assessment concludes that they are sufficient to address the non-compliance(s), accept these.
 - (3) Where an organisation fails to submit an acceptable corrective action plan, or to perform the corrective action within the time period accepted or extended by the competent authority, the finding shall be raised to a level 1 finding and action taken as laid down in (d)(1).
 - (4) The competent authority shall record all findings it has raised or that have been communicated to it and, where applicable, the enforcement measures it has applied, as well as all corrective actions and date of action closure for findings.
- (da) By way of derogation from paragraphs (a) to (d), in the case of DTOs, if during oversight or by any other means the competent authority finds evidence that indicates DTO non-compliance with the essential requirements set out in Annex IV to [Regulation \(EU\) 2018/1139](#), with the requirements of Annex I (Part-FCL) and Annex VIII (Part-DTO) to this Regulation, or with the requirements of Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), the competent authority shall:
- (1) raise a finding, record it, communicate it in writing to the representative of the DTO and determine a reasonable period of time within which the DTO is to take the steps specified in point [DTO.GEN.150](#) of Annex VIII (Part-DTO);
 - (2) take immediate and appropriate action to limit or prohibit the training activities affected by the non-compliance until the DTO has taken the corrective action referred to in point (1), where any of the following situations occurs:
 - (i) a safety problem has been identified;
 - (ii) the DTO fails to take corrective action in accordance with point [DTO.GEN.150](#);
 - (3) in respect of the training programmes referred to in point [DTO.GEN.230\(c\)](#) of Annex VIII (Part-DTO), limit, suspend or revoke the approval of the training programme;
 - (4) take any further enforcement measures necessary in order to ensure the termination of the non-compliance and, where relevant, remedy the consequences thereof.
- (e) Without prejudice to any additional enforcement measures, if the authority of a Member State that acts in accordance with point [ARA.GEN.300\(d\)](#) identifies any non-compliance with the

essential requirements set out in Annex IV to [Regulation \(EU\) 2018/1139](#), with the requirements of Annex I (Part-FCL), Annex VII (Part-ORA) and Annex VIII (Part-DTO) to this Regulation, or with the requirements of Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#) by an organisation certified by, or having made a declaration to, the competent authority of another Member State or the Agency, it shall inform that competent authority of that non-compliance.

GM1 ARA.GEN.350 Findings and corrective actions – organisations

ED Decision 2012/006/R

TRAINING

For a level 1 finding it may be necessary for the competent authority to ensure that further training by the organisation is carried out and audited by the competent authority before the activity is resumed, dependent upon the nature of the finding.

GM1 ARA.GEN.350(e) Findings and corrective actions – organisations

ED Decision 2018/009/R

LEVELS OF FINDINGS ISSUED TO A DTO

Part-ARA requirements do not require competent authorities to categorise findings issued to a DTO. As a consequence, point [ARA.GEN.350\(e\)](#) does not require competent authorities to provide other competent authorities with an indication of the level of the findings issued to a DTO. However, point [ARA.GEN.350\(e\)](#) must not be understood as a prohibition for competent authorities to inform other competent authorities about the level of a finding in such a case, if such finding levels are used by that competent authority on a voluntary basis.

ARA.GEN.355 Findings and enforcement measures – persons

Regulation (EU) No 290/2012

- (a) If, during oversight or by any other means, evidence is found by the competent authority responsible for oversight in accordance with [ARA.GEN.300\(a\)](#) that shows a non-compliance with the applicable requirements by a person holding a licence, certificate, rating or attestation issued in accordance with Regulation (EC) No 216/2008 and its Implementing Rules, the competent authority shall raise a finding, record it and communicate it in writing to the licence, certificate, rating or attestation holder.
- (b) When such finding is raised, the competent authority shall carry out an investigation. If the finding is confirmed, it shall:
 - (1) limit, suspend or revoke the licence, certificate, rating or attestation as applicable, when a safety issue has been identified; and
 - (2) take any further enforcement measures necessary to prevent the continuation of the non-compliance.
- (c) Where applicable, the competent authority shall inform the person or organisation that issued the medical certificate or attestation.
- (d) Without prejudice to any additional enforcement measures, when the authority of a Member State acting under the provisions of [ARA.GEN.300\(d\)](#) finds evidence showing a non-compliance

with the applicable requirements by a person holding a licence, certificate, rating or attestation issued by the competent authority of any other Member State, it shall inform that competent authority.

- (e) If, during oversight or by any other means, evidence is found showing a non-compliance with the applicable requirements by a person subject to the requirements laid down in Regulation (EC) No 216/2008 and its Implementing Rules and not holding a licence, certificate, rating or attestation issued in accordance with that Regulation and its Implementing Rules, the competent authority that identified the non-compliance shall take any enforcement measures necessary to prevent the continuation of that non-compliance.

GM1 ARA.GEN.355(b)(1) Limitation, suspension or revocation of licences, ratings, certificates or attestations

ED Decision 2018/009/R

ENFORCEMENT MEASURES IN CASE OF NON-COMPLIANCE WITH PART-FCL

If the holder of a licence, rating, certificate or attestation does not or no longer comply with the applicable requirements, the competent authority, when acting in accordance with point [ARA.GEN.355\(b\)](#), should take enforcement measures which should be commensurate with the nature of the non-compliance. For example, if the training required for the issuing of the pilot licence was not fully completed as required, the competent authority may decide, subject to the amount and nature of the missing training elements, to suspend the licence in accordance with point [ARA.FCL.250](#) until the missing training elements and a new skill test have been completed rather than revoking the licence.

GM1 ARA.GEN.355(e) Findings and enforcement measures – persons

ED Decision 2018/009/R

This provision is necessary to ensure that enforcement measures will be taken also in cases where the competent authority may not act on the licence, certificate or attestation. The type of enforcement measure will depend on the applicable national law and may include for example the payment of a fine or the prohibition from exercising.

It covers two cases:

- (a) persons subject to the requirements laid down in Regulation (EC) No 216/2008 and its Implementing Rules who are not required to hold a licence, certificate or attestation - for example general medical practitioners (GMPs); and
- (b) persons who are required to hold a licence, rating, certificate or attestation, but who do not hold the appropriate licence, rating, certificate or attestation as required for the activity they perform.

ARA.GEN.360 Change of competent authority

Regulation (EU) 2024/2076

- (a) Upon receiving a licence holder's request for a change of competent authority as specified in point [FCL.015\(e\)](#) of Annex I (Part-FCL), point BFCL.015(f) of Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) or point SFCL.015(f) of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), the receiving competent authority shall, without

undue delay, request the competent authority of the licence holder to transfer, without undue delay, all of the following:

- (1) a verification of the licence;
 - (2) copies of the licence holder's medical records kept by that competent authority in accordance with points [ARA.GEN.220](#) and [ARA.MED.150](#). The medical records shall be transferred in accordance with point [MED.A.015](#) of Annex IV (Part-MED) and shall include a summary of the relevant medical history of the applicant, verified and signed by the medical assessor.
- (b) The transferring competent authority shall keep the licence holder's original licensing and medical records in accordance with points [ARA.GEN.220](#), [ARA.FCL.120](#) and [ARA.MED.150](#).
- (c) The receiving competent authority shall, without undue delay, reissue the licence and medical certificate provided that it has received and processed all documents specified in point (a). Upon the reissuance of the licence and medical certificate, the receiving competent authority shall immediately request the licence holder to surrender to it the licence issued by the transferring competent authority and the associated medical certificate.
- (d) The receiving competent authority shall immediately notify the transferring competent authority once it has reissued the licence and medical certificate to the licence holder and the licence holder has surrendered the licence and medical certificate pursuant to point (c). Until such a notification is received, the transferring competent authority remains responsible for the licence and the medical certificate originally issued to that licence holder.
- (e) When a competent authority receives an medical certificate holder's request for a change of competent authority, as specified in the requirements referred to in point (a), the procedure specified in points (a) to (d) shall apply.

AMC1 ARA.GEN.360(a) Change of competent authority

ED Decision 2020/005/R

When transferring the summary of the applicant's relevant medical history and copies of medical records to the receiving competent authority in accordance with point [ARA.GEN.360\(a\)](#), the transferring competent authority should include at least all of the following:

- (a) copies of:
- (1) the most recent aeromedical report containing the detailed results of the aeromedical examinations and assessments that are required for the class of medical certificate;
 - (2) the application form, examination form, and medical certificate issued;
 - (3) the most recent electrocardiogram (ECG), ophthalmological and ear-nose-throat (ENT), including audiometry, examination reports, as applicable for the class of medical certification;
 - (4) the initial medical examination or the supporting documents for the last medical-file transfer between licensing authorities; where this is not available, a copy of the medical report from the last three aeromedical examinations should be transferred as an alternative;
 - (5) the mental health assessment, as applicable for the class of medical certificate; and
 - (6) any other relevant medical documentation; and

- (b) the ‘Summary of medical history’ form of [AMC1 ARA.GEN.360\(a\)\(2\)](#), filled in and signed by the medical assessor.

AMC1 ARA.GEN.360(a)(1) Change of competent authority

ED Decision 2020/005/R

LICENCE VERIFICATION FORM

In this form, ‘issuing competent authority of the license’ means the ‘transferring competent authority’ of [ARA.GEN.360](#).

LICENCE VERIFICATION FORM				
It is required that this form is filled in and signed by the issuing competent authority of the licence being transferred.				
ITEM	DESCRIPTION			
1	State of licence(s) issue		Country	
2	Title of licences/certificates (including restriction(s)) and corresponding licences/certificates numbers ^{1*}		e.g. PPL(A) — UN country code.FCL.xxx — no valid ratings or SPL — UN country code.FCL.xxx	
3	Licence issue date and expiry date (if applicable)		Issue PPL(A): xx/xx/xxxx Issue SPL: xx/xx/xxxx	
4	Full name (Last and first names)		LAST NAME 1, LAST NAME 2, etc. First name 1, First name 2, etc.	
5	Date of birth (dd/mm/yyyy)		xx/xx/xxxx	
6	Address (as on the licence)			
7	Contact details: email and phone number.		e.g. example@example.eu +(country code) xxxxxxxxxxx	
8	Nationality		Country	
9	Issuing authority (conditions under which the licence was issued, where necessary)		Country and authority	
10	Valid and non-expired ratings/privileges and certificates held (Type/class/instrument/additional ratings and instructor/examiner certificates) Note: indicate all applicable restrictions and extensions.	Ratings and certificates	Valid until (dd/mm/yyyy)	
		e.g. TMG (Sailplane)	xx/xx/xxxx	
		e.g. FI (Sailplane) with extensions for TMG and FI	xx/xx/xxxx	
11	Expired ratings and certificates held (Type/class/instrument/additional ratings and instructor/examiner certificates) Note: indicate all applicable restrictions and extensions.	Ratings and certificates	Valid until (dd/mm/yyyy)	
		e.g. TMG (Aeroplane)	xx/xx/xxxx	
12	Remarks, i.e. special endorsements relating to limitations, restrictions, or endorsements for privileges (e.g. language proficiency level and validity (English, others))	Special endorsements		
		Language	Level	Validity (dd/mm/yyyy)
13	Details on completion of theoretical-knowledge or flight instruction, theoretical-knowledge examination or		e.g. IR theory valid until xx/xx/xxxx	

¹ Indicate all licences and certificates held. Indicate the certificate(s) if you do not hold a valid licence anymore.

	skill test in other Member States, if applicable (e.g validity of the ATPL theoretical knowledge)	
14	Past or pending enforcement action ¹	Yes ☐ No ☐ (If yes, please give details on a separate page.)
I, _____ certify that the details entered on this information form are true, complete, and correct. For any comments, please use the space provided below or on the next page, and tick here: ☐		
Authority: _____ Contact details: _____ Position: _____ Signature: _____ Stamp/seal: _____ Date: _____		
Comments: _____ 		

AMC1 ARA.GEN.360(a)(2) Change of competent authority

ED Decision 2020/005/R

SUMMARY OF MEDICAL HISTORY — FORM FOR THE TRANSFER OF MEDICAL RECORDS

SUMMARY OF MEDICAL HISTORY — FORM FOR THE TRANSFER OF MEDICAL RECORDS		
MEDICAL DETAILS IN CONFIDENCE		
Item	Description	
1	State of licence(s) issue	<i>Country</i>
2	Title of licence(s)/certificate(s) and corresponding serial number of licence(s) held (or national medical reference number)	<i>e. g. PPL(A) — UN country code.FCL.xxx or SPL — UN country code.FCL.xxx</i>
3	Full name (Last and first names)	<i>LAST NAME 1, LAST NAME 2, etc. First name 1, First name 2, etc.</i>
4	Date of birth (dd/mm/yyyy)	<i>xx/xx/xxxx</i>
5	Address	
6	Contact details: email; and phone number.	<i>e.g.</i> <i>(a) example@example.eu</i> <i>(b) +(country code) xxxxxxxxxxxx</i>
7	Nationality	<i>Country</i>
8	Issuing authority	<i>Country and authority</i>
9	Initial medical certificate:	Date of issue <i>xx/xx/xxxx</i>
		Date of examination <i>xx/xx/xxxx</i>
		Type of certificate (Joint Aviation Authorities (JAR), Part-Med or national)
		Class
10	Dates of last three revalidation/renewal examinations (if any)	

¹ Item 14: specify if there is a current investigation into the medical certificate and licence, or suspension or revocation thereof.

11	Limitations (if any)	
12	Comments on any relevant aspect of the applicant's medical history or examination (if applicable, please enclose reports) Please enclose at least the latest examination report and electrocardiogram (ECG). In addition, where applicable for the class of medical certification, please enclose the latest ophthalmological, ear-nose-throat (ENT), and mental health assessment reports.	
13	Past or pending enforcement action ¹	Yes ☐ No ☐ (If yes, please give details on a separate page.)

If there is insufficient space on this form for any information, please use additional pages.

CERTIFICATION		
I, Dr _____, as medical assessor of the (NAA name) _____, certify that the details given above and on any additional pages included are true, complete, and correct.		
Date	Signature	Licensing authority and stamp/seal

GM1 ARA.GEN.360 Change of competent authority

ED Decision 2020/005/R

APPLICATION FORM FOR CHANGE OF COMPETENT AUTHORITY

In this form, 'current competent authority' means the 'transferring competent authority' of [ARA.GEN.360](#), and 'future competent authority' means the 'receiving competent authority' of [ARA.GEN.360](#).

APPLICATION FORM FOR CHANGE OF COMPETENT AUTHORITY		
Applicant details:	Full name (Last and first names)	<i>LAST NAME 1, LAST NAME 2, etc.</i> <i>First name 1, First name 2, etc.</i>
	Title of licence(s)/certificate(s) (including restriction(s)) and corresponding licence(s)/certificate(s) number(s) ²	<i>e.g. PPL(A) — UN country code.FCL.xxx</i> <i>e.g. SPL — UN country code.FCL.xxx</i>
	Current competent authority	<i>Country and authority</i>
	Future competent authority	<i>Country and authority</i>

¹ Item 13: specify if there is a current investigation into the medical certificate and licence, or suspension or revocation thereof.

² Indicate all licences and certificates currently held. Indicate only the related certificate(s) if you do not hold a valid licence anymore (e.g. SFI(A)).

I, _____ (last name, first name) hereby apply for a change of competent authority from my current competent authority to the future competent authority. To that end, I consent to a transfer of medical records, including the transfer of medical records and associated exchange of information between the current and future competent authorities. I apply for transfer of all my licences issued in accordance with Regulations (EU) No 1178/2011, (EU) 2018/395, and (EU) 2018/1976 within the different categories.

I will immediately surrender my current licences/certificates and medical certificate to the future competent authority upon receiving the 'new' licences/certificates and medical certificate.

I understand that the current competent authority remains my competent authority until I have received the new licences/certificates and medical certificate, as applicable, issued by the future competent authority.

I hereby declare that I have not submitted any other request to another competent authority than the future competent authority as indicated above.

I have fully reviewed the *[please insert reference to the current competent authority's relevant information material]* and have submitted all the necessary paperwork for my application to be considered.

I declare that the information provided on this application form is true, complete, and correct.

Any incorrect information on this form or non-compliance with the essential requirements of Annex IV to the Basic Regulation or with the requirements of Regulations (EU) No 1178/2011, (EU) 2018/395, and (EU) 2018/1976 could disqualify the applicant from having his records transferred from the current to the future competent authority.

Signature:

Date:

GM2 ARA.GEN.360 Change of competent authority

ED Decision 2020/005/R

LICENCE VERIFICATION

The licence verification includes the verification of all associated privileges, ratings, certificates, and endorsements that were obtained in accordance with the technical requirements of Regulations (EU) No 1178/2011, (EU) 2018/395, and (EU) 2018/1976. This means that for example, senior examiner privileges are not included.

AVAILABLE RECORDS

Available medical records are all medical records of the licence holder that are related to the history of the medical certificate.

RECORDS

Original licensing and medical records are the original records of the licence holder or electronic records kept by the competent authority.

VALIDITY PERIODS

When reissuing the licence(s) and medical certificate(s), the receiving competent authority should ensure that the validity periods and limitations (if any) are in accordance with the ones of the licence(s) and medical certificate(s) transferred.

PROCESSING

Processing all documents means that the receiving competent authority checks the completeness, and correctness of all the information provided by the transferring competent authority and asks the transferring competent authority for clarification, if needed. If by any means, the receiving competent authority becomes aware of non-compliance with the essential requirements of Annex IV to the Basic Regulation or with the requirements of Regulations (EU) No 1178/2011, (EU) 2018/395, and (EU)

2018/1976 during the processing of the documents, it should reject the application for change of competent authority and inform the transferring competent authority in accordance with its national administration rules.

GM3 ARA.GEN.360 Change of competent authority

ED Decision 2020/005/R

The competent authority can establish and implement its administrative procedures as it considers appropriate. The following practical guidance is considered best practice that may facilitate the work of, and coordination between, competent authorities.

CASES OF SUSPENSION, REVOCATION, OR CURRENT INVESTIGATION

In case of suspension of a licence or medical certificate, the competent authority responsible for the suspension is the only one entitled to remove the suspension. Therefore, a licence holder with a suspended licence or medical certificate cannot apply for change of competent authority until the suspension is revoked.

In case of revocation of a licence, the licence holder can apply for change of competent authority. The licence holder does not immediately receive a new licence after the change of competent authority, but is able to apply for a new licence to the new authority after all necessary requirements of Annex I (Part-FCL) to Regulation (EU) No 1178/2011 and/or Annex III (Part-BFCL) to Regulation (EU) 2018/395 and/or Annex III (Part-SFCL) to Regulation (EU) 2018/1976 are met. However, the licence holder may immediately receive a medical certificate from the receiving competent authority, if applicable.

In case of revocation of a medical certificate, the certificate holder can apply for change of competent authority. The certificate holder does not immediately receive a new licence after the change of competent authority, but is able to apply for a new certificate and licence to the new authority after all necessary requirements of Annexes I (Part-FCL) and IV (Part-MED) to Regulation (EU) No 1178/2011 and/or Annex III (Part-BFCL) to Regulation (EU) 2018/395 and/or Annex III (Part-SFCL) to Regulation (EU) 2018/1976 are met.

In case of an ongoing investigation that is based on evidence of non-compliance, the licence holder cannot immediately apply for change of competent authority. Sufficient time to investigate the case should be provided to reach a conclusion whether or not the licence or medical certificate must be suspended or revoked before the licence holder can apply for change of competent authority.

SUBPART FCL – SPECIFIC REQUIREMENTS RELATING TO FLIGHT CREW LICENSING

SECTION I – GENERAL

ARA.FCL.120 Record-keeping

Regulation (EU) No 1178/2011

In addition to the records required in [ARA.GEN.220\(a\)](#), the competent authority shall include in its system of record-keeping results of theoretical knowledge examinations and the assessments of pilots' skills.

SECTION II – LICENCES, RATINGS AND CERTIFICATES

ARA.FCL.200 Procedure for issue, revalidation or renewal of a licence, rating or certificate

Regulation (EU) 2024/2076

- (a) Issue of licences and ratings. The competent authority shall issue a pilot licence and associated ratings, using the form as established in [Appendix I](#) to this Part.

If a pilot intends to fly outside Union territory on an aircraft registered in a Member State other than the Member State that issued the flight crew licence, the competent authority shall:

- (1) add the following remark on the flight crew licence under item XIII: “This licence is automatically validated as per the ICAO attachment to this licence”; and
 - (2) make the ICAO attachment available to the pilot in print or electronic format.
- (b) Issue of instructor and examiner certificates. The competent authority shall issue an instructor or examiner certificate as:
- (1) an endorsement of the relevant privileges in the pilot licence as established in [Appendix I](#) to this Part; or
 - (2) a separate document, in a form and manner specified by the competent authority.
- (c) Endorsement of licences by examiners. Before specifically authorising an examiner to revalidate or renew ratings or certificates, the competent authority shall develop appropriate procedures.
- (d) Endorsement of licence by instructors. Before specifically authorising certain instructors to revalidate an SEP aeroplane class rating, a TMG class rating or a type rating for a single-engine helicopter up to a MTOM of 3 175 kg, the competent authority shall develop appropriate procedures.
- (e) Instructors for FI(B) or FI(S) certificates: The competent authority shall develop appropriate procedures for the conduct of the training flights under supervision specified in:
- (1) points BFCL.315(a)(4)(ii) and BFCL.360(a)(2) of Annex III (Part-BFCL) to [Regulation \(EU\) 2018/395](#); and
 - (2) points SFCL.315(a)(7)(ii) and SFCL.360(a)(2) of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#).

GM1 ARA.FCL.200 Procedure for issue, revalidation or renewal of a licence, rating or certificate

ED Decision 2025/002/R

LICENCE ENDORSEMENTS

- (a) General

This GM provides guidance on how to make endorsements in pilot licences, in accordance with the applicable requirements of Annex I (Part-FCL) and following the licence format set out in [Appendix I](#) to Annex VI (Part-ARA).

- (b) Endorsement for licence privileges

For endorsing extra privileges of the holder of a PPL(A) on an MPL in accordance with point [FCL.405.A\(b\)\(1\)](#) of Part-FCL, the phrase ‘PPL(A) privileges included’, followed by the date of endorsement of these PPL(A) privileges, should be endorsed in Section XIII (‘Remarks’) as shown below:

XIII Remarks:

PPL(A) privileges included (date of endorsement: DD.MM.YYYY)

(c) *Reserved*

(d) Endorsement for class and type rating privileges

Class and type ratings should be endorsed as set out in the EASA class and type rating endorsement lists, as published on the EASA website. Remarks and restrictions to class and type ratings (relevant line in SECTION XII, page 4 of the licence format) should be endorsed as follows:

Table 1 — Aeroplanes

<i>Reference</i>	<i>Requirement</i>	<i>Endorsement</i>
FCL.725(d)(2) Part-FCL Appendix 9 , B(5)(j) and (6)(h)	Restriction to multi-pilot operation in a single-pilot aeroplane	MPO only
FCL.720.A(c)	Cruise relief co-pilot restriction	CRCP only
FCL.720.A(d)	OSD restriction to flights with instructor	With instructor only

Table 2 — Helicopters

<i>Reference</i>	<i>Requirement</i>	<i>Endorsement</i>
FCL.725(d)(2)	Restriction to multi-pilot operation in a single-pilot helicopter	MPO only
FCL.720.H(b)	Co-pilot restriction for graduates from integrated training with less than 70 hours of PIC experience	Co-pilot only

(e) Endorsements for additional ratings

The night rating as per point [FCL.810](#) is applicable exclusively for holders of an LAPL(A), PPL(A), PPL(H) and PPL(As) and needs to be endorsed solely on these licences. For pilot licences that automatically include NVFR privileges (CPL, MPL, ATPL), there is no need to separately endorse NVFR privileges.

AMC1 ARA.FCL.200(a)(1) Procedure for issue, revalidation or renewal of a licence, rating or certificate

ED Decision 2025/002/R

REMARK ON THE LICENCE

When issuing the licence with the remark on the licence item XIII: ‘This licence is automatically validated as per the ICAO attachment to this licence’, the competent authority should provide the holder of the licence with the ICAO attachment.

AMC1 ARA.FCL.200(a)(2) Procedure for issue, revalidation or renewal of a licence, rating or certificate

ED Decision 2025/002/R

ICAO ATTACHMENT

The layout of the ICAO attachment in electronic or paper format is the following:



EUROPEAN UNION ICAO attachment to automatically validate licences (Issue 1) issued in accordance with Commission Regulation (EU) No 1178/2011
1. The licence is automatically validated by all the ICAO Contracting States listed in point (2) under an agreement registered with ICAO. The ICAO Registration Numbers are: — 5950 (EU) — 5951 (EU plus Switzerland); and — 5952 (EU plus Norway and Iceland).
2. The ICAO Contracting States that automatically validate this licence are: [Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland.] * Please select the applicable ICAO Contracting State(s)
[Competent authority] European Aviation Safety Agency Date of issue: _____

EASA Form 207 – Issue 1

ARA.FCL.205 Monitoring of examiners

Regulation (EU) No 245/2014

- (a) The competent authority shall develop an oversight programme to monitor the conduct and performance of examinerstaking into account:

- (1) the number of examiners it has certified; and
- (2) the number of examiners certified by other competent authorities exercising their privileges within the territory where the competent authority exercises oversight.
- (b) The competent authority shall maintain a list of examiners it has certified. The list shall state the privileges of the examiners and be published and kept updated by the competent authority.
- (c) The competent authority shall develop procedures to designate examiners for the conduct of skill tests.

AMC1 ARA.FCL.205 Monitoring of examiners

ED Decision 2012/006/R

QUALIFICATION OF INSPECTORS

Inspectors of the competent authority supervising examiners should ideally meet the same requirements as the examiners being supervised. However, it is unlikely that they could be so qualified on the large variety of types and tasks for which they have a responsibility and, since they normally only observe training and testing, it is acceptable if they are qualified for the role of an inspector.

AMC2 ARA.FCL.205 Monitoring of examiners

ED Decision 2021/002/R

EBT PROGRAMME

The operator's competent authority should include the EBT manager(s) in the programme of monitoring of examiners even if they hold an examiner certificate issued by another competent authority. At the discretion of the competent authority, this may also include an inspection of training delivery within the EBT programme.

GM1 ARA.FCL.205 Monitoring of examiners

ED Decision 2021/002/R

EBT PROGRAMME — INSPECTION OF TRAINING DELIVERY

When the authority conducts an inspection of the FCL requirements (e.g. training delivery), it is advisable that the inspector of the competent authority follow the content of AMC1 ARO.OPS.226(a). This inspection may be combined with the oversight required in ARO.OPS.226 of [Regulation \(EU\) No 965/2012](#).

ARA.FCL.210 Information for examiners

Regulation (EU) No 245/2014

- (a) The competent authority shall notify the Agency of the national administrative procedures, requirements for protection of personal data, liability, accident insurance and fees applicable in its territory, which shall be used by examiners when conducting skill tests, proficiency checks or assessments of competence of an applicant for which the competent authority is not the same that issued the examiner's certificate.
- (b) To facilitate dissemination and access to the information received from competent authorities under (a), the Agency shall publish this information according to a format prescribed by it.

- (c) The competent authority may provide examiners it has certified and examiners certified by other competent authorities exercising their privileges in their territory with safety criteria to be observed when skill tests and proficiency checks are conducted in an aircraft.

ARA.FCL.215 Validity period

Regulation (EU) No 290/2012

- (a) When issuing or renewing a rating or certificate, the competent authority or, in the case of renewal, an examiner specifically authorised by the competent authority, shall extend the validity period until the end of the relevant month.
- (b) When revalidating a rating, an instructor or an examiner certificate, the competent authority, or an examiner specifically authorised by the competent authority, shall extend the validity period of the rating or certificate until the end of the relevant month.
- (c) The competent authority, or an examiner specifically authorised for that purpose by the competent authority, shall enter the expiry date on the licence or the certificate.
- (d) The competent authority may develop procedures to allow privileges to be exercised by the licence or certificate holder for a maximum period of 8 weeks after successful completion of the applicable examination(s), pending the endorsement on the licence or certificate.

ARA.FCL.220 Procedure for the re-issue of a pilot licence

Regulation (EU) No 290/2012

- (a) The competent authority shall re-issue a licence whenever necessary for administrative reasons and:
- (1) after initial issue of a rating; or
 - (2) when paragraph XII of the licence established in Appendix I to this Part is completed and no further spaces remain.
- (b) Only valid ratings and certificates shall be transferred to the new licence document.

ARA.FCL.250 Limitation, suspension or revocation of licences, ratings and certificates

Regulation (EU) 2020/359

- (a) The competent authority shall limit, suspend or revoke as applicable a pilot licence and associated ratings or certificates in accordance with [ARA.GEN.355](#) in, but not limited to, the following circumstances:
- (1) obtaining the pilot licence, rating or certificate by falsification of submitted documentary evidence;
 - (2) falsification of the logbook and licence or certificate records;
 - (3) the licence holder no longer complies with the applicable requirements of Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) or Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#);
 - (4) exercising the privileges of a licence, rating or certificate when adversely affected by alcohol or drugs;
 - (5) non-compliance with the applicable operational requirements;

- (6) evidence of malpractice or fraudulent use of the certificate; or
- (7) unacceptable performance in any phase of the flight examiner's duties or responsibilities.
- (b) The competent authority may also limit, suspend or revoke a licence, rating or certificate upon the written request of the licence or certificate holder.
- (c) All skill tests, proficiency checks or assessments of competence conducted during suspension or after the revocation of an examiner's certificate will be invalid.

SECTION III – THEORETICAL KNOWLEDGE EXAMINATIONS

ARA.FCL.300 Examination procedures

Regulation (EU) 2024/2076

- (a) The competent authority shall put in place the necessary arrangements and procedures to allow applicants to take theoretical knowledge examinations in accordance with the applicable requirements of Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) or Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#).
- (b) In the case of the ATPL, MPL, commercial pilot licence (CPL), and instrument ratings, those procedures shall comply with all of the following:
 - (1) Examinations shall be done in written or computer-based form.
 - (2) Questions for an examination shall be selected by the competent authority, according to a common method which allows coverage of the entire syllabus in each subject, from the European Central Question Bank (ECQB). The ECQB is a database of multiple choice questions held by the Agency.
 - (3) The examination in communications may be provided separately from those in other subjects.
- (c) The competent authority shall inform applicants of the languages available for examinations.
- (d) The competent authority shall establish appropriate procedures to ensure the integrity of the examinations.
- (e) If the competent authority finds that the applicant is not complying with the examination procedures during the examination, this shall be assessed with a view to failing the applicant, either in the examination of a single subject or in the examination as a whole.
- (f) The competent authority shall ban applicants who are proven to be cheating from taking any further examination for a period of at least 12 months from the date of the examination in which they were found cheating.

AMC1 ARA.FCL.300 Examination procedures

ED Decision 2019/017/R

GENERAL

- (a) The competent authority should provide suitable facilities for the conduct of examinations.
- (b) The content of the examination papers should retain a confidential status until the end of the examination session.
- (c) The identity of the applicant should be confirmed before an examination is taken.
- (d) Examination applicants should be seated in a way so that they cannot read each other's examination papers. They should not speak to any person other than the invigilators.
- (e) All examination papers, associated documents and additional papers handed out to the applicants for the examination should be handed back to the invigilator at the end of the examination.

- (f) Only the examination paper, specific documentation and tools needed for the examination should be available to the applicant during the examination.
- (g) Applicants may use the following equipment during an examination:
 - (1) a scientific, non-programmable, non-alphanumeric calculator without specific aviation functions;
 - (2) mechanical navigation slide-rule (DR calculator);
 - (3) protractor;
 - (4) compasses and dividers;
 - (5) ruler.
- (h) Applicants may use a translation dictionary at the discretion of the competent authority.
- (i) Except equipment specified above, applicant(s) should not use any electronic equipment during the examination(s).

AMC1 ARA.FCL.300(b) Examination procedures

ED Decision 2025/002/R

THEORETICAL KNOWLEDGE EXAMINATIONS FOR PROFESSIONAL LICENCES AND INSTRUMENT RATINGS

With regard to the IR(A), CBIR(A) and BIR, these tables apply to theoretical knowledge examinations for applicants who have completed the appropriate elements of theoretical knowledge instruction of a modular training course for the IR(A) according to Appendix 6 Section A, for the CBIR(A) according to Appendix 6 Section Aa, and for the BIR according to [FCL.835](#).

Subject 010 — AIR LAW										
Theoretical knowledge examination										
Exam length, total number of questions, and distribution of questions										
	ATPL(A)	CPL(A)	ATPL(H)/ IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	1:00	0:45	1:00	0:45	0:45	0:45	0:30	*	*	*
Distribution of questions with regard to the topics of the syllabus										
010 01	02	02	02	02	02	XX	XX	XX	XX	XX
010 02	01	01	01	01	01	XX	XX	XX	XX	XX
010 03	XX	XX	XX	XX	XX	XX	XX	XX	XX	XX
010 04	01	01	01	01	01	01	01	XX	XX	XX
010 05	09	09	09	09	09	06	05	XX	XX	02
010 06	10	05	10	05	05	08	06	XX	06	XX
010 07	06	04	06	04	04	06	03	XX	XX	06
010 08	01	01	01	01	01	02	01	XX	01	XX
010 09	08	05	08	05	05	07	02	XX	02	XX
010 10	01	01	01	01	01	XX	XX	XX	XX	XX
010 11	01	01	01	01	01	XX	XX	XX	XX	XX
010 12	02	02	02	02	02	XX	XX	XX	XX	XX
010 13	02	01	02	01	01	XX	XX	XX	XX	XX
Total number of questions	44	33	44	33	33	30	18	XX	09	08

Subject 021 — AIRCRAFT GENERAL KNOWLEDGE — AIRFRAME/SYSTEMS/POWER PLANT

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H)/ IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	2:00	1:30	2:00	2:00	1:30	XX	XX	XX	XX	XX
Distribution of questions with regard to the topics of the syllabus										
021 01	02	01	02	02	01	XX	XX	XX	XX	XX
021 02	02	02	02	02	01	XX	XX	XX	XX	XX
021 03	05	03	03	03	02	XX	XX	XX	XX	XX
021 04	05	04	03	03	02	XX	XX	XX	XX	XX
021 05	08	06	05	05	04	XX	XX	XX	XX	XX
021 06	05	02	02	02	02	XX	XX	XX	XX	XX
021 07	02	02	02	02	02	XX	XX	XX	XX	XX
021 08	05	03	05	05	03	XX	XX	XX	XX	XX
021 09	15	11	14	14	09	XX	XX	XX	XX	XX
021 10	06	11	06	06	09	XX	XX	XX	XX	XX
021 11	19	11	17	17	09	XX	XX	XX	XX	XX
021 12	03	02	03	03	02	XX	XX	XX	XX	XX
021 13	03	02	XX	XX	XX	XX	XX	XX	XX	XX
021 14	XX	XX	01	01	01	XX	XX	XX	XX	XX
021 15	XX	XX	04	04	03	XX	XX	XX	XX	XX
021 16	XX	XX	06	06	05	XX	XX	XX	XX	XX
021 17	XX	XX	05	05	05	XX	XX	XX	XX	XX
Total number of questions	80	60	80	80	60	XX	XX	XX	XX	XX

Subject 022 — AIRCRAFT GENERAL KNOWLEDGE — INSTRUMENTATION

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H)/ IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	1:30	1:00	1:30	1:30	1:00	0:30	0:20	*	*	*
Distribution of questions with regard to the topics of the syllabus										
022 01	03	03	03	03	03	XX	XX	XX	XX	XX
022 02	08	08	09	09	08	04	04	03	XX	XX
022 03	02	02	03	03	02	01	01	01	XX	XX
022 04	04	04	06	06	04	02	02	03	XX	XX
022 05	03	XX	03	03	XX	XX	XX	XX	XX	XX
022 06	11	09	XX	XX	XX	02	XX	XX	XX	XX
022 07	XX	XX	14	14	11	XX	XX	XX	XX	XX
022 08	03	02	XX	XX	XX	01	XX	XX	XX	XX
022 09	04	XX	XX	XX	XX	XX	XX	XX	XX	XX
022 10	02	XX	XX	XX	XX	XX	XX	XX	XX	XX
022 11	06	01	06	06	XX	03	01	XX	XX	01
022 12	08	07	09	09	07	03	XX	XX	XX	XX
022 13	04	03	05	05	04	03	03	04	XX	XX
022 14	01	01	01	01	01	XX	XX	XX	XX	XX
022 15	01	XX	01	01	XX	01	01	XX	XX	XX
Total number of questions	60	40	60	60	40	20	12	11	XX	01

Subject 031 — FLIGHT PERFORMANCE AND PLANNING — MASS AND BALANCE

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H)/ IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	1:15	1:00	1:15	1:15	1:00	XX	XX	XX	XX	XX
Distribution of questions with regard to the topics of the syllabus										
031 01	01	01	01	01	01	XX	XX	XX	XX	XX
031 02	08	07	09	09	07	XX	XX	XX	XX	XX
031 03	XX	XX	XX	XX	XX	XX	XX	XX	XX	XX
031 04	05	04	04	04	04	XX	XX	XX	XX	XX
031 05	09	07	09	09	07	XX	XX	XX	XX	XX
031 06	02	02	02	02	02	XX	XX	XX	XX	XX
Total number of questions	25	21	25	25	21	XX	XX	XX	XX	XX

Subject 032 — FLIGHT PERFORMANCE AND PLANNING — PERFORMANCE (AEROPLANES)

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	2:00	1:15	XX	XX	XX	XX	XX	XX	XX	XX
Distribution of questions with regard to the topics of the syllabus										
032 01	12	12	XX	XX	XX	XX	XX	XX	XX	XX
032 02	06	09	XX	XX	XX	XX	XX	XX	XX	XX
032 03	03	06	XX	XX	XX	XX	XX	XX	XX	XX
032 04	18	01	XX	XX	XX	XX	XX	XX	XX	XX
032 05	06	XX	XX	XX	XX	XX	XX	XX	XX	XX
Total number of questions	45	28	XX	XX	XX	XX	XX	XX	XX	XX

Subject 033 — FLIGHT PERFORMANCE AND PLANNING — FLIGHT PLANNING AND MONITORING

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	2:00	1:30	2:00	1:30	1:30	1:15	1:00	*	*	*
Distribution of questions with regard to the topics of the syllabus										
033 01	06	11	07	11	11	XX	XX	XX	XX	XX
033 02	13	XX	13	XX	XX	13	12	XX	07	XX
033 03	11	10	10	10	10	05	05	XX	XX	01
033 04	06	06	06	06	06	05	04	XX	XX	02
033 05	01	01	01	01	01	01	01	XX	XX	01
033 06	05	05	05	05	05	03	XX	XX	XX	XX
Total number of questions	42	33	42	33	33	27	22	XX	07	04

Subject 034 — FLIGHT PERFORMANCE AND PLANNING — PERFORMANCE (HELICOPTERS)

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	XX	XX	1:15	1:15	0:45	XX	XX	XX	XX	XX
Distribution of questions with regard to the topics of the syllabus										
034 01	XX	XX	12	12	11	XX	XX	XX	XX	XX
034 02	XX	XX	07	07	09	XX	XX	XX	XX	XX
034 03	XX	XX	03	03	XX	XX	XX	XX	XX	XX
034 04	XX	XX	13	13	XX	XX	XX	XX	XX	XX
Total number of questions	XX	XX	35	35	20	XX	XX	XX	XX	XX

Subject 040 — HUMAN PERFORMANCE

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	1:30	1:00	1:30	1:30	1:00	1:00	0:30	*	*	*
Distribution of questions with regard to the topics of the syllabus										
040 01	04	03	04	04	03	03	01	01	XX	XX
040 02	24	18	24	24	18	18	08	07	XX	XX
040 03	20	14	20	20	14	14	07	08	XX	XX
Total number of questions	48	35	48	48	35	35	16	16	XX	XX

Subject 050 — METEOROLOGY

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	2:00	1:30	2:00	2:00	1:30	1:30	0:50	*	*	*
Distribution of questions with regard to the topics of the syllabus										
050 01	10	08	10	10	08	08	04	02	XX	XX
050 02	10	06	10	10	06	06	03	02	XX	XX
050 03	03	03	03	03	03	03	01	01	XX	XX
050 04	08	06	08	08	06	06	05	05	XX	XX
050 05	02	02	02	02	02	02	02	03	XX	XX
050 06	07	06	07	07	06	06	04	XX	XX	04
050 07	06	02	06	06	02	02	01	XX	XX	01
050 08	08	03	08	08	03	03	01	XX	XX	XX
050 09	14	13	14	14	13	13	08	XX	XX	10
050 10	16	14	16	16	14	14	06	XX	06	XX
Total number of questions	84	63	84	84	63	63	35	13	06	15

Subject 061 — GENERAL NAVIGATION

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	2:15	2:00	2:15	2:15	2:00	XX	XX	XX	XX	XX

Distribution of questions with regard to the topics of the syllabus

061 01	28	22	28	28	22	XX	XX	XX	XX	XX
061 02	07	06	07	07	06	XX	XX	XX	XX	XX
061 03	05	05	05	05	05	XX	XX	XX	XX	XX
061 04	12	09	12	12	09	XX	XX	XX	XX	XX
061 05	03	03	03	03	03	XX	XX	XX	XX	XX
Total number of questions	55	45	55	55	45	XX	XX	XX	XX	XX

Subject 062 — RADIO NAVIGATION

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	1:30	0:30	1:30	1:00	0:30	1:00	0:40	*	*	*

Distribution of questions with regard to the topics of the syllabus

062 01	05	04	05	05	04	03	XX	XX	XX	XX
062 02	22	11	22	20	11	22	13	XX	06	XX
062 03	11	02	11	08	02	05	03	XX	02	XX
062 04	XX	XX	XX	XX	XX	XX	XX	XX	XX	XX
062 05	XX	XX	XX	XX	XX	XX	XX	XX	XX	XX
062 06	15	05	15	11	05	08	04	XX	04	XX
062 07	13	XX	13	XX	XX	06	04	XX	07	XX
Total number of questions	66	22	66	44	22	44	24	XX	19	XX

Subject 070 — OPERATIONAL PROCEDURES

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	1:15	1:00	1:15	1:00	1:00	XX	XX	XX	XX	XX
Distribution of questions with regard to the topics of the syllabus										
071 01	17	09	10	08	08	XX	XX	XX	XX	XX
071 02	24	20	22	17	17	XX	XX	XX	XX	XX
071 03	XX	XX	06	05	05	XX	XX	XX	XX	XX
071 04	01	01	02	02	02	XX	XX	XX	XX	XX
Total number of questions	42	30	40	32	32	XX	XX	XX	XX	XX

Subject 081 — PRINCIPLES OF FLIGHT (AEROPLANES)

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	1:30	1:15	XX	XX	XX	XX	XX	XX	XX	XX
Distribution of questions with regard to the topics of the syllabus										
081 01	14	13	XX	XX	XX	XX	XX	XX	XX	XX
081 02	04		XX	XX	XX	XX	XX	XX	XX	XX
081 03	09	07	XX	XX	XX	XX	XX	XX	XX	XX
081 04	04	03	XX	XX	XX	XX	XX	XX	XX	XX
081 05	03	03	XX	XX	XX	XX	XX	XX	XX	XX
081 06	03	03	XX	XX	XX	XX	XX	XX	XX	XX
081 07	04	04	XX	XX	XX	XX	XX	XX	XX	XX
081 08	05	04	XX	XX	XX	XX	XX	XX	XX	XX
Total number of questions	46	37	XX	XX	XX	XX	XX	XX	XX	XX

Subject 082 — PRINCIPLES OF FLIGHT (HELICOPTERS)

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	XX	XX	1:15	1:15	1:15	XX	XX	XX	XX	XX
Distribution of questions with regard to the topics of the syllabus										
082 01	XX	XX	06	06	06	XX	XX	XX	XX	XX
082 02	XX	XX	03	03	03	XX	XX	XX	XX	XX
082 03	XX	XX	01	01	01	XX	XX	XX	XX	XX
082 04	XX	XX	11	11	11	XX	XX	XX	XX	XX
082 05	XX	XX	08	08	08	XX	XX	XX	XX	XX
082 06	XX	XX	04	04	04	XX	XX	XX	XX	XX
082 07	XX	XX	06	06	06	XX	XX	XX	XX	XX
082 08	XX	XX	03	03	03	XX	XX	XX	XX	XX
Total number of questions	XX	XX	42	42	42	XX	XX	XX	XX	XX

Subject 090 - COMMUNICATION

Theoretical knowledge examination

Exam length, total number of questions, and distribution of questions

	ATPL(A)	CPL(A)	ATPL(H) /IR	ATPL(H)	CPL(H)	IR(A) and IR(H)	CB- IR(A)	BIR M01	BIR M02	BIR M03
Time allowed (hours)	1:00	1:00	1:00	1:00	1:00	1:00	1:00	*	*	*
Distribution of questions with regard to the topics of the syllabus										
090 01	04	04	04	04	04	04	04	XX	XX	02
090 02	17	17	17	17	17	17	17	XX	XX	06
090 03	02	02	02	02	02	02	02	XX	XX	01
090 04	04	04	04	04	04	04	04	XX	XX	02
090 05	03	03	03	03	03	03	03	XX	XX	03
090 06	02	02	02	02	02	02	02	XX	XX	01
090 07	02	02	02	02	02	02	02	XX	XX	01
Total number of questions	34	34	34	34	34	34	34	XX	XX	16

*Refer to [AMC2 ARA.FCL.300\(b\)](#)

AMC2 ARA.FCL.300(b) Examination procedures

ED Decision 2020/018/R

TOTAL TIME ALLOWED AND TOTAL NUMBER OF QUESTIONS FOR BIR EXAMINATION

- (a) The total number of questions for the examination for Module 1 is 40 questions and the total time allowed for the Module 1 examination is 1:20 hours.
- (b) The total number of questions for the examination for Module 2 is 41 questions and the total time allowed for the Module 2 examination is 1:30 hours.
- (c) The total number of questions for the examination for Module 3 is 44 questions and the total time allowed for the Module 3 examination is 1:30 hours.

GM1 ARA.FCL.300(b) Examination procedures

ED Decision 2025/002/R

DETAILED THEORETICAL KNOWLEDGE EXAMINATION STRUCTURE FOR THE BIR

The tables below provide a detailed structure for the examination papers for the BIR theoretical knowledge examination, based on the relevant content of [AMC1 ARA.FCL.300\(b\)](#) and [AMC2 ARA.FCL.300\(b\)](#), and the detailed syllabus set out in [AMC1 FCL.310](#); [FCL.515\(b\)](#); [FCL.615\(b\)](#); [FCL.835\(d\)](#). Each of the BIR training modules should be examined through one examination paper. The tables indicate the allocation of Subject / topic elements to the examination papers, and how much time should be allocated to the Subject / topic elements within each examination paper. Subject 050 has elements in all three examination papers, Subjects 010, 022 and 033 are split across two examination papers, while Subjects 040, 062 and 090 are covered in one of the examination papers.

Table 1: Module 1 BIR

Subject / topic elements	Number of questions	Time (hours:minutes)
022 02	3	0:20
022 03	1	
022 04	3	
022 13	4	
040 01	1	0:30
040 02	7	
040 03	8	
050 01	2	0:30
050 02	2	
050 03	1	
050 04	5	
050 05	3	
	40	1:20

Table 2: Module 2 BIR

Subject / topic elements	Number of questions	Time (hours:minutes)
010 06	6	0:15
010 08	1	
010 09	2	
033 02	7	0:20

Subject / topic elements	Number of questions	Time (hours:minutes)
050 10	6	0:15
062 02	6	0:40
062 03	2	
062 06	4	
062 07	7	
	41	1:30

Table 3: Module 3 BIR

Subject / topic elements	Number of questions	Time (hours:minutes)
010 05	2	0:15
010 07	6	
022 11	1	0:03
033 03	1	0:10
033 04	2	
033 05	1	
050 06	4	0:30
050 07	1	
050 09	10	
090 01	2	0:32
090 02	6	
090 03	1	
090 04	2	
090 05	3	
090 06	1	
090 07	1	
	44	1:30

GM2 ARA.FCL.300(b) Examination procedures

ED Decision 2025/002/R

CONSIDERATION OF CPL THEORETICAL KNOWLEDGE CREDITS FOR THE BIR EXAMINATION PAPERS

- Point [FCL.035](#) and [Appendix 1](#) to Part-FCL establish credits with regard to the theoretical knowledge examinations for applicants for a BIR, when they already hold a CPL(A) or have passed the theoretical knowledge examinations for the CPL(A). In the case of applicants for a BIR who have passed the relevant theoretical knowledge examinations for a CPL(A), [Appendix 1](#) to Part-FCL includes credits for Subjects 040 Human Performance, 050 Meteorology, and 090 Communication. With regard to the latter, point [FCL.035\(b\)\(6\)](#) includes specific crediting arrangements for cases where applicants, prior to the introduction of Subject 090 Communication, have completed either Subject 091 VFR Communication or Subject 092 IFR Communication (refer to [GM1 FCL.035\(b\)\(6\)\(ii\)](#) for related guidance).
- Competent authorities should determine the areas of theoretical knowledge for which these applicants will need to take theoretical knowledge examinations. For that purpose, and for designing BIR examination papers that consider specific credits to be granted to individual applicants, competent authorities can use the tables set out in [GM1 ARA.FCL.300\(b\)](#).

SUBPART CC – SPECIFIC REQUIREMENTS RELATING TO CABIN CREW

SECTION I – CABIN CREW ATTESTATIONS

ARA.CC.100 Procedures for cabin crew attestations

Regulation (EU) No 1178/2011

- (a) The competent authority shall establish procedures for the issue, record-keeping and oversight of cabin crew attestations in accordance with [ARA.GEN.315](#), [ARA.GEN.220](#) and [ARA.GEN.300](#) respectively.
- (b) Cabin crew attestations shall be issued, using the format and specifications established in Appendix II to this Part, either
 - (1) by the competent authority; and/or, if so decided by a Member State
 - (2) by an organisation approved to do so by the competent authority.
- (c) The competent authority shall make publicly available:
 - (1) which body(ies) issue cabin crew attestations in their territory; and
 - (2) if organisations are approved to do so, the list of such organisations.

ARA.CC.105 Suspension or revocation of cabin crew attestations

Regulation (EU) No 290/2012

The competent authority shall take measures in accordance with [ARA.GEN.355](#), including the suspension or revocation of a cabin crew attestation, at least in the following cases:

- (a) non-compliance with Part-CC or with the applicable requirements of Part-ORO and Part-CAT, where a safety issue has been identified;
- (b) obtaining or maintaining the validity of the cabin crew attestation by falsification of submitted documentary evidence;
- (c) exercising the privileges of the cabin crew attestation when adversely affected by alcohol or drugs; and
- (d) evidence of malpractice or fraudulent use of the cabin crew attestation.

SECTION II – ORGANISATIONS PROVIDING CABIN CREW TRAINING OR ISSUING CABIN CREW ATTESTATIONS

ARA.CC.200 Approval of organisations to provide cabin crew training or to issue cabin crew attestations

Regulation (EU) No 1178/2011

- (a) Before issuing an approval to a training organisation or a commercial air transport operator to provide cabin crew training, the competent authority shall verify that:
- (1) the conduct, the syllabi and associated programmes of the training courses provided by the organisation comply with the relevant requirements of Part-CC;
 - (2) the training devices used by the organisation realistically represent the passenger compartment environment of the aircraft type(s) and the technical characteristics of the equipment to be operated by the cabin crew; and
 - (3) the trainers and instructors conducting the training sessions are suitably experienced and qualified in the training subject covered.
- (b) If in a Member State organisations may be approved to issue cabin crew attestations, the competent authority shall only grant such approvals to organisations complying with the requirements in (a). Before granting such an approval, the competent authority shall:
- (1) assess the capability and accountability of the organisation to perform the related tasks;
 - (2) ensure that the organisation has established documented procedures for the performance of the related tasks, including for the conduct of examination(s) by personnel who are qualified for this purpose and free from conflict of interest, and for the issue of cabin crew attestations in accordance with [ARA.GEN.315](#) and [ARA.CC.100\(b\)](#); and
 - (3) require the organisation to provide information and documentation related to the cabin crew attestations it issues and their holders, as relevant for the competent authority to conduct its record-keeping, oversight and enforcement tasks.

AMC1 ARA.CC.200(b)(2) Approval of organisations to provide cabin crew training or to issue cabin crew attestations

ED Decision 2012/006/R

PERSONNEL CONDUCTING EXAMINATIONS

For any element being examined for the issue of a cabin crew attestation as required in Part-CC, the person who delivered the associated training or instruction should not also conduct the examination. However, if the organisation has appropriate procedures in place to avoid conflict of interest regarding the conduct of the examination and/or the results, this restriction need not apply.

SUBPART ATO – SPECIFIC REQUIREMENTS RELATED TO APPROVED TRAINING ORGANISATIONS (ATOs)

SECTION I – GENERAL

ARA.ATO.105 Oversight Programme

Regulation (EU) No 1178/2011

The oversight programme for ATOs shall include the monitoring of course standards, including the sampling of training flights with students, if appropriate to the aircraft used.

AMC1 ARA.ATO.105 Oversight programme

ED Decision 2012/006/R

GENERAL

- (a) The audit or inspection of an ATO should be conducted on the basis of checking the facility for compliance, interviewing personnel and sampling any relevant training course for its conduct and standard.
- (b) In addition to the items required in [AMC1 ARA.GEN.310\(a\)](#), such an audit or inspection should focus on:
 - (1) information on flight instructors, validity of licences, certificates, ratings and log books;
 - (2) evidence of sufficient funding;
 - (3) training aircraft in use, including their registration, associated documents and maintenance records;
 - (4) aerodromes, operating sites and associated facilities;
 - (5) facilities with regard to their adequacy to the courses being conducted and number of students;
 - (6) FSTDs, including their qualification certificates, associated documents and maintenance records;
 - (7) documentation, in particular documents related to courses, information on the updating system, and training and operations manual(s);
 - (8) training records and checking forms; and
 - (9) flight instruction, including pre-briefing, actual flight and debriefing.

ARA.ATO.110 Approval of minimum equipment lists

Regulation (EU) 2020/359

When the competent authority receives an application for approval of a minimum equipment list under points ORO.MLR.105 of Annex III (Part-ORO) and NCC.GEN.101 of Annex VI (Part-NCC) to [Regulation \(EU\) No 965/2012](#), it shall act in accordance with point ARO.OPS.205 of Annex II (Part-ARO) to that Regulation.

ARA.ATO.120 Record-keeping

Regulation (EU) No 290/2012

In addition to the records required in [ARA.GEN.220](#), the competent authority shall include in its system of record-keeping details of courses provided by the ATO, and if applicable, records relating to FSTDs used for training.

AMC1 ARA.ATO.120 Record-keeping

ED Decision 2012/006/R

FSTDs

Records relating to FSTDs should include, as a minimum:

- (a) the application for an FSTD qualification;
- (b) the FSTD qualification certificate including any changes;
- (c) a copy of the evaluation programme listing the dates when evaluations are due and when evaluations were carried out;
- (d) initial and recurrent evaluation records;
- (e) copies of all relevant correspondence;
- (f) details of any exemption and enforcement actions; and
- (g) any report from other competent authorities relating to initial and recurrent evaluations.

SUBPART FSTD – SPECIFIC REQUIREMENTS RELATED TO THE QUALIFICATION OF FLIGHT SIMULATION TRAINING DEVICES (FSTDs)

SECTION I – GENERAL

ARA.FSTD.100 Initial evaluation procedure

Regulation (EU) No 1178/2011

- (a) Upon receiving an application for an FSTD qualification certificate, the competent authority shall:
 - (1) evaluate the FSTD submitted for initial evaluation or for upgrading against the applicable qualification basis;
 - (2) assess the FSTD in those areas that are essential to completing the flight crew member training, testing and checking process, as applicable;
 - (3) conduct objective, subjective and functions tests in accordance with the qualification basis and review the results of such tests to establish the qualification test guide (QTG); and
 - (4) verify if the organisation operating the FSTD is in compliance with the applicable requirements. This does not apply to the initial evaluation of basic instrument training devices (BITDs).
- (b) The competent authority shall only approve the QTG after completion of the initial evaluation of the FSTD and when all discrepancies in the QTG have been addressed to the satisfaction of the competent authority. The QTG resulting from the initial evaluation procedure shall be the master QTG (MQTG), which shall be the basis for the FSTD qualification and subsequent recurrent FSTD evaluations.
- (c) Qualification basis and special conditions.
 - (1) The competent authority may prescribe special conditions for the FSTD qualification basis when the requirements of ORA.FSTD.210(a) are met and when it is demonstrated that the special conditions ensure an equivalent level of safety to that established in the applicable certification specification.
 - (2) When the competent authority, if other than the Agency, has established special conditions for the qualification basis of an FSTD, it shall without undue delay notify the Agency thereof. The notification shall be accompanied by a full description of the special conditions prescribed, and a safety assessment demonstrating that an equivalent level of safety to that established in the applicable Certification Specification is met.

AMC1 ARA.FSTD.100(a)(1) Initial evaluation procedure

ED Decision 2012/006/R

ASSESSMENT PROCESS LEADING TO THE ISSUE OF AN FSTD QUALIFICATION

-
- (a) FSTDs require evaluation leading to qualification. The required process should be accomplished in two distinct steps. First, a check should be made to determine whether or not the FSTD complies with the applicable requirements. When making this check, the competent authority should ensure that accountability for the issue of an FSTD qualification is clearly defined. In all cases an individual department manager of the competent authority should be appointed under whose personal responsibility the issue of an FSTD qualification is to be considered. The second step should be the grant (or refusal) of an FSTD qualification.
- (b) When checking compliance with the applicable requirements, the competent authority should ensure that the following steps are taken:
- (1) Once an FSTD is contracted to be built, the organisation that is to operate the FSTD should ensure that the regulatory standard upon which the FSTD will eventually be qualified against is acceptable to the competent authority. This should be the current applicable version of CS-FSTD(A) or CS-FSTD(H) at the time of application.
 - (2) A written application for an FSTD qualification should be submitted, in a format according to ORA.FSTD.200, at least 3 months before the date of intended operation. However, the qualification test guide (QTG) may be submitted later, but not less than 30 days before the date of intended evaluation. The application form should be printed in English and any other language(s) of the competent authority's choosing.
 - (3) An individual should be nominated by the department manager of the competent authority to oversee, and become the focal point for, all aspects of the FSTD qualification process, and to coordinate all necessary activity. The nominated person should be responsible to the department manager for confirming that all appropriate evaluations/inspections are made.
 - (4) The ability of the applicant to secure, in compliance with the applicable requirements and certification specifications, the safe and reliable operation and proper maintenance of the FSTD should be assessed.
 - (5) The applicant's proposed compliance monitoring system should be scrutinised with particular regard to the allocated resources. Care should be taken to verify that the system is comprehensive and likely to be effective.
 - (6) The competent authority should inform the applicant of its final decision concerning the qualification within 14 days of completion of the evaluation process irrespective of any temporary qualification issued.
 - (7) On completion of the evaluation process, the application, together with a written recommendation and evidence of the result of all evaluations or assessments, should be presented to the nominated person responsible for FSTD qualification. The presentation should be made by the person with overall responsibility, nominated in accordance with (b)(3).
 - (8) The department manager of the competent authority should only issue an FSTD qualification certificate if he/she is completely satisfied that all requirements have been met. If he/she is not satisfied, the applicant should be informed in writing of the improvements that are required in order to satisfy the competent authority.
 - (9) If an application for an FSTD qualification is refused, the applicant should be informed of such rights of appeal as exist under national regulations.

AMC2 ARA.FSTD.100(a)(1) Initial evaluation procedure

ED Decision 2012/006/R

GENERAL

- (a) During initial and recurrent FSTD evaluations it should be necessary for the competent authority to conduct an appropriate sample of the objective and subjective tests described in Part-ORA and detailed in CS-FSTD(A) and CS-FSTD(H), as applicable. There may be occasions when all tests cannot be completed – for example during recurrent evaluations on a convertible FSTD – but arrangements should be made for all tests to be completed within a reasonable time.
- (b) Following an evaluation, it is possible that a number of defects are identified. Generally, these defects should be rectified and the competent authority notified of such action within 30 days. Serious defects, which affect flight crew training, testing and checking, could result in an immediate downgrading of the qualification level I. If any defect remains unattended without good reason for a period greater than 30 days, subsequent downgrading may occur or the FSTD qualification could be revoked.
- (c) For the evaluation of an FSTD the standard form as mentioned in [AMC5 ARA.FSTD.100\(a\)\(1\)](#) should be used.

AMC3 ARA.FSTD.100(a)(1) Initial evaluation procedure

ED Decision 2012/006/R

INITIAL EVALUATION

- (a) The main focus of objective testing is the QTG. Well in advance of the evaluation date, the aircraft manufacturer and the competent authority should agree on the content and acceptability of the validation tests contained in the QTG data package. This will ensure that the content of the QTG is acceptable to the competent authority and avoid time being wasted during the initial qualification. The acceptability of all tests depends upon their content, accuracy, completeness and recency of the results.
- (b) Much of the time allocated to objective tests depends upon the speed of the automatic and manual systems set up to run each test and whether or not special equipment is required. The competent authority should not necessarily warn the organisation operating an FSTD of the sample validation tests which should be run on the day of the evaluation, unless special equipment is required.
- (c) The FSTD cannot be used for subjective tests while part of the QTG is being run. Therefore, sufficient time (at least 8 consecutive hours) should be set aside for the examination and running of the QTG.
- (d) The subjective tests for the evaluation can be found in CS-FSTD(A) or CS-FSTD(H), and a suggested subjective test profile is described in [AMC1 ARA.FSTD.100\(a\)\(3\)](#). Essentially, 1 working day should be required for the subjective test routine, which effectively denies use of the FSTD for any other purpose.
- (e) To ensure adequate coverage of subjective and objective tests and to allow for cost effective rectification and re-test before departure of the inspection team, adequate time (up to 3 consecutive days) should be dedicated to an initial evaluation of an FSTD.

AMC4 ARA.FSTD.100(a)(1) Initial evaluation procedure

ED Decision 2012/006/R

COMPOSITION OF THE EVALUATION TEAM

- (a) The competent authority should appoint a technical team to evaluate an FSTD in accordance with a structured routine to gain a qualification level. The team should normally consist of at least the following personnel:
- (1) A technical FSTD inspector of the competent authority, or an accredited inspector from another competent authority, qualified in all aspects of flight simulation hardware, software and computer modelling or, exceptionally, a person designated by the competent authority with equivalent qualifications; and
 - (2) One of the following:
 - (i) a flight inspector of the competent authority, or an accredited inspector from another competent authority, who is qualified in flight crew training procedures and holds a valid type rating on the aeroplane/helicopter (or for flight navigation procedures trainer (FNPT) and basic instrument training device (BITD), class rated on the class of aeroplane/type of helicopter) being simulated; or
 - (ii) a flight inspector of the competent authority who is qualified in flight crew training procedures, assisted by a type rating instructor holding a valid type rating on the aeroplane/helicopter (or for FNPT and BITD, class rated on the class of aeroplane/type of helicopter) being simulated; or, exceptionally,
 - (iii) a person designated by the competent authority who is qualified in flight crew training procedures and holds a valid type rating on the aeroplane/helicopter (or for FNPT and BITD, class rated on the class of aeroplane/type of helicopter) being simulated and sufficiently experienced to assist the technical team. This person should fly out at least part of the functions and subjective test profiles.
 - (3) Where a designee is used as a substitute for one of the competent authority's inspectors, the other person shall be a properly qualified inspector of the competent authority or an accredited inspector from another Member State's competent authority.
- (b) For a flight training device (FTD) level 1 and FNPT Type I, one suitably qualified inspector may combine the functions in (a)(1) and (a)(2).
- (c) For a BITD this team should consist of an inspector from a competent authority and one from another competent authority, including the manufacturer's competent authority, if applicable.
- (d) Additionally, the following persons should be present:
- (1) for a full flight simulator (FFS), FTD and FNPT a type or class rated instructor from the ATO operating an FSTD or from the main FSTD user;
 - (2) for all types, sufficient FSTD support staff to assist with the running of tests and operation of the instructor's station.

AMC5 ARA.FSTD.100(a)(1) Initial evaluation procedure

ED Decision 2012/006/R

FSTD EVALUATION REPORT FOR INITIAL AND RECURRENT EVALUATION

FSTD Evaluation Report

Date:.....

[competent authority]
FSTD EVALUATION REPORT

[Member State] FSTD code (if applicable):

EASA FSTD code (if applicable):

Aircraft type and variant:

Class of aeroplane / type of helicopter:

Engine fit(s) simulated:

Contents

1. Flight simulation training device (FSTD) characteristics
2. Evaluation details
3. Supplementary information
4. Training, testing and checking considerations
5. Classification of items
6. Results
7. Evaluation team

The conclusions presented are those of the evaluation team. The competent authority reserves the right to change these after internal review.

1. Flight simulation training device (FSTD)										
(a) Organisation operating the FSTD:										
(b) FSTD Location:										
(c) FSTD Identification (Member State FSTD code / EASA FSTD Code):										
(d) FSTD Manufacturer and FSTD Identification serial number:										
(e) First entry into service (month/year):										
(f) Visual system (manufacturer and type):										
(g) Motion system (manufacturer and type) :										
(h) Aircraft type and variant:										
(i) Engine fit(s):										
(k) Engine instrumentation: Flight instrumentation:										
2. Evaluation details										
(a) Date of evaluation:						(b) Date of previous evaluation:				
(c) Type of evaluation: initial recurrent special										
(d) FSTD Qualification Level recommended:										
	FFS	A	B	C	D	AG	BG	CG	DG	SC
	FTD	1	2	3						
FNPT	I	II	III	MCC						
BITD										

Technical criteria primary reference document:				
Validation data roadmap (VDR) ID-No.:				
3. Supplementary information				
Company representative(s) (FSTD operator, Main FSTD user)				
FSTD seats available				
Visual databases used during evaluation				
Other				
4. Training, testing and checking considerations				
CAT I	RVR	m	DH	ft
CAT II	RVR	m	DH	ft
CAT III (lowest minimum)	RVR	m	DH	ft
LVTO	RVR	m		
Recency				
IFR-training/check				
Type rating				
Proficiency checks				
Autocoupled approach				
Autoland/Roll out guidance				
ACAS I / II				
Windshear warning system/predictive windshear				
WX-Radar				
HUD/HUGS				
FANS				
GPWS/EGPWS				
ETOPS capability				
RNP APCH LNAV				
RNP APCH LNAV/VNAV				
RNP APCH LPV				
RNP AR APCH				
Other				

5. Classification of items

UNACCEPTABLE

An item that fails to comply with the required standard and, therefore, affects the level of qualification or the qualification itself. If these items will not be corrected or clarified within a given time limit, the (*competent authority*) should have to vary, limit, suspend or revoke the FSTD qualification.

RESERVATION

An item where compliance with the required standard is not clearly proven and the issue will be reserved for a later decision. Resolution of these items will require either:

1. a *competent authority* policy ruling; or
2. additional substantiation.

UNSERVICEABILITY

A device that is temporarily inoperative or performing below its nominal level.

LIMITATION

An item that prevents the full usage of the FSTD according to the training, testing and checking considerations due to the unusable devices, systems or parts thereof.

RECOMMENDATION FOR IMPROVEMENT

An item that meets the required standard, but where considerable improvement is strongly recommended.

COMMENT

Self-explanatory

Period of Rectification

As set out in [AMC2 ARA.FSTD.100\(a\)\(1\)](#) point (b):

Following an evaluation, it is possible that a number of defects are identified. Generally, these defects should be rectified and the competent authority notified of such action within 30 days. Serious defects, which affect flight crew training, testing and checking, could result in an immediate downgrading of the qualification level, or if any defect remains unattended without good reason for a period greater than 30 days, subsequent downgrading may occur or the FSTD qualification could be revoked.

6. Results

6.1 Subjective/Functional

A Unacceptable	
1	
B Reservation	
1	
C Unserviceability	
1	
D Restriction	

1	
E Recommendation for improvement	
1	
F Comment	
1	

6.2 Objective

A Unacceptable	
1	
B Reservation	
1	
E Recommendation for improvement	
1	
F Comment	
1	

7. Evaluation Team

Name	Position	Organisation	Signature
	Technical Inspector or person designated by the competent authority		
	Flight Inspector or person designated by the competent authority		
		[FSTD User]	
		[Organisation operating the FSTD]	

Signed: For the competent authority

GM1 ARA.FSTD.100(a)(1) Initial evaluation procedure

ED Decision 2012/006/R

INITIAL EVALUATION

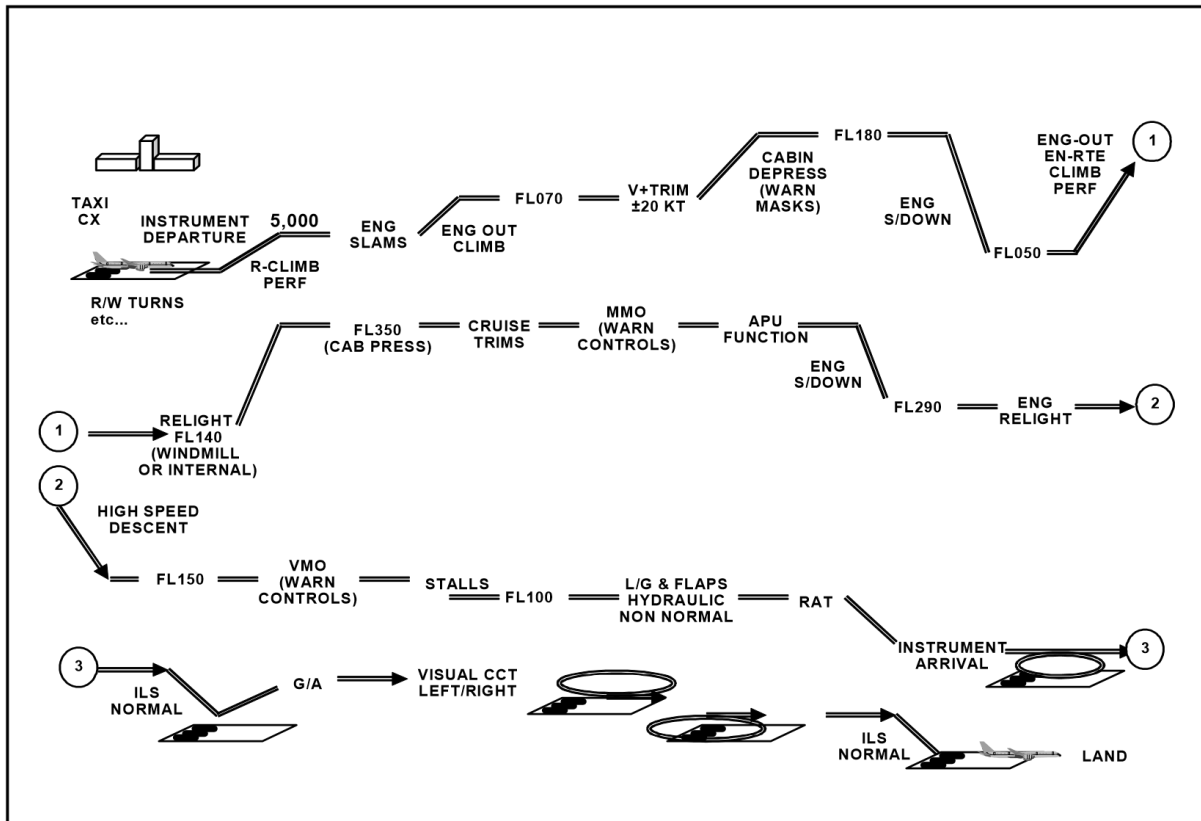
A useful explanation of how the validation tests should be run is contained in the 'RAeS Aeroplane Flight Simulator Evaluation Handbook' (February 1995 or as amended) produced in support of the ICAO Doc 9625, 'Manual of Criteria for the Qualification of Flight Simulators'.

AMC1 ARA.FSTD.100(a)(3) Initial evaluation procedure

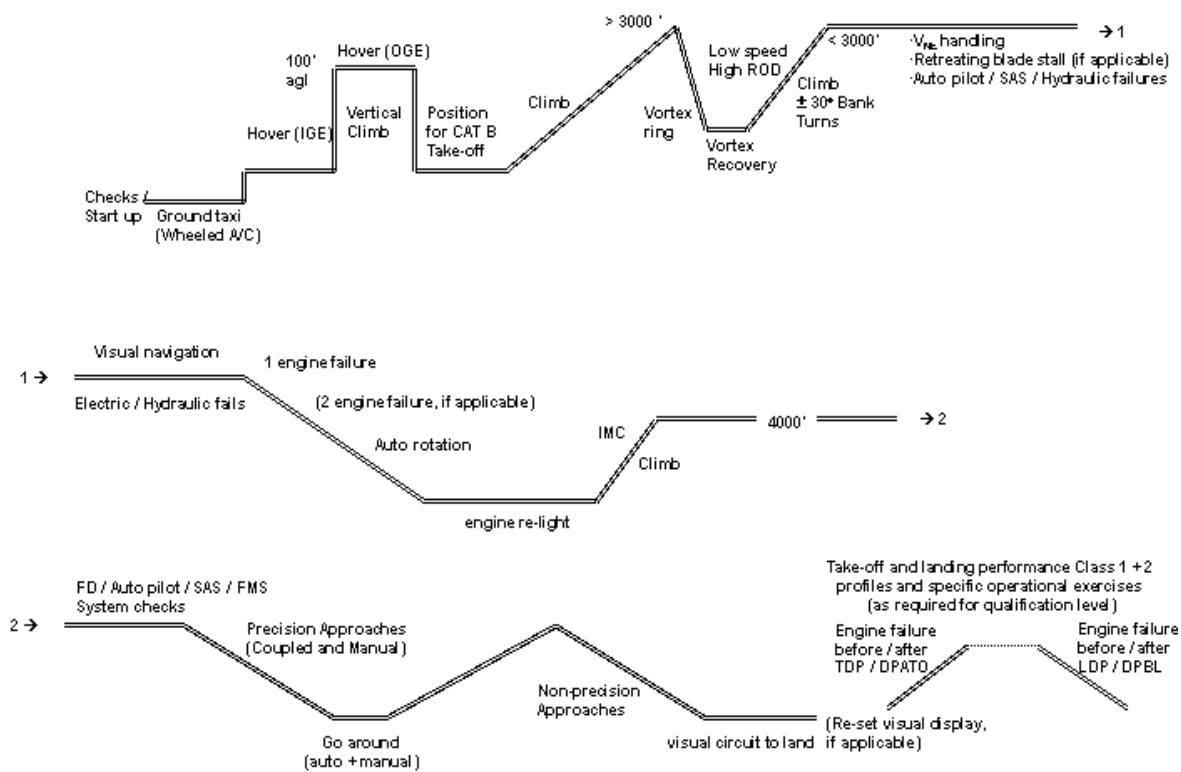
ED Decision 2012/006/R

FUNCTIONS AND SUBJECTIVE TESTS – SUGGESTED TEST ROUTINE

- (a) During initial and recurrent evaluations of an FSTD, the competent authority should conduct a series of functions and subjective tests that together with the objective tests complete the comparison of the FSTD with the aircraft, the class of aeroplane or type of helicopter.
- (b) Functions tests verify the acceptability of the simulated aircraft systems and their integration. Subjective tests verify the fitness of the FSTD in relation to training, checking and testing tasks.
- (c) The FSTD should provide adequate flexibility to permit the accomplishment of the desired and required tasks while maintaining an adequate perception by the flight crew that they are operating in a real aircraft environment. Additionally, the instructor operating station (IOS) should not present an unnecessary distraction from observing the activities of the flight crew whilst providing adequate facilities for the tasks.
- (d) It is important that both the competent authority and the organisation operating an FSTD understand what to expect from the routine of FSTD functions and subjective tests. Part of the subjective tests routine for an FSTD should involve an uninterrupted fly-out (except for FTD level 1) comparable with the duration of typical training sessions in addition to assessment of flight freeze and repositioning. An example of such a profile is to be found under points (f) and (g) (for BITD point (h)).
- (e) The competent authorities, and organisations operating FSTD, who are unfamiliar with the evaluation process should contact the Agency or the competent authority of another Member State with adequate expertise in this field.
- (f) Typical test profile for an FSTD aeroplane:



(g) Typical test profile for an FSTD helicopter:



(h) Typical subjective test profile for BITDs (approximately 2 hours) - items and altitudes, as applicable:

- (1) instrument departure, climb performance,
- (2) level-off at 4 000 ft,
- (3) fail engine (if applicable),
- (4) engine out climb to 6 000 ft (if applicable),
- (5) engine out cruise performance (if applicable), restart engine,
- (6) all engine cruise performance with different power settings,
- (7) descent to 2 000 ft,
- (8) all engine performance with different configurations, followed by instrument landing system (ILS) approach,
- (9) all engine go-around,
- (10) non-precision approach,
- (11) go-around with engine failure (if applicable),
- (12) engine out ILS approach (if applicable),
- (13) go-around engine out (if applicable),

- (14) non-precision approach engine out (if applicable), followed by go-around,
- (15) restart engine (if applicable),
- (16) climb to 4 000 ft,
- (17) manoeuvring,
- (18) normal turns left and right,
- (19) steep turns left and right,
- (20) acceleration and deceleration within operational range,
- (21) approaching to stall in different configurations,
- (22) recovery from spiral dive,
- (23) auto flight performance (if applicable),
- (24) system malfunctions,
- (25) approach.

GM1 ARA.FSTD.100(a)(3) Initial evaluation procedure

ED Decision 2012/006/R

GENERAL

A useful explanation of functions and subjective tests and an example of subjective test routine checklist may be found in the 'RAeS Airplane Flight Simulator Evaluation Handbook' Volume II (February 1995 or as amended) produced in support of ICAO Doc 9625, 'Manual of Criteria for the Qualification of Flight Simulators'.

ARA.FSTD.110 Issue of an FSTD qualification certificate

Regulation (EU) No 290/2012

- (a) After completion of an evaluation of the FSTD and when satisfied that the FSTD meets the applicable qualification basis in accordance with ORA.FSTD.210 and that the organisation operating it meets the applicable requirements to maintain the qualification of the FSTD in accordance with ORA.FSTD.100, the competent authority shall issue the FSTD qualification certificate of unlimited duration, using the form as established in Appendix IV to this Part.

AMC1 ARA.FSTD.110 Issue of an FSTD qualification certificate

ED Decision 2012/006/R

BASIC INSTRUMENT TRAINING DEVICE (BITD)

- (a) The competent authority should only grant a BITD qualification for the BITD model to a BITD manufacturer following satisfactory completion of an evaluation.
- (b) This qualification should be valid for all serial numbers of this model without further technical evaluation.
- (c) The BITD model should be clearly identified by a BITD model number. A running serial number should follow the BITD model identification number.

- (d) The competent authority should establish and maintain a list of all BITD qualifications it has issued, containing the number of the BITD model with a reference to the hardware and software configuration.

ARA.FSTD.115 Interim FSTD qualification

Regulation (EU) No 1178/2011

- (a) In the case of the introduction of new aircraft programmes, when compliance with the requirements established in this Subpart for FSTD qualification is not possible, the competent authority may issue an interim FSTD qualification level.
- (b) For full flight simulators (FFS) an interim qualification level shall only be granted at level A, B or C.
- (c) This interim qualification level shall be valid until a final qualification level can be issued and, in any case, shall not exceed 3 years.

AMC1 ARA.FSTD.115 Interim FSTD qualification

ED Decision 2012/006/R

NEW AIRCRAFT FFS / FTD QUALIFICATION – ADDITIONAL INFORMATION

- (a) Aircraft manufacturers' final data for performance, handling qualities, systems or avionics are seldom available until well after a new or derivative aircraft has entered service. Because it is often necessary to begin flight crew training and certification several months prior to the entry of the first aircraft into service, it may be necessary to use aircraft manufacturer-provided preliminary data for interim qualification of FSTDs. This is consistent with the possible interim approval of operational suitability data (OSD) relative to FFS in the type certification process under Part-21.
- (b) In recognition of the sequence of events that should occur and the time required for final data to become available, the competent authority may accept the use of certain partially validated preliminary aircraft and systems data, and early release ('red label') avionics in order to permit the necessary programme schedule for training, certification and service introduction.
- (c) Organisations seeking qualification based on preliminary data should, however, consult the competent authority as soon as it is known that special arrangements will be necessary, or as soon as it is clear that preliminary data will need to be used for FSTD qualification. Aircraft and FSTD manufacturers should also be made aware of the needs and agree on the data plan and FSTD qualification plan. There should be periodic meetings to keep the interested parties informed of the project's status.
- (d) The precise procedure to be followed to gain competent authority acceptance to use preliminary data should vary from case to case and between aircraft manufacturers. Each aircraft manufacturer's new aircraft development and test programme is designed to suit the needs of the particular project and may not contain the same events or sequence of events as another manufacturer's programme or even the same manufacturer's programme for a different aircraft. Hence, there cannot be a prescribed invariable procedure for acceptance to use preliminary data. Instead there should be a statement describing the final sequence of events, data sources, and validation procedures agreed by the FSTD operator, the aircraft manufacturer, the FSTD manufacturer and the competent authority. The approval by the Agency of the definition of scope of the aircraft validation source data to support the objective

qualification as part of the OSD can also be an interim approval in case of preliminary data. The preliminary data to be used should be based on this interim approval.

- (e) There should be assurance that the preliminary data are the manufacturer's best representation of the aircraft and reasonable certainty that final data will not deviate to a large degree from these preliminary, but refined, estimates. First of all there should be an interim approval of OSD relative to flight simulators in the type certification process under Part-21. Furthermore, the data derived from these predictive or preliminary techniques should be validated by available sources including, at least, the following:
 - (1) *Manufacturer's engineering report.* Such reports explain the predictive method used and illustrate past successes of the method on similar projects. For example, the manufacturer could show the application of the method to an earlier aircraft model or predict the characteristics of an earlier model and compare the results to final data for that model.
 - (2) *Early flight tests results.* Such data will often be derived from aircraft certification tests, and should be used to maximum advantage for early FSTD validation. Certain critical tests, which would normally be done early in the aircraft certification programme, should be included to validate essential pilot training and certification manoeuvres. These include cases in which a pilot is expected to cope with an aircraft failure mode, including engine failures. The early data available will, however, depend on the aircraft manufacturer's flight test programme design and may not be the same in each case. However it is expected that the flight test programme of the aircraft manufacturer includes provisions for generation of very early flight tests results for FSTD validation.
- (f) The use of preliminary data is not indefinite. The aircraft manufacturer's final data should be available within 6 months after the aircraft's first 'service entry' or as agreed by the competent authority, the organisation and the aircraft manufacturer, but usually not later than 1 year. When an organisation applies for an interim qualification using preliminary data, the organisation and the competent authority should agree upon the update programme. This should normally specify that the final data update will be installed in the FSTD within a period of 6 months following the final data release unless special conditions exist and a different schedule agreed. The FSTD performance and handling validation would then be based on data derived from flight tests. Initial aircraft systems data should be updated after engineering tests. Final aircraft systems data should also be used for FSTD programming and validation.
- (g) FSTD avionics should stay essentially in step with aircraft avionics (hardware and software) updates. The permitted time lapse between aircraft and FSTD updates is not a fixed time but should be minimal. It may depend on the magnitude of the update and whether the QTG and pilot training and certification are affected. Permitted differences in aircraft and FSTD avionics versions and the resulting effects on FSTD qualification should be agreed between the organisation and the competent authority. Consultation with the FSTD manufacturer is desirable throughout the agreement of the qualification process.
- (h) The following describes an example of the design data and sources which might be used in the development of an interim qualification plan:
 - (1) The plan should consist of the development of a QTG based upon a mix of flight test and engineering simulation data. For data collected from specific aircraft flight tests or other flights, the required designed model and data changes necessary to support an acceptable proof of match (POM) should be generated by the aircraft manufacturer.

- (2) In order that the two sets of data are properly validated, the aircraft manufacturer should compare their simulation model responses against the flight test data, when driven by the same control inputs and subjected to the same atmospheric conditions as were recorded in the flight test. The model responses should result from a simulation where the following systems are run in an integrated fashion and are consistent with the design data released to the FSTD manufacturer:
- (i) propulsion,
 - (ii) aerodynamics,
 - (iii) mass properties,
 - (iv) flight controls,
 - (v) stability augmentation,
 - (vi) brakes and landing gear.
- (i) For the qualification of FSTD of new aircraft types, it may be beneficial that the services of a suitably qualified test pilot are used for the purpose of assessing handling qualities and performance evaluation.

GM1 ARA.FSTD.115 Interim FSTD qualification

ED Decision 2012/006/R

NEW AIRCRAFT FFS/FTD QUALIFICATION – ADDITIONAL INFORMATION

- (a) A description of aircraft manufacturer-provided data needed for flight simulator modelling and validation is to be found in the IATA Document *Flight Simulator Design and Performance Data Requirements* (Edition 6 2000 or as amended).
- (b) The proof of match should meet the relevant tolerances in AMC1 CS-FSTD(A).300 respectively AMC1 CS-FSTD(H).300.

ARA.FSTD.120 Continuation of an FSTD qualification

Regulation (EU) 2024/2076

- (a) The competent authority shall continuously monitor the organisation operating the FSTD, as part of the oversight programme, to verify that the following conditions are met:
 - (1) the complete set of tests in the master QTG is rerun progressively over a 12-month period;
 - (2) the results of recurrent evaluations continue to comply with the qualification basis and are dated and retained;
 - (3) a configuration control system is in place to ensure the continued integrity of the hardware and software of the qualified FSTD.
- (b) The competent authority shall conduct recurrent evaluations of the FSTD in accordance with the procedures detailed in point [ARA.FSTD.100](#). Those evaluations shall take place:
 - (1) every year, in the case of a full flight simulator (FFS), flight training device (FTD) or flight and navigation procedures trainer (FNPT). The start for each recurrent 12-month period is the end of the month of the initial qualification unless another date is agreed between

- the competent authority and the organisation operating the FSTD. Each FSTD recurrent evaluation shall take place within a period of 60 days before and 30 days after the start of each recurrent 12-month period;
- (2) every 3 years, in the case of a BITD.
- (c) The competent authority may extend the recurrent evaluation period of an FSTD specified in point [ARA.FSTD.120\(b\)\(1\)](#) to a maximum of 36 months, provided that all of the following apply:
- (1) during the preceding 36 months, the organisation operating that FSTD complies with the criteria specified in points [ARA.GEN.305\(c\)\(1\)](#) to (c)(4);
 - (2) the FSTD has been subject to an initial and at least one recurrent evaluation that have established its continuous compliance with the qualification basis;
 - (3) the competent authority performs an audit of the elements of the management system of the organisation, as specified in points [ORA.GEN.200\(a\)\(3\)](#) and (a)(6) of Annex VII, every 12 months;
 - (4) the organisation has developed procedures to conduct the tasks specified in point [ORA.FSTD.225\(b\)](#) of Annex VII.

AMC1 ARA.FSTD.120 Continuation of an FSTD qualification

ED Decision 2012/006/R

GENERAL

- (a) *Objective Testing.* During recurrent evaluations, the competent authority should wish to see evidence of the successful running of the QTG between evaluations. The competent authority should select a number of tests to be run during the evaluation, including those that may be cause for concern. Again adequate notification would be given when special equipment is required for the test.
- (b) Essentially the time taken to run the objective tests depends upon the need for special equipment, if any, and the test system, and the FSTD cannot be used for subjective tests or other functions whilst testing is in progress.
- (c) For a modern FSTD incorporating an automatic test system, four hours would normally be required. FSTDs that rely upon manual testing may require a longer period of time.
- (d) *Subjective Testing.* Essentially the same subjective test routine should be flown as per the profile described in [AMC1 ARA.FSTD.100\(a\)\(3\)](#) with a selection of the subjective tests taken from CS-FSTD(A) or CS-FSTD(H), as appropriate.
- (e) Normally, the time taken for recurrent subjective testing is about 4 hours, and the FSTD should not perform other functions during this time.
- (f) To ensure adequate coverage of subjective and objective tests during a recurrent evaluation, a total of 8 hours should be allocated, (4 hours for a BITD). However, it should be remembered that any FSTD deficiency that arises during the evaluation could necessitate the extension of the evaluation period.

AMC2 ARA.FSTD.120 Continuation of an FSTD qualification

ED Decision 2012/006/R

COMPOSITION OF THE EVALUATION TEAM

- (a) The composition of the evaluation team for a recurrent evaluation should be the same as for the initial evaluation (see [AMC4 ARA.FSTD.100\(a\)\(1\)](#)).

On a case-by-case basis (except for BITD), when a specific FSTD in operation by a specific organisation is being evaluated, the competent authority may reduce the evaluation team to:

- (1) the competent authority's flight inspector; and
 - (2) a type rated instructor (or class rated instructor for FNPT) from a main FSTD user.
- (b) Evaluations with a reduced evaluation team in line with (a) may only take place if:
- (1) this composition is not being used prior to the second recurrent evaluation;
 - (2) such an evaluation is followed by an evaluation with a full competent authority evaluation team;
 - (3) the competent authority's flight inspector performs some spot checks in the area of objective testing;
 - (4) no major change or upgrading has been applied since the directly preceding evaluation;
 - (5) no relocation of the FSTD has taken place since the last evaluation;
 - (6) a system is established enabling the competent authority to monitor and analyse the status of the FSTD on a continuous basis; and
 - (7) the FSTD hardware and software has been working reliably for the previous years. This should be reflected in the number and kind of discrepancies (technical log entries) and the results of the compliance monitoring system audits.
- (c) In the case of a BITD, the recurrent evaluation may be conducted by one suitably qualified flight inspector only, in conjunction with the inspection of any ATO, using the BITD.

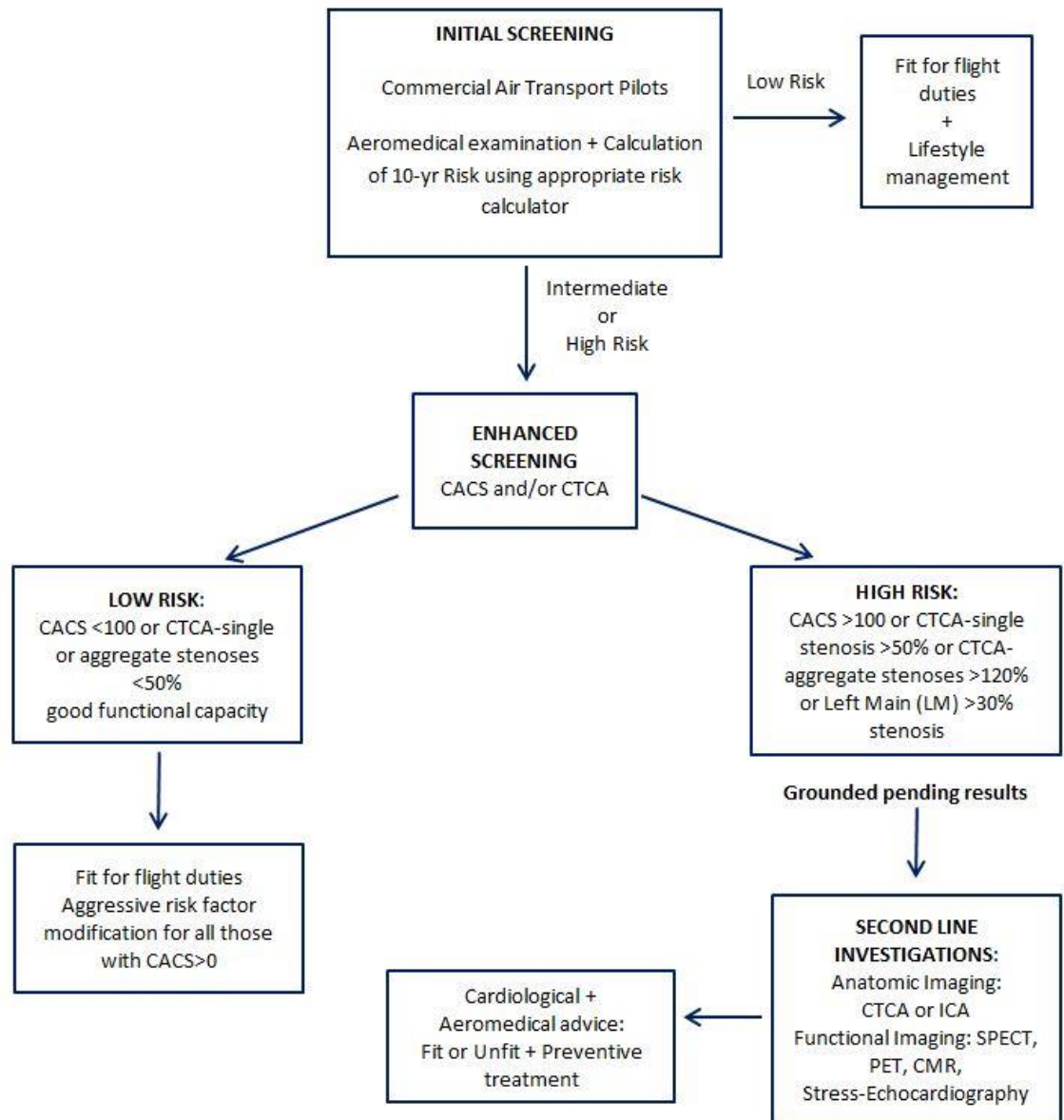
AMC3 ARA.FSTD.120 Continuation of an FSTD qualification

ED Decision 2025/002/R

EXTENDED EVALUATION PERIOD — GENERAL

- (a) The competent authority should determine the length of the extended recurrent evaluation period of an FSTD, taking into consideration the information listed in point (a) of [AMC2 ORA.FSTD.225\(b\)](#). The extended evaluation period should be 24 or 36 months.
- (b) The extension of the recurrent evaluation period should be driven by a risk-based approach as part of the continuous oversight. As such, the performance of the organisation, as well as the performance of the device, should be monitored.
- (c) Once the competent authority has granted an extended recurrent evaluation period, it may plan all subsequent recurrent evaluations of the relevant FSTD with the extended time interval of 24 or 36 months, as applicable, until it decides to revert again to a shorter recurrent evaluation period, should the conditions of point [ARA.FSTD.120\(c\)](#) be no longer fulfilled.

- (d) The audit of the management system elements, described in point [ARA.FSTD.120\(c\)\(3\)](#), may be performed remotely, at the discretion of the competent authority, considering the performance of the organisation.



ARA.FSTD.130 Changes

Regulation (EU) No 1178/2011

- (a) Upon receipt of an application for any changes to the FSTD qualification certificate, the competent authority shall comply with the applicable elements of the initial evaluation procedure requirements as described in [ARA.FSTD.100\(a\) and \(b\)](#).
- (b) The competent authority may complete a special evaluation following major changes or when an FSTD appears not to be performing at its initial qualification level.

- (c) The competent authority shall always conduct a special evaluation before granting a higher level of qualification to the FSTD.

AMC1 ARA.FSTD.130 Changes

ED Decision 2012/006/R

GENERAL

- (a) The organisation operating an FSTD who wishes to modify, upgrade, de-activate or relocate its FSTD should notify the competent authority. When considering applications for a change of the existing FSTD qualification level, the competent authority should ensure that accountability for the change is clearly defined.
- (b) An individual department manager of the competent authority should be appointed under whose personal authority an FSTD qualification may be changed.
- (c) The written application for a change, including appropriate extracts from the qualification test guide indicating proposed amendments should be submitted in a format and manner as specified by the competent authority. This application should be submitted no later than 30 days before the date of intended change, unless otherwise agreed with the competent authority.
- (d) On receipt of an application for a change of the existing FSTD qualification level, the competent authority should conduct such evaluations and inspections as are necessary to ensure that the full implications of the request have been addressed by the organisation operating the FSTD.
- (e) During the processing of a change request, the continued adequacy of the compliance monitoring should be reviewed.
- (f) When the request has been considered and examined, the competent authority should decide on the depth of inspection of the FSTD that is required.
- (g) The department manager, if satisfied that the organisation operating the FSTD remains competent and the qualification level of the FSTD can be maintained, should issue revised FSTD qualification documentation, as appropriate.
- (h) The competent authority should inform the organisation operating the FSTD of its decision within 30 days of receipt of all documentation where no evaluation is required, or within 14 days of any subsequent evaluation.
- (i) Such documentation includes the appropriate extracts from the QTG amended, when necessary, to the competent authority's satisfaction.

GM1 ARA.FSTD.130 Changes

ED Decision 2012/006/R

QUALIFICATION OF NEW TECHNOLOGY OR SYSTEMS

Where an update to an FSTD involves a change of technology or the addition of a new system or equipment that is not covered by the qualification basis used for the existing qualification, an evaluation of such changes may not be possible using this original qualification basis. For these cases, the specific changes can be qualified by using newer Certification Specifications, new AMCs or alternative means of compliance, that apply to these changes, without affecting the overall qualification of the FSTD. This approach should be documented.

ARA.FSTD.135 Findings and corrective actions – FSTD qualification certificate

Regulation (EU) No 290/2012

The competent authority shall limit, suspend or revoke, as applicable, an FSTD qualification certificate in accordance with [ARA.GEN.350](#) in, but not limited to, the following circumstances:

- (a) obtaining the FSTD qualification certificate by falsification of submitted documentary evidence;
- (b) the organisation operating the FSTD can no longer demonstrate that the FSTD complies with its qualification basis; or
- (c) the organisation operating the FSTD no longer complies with the applicable requirements of Part-ORA.

AMC1 ARA.FSTD.135 Findings and corrective actions – FSTD qualification certificate

ED Decision 2012/006/R

GENERAL

- (a) The competent authority's inspection and monitoring process should confirm the competent authority's continued confidence in the effectiveness of the compliance monitoring system of the organisation operating an FSTD, and its ability to maintain an adequate standard.
- (b) If the competent authority is not satisfied, the organisation operating an FSTD should be informed in writing of the details of the conduct of its operation which are causing the competent authority concern. The competent authority should require corrective action to be taken within a specified period (see [AMC2 ARA.FSTD.100\(a\)\(1\)](#) point (b)).
- (c) In the event that an organisation operating an FSTD fails, in spite of warning and advice, to satisfy the competent authority's concerns, a final written warning should, whenever possible, be given to the organisation together with a firm date by which specified action to satisfy the competent authority should be taken. It should be made clear that failure to comply may result in enforced limitation or suspension of the FSTD's qualification.
- (d) Circumstances may, however, preclude recourse to the process described under (a) to (c). In such cases the competent authority's duty to preserve quality of training, testing and checking is of paramount importance and therefore the competent authority may immediately limit or suspend any FSTD qualification which it has issued.

AMC2 ARA.FSTD.135 Findings and corrective actions – FSTD qualification certificate

ED Decision 2012/006/R

SUSPENSION AND LIMITATION

- (a) When a decision has been taken to suspend, or limit, an FSTD qualification certificate, the organisation operating an FSTD should be informed immediately by the quickest available means.
- (b) In the event of full suspension of an FSTD qualification certificate, the organisation operating an FSTD should be instructed that the FSTD concerned cannot be used for any credited training,

testing or checking. The "quickest available means" will in most situations mean the use of a facsimile or email message.

- (c) This should be followed by a formal letter giving notice of suspension, or limitation, restating the requirement to cease operations as applicable, and also setting out the conditions on which suspension may be lifted.
- (d) If it becomes apparent to the competent authority that all operations have ceased over a period in excess of 6 months, the competent authority should consider opening the warning process described in [AMC1 ARA.FSTD.135](#), points (a) to (d).
- (e) The FSTD qualification certificate should not remain suspended indefinitely. Further steps may be taken by the organisation operating an FSTD to reinstate the FSTD qualification or, in default, should be taken by the competent authority to revoke the FSTD qualification certificate. Should an organisation operating an FSTD wish to dispute the suspension of its FSTD's qualification certificate, it should be informed of such rights of appeal as exist under national regulations. If an appeal is lodged, the FSTD qualification may remain suspended until the appeal process is complete.
- (f) Suspension of an FSTD qualification certificate may be lifted on appeal or if the organisation operating an FSTD restores the FSTD to its previously acceptable standard.
- (g) In neither case should operations be permitted to restart until it has been demonstrated that the cause of the suspension or limitation has been rectified. The competent authority may require a special evaluation depending on the severity of the problem.
- (h) The competent authority should issue a formal notice of the lifting of suspension before the organisation operating an FSTD is permitted to resume use of an FSTD.

AMC3 ARA.FSTD.135 Findings and corrective actions – FSTD qualification certificate

ED Decision 2012/006/R

REVOCATION

- (a) The competent authority should give the organisation operating an FSTD notice that it intends to revoke the FSTD qualification followed by a formal letter of revocation.
- (b) Should an organisation operating an FSTD wish to dispute this revocation, it should be informed of such rights of appeal as exist under applicable regulations. Once revoked, there can be no further activities under the terms of the FSTD qualification.

ARA.FSTD.140 Record keeping

Regulation (EU) No 1178/2011

In addition to the records required in [ARA.GEN.220](#), the competent authority shall keep and update a list of the qualified FSTDs under its supervision, the dates when evaluations are due and when such evaluations were carried out.

SUBPART AeMC – SPECIFIC REQUIREMENTS RELATING TO AERO-MEDICAL CENTRES (AeMCs)

SECTION I – GENERAL

ARA.AeMC.110 Initial certification procedure

Regulation (EU) No 1178/2011

The certification procedure for an AeMC shall follow the provisions laid down in [ARA.GEN.310](#).

ARA.AeMC.150 Findings and corrective actions – AeMC

Regulation (EU) No 1178/2011

Without prejudice to [ARA.GEN.350](#), level 1 findings include, but are not limited to, the following:

- (a) failure to nominate a head of the AeMC;
- (b) failure to ensure medical confidentiality of aero-medical records; and
- (c) failure to provide the competent authority with the medical and statistical data for oversight purposes.

SUBPART MED – SPECIFIC REQUIREMENTS RELATING TO AERO-MEDICAL CERTIFICATION

SECTION I – GENERAL

ARA.MED.120 Medical assessors

Regulation (EU) 2024/2076

The competent authority shall appoint one or more medical assessor(s) to undertake the aero-medical tasks described in this Regulation. The medical assessor shall be licensed and qualified in medicine and have the following:

- (a) postgraduate work experience in clinical medicine;
- (b) specific knowledge and experience in aviation medicine and aero-medical practice;
- (c) specific training in aero-medical certification.

AMC1 ARA.MED.120 Medical assessors

ED Decision 2025/002/R

EXPERIENCE AND KNOWLEDGE

Medical assessors should:

- (a) have considerable experience of aero-medical practice, having held AME privileges and having undertaken a minimum of 200 class 1 or class 3 medical examinations or equivalent;
- (b) undergo specific training on the regulatory processes and aero-medical certification of referred cases; and
- (c) maintain their medical professional competence in aviation medicine. The following should count towards maintaining medical professional competence:
 - (1) undertaking regular refresher training;
 - (2) participating in international aviation medicine conferences;
 - (3) undertaking research activities, including publication of results of the research.

AMC2 ARA.MED.120 Medical assessors

ED Decision 2025/002/R

TASKS

Medical assessors' tasks include:

- (a) approving and overseeing lectures in basic, advanced and refresher training courses for aero-medical examiners (AMEs) and aero-medical centres (AeMCs). Medical assessors may also deliver lectures during those training courses provided that a procedure is in place to avoid conflict of interest;
- (b) carrying out supervision and audits of AeMCs, AMEs and AME training facilities;

- (c) performing the aero-medical assessment of applicants for, or holders of, medical certificates in the case of consultation, referral or secondary review, or when medical certificates have been issued by non-compliant AMEs;
- (d) certifying and overseeing AeMCs and AMEs, including review of medical files submitted by them to the competent authority;
- (e) managing medical files including transfers of medical files in the case of a change of state of licence issue;
- (f) assisting AMEs and AeMCs, on their request, regarding aero-medical fitness assessments in borderline and difficult cases or cases not regulated in Part-MED or Part-ATCO.MED of Regulation (EU) 2015/340, as applicable; and
- (g) issuing a medical certificate if a case is referred or if corrections to the information of a medical certificate are necessary.

AMC3 ARA.MED.120 Medical assessors

ED Decision 2025/002/R

DELEGATION OF MEDICAL ASSESSOR TASKS

The medical assessor may delegate certain tasks to other staff designated by the competent authority or other persons contracted by the competent authority. The competent authority should ensure that such person has relevant training and experience for the delegated task and that the entire process is properly documented.

GM1 ARA.MED.120 Medical assessors

ED Decision 2025/002/R

DELEGATION OF MEDICAL ASSESSOR TASKS

Properly qualified medical assessors are essential for maintaining flight safety and an efficient and functional aero-medical system. Medical assessors, like any inspector of the competent authority, should, by their qualifications and competencies, command the professional respect of the personnel and organisations they inspect, authorise or oversee. These guidelines aim to establish possible solutions to optimise the use of qualified medical assessors as well as temporary solutions until properly qualified medical assessors are readily available. These guidelines should be interpreted and implemented only to the extent that they provide for sound and effective oversight in accordance with the principles of safety risk management.

For all of the medical assessor tasks, the support staff may provide administrative support with regard to the paperwork and preparation work. Furthermore, some tasks may be partially delegated to other staff members of the competent authority or other persons contracted by the competent authority. The medical assessor should select to whom the tasks are delegated based on their qualifications in order to ensure that the entire performance is in line with the applicable provision both in the field of aviation and in the medical field and is properly documented. The compliance monitoring system of the competent authority should ensure that delegation of certain tasks has no negative impact on flight safety and data protection.

In order to maintain their medical proficiency, medical assessors may act as an AME subject to a proper procedure being in place to avoid conflict of interest.

The following steps may be considered when required:

(a) Employment of a not fully qualified medical assessor

When recruiting a fully qualified medical assessor is not possible, the competent authority may employ a medical doctor to be trained and nominated as a medical assessor once the training is finalised. The performance of these doctors should be supervised by a qualified medical assessor from the pool of experts.

(b) Assignment of the role of a team member to qualified inspectors (e.g. assessing the SMS system of an AeMC)

In this context, the qualified inspectors performing duties within the inspection/oversight team are expected to document their work and to report to the medical assessor as the accountable person for the process.

(c) Use of appropriately qualified medical assessors or AMEs from a pool of experts

The use of AMEs or medical assessors from a pool of experts should be limited to the sharing of experts to cover unplanned activity or temporary/transitional shortage of expertise rather than a consistent long-term use.

The following types of pools of experts may be considered:

- qualified AMEs;
- medical assessors from the NCAs of other Member States or from EASA;
- medical assessors/AMEs from military aviation.

The following issues should be assessed and the associated risks mitigated in the case of using a pool of experts:

- assessment and oversight of the expert's performance as well as enforcement in case of non-compliance;
- authorisation of the expert to access medical practices, investigate, conduct interviews and collect evidence;
- financial, contracting and administrative aspects;
- recurrent training on administrative procedures;
- ability of the nominated expert to write reports and findings;
- avoidance of conflict of interest;
- sustainability (i.e. to avoid relying permanently on the pool of experts);
- commercial sensitivity of AMEs/AeMCs, cultural issues;
- data protection issues;
- language barriers;
- recognition between Member States, including the right to practice medicine in a different State and medical indemnity/liability insurance.

Bilateral sharing of experts is convenient when:

- the requesting authority is aware of the resources available in the resource provider;
- the agreement between the NCAs exists or is easy to establish;
- the planning for the availability of the resources can easily be managed.

Whether the sharing of medical assessors is concluded directly between two NCAs or through a sharing platform, sustainability can only be ensured if all stakeholders are willing to consider global optimisation as a priority. The challenge is that the management system of each NCA may systematically reduce its resources so that all qualified medical assessors are occupied at all times. Such planning strategy does not provide any extra margin for contingencies and may easily drift towards understaffing. It is always difficult to swiftly adjust the number of permanently employed experts to the short-term oversight needs. Therefore, while attempting to 'optimise' its own resources, each NCA may rely more and more on the experts from other NCAs and further reduce its staff. While this may work for a limited period of time, in the long run the sharing of experts may simply become impossible as all NCAs will be requesting qualified medical assessors while no NCA would be able to provide any. A similar reasoning applies when experts from the industry are shared.

The concept of sharing implies availability of resources. Availability means extra capacity. Therefore all stakeholders involved in the sharing are expected to coordinate their staffing strategies globally. This ensures global optimisation by reallocating resources so that no expert is underused and that the costs are shared based on the level of support obtained. Additionally, it is expected that activity planning is coordinated among all involved stakeholders.

ARA.MED.125 Referral to the licensing authority

Regulation (EU) 2024/2076

When an AeMC or aero-medical examiner (AME) has referred the decision on the fitness of an applicant to the medical assessor of the licensing authority, the following steps shall be taken:

- (a) the medical assessor or medical staff designated by the medical assessor shall evaluate the relevant medical documentation and request further medical documentation, examinations and tests where necessary;
- (b) the medical assessor shall determine the applicant's fitness for the issuance of a medical certificate with one or more limitation(s) if necessary;
- (c) the medical assessor shall inform the AeMC or AME of the decision;
- (d) in case the applicant is assessed as fit, the medical assessor shall issue, if appropriate, the medical certificate or delegate the issuance to the AeMC or AME that referred the respective applicant.

AMC1 ARA.MED.125 Referral to the licensing authority

ED Decision 2025/002/R

- (a) The aero-medical section of the licensing authority should supply the AeMC or AME with all necessary information that led to the decision on aero-medical fitness.
- (b) The aero-medical section of the licensing authority should ensure that borderline and difficult cases or those not regulated in Part-MED or Part-ATCO.MED, as applicable, are evaluated on a common basis.
- (c) Each competent authority should define the time limit for the assessment of referred cases in their procedure regarding the management of referrals.

ARA.MED.126 Limitation, suspension or revocation of medical certificates

Regulation (EU) 2024/2076

- (a) The licensing authority shall establish a procedure to limit, suspend or revoke a medical certificate.
- (b) The licensing authority shall limit, suspend or revoke a medical certificate if there is evidence that:
 - (1) a medical certificate is falsified or obtained by a false declaration or false evidence;
 - (2) a medical certificate is used in violation of the provisions of point [MED.A.020](#) of Annex IV;
 - (3) the holder of a medical certificate is no longer compliant with Annex IV (Part-MED);
- (c) The licensing authority may also suspend or revoke a medical certificate upon the written request of the holder of a medical certificate.
- (d) In case of limitation, suspension or revocation of a medical certificate, the licensing authority shall inform the issuing AME or AeMC about the reason for limitation, suspension or revocation.
- (e) In case of suspension or revocation of a medical certificate, the licensing authority shall ensure that the provisions of point [MED.A.046](#) of Annex IV (Part-MED) are complied with.
- (f) The licensing authority shall establish a procedure for reinstating a medical certificate.

ARA.MED.128 Consultation procedure

Regulation (EU) 2024/2076

The competent authority shall establish a consultation procedure for the AeMCs and AMEs in accordance with Annex IV (Part-MED).

AMC1 ARA.MED.128 Consultation procedure

ED Decision 2025/002/R

This procedure should include at least a summary of the consultation process and related documentation.

ARA.MED.130 Medical certificate format

Regulation (EU) 2024/2076

The medical certificate shall conform to the following specifications:

- (a) Content
 - (1) State where the pilot licence has been issued or applied for (I),
 - (3) Medical certificate number commencing with the UN country code of the State where the pilot licence has been issued or applied for and followed by a code of numbers and/or letters in Arabic numerals and Latin script (III)
 - (4) Name of holder (IV),
 - (5) Nationality of holder (VI),

- (6) Date of birth of holder: (dd/mm/yyyy) (IVa)
 - (7) Signature of holder (VII),
 - (8) Limitation(s) (XIII),
 - (9) Expiry date of the medical certificate (IX) for:
 - (i) Class 1,
 - (ii) Class 1 single-pilot commercial operations carrying passengers,
 - (iii) Class 2,
 - (iv) LAPL
 - (10) Date of medical examination
 - (11) Date of last and next electrocardiogram
 - (12) Date of last and next audiogram
 - (12a) Date of last and next ophthalmological examination
 - (13) Date of issue and signature of the AME or medical assessor that issued the certificate. GMP may be added to this field if they have the competence to issue medical certificates under the national law of the Member State where the licence is issued.
 - (14) Seal or stamp (XI)
 - (15) Other information
- (b) The paper or other material used shall prevent or readily show any alterations or erasures. Any entries or deletions to the form shall be clearly authorised by the licensing authority.
 - (c) Language: Medical certificates shall be written in the national language(s) and in English and such other languages as the competent authority deems appropriate.
 - (d) All dates on the medical certificate shall be written in a dd/mm/yyyy format.

AMC1 ARA.MED.130 Medical certificate format

ED Decision 2025/002/R

STANDARD EASA MEDICAL CERTIFICATE FORMAT

The format of the medical certificate should be as shown below.

Competent authority name and logo (English and any language(s) determined by the competent authority) EUROPEAN UNION (English only) MEDICAL CERTIFICATE pertaining to a Part-FCL licence (English and any language(s) determined by the competent authority) Issued in accordance with Part-MED This medical certificate complies with ICAO standards, except for the LAPL medical certificate (English and any language(s) determined by the competent authority)	Requirements ‘European Union’ to be deleted for non-EU Member States Size of each page shall be one eighth A4
---	--

I	National language(s)/ Authority that issued or is to issue the pilot licence:
III	National language(s)/Certificate number
IV	National language(s)/ Last and first name of holder:
XIV	National language(s)/Date of birth: (dd/mm/yyyy)
VI	National language(s)/Nationality(ies):
VII	National language(s)/ Signature of holder:
2	

XIII	National language(s)/Limitations: Code: Operational remark:
X	National language(s)/* Date of issue: (dd/mm/yyyy) Name and signature of issuing AME/medical assessor /GMP:
XI	National language(s)/Seal or Stamp:
3	

IX Nat. lang(s)/ Expiry date of this certificate	Class 1 (dd/mm/yyyy or 'N/A')			
	Class 1 single-pilot commercial operations carrying passengers (dd/mm/yyyy or 'N/A')			
	Class 2 (dd/mm/yyyy or 'N/A')			
	LAPL (dd/mm/yyyy)			
Nat. lang(s)/Examination date: (dd/mm/yyyy)				
Type of examination	Last	Next		
		Class 1	Class 2	LAPD
ECG				
Audiogram (For class 1, and for class 2 with IR or en route IR)				
Ophthalmological examination				
Other information				

MED.A.020 Decrease in medical fitness

(a) Licence holders shall not exercise the privileges of their licence and related ratings or certificates, and student pilots shall not fly solo, at any time when they:

- (1) are aware of any decrease in their medical fitness that might render them unable to safely exercise those privileges;
- (2) take or use any prescribed or non-prescribed medication that is likely to interfere with the safe exercise of the privileges of the applicable licence; or
- (3) receive any medical, surgical or other treatment that is likely to interfere with the safe exercise of the privileges of the applicable licence.

(b) In addition, licence holders shall, without undue delay and before exercising the privileges of their licence, seek aero-medical advice from the AeMC, AME or GMP, as applicable, when they:

- (1) have undergone a surgical operation or invasive procedure;
- (2) have commenced the regular use of any medication;
- (3) have suffered any significant personal injury involving incapacity to function as a member of the flight crew;
- (4) have been suffering from any significant illness involving incapacity to function as a member of the flight crew;
- (5) are pregnant;
- (6) have been admitted to hospital or medical clinic; or
- (7) first require correcting lenses.

4

* Date of issue is the date the certificate is issued and signed

ARA.MED.135 Aero-medical forms

Regulation (EU) 2024/2076

The competent authority shall provide the AMEs with the format for the following documents:

- (a) the application form for a medical certificate;
- (b) the examination report form for class 1 and class 2 applicants;
- (c) the examination report form for light aircraft pilot licence (LAPL) applicants.

AMC1 ARA.MED.135(a) Aero-medical forms

ED Decision 2019/002/R

APPLICATION FORM FOR A MEDICAL CERTIFICATE

The form referred to in point [ARA.MED.135\(a\)](#) should reflect the information indicated in the following form and corresponding instructions for completion.

LOGO

CIVIL AVIATION ADMINISTRATION / MEMBER STATE

APPLICATION FORM FOR A MEDICAL CERTIFICATE

Complete this page fully and in block capitals - Refer to instructions pages for details.

MEDICAL IN CONFIDENCE

(1) State of licence issue:		(2) Medical certificate applied for: class 1 ☐ class 2 ☐ LAPL ☐ class 3 ☐	
(3) Surname:		(4) Previous surname(s):	(12) Application Initial ☐ Revalidation/Renewal ☐
(5) Forenames:		(6) Date of birth (dd/mm/yyyy):	(7) Sex Male ☐ Female ☐
(8) Place and country of birth:		(9) Nationality:	(14) Type of licence applied for:
(10) Permanent address: Country: Telephone No.: Mobile No.: e-mail:		(11) Postal address (if different) Country: Telephone No.:	(15) Occupation (principal)
			(16) Employer
			(17) Last medical examination Date: Place: Completed: No ☐ Yes ☐
(18) Aviation licence(s) held (type): Licence number: State of issue:		(19) Any Limitations on Licence/ Medical Certificate No ☐ Yes ☐ Details:	
(20) Have you ever had an aviation medical certificate denied, suspended or revoked by any licensing authority? No ☐ Yes ☐ Date: Country: Details:		(21) Flight time hours total:	(22) Flight time hours since last medical:
		(23) Aircraft class /type(s) presently flown:	
(24) Any aviation accident medical event whilst exercising the privileges of the licence since the last medical examination? No ☐ Yes ☐ Date: Place: Details:		(25) Current/intended pilot activity: Commercial ☐ Non-commercial ☐ Other Single-pilot ☐ Multi-pilot ☐	
		(26) Current/intended ATC activity: ADI ☐ APS ☐ ACS ☐ ADV ☐ APP ☐ ACP ☐	
(27) Do you drink alcohol? ☐ No ☐ Yes, state average weekly amount: Do you use drugs? ☐ No ☐ Yes, state the type:		(28) Do you currently use any medication? No ☐ Yes ☐ State drug, dose, date started and why:	
(29) Do you smoke tobacco? ☐ No, never ☐ No, date stopped: ☐ Yes, state type and amount:			

General and medical history: Do you have, or have you ever had, any of the following? (Please tick a response for each question). If yes, give details in the remarks section (30).

	Yes	No		Yes	No		Yes	No	Family history of:	Yes	No
101 Eye trouble/eye operation			112 Nose, throat or speech disorder			123 Malaria or other tropical disease			170 Heart or vascular disease		
102 Spectacles and/or contact lenses ever worn			113 Head injury or concussion			124 A positive HIV test			171 High blood pressure		
			114 Frequent or severe headaches			125 Sexually transmitted disease			172 High cholesterol level		
103 Spectacle/contact lens prescriptions change since last medical exam.			115 Dizziness or fainting spells			126 Sleep disorder/apnoea syndrome			173 Epilepsy		
			116 Unconsciousness for any reason			127 Musculoskeletal illness/impairment			174 Mental illness or suicide		

104 Hay fever, other allergy		117 Neurological disorders; stroke, epilepsy, seizure, paralysis, etc		128 Any other illness or injury		175 Diabetes	
105 Asthma, lung disease				129 Admission to hospital		176 Tuberculosis	
106 Heart or vascular trouble		118 Psychological/psychiatric trouble of any sort		130 Visit to medical practitioner or mental health specialist since last medical examination		177 Allergy/asthma/eczema	
107 High or low blood pressure						178 Inherited disorders	
108 Kidney stone or blood in urine		119 misuse of psychoactive substances		131 Refusal of life insurance		179 Glaucoma	
109 Diabetes, hormone disorder		120 Attempted suicide or self-harm		132 Refusal of aviation licence			
110 Stomach, liver or intestinal trouble		121 Motion sickness requiring medication		133 Medical rejection from or for military service		Females only:	
						150 Gynaecological, menstrual problems	
111 Deafness, ear disorder		122 Anaemia / Sickle cell trait/other blood disorders		134 Award of pension or compensation for injury or illness		151 Are you pregnant?	
(30) Remarks:							
(31) Declaration: I hereby declare that I have carefully considered the statements made above and to the best of my belief they are complete and correct and that I have not withheld any relevant information or made any misleading statements. I understand that, if I have made any false or misleading statements in connection with this application, or fail to release the supporting medical information, the licensing authority may refuse to grant me a medical certificate or may withdraw any medical certificate granted, without prejudice to any other action applicable under national law. I hereby declare that I have been informed and I understand that all information provided to my AME contained in this report and its attachments and all information which is provided to my licensing authority and that relates to me, may be released to the medical assessor of my licensing authority, other health professionals and medical administration staff as part of the aero-medical assessment process and to the medical assessor of the competent authority of my AME, recognising that these documents or electronically stored data are to be used for the completion of an aero-medical assessment and for oversight purposes, providing that I or my physician may have access to them in accordance with national law. Medical confidentiality will be respected at all times. NOTIFICATION OF DISCLOSURE OF PERSONAL DATA: I hereby declare that I have been informed and I understand that the data contained in my medical certificate in accordance with point ARA.MED.130, or point ATCO.AR.F.005 of Regulation (EU) 2015/340 if applicable, may be electronically stored and made available to my AME in order to provide historical data required in point MED.A.035(b)(2)(ii)/(iii) or, if applicable, points ATCO.MED.A.035(b)(2)(ii) or ATCO.MED.A.035(b)(2)(iii), and to the medical assessors of the competent authorities of the Member States in order to facilitate the enforcement of point ARA.MED.150(c)(4). ----- Date Signature of applicant Signature of AME/(GMP)/ (medical assessor)							

INSTRUCTIONS FOR COMPLETION OF THE APPLICATION FORM FOR A MEDICAL CERTIFICATE

This application form and all attached report forms will be transmitted to the licensing authority. Medical confidentiality shall be respected at all times.

The applicant should personally complete, in full, all questions (sections) on the application form. Writing should be legible and in block capitals, using a ball-point pen. Completion of this form by typing/printing is also acceptable. If more space is required to answer any questions, a plain sheet of paper should be used, bearing the applicant's name and signature, and the date of signing. The following numbered instructions apply to the numbered headings on the application form for a medical certificate.

Failure to complete the application form in full, or to write legibly, may result in non-acceptance of the application form. The making of false or misleading statements or the withholding of relevant information in respect of this application may result in criminal prosecution, denial of this application and/or withdrawal of any medical certificate(s) granted.

1. LICENSING AUTHORITY:	17. LAST APPLICATION FOR A MEDICAL CERTIFICATE:
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State name of country that has issued the pilot or ATCO licence or where a licence has not been issued, the country where the applicant intends to apply for a licence.	State date (day, month, year) and place (town, country) Initial applicants state 'NONE'.
2. MEDICAL CERTIFICATE APPLIED FOR:	18. LICENCE(S) HELD (TYPE):
Tick appropriate box representing the type of medical certificate applied for, e.g. class 1, class 2, class 3 or LAPL.	State type of licence(s) held. Enter licence number and State of issue. If no licences are held, state 'NONE'.
3. SURNAME: State surname/family name.	19. ANY LIMITATIONS ON THE LICENCE(S)/MEDICAL CERTIFICATE: Tick appropriate box and give details of any limitations on your licence(s)/medical certificate, e.g. vision, colour vision, safety pilot, etc.
4. PREVIOUS SURNAME(S): If your surname or family name has changed for any reason, state previous name(s).	20. MEDICAL CERTIFICATE DENIAL, SUSPENSION OR REVOCATION: Tick 'YES' box if you have ever had a medical certificate denied, suspended or revoked. If 'YES', state date (dd/mm/yyyy) and country where it occurred.
5. FORENAME(S): State first and middle names (maximum three).	21. FLIGHT TIME TOTAL: State total number of hours flown.
6. DATE OF BIRTH: Specify in order dd/mm/yyyy.	22. FLIGHT TIME SINCE LAST MEDICAL: State number of hours flown since your last medical examination.
7. SEX: Tick appropriate box.	23. AIRCRAFT CLASS/TYPE(S) PRESENTLY FLOWN: State name of principal aircraft flown, e.g. Boeing 737, Cessna 150, etc.
8. PLACE AND COUNTRY OF BIRTH: State town and country of birth.	24. ANY AVIATION ACCIDENT OR MEDICAL EVENT WHILST EXERCISING THE PRIVILEGES OF THE LICENCE SINCE THE LAST MEDICAL EXAMINATION: If 'YES' box ticked, state date (dd/mm/yyyy) and country of occurrence and provide details.
9. NATIONALITY: State name of country of citizenship	25. CURRENT/INTENDED PILOT ACTIVITY: Please tick the appropriate box regarding the current/intended activity during the following certification period: - Commercial, non-commercial or other (for other, please specify the type of operation) - Single-pilot or multi-pilot
10. PERMANENT ADDRESS: State permanent postal address and country. Enter telephone area code as well as telephone number.	26. CURRENT/INTENDED ATC ACTIVITY:: Please tick the appropriate box regarding the current/intended activity during the following certification period e.g. ADI, APS, ACS.
11. POSTAL ADDRESS (IF DIFFERENT): If different from permanent address, state full current postal address including telephone number and area code. If the same, enter 'SAME'.	27. DO YOU DRINK ALCOHOL OR USE DRUGS? Tick applicable box. If yes, state weekly alcohol consumption e.g. 2 of litres beer.
12. APPLICATION: Tick appropriate box.	28. DO YOU CURRENTLY USE ANY MEDICATION?: If 'YES', give full details - name, how much you take and when, etc. Include any non-prescription medication.
13. MEDICAL CERTIFICATE/EAMR ID NUMBER: State medical certificate number allocated to you by the licensing authority/EAMR ID unique number Initial applicants enter 'NONE'.	29. DO YOU SMOKE TOBACCO? Tick applicable box. Current smokers state type (cigarettes, cigars, pipe) and amount (e.g. 2 cigars daily; pipe – 1 oz. weekly)
14. TYPE OF LICENCE APPLIED FOR: State type of licence applied for from the following list: Airline Transport Pilot Licence* Multi-Pilot Licence* Commercial Pilot Licence/Instrument Rating* Commercial Pilot Licence* Private Pilot Licence/Instrument Rating* Private Pilot Licence* Sailplane Pilot Licence	30. GENERAL AND MEDICAL HISTORY All items under this heading, from number 101 to 179 inclusive, should have the answer 'YES' or 'NO' ticked. You should tick 'YES' if you have ever had the condition in your life and describe the condition and approximate date in the remarks section (30). All questions asked are medically important even though this may not be readily apparent. Items numbered 170 to 179 relate to immediate family history, whereas items numbered 150 to 151 should be answered by female applicants only. Do not report occasional common illnesses such as colds.

Balloon Pilot Licence Light Aircraft Pilot Licence* Air Traffic Controller Licence Other – Please specify *Please specify whether Fixed Wing / Rotary Wing / Both	
15. OCCUPATION (PRINCIPAL): Indicate your principal employment.	
16. EMPLOYER: If principal occupation is pilot, then state employer's name or if self-employed, state 'self'.	31. DECLARATION AND NOTIFICATION OF DISCLOSURE OF PERSONAL DATA: Do not sign or date these declarations until indicated to do so by the AME/GMP who will act as witness and sign accordingly.

AMC1 ARA.MED.135(b);(c) Aero-medical forms

ED Decision 2025/002/R

MEDICAL EXAMINATION REPORT FORMS

The forms referred to in points ARA.MED.135(b) and (c) should reflect the information indicated in the following forms and corresponding instructions for completion.

MEDICAL EXAMINATION REPORT FORM FOR CLASS 1,2 & 3 APPLICANTS

MEDICAL IN CONFIDENCE

(201) Examination category Initial ☐ Revalidation ☐ Renewal ☐ Special referral ☐	(202) Height (cm)	(203) Weight (kg)	(204) Colour eye	(205) Colour hair	(206) Blood pressure-seated (mmHg) Systolic Diastolic	(207) Pulse - resting Rate (bpm) Rhythm: regular ☐ irregular ☐
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Clinical exam: Check each item Normal/Abnormal

(208) Head, face, neck, scalp	(218) Abdomen, hernia, liver, spleen
(209) Mouth, throat, teeth	(219) Anus, rectum
(210) Nose, sinuses	(220) Genito-urinary system
(211) Ears, drums, eardrum motility	(221) Endocrine system
(212) Eyes - orbit & adnexa; visual fields	(222) Upper & lower limbs, joints
(213) Eyes - pupils and optic fundi	(223) Spine, other musculoskeletal
(214) Eyes - ocular motility; nystagmus	(224) Neurologic - reflexes, etc.
(215) Lungs, chest, breasts	(225) Mental health
(216) Heart	(226) Skin, identifying marks and lymphatics
(217) Vascular system	(227) General systemic

(228) Notes: Describe every abnormal finding. Enter applicable item number before each comment.

Visual acuity

(229) Distant vision at 5m/6m (236) Pulmonary function (237) Haemoglobin

	Uncorrected	Spectacles	Contact lenses
Right eye	Corr. to		
Left eye	Corr. to		
Both eyes	Corr. to		

FEV ₁ /FVC _____ %	_____ (unit)
Normal ☐ Abnormal ☐	Normal ☐ Abnormal ☐

(236a) OSA screening

Applicant at risk of OSA: Yes ☐ No ☐
Specify if applicant undergoes treatment for OSA:

(235) Urinalysis Normal ☐ Abnormal ☐

Glucose	Protein	Blood	Other
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Accompanying reports

	Not performed	Normal	Abnormal/Comment
(238) ECG			
(239) Audiogram			
(240) Ophthalmology			

(230) Intermediate vision N14 at 100 cm	Uncorrected		Corrected	
	Yes	No	Yes	No
Right eye				
Left eye				
Both eyes				

(231) Near vision N5 at 30-50 cm	Uncorrected		Corrected	
	Yes	No	Yes	No

Right eye				
Left eye				
Both eyes				

(232) Spectacles		(233) Contact lenses		
Yes ☐ No ☐		Yes ☐ No ☐		
Type:		Type:		
Refraction	Sph	Cyl	Axis	Add
Right eye				
Left eye				

(313) Colour perception		Normal ☐ Abnormal ☐
Pseudo-isochromatic plates	Type: Ishihara (24 plates)	
No of plates:	No of errors:	

(234) Hearing		(when 239/241 not performed)		Right ear	Left ear
Conversational voice test (2m)		Yes ☐	No ☐	Yes ☐	No ☐
with back turned to examiner		Yes ☐	No ☐	Yes ☐	No ☐
Audiometry					
Hz	500	1000	2000	3000	
Right					
Left					

(241) ORL (ENT)			
(242) Blood lipids			
(243) Pulmonary function			
(244) Other (what?)			

(247) AME recommendation:		
Name of applicant:	Date of birth:	Reference number:
☐ Fit for class: ----- ☐ Medical certificate issued by undersigned (copy attached) for class: ----- ☐ Unfit for class: ----- ☐ Deferred for further evaluation. If yes, why and to whom?		
(248) Comments, limitations		

(249) AME declaration:		
I hereby certify that I/my AME group have personally examined the applicant named on this medical examination report and that this report with any attachment embodies my findings completely and correctly.		
(250) Place and date:	AME name and address:	AME certificate No.:
AME signature:	E-mail:	
	Telephone No.:	
	Telefax No.:	

Shaded areas do not require completion

MEDICAL EXAMINATION REPORT FORM FOR LAPL APPLICANTS

MEDICAL IN CONFIDENCE

(201) Examination category	(202) Height (cm)	(203) Weight (kg)	(204) Colour eye	(205) Colour hair	(206) Blood pressure-seated (mmHg)	(207) Pulse - resting
Initial ☐					Systolic	Rate (bpm)
Revalidation ☐ Renewal ☐					Diastolic	Rhythm: regular ☐ irregular ☐
Special referral ☐						

Clinical exam: Check each item NormalAbnormal NormalAbnormal

(208) Head, face, neck, scalp		(218) Abdomen, hernia, liver, spleen	
(209) Mouth, throat, teeth		(219) Anus, rectum	
(210) Nose, sinuses		(220) Genito-urinary system	
(211) Ears, drums, eardrum motility		(221) Endocrine system	
(212) Eyes - orbit & adnexa; visual fields		(222) Upper & lower limbs, joints	
(213) Eyes - pupils and optic fundi		(223) Spine, other musculoskeletal	
(214) Eyes - ocular motility; nystagmus		(224) Neurologic - reflexes, etc.	
(215) Lungs, chest, breasts		(225) Psychiatric	
(216) Heart		(226) Skin, identifying marks and lymphatics	
(217) Vascular system		(227) General systemic	

(228) Notes: Describe every abnormal finding. Enter applicable item number before each comment.

Visual acuity

(229) Distant vision at 5m/6m (236) Pulmonary function (237) Haemoglobin

	Uncorrected		Spectacles	Contact lenses
Right eye		Corr. to		
Left eye		Corr. to		
Both eyes		Corr. to		

FEV ₁ /FVC _____ %	_____ (unit)
Normal ☐ Abnormal ☐	Normal ☐ Abnormal ☐

(235) Urinalysis Normal ☐ Abnormal ☐

(230) <i>Intermediate vision</i> N14 at 100 cm	Uncorrected		Corrected	
	Yes	No	Yes	No
Right eye				
Left eye				
Both eyes				

(231) <i>Near vision</i> N5 at 30-50 cm	Uncorrected		Corrected	
	Yes	No	Yes	No
Right eye				
Left eye				
Both eyes				

(232) Spectacles		(233) Contact lenses		
Yes ☐ No ☐		Yes ☐ No ☐		
Type:		Type:		
Refraction	Sph	Cyl	Axis	Add
Right eye				
Left eye				

(233) Colour perception	Normal ☐ Abnormal ☐
Pseudo-isochromatic plates	Type: Ishihara (24 plates)
No of plates:	No of errors:

(234) Hearing				
(when 239/241 not performed)				
	Right ear		Left ear	
Conversational voice test (2m)	Yes ☐	Yes ☐	Yes ☐	Yes ☐
with back turned to examiner	No ☐	No ☐	No ☐	No ☐
Audiometry				
Hz	500	1000	2000	3000
Right				
Left				

Glucose	Protein	Blood	Other
Accompanying reports			
	Not performed	Normal	Abnormal/Comment
(238) ECG			
(239) Audiogram			
(240) Ophthalmology			
(241) ORL (ENT)			
(242) Blood lipids			
(243) Pulmonary function			
(244) Other (what?)			

(247) AME/GMP recommendation:		
Name of applicant:	Date of birth:	Reference number:
----- ☐ Fit for medical certificate for LAPL ☐ Medical certificate issued by undersigned (copy attached) for LAPL ☐ Unfit for class: ----- ☐ Deferred for further evaluation. If yes, why and to whom?		
(248) Comments, limitations		

(249) AME/GMP declaration:		
I hereby certify that I have personally examined the applicant named on this medical examination report and that this report with any attachment embodies my findings completely and correctly.		
(250) Place and date:	AME name and address:	AME certificate No./GMP identification No.:
AME/GMP signature:	E-mail:	
	Telephone No.:	
	Telefax No.:	

INSTRUCTIONS FOR COMPLETION OF THE MEDICAL EXAMINATION REPORT FORMS

The AME performing the examination should verify the identity of the applicant.

All questions (sections) on the medical examination report form should be completed in full. If an otorhinolaryngology examination report form is attached, then questions 209, 210, 211, and 234 may be omitted. If an ophthalmology examination report form is attached, then questions 212, 213, 214, 229, 230, 231, 232, and 233 may be omitted.

Writing should be legible and in block capitals using a ball-point pen. Completion of this form by typing/printing is also acceptable. If more space is required to answer any question, a plain sheet of paper should be used, bearing the applicant's name, the AME's name and signature, and the date of signing. The following numbered instructions apply to the numbered headings on the medical examination report form.

Failure to complete the medical examination report form in full, as required, or to write legibly, may result in non-acceptance of the application in total and may lead to withdrawal of any medical certificate issued. The making of false or misleading statements or the withholding of relevant information by an AME may result in criminal prosecution, denial of an application or withdrawal of any medical certificate(s) granted.

Shaded areas do not require completion for the medical examination report form for the LAPL.

201 EXAMINATION CATEGORY – Tick appropriate box.

Initial – Initial examination for either LAPL, class 1, 2 or 3; also initial examination for upgrading from LAPL to class 2, or from class 2 to 1 (insert ‘upgrading’ in box 248).

Renewal/Revalidation – Subsequent ROUTINE examinations.

Extended Renewal/Revalidation – Subsequent ROUTINE examinations, which include comprehensive ophthalmological and otorhinolaryngology examinations.

- 202 HEIGHT – Measure height, without shoes, in centimetres to nearest cm.
- 203 WEIGHT – Measure weight, in indoor clothes, in kilograms to nearest kg.
- 204 COLOUR EYE – State colour of applicant’s eyes from the following list: brown, blue, green, hazel, grey, multi.
- 205 COLOUR HAIR – State colour of applicant’s hair from the following list: brown, black, red, fair, bald.
- 206 BLOOD PRESSURE – Blood pressure readings should be recorded as Phase 1 for Systolic pressure and Phase 5 for Diastolic pressure. The applicant should be seated and rested. Recordings in mm Hg.
- 207 PULSE (RESTING) – The pulse rate should be recorded in beats per minute and the rhythm should be recorded as regular or irregular. Further comments if necessary may be written in section 228, 248 or separately.
- 208 to 227 inclusive constitute the general clinical examination, and each of the boxes should be marked (with a tick) as normal or abnormal.
- 208 HEAD, FACE, NECK, SCALP – To include appearance, range of neck and facial movements, symmetry, etc.
- 209 MOUTH, THROAT, TEETH – To include appearance of buccal cavity, palate motility, tonsillar area, pharynx and also gums, teeth and tongue.
- 210 NOSE, SINUSES – To include appearance and any evidence of nasal obstruction or sinus tenderness on palpation.
- 211 EARS, DRUMS, EARDRUM MOTILITY – To include otoscopy of external ear, canal, tympanic membrane. Eardrum motility by Valsalva manoeuvre or by pneumatic otoscopy.
- 212 EYES – ORBIT AND ADNEXA; VISUAL FIELDS – To include appearance, position and movement of eyes and their surrounding structures in general, including eyelids and conjunctiva. Visual fields check by campimetry, perimetry or confrontation.
- 213 EYES – PUPILS AND OPTIC FUNDI – To include appearance, size, reflexes, red reflex and fundoscopy. Special note of corneal scars.
- 214 EYES – OCULAR MOTILITY, NYSTAGMUS – To include range of movement of eyes in all directions; symmetry of movement of both eyes; ocular muscle balance; convergence; accommodation; signs of nystagmus.
- 215 LUNGS, CHEST, BREASTS – To include inspection of chest for deformities, operation scars, abnormality of respiratory movement, auscultation of breath sounds. Physical examination of female applicant’s breasts should only be performed with informed consent.
- 216 HEART – To include apical heartbeat, position, auscultation for murmurs, carotid bruits, palpation for trills.
- 217 VASCULAR SYSTEM – To include examination for varicose veins, character and feel of pulse, peripheral pulses, evidence of peripheral circulatory disease.
- 218 ABDOMEN, HERNIA, LIVER, SPLEEN – To include inspection of abdomen; palpation of internal organs; check for inguinal hernias in particular.
- 219 ANUS, RECTUM – Examination only or clinical indication following an informed consent.

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- 220 GENITO-URINARY SYSTEM – To include renal palpation; inspection palpation male/female reproductive organs only on clinical indication following an informed consent.
- 221 ENDOCRINE SYSTEM – To include inspection, palpation for evidence of hormonal abnormalities/imbalance; thyroid gland.
- 222 UPPER AND LOWER LIMBS, JOINTS – To include full range of movements of joints and limbs, any deformities, weakness or loss. Evidence of arthritis.
- 223 SPINE, OTHER MUSCULOSKELETAL – To include range of movements, abnormalities of joints.
- 224 NEUROLOGIC – REFLEXES ETC. To include reflexes, sensation, power, vestibular system – balance, romberg test, etc.
- 225 MENTAL HEALTH – To include appearance, appropriate mood/thought, unusual behaviour.
- 226 SKIN, IDENTIFYING MARKS AND LYMPHATICS – To include inspection of skin; inspection, palpation for lymphadenopathy, etc. Briefly describe scars, tattoos, birthmarks, etc. which could be used for identification purposes.
- 227 GENERAL SYSTEMIC – All other areas, systems and nutritional status.
- 228 NOTES – Any notes, comments or abnormalities to be described – extra notes if required on separate sheet of paper, signed and dated.
- 229 DISTANT VISION AT 5/6 METRES – Each eye to be examined separately and then both together. First without correction, then with spectacles (if used) and lastly with contact lenses, if used. Record visual acuity in appropriate boxes. Visual acuity to be tested at either 5 or 6 metres with the appropriate chart for the distance.
- 230 INTERMEDIATE VISION AT 100 CM – Each eye to be examined separately and then both together. First without correction, then with spectacles if used and lastly with contact lenses if used. Record visual acuity in appropriate boxes as ability to read N14 at 100 cm (Yes/No).
- 231 NEAR VISION AT 30-50 CM. – Each eye to be examined separately and then both together. First without correction, then with spectacles if used and lastly with contact lenses, if used. Record visual acuity in appropriate boxes as ability to read N5 at 30-50 cm (Yes/No).
- Note: Bifocal contact lenses and contact lenses correcting for near vision only are not acceptable.
- 232 SPECTACLES – Tick appropriate box signifying whether spectacles are or are not worn by applicant. If used, state whether unifocal, bifocal, varifocal or look-over.
- 233 CONTACT LENSES – Tick appropriate box signifying if contact lenses are or are not worn. If worn, state type from the following list; hard, soft, gas-permeable or disposable.
- 313 COLOUR PERCEPTION – Tick appropriate box signifying if colour perception is normal or not. If abnormal; state number of plates of the first 15 of the pseudo-isochromatic plates (Ishihara 24 plates) have not been read correctly.
- 234 HEARING – Tick appropriate box to indicate hearing level ability as tested separately in each ear at 2 m.
- 235 URINALYSIS – State whether result of urinalysis is normal or not by ticking appropriate box. If no abnormal constituents, state NIL in each appropriate box.
- 236 PULMONARY FUNCTION – When required or on indication, state actual FEV1/FVC value obtained in % and state whether normal or not with reference to height, age, sex and race.
- 236(a) OSA screening: Determine the risk of OSA using appropriate diagnostic tool.
- 237 HAEMOGLOBIN – Enter actual haemoglobin test result and state units used. Then state whether normal value or not, by ticking appropriate box.
- 238 to 244 inclusive: ACCOMPANYING REPORTS – One box opposite each of these sections must be ticked. If the test is not required and has not been performed, then tick the NOT PERFORMED box. If the test has been
-

247 AME RECOMMENDATION – The applicant’s name, date of birth and reference number, should be entered here in block capitals. The applicable class of medical certificate should be indicated by a tick in the appropriate box. If a fit assessment is recommended and a medical certificate has been issued, this should be indicated in the appropriate box. An applicant may be recommended as fit for a lower class of medical certificate (e.g. class 2), but also be deferred or recommended as unfit for a higher class of medical certificate (e.g. class 1). If an unfit recommendation is made, applicable Part-MED point references should be entered. If an applicant is deferred for further evaluation, the reason and the doctor or licensing authority to whom the applicant is referred should be indicated.

248 COMMENTS, LIMITATIONS, ETC. – The AME’s findings and assessment of any abnormality in the history or examination, should be entered here. The AME should also state any limitation required.

249 AME DETAILS – The AME should sign the declaration, complete his or her name and address in block capitals, contact details and lastly stamp the relevant section with his or her designated AME stamp incorporating his or her AME number. The GMP identification no. is the number provided by the national medical system.

250 PLACE AND DATE – The place (town or city) and the date of examination should be entered here. The date of examination is the date of the general examination and not the date of finalisation of the form. If the medical examination report is finalised on a different date, the date of finalisation should be entered in section 248 as ‘Report finalised on’.

ED Decision 2025/002/R

MEDICAL IN CONFIDENCE

Spectacles	Contact lenses
1. The patient is wearing spectacles. 2. The patient is wearing contact lenses.	1. The patient is wearing contact lenses.

(304) Eyes, external & eyelids			
(305) Eyes, Exterior (slit lamp, ophth.)			
(306) Eye position and movements			
(307) Visual fields (confrontation)			
(308) Pupillary reflexes			
(309) Fundi (Ophthalmoscopy)			
(310) Convergence	cm		
(311) Accommodation	D		

(312) Ocular muscle balance (in prisme dioptres)			
Distant at 5m/6m	Near at 30-50 cm		
Ortho	Ortho		
Eso	Eso		
Exo	Exo		
Hyper	Hyper		
Cyclo	Cyclo		
Tropia Yes No Phoria Yes No			
Fusional reserve testing		Not performed	Normal Abnormal

(313) Colour perception	
Pseudo-Isochromatic plates	Type: Ishihara (24 plates)
No of plates:	No of errors:
Advanced colour perception testing indicated Yes No	
Method:	
Class 1&2 Colour SAFE Colour UNSAFE	
For ATCOs Normal trichromat Yes ☐ No ☐	

(317) Refraction				
	Sph	Cylinder	Axis	Near (add)
Right eye				
Left eye				
Actual refraction examined Spectacles prescription based				

(318) Spectacles		(319) Contact lenses	
Yes ☐ No ☐	Yes ☐ No ☐		
Type:	Type:		

(320) Intra-ocular pressure	
Right (mmHg)	Left (mmHg)
Method Normal ☐ Abnormal ☐	

(321) Ophthalmological remarks:

(322) Examiner's declaration:

I hereby certify that I have personally examined or assessed the eye specialist's examination report of the applicant named in this medical examination report and that this report with any attachment embodies the findings completely and correctly.

(323) Place and date:	Name and address: (block capitals)	AME or eye specialist stamp with No.:
AME or eye specialist signature:	E-mail:	
	Telephone No.: Telefax No.:	

INSTRUCTIONS FOR COMPLETION OF THE OPHTHALMOLOGY EXAMINATION REPORT FORM

Writing should be legible and in block capitals using a ball-point pen. Completion of this form by typing or printing is also acceptable. If more space is required to answer any question, a plain sheet of paper should be used, bearing the applicant's name, the name and signature of the AME or ophthalmology specialist performing the examination and the date of signing. The following numbered instructions apply to the numbered headings on the ophthalmology examination report form.

Failure to complete the medical examination report form in full, as required, or to write legibly may result in non-acceptance of the application in total and may lead to withdrawal of any medical certificate issued. The making of false or misleading statements or the withholding of relevant information by an examiner may result in criminal prosecution, denial of an application or withdrawal of any medical certificate granted.

The AME or ophthalmology specialist performing the examination should verify the identity of the applicant. The applicant should then be requested to complete the sections 1, 2, 3, 4, 5, 6, 7, 12 and 13 on the form and then sign and date the consent to release of medical information (section 301) with the examiner countersigning as witness.

302 EXAMINATION CATEGORY – Tick appropriate box.

- Initial – Initial examination for either class 1 or 2; also initial examination for upgrading from class 2 to 1 (insert 'upgrading' in section 303).
- Renewal/Revalidation – Subsequent comprehensive ophthalmological examinations (due to refractive error).
- Special referral – NON-ROUTINE examination for assessment of an ophthalmological symptom or finding.
- 303 OPTHALMOLOGICAL HISTORY – Detail here any history of note or reasons for special referral.
- 304 to 309 inclusive: CLINICAL EXAMINATION – These sections together cover the general clinical examination and each of the sections should be marked (with a tick) as normal or abnormal. Any abnormal findings or comments on findings should be entered in section 321.
- 310 CONVERGENCE – Enter near point of convergence in cm, as measured using RAF near point rule or equivalent. Tick whether normal or abnormal. Any abnormal findings or comments on findings should be entered in section 321.
- 311 ACCOMMODATION – Enter measurement recorded in dioptres using RAF near point rule or equivalent. Tick whether normal or abnormal. Any abnormal findings or comments on findings should be entered in section 321.
- 312 OCULAR MUSCLE BALANCE – Ocular muscle balance is tested at distant 5 or 6 m and near at 30-50 cm and results recorded. Presence of tropia or phoria must be entered accordingly and also whether fusional reserve testing was NOT performed and if performed whether normal or not.
- 313 COLOUR PERCEPTION – Enter type of pseudo-isochromatic plates (Ishihara) as well as number of plates presented with number of errors made by examinee. 15 plates should normally be presented from the 24-plate series, in random order. State whether advanced colour perception testing is indicated and what methods used (CAD or anomaloscopy) and finally whether judged to be colour safe or unsafe. Advanced colour perception testing is usually only required for initial assessment, unless indicated by change in applicant's colour perception. Class 3 applicants are required to demonstrate normal trichromacy which cannot be done by using only pseudo-isochromatic plates, therefore, in their case, advanced colour perception testing is needed as default at the initial examination or whenever there is a clinical indication.
- 314–316 VISUAL ACUITY TESTING AT 5 m/6 m, 1 m and 30-50 cm – Record actual visual acuity obtained in appropriate boxes. If correction not worn nor required, put line through corrected vision boxes. Distant visual acuity to be tested at either 5 m or 6 m with the appropriate chart for that distance.
- 317 REFRACTION – Record results of refraction. Indicate also whether for class 2 applicants, refraction details are based upon spectacle prescription.
- 318 SPECTACLES – Tick appropriate box signifying if spectacles are or are not worn by applicant. If used, state whether unifocal, bifocal, varifocal or look-over.
- 319 CONTACT LENSES – Tick appropriate box signifying if contact lenses are or are not worn. If worn, state type from the following list; hard, soft, gas-permeable, disposable.
- 320 INTRA-OCULAR PRESSURE – Enter intra-ocular pressure recorded for right and left eyes and indicate whether normal or not. Also indicate method used – applanation, air etc.
- 321 OPTHALMOLOGICAL REMARKS AND RECOMMENDATION – Enter here all remarks, abnormal findings and assessment results. Also enter any limitations recommended. If there is any doubt about findings or recommendations, the examiner may contact the medical assessor of the licensing authority for advice before finalising the report form.
- 322 OPTHALMOLOGY EXAMINER'S DETAILS – The ophthalmology examiner must sign the declaration, complete his or her name and address in block capitals, contact details and lastly stamp the report with his or her designated stamp incorporating his or her AME or specialist number.
- 323 PLACE AND DATE – Enter the place (town or city) and the date of examination. The date of examination is the date of the clinical examination and not the date of finalisation of form. If the ophthalmology

examination report is finalised on a different date, enter date of finalisation on section 321 as 'Report finalised on'.

OTORHINOLARYNGOLOGY (ENT) EXAMINATION REPORT FORM

Complete this page fully and in block capitals – Refer to instructions for completion.

MEDICAL IN CONFIDENCE

Applicant's details

(1) State applied to:	(2) Medical certificate applied for: class 1 ☐ class 2 ☐ class 3 ☐		
(3) Surname:	(4) Previous surname(s):		(12) Application: Initial ☐ Revalidation/Renewal ☐
(5) Forename(s):	(6) Date of birth:	(7) Sex: Male ☐ Female ☐	(13) Reference number:
(401) Consent to release of medical information: I hereby declare that I have been informed and I understand that all information provided to my AME, contained in this report and its attachments, may be released to the medical assessor of the my licensing authority and to the medical assessor of the competent authority of my AME, recognising that these documents or electronically stored data are to be used for completion of a medical assessment and for oversight purposes, providing that I or my physician may have access to them in accordance with national law. Medical confidentiality will be respected at all times. ----- Date Signature of applicant			

(402) Examination category:	(403) Otorhinolaryngological (ENT) history:
Initial ☐	
Special referral ☐	

Clinical examination

Check each item	Normal	Abnormal
(404) Head, face, neck, scalp		
(405) Buccal cavity, teeth		
(406) Pharynx		
(407) Nasal passages and naso-pharynx (incl. anterior rhinoscopy)		
(408) Vestibular system incl. Romberg test		
(409) Speech		
(410) Sinuses		
(411) Ext acoustic meati, tympanic membranes		
(412) Pneumatic otoscopy		
(413) Tympanometry including Valsalva manoeuvre (initial or if clinically indicated)		

(419) Pure tone audiometry dB HL (hearing level)

Hz	Right ear	Left ear
250		
500		
1000		
2000		
3000		
4000		
6000		
8000		

(420) Audiogram

o = Right --- = Air
x = Left..... = Bone

dB/HL									
-10									
0									
10									
20									
30									
40									
50									
60									
70									
80									
90									
100									
110									
120									
Hz	250	500	1000	2000	3000	4000	6000	8000	

Additional testing (if indicated)

	Not performed	Normal	Abnormal
(414) Speech discrimination test with/without hearing aids, as applicable			
(415) Posterior rhinoscopy			
(416) ENG; spontaneous and positional nystagmus			
(417) Caloric test or vestibular rotation test			
(418) Mirror or fibre laryngoscopy			

(421) Otorhinolaryngology remarks:

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(422) Examiner's declaration:

I hereby certify that I have personally examined or assessed the ENT specialist's examination report of the applicant named in this medical examination report and that this report with any attachment embodies the findings completely and correctly.		
(423) Place and date:	Name and address: (block capitals)	AME or ENT specialist stamp with No:
AME or ENT specialist signature:	E-mail: Telephone No.: Telefax No.:	

INSTRUCTIONS FOR COMPLETION OF THE OTORHINOLARYNGOLOGY (ENT) EXAMINATION REPORT FORM

Writing should be legible and in block capitals using a ball-point pen. Completion of this form by typing or printing is also acceptable. If more space is required to answer any question, a plain sheet of paper should be used, bearing the applicant's name, the name and signature of the AME or otorhinolaryngology specialist performing the examination and the date of signing. The following numbered instructions apply to the numbered headings on the otorhinolaryngology examination report form.

Failure to complete the medical examination report form in full, as required, or to write legibly may result in non-acceptance of the application in total and may lead to withdrawal of any medical certificate issued. The making of false or misleading statements or the withholding of relevant information by an examiner may result in criminal prosecution, denial of an application or withdrawal of any medical certificate granted.

The AME or otorhinolaryngology specialist performing the examination should verify the identity of the applicant. The applicant should then be requested to complete the sections 1, 2, 3, 4, 5, 6, 7, 12 and 13 on the form and then sign and date the consent to release of medical information (section 401) with the examiner countersigning as witness.

402 EXAMINATION CATEGORY – Tick appropriate box.

Initial – Initial examination for class 1; also initial examination for upgrading from class 2 to 1 (insert 'upgrading' in section 403)

Special Referral – NON-ROUTINE examination for assessment of an ORL (ENT) symptom or finding

403 OTORHINOLARYNGOLOGICAL (ENT) HISTORY – Detail here any history of note or reasons for special referral.

404-413 inclusive: CLINICAL EXAMINATION – These sections together cover the general clinical examination and each of the sections should be marked (with a tick) as normal or abnormal. Any abnormal findings or comments on findings should be entered in section 421.

414-418 inclusive: ADDITIONAL TESTING – These tests are only required to be performed if indicated by history or clinical findings and are not routinely required. For each test one of the boxes must be completed – if the test is not performed then tick that box – if the test has been performed then tick the appropriate box for a normal or abnormal result. All remarks and abnormal findings should be entered in section 421.

419 PURE TONE AUDIOMETRY – Complete figures for dB HL (hearing level) in each ear at all listed frequencies.

420 AUDIOGRAM – Complete audiogram from figures as listed in section 419.

421 OTORHINOLARYNGOLOGY (ENT) REMARKS AND RECOMMENDATION – Enter here all remarks, abnormal findings and assessment results. Also enter any limitations recommended. If there is any doubt about findings or recommendations the examiner may contact the medical assessor of the licensing authority for advice before finalising the report form.

422 OTORHINOLARYNGOLOGY (ENT) EXAMINER'S DETAILS – The otorhinolaryngology (ENT) examiner must sign the declaration, complete his or her name and address in block capitals, contact details and lastly stamp the report with his or her designated stamp incorporating his or her AME or specialist number.

423 PLACE AND DATE – Enter the place (town or city) and the date of examination. The date of examination is the date of the clinical examination and not the date of finalisation of form. If the ORL (ENT)

examination report is finalised on a different date, enter date of finalisation in section 421 as 'Report finalised on'.

ARA.MED.145 GMP notification to the competent authority

Regulation (EU) 2024/2076

The competent authority, when applicable, shall establish a notification process for general medical practitioners (GMPs) to ensure that the GMP is aware of the applicable requirements laid down in this Regulation.

ARA.MED.150 Record-keeping

Regulation (EU) 2024/2076

- (a) In addition to the records required in point [ARA.GEN.220](#), the competent authority shall include in its system of record-keeping, details of aero-medical examinations, and assessments submitted by AMEs, AeMCs or GMPs.
- (b) All aero-medical records of applicants/licence holders shall be kept for a minimum period of 10 years after the expiry date of their last medical certificate.
- (c) For the purpose of aero-medical assessments and standardisation, aero-medical records shall be made available after written consent of the applicant/licence holder to the following entities:
 - (1) an AeMC, AME or GMP for the purpose of completion of an aero-medical assessment;
 - (2) a medical review board that may be established by the competent authority for secondary review of borderline cases;
 - (3) relevant medical specialists for the purpose of completion of an aero-medical assessment;
 - (4) the medical assessor of the competent authority of another Member State for the purpose of cooperative oversight;
 - (5) the applicant/licence holder concerned upon their written request;
 - (6) the Agency for standardisation purposes, in a manner that ensures that medical confidentiality is respected at all times.
- (d) The competent authority may make aero-medical records available for other purposes than those mentioned in point (c) in accordance with Regulation (EU) 2016/679.
- (e) The competent authority shall maintain a list of:
 - (1) AeMCs and AMEs that it has certified;
 - (2) AMEs certified by other competent authorities exercising their privileges in its territory and to whom it has provided a briefing in accordance with point [MED.D.001\(f\)\(3\)](#) of Annex IV (Part-MED);
 - (3) GMPs exercising their privileges in accordance with point [MED.A.040](#) of Annex IV (Part-MED), where applicable;
 - (4) OHMPs having notified the competent authority of their intention to perform cabin crew aero-medical assessments in accordance with points [MED.C.005\(c\)](#) and [MED.D.040](#) of Annex IV (Part-MED), where applicable.

The list shall state the privileges of the persons and organisations specified in points (1) to (4) of the first paragraph and shall be published and kept updated by the competent authority.

- (f) The competent authority shall analyse the health data of pilots above the age of 60, especially of those involved in single-pilot HEMS operations, and report such health data in an anonymised and aggregated manner to EASA on a yearly basis.

AMC1 ARA.MED.150 Record-keeping

ED Decision 2012/006/R

RELEASE OF AERO-MEDICAL RECORDS

In accordance with Directive 95/46/EC as implemented under national law, aero-medical records may also be released:

- (a) upon written request of the applicant, to management of the competent authority, for review in response to a complaint;
- (b) to research institutes for the purpose of scientific research, with assurance of de-identification prior to publication;
- (c) to any investigation body (accident, security, police), when required under national law; and
- (d) for any other circumstances, as required under national law.

AMC1 ARA.MED.150(f) Record-keeping

ED Decision 2025/002/R

REPORTING HEALTH DATA OF PILOTS ABOVE THE AGE OF 60

For pilots above the age of 60 the competent authorities performing the analysis of health data should report in an aggregated manner to EASA at least the following data:

- (a) number and proportion of pilots above the age of 60 assessed as unfit, as well as the most common medical conditions that triggered unfitness and the age distribution;
- (b) proportion of incapacitation (partial and total) events among this category of pilots and the most common medical and, if applicable, the operational conditions that triggered incapacitation;
- (c) the proportion of pilots above the age of 60 who did not revalidate their medical certificate;
- (d) any safety concerns based on the trends identified as a result of the data analysis.

ARA.MED.160 Exchange of information on medical certificates through a central repository.

Regulation (EU) 2019/27

- (a) The Agency shall establish and manage a central repository, the European Aero-Medical Repository (EAMR).
- (b) For the purposes of medical certification and oversight of applicants for and holders of class 1 medical certificates and for the oversight of AMEs and AeMCs, the persons referred to in point (c) shall exchange the following information through EAMR:

- (1) basic data of the applicant for or holder of a class 1 medical certificate: licensing authority; surname and forename; date of birth; nationality; email address and the number of one or more identification documents (national identity card or passport) as provided by the applicant;
 - (2) class 1 medical certificate data: date of the medical examination or, in case the medical examination is not finalised, the date of initiation of the medical examination; dates of issuing and of expiration of the class 1 medical certificate; place of the examination; status of limitations; status of that certificate (new, released, suspended or revoked); unique reference number of the medical assessor of the licensing authority; AME or AeMC issuing that certificate and of its competent authority.
- (c) For the purposes of point (b), the following persons shall have access to EAMR and the information contained therein:
- (1) medical assessors of the licensing authority of the applicant for or holder of a class 1 medical certificate, as well as any other duly authorised personnel of that authority in charge of creating or managing the record of that applicant or holder as required by this Regulation;
 - (2) AMEs and any duly authorised personnel of AeMCs to whom that applicant or holder has provided a declaration in accordance with point MED.A.035(b)(2);
 - (3) any duly authorised personnel of the competent authority responsible for the oversight of AMEs or AeMCs conducting aero-medical assessments of those applicants or holders.

In addition, the Agency and national competent authorities may grant access to EAMR and the information contained therein to other persons, where necessary for the purposes of ensuring the proper functioning of EAMR, in particular its technical maintenance. In that case, the Agency or the national competent authority concerned shall ensure that those persons are duly authorised and qualified, that their access remains limited to what is necessary for the purposes for which they have been granted access and that they have received prior training on the applicable personal data protection legislation and related safeguards. Whenever a competent authority grants a person such access, it shall inform the Agency beforehand.

- (d) The licensing authorities, AMEs and AeMCs referred to in point (c) shall, each time immediately upon having examined an applicant for or a holder of a class 1 medical certificate, enter the data referred to in point (b) into EAMR or update that data where necessary.
- (e) Where the data constitutes personal data as defined in point a of Article 2 of Regulation (EC) No 45/2001¹, they shall, each time when entering or updating that data, inform, ex ante, the applicant for or holder of the class 1 certificate thereof.
- (f) The Agency shall ensure the integrity and security of EAMR and the information contained therein by appropriate information technology infrastructure. It shall establish and apply, in consultation with the national competent authorities, the protocols and technological measures necessary to ensure that any access to EAMR and the information contained therein is lawful and secure.
- (g) The Agency shall ensure that any information contained in EAMR is deleted after a period of 10 years. That period shall be calculated from the date of expiration of the last class 1 certificate

¹ Regulation (EC) No 45/2001 of the European Parliament and of the Council of 18 December 2000 on the protection of individuals with regard to the processing of personal data by the Community institutions and bodies and on the free movement of such data (OJ L 8, 12.1.2001, p. 1).

issued in respect of the applicant or holder concerned, or from the date of the last entry or update of data in respect of that applicant or holder, whichever date is later.

- (h) The Agency shall ensure that applicants for or holders of class 1 medical certificates can access any information relating to them contained in EAMR and that they are informed that they can request that information to be rectified or deleted. The licensing authorities shall assess such requests and, where they consider that the information concerned is incorrect or not necessary for the purposes specified in point (b), ensure that the information is rectified or deleted.'

AMC1 ARA.MED.160(b) Exchange of information on medical certificates

ED Decision 2019/002/R

DATA CATEGORIES

For the purpose of the EAMR, the information processed is divided into two categories as follows:

Category 1: Basic applicant data as described in [ARA.MED.160\(b\)\(1\)](#)

Category 2: Medical certificate data as described in [ARA.MED.160\(b\)\(2\)](#)

Typically, the following information should not be recorded:

- Reasons for which a medical certificate has not been issued
Only the fact that no certificate has been issued should be indicated. Any need for further clarification on whether the certificate has not been issued because of medical reasons, administrative matters or interruption of the medical assessment process before reaching the conclusion should be addressed, outside the scope of the EAMR, by the medical assessor of the licensing authority associated with the applicant's class 1 medical certificate.
- Details of the limitations associated with a given medical certificate
Only a 'Yes/No' status on the existence of such a limitation should be recorded. Any need for further clarification on the limitation(s) should be addressed, outside the scope of the EAMR, by the medical assessor of the licensing authority associated with the applicant's class 1 medical certificate.

AMC1 ARA.MED.160(c) Exchange of information on medical certificates

ED Decision 2019/002/R

ROLE OF THE COMPETENT AUTHORITIES

Each competent authority should:

- (a) designate its EAMR administrator;
- (b) ensure control and oversight of all personnel managing or using the EAMR.

AMC2 ARA.MED.160(c) Exchange of information on medical certificates

ED Decision 2019/002/R

RESTRICTED ACCESS TO INFORMATION

Each competent authority should restrict access to personal data in the EAMR on need-to-know basis as follows:

Category as determined by AMC2 ARA.MED.160(a)	Restricted access
Category 1	(a) to relevant authorised administrative personnel of the licensing authority, to the extent needed to create and manage the applicant's record for licensing purposes, as required by Commission Regulation (EU) No 1178/2011.
Category 1 & 2	(b) to the AeMC(s) or the AME(s) to whom the applicant submits a declaration in accordance with MED.A.035(b)(2) for a class 1 medical certificate, to the extent needed to verify their previous medical certificate history, as required by Commission Regulation (EU) No 1178/2011; (c) to the medical assessor(s) of the licensing authority and the competent authority(ies) exercising oversight on the AeMC(s) or the AME(s) to whom the application for a class 1 medical certificate is submitted, to the extent needed to ensure proper implementation of Commission Regulation (EU) No 1178/2011.

AMC3 ARA.MED.160(c) Exchange of information on medical certificates

ED Decision 2019/002/R

USE OF THE EAMR

The competent authority should ensure that:

- (a) all personnel accessing the EAMR are trained and proficient in using the system and having the necessary knowledge for implementing the applicable data protection legislation;
- (b) the oversight of persons and organisations, subject to Regulation (EU) No 2018/1139 and its implementing rules, includes the assessment of compliance with the provisions applicable to the use and functioning of the EAMR.

AMC1 ARA.MED.160(d) Exchange of information on medical certificates

ED Decision 2019/002/R

APPLICANT'S RECORD

Each competent authority should ensure that:

- (a) for each applicant for a class 1 medical certificate, a unique personal record is created in the EAMR, containing the category 1 personal data listed in [ARA.MED.160\(b\)\(1\)](#). This record is referred to as the 'applicant's record';
- (b) the applicant's record is managed in accordance with the applicable regulation (typically for inserting, updating, viewing, validating data, etc.).
- (c) an applicant is granted the right to obtain, without undue delay, the rectification of inaccurate personal data concerning them and, taking into account the purposes of the EAMR, the

applicant is granted the right to have incomplete personal data completed. Such corrections should also be mirrored in the associated records kept in accordance with [ARA.MED.150](#).

- (d) the data recorded in the EAMR is complete as relevant for the purpose of the EAMR as described in [AMC1 ARA.MED.160\(b\)](#).

AMC1 ARA.MED.160(d) Exchange of information on medical certificates

ED Decision 2019/002/R

RECOVERY FROM UNSERVICEABILITY

The competent authority should ensure that class 1 medical certificates issued or amended without being properly recorded in the EAMR, due to unserviceability of the system, are entered in the EAMR without undue delay when the system recovers.

AMC1 ARA.MED.160(h) Exchange of information on medical certificates

ED Decision 2019/002/R

INFORMATION OF APPLICANTS

The competent authority should ensure at least the following:

- (a) At the time of the creation of the applicant's record at the latest, the applicants should be informed:
- (1) that their personal data as listed in [ARA.MED.160\(b\)\(1\)](#) will be lawfully processed in a European central repository, in accordance with Article 72 of Regulation (EU) 2018/1139 and [ARA.GEN.200\(c\)](#) and [ARA.MED.160](#) of Commission Regulation (EU) No 1178/2011.
 - (2) that the purpose of the processing is to verify that the information, as regards their previous medical certificates, provided in their declaration submitted in accordance with MED.A.035(b)(2), is consistent with the records available to all competent authorities in accordance with [ARA.MED.150](#);
 - (3) of the contact details of the data protection officer as applicable;
 - (4) that the period for which the personal data will be stored is determined in accordance with [ARA.MED.160\(g\)](#);
 - (5) of the existence of their right to request access to, and rectification of personal data;
 - (6) of the contact details of the data controller;
 - (7) of their right to lodge a complaint with the competent data protection authority in accordance with the applicable data protection legislation;
 - (8) that it is ensured that access to personal data contained in the EAMR is restricted to authorised personnel in accordance with Commission Regulation (EU) No 1178/2011.
- (b) When applying for a class 1 medical certificate, the applicants should be informed that the category 2 data of their medical certificate, as listed in [ARA.MED.160\(b\)\(2\)](#), will be processed to verify that the information provided in their declaration, as regards their previous medical certificates, is consistent with the information available in the EAMR.

SECTION II – AERO-MEDICAL EXAMINERS (AMEs)

ARA.MED.200 Procedure for the issue, revalidation, renewal or change of an AME certificate

Regulation (EU) 2024/2076

Without prejudice to the provisions laid down in point [ARA.GEN.315](#), all of the following shall apply:

- (a) the competent authority shall ensure that before the issue, revalidation, renewal or extension of privileges of an AME certificate, applicants demonstrate their aero-medical competency in accordance with points [MED.D.030](#) (a)(6) and (b)(5) of Annex IV;
- (b) the competent authority shall have a procedure in place to ensure that, before issuing the AME certificate, it has the evidence that the AME practice is equipped and the appropriate processes are in place to perform aero- medical examinations within the scope of the AME certificate applied for. In the case of multiple AME practice locations, all of them shall be specified on the AME certificate;
- (c) for applicants referred to in point [MED.D.020](#)(aa) of Annex IV, the competent authority may accept an aviation medicine training course completed by an applicant outside the territories for which Member States are responsible under the Chicago Convention, provided that the competent authority has done all of the following:
 - (1) assessed and verified that the course syllabus is equivalent to the aviation medicine training courses available in the Member States;
 - (2) provided to the applicant a specific training module on the aero-medical requirements detailed in Annex IV (Part-MED);
- (d) when satisfied that the AME is in compliance with the applicable requirements, the competent authority shall issue, revalidate, renew or change the AME certificate for a period not exceeding 3 years, using the form established in Appendix VII.

AMC1 ARA.MED.200 Procedure for the issue, revalidation, renewal or change of an AME certificate

ED Decision 2025/002/R

INSPECTION OF THE AME PRACTICE

Upon request for issue, revalidation, renewal or change of an AME certificate, the competent authority should conduct an inspection of the AME practice to verify compliance with point [ARA.MED.200](#).

For applicants for an AME certificate with the privileges of class 2 medical certification only, a virtual inspection of the AME premises may be acceptable. In case of concerns regarding compliance with this Regulation, an on-site inspection should be conducted.

AMC2 ARA.MED.200 Procedure for the issue, revalidation, renewal or change of an AME certificate

ED Decision 2025/002/R

The competent authority should implement a procedure to verify:

- (a) for the initial issue or extension of an AME certificate, evidence of successful completion of an approved aviation medicine training course in accordance with the privileges of the AME certificate applied for;
- (b) for revalidation and renewal of an AME certificate, evidence of refresher training and maintenance of aero-medical competency.

ARA.MED.240 General medical practitioners (GMPs) exercising the privileges in accordance with point MED.A.040 of Annex IV (Part-MED)

Regulation (EU) 2024/2076

The competent authority of a Member State shall notify the Agency and competent authorities of other Member States if aero-medical examinations for the LAPL can be carried out on its territory by GMPs.

ARA.MED.245 Continuing oversight of AMEs and GMPs

Regulation (EU) 2024/2076

When developing the continuing oversight programme referred to in point [ARA.GEN.305](#), the competent authority shall take into account:

- (1) the number of AMEs and GMPs exercising their privileges within the territory where the competent authority exercises oversight;
- (2) the number of AMEs certified by competent authorities of other Member States exercising their privileges within the territory where the competent authority exercises oversight;
- (3) a risk-based assessment of the AMEs' and GMPs' activity.

ARA.MED.246 Cooperative oversight of AMEs and AeMCs

Regulation (EU) 2024/2076

Without prejudice to the provisions laid down in point [ARA.GEN.300](#)(e):

- (a) where an AME or AeMC carries out their activity in more than one Member State, the competent authority that certified the AME or AeMC shall have a procedure in place to ensure the exchange of information in accordance with point [ARA.GEN.200](#)(c) and points [ARA.GEN.300](#)(d) and (e) with the competent authority of the other Member State(s) where the AME or AeMC carries out their activity. The procedure shall be agreed upon by the competent authorities involved;
- (b) in the case mentioned in point (a), the competent authority of the other Member State(s) where the AME or AeMC carries out their activity shall share all information relevant to the oversight of the AME or AeMC with the competent authority certifying the AME or AeMC.

AMC1 ARA.MED.246 Cooperative oversight of AMEs and AeMCs

ED Decision 2025/002/R

The cooperative oversight procedure may include oversight tasks to be undertaken by the competent authority of the Member State where the AME/AeMC has its secondary place of business.

The results of the oversight should be shared among the competent authorities of the Member States involved.

AMC1 ARA.MED.250(a) Limitation, suspension or revocation of an AME certificate

ED Decision 2025/002/R

- (a) The competent authority should consider, as part of the assessment of compliance, the compliance with the applicable implementing rules and acceptable means of compliance, as well as the national procedures in place to implement the respective requirements.
- (b) The competent authority should consider the level of aero-medical competency as one of the criteria for continuing certification.

ARA.MED.250 Limitation, suspension or revocation of an AME certificate

Regulation (EU) 2024/2076

- (a) The competent authority shall limit, suspend or revoke an AME certificate in the following circumstances:
 - (1) the AME does not comply with applicable requirements;
 - (2) failure to meet the criteria for certification or continuing certification;
 - (3) deficiency of aero-medical record-keeping or submission of incorrect data or information;
 - (4) falsification of medical records, certificates or documentation;
 - (5) concealment of facts appertaining to an application for, or holder of, a medical certificate or false or fraudulent statements or representations to the competent authority;
 - (6) failure to correct findings from audit of the AME practice;
 - (7) at the request of the certified AME;
 - (8) any operational context of the AME that may have a direct or indirect negative impact on flight safety.
- (b) The certificate of an AME shall be considered invalid in either of the following circumstances and the competent authority shall immediately revoke it:
 - (1) revocation of medical licence to practice; or
 - (2) removal from the Medical Register.
- (c) The competent authority shall have a process in place for retrieval of the revoked AME certificates, shall update the AME list, and inform the competent authorities of the other Member States accordingly.

ARA.MED.255 Enforcement measures

Regulation (EU) 2024/2076

If, during oversight or by any other means, evidence is found showing a non-compliance of an AeMC, an AME or a GMP, the competent authority shall have a process to review the medical certificates issued by that AeMC, AME or GMP and may render them invalid, where required, to ensure flight safety.

For medical certificates issued to applicants who have a licensing authority different from the competent authority that issued the AME certificate, that competent authority shall inform and exchange relevant information with the medical assessor of the licensing authority of the affected medical certificate holder.

SECTION III – MEDICAL CERTIFICATION

ARA.MED.315 Review of examination reports

Regulation (EU) 2024/2076

The licensing authority shall have a process in place for the medical assessor to take the following steps:

- (a) review examination and assessment reports received from the AeMCs, AMEs and GMPs and inform them of any inconsistencies, mistakes or errors made in the assessment process;
- (aa) take the appropriate corrective actions for any inconsistencies, mistakes or errors identified;
- (b) assist AMEs and AeMCs on their request regarding their decision on aero-medical fitness in borderline and complex cases.

AMC1 ARA.MED.315(a) Review of examination reports

ED Decision 2025/002/R

GENERAL

- (a) The process to review examination and assessment reports received from AeMCs, AMEs and GMPs should aim to check all reports.
- (b) The aero-medical section of the licensing authority should implement a performance assessment process for AMEs to take account of the proportion of inconsistencies or errors found, and adapt the sample size accordingly and consider corrective action(s).
- (c) The aero-medical section of the licensing authority should implement a medical review process of all examination and assessment reports received from AeMCs, AMEs and GMPs certified by the competent authority of another Member State.

ARA.MED.325 Secondary review procedure

Regulation (EU) 2024/2076

The competent authority shall establish a procedure for the review of borderline and complex cases and cases where an applicant requests a review in accordance with the applicable medical requirements and accredited medical conclusion as defined in point [MED.A.010](#) of Annex IV (Part-MED).

AMC1 ARA.MED.325 Secondary review procedure

ED Decision 2025/002/R

- (a) The secondary review procedure should specify:
 - (1) the establishment of a review board and its composition;
 - (2) how potential conflict of interest should be managed;
 - (3) how the accredited medical conclusions of the review board will be implemented.
- (b) The composition of the review board should be decided by the aero-medical section of the licensing authority. It may be preceded by the advice of the medical assessor and may consist of, but not be limited to:

- (1) clinical medical experts according to the case;
- (2) other technical experts according to the case;
- (3) aviation medicine experts;
- (4) AME(s) with privileges according to the class on the medical certificate in question, other than the AME(s) involved in the assessment of the fitness of the applicant.

ARA.MED.330 Special medical circumstances

Regulation (EU) 2024/2076

- (a) When new medical technology, medication or procedures are identified that may justify a fit assessment of applicants otherwise not in compliance with the requirements, research may be carried out to gather evidence on the safe exercise of the privileges of the licence.
- (b) In order to undertake research, a competent authority, in cooperation with at least one other competent authority, may develop and evaluate a medical assessment protocol based on which these competent authorities may issue a defined number of pilot medical certificates with appropriate limitations.
- (c) AeMCs and AMEs may only issue medical certificates on the basis of a research protocol if instructed to do so by the competent authority.
- (d) The protocol shall be agreed between the competent authorities concerned and shall include as a minimum:
 - (1) a risk assessment;
 - (2) a literature review and evaluation to provide evidence that issuing a medical certificate based on the research protocol would not jeopardise the safe exercise of the privileges of the licence;
 - (3) detailed selection criteria for pilots to be admitted to the protocol;
 - (4) the limitations that will be endorsed on the medical certificate;
 - (5) the monitoring procedures to be implemented by the competent authorities concerned;
 - (6) the determination of end points for terminating the protocol.
- (e) The protocol shall be compliant with relevant ethical principles.
- (f) The exercise of licence privileges by licence holders with a medical certificate issued on the basis of the protocol shall be restricted to flights in aircraft registered in the Member States involved in the research protocol. This restriction shall be indicated on the medical certificate.

- (g) The participating competent authorities shall:
- (1) provide the Agency with:
 - (i) the research protocol before implementation;
 - (ii) the details and qualifications of the nominated focal point of each participating competent authority;
 - (iii) documented reports of regular evaluations of its effectiveness;
 - (2) provide the AeMCs and AMEs within their jurisdiction with details of the protocol before implementation for their information.

[applicable until 12 February 2025 - Regulation (EU) 2015/445]

SUBPART DTO – SPECIFIC REQUIREMENTS RELATING TO DECLARED TRAINING ORGANISATIONS (DTOs)

ARA.DTO.100 Declaration to the competent authority

Regulation (EU) 2020/359

- (a) Upon receiving a declaration from a DTO, the competent authority shall verify that the declaration contains all the information specified in point [DTO.GEN.115](#) of Annex VIII (Part-DTO) and acknowledge receipt of the declaration, including the assignment of an individual DTO reference number to the representative of the DTO.
- (b) If the declaration does not contain the required information or contains information that indicates a non-compliance with the essential requirements set out in Annex IV to [Regulation \(EU\) 2018/1139](#), with the requirements of Annex I (Part-FCL) and Annex VIII (Part-DTO) to this Regulation, or with the requirements of Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), the competent authority shall act in accordance with point [ARA.GEN.350\(da\)](#).

AMC1 ARA.DTO.100(a) Declaration to the competent authority

ED Decision 2018/009/R

ACKNOWLEDGEMENT OF RECEIPT OF THE DECLARATION

The competent authority should acknowledge receipt of the declaration to the DTO in writing within 10 working days.

GM1 ARA.DTO.100(a) Declaration to the competent authority

ED Decision 2018/009/R

ASSIGNMENT OF AN INDIVIDUAL DTO REFERENCE NUMBER

It is recommended to create DTO reference numbers by commencing with the UN country code of the State of the competent authority to which the declaration is sent, followed by the term ‘.DTO.’ and a consecutive numbering (example: AT.DTO.001).

GM2 ARA.DTO.100(a) Declaration to the competent authority

ED Decision 2018/009/R

The verification made by the competent authority upon receipt of the declaration does not imply an inspection. The aim is to check whether the declaration complies with the applicable requirements.

ARA.DTO.105 Changes to declarations

Regulation (EU) 2018/1119

Upon receiving a notification of a change to the information contained in the declaration of a DTO, the competent authority shall act in accordance with point [ARA.DTO.100](#).

ARA.DTO.110 Verification of compliance of the training programme

Regulation (EU) 2020/359

- (a) Upon receiving the training programmes of a DTO, and any changes thereto, notified to it in accordance with point [DTO.GEN.115\(c\)](#) of Annex VIII (Part-DTO) or the application for approval of the training programmes of a DTO submitted to it in accordance with point [DTO.GEN.230\(c\)](#) of that Annex, the competent authority shall verify the compliance of those training programmes with the requirements of Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), as applicable.
- (b) When satisfied that the DTO training programme, and any subsequent changes thereto, are in compliance with those requirements, the competent authority shall inform the representative of the DTO thereof in writing or, in the case referred to in point [DTO.GEN.230\(c\)](#) of Annex VIII (Part-DTO), approve the training programme. For such approval it shall use the form contained in [Appendix VIII](#) to this Annex (Part-ARA).
- (c) In case of any non-compliance, the competent authority shall act in accordance with point [ARA.GEN.350\(da\)](#) or, in the case referred to in point [DTO.GEN.230\(c\)](#) of Annex VIII (Part-DTO), reject the application for approval of the training programme.

AMC1 ARA.DTO.110 Verification of compliance of the training programme(s)

ED Decision 2018/009/R

Without prejudice to national provisions on administrative procedures, and unless the training programme has already been verified for Part-FCL compliance (AMC1 DTO.GEN.115(c)), when receiving an initial declaration, the competent authority should verify the compliance of the training programme(s) attached to that declaration within 6 months from the time it acknowledged receipt of the declaration in accordance with point [ARA.DTO.100\(a\)](#).

APPENDICES TO ANNEX VI

Appendix I to ANNEX VI (Part-ARA) – Flight crew licence

Regulation (EU) 2025/134

The flight crew licence issued by a Member State in accordance with Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#) shall conform to the following specifications:

- (a) Content. The item number shown shall always be printed in association with the item heading. Items I to XI are the “permanent” items and items XII to XIV are the “variable” items which may appear on a separate or detachable part of the main form. Any separate or detachable part shall be clearly identifiable as part of the licence.
 - (1) Permanent items:
 - (I) State of licence issue;
 - (II) title of licence;
 - (III) serial number of the licence commencing with the UN country code of the State of licence issue and followed by ‘FCL’, ‘BFCL’ or ‘SFCL’, as applicable, and a code of numbers and/or letters in Arabic numerals and in Latin script;
 - (IV) name of holder (in Latin script, even if the script of the national language(s) is other than Latin);
 - (IVa) date of birth;
 - (V) holder's address;
 - (VI) nationality of holder;
 - (VII) signature of holder;
 - (VIII) competent authority and, where necessary, conditions under which the licence was issued;
 - (IX) certification of validity and authorisation for the privileges granted;
 - (X) signature of the officer issuing the licence and the date of issue; and
 - (XI) seal or stamp of the competent authority.
 - (2) Variable items:
 - (XII) ratings, certificates and, in the case of balloons and sailplanes, privileges: class, type, instructor certificates, etc., with dates of expiry, as applicable. Radio telephony (R/T) privileges may appear on the licence or on a separate certificate;
 - (XIII) remarks: i.e. special endorsements relating to limitations and endorsements for privileges, including endorsements of language proficiency, and remarks on the automatic validation of the licence; and
 - (XIV) any other details required by the competent authority (e.g. place of birth/place of origin).
- (b) Material. The paper or other material used will prevent or readily show any alterations or erasures. Any entries or deletions to the form will be clearly authorised by the competent authority.

- (c) Language. Licences shall be written in the national language(s) and in English and such other languages as the competent authority deems appropriate.

Cover page

Competent Authority name and logo (English and any language(s) determined by the competent authority)	Requirements
EUROPEAN UNION (English only)	
FLIGHT CREW LICENCE (English and any language(s) determined by the competent authority)	"European Union" to be deleted for non-EU Member States
Issued in accordance with Part-FCL/Part-BFCL/Part-SFCL (non-applicable terms to be deleted)	Size of each page shall be one eighth A4
This licence complies with ICAO standards, except for the LAPL and BIR privileges or when accompanied by an LAPL medical certificate <i>[applicable until 17 February 2026 - Regulation (EU) 2024/2076]</i>	
This licence complies with ICAO standards, except for the LAPL, GPL and BIR privileges, or when accompanied by an LAPL medical certificate. <i>[applicable from 18 February 2026 - Regulation (EU) 2025/134]</i>	
(English and any language(s) determined by the competent authority)	
EASA Form 141 Issue 2 <i>[applicable until 17 February 2026 - Regulation (EU) 2024/2076]</i>	
EASA Form 141 Issue 3 <i>[applicable from 18 February 2026 - Regulation (EU) 2025/134]</i>	

Page 2

I	State of issue		Requirements
III	Licence number		Serial number of the licence will always commence with the UN country code of the State of the licence issue, followed by "FCL.", "BFCL." or "SFCL.", as applicable.
IV	Last and first name of holder		
IVa	Date of birth (see instructions)		Standard date format is to be used, dd/mm/yyyy in full.
XIV	Place of birth		
V	Address of holder: Street, town, area, postal code		
VI	Nationality		
VII	Signature of holder		
VIII	Issuing competent authority E.g. This CPL(A) has been issued on the basis of an ATPL issued by (third country)		
X	Signature of issuing officer and date		
XI	Seal or stamp of issuing competent authority		

Page 3

II	Title of the licence, date of initial issue and country code		Abbreviations used will be as those used in Part-FCL (e.g. PPL(H), ATPL(A), etc.), Part-BFCL and Part-SFCL Standard format is to be used, dd/mm/yyyy in full.
IX	Validity: The privileges of the licence shall be exercised only if the holder has a valid medical certificate for the required privilege. A document containing a photo shall be carried for the purposes of identification of the licence holder.		This document is not specified, but a passport would suffice when outside the State of licence issue.
XII	Radiotelephony privileges: The holder of this licence has demonstrated competence to operate R/T equipment on board aircraft in (specify the language(s))		
XIII	Remarks: Language Proficiency: (language(s)/level/validity date)		All additional licensing information required and privileges established by ICAO, EC or EU Directives/Regulations to be entered here. Language proficiency endorsement(s), level and validity date shall be included. In case of LAPL: LAPL not issued in accordance with ICAO standards In case of SPL, except for the cases referred to in Point (a) of Article 3b(2) of Commission Implementing Regulation (EU) 2018/1976 : Privileges for aerobatic and sailplane cloud flying as well as for launching methods to be exercised in accordance with points SFCL.155, SFCL.200 and SFCL.215 of

			Annex III (Part-SFCL) to Commission Implementing Regulation (EU) 2018/1976 , as applicable.
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Additional pages — Requirements:

Pages 1, 2, and 3 of the licence shall be in accordance with the format laid down in the model in this point. The competent authority shall include additional customized pages containing tables which shall contain at least the following information:

- Ratings, certificates, endorsements and privileges;
- Expiry dates of the ratings, the instructor and examiner certificate privileges;
- Dates of the test or check;
- Remarks and restrictions (operational limitations);
- Fields for the examiner and/or instructor certificate number and signature, as applicable;
- Abbreviations.

These additional pages are intended for use by the competent authority, or by specifically authorised instructors or examiners.

Initial issues of ratings or certificates shall be entered by the competent authority. Revalidation or renewal of ratings or certificates may be entered by the competent authority or by specifically authorised instructors or examiners.

Operational limitations shall be entered in “Remarks and Restrictions” against the appropriate restricted privilege, e.g. IR skill test taken with co-pilot, restricted instruction privileges to 1 aircraft type.

Ratings that are not validated may be removed from the licence by the competent authority.

AMC1 to Appendix I to ANNEX VI (Part-ARA) – Flight crew licence

ED Decision 2018/011/R

In case of using privileges outside the Union territory to which the Treaty applies on an aircraft registered in a Member State other than the one that issued the flight crew licence, the following remark should be added to licence item XIII: ‘This licence is automatically rendered valid as per the ICAO attachment to this licence.

Appendix II to ANNEX VI (Part-ARA) – Standard EASA format for cabin crew attestations

Regulation (EU) 2015/445

Cabin crew attestations issued in accordance with Part-CC in a Member State shall conform to the following specifications:

1. CABIN CREW ATTESTATION Issued in accordance with Part-CC 2. Reference number: 3. State of issue: 4. Full name of holder: 5. Date and place of birth: 6. Nationality: 7. Signature of holder: 8. Competent authority: 9. Issuing body: <i>Official seal, Stamp or Logo</i> 10. Signature of issuing officer: 11. Date of issue: 12. The holder may only exercise the privileges to act as cabin crew on aircraft engaged in commercial air transport operations if he/she complies with the requirements in Part-CC for continuous fitness and valid aircraft type qualifications. EASA Form 142 Issue 1
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Instructions:

- (a) The cabin crew attestation shall include all items specified in EASA Form 142 in accordance with items 1 - 12 as listed and described below.
- (b) Size shall be either 105mm × 74mm (one-eighth A4) or 85mm × 54mm, and the material used shall prevent or readily show any alterations or erasures.
- (c) The document shall be printed in English and such other languages as the competent authority deems appropriate.
- (d) The document shall be issued by the competent authority or by an organisation approved to issue cabin crew attestations. In that latter case reference to the approval by the competent authority of the Member State shall be stated.
- (e) The cabin crew attestation is recognised in all Member States and it is not necessary to exchange the document when working in another Member State.

Item 1: The title "CABIN CREW ATTESTATION" and the reference to Part-CC.

Item 2: Attestation reference number shall commence with the UN country code of the Member State followed by at least the two last numbers of the year of issue and an individual reference/number according to a code established by the competent authority (e.g. BE-08-xxxx).

Item 3: The Member State where the attestation is issued.

Item 4: The full name (surname and first name) stated in the official identity document of the holder.

Items 5 and 6: Date and place of birth as well as nationality as stated in the official identity document of the holder.

Item 7: The signature of the holder.

Item 8: Identification details of the competent authority of the Member State where the attestation is issued shall be entered and shall provide the full name of the competent authority, postal address, and official seal, stamp or logo as applicable.

Item 9: If the competent authority is the issuing body, the term “competent authority” and official seal, stamp or logo shall be entered. In this case only, the competent authority may determine if its official seal, stamp or logo shall also be entered under Item 8.

In the case of an approved organisation, identification details shall be entered and shall at least provide the full name of the organisation, postal address and if applicable, the logo and:

- (a) in the case of a commercial air transport operator, the air operator certificate (AOC) number and detailed reference to the approvals by the competent authority to provide cabin crew training and to issue attestations; or
- (b) in the case of an approved training organisation, the reference number of the relevant approval by the competent authority.

Item 10: The signature of the officer acting on behalf of the issuing body.

Item 11: Standard date format shall be used: i.e. day/month/year in full (e.g. 22/02/2008).

Item 12: The same sentence in English and its full and precise translation into such other languages as the competent authority deems appropriate.

Appendix III to ANNEX VI (Part-ARA) – Certificate for approved training organisations (ATOs)

Regulation (EU) 2024/2076

European Union (*)
Competent Authority

APPROVED TRAINING ORGANISATION CERTIFICATE

[CERTIFICATE NUMBER/REFERENCE]

Pursuant to [Commission Regulation \(EU\) No 1178/2011](#) [and [Commission Regulation \(EU\) 2018/395/Commission Implementing Regulation \(EU\) 2018/1976](#) (ADJUST AS APPLICABLE)] and subject to the conditions specified below, the [Competent Authority] hereby certifies

[NAME OF THE TRAINING ORGANISATION]

[ADDRESS OF THE TRAINING ORGANISATION]

as a Part-ORA certified training organisation with the privilege to provide Part-FCL training courses, including the use of FSTDs, as listed in the attached training course approval/Part-BFCL training courses/Part-SFCL training courses [ADJUST AS APPLICABLE].

CONDITIONS:

This certificate is limited to the privileges and the scope of providing the training courses, including the use of FSTDs, as listed in the attached training course approval.

This certificate is valid whilst the approved organisation remains in compliance with Part-ORA, Part-FCL, Part-BFCL, Part- SFCL [ADJUST AS APPLICABLE] and other applicable regulations.

Subject to compliance with the foregoing conditions, this certificate shall remain valid unless it has been surrendered, superseded, limited, suspended or revoked.

Date of issue:

Signed:

[Competent Authority]

(*) 'European Union' to be deleted for non-EU Member States or EASA.

APPROVED TRAINING ORGANISATION CERTIFICATE**TRAINING COURSE APPROVAL**

Attachment to ATO Certificate Number:

[CERTIFICATE NUMBER/REFERENCE]

[NAME OF THE TRAINING ORGANISATION]

has obtained the privilege to provide and conduct the following Part-FCL/Part-BFCL/Part-SFCL [ADJUST AS APPLICABLE] training courses and to use the following FSTDs:

Training course	FSTD(s) used, including letter code ⁽¹⁾
⁽¹⁾ as indicated on the qualification certificate	

This training course approval is valid as long as:

- (a) the ATO certificate has not been surrendered, superseded, limited, suspended or revoked; and
- (b) all operations are conducted in compliance with Part-ORA, Part-FCL, Part-BFCL, Part-SFCL [ADJUST AS APPLICABLE], other applicable regulations, and, when relevant, with the procedures in the organisation's documentation as required by Part-ORA.

Date of issue:

Signed: [Competent Authority]

For the Member State/EASA

EASA FORM 143 Issue 3 – page 2/2

Appendix IV to ANNEX VI (Part-ARA) – Flight simulation training device qualification certificate

Regulation (EU) 2024/2076

Introduction

EASA Form 145 shall be used for the FSTD qualification certificate. This document shall contain the FSTD Specification including any limitation(s) and special authorisation(s) or approval(s) as appropriate to the FSTD concerned. The qualification certificate shall be printed in English and in any other language(s) determined by the competent authority.

Convertible FSTDs shall have a separate qualification certificate for each aircraft type. Different engine and equipment fit on one FSTD shall not require separate qualification certificates. All qualification certificates shall carry a serial number prefixed by a code in letters, which shall be specific to that FSTD. The letter code shall be specific to the competent authority of issue.

European Union (*)**Competent Authority****FLIGHT SIMULATION TRAINING DEVICE QUALIFICATION CERTIFICATE**

REFERENCE:

Pursuant to Commission Regulation (EU) No 1178/2011 and subject to the conditions specified below, the [competent authority] hereby certifies that

FSTD [TYPE AND LETTER CODE]

located at [NAME and ADDRESS OF THE ORGANISATION]

has satisfied the qualification requirements prescribed in Part-OR, subject to the conditions of the attached FSTD specification

This qualification certificate shall remain valid subject to the FSTD and the holder of the qualification certificate remaining in compliance with the applicable requirements of Part-OR, unless it has been surrendered, superseded, suspended or revoked.

Date of issue:

Signed:

(*) 'European Union' to be deleted for non-EU Member States or EASA.

EASA Form 145 – Issue 2 – page 1/2

[competent authority]

FSTD QUALIFICATION CERTIFICATE: [Reference]

FSTD SPECIFICATIONS

- A. Type or variant of aircraft:
- B. FSTD qualification level:
- C. Primary reference document:
- D. Visual system:
- E. Motion system:
- F. Engine fit:
- G. Instrument fit:
- H. ACAS fit:
- I. Windshear:
- J. Additional capabilities:
- K. Restrictions or limitations:

L. Guidance information for training, testing and checking considerations

CAT I	RVR	m	DH	ft	
CAT II	RVR	m	DH	ft	
CAT III	RVR	m	DH	ft	
(lowest minimum)					
LVTO	RVR	m			
Recency					
IFR-training/check					
					/
Type rating					
Proficiency checks					
Autocoupled approach					
Autoland/roll out guidance					
					/
ACAS I/II					
					/
Windshear warning system/predictive windshear					
					/
WX-radar					
HUD/HUGS					
					/
FANS					
GPWS/EGPWS					
					/
ETOPS capability					
GPS					
Other					

Date of issue:

Signed:

For the Member State/EASA

Appendix V to ANNEX VI (Part-ARA) – Certificate for Aeromedical Centres (AeMCs)

Regulation (EU) 2024/2076

CERTIFICATE FOR AERO-MEDICAL CENTRES (AeMCs)

European Union ⁽¹⁾**Competent Authority**

AERO-MEDICAL CENTRE CERTIFICATE

REFERENCE:

Pursuant to Commission Regulation (EU) No 1178/2011 and Regulation (EU) 2015/340 ⁽²⁾ and subject to the conditions specified below, the [competent authority] hereby certifies

[NAME OF THE ORGANISATION]

[ADDRESS OF THE ORGANISATION]

as a Part-ORA certified Aero-medical centre with the privileges and the scope of activities as listed in the attached terms of approval.

CONDITIONS:

1. This certificate is limited to that specified in the scope of approval section of the approved organisation manual;
2. This certificate requires compliance with the procedures specified in the organisation documentation as required by Part- ORA.
3. This certificate shall remain valid subject to compliance with the requirements of Part-ORA unless it has been surrendered, superseded, suspended or revoked.

Date of issue: dd/mm/yyyy:

Signature: [Competent Authority]:

¹ 'European Union' to be deleted for non-EU Member States or EASA.

² Reference to Regulation (EU) No 1178/2011 or Regulation (EU) 2015/340 shall be modified as per the certificate's scope.

EASA Form 146 Issue 2

AERO-MEDICAL CENTRE CERTIFICATE

Attachment ⁽¹⁾ to AeMC certificate number:

PRIVILEGES AND SCOPE

[Name of the organisation] has obtained the privilege(s) to undertake aero-medical examinations and assessments for the issuance of medical certificates and medical reports as stated in the table below and to issue these medical certificates and medical reports for:

	Initial/revalidation/renewal	Date of issue
Class 1		
Class 2/LAPL/Cabin Crew		
Class 3 ⁽²⁾		

Date of issue: dd/mm/yyyy:

Signature: [Competent Authority]:

¹ Class 3 shall be added only for the AeMCs approved to perform class 3 aero-medical examinations.

² Class 3 shall be added only for the AeMCs approved to perform class 3 aero-medical examinations.

Appendix VI to ANNEX VI (Part-ARA)

Regulation (EU) No 245/2014

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Appendix VII to ANNEX VI (Part-ARA) – Certificate for Aeromedical Examiners (AMEs)

Regulation (EU) 2024/2076

CERTIFICATE FOR AERO-MEDICAL EXAMINERS (AMEs)

European Union ⁽¹⁾**Competent Authority**

AERO-MEDICAL EXAMINER CERTIFICATE

CERTIFICATE NUMBER/REFERENCE:

Pursuant to Commission Regulation (EU) No 1178/2011 and Regulation (EU) 2015/340 ⁽²⁾ and subject to the conditions specified below, the [competent authority] hereby certifies

[NAME OF THE AERO-MEDICAL EXAMINER]

[PRACTICE ADDRESS(ES) OF THE AERO-MEDICAL EXAMINER]

as aero-medical examiner

CONDITIONS:

1. This certificate is limited to the privileges specified in the attachment to this AME certificate;
2. This certificate requires compliance with the implementing rules and procedures specified in Part-MED/Part ATCO.MED ⁽³⁾.
3. This certificate shall remain valid from [dd/mm/yyyy] until [dd/mm/yyyy ⁽⁴⁾] subject to compliance with the requirements of Part-MED/Part ATCO.MED (2) unless it has been surrendered, superseded, suspended or revoked.

Date of issue: dd/mm/yyyy

Signature: [Competent Authority]

¹ 'European Union' to be deleted for non-EU Member States

² Reference to Regulation (EU) No 1178/2011 or Regulation (EU) 2015/340 as well as Part-MED and Part ATCO.MED shall be modified as per the certificate's scope.

³ Reference to Regulation (EU) No 1178/2011 or Regulation (EU) 2015/340 as well as Part-MED and Part ATCO.MED shall be modified as per the certificate's scope.

⁴ Expiry date format: day/month/year.

EASA Form 148 Issue 2

AERO-MEDICAL EXAMINER CERTIFICATE

Attachment ⁽¹⁾ to AME certificate number:

PRIVILEGES AND SCOPE

[Name and academic title of the aero-medical examiner] has obtained the privilege(s) to undertake aero-medical examinations and assessments for the issuance of medical certificates and medical reports as stated in the table below and to issue these medical certificates and medical reports for:

Class 1 revalidation/renewal	[valid until]/[Not Applicable]
Class 2/LAPL/Cabin crew Initial/revalidation/renewal	[valid until]
Class 3 ⁽²⁾ revalidation/renewal	[valid until]/[Not Applicable]

Date of issue: dd/mm/yyyy

Signature: [Competent Authority]

¹ This attachment may be issued as part of the AME certificate or as a separate document.

² Class 3 shall be added only for the AMEs approved to perform class 3 aero-medical examinations.

Appendix VIII to ANNEX VI (Part-ARA) – Training programme approval

Regulation (EU) 2020/2193

Training programme approval

for a declared training organisation (DTO)

European Union (*)

Competent authority

Issuing authority:		
Name of DTO:		
DTO reference number:		
Training programme(s) approved:	Doc reference:	Remarks:
Examiner standardisation – FE(S), FE(B) (**)		
Examiner refresher course – FE(S), FE(B) (**)		
The above-mentioned training programme(s) has (have) been verified by the above-mentioned competent authority and found to be in compliance with the requirements of Annex I (Part-FCL) to Commission Regulation (EU) No 1178/2011 , Annex III (Part-BFCL) to Commission Regulation (EU) 2018/395 and Annex III (Part-SFCL) to Commission Implementing Regulation (EU) 2018/1976 .		
Date of issue:		
Signed: [competent authority]		

(*) 'European Union' to be deleted for non-EU Member States.

(**) To be adjusted as applicable.

ANNEX VII (PART-ORA)

SUBPART GEN – GENERAL REQUIREMENTS

SECTION I – GENERAL

ORA.GEN.105 Competent authority

Regulation (EU) No 1178/2011

- (a) For the purpose of this Part, the competent authority exercising oversight over:
- (1) organisations subject to a certification obligation shall be:
 - (i) for organisations having their principal place of business in a Member State, the authority designated by that Member State;
 - (ii) for organisations having their principal place of business located in a third country, the Agency;
 - (2) FSTDs shall be:
 - (i) the Agency, for FSTDs:
 - located outside the territory of the Member States, or,
 - located within the territory of the Member States and operated by organisations having their principal place of business located in a third country,
 - (ii) for FSTDs located within the territory of the Member States and operated by organisations having their principal place of business in a Member State, the authority designated by the Member State where the organisation operating it has its principle place of business, or the Agency, ifso requested by the Member State concerned.
- (b) When the FSTD located outside the territory of the Member States is operated by an organisation certified by a Member State, the Agency shall qualify this FSTD in coordination with the Member State that has certified the organisation that operates such FSTD.

ORA.GEN.115 Application for an organisation certificate

Regulation (EU) No 1178/2011

- (a) The application for an organisation certificate or an amendment to an existing certificate shall be made in a form and manner established by the competent authority,taking into account the applicable requirements of Regulation (EC) No 216/2008 and its Implementing Rules.
- (b) Applicants for an initial certificate shall provide the competent authority with documentation demonstrating how they will comply with the requirements established in Regulation (EC) No 216/2008 and its Implementing Rules. Such documentation shall include a procedure describing how changes not requiring prior approval will be managed and notified to the competent authority.

ORA.GEN.120 Means of compliance

Regulation (EU) No 290/2012

- (a) Alternative means of compliance to the AMC adopted by the Agency may be used by an organisation to establish compliance with Regulation (EC) No 216/2008 and its Implementing Rules.
- (b) When an organisation wishes to use an alternative means of compliance, it shall, prior to implementing it, provide the competent authority with a full description of the alternative means of compliance. The description shall include any revisions to manuals or procedures that may be relevant, as well as an assessment demonstrating that Regulation (EC) No 216/2008 and its Implementing Rules are met.

The organisation may implement these alternative means of compliance subject to prior approval by the competent authority and upon receipt of the notification as prescribed in ARA.GEN.120(d).

AMC1 ORA.GEN.120(a) Means of compliance

ED Decision 2012/007/R

DEMONSTRATION OF COMPLIANCE

In order to demonstrate that the Implementing Rules are met, a risk assessment should be completed and documented. The result of this risk assessment should demonstrate that an equivalent level of safety to that established by the Acceptable Means of Compliance (AMC) adopted by the Agency is reached.

ORA.GEN.125 Terms of approval and privileges of an organisation

Regulation (EU) No 1178/2011

A certified organisation shall comply with the scope and privileges defined in the terms of approval attached to the organisation's certificate.

AMC1 ORA.GEN.125 Terms of approval and privileges of an organisation

ED Decision 2012/007/R

MANAGEMENT SYSTEM DOCUMENTATION

The management system documentation should contain the privileges and detailed scope of activities for which the organisation is certified, as relevant to the applicable requirements. The scope of activities defined in the management system documentation should be consistent with the terms of approval.

ORA.GEN.130 Changes to organisations

Regulation (EU) No 1178/2011

- (a) Any change affecting:
 - (1) the scope of the certificate or the terms of approval of an organisation; or
 - (2) any of the elements of the organisation's management system as required in [ORA.GEN.200\(a\)\(1\) and \(a\)\(2\)](#),

shall require prior approval by the competent authority.

- (b) For any changes requiring prior approval in accordance with Regulation (EC) No 216/2008 and its Implementing Rules, the organisation shall apply for and obtain an approval issued by the competent authority. The application shall be submitted before any such change takes place, in order to enable the competent authority to determine continued compliance with Regulation (EC) No 216/2008 and its Implementing Rules and to amend, if necessary, the organisation certificate and related terms of approval attached to it.

The organisation shall provide the competent authority with any relevant documentation.

The change shall only be implemented upon receipt of formal approval by the competent authority in accordance with ARA.GEN.330.

The organisation shall operate under the conditions prescribed by the competent authority during such changes, as applicable.

- (c) All changes not requiring prior approval shall be managed and notified to the competent authority as defined in the procedure approved by the competent authority in accordance with ARA.GEN.310(c).

AMC1 ORA.GEN.130 Changes to organisations

ED Decision 2012/007/R

APPLICATION TIME FRAMES

- (a) The application for the amendment of an organisation certificate should be submitted at least 30 days before the date of the intended changes.
- (b) In the case of a planned change of a nominated person, the organisation should inform the competent authority at least 10 days before the date of the proposed change.
- (c) Unforeseen changes should be notified at the earliest opportunity, in order to enable the competent authority to determine continued compliance with the applicable requirements and to amend, if necessary, the organisation certificate and related terms of approval.

GM1 ORA.GEN.130(a) Changes to organisations

ED Decision 2017/022/R

GENERAL

- (a) Typical examples of changes requiring prior approval which may affect the certificate or the terms of approval are listed below:
 - (1) the name of the organisation;
 - (2) the organisation's principal place of business;
 - (3) the organisation's scope of activities;
 - (4) additional locations of the organisation;
 - (5) the accountable manager;
 - (6) any of the persons referred to in [ORA.GEN.210\(a\) and \(b\)](#);
 - (7) the organisation's documentation as required by this Part, safety policy and procedures;
 - (8) the facilities.

- (b) Prior approval by the competent authority is required for any changes to the organisation's procedure describing how changes not requiring prior approval will be managed and notified to the competent authority.
- (c) Changes requiring prior approval may only be implemented upon receipt of formal approval by the competent authority.

GM2 ORA.GEN.130(a) Changes to organisations

ED Decision 2012/007/R

CHANGE OF NAME OF THE ORGANISATION

A change of name requires the organisation to submit a new application as a matter of urgency.

Where this is the only change to report, the new application can be accompanied by a copy of the documentation previously submitted to the competent authority under the previous name, as a means of demonstrating how the organisation complies with the applicable requirements.

GM1 ORA.GEN.130(c) Changes to organisations

ED Decision 2017/022/R

GENERAL

Typical examples of changes not requiring prior approval are to the following items:

- (a) medical equipment (e.g. electrocardiograph (ECG), ophthalmoscope);
- (b) flight simulation training device (FSTD) operator's technical personnel;
- (c) change in schedule of preventive maintenance; and
- (d) list of instructors.

It is recommended that all information on changes not requiring prior approval be included as annexes to the approved training organisation (ATO)'s, FSTD operator's, as well as aeromedical centre's documentation

ORA.GEN.135 Continued validity

Regulation (EU) No 1178/2011

- (a) The organisation's certificate shall remain valid subject to:
 - (1) the organisation remaining in compliance with the relevant requirements of Regulation (EC) No 216/2008 and its Implementing Rules, taking into account the provisions related to the handling of findings as specified under [ORA.GEN.150](#);
 - (2) the competent authority being granted access to the organisation as defined in [ORA.GEN.140](#) to determine continued compliance with the relevant requirements of Regulation (EC) No 216/2008 and its Implementing Rules; and
 - (3) the certificate not being surrendered or revoked.
- (b) Upon revocation or surrender the certificate shall be returned to the competent authority without delay.

ORA.GEN.140 Access

Regulation (EU) No 290/2012

For the purpose of determining compliance with the relevant requirements of Regulation (EC) No 216/2008 and its Implementing Rules, the organisation shall grant access to any facility, aircraft, document, records, data, procedures or any other material relevant to its activity subject to certification, whether it is contracted or not, to any person authorised by:

- (a) the competent authority defined in [ORA.GEN.105](#); or
- (b) the authority acting under the provisions of ARA.GEN.300(d), ARA.GEN.300(e) or ARO.RAMP.

ORA.GEN.150 Findings

Regulation (EU) No 1178/2011

After receipt of notification of findings, the organisation shall:

- (a) identify the root cause of the non-compliance;
- (b) define a corrective action plan; and
- (c) demonstrate corrective action implementation to the satisfaction of the competent authority within a period agreed with that authority as defined in ARA.GEN.350(d).

AMC1 ORA.GEN.150(b) Findings

ED Decision 2012/007/R

GENERAL

The corrective action plan defined by the organisation should address the effects of the non-conformity, as well as its root-cause.

GM1 ORA.GEN.150 Findings

ED Decision 2012/007/R

GENERAL

- (a) Corrective action is the action to eliminate or mitigate the root cause(s) and prevent recurrence of an existing detected non-compliance or other undesirable condition or situation.
- (b) Proper determination of the root cause is crucial for defining effective corrective actions.

ORA.GEN.155 Immediate reaction to a safety problem

Regulation (EU) No 1178/2011

The organisation shall implement:

- (a) any safety measures mandated by the competent authority in accordance with ARA.GEN.135(c); and
- (b) any relevant mandatory safety information issued by the Agency, including airworthiness directives.

ORA.GEN.160 Occurrence reporting

Regulation (EU) No 70/2014

- (a) As part of its management system, the organisation shall establish and maintain an occurrence-reporting system, including mandatory and voluntary reporting. For organisations having their principal place of business in a Member State, that system shall meet the requirements of [Regulation \(EU\) No 376/2014](#) and [Regulation \(EU\) 2018/1139](#) as well as the delegated and implementing acts adopted on the basis of those Regulations.
- (b) The organisation shall report to the competent authority and, in case of aircraft not registered in a Member State, the State of Registry any safety-related event or condition that endangers or, if not corrected or addressed, could endanger an aircraft, its occupants or any other person, and in particular any accident or serious incident.
- (c) Without prejudice to point (b), the organisation shall report to the competent authority and the design approval holder of the aircraft any incident, malfunction, technical defect, exceeding of technical limitations, occurrence that would highlight inaccurate, incomplete or ambiguous information, contained in data established in accordance with [Regulation \(EU\) No 748/2012](#), or other irregular circumstance that has or may have endangered an aircraft, its occupants or any other person and has not resulted in an accident or serious incident.
- (d) Without prejudice to [Regulation \(EU\) No 376/2014](#) and the delegated and implementing acts adopted on the basis thereof, reports in accordance with point (c) shall:
 - (1) be made as soon as practicable, but in any case no later than 72 hours after the organisation has identified the event or condition to which the report relates unless exceptional circumstances prevent this;
 - (2) be made in a form and manner established by the competent authority, as defined in point [ORA.GEN.105](#);
 - (3) contain all pertinent information about the condition known to the organisation.
- (e) For organisations not having their principal place of business in a Member State:
 - (1) initial mandatory reports shall:
 - (i) appropriately safeguard the confidentiality of the identity of the reporter and of the persons mentioned in the report;
 - (ii) be made as soon as practicable, but in any case, no later than 72 hours after the organisation has become aware of the occurrence unless exceptional circumstances prevent this;
 - (iii) be made in a form and manner established by the Agency;
 - (iv) contain all pertinent information about the condition known to the organisation;
 - (2) where relevant, a follow-up report providing details of actions the organisation intends to take to prevent similar occurrences in the future shall be made as soon as those actions have been identified; those follow-up reports shall:
 - (i) be sent to relevant entities initially reported to in accordance with points (b) and (c);
 - (ii) be made in a form and manner established by the Agency.

AMC1 ORA.GEN.160 Occurrence reporting system

ED Decision 2025/002/R

GENERAL

In addition to occurrence reporting in accordance with Regulation (EU) No 376/2014 the organization should report all occurrences defined in AMC 20-8.

SECTION II – MANAGEMENT

ORA.GEN.200 Management system

Regulation (EU) 2020/2193

- (a) The organisation shall establish, implement and maintain a management system that includes:
- (1) clearly defined lines of responsibility and accountability throughout the organisation, including a direct safety accountability of the accountable manager;
 - (2) a description of the overall philosophies and principles of the organisation with regard to safety, referred to as the safety policy;
 - (3) the identification of aviation safety hazards entailed by the activities of the organisation, their evaluation and the management of associated risks, including taking actions to mitigate the risk and verify their effectiveness;
 - (4) maintaining personnel trained and competent to perform their tasks;
 - (5) documentation of all management system key processes, including a process for making personnel aware of their responsibilities and the procedure for amending this documentation;
 - (6) a function to monitor compliance of the organisation with the relevant requirements. Compliance monitoring shall include a feedback system of findings to the accountable manager to ensure effective implementation of corrective actions as necessary; and
 - (7) any additional relevant requirements prescribed in [Regulation \(EU\) 2018/1139](#) and in [Regulation \(EU\) No 376/2014](#) as well as in the delegated and implementing acts adopted on the basis thereof.
- (b) The management system shall correspond to the size of the organisation and the nature and complexity of its activities, taking into account the hazards and associated risks inherent in these activities.
- (c) Notwithstanding point (a), in an organisation providing training only for the LAPL, PPL, SPL or BPL and the associated ratings or certificates, safety risk management and compliance monitoring defined in points (a)(3) and (a)(6) may be accomplished by an organisational review, to be performed at least once every calendar year. The competent authority shall be notified about the results of this review by the organisation without undue delay.

AMC1 ORA.GEN.200(a)(1);(2);(3);(5) Management system

ED Decision 2012/007/R

NON-COMPLEX ORGANISATIONS - GENERAL

- (a) Safety risk management may be performed using hazard checklists or similar risk management tools or processes, which are integrated into the activities of the organisation.
- (b) The organisation should manage safety risks related to a change. The management of change should be a documented process to identify external and internal change that may have an adverse effect on safety. It should make use of the organisation's existing hazard identification, risk assessment and mitigation processes.

- (c) The organisation should identify a person who fulfils the role of safety manager and who is responsible for coordinating the safety management system. This person may be the accountable manager or a person with an operational role in the organisation.
- (d) Within the organisation, responsibilities should be identified for hazard identification, risk assessment and mitigation.
- (e) The safety policy should include a commitment to improve towards the highest safety standards, comply with all applicable legal requirements, meet all applicable standards, consider best practices and provide appropriate resources.
- (f) The organisation should, in cooperation with other stakeholders, develop, coordinate and maintain an emergency response plan (ERP) that ensures orderly and safe transition from normal to emergency operations and return to normal operations. The ERP should provide the actions to be taken by the organisation or specified individuals in an emergency and reflect the size, nature and complexity of the activities performed by the organisation.

AMC1 ORA.GEN.200(a)(1) Management system

ED Decision 2012/007/R

COMPLEX ORGANISATIONS - ORGANISATION AND ACCOUNTABILITIES

The management system of an organisation should encompass safety by including a safety manager and a safety review board in the organisational structure.

- (a) Safety manager
 - (1) The safety manager should act as the focal point and be responsible for the development, administration and maintenance of an effective safety management system.
 - (2) The functions of the safety manager should be to:
 - (i) facilitate hazard identification, risk analysis and management;
 - (ii) monitor the implementation of actions taken to mitigate risks, as listed in the safety action plan;
 - (iii) provide periodic reports on safety performance;
 - (iv) ensure maintenance of safety management documentation;
 - (v) ensure that there is safety management training available and that it meets acceptable standards;
 - (vi) provide advice on safety matters; and
 - (vii) ensure initiation and follow-up of internal occurrence / accident investigations.
- (b) Safety review board
 - (1) The Safety review board should be a high level committee that considers matters of strategic safety in support of the accountable manager's safety accountability.
 - (2) The board should be chaired by the accountable manager and be composed of heads of functional areas.
 - (3) The safety review board should monitor:
 - (i) safety performance against the safety policy and objectives;
 - (ii) that any safety action is taken in a timely manner; and

- (iii) the effectiveness of the organisation's safety management processes.
- (c) The safety review board should ensure that appropriate resources are allocated to achieve the established safety performance.
- (d) The safety manager or any other relevant person may attend, as appropriate, safety review board meetings. He/she may communicate to the accountable manager all information, as necessary, to allow decision making based on safety data.

GM1 ORA.GEN.200(a)(1) Management system

ED Decision 2012/007/R

SAFETY MANAGER

- (a) Depending on the size of the organisation and the nature and complexity of its activities, the safety manager may be assisted by additional safety personnel for the performance of all safety management related tasks.
- (b) Regardless of the organisational set-up it is important that the safety manager remains the unique focal point as regards the development, administration and maintenance of the organisation's safety management system.

GM2 ORA.GEN.200(a)(1) Management system

ED Decision 2012/007/R

COMPLEX ORGANISATIONS - SAFETY ACTION GROUP

- (a) A safety action group may be established as a standing group or as an ad-hoc group to assist or act on behalf of the safety review board.
- (b) More than one safety action group may be established depending on the scope of the task and specific expertise required.
- (c) The safety action group should report to and take strategic direction from the safety review board and should be comprised of managers, supervisors and personnel from operational areas.
- (d) The safety action group should:
 - (1) monitor operational safety;
 - (2) resolve identified risks;
 - (3) assess the impact on safety of operational changes; and
 - (4) ensure that safety actions are implemented within agreed timescales.
- (e) The safety action group should review the effectiveness of previous safety recommendations and safety promotion.

AMC1 ORA.GEN.200(a)(2) Management system

ED Decision 2012/007/R

COMPLEX ORGANISATIONS - SAFETY POLICY

- (a) The safety policy should:
 - (1) be endorsed by the accountable manager;

- (2) reflect organisational commitments regarding safety and its proactive and systematic management;
 - (3) be communicated, with visible endorsement, throughout the organisation; and
 - (4) include safety reporting principles.
- (b) The safety policy should include a commitment:
- (1) to improve towards the highest safety standards;
 - (2) to comply with all applicable legislation, meet all applicable standards and consider best practices;
 - (3) to provide appropriate resources;
 - (4) to enforce safety as one primary responsibility of all managers; and
 - (5) not to blame someone for reporting something which would not have been otherwise detected.
- (c) Senior management should:
- (1) continually promote the safety policy to all personnel and demonstrate their commitment to it;
 - (2) provide necessary human and financial resources for its implementation; and
 - (3) establish safety objectives and performance standards.

GM1 ORA.GEN.200(a)(2) Management system

ED Decision 2012/007/R

SAFETY POLICY

The safety policy is the means whereby the organisation states its intention to maintain and, where practicable, improve safety levels in all its activities and to minimise its contribution to the risk of an aircraft accident as far as is reasonably practicable.

The safety policy should state that the purpose of safety reporting and internal investigations is to improve safety, not to apportion blame to individuals.

AMC1 ORA.GEN.200(a)(3) Management system

ED Decision 2012/007/R

COMPLEX ORGANISATIONS - SAFETY RISK MANAGEMENT

- (a) Hazard identification processes
- (1) Reactive and proactive schemes for hazard identification should be the formal means of collecting, recording, analysing, acting on and generating feedback about hazards and the associated risks that affect the safety of the operational activities of the organisation.
 - (2) All reporting systems, including confidential reporting schemes, should include an effective feedback process.
- (b) Risk assessment and mitigation processes
- (1) A formal risk management process should be developed and maintained that ensures analysis (in terms of likelihood and severity of occurrence), assessment (in terms of tolerability) and control (in terms of mitigation) of risks to an acceptable level.

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- (2) The levels of management who have the authority to make decisions regarding the tolerability of safety risks, in accordance with (b)(1), should be specified.
 - (c) Internal safety investigation
 - (1) The scope of internal safety investigations should extend beyond the scope of occurrences required to be reported to the competent authority.
 - (d) Safety performance monitoring and measurement
 - (1) Safety performance monitoring and measurement should be the process by which the safety performance of the organisation is verified in comparison to the safety policy and objectives.
 - (2) This process should include:
 - (i) safety reporting;
 - (ii) safety studies, that is, rather large analyses encompassing broad safety concerns;
 - (iii) safety reviews including trends reviews, which would be conducted during introduction and deployment of new technologies, change or implementation of procedures, or in situations of structural change in operations;
 - (iv) safety audits focussing on the integrity of the organisation's management system, and periodically assessing the status of safety risk controls; and
 - (v) safety surveys, examining particular elements or procedures of a specific operation, such as problem areas or bottlenecks in daily operations, perceptions and opinions of operational personnel and areas of dissent or confusion.
 - (e) The management of change

The organisation should manage safety risks related to a change. The management of change should be a documented process to identify external and internal change that may have an adverse effect on safety. It should make use of the organisation's existing hazard identification, risk assessment and mitigation processes.
 - (f) Continuous improvement

The organisation should continuously seek to improve its safety performance. Continuous improvement should be achieved through:

 - (1) proactive and reactive evaluations of facilities, equipment, documentation and procedures through safety audits and surveys;
 - (2) proactive evaluation of individuals' performance to verify the fulfilment of their safety responsibilities; and
 - (3) reactive evaluations in order to verify the effectiveness of the system for control and mitigation of risk.
 - (g) The emergency response plan (ERP)
 - (1) An ERP should be established that provides the actions to be taken by the organisation or specified individuals in an emergency. The ERP should reflect the size, nature and complexity of the activities performed by the organisation.
 - (2) The ERP should ensure:
 - (i) an orderly and safe transition from normal to emergency operations;

- (ii) safe continuation of operations or return to normal operations as soon as practicable; and
- (iii) coordination with the emergency response plans of other organisations, where appropriate.

GM1 ORA.GEN.200(a)(3) Management system

ED Decision 2012/007/R

INTERNAL OCCURRENCE REPORTING SCHEME

- (a) The overall purpose of the scheme is to use reported information to improve the level of safety performance of the organisation and not to attribute blame.
- (b) The objectives of the scheme are to:
 - (1) enable an assessment to be made of the safety implications of each relevant incident and accident, including previous similar occurrences, so that any necessary action can be initiated; and
 - (2) ensure that knowledge of relevant incidents and accidents is disseminated, so that other persons and organisations may learn from them.
- (c) The scheme is an essential part of the overall monitoring function and it is complementary to the normal day-to-day procedures and 'control' systems and is not intended to duplicate or supersede any of them. The scheme is a tool to identify those instances where routine procedures have failed.
- (d) All occurrence reports judged reportable by the person submitting the report should be retained as the significance of such reports may only become obvious at a later date.

GM3 ORA.GEN.200(a)(3) Management system

ED Decision 2013/008/R

APPROVED TRAINING ORGANISATIONS - RISK MANAGEMENT OF FLIGHT OPERATIONS WITH KNOWN OR FORECAST VOLCANIC ASH CONTAMINATION

- (a) Responsibilities

The ATO is responsible for the safety of its operations, including within an area with known or forecast volcanic ash contamination.

The ATO should complete this assessment of safety risks related to known or forecast volcanic ash contamination as part of its management system before initiating operations into airspace forecast to be or aerodromes/operating sites known to be contaminated with volcanic ash.

This process is intended to ensure the ATO takes into account the likely accuracy and quality of the information sources it uses in its management system and to demonstrate its own competence and capability to interpret data from different sources in order to achieve the necessary level of data integrity reliably and correctly resolve any conflicts among data sources that may arise.

In order to decide whether or not to operate into airspace forecast to be or aerodromes/operating sites known to be contaminated with volcanic ash, the ATO should make use of the safety risk assessment within its management system as required by [ORA.GEN.200](#).

The ATO's safety risk assessment should take into account all relevant data including data from the type certificate holders (TCHs) regarding the susceptibility of the aircraft they operate to

volcanic cloud-related airworthiness effects, the nature and severity of these effects and the related pre-flight, in-flight and post-flight precautions to be observed by the ATO.

The ATO should ensure that personnel required to be familiar with the details of the safety risk assessments receives all relevant information (both pre-flight and in-flight) in order to be in a position to apply appropriate mitigation measures as specified by the safety risk assessments.

(b) Procedures

The ATO should have documented procedures for the management of operations into airspace forecast to be or aerodromes/operating sites known to be contaminated with volcanic ash.

These procedures should ensure that, at all times, flight operations remain within the accepted safety boundaries as established through the management system allowing for any variations in information sources, equipment, operational experience or organisation. Procedures should include those for flight crew and any other relevant personnel such that they are in a position to evaluate correctly the risk of flights into airspace forecast to be contaminated by volcanic ash and to plan accordingly.

Continuing airworthiness personnel should be provided with procedures allowing them to correctly assess the need for and to execute relevant maintenance or continuing airworthiness interventions.

The ATO should retain sufficient qualified and competent staff to generate well supported operational risk management decisions and ensure that its staff are appropriately trained and current. It is recommended that the ATO make the necessary arrangements for its relevant staff to take up opportunities to be involved in volcanic ash exercises conducted in their areas of operation.

(c) Volcanic activity information and the ATO's potential response

Before and during operations, information valuable to the ATO is generated by various volcano agencies worldwide. The ATO's risk assessment and mitigating actions need to take account of and respond appropriately to the information likely to be available during each phase of the eruptive sequence from pre-eruption through to end of eruptive activity. It is nevertheless noted that eruptions rarely follow a deterministic pattern of behaviour. A typical ATO's response may consist of the following:

(1) Pre-eruption

The ATO should have in place a robust mechanism for ensuring that it is constantly vigilant for any alerts of pre-eruption volcanic activity relevant to its operations. The staff involved need to understand the threat to safe operations that such alerts represent.

An ATO whose areas of activity include large, active volcanic areas for which immediate International Airways Volcano Watch (IAVW) alerts may not be available, should define its strategy for capturing information about increased volcanic activity before pre-eruption alerts are generated. For example, an ATO may combine elevated activity information with information concerning the profile and history of the volcano to determine an operating policy, which could include re-routing or restrictions at night. This would be useful when dealing with the 60% of volcanoes which are unmonitored.

Such an ATO should also ensure that its crews are aware that they may be the first to observe an eruption and so need to be vigilant and ready to ensure that this information is made available for wider dissemination as quickly as possible.

(2) Start of an eruption

Given the likely uncertainty regarding the status of the eruption during the early stages of an event and regarding the associated volcanic cloud, the ATO's procedures should include a requirement for crews to initiate re-routes to avoid the affected airspace.

The ATO should ensure that flights are planned to remain clear of the affected areas and that consideration is given to available aerodromes/operating sites and fuel requirements.

It is expected that the following initial actions will be taken by the ATO:

- (i) determine if any aircraft in flight could be affected, alert the crew and provide advice on re-routing as required;
 - (ii) alert management;
 - (iii) for flight departures, brief flight crew and revise flight and fuel planning in accordance with the safety risk assessment;
 - (iv) alert flight crew to the need for increased monitoring of information (e.g. special air report (AIREP), volcanic activity report (VAR), significant weather information (SIGMET), NOTAMs and company messages);
 - (v) initiate the gathering of all data relevant to determining the risk; and
 - (vi) apply mitigations identified in the safety risk assessment.
- (3) On-going eruption

As the eruptive event develops, the ATO can expect the responsible Volcanic Ash Advisory Centre (VAAC) to provide volcanic ash advisory messages (VAA/VAGs) defining, as accurately as possible, the vertical and horizontal extent of areas and layers of volcanic clouds. As a minimum, the ATO should monitor, and take account of, this VAAC information as well as of relevant SIGMETs and NOTAMs.

Other sources of information are likely to be available such as VAR/AIREPs, satellite imagery and a range of other information from State and commercial organisations. The ATO should plan its operations in accordance with its safety risk assessment taking into account the information that it considers accurate and relevant from these additional sources.

The ATO should carefully consider and resolve differences or conflicts among the information sources, notably between published information and observations (pilot reports, airborne measurements, etc.).

Given the dynamic nature of the volcanic hazards, the ATO should ensure that the situation is monitored closely and operations adjusted to suit changing conditions.

The ATO should be aware that, depending on the State concerned the affected or danger areas may be established and presented in a different way than the one currently used in Europe as described in EUR Doc 019-NAT Doc 006.

The ATO should require reports from its crews concerning any encounters with volcanic emissions. These reports should be passed immediately to the appropriate air traffic services (ATS) unit and to the ATO's competent authority.

For the purpose of flight planning, the ATO should treat the horizontal and vertical limits of the temporary danger area (TDA) or airspace forecast to be contaminated by volcanic ash as applicable, to be over-flown as it would mountainous terrain, modified in accordance with its safety risk assessment. The ATO should take account of the risk of

cabin depressurisation or engine failure resulting in the inability to maintain level flight above a volcanic cloud. Additional minimum Equipment List (MEL) provisions, if applicable, should be considered in consultation with the TCHs.

Flying below a volcanic ash contaminated airspace should be considered on a case by case basis. It should only be planned to reach or leave an aerodrome/operating site close to the boundary of this airspace or where the ash contamination is very high and stable. The establishment of Minimum Sector Altitude (MSA) and the availability of aerodromes/operating sites should be considered.

(d) Safety risk assessment

When directed specifically at the issue of intended flight into airspace forecast to be or aerodromes/operating sites known to be contaminated with volcanic ash, the process should involve the following:

(1) Identifying the hazards

The generic hazard, in the context of this document, is airspace forecast to bear aerodromes/operating sites known to be contaminated with volcanic ash, and whose characteristics are harmful to the airworthiness and operation of the aircraft.

This GM is referring to volcanic ash contamination since it is the most significant hazard for flight operations in the context of a volcanic eruption. Nevertheless, it might not be the only hazard and therefore the operator should consider additional hazards which could have an adverse effect on aircraft structure or passengers safety such as gases.

Within this generic hazard, the ATO should develop its own list of specific hazards taking into account its specific aircraft, experience, knowledge and type of operation, and any other relevant data stemming from previous eruptions.

(2) Considering the severity and consequences of the hazard occurring (i.e. the nature and actual level of damage expected to be inflicted on the particular aircraft from exposure to that volcanic ash cloud).

(3) Evaluating the likelihood of encountering volcanic ash clouds with characteristics harmful to the safe operation of the aircraft.

For each specific hazard within the generic hazard, the likelihood of adverse consequences should be assessed, either qualitatively or quantitatively.

(4) Determining whether the consequent risk is acceptable and within the ATO's risk performance criteria.

At this stage of the process, the safety risks should be classified as acceptable or unacceptable. The assessment of tolerability will be subjective, based on qualitative data and expert judgement, until specific quantitative data are available in respect of a range of parameters.

(5) Taking action to reduce the safety risk to a level that is acceptable to the ATO's management.

Appropriate mitigation for each unacceptable risk identified should then be considered in order to reduce the risk to a level acceptable to the ATO's management.

(e) Procedures to be considered when identifying possible mitigations actions

When conducting a volcanic ash safety risk assessment, the ATO should consider the following non-exhaustive list of procedures and processes as mitigation:

(1) Type certificate holders

Obtaining advice from the TCHs and other engineering sources concerning operations in potentially contaminated airspace and/or aerodromes/operating sites contaminated by volcanic ash.

This advice should set out:

- (i) the features of the aircraft that are susceptible to airworthiness effects related to volcanic ash;
- (ii) the nature and severity of these effects;
- (iii) the effect of volcanic ash on operations to/from contaminated aerodromes/operating sites, including the effect on take-off and landing aircraft performance;
- (iv) the related pre-flight, in-flight and post-flight precautions to be observed by the ATO including any necessary amendments to aircraft operating manuals, aircraft maintenance manuals, master minimum equipment list/dispatch deviation or equivalents required to support the ATO; and
- (v) the recommended inspections associated with inadvertent operations in volcanic ash contaminated airspace and operations to/from volcanic ash contaminated aerodromes/operating sites; this may take the form of instructions for continuing airworthiness or other advice.

(2) ATO/contracted organisations' personnel

Definition of procedures for flight planning and operations ensuring that:

- (i) flight crews are in a position to evaluate correctly the risk of encountering volcanic ash contaminated airspace, or aerodromes/operating sites, and can plan accordingly;
- (ii) flight planning and operational procedures enable crews to avoid areas and aerodromes/operating sites with unacceptable volcanic ash contamination;
- (iii) flight crew are aware of the possible signs of entry into a volcanic ash cloud and execute the associated procedures;
- (iv) continuing airworthiness personnel are able to assess the need for, and to execute, any necessary maintenance or other required interventions; and
- (v) crews are provided with appropriate aircraft performance data when operating to/from aerodromes/operating sites contaminated with volcanic ash.

(3) Provision of enhanced flight watch

This should ensure:

- (i) close and continuous monitoring of VAA, VAR/AIREP, SIGMET, NOTAM and ASHTAM and other relevant information, and information from crews, concerning the volcanic ash cloud hazard;
- (ii) access to plots of the affected areas from SIGMETs, NOTAMs and other relevant information for crews; and
- (iii) communication of the latest information to crews in a timely fashion.

(4) Flight planning

Flexibility of the process to allow re-planning at short notice should conditions change.

(5) Departure, destination and alternate aerodromes

For the airspace to be traversed, or the aerodromes/operating sites in use, parameters to evaluate and take account of:

- (i) the probability of contamination;
- (ii) any additional aircraft performance requirements;
- (iii) required maintenance considerations;
- (iv) fuel requirements for re-routeing and extended holding.

(6) Routing policy

Parameters to evaluate and take account of:

- (i) the shortest period in and over the forecast contaminated area;
- (ii) the hazards associated with flying over the contaminated area;
- (iii) drift down and emergency descent considerations;
- (iv) the policy for flying below the contaminated airspace and the associated hazards.

(7) Diversion policy

Parameters to evaluate and take account of:

- (i) maximum allowed distance from a suitable aerodrome/operating site;
- (ii) availability of aerodromes/operating sites outside the forecast contaminated area;
- (iii) diversion policy after an volcanic ash encounter.

(8) Minimum equipment list

Additional provisions in the MEL, if applicable, for dispatching aircraft with unserviceabilities that might affect the following non-exhaustive list of systems:

- (i) air conditioning packs;
- (ii) engine bleeds;
- (iii) pressurisation system;
- (iv) electrical power distribution system;
- (v) air data system;
- (vi) standby instruments;
- (vii) navigation systems;
- (viii) de-icing systems;
- (ix) engine driven generators;
- (x) auxiliary power unit (APU);
- (xi) airborne collision avoidance system (ACAS);
- (xii) terrain awareness warning system (TAWS);
- (xiii) autoland systems;
- (xiv) provision of crew oxygen;

- (xv) supplemental oxygen for passengers.
- (9) Standard operating procedures

Crew training to ensure they are familiar with normal and abnormal operating procedures and particularly any changes regarding but not limited to:

 - (i) pre-flight planning;
 - (ii) in-flight monitoring of volcanic ash cloud affected areas and avoidance procedures;
 - (iii) diversion;
 - (iv) communications with ATC;
 - (v) in-flight monitoring of engine and systems potentially affected by volcanic ash cloud contamination;
 - (vi) recognition and detection of volcanic ash clouds and reporting procedures;
 - (vii) in-flight indications of a volcanic ash cloud encounter;
 - (viii) procedures to be followed if a volcanic ash cloud is encountered;
 - (ix) unreliable or erroneous airspeed;
 - (x) non-normal procedures for engines and systems potentially affected by volcanic ash cloud contamination;
 - (xi) engine-out and engine relight;
 - (xii) escape routes; and
 - (xiii) operations to/from aerodromes/operating sites contaminated with volcanic ash.
- (10) Provision for aircraft technical log

This should ensure:

 - (i) Systematic entry in the aircraft continuing airworthiness records or aircraft log if available related to any actual or suspected volcanic ash encounter whether in-flight or at an aerodrome/operating site; and
 - (ii) Checking, prior to flight, of the completion of maintenance actions related to an entry in the continuing airworthiness records or aircraft log if available for a volcanic ash cloud encounter on a previous flight.
- (11) Incident reporting

Crew requirements for:

 - (i) reporting an airborne volcanic ash cloud encounter (VAR);
 - (ii) post-flight volcanic ash cloud reporting (VAR);
 - (iii) reporting non encounters in airspace forecast to be contaminated; and
 - (iv) filing a mandatory occurrence report in accordance with [ORA.GEN.160](#).
- (12) Continuing airworthiness procedures

Procedures when operating in or near areas of volcanic ash cloud contamination:

 - (i) enhancement of vigilance during inspections and regular maintenance and appropriate adjustments to maintenance practices;

- (ii) definition of a follow-up procedure when a volcanic ash cloud encounter has been reported or suspected;
- (iii) thorough investigation for any sign of unusual or accelerated abrasions or corrosion or of volcanic ash accumulation;
- (iv) reporting to TCHs and the relevant authorities observations and experiences from operations in areas of volcanic ash cloud contamination;
- (v) completion of any additional maintenance recommended by the TCH or by the competent authority.

(f) Reporting

The ATO should ensure that reports are immediately submitted to the nearest ATS unit using the VAR/AIREP procedures followed up by a more detailed VAR on landing together with, as applicable, a report as defined in Regulation (EU) No 996/2010 and Directive 2003/42/EC, and an aircraft technical log entry for:

- (1) any incident related to volcanic clouds;
- (2) any observation of volcanic ash activity and
- (3) anytime that volcanic ash is not encountered in an area where it was forecast to be.

(g) Additional guidance

Further guidance on volcanic ash safety risk assessment is given in ICAO Doc. 9974 (Flight safety and volcanic ash – Risk management of flight operations with known or forecast volcanic ash contamination).

GM4 ORA.GEN.200(a)(3) Management system

ED Decision 2013/008/R

SAFETY RISK ASSESSMENT – RISK REGISTER

The results of the assessment of the potential adverse consequences or outcome of each hazard may be recorded by the ATO in a risk register, an example of which is provided below.

Hazard		Incident Sequence Description	Existing Controls	Outcome (Pre-Mitigation)			Additional Mitigation required	Outcome (Post-Mitigation)			Actions and Owners	Monitoring and Review Requirements
No	Description			Severity	Likelihood	Risk		Severity	Likelihood	Risk		

AMC1 ORA.GEN.200(a)(4) Management system

ED Decision 2012/007/R

TRAINING AND COMMUNICATION ON SAFETY

- (a) Training
 - (1) All personnel should receive safety training as appropriate for their safety responsibilities.
 - (2) Adequate records of all safety training provided should be kept.
- (b) Communication
 - (1) The organisation should establish communication about safety matters that:
 - (i) ensures that all personnel are aware of the safety management activities as appropriate for their safety responsibilities;
 - (ii) conveys safety critical information, especially relating to assessed risks and analysed hazards;
 - (iii) explains why particular actions are taken; and
 - (iv) explains why safety procedures are introduced or changed.
 - (2) Regular meetings with personnel where information, actions and procedures are discussed may be used to communicate safety matters.

GM1 ORA.GEN.200(a)(4) Management system

ED Decision 2012/007/R

TRAINING AND COMMUNICATION ON SAFETY

The safety training programme may consist of self-instruction via a media (newsletters, flight safety magazines), class-room training, e-learning or similar training provided by training service providers.

AMC1 ORA.GEN.200(a)(5) Management system

ED Decision 2012/007/R

ORGANISATION'S MANAGEMENT SYSTEM DOCUMENTATION

- (a) The organisation's management system documentation should at least include the following information:
 - (1) a statement signed by the accountable manager to confirm that the organisation will continuously work in accordance with the applicable requirements and the organisation's documentation as required by this Part;
 - (2) the organisation's scope of activities;
 - (3) the titles and names of persons referred to in [ORA.GEN.210\(a\) and \(b\)](#);
 - (4) an organisation chart showing the lines of responsibility between the persons referred to in [ORA.GEN.210](#);
 - (5) a general description and location of the facilities referred to in [ORA.GEN.215](#);
 - (6) procedures specifying how the organisation ensures compliance with the applicable requirements;
 - (7) the amendment procedure for the organisation's management system documentation.

- (b) The organisation's management system documentation may be included in a separate manual or in (one of) the manual(s) as required by the applicable Subpart(s). A cross reference should be included.

AMC2 ORA.GEN.200(a)(5) Management system

ED Decision 2025/002/R

COMPLEX ORGANISATIONS – ORGANISATION'S SAFETY MANAGEMENT MANUAL

- (a) The safety management manual (SMM) should be the key instrument for communicating the approach to safety for the whole of the organisation. The SMM should document all aspects of safety management, including the safety policy, objectives, procedures and individual safety responsibilities.
- (b) The contents of the safety management manual should include all of the following:
- (1) scope of the safety management system;
 - (2) safety policy and objectives;
 - (3) safety accountability of the accountable manager;
 - (4) safety responsibilities of key safety personnel;
 - (5) documentation control procedures;
 - (6) hazard identification and risk management schemes;
 - (7) safety action planning;
 - (8) safety performance monitoring;
 - (9) incident investigation and reporting;
 - (10) emergency response planning;
 - (11) management of change (including organisational changes with regard to safety responsibilities);
 - (12) safety promotion.
- (c) The SMM may be contained in (one of) the manual(s) of the organisation.

GM1 ORA.GEN.200(a)(5) Management system

ED Decision 2012/007/R

ORGANISATION'S MANAGEMENT SYSTEM DOCUMENTATION

- (a) It is not required to duplicate information in several manuals. The information may be contained in any of the organisation manuals (e.g. operations manual, training manual), which may also be combined.
- (b) The organisation may also choose to document some of the information required to be documented in separate documents (e.g. procedures). In this case, it should ensure that manuals contain adequate references to any document kept separately. Any such documents are then to be considered an integral part of the organisation's management system documentation.

AMC1 ORA.GEN.200(a)(6) Management system

ED Decision 2012/007/R

COMPLIANCE MONITORING - GENERAL

(a) Compliance monitoring

The implementation and use of a compliance monitoring function should enable the organisation to monitor compliance with the relevant requirements of this Part and other applicable Parts.

- (1) The organisation should specify the basic structure of the compliance monitoring function applicable to the activities conducted.
- (2) The compliance monitoring function should be structured according to the size of the organisation and the complexity of the activities to be monitored.

(b) Organisations should monitor compliance with the procedures they have designed to ensure safe activities. In doing so, they should as a minimum, and where appropriate, monitor:

- (1) privileges of the organisation;
- (2) manuals, logs, and records;
- (3) training standards;
- (4) management system procedures and manuals.

(c) Organisational set up

- (1) To ensure that the organisation continues to meet the requirements of this Part and other applicable Parts, the accountable manager should designate a compliance monitoring manager. The role of the compliance monitoring manager is to ensure that the activities of the organisation are monitored for compliance with the applicable regulatory requirements, and any additional requirements as established by the organisation, and that these activities are being carried out properly under the supervision of the relevant head of functional area.
- (2) The compliance monitoring manager should be responsible for ensuring that the compliance monitoring programme is properly implemented, maintained and continually reviewed and improved.
- (3) The compliance monitoring manager should:
 - (i) have direct access to the accountable manager;
 - (ii) not be one of the other persons referred to in [ORA.GEN.210\(b\)](#);
 - (iii) be able to demonstrate relevant knowledge, background and appropriate experience related to the activities of the organisation; including knowledge and experience in compliance monitoring; and
 - (iv) have access to all parts of the organisation, and as necessary, any contracted organisation.
- (4) In the case of a non-complex organisation, this task may be exercised by the accountable manager provided he/she has demonstrated having the related competence as defined in (c)(3)(iii).
- (5) In the case the same person acts as compliance monitoring manager and as safety manager, the accountable manager, with regards to his/her direct accountability for

safety, should ensure that sufficient resources are allocated to both functions, taking into account the size of the organisation and the nature and complexity of its activities.

- (6) The independence of the compliance monitoring function should be established by ensuring that audits and inspections are carried out by personnel not responsible for the function, procedure or products being audited.
- (d) Compliance monitoring documentation
 - (1) Relevant documentation should include the relevant part(s) of the organisation's management system documentation.
 - (2) In addition, relevant documentation should also include the following:
 - (i) terminology;
 - (ii) specified activity standards;
 - (iii) a description of the organisation;
 - (iv) the allocation of duties and responsibilities;
 - (v) procedures to ensure regulatory compliance;
 - (vi) the compliance monitoring programme, reflecting:
 - (A) schedule of the monitoring programme;
 - (B) audit procedures;
 - (C) reporting procedures;
 - (D) follow-up and corrective action procedures; and
 - (E) recording system.
 - (vii) the training syllabus referred to in (e)(2);
 - (viii) document control.
- (e) Training
 - (1) Correct and thorough training is essential to optimise compliance in every organisation. In order to achieve significant outcomes of such training, the organisation should ensure that all personnel understand the objectives as laid down in the organisation's management system documentation.
 - (2) Those responsible for managing the compliance monitoring function should receive training on this task. Such training should cover the requirements of compliance monitoring, manuals and procedures related to the task, audit techniques, reporting and recording.
 - (3) Time should be provided to train all personnel involved in compliance management and for briefing the remainder of the personnel.
 - (4) The allocation of time and resources should be governed by the volume and complexity of the activities concerned.

GM1 ORA.GEN.200(a)(6) Management system

ED Decision 2012/007/R

COMPLIANCE MONITORING - GENERAL

- (a) The organisational set-up of the compliance monitoring function should reflect the size of the organisation and the nature and complexity of its activities. The compliance monitoring manager may perform all audits and inspections himself/herself or appoint one or more auditors by choosing personnel having the related competence as defined in [AMC1 ORA.GEN.200\(a\)\(6\)](#) point (c)(3)(iii), either from within or outside the organisation.
- (b) Regardless of the option chosen it must be ensured that the independence of the audit function is not affected, in particular in cases where those performing the audit or inspection are also responsible for other functions within the organisation.
- (c) In case external personnel are used to perform compliance audits or inspections:
 - (1) any such audits or inspections are performed under the responsibility of the compliance monitoring manager; and
 - (2) the organisation remains responsible to ensure that the external personnel has relevant knowledge, background and experience as appropriate to the activities being audited or inspected; including knowledge and experience in compliance monitoring.
- (d) The organisation retains the ultimate responsibility for the effectiveness of the compliance monitoring function in particular for the effective implementation and follow-up of all corrective actions.

GM2 ORA.GEN.200(a)(6) Management system

ED Decision 2012/007/R

COMPLEX ORGANISATIONS - COMPLIANCE MONITORING PROGRAMME FOR ATOs

- (a) Typical subject areas for compliance monitoring audits and inspections for approved training organisations (ATOs) should be the following:
 - (1) facilities;
 - (2) actual flight and ground training;
 - (3) technical standards.
- (b) ATOs should monitor compliance with the training and operations manuals they have designed to ensure safe and efficient training. In doing so, they should, where appropriate, additionally monitor the following:
 - (1) training procedures;
 - (2) flight safety;
 - (3) flight and duty time limitations, rest requirements and scheduling;
 - (4) aircraft maintenance/operations interface.

GM3 ORA.GEN.200(a)(6) Management system

ED Decision 2012/007/R

AUDIT AND INSPECTION

- (a) 'Audit' means a systematic, independent and documented process for obtaining evidence and evaluating it objectively to determine the extent to which requirements are complied with.

- (b) 'Inspection' means an independent documented conformity evaluation by observation and judgement accompanied as appropriate by measurement, testing or gauging, in order to verify compliance with applicable requirements.

AMC1 ORA.GEN.200(b) Management system

ED Decision 2020/005/R

SIZE, NATURE AND COMPLEXITY OF THE ACTIVITY

- (a) An organisation should be considered as complex when it has a workforce of more than 20 full time equivalents (FTEs) involved in the activity subject to Regulation (EC) No 216/2008¹ and its Implementing Rules.
- (b) Organisations with up to 20 full time equivalents (FTEs) involved in the activity subject to Regulation (EC) No 216/2008 and its Implementing Rules, may also be considered complex based on an assessment of the following factors:
- (1) in terms of complexity, the extent and scope of contracted activities subject to the approval;
 - (2) in terms of risk criteria, whether any of the following are present:
 - (i) operations requiring the following specific approvals: performance based navigation (PBN), low visibility operation (LVO), extended range operations with two-engined aeroplanes (ETOPS), helicopter hoist operation (HHO), helicopter emergency medical service (HEMS), night vision imaging system (NVIS) and dangerous goods (DG);
 - (ii) different types of aircraft used;
 - (iii) the environment (offshore, mountainous area etc.);
- (c) Regardless of the criteria mentioned in (a) and (b), the following organisations should always be considered as non-complex:
- (1) Approved Training Organisations (ATOs) only providing training for the light aircraft pilot licence (LAPL), private pilot licence (PPL), sailplane pilot licence (SPL) or balloon pilot licence (BPL) and the associated ratings and certificates;
 - (2) Aero-Medical Centres (AeMCs).
- (d) Regardless of the criteria mentioned in (a) and (b), the organisations that provide training in the following areas should always be considered as complex:
- (1) full flight simulators (FFSs); or
 - (2) multi-pilot (MP) type rating; or
 - (3) zero-flight-time training (ZFTT); or
 - (4) complex aircraft; or
 - (5) different categories of aircraft; or
 - (6) instructor certificates for point (2) and (4) aircraft; or

¹ Regulation (EC) No 216/2008 of the European Parliament and of the Council of 20 February 2008 on common rules in the field of civil aviation and establishing a European Aviation Safety Agency, and repealing Council Directive 91/670/EEC, Regulation (EC) No 1592/2002 and Directive 2004/36/EC. OJ L 79, 19.3.2008, p. 1.

- (7) two or more aerodromes/operating sites.

AMC1 ORA.GEN.200(c) Management system

ED Decision 2015/011/R

ATOs PROVIDING TRAINING ONLY FOR THE LAPL, PPL, SPL AND BPL AND THE ASSOCIATED RATINGS OR CERTIFICATES – ORGANISATIONAL REVIEW

- (a) The primary objective of the organisational review is to enable the organisation to ensure that its management system remains effective by verifying that it:
- (1) has continually identified its aviation safety hazards;
 - (2) has effectively mitigated the associated risks; and
 - (3) monitors compliance with the applicable requirements.
- (b) Safety risk management should:
- (1) be performed using internal safety or occurrence reports, hazard checklists, risk registers or similar risk management tools or processes, integrated into the activities of the organisation;
 - (2) in particular address safety risks related to a change; making use of the existing hazard identification, risk assessment and mitigation tools or processes; and
 - (3) include provisions for emergency response or a formal Emergency Response Plan (ERP).
- (c) As part of the management system documentation required by [ORA.GEN.200\(a\)\(5\)](#), the organisation should describe the organisational review programme and related responsibilities. Persons responsible for the organisational review should have a thorough knowledge of the applicable requirements and of the organisation's procedures.
- (d) The status of all corrective and risk mitigation actions should be monitored by the person responsible for the organisational review programme and implemented within a specified time frame. Action closure should be recorded by the person responsible for the organisational review programme, along with a summary of the action taken.
- (e) The results of the organisational review, including all non-compliance findings and new risks identified during the review, should be presented to the accountable manager and the person or group of persons nominated in accordance with [ORA.GEN.210\(b\)](#) prior to notification to the competent authority. All level 1 findings in the sense of ARA.GEN.350 should be immediately notified to the competent authority and all necessary actions immediately taken.
- (f) Based on the results of the organisational review, the accountable manager should determine the need for and initiate, as appropriate, further actions to address deficiencies in or further improve the organisation's management system.

GM1 ORA.GEN.200(c) Management system

ED Decision 2015/011/R

ATOs PROVIDING TRAINING ONLY FOR THE LAPL, PPL, SPL OR BPL AND THE ASSOCIATED RATINGS OR CERTIFICATES – ORGANISATIONAL REVIEW PROGRAMME

- (a) The organisational review programme may consist of:
- (1) checklist(s) covering all items necessary to be addressed in order to ensure that the organisation identified its aviation safety hazards, effectively mitigates the associated

risks and ensures effective compliance with the applicable requirements. These should address all procedures described in the management system documentation and training manual; and

- (2) a schedule for the accomplishment of the different checklist items, with each item being checked at least once within any 12-month period. The organisation may choose to conduct one full review annually or to conduct several partial reviews.
- (b) Performance of organisational reviews:
- Each review item may be addressed using an appropriate combination of:
- (1) review of training records, training documentation;
 - (2) review of internal safety reports (e.g. notified difficulties in using current procedures and training material, etc.);
 - (3) review of the risk register and hazard checklists, as applicable;
 - (4) sample check of training courses;
 - (5) witnessing of examinations, as appropriate;
 - (6) interview of the personnel involved; and
 - (7) review of the feedback provided by students and customers.
- (c) It is recommended that internal safety reports and occurrence reports be reviewed on a continual basis with the aim of identifying possible corrective and risk mitigation actions.

GM2 ORA.GEN.200(c) Management system

ED Decision 2015/011/R

ATOs PROVIDING TRAINING ONLY FOR THE LAPL, PPL, SPL OR BPL AND THE ASSOCIATED RATINGS OR CERTIFICATES – ORGANISATIONAL REVIEW ITEMS

The following provides a list of typical items for an organisational review checklist, to be adapted as necessary to cover all relevant procedures described in the management system documentation and training manual:

- (a) Terms of approval
- Check that:
- (1) no training has been performed outside the terms of approval;
 - (2) changes not requiring prior approval have been properly managed.
- (b) Training syllabi and course material
- Check that:
- (1) training syllabi and course materials are in compliance with the applicable requirements, as last amended;
 - (2) training practices are in compliance with the documentation; and
 - (3) instructor training practices are standardised.
- (c) Training equipment and tools
- Check that all equipment and tools other than aircraft and FSTDs are present and meet the criteria defined in the training manual.

(d) Facilities

Check that the facilities meet the criteria defined in the training manual.

(e) Training aircraft and FSTDs

Check that the training aircraft and FSTDs meet the criteria defined in the training manual.

(f) Personnel

Check that:

- (1) the current accountable manager and other nominated persons are correctly identified;
- (2) the organisation chart accurately indicates lines of responsibility and accountability throughout the organisation;
- (3) the organisation remains in compliance with the applicable requirements, in case the number of personnel has decreased or if the activity has increased;
- (4) the qualification of all new personnel (or personnel with new functions) has been appropriately assessed;
- (5) staff involved in any safety management-related processes and tasks has been properly trained; and
- (6) staff has been trained, as necessary, to cover changes in regulations, in competent authority publications, in the organisation, its management system documentation and in associated procedures, etc.

(g) Contracted activities (In case the organisation has contracted activities):

- (1) Check that new providers have been assessed prior to the establishment of any contract;
- (2) For existing providers approved for such activities: check the authorisation and approval status of the contracted organisation; and
- (3) For existing providers not approved for such activities: check that the service provided conforms to the applicable requirements of this Part.

(h) Training and communication on safety

Check that:

- (1) all personnel are aware of safety management policies, processes and tasks;
- (2) safety-related documentations and publications are available; and
- (3) safety-critical information derived from internal safety or occurrence reporting and hazard identification have been timely communicated to all staff concerned.

(i) Management system documentation

Check that:

- (1) the documentation is adequate and updated;
- (2) staff are aware of the safety policy; and
- (3) staff can easily access such documentation when needed.

(j) Record-keeping

Check that:

- (1) the records cover all the training activities and management system processes; and

- (2) minimum record-keeping periods (random checks) are complied with.
- (k) Emergency response provisions or ERP
Check that:
 - (1) emergency response information is up to date and readily available; and
 - (2) all staff are aware of emergency response information or the ERP, as applicable (random checks).
- (l) Internal safety or occurrence reporting procedures
 - (1) Check the number of reports received since the last review;
 - (2) Check that:
 - (i) internal reporting and external occurrence reporting are performed in accordance with reporting procedures;
 - (ii) the safety or occurrence reports are analysed; and
 - (iii) feedback is provided to reporters.
- (m) Other risk management tools or processes implemented
 - (1) As applicable, check that:
 - (i) records of hazards and risks are assessed; in particular following analysis of safety or occurrence reports and when significant changes occur (regulations, personnel, training aircraft, training courses, etc.);
 - (ii) the risks are assessed and the risk mitigation actions followed up and recorded;
 - (iii) any risk that has been found acceptable is duly justified; and
 - (iv) the assumptions made for the risk assessment remain valid;
 - (2) Verify the effectiveness of all risk mitigation actions initiated since the last organisational review.

ORA.GEN.200A Information security management system

Regulation (EU) 2023/203

In addition to the management system referred to in point [ORA.GEN.200](#), the organisation shall establish, implement and maintain an information security management system in accordance with [Implementing Regulation \(EU\) 2023/203](#) in order to ensure the proper management of information security risks which may have an impact on aviation safety.

[applicable from 22 February 2026 — Implementing Regulation (EU) 2023/203]

ORA.GEN.205 Contracted activities

Regulation (EU) No 290/2012

- (a) Contracted activities include all activities within the organisation's scope of approval that are performed by another organisation either itself certified to carry out such activity or if not certified, working under the contracting organisation's approval. The organisation shall ensure that when contracting or purchasing any part of its activity, the contracted or purchased service or product conforms to the applicable requirements.

- (b) When the certified organisation contracts any part of its activity to an organisation that is not itself certified in accordance with this Part to carry out such activity, the contracted organisation shall work under the approval of the contracting organisation. The contracting organisation shall ensure that the competent authority is given access to the contracted organisation, to determine continued compliance with the applicable requirements.

AMC1 ORA.GEN.205 Contracted activities

ED Decision 2012/007/R

RESPONSIBILITY WHEN CONTRACTING ACTIVITIES

- (a) The organisation may decide to contract certain activities to external organisations.
- (b) A written agreement should exist between the organisation and the contracted organisation clearly defining the contracted activities and the applicable requirements.
- (c) The contracted safety related activities relevant to the agreement should be included in the organisation's safety management and compliance monitoring programmes.
- (d) The organisation should ensure that the contracted organisation has the necessary authorisation or approval when required, and commands the resources and competence to undertake the task.

GM1 ORA.GEN.205 Contracted activities

ED Decision 2012/007/R

RESPONSIBILITY WHEN CONTRACTING ACTIVITIES

- (a) Regardless of the approval status of the contracted organisation, the contracting organisation is responsible to ensure that all contracted activities are subject to hazard identification and risk management as required by [ORA.GEN.200\(a\)\(3\)](#) and to compliance monitoring as required by [ORA.GEN.200\(a\)\(6\)](#).
- (b) When the contracted organisation is itself certified to carry out the contracted activities, the organisation's compliance monitoring should at least check that the approval effectively covers the contracted activities and that it is still valid.
- (c) If the organisation requires the contracted organisation to conduct an activity which exceeds the contracted organisation's terms of approval, this will be considered as the contracted organisation working under the approval of the contracting organisation.

ORA.GEN.210 Personnel requirements

Regulation (EU) No 290/2012

- (a) The organisation shall appoint an accountable manager, who has the authority for ensuring that all activities can be financed and carried out in accordance with the applicable requirements. The accountable manager shall be responsible for establishing and maintaining an effective management system.
- (b) A person or group of persons shall be nominated by the organisation, with the responsibility of ensuring that the organisation remains in compliance with the applicable requirements. Such person(s) shall be ultimately responsible to the accountable manager.
- (c) The organisation shall have sufficient qualified personnel for the planned tasks and activities to be performed in accordance with the applicable requirements.

- (d) The organisation shall maintain appropriate experience, qualification and training records to show compliance with paragraph (c).
- (e) The organisation shall ensure that all personnel are aware of the rules and procedures relevant to the exercise of their duties.

ORA.GEN.215 Facility requirements

Regulation (EU) No 1178/2011

The organisation shall have facilities allowing the performance and management of all planned tasks and activities in accordance with the applicable requirements.

AMC1 ORA.GEN.215 Facility requirements

ED Decision 2012/007/R

ATOs PROVIDING TRAINING FOR the CPL, MPL AND ATPL AND THE ASSOCIATED RATINGS AND CERTIFICATES

- (a) For ATOs providing flight training, the following flight operations accommodation should be available:
 - (1) an operations room with facilities to control flying operations;
 - (2) a flight planning room with the following facilities:
 - (i) appropriate current maps and charts;
 - (ii) current aeronautical information service (AIS) information;
 - (iii) current meteorological information;
 - (iv) communications to air traffic control (ATC) and the operations room;
 - (v) any other flight safety related material.
 - (3) adequate briefing rooms/cubicles of sufficient size and number;
 - (4) suitable offices for the supervisory personnel and room(s) to allow flight instructors to write reports on students, complete records and other related documentation;
 - (5) furnished crew-room(s) for instructors and students.
- (b) For ATOs providing theoretical knowledge training, the following facilities for theoretical knowledge instruction should be available:
 - (1) adequate classroom accommodation for the current student population;
 - (2) suitable demonstration equipment to support the theoretical knowledge instruction;
 - (3) a radiotelephony training and testing facility;
 - (4) a reference library containing publications giving coverage of the syllabus;
 - (5) offices for the instructional personnel.

AMC2 ORA.GEN.215 Facility requirements

ED Decision 2012/007/R

ATOs PROVIDING TRAINING FOR THE LAPL, PPL, SPL OR BPL AND THE ASSOCIATED RATINGS AND CERTIFICATES

- (a) The following flight operations accommodation should be available:

- (1) a flight planning room with the following facilities:
 - (i) appropriate current aviation maps and charts;
 - (ii) current AIS information;
 - (iii) current meteorological information;
 - (iv) communications to ATC (if applicable);
 - (v) any other flight safety related material.
 - (2) adequate briefing room(s)/cubicles of sufficient size and number;
 - (3) suitable office(s) to allow flight instructors to write reports on students, complete records and other related documentation;
 - (4) suitable rest areas for instructors and students, where appropriate to the training task;
 - (5) in the case of ATOs providing training for the BPL or LAPL(B) only, the flight operations accommodation listed in (a)(1) to (a)(4) may be replaced by other suitable facilities when operating outside aerodromes.
- (b) The following facilities for theoretical knowledge instruction should be available:
- (1) adequate classroom accommodation for the current student population;
 - (2) suitable demonstration equipment to support the theoretical knowledge instruction;
 - (3) suitable office(s) for the instructional personnel.
- (c) A single room may be sufficient to provide the functions listed in (a) and (b).

ORA.GEN.220 Record-keeping

Regulation (EU) No 1178/2011

- (a) The organisation shall establish a system of record-keeping that allows adequate storage and reliable traceability of all activities developed, covering in particular all the elements indicated in [ORA.GEN.200](#).
- (b) The format of the records shall be specified in the organisation's procedures.
- (c) Records shall be stored in a manner that ensures protection from damage, alteration and theft.

AMC1 ORA.GEN.220(b) Record-keeping

ED Decision 2012/007/R

GENERAL

- (a) The record-keeping system should ensure that all records are accessible whenever needed within a reasonable time. These records should be organised in a way that ensures traceability and retrievability throughout the required retention period.
- (b) Records should be kept in paper form or in electronic format or a combination of both. Records stored on microfilm or optical disc format are also acceptable. The records should remain legible throughout the required retention period. The retention period starts when the record has been created or last amended.
- (c) Paper systems should use robust material which can withstand normal handling and filing. Computer systems should have at least one backup system which should be updated within 24

hours of any new entry. Computer systems should include safeguards against the ability of unauthorised personnel to alter the data.

- (d) All computer hardware used to ensure data backup should be stored in a different location from that containing the working data and in an environment that ensures they remain in good condition. When hardware or software changes take place, special care should be taken that all necessary data continues to be accessible at least through the full period specified in the relevant Subpart. In the absence of such indication, all records should be kept for a minimum period of 5 years.

GM1 ORA.GEN.220(b) Record-keeping

ED Decision 2012/007/R

RECORDS

Microfilming or optical storage of records may be carried out at any time. The records should be as legible as the original record and remain so for the required retention period.

SUBPART ATO – APPROVED TRAINING ORGANISATIONS

SECTION I – GENERAL

ORA.ATO.100 Scope

Regulation (EU) No 1178/2011

This Subpart establishes the requirements to be met by organisations providing training for pilot licences and associated ratings and certificates.

GM1 ORA.ATO.100 Scope

ED Decision 2012/007/R

The content of this Section contains the requirements applicable to all ATOs providing training for pilot licences and the associated ratings and certificates.

It is applicable to ATOs providing training for:

- (a) the LAPL, PPL, SPL and BPL and the associated ratings and certificates; and
- (b) the commercial pilot licence (CPL), multi-crew pilot licence (MPL) and airline transport pilot licence (ATPL) and the associated ratings and certificates.

ORA.ATO.105 Application

Regulation (EU) No 290/2012

- (a) Applicants for the issue of a certificate as an approved training organisation (ATO) shall provide the competent authority with:
 - (1) the following information:
 - (i) name and address of the training organisation;
 - (ii) date of intended commencement of activity;
 - (iii) personal details and qualifications of the head of training (HT), the flight instructor(s), flight simulation training instructors and the theoretical knowledge instructor(s);
 - (iv) name(s) and address(es) of the aerodromes(s) and/or operating site(s) at which the training is to be conducted;
 - (v) list of aircraft to be operated for training, including their group, class or type, registration, owners and category of the certificate of airworthiness, if applicable
 - (vi) list of flight simulation training devices (FSTDs) that the training organisation intends to use, if applicable;
 - (vii) the type of training that the training organisation wishes to provide and the corresponding training programme; and
 - (2) the operations and training manuals.
- (b) Flight test training organisations. Notwithstanding (a)(1)(iv) and (v), training organisations providing flight test training shall only need to provide:

- (1) the name(s) and address(es) of the main aerodromes and/or operating site(s) at which the training is to be conducted; and
 - (2) a list of the types or categories of aircraft to be used for flight test training.
- (c) In the case of a change to the certificate, applicants shall provide the competent authority with the relevant parts of the information and documentation referred to in (a).

AMC1 ORA.ATO.105 Application

ED Decision 2012/007/R

APPLICATION FORM

APPLICATION FORM FOR AN ATO CERTIFICATE		
N°	Question	Supplementary information
1.	Name of training organisation under which the activity is to take place	address, fax number, e-mail, URL
2.	Training courses offered	theory and/or flight training
3.	Name of head of training	type and number of licence full/part-time
4.	Name of chief flight instructor	as (3)
5.	Name of chief theoretical knowledge instructor	as (3)
6.	Name of flight instructor(s), where applicable	as (3)
7.	Aerodrome(s) / operating site(s) to be used	IFR approaches, if applicable night flying, if applicable air traffic control flight testing facilities, if applicable data reply facilities, if applicable
8.	Flight operations accommodation	location, number and size of rooms
9.	Theoretical instruction facilities	location, number and size of rooms
10.	Description of training devices (as applicable)	FFS, FNPT I, II and III, FTD 1, 2 and 3, and 3, and BITD
11.	Description of aircraft	Class/type(s) of aircraft registration of aircraft IFR equipped, if applicable Flight test instrumentation, if applicable
12.	Proposed administration and manuals: (submit with application if required)	(a) course programmes (b) training records (c) operations manual (d) training manual
13.	Details of proposed compliance monitoring system	

Note 1: If answers to any of the above questions are incomplete, the applicant should provide full details of alternative arrangements separately.

Note 2: instrument flight rules (IFR), full flight simulator (FFS), flight and navigation procedures trainer (FNPT), flight training device (FTD), basic instrument training device (BITD)

I, (name), on behalf of (name of training organisation) certify that all the above named persons are in compliance with the applicable requirements and that all the above information given is complete and correct. (Date) (Signature)

ORA.ATO.110 Personnel requirements

Regulation (EU) 2020/359

- (a) An HT shall be nominated. The HT shall have extensive experience as an instructor in the areas relevant for the training provided by the ATO and shall possess sound managerial capability.
- (b) The HT's responsibilities shall include:
 - (1) ensuring that the training provided is in compliance with Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#), Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), as applicable, and, in the case of flight test training, that the relevant requirements of Annex I (Part 21) to [Commission Regulation \(EU\) No 748/2012](#) and the training programme have been established;
 - (2) ensuring the satisfactory integration of flight training in an aircraft or a flight simulation training device (FSTD) and theoretical knowledge instruction; and
 - (3) supervising the progress of individual students.
- (c) Theoretical knowledge instructors shall have:
 - (1) practical background in aviation in the areas relevant for the training provided and have undergone a course of training in instructional techniques; or
 - (2) previous experience in giving theoretical knowledge instruction and an appropriate theoretical background in the subject on which they will provide theoretical knowledge instruction.
- (d) Flight instructors and flight simulation training instructors shall hold the qualifications required by Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#) for the type of training they provide.

AMC1 ORA.ATO.110(b) Personnel requirements

ED Decision 2012/007/R

HEAD OF TRAINING

The nominated head of training (HT) should have the overall responsibility to ensure that the training is in compliance with the appropriate requirements. In an ATO providing training courses for different aircraft categories, the HT shall be assisted by one or more nominated deputy HT(s) for certain flight training courses.

AMC1 ORA.ATO.110(c) Personnel requirements

ED Decision 2012/007/R

THEORETICAL KNOWLEDGE INSTRUCTORS

Theoretical knowledge instructors should, before appointment, prove their competency by giving a test lecture based on material they have developed for the subjects they are to teach.

GM1 ORA.ATO.110(d) Personnel requirements

ED Decision 2025/002/R

- (a) Before allowing an FI that is experienced as specified in point [FCL.905.FI\(h\)\(3\)](#) to supervise SPIC flights during an IR training course, the ATO should consider at least the following factors:
 - (1) the experience of the FI;
 - (2) the experience of the student pilot;
 - (3) the nature and complexity of the SPIC flight to be performed;
 - (4) the complexity and characteristics of the training aircraft; and
 - (5) the prevailing weather conditions.
- (b) The ATO should identify the hazards related to the SPIC supervision and apply appropriate mitigation measures to reduce the associated risks.

ORA.ATO.120 Record-keeping

Regulation (EU) 2018/1119

The following records shall be kept throughout the course and for a period of three years after the completion of the training:

- (a) details of ground, flight, and simulated flight training given to individual students;
- (b) detailed and regular progress reports from instructors including assessments, and regular progress flight tests and ground examinations; and
- (c) information on the licences and associated ratings and certificates of the students, including the expiry dates of medical certificates and ratings.

AMC1 ORA.ATO.120(a);(b) Record-keeping

ED Decision 2012/007/R

ATOs PROVIDING TRAINING ONLY FOR THE LAPL, PPL, SPL OR BPL AND THE ASSOCIATED RATINGS AND CERTIFICATES

The details of ground, flight and flight instruction by using FSTD given to a specific individual student and the detailed progress reports from instructors may be kept also in a student's progress card. This progress card should contain all the exercises of the training syllabus. The instructor should sign this card if a certain exercise has been completed or a specific assessment has been conducted.

ORA.ATO.125 Training programme

Regulation (EU) 2020/359

- (a) A training programme shall be developed for each type of course offered.
- (b) The training programme shall comply with the requirements of Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#), Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), as applicable, and, in the case of flight test training, the relevant requirements of Annex I (Part 21) to [Commission Regulation \(EU\) No 748/2012](#).

AMC1 ORA.ATO.125 Training programme

ED Decision 2012/007/R

GENERAL

Flight training in an FSTD and theoretical knowledge instruction should be phased in such a manner as to ensure that students are able to apply to flight exercises the knowledge gained on the ground. Arrangements should be made so that problems encountered during instruction can be resolved during subsequent training.

AMC2 ORA.ATO.125 Training programme

ED Decision 2025/002/R

TYPE RATING COURSES – AEROPLANES

(a) Introduction

- (1) When developing the training programme for a type rating course, in addition to complying with the standards included in the operational suitability data (OSD), as established in accordance with Regulation (EC) 1702/2003¹ for the applicable type, the ATO should also follow any further recommendations contained therein.
- (2) The type rating course should, as far as possible, provide for a continual process of ground, FSTD and flight training to enable the student to assimilate the knowledge and skills required to operate a specific aircraft type safely and efficiently. The student's ability to do this should be determined by the demonstration of a satisfactory level of theoretical knowledge of the aircraft determined by progressive checking of knowledge and examination, progressive assessment by the ATO during flight training and the successful completion of a practical skill test with an examiner.
- (3) The type rating course should normally be conducted as a single, fulltime course of study and training. However, in the situation where the course is intended to enable a pilot to fly a further aircraft type while continuing to fly a current type, such as to enable mixed fleet flying with the same operator, some elements of the theoretical knowledge course conducted by self-study may be undertaken while the student continues to fly the current type.

(b) Variants

- (1) Familiarisation: Where an aeroplane type rating also includes variants of the same aircraft type requiring familiarisation, the additional familiarisation element may be included in the theoretical knowledge training of the initial type rating course.
- (2) Differences training: Where an aeroplane type rating also includes variants of the same aircraft type for which difference training is required, the initial training course should be directed towards a single variant. Additional training to operate other variants within the same type rating should be completed after successful completion of the initial type rating course. However, elements of this differences training may be undertaken at appropriate stages of the initial course, with the agreement of the competent authority.

(c) Programme of theoretical knowledge and flight training

¹ Commission Regulation (EC) No 1702/2003 of 24 September 2003 laying down implementing rules for the airworthiness and environmental certification of aircraft and related products, parts and appliances, as well as for the certification of design and production organisations (Part 21) (OJ L 243, 27.9.2003, p. 6). Regulation as last amended by Regulation (EC) No 1194/2009 (OJ L 321, 8.12.2009, p. 5).

- (1) The training programme should specify the time allocated to theoretical knowledge training, FSTD training and, if not approved for zero flighttime training (ZFTT), the aeroplane. The initial type rating course should be programmed on the basis that the student has the minimum licensing and experience requirements for entry to the course. For a first type rating on a multi-pilot aeroplane (MPA), the course should also provide for consolidation and type-specific training in those elements of basic multi-crew cooperation (MCC) training relevant to the type or variant.
- (2) If the ATO wishes to provide a training course that includes credit for previous experience on similar types of aircraft, such as those with common systems or operating procedures with the new type, the entry requirements to such courses should be specified by the ATO and should define the minimum level of experience and qualification required of the flight crew member.
- (3) The ATO is permitted to contract elements of training to a third party training provider. In such cases the contracted organisation should normally be approved to conduct such training. When the contracted organisation is not an ATO, the competent authority should, within the approval process of the ATO, include the contracted organisation and be satisfied that the standard of training intended to be given meets the requirements. The other obligations of the ATO, such as student progress monitoring and an adequate management system, can be exercised by the ATO seeking approval and which retains responsibility for the whole course.

GROUND TRAINING

(d) Syllabus

The ground training syllabus should provide for the student to gain a thorough understanding of the operation, function and, if appropriate, abnormal and emergency operation of all aircraft systems. This training should also include those systems essential to the operation of the aircraft, such as 'fly-by-wire' flight control systems, even if the flight crew have little or no control of their normal or abnormal operation.

(e) Theoretical knowledge instruction

The theoretical knowledge instruction training should meet the general objectives of (but not be limited to) giving the student:

- (1) a thorough knowledge of the aircraft structure, powerplant and systems, and their associated limitations, including mass and balance, aircraft performance and flight planning considerations;
- (2) a knowledge of the positioning and operation of the cockpit controls and indicators for the aircraft and its systems;
- (3) an understanding of system malfunctions, their effect on aircraft operations and interaction with other systems; and
- (4) the understanding of normal, abnormal and emergency procedures.

(f) Facilities and training aids

The ATO should provide adequate facilities for classroom instruction and have available appropriately qualified and experienced instructors. Training aids should enable students to gain practical experience of the operation of systems covered by the theoretical knowledge syllabus and, in the case of multi-pilot aeroplanes, enable such practical application of the

knowledge to be carried out in a multi-crew environment. Facilities should be made available for student self-study outside the formal training programme.

(g) Computer-based training (CBT)

CBT provides a valuable source of theoretical instruction, enabling the students to progress at their own pace within specified time limits. Many such systems ensure that syllabus subjects are fully covered and progress can be denied until a satisfactory assimilation of knowledge has been demonstrated. Such systems may allow self-study or distance learning, if they incorporate adequate knowledge testing procedures. When CBT is used as part of the theoretical knowledge instruction phase, the student should also have access to a suitably qualified instructor able to assist with areas of difficulty for the student.

(h) Self-study and distance learning

Elements of the theoretical knowledge syllabus may be adequately addressed by distance learning, if approved, or self-study, particularly when utilising CBT. Progress testing, either by self-assessed or instructor-evaluated means should be included in any self-study programme. If self-study or distance learning is included in the theoretical knowledge training, the course should also provide for an adequate period of supervised consolidation and knowledge testing.

(i) Progress tests and final theoretical knowledge examination

- (1) The theoretical knowledge training programme should provide for progressive testing of the assimilation of the required knowledge. This testing process should also provide for retesting of syllabus items so that a thorough understanding of the required knowledge is assured. This should be achieved by intervention by a qualified instructor or, if using CBT with a self-testing facility, and by further testing during the supervised consolidation phase of the ground course.
- (2) The final theoretical knowledge examination should cover all areas of the theoretical knowledge syllabus. The final examination should be conducted as a supervised written (including computer-based) knowledge test without reference to course material. The pass mark of 75% assumes the achievement of satisfactory levels of knowledge during the progressive phase tests of the course. The student should be advised of any areas of lack of knowledge displayed during the examination and, if necessary, given remedial instruction. A successful pass of the theoretical knowledge course and final examination should be a pre-requisite for progression to the flight training phase of the type rating course, unless otherwise determined in the OSD established in accordance with Regulation (EC) 1702/2003.

FLIGHT TRAINING

(j) Flight simulation training devices (FSTDs)

A type rating course for a multi-pilot aeroplane should include FSTD training.

The amount of training required when using FSTDs will depend on the complexity of the aeroplane concerned, and to some extent on the previous experience of the pilot. Except for those courses giving credit for previous experience (c.2.), a minimum of 32 hours of FSTD training should be programmed for a crew of a multipilot aeroplane, of which at least 16 hours should be in an FFS operating as a crew. FFS time may be reduced if other qualified FSTDs used during the flight training programme accurately replicate the cockpit environment, operation and aeroplane response. Such FSTDs may typically include flight management computer (FMC) training devices using hardware and computer programmes identical to those of the aeroplane.

(k) Aeroplane training with FFS

- (1) with the exception of courses approved for ZFTT, certain training exercises normally involving take-off and landing in various configurations should be completed in the aeroplane rather than in an FFS. Unless otherwise specified in the OSD established in accordance with Regulation (EU) No 748/2012 this take-off and landing training should include:
 - (A) at least four landings in the case of MPAs (or single-pilot high performance complex aeroplanes (SP HPAs)) where the student pilot has more than 500 hours of MPA experience (or SPA experience) in aeroplanes of similar size and performance or, in all other cases, at least six landings;
 - (B) at least one full-stop landing; and
 - (C) one go-around with all engines operating.

This aeroplane training may be completed after the student pilot has completed the FSTD training and has successfully undertaken the type rating skill test, provided it does not exceed 2 hours of the flight training course.
- (2) courses approved for ZFTT
 - (i) During the specific simulator session before line flying under supervision (LIFUS), consideration should be given to varying conditions, for example:
 - (A) runway surface conditions;
 - (B) runway length;
 - (C) flap setting;
 - (D) power setting;
 - (E) crosswind and turbulence conditions; and
 - (F) maximum take-off mass (MTOM) and maximum landing mass (MLM).
 - (ii) At least one landing should be conducted as full-stop landing. The session should be flown in normal operation. Special attention should be given to the taxiing technique.
 - (iii) A training methodology should be agreed with the competent authority that ensures the trainee is fully competent with the exterior inspection of the aeroplane before conducting such an inspection un-supervised.
 - (iv) The LIFUS should be performed as soon as possible after the specific FFS session.
 - (v) The licence endorsement should be entered on the licence after the skill test, but before the first four take-offs and landings in the aeroplane. At the discretion of the competent authority, provisional or temporary endorsement and any restriction should be entered on the licence.
 - (vi) Where a specific arrangement exists between the ATO and the commercial air transport operator, the operator proficiency check (OPC) and the ZFTT specific details should be conducted using the operator's standard operating procedures (SOPs).
- (3) All training exercises should be designed to remain within the training envelope as determined by the ATO (Note: Further guidance regarding the training envelope can be found in [GM1 ORA.ATO.125](#) point (f)).

(I) Aeroplane without FFS

- (1) Flight training conducted solely in an aeroplane without the use of FSTDs cannot cover the crew resource management (CRM) and multicrew cockpit (MCC) aspects of MPA flight training, and for safety reasons cannot cover all emergency and abnormal aircraft operation required for the training and skill test. In such cases, the ATO should demonstrate to the competent authority that adequate training in these aspects can be achieved by other means. For training conducted solely on an MPA where two pilots are trained together without the use of an FSTD, a minimum of 8 hours of flight training as pilot flying (PF) for each pilot should normally be required. For training on a single-pilot aeroplane, 10 hours of flight training should normally be required. It is accepted that for some relatively simple single or multi-engine aircraft without systems such as pressurisation, flight management system (FMS) or electronic cockpit displays, this minimum may be reduced.
- (2) Aeroplane training normally involves an inherent delay in achieving an acceptable flight situation and configuration for training to be carried out in accordance with the agreed syllabus. These could include ATC or other traffic delay on the ground prior to take-off, the necessity to climb to height or transit to suitable training areas and the unavoidable need to physically reposition the aircraft for subsequent or repeat manoeuvres or instrument approaches. In such cases it should be ensured that the training syllabus provides adequate flexibility to enable the minimum amount of required flight training to be carried out.

- (Ia) Additional UPRT training as per point FCL.725.A(c) UPRT as per point FCL.725.A(c) should include the elements and components in table 1.

Table 1: Elements and respective components of upset prevention training

Elements and components		TK instruction	FSTD/ Aeroplane training
A.	Aerodynamics		
1.	General aerodynamic characteristics	•	
2.	Aeroplane certification and limitations	•	
3.	Aerodynamics (high and low altitudes)	•	•
4.	Aeroplane performance (high and low altitudes)	•	•
5.	AoA and stall awareness	•	•
6.	Stick shaker or other stall-warning device activation (as applicable)	•	•
7.	Stick pusher (as applicable)	•	•
8.	Mach effects (if applicable to the aeroplane type)	•	•
9.	Aeroplane stability	•	•
10.	Control surface fundamentals	•	•
11.	Use of trims	•	•
12.	Icing and contamination effects	•	•
13.	Propeller slipstream (as applicable)	•	•
B.	Causes of and contributing factors to upsets		
1.	Environmental	•	
2.	Pilot-induced	•	
3.	Mechanical (aeroplane systems)	•	

Elements and components		TK instruction	FSTD/ Aeroplane training
C.	Safety review of accidents and incidents relating to aeroplane upsets		
1.	Safety review of accidents and incidents relating to aeroplane upsets	•	
D.	G-load awareness and management		
1.	Positive/negative/increasing/decreasing G-loads	•	•
2.	Lateral G awareness (sideslip)	•	•
3.	G-load management	•	•
E.	Energy management		
1.	Kinetic energy vs potential energy vs effect of thrust-drag ratio on the total energy	•	•
F.	Flight path management		
1.	Relationship between pitch, power and performance	•	•
2.	Performance and effects of differing power plants (if applicable)	•	•
3.	Manual and automation inputs for guidance and control	•	•
4.	Type-specific characteristics	•	•
5.	Management of go-arounds from various stages during the approach	•	•
6.	Automation management	•	•
7.	Proper use of rudder	•	•
G.	Recognition		
1.	Type-specific examples of physiological, visual and instrument clues during developing and developed upsets	•	•
2.	Pitch/power/roll/yaw	•	•
3.	Effective scanning (effective monitoring)	•	•
4.	Type-specific stall protection systems and cues	•	•
5.	Criteria for identifying stalls and upsets	•	•
H.	System malfunction (including immediate handling and subsequent operational considerations, as applicable)		
1.	Flight control defects	•	•
2.	Engine failure (partial or full)	•	•
3.	Instrument failures	•	•
4.	Loss of reliable airspeed (see also point (lb) of this AMC)	•	•
5.	Automation failures	•	•
6.	Fly-by-wire (FBW) protection degradations	•	•
7.	Stall protection system failures including icing alerting systems	•	•

- (lb) Flight path management (manual or automatic, as appropriate) during unreliable airspeed indication and other failures at high altitude in aeroplanes with a maximum cruising altitude above FL300

The following training elements should be integrated into type rating training courses for aeroplanes with a maximum cruising altitude above FL300:

Element	TK instruction	FSTD / Aeroplane training
Basic flight physics principles concerning flight at high altitude, with a particular emphasis on the relative proximity of the critical Mach number and the stall, pitch behaviour, and an understanding of the reduced stall angle of attack when compared with low altitude flight.	•	•
Interaction of the automation (autopilot, flight director, auto-throttle/auto-thrust) and the consequences of failures inducing disconnection of the automation.	•	•
Consequences of an unreliable airspeed and other failures indication at high altitude and the need for the flight crew to promptly identify the failure and react with appropriate (minimal) control inputs to keep the aircraft in a safe envelope.	•	•
Degradation of FBW flight control laws/modes and its consequence on aircraft stability and flight envelope protections, including stall warnings.	•	•
Practical training, using appropriate simulators, on manual handling at high altitude in normal and in non-normal flight control laws/modes, with particular emphasis on pre-stall buffet, the reduced stall angle of attack when compared with low altitude flight, and the effect of pitch inputs on the aircraft trajectory and energy state.		•
The requirement to promptly and accurately apply the stall recovery procedure, as provided by the aircraft manufacturer, at the first indication of an impending stall. Differences between high-altitude and low-altitude stalls must be addressed.	•	•
Procedures for taking over and transferring manual control of the aircraft, especially for FBW aeroplanes with independent side-sticks.	•	•
Task sharing and crew coordination in high workload/stress conditions with appropriate call-out and acknowledgement to confirm changes to the aircraft flight control law/mode.	•	•

SKILL TEST

- (m) Upon completion of the flight training, the pilot will be required to undergo a skill test with an examiner to demonstrate adequate competency of aircraft operation for issue of the type rating. The skill test should be separate from the flight training syllabus, and provision for it cannot be included in the minimum requirements or training hours of the agreed flight training programme. The skill test may be conducted in an FFS, the aeroplane or, in exceptional circumstances, a combination of both.

COURSE COMPLETION CERTIFICATE

- (n) The HT, or a nominated representative, should certify that all training has been carried out before an applicant undertakes a skill test for the type rating to be included in the pilot's licence. If an ATO is unable to provide certain elements of the training that is required to be carried out on an aircraft the ATO may issue such a certificate confirming the completion of the ground training or the training in an FSTD.

AMC3 ORA.ATO.125 Training programme

ED Decision 2012/007/R

TYPE RATING COURSES – HELICOPTERS

- (a) Introduction

- (1) when developing the training programme for a type rating course, in addition to complying with the standards included in the OSD as established in accordance with Regulation (EC) 1702/2003 for the applicable type, the ATO should also follow any further recommendations contained therein.
 - (2) the course should, as far as possible, provide for integrated ground, FSTD and flight training designated to enable the student to operate safely and qualify for the grant of a type rating. The course should be directed towards a helicopter type, but where variants exist, all flying and ground training forming the basis of the course should relate to a single variant.
- (b) Variants
- (1) Familiarisation training: where a helicopter type rating also includes variants of the same aircraft type requiring familiarisation training, the additional familiarisation training may be included in the theoretical knowledge training of the initial type rating course.
 - (2) Differences training: where a helicopter type rating also includes variants of the same aircraft type for which difference training is required, the initial training course should be directed towards a single variant. Additional training to operate other variants within the same type rating should be completed after successful completion of the initial type rating course, although elements of this differences training may be undertaken at appropriate stages of the initial course, with the agreement of the competent authority.
- (c) Training in helicopter and FSTDs
- The training programme should specify the amounts of flight training in the helicopter type and in FSTDs (FFSs, flight training devices (FTDs), or other training devices (OTDs)). Where a suitable FFS is geographically remote from the normal training base, the competent authority may agree to some additional training being included in the programme at a remote facility.
- (d) Skill test
- The content of the flight training programme should be directed towards the skill test for that type. The practical training given in Part-FCL should be modified as necessary.
- The skill test may be completed in a helicopter, in an FFS or partially in a helicopter and in an FSTD. The use of an FSTD for skill tests is governed by the level of approval of the flight simulator and the previous experience of the candidate. Where an FSTD is not available, abnormal operations of systems should not be practised in a helicopter other than as allowed for in the skill test form for the type.
- (e) Phase progress tests and final theoretical knowledge examination
- Prior to the final theoretical knowledge examination covering the whole syllabus, the training programme should provide for phase progress tests associated with each phase of theoretical knowledge instruction. The phase progress tests should assess the candidate's knowledge on completion of each phase of the training programme.
- (f) Facilities: ground school equipment, training facilities and aids
- The ATO should provide, as a minimum, facilities for classroom instruction. Additional classroom training aids and equipment including, where appropriate, computers, should reflect the content of the course and the complexity of the helicopter. For multi-engine and multi-pilot helicopters, the minimum level of ground training aids should include equipment that provides a realistic cockpit working environment. Task analysis and the latest state-of-the-art training

technology is encouraged and should be fully incorporated into the training facilities wherever possible. Facilities for self and supervised testing should be available to the student.

(g) Training devices

An FTD or OTD may be provided to supplement classroom training in order to enable students to practice and consolidate theoretical instruction. Where suitable equipment is not available, or is not appropriate, a helicopter or flight simulator of the relevant variant should be available. If an FTD represents a different variant of the same helicopter type for which the student is being trained, then differences or familiarisation training is required.

(h) Computer-based training (CBT)

Where CBT aids are used as a training tool, the ATO should ensure that a fully qualified ground instructor is available at all times when such equipment is being used by course students. Other than for revision periods, CBT lessons should be briefed and debriefed by a qualified ground instructor.

(i) Theoretical knowledge instruction

The theoretical knowledge instruction training should meet the general objectives of giving the student:

- (1) a thorough knowledge of the helicopter structure, transmissions, rotors and equipment, powerplant and systems, and their associated limitations;
- (2) a knowledge of the positioning and operation of the cockpit controls and indicators for the helicopter and its systems;
- (3) a knowledge of performance, flight planning and monitoring, mass and balance, servicing and optional equipment items;
- (4) an understanding of system malfunctions, their effect on helicopter operations and interaction with other systems; and
- (5) the understanding of normal, abnormal and emergency procedures and giving the student the understanding of potential control problems near the edge of the handling envelope. In particular, the phenomenon of 'servo transparency' (also known as 'jack stall') should be covered for those helicopter types where it is a known problem.

The amount of time and the contents of the theoretical instruction will depend on the complexity of the helicopter type involved and, to some extent, on the previous experience of the student.

(j) Flight training

(1) FSTDs

The level of qualification and the complexity of the type will determine the amount of practical training that may be accomplished in an FSTD, including completion of the skill test. Prior to undertaking the skill test, a student should demonstrate competency in the skill test items during the practical training.

(2) Helicopter (with FSTD)

With the exception of courses approved for ZFTT, the amount of flight time in a helicopter should be adequate for completion of the skill test.

(3) Helicopters (without FSTD)

Whenever a helicopter is used for training, the amount of flight time practical training should be adequate for the completion of the skill test. The amount of flight training will depend on the complexity of the helicopter type involved and, to some extent, on the previous experience of the applicant.

AMC4 ORA.ATO.125 Training programme

ED Decision 2012/007/R

FLIGHT TEST TRAINING COURSES – AEROPLANES AND HELICOPTERS

- (a) Introduction
 - (1) The flight test training course should, as far as possible, provide for a continuous process of ground and flight training to enable the student to assimilate the knowledge and skills required to conduct flight testing safely and efficiently. The student's ability to do this should be determined by the demonstration of a satisfactory level of theoretical knowledge of flight testing determined by progressive checking of knowledge and examination and progressive assessment by the ATO during flying training. There should be no difference in the level of knowledge or competency required of the student, irrespective of the intended role of the student as test pilot or other flight test personnel (for example, flight test engineer) within the flight crew.
 - (2) The flight test training course should normally be conducted as a single, full-time course of study and training.
- (b) Programme of theoretical knowledge and flight training
 - (1) The training programme should specify the time allocated to theoretical knowledge training and flying training.
 - (2) If the ATO wishes to provide a flight test training course that includes credit for previous experience on flight testing activity, the entry requirements to such courses should be specified by the ATO and should define the minimum level of experience and qualification required of the flight test crew member.

GROUND TRAINING

- (c) Syllabus
 - (1) The ground training syllabus should provide for the student to gain a thorough understanding of flight testing techniques.
- (d) Theoretical knowledge instruction
 - (1) The theoretical knowledge instruction training should give the student a thorough knowledge of the academic requirements of flight testing.
- (e) Facilities and training aids
 - (1) The ATO should provide adequate facilities for classroom instruction and have available appropriately qualified and experienced instructors. Training aids should enable students to gain practical experience of flight testing covered by the theoretical knowledge syllabus and enable such practical application of the knowledge to be carried out in a multicrew environment. Facilities should be made available for student selfstudy outside the formal training programme.
- (f) Computer-based training (CBT)

- (1) CBT provides a valuable source of theoretical instruction, enabling the student to progress at his/her own pace within specified time limits. Many such systems ensure that syllabus subjects are fully covered and progress can be denied until a satisfactory assimilation of knowledge has been demonstrated. Such systems may allow self-study or distance learning, if they incorporate adequate knowledge testing procedures. When CBT is used as part of the theoretical knowledge instruction phase, the student should also have access to a suitably qualified instructor able to assist with areas of difficulty for the student.
- (g) Self-study and distance learning
 - (1) Elements of the theoretical knowledge syllabus may be adequately addressed by distance learning, if approved, or self-study, particularly when utilising CBT. Progress testing, either by self-assessed or instructor-evaluated means, should be included in any self-study programme. If self-study or distance learning is included in the theoretical knowledge training, the course should also provide for an adequate period of supervised consolidation and knowledge testing prior to the commencement of flight training.
- (h) Progress tests and final theoretical knowledge examination
 - (1) The theoretical knowledge training programme should provide for progressive testing of the assimilation of the required knowledge. This testing process should also provide for retesting of syllabus items so that a thorough understanding of the required knowledge is assured. This should be achieved by intervention by a qualified instructor or, if using CBT with a self-testing facility, and by further testing during the supervised consolidation phase of the ground course.
 - (2) The theoretical knowledge examinations should cover all areas of the theoretical knowledge syllabus. The examinations should be conducted as supervised written or oral knowledge tests without reference to course material. The pass mark (as defined by the ATO) assumes the achievement of satisfactory levels of knowledge during the progressive phase tests of the course. The student should be advised of any areas of lack of knowledge displayed during the examination and, if necessary, given remedial instruction.

FLIGHT TRAINING

- (i) Aeroplane and helicopter training
 - (1) It is widely accepted that flying training normally involves inherent delay in achieving an acceptable flight situation and configuration for training to be carried out in accordance with the agreed syllabus. These could include ATC or other traffic delay on the ground prior to take off, the necessity to climb to height or transit to suitable training areas and the unavoidable need to physically reposition the aircraft for subsequent or repeat manoeuvres or instrument approaches. In such cases it should be ensured that the training syllabus provides adequate flexibility to enable the minimum amount of required flight training to be carried out.

FINAL IN-FLIGHT EXERCISE

- (j) Upon completion of the flight test training, the test pilot or flight test engineer will be required to undergo in-flight exercise with a flight test instructor (FTI) to demonstrate adequate competency of flight testing for issue of the flight test rating. The final in-flight exercise must be conducted in an appropriate aeroplane or helicopter (as applicable).

COURSE COMPLETION CERTIFICATE

- (k) The HT is required to certify that the applicant has successfully completed the training course.

GM1 ORA.ATO.125 Training programme

ED Decision 2019/005/R

UPSET PREVENTION AND RECOVERY TRAINING (UPRT)

(a) General

The objective of the UPRT is to ensure that pilots are competent to prevent or recover from a developing or developed aeroplane upset. Prevention training prepares pilots to avoid upsets whereas recovery training prepares pilots to prevent an accident once an upset condition has developed.

(b) Human factors

Threat and Error Management (TEM) and Crew Resource Management (CRM) principles should be integrated into the UPRT. In particular, the surprise and startle effect as well as the importance of resilience development should be emphasised.

Training should also emphasise that an actual upset condition may expose pilots to significant physiological and psychological challenges, such as visual illusions, spatial disorientation and unusual G-forces, with the objective of developing strategies to deal with such challenges.

(c) Development of training scenarios

During the development of training scenarios, the ATO should ensure that all of the following is avoided:

- (a) negative training and negative transfer of training; and
- (b) training utilising predictive scenarios.

Please refer to Revision 2 of the Airplane Upset Recovery Training Aid (AURTA) for further guidance on the development of training scenarios.

(d) Additional guidance

Specific guidance to the UPRT elements and exercises is available in:

- (1) the latest revision of the ICAO Doc 10011 'Manual on Aeroplane Upset Prevention and Recovery Training';
- (2) Revision 3 of the Airplane Upset Prevention and Recovery Training Aid (AUPRTA); and
- (3) the Flight Safety Foundation publication 'A Practical Guide for Improving Flight Path Monitoring', November 2014.

(e) Training platform

- (1) When designing a training course, ATOs should select aeroplanes that are suitable for all the required training exercises. Where certain exercises require particular capabilities, then an ATO may consider the use of different aeroplanes for different exercises. Examples include basic UPRT or instrument flight training and the advanced UPRT course.
- (2) For basic UPRT training conducted during the CPL or ATP courses, it is not anticipated that aerobatic category aeroplanes will be required or that aircraft need to be certificated for intentional spins. Aeroplanes with a maximum bank angle limitation may not be suitable for exercises such as steep turns or recovery from spiral dive.

- (3) For the advanced UPRT course (FCL.745), the use of an aeroplane certificated in the aerobatic category will provide the greatest safety margin. Aeroplanes certificated in the normal or utility category may also be suitable provided the exercises used during the training take into account the capabilities of the aeroplane and are planned to remain within the training envelope for the aeroplane, as determined by the ATO (see point (f)).
- (f) Training envelope
- The training envelope is the envelope within which all training exercises will be carried out. It should be specified by the ATO in terms of the range of attitudes, speed and g-loads that can be used for training, taking into account:
- (1) the training environment;
 - (2) the capabilities of the instructors; and
 - (3) in the case of training in FSTDs, the limitations of the FSTD (as per GM3 FCL.010 for the FSTD training envelope); and
 - (4) in the case of training in aeroplanes, the capabilities and certification of the aircraft, while considering a margin of safety in order to ensure that unintentional deviations from the training envelope will not exceed aircraft limitations. Different training envelopes may be specified for different aeroplane types even within a single training course.

ORA.ATO.130 Training manual and operations manual

Regulation (EU) No 1178/2011

- (a) The ATO shall establish and maintain a training manual and operations manual containing information and instructions to enable personnel to perform their duties and to give guidance to students on how to comply with course requirements.
- (b) The ATO shall make available to staff and, where appropriate, to students the information contained in the training manual, the operations manual and the ATO's approval documentation.
- (c) In the case of ATOs providing flight test training, the operations manual shall comply with the requirements for the flight test operations manual, as established in Part-21.
- (d) The operations manual shall establish flight time limitation schemes for flight instructors, including the maximum flying hours, maximum flying duty hours and minimum rest time between instructional duties in accordance with Part-ORO.

ORA.ATO.135 Training aircraft and FSTDs

Regulation (EU) 2019/1747

- (a) The ATO shall use an adequate fleet of training aircraft or FSTDs appropriately equipped for the training courses provided. The fleet of aircraft shall be composed of aircraft that comply with all requirements defined in Regulation (EU) 2018/1139. Aircraft that fall under points (a), (b), (c) or (d) of Annex I to Regulation (EU) 2018/1139, may be used for training if all of the following conditions are met:
 - (1) during an evaluation process the competent authority has confirmed a level of safety comparable to the one defined by all essential requirements laid down in Annex II to Regulation (EU) 2018/1139;
 - (2) the competent authority has authorised the use of the aircraft for training in the ATO.

- (b) The ATO shall only provide training in FSTDs when it demonstrates to the competent authority:
 - (1) the adequacy between the FSTD specifications and the related training programme;
 - (2) that the FSTDs used comply with the relevant requirements of Part-FCL;
 - (3) in the case of full flight simulators (FFSs), that the FFS adequately represents the relevant type of aircraft; and
 - (4) that it has put in place a system to adequately monitor changes to the FSTD and to ensure that those changes do not affect the adequacy of the training programme.
- (c) If the aircraft used for the skill test is of a different type to the FFS used for the visual flight training, the maximum credit shall be limited to that allocated for flight and navigation procedures trainer II (FNPT II) for aeroplanes and FNPT II/III for helicopters in the relevant flight training programme.
- (d) Flight test training organisations. Aircraft used for flight test training shall be appropriately equipped with flight testing instrumentation, according to the purpose of the training.

AMC1 ORA.ATO.135 Training aircraft and FSTDs

ED Decision 2025/002/R

ALL ATOs, EXCEPT THOSE PROVIDING FLIGHT TEST TRAINING

- (a) The number of training aircraft may be affected by the availability of FSTDs.
- (b) Each training aircraft should be:
 - (1) equipped as required in the training specifications concerning the course in which it is used;
 - (2) except in the case of balloons or single-seat aircraft, fitted with primary flight controls that are instantly accessible by both the student and the instructor (for example dual flight controls or a centre control stick). Swing-over flight controls should not be used.
- (c) The fleet should include, as appropriate to the courses of training:
 - (1) aircraft suitably equipped to simulate instrument meteorological conditions (IMC) and for the instrument flight training required. For flight training and testing for the instrument rating and the basic instrument rating (BIR), an adequate number of IFR-certificated aircraft should be available;
 - (2) in the case of aeroplanes and sailplanes, aircraft suitable for demonstrating stalling and spin avoidance;
 - (3) for the flight instructor (FI) training courses on aeroplanes and sailplanes, aircraft suitable for spin recovery at the developed stage;
 - (4) in the case of helicopters, helicopters suitable for autorotation demonstration;
 - (5) in the case of a non-complex ATO, one aircraft fulfilling all the required characteristics for a training aircraft might be sufficient;
 - (6) each FSTD should be equipped as required in the training specifications concerning the course in which it is used.

AMC2 ORA.ATO.135 Training aircraft and FSTDs

ED Decision 2020/005/R

EVALUATION PROCESS

Two cases for the evaluation process of Annex-I aircraft are distinguished:

- (a) Annex-I aircraft that hold an ICAO-level certificate of airworthiness (CoA)
 - (1) To support the evaluation process performed by the competent authority and provide the competent authority with sufficient data related to the aircraft in question, an instructor who is qualified in accordance with Annex I (Part-FCL) to Regulation (EU) No 1178/2011 and nominated by the head of training (HT) of the ATO should assess that the aircraft is appropriately equipped and suitable for the training courses provided. The result of this assessment should be submitted to the competent authority and may be included already in the application for the authorisation.
 - (2) During the evaluation process, the competent authority should consider aircraft that hold a CoA issued in accordance with Annex 8 to the Chicago Convention to provide a level of safety comparable to that required by Annex II to the Basic Regulation, unless the competent authority determines that the airworthiness requirements used for certification of the aircraft, or the service experience, or the safety system of the State of design, do not provide for a comparable level of safety.
- (b) Annex-I aircraft that do not hold an ICAO-level CoA

Before the inclusion of these aircraft in the fleet of an ATO and their use in training to obtain Part-FCL licences and ratings, the ATO should apply for the authorisation to the competent authority that should perform the evaluation process in the following order:

- (1) Initial assessment by the competent authority and criteria taken into consideration

The competent authority should take into account the following criteria (non-exhaustive list):

 - (i) national airworthiness requirements based on which the aircraft CoA was issued;
 - (ii) aircraft similarities to a certified variant;
 - (iii) aircraft with a satisfactory in-service experience as training aircraft;
 - (iv) simple and conventional aircraft design;
 - (v) aircraft that does not have hazardous design features or details, judging by experience; and
 - (vi) operable aircraft systems, equipment, and appliances that do not require exceptional skills or strength.
- (2) Additional assessment by a qualified instructor

To support the evaluation process performed by the competent authority and provide the competent authority with sufficient data related to the aircraft in question, after the positive initial assessment by the competent authority as per point (1), an instructor who is qualified in accordance with Part-FCL and nominated by the HT of the ATO should show through an evaluation report that the aircraft is appropriately equipped and suitable for the training courses provided. That evaluation report should consider all of the following criteria:

- (i) the aircraft should be safely controllable and manoeuvrable under all anticipated operating conditions, including after failure of one or more propulsion systems;
- (ii) the aircraft should allow for a smooth transition from one flight phase to another without requiring exceptional piloting skills, alertness, strength, or workload under any probable operating conditions;
- (iii) the aircraft should have sufficient stability to ensure that the demands made on the pilot are not excessive, considering the phase and duration of flight; and
- (iv) the assessment should take into account control forces, flight deck environment, pilot workload, and other human factors (HF) considerations, depending on the phase and duration of flight.

Subject to a positive evaluation report as per point (2), the competent authority should issue the authorisation.

ORA.ATO.140 Aerodromes and operating sites

Regulation (EU) No 1178/2011

When providing flight training on an aircraft, the ATO shall use aerodromes or operating sites that have the appropriate facilities and characteristics to allow training of the manoeuvres relevant, taking into account the training provided and the category and type of aircraft used.

AMC1 ORA.ATO.140 Aerodromes and operating sites

ED Decision 2012/007/R

GENERAL

- (a) Except in the case of balloons, the base aerodrome or operating site and any alternative base aerodromes at which flight training is being conducted should have at least the following facilities:
 - (1) at least one runway or final approach and take-off area (FATO) that allows training aircraft to make a normal take-off or landing within the performance limits of all the aircraft used for the training flights.
 - (2) a wind direction indicator that is visible at ground level from the ends of each runway or at the appropriate holding points;
 - (3) adequate runway electrical lighting if used for night training;
 - (4) an air traffic service, except for uncontrolled aerodromes or operating sites where the training requirements may be satisfied safely by another acceptable means of air-to-ground communication.
- (b) Except in the case of ATOs providing flight test training, in addition to (a), for helicopters, training sites should be available for:
 - (1) confined area operation training;
 - (2) simulated engine off autorotation; and
 - (3) sloping ground operation.
- (c) In the case of balloons, the take-off sites used by the ATO should allow a normal take-off and clearing of all obstacles in the take-off flight path by at least 50 ft.

ORA.ATO.145 Pre-requisites for training

Regulation (EU) No 70/2014

- (a) The ATO shall ensure that the students meet all the pre-requisites for training established in Part-Medical, Part- FCL, and, if applicable, as defined in the mandatory part of the operational suitability data established in accordance with Regulation (EU) No 748/2012.
- (b) In the case of ATOs providing flight test training, the students shall meet all the pre-requisites for training established in accordance with Regulation (EU) No 748/2012.

AMC1 ORA.ATO.145 Pre-requisites for training

ED Decision 2012/007/R

ENTRANCE REQUIREMENTS

ATOs providing training for other than the LAPL, PPL, SPL or BPL and the associated ratings and certificates should establish entrance requirements for students in their procedures. The entrance requirements should ensure that the students have enough knowledge, particularly of physics and mathematics, to be able to follow the courses.

ORA.ATO.150 Training in third countries

Regulation (EU) No 290/2012

When the ATO is approved to provide training for the instrument rating (IR) in third countries:

- (a) the training programme shall include acclimatisation flying in one of the Member States before the IR skilltest is taken; and
- (b) the IR skill test shall be taken in one of the Member States.

SECTION II – ADDITIONAL REQUIREMENTS FOR ATOs PROVIDING TRAINING FOR CPL, MPL AND ATPL AND THE ASSOCIATED RATINGS AND CERTIFICATES

ORA.ATO.210 Personnel requirements

Regulation (EU) No 1178/2011

- (a) Head of training (HT). Except in the case of ATOs providing flight test training, the nominated HT shall have extensive experience in training as an instructor for professional pilot licences and associated ratings or certificates.
- (b) Chief flight instructor (CFI). The ATO providing flight instruction shall nominate a CFI who shall be responsible for the supervision of flight and flight simulation training instructors and for the standardisation of all flight instruction and flight simulation instruction. The CFI shall hold the highest professional pilot licence and associated ratings related to the flight training courses conducted and hold an instructor certificate with the privilege to instruct for at least one of the training courses provided.
- (c) Chief theoretical knowledge instructor (CTKI). The ATO providing theoretical knowledge instruction shall nominate a CTKI who shall be responsible for the supervision of all theoretical knowledge instructors and for the standardisation of all theoretical knowledge instruction. The CTKI shall have extensive experience as a theoretical knowledge instructor in the areas relevant for the training provided by the ATO.

AMC1 ORA.ATO.210 Personnel requirements

ED Decision 2025/002/R

GENERAL

- (a) The management structure should ensure supervision of all grades of personnel by persons having the experience and qualities necessary to ensure the maintenance of high standards. Details of the management structure, indicating individual responsibilities, should be included in the ATOs operations manual.
- (b) The ATO should demonstrate to the competent authority that an adequate number of qualified, competent staff is employed.
- (c) In the case of an ATO offering integrated courses, the head of training (HT), the chief flying instructor (CFI) and the chief theoretical-knowledge instructor (CTKI) should be employed full-time or part-time, depending upon the scope of training offered. The three positions of HT, CFI, and CTKI should not be combined but should be filled by three persons.
- (d) In all other cases the positions of HT, CFI and CTKI may be combined and filled by one or two persons with extensive experience in the training conducted by the training organisation, full-time or part-time, depending upon the scope of training offered, provided that the organisation has demonstrated to the competent authority the adequacy of such personnel set-up for its organisation (point [ORA.GEN.200\(b\)](#)).
- (e) In the case of an ATO that provides flight training only, no CTKI function is required in the ATO. In the case of an ATO that provides theoretical-knowledge instruction only, no CFI function is required in the ATO.
- (f) The ratio of all students to all flight instructors, excluding the HT, should not exceed 6:1.

- (g) Classes in ground subjects that require maximal supervision or intensive practical work should not include more than 28 students.

THEORETICAL KNOWLEDGE INSTRUCTORS

- (h) The theoretical knowledge instruction for type or class ratings should be conducted by instructors holding the appropriate type or class rating, or having appropriate experience in aviation and knowledge of the aircraft concerned.
- (i) For this purpose, a flight engineer, a maintenance engineer or a flight operations officer should be considered as having appropriate experience in aviation and knowledge of the aircraft concerned.

AMC2 ORA.ATO.210 Personnel requirements*ED Decision 2012/007/R***QUALIFICATION OF HEAD OF TRAINING AND CHIEF FLIGHT INSTRUCTOR**

- (a) Head of training (HT)

The nominated HT should hold or have held in the 3 years prior to first appointment as HT, a professional pilot licence and associated ratings or certificates issued in accordance with Part-FCL, related to the flight training courses provided.

- (b) Chief flight instructor (CFI)

- (1) The CFI may delegate standardisation and supervision to the flight instructors. In all cases it is the CFI who is ultimately responsible for ensuring quality and standards.
- (2) The CFI should, except in the case of ATOs providing flight test training, have completed 1 000 hours of flight time as pilot-in-command (PIC).

At least 500 of those hours should be on flying instructional duties related to the flying courses provided, of which 200 hours may be instrument ground time.

ORA.ATO.225 Training programme*Regulation (EU) No 1178/2011*

- (a) The training programme shall include a breakdown of flight and theoretical knowledge instruction, presented in a week-by-week or phase layout, a list of standard exercises and a syllabus summary.
- (b) The content and sequence of the training programme shall be specified in the training manual.

ORA.ATO.230 Training manual and operations manual*Regulation (EU) No 290/2012*

- (a) The training manual shall state the standards, objectives and training goals for each phase of training that the students are required to comply with and shall address the following subjects:
- training plan,
 - briefing and air exercises,
 - flight training in an FSTD, if applicable,
 - theoretical knowledge instruction.

- (b) The operations manual shall provide relevant information to particular groups of personnel, as flight instructors, flight simulation training instructors, theoretical knowledge instructors, operations and maintenance personnel, and shall include general, technical, route and staff training information.

AMC1 ORA.ATO.230(a) Training manual and operations manual

ED Decision 2018/001/R

TRAINING MANUAL

Training manuals for use at an ATO to conduct integrated or modular flight training courses should include the following:

- (a) The training plan:

(1) The aim of the course (ATP, CPL/IR, CPL, etc. as applicable)	A statement of what the student is expected to do as a result of the training, the level of performance, and the training constraints to be observed.
(2) Pre-entry requirements	(i) Minimum age, educational requirements (including language), medical requirements; (ii) Any individual Member State requirements.
(3) Credits for previous experience	To be obtained from the competent authority before training begins.
(4) Training syllabi	As applicable, the flying syllabus (single-engine or multiengine, as applicable), the flight simulation training syllabus and the theoretical knowledge training syllabus.
(5) The time scale and scale, in weeks, for each syllabus	Arrangements of the course and the integration of syllabi time.
(6) Training programme	(i) The general arrangements of daily and weekly programmes for flying, theoretical knowledge training and training in FSTDs, if applicable; (ii) Bad weather constraints; (iii) Programme constraints in terms of maximum student training times, (flying, theoretical knowledge, on FSTDs), for example per day, week or month; (iv) Restrictions in respect of duty periods for students; (v) Duration of dual and solo flights at various stages; (vi) Maximum flying hours in any day or night; (vii) Maximum number of training flights in any day or night; (viii) Minimum rest period between duty periods.
(7) Training records	(i) Rules for security of records and documents; (ii) Attendance records; (iii) The form of training records to be kept; (iv) Persons responsible for checking records and students' log books; (v) The nature and frequency of record checks; (vi) Standardisation of entries in training records; (vii) Rules concerning log book entries.
(8) Safety training	(i) Individual responsibilities; (ii) Essential exercises; (iii) Emergency drills (frequency); (iv) Dual checks (frequency at various stages); (v) Requirement before first solo day, night or navigation etc. if applicable.

(9) Assessments, tests and examinations	(i) Flying: (A) progress checks; (B) skill tests. (ii) Theoretical knowledge: (A) progress tests; (B) theoretical knowledge examinations. (C) Area 100 KSA assessments. (iii) Authorisation for test; (iv) Rules concerning refresher training before retest; (v) Test and assessment reports and records; (vi) Procedures for examination paper preparation, type of question and assessment, standard required for 'pass'; (vii) Procedure for question analysis and review and for raising replacement papers; (viii) Examination resit procedures.
(10) Training effectiveness	(i) Individual responsibilities; (ii) General assessment; (iii) Liaison between departments; (iv) Identification of unsatisfactory progress (individual students); (v) Actions to correct unsatisfactory progress; (vi) Procedure for changing instructors; (vii) Maximum number of instructor changes per student; (viii) Internal feedback system for detecting training deficiencies; (ix) Procedure for suspending a student from training; (x) Discipline; (xi) Reporting and documentation.
(11) Standards and level of performance at various stages	(i) Individual responsibilities; (ii) Standardisation; (iii) Standardisation requirements and procedures; (iv) Application of test criteria.

(b) Briefing and air exercises:

(1) Air exercise	A detailed statement of the content specification of all the air exercises to be taught, arranged in the sequence to be flown with main and subtitles.
(2) Air exercise reference list	An abbreviated list of the above exercises giving only main and subtitles for quick reference, and preferably in flip-card form to facilitate daily use by instructors.
(3) Course structure: phase of training	A statement of how the course will be divided into phases, indication of how the above air exercises will be divided between the phases and how they will be arranged to ensure that they are completed in the most suitable learning sequence and that essential (emergency) exercises are repeated at the correct frequency. Also, the syllabus hours for each phase and for groups of exercises within each phase should be stated and when progress tests are to be conducted, etc.
(4) Course structure: integration of syllabi	The manner in which theoretical knowledge and flight training in an aircraft or an FSTD will be integrated so that as the flying training exercises are carried out students will be able to apply the knowledge gained from the associated theoretical knowledge instruction and flight training.

(5) Student progress	The requirement for student progress and include a brief but specific statement of what a student is expected to be able to do and the standard of proficiency he/she must achieve before progressing from one phase of air exercise training to the next. Include minimum experience requirements in terms of hours, satisfactory exercise completion, etc. as necessary before significant exercises, for example night flying.
(6) Instructional methods	The ATO requirements, particularly in respect of pre- and postflying briefing, adherence to syllabi and training specifications, authorisation of solo flights, etc.
(7) Progress tests	The instructions given to examining staff in respect of the conduct and documentation of all progress tests.
(8) Glossary of terms	Definition of significant terms as necessary.
(9) Appendices	(i) Progress test report forms; (ii) Skill test report forms; (iii) ATO certificates of experience, competence, etc. as required.

(c) Flight training in an FSTD, if applicable: Structure generally as for (b)

(d) Theoretical knowledge instruction:

(1) Structure of the theoretical knowledge course	A statement of the structure of the course, including the general sequence of the topics to be taught in each subject, the time allocated to each topic, the breakdown per subject and an example of a course schedule. Distance learning courses should include instructions of the material to be studied for individual elements of the course.
(2) Lesson plans	A description of each lesson or group of lessons including teaching materials, training aids, progress test organisation and inter-connection of topics with other subjects.
(3) Teaching materials	Specification of the training aids to be used (for example study materials, course manual references, exercises, self-study materials, demonstration equipment).
(4) Student progress	The requirement for student progress, including a brief but specific statement of the standard that must be achieved and the mechanism for achieving this, before application for theoretical knowledge examinations.
(5) Progress testing	The organisation of progress testing in each subject, including topics covered, evaluation methods and documentation.
(6) Review procedure	The procedure to be followed if the standard required at any stage of the course is not achieved, including an agreed action plan with remedial training if required.
(7) Appendices	(i) Examples of Area 100 KSA summative assessments; (ii) Area 100 KSA mental maths test example.

AMC2 ORA.ATO.230(a) Training manual and operations manual

ED Decision 2018/001/R

THEORETICAL KNOWLEDGE COURSE DESIGN REQUIREMENTS

An ATO that delivers theoretical knowledge instruction for professional pilot licences should ensure that:

- (a) the courses are designed and developed using the instructional systems design (ISD) methodology, which is supported by a robust and effective management system;
- (b) the courses include a standardised and dynamic assessment and testing system;
- (c) instructors that deliver KSA instruction have received appropriate training covering at least learning styles, teaching methods, facilitation techniques, threat and error management (TEM), the applicable competencies, and the content of the subject(s) and exercises that they are to deliver;
- (d) the recurrent training of instructors is conducted at least annually;
- (e) the instructors that are responsible for assessing Area 100 KSA have received appropriate training regarding the assessment(s) that they are to conduct, and are to be standardised to ensure that the assessment grades awarded are consistent across the ATO; this standardisation should include at least familiarisation with the performance indicators, the ATO's word pictures for grading, and the ATO's debriefing system; and
- (f) recurrent standardisation training is conducted at least annually to ensure continued inter-rater reliability.

AMC3 ORA.ATO.230(a) Training manual and operations manual

ED Decision 2018/001/R

AREA KSA 100 02 AND 100 03 LEARNING OBJECTIVES, ASSESSMENTS AND RECORDS

- (a) An ATO that delivers theoretical knowledge instruction for professional pilot licences should ensure that for the learning objectives (LOs) in topics 100 02 and 100 03 of Area 100 KSA there are at least two summative assessments and at least one formative assessment. The summative assessments are to be documented in the student's training records. Both the summative assessments and the formative assessment(s) should be debriefed.
- (b) The formative assessment(s) should:
 - (1) be designed such that the student has the opportunity to ask questions and develop competencies in most of the LOs in 100 02 and 100 03 of Area 100 KSA;
 - (2) be conducted during the training; the ATO may in addition conduct a formative evaluation (continuous assessment) over a specified phase of the course; and
 - (3) be conducted by an instructor that is trained to deliver the formative assessment.
- (c) The summative assessments should:
 - (1) be designed so that they collectively give the student the opportunity to demonstrate competency in all LOs in 100 02 and 100 03 of Area 100 KSA; each individual summative assessment may address some of the LOs in 100 02 and 100 03 of Area 100 KSA;
 - (2) be satisfactorily completed before the student is recommended by the ATO for their first attempt to take the final theoretical knowledge examination paper, and the outcome of the assessments should be included in the student's training record;
 - (3) require that for a student to be considered that they have achieved a 'Satisfactory' standard, they:
 - (i) meet at least 35 % (which defines the term 'some' used in the word pictures) of the indicators relevant to the assessment exercise, in each competency;

- (ii) have an overall positive effect on the outcome or completion of the exercise without any external input from the instructor, or where the assessment requires the instructor to facilitate the exercise, without the instructor providing any knowledge or corrective input to assist in the completion of the exercise; and
 - (4) be conducted by an instructor that is trained to deliver the summative assessments.
- (d) The training manual should include the following elements regarding the theoretical knowledge training and assessment of the LOs in topics 100 02 and 100 03 of Area 100 KSA:
 - (1) the positions, or range of positions, of the formative assessment exercise(s) and summative assessment exercises in the training programme;
 - (2) a description of the summative assessments, including a matrix that shows which Area 100 KSA LOs are covered in each exercise;
 - (3) the grading system of the Area 100 KSA summative assessment and a description of the ATO's minimum required standard;
 - (4) the template for the information about Area 100 KSA to be included in the student's training record, which should include at least the dates and result ('Pass' or 'Fail') of the summative assessments and the date and score of the mental maths test;
 - (5) the method of assessment debrief for each summative and formative assessment;
 - (6) for a student who performs below the satisfactory standard in a summative assessment(s), the method to further develop the student's competencies and how to conduct the reassessment.
- (e) Access to the information on Area 100 KSA kept in the student's training records should be restricted to the student and authorised ATO personnel, and should not be disclosed outside the ATO. The information on the record should first be de-identified before it is used to support course design improvements.

AMC4 ORA.ATO.230(a) Training manual and operations manual

ED Decision 2018/001/R

AREA 100 04 LEARNING OBJECTIVES: MENTAL MATHS TEST AND RECORDS

- (a) An ATO that delivers theoretical knowledge instruction for professional pilot licences should ensure that at least one KSA mental maths test is conducted and the outcome(s) documented in the student's training records.
- (b) The mental maths test(s) may be written or oral in format and should, where possible, be scenario-based, with at least two questions per LO in topic 100 04 of Area 100 KSA.
- (c) The minimum score to pass the Area 100 KSA mental maths test(s) should be 75 % of the marks allocated to a test. However, a higher pass mark may be defined by the ATO.
- (d) The mental maths test(s) should be satisfactorily completed before the student is recommended by the ATO for their first attempt to take their final theoretical knowledge examination paper.

GM1 ORA.ATO.230(a) Training manual and operations manual

ED Decision 2018/001/R

ASSESSMENT OF STUDENTS IN AREA 100 KSA

- (a) The Area 100 KSA formative assessment(s) and summative assessments may include but not be limited to: written planning exercises combining multiple subjects; practical exercises using training devices (if available); scenario-based oral board (viva voce); scenario-based communications exercises; written assignments or project work; and preparation and delivery of group or individual presentations.
- (b) The format of formative and summative assessment debriefs should be effective, highlighting the student's strengths and weaknesses and enabling future improvement.

GM2 ORA.ATO.230(a) Training manual and operations manual

ED Decision 2018/001/R

AREA 100 KSA WORD PICTURES

- (a) 'Word pictures' are a proven assessment tool that standardises pilot core competencies, and can be used to assess student's competency in the Area 100 KSA LOs in topics 100 02 and 100 03. Word pictures describe the student's performance. Each word picture is associated with a numerical grade; within the range of grades, the minimum acceptable standard is defined. Additionally, a word picture describing performance that falls below the minimum satisfactory standard should be included in the range, as well as additional word pictures that relate to grades which exceed this minimum satisfactory standard.

Word pictures enable the standardisation of the assessment performance and facilitate inter-rater reliability within an ATO.

- (b) This GM provides two examples of word pictures.
- (c) The most commonly used word pictures are shown in Section A below. They are based on performance indicators, which explain what the student should demonstrate in order to attain the specific Area 100 KSA LOs that are addressed by the assessment exercise. Word pictures are formed of elements that contain the following:
 - (1) HOW MANY of the performance indicators were observed and, where relevant, HOW OFTEN;
 - (2) HOW WELL the competency was demonstrated in the assessment exercise to have an overall positive effect on the outcome or completion of the assessment exercise;
 - (3) the level of success in the OUTCOME of the assessment exercise.
- (d) An ATO could establish its own set of word picture descriptions as long as they are comparable in the grading of each competency, similar to the 'Communication' and 'Application of knowledge, UPRT and resilience' word pictures example in Section B below.
- (e) The advantage of word pictures is that they provide meaningful and standard data to enable identification of individual, crew, class, instructor and ATO trends, which can be analysed in order to provide feedback for further improvement or development.
- (f) An ATO should ensure that the detailed information obtained through its grading in Area 100 KSA is de-identified before using it to support course improvement.

SECTION A — EXAMPLE 1

AREA 100 KSA WORD PICTURE GRADE LEVELS (USING INDICATORS)

- (g) The example shown below in this Section contains the most commonly used word pictures, which are formed of elements that contain the following:

- (1) HOW MANY of the performance indicators in the table further below relevant to that summative assessment were observed in that competency (as a percentage);
 - (2) HOW WELL the competency was demonstrated in the assessment; and
 - (3) the level of success in the OUTCOME of the summative assessment.
- (h) In order to satisfactorily complete an Area KSA 100 summative assessment, the student should reach at least the minimum satisfactory level in each competency covered by that assessment. In case the student fails to reach the minimum satisfactory level in each competency, the student should repeat the summative assessment or another summative assessment that covers the competency(ies) where performance was previously assessed as unsatisfactory.

Table 1: Example generic competency framework that can be applied for assessing the student's level of performance

Competency	Level 1 Unsatisfactory	Level 2 Satisfactory	Level 3 Good	Level 4 Very good	Level 5 Excellent
General description of each competency level. To be applied to each individual competency in LOs 100 02 and 100 03 of Area 100 KSA.	The student's performance in this competency was ineffective or inadequate, which in relation to this competency had a neutral or negative effect on others or on the outcome of the exercise. The student showed none or few of the relevant performance indicators in this competency.	The student's performance in this competency was satisfactory, which had a slightly positive effect on the satisfactory outcome of the exercise, and in group situations had a slightly positive effect on others. The student showed at least some* of the relevant performance indicators in this competency.	The student's performance in this competency was effective, which in the case of an exercise where the student is the only participant, significantly contributed to a good outcome. In group situations, the student's contribution had a good effect on others and significantly contributed to the overall outcome of the exercise. The student showed most of the relevant performance indicators to a good standard.	The student's performance in this competency was highly effective, which in the case of an exercise where the student is the only participant, significantly enhanced the very good outcome. In group situations, the student's contribution had a very good effect on others and significantly enhanced the overall outcome of the exercise. The student showed most or all of the relevant performance indicators to a very good standard.	The student's performance in this competency was exemplary, which in the case of an exercise where the student is the only participant, had an outstanding effect on the excellent outcome of the exercise. In group situations, the student's contribution had an excellent effect on others and had an outstanding effect on the overall outcome of the exercise. The student showed all of the relevant performance indicators to an excellent standard.

* 'Some' is defined as showing at least 35 % of the performance indicators in that competency, which were relevant to that exercise.

AREA 100 KSA ASSESSMENT PERFORMANCE INDICATORS

- (i) The performance indicators that relate to the LOs in topics 100 02 and 100 03 of Area 100 KSA can be used to form a word picture.

Table 2: Performance indicators relevant to the LOs in topics 100 02 to 100 03 of Area 100 KSA

Competency	Competency description	Indicators
Communication	Demonstrates effective oral, non-verbal and written communication skills in classroom exercise and assessment situations	— Ensures the recipient is ready and prepared to receive the information. — Selects appropriately what, when, how and with whom to communicate. — Conveys messages clearly, accurately and concisely. — Confirms that the recipient correctly understands important information. — Listens actively and demonstrates understanding of the information they receive. — Asks relevant and effective questions. — Adheres to standard radio-telephony phraseology. — Accurately reads, interprets, constructs and responds to given documentation in English. — Correctly interprets non-verbal communication. — Uses eye contact, body language and gestures that are consistent with and support verbal messages.
Leadership and teamwork	Displays effective leadership and teamwork.	— Creates an atmosphere of open communication and encourages team participation. — Uses initiative and gives directions when required. — Admits mistakes and takes responsibility. — Anticipates and responds appropriately to others' needs. — Carries out instructions when directed. — Communicates relevant concerns and intentions. — Gives and receives feedback constructively. — Demonstrates empathy and shows respect and tolerance for others. — Engages others in planning and allocates activities fairly and appropriately according to abilities. — Addresses and resolves conflicts and disagreements in a constructive manner. — Projects self-control.
Problem-solving and decision-making	Accurately identifies risks and resolves problems. Uses the appropriate decision-making processes.	— Seeks accurate and adequate information from appropriate sources. — Identifies and verifies what and why things have gone wrong. — Employs proper problem-solving strategies. — Perseveres in working through problems.

		— Uses appropriate decision-making processes in a timely manner. — Sets priorities appropriately. — Identifies and considers options effectively. — Monitors, reviews, and adapts decisions as required. — Identifies and manages risks effectively.
Situation awareness	Perceives and comprehends all the relevant information available, anticipates what could happen that could affect the exercise or situations discussed in the classroom, and gives effective solutions to resolve the situation.	— Identifies and assesses accurately the general environment as it may affect the operation. — Identifies and manages threats, errors, and undesirable aircraft states.
Workload management	Manages available resources or time to efficiently prioritise and complete or perform tasks in a timely manner.	— Maintains self-control. — Plans, prioritises and schedules tasks effectively. — Manages time efficiently when carrying out tasks. — Offers and accepts assistance, delegates when necessary, and asks for help early. — Manages and recovers from interruptions, distractions, variations, and failures effectively.
Application of knowledge, UPRT and resilience	Demonstrates correct and deep understanding of the subject(s), and is able to effectively relate this knowledge between subjects and apply the knowledge for effective threat and error management (TEM).	— Correctly completes pre-flight planning in the practical exercise. — Demonstrates KSA and TEM relating to phases of flight in the ground training environment. — Correctly and effectively applies knowledge to identify and manage threats and errors that could lead to a potential upset in scenario situations. — Recognises potential upset ‘threats’ and suggests effective ‘threat management’ in scenario situations. — Recognises potential upset ‘errors’ and suggests effective ‘error management’ in scenario situations. — Identifies the causes of and contributing factors to upsets in aircraft accident and incident reviews and in reported recovered situations or scenarios. — Is resilient, i.e. recognises and adapts to disruptions during scenarios and other exercises. — Identifies the signs of stress and discusses the effects of stress, fatigue and aviation lifestyle on situation awareness, including how to cope with these in order to maintain situation awareness.

SECTION B — EXAMPLE 2

AREA 100 KSA WORD PICTURES (USING DESCRIPTIONS)

- (j) An ATO may devise its own word pictures for each of the competencies, to be used alongside or combined with those given in EXAMPLE 1.

A descriptive word picture typically includes descriptive examples that the ATO's instructors could readily identify and then equate to given competency levels. Below are two examples (for 'Communication' and 'Application of knowledge, UPRT and resilience').

Table 3: Example word pictures using descriptions

Competency	Level 1 Unsatisfactory	Level 2 Satisfactory	Level 3 Good	Level 4 Very good	Level 5 Excellent
Communication	The student's performance in communication had a neutral or negative effect on the exercise or situation. The student may not have contributed to the exercise or the communication was unclear or insufficient. The student may have occasionally interrupted others, not listened, or showed frustration or inappropriate nonobjective communication. The student may have asked unrelated or unclear questions, or provided insufficient clarity in directions or comments for others to understand. The student may have written without structure or clarity.	The student's written or oral communication was sufficient to convey the intent of the exercise. In oral communication the listener may have rarely needed to ask for clarification which the student then positively and clearly provided. The student listened to instructions but may have occasionally been reticent and hesitant to ask questions or make comments. The student may have rarely shown underconfidence or passiveness during the exercise. However, overall their communication was sufficient to ensure a positive outcome of the exercise.	The student's written or oral communication was good. Explanations, discussions, directions and comments were well-structured and clear. The student listened to others actively and when unsure asked appropriate questions to seek clarification. The student showed appropriate confidence and open body language.	The student's written or oral communication was consistently very good. All communication was clear, concise and well-structured, which ensured a very effective outcome. In group situations, the student's ability to interpret others' body language, and the use of body language to ensure a positive outcome, was very effective. At all times the student was calm, engaged and confident.	The student's communication skills were exemplary. At all times the student observed others and ensured that their own communication was extremely effective. In group situations, the student's communication enabled all members to contribute to their greatest ability whilst also ensuring an excellent outcome of the exercise. In group situations, when appropriate, the student proactively and subtly managed the group's mood or motivated the group members appropriately.
Competency	Level 1 Unsatisfactory	Level 2 Satisfactory	Level 3 Good	Level 4 Very good	Level 5 Excellent

Application of knowledge, UPRT and resilience	The student's knowledge was at times insufficient or incorrect, which had an adverse effect on the exercise. The student displayed limited ability to relate knowledge between subjects or to apply knowledge to scenarios, exercises or in answers to questions.	The student had the minimum acceptable level of knowledge to complete the exercise to a satisfactory standard. The student occasionally demonstrated the ability to relate knowledge between subjects. The student could identify some threats or errors, and when presented with a threat or error could in most situations suggest at least one possible effective method of mitigation.	The student demonstrated a good level of knowledge with the ability to relate this knowledge effectively between subjects and in scenario exercises or situations. The student identified many threats and errors, and when presented with threats or errors used their knowledge effectively to suggest appropriate mitigations and actions.	The student demonstrated a very good level of knowledge, and correctly and readily related this knowledge across subjects and in scenario situations. The student identified most threats and errors, and immediately used their knowledge to manage them effectively.	The student had an excellent level of understanding which they immediately and appropriately applied across subjects and to the exercise or scenario situation. The student identified all actual threats and errors in scenario situations, anticipated some possible threats and errors (what if's), and used their knowledge to manage them efficiently and very effectively.
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GM3 ORA.ATO.230(a) Training manual and operations manual

ED Decision 2018/001/R

AREA 100 KSA EXERCISES AND ASSESSMENTS

Exercises and assessments are to be interwoven into the theoretical knowledge training, utilising a range of learning styles; they should address subject or cross-subject topics, with the application of threat and error management (TEM) and, where possible, be scenario-based. The exercises and assessments do not need to be confined to a classroom.

- Area 100 KSA exercises may be of short duration within a lesson, and the student's performance in the exercises does not need to be recorded, although the main subject and KSA learning points are likely to be discussed (or for distance learning, reviewed) within a post exercise debrief or lesson summary. To allow for flexibility and development, the exercises do not need to be specified in the training plan.
- When a single formative assessment is specified in the training plan, it is likely to be extensive as it will cover many of the LOs in Area KSA 100 02 and 100 03. Alternatively, an ATO may specify a number of shorter-duration formative assessments that each covers a narrower range of LOs, and these may build in terms of content difficulty.
- The exercises and formative and summative assessments may include but not be limited to: scenario planning exercises combining multiple subjects; practical exercises using training devices (where available); oral communication exercises; written assignments and/or project

work; discussions; the preparation and delivery of group or individual presentations and discussions; and enable scenario-based content and individual, pair or group situation(s).

- (d) The type of assessment and the environment should be recorded in the ATO's training plan.

GM4 ORA.ATO.230(a) Training manual and operations manual

ED Decision 2018/001/R

AREA 100 KSA INSTRUCTION AND ASSESSMENT TRAINING

- (a) The following material has been developed to provide additional guidance to organisations to help them develop an effective KSA 100 instruction and assessment training programme that satisfies the provisions in AMC2 ORA.ATO.230 (c) to (f).
- (b) An ATO should ensure that an instructor who conducts the Area 100 KSA formative assessment(s) has received adequate training to be familiar with the:
- (1) relevant competencies and performance indicators;
 - (2) Area 100 KSA Learning Objectives (LOs);
 - (3) formative assessment(s) that they will conduct including: the applicable LOs, purpose and content of the assessment(s) and position(s) in the training plan, assessment resources, assessment environment;
 - (4) Area 100 KSA grading system, including familiarisation with the performance indicators and the ATO's word pictures; and
 - (5) student debrief methods and procedure.
- (c) An ATO should ensure that an instructor who conducts the Area 100 KSA summative assessments has received adequate training to be familiar with:
- (1) the summative assessments that they will conduct including: the applicable LOs, purpose and content of the exercise(s) and position(s) in the training plan, assessment resources, assessment environment, and the minimum acceptable level;
 - (2) the assessment feedback, evaluation and development process; and
 - (3) KSA candidate appeal procedure.
- (d) An Area 100 KSA instruction and assessment course should include practical training on the conduct of an assessment, including grading to achieve inter-rater reliability, and the debrief under supervision.

GM5 ORA.ATO.230(a) Training manual and operations manual

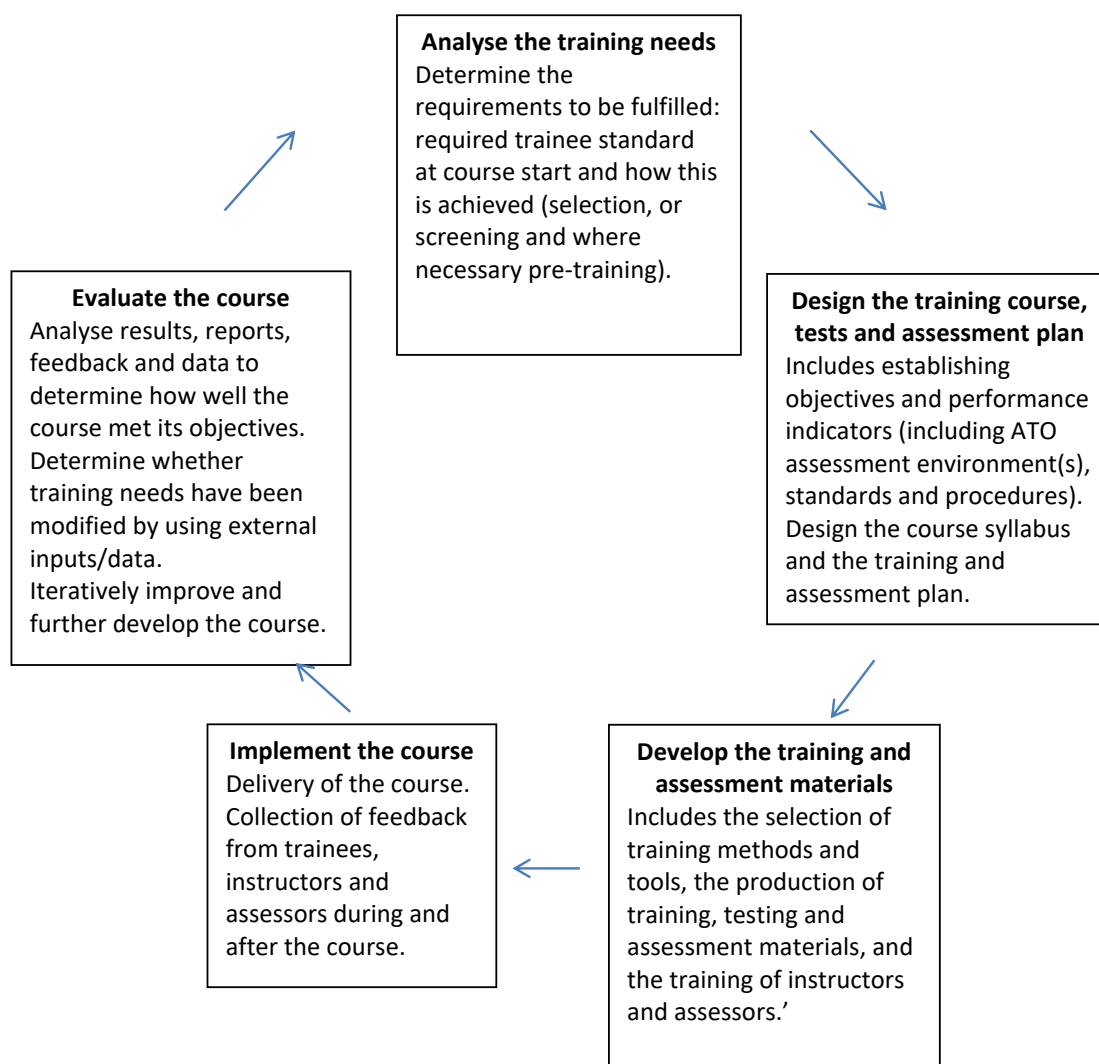
ED Decision 2018/001/R

INSTRUCTIONAL SYSTEMS DESIGN

- (a) The instructional systems design (ISD) provides a systematic and iterative process for course design based on educational best practices. There are several effective ISD models in use today, with the analyse, design, develop, implement and evaluate (ADDIE) framework being generic to all.

The purpose of using ISD to design training courses is to facilitate the students' efficient and effective acquisition of knowledge and skills based on current training needs.

- (b) To provide evidence of the effective use of the ISD methodology in the design and continued development of their course(s), an ATO may use documentation and records that relate to the ISD phases.
- (d) ADDIE model example. The ‘analysis’, ‘design’, ‘development’, ‘implementation’ and ‘evaluation’ phases of the ADDIE model are shown below with brief phase descriptions.



AMC1 ORA.ATO.230(b) Training manual and operations manual

ED Decision 2025/002/R

ALL ATOs, EXCEPT THOSE PROVIDING FLIGHT TEST TRAINING

OPERATIONS MANUAL

The operations manual for use at an ATO conducting integrated or modular flight training courses should include the following:

- (a) General:

- (1) a list and description of all volumes in the operations manual;
 - (2) administration (function and management);
 - (3) responsibilities (all management and administrative staff);
 - (4) student discipline and disciplinary action;
 - (5) approval or authorisation of flights;
 - (6) preparation of flying programme (restriction of numbers of aircraft in poor weather);
 - (7) command of aircraft;
 - (8) responsibilities of the PIC;
 - (9) carriage of passengers;
 - (10) aircraft documentation;
 - (11) retention of documents;
 - (12) flight crew qualification records (licences and ratings);
 - (13) revalidation (medical certificates and ratings);
 - (14) flight duty period and flight time limitations (flying instructors);
 - (15) flight duty period and flight time limitations (students);
 - (16) rest periods (flight instructors);
 - (17) rest periods (students);
 - (18) pilots' log books;
 - (19) flight planning (general);
 - (20) safety (general): equipment, radio listening watch, hazards, accidents and incidents (including reports), safety pilots etc..
- (b) Technical:
- (1) aircraft descriptive notes;
 - (2) aircraft handling (including checklists, limitations, maintenance and technical logs, in accordance with relevant requirements, etc.);
 - (3) emergency procedures;
 - (4) radio and radio navigation aids;
 - (5) allowable deficiencies (based on the master minimum equipment list (MMEL), if available).
- (c) Route:
- (1) performance (legislation, take-off, route, landing etc.);
 - (2) flight planning (fuel, oil, minimum safe altitude, navigation equipment etc.);
 - (3) loading (load sheets, mass, balance and limitations);
 - (4) weather minima (flying instructors);
 - (5) weather minima (students – at various stages of training);
 - (6) training routes or areas.

(d) Personnel training

- (1) appointments of persons responsible for standards/competence of flight personnel;
- (2) initial training;
- (3) refresher training;
- (4) standardisation training;
- (5) checks to verify that training personnel is proficient with regard to all internal ATO standards, processes and procedures;
- (6) upgrading training;
- (7) ATO personnel standards evaluation.

SECTION III – ADDITIONAL REQUIREMENTS FOR ATOS PROVIDING SPECIFIC TYPES OF TRAINING

CHAPTER 1 – DISTANCE LEARNING COURSE

ORA.ATO.300 General

Regulation (EU) No 1178/2011

The ATO may be approved to conduct modular course programmes using distance learning in the following cases:

- (a) modular courses of theoretical knowledge instruction;
- (b) courses of additional theoretical knowledge for a class or type rating; or
- (c) courses of approved pre-entry theoretical knowledge instruction for a first type rating for a multi-engined helicopter.

AMC1 ORA.ATO.300 General

ED Decision 2025/002/R

DISTANCE LEARNING

- (a) A variety of methods is open to ATOs to present course material. It is, however, necessary for ATOs to maintain comprehensive records in order to ensure that students make satisfactory academic progress and meet the time constraints laid down in Part-FCL for the completion of modular courses.
- (b) The following are given as planning guidelines for ATOs developing the distance learning element of modular courses:
 - (1) an assumption that a student will study for at least 15 hours per week;
 - (2) an indication throughout the course material of what constitutes a week's study;
 - (3) a recommended course structure and order of teaching;
 - (4) one progress test for each subject for every 15 hours of study, which should be submitted to the ATO for assessment. Additional self-assessed progress tests should be completed at intervals of five to 10 study hours;
 - (5) appropriate contact times throughout the course when a student can have access to an instructor by telephone, fax, email or the Internet;
 - (6) measurement criteria to determine whether a student has satisfactorily completed the appropriate elements of the course to a standard that, in the judgement of the HT, or CTKI, will enable them to be entered for the Part-FCL theoretical examinations with a good prospect of success;
 - (7) if the ATO provides the distance learning by help of IT solutions, for example the Internet, instructors should monitor students' progress by appropriate means.
- (c) Where an assessment (e.g. planning, written, scenario or practical exercise, or other assessment) is conducted outside the classroom via distance learning, the ATO should have a procedure or process in place to establish that the student themselves have completed the

assessment and that the assessment method(s) for that particular exercise has (have) been effective.

ORA.ATO.305 Classroom instruction

Regulation (EU) No 1178/2011

- (a) An element of classroom instruction shall be included in all subjects of modular distance learning courses.
- (b) The amount of time spent in actual classroom instruction shall not be less than 10 % of the total duration of the course.
- (c) To this effect, classroom accommodation shall be available either at the principal place of business of the ATO or within a suitable facility elsewhere.

AMC1 ORA.ATO.305(b) Classroom instruction

ED Decision 2017/022/R

Classroom instruction delivered by an instructor to a student may include videoconferencing appropriate to the task if the necessary level of communication is ensured and appropriate equipment and tools are available.

ORA.ATO.310 Instructors

Regulation (EU) No 1178/2011

All instructors shall be fully familiar with the requirements of the distance learning course programme.

CHAPTER 2 – ZERO FLIGHT-TIME TRAINING

ORA.ATO.330 General

Regulation (EU) No 1178/2011

- (a) Approval for zero flight-time training (ZFTT), as specified in Part-FCL, shall only be given to ATOs that also have the privileges to conduct commercial air transport operations or ATOs having specific arrangements with commercial air transport operators.
- (b) Approval for ZFTT shall only be given if the operator has at least 90 days of operational experience on the aeroplane type.
- (c) In the case of ZFTT provided by an ATO having a specific arrangement with an operator, the 90 days of operational experience requirements will not apply if the type rating instructor (TRI(A)) involved in the additional take-offs and landings, as required in Part-ORO, has operational experience on the aeroplane type.

AMC1 ORA.ATO.330 General

ED Decision 2012/007/R

INITIAL APPROVAL

For an initial approval to conduct ZFTT, the operator should have held an air operator's certificate for commercial air transport for at least 1 year. This period may be reduced where the operator and the ATO have experience of type rating training.

ORA.ATO.335 Full flight simulator

Regulation (EU) No 290/2012

- (a) The FFS approved for ZFTT shall be serviceable according to the management system criteria of the ATO.
- (b) The motion and the visual system of the FFS shall be fully serviceable, in accordance with the applicable certification specifications for FSTD as mentioned in [ORA.FSTD.205](#).

CHAPTER 3 – MULTI-CREW PILOT LICENCE (MPL) COURSES

ORA.ATO.350 General

Regulation (EU) No 1178/2011

The privileges to conduct MPL integrated training courses and MPL instructor courses shall only be given to the ATO if it also has the privilege to conduct commercial air transport operations or a specific arrangement with a commercial air transport operator.

CHAPTER 4 – FLIGHT TEST TRAINING

ORA.ATO.355 Flight test training organisations

Regulation (EU) No 1178/2011

- (a) The ATO that has been approved to provide flight test training for the issue of a category 1 or 2 flight test rating in accordance with Part-FCL may have its privileges extended to providing training for other categories of flight tests and other categories of flight test personnel, provided that:
 - (1) the relevant requirements of Part-21 are met; and
 - (2) a specific arrangement exists between the ATO and the Part-21 organisation that employs, or intends to employ, such personnel.
- (b) The training records shall include the written reports by the student, as required by the training programme, including, where applicable, data processing and analysis of recorded parameters relevant to the type of flight test.

SUBPART FSTD – REQUIREMENTS FOR ORGANISATIONS OPERATING FLIGHT SIMULATION TRAINING DEVICES (FSTDs) AND THE QUALIFICATION OF FSTDs

SECTION I – REQUIREMENTS FOR ORGANISATIONS OPERATING FSTDs

ORA.FSTD.100 General

Regulation (EU) No 1178/2011

- (a) The applicant for an FSTD qualification certificate shall demonstrate to the competent authority that it has established a management system in accordance with ORA.GEN Section II. This demonstration shall ensure that the applicant has, directly or through contract, the capability to maintain the performance, functions and other characteristics specified for the FSTD's qualification level and to control the installation of the FSTD.
- (b) If the applicant is the holder of a qualification certificate issued in accordance with this Part, the FSTD specifications shall be detailed:
 - (1) in the terms of the ATO certificate; or
 - (2) in the case of an AOC holder, in the training manual.

AMC1 ORA.FSTD.100 General

ED Decision 2012/007/R

COMPLIANCE MONITORING PROGRAMME – ORGANISATIONS OPERATING FSTDs

- (a) Introduction.
 - (1) The purpose of this AMC is to provide additional and specific information to an organisation operating FSTDs on how to establish a compliance monitoring programme (CMP) that enables compliance with the applicable requirements.
- (b) Compliance monitoring programme
 - (1) Typical subject areas for inspections are the following:
 - (i) actual FSTD operation;
 - (ii) maintenance;
 - (iii) technical standards
 - (iv) FSTD safety features.
- (c) Audit scope
 - (1) Organisations operating FSTDs are required to monitor compliance with the procedures they have designed to ensure specified performance and functions. In doing so they should as a minimum, and where appropriate, monitor the following:
 - (i) organisation;
 - (ii) plans and objectives;

- (iii) maintenance procedures;
- (iv) FSTD qualification level;
- (v) supervision;
- (vi) FSTD technical status;
- (vii) manuals, logs and records;
- (viii) defect deferral;
- (ix) personnel training;
- (x) aircraft modifications;
- (xi) FSTD configuration management.

AMC2 ORA.FSTD.100 General

ED Decision 2012/007/R

COMPLIANCE MONITORING PROGRAMME – ORGANISATIONS OPERATING FSTDs

One acceptable means of measuring FSTD performance is contained in ARINC report 433-1 (December 14th, 2007 or as amended) *Standard Measurements for Flight Simulation Quality*.

AMC3 ORA.FSTD.100 General

ED Decision 2012/007/R

COMPLIANCE MONITORING PROGRAMME – ORGANISATIONS OPERATING BASIC INSTRUMENT TRAINING DEVICES (BITDs)

- (a) The compliance monitoring programme together with a statement acknowledging completion of a periodic review by the accountable manager should include the following:
 - (1) a maintenance facility that provides suitable BITD hardware and software test and maintenance capability;
 - (2) a recording system in the form of a technical log in which defects, deferred defects and development work are listed, interpreted, actioned and reviewed within a specified time scale; and
 - (3) planned routine maintenance of the BITD and periodic running of the qualification test guide (QTG) with adequate manning to cover BITD operating periods and routine maintenance work.
- (b) A planned audit schedule and a periodic review should be used to verify that corrective action was carried out and that it was effective. The auditor should have adequate knowledge of BITDs.

GM1 ORA.FSTD.100 General

ED Decision 2012/007/R

COMPLIANCE MONITORING – ORGANISATIONS OPERATING FSTDs – GENERAL

- (a) The concept of compliance monitoring (CM) is a fundamental requirement for organisations operating FSTDs. An effective CM function is vitally important in supporting operation of the devices, in a structured way, to ensure they remain in compliance with the technical standards

of CS-FSTD(A) and CS-FSTD(H) and continue to be effective training tools. An effective CM function is also essential to support any level of extended recurrent evaluation period as permitted by [ORA.FSTD.225\(b\)](#).

- (b) The following guidance has been developed to provide additional material to help both organisations operating FSTDs and competent authorities in developing effective CM that satisfy the applicable requirements and ensure the highest standards of training are maintained.
- (c) Additional GM provide a compliance checklist for organisations operating FSTDs ([GM2 ORA.FSTD.100](#)) and guidance detailing the preparation for an evaluation by the competent authority ([GM3 ORA.FSTD.100](#)). The compliance checklist should be used by the competent authorities as a standardised checklist for the elements that are expected in the CM function of an organisation operating FSTDs. The organisation should complete as a minimum the second column of the checklist by providing appropriate manual or procedure references for each of the identified elements of the CM function. Additional information can be provided in the third column to aid assessment of the checklist as appropriate. This would then be provided to the competent authority. Use of this checklist should assist in ensuring a consistent approach by the competent authorities and also provide organisations operating FSTDs with additional guidance on all the elements of a CM function that the competent authorities will expect. The guidance is provided to help organisations operating FSTDs to prepare for authority visits.
- (d) The documentation of the CM may be electronic, provided the necessary controls can be demonstrated. This should include control of any paper copies that may be downloaded for use by individuals. It is recommended that any such copies are automatically designated as uncontrolled as part of the download process. Whilst electronic signatures on master documents may be accepted, with appropriate protections, a hardcopy master of the CM manual should be provided, with wet-ink signatures to be held by the applicant.
- (e) It should be recognised that whatever CM is developed, it will not be effective unless it becomes an integral part of the way in which the organisation works. It includes both the necessary procedures for maintaining compliance with all the applicable requirements and a compliance monitoring programme (CMP) to monitor the execution of these procedures. A successful CM will ensure that the highest training tool is available at all times. If the CM is viewed as an add-on to existing processes it will become a burden and it will never be wholly effective. It should also be noted that compliance control or inspection is only a small part of a CM. If the CM is working effectively, inspections such as fly-outs should become routine revealing little beyond day-to-day unserviceabilities. Systematic defects should be captured by the CMP.
- (f) The competent authority should be satisfied that the accountable manager is able to adequately provide the required level of resources to properly support the FSTD. Detailed knowledge of FSTD requirement standards are not necessary, only sufficient to understand his/her responsibility for ensuring the FSTD is properly supported. The assessment of the compliance monitoring manager should concentrate on establishing that the nominee has sufficient knowledge and experience of both CM management and FSTD operations to operate a compliance monitoring system (CMS) within an organisation operating FSTDs. This is likely to require experience of working in the compliance monitoring field and sufficient knowledge of FSTDs and the technical standards with which they should comply.

- (g) If an organisation operating FSTDs is certified under any international quality standard it should assure that it fully covers the applicable organisation requirements of Part-ORA and the qualification basis.
- (h) For small organisations, it is perfectly acceptable to combine the roles of compliance monitoring manager and accountable manager. For other organisations that hold multiple certificates and may cover multiple sites, it is advantageous to have a common CM function with an overall compliance monitoring manager. However, it is essential, particularly where sites may be significantly separated geographically, that there is a nominated representative at each site and possibly for each certificate. These representatives should hold the delegated responsibility of the CM manager for the day-to-day CM role at their site and in their function and have the necessary direct reporting line to the overall CM manager. It will also be necessary to ensure that local representatives are also acceptable to the local competent authority. In many cases the local representatives may perform other functions in addition to this role. This is acceptable provided the necessary independence of any compliance monitoring activity is maintained.
- (i) CM, as a whole, begins with the requirements with which the system seeks to comply. These include both the technical standards, in this case the relevant parts of CS-FSTD(A)/(H) plus any other specific standards, for example health and safety regulations, and the compliance monitoring objectives, such as defect rates and rectification intervals and FSTD reliability targets. The CM should define the process by which these standards are made available to those who require them.
- (j) The next part of CM is that part which defines the day-to-day procedures or working practices by which the standards will be achieved. These procedures should include as a minimum defect reporting systems, defect rectification processes, tracking mechanisms, preventative maintenance programmes, spares handling, equipment calibration and configuration management of the device. They should include checks to assess the compliance of the performed actions. These procedures and standards should be made readily available to anybody involved in the maintenance and day-to-day operation of the FSTD.
- (k) The third part of CM is the method by which the organisation operating an FSTD confirms the device is maintained in compliance with the defined standards and is being operated in accordance with the defined procedures. This is the compliance monitoring programme (CMP) and includes the audit methods, reporting and corrective action procedures and feedback, management reviews and schedules for audits of all aspects of the FSTD operation.
- (l) Across all aspects of CM, and most important to it, are the people. CM includes the definition of the responsibilities of all staff and should include a declaration of the minimum levels of resource proposed for the direct support of the FSTD plus the levels of support and managerial staff proposed. The levels of resource can be affected by factors such as local health and safety regulations, existence of weekend and/or night usage of the device(s), etc. CM also includes definition of the skills and experience required for staff and leads to definition of any required training programmes. Training needs cover both technical training and audit training, including QTG running and checking and fly-out techniques for flight crew.
- (m) The documentation of CM may be provided in any number of documents provided there are appropriate cross-references in all documents such that the system is fully traceable in both directions from end to end. For all but small organisations at least two documents would be expected:

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- (1) Firstly, a CM manual containing the policy, terminology, organisational charts and responsibilities, an overview of all processes, within the system, including those for maintaining regulatory compliance such as QTG running and fly-outs (function and subjective testing), CMP including the audit schedule and audit procedures including reporting and corrective action procedures. In addition, the CM manual should include, either directly or by reference, the identification of skills and experience and associated training.
- (2) Secondly, a procedures manual containing, as a minimum, software and hardware control procedures, configuration control procedures including, for example, control of training loads, updates to visual models, navigation and instructor operation station (IOS) databases, QTG running and checking procedures, fly-out procedures, maintenance procedures including both defect rectification and preventative maintenance processes. Any standard forms and checklists should also be included.
- (n) The CM documentation also includes all records such as technical logs, QTG runs, fly-out reports and maintenance job cards.
- (o) For organisations with several certificates, separate and modular procedures manuals with a single CM manual covering all approvals, may be acceptable.
- (p) It is important to understand the difference between compliance assurance and compliance control. An effective CM will contain elements of both. Compliance control is normally done by inspection of the product; it provides confirmation at the time of the inspection that the product conforms to a defined standard.
- (q) The compliance assurance element is essential to ensure the standard is maintained throughout the periods between product (FSTD) inspections. Within a CMP, the processes are defined that are necessary to provide confidence that the FSTD(s) is/are being supported and maintained to the highest possible standard and in compliance with the relevant requirements. A programme of internal audits is then set in place to confirm that the processes are being followed and are effective. The competent authority would normally oversee a certified organisation by process and system audit, however, in the case of FSTDs, authority oversight includes an inspection element in the form of the recurrent FSTD evaluation.
- (r) In addition to the normal process and system audits, the compliance assurance audit schedule should include the schedule for each FSTD for fly-outs and QTG running through the audit year.
- (s) The audit procedure should include, at least, the following: statement of scope, planning, initiation of audit, collection of evidence, analysis, reporting of findings, identification and agreement of corrective actions and feedback, including reporting significant findings to the competent authority, where appropriate. The review of published material could include, in addition to the CM and procedures manuals, QTG records, fly-out reports, technical log sheets, maintenance records and configuration control records.
- (t) In addition to basic knowledge of FSTD requirements and operation, it is expected that auditors have received training in CM and audit techniques.
- (u) The routine fly-outs of the device are a specialised part of the audit programme. It is essential that the pilots tasked with carrying out these fly-outs are adequately experienced. They would be expected to be type rating instructor/examiner (TRI/TRE) qualified on the type, and should have experience of simulator evaluations carried out by the competent authority. The assignment of such pilots can present difficulties, particularly for the independent organisation
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operating FSTDs not directly associated with an airline. It is vital for the organisation to ensure their users are aware of the importance of the fly-outs as part of the continued qualification of the device and the need to assist in the provision of suitably qualified pilots to carry them out. It is worth noting that simulator users are required to satisfy themselves that the training devices they use are assessed for continued suitability, as part of their own CMP. Involvement in fly-outs assists in meeting this need.

- (v) Whilst it is accepted that the number of audits required in an organisation with a single device will be significantly less than those in larger organisations with multiple devices, the CMP should still meet the same criteria, and cover all aspects of the operation within a 12 month period. The independence of the audit personnel should be maintained at all times. The audit programme, whether by full audit or by using a checklist system should still be sufficiently comprehensive to provide the necessary level of confidence that the device is maintained and operated to the highest possible standard. This includes monitoring and review of corrective actions and feedback processes.
- (w) The successful use of sub-contractors who play a significant role in the provision of services, such as maintenance or engineering services, to an organisation operating FSTDs is reliant on the sub-contractor operating under the CM of the organisation. All requirements that an organisation is expected to meet are equally applicable to his/her sub-contractor. It is the organisation's responsibility to ensure that the sub-contractor complies with its CM.
- (x) It is essential that a proper understanding of the CM and how it applies to each and every staff member is provided by appropriate training to all, not just those directly involved in operating the CM, such as the accountable manager, the CM manager, representatives and the auditors. The training given to those directly involved in CM should cover the CM, audit techniques and applicable technical standards. CM familiarisation training should be an integral part of any induction training and recurrent training. Update training on technical standards for audit personnel, is also of particular importance.
- (y) Any effective CM will include measurement of its effectiveness. The organisation should develop performance measures that can be monitored against objectives. Such measures, often referred to as metrics, should be reviewed by the competent authority as part of its oversight of the CM within the organisation and during recurrent evaluations. In addition they should form part of the data reviewed during scheduled management reviews as part of the CM.
- (z) ARINC 433 provides good guidance on FSTD compliance measurement. Metrics should monitor not only individual FSTD performance but, for larger organisations, how each FSTD is performing within the fleet. It is also recommended that metrics data be shared, regularly, with the FSTD manufacturers to allow monitoring for generic problems such as design issues, which may be best addressed with a fleet-wide solution.

GM2 ORA.FSTD.100 General

ED Decision 2012/007/R

COMPLIANCE MONITORING – ASSESSMENT FOR ORGANISATIONS OPERATING FSTDs

COMPLIANCE MONITORING ASSESSMENT FOR ORGANISATIONS OPERATING FSTDs	
Organisation:	
Site Assessed:	

Date of Assessment:			
Accountable Manager:			
Compliance Monitoring Manager:			
Number and Type of FSTDs:			
CM Manual Reference:			
Audit Area	CM/Proc Ref	Comments	Satisfactory Y/N
1. ACCOUNTABLE MANAGER			
Has an accountable manager (AM) with overall responsibility for compliance monitoring (CM) been nominated?			
Does the accountable manager have corporate authority to ensure all necessary activities can be financed and carried out to the standard required by the competent authority?			
Has a formal written compliance policy statement been established, included in the CM manual and signed by the accountable manager?			
2. COMPLIANCE MONITORING MANAGER			
Has a compliance monitoring manager (CM manager) been nominated?			
Are the posts of CM manager and AM combined? If so, is the independence of compliance audits assured?			
Does the CM manager have overall responsibility and authority to: a) verify that standards are met; and b) ensure that the compliance monitoring programme is established, implemented and maintained?			
Does the CM manager have direct access to the AM?			
Does the CM manager have access to all parts of the organisation operating an FSTD and as necessary any sub-contractor's organisation?			
3. COMPLIANCE MONITORING (CM)			
Has CM been established by the operator?			
Is CM properly documented? (see Section 4)			
Is the CM structured according to the size and complexity of the operator?			
Does the CM include the following as a minimum: a) monitoring of compliance with required technical standards; b) identification of corrective actions and person responsible for rectification;		a) b)	

c) a feedback system to accountable manager to ensure corrective action are promptly addressed;		c)	
d) reporting of significant noncompliances to the competent authority;		d)	
e) a compliance monitoring programme to verify continued compliance with applicable requirements, standards and procedures.		e)	
Is the CM structured according to the size and complexity of the operator?			
Are the responsibilities of the CM manager defined to include, as a minimum:			
a) monitoring of corrective action programme;		a)	
b) ensuring that the corrective actions contain the necessary elements;		b)	
c) providing management with an independent assessment of corrective action, implementation and completion;		c)	
d) evaluation of the effectiveness of the corrective action programme.		d)	
Are adequate financial, material and human resources in place to support CM?			
Are management evaluations/reviews of CM held at least quarterly?			
Does the management evaluation ensure that the CMS is working effectively and is it comprehensive and well documented?			
Does the compliance monitoring programme identify the processes necessary and the persons within the organisation who have the training, experience, responsibility and authority to carry out the following:			
a) schedule and perform quality inspections and audits, including unscheduled audits when required;		a)	
b) identify and record any concerns or findings, and the evidence necessary to substantiate such concerns or findings;		b)	
c) initiate or recommend solutions to concerns or findings through designated reporting channels;		c)	
d) verify the implementation of solutions within specific timescales.		d)	
Is there sufficient auditor resource available and can their required level of independence be demonstrated?			
Do the auditors report directly to the compliance monitoring manager?			

Does the defined audit schedule cover the following areas, within each 12 month period?		a) b) c) d) e) f) g) h) i) j)	
a) organisation		a)	
b) plans and objectives		b)	
c) maintenance procedures		c)	
d) FSTD qualification level;		d)	
e) supervision		e)	
f) FSTD technical status		f)	
g) manuals, logs and records		g)	
h) defect deferral		h)	
i) personnel training		i)	
j) aircraft and simulator configuration management, including Airworthiness Directives		j)	
How are audit noncompliances recorded?			
Are procedures in place to ensure that corrective actions are taken in response to findings?			
Are records of the compliance monitoring programme:			
a) accurate		a)	
b) complete and		b)	
c) readily accessible?		c)	
Is there an acceptable and effective procedure for providing a briefing on the CM to all personnel?			
Is there an acceptable and effective procedure for ensuring that all those responsible for managing the CM receive training covering:			
a) an introduction to the concept of the CM;		a)	
b) compliance management;		b)	
c) the concept of compliance assurance;		c)	
d) CM manuals;		d)	
e) audit techniques;		e)	
f) reporting and recording;		f)	
g) how the CM supports continuous improvement within the organisation.		g)	
Are suitable training records maintained?			
Are activities within the CM sub-contracted out to external agencies?			
Do written agreements exist between the organisation and the sub-contractor clearly defining the services and standard to be provided?			
Are the procedures in place to ensure that the necessary authorisations/ approval when required are held by a sub-contractor?			
Are the procedures in place to establish that the subcontractor has the necessary technical competence?			

4. CM MANUAL			
What is the current status of the CM manual – amendment and issue date?			
Is there a procedure in place to control copies and the distribution of the CM manual?			
Is the CM manual signed by the accountable manager and the compliance monitoring manager?			
Does the CM manual include, either directly or by reference to other documents, the following: a) a description of the organisation; b) reference to appropriate FSTD technical standards; c) allocation of duties and responsibilities; d) audit procedures; e) reporting procedures; f) follow-up and corrective action procedures; g) document retention policy; h) training records		a) b) c) d) e) f) g) h)	
Is there a document retention policy covering: a) audit schedules; b) inspection and audit reports; c) responses to findings; d) corrective action reports; e) follow-up and closure reports; f) management evaluation reports.		a) b) c) d) e) f)	
Does the CM manual include, either directly or by reference to other documents, the following procedures for day to day operation of the FSTD: a) defect reporting systems; b) defect rectification processes; c) tracking mechanisms; d) preventative maintenance programmes; e) spares handling; f) equipment calibration; g) configuration management of the device including visual, IOS and navigation databases; h) configuration control system to ensure the continued integrity of the hardware and software qualified; i) QTG running and function and subjective tests.		a) b) c) d) e) f) g) h) i)	

Does the CM manual include, either directly or by reference to other documents, procedures for notification of the competent authorities of the following: a) any change in the organisation including company name, location, management; b) major changes to a qualified device; c) deactivation or relocation of a qualified device; d) major failures of a qualified device; e) major safety issue associated with the installation.		a) b) c) d) e)	
Does the CM manual define acceptable and effective procedures to ensure compliance with applicable health and safety regulations, including: a) safety briefings; b) fire/smoke detection and suppression; c) protection against electrical, mechanical, hydraulic and pneumatic hazards; d) other items as defined in AMC1 ORA.FSTD.115		a) b) c) d)	
Does the CM manual include acceptable and effective procedures for regularly checking FSTD safety features such as emergency stops and emergency lighting, and are such tests recorded?			
5. COMPLIANCE MEASURES			
Have compliance monitoring objectives been developed from the policy statement, and included either directly or by reference in the CMS manual?			
Does the CMS include processes to produce and review appropriate metrics data?			
Do these compliance measures track the following: a) FSTD availability; b) numbers of defects; c) open defects; d) defect closure rates; e) training session interrupt rates; f) training session compliance rating.		a) b) c) d) e) f)	
Do the compliance measures support the compliance objectives?			

Required actions/Comments

Signature:.....

Date:.....

GM3 ORA.FSTD.100 General

ED Decision 2012/007/R

**COMPLIANCE MONITORING SYSTEM – GUIDANCE FOR ORGANISATIONS OPERATING FSTDs TO PREPARE FOR
A COMPETENT AUTHORITY EVALUATION****(a) Introduction**

The following material provides guidance on what is expected by the competent authorities to support the discussion during the preliminary briefing, which is a first step of any initial or recurrent evaluation of an FSTD carried out by a competent authority.

This document has been developed as well to standardise working methods throughout Member States and to develop effective CM spot checks to satisfy the applicable requirements and therefore to ensure the highest standards of training are attained.

(b) Document form

Different document forms can be considered. Nevertheless, it appears that the best solution is a dossier, which includes all the information required by the competent authority to perform an evaluation.

(c) Contents of the dossier for an initial evaluation:

- (1) type of FSTD and qualification level requested;
- (2) evaluation agenda: including date of evaluation, name of people involved for the competent authority, contact details for the FSTD operator, schedules for the subjective flight profile, QTG rerun;
- (3) FSTD identification and detailed technical specification including, type of FSTD, manufacturer, registration number, date of entry into service, host computer, visual system, motion system, type of IOS, simulated version(s), standards of all the aircraft computers, if applicable. Manuals needed for an evaluation (e.g. flight manuals, system manuals, acceptance test manual, IOS user manual etc. – if applicable) could already be provided as part of the dossier in an electronic format;
- (4) planned modifications;
- (5) subjective open defect(s);
- (6) airport visual databases including for each visual scene, name of the airport, IATA and ICAO codes, type of visual scene (specific or generic), additional capabilities (e.g. snow model, WGS 84 compliance, enhanced ground proximity warning system (EGPWS)); and

- (7) QTG status: the list should include for each QTG test available the status of the tests following the FSTD operator and competent authority reviews.
- (d) Contents of the dossier for a recurrent evaluation:
 - (1) type of FSTD and qualification level requested;
 - (2) evaluation agenda, including date of evaluation, name of people involved for the competent authority, contact details for the operator, schedules for the subjective flight profile, QTG rerun and QTG review;
 - (3) FSTD identification, including type of FSTD, manufacturer, registration number, date of entry into service, host computer, visual system, motion system, type of IOS, simulated version(s), standards of all the aircraft computers, if applicable;
 - (4) status of items raised during the last evaluation and date of closure;
 - (5) reliability data: training hours month by month during the past year, numbers of complaints mentioned in the technical log, training hours lost, availability rate;
 - (6) operational data: a list of FSTD users over the previous 12 months should be provided, with number of training hours;
 - (7) failure tabulation including categorisation of failures (by ATA chapter and Pareto diagram, ARINC classification);
 - (8) details of main failures leading to training interruption or multiple occurrences of some failures;
 - (9) hardware and/or software updates or changes since last evaluation and planned hardware and/or software updates or changes;
 - (10) subjective open defect(s);
 - (11) airport visual databases including for each visual scene, name of the airport, ATA and ICAO codes, type of visual scene (specific or generic), additional capabilities (snow model, WGS 84 compliance, EGPWS);
 - (12) QTG status: the list should include for each QTG test available, the date of run during the past year, any comment, and the status of the tests; and
 - (13) results of scheduled internal audits and additional quality inspections (if any) since last evaluation and a summary of actions taken.

ORA.FSTD.105 Maintaining the FSTD qualification

Regulation (EU) No 290/2012

- (a) In order to maintain the qualification of the FSTD, an FSTD qualification certificate holder shall run the complete set of tests contained within the master qualification test guide (MQTG) and functions and subjective tests progressively over a 12-month period.
- (b) The results shall be dated, marked as analysed and evaluated, and retained in accordance with [ORA.FSTD.240](#), in order to demonstrate that the FSTD standards are being maintained.
- (c) A configuration control system shall be established to ensure the continued integrity of the hardware and software of the qualified FSTD.

ORA.FSTD.110 Modifications

Regulation (EU) No 1178/2011

- (a) The holder of an FSTD qualification certificate shall establish and maintain a system to identify, assess and incorporate any important modifications into the FSTDs it operates, especially:
 - (1) any aircraft modifications that are essential for training, testing and checking, whether or not enforced by an airworthiness directive; and
 - (2) any modification of an FSTD, including motion and visual systems, when essential for training, testing and checking, as in the case of data revisions.
- (b) Modifications of the FSTD hardware and software that affect handling, performance and systems operation or any major modifications of the motion or visual system shall be evaluated to determine the impact on the original qualification criteria. The organisation shall prepare amendments for any affected validation tests. The organisation shall test the FSTD to the new criteria.
- (c) The organisation shall inform the competent authority in advance of any major changes to determine if the tests carried out are satisfactory. The competent authority shall determine if a special evaluation of the FSTD is necessary prior to returning it to training following the modification.

AMC1 ORA.FSTD.110 Modifications

ED Decision 2012/007/R

GENERAL

- (a) The FSTD, where applicable, should be maintained in a configuration that accurately represents the aircraft being simulated. This may be a specific aircraft tail number or may be a representation of a common standard.
- (b) Users of the device should always establish a differences list for any device they intend to use, and to identify how any differences should be covered in training. In order to ensure each device is maintained in the appropriate configuration, the organisation operating an FSTD should have a system that ensures that all relevant airworthiness directives (ADs) are introduced where applicable on affected FSTDs.
- (c) ADs from both the State of Design of the aircraft and the State where the FSTD is located should be monitored. ADs from the State of Design of an aircraft are usually automatically applicable, unless specifically varied by the aircraft's State of Registry.
- (d) Where appropriate, ADs issued by States where users of the device have aircraft registered should also be monitored. In addition to ADs, the FSTD operator should also put in place processes that ensure all aircraft modifications are reviewed for any effect on training, testing and checking. This can be achieved by reviewing the aircraft manufacturer's service bulletins and may require a specific link to the aircraft manufacturer to be developed. In practice this link is often established through aircraft operators who use the device.
- (e) Organisations operating FSTDs should notify the competent authority of major changes.
- (f) This does not imply that the competent authority will always wish to directly evaluate the change. The competent authority should be mindful of the potential burden placed on the

organisation by a special evaluation and should always consider that burden when deciding if such an evaluation is necessary.

- (g) The organisation operating FSTDs should have an internal acceptance process for modifications, to be used when implementing all modifications, even if the competent authority has made a decision to carry out an evaluation.

GM1 ORA.FSTD.110 Modifications

ED Decision 2012/007/R

EXAMPLES OF MAJOR MODIFICATIONS

The following are examples of modifications that should be considered as major. This list is not exhaustive and modifications need to be classified on a case-by-case basis:

- (a) any change that affects the QTG;
- (b) introduction of new standards of equipment such as flight management and guidance computer (FMGC) and updated aerodynamic data packages;
- (c) re-hosting of the FSTD software;
- (d) introduction of features that model new training scenarios; e.g. airborne collision avoidance system (ACAS), EGPWS;
- (e) aircraft modifications that could affect the FSTD qualification; and
- (f) FSTD hardware or software modifications that could affect the handling qualities, performance or system representation.

ORA.FSTD.115 Installations

Regulation (EU) No 1178/2011

- (a) The holder of an FSTD qualification certificate shall ensure that:
 - (1) the FSTD is housed in a suitable environment that supports safe and reliable operation;
 - (2) all FSTD occupants and maintenance personnel are briefed on FSTD safety to ensure that they are aware of all safety equipment and procedures in the FSTD in case of an emergency; and
 - (3) the FSTD and its installations comply with the local regulations for health and safety.
- (b) The FSTD safety features, such as emergency stops and emergency lighting, shall be checked at least annually and recorded.

AMC1 ORA.FSTD.115 Installations

ED Decision 2012/007/R

MINIMUM ELEMENTS FOR SAFE OPERATION

- (a) Introduction
 - (1) This AMC identifies those elements that are expected to be addressed, as a minimum, to ensure that the FSTD installation provides a safe environment for the users and operators of the FSTD under all circumstances.

(b) Expected elements

- (1) Adequate fire/smoke detection, warning and suppression arrangements should be provided to ensure safe passage of personnel from the FSTD.
- (2) Adequate protection should be provided against electrical, mechanical, hydraulic and pneumatic hazards, including those arising from the control loading and motion systems, to ensure maximum safety of all persons in the vicinity of the FSTD.
- (3) Other areas that should be addressed include the following:
 - (i) a two-way communication system that remains operational in the event of a total power failure;
 - (ii) emergency lighting;
 - (iii) escape exits and escape routes;
 - (iv) occupant restraints (seats, seat belts etc.);
 - (v) external warning of motion and access ramp or stairs activity;
 - (vi) danger area markings;
 - (vii) guard rails and gates;
 - (viii) motion and control loading emergency stop controls accessible from either pilot or instructor seats;
 - (ix) a manual or automatic electrical power isolation switch.

GM1 ORA.FSTD.115 Installations

ED Decision 2012/007/R

GENERAL

- (a) The intent of [ORA.FSTD.115](#) is to establish that the organisation operating an FSTD has all the necessary procedures in place to ensure that the FSTD installation remains in compliance with all requirements affecting the safety of the device and its users.
- (b) Based on experience, the competent authority should pay particular attention to the quality of safety briefings on the FSTD provided to users and instructors, and to the execution of regular checks on the FSTD safety features.
- (c) It is recognised that certain checks, such as that of the emergency stop, can have adverse impact on the FSTD if carried out in full.
- (d) It is acceptable to develop a procedure that protects elements of the device by shutting them down in advance, in a more controlled manner, provided it can be shown that the procedure still demonstrates the whole device can be shut down by the operation of a single emergency stop button, when required.

ORA.FSTD.120 Additional equipment

Regulation (EU) No 1178/2011

Where additional equipment has been added to the FSTD, even though not required for qualification, it shall be assessed by the competent authority to ensure that it does not adversely affect the quality of training.

SECTION II – REQUIREMENTS FOR THE QUALIFICATION OF FSTDs

ORA.FSTD.200 Application for FSTD qualification

Regulation (EU) No 1178/2011

- (a) The application for an FSTD qualification certificate shall be made in a form and manner established by the competent authority:
- (1) in the case of basic instrument training devices (BITDs), by the BITD manufacturer;
 - (2) in all other cases, by the organisation intending to operate the FSTD.
- (b) Applicants for an initial qualification shall provide the competent authority with documentation demonstrating how they will comply with the requirements established in this Regulation. Such documentation shall include the procedure established to ensure compliance with [ORA.GEN.130](#) and [ORA.FSTD.230](#).

AMC1 ORA.FSTD.200 Application for FSTD qualification

ED Decision 2012/007/R

LETTER OF APPLICATION FOR INITIAL QUALIFICATION OF AN FSTD; EXCEPT BASIC INSTRUMENT TRAINING DEVICE (BITD)

A sample of letter of application is provided overleaf.

Part A

To be submitted not less than 3 months prior to requested qualification date

(Date)
(Office – Competent Authority)
(Address)
(City)
(Country)

Type of FSTD	Aircraft Type/class	Qualification Level Sought				
		A	B	C	D	Sp./Cat
Full Flight Simulator						
FFS						
Flight Training Device		1	2	3		
FTD						
Flight and Navigation Procedures Trainer		I	II	III	II MCC	III MCC
FNPT						

Interim Qualification Level requested: YES/NO

Dear,

<Name of Applicant> requests the evaluation of its flight simulation training device <operator's identification of the FSTD> for qualification. The <FSTD manufacturer's name> FSTD with its <visual system and manufacturer's name, if applicable> visual system.

Evaluation is requested for the following configurations and engine fits as applicable:

e.g. 767 PW/GE and 757RR

1.....

2.....

3.....

Dates requested are: <date(s)> and the FSTD will be located at <place>.

The objective tests of the QTG will be submitted by <date> and in any event not less than 30 days before the requested evaluation date unless otherwise agreed with the competent authority.

Comments:

.....
.....

Signed

.....

Print name:

Position/appointment held:

Email address:

Telephone number:

Part B

To be completed with attached QTG results

(Date)

We have completed tests of the FSTD and declare that it meets all applicable requirements except as noted below.

The following QTG tests still have to be provided:

Tests	Comments

(Add boxes as required)

It is expected that they will be completed and submitted 3 weeks prior to the evaluation date.

Signed

.....

Print name:

Position/appointment held:

E-mail address:

Telephone number:

Part C**To be completed not less than 7 days prior to initial evaluation**

(Date)

The FSTD has been assessed by the following evaluation team:

(Name) Qualification

(Name) Qualification

(Name) Qualification

(Name) Pilot's Licence Nr

(Name) Flight Engineer's Licence Nr (if applicable)

☐ FFS/FTD: This team attests that the <type of FSTD> conforms to the aeroplane flight deck/helicopter cockpit configuration of <name of aircraft operator (if applicable), type of aeroplane/helicopter> aeroplane/helicopter within the requirements for <type of FSTD and level> and that the simulated systems and subsystems function equivalently to those in that aeroplane/helicopter. The pilot of this evaluation team has also assessed the performance and the flying qualities of the FSTD and finds that it represents the designated aeroplane/helicopter.

☐ FNPT: This team attest(s) that the <type of FSTD> represents the flight deck or cockpit environment of a <aeroplane/helicopter or class of aeroplane/type of helicopter> within the requirements for <type of FSTD and level> and that the simulated systems appear to function as in the class of aeroplane/type of helicopter. The pilot of this evaluation team has also assessed the performance and the flying qualities of the FSTD and finds that it represents the designated class of aeroplane/type of helicopter.

(Additional comments as required)

.....

.....

.....

Signed

.....

Print name:

Position/appointment held:

E-mail address:

Telephone number:

GM1 ORA.FSTD.200 Application for FSTD qualification

ED Decision 2012/007/R

USE OF FOOTPRINT TESTS IN QUALIFICATION TEST SUBMISSION

(a) Introduction

- (1) Recent experience during initial qualification of some FFSs has required acceptance of increasing numbers of footprint tests. This is particularly true for FFSs of smaller or older aircraft types, where there may be a lack of aircraft flight test data. However, the large number of footprint tests offered in some QTGs has given rise to concern.
- (2) This guidance is applicable to FFS aeroplane, FTD aeroplane, FFS helicopter and FTD helicopter qualifications.

(b) Terminology

- (1) Footprint test - footprint test data are derived from a subjective assessment carried out on the actual FSTD requiring qualification. The assessment and validation of these data are carried out by a pilot appointed by the competent authority. The resulting data are the footprint validation data for the FSTD concerned.

(c) Recommendation

- (1) It is permitted to use footprint data where flight test data is not available. Only when all other alternative possible sources of data have been thoroughly reviewed without success may a footprint test be acceptable, subject to a case-by-case review with the competent authorities concerned, and taking into consideration the level of qualification sought for the FSTD.
- (2) Footprint test data should be:
 - (i) constructed with initial conditions and FFS set up in the appropriate configuration (e.g. correct engine rating) for the required validation data;
 - (ii) a manoeuvre representative of the particular aircraft being simulated;
 - (iii) manually flown out by a type rated pilot who has current experience on type* and is deemed acceptable by the competent authority**;
 - (iv) constructed from validation data obtained from the footprint test manoeuvre and transformed into an automatic test;
 - (v) an automatic test run as a fully integrated test with pilot control inputs; and
 - (vi) automatically run for the initial qualification and recurrent evaluations.

* In this context, 'current' refers to the pilot experience on the aircraft and not to the Part-FCL standards.

** The same pilot should sign off the complete test as being fully representative.
- (3) A clear rationale should be included in the QTG for each footprint test. These rationales should be added to and clearly recorded within the validation data roadmap (VDR) in accordance with and as defined in Appendix 2 to AMC1-CS-FSTD(A).300.
- (4) Where the number of footprint tests is deemed by the competent authority to be excessive, the maximum level of qualification may be affected. The competent authority should review each area of validation test data where the use of footprint tests as the

basis for the validation data is proposed. Consideration should be given to the extent to which footprint tests are used in any given area.

For example, it would be unacceptable if all or the vast majority of takeoff tests were proposed as footprint tests, with little or no flight test data being presented. It should be recognised, therefore, that it may be necessary for new flight test data to be gathered if the use of footprint tests becomes excessive, not just overall, but also in specific areas.

- (5) For recurrent evaluation purposes an essential match is to be expected. Validation tests using footprint data which do not provide an essential match should be justified to the satisfaction of the competent authority.

The competent authority should be consulted at the point of definition of the aircraft data for qualification prior to the procurement of the device if footprint tests need to be used.

ORA.FSTD.205 Certification specifications for FSTDs

Regulation (EU) No 1178/2011

- (a) The Agency shall issue, in accordance with Article 19 of Regulation (EC) No 216/2008, Certification Specifications as standard means to show compliance of FSTDs with the Essential Requirements of Annex III to Regulation (EC) No 216/2008.
- (b) Such Certification Specifications shall be sufficiently detailed and specific to indicate to applicants the conditions under which qualifications will be issued.

ORA.FSTD.210 Qualification basis

Regulation (EU) No 70/2014

- (a) The qualification basis for the issuance of an FSTD qualification certificate shall consist of:
- (1) the applicable Certification Specifications established by the Agency that are effective on the date of the application for the initial qualification;
 - (2) the aircraft validation data defined by the mandatory part of the operational suitability data as approved under Regulation (EU) No 748/2012, if applicable; and
 - (3) any special conditions prescribed by the competent authority if the related Certification Specifications do not contain adequate or appropriate standards for the FSTD because the FSTD has novel or different features to those upon which the applicable Certification Specifications are based.
- (b) The qualification basis shall be applicable for future recurrent qualifications of the FSTD, unless it is re-categorised.

ORA.FSTD.225 Duration and continued validity

Regulation (EU) 2024/2076

- (a) The FSTD qualification certificate shall remain valid subject to the following conditions:
- (1) the FSTD and the operating organisation remaining in compliance with the applicable requirements;

- (2) the competent authority being granted access to the organisation as defined in point [ORA.GEN.140](#) to determine continued compliance with the relevant requirements of Regulation (EU) 2018/1139 and the implementing and delegated acts adopted on the basis thereof;
- (3) the qualification certificate not being surrendered or revoked.
- (b) If the competent authority has extended the recurrent evaluation period for an FSTD in accordance with point [ARA.FSTD.120](#)(c) of Annex VI (Part-ARA), the organisation shall assign a person or group of persons with adequate experience who shall do all of the following within a period of 60 days before and 30 days after the start of each recurrent 12-month period in accordance with point [ARA.FSTD.120](#)(b)(1) of Annex VI:
 - (1) review the regular reruns of the complete tests in the master QTG;
 - (2) conduct the relevant functions and subjective tests;
 - (3) send a report of the results to the competent authority.
- (c) A BITD qualification shall remain valid subject to regular evaluation for compliance with the applicable qualification basis by the competent authority in accordance with point [ARA.FSTD.120](#) of Annex VI.
- (d) Upon surrender or revocation, the FSTD qualification certificate shall be returned to the competent authority.

AMC1 ORA.FSTD.225(b) Duration and continued validity

ED Decision 2025/002/R

EXPERIENCE OF THE ASSIGNED PERSON OR GROUP OF PERSONS

- (a) A person or a group of persons assigned in accordance with point [ORA.FSTD.225](#)(b) should:
 - (1) be qualified in flight simulation hardware, software, computer modelling or equivalent with sufficient technical knowledge and experience to understand and analyse objective test results, and assess that the FSTD complies with the applicable certification specifications;
 - (2) be qualified to assess the performance, handling and features of the FSTD, hold or have held an instructor certificate and meet either of the following:
 - (i) hold(s) or has (have) held a type/class rating on the aircraft being simulated; or
 - (ii) is (are) assisted by an instructor that holds or has held a rating as specified in point (2)(i), with sufficient flying experience in that type/class to support the evaluation team;
 - (3) have been involved at least in an initial evaluation or one recurrent evaluation carried out by a competent authority within the past 36 months.
- (b) The organisation that operates the FSTD should maintain the list of assigned person(s) qualified to perform the task.

AMC2 ORA.FSTD.225(b) Duration and continued validity

ED Decision 2025/002/R

EXTENDED RECURRENT EVALUATION PERIOD — ORGANISATION'S DOCUMENTATION

- (a) For the purpose of extending the recurrent evaluation period of an FSTD, the organisation that operates the FSTD should provide the following to the competent authority:
- (1) a declaration that the organisation that operates the FSTD fulfils the criteria of point [ARA.FSTD.120\(c\)\(1\)](#);
 - (2) the procedures in accordance with point [ARA.FSTD.120\(c\)\(4\)](#);
 - (3) the qualification and experience of the person(s) assigned in accordance with point [ORA.FSTD.225\(b\)](#);
 - (4) the FSTD performance evaluation metrics developed in accordance with AMC2 [ORA.FSTD.100](#) for the previous 2 years;
 - (5) the evaluation reports together with the follow-up reports for the previous 2 years.
- (b) The assigned person or group of persons should provide a dossier with the same content as for a recurrent evaluation for information to the competent authority 14 days before performing the tasks under point [ORA.FSTD.225\(b\)](#).
- Note:* Further guidance on the content of the dossier for a recurrent evaluation can be found in [GM3 ORA.FSTD.100](#), point(d).
- (c) The assigned person or group of persons should prepare a report following the objective, functions and subjective testing with results, including any items identified during the conduct of the tasks described in point [ORA.FSTD.225\(b\)](#). The report should be submitted to the competent authority within the period specified in point [ORA.FSTD.225\(b\)](#).

GM1 ORA.FSTD.225(b) Duration and continued validity

ED Decision 2025/002/R

INDEPENDENCE OF THE ASSIGNED PERSON(S)

An effective compliance monitoring function is necessary to support the implementation of the extended evaluation period. It is essential that an appropriate level of independence be maintained, and the tasks referred to in point [ORA.FSTD.225\(b\)](#) should not be undertaken by a person that, within the previous 12-month recurrent period, was involved in the tasks referred to in point [ORA.FSTD.105\(b\)](#) for the relevant FSTD being evaluated.

GM2 ORA.FSTD.225(b) Duration and continued validity

ED Decision 2025/002/R

PROCESS TO CONDUCT THE FUNCTIONS AND SUBJECTIVE TESTS

When conducting the functions and subjective tests, especially for a complex motor-powered aircraft, the assigned person or group of persons, when being qualified in accordance with point (a)(2)(i) of [AMC1 ORA.FSTD.225\(b\)](#), should be supported by a pilot who holds or has held a rating on the same or similar type as the aircraft being simulated.

ORA.FSTD.230 Changes to the qualified FSTD

Regulation (EU) No 1178/2011

- (a) The holder of an FSTD qualification certificate shall inform the competent authority of any proposed changes to the FSTD, such as:
 - (1) major modifications;
 - (2) relocation of the FSTD; and
 - (3) any de-activation of the FSTD.
- (b) In case of an upgrade of the FSTD qualification level, the organisation shall apply to the competent authority for an upgrade evaluation. The organisation shall run all validation tests for the requested qualification level. Results from previous evaluations shall not be used to validate FSTD performance for the current upgrade.
- (c) When an FSTD is moved to a new location, the organisation shall inform the competent authority before the planned activity along with a schedule of related events.

Prior to returning the FSTD to service at the new location, the organisation shall perform at least one third of the validation tests, and functions and subjective tests to ensure that the FSTD performance meets its original qualification standard. A copy of the test documentation shall be retained together with the FSTD records for review by the competent authority.

The competent authority may perform an evaluation of the FSTD after relocation. The evaluation shall be in accordance with the original qualification basis of the FSTD.

- (d) If an organisation plans to remove an FSTD from active status for prolonged periods, the competent authority shall be notified and suitable controls established for the period during which the FSTD is inactive.

The organisation shall agree with the competent authority a plan for the de-activation, any storage and re-activation to ensure that the FSTD can be restored to active status at its original qualification level.

AMC1 ORA.FSTD.230(b) Changes to the qualified FSTD

ED Decision 2012/007/R

UPDATING AND UPGRADING EXISTING FSTDs

- (a) An update is a result of a change to the existing device where it retains its existing qualification level. The change may be certified through a recurrent inspection or an extra inspection if deemed necessary by the competent authority according to the applicable requirements in effect at the time of initial qualification.
- (b) If such a change to an existing device would imply that the performance of the device could no longer meet the requirements at the time of initial qualification, but that the result of the change would, in the opinion of the competent authority, clearly mean an improvement to the performance and training capabilities of the device altogether, then the competent authority might accept the proposed change as an update while allowing the device to retain its original qualification level.
- (c) An upgrade is defined as the raising of the qualification level of a device, or an increase in training credits, which can only be achieved by undergoing an initial qualification according to the latest applicable requirements.

- (d) As long as the qualification level of the device does not change, all changes made to the device should be considered to be updates pending approval by the competent authority.
- (e) An upgrade, and consequent initial qualification according to the latest applicable requirements, is only applicable when the organisation requests another qualification level (reclassification) for the FSTD.

ORA.FSTD.235 Transferability of an FSTD qualification

Regulation (EU) No 290/2012

- (a) When there is a change of the organisation operating an FSTD, the new organisation shall inform the competent authority in advance in order to agree upon a plan of transfer of the FSTD.
- (b) The competent authority may perform an evaluation in accordance with the original qualification basis of the FSTD.
- (c) When the FSTD no longer complies with its initial qualification basis, the organisation shall apply for a new FSTD qualification certificate.

ORA.FSTD.240 Record-keeping

Regulation (EU) No 1178/2011

The holder of an FSTD qualification certificate shall keep records of:

- (a) all documents describing and proving the initial qualification basis and level of the FSTD for the duration of the FSTD's lifetime; and
- (b) any recurrent documents and reports related to each FSTD and to compliance monitoring activities for a period of at least 5 years.

AMC1 ORA.FSTD.240 Record-keeping

ED Decision 2012/007/R

FSTD RECORDS

- (a) FSTD records to be kept should include the following:
 - (1) for the lifetime of the device:
 - (i) the master QTG (MQTG) of the initial evaluation;
 - (ii) the qualification certificate of the initial evaluation; and
 - (iii) the initial evaluation report;
 - (2) for a period of at least 5 years (in paper or electronic format):
 - (i) recurrent QTG runs;
 - (ii) recurrent evaluation reports;
 - (iii) reports of internal functions and subjective testing;
 - (iv) technical log;
 - (v) CMS report;
 - (vi) audit schedule;

- (vii) evaluation programme;
- (viii) management evaluation reports;
- (ix) obsolete procedures and forms.

SUBPART AeMC – AERO-MEDICAL CENTRES

SECTION I – GENERAL

ORA.AeMC.105 Scope

Regulation (EU) 2024/2076

This Subpart establishes the additional requirements to be met by an organisation to qualify for the issue or continuation of an approval as an aero-medical centre (AeMC) to:

- (a) provide aero-medical expertise and practical training for AMEs;
- (b) issue medical certificates and cabin crew medical reports, including initial class 1 medical certificates, or class 3 medical certificates in accordance with Commission Regulation (EU) 2015/340 ⁽¹⁾, or both, as applicable.

GM1 ORA.AeMC.105 Scope

ED Decision 2025/002/R

It is recommended that AeMCs provide support to regional AME peer groups in order to enhance professional expertise.

ORA.AeMC.115 Application

Regulation (EU) 2024/2076

Applicants for an AeMC certificate shall:

- (a) comply with MED.D.005; and
- (b) in addition to the documentation for the approval of an organisation required in point [ORA.GEN.115](#), provide details of activities that are contracted to designated hospitals or medical institutes for the purpose of specialist medical examinations.

AMC1 ORA.AeMC.115 Application

ED Decision 2025/002/R

GENERAL

- (a) The documentation for the approval of an AeMC should include the names and qualifications of all medical staff and of supporting specialist consultants, and a list of medical and technical facilities for initial class 1 and class 3 aero-medical examinations, as applicable according to the scope of the AeMC approval.
- (b) Medical staff should be sufficient to perform the standard required medical examinations to be performed within the organisation of the AeMC.

¹ Commission Regulation (EU) 2015/340 of 20 February 2015 laying down technical requirements and administrative procedures relating to air traffic controllers' licences and certificates pursuant to Regulation (EC) No 216/2008 of the European Parliament and of the Council, amending Commission Implementing Regulation (EU) No 923/2012 and repealing Commission Regulation (EU) No 805/2011 (OJ L 63, 6.3.2015, p. 1, ELI: <http://data.europa.eu/eli/reg/2015/340/oj>).

- (c) The standard required medical examinations should at least encompass the following specialties: ophthalmology including colour vision, otorhinolaryngology, cardiology and mental health.
- (d) Contracted activities with designated hospitals or medical institutes for the purpose of additional specialist medical examinations include clinical attachments or liaison with hospitals, medical institutions and/or specialists.

ORA.AeMC.120 AeMC certificate

Regulation (EU) 2024/2076

An organisation holding an AeMC certificate shall not, at any time, hold more than one AeMC certificate issued with the same scope in accordance with Regulation (EU) 2018/1139 and the implementing and delegated acts adopted on the basis thereof.

ORA.AeMC.135 Continued validity

Regulation (EU) 2024/2076

The AeMC certificate shall be issued for an unlimited duration. It shall remain valid subject to the holder and the aero-medical examiners of the organisation complying with the following conditions:

- (a) complying with point [MED.D.030](#) of Annex IV (Part-MED) to this Regulation or point ATCO.MED.C.025 of Annex IV (Part-ATCO.MED) to Regulation (EU) 2015/340, as applicable;
- (b) ensuring their continued experience by performing an adequate number of class 1 medical examinations, or class 3 medical examinations in accordance with Regulation (EU) 2015/340, or equivalent military aero- medical examinations, as appropriate, every year.

AMC1 ORA.AeMC.135 Continued validity

ED Decision 2025/002/R

EXPERIENCE

- (a) A total of at least 200 class 1, class 3 or equivalent military aero-medical examinations and assessments should be performed at an AeMC every year.
- (b) In Member States where the number of aero-medical examinations and assessments mentioned in point (a) cannot be reached due to a low number of professional pilots or ATCOs, a proportionate number, as defined by the competent authority, of class 1 or class 3 aero-medical examinations and assessments should be performed.
- (c) In these cases, the continued experience of the AMEs in an AeMC may also be ensured by them performing aero-medical examinations and assessments for:
 - (1) class 2 medical certificates as established in Part-MED; and/or
 - (2) third-country class 1 , class 3 or equivalent military medical certificates.
- (d) Aero-medical research including publication in peer-reviewed journals may also be accepted as contributing to the continued experience of the AMEs in an AeMC.

ORA.AeMC.160 Reporting

Regulation (EU) 2024/2076

The AeMC shall provide the competent authority with statistical reports regarding the aero-medical assessments of applicants, including reports of the drugs and alcohol screening performed in accordance with point [MED.B.055](#)(b) of Annex IV (Part-MED) and any health risk factors or trends identified during the aero-medical assessments.

SECTION II – MANAGEMENT

ORA.AeMC.200 Management system

Regulation (EU) 2024/2076

The AeMC shall establish and maintain a management system that includes the items addressed in [ORA.GEN.200](#) and, in addition, processes:

- (a) for medical certification in compliance with Part-MED;
- (aa) to facilitate cooperation between the AMEs and other medical experts of the AeMC; and
- (b) to ensure medical confidentiality at all times.

AMC1 ORA.AeMC.200 Management system

ED Decision 2025/002/R

- (a) In order to maintain personnel trained and competent to perform their tasks as specified in [ORA.GEN.200\(a\)\(4\)](#), the management system should ensure that each AME performs a sufficient number of aero-medical examinations and assessments to meet the professional standards of an AeMC. The required activity of each AME should be specified in the management system.
- (b) The management system should encompass regular exchange of professional expertise including case analysis.

GM1 ORA.AeMC.200 Management system

ED Decision 2012/007/R

RESEARCH

If aero-medical research is conducted at an AeMC, its management system should include processes to conduct that research and publish the results.

GM2 ORA.AeMC.200 Management system

ED Decision 2025/002/R

The assessment of the AeMC's management system by a national health authority may be a part of the AeMC's overall management system.

ORA.AeMC.205 Contracted activities

Regulation (EU) 2024/2076

Notwithstanding point [ORA.GEN.205](#), all of the following shall apply:

- (a) minimum required aero-medical examinations shall be performed within the organisation of the AeMC, in accordance with the scope and privileges defined in the terms of approval attached to the AeMC's certificate;
- (b) additional medical examinations and investigations may be performed by contracted individual experts or organisations. The organisation shall ensure that when contracting any part of its activity, the contracted service or product conforms to the applicable requirements.

AMC1 ORA.AeMC.205(a) Contracted activities

ED Decision 2025/002/R

In addition to the documentation for the approval of an organisation listed in [AMC1 ORA.GEN.205](#), the AeMC should provide a written declaration of their subcontractor that contracted examinations or assessments will be performed on the basis of the requirements of Regulation (EU) No 1178/2011 and associated AMC and GM.

ORA.AeMC.210 Personnel requirements

Regulation (EU) 2024/2076

- (a) The AeMC shall have the following in its staff:
 - (1) an aero-medical examiner (AME) nominated as head of the AeMC, with privileges to issue class 1 medical certificates, or class 3 medical certificates in accordance with Regulation (EU) 2015/340, as applicable, in accordance with the scope defined in the terms of approval attached to the AeMC's certificate and sufficient experience in aviation medicine to exercise his or her duties;
 - (2) at least one additional qualified AME with privileges to issue class 1 medical certificates, or class 3 medical certificates in accordance with Regulation (EU) 2015/340, as applicable, in accordance with the scope defined in the terms of approval attached to the AeMC's certificate privileges, and other technical staff;
 - (3) available medical experts.
- (b) The head of the AeMC shall be responsible for:
 - (1) coordinating the assessment of examination results;
 - (2) signing reports, certificates, and initial class 1 medical certificates and class 3 medical certificates in accordance with Regulation (EU) 2015/340.

AMC1 ORA.AeMC.210 Personnel requirements

ED Decision 2025/002/R

GENERAL

An aero-medical examiner (AME) should have held AME privileges, as applicable in accordance with the scope defined in the terms of approval attached to the AeMC's certificate and have performed at least 200 aero-medical examinations and assessments for a class 1, class 3 or equivalent military medical certificate before being nominated as head of an AeMC.

ORA.AeMC.215 Facility requirements

Regulation (EU) No 1178/2011

The AeMC shall be equipped with medico-technical facilities adequate to perform aero-medical examinations necessary for the exercise of the privileges included in the scope of the approval.

AMC1 ORA.AeMC.215 Facility requirements

ED Decision 2025/002/R

MEDICAL-TECHNICAL FACILITIES

The medical-technical facilities of an AeMC should consist of the equipment of a general medical practice and, in addition, of equipment for:

(a) Cardiology

Facilities to perform:

- (1) 12-lead resting ECG;

(b) Ophthalmology

Facilities for the examination of:

- (1) near, intermediate and distant vision;
- (2) external eye, anatomy, media and funduscopy;
- (3) ocular motility;
- (4) binocular vision;
- (5) colour vision (anomaloscopy or equivalent);
- (6) visual fields;
- (7) refraction
- (8) heterophoria; and
- (9) contrast sensitivity, including mesopic conditions with and without glare.

(c) Hearing

- (1) pure-tone audiometer

(d) Otorhinolaryngology (ENT)

Facilities for the clinical examination of mouth and throat and:

- (1) otoscopy;
- (2) rhinoscopy;
- (3) tympanometry or equivalent; and
- (4) clinical assessment of vestibular system.

(e) Examination of pulmonary function

- (1) spirometry

(f) The following facilities should be available at the AeMC or arranged through contracted activities:

- (1) clinical laboratory facilities; and
- (2) ultrasound of the abdomen;
- (3) exercise ECG;
- (4) 24-hour heart rhythm monitoring;
- (5) 24-hour blood pressure monitoring; and
- (6) mental health assessment including psychometric testing.

ORA.AeMC.220 Record-keeping

Regulation (EU) No 1178/2011

In addition to the records required in [ORA.GEN.220](#), the AeMC shall:

- (a) maintain records with details of medical examinations and assessments performed for the issue, revalidation or renewal of medical certificates and their results, for a minimum period of 10 years after the last examination date; and
- (b) keep all medical records in a way that ensures that medical confidentiality is respected at all times.

ANNEX VIII (PART-DTO)

DTO.GEN.100 General

Regulation (EU) 2018/1119

In accordance with the second subparagraph of Article 10a(1), this Annex (Part-DTO) sets out the requirements applicable to pilot training organisations providing the training referred to in point [DTO.GEN.110](#) on the basis of an declaration made in accordance with point [DTO.GEN.115](#).

DTO.GEN.105 Competent authority

Regulation (EU) 2018/1119

For the purpose of this Annex (Part-DTO), the competent authority in respect of a DTO shall be the authority designated by the Member State on the territory of which the DTO has its principal place of business.

DTO.GEN.110 Scope of the training

Regulation (EU) 2025/134

- (a) A DTO shall be entitled to provide the following training, provided that the DTO has submitted a declaration in accordance with point [DTO.GEN.115](#):
- (1) for aeroplanes:
 - (a) theoretical knowledge instruction for LAPL(A) and PPL(A);
 - (b) flight instruction for LAPL(A) and PPL(A);
 - (c) training towards class rating for SEP(land), SEP(sea) and TMG;
 - (d) training towards additional ratings: night, aerobatics, mountain, sailplane and banner towing;
 - (2) for helicopters:
 - (a) theoretical knowledge instruction for LAPL(H) and PPL(H);
 - (b) flight instruction for LAPL(H), PPL(H);
 - (c) single-engine type rating for helicopters for which the maximum certified seat configuration does not exceed five seats;
 - (d) training towards night rating;
 - (3) for sailplanes, in accordance with the requirements of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#):
 - (a) theoretical knowledge instruction for the SPL;
 - (b) flight instruction for the SPL;
 - (c) training towards extension of privileges to sailplanes or TMGs in accordance with point SFCL.150;
 - (d) training towards additional launching methods in accordance with point SFCL.155;

- (e) training towards additional ratings and privileges: basic aerobatic and advanced aerobatic privileges, sailplane and banner towing rating, TMG night rating, and sailplane cloud flying privileges;
 - (f) training towards flight instructor certificate for sailplanes (FI(S));
 - (g) FI(S) refresher course;
 - (4) for balloons, in accordance with the requirements of Annex III (Part-BFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#):
 - (a) theoretical knowledge instruction for the BPL;
 - (b) flight instruction for the BPL;
 - (c) training towards class or group extension in accordance with point BFCL.150;
 - (d) training towards additional ratings: tethered hot-air balloon flight, night, and commercial operation rating;
 - (g) training towards flight instructor certificate for balloons (FI(B));
 - (h) FI(B) refresher course.
 - (5) for gyroplanes:
 - (a) theoretical knowledge instruction for the GPL;
 - (b) flight instruction for the GPL;
 - (c) training towards privileges for single-pilot gyroplane classes and types;
 - (d) training towards night rating
- [applicable from 18 February 2026 - Regulation (EU) 2025/134]*
- (b) A DTO shall be entitled to also provide the examiner courses referred to in points BFCL.430 and BFCL.460(b)(1) of Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) for FE(B), as well as in points SFCL.430 and SFCL.460(b)(1) of Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#) for FE(S), provided that the DTO has submitted a declaration in accordance with point [DTO.GEN.115](#) and the competent authority has approved the training programme in accordance with point [DTO.GEN.230\(c\)](#).

GM1 DTO.GEN.110 Scope

ED Decision 2018/009/R

Point [DTO.GEN.110](#) lists all the training activities that are regulated by Part-FCL and which can be conducted at a DTO. However, for some of the training activities mentioned, Part-FCL (points FCL.130.S, FCL.130.B, FCL.225.B(b), FCL.810(c)) does not require the involvement of a training organisation at all. In this regard, point [DTO.GEN.110](#) does not constitute an obligation for these training activities to be undertaken at a DTO only.

DTO.GEN.115 Declaration

Regulation (EU) 2020/359

- (a) Prior to providing any of the training specified in point [DTO.GEN.110](#), an organisation intending to provide such training shall submit a declaration to the competent authority. The declaration shall contain at least the following information:
 - (1) the name of the DTO;

- (2) contact details of the DTO's principal place of business and, where applicable, the contact details of the aerodromes and the operating sites of the DTO;
 - (3) names and contact details of the following persons:
 - (i) the representative of the DTO;
 - (ii) the head of training of the DTO; and
 - (iii) all deputy heads of training, if required by point [DTO.GEN.250\(b\)\(1\)](#);
 - (4) the type of training, as specified in point [DTO.GEN.110](#), provided at each aerodrome and/or operating site;
 - (5) a list of all aircraft and FSTDs to be used for the training, if applicable;
 - (6) the date of intended commencement of the training;
 - (7) a statement confirming that the DTO has developed a safety policy and will apply that policy during all training activities covered by the declaration, in accordance with point [DTO.GEN.210\(a\)\(1\)\(ii\)](#);
 - (8) a statement that confirms that the DTO complies and will, during all training activities covered by the declaration, continue to comply with the essential requirements set out in Annex IV to [Regulation \(EU\) 2018/1139](#), with the requirements of Annex I (Part-FCL) and Annex VIII (Part-DTO) to this Regulation and with the requirements of Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#).
- (b) The declaration, and any subsequent changes thereto, shall be made using the form contained in [Appendix 1](#).
- (c) A DTO shall, together with the declaration, submit to the competent authority the training programme or programmes, which it uses or intends to use to provide the training, as well as its application for approval of the training programme or programmes where such approval is required in accordance with point [DTO.GEN.230\(c\)](#).
- (d) By derogation from point (c), an organisation which holds an approval issued in accordance with Subpart ATO of Annex VII (Part-ORA) may, together with the declaration, only submit the reference to the already approved training manual or manuals.

GM1 DTO.GEN.115(a) Declaration

ED Decision 2018/009/R

SUBMISSION OF THE DECLARATION

The DTO should submit the declaration ([Appendix 1](#) to Part-DTO), and any attachment(s) thereto, in a manner established by the competent authority.

GM2 DTO.GEN.115(a) Declaration

ED Decision 2018/009/R

RESPONSIBILITY OF THE DTO FOR THE SUCCESSFUL SUBMISSION OF THE DECLARATION

It is the responsibility of the DTO to successfully submit the declaration to the competent authority. If the DTO does not receive the acknowledgement of receipt of the declaration from the competent authority pursuant to point ARA.DTO.100 within a reasonable period of time following the submission

of the declaration, the DTO should contact the competent authority to investigate whether or not the submission of the declaration has been successful.

AMC1 DTO.GEN.115(a)(2) Declaration

ED Decision 2018/009/R

LIST OF AERODROMES AND OPERATING SITES OF THE DTO

Except for DTOs that provide training for balloons, the list of aerodromes and operating sites on the declaration should contain at least those aerodromes and operating sites where the DTO, either permanently or temporarily (e.g. for training camps), conducts its training activities, where its training aircraft are based and where it has its facilities, as required by Part-DTO.

Aerodromes and operating sites that solely serve as destinations for cross-country training flights do not need to be listed on the declaration.

AMC1 DTO.GEN.115(a)(5) Declaration

ED Decision 2018/009/R

LIST OF AIRCRAFT AND FLIGHT SIMULATION TRAINING DEVICES (FSTDs)

- (a) The list on the declaration of aircraft used by the DTO should contain at least the models used for training (e.g. Cessna 152, Piper PA 28, Robinson R22, etc.). It is not necessary to list on the declaration each individual aircraft with its registration mark.
- (b) The list on the declaration of FSTDs used by the DTO should contain the references to the FTSD qualification certificates.

AMC1 DTO.GEN.115(c) Declaration

ED Decision 2018/009/R

SUBMISSION OF TRAINING PROGRAMMES WITH THE DECLARATION

Except for training programmes for examiner standardisation or refresher courses, a DTO may include in the declaration only a reference to a training programme if this training programme:

- (a) has already been verified for Part-FCL compliance by the competent authority; or
- (b) has been developed by the competent authority as a standard training programme, if applicable.

DTO.GEN.116 Notification of changes and cessation of training activities

Regulation (EU) 2018/1119

A DTO shall notify the competent authority without undue delay of the following:

- (a) any changes to the information contained in the declaration specified in point [DTO.GEN.115\(a\)](#) and to the training programme or programmes or the approved training manual or manuals referred to in points [DTO.GEN.115\(c\)](#) and [\(d\)](#) respectively;
- (b) the cessation of some or all training activities covered by the declaration.

DTO.GEN.135 Termination of entitlement to provide training

Regulation (EU) 2024/2076

- (a) A DTO shall no longer be entitled to provide some or all of the training specified in its declaration on the basis of that declaration, where one of the following occurs:
 - (1) the DTO has notified the competent authority of the cessation of some or all of the training activities covered by the declaration in accordance with point [DTO.GEN.116\(b\)](#);
 - (2) the DTO has not provided the training for more than 36 consecutive months.
- (b) A DTO shall return approval certificates in accordance with point [DTO.GEN.230\(c\)](#) to the competent authority without delay:
 - (1) in case of cessation of training activities in accordance with point (a);
 - (2) in case of revocation in accordance with point [ARA.GEN.350\(da\)\(3\)](#) of Annex VI (Part-ARA).

DTO.GEN.140 Access

Regulation (EU) 2018/1119

For the purpose of determining whether a DTO is acting in compliance with its declaration, the DTO shall grant access at any time to any facility, aircraft, document, records, data, procedures or any other material relevant to its training activities covered by the declaration, to any person authorised by the competent authority

DTO.GEN.150 Findings

Regulation (EU) 2018/1119

After the competent authority has communicated a finding to a DTO in accordance with point ARA.GEN.350(da)(1), the DTO shall take the following steps within the time period determined by the competent authority:

- (a) identify the root cause of the non-compliance;
- (b) take the necessary corrective action to terminate the non-compliance and, where relevant, remedy the consequences thereof;
- (c) inform the competent authority about the corrective action it has taken.

DTO.GEN.155 Reaction to a safety problem

Regulation (EU) 2018/1119

As a reaction to a safety problem, a DTO shall implement:

- (a) the safety measures mandated by the competent authority in accordance with point ARA.GEN.135(c);
- (b) the relevant mandatory safety information issued by the Agency, including airworthiness directives.

DTO.GEN.210 Personnel requirements

Regulation (EU) 2020/359

- (a) A DTO shall designate:

-
- (1) a representative, who shall be responsible and duly authorised to do at least the following:
 - (i) ensure compliance of the DTO and its activities with the applicable requirements and with its declaration;
 - (ii) develop and establish a safety policy which ensures that the DTO's activities are carried out safely, ensure that the DTO adheres to that safety policy and take the necessary measures in order to achieve the objectives of that safety policy;
 - (iii) promote safety within the DTO;
 - (iv) ensure the availability of sufficient resources within the DTO so that the activities referred to in points (i), (ii) and (iii) can be carried out in an effective manner.
 - (2) a head of training, who shall be responsible and qualified to ensure at least the following:
 - (i) that the training provided complies with the requirements of Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Regulation \(EU\) 2018/1976](#) and with the DTO's training programme;
 - (ii) the satisfactory integration of flight training in an aircraft or a flight simulation training device (FSTD) and theoretical knowledge instruction;
 - (iii) the supervision of the progress of students;
 - (iv) in the case referred to in point [DTO.GEN.250\(b\)](#), the supervision of the deputy head or heads of training.
- (b) A DTO may designate a single person as its representative and its head of training.
- (c) A DTO shall not designate a person as its representative or its head of training if there are objective indications that he or she cannot be trusted to carry out the tasks listed in point (a) in a manner which safeguards and furthers aviation safety. The fact that a person has been subject to an enforcement measure taken in accordance with point ARA.GEN.355 in the past three years shall be deemed to constitute such an objective indication, unless that person can demonstrate that the finding leading to that measure, by reason of its nature, scale or impact on aviation safety, is not such as to indicate that he or she cannot be trusted to carry out those tasks in that manner.
- (d) A DTO shall ensure that its theoretical knowledge instructors have either of the following qualifications:
- (1) practical background in aviation in the areas relevant for the training provided and have undergone a course of training instructional techniques;
 - (2) previous experience in giving theoretical knowledge instruction and an appropriate theoretical background in the subject on which they will provide theoretical knowledge instruction.
- (e) Flight instructors and flight simulation training instructors shall hold the qualifications required by Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#) for the type of training they provide.

GM1 DTO.GEN.210(a)(1)(i) Personnel requirements

ED Decision 2018/009/R

OCCURRENCE-REPORTING SYSTEM COMPLIANT WITH REGULATION (EU) No 376/2014

The following list provides an overview of the main elements of the occurrence-reporting system that is compliant with Regulation (EU) No 376/2014 and provides references to the relevant articles of that Regulation.

- (a) Occurrence-reporting system that caters for both mandatory and voluntary reporting (cf. Articles 4 and 5).

Note 1: The mandatory reporting system established under Regulation (EU) No 376/2014 is also intended for the reporting of those additional items that qualify for mandatory reporting and are defined in the EASA implementing rules.

Note 2: The voluntary reporting system is intended to facilitate the collection of details of occurrences that may not be captured by the mandatory system and of other safety-related information which is perceived by the reporting organisation as an actual or potential hazard to aviation safety.

- (b) Designation of one or more persons to independently handle the collection, evaluation, processing, analysis and storage of details of occurrences with regard to data collection and hazard identification (cf. Article 6(1)).

Note 1: In agreement with their competent authority, small-sized organisations may make use of simplified mechanisms to ensure the collection, evaluation, processing, analysis and storage of details of occurrences, possibly by sharing those tasks with other similar organisations.

Note 2: An existing internal safety-reporting scheme, which collects safety-relevant data, proposals and information, including data, proposals and information on potential safety issues that have not resulted in any occurrence, may serve as a basis for the mandatory and voluntary occurrence-reporting system. From this pool of safety relevant information and data collected internally, the organisation will determine whether a mandatory report is required or whether a voluntary report may be adequate.

- (c) Reporting details of occurrences collected under the mandatory scheme as soon as possible and in any event no later than 72 hours after becoming aware of the occurrence (cf. Article 4(8) & (9)).

Note 1: The reference to 'becoming aware of' an occurrence implies that a person in the organisation identifies the occurrence as falling into the category of a mandatory occurrence report — usually through being involved in the occurrence or witnessing it, but also on review or investigation of information reported to the organisation's safety reporting scheme. In the case of design or production organisations, the 72-hour period starts at the point when the unsafe condition is identified.

In the case of automated data collection systems, the 72-hour period starts when the person that is responsible for the analysis of the data detected the reportable occurrence.

Note 2: The 72-hour period does not apply to the reporting of details of occurrences which may involve an actual or potential aviation safety risk and safety-related information collected under the voluntary scheme. These are to be reported in a timely manner (cf. Article 5(5) & (6)).

- (d) Establishment of data quality checking processes to ensure that the information initially collected and the data stored in the database(s) are consistent (cf. Article 7(3)).

Note: It is understood that data quality checking processes should address four main areas:

- errors in data entry;
- completeness of data, specially referring to mandatory data;
- proper use of the ADREP¹ taxonomy;
- improve data consistency, notably between the information collected initially and the report stored in the database (cf. Article 7(3)).

- (e) Storage of occurrence reports that qualify for mandatory and voluntary reporting in one or more databases (cf. Article 6(5)) using standardised formats to facilitate information exchange and which are compatible with the ECCAIRS² software and ADREP taxonomy (cf. Article 7(4)).

Note: Organisations that are able to report through an ECCAIRS software compatible reporting system provided by their competent authority are deemed to be automatically compliant with the reporting system requirements in Article 7(4) and do not need to have their own ECCAIRS software compatible reporting system.

- (f) Application of the safety policy (cf. [AMC1 DTO.GEN.210\(a\)\(1\)\(ii\)](#)) to occurrences:
- (1) identification of the safety hazards that are associated with identified occurrences or groups of occurrences reported to the competent authority (cf. Article 13(1));
 - (2) analysis of the related risks in terms of probability and severity of the outcome, as well as assessment of the risks in terms of tolerability;
 - (3) based on the result of the risk assessment: determination of the need for mitigation action, as required for improving aviation safety (cf. Article 13(2)); and
 - (4) monitoring the timely implementation and effectiveness of any mitigation action required (cf. Article 13(2)).
- (g) In addition to the actions required under paragraph (6) above, where the organisation identifies an actual or potential aviation safety risk as a result of the analysis of occurrences or group of occurrences:
- (1) transmission of the following information to the competent authority within 30 days from the date of notification of the occurrence to the authority (cf. Article 13(4)):
 - (i) the preliminary results of the risk assessment performed; and
 - (ii) any preliminary mitigation action to be taken.
 - (2) where required, transmission of the final results of the risk analysis to the competent authority as soon as they are available and, in principle, no later than 3 months from the date of notification of the occurrence to the authority (cf. Article 13(4)).

Note: The legal obligation to provide the initial results of the analysis of the occurrence, follow-up reports and final results lies with the other organisation that issued the initial report. Where an organisation receives a copy of a report from another organisation that initially reported the occurrence to the competent authority, depending on its contribution to the actual or potential aviation safety risk underlying the occurrence, it may however be required to perform its own analysis of the issue reported and to provide a follow-up report to the competent authority.

¹ The ICAO Accident/Incident Data Reporting (ADREP) system.

² European Coordination Centre for Accident and Incident Reporting Systems.

- (h) Safety policy and just culture: Consultation of staff representatives to ensure mutual agreement on and adoption of the rules describing how 'just culture' principles are guaranteed and implemented within the organisation.

Note 1: The purpose of those rules is to ensure that employees and contracted personnel that report or are mentioned in occurrence reports, both mandatory or voluntary, are not subject to any prejudice by their employer or any other organisation for which the services are provided on the basis of the information supplied by the reporter (cf. Article 16(9)), unless an exception applies (cf. Article 16(10)).

Note 2: Staff representatives may be nominated either by the trade union(s) or by the staff themselves.

- (i) Ensuring that employees and contracted personnel are regularly provided with information concerning the analysis of, and follow-up on, occurrences for which mitigation action is taken (cf. Article 13(3), while ensuring that only disidentified information is disseminated.
- (j) Ensuring that personal details are made available to staff of their organisation, other than the persons designated in accordance with paragraph (2), only where absolutely necessary to investigate occurrences with a view to enhancing aviation safety.
- (k) Ensuring that reports addressed to the competent authority contain at least the information listed in Annex I to Regulation (EU) No 376/2014.

AMC1 DTO.GEN.210(a)(1)(ii) Personnel requirements

ED Decision 2025/002/R

SAFETY POLICY

- (a) The safety policy should define, in relation to the DTO training programme, at least the means and methods used for:
- (1) hazard identification;
 - (2) risk assessment; and
 - (3) effectiveness of the mitigation measures (implementation and follow-up).
- (b) The safety policy should additionally include the procedures required for occurrence reporting pursuant to Regulation (EU) No 376/2014 (see [GM1 DTO.GEN.210\(a\)\(1\)\(i\)](#)).

AMC1 DTO.GEN.210(a)(2) Personnel requirements

ED Decision 2018/009/R

QUALIFICATION AND EXPERIENCE OF THE HEAD OF TRAINING (HT)

- (a) The HT should, with regard to the size and the training scope of the DTO, possess sufficient managerial capabilities in order to discharge their responsibilities, and should:
- (1) in the case of a DTO that provides training in aircraft or FSTDs, hold an unrestricted instructor certificate in accordance with Part-FCL with instructional privileges that are relevant to the training provided by the DTO, including sufficient experience as necessary;
 - (2) in the case of a DTO that provides theoretical knowledge training only, have appropriate experience in aviation and knowledge relevant to the training provided.

- (b) At a DTO that provides training courses for different aircraft categories, the HT should be assisted by one or more nominated deputy HTs qualified in accordance with paragraph (a) and with regard to the other category or categories of aircraft.

GM1 DTO.GEN.210(a)(2) Personnel requirements

ED Decision 2025/002/R

SUFFICIENT EXPERIENCE OF THE HEAD OF TRAINING (HT)

‘Sufficient experience’, as per [AMC1 DTO.GEN.210\(a\)\(2\)](#) point (a)(1), means that the HT should have gained the required experience as an instructor in order to have the capacity to administer the particular training activity of the DTO in question. The following factors should be taken into consideration for determining the experience required:

- (a) training scope of the DTO, including specific training courses (e.g. aerobatic rating, sailplane cloud flying rating, examiner courses for sailplanes and balloons);
- (b) location of the DTO training area (e.g. mountains, sea, congested airspace);
- (c) size of the DTO (volume of activity, number of training aerodromes and operating sites);
- (d) use of FSTDs;
- (e) training aircraft models used by the DTO.

GM1 DTO.GEN.210(c) Personnel requirements

ED Decision 2025/002/R

CIRCUMSTANCES UNDER WHICH A PERSON CANNOT BE TRUSTED TO CARRY OUT THE TASKS OF A REPRESENTATIVE OR A HEAD OF TRAINING (HT) IN A MANNER WHICH SAFEGUARDS AND FURTHERS AVIATION SAFETY

Examples of objective indications that a person cannot be trusted to carry out the tasks of a representative or an HT in a manner which safeguards and furthers aviation safety.

If that person, within the last 3 years preceding their nomination as representative or HT, in a declaration in accordance with Part-DTO:

- (a) holds or has held a pilot licence and that licence and/or any associated ratings, certificates or authorisations have been subject to limitation, suspension or revocation;
- (b) has knowingly and deliberately been responsible for committing any non-compliance with the Basic Regulation and its implementing rules.

AMC1 DTO.GEN.210(d);(e) Personnel requirements

ED Decision 2018/009/R

DTO INSTRUCTORS

In order to ensure and monitor that instructors maintain their required qualification, DTOs should permanently keep a list of all instructors, including information on their instructional privileges as well as on the validity periods of their licences, ratings and certificates, including their medical certificates.

GM1 DTO.GEN.210(d);(e) Personnel requirements

ED Decision 2018/009/R

RESOURCES (INSTRUCTORS)

- (a) The ratio of all students to flight instructors should allow maintaining the quality and safety of the training provided.
- (b) Class numbers in ground subjects involving a high degree of supervision or practical work should not exceed 28 students.

DTO.GEN.215 Facility requirements

Regulation (EU) 2018/1119

A DTO shall have facilities in place allowing the performance and management of all its activities in accordance with the essential requirements of Annex III to Regulation (EC) No 216/2008 and with the requirements of this Annex (Part-DTO).

AMC1 DTO.GEN.215 Facility requirements

ED Decision 2018/009/R

- (a) The facilities of a DTO should comprise:
 - (1) flight planning facilities providing access to at least:
 - (i) appropriate and current aviation maps and charts;
 - (ii) current aeronautical information service (AIS) information;
 - (iii) current meteorological information;
 - (iv) communications to air traffic control (ATC) (if applicable);
 - (v) any other flight-safety-related material;
 - (2) adequate briefing facilities of sufficient size and number;
 - (3) suitable office(s) to allow flight instructors to write reports on students, complete records and other related documentation, as appropriate;
 - (4) suitable rest areas for instructors and students, where appropriate to the training task;
 - (5) in the case of DTOs that provide training for BPL or LAPL(B) only, the flight operations accommodation listed in (a)(1) to (a)(4) may be replaced by other suitable facilities when operating outside aerodromes.
- (b) The following facilities for theoretical knowledge instruction should be available:
 - (1) adequate classroom accommodation for the current student population;
 - (2) suitable demonstration equipment to support the theoretical knowledge instruction;
 - (3) suitable office(s) for the instructional personnel.

DTO.GEN.220 Record-keeping

Regulation (EU) 2018/1119

- (a) A DTO shall keep for each individual student the following records throughout the training course and for three years after completion of the last training session:
 - (1) details of ground, flight and simulated flight training;
 - (2) information on individual progress;

- (3) information on the licences and associated ratings relevant to the training provided, including expiry dates of ratings and medical certificates.
- (b) A DTO shall keep the report on the annual internal review and the activity report referred to in point [DTO.GEN.270\(a\) and \(b\)](#) respectively for three years from the date at which the DTO established those reports.
- (c) A DTO shall keep its training programme for three years from the date at which it provided the last training course in accordance with that programme.
- (d) A DTO shall, in accordance with the applicable law on the protection of personal data, store the records referred to in point (a) in a manner that ensures protection by appropriate tools and protocols and take the necessary measures to restrict the access to those records to persons who are duly authorised to access them.

AMC1 DTO.GEN.220 Record-keeping

ED Decision 2018/009/R

Training records should be kept in a paper or electronic version by the DTO where the candidate is undertaking their training.

DTO.GEN.230 DTO training programme

Regulation (EU) 2020/359

- (a) A DTO shall establish a training programme for each of the trainings specified in point [DTO.GEN.110](#) which the DTO provides.
- (b) The training programmes shall comply with the requirements of Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), as applicable.
- (c) A DTO shall be entitled to provide the training referred to in point [DTO.GEN.110\(b\)](#) only when its training programme for that training, and any changes thereto, have been issued by the competent authority, upon application by the DTO, with an approval in accordance with point [ARA.DTO.110](#), confirming that the training programme and any changes thereto comply with the requirements of Annex I (Part-FCL), Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#), as applicable. A DTO shall apply for such approval through the submission of its declaration in accordance with point [DTO.GEN.115](#).
- (d) Point (c) shall not apply to an organisation also holding an approval issued in accordance with Subpart ATO of Annex VII (Part-ORA) that includes privileges for that training.

AMC1 DTO.GEN.230 DTO training programme

ED Decision 2018/009/R

- (a) The DTO training programme should include at least the following information:
 - (1) the aim of the course;
 - (2) crediting of previous experience and pre-entry requirements (including appropriate procedures for students that wish to complete their training after having started at a different training organisation);

- (3) a list of all air and FSTD exercises to be taught, including a description of the objective of each exercise;
 - (4) a syllabus summary;
 - (5) structure and content of the theoretical knowledge instruction;
 - (6) structure of the entire course and integration of theoretical knowledge instruction, FSTD and flight training;
 - (7) student progress checks for theoretical knowledge and flight training, as appropriate.
- (b) When developing the training programme for a type rating course, in addition to complying with the mandatory training elements included in the operational suitability data (OSD), as established in accordance with Regulation (EU) No 748/2012¹, the DTO should also follow any further recommendations (i.e. acceptable means of compliance (AMC)) contained therein.

DTO.GEN.240 Training aircraft and FSTDs

Regulation (EU) 2019/1747

- (a) A DTO shall use an adequate fleet of training aircraft or FSTDs appropriately equipped for the training course provided. The fleet of aircraft shall be composed of aircraft that comply with all requirements defined in Regulation (EU) 2018/1139. Aircraft that fall under points (a), (b), (c) or (d) of Annex I to Regulation (EU) 2018/1139, may be used for training if all of the following conditions are met:
- (1) during an evaluation process the competent authority has confirmed a level of safety comparable to the one defined by all essential requirements laid down in Annex II to Regulation (EU) 2018/1139;
 - (2) the competent authority has authorised the use of the aircraft for training in the DTO.
- (b) A DTO shall establish and keep up-to-date a list of all aircraft, including their registration marks, used for the training it provides.

AMC1 DTO.GEN.240 Training aircraft and FSTDs

ED Decision 2018/009/R

- (a) The number of training aircraft may be affected by:
- (1) the availability of FSTDs; and
 - (2) the number of aerodromes and operating sites of the DTO (cf. [AMC1 DTO.GEN.115\(a\)\(2\)](#)).
- (b) Each training aircraft should be:
- (1) equipped as required in the training specifications concerning the exercise for which it is used;
 - (2) except in the case of balloons or single-seat aircraft, fitted with primary flight controls that are instantly accessible by both the student and the instructor (for example, dual flight controls or a centre control stick); swing-over flight controls should not be used.
- (c) The fleet should include, as appropriate to the training courses:

¹ Commission Regulation (EU) No 748/2012 of 3 August 2012 laying down implementing rules for the airworthiness and environmental certification of aircraft and related products, parts and appliances, as well as for the certification of design and production organisations (OJ L 224, 21.8.2012, p. 1), as amended.

- (1) in the case of aeroplanes and sailplanes, aircraft suitable for demonstrating stalling and spin avoidance;
 - (2) in the case of helicopters, helicopters suitable for autorotation demonstration;
 - (3) FSTDs; each FSTD should be equipped as required in the training specifications concerning the course for which it is used.
- (d) One single aircraft that has all the required characteristics of a training aircraft mentioned in (b) and (c) above may be sufficient.

GM1 DTO.GEN.240 Training aircraft and FSTDs

ED Decision 2018/009/R

The DTO is required to use an adequate fleet of training aircraft. However, a DTO is not required to own the aircraft used. In any case, the DTO has the responsibility to use airworthy and appropriately equipped, certified and insured aircraft and FSTDs, as relevant to the particular training exercise.

AMC3 DTO.GEN.240 Training aircraft and FSTDs

ED Decision 2020/005/R

EVALUATION PROCESS

Two cases for the evaluation process of Annex-I aircraft are distinguished:

- (a) Annex-I aircraft that hold an ICAO-level certificate of airworthiness (CoA)
 - (1) To support the evaluation process performed by the competent authority and provide the competent authority with sufficient data related to the aircraft in question, an instructor who is qualified in accordance with Annex I (Part-FCL) to Regulation (EU) No 1178/2011 and nominated by the head of training (HT) of the DTO should assess that the aircraft is appropriately equipped and suitable for the training courses provided. The result of this assessment should be submitted to the competent authority and may be included already in the application for the authorisation.
 - (2) During the evaluation process, the competent authority should consider aircraft that hold a CoA issued in accordance with Annex 8 to the Chicago Convention to provide a level of safety comparable to that required by Annex II to the Basic Regulation, unless the competent authority determines that the airworthiness requirements used for certification of the aircraft, or the service experience, or the safety system of the State of design, do not provide for a comparable level of safety.
- (b) Annex-I aircraft that do not hold an ICAO-level CoA

Before the inclusion of these aircraft in the fleet of an DTO and their use in training to obtain Part-FCL licences and ratings, the DTO should apply for the authorisation to the competent authority that should perform the evaluation process in the following order:

 - (1) Initial assessment by the competent authority and criteria taken into consideration

The competent authority should take into account the following criteria (non-exhaustive list):

 - (i) national airworthiness requirements based on which the aircraft CoA was issued;
 - (ii) aircraft similarities to a certified variant;
 - (iii) aircraft with a satisfactory in-service experience as training aircraft;

- (iv) simple and conventional aircraft design;
 - (v) aircraft that does not have hazardous design features or details, judging by experience; and
 - (vi) operable aircraft systems, equipment, and appliances that do not require exceptional skills or strength.
- (2) Additional assessment by a qualified instructor

To support the evaluation process performed by the competent authority and provide the competent authority with sufficient data related to the aircraft in question, after the positive initial assessment by the competent authority as per point (1), an instructor who is qualified in accordance with Part-FCL and nominated by the HT of the DTO should show through an evaluation report that the aircraft is appropriately equipped and suitable for the training courses provided. That evaluation report should consider all of the following criteria:

- (i) the aircraft should be safely controllable and manoeuvrable under all anticipated operating conditions, including after failure of one or more propulsion systems;
- (ii) the aircraft should allow for a smooth transition from one flight phase to another without requiring exceptional piloting skills, alertness, strength, or workload under any probable operating conditions;
- (iii) the aircraft should have sufficient stability to ensure that the demands made on the pilot are not excessive, considering the phase and duration of flight; and
- (iv) the assessment should take into account control forces, flight deck environment, pilot workload, and other human factors (HF) considerations, depending on the phase and duration of flight.

Subject to a positive evaluation report as per point (2), the competent authority should issue the authorisation.

DTO.GEN.250 Aerodromes and operating sites

Regulation (EU) 2018/1119

- (a) When providing flight training on an aircraft, a DTO shall only use aerodromes or operating sites that have the appropriate facilities and characteristics to allow training of the relevant manoeuvres, taking into account the training provided and the category and type of aircraft used.
- (b) When a DTO uses more than one aerodrome to provide any of the training specified in point [DTO.GEN.110\(a\)\(1\) and \(2\)](#), it shall:
 - (1) for each additional aerodrome, designate a deputy head of training, who shall be responsible for the tasks referred to in point [DTO.GEN.210\(a\)\(2\)\(i\) to \(iii\)](#) on that aerodrome; and
 - (2) ensure the availability of sufficient resources to safely operate on all aerodromes, in compliance with the requirements of this Annex (Part-DTO).

AMC1 DTO.GEN.250 Aerodromes and operating sites

ED Decision 2018/009/R

GENERAL

- (a) Except in the case of balloons, the base aerodrome or operating site and any other aerodromes or operating sites at which flight training is being conducted should have at least the following facilities:
 - (1) at least one runway or final approach and take-off area (FATO) that allows training aircraft to make a normal take-off or landing within the performance limits of all the aircraft used for the training flights at that aerodrome or operating site;
 - (2) a wind direction indicator that is visible at ground level from the ends of each runway or at the appropriate holding points;
 - (3) adequate runway electrical lighting, if used for night training;
 - (4) an air traffic service (ATS), except for uncontrolled aerodromes or operating sites where the training requirements may be satisfied safely by another acceptable means of communication.
- (b) In addition to (a), for helicopters, training sites should be available for:
 - (1) confined area operation training;
 - (2) simulated engine-off autorotation; and
 - (3) sloping ground operation.
- (c) In the case of balloons, the take-off sites used by the DTO should allow a normal take-off and clearing of all obstacles in the take-off flight path by at least 50 ft.
- (d) By way of derogation from paragraphs (a) to (c) above, for training that needs to take place in a specific environment (training for mountain rating, training on seaplanes), the training sites used should have the characteristics and facilities that are necessary to ensure a safe conduct of the training.

AMC1 DTO.GEN.250(b) Aerodromes and operating sites

ED Decision 2018/009/R

SUFFICIENT RESOURCES OF A DTO THAT PROVIDES TRAINING FOR AEROPLANES OR HELICOPTERS AT MORE THAN ONE AERODROME OR OPERATING SITE

- (a) Deputy heads of training should meet the same qualification requirements as set out in [AMC1 DTO.GEN.210\(a\)\(2\)](#) for the head of training (HT).
- (b) The DTO should have the necessary number of instructors (point [DTO.GEN.210\(d\) and \(e\)](#)) as well as the necessary number of training aircraft (point [DTO.GEN.240](#)) in place to ensure proper training at all aerodromes and operating sites.
- (c) At each aerodrome or operating site of the DTO, the DTO should have in place the facilities (point [DTO.GEN.215](#)) as appropriate for the type of training carried out at each aerodrome or operating site.

DTO.GEN.260 Theoretical knowledge instruction

Regulation (EU) 2018/1119

- (a) When providing theoretical knowledge instruction, a DTO may use on-site instruction or distance learning.
- (b) A DTO shall monitor and record the progress of any student undergoing theoretical knowledge instruction.

DTO.GEN.270 Annual internal review and annual activity report

Regulation (EU) 2018/1119

A DTO shall take the following steps:

- (a) conduct an annual internal review of the tasks and responsibilities specified in point [DTO.GEN.210](#) and establish a report on that review;
- (b) establish an annual activity report;
- (c) submit the report on the annual internal review and the annual activity report to the competent authority by the date determined by the competent authority.

AMC1 DTO.GEN.270(a) Annual internal review and annual activity report

ED Decision 2025/002/R

ANNUAL INTERNAL REVIEW

The annual internal review should consist of a comprehensive assessment of whether the DTO effectively carries out the tasks and responsibilities pursuant to point [DTO.GEN.210](#). Specific emphasis should be given to the following:

- (a) availability of sufficient resources;
- (b) conduct of training in accordance with the requirements of Part-FCL and Part-DTO, with the DTO training programme(s) and with the DTO's safety policy;
- (c) random checks of training records and course completion certificates issued by the DTO;
- (d) assessment of the training programme(s) for its (their) adequacy and currency;
- (e) training aircraft including their documents and maintenance records;
- (f) aerodromes and operating sites, including associated facilities;
- (g) evaluation of both adequacy and effectiveness of the follow-up, corrective and, as applicable, remedial action taken after non-compliances that have been detected internally or that have been subject to findings as per point [DTO.GEN.150](#);
- (h) assessment of the safety policy including its means and methods as defined in AMC1 [DTO.GEN.210](#)(a)(1)(ii) for its adequacy and currency;
- (i) assessment of the effectiveness of the implementation of the mitigation measures, as foreseen in the DTO's safety policy.

AMC1 DTO.GEN.270(b) Annual internal review and annual activity report

ED Decision 2018/009/R

ANNUAL ACTIVITY REPORT

- (a) With regard to the past calendar year, the annual activity report should contain at least lists of:
 - (1) all training courses and refresher trainings actually provided;
 - (2) names of all flight, synthetic flight and theoretical knowledge instructors involved in the provision of training, including, in the case of DTOs for aeroplanes, helicopters and

sailplanes, information on the aerodromes and operating sites of the DTO where it has mainly been providing training;

- (3) number of students per training course;
- (4) all training aircraft and FSTDs used, including registration marks and FSTD qualification letter codes (as applicable), including, with regard to each aircraft, information on:
 - (i) the training courses for which the aircraft has been used; and
 - (ii) the aerodromes of the DTO where the aircraft has been mainly used;
- (5) all occurrences, accidents and incidents that occurred during the training courses; and
- (6) any other information that is deemed relevant by the DTO.

AMC1 DTO.GEN.270(c) Annual internal review and annual activity report

ED Decision 2018/009/R

SUBMISSION OF ANNUAL INTERNAL REVIEW AND ANNUAL ACTIVITY REPORT TO THE COMPETENT AUTHORITY

The annual internal review and the annual activity report for each past calendar year should be submitted to the competent authority within a time frame agreed between the DTO and the competent authority.

GM1 DTO.GEN.270(c) Annual internal review and annual activity report

ED Decision 2018/009/R

SUBMISSION OF ANNUAL INTERNAL REVIEW AND ANNUAL ACTIVITY REPORT TO THE COMPETENT AUTHORITY

It is recommended that the competent authority and the DTO agree on the regular time frames; for example, to agree that the annual internal review and annual activity report for the past calendar year should be submitted during the first quarter of each year.

APPENDIX TO ANNEX VIII

Appendix 1 to Annex VIII (Part-DTO)

Regulation (EU) 2020/359

DECLARATION pursuant to Commission Regulation (EU) No 1178/2011	
☐ Initial declaration ☐ Notification of changes ⁽¹⁾ – DTO reference number:	
1. Declared training organisation (DTO) Name:	
2. Place(s) of business Contact details (address, phone, email) of the DTO's principal place of business:	
3. Personnel Name and contact details (address, phone, email) of the DTO's representative: Name and contact details (address, phone, email) of the DTO's head of training and, if applicable, of the DTO's deputy head(s) of training:	
4. Training scope List of all training provided: List of all training programmes used to provide the training (documents to be attached to this declaration) or, in the case referred to in point DTO.GEN.230(d) of Annex VIII (Part-DTO) to Regulation (EU) No 1178/2011, the reference to all approved training manuals used to provide the training:	
5. Training aircraft and FSTDs List of aircraft used for the training: List of qualified FSTDs used for the training (if applicable, including letter code as indicated on the qualification certificate):	
6. Aerodrome(s) and the operating site(s) Contact details (address, phone, email) of all aerodromes and operating sites used by the DTO to provide the training:	
7. Date of intended commencement of training:	
8. Application for approval of examiner standardisation courses and refresher seminars (if applicable) ☐ The DTO hereby applies for approval of the above-mentioned training programme(s) for examiner courses for sailplanes or balloons in accordance with points DTO.GEN.110(b) and DTO.GEN.230(c) of Annex VIII (Part-DTO) to Regulation (EU) No 1178/2011.	
9. Statement The DTO has developed a safety policy in accordance with Annex VIII (Part-DTO) to Commission Regulation (EU) No 1178/2011 , and in particular with point DTO.GEN.210(a)(1)(ii) thereof, and will apply that policy during all training activities covered by the declaration. The DTO complies and will, during all training activities covered by the declaration, continue to comply with the essential requirements set out in Annex IV to Regulation (EU) 2018/1139 , with the requirements	

¹ In the case of changes, only point 1 and those fields containing changes need to be completed.

of Annex I (Part-FCL) and Annex VIII (Part-DTO) to [Commission Regulation \(EU\) No 1178/2011](#), and with the requirements of Annex III (Part-BFCL) to [Commission Regulation \(EU\) 2018/395](#) and Annex III (Part-SFCL) to [Commission Implementing Regulation \(EU\) 2018/1976](#).

We confirm that all information contained in this declaration, including its annexes (if applicable), is complete and correct.

Name, date and signature of the representative of the DTO

Name, date and signature of the head of training of the DTO

COMMISSION DELEGATED REGULATION (EU) 2020/723

Commission Delegated Regulation (EU) 2020/723

of 4 March 2020

laying down detailed rules with regard to the acceptance of third-country certification of pilots and amending Regulation (EU) No 1178/2011

Regulation (EU) 2020/723

THE EUROPEAN COMMISSION,

Having regard to the Treaty on the Functioning of the European Union,

Having regard to [Regulation \(EU\) 2018/1139](#) of the European Parliament and of the Council of 4 July 2018 on common rules in the field of civil aviation and establishing a European Union Aviation Safety Agency, and amending Regulations [\(EC\) No 2111/2005](#), [\(EC\) No 1008/2008](#), [\(EU\) No 996/2010](#), [\(EU\) No 376/2014](#) and Directives [2014/30/EU](#) and [2014/53/EU](#) of the European Parliament and of the Council, and repealing Regulations [\(EC\) No 552/2004](#) and [\(EC\) No 216/2008](#) of the European Parliament and of the Council and [Council Regulation \(EEC\) No 3922/91](#)¹, and in particular Article 68(3) thereof,

Whereas:

- (1) With the adoption of [Regulation \(EU\) 2018/1139](#) and in particular its Article 68, the Commission is now empowered to adopt delegated acts with regard to the acceptance of certificates and other documentation attesting compliance with civil aviation rules issued in accordance with the laws of a third country, whilst ensuring an equivalent level of safety to that provided for in [Regulation \(EU\) 2018/1139](#).
- (2) The main objective of this Regulation is to bring the current legal framework into line with [Regulation \(EU\) 2018/1139](#) and therefore the content of [Article 8](#), [Annex III](#) and related provisions of [Commission Regulation \(EU\) No 1178/2011](#)² concerning the acceptance of third country certificates should be transferred into a delegated act. Furthermore, those provisions should now also include rules on acceptance of third-country sailplanes and balloons certificates.
- (3) [Regulation \(EU\) No 1178/2011](#) lists conditions for the acceptance of licences from third countries. Under certain conditions, a pilot licence issued by a third country may currently be accepted by Member States or the holder of such a third country licence can obtain credit when applying for a licence in accordance with [Regulation \(EU\) No 1178/2011](#). Such credit is currently determined based on a recommendation from an approved training organisation.
- (4) The rules applicable to training organisations providing training for certain non-commercial pilot licences and ratings have been simplified and the declared training organisation ('DTO') has been introduced, pursuant to [Commission Regulation \(EU\) 2018/1119](#)³. Therefore, the rules concerning the acceptance of licences from third countries should be updated in order to permit

¹ [OJ L 212, 22.8.2018, p. 1.](#)

² Commission Regulation (EU) No 1178/2011 of 3 November 2011 laying down technical requirements and administrative procedures related to civil aviation aircrew pursuant to Regulation (EC) No 216/2008 of the European Parliament and of the Council ([OJ L 311, 25.11.2011, p. 1](#)).

³ Commission Regulation (EU) 2018/1119 of 31 July 2018 amending Regulation (EU) No 1178/2011 as regards declared training organisations ([OJ L 204, 13.8.2018, p. 13](#)).

DTOs to give credit to holders of third country licences who apply for a licence issued under the Union legal framework.

- (5) Article 8 of [Regulation \(EU\) No 1178/2011](#) as well Annex III to that Regulation, currently containing requirements for the acceptance of licences from third countries, should therefore be deleted,

HAS ADOPTED THIS REGULATION:

SECTION 1 — GENERAL PROVISIONS

Article 1 — Scope

Regulation (EU) 2020/723

This Regulation lays down the detailed rules for the conditions for the acceptance of pilot licences and associated ratings, privileges or certificates, as well as associated medical certificates issued in accordance with laws of third countries.

Article 2 — Definitions

Regulation (EU) 2020/723

1. The definitions contained in [Regulation \(EU\) No 1178/2011](#), [Commission Regulation \(EU\) 2018/395](#)¹ and [Commission Implementing Regulation \(EU\) 2018/1976](#)² shall apply for the purposes of this Regulation.
2. In addition, for the purposes of this Regulation, ‘manufacturer flights’ means the flights referred to in Article 6(3) of [Commission Regulation \(EU\) No 965/2012](#)³.

Article 3 — Acceptance of licences from third countries

Regulation (EU) 2020/723

Without prejudice to international agreements concluded between the Union and a third country in accordance with point (a) of Article 68(1) of [Regulation \(EU\) 2018/1139](#), Member States may:

- (a) in accordance with Section 2 of this Regulation accept pilot licences and associated ratings, privileges or certificates, as well as associated medical certificates issued in accordance with laws of third countries;
- (b) in accordance with [Article 3](#) of [Regulation \(EU\) No 1178/2011](#), Article 3a of [Regulation \(EU\) 2018/395](#) or Article 3a of [Implementing Regulation \(EU\) 2018/1976](#), as applicable, issue equivalent licences to applicants who already hold an equivalent licence, rating, privilege or certificate issued in accordance with Annex 1 to the Convention on International Civil Aviation, signed on 7 December 1944 in Chicago (‘the Chicago Convention’) by a third country, provided that those applicants comply with the requirements of Section 3 and taking account of any credit based on a recommendation from an approved training organisation or a declared training organisation;
- (c) give full credits as regards the requirements to undergo a training course prior to undertaking the theoretical knowledge examinations and the skill test to holders of an airline transport pilots licence (‘ATPL’) issued by or on behalf of a third country in accordance with Annex 1 to the Chicago Convention provided that those holders have completed the experience requirements for the issue of an ATPL in the relevant aircraft category as set out in Subpart F of Annex I to [Regulation \(EU\) No 1178/2011](#) and provided that the third country licence contains a valid type rating for the aircraft to be used for the ATPL skill test;

¹ Commission Regulation (EU) 2018/395 of 13 March 2018 laying down detailed rules for the operation of balloons pursuant to Regulation (EC) No 216/2008 of the European Parliament and of the Council ([OJ L 71, 14.3.2018, p. 10](#)).

² Commission Implementing Regulation (EU) 2018/1976 of 14 December 2018 laying down detailed rules for the operation of sailplanes pursuant to Regulation (EU) 2018/1139 of the European Parliament and of the Council ([OJ L 326, 20.12.2018, p. 64](#)).

³ Commission Regulation (EU) No 965/2012 of 5 October 2012 laying down technical requirements and administrative procedures related to air operations pursuant to Regulation (EC) No 216/2008 of the European Parliament and of the Council ([OJ L 296, 25.10.2012, p. 1](#)).

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- (d) issue aeroplane or helicopter type ratings to holders of licences issued in accordance with [Regulation \(EU\) No 1178/2011](#) that comply with the requirements established by a third country for the issue of such ratings; those ratings shall be restricted to aircraft registered in that third country, but this restriction may be removed when the pilot complies with the requirements in [Article 10](#) to this Regulation.

SECTION 2 — VALIDATION OF LICENCES

Article 4 — General provisions for validation of licences

Regulation (EU) 2020/723

1. A competent authority of a Member State may validate a pilot licence issued by a third country in compliance with the requirements of Annex 1 to the Chicago Convention.
2. For the purposes of the provisions set out in this Regulation, the competent authority of the Member State shall be the following:
 - (a) for pilots residing within the territory of the Union — a competent authority of the Member State of a place where a pilot resides or is established;
 - (b) for pilots not residing in the territory of the Union — a competent authority of the Member State where the operator for which they are flying or intend to fly has its principal place of business, or where the aircraft on which they are flying or intend to fly is registered.
3. The validation of a licence shall have a validity period, which does not exceed one year, and its privileges shall only be exercised as long as the licence remains valid.

The competent authority that validated the licence may extend the validity only once and only by a maximum of one year, if during the validity period the pilot has applied for a licence in accordance with Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#) or is undergoing training for the issuance of such a licence. In that last case, the extension shall cover the period of time necessary for the licence to be issued in accordance with Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#).
4. The holders of a licence validated by a Member State shall exercise their privileges in accordance with the requirements stated in Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#).

Article 5 — Pilot licences for commercial air transport and other commercial activities

Regulation (EU) 2020/723

For the validation of pilot licences for commercial air transport and other commercial activities, the holders shall comply with the following requirements, as applicable, for the privileges sought:

- (a) complete, as a skill test, the type or class rating revalidation requirements of Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#), as relevant to the privileges of the licence held;
- (b) demonstrate knowledge of the relevant parts of the operational requirements and Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#);
- (c) demonstrate language proficiency in accordance with Point [FCL.055](#) of Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#);
- (d) hold a valid Class 1 medical certificate, issued in accordance with Annex IV (Part-MED) to [Regulation \(EU\) No 1178/2011](#);
- (e) in the case of aeroplanes, in addition to the requirements in points (a) to (d), comply with the experience requirements set out in table 1 in the [Annex](#) to this Regulation;
- (f) in the case of helicopters, in addition to the requirements in points (a) to (d), comply with the experience requirements set out in table 2 in the [Annex](#) to this Regulation.

Article 6 — Pilot licences for non-commercial activities with an instrument rating

Regulation (EU) 2020/723

For the validation of private pilot licences with an instrument rating, or Commercial Pilot Licences ('CPL') and Airline Transport Pilot Licences ('ATPL') with an instrument rating where the pilot intends only to exercise private pilot privileges, holders shall comply with all of the following requirements:

- (a) complete the skill test for instrument rating and the type or class ratings relevant to the privileges of the licence held, in accordance with [Appendix 7](#) and [Appendix 9](#) of Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#);
- (b) demonstrate knowledge of Air Law, Aeronautical Weather Codes, Flight Planning and Performance (IR) and Human Performance;
- (c) demonstrate language proficiency in accordance with [FCL.055](#) of Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#);
- (d) hold at least a valid Class 2 medical certificate issued in accordance with Annex 1 to the Chicago Convention;
- (e) have a minimum experience of at least 100 hours of instrument flight time as pilot in command ('PIC') in the relevant category of aircraft.

Article 7 — Pilot licences for non-commercial activities without an instrument rating

Regulation (EU) 2020/723

For the validation of private pilot licences, or CPL and ATPL licences without an instrument rating where the pilot intends only to exercise private pilot privileges, holders shall comply with all of the following requirements:

- (a) demonstrate knowledge of Air Law and Human Performance;
- (b) pass the private pilot licence ('PPL') skill test as set out in point [FCL.235](#) of Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#);
- (c) fulfil the relevant requirements of Subpart H of Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#), for the issuance of a type or class rating as relevant to the privileges of the licence held;
- (d) hold at least a Class 2 medical certificate issued in accordance with Annex 1 to the Chicago Convention;
- (e) demonstrate language proficiency in accordance with [FCL.055](#) of Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#);
- (f) have a minimum experience of at least 100 hours as pilot in the relevant category of aircraft.

Article 8 — Validation of pilot licences for specific tasks of limited duration

Regulation (EU) 2020/723

1. Notwithstanding the provisions of the Articles above, in the case of manufacturer flights, a competent authority of a Member State may accept a licence issued in accordance with Annex 1 to the Chicago Convention by a third country for a maximum of 12 months for specific tasks of

limited duration, such as instruction flights for initial entry into service, demonstration, ferry or test flights, provided that the applicant complies with the following requirements:

- (a) holds an appropriate licence and medical certificate and associated ratings or qualifications issued in accordance with Annex 1 to the Chicago Convention;
- (b) is employed, directly or indirectly, by an aircraft manufacturer or by an aviation authority.

In this case, the privileges of the holder shall be limited by the competent authority to performing flight instruction and testing for initial issue of type ratings, the supervision of initial line flying by the operators' pilots, delivery or ferry flights, initial line flying, flight demonstrations or test flights, as appropriate to the tasks foreseen under this paragraph.

2. By way of derogation from Articles 4 to 7, a competent authority of a Member States may, for competition flights or display flights of limited duration, validate a licence issued by a third country allowing the holder to exercise the privileges of a PPL as specified in Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#), a Balloon Pilot Licence (BPL) as specified in Annex III (Part-BFCL) to [Regulation \(EU\) 2018/395](#) or an Sailplane Pilot Licence (SPL) as specified in Annex III (Part-SFCL) to [Implementing Regulation \(EU\) 2018/1976](#), provided that all of the following requirements are complied with:

- (a) prior to the event, the organiser of the competition or display flights provides the competent authority with adequate evidence on how it will ensure that the pilot will be familiarised with the relevant safety information and manage any risk associated with the flights;
- (b) the applicant holds an appropriate licence and medical certificate and associated ratings or qualifications issued in accordance with Annex 1 to the Chicago Convention.

3. By way of derogation from the provisions of Articles [4](#) to [7](#), a competent authority of a Member State may validate a licence which is equivalent to one of those referred to in paragraph 2 and issued in compliance with the requirements of Annex 1 to the Chicago Convention by a third country for a maximum of 28 days per calendar year for specific non-commercial tasks, provided that the applicant complies with all of the following requirements:

- (a) holds an appropriate licence and medical certificate and associated ratings or qualifications issued in accordance with Annex 1 to the Chicago Convention;
- (b) has completed at least one acclimatisation flight with a qualified instructor prior to carrying out the specific tasks of limited duration.

SECTION 3 — CONVERSION OF LICENCES

Article 9 — Conditions for conversion of licences

Regulation (EU) 2020/723

1. The competent authority of a Member State may convert a licence for the relevant aircraft category into a PPL in accordance with Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#) with a single-pilot class or type rating, a BPL in accordance with Annex III (Part-BFCL) to [Regulation \(EU\) 2018/395](#) or an SPL in accordance with Annex III (Part-SFCL) to [Implementing Regulation \(EU\) 2018/1976](#), where the original licence is issued in compliance with the requirements of Annex 1 to the Chicago Convention by a third country and the licence is, alternatively:
 - (a) an equivalent licence to the licences referred to in paragraph 1;
 - (b) a CPL or an ATPL.
2. The holder of the licence to be converted shall comply with the following minimum requirements for the relevant aircraft category:
 - (a) pass a written examination in Air Law and Human Performance;
 - (b) pass the PPL, BPL or SPL skill test, as relevant, in accordance with Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#), Annex III (Part-BFCL) to [Regulation \(EU\) 2018/395](#) or Annex III (Part-SFCL) to [Implementing Regulation \(EU\) 2018/1976](#);
 - (c) fulfil the requirements for the issue of the relevant class or type rating, in accordance with Subpart H;
 - (d) hold a medical certificate, as required and issued in accordance with Annex IV (Part-MED) to [Regulation \(EU\) No 1178/2011](#);
 - (e) demonstrate language proficiency in accordance with [FCL.055](#) of Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#);
 - (f) have completed at least 100 hours of flight time as a pilot.

SECTION 4 — ACCEPTANCE OF CLASS AND TYPE RATINGS

Article 10 — Conditions for acceptance of class and type ratings

Regulation (EU) 2020/723

A valid class or type rating contained in a licence issued by a third country may be inserted in a licence issued in accordance with Annex I (Part-FCL) to [Regulation \(EU\) No 1178/2011](#), provided that the applicant:

- (a) complies with the experience requirements and the prerequisites for the issue of the applicable type or class rating in accordance with Part-FCL;
- (b) passes the relevant skill test for the issue of the applicable type or class rating in accordance with Part-FCL;
- (c) is in current flying practice;
- (d) has no less than:
 - (i) for aeroplane class ratings, 100 hours of flight experience as a pilot in that class;
 - (ii) for aeroplane type ratings, 500 hours of flight experience as a pilot in that type;
 - (iii) for single-engine helicopters with a maximum certificated take-off mass of up to 3 175 kg, 100 hours of flight experience as a pilot in that type;
 - (iv) for all other helicopters, 350 hours of flight experience as a pilot in that type.

Article 11 — Amendments to Regulation (EU) No 1178/2011

Regulation (EU) 2020/723

Regulation (EU) No 1178/2011 is amended as follows:

- (a) Article 8 is deleted;
- (b) Annex III is deleted.

Article 12 — Entry into force and application

Regulation (EU) 2020/723

This Regulation shall enter into force on the twentieth day following that of its publication in the *Official Journal of the European Union*.

This Regulation shall be binding in its entirety and directly applicable in all Member States.

Done at Brussels, 4 March 2020.

For the Commission

The President

Ursula VON DER LEYEN

ANNEX TO DELEGATED REGULATION (EU) 2020/723

Conditions for acceptance of licences issued by or on behalf of third countries

Regulation (EU) 2020/723
Table 1
Experience requirements for aeroplanes

Licence held	Total flying hours experience	Privileges	
(1)	(2)	(3)	
ATPL(A)	> 1 500 hours as PIC on multi-pilot aeroplanes	Commercial air transport in multi-pilot aeroplanes as PIC	(a)
ATPL(A) or CPL(A)/IR ¹	> 1 500 hours as PIC or co-pilot on multi-pilot aeroplanes according to operational requirements	Commercial air transport in multi-pilot aeroplanes as co-pilot	(b)
MPL	> 1 500 hours as co-pilot on multi-pilot aeroplanes according to operational requirements	Commercial air transport in multi-pilot aeroplanes as co-pilot	(ba)
CPL(A)/IR	> 1 000 hours as PIC in commercial air transport since gaining an IR	Commercial air transport in single-pilot aeroplanes as PIC	(c)
CPL(A)/IR	> 1 000 hours as PIC or as co-pilot in single-pilot aeroplanes according to operational requirements	Commercial air transport in single-pilot aeroplanes as co-pilot according to operational requirements	(d)
ATPL(A), CPL(A)/IR, CPL(A)	> 700 hours in aeroplanes, including 200 hours in the activity role for which acceptance is sought, and 50 hours in that role in the last 12 months	Exercise of privileges in aeroplanes in operations other than commercial air transport	(e)
CPL(A)	> 1 500 hours as PIC in commercial air transport including 500 hours on seaplane operations	Commercial air transport in single-pilot aeroplanes as PIC	(f)

Table 2
Experience requirements for helicopters

Licence held	Total flying hours experience	Privileges	
(1)	(2)	(3)	
ATPL(H) valid IR	> 1 000 hours as PIC on multi-pilot helicopters	Commercial air transport in multi-pilot helicopters as PIC in VFR and IFR operations	(a)
ATPL(H) no IR privileges	> 1 000 hours as PIC on multi-pilot helicopters	Commercial air transport in multi-pilot helicopters as PIC in VFR operations	(b)

¹ CPL(A)/IR holders on multi-pilot aeroplanes shall have demonstrated ICAO ATPL(A) level knowledge before acceptance.

Licence held	Total flying hours experience	Privileges	
ATPL(H) valid IR	> 1 000 hours as pilot on multi-pilot helicopters	Commercial air transport in multi-pilot helicopters as co-pilot in VFR and IFR operations	(c)
ATPL(H) no IR privileges	> 1 000 hours as pilot on multi-pilot helicopters	Commercial air transport in multi-pilot helicopters as co-pilot in VFR operations	(d)
CPL(H)/IR ¹	> 1 000 hours as pilot on multi-pilot helicopters	Commercial air transport in multi-pilot helicopters as co-pilot	(e)
CPL(H)/IR	> 1 000 hours as PIC in commercial air transport since gaining an IR	Commercial air transport in single-pilot helicopters as PIC	(f)
ATPL(H) with or without IR privileges, CPL(H)/IR, CPL(H)	> 700 hours in helicopters other than those certificated under CS-27/29 or equivalent, including 200 hours in the activity role for which acceptance is sought, and 50 hours in that role in the last 12 months	Exercise of privileges in helicopters in operations other than commercial air transport	(g)

¹ CPL(H)/IR holders on multi-pilot helicopters shall have demonstrated ICAO ATPL(H) level knowledge before acceptance.